ENERGY TRANSFER

July 1, 2026

Ms. Debbie-Anne A. Reese, Secretary
Federal Energy Regulatory Commission
888 First Street, N.E.
Washington, D.C. 20426

Re: ETC Tiger Pipeline, LLC
FERC Docket No. CP26-__-000
Franklin Farms Project

Dear Ms. Reese:

ETC Tiger Pipeline, LLC ("ETC Tiger") hereby submits for filing with the Federal Energy Regulatory Commission ("Commission" or "FERC") an abbreviated application pursuant to Section 7(c) of the Natural Gas Act, as amended, and Parts 157 and 284 of the Commission's regulations, for a certificate of public convenience and necessity authorizing ETC Tiger to construct, own, operate, and maintain the proposed Franklin Farms ("Project") facilities in Richland Parish, Louisiana. The Project will enable ETC Tiger to deliver up to 1,000,000 million British thermal units per day ("MMBtu/d") off the proposed Franklin Farms Lateral.

This FERC filing is being submitted under the following volumes:

- Volume I – contains **Public Information**
- Volume II – contains **Critical Energy Infrastructure Information ("CEII")**
- Volume III – contains **Privileged Information**

ETC Tiger respectfully requests that only the information submitted as **Public** in Volume I be placed on the internet as "Public Information". In addition, ETC Tiger respectfully requests that the **CEII** information submitted in Volume II identified as "**CUI/Critical Energy Infrastructure Information – Do Not Release**", and the **Privileged** information submitted in Volume III identified as "**CUI/Privileged Information – Do Not Release**" be accorded special treatment pursuant to 18 CFR § 388.112.

ETC Tiger anticipates a shortened Commission review of the Franklin Farms Project because a complete environmental report is attached to the application.

For the reasons stated above, ETC Tiger respectfully requests that the Commission review this certificate application and promptly issue an order by April 30, 2027, so that the Project may be constructed and placed in service by February 1, 2028.

This filing is being submitted electronically to the Commission's eFiling website pursuant to the Commission's Order No. 703, Filing via the Internet Guidelines, issued on November 15, 2007, in FERC

Docket No. RM07-16-000. Any questions or comments regarding this filing should be directed to the undersigned at (713) 989-2605.

Respectfully submitted,

/s/ Blair Lichtenwalter

_____

Blair Lichtenwalter, Sr. Director
Regulatory Affairs

cc: Ms. Pamela Boudreau – FERC Office of Energy Projects
Ms. Gertrude Fernandez Johnson – FERC Office of Energy Projects

ETC Tiger Pipeline, LLC

Volume I

Public Information

UNITED STATES OF AMERICA

BEFORE THE

FEDERAL ENERGY REGULATORY COMMISSION

In the Matter of	§	
	§	Docket No. CP26- _____ -000
ETC Tiger Pipeline, LLC	§	

ABBREVIATED APPLICATION OF ETC TIGER PIPELINE, LLC FOR A CERTIFICATE OF PUBLIC CONVENIENCE AND NECESSITY

Dated: July 1, 2026

UNITED STATES OF AMERICA

BEFORE THE

FEDERAL ENERGY REGULATORY COMMISSION

In the Matter of	§	
	§	Docket No. CP26- _____ -000
ETC Tiger Pipeline, LLC	§	

ABBREVIATED APPLICATION OF ETC TIGER PIPELINE, LLC FOR A CERTIFICATE OF PUBLIC CONVENIENCE AND NECESSITY

ETC Tiger Pipeline, LLC ("ETC Tiger") hereby files this Abbreviated Application ("Application") with the Federal Energy Regulatory Commission ("Commission") pursuant to Section 7(c) of the Natural Gas Act ("NGA"),[1] as amended, and Parts 157 and 284 of the Commission's regulations,[2] for a certificate of public convenience and necessity authorizing ETC Tiger to construct a new pipeline lateral with a capacity of 1,000,000 Dth/day ("Franklin Farms Lateral") and related facilities in Richland Parish, Louisiana, located at approximate milepost 163.87 of its mainline ("Franklin Farms Project" or the "Project"). ETC Tiger and Entergy Louisiana, LLC ("Entergy") entered into a precedent agreement whereby ETC Tiger will provide Entergy firm natural gas transportation service in an amount of 250,000 Dth/day of mainline capacity, and 1,000,000 Dth/day of capacity on the Franklin Farms Lateral, from receipt points on ETC Tiger's system and at points of interconnection between the Franklin Farms Lateral and Gulf Run Transmission, LLC, Gulf South Pipeline Company, LLC, Gulf Crossing Pipeline Company, LLC, and Midcontinent Express Pipeline LLC, to delivery points on the Franklin Farms Lateral. The Project will enable natural gas deliveries to generation facilities that Entergy is constructing to serve a new artificial intelligence data center, as more fully described herein.

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1 15 U.S.C. § 717f(c).

2 18 C.F.R. Parts 157 and 284.

In support of this Application, ETC Tiger respectfully states as follows:

I.
INFORMATION REGARDING THE APPLICANT

The exact legal name of the applicant is ETC Tiger Pipeline, LLC. ETC Tiger is a limited liability company that is organized under the laws of the state of Delaware, with its principal offices located at 1300 Main Street, Houston, Texas, 77002. ETC Tiger is a wholly owned subsidiary of Energy Transfer Interstate Holdings LLC, which is wholly owned by Energy Transfer LP, a Delaware limited partnership traded on the New York Stock Exchange under the ticker symbol ET.

ETC Tiger is a "natural gas company" as that term is defined in Section 2 of the NGA and is engaged in the transportation of natural gas in interstate commerce subject to the Commission's jurisdiction. ETC Tiger provides open-access transportation services pursuant to Part 284 of the Commission's regulations, and under its blanket certificates issued in Docket No. CP09-460-000.[3] ETC Tiger owns and operates an interstate natural gas transmission system that provides access to natural gas supplies primarily from the Haynesville Shale in East Texas and Western Louisiana. ETC Tiger serves markets off the east end of its system in Louisiana, including several large pipeline interconnects in the Perryville, Louisiana area.

[3] *ETC Tiger Pipeline, LLC*, 131 FERC ¶ 61,010 (2010).

II.
CORRESPONDENCE AND COMMUNICATIONS

The names, titles, mailing addresses, telephone numbers and email addresses of those persons to whom all communications concerning this Application are to be directed:

Mr. Blair Lichtenwalter[4] [5]
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Houston, Texas 77002
(713) 989-2605
Blair.Lichtenwalter@energytransfer.com

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III.
DESCRIPTION OF THE PROJECT

ETC Tiger and Entergy entered into a precedent agreement ("Precedent Agreement") pursuant to which, subject to the satisfaction of certain conditions precedent, ETC Tiger and Entergy will execute a transportation agreement pursuant to Rate Schedule FTS ("FTS Agreement") whereby Entergy will take quantities of firm transportation service on ETC Tiger. Under the FTS Agreement, Entergy will receive firm transportation service in an amount of 250,000 Dth/day of mainline capacity, and 1,000,000 Dth/day of lateral capacity on the newly constructed Franklin Farms Lateral, for a primary term of twenty years, with primary receipt points at Houston Pipe Line Carthage and Deadwood, in Panola County, Texas, and

[4] Designated to receive service pursuant to Rule 2010 of the Commission's Rules of Practice and Procedure. 18 C.F.R. § 385.2010. ETC Tiger respectfully requests that the Commission waive Rule 385.203(b)(3) in order to allow ETC Tiger to include each of the designated representatives on the official service list. *Id.* at § 385.203(b)(3).

[5] Designated as the responsible ETC Tiger official under Rule 154.7(a)(2) of the Commission's regulations. *Id.* at § 154.7(a)(2).

receipt points along the Franklin Farms Lateral, with primary delivery points located on the Franklin Farms Lateral.[6] Entergy's FTS Agreement represents the capacity of the Project. The FTS Agreement will become effective the later of February 1, 2028, or the first day of the calendar month following the day on which ETC Tiger is able to provide firm transportation service from the primary points of receipt to the primary points of delivery.

As the sole shipper for the Project, Entergy will use the Project facilities to supply natural gas to its power generation it is constructing to specifically provide power for the Richland Parish Data Center, as well as be connected to the larger electric grid to possibly supply power to other customers. The Richland Parish Data Center is Meta's $27 billion artificial intelligence data center that is also referred to as the Hyperion project.[7] Hyperion is Meta's largest multi-gigawatt AI training cluster that will employ a peak construction force of 5,000 people and, once completed, Hyperion will have 500 operational jobs.

Meta's Hyperion project is critical to the Trump Administration's and the Commission's goal of securing the nation's "energy dominance," which "is crucial to our economic and national security," as well as ensuring "that we win the artificial intelligence data race . . . so that American data does not go abroad."[8] In testimony before Congress, Chairman Swett characterized the current situation as America being "in a global energy arms race created by the rise of technology, its tremendous load growth, and our push to onshore manufacturing and jobs" and "I want the United States to lead that race," taking this "moment to cement America's energy leadership."[9] Under that leadership, Chairman Swett announced that one of the Commission's top priorities "is to ensure that our country can connect and power data centers as quickly and

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[6] Pursuant to a one-time option valid until June 30, 2026, Entergy had the option to increase its mainline firm transportation quantity to 500,000 Dth/day. Such option was not exercised.

[7] The largest Meta data center yet brings big impact to Louisiana (*available at* https://datacenters.atmeta.com/richland-parish-data-center/).

[8] Chairman Laura V. Swett, Opening Remarks at the Commission's November 2025 Meeting (Nov. 20, 2025); Winning the Race, America's AI Action Plan (July 2025) (*available at* https://www.whitehouse.gov/wp-content/uploads/2025/07/Americas-AI-Action-Plan.pdf) (recognizing the construction of "American AI Infrastructure" as the second pillar of America's AI Action Plan.

[9] Written Testimony of Laura V. Swett Chairman, Federal Energy Regulatory Commission, Before the U.S. House of Representatives, Committee on Energy and Commerce, Subcommittee on Energy, at 1-2 (Feb. 3, 2026) https://www.ferc.gov/news-events/news/testimony-laura-v-swett-chairman-federal-energy-regulatory-commission-us-house.

as durably as possible."[10] Chairman Swett testified before Congress that she is "prioritizing the ability of Americans to build, connect and operate data centers in our country . . . protecting our data and sheltering it from our foreign adversaries, which is paramount to maintaining our national and economic security [with] the benefit of more American jobs" building such data centers.[11]

The Trump Administration has championed the need to win the AI race by building out data centers. President Trump declared in his *Accelerating Federal Permitting of Data Center Infrastructure* Executive Order that his "Administration has inaugurated a golden age for American manufacturing and technological dominance" and that "[w]e will pursue bold, large-scale industrial plans to vault the United States further into the lead on critical manufacturing processes and technologies that are essential to national security, economic prosperity, and scientific leadership," including "artificial intelligence (AI) data centers and [energy] infrastructure that powers them," such as natural gas pipelines and laterals.[12] President Trump provided more details in "Winning the Race, America's AI Action Plan", explaining that, like the previous space race, America and its allies will win the race to achieve global dominance in AI because it is imperative that we do so in order to "set global AI standards and reap broad economic and military benefits."[13] To win the AI race, America needs to "build and maintain vast AI infrastructure and the energy to power it" and "[t]o do that, [the Administration] will continue to reject radical climate dogma and bureaucratic red tape", which will allow the nation to "Build, Baby, Build!"[14] The expansion of ETC Tiger's natural gas infrastructure set out in this Application will allow natural gas to be delivered to Entergy's new electric generator facilities that will generate electricity for the Hyperion artificial intelligence data center in Richland Parish, Louisiana. ETC Tiger's Franklin Farms Project is precisely the type of critical energy infrastructure that the Commission and the Administration have identified as essential to maintaining America's competitive edge in the global AI and data center race.

[10] *See, e.g.*, Chairman Laura V. Swett, Opening Remarks at FERC's November 2025 Open Meeting (Nov. 20, 2025).
[11] Testimony of Laura V. Swett Chairman, Federal Energy Regulatory Commission, Before the U.S. House of Representatives, Committee on Energy and Commerce, Subcommittee on Energy (Feb. 3, 2026).
[12] Exec. Order No. 14,318, 90 Fed. Reg. 35,385 (2025).
[13] Winning the Race, America's AI Action Plan at 1 (July 2025).
[14] *Id.*

The Franklin Farms Lateral will run in a generally south-north direction originating at a point of receipt on the ETC Tiger mainline at milepost 163.87 located south of Interstate I-20 in Richland Parish, Louisiana. The 36-inch Franklin Farms Lateral is approximately 13.24 miles in length and the lateral and all Project related facilities are on private land. From its origin at the ETC Tiger mainline and interconnects with Gulf Run Pipeline and Gulf South Pipeline, the lateral proceeds north, crossing underneath I-20, and continues north to interconnects with Midcontinent Express Pipeline and Gulf Crossing Pipeline, and then to delivery interconnects that will provide natural gas to Entergy's generating facilities CCCT 1&2 (Franklin Farms), CCCT 3&4 (Bee Bayou) and CCCT 5&6 (Trio Farm).[15] Additionally, ETC Tiger will construct launcher and receiver facilities at or near measurement locations to ensure the necessary operation and maintenance of the lateral. The Project does not include the construction of any compression facilities. To ensure receipt and deliveries of the transportation quantities at the required pressures, ETC Tiger is proposing to construct, modify, install, own, and operate the following facilities:

- Franklin Farms Lateral consisting of approximately 13.24 miles of 36-inch pipeline with a capacity of 1,000,000 Dth/d.
- Franklin Farms Delivery Meter with a capacity of up to 320,000 Dth/d. ETC Tiger will install, construct and operate the following at this meter site: two new 10-inch meter runs, two new 10-inch control valve runs, one new RTU/GC building, two new filter separators with one tank, one 36-inch receiver barrel, and approximately 60 feet of 20-inch interconnect piping from the new Franklin Farms Lateral.
- Bee Bayou Delivery Meter with a capacity of up to 320,000 Dth/d. ETC Tiger will install, construct and operate the following at this meter site: two new 10-inch meter runs, two new 10-inch control valve runs, one new RTU/GC building, two new filter separators with one tank, one 36-inch mainline valve, and approximately 47 feet of 20-inch interconnect piping from the new Franklin Farms Lateral.
- Trio Farm Delivery Meter with a capacity of up to 320,000 Dth/d. ETC Tiger will install, construct and operate the following at this meter site: two new 10-inch meter runs, two new 10-inch control valve runs, one new RTU/GC building, two new filter separators with one tank, one 20-inch receiver barrel, and approximately 50 feet of 20-inch interconnect piping from the new FF Trio Farm Lateral.
- 1.75 miles of 20-inch FF Trio Farm Lateral to provide service to the Trio Farm Delivery Meter.
- MEP Receipt Meter with a capacity of up to 500,000 Dth/d. ETC Tiger will install, construct and operate the following at this meter site: two new 12-inch meter runs, two new 10-inch control valve

[15] CCCT refers to the Combined Cycle Combustion Turbines being installed by Entergy.

runs, two new RTU/GC building, one new filter separator with tank, one new 36-inch mainline valve, approximately 262 feet of 24-inch interconnect piping from the existing MEP pipeline, and approximately 30 feet of 24-inch interconnect piping to the new Franklin Farms Lateral. Midcontinent Express Pipeline will install, construct, and operate a hot tap and interconnect piping pursuant to its blanket certificate.

- Gulf South Pipeline Receipt Meter with a capacity of up to 500,000 Dth/d. ETC Tiger will install, construct and operate the following at this meter site: two new 12-inch meter runs, two new 10-inch control valve runs, two new RTU/GC building, one new filter separator with tank, approximately 405 feet of 24-inch interconnect piping from the existing Gulf South pipeline, and approximately 160 feet of 24-inch interconnect piping to the new Franklin Farms Lateral. Gulf South Pipeline will install, construct, and operate a hot tap and interconnect piping pursuant to its blanket certificate.

- Gulf Crossing Pipeline receipt meter with a capacity of up to 500,000 Dth/day. ETC Tiger will install, construct and operate the following at this meter site: two new 12-inch meter runs, two new 10-inch control valve runs, two new RTU/GC building, one new filter separator with tank, approximately 80 feet of 24-inch interconnect piping from the existing Gulf Crossing Pipeline, and approximately 105 feet of 24-inch interconnect piping to the new Franklin Farms Lateral. Gulf Crossing Pipeline will install, construct, and operate a hot tap and interconnect piping pursuant to its blanket certificate.

- Gulf Run Pipeline Bi-directional Meter with a capacity of up to 500,000 Dth/d. ETC Tiger will install, construct and operate the following at this meter site: one new 36-inch launcher barrel, and three new 10-inch control valve runs. Gulf Run Pipeline will install, construct and operate the following facilities pursuant to its blanket certificate: two new 12-inch meter runs, two new 10-inch control valve runs, one new 20-inch flow switching skid, one new RTU/GC building, one new filter separator with tank, approximately 655 feet of 20-inch interconnect piping from the existing Gulf Run pipeline, and approximately 235 feet of 24-inch interconnect piping to the new Franklin Farms Lateral.

- Three mainline valve facilities at multiple locations to isolate gas flow from either direction on the Franklin Farms Lateral allowing for operational flexibility to the extent a portion of the Franklin Farms Lateral necessitates isolation.

- FF Interconnect Site between the Franklin Farms Lateral and the FF Trio Farm Lateral. ETC Tiger will install, construct and operate the following at this interconnect site: one new 20-inch launcher barrel, one new 36-inch mainline valve, and one new RTU rack.

- Construct, install, and modify other appurtenant facilities as needed.

Construction of the Project is expected to be completed by February 1, 2028, following receipt of applicable permits and clearances. The Project facilities will enable ETC Tiger to meet its contractual obligations to Entergy and will have no impact on service to ETC Tiger's existing customers.

IV.
PRECEDENT AGREEMENT

ETC Tiger and Entergy entered into a Precedent Agreement, pursuant to which ETC Tiger and Entergy will, subject to the satisfaction of certain conditions precedent, execute the FTS Agreement whereby Entergy will take quantities of firm transportation service from ETC Tiger. A copy of the Precedent Agreement, which has been marked as "*CUI/Privileged Information – Do Not Release*" is included in Exhibit I hereto, and ETC Tiger respectfully requests that the Precedent Agreement be accorded privileged and confidential treatment pursuant to 18 CFR § 388.112. The FTS Agreement will have a primary term of twenty years commencing on the in-service date of the Project, which is anticipated to be February 1, 2028, and will have fixed negotiated rates. Entergy is not an affiliate of ETC Tiger and the FTS Agreement represents the capacity of the Project.

The Commission permits pipelines to include non-conforming provisions in their service agreements, subject to the Commission's review and approval. Consistent with the Commission's Policy Statement on Creditworthiness for Interstate Natural Gas Pipelines and Order Withdrawing Rulemaking Proceeding,[16] the precedent agreement filed herein includes a credit agreement that requires the project shipper to meet certain objective creditworthiness standards, or to provide ETC Tiger with appropriate credit support, including no more than 18 months of reservation charges. The service agreements attached to the precedent agreement also contain a provision permitting termination of the service agreements in the event that the Project enters into partial in-service but then does not subsequently enter into full in-service by an outside date. Such a termination provision makes it possible to have interim, partial in-service and the ability to terminate the service agreements if full in-service is not achieved by a certain date.

[16] *Policy Statement on Creditworthiness Issues for Interstate Natural Gas Pipelines and Order Withdrawing Rulemaking Proceeding* FERC Stats. & Regs., Regs. Preambles 2001-2005 ¶ 31,191, at PP 17-20 (2005) (permitting larger collateral requirements from initial shippers in cases of new construction and requiring issues relating to such collateral to be determined in precedent agreements).

These non-conforming provisions are permissible because they do not present a substantial risk of undue discrimination. The Commission has consistently pre-approved non-conforming provisions in certificate proceedings where the provisions are necessary to address the unique circumstances involved with the construction of new infrastructure and do not affect the operational conditions of providing service.[17] Therefore, ETC Tiger respectfully requests that the Commission review and approve this non-conforming provision in this certificate proceeding.

V.
RATES, COST AND FINANCING

ETC Tiger will charge customers for transportation service using the Project facilities under its existing, currently effective rates pursuant to its FERC NGA Gas Tariff ("Tariff"). ETC Tiger is not proposing at this time any new or revised rates or fuel charges related to this Project. Under the Commission's Alternative Rate Policy Statement, if a pipeline enters into negotiated rate agreements, the pipeline must provide recourse rates as an alternative.[18] Details of the negotiated rate authority under which shippers make these elections are contained in ETC Tiger's Rate Schedule FTS, Section 5, and the General Terms and Conditions, Section 30 that explains the discounting provisions applicable to ETC Tiger's maximum recourse rates. ETC Tiger estimates that the Project facilities will cost approximately $137,388,622, as shown on Exhibit K.

VI.
OPEN SEASON

ETC Tiger held a binding Open Season effective October 29, 2025, 9:00 AM Central Time, through October 31, 2025, 4:00 PM Central Time, pursuant to Section 2 of the General Terms and Conditions of its Tariff [with respect to the new capacity on the Franklin Farms Lateral and the unsubscribed capacity on the mainline]. In particular, ETC Tiger solicited whether any parties had an interest in 250,000 MMBtu of firm

[17] *See, e.g., Natural Gas Pipeline Co. of America, LLC*, 189 FERC ¶ 61,131, at P 27 (2024) (Commission approved nonconforming provision requiring project shippers to provide 24 months of credit support)."

[18] *Alternatives to Traditional Cost-of-Service Ratemaking for Natural Gas Pipelines and Regulation of Negotiated Transportation Services of Natural Gas Pipelines*, 74 FERC ¶ 61,076 (1996), *reh'g and clarification denied*, 75 FERC ¶ 61,024 (1996).

transportation capacity for receipts from "Unsubscribed Capacity" to deliveries to "Unsubscribed Capacity". No parties submitted requests for capacity during the Open Season, and capacity was awarded to Entergy.

VII.
CERTIFICATE POLICY STATEMENT AND PUBLIC CONVENIENCE AND NECESSITY

The Commission Policy Statement[19] on certificating new pipeline construction establishes criteria for determining whether there is a need for a proposed project and whether the proposed project will serve the public interest. The Commission Policy Statement explains that in deciding to authorize the construction of major new pipeline facilities, the Commission balances the public benefits against the potential adverse consequences. The Commission gives appropriate consideration to the enhancement of competitive transportation alternatives, the possibility of overbuilding, subsidization by existing customers, the applicant's responsibility for unsubscribed capacity, the avoidance of unnecessary disruptions of the environment, and the unneeded exercise of eminent domain when evaluating new pipeline construction.

Under the Commission Policy Statement, the threshold requirement for existing pipelines proposing new projects is that the pipeline must be prepared to financially support the project without relying on subsidization from existing customers. The next step is to determine whether the applicant has made efforts to minimize any adverse effect the project might have on the applicant's existing customers, existing pipelines in the market and their captive customers, or landowners and communities affected by the route of the new pipeline. If residual adverse effects on these interest groups are identified, after efforts have been made to minimize them, the Commission evaluates the project by balancing the evidence of public benefits to be achieved against the residual adverse effects. As discussed below, the Project meets each of the Commission's objectives and criteria established in the Commission Policy Statement.

[19] *Certification of New Interstate Natural Gas Pipeline Facilities*, 88 FERC ¶ 61,227 (1999); *Order Clarifying Statement of Policy*, 90 FERC ¶ 61,128 (2000); *Order Further Clarifying Statement of Policy*, 92 FERC ¶ 61,094 (2000).

A. Impact on Existing Shippers – No Subsidization

The Project will consist of a new pipeline lateral on the ETC Tiger system and the financial terms of the Project are outlined in the FTS Agreement between ETC Tiger and Entergy. This Project will have no financial impact to other existing customers, as shown on Exhibit N. ETC Tiger's Project meets the Commission's long-standing requirement of a demonstration of market support. Entergy has executed a Precedent Agreement pursuant to which, subject to the satisfaction of certain conditions precedent set forth in the Precedent Agreement, Entergy will execute the FTS Agreement for firm transportation service for quantities of 250,000 Dth/d of mainline capacity and 1,000,000 Dth/d of Franklin Farms Lateral capacity from the Effective Date through the primary term of twenty years. As noted above, Entergy is not an affiliate of ETC Tiger and the FTS Agreement represents the capacity of the Project.

ETC Tiger proposes utilizing a negotiated rate with service provided under Rate Schedule FTS for cost recovery. ETC Tiger is seeking rolled-in rate treatment. ETC Tiger also is proposing to utilize its current system fuel rates set out in its Tariff. The Project satisfies the economic threshold requirement as it will not rely on subsidies from ETC Tiger's existing shippers. Therefore, no adverse impacts on existing shippers will take place from the construction or from the costs of operation and maintenance of the Project.

B. No Adverse Impact on Existing Pipelines and Their Captive Customers

The Project will provide shippers access to expanded capacity for receipts from ETC Tiger's west end near the Texas-Louisiana border to deliveries on the proposed lateral located in Richland Parish, Louisiana. This Project will have no adverse impact to any existing pipelines and their captive customers because: (1) no existing service by any other pipeline system will be displaced; and (2) no other nearby existing pipeline system has the available capacity to transport the shipper requirements along this path. More generally, no adverse impact on competing pipelines and their captive customers will result because: (1) the ETC Tiger pipeline is an open-access pipeline providing nondiscriminatory service in a competitive market; and (2) construction and operation of the Project will serve to further enhance competition in the market by providing additional competitive service options.

C. Impact on Landowners and Communities Has Been Minimized

ETC Tiger's construction activities for the Project will occur within new easements. ETC Tiger is in the process of obtaining easements from landowners for the Project facilities. ETC Tiger does not anticipate the use of eminent domain for this Project.

D. Benefits Associated with the Project Outweigh Any Adverse Impacts

There are several public benefits associated with the Project. First, it will provide for additional natural gas infrastructure in Richland Parish, Louisiana, that will contribute to stimulating the Louisiana economy. Secondly, it will provide much needed markets for natural gas produced in East Texas and Western Louisiana. The public benefits of the Project far outweigh any potential minor or temporary adverse impacts.

E. The Project Is Required by the Public Convenience and Necessity

In determining whether a proposed project is required by the public convenience and necessity, the Commission considers whether the proposal meets the criteria set forth in the Commission Policy Statement. As discussed above, the Project is consistent with the objectives and criteria of the Commission Policy Statement. Furthermore, the Project will provide benefits to all sectors of the natural gas market by providing shippers with additional access to markets resulting in enhanced market competition, reduced price volatility and lower prices.

For these reasons, and consistent with the criteria set forth in the Commission Policy Statement, ETC Tiger respectfully submits that the Project is in the present and future public convenience and necessity, and that the authorizations requested herein should be granted promptly.

VIII.
STAKEHOLDER AND LANDOWNER OUTREACH

ETC Tiger will comply with the landowner notification requirements under Section 157.6(d) of the Commission's regulations[20] and shall make a good faith effort to notify all affected landowners, towns, communities, and local, state and federal agencies involved in the Project. A list of the affected

[20] 18 C.F.R. § 157.6(d).

landowner(s) is included with the Environmental Report under Appendix E and is designated as **CUI/Privileged Information – Do Not Release**. Affected landowners include owners that are directly crossed-by, and owners of land that abut the Project's right-of-way. ETC Tiger will also serve all affected landowners and other parties specified in the Commission's Order Nos. 609 and 609-A, with the required landowner notification. Such notice will be served by United States Postal Service first class mail, within three business days following the date the Commission issues a Notice of Application. ETC Tiger will publish the information required in Section 157.6(d) twice in a daily or weekly newspaper of general circulation in the areas where the Project is located. It is ETC Tiger's intent to work with all affected stakeholders and to address their concerns to minimize environmental impacts caused by the Project.

Additionally, a complete (PUBLIC) copy of the application will be placed in the following public library:

Richland Parish Library
Richland Parish, Louisiana
1410 Louisa Street
Rayville, Louisiana 71269
318-728-4806

IX.
ENVIRONMENTAL IMPACT AND COMPLIANCE

Pursuant to Sections 157.14 and 380.12 of the Commission's regulations, an analysis of the environmental impacts for the Project are included in the Environmental Report under Exhibit F-1. Included in the Environmental Report, among other things, is Resource Report 1, Table 1.7.1, Federal and State Environmental Permits, Approvals, and Consultations, which provides a current status of Federal and state environmental clearances.

The Project will incorporate proven construction techniques and mitigation procedures and will result in minimal impact on the quality of human health, the environment, landowners, or surrounding communities. ETC Tiger proposes to construct the Project in accordance with the Commission's *Upland Erosion Control, Revegetation and Maintenance Plan (2013) and Wetland and Waterbody Construction and Mitigation Procedures (2013)*. ETC Tiger has concluded from its environmental review that the Project

construction will not individually or cumulatively have a significant effect on the quality of human health, the environment, or landowners. ETC Tiger agrees to comply with all mitigation requirements stated in the environmental clearances from Federal, state, and local agencies for the Project.

X.
CERTIFICATIONS

Pursuant to the Natural Gas Pipeline Safety Act of 1968,[21] and Section 157.14(a)(10)(vi) of the Commission's regulations, ETC Tiger certifies that the facilities proposed herein will be designed, constructed, tested, operated, replaced, and maintained to conform with or exceed the requirements of Title 49 Part 192 of the Code of Federal Regulations ("49 CFR Part 192"), or any superseding Federal or state safety code applicable to natural gas transmission pipelines.[22] These regulations are intended to ensure adequate protection for the public and to prevent natural gas facility accidents and failures. 49 CFR Part 192 specifies material selection and qualification, minimum design requirements, and protection from internal, external, and atmospheric corrosion.

[21] Pub. L. No. 90-481, 82 Stat. 720 (1968) (codified as amended at 49 U.S.C. §§ 60101–60140).

[22] The United States Department of Transportation ("USDOT") has exclusive authority to promulgate safety and design standards for pipelines and transportation facilities under the Natural Gas Pipeline Safety Act. The USDOT pipeline standards are published in 49 CFR Parts 190–199.

XI.

DESCRIPTION OF EXHIBITS

This is an Abbreviated Application pursuant to Part 157 of the Commission's regulations.[23] The following exhibits are attached, incorporated by reference, or omitted for the reasons indicated. To the extent any required exhibits have been omitted, ETC Tiger requests that the Commission treat the omitted material as inapplicable or otherwise unnecessary to fully disclose the nature of the Project as proposed herein.

NOTICE **NOTICE OF APPLICATION**

Submitted herein is a notice prepared in conformity with Sections 2.1 and 157.6(b)(7) of the Commission's regulations,[24] suitable for publication in the *Federal Register*.

EXHIBIT A **ARTICLES OF INCORPORATION AND BYLAWS OR OTHER SIMILAR DOCUMENTS**

Omitted. ETC Tiger filed Exhibit A in Docket No. CP09-460-000, filed August 31, 2009, which is incorporated herein by this reference.

EXHIBIT B **STATE AUTHORIZATION**

Omitted. ETC Tiger filed Exhibit B in Docket No. CP09-460-000, filed August 31, 2009, which is incorporated herein by this reference.

EXHIBIT C **COMPANY OFFICIALS**

Submitted herein.

EXHIBIT D **SUBSIDIARIES AND AFFILIATIONS**

ETC Tiger is 100% owned by Energy Transfer Interstate Holdings LLC. Energy Transfer Interstate Holdings LLC is 100% owned by Energy Transfer LP, a Delaware limited partnership traded on the New York Stock Exchange under the ticker symbol ET.

EXHIBIT E **OTHER PENDING APPLICATIONS AND FILINGS**

There are no pending applications or filings at the Commission that would affect the outcome of this proceeding.

---

23 18 C.F.R. at Part 157.

24 *Id.* at §§ 2.1, 157.6(b)(7).

EXHIBIT F **LOCATION OF FACILITIES**

Submitted herein. In addition, under Exhibit F-1, Environmental Report under Appendix A are Project Location Maps, (USGS Quad Map, and Aerial) of the proposed sites.

EXHIBIT F-1 **ENVIRONMENTAL REPORT**

Submitted herein is the Environmental Report for the Project. The information submitted under Appendix B is designated as **CUI/Critical Energy Infrastructure Information – Do Not Release** and placed under Volume II. The information submitted under Appendix E is designated as **CUI/Privileged Information – Do Not Release** and placed under Volume III.

EXHIBIT G **FLOW DIAGRAM SHOWING DAILY DESIGN CAPACITY, AND REFLECTING OPERATION WITH AND WITHOUT PROPOSED FACILITIES ADDED**

Submitted herein is a Flow Diagram under Exhibit G. The Flow Diagram, and flow modeling files are designated as **CUI/Critical Energy Infrastructure Information – Do Not Release** and are placed under Volume II.

EXHIBIT G-I **FLOW DIAGRAM REFLECTING MAXIMUM CAPABILITIES**

Exhibit G-I is reflected in Exhibit G designated as **CUI/Critical Energy Infrastructure Information – Do not Release** and placed under Volume II.

EXHIBIT G-II **FLOW DIAGRAM DATA**

Exhibit G-II is reflected in Exhibit G designated as **CUI/Critical Energy Infrastructure Information – Do not Release** and placed under Volume II.

EXHIBIT H **TOTAL GAS SUPPLY DATA**

Omitted. Not Applicable.

EXHIBIT I **MARKET DATA**

Submitted herewith under separate cover. The Precedent Agreement has been designated as Privileged Information and is contained in Volume III.

EXHIBIT J **FEDERAL AUTHORIZATIONS**

Submitted herein under Exhibit F-1, Environmental Report is Table 1.7-1, Federal and State Environmental Permits, Approvals, and Consultations, which discusses the current status of Federal and State authorizations.

EXHIBIT K **COST OF FACILITIES**

Submitted herein.

EXHIBIT L **FINANCING**

Omitted. ETC Tiger will finance the construction with funds generated internally, through borrowings, bond offerings, and/or equity offerings.

EXHIBIT M **CONSTRUCTION, OPERATION, AND MANAGEMENT**

Omitted. ETC Tiger and/or independent contractors will accomplish the construction. The employees of ETC Tiger in the ordinary course of business will carry out operation and maintenance of the Project facilities.

EXHIBIT N **REVENUES, EXPENSES, AND INCOME**

Submitted herein.

EXHIBIT O **DEPRECIATION AND DEPLETION**

Omitted. This application does not propose any changes to ETC Tiger's depreciation and depletion rates as approved in the Commission's April 7, 2010 Order Issuing Certificates.[25]

EXHIBIT P **TARIFF**

Omitted. This application does not propose any changes to ETC Tiger's currently effective Tariff, including Currently Effective Rates, Rate Schedules, or General Terms and Conditions.

EXHIBIT Z-1 **FORM OF PROTECTIVE AGREEMENT**

Submitted herein is the Form of Protective Agreement governing the use of all Protected Materials produced by, or on behalf of, ETC Tiger to Participants hereunder.

XII.
REQUEST FOR SHORTENED PROCEDURES

ETC Tiger respectfully requests that the Commission grant this Application based upon this submission and without a trial-type evidentiary hearing in accordance with Rules 801 and 802 of the Commission's Rules of Practice and Procedure.[26] In anticipation that this proceeding will be non-contested, ETC Tiger requests that the intermediate decision procedure be omitted, and waives oral hearing and opportunity for filing exceptions to the decision of the Commission.

[25] *ETC Tiger Pipeline, LLC*, 131 FERC ¶ 61,010 (2010).
[26] 18 CFR §§ 385.801, 385.802.

XIII.

CONCLUSION

For the foregoing reasons, ETC Tiger respectfully requests that the Commission grant the instant Application for issuance of a certificate of public convenience and necessity authorizing ETC Tiger to construct, own, and operate the Franklin Farms Project in Richland Parish, Louisiana in order for ETC Tiger to provide natural gas transportation service to generation facilities that Entergy is constructing to serve a new artificial intelligence data center. ETC Tiger respectfully requests that these authorizations be promptly granted on or before April 30, 2027, so that the facilities may be constructed and placed in service by February 1, 2028.

Respectfully submitted,

/s/ Blair Lichtenwalter

_____

Blair Lichtenwalter,
Senior Director, Certificates
ETC Tiger Pipeline, LLC

VERIFICATION STATEMENT OF
ETC TIGER PIPELINE, LLC

Blair Lichtenwalter, Senior Director of Regulatory Affairs, for ETC Tiger Pipeline, LLC hereby certifies in accordance with 18 C.F.R. § 385.2005(a) that I have read the above and foregoing Application of ETC Tiger Pipeline, LLC, and know its contents; the contents are true as stated, and correct to the best of my knowledge, information, and belief; and that I possess full power and authority to sign the Application on behalf of ETC Tiger Pipeline, LLC.

Name: _____
Blair Lichtenwalter, Senior Director

Subscribed and sworn to before me this 1st day of July 2026.

Name: _____

Title: Notary Public in the State of Texas

My Commission Expires: June 24, 2030

BETTY J PEUGH
Notary ID #8423226
My Commission Expires
June 24, 2030

NOTICE OF APPLICATION

UNITED STATE OF AMERICA
Before the
FEDERAL ENERGY REGULATORY COMMISSION

In the Matter of	§	
	§	Docket No. CP26- _____ -000
ETC Tiger Pipeline, LLC	§	

NOTICE OF APPLICATION

Take notice that on July 1, 2026, ETC Tiger Pipeline, LLC (ETC Tiger) located at 1300 Main Street, Houston, TX 77002, filed an abbreviated application in Docket No. CP26-___-000 pursuant to Section 7(c) of the Natural Gas Act (NGA), and Parts 157 and 284 of the Commission's regulations requesting the issuance of a certificate of public convenience and necessity to construct a new pipeline lateral in Richland Parish, Louisiana, located at approximate milepost 163.87 of its mainline, in order for ETC Tiger to provide firm transportation service pursuant to ETC Tiger's Rate Schedule FTS in an amount of 1,000,000 Dth/day.

ETC Tiger's filing is available for review at the Commission in the Public Reference Room or may be viewed on the Commission's website at http://www.ferc.gov using the "eLibrary" link. Enter the docket number excluding the last three digits in the docket number filed to access the document. For assistance, please contact FERC Online Support at FERCOnlineSupport@ferc.gov or toll free at (866)208-3676, or for TTY, contact (202) 502-8659.

Any questions regarding the application should be directed to Blair Lichtenwalter, Senior Director, Regulatory Affairs, ETC Tiger Pipeline, LLC, 1300 Main Street, Houston, Texas 77002, at (713) 989-2605.

Pursuant to section 157.9 of the Commission's rules, 18 CFR 157.9, within 90 days of this Notice the Commission staff will either: complete its environmental assessment (EA) and place it into the Commission's public record (eLibrary) for this proceeding; or issue a Notice of Schedule for Environmental Review. If a Notice of Schedule for Environmental Review is issued, it will indicate, among other milestones, the anticipated date for the Commission staff's issuance of the EA for this proposal. The filing of the EA in the Commission's public record for this proceeding or the issuance of a Notice of Schedule will serve to notify federal and state agencies of the timing for the completion of all necessary reviews, and the subsequent need to complete all federal authorizations within 90 days of the date of issuance of the Commission staff's EA.

There are two ways to become involved in the Commission's review of this project. First, any person wishing to obtain legal status by becoming a party to the proceedings for this project should, on or before the comment date stated below, file with the Federal Energy Regulatory Commission, 888 First Street, NE, Washington, DC 20426, a motion to intervene in accordance with the requirements of the Commission's Rules of Practice and Procedure (18 CFR 385.214 or 385.211) and the Regulations under the NGA (18 CFR 157.10). A person obtaining party status will be placed on the service list maintained by the Secretary of the Commission and will receive copies of all documents filed by the applicant and by all other parties. A party must submit 5

copies of filings made with the Commission and must mail a copy to the applicant and to every other party in the proceeding. Only parties to the proceeding can ask for court review of Commission orders in the proceeding.

However, a person does not have to intervene in order to have comments considered. The second way to participate is by filing with the Secretary of the Commission, as soon as possible, an original and two copies of comments in support of or in opposition to this project. The Commission will consider these comments in determining the appropriate action to be taken, but the filing of a comment alone will not serve to make the filer a party to the proceeding. The Commission's rules require that persons filing comments in opposition to the project provide copies of their protests only to the party or parties directly involved in the protest.

Persons who wish to comment only on the environmental review of this project should submit an original and two copies of their comments to the Secretary of the Commission. Environmental commenters will be placed on the Commission's environmental mailing list, will receive copies of the environmental documents, and will be notified of meetings associated with the Commission's environmental review process. Environmental commenters will not be required to serve copies of filed documents on all other parties. However, the non-party commenters will not receive copies of all documents filed by other parties or issued by the Commission (except for the mailing of environmental documents issued by the Commission) and will not have the right to seek court review of the Commission's final order.

The Commission strongly encourages electronic filings of comments, protests and interventions in lieu of paper using the "eFiling" link at http://www.ferc.gov. Persons unable to file electronically should submit an original and 5 copies of the protest or intervention to the Federal Energy Regulatory Commission, 888 First Street, NE, Washington, DC 20426.

Comment Date:

Debbie-Anne Reese,
Secretary

Exhibit C
Company Officials

ETC TIGER PIPELINE, LLC
COMPANY OFFICIALS

MANAGERS

Name	Title
Long, Thomas E.	Manager
McCrea, Marshall S. III	Manager

OFFICERS

Name	Title
Arthur, Adam Y.	Executive Vice President – Crude Oil
Aube, Sonia	Vice President – Administration & Assistant Secretary
Beebe, Brian	Executive Vice President – Gas Gathering & Optimization
Bennett, Robert B.	Vice President
Billman, Richard J.	Senior Vice President – Business Development
Bramhall, Dylan A.	Executive Vice President & Group Chief Financial Officer
Cone, Tina E.	Vice President – Financial Planning & Analysis
Devarpiran, Sankar R.	Group Senior Vice President, Engineering & Special Projects
Dolle, Justin K.	Vice President – Financial Reporting
Flavin, Patrick S.	Group Senior Vice President – Measurement
Grossman, Scott B.	Deputy General Counsel
Hardy, Samuel B. IV	Deputy General Counsel – Head of Litigation
Harrison, Peggy J.	Senior Manager – Corporate Governance & Assistant Secretary
Hayse, Ashton	Vice President & Treasurer
Healy, William J.	Deputy General Counsel and Secretary
Hefty, Christopher	Group Senior Vice President – Strategic Development
Herrscher, Roger Barton	Executive Vice President – NGLs, Refined Products & Petrochemicals
Hotte, Steve J.	Group Senior Vice President & Chief Information Officer
Hunt, Jonathan Ashmore	Senior Vice President - Operations
Kerrigan, Robert M. III	Group Senior Vice President – Human Resources & Administration

Krebs, Darryl	Vice President - Tax
Long, Thomas E.	Co-Chief Executive Officer
McCrea, Marshall S. III	Co-Chief Executive Officer
Morgan, Michael T.	Senior Vice President – Construction & Project Execution
Payne, Christina	Senior Vice President – Crude Business Development
Pieper, Keegan	Deputy General Counsel
Rabun, Jonathan Tyler	Senior Vice President – Crude Oil Trading & Marketing
Schultz, Nicholas J.	Vice President – Power Optimization & DDT
Stamm, Todd	Group Senior Vice President, Operations
Street, Jennifer	Group Senior Vice President – Operations Services
Sturrock, Troy	Group Senior Vice President & Controller
Taliaferro, Kevin	Senior Director – Right-of-Way
Vedral, Mark L.	Senior Director – Right-of-Way
Walker, Michael L.	Senior Vice President – Commercial Gathering & Processing
Wentworth, Daniel	Group Senior Vice President – Operations & Engineering
Whitehurst, Bradford D.	Executive Vice President – Tax & Corporate Initiatives
Wright, James M. Jr.	Executive Vice President – General Counsel, Chief Compliance Officer & Assistant Secretary

The addresses for the Managers and Officers is:

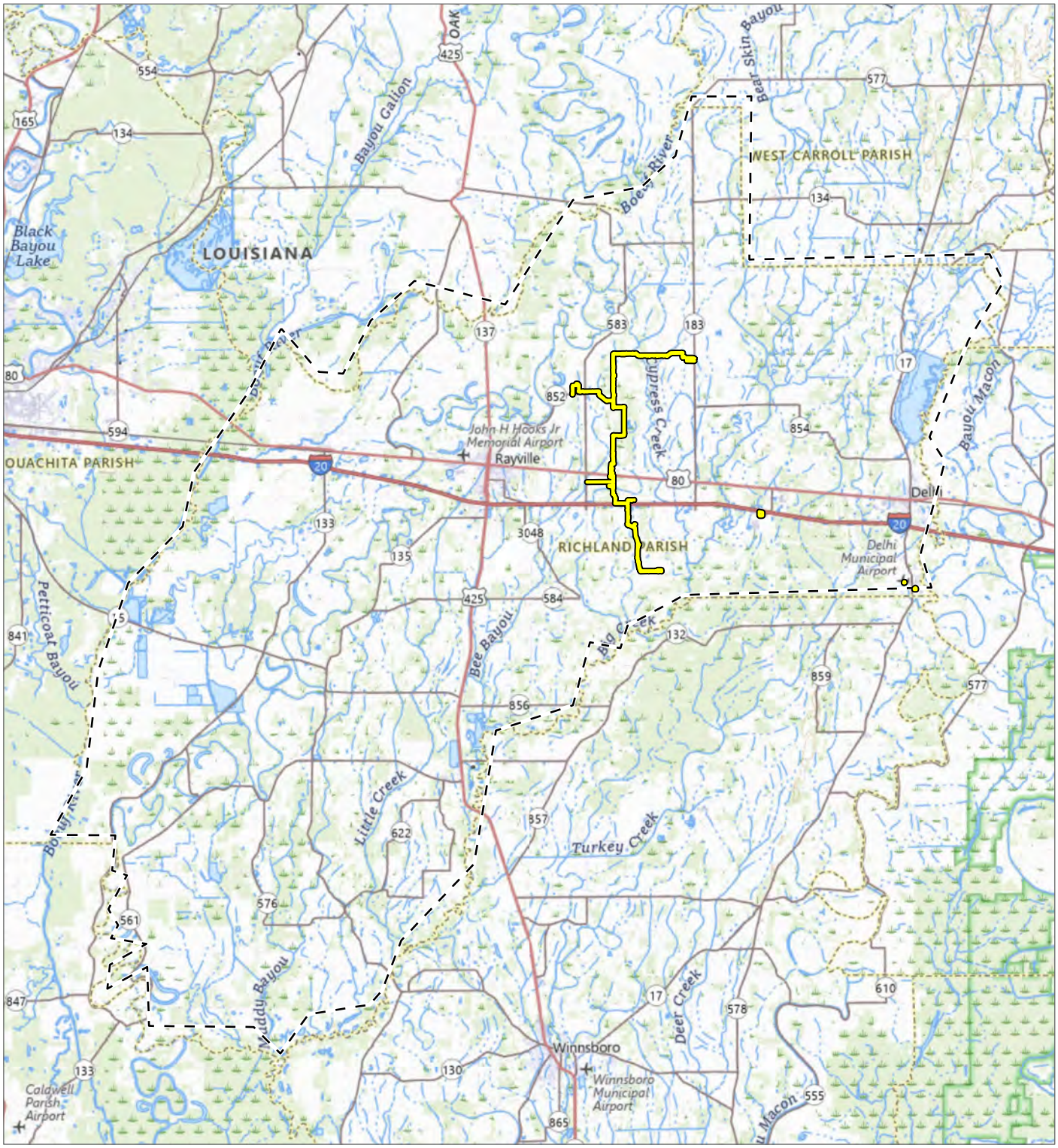
Physical Address
ETC Tiger Pipeline, LLC
1300 Main St.
Houston, Texas 77002

Mailing Address
ETC Tiger Pipeline, LLC
P. O. Box 4967
Houston, Texas 77002-4967

Exhibit F

ETC Tiger Pipeline, LLC

Project Map



PROJECT BOUNDARY

RICHLAND PARISH

0 14,000 28,000

FEET

1:336,000 1" = 28,000'

BASE MAP: USGS "THE NATIONAL MAP" ONLINE SERVICE LAYER;
DATA SOURCES: TRC, USGS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

GENERAL PROJECT LOCATION MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 1.B.1

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC APRX; LAYOUT NAME: 1.B.1 GENERAL PROJECT LOCATION

Exhibit F-1

ETC Tiger Pipeline, LLC

Environmental Report

The Environmental Report is submitted under three Volumes:

- Volume I– Public information
- Volume II – CUI/Critical Energy Infrastructure Information ("CUI/CEII")
- Volume III – CUI/Privileged Information

ETC Tiger Pipeline, LLC
Franklin Farms Project

RESOURCE REPORT NO. 1
General Project Description

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 1 FILING REQUIREMENTS	
Information	**Section Reference**
1. Provide a detailed description and location map of the project facilities (§ 380.12(c)(1)) • Include all pipeline and aboveground facilities. • Include support areas for construction or operation. • Identify facilities to be abandoned.	Section 1.1, Appendix A
2. Describe any non-jurisdictional facilities that would be built in association with the project (§ 380.12(c)(2)) • Include auxiliary facilities (See § 2.55(a)). • Describe the relationship to the jurisdictional facilities. • Include ownership, land requirements, gas consumption, megawatt size, construction status, and an update of the latest status of federal, state, and local permits/approvals. • Include the length and diameter of any interconnecting pipeline. • Apply the four-factor test to each facility. (see § 380.12(c)(2)(ii))	Section 1.9
3. Provide current, original United States Geological Survey (USGS) 7.5-minute series topographic maps with mileposts showing the project facilities (§ 380.12(c)(3)) • Maps of equivalent details are acceptable if legible (check with staff). • Show locations of all linear project elements, and label them. • Show locations of all significant aboveground facilities and label them.	Appendix A-2
4. Provide aerial images or photographs or alignment sheets based on these sources with mileposts showing the project facilities. (§ 380.12(c)(3)) • No more than 1-year old • Scale no smaller than 1:6,000	Appendix A-3, Appendix A-7
5. Provide plot/site plans of compressor stations showing the location of the nearest noise-sensitive areas (NSA) within 1 mile. (§ 380.12(c)(3,4)) • Scale no smaller than 1:3,600 • Show reference to topographic maps and aerial alignments provided above.	Not Applicable
6. Describe construction and restoration methods. (§ 380.12(c)(6))	Section 1.3
7. Identify the permits required for construction across surface waters. (§ 380.12(c)(9)) • Include the status of all permits. • For construction in the federal offshore area be sure to include consultation with the MMS. File with the MMS for rights-of-way grants at the same time or before you file with the FERC.	Section 1.7
8. Provide the names and addresses of all affected landowners as required and certify that all affected landowners will be notified. • Affected landowners are defined in § 157.6(d)(2). • Provide an electronic copy directly to the environmental staff.	Appendix E
Additional Information Often Missing and Resulting in Data Requests	
Describe all authorizations required to complete the proposed action and the status of applications for such authorizations.	Section 1.7
Provide plot/site plans of all other aboveground facilities that are not completely within the right-of-way.	Appendix B-2
Provide detailed typical construction right-of-way cross-section diagrams showing information such as widths and relative locations of existing rights-of-way, new	Appendix B-1

RESOURCE REPORT NO. 1 FILING REQUIREMENTS	
Information	**Section Reference**
permanent rights-of-way, and temporary construction rights-of-way. See Resource Report 8 – Land Use, Recreation, and Aesthetics.	
Summarize the total acreage of land affected by construction and operation of the project.	Section 1.2
Describe the cathodic protection system; include associated land requirements as appropriate.	Section 1.2.1.1
If Resource Report 5 - Socioeconomics is not provided, provide the start and end dates of construction, the number of pipeline spreads that would be used, and the workforce per spread.	Not Applicable
Send two (2) additional copies of topographic maps and aerial images/photographs directly to the environmental staff of the Office of Energy Projects.	Appendix A
Provide an electronic copy of the landowner list directly to Commission environmental staff. Check with Commission staff for required format.	Appendix E-1

RESOURCE REPORT NO. 1
GENERAL PROJECT DESCRIPTION

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Appendix A-1 – Project Overview Map
Appendix A-2 – USGS Topographic Maps
Appendix A-3 – Aerial Mapping
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Appendix A-5 – Soils Maps
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B Typical and Site-Specific Detail Drawings

Appendix B-1 – Project Typicals for Construction
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C Project Construction and Mitigation Plans

Appendix C-1 – FERC *Upland Erosion Control, Revegetation, and Maintenance Plan* (2013)
Appendix C-2 – FERC *Wetland and Waterbody Construction and Mitigation Procedures* (2013)
Appendix C-3 – *Spill Prevention and Response Plan*
Appendix C-4 – *Unanticipated Discovery Plan*
Appendix C-5 – *Fugitive Dust Control Plan*
Appendix C-6 – *Noxious and Invasive Plant Management Plan*
Appendix C-7 – *Environmental Complaint Resolution Procedures*

D Agency and Stakeholder Correspondence

Appendix D-1– Agency Correspondence (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")
Appendix D-2 – SHPO Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")
Appendix D-3 – Native American Organizations and Tribes Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

E Landowner Information

Appendix E-1 – Landowner/Stakeholder List (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE"*)

F Wetland and Waterbody Delineation Report and Listed Species Report

Appendix F-1 – Wetland and Waterbody Delineation Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")
Appendix F-2 – Listed Species Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

G Cultural Resource Assessment Survey Reports (*Submitted as* "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

H Air Quality Calculations (Construction Emissions)

I Sound Study Report

ABBREVIATIONS AND ACRONYMS

APE	area of potential effect
ATWS	additional temporary workspace
BMPs	best management practices
CEQ	Council on Environmental Quality
Certificate	Certificate of Public Convenience and Necessity
CFR	Code of Federal Regulations
CWA	Clean Water Act
Dth/day	dekatherms per day
EI	Environmental Inspector
EIS	Environmental Impact Statement
ETC Tiger	ETC Tiger Pipeline, LLC
EPA	United States Environmental Protection Agency
FERC or Commission	Federal Energy Regulatory Commission
FERC Plan	*Upland Erosion, Control, Revegetation, and Maintenance Plan* (2013)
FERC Procedures	*Wetland and Waterbody Construction and Mitigation Procedures* (2013)
FF	Franklin Farms
GC	gas chromatograph
Gulf Crossing	Gulf Crossing Pipeline Company, LLC
Gulf Run	Gulf Run Transmission, LLC
Gulf South	Gulf South Pipeline Company, LLC
HUC	hydrologic unit code
LAC	Louisiana Administrative Code
LDEQ	Louisiana Department of Environmental Quality
LDWF	Louisiana Department of Wildlife and Fisheries
MEP	Midcontinent Express Pipeline LLC
MP	milepost
N/A	not applicable
NEPA	National Environmental Policy Act
NGA	Natural Gas Act
NHPA	National Historic Preservation Act
NPDES	National Pollutant Discharge Elimination System
NWI	National Wetlands Inventory
NWP	Nationwide Permit
OD	outer diameter
PAR	permanent access road
PCN	Pre-Construction Notification
Project	Franklin Farms Project
RHA	Rivers and Harbors Act
ROI	region of interest
ROW	right-of-way
SHPO	State Historical Preservation Office
SPAR Plan	Spill Prevention and Response Plan
TAR	temporary access road
USACE	United States Army Corps of Engineers
USDOE	United States Department of Energy

USDOT	United States Department of Transportation
USFWS	United States Fish & Wildlife Service
USGS	United States Geological Survey
WQC	Water Quality Certification

1.0 GENERAL PROJECT DESCRIPTION

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, an interconnect site, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project in June 2027, and following the receipt of all applicable approvals, anticipating construction activities to begin in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

Resource Report No. 1 describes the purpose and need for the Project, proposed facilities for the Project, construction and restoration procedures, future plans, authorizations that must be obtained, affected landowners and other stakeholders, and the approach to cumulative impacts analysis.

1.1 PROPOSED FACILITIES

1.1.1 Purpose and Need

The proposed Project is designed to provide 250,000 dekatherms per day (Dth/day) of firm transportation service mainline capacity and 1,000,000 Dth/day of capacity on a newly constructed lateral to meet the increasing market demand for industrial customers in Richland Parish, Louisiana. ETC Tiger has executed a Precedent Agreement with Entergy Louisiana, LLC for the full capacity of the Project. The Project will supply natural gas to Entergy Louisiana, LLC's new Combined Cycle Combustion Turbine generating facilities which will be utilized for power generation to support a data center campus in Richland Parish, Louisiana. In support of powering artificial intelligence data centers consistent with the current Administration and FERC priorities, the Project expands natural gas infrastructure which will allow greater amounts of natural gas to be delivered to new electric generator facilities, such as the natural gas-fired turbines that will generate electricity for the new data center in Richland Parish, Louisiana. The proposed Project will enable ETC Tiger to meet its contractual obligations to Entergy Louisiana, LLC, and to continue fulfilling existing contractual obligations thereby ensuring that the Project has no impact on service to ETC Tiger's existing customers.

1.1.2 Location and Description of Facilities

ETC Tiger is proposing to construct the facilities and appurtenances listed in **Table 1.1-1**. **Appendix A** includes a Project overview map (**Appendix A-1**), United States Geological Survey (USGS) topographic maps (**Appendix A-2**), aerial mapping (**Appendix A-3**), National Wetlands Inventory (NWI) maps (**Appendix A-4**), soils maps (**Appendix A-5**), habitat maps (**Appendix A-6**), and alignment sheets (**Appendix A-7**) which depict the location of the proposed Project facilities. The Project will consist of the construction of the facilities presented within **Table 1.1-1**.

TABLE 1.1-1
Proposed Project Facilities

Facility [a]	Location (Parish, State)	Approximate Milepost	Proposed Work Description
Pipeline			
FF Lateral	Richland Parish, LA	0.0 – 13.2	Install approximately 13.2 miles of 36-inch OD pipeline.
FF Trio Farm Lateral	Richland Parish, LA	0.0 – 1.7	Install approximately 1.7 miles of 20-inch OD pipeline.
Aboveground Facilities			
FF Gulf Run Bi Di Meter Station	Richland Parish, LA	0.0 (Franklin Farms Lateral)	Construct one new receipt point providing a maximum rate of 500,000 million British thermal units per day (MMbtu/d) to the Franklin Farms Lateral. ETC Tiger will install, construct, and operate: • two new 12-inch meter runs; • two new 10-inch control valve runs; • one new 20-inch flow switching skid; • one new remote terminal unit (RTU)/gas chromatograph (GC) building; • one new Filter Separator with tank; • approximately 655 feet of 20-inch interconnect piping from the existing Gulf Run Pipeline; • approximately 450 feet of 24-inch interconnect piping from the existing Tiger Pipeline; • approximately 235 feet of 24-inch interconnect piping to the new Franklin Farm Lateral; • one new 36-inch launcher barrel; and • three new 10-inch control valve runs. Gulf Run will install, construct, and operate: • Hot Tap and interconnect piping under their blanket authorization with FERC.
FF Gulf South Receipt Meter Station	Richland Parish, LA	0.0 (Franklin Farms Lateral)	Construct one new receipt point providing a maximum rate of 500,000 MMbtu/d to the Franklin Farms Lateral. ETC Tiger will install, construct, and operate: • two new 12-inch meter runs; • two new 10-inch control valve runs; • one new RTU/GC building; • one new Gulf South RTU/GC building; • one new Filter Separator with tank; • approximately 405 feet of 24-inch interconnect piping from the existing Gulf South pipeline; and • approximately 160 feet of 24-inch interconnect piping to the new Franklin Farm Lateral. Gulf South will install, construct, and operate: • Hot Tap and interconnect piping under their blanket authorization with FERC.
FF Bee Bayou Delivery Meter Station	Richland Parish, LA	4.5 (Franklin Farms Lateral)	Construct one new delivery point providing a maximum rate of 320,000 MMbtu/d from the Franklin Farms Lateral. ETC Tiger will install, construct, and operate:

TABLE 1.1-1
Proposed Project Facilities

Facility [a]	Location (Parish, State)	Approximate Milepost	Proposed Work Description
			• two new 10-inch meter runs; • two new 10-inch control valve runs; • one new RTU/GC building; • two new Filter Separators with tank; • one 36-inch mainline valve; and • approximately 47 feet of 20-inch interconnect piping from the new Franklin Farm Lateral.
FF Interconnect Site	Richland Parish, LA	8.7 (Franklin Farms Lateral); 0.0 (FF Trio Farm Lateral)	Construct one new interconnect point providing a maximum rate of 320,000 MMbtu/d from the Franklin Farms Lateral to the FF Trio Farm Lateral. ETC Tiger will install, construct, and operate: • one new 20-inch launcher barrel; • one new 36-inch mainline valve; and • one new RTU rack.
FF Gulf Crossing Receipt Meter Station	Richland Parish, LA	9.7 (Franklin Farms Lateral)	Construct one new receipt point providing a maximum rate of 500,000 MMbtu/d to the Franklin Farms Lateral. ETC Tiger will install, construct, and operate: • two new 12-inch meter runs; • two new 10-inch control valve runs; • one new RTU/GC building; • one new Gulf Crossing RTU/GC building; • one new Filter Separator with tank; • approximately 80 feet of 24-inch interconnect piping from the existing Gulf Crossing pipeline; and • approximately 105 feet of 24-inch interconnect piping to the new Franklin Farm Lateral. Gulf Crossing will install, construct, and operate: • Hot Tap and interconnect piping under their blanket authorization with FERC.
MEP Tap Site	Richland Parish, LA	10.8 (Franklin Farms Lateral)	ETC Tiger will acquire one tap site at the MEP tie-in along the Franklin Farms Lateral. MEP will install construct, and operate this facility within ETC Tiger's proposed MEP Tap Site area under their blanket authorization with FERC.
FF MEP Receipt Meter Station	Richland Parish, LA	10.9 (Franklin Farms Lateral)	Construct one new receipt point providing a maximum rate of 500,000 MMbtu/d to the Franklin Farms Lateral. ETC Tiger will install, construct, and operate: • two new 12-inch meter runs; • two new 10-inch control valve runs; • one new RTU/GC building; • one new MEP RTU/GC building; • one new Filter Separator with tank; • one new 36-inch mainline valve;

TABLE 1.1-1
Proposed Project Facilities

Facility [a]	Location (Parish, State)	Approximate Milepost	Proposed Work Description
			• approximately 262 feet of 24-inch interconnect piping from the existing MEP pipeline; and • approximately 30 feet of 24-inch interconnect piping to the new Franklin Farm Lateral.
Franklin Farms Delivery Meter Station	Richland Parish, LA	13.2 (Franklin Farms Lateral)	Construct one new delivery point providing a maximum rate of 320,000 MMbtu/d from the Franklin Farms Lateral. ETC Tiger will install, construct, and operate: • two new 10-inch meter runs; • two new 10-inch control valve runs; • one new RTU/GC building; • two new Filter Separators with tank; • one new 36-inch receiver barrel; and • approximately 60 feet of 20-inch interconnect piping from the new Franklin Farm Lateral.
FF Trio Farm Delivery Meter Station	Richland Parish, LA	1.7 (FF Trio Farm Lateral)	Construct one new delivery point providing a maximum rate of 320,000 MMbtu/d from the FF Trio Farm Lateral. ETC Tiger will install, construct, and operate: • two new 10-inch meter runs; • two new 10-inch control valve runs; • one new RTU/GC building; • two new Filter Separators with tank; • one 20-inch receiver barrel; and • approximately 50 feet of 20-inch interconnect piping from the new FF Trio Farm Lateral pipeline.
Contractor Yards			
Contractor Yards	Richland Parish, LA	N/A	The Project will require three contractor yards for storage of construction equipment and pipe as well as for temporary office buildings and parking.
Access Roads [b]			
Temporary and Permanent Access Roads	Richland Parish, LA	N/A	The Project will require temporary and permanent access roads for construction and operation.

Notes:

[a] Acronyms used in the proposed Project facility names are defined as follows: Franklin Farms (FF); Gulf Run Transmission, LLC (Gulf Run); Gulf South Pipeline Company, LLC (Gulf South); Gulf Crossing Pipeline Company, LLC (Gulf Crossing); Midcontinent Express Pipeline, LLC (MEP).

[b] Temporary access roads will be returned to pre-construction conditions or per landowner agreement following construction. Access roads are detailed in Table 1.2-2.

1.1.3 Pipeline Facilities

1.1.3.1 Franklin Farms Lateral

The Project involves constructing the Franklin Farms Lateral, which begins at MP 0.0 at the interconnection with the existing Gulf Run Pipeline and existing Tiger Pipeline in Richland Parish, Louisiana. The pipeline will extend north from the interconnection at the proposed FF Gulf Run Bi Di Meter Station and the proposed FF Gulf South Receipt Meter Station to the termination point (approximate MP 13.2) at the proposed Franklin Farms Delivery Meter Station.

1.1.3.2 FF Trio Farm Lateral

The Project involves constructing the FF Trio Farm Lateral beginning at MP 0.0 at the interconnect with the proposed Franklin Farms Lateral (at MP 8.7 on the Franklin Farms Lateral) in Richland Parish, Louisiana. The pipeline will extend west from the proposed FF Interconnect Site to the termination point (approximate MP 1.7) at the proposed FF Trio Farm Delivery Meter Station.

1.1.4 Aboveground Facilities

1.1.4.1 Meter Stations

The Project involves constructing seven new meter stations along the Franklin Farms Lateral and the FF Trio Farm Lateral. Details regarding each meter station, location and total capacity are described in **Table 1.1-1**. Plot Plans showing the meter stations are provided in **Appendix B-2**.

The FF Bee Bayou Delivery Meter Station, Franklin Farms Delivery Meter Station, and FF Trio Farm Delivery Meter Station will be previously disturbed (grading and /or graveling) by Entergy Louisiana, LLC prior to ETC Tiger commencing construction in these Project areas. Entergy Louisiana, LLC will be conducting work in these areas to support their energy development projects as further discussed in **Section 1.9** and **Section 1.10**. Entergy Louisiana, LLC will temporarily stabilize these areas and ETC Tiger will perform final restoration of the temporarily impacted areas within the FF Bee Bayou Delivery Meter Station, Franklin Farms Delivery Meter Station, and FF Trio Farm Delivery Meter Station once construction is complete.

It is anticipated that the entities for the following meter sites: FF Gulf South Receipt Meter Station; FF Gulf Crossing Receipt Meter Station; and FF Gulf Run Bi Di Meter Station, will construct the meter facilities to tie their respective existing pipelines into the Franklin Farms Lateral within the proposed meter site area. It is anticipated that these pipeline companies (Gulf Run Transmission, LLC [Gulf Run]; Gulf South Pipeline Company, LLC [Gulf South]; Gulf Crossing Pipeline Company, LLC [Gulf Crossing]) will conduct the meter site work under their blanket certification with FERC and that ETC Tiger will reimburse these entities for their costs.

1.1.4.2 FF Interconnect Site

The Project involves constructing a new interconnect site between the Franklin Farms Lateral at MP 8.7 and the FF Trio Farm Lateral at MP 0.0. The location of the FF Interconnect Site is shown in **Appendix A-7.**

1.1.4.3 MEP Tap Site

The Project involves acquiring one tap site, the MEP Tap Site, along the Franklin Farms Lateral. It is anticipated that Midcontinent Express Pipeline, LLC (MEP) will construct the tap site to tie their existing

pipeline into the Franklin Farms Lateral within the proposed MEP Tap Site area. It is anticipated that MEP will conduct the tap site work under their blanket certification with FERC and that ETC Tiger will reimburse these entities for their costs.

MEP is anticipated to construct the MEP Tap Site after ETC Tiger has installed the Franklin Farms Lateral. Therefore, ETC Tiger will not be able to complete final restoration for the areas that overlap with the MEP Tap Site; however, ETC Tiger will temporarily stabilize the areas that overlap. MEP will perform final restoration of temporary impacted areas for the MEP Tap Site once construction is completed.

The location of the MEP Tap Site is shown in **Appendix A-7.**

1.2 LAND REQUIREMENTS

Table 1.2-1 summarizes land requirements for construction and operation of the Project. Typical and detailed site-specific drawings of the Project facilities are provided in **Appendix B**. Current land uses of all areas affected by the Project are described in Resource Report No. 8 (*Land Use, Recreation and Aesthetics*). A description of the land requirements for the proposed Project facilities is provided in the following sections.

TABLE 1.2-1
Land Requirements for the Project

Facility	Land Affected (Acres) [a]		
	Temporary Impacts	Permanent Impacts	Total Workspace for Construction [b]
Pipeline Facilities [c]			
Franklin Farms Lateral	106.9	78.5	185.4
FF Trio Farm Lateral	7.9	10.5	18.3
Pipeline Facilities Subtotal	**114.7**	**89.0**	**203.7**
Aboveground Facilities [c]			
FF Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station [d]	2.3	3.6	6.0
FF Bee Bayou Delivery Meter Station	0.5	1.0	1.5
FF Interconnect Site	0.2	0.9	1.0
FF Gulf Crossing Receipt Meter Station	0.3	1.4	1.7
MEP Tap Site	0.0	0.1	0.1
FF MEP Receipt Meter Sation	0.2	1.2	1.4
Franklin Farms Delivery Meter Station	0.6	1.0	1.6
FF Trio Farm Delivery Meter Station	0.6	1.4	2.0
Aboveground Facility Subtotal	**4.6**	**10.7**	**15.4**
Other Work Areas			
Contractor Yards	14.8	0.0	14.8

TABLE 1.2-1
Land Requirements for the Project

Facility	Land Affected (Acres) [a]		
	Temporary Impacts	Permanent Impacts	Total Workspace for Construction [b]
Access Roads	7.7 [e]	7.3 [f]	14.9
Other Work Areas Subtotal	**22.5**	**7.3**	**29.8**
GRAND TOTAL	**141.9**	**107.0**	**248.9**

Notes:
[a] The numbers in this table have been rounded for presentation purposes. As a result, the totals may not reflect the sum of the addends.
[b] The land affected during construction includes both temporary and permanent workspace.
[c] Temporary impact acreages include the Additional Temporary Workspace (ATWS) that is listed in Resource Report No. 8, Table 8.1-4.
[d] The FF Gulf Run Bi Di Meter Station and the FF Gulf South Receipt Meter Station share temporary and permanent workspace.
[e] Temporary access roads will be returned to pre-construction conditions or per landowner agreement following construction.
[f] Access roads FFL-PAR-0.15.000 and FFL-PAR-0.17.000 are located partially on top of the permanent easement for the Franklin Farms Lateral; thus, the operational ROW is shared for these workspaces. For presentation purposes, the impact from FFL-PAR-0.15.000 and FFL-PAR-0.17.000 are inclusive of operational impacts from the Franklin Farms Lateral for the area of overlap.

1.2.1 Pipeline Facilities

Land requirements for the proposed Franklin Farms Lateral and FF Trio Farm Lateral are summarized in **Table 1.2-1**, and a description of the land requirements is provided in the following sections. Construction right-of-way (ROW) limits for the proposed Franklin Farms Lateral and FF Trio Farm Lateral are shown on aerial alignment sheets included in **Appendix A-7**. **Appendix B-1** includes the typical ROW cross-section drawings for the proposed Franklin Farms Lateral and FF Trio Farm Lateral.

1.2.1.1 Permanent (Operational) Right-of-Way

ETC Tiger will need to acquire new permanent ROW for the Franklin Farms Lateral and the FF Trio Farm Lateral. The new permanent ROW for both the Franklin Farms Lateral and the FF Trio Farm Lateral is 50 feet wide. Land use requirements within the new permanent pipeline ROW are provided in Table 8.1-1 of Resource Report No. 8. The predominant land uses affected by the pipeline's new permanent ROW are also discussed in Resource Report No. 8. ETC Tiger's new permanent pipeline ROW along with its location, orientation, and dimensions, is specifically identified on the aerial alignment sheets provided in **Appendix A-7**. Typical drawings of the pipeline permanent ROW are provided in **Appendix B-1**.

Cathodic Protection

ETC Tiger proposes to install cathodic protection and alternating current mitigation systems, such as rectifiers, test leads, and anode beds within the permanent ROW of the Franklin Farms Lateral and the FF Trio Farm Lateral. The cathodic protection and alternating current mitigation systems will be installed to prevent external corrosion of the new pipelines. All areas disturbed during the installation of cathodic protection and alternating current mitigation systems will be limited to the construction ROW.

1.2.1.2 Temporary (Construction) Right-of-Way

In addition to the new permanent ROW, construction of the proposed Franklin Farms Lateral and the FF Trio Farm Lateral will require temporary (construction) workspace to facilitate construction equipment, referred to as the construction ROW. Installation of the proposed Franklin Farms Lateral will typically require use of a 100-foot-wide construction ROW while the proposed FF Trio Farm Lateral will typically require use of a 75-foot-wide construction ROW. The proposed construction ROW widths for the Franklin Farms Lateral and the FF Trio Farm Lateral are required to provide a safe work site, promote effective implementation of erosion control measures, and provide adequate work area for topsoil segregation.

Conventional overland construction of large diameter pipelines requires larger construction equipment to meet the demands of stringing and laying larger, heavier pipe sections. Large diameter pipe also requires wider and deeper trenches, resulting in the need to expand typical construction corridors from the conventional 75 feet to 100 feet width depending on soil stability and terrain; therefore, a 100-foot-wide construction ROW is proposed for the Franklin Farms Lateral. In wetland areas, the width of the construction corridor will be reduced to 75 feet in accordance with the FERC *Wetland and Waterbody Construction and Mitigation Procedures* (FERC Procedures, May 2013 version), as feasible.

Land use requirements within the construction ROW are provided in Table 8.1-1 of Resource Report No. 8. The predominant land uses, which will be temporarily affected during construction, are also discussed in Resource Report No. 8. ETC Tiger's pipeline temporary (construction) ROW, along with its location, orientation, and dimensions, is specifically identified on the aerial alignment sheets provided in **Appendix A-7**. Typical drawings of the pipeline construction ROW are provided in **Appendix B-1**.

1.2.1.3 Additional Temporary Workspace

In addition to the temporary (construction) ROW, ETC Tiger will require ATWS to facilitate Franklin Farms Lateral and the FF Trio Farm Lateral construction at:

- Road, wetland, waterbody, railroad, and foreign utility crossings;
- Pipeline fitting and installation;
- Points of inflection;
- Aboveground facilities;
- Topsoil segregation in active agricultural lands;
- Equipment and material storage locations; and
- Equipment turnarounds.

The extent of ATWS is determined on a site-specific basis. The ATWS areas are restricted to the minimum size necessary to safely construct the proposed Franklin Farms Lateral and the FF Trio Farm Lateral. Justifications for ATWS required for the Project are detailed in Table 8.1-4 of Resource Report No. 8.

In the case of wetlands and waterbodies, ETC Tiger has attempted to locate ATWS in accordance with the 50-foot setback requirement contained in the FERC Procedures. In certain instances, the 50-foot setback could not be maintained due to construction limitations, such as trenchless road crossing workspace requirements. In those cases, ETC Tiger is requesting deviations from the FERC Procedures. **Table 1.3-3** identifies the ATWS locations where these deviations are requested as well as the justification for the request.

1.2.2 Aboveground Facilities

Land requirements for the proposed aboveground facilities are summarized in **Table 1.2-1**. Construction workspace requirements for the proposed facilities are shown on the Project mapping included in **Appendix A**. A description of the land requirements for the proposed aboveground facilities is provided in the following sections.

1.2.2.1 Meter Stations

Plot Plans showing the seven proposed meter stations are provided in **Appendix B-2**. The dimensions of each meter station are depicted on alignment sheets provided in **Appendix A-7**. The new meter stations will be graveled and fenced.

The FF Gulf Run Bi Di Meter Station and the FF Gulf South Receipt Meter Station are both located at the beginning of the Franklin Farms Lateral at approximately MP 0.0. These two meter stations will share temporary and permanent workspace.

1.2.2.2 FF Interconnect Site

The dimensions of the FF Interconnect Site are depicted on alignment sheets provided in **Appendix A-7**.

1.2.2.3 MEP Tap Site

MEP will construct the MEP Tap Site to tie into MEP's existing pipeline and the Franklin Farms Lateral. ETC Tiger will acquire the MEP Tap Site following completion of work by MEP. The dimensions of the MEP Tap Site are depicted on alignment sheets provided in **Appendix A-7**.

1.2.2.4 Access Roads

Temporary access roads (TAR) used for construction crews and the delivery of materials and equipment to the Project will be via existing private roads or new temporary access points off public roads. TARs will be returned to pre-construction conditions or per landowner agreement following construction. New permanent access roads (PAR) will be installed to access Project facilities. FFL-PAR-0.15.000 and FFL-PAR-0.17.000 are partially located on top of the permanent easement for the Franklin Farms Lateral resulting in less environmental impacts than if these features were not overlapped. The proposed PARs vary in width from 30 to 60 feet to accommodate large equipment needed to construct the Project and for maintenance activities during operation. Additionally, the PARs proposed to be 60-feet are designed for bi-directional access which is critical to allow for reliable access during construction and operation for safety purposes, and the bi-directional nature of the PARs will allow for more efficient construction which will reduce the overall Project construction duration.

Details of the planned permanent and temporary access roads are provided in **Table 1.2-2**. Access roads that will be used during construction are depicted on the Project mapping included in **Appendix A**.

TABLE 1.2-2
Project Access Roads

Project Facility	Access Road Name	Milepost	Access Road Type	Width (feet)	Length (feet)	Acres	Current Land Use	Improvements
FF Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station	FFL-PAR-002.000	0.0 (Franklin Farms Lateral)	Permanent	60	56.5	0.08	Developed, Agricultural Land, Surface Water	Construct new permanent road for operations
FF Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station	FFL-PAR-002.100	0.00 (Franklin Farms Lateral)	Permanent	60	56.5	0.08	Developed, Agricultural Land, Surface Water	Construct new permanent road for operations
FF Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station	FFL-TAR-001.000	0.00 (Franklin Farms Lateral)	Temporary	60	38.4	0.05	Developed, Agricultural Land, Surface Water	Construct new temporary construction entrance
FF Interconnect Site	TFL-PAR-001.000	0.00 (FF Trio Farm Lateral)	Permanent	30	1392.0	0.98	Agricultural Land, Developed, Surface Water, Palustrine Forested Wetland (PFO)	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-003.000	0.33 (Franklin Farms Lateral)	Temporary	60	30.6	0.04	Developed, Agricultural Land, Surface Water	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-004.000	1.55 (Franklin Farms Lateral)	Temporary	30	304.3	0.07	Developed, Open Land, Surface Water	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-005.000	1.59 (Franklin Farms Lateral)	Temporary	30	234.1	0.29	Developed, Open Land	Grade and Gravel
FF Trio Farm Delivery Meter Station	TFL-PAR-002.000	1.75 (FF Trio Farm Lateral)	Permanent	30	2618.9	1.81	Developed, Agricultural Land, Surface Water	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-006.000	2.27 (Franklin Farms Lateral)	Temporary	30	812.4	0.57	Developed, Agricultural Land	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-007.000	3.31 (Franklin Farms Lateral)	Temporary	30	1680.9	1.17	Open Land, Developed, Palustrine Emergent Wetland (PEM)	Grade, Gravel and Mat
FF Bee Bayou Delivery Meter Station	FFL-PAR-008.000	4.55 (Franklin Farms Lateral)	Permanent	30	610.3	0.43	Agricultural Land Developed, Surface Water	Construct new permanent road for operations
Franklin Farms Lateral	FFL-TAR-009.000	4.81 (Franklin Farms Lateral)	Temporary	30	5049.8	3.48	Agricultural Land	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-010.000	4.81 (Franklin Farms Lateral)	Temporary	30	48.7	0.03	Agricultural Land, Developed, Surface Water	Grade and Gravel

TABLE 1.2-2
Project Access Roads

Project Facility	Access Road Name	Milepost	Access Road Type	Width (feet)	Length (feet)	Acres	Current Land Use	Improvements
Franklin Farms Lateral	FFL-TAR-011.000	4.99 (Franklin Farms Lateral)	Temporary	30	80.1	0.06	Agricultural Land, Developed, Open Land, Surface Water	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-012.000	5.42 (Franklin Farms Lateral)	Temporary	30	1697.0	1.18	Agricultural Land, Developed, Open Land, Surface Water	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-013.000	5.99 (Franklin Farms Lateral)	Temporary	30	355.1	0.24	Forested/Woodland, Agricultural Land, Open Land, Surface Water, PFO Wetland	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-014.000	6.72 (Franklin Farms Lateral)	Temporary	30	166.0	0.11	Developed	Grade and Gravel
FF Gulf Crossing Receipt Meter Station	FFL-PAR-015.000	9.47 (Franklin Farms Lateral)	Permanent	30	882.1	0.62	Agricultural Land, Open Land	Construct new permanent road for operations
Franklin Farms Lateral	FFL-TAR-016.000	10.51 (Franklin Farms Lateral)	Temporary	60	44.3	0.06	Developed, Agricultural Land, Surface Water	Grade and Gravel
FF MEP Receipt Meter Station	FFL-PAR-017.000	10.81 (Franklin Farms Lateral)	Permanent	30	166.9	0.11	Developed, Agricultural Land, Surface Water	Construct new permanent road for operations
FF MEP Receipt Meter Station	FFL-PAR-018.000	10.85 (Franklin Farms Lateral)	Permanent	60	26.3	0.04	Developed, Agricultural Land, Surface Water	Construct new permanent road for operations
Franklin Farms Lateral	FFL-TAR-019.000	10.95 (Franklin Farms Lateral)	Temporary	30	44.2	0.03	Developed, Agricultural Land, Surface Water	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-020.000	11.19 (Franklin Farms Lateral)	Temporary	30	88.9	0.07	Developed, Agricultural Land, Surface Water	Grade and Gravel
Franklin Farms Lateral	FFL-TAR-021.000	11.31 (Franklin Farms Lateral)	Temporary	60	34.1	0.05	Developed, Forested/Woodland, Surface Water	Construct new temporary construction entrance
Franklin Farms Lateral	FFL-TAR-022.000	12.94 (Franklin Farms Lateral)	Temporary	60	24.1	0.03	Developed, Open Land,	None
Franklin Farms Lateral	FFL-TAR-023.000	13.04 (Franklin Farms Lateral)	Temporary	60	34.1	0.05	Developed	None
Franklin Farms Delivery Meter Station	FFL-PAR-024.000	13.24 (Franklin Farms Lateral)	Permanent	30	2402.2	1.65	Developed	None

TABLE 1.2-2
Project Access Roads

Project Facility	Access Road Name	Milepost	Access Road Type	Width (feet)	Length (feet)	Acres	Current Land Use	Improvements
Franklin Farms Delivery Meter Station	FFL-PAR-025.000	13.24 (Franklin Farms Lateral)	Permanent	30	2197.7	1.50	Developed	None
Contractor Yard 2	FFL-TAR-026.000	N/A	Temporary	40	25.5	0.02	Open Land	Grade and Gravel
Contractor Yard 1	FFL-TAR-027.000	N/A	Temporary	60	17.2	0.02	Open Land	Grade and Gravel
Contractor Yard 1	FFL-TAR-028.000	N/A	Temporary	60	17.8	0.02	Open Land, Surface Water	Grade and Gravel

Notes:
N/A = not applicable; PEM = palustrine emergent wetland; PFO = palustrine forested wetland

1.2.3 Contractor Yards

ETC Tiger intends to use three contractor yards in Richland Parish, Louisiana. The contractor yards will be allowed to revert to previous land uses following construction and per landowner agreements. One PEM wetland (PEM-T3F-06) is located within Contractor Yard 3; however, ETC Tiger will avoid all wetland impacts within the proposed contractor yard; therefore, no wetland impacts are anticipated at this location.

1.3 CONSTRUCTION PROCEDURES

The Project will be constructed in compliance with applicable federal and state regulations and guidelines, and the specific requirements of the necessary permits (see **Section 1.7**). These regulations ensure adequate protection for the public and prevent natural gas facility accidents and failures. Key federal requirements and guidelines include:

- 18 Code of Federal Regulations (CFR) Part 380 – *The FERC Regulations Implementing the National Environmental Policy Act* (NEPA) (including Section 380.15 - *Siting and Maintenance Requirements*); and
- 49 CFR Part 192 – *Transportation of Natural Gas and Other Gas by Pipeline: Minimum Federal Safety Standards*.

In addition to the key federal requirements and guidelines listed above, ETC Tiger will implement the following plans and procedures (included in **Appendix C**) to ensure adequate protection of environmental and cultural resources during construction.

- The FERC *Upland Erosion Control, Revegetation, and Maintenance Plan* (FERC Plan, May 2013 Version) provided in **Appendix C-1**;
- The FERC Procedures with requested deviations (see **Table 1.3-2** and **Table 1.3-3**) provided in **Appendix C-2**;
- ETC Tiger's *Spill Prevention and Response Plan* (SPAR Plan) provided in **Appendix C-3**;
- ETC Tiger's *Unanticipated Discovery Plan* (i.e., cultural resources) provided in **Appendix C-4**;
- ETC Tiger's *Fugitive Dust Control Plan* provided in **Appendix C-5**;
- ETC Tiger's *Noxious and Invasive Plant Management Plan* provided in **Appendix C-6***; and*
- ETC Tiger's *Environmental Complaint Resolution Procedures* provided in **Appendix C-7**.

The following sections identify the general construction procedures for routine (conventional) pipeline construction for the Project, as well as the specialized construction techniques that will be utilized in environmentally sensitive areas. Construction typicals are provided in **Appendix B-1**.

1.3.1 Pipeline

As described in the following sections, ETC Tiger will use a combination of conventional and specialized construction procedures to complete the proposed Franklin Farms Lateral and FF Trio Farm Lateral installation. The construction process will be coordinated to minimize the total time a tract of land is disturbed, reducing erosion potential, and loss of normal land use. **Table 1.3-1** depicts the construction methods by milepost for the Franklin Farms Lateral and FF Trio Farm Lateral, respectively.

TABLE1.3-1
Construction Methods by Milepost for Project Pipeline Laterals

Construction Type	Length (Feet)	Milepost		Length (Miles)	Bore Crossing
		Begin	End		
Franklin Farms Lateral					
Conventional	3,119	0.00	0.59	0.59	N/A
Horizontal Bore	100	0.59	0.61	0.02	Mengle Road
Conventional	5,281	0.61	1.61	1.00	N/A
Horizontal Bore	80	1.61	1.62	0.02	Pony Greer Road
Conventional	6,153	1.62	2.79	1.17	N/A
Horizontal Bore	80	2.79	2.81	0.02	Ollie Morris Road
Conventional	2,294	2.81	3.24	0.43	N/A
Horizontal Bore	360	3.24	3.31	0.07	Interstate 20
Conventional	9,458	3.31	5.10	1.79	N/A
Horizontal Bore	300	5.10	5.16	0.06	Railroad and Highway 80
Conventional	22,543	5.16	9.43	4.27	N/A
Horizontal Bore	280	9.43	9.48	0.05	Pipeline Corridor and Smalling Road
Conventional	15,810	9.48	12.47	2.99	N/A
Horizontal Bore	80	12.47	12.49	0.02	Jaggers Ln
Conventional	3,251	12.49	13.10	0.62	N/A
Horizontal Bore	380	13.10	13.18	0.07	Entergy Laydown Yard
Conventional	342	13.18	13.24	0.06	N/A
FF Trio Farm Lateral					
Conventional	6,262	0.00	1.19	1.19	N/A
Horizontal Bore	120	1.19	1.21	0.02	Highway 583
Conventional	2,869	1.21	1.75	0.54	N/A

Notes:
N/A = not applicable

1.3.1.1 Conventional Pipeline Installation Procedures

Conventional pipeline installation, also referred to as standard open-cut construction, generally occurs in a linear sequence defined by clearing, grading, trenching, stringing, lowering in, backfilling, testing, clean-up, and restoration. A brief description of each of the main activities is presented in the following sections.

Clearing and Grading
Prior to construction activities, ETC Tiger will survey and stake the limits of the approved workspace to clearly identify Project limits. No work or ground disturbing activities will take place outside of the approved Project footprint without prior approval from the FERC, other permitting agencies, and landowner approvals, as required. Pipeline installation will begin with clearing and grading of the construction ROW and ATWS areas. ETC Tiger will perform necessary brush and tree clearing within the Project construction ROW. Timber and other vegetative debris may be disposed of in one or more of the following ways:

- Chipping slash and brush and leaving the chips on the ROW (i.e., erosion control mulch);
- Offsite disposal in approved disposal areas, and only if the preceding options are not viable.

Existing fences will be cut and braced as needed along the ROW. Temporary fencing, safety fencing, and/or gates will be erected/relocated as needed and in accordance with permits and landowner agreements. The ROW will be graded as necessary to create a reasonably level working surface to allow safe passage and operation of equipment. Temporary erosion and sediment controls will be installed in accordance with the FERC Plan, FERC Procedures, permits, and agreements.

Pipeline Trenching and Dewatering

Following clearing and grading, a trench will be excavated for installation of the pipe. The trench will be excavated to a sufficient depth to allow a minimum of three feet of soil cover between the top of the pipe and the final land surface after backfilling in non-cropland. In croplands, there will be a minimum of five feet of cover. If necessary, trench dewatering will be performed in accordance with the FERC Plan, FERC Procedures, and any applicable permit requirements. Best management practices (BMP) such as filter bags or silt fence/hay bale structures will be used to control erosion and siltation upon discharge.

Pipe Stringing and Lowering

The pipe segments will be temporarily placed or strung alongside the trench, typically on skids, where they will be bent as necessary, welded together, inspected, and the joints coated in preparation for lowering. The prepared sections of pipe will be lifted off temporary supports and lowered into the trench by side-boom tractors. Prior to lowering the pipe, the trench will be inspected to ensure that it is free of rocks and other debris that could damage the pipe or the coating. Before lowering the pipe into the trench, the pipe and trench configurations will be inspected to ensure compatibility.

Pipeline Padding and Backfilling

After the pipe is lowered into the trench, the trench will be backfilled. Previously excavated materials will be pushed back into the trench using bladed equipment or backhoes. Where the previously excavated material contains large rocks or other materials that could damage the pipe or coating, clean fill or a protective coating will be placed around the pipe prior to backfilling. Following backfilling, a small crown of material will be left to account for future soil settling that may occur.

Fugitive Dust Control

ETC Tiger will implement measures to minimize the effects of fugitive dust as required. Typical practices to minimize fugitive dust include watering exposed soil surfaces, applying temporary mulch, and expediting restoration and revegetation activities. ETC Tiger's *Fugitive Dust Control Plan* is included in **Appendix C-5**.

Hydrostatic Testing and Final Tie-in

Following backfilling of the trench, the pipeline and aboveground piping systems will be hydrostatically tested to ensure it can operate at the designed pressure. Hydrostatic testing will be conducted in accordance with the requirements of United States Department of Transportation (USDOT) pipeline safety regulations, 49 CFR Part 192, ETC Tiger testing specifications, and applicable permits. The water in the pipe will be pressurized and held for a minimum of eight hours. Any loss of pressure that cannot be attributed to other factors, such as temperature changes, will be investigated. Any leaks detected will be repaired and the segment retested.

Estimated hydrostatic test water requirements for the proposed Project facilities are provided in Table 2.1-3 of Resource Report No. 2 (*Water Use and Quality*). Hydrostatic test-water will be obtained from commercial/municipal sources and/or identified existing wells within the Project area. Further information regarding existing wells is within Resource Report No. 2. Test water will contact only new pipe, and no chemicals or biocides will be added.

After testing, hydrostatic test water will be discharged in accordance with the FERC Plan and FERC Procedures. This includes discharging test water into a well vegetated upland area and filtering the water through a filter bag and/or erosion control barrier. ETC Tiger will obtain and comply with all applicable state permits prior to the discharge of test water. Once a segment of pipe has been successfully tested and dried, the test cap and manifold will be removed, and the pipe will be connected to the remainder of the pipeline.

Cleanup, Restoration and Revegetation

Cleanup activities will commence in the construction work area immediately following construction completion in accordance with the FERC Plan and FERC Procedures. Every reasonable effort will be made to complete final cleanup (including final grading and installation of permanent erosion control devices) within timeframes required by permits, in accordance with landowner requests, or in compliance with the FERC Plan and FERC Procedures. If seasonal or other weather conditions prevent compliance with the recommended time frames, ETC Tiger will maintain temporary erosion controls (i.e., temporary slope breakers, sediment barriers, and/or mulch) until conditions allow completion of cleanup.

All disturbed areas will be finish-graded and any remaining construction debris will be collected and properly disposed of in compliance with applicable regulations. Contours will be restored to pre-existing conditions as closely as possible. Topsoil will be spread over the surface of the construction work area and permanent erosion controls will be installed. Revegetation measures will be implemented in accordance with the FERC Plan and FERC Procedures. Periodic inspections of the ROW will be conducted throughout restoration in accordance with the FERC Plan.

Post-Construction Monitoring

ETC Tiger will monitor all areas disturbed by construction to ensure successful revegetation of the ROW and to identify any problem areas, as required by the FERC Plan and FERC Procedures, and in compliance with all applicable permit requirements.

1.3.1.2 Wetland Construction Methods

A summary of wetlands affected by construction and operation of the Project is provided in Table 2.3-1 of Resource Report No. 2. No impacts to wetlands are anticipated from construction of the proposed aboveground facilities. Temporary and permanent impacts are expected from construction and operation of the proposed pipelines.

Crossings of jurisdictional wetlands will be conducted in accordance with applicable regulations, U.S. Army Corps of Engineers (USACE) permits (compatible with the Clean Water Act (CWA) Section 404 guidelines), and the FERC Procedures (with requested alternative measures). Overall, the wetland crossing methods and mitigation measures identified in the FERC Procedures are designed to minimize the extent and duration of construction-related disturbance within wetlands.

ETC Tiger will limit the use of construction equipment in wetlands areas to that needed to clear the ROW, dig the trench, fabricate the pipe, install the pipe, backfill the trench, and restore contours within the ROW. Where feasible, ETC Tiger will utilize a 75-foot-wide construction ROW through wetlands as shown on the aerial alignment sheets (**Appendix A-7**). The entire construction ROW will be cleared of vegetation and then graded, where necessary, to create a level and safe working surface for construction equipment. Vegetation will be removed by mechanical cutting or by hand. Stumps will be removed along the pipeline ditch and from the construction work area, as necessary, for safe installation of the pipeline. ETC Tiger will utilize low ground weight equipment or operate standard equipment off timber equipment mats where rutting could occur to prevent mixing of wetland topsoil with subsoil.

Site-specific crossing procedures used to install the pipeline across wetlands will vary based on the level of soil stability and saturation encountered during construction. A typical wetland crossing plan is provided in **Appendix B-1**. Construction procedures used to cross unsaturated "dry" wetlands will be similar to those used in dry, upland areas. During construction, to prevent mixing of the soil horizons, topsoil segregation will be performed in unsaturated wetlands areas. A maximum of the upper 12 inches of topsoil will be excavated and segregated from subsoil trench spoil, except in areas where standing water is present or soils are saturated. In accordance with the FERC Procedures, ETC Tiger will minimize the length of time that topsoil is segregated and the trench is left open in wetlands.

ATWS will be required to stage construction equipment, fabricate the pipeline, cross features, construct aboveground facilities, and store construction materials. Except as requested in **Table 1.3-3**, and approved by the Commission, the ATWS will be located at least 50 feet away from the edge of each wetland. In addition, the refueling and lubricating of equipment will take place in upland areas that are at least 100 feet from the edge of wetlands.

Once the pipeline is successfully installed, ETC Tiger will restore segregated topsoil during backfilling and restore grade to preconstruction conditions or better. Implementation of proper topsoil segregation, where necessary, will help ensure post-construction revegetation success, thereby minimizing the potential for erosion due to lack of vegetative cover. Temporary and permanent erosion and sedimentation control measures will be installed and maintained as required by permit conditions until revegetation and stabilization of each wetland crossing is complete. ETC Tiger will install trench breakers, as necessary, to maintain wetlands hydrology. ETC Tiger will remove construction debris and cleared vegetative material from wetland areas for proper disposal.

Areas temporarily disturbed during construction will be restored to their pre-existing contours, to the extent practicable, and allowed to naturally revegetate. Revegetation measures will be implemented in accordance with the FERC Procedures. ETC Tiger will monitor the ROW, as required by the FERC Procedures and further described in Resource Report No. 2 and Resource Report No. 7 (*Soils*).

1.3.1.3 Waterbody Construction Methods

A complete list of waterbodies affected by construction and operation of the Project is provided in Table 2.2-2 of Resource Report No. 2. During construction, ETC Tiger will adhere to the BMPs contained in the FERC Procedures which are designed to avoid and minimize impacts to waterbodies. This includes restricting the duration of in-water construction to the shortest time possible to allow pipeline construction across a waterbody.

During the clearing and grading phase of construction, temporary bridges will be installed across waterbodies in accordance with the FERC Procedures to allow construction equipment and personnel to cross. The bridges may include clean rock fill over culverts, timber mats supported by flumes, railcar flatbeds, flexi-float apparatuses, or other types of spans. Temporary bridges will be constructed to minimize the potential for sediment, grease, oil, or other pollutants to enter the waterbody. Construction equipment will be required to use the bridges, except that the clearing and bridge installation crews will be allowed one pass through waterbodies before bridges are installed, unless such crossings are restricted by the Project permits. The temporary bridges will be removed when construction and restoration activities are complete.

ATWS will be required for waterbody crossings to stage construction equipment, fabricate the pipeline, and store construction materials. ATWS will be located at least 50 feet away from the water's edge at each waterbody, as feasible, except as requested in **Table 1.3-3**, and approved by the Commission. The refueling and lubricating of equipment will also take place in upland areas that are at least 100 feet from the edge of the waterbody and adjacent wetlands. However, there will be certain instances where equipment refueling and lubricating may be necessary in or near waterbodies. For example, stationary equipment may need to

be operated continuously on the banks of waterbodies for the duration of the crossing and may require refueling in place. In these instances, stationary equipment within the 100-foot buffer will be placed in appropriately sized secondary containment, as described in the FERC Procedures and ETC Tiger's *SPAR Plan* (see **Appendix C-3**). ETC Tiger's *SPAR Plan* addresses the proper handling of fuel and other materials associated with the Project near sensitive resource areas and as required by the FERC Procedures, will be made available at each construction spread.

Open-Cut Method

The open-cut crossing method will involve trenching through the waterbody while water continues to flow through the trenching area. A typical open-cut waterbody crossing plan is provided in **Appendix B-1**.

Clearing adjacent to waterbodies will involve the removal of trees and brush from the construction ROW and ATWS areas. Woody vegetation within the construction ROW will be cleared to the edge of each waterbody. Initial grading of the herbaceous strip will be limited to the extent needed to create a safe approach to the waterbody and to install temporary bridges. During clearing, sediment barriers will be installed and maintained across the ROW adjacent to waterbodies and within ATWS to minimize the potential for sediment runoff. Silt fence or equivalent BMPs located across the working side of the ROW will be removed during the day when vehicle traffic is present and will be replaced each night.

Prior to initiating construction across the waterbody, the crossing section of pipeline will be fabricated (i.e., bent, welded, and coated) in adjacent ATWS areas. Excavators will then be used to excavate a trench in the flowing waterbody from one or both banks of the waterbody. Where the waterbody is too wide to excavate the trench from the banks, equipment may operate from within the waterbody with approval from the appropriate regulatory agencies. Equipment operating within the waterbody will be limited to that needed to construct the crossing. Spoil excavated from the trench will be placed on the bank above the high water mark (at least 10 feet from the edge of the waterbody) for use as backfill. The prefabricated segment of pipeline will then be placed into the trench using side-boom tractors or similar equipment. Concrete or sandbag weights will be utilized, as necessary, to provide negative buoyancy for the pipeline. During these operations, flow will be maintained at the crossing as specified in the FERC Procedures.

After the pipeline is installed across a waterbody, the trench will be backfilled with native material excavated from the trench. The streambed profile will be restored to pre-construction contours and grade conditions to prevent scouring. The stream banks will then be restored as near as practicable to preconstruction conditions or will be restored to a stable angle of repose if pre-construction stream bank conditions were unstable. Excavated material not required for backfill will be removed and disposed of at approved upland disposal sites. Once the stream banks have been reestablished, the banks will be stabilized using seeding, vegetation planting, installation of erosion control blankets, or installation of riprap materials, as appropriate and where agency approval for installation of riprap has been received. Temporary erosion controls will be installed immediately following bank restoration. The waterbody crossing area will be inspected and maintained until restoration of vegetation is complete.

Flume

The flume crossing method is an alternative to the open-cut method in which water flow is temporarily directed through one or more flume pipes placed over the excavation area. Temporary dams (consisting of sandbags, bladders, or other impervious materials) are installed upstream and downstream of the proposed crossing and are used to divert water into the flume(s). The use of the flume allows trenching and pipeline installation to occur primarily under dry conditions without significant disruption of water flow.

Dam and Pump

The dam and pump crossing method is an alternative to the open-cut method that allows trenching and pipeline installation to occur under relatively dry conditions with minimal impact on water flow. This

method involves the temporary installation of dams (consisting of sandbags, bladders, or other impervious materials) upstream and downstream of the proposed crossing. Pumps are then used to dewater the excavation area and to transport the water flow around the construction work area.

1.3.1.4 Horizontal Bore

Horizontal bores consists of installing a pipe along a horizontal path beneath a surface feature from excavated pits on each end. This type of bore is commonly referred to as "dry" bores due to the construction method not requiring any fluid or slurry to aid in the installation. Depending on site specific conditions and requirements, Horizontal bore installation can be guided or unguided with a typical steering tolerance of less than one foot for guided bores.

1.3.1.5 Railroad and Road Crossings

ETC Tiger will use trenchless construction methods, such as conventional bore or direct pipe, for paved road and railroad crossings in accordance with state, county, and local permits. Unpaved roads and driveways will typically be open-cut while railroads will typically be conventional bore. Constructing the Project across public and private roadways and railroads will be based on ETC Tiger's specifications, site conditions, as well as USDOT regulations, including 49 CFR Part 192 requirements regarding depth of cover. A list of road and railroad crossings associated with the Project is provided in **Table 1.3-1** and additional information is provided in Table 8.3-2 of Resource Report No. 8.

For unpaved roads that will be open-cut, at least one traffic lane will be maintained, except for brief periods essential to laying the new pipeline. If Project construction crosses roads that access private residences or businesses and no alternate entrance exists, measures will be taken to maintain passage for landowners during construction. Attempts will be made to avoid peak traffic time periods during construction that temporarily closes roads.

Most paved roads and railroads will be crossed via a bore. A conventional bore is performed by excavating a pit for the auger/bore machine on one side of the road being crossed and a receiving pit on the other side of the road. The bore machine jacks the pipe along the alignment while simultaneously excavating the soil with a rotating cutting head or auger. The installed segment is then tied into the remainder of the pipeline. A typical conventional bored road and railroad crossing plan is included in **Appendix B-1.**

1.3.1.6 Residential Construction

Residences within 50 feet of construction work areas are identified in Table 8.2-1 of Resource Report No. 8. Where residences are located within 50 feet of the edge of the construction ROW, ETC Tiger has reduced the construction ROW as practicable to minimize impacts on property owners. No residences are located less than 25 feet from the construction work area.

To minimize impacts on residences, ETC Tiger will stake the limits of disturbance and the centerline of the pipeline and will coordinate with property owners before and throughout the construction process. Owners of residential property crossed by the pipelines will be contacted to notify them of the approximate time that construction will take place on their property and to keep them informed about any special construction activities that may concern them. Safety fencing will be installed along the boundaries of the active construction ROW including temporary workspace areas and pipe and contractor yards as necessary and special care will be taken to control noise and dust. Resource Report No. 8 provides additional details on the general measures to minimize construction-related impacts on all residences and other structures located within 50 feet of the construction ROW.

ETC Tiger has developed its *Environmental Complaint Resolution Procedures* to address concerns or complaints received from affected landowners during construction of the Project (**Appendix C-7**). Affected landowners will be provided with a copy of the *Environmental Complaint Resolution Procedures* prior to the commencement of construction. ETC Tiger is committed to working closely with affected landowners to ensure that any concerns or complaints are resolved satisfactorily and in a timely manner.

1.3.1.7 Active Cropland Construction

To avoid and minimize effects on agricultural lands, ETC Tiger will adhere to the FERC Plan during construction in agricultural land. ETC Tiger will utilize mitigation measures such as topsoil segregation and/or topsoil compaction testing. ETC Tiger will utilize topsoil segregation as required by landowner agreements or the Natural Resource Conservation Service District, as appropriate based on site-specific conditions. ETC Tiger will coordinate with landowners and local agencies, as appropriate, to identify agricultural drainage systems. ETC Tiger will repair or replace drainage tiles that are damaged during Project construction.

1.3.2 Aboveground Facilities

The proposed aboveground facilities will be constructed in compliance with the same federal regulations and guidelines as the pipeline facilities, and in accordance with the specific requirements of applicable federal and state approvals. ETC Tiger will conduct the aboveground facility construction activities in accordance with the FERC Plan, FERC Procedures, and ETC Tiger's SPAR Plan, where appropriate.

ETC Tiger will install the proposed seven new meter stations and FF Interconnect Site in upland areas within the permanent ROW. At each site, ETC Tiger will install and maintain a gravel pad, as well as a chain-link-fence enclosure around the new facility.

Aboveground valve and piping will be installed using pipe supports. All aboveground piping surfaces will be cleaned and painted in accordance with ETC Tiger painting specifications. All paint inspection and cleanup will be conducted in accordance with regulatory requirements. Piping installed below grade will be coated for corrosion protection prior to backfilling, and an auxiliary cathodic protection system will be installed to protect ETC Tiger's underground piping.

Prior to placing the facilities in-service, the completed above- and below-ground gas piping systems will be hydrostatically tested in accordance with the requirements of USDOT pipeline safety regulations, 49 CFR Part 192, ETC Tiger testing specifications, and applicable permits. This testing will assure that the above and below-ground gas piping systems are free from leaks and will provide the required margin of safety for operation at the anticipated natural gas pressures. Hydrostatic test-water will be obtained from identified wells within the Project area. Further information regarding these wells is within Resource Report No. 2.

1.3.3 Proposed Site-Specific Deviations to the FERC Plan and Procedures

ETC Tiger is committed to constructing the Project in accordance with the FERC Plan and FERC Procedures to the extent feasible. However, there are places where the topography, ROW, and natural conditions make it impractical to implement some of the measures specified in the FERC Procedures. **Table 1.3-2** presents the site-specific deviations to the FERC Procedures identified to date and includes the justification for why each deviation is necessary as well as any proposed equal compliance measures.

TABLE 1.3-2
Requested Deviations from FERC Procedures for ATWS within 50 feet of Wetlands and Waterbodies

ATWS ID	Project Facility	Milepost	Area (Acres)	Average Width (feet)	Average Length (feet)	Requested Deviation	Equal Compliance Measure	Justification
FFL-ATWS-0039	Franklin Farms Lateral	3.31	0.12	Varies	Varies	VI.B.1.a	The wetland shall be matted and protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for the safe movement/offloading of equipment and materials north of Interstate 20, approximately 0.01 acres of a wetland (PEM-T2F-10) will be impacted by FFL-ATWS-0039. The proposed ATWS location was identified to reduce the impact to the public by placing the offloading site in a location to allow the construction contractor to conduct activities away from Mengle Road and the single-family home.
FFL-ATWS-0040	Franklin Farms Lateral	3.31	0.88	100	385	VI.B.1.a	The wetland shall be matted and protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for the safe movement/offloading of equipment and materials north of Interstate 20, approximately 0.01 acres of a wetland (PEM-T2F-10) will be impacted by FFL-ATWS-0040. The proposed ATWS location was identified to reduce the impact to the public by placing the offloading site in a location to allow the construction contractor to conduct activities away from Mengle Road and the single-family home.
FFL-ATWS-0067	Franklin Farms Lateral	6.91	0.18	50	155	VI.B.1.a	The wetland shall be matted and protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for safe construction operations for the pipeline installation of two stream crossings (S-T1F-02 and S-T1F-04), FFL-ATWS-0067 is placed within a wetland (PFO-T1F-01).
FFL-ATWS-0068	Franklin Farms Lateral	6.96	0.10	50	89	V.B.2.a, VI.B.1.a	The wetlands and streams shall be protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for safe construction operations for the pipeline installation of two stream crossings (S-T1F-02 and S-T1F-04), FFL-ATWS-0068 is placed within a wetland (PFO-T1F-01) and within 50 feet of streams S-T1F-02 and S-T1F-03.
FFL-ATWS-0069	Franklin Farms Lateral	7.02	0.19	50	168	VI.B.1.a	The wetland shall be matted and protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for safe construction operations for the pipeline installation of one stream crossing (S-T1F-04), FFL-ATWS-0069 is placed within a wetland (PFO-T1F-01).

TABLE 1.3-2
Requested Deviations from FERC Procedures for ATWS within 50 feet of Wetlands and Waterbodies

ATWS ID	Project Facility	Milepost	Area (Acres)	Average Width (feet)	Average Length (feet)	Requested Deviation	Equal Compliance Measure	Justification
FFL-ATWS-0070	Franklin Farms Lateral	7.09	0.12	25	202	VI.B.1.a	The wetland shall be matted and protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for safe construction operations for the pipeline fitting installation, FFL-ATWS-0070 is placed within a wetland (PFO-T1F-01).
FFL-ATWS-0071	Franklin Farms Lateral	7.13	0.19	50	164	VI.B.1.a	The wetland shall be matted and protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for safe construction operations for the pipeline installation of one stream crossing (S-T1F-04), FFL-ATWS-0071 is placed within a wetland (PFO-T4F-03).
FFL-ATWS-0072	Franklin Farms Lateral	7.19	0.19	50	165	VI.B.1.a	The wetland shall be matted and protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for safe construction operations for the pipeline installation of one stream crossing (S-T1F-04), FFL-ATWS-0072 is placed within a wetland (PFO-T4F-03).
FFL-ATWS-0079	Franklin Farms Lateral	9.41	0.17	50	150	VI.B.1.a	The wetland shall be matted and protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for safe construction operations of the horizontal bore crossing the existing pipeline corridor and Smalling Road FFL-ATWS-0079 is placed within a wetland (PFO-T1F-06).
FFL-ATWS-0114	Franklin Farms Lateral	12.50	0.13	Varies	Varies	VI.B.1.a	The wetland shall be matted and protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for safe construction operations of the horizontal bore crossing Jaggers Ln. and the pipeline fitting installation, the temporary workspace is not reduced and FFL-ATWS-0114 impacts less than 0.01 acres of a wetland (PFO-T2F-05).
FFL-ATWS-0115	Franklin Farms Lateral	12.55	0.12	25	207	V.B.2.a, VI.B.1.a	The wetlands and stream shall be protected utilizing BMPs in close collaboration with the on-site environmental inspector.	To allow for safe construction operations of the horizontal bore crossing Jaggers Ln. and the pipeline fitting installation, FFL-ATWS-0115 is placed within 50-feet of two wetlands (PFO-T2F-05 and PFO-T2F-07) and one stream (S-T2F-09).

1.3.4 Construction Schedule and Workforce

ETC Tiger anticipates commencing construction July 2027, subject to the receipt of necessary permits and approvals. ETC Tiger also anticipates that construction will take six months to complete. Restoration activities will continue after the Project is placed in-service and will continue until disturbed areas are stabilized in accordance with the FERC Plan, FERC Procedures, and applicable permit requirements. **Table 1.3-4** details the anticipated construction schedule and peak workforce for the Project facilities.

TABLE 1.3-3
Construction Schedule and Workforce by Project Facility

Facility	Anticipated Start Date	Anticipated Completion Date	Peak Work Force
Franklin Farms Lateral	July 2027	January 2028	250
FF Trio Farm Lateral	July 2027	January 2028	100
FF Gulf Run Bi Di Meter Station	July 2027	November 2027	60
FF Gulf South Receipt Meter Station	July 2027	November 2027	60
FF Bee Bayou Delivery Meter Station	July 2027	November 2027	60
FF Gulf Crossing Receipt Meter Station	September 2027	January 2028	60
FF MEP Receipt Meter Station	September 2027	January 2028	60
Franklin Farms Delivery Meter Station	July 2027	November 2027	60
FF Trio Farm Delivery Meter Station	September 2027	January 2028	60

ETC Tiger anticipates construction of the Project will be accomplished using one construction spread. It is anticipated that ETC Tiger will not need to hire any permanent workers for operation of the pipeline facilities.

Construction is anticipated to occur Monday through Saturday from 6 AM through 10 PM. Construction will only occur 24 hours per day and/or on Sundays or holidays, as necessary. Hydrostatic testing is anticipated to be 24-hours per day during critical activities.

1.4 ENVIRONMENTAL COMPLIANCE, TRAINING AND INSPECTION

Prior to construction, ETC Tiger will provide its contractors with copies of specifications, an "approved for construction" Construction Drawing Package, and all environmental permits, certificates, and/or clearances associated with the Project so that construction of the proposed facilities will comply with the measures identified in this Environmental Report and all applicable permitting agencies. Additionally, ETC Tiger will conduct environmental training for its field construction personnel and construction contractor's personnel prior to and during construction of the Project. This training will focus on implementation of the FERC Plan, FERC Procedures, and other Project-specific permit conditions and mitigation measures, as appropriate.

In accordance with the FERC Plan, ETC Tiger will employ at least one Environmental Inspector (EI) during construction. The EI's duties will include, but are not limited to, ensuring compliance with all environmental conditions. The EI will have peer status with any/all other inspectors; will be present throughout construction and restoration; and will have the authority to enforce permit and FERC environmental conditions, to issue stop-activity orders, and impose corrective actions to maintain environmental compliance.

1.5 OPERATION AND MAINTENANCE

ETC Tiger will continue to operate and maintain its facilities in compliance with USDOT regulations, 49 CFR Part 192, and the maintenance provisions of the FERC Plan as applicable to ensure the integrity and safe operation of the facilities and to maintain firm natural gas transportation service.

Operations and maintenance activities include vegetation management, pipeline inspections, and cleaning and pipeline repairs and maintenance. ETC Tiger's personnel will continue to perform routine checks of the facilities, including calibration, inspection, and scheduled/preventative maintenance of equipment. Corrective actions will be taken regarding any identified problem.

1.6 FUTURE PLANS AND ABANDONMENT

ETC Tiger currently has no plans for future expansion or abandonment of the Project facilities described in this application. At the end of the useful life of the pipeline facilities, ETC Tiger will obtain the necessary authorizations for abandonment.

1.7 PERMITS AND APPROVALS

Table 1.7-1 lists the required permits and approvals, dates for submittal of applications, and anticipated date of receipt for the Project. ETC Tiger will obtain all necessary permits, licenses, clearances, and approvals related to construction and operation of the Project. Agency and stakeholder correspondence is included in **Appendix D**.

TABLE 1.7-1
Federal and State Environmental Permits, Approvals and Consultations

Agency	Permit/Approval/Consultation/Coordination	Submittal Date (Anticipated)	Approval Date (Anticipated)
Federal			
FERC	Certificate of Public Convenience and Necessity under Section 7(c) of the NGA	Certificate Application filed herewith	(June 2027)
USACE Vicksburg District	Section 404 CWA Nationwide Permit (NWP) 12	Pre-Construction Notification (PCN) (June 2026)	(October 2026)
United States Fish and Wildlife Service (USFWS), Louisiana Ecological Services Field Office	Consultations under Section 7 of the Endangered Species Act, the Migratory Bird Treaty Act, and the Fish and Wildlife Coordination Act	June 8, 2026	(October 2026)
State of Louisiana			

TABLE 1.7-1
Federal and State Environmental Permits, Approvals and Consultations

Agency	Permit/Approval/Consultation/Coordination	Submittal Date (Anticipated)	Approval Date (Anticipated)
Louisiana Department of Wildlife and Fisheries (LDWF) Wildlife Diversity Program	LDWF State Protection Status Coordination	June 8, 2026	(August 2026)
Louisiana Department of Environmental Quality (LDEQ)	Section 401 CWA Water Quality Certification (WQC)	Concurrent with PCN NWP 12 authorization	Concurrent with PCN NWP 12 authorization, LDEQ issued WQC for NWP 12
	Louisiana Pollutant Discharge Elimination System (LPDES) General Permit for Hydrostatic Test and Vessel Testing Wastewater (LAG6700000))	Notice of Intent (NOI) to be Submitted 48 Hours Prior to Discharge	48 Hours After NOI Submittal
	Storm Water General Permit for Large Construction Activities (LAR100000)	Exempt per Louisiana Administrative Code (LAC) 33:IX.25.11.A.2.	Exempt per LAC 33:IX.25.11.A.2.
Louisiana - State Historic Preservation Office (SHPO)	Consultation for cultural resources under Section 106 of the NHPA	Scope of Work Submitted February 2, 2026 Phase I Cultural Resources Survey Submitted on April 24, 2026 Amended Phase I Cultural Resources Survey Submitted on June 10, 2026	Scope of Work Approval February 4, 2026 Phase I Cultural Resources Survey/Amendment Concurrence (July 2026)
Local			
Richland Parish	Floodplain Permit	Prior to Construction	Prior to Construction
Tribes			
Federal and State of Louisiana Recognized Tribes	Coordination for cultural resources under Section 106 of the NHPA	June 11, 2026	Ongoing

1.8 AFFECTED LANDOWNERS

A list of affected landowner and stakeholder names and addresses is included in **Appendix E-1** and is submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*". ETC Tiger will comply with 18 CFR Section 157.6(d) regarding notification to landowners, local governments and regulatory agencies pursuant to Orders 609 and 609-A with the required landowner notification.

1.9 NON-JURISDICTIONAL FACILITIES

The Project will require minor third-party facilities to provide utilities for the Project facilities as detailed in **Table 1.9-1**. These will be completed by local utility providers. It is ETC Tiger's understanding that any non-jurisdictional power service to be constructed to support the proposed Project will be subject to state and local jurisdictions and that these utility providers will obtain all required permits and approvals prior to initiating construction of the non-jurisdictional facilities.

TABLE 1.9-1
Non-Jurisdictional Facilities Associated with the Project

Facility	Associated Pipeline	Milepost	Non-Jurisdictional Facility
FF Gulf Run Bi Di Meter Station	Franklin Farms Lateral	0.0	120/240V - 100 A Service
FF Gulf South Receipt Meter Station	Franklin Farms Lateral	0.0	120/240V - 100 A Service
FF Bee Bayou Delivery Meter Station	Franklin Farms Lateral	4.5	120/240V - 100 A Service
FF Interconnect Site	Franklin Farms Lateral/ FF Trio Farm Lateral	8.7 / 0.0	120/240V - 100 A Service
FF Gulf Crossing Receipt Meter Station	Franklin Farms Lateral	9.7	120/240V - 100 A Service
FF MEP Receipt Meter Station	Franklin Farms Lateral	10.8	120/240V - 100 A Service
Franklin Farms Delivery Meter Station	Franklin Farms Lateral	13.2	120/240V - 100 A Service
FF Trio Farm Delivery Meter Station	FF Trio Farm Lateral	1.7	120/240V - 100 A Service

In addition, the proposed Project will be providing natural gas to Entergy Louisiana, LLC's Franklin Farms Power Stations 1 and 2. This facility will include construction of a pair of new combined-cycle combustion turbine generation facilities to provide 754 MW each for a total of 1,500 MW of natural gas generation capacity to support the Hyperion Data Center, and the facility will overlap with Project workspaces. The Entergy Franklin Farms Power Stations 1 and 2 are expected to be operational in 2028. Given the proximity of the Entergy Franklin Farms Power Stations 1 and 2 to the proposed Project, environmental impacts associated with construction and operation of this non-jurisdictional facility are identified and are discussed in the applicable cumulative impacts sections of the resource reports in this Environmental Report (refer to **Section 1.10** of this Resource Report).

The federal government will not provide financial support, nor are federal lands utilized for any of the above-described non-jurisdictional activities; as a result, there is no federal oversight or responsibility associated with these utilities. For these reasons, environmental review of these non-jurisdictional facilities by the Commission is not warranted.

1.10 CUMULATIVE IMPACTS

Cumulative impacts may result when the environmental impacts associated with the proposed Project are added to the temporary (construction-related) or permanent (operation-related) impacts associated with

past, present, or reasonably foreseeable future projects within the same resource impact analysis area, referred to as the Region of Interest (ROI). This cumulative impact assessment includes impacts associated with the proposed Project, pursuant to Section 380.12(b)(3) of the Commissions regulations.

The purpose of this cumulative impact analysis is to identify and describe the potential cumulative impacts that could result from implementation of multiple projects located in the same resource-specific ROI as the proposed Project. This analysis uses an approach consistent with the methodologies set forth in the Council on Environmental Quality's (CEQ) 2005 Guidance on the Consideration of Past Actions in Cumulative Effects Analysis (CEQ, June 2005). The analysis focuses on resource areas or issues where the incremental contribution of other projects could be potentially significant when added to the potential impacts of the proposed Project. The actions considered in this cumulative impact analysis may vary from the proposed Project in nature, magnitude, and duration; however, the action must meet the following three criteria to be included in the cumulative impacts analysis:

- Impacts a resource area potentially affected by the proposed Project;
- Causes this impact within all, or part of, a resource-specific ROI (see **Section 1.11.1**); and
- Causes this impact within all, or part of, the time span of potential impacts from the proposed Project.

1.10.1 Region of Influence

Selection of projects to be included in the cumulative impact analysis was dependent on their being located within the ROI in which impacts may have the potential to be cumulative. The actual geographic extent of the ROI considered varied by the affected resource, resulting in resource-specific ROIs. The CEQ recommends setting the geographic scope of a cumulative analysis based on the natural boundaries of the resource affected, as opposed to jurisdictional boundaries. Further, FERC advises that the spatial migration of impacts away from construction and/or operation work areas should be considered. When doing so, mitigation measures (e.g., FERC Plan and FERC Procedures) should also be taken into consideration. In accordance with this guidance, the general ROI (geographic scope) that was considered per individual resource type is provided in **Table 1.10-1**.

TABLE 1.10-1
Resource-Specific Determination of Region of Influence (Geographic Scope) for Cumulative Impacts

Resource(s)	ROI (Geographic Scope)
Groundwater	Hydrologic Unit Code (HUC) 12 Watershed [a] (Congo Creek-Big Creek: 080500011002; Cow Bayou-Big Creek: 080500011004; Cypress Bayou-Bayou Macon: 080500020505; Cypress Creek: 080500011003; Dry Fork Creek-Bee Bayou: 080500011005; Turkey Creek: 080500011007)
Surface Water Resources	HUC 12 Watershed
Wetlands	HUC 12 Watershed
Fisheries and Wildlife [b]	HUC 12 Watershed
Vegetation	HUC 12 Watershed
Cultural Resources	Overlapping impacts within the Area of Potential Effect (APE)
Geology and Soils	Overlapping impacts within the Project footprint
Land Use and Recreation	1-mile radius

TABLE 1.10-1
Resource-Specific Determination of Region of Influence (Geographic Scope) for Cumulative Impacts

Resource(s)	ROI (Geographic Scope)
Visual	For aboveground facilities, the distance that the tallest feature at the planned facility would be visible from neighboring communities. For pipelines, a distance of 0.25-miles and existing visual access points (e.g., road crossings).
Noise	0.25-mile from Project workspace
Socioeconomics	Within the same county (Jefferson, Monroe)
Air Quality	Construction: 0.25-mile, Operation: 1-mile (M&R station)

Notes:
ROI=Region of Interest
[a] HUC 12 is considered a local sub-watershed level that captures tributary systems.
[b] Includes federal and state-listed threatened and endangered species.

1.10.2 Identification of Past, Present, and Reasonably Foreseeable Projects

The current environment of the proposed Project area reflects a mixture of natural processes and human influences (e.g., existing pipelines, roadways). The CEQ issued an interpretive memorandum on June 24, 2005, regarding analysis of past actions, which stated: "*agencies can conduct an adequate cumulative effects analysis by focusing on the current aggregate effects of past actions without delving into the historical details of individual past actions*". To understand the contribution of past actions to the cumulative impacts, the analysis provided below relies on current environmental conditions as a proxy for the impacts of past actions. Existing conditions reflect the aggregate impact of all prior human actions and natural events that have affected the environment and might contribute to cumulative effects. Thus, the overview of anticipated cumulative impacts provided in this analysis is based on the environmental baseline described in the resource-specific sections in this report and the projects included in the analysis represent recent past (in development since 2025), present, or reasonably foreseeable actions.

ETC Tiger identified recent past (in development since 2025), present, and reasonably foreseeable projects from a review of the publicly available sources including but not limited to the following:

- Federal Agencies
    - Federal Infrastructure Projects Permitting Dashboard (Permitting Dashboard, 2026);
    - The FERC eLibrary system, pending applications and recently approved (2015-Present) projects (FERC, 2026a, 2026b);
    - USACE Jurisdictional Determinations and Permit Decisions website (USACE, 2026);
    - USACE Vicksburg Public Notices (2025-present) (USACE Vicksburg District, 2026);
    - United States Department of Energy (USDOE), Active NEPA Projects (USDOE, 2026); and
    - United States Environmental Protection Agency (EPA) Environmental Impact Statement (EIS) database (EPA, 2026).
- State Agencies
    - Louisiana Department of Transportation (2026);
    - Louisiana Department of Conservation and Energy (2026); and

- Louisiana Department of Environmental Quality (2026).

Projects either pending or approved in 2025 or later were screened within these databases. Individual projects were then assessed to determine if they were located within a resource-specific ROI. If a project was determined to be located within a resource-specific ROI, it was considered to have the potential to contribute to cumulative effects for that resource.

1.10.3 Projects Potentially Contributing to Cumulative Effects

A summary of the recent past, present, and reasonably foreseeable projects identified with the potential for cumulative impacts with the proposed Project is provided in **Table 1.10-2**. The individual projects in **Table 1.10-2** were reviewed to determine if they warranted being carried forward in the analysis. The general location of the identified projects in relation to the proposed Project area is depicted in **Appendix A-8**. A discussion of potential resource-specific cumulative impacts is provided in **Section 1.10.4**.

TABLE 1.10-2
Projects Potentially Contributing to Cumulative Impacts to the Project

ID No.	Project Name (Sponsor)	Description	Location Relative to the Project		Status		Driver for Cumulative Effect	Carried Forward for Analysis	Source
			Distance (miles)/Facility	Direction	Past, Present, Reasonably Foreseeable	Timeline and Permitting			
1	Oak Grove Enhancement Project (ANR Pipeline Company)	Construction of 34.1 miles of new 30-inch-diameter natural gas pipeline to replace 33.6 miles of existing 30-inch-diameter natural gas pipeline in Richland and West Carroll Parishes, Louisiana.	0 / Contractor Yard 3	Overlaps Project contractor yard	Past	Construction complete as of November 2025	Groundwater, Surface water resources, Wetlands, Fisheries and wildlife, Vegetation, Cultural Resources, Geology and Soils, Land use and recreation, Visual, Noise, Socioeconomics, Air quality - construction	No, the construction window does not overlap with the proposed Project. The proposed Project utilizes the same contractor yard area as the Oak Grove Enhancement; however, the Oak Grove Enhancement contractor yard is no longer being utilized as construction is now complete. For the proposed Project, the contractor yard will be returned to pre-construction conditions after Project construction. Additionally, all other Oak Grove Enhancement Project components are located further than 1-mile away from the proposed Project.	FERC, 2026b; FERC 2026c
2	Titanium Transmission Line Project (Entergy Louisiana, LLC)	Construction of a new 500 kV switchyard (Car Gas Road Switchyard) and two new 500 kV transmission lines, each approximately 27 miles in length, to connect the Switchyard to the Smalling Substation. The new infrastructure will meet the anticipated energy demands of a new 2,640 MW hyperscale data center (Hyperion Data Center).	0 / Franklin Farms Lateral	Crosses Project	Present	Construction anticipated to begin Summer 2026 and finish at the end of 2027	Groundwater, Surface water resources, Wetlands, Fisheries and wildlife, Vegetation, Cultural Resources, Geology and Soils, Land use and recreation, Visual, Noise, Socioeconomics, Air quality - construction	Yes, the construction window could overlap with the proposed Project, and it could result in similar impacts.	Entergy Louisiana LLC, 2026
3	Smalling Substation (Entergy Louisiana, LLC)	The Smalling Substation will cover about 55 acres and will connect to high-voltage power lines that bring energy from across the region and deliver it to the Hyperion Data Center. Entergy will construct eight 230 kV transmission lines from the Smalling Substation to six substations located on the Hyperion Data Center.	0.8 / Franklin Farms Delivery Meter Station	S	Present	Construction began June 2025. Unknown anticipated completion date.	Groundwater, Surface water resources, Wetlands, Fisheries and wildlife, Vegetation, Land use and recreation, Visual, Socioeconomics, Air quality - operation	Yes, the construction window could overlap with the proposed Project, and it could result in similar impacts.	Entergy Louisiana LLC, 2025a
4	Franklin Farms Power Station 1 & 2 (Entergy Louisiana, LLC)	The construction of a pair of new combined-cycle combustion turbine generation facilities to provide 754 MW each for a total of 1,500 MW of natural gas generation capacity to support the Hyperion Data Center.	0 / Franklin Farms Lateral	Crosses Project	Present	Construction began December 2025 and is anticipated to be complete in 2028.	Groundwater, Surface water resources, Wetlands, Fisheries and wildlife, Vegetation, Cultural Resources, Geology and Soils, Land use and recreation, Visual, Noise, Socioeconomics, Air quality - construction; Air quality - operation	Yes, the construction window could overlap with the proposed Project, and it could result in similar impacts.	Entergy Louisiana LLC, 2025b
5	Hyperion Data Center (Meta Platforms, Inc.)	Construction of a hyperscale data center campus with nine buildings planned. The property is over 2,250 acres and is in Richland Parish. The data center campus will be 4 million square feet and would total more than 2 gigawatts at full build-out. The Hyperion Data Center will be used to train Llama AI models.	0.8 / Franklin Farms Delivery Meter Station	S	Present	Construction ongoing and anticipated to be complete in 2030	Groundwater, Surface water resources, Wetlands, Fisheries and wildlife, Vegetation, Land use and recreation, Visual, Socioeconomics, Air quality - operation	Yes, the construction window could overlap with the proposed Project, and it could result in similar impacts.	Louisiana Economic Development, 2024
6	Workforce Housing - Holly Ridge, LA (Mammoth Industries, Inc)	Development of an approximate 130-acre workforce housing complex to support construction of the Hyperion Data Center.	0.8 / Franklin Farms Delivery Meter Station	S	Past	Construction began April 2025 and was completed in November 2025	Groundwater, Surface water resources, Wetlands, Fisheries and wildlife, Vegetation, Land use and recreation, Visual, Socioeconomics, Air quality - operation	No, the construction window does not overlap with the proposed Project. Additionally, construction and operation of this project is anticipated to have negligible to minor environmental effects since work will adhere to federal, state, and local environmental permits; thus, it is unlikely to lead to significant cumulative impacts with the scope of the proposed Project.	Mammoth Industries, Inc., 2025

TABLE 1.10-2
Projects Potentially Contributing to Cumulative Impacts to the Project

ID No.	Project Name (Sponsor)	Description	Location Relative to the Project		Status		Driver for Cumulative Effect	Carried Forward for Analysis	Source
			Distance (miles)/Facility	Direction	Past, Present, Reasonably Foreseeable	Timeline and Permitting			
7	LA 133: LA 15 - US 80 (Louisiana DOT)	Existing roadway improvement project for drainage structures, pavement patching, and asphalt concrete overlay.	9.7 / FEL-TAR-009.000	W	Past	Work started November 2024, Open to traffic June 2025	Socioeconomics	No, the construction window does not overlap with the proposed Project. Additionally, construction and operation of this project is anticipated to have negligible socioeconomic effects since work will be done on an existing roadway and is unlikely to lead to significant cumulative impacts with the scope of the proposed Project.	Louisiana Department of Transportation, 2026
8	LA 15: Boeuf River Bridge Replacement (Louisiana DOT)	Bridge replacement project over the Boeuf River.	12.5 / Franklin Farms Lateral	SW	Past	Work started September 2021, Open to traffic January 2025	Socioeconomics	No, the construction window does not overlap with the proposed Project. Additionally, construction and operation of this project is anticipated to have negligible socioeconomic effects since work will be done on an existing roadway and is unlikely to lead to significant cumulative impacts with the scope of the proposed Project.	Louisiana Department of Transportation, 2026
9	LA 132: Bridges near Mangham (Louisiana DOT)	Bridge replacements along LA 132.	10.8 / Franklin Farms Lateral	SW	Past	Work started October 2023, Open to traffic March 2025	Socioeconomics	No, the construction window does not overlap with the proposed Project. Additionally, construction and operation of this project is anticipated to have negligible socioeconomic effects and is unlikely to lead to significant cumulative impacts with the scope of the proposed Project.	Louisiana Department of Transportation, 2026
10	D05 Rumble Strips and Striping Ph 1 (Louisiana DOT)	This project involves installing rumble strips, striping, and related work along several existing roadways.	Adjacent / Contractor Yard 3	S	Past	Work started March 2024, Open to traffic October 2025	Groundwater, Surface water resources, Wetlands, Fisheries and wildlife, Vegetation, Land use and recreation, Visual, Noise, Socioeconomics, Air quality - construction	No, the construction window does not overlap with the proposed Project. Additionally, construction and operation of this project is anticipated to have negligible environmental effects since work will be done on an existing roadway and none with the potential to lead to significant cumulative impacts with the scope of the proposed Project.	Louisiana Department of Transportation, 2026
11	D05 Rumble Strips and Striping Ph 2 (Louisiana DOT)	This project involves installing rumble strips, striping, and related work along several existing roadways.	4.8 / Franklin Farms Lateral	W	Present	Work started February 2025, Anticipated completion July 2026	Groundwater, Socioeconomics	No, the construction window is not anticipated to overlap with the proposed Project. Additionally, construction and operation of this project is anticipated to have negligible groundwater and socioeconomic effects since work will be done on an existing roadway and none with the potential to lead to significant cumulative impacts with the scope of the proposed Project.	Louisiana Department of Transportation, 2026
12	McLemore Road Over Bee Bayou (Louisiana DOT)	Bridge replacement project over the Bee Bayou.	4.1 / Franklin Farms Lateral	W	Present	Work started February 2022, Anticipated completion June 2026	Groundwater, Socioeconomics	No, the construction window is not anticipated to overlap with the proposed Project. Additionally, construction and operation of this project is anticipated to have negligible groundwater and socioeconomic effects since work will be done on an existing roadway and none with the potential to lead to significant cumulative impacts with the scope of the proposed Project.	Louisiana Department of Transportation, 2026
13	LA 583: Sumlin Ranch Road - LA 134 (Louisiana DOT)	Existing roadway improvement project for asphalt surface treatment, patch, mill and overlay.	0 / FF Trio Farm Lateral	Crosses Project	Present	Work started August 2025, Anticipated completion June 2026	Groundwater, Surface water resources, Wetlands, Fisheries and wildlife, Vegetation, Cultural Resources, Geology and Soils, Land use and recreation, Visual, Noise, Socioeconomics, Air quality - construction	No, the construction window is not anticipated to overlap with the proposed Project. Additionally, construction and operation of this project is anticipated to have negligible environmental effects since work will be done on an existing roadway and none with the potential to lead to significant cumulative impacts with the scope of the proposed Project.	Louisiana Department of Transportation, 2026

1.10.4 Resource-Specific Cumulative Impacts

As described in Section 1.10.1, the ROI evaluated in this cumulative impact analysis varies by resource. Of the projects detailed in **Table 1.10-2**, four projects were carried forward for cumulative impacts analysis including the Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center. The analysis of these four projects for potential cumulative impacts with the proposed Project is provided in **Table 1.10-3**.

Given the limited scope of the proposed Project, ETC Tiger believes the Project will not result in any significant cumulative impacts. As discussed in this Environmental Report, the proposed Project will not result in any significant direct effects on waterbodies; wetlands; threatened, endangered, and candidate species; fisheries; soils; vegetation; cultural resources; geology; land use, recreation, and aesthetics; or air and noise emissions. A basic assumption of this cumulative impacts analysis is that if there are no Project-related impacts for a resource, then there would be no cumulative impacts for that particular resource. Similarly, if the Project activities are determined to have a negligible impact on a resource, the potential cumulative impacts resulting from the Project would also be negligible on that particular resource.

TABLE 1.10-3
Resource-Specific Determination of the Proposed Franklin Farms Project's Cumulative Impacts

Resource	Description of Potential Cumulative Impacts	Cumulative Impact Determination
Geology and Soils	Any effects to geology and soils will be highly localized, limited primarily to the proposed Project footprint, and limited to the period of active construction. Cumulative impacts on soils may include loss of excavated soil through water and wind erosion, soil compaction from movement of construction equipment, mixing of topsoil and subsoil, and would only occur if other geographically overlapping projects were constructed at the same time (and place) as the Project, and the exposure of soils to erosion and sedimentation occurs. The only projects identified in the geographic ROI that could result in cumulative effects to geology and soils with the proposed Project are the Titanium Transmission Line Project and the Franklin Farms Power Station 1 & 2 which cross the proposed Project area and overlap in construction schedule.	The proposed Project will adhere to the BMPs in FERC's Plan, FERC's Procedures, and ETC Tiger's *Fugitive Dust Control Plan*. While the Entergy projects cross the proposed Project area, it is unlikely that active construction will be occurring concurrently for both projects. Further, the Titanium Transmission Line Project and the Franklin Farms Power Station 1 & 2 projects are anticipated to have similar BMPs in place to protect geology and soils. Thus, the proposed Project is not anticipated to result in significant cumulative effects to geology and soils.
Groundwater	Potential cumulative impacts on groundwater may occur through construction activities, including clearing and grading, shallow excavations, dewatering, contamination through fuel and other hazardous material spills, and groundwater withdrawal. Cumulative impacts may also occur during operation, including changes in near-surface hydrology from the addition of impervious surfaces. The groundwater geographic ROI for cumulative effects corresponds to the HUC 12 watersheds. All four projects, Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center, are within a HUC 12 watershed that the proposed Project is in.	Potential affects to groundwater during construction of the proposed Project will be minimized with adherence to the BMPs in FERC's Plan, FERC's Procedures, and ETC Tiger's SPAR Plan. Groundwater withdrawal will be in accordance with state and local regulations. Further, the Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center are anticipated to have similar BMPs in place to protect groundwater and are expected to adhere to state and local regulations for groundwater use. Due to the limited scope of proposed Project construction within the HUC 12 watersheds, it is not anticipated to result in significant cumulative effects to groundwater.
Surface Waters, Fisheries, and Aquatic Resources	Potential cumulative impacts to surface water resources, fisheries, and aquatic resources could occur due to increased sedimentation and turbidity from surface water runoff, surface water use, and water contamination from accidental spills of construction-related fluids (e.g., oil, gasoline, or hydraulic fluids). Cumulative impacts may also occur due to changes in near-surface hydrology from the addition of impervious surfaces at aboveground facilities. The geographic ROI for cumulative effects to surface water resources corresponds to the HUC 12 watershed. All four projects, Titanium Transmission Line Project, Smalling Substation, Franklin Farms	Surface waters will not be used for Project hydrostatic test water. Project hydrostatic test water will be discharged in accordance with the FERC Plan and Procedures and will be discharged into a well vegetated area with filter bags and/or erosion control barrier to protect water quality. During operation of the proposed Project there will be no surface water use. While surface waters will be crossed using the open cut method by the proposed Project, ETC Tiger will utilize BMPs contained in the FERC Procedures which are designed to avoid and minimize impacts to waterbodies. This includes restricting the duration of in-water construction to the shortest time possible to allow pipeline construction across a waterbody. Additionally, the Project will adhere to

TABLE 1.10-3
Resource-Specific Determination of the Proposed Franklin Farms Project's Cumulative Impacts

Resource	Description of Potential Cumulative Impacts	Cumulative Impact Determination
	Power Station 1 & 2, and Hyperion Data Center, are within a HUC 12 watershed that the proposed Project is in.	Section 401 and 404 of the Clean Water Act and will implement ETC Tiger's SPAR Plan to protect water resources. Thus, the proposed Project is not anticipated to result in significant cumulative effects to surface waters, fisheries, or aquatic resources. All four projects, Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center, are anticipated to have similar BMPs in place to protect surface waters. Further, ETC Tiger expects that all four projects will adhere to Section 401 and 404 of the Clean Water Act, as applicable, to protect water resources. Resultingly, cumulative impacts are not anticipated.
Wetlands	The geographic ROI for cumulative effects to wetlands corresponds to the HUC 12 watersheds. Potential cumulative impacts to wetlands could occur due to permanent conversion of wetland types (palustrine forested (PFO) to palustrine emergent (PEM)) or permanent wetland impacts. All four projects, Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center, are within a HUC 12 watershed that the proposed Project is in.	The Project aboveground facilities will not impact any wetlands; however temporary and permanent wetland impacts are expected from construction and operation of the proposed Franklin Farms Lateral, Trio Farm lateral, and PARs. The contractor yards and TARs will have temporary impacts on wetlands. ETC Tiger will adhere to the USACE permit requirements regarding any compensatory mitigation for unavoidable impacts to wetlands. ETC Tiger plans to purchase compensatory wetland mitigation credits from a mitigation bank, as required. Further, ETC Tiger will avoid and minimize potential impacts by following the BMPs described in the FERC Plan and FERC Procedures. Thus, the proposed Project is not anticipated to result in significant cumulative effects to wetlands. All four projects, Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center, are anticipated to adhere to Section 404 of the Clean Water Act, as applicable, to mitigate for wetland impacts if applicable. Additionally, ETC Tiger expects that all four projects will utilize BMPs to avoid or minimize wetland impacts. Resultingly, cumulative impacts are not anticipated.
Vegetation, Wildlife, and Threatened and Endangered Species	The geographic ROI for cumulative effects to vegetation, wildlife, and protected species corresponds to the HUC 12 watersheds. Potential cumulative impacts to vegetation, wildlife and threatened and endangered species include loss of habitat and displacement of wildlife. All four projects, Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and	Construction and operation of the Project is not anticipated to have a significant impact on vegetation as most of the proposed construction activities will occur in areas of herbaceous vegetation which will be allowed to naturally revegetate from existing rootstocks and seedbanks following construction. Construction and operation of the Project is not expected to adversely affect the distribution or regional abundance of wildlife species given the short duration of the disturbance and the amount and distribution of similar habitat types available in

TABLE 1.10-3
Resource-Specific Determination of the Proposed Franklin Farms Project's Cumulative Impacts

Resource	Description of Potential Cumulative Impacts	Cumulative Impact Determination
	Hyperion Data Center, are within a HUC 12 watershed that the proposed Project is in.	the immediate Project area. With adherence to tree clearing outside of sensitive bat seasons and implementation of the BMPs in the FERC Plan, FERC Procedures, and ETC Tiger's SPAR Plan, it is anticipated that the Project construction will either have "*No Effect*" or "*May Affect, but is not Likely to Adversely Affect*" federal and state protected species with potential to occur in the Project area. Thus, construction and operation of the proposed Project would not contribute to cumulative impacts on vegetation, wildlife, or protected species. All four projects, Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center, are located in similar habitats as the proposed Project with abundant similar available habitat in the immediate area. Further, the proponents of the Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center are subject to the requirements under the Endangered Species Act and must consult with USFWS and state agencies. Therefore, cumulative impacts are not anticipated.
Cultural Resources	The geographical ROI for cultural resources is overlapping impacts within the APE. ETC Tiger conducted cultural resource surveys to identify resources within the APE. The Titanium Transmission Line Project and the Franklin Farms Power Station 1 & 2 cross the proposed Project area and overlap the APE.	As a result of Project cultural resource investigations, a finding of no historic properties affected and that no further cultural resources investigations are necessary are recommended. ETC Tiger has developed a Project-specific plan (*Unanticipated Discovery Plan*) to address unanticipated discoveries of cultural resources and human remains during construction of the proposed Project. With adherence to the procedures in ETC Tiger's *Unanticipated Discovery Plan*, the Project will not contribute to cumulative effects on cultural resources. ETC Tiger anticipates that the Titanium Transmission Line Project and the Franklin Farms Power Station 1 & 2 will also conduct cultural investigations and will also prepare a similar plan to address unanticipated discoveries of cultural resources. Thus, cumulative impacts are not anticipated.
Land Use, Recreation, and Visual Resources	The geographical ROI for land use, recreation, and visual resources is considered to occur within a 1-mile radius of the Project. All four projects, Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center, are within a 1-mile of the proposed Project.	ETC Tiger does not anticipate that installing the proposed Project facilities will significantly affect land use based on (1) the minor amount of land necessary to install the facilities, (2) the current use of these lands, and (3) except for a very limited amount of area, current land uses will be allowed to resume following construction. The Project pipelines will not have an impact on the nearby viewshed upon the completion of construction activities and commencement of operation, as the pipelines will be underground and no longer visible. Many of

TABLE 1.10-3
Resource-Specific Determination of the Proposed Franklin Farms Project's Cumulative Impacts

Resource	Description of Potential Cumulative Impacts	Cumulative Impact Determination
		the residences in the Project vicinity are a significant distance from the proposed meter stations and/or screened by vegetative buffers; therefore, the proposed Project would not contribute to any significant cumulative impacts land use, recreation, or visual resources. All four projects, Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center, will be located near each other in an area that has been selected for development by state and local agencies. Therefore, while permanent land use and visual impacts will occur, this will be consistent with approvals for these projects. Thus, cumulative impacts are not anticipated.
Socioeconomics	The ROI for socioeconomics is the Parish in which the Project occurs in. All four projects, Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center, are located in Richland Parish. Richland Parish has the necessary infrastructure to provide public services and utilities to support these projects. Simultaneous construction with the proposed Project could result in temporary cumulative impacts on road traffic. Increased use of local roadways from multiple projects could also accelerate degradation of roadways and require early replacement of road surfaces.	Due to the limited scope of the proposed Project, it is not anticipated to contribute to any significant cumulative impacts to local populations, employment, housing, or public services. Additionally, long-term positive impacts from the Project's operation will arise from the increased property tax base. Given the relatively small construction workforce, limited duration of construction activities required, and use of roads outside of peak hours, cumulative impacts to traffic from proposed Project construction are not anticipated to be significant. Further, the proposed Project will not contribute to any long-term cumulative impact on traffic or transportation infrastructure because additional (new) permanent employees will not be required for operation of the proposed Project facilities. None of the projects identified, Titanium Transmission Line Project, Smalling Substation, Franklin Farms Power Station 1 & 2, and Hyperion Data Center, are anticipated to have a significant, negative socioeconomic impact on the local community and will likely be beneficial in terms of employment and public services. Additionally, the Hyperion Data Center has provided temporary on-site housing to non-local workers therefore impacts to housing are not expected. Thus, cumulative impacts are not anticipated.
Air Quality	The geographical ROI for air quality during construction is considered to be within 0.25-mile of the Project while the ROI for air quality during operation is considered to occur within a 1-mile radius of the Project meter stations. The projects identified in the geographic ROI that could result in cumulative effects to air quality during construction are the Titanium Transmission Line Project and the	Project air emissions during construction will be limited to vehicle and construction equipment emissions and dust which will be localized. ETC Tiger will implement the *Fugitive Dust Control Plan* to minimize air impacts during construction. Once constructed, the proposed Project is not anticipated to result in operational air emissions. Thus, the Project is anticipated to have negligible air impacts during construction and no operational air impacts.

TABLE 1.10-3
Resource-Specific Determination of the Proposed Franklin Farms Project's Cumulative Impacts

Resource	Description of Potential Cumulative Impacts	Cumulative Impact Determination
	Franklin Farms Power Station 1 & 2 while the projects identified in the geographic ROI that could result in cumulative effects to air quality during operation are the Hyperion Data Center, Smalling Substation, and Franklin Farms Power Station 1 & 2.	Similar to the proposed Project, ETC Tiger expects that the Titanium Transmission Line Project and the Franklin Farms Power Station 1 & 2 will implement fugitive dust control measures such as BMPs for maintaining roadways and spraying water as needed for dust suppression; thus, cumulative impacts for air quality during construction are not anticipated to be significant. For operational air impacts, ETC Tiger expects that Hyperion Data Center, Smalling Substation, and Franklin Farms Power Station 1 & 2 will adhere to State of Louisiana air quality requirements; however, since the proposed Project will not have operational air quality impacts, no cumulative impacts will occur.
Noise	The geographical ROI for noise is considered to occur within a 0.25-mile radius of the Project. The only projects identified in the geographic ROI that could result in cumulative effects to noise with the proposed Project are the Titanium Transmission Line Project and the Franklin Farms Power Station 1 & 2. It is anticipated that these projects could have similar cumulative effects on noise during construction. Once constructed, the proposed Project is anticipated to result in a negligible amount of noise.	Project construction activities will result in temporary increases in ambient noise levels, typically during daylight hours unless nightwork is required. Because noise attenuates rapidly from the source, ETC Tiger anticipates that the greatest noise impacts will be localized in the immediate vicinity of the workspaces and that noise attenuation would further increase due to the topography, vegetation, and structures between the construction sites and the noise sensitive areas. Cumulative construction noise impacts depend on the overall timing of construction for the Titanium Transmission Line Project and the Franklin Farms Power Station 1 & 2 projects and the type of daily construction activities occurring at each project site. Due to the limited workspace where these projects overlap with the proposed Project, ETC Tiger does not anticipate that the area of overlap will be under construction by the projects simultaneously. Further, similar to the proposed Project, ETC Tiger expects that all projects will adhere to necessary federal and local noise regulations. Due to the limited scope of the proposed Project, it is anticipated that any temporary cumulative impacts of construction noise, should there be simultaneous construction with the other projects, will not be significant.

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ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 2
Water Use and Quality

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 2 FILING REQUIREMENTS	
Information	**Section Reference**
1. Identify all perennial surface waterbodies crossed by the Project and their water quality classification. (§380.12(d)(1)). • Identify by milepost • Indicate if potable water intakes are within 3 miles downstream of the crossing.	Section 2.2.2
2. Identify all waterbody crossings that may have contaminated waters or sediments. (§380.12(d)(1)). • Identify by milepost • Include offshore sediments.	Section 2.2.2, Section 2.2.3.3
3. Identify watershed areas, designated surface water protection areas, and sensitive waterbodies crossed by the Project. (§380.12(d)(1)). • Identify by milepost	Section 2.2.1, Section 2.2.2 Section 2.2.6.3
4. Provide a table (based on NWI maps if delineations have not been done) identifying all wetlands, by milepost and length, crossed by the Project (including abandoned pipeline), and the total acreage and acreage of each wetland type that would be affected by construction. (§380.12(d)(1 & 4)).	Section 2.3.1
5. Discuss construction and restoration methods proposed for crossing wetlands, and compare them to staff's Wetland and Waterbody Construction and Mitigation Procedures. (§380.12(d)(2)).	Section 2.3.2 Section 2.3.3
6. Describe the proposed waterbody construction, impact mitigation, and restoration methods to be used to cross surface waters and compare to the staff's Wetland and Waterbody Construction and Mitigation Procedures. (§380.12(d)(2)). • Although the Procedures do not apply offshore, the first part of this requirement does apply. Be sure to include effects of sedimentation, etc. This information is needed on a mile-by-mile basis and will require completion of geophysical and other surveys before filing. (See also Resource Report 3.)	Section 2.2.6 Section 2.3.4
7. Provide original National Wetlands Inventory (NWI) maps or the appropriate state wetland maps, if NWI maps are not available, that show all proposed facilities and include milepost locations for proposed pipeline routes. (§ 380.12(d)(4)).	Appendix A-4
8. Identify all U.S. Environmental Protection Agency (EPA) or state-designated aquifers crossed. (§ 380.12(d)(9)). • Identify the location of known public and private groundwater supply wells or springs within 150 feet of construction.	Section 2.1
Additional Information Often Missing and Resulting in Data Requests	
Identify proposed mitigation for impacts on groundwater resources.	Section 2.1.7
Discuss the potential for blasting to affect water wells, springs, and wetlands, and associated mitigation.	Section 2.1.7.2

RESOURCE REPORT NO. 2 FILING REQUIREMENTS	
Information	**Section Reference**
Identify all sources of water required for construction (e.g., hydrostatic testing, dust suppression, horizontal directional drills [HDD]), the quantity of water required, and methods for withdrawal. Identify the treatment of discharge, discharge volumes, rates, and locations, and any waste products generated.	Section 2.1.7
Identify operating water requirements for proposed liquefied natural gas facilities, including the water use, source(s), and volumes.	Section 2.1.7
If underground storage of natural gas is proposed, identify how water produced from the storage field will be disposed.	Not applicable
If salt caverns are proposed for storage of natural gas, identify the source locations, the quantity required, the method and rate of water withdrawal, and disposal methods.	Not applicable
Provide a site-specific construction plan for each proposed HDD crossing in accordance with section V.B.6.d of the Federal Energy Regulatory Commission's Wetland and Waterbody Construction and Mitigation Procedures.	Not applicable
Provide a site-specific construction plan for crossing each waterbody greater than 100 feet wide. Include a discussion on the feasibility of a trenchless crossing method.	Not applicable
Identify mitigation measures to avoid impacts on springs; especially those used for drinking water or livestock.	Section 2.1.7.1
Identify mitigation measures to ensure that public or private water supplies are returned to their former capacity or replaced in the event of damage resulting from construction.	Section 2.1.7.1
In addition to identifying perennial surface waterbodies crossed or affected by the project, also identify intermittent and ephemeral waterbodies.	Section 2.2.2
Show the locations of wetlands and waterbodies relative to the construction and permanent rights-of-way and additional temporary workspaces on mile posted alignment sheets or aerial photography.	Appendix A-7
If wetlands would be filled or permanently lost or altered, describe proposed measures to compensate for permanent wetland losses. Include copies of any compensatory mitigation plans and discuss the status of agency consultations/approvals.	Section 2.3.4
Describe measures to avoid or minimize impacts on forested wetlands. If impacts are unavoidable, describe proposed measures to restore forested wetlands following construction.	Section 2.3.2
Describe techniques to be used to minimize turbidity and sedimentation impacts associated with offshore trenching, if applicable.	Not applicable
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RESOURCE REPORT NO. 2
WATER USE AND QUALITY

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Appendix D-3 – Native American Organizations and Tribes Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

E Landowner Information

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F Wetland and Waterbody Delineation Report and Listed Species Report

Appendix F-1 – Wetland and Waterbody Delineation Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

Appendix F-2 – Listed Species Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

G Cultural Resource Assessment Survey Report (*Submitted as* "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

H Air Quality Calculations (Construction Emissions)

I Sound Study Report

ABBREVIATIONS AND ACRONYMS

ATWS	additional temporary workspace
BMP	best management practice
Certificate	Certificate of Public Convenience and Necessity
CWA	Clean Water Act
DDT	dichlorodiphenyltrichloroethane
DO	dissolved oxygen
EI	environmental inspector
EPA	Environmental Protection Agency
ETC Tiger	ETC Tiger Pipeline LLC
FEMA	Federal Emergency Management Agency
FERC or Commission	Federal Energy Regulatory Commission
FERC Plan	FERC Upland Erosion Control, Revegetation, and Maintenance Plan
FERC Procedures	FERC Wetland and Waterbody Construction and Mitigation Procedures
FF	Franklin Farms
HUC	hydrologic unit code
ID	identification
LAC	Louisiana Administrative Code
LDCE	Louisiana Department of Conservation and Energy
LDEQ	Louisiana Department of Environmental Quality
mg/L	milligrams per liter
MP	milepost
N/A	not applicable
NEPA	National Environmental Policy Act
NGA	Natural Gas Act
NHD	National Hydrography Dataset
NLCD	National Land Cover Dataset
NRCS	Natural Resources Conservation Service
NWI	National Wetlands Inventory
PAR	permanent access road
PEM	Palustrine Emergent Wetland
PFO	Palustrine Forested Wetland
PSS	Palustrine Scrub Shrub
Project	Franklin Farms Project
PSS	Palustrine Scrub Shrub Wetland
ROW	right-of-way
SMCL	secondary maximum contaminant levels
SONRIS	Strategic Online Natural Resources Information System
SPAR Plan	Spill Prevention and Response Plan
TAR	temporary access road
TDS	total dissolved solids
TMDL	total maximum daily load
TSS	total suspended solids

TWS	temporary workspace
USACE	United States Army Corps of Engineers
USFWS	United States Fish and Wildlife Service
USGS	United States Geological Survey

2.0 WATER USE AND QUALITY

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, an interconnect site, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project in June 2027, and following the receipt of all applicable approvals, anticipating anticipate beginning construction activities beginning in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

Resource Report No. 2 describes the existing hydrology, water quality, and use of the regional aquifers underlying the Project; the potential for impacts on water resources; and the proposed measures to mitigate any identified impacts to water resources.

To avoid and minimize impacts to water use and quality during construction and operation of the Project, ETC Tiger will implement the FERC *Upland Erosion Control, Revegetation and Maintenance Plan*, May 2013 (FERC Plan, **Appendix C-1**), the FERC *Wetland and Waterbody Construction and Mitigation Procedures*, May 2013 (FERC Procedures, **Appendix C-2**), with proposed site-specific exceptions, and ETC Tiger's *Spill Prevention and Response Plan* (SPAR Plan) (**Appendix C-3**).

2.1 GROUNDWATER RESOURCES

2.1.1 AQUIFERS

The proposed Project area in Louisiana is in the Mississippi River Valley alluvial aquifer, underlain by the Mississippi embayment aquifer system. The Mississippi River Valley alluvial aquifer is the largest alluvial aquifer within the surficial aquifer system that extends across parts of eastern Arkansas, northeastern Louisiana, and northwestern Mississippi. This is a freshwater aquifer composed of sand, gravel, silt, and minor clay deposits of Quaternary age that served as a major drainageway for glacial meltwater, depositing considerable amounts of course-grained detritus that have infilled with alluvial deposits through the Pleistocene and Holocene. Therefore, the alluvial valley includes two hydrogeologic units: an upper confining unit of silt, clay, and fine sand, and a lower layer of coarse sand and gravel. The aquifer ranges from approximately 25 to more than 150 feet thick with eastern Richland Parish generally 75 to 150 feet in thickness. Well withdrawals have replaced natural flow into the Mississippi River and its tributaries as the primary means of groundwater discharge within the aquifer. Wells almost anywhere within the Mississippi River Valley alluvial aquifer can yield 500 gallons per minute with some irrigation wells yielding 1,000 to 5,000 gallons per minute. However, withdrawal amounts have declined over recent decades with an increase in uptake from wells for agricultural use (Renken, 1998).

The Mississippi embayment aquifer system underlies most of northern and central Louisiana, southern Mississippi, and southern Alabama, extending along the Mississippi River as far north as southern Illinois, Missouri, and western Kentucky. The aquifer system comprises six aquifers and three regional confining units characterized by bands of poorly consolidated to unconsolidated sand, silt, and clay. The Project, along with most of Richland Parish, lies within the Upper Claiborne aquifer within this system. The thickness of the aquifer is generally between 2,000 and 4,000 feet within Richland Parish. Well yields within the aquifer range from 100 to 300 gallons per minute, though large withdrawals have resulted in a long-term decline in water levels (Renken, 1998).

2.1.2 SOLE SOURCE AQUIFERS

The United States Environmental Protection Agency (EPA) defines a sole source aquifer as one where:

- The aquifer supplies at least 50 percent of the drinking water for its service area; and
- There are no reasonably available alternative drinking water sources should the aquifer become contaminated.

The Sole Source Aquifer Program is authorized by Section 1424(e) of the Safe Drinking Water Act of 1974. Designating an aquifer as a sole source aquifer provides EPA with the authority to review federal financially assisted projects (e.g., highway improvements, public water supply wells) planned for the area to determine their potential for contaminating the aquifer. Proposed projects that are funded entirely by the private sector, such as this Project, are not subject to EPA review (EPA, 2026a).

None of the Project area overlies an EPA-designated sole source aquifer (EPA, 2026b,c).

2.1.3 GROUNDWATER LEVELS

Available United States Geological Survey (USGS) and Louisiana Department of Conservation and Energy (LDCE) Strategic Online Natural Resources Information System (SONRIS) depth to groundwater data for the four nearest USGS monitoring stations to the Project with available data are provided in **Table 2.1-1**.

TABLE 2.1-1
Groundwater Level Measurements

USGS Well No.	Location				Depth to Groundwater (feet)
	Hydrologic Unit	Latitude	Longitude	Approximate Distance from the Pipeline (miles)	
323029091430001	08050001	32° 30' 29”	-91° 43' 0"	0.9	20
323030091554801	08050001	32° 30' 30”	-91° 55' 48"	13.1	80
324508091252302	08050001	32° 45' 8”	-91° 25' 24”	19.6	36
325906091391201	08040205	38° 12’38”	-90° 16’00”	31.1	21

Source:
USGS, 2026a; LDCE, 2026 (Based on three or more years of daily mean data)

2.1.4 PUBLIC AND PRIVATE WATER SUPPLY WELLS AND SPRINGS

Publicly available data from the SONRIS database was assessed to determine the presence of groundwater resources (i.e., springs, public water supply wells, and private water supply wells) located within 150 feet of the proposed Project area. Groundwater resources identified within 150 feet of the Project area are listed in **Table 2.1-2**. The groundwater resources identified in **Table 2.1-2** will be identified in the field prior to construction of the Project. According to SONRIS data, there are no public water systems, commercial water systems, or groundwater areas of concern within 150 feet of the Project (LDCE, 2026).

TABLE 2.1-2
Public and Private Water Wells Identified within 150 feet of the Franklin Farms Project

Well Number	Location (Milepost)	Route/Site	Type	Status	Offset Distance to Work Area (Feet)	Offset distance to Pipeline (Feet)	Direction of Offset
WW - 1	N/A	Franklin Farms (FF) Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station	Irrigation	Active	Within Site	N/A	N/A
WW - 2	1.4	Franklin Farms Lateral	Irrigation	Active	23	58	East
WW - 3	3.0	Franklin Farms Lateral	Irrigation	Inactive	Within Work Area	46	East
WW - 4	6.3	Franklin Farms Lateral	Irrigation	Active	17	52	East
WW - 5	6.7	Franklin Farms Lateral	Irrigation	Inactive	33	68	North
WW - 6	8.7	Franklin Farms Lateral	Irrigation	Inactive	3	38	West
WW - 7	9.0	Franklin Farms Lateral	Irrigation	Active	28	63	West
WW - 8	10.5	Franklin Farms Lateral	Irrigation	Active	8	43	Northwest
WW - 9	11.2	Franklin Farms Lateral	Irrigation	Active	8	75	North

Source: LDCE, 2026.

2.1.5 GROUNDWATER QUALITY

The quality of water within the Mississippi River Valley alluvial aquiver is generally suitable for most uses (Renken, 1998). Total dissolved solids (TDS) concentrations are typically less than 500 milligrams per liter (mg/L), though TDS concentrations in the portion of Richland Parish containing the Project range from 500 to 1,000 mg/L. Water from Mississippi River alluvial aquifer wells in Richland Parish is generally within the EPA Secondary Maximum Contaminant Levels (SMCL) and can be moderately to very hard and contain large concentrations of iron and manganese (USGS 1975, 2019).

The Project is not located within an area underlain by soluble rocks and sinkholes (USGS 2020), and karst topography is likely not present. Karst features are important to groundwater quality as they provide a vulnerability for allowing contaminants (e.g., nitrates, pesticides) to move rapidly from the land surface into the aquifer (Berndt et al. 2014). Additional details concerning the potential for karst features within the Project area are provided in Resource Report No. 6 (*Geological Resources)*.

2.1.6 POTENTIALLY CONTAMINATED GROUNDWATER

Information regarding potentially contaminated groundwater in the Project vicinity was obtained from the following sources:

- EPA National Environmental Policy Act (NEPA) Assist Webmapper Tool (EPA, 2026c);
- EPA Cleanups in My Community Webmapper Tool (EPA, 2026d); and
- EPA EnviroMapper (EPA, 2026e).

Based on review of the EPA databases, no superfund sites, brownfields, toxic releases, or hazardous waste facilities are within the Project. Information regarding landfills and potential hazardous waste facilities within 0.25 miles of the Project is included in Section 8.3.5 of Resource Report No. 8 (*Land Use)* and depicted in **Appendix A-9**.

2.1.7 CONSTRUCTION IMPACTS AND MITIGATION

Project construction is not anticipated to adversely impact groundwater quality or supply. Estimated water use requirements for Project construction are identified in **Table 2.1-3**. Water required to construct the pipeline will be acquired from select water sources as identified below.

TABLE 2.1-3
Water Use for Construction

Facility	Description	Length (Miles)	Hydro Test		Dust Suppression	
			Volume (gallons)	Source	Volume (gallons)	Source
Franklin Farms Lateral	Pre-Test - Interstate 20	0.1	18,000	WW #1	N/A	N/A
	Pre-Test - Railroad and Highway 80	0.1	15,000	WW #1	N/A	N/A
	0.5 DF Test - Bee Bayou Facility	0.8	200,000	WW #1	N/A	N/A
	0.5 DF Test - Franklin Farms Facility	0.8	200,000	WW #9	N/A	N/A
	Mainline Test	13.2	3,500,000	WW #1	N/A	N/A
	Dust Suppression	N/A	N/A	N/A	345600	WW #1
					345600	WW #4
					345600	WW #7
					691200	WW #9
FF Trio Farm Lateral	0.5 DF Test - Trio Farms Facility	0.3	26,000	WW #7	N/A	N/A
	Mainline Test	1.8	144,000	WW #7	N/A	N/A
	Dust Suppression	N/A	N/A	N/A	192,000	WW #7
FF Gulf Run Bi Di Meter Station	Hydro test	N/A	25,000	WW #1	N/A	N/A
	Dust Suppression	N/A	N/A	N/A	576,000	WW #1
FF Gulf South Receipt Meter Station	Hydro test	N/A	12,000	WW #1	N/A	N/A
	Dust Suppression	N/A	N/A	N/A	576,000	WW #1
FF Bee Bayou Delivery Meter Station	Hydro test	N/A	15,000	WW #4	N/A	N/A
	Dust Suppression	N/A	N/A	N/A	576,000	WW #4
FF Gulf Crossing Receipt Meter Station	Hydro test	N/A	15,000	WW #7	N/A	N/A
	Dust Suppression	N/A	N/A	N/A	288,000	WW #7
FF MEP Receipt Meter Station	Hydro test	N/A	15,000	WW #9	N/A	N/A
	Dust Suppression	N/A	N/A	N/A	288,000	WW #9
Franklin Farms Delivery Meter Station	Hydro test	N/A	15,000	WW #9	N/A	N/A
	Dust Suppression	N/A	N/A	N/A	576,000	WW #9
FF Trio Farm Delivery Meter Station	Hydro test	N/A	12,500	WW #7	N/A	N/A
	Dust Suppression	N/A	N/A	N/A	576,000	WW #7

TABLE 2.1-3
Water Use for Construction

Facility	Description	Length (Miles)	Hydro Test		Dust Suppression	
			Volume (gallons)	Source	Volume (gallons)	Source
FF Interconnect Site	Hydro test	N/A	14,000	WW #7	N/A	N/A
	Dust Suppression	N/A	N/A	N/A	N/A	N/A

Although the Project anticipates utilizing water from existing local groundwater wells, the Project does not anticipate having any adverse impacts on these wells. ETC Tiger will offer landowners pre- and post-construction testing of all potable wells within 150 feet of the construction workspace and will test for yield and turbidity parameters. For any significant differences in the well yield and water quality between pre- and post-construction sampling that cannot be attributed to naturally occurring conditions (e.g., seasonal groundwater level fluctuations), ETC Tiger will provide an alternate source of potable water for the landowner until the water supply and/or well is repaired or replaced.

Clearing and grading the construction right-of-way (ROW) could affect local overland water flow and recharge. Operating heavy construction equipment could also cause localized areas of soil compaction, reducing the soil's ability to absorb water. In addition, dewatering the pipeline trench, if necessary, could temporarily lower the water table. If these minor impacts were to occur, they will only occur within discrete areas, will be temporary in nature, and will not significantly affect groundwater resources or change groundwater flow patterns. Pipeline construction activities within any one location will typically be completed within several days, and any localized lowering of groundwater from dewatering is expected to be temporary and minor. To recharge the aquifer and prevent silt laden waters from flowing into streams and wetlands, ETC Tiger proposes to discharge all water from dewatering activities into an adjacent upland area and through sediment control filter bags or silt fence/hay bale structures.

Construction of the Project will require trenching and backfilling to a depth of approximately six to eight feet below the ground surface. Trenching and backfilling could potentially cause minor localized fluctuations in groundwater levels and/or increase turbidity within the zone of shallow groundwater adjacent to the trench. Any shallow groundwater disturbance will be temporary in nature and localized to the immediate area of the trenching and backfilling activity and will not affect the overall quality of groundwater in the Project area.

2.1.7.1 Public and Private Wells and Springs

Based on data from the SONRIS database, ETC Tiger has identified drinking water supplies located within 150 feet of the proposed Project area (see **Table 2.1-2**) (LDCE, 2026). The location of these water supplies will be confirmed in the field prior to construction.

ETC Tiger will offer landowners pre- and post-construction testing of all potable wells within 150 feet of the construction workspace and will test for yield and turbidity parameters. For any significant differences in the well yield and water quality between pre- and post-construction sampling that cannot be attributed to naturally occurring conditions (e.g., seasonal groundwater level fluctuations), ETC Tiger will provide an alternate source of potable water for the landowner until the water supply and/or well is repaired or replaced.

2.1.7.2 Blasting

Blasting is not anticipated to be utilized for this Project.

2.1.7.3 Horizontal Directional Drill

ETC Tiger does not propose the use of the horizontal directional drill method for Project construction.

2.1.7.4 Spills

Spills or leaks of hazardous liquids have the potential for long-term impacts on groundwater resources, especially in areas where there is a high susceptibility to surface contamination. Factors in determining groundwater susceptibility to contamination include the type of underlying bedrock, depth to bedrock, depth to the water table, and characteristics of soils and surficial deposits. Of these, soil and surficial deposit

characteristics are considered the most important factors in determining how susceptible an area is to groundwater contamination. Areas with sand and gravel are considered more susceptible to groundwater contamination than areas consisting of silt and clay, mainly due to the permeability of the material.

Impacts to groundwater resources will be avoided and minimized with the use of Best Management Practices (BMP) and protective measures in accordance with the FERC Plan, FERC Procedures, and ETC Tiger's SPAR Plan. ETC Tiger's SPAR Plan was specifically developed to avoid and minimize potential contamination if a spill were to occur. Spill-related impacts from construction are mainly associated with fuel storage, equipment refueling, and equipment maintenance. These potential impacts will be avoided or greatly reduced by regulating fuel storage and refueling activities, and by requiring immediate cleanup should a spill or leak occur. ETC Tiger's SPAR Plan describes the preventative measures to be used to avoid spills and leaks in detail, as well as the mitigative measures to be utilized to minimize potential impacts should a spill or leak occur. This includes prohibiting refueling and storage of hazardous materials, including chemicals, fuels, and lubricating oils within 200 feet of private wells and 500 feet of community and municipal wells. In addition, no potentially hazardous materials, other than essential equipment fuels (e.g., gasoline and diesel) or standard lubricants (e.g., engine oils and grease) will be transported along the ROW or into a construction area without Environmental Inspector (EI) coordination and approval.

2.1.8 OPERATION IMPACTS AND MITIGATION

Project operations are not anticipated to adversely impact either groundwater quality or supply. The Project does not include any new permanent aboveground facilities that will generate long-term groundwater use. Further, during operations, impacts to groundwater resources will be avoided or minimized with the use of BMPs and protective measures in accordance with the FERC Plan, FERC Procedures, and ETC Tiger's SPAR Plan.

2.2 SURFACE WATER RESOURCES

This section assesses surface water resources, focusing on waterbodies; wetlands are discussed separately in Section 2.3. Information in this section includes the existing surface water resources that could potentially be affected by construction and operation of the proposed Project, the presence of any sensitive or protected waters, and proposed methods to minimize and mitigate any effects to surface waters.

2.2.1 WATERSHEDS

Watersheds form natural boundaries within a land mass and are geographically focused and hydrologically defined. Watersheds are delineated by USGS using a nationwide system that divides the country using a hierarchical hydrologic unit code (HUC) consisting of two additional digits for each level in the hydrologic unit system (USGS, 2013). The Project area is located within the 12-digit HUCs in **Table 2.2-1**.

TABLE 2.2-1
Watersheds Crossed by the Franklin Farms Project

Facility	Watershed	Milepost (MP)		Laterals Crossed
		Begin	End	
Pipeline				
Contractor Yard 3	Congo Creek-Big Creek (080500011002)	N/A	N/A	N/A
PAR, TAR, Perm Easement, ATWS, Temp Workspace, Contractor Yard 3	Cow Bayou-Big Creek (080500011004)	0	8.4	Franklin Farms
TAR, Contractor Yards 1 and 2	Cypress Bayou-Bayou Macon (080500020505)	N/A	N/A	N/A

TABLE 2.2-1
Watersheds Crossed by the Franklin Farms Project

Facility	Watershed	Milepost (MP)		Laterals Crossed
		Begin	End	
PAR, TAR, Perm Easement, ATWS, Temp Workspace	Cypress Creek (080500011003)	11.1	13.2	Franklin Farms
PAR, TAR, Perm Easement, ATWS, Temp Workspace	Dry Fork Creek-Bee Bayou (080500011005)	8.5 (Trio Farms: 0)	11.1 (Trio Farms: 1.7)	Franklin Farms, Trio Farms
Contractor Yard 3	Turkey Creek (080500011007)	N/A	N/A	N/A
Aboveground facilities				
FF Bee Bayou Delivery Meter Station, FF Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station	Cow Bayou-Big Creek (080500011004)	0	8.4	N/A
Franklin Farms Delivery Meter Station	Cypress Creek (080500011003)	11.1	13.2	N/A
FF Gulf Crossing Receipt Meter Station, FF Trio Farm Delivery Meter Station, FF MEP Receipt Meter Station, FF Interconnect Site	Dry Fork Creek-Bee Bayou (080500011005)	8.5 (Trio Farms: 0)	11.1 (Trio Farms: 1.7)	N/A

NOTES:
PAR = permanent access roads; TAR = temporary access roads; ATWS = additional temporary workspace; N/A = not applicable

2.2.2 WATERBODIES

ETC Tiger conducted field surveys of waterbodies within the Project area in February, March, and May 2026. The surveys were conducted in accordance with the *1987 Corps of Engineers Wetland Delineation Manual*; and the *Regional Supplement to the Corps of Engineers Wetland Delineation Manual: Atlantic and Gulf Coastal Plain Region* (Version 2.0). Prior to the surveys, a desktop review was conducted using the following publicly available information:

- Recent and historical aerial imagery;
- USGS 7.5-minute Series Topographic Maps (USGS, 2026b);
- USGS National Land Cover Database (NLCD) (USGS, 2025);
- USGS National Hydrography Dataset (NHD) (USGS, 2026b);
- Natural Resource Conservation Service (NRCS) Web Soil Survey (NRCS, 2026);
- U.S. Fish and Wildlife Service (USFWS) National Wetlands Inventory (NWI) data (USFWS, 2026); and
- Federal Emergency Management Agency (FEMA) National Floor Hazard Layer data and Flood Insurance Rate Maps (FEMA, 2026).

The survey included evaluating an approximate 596-acre area overlying the Project centerline. In addition, the entire footprint of the proposed workspace for the proposed stations, aboveground facilities, contractor yards, and access roads was surveyed. A copy of the field survey report is provided in **Appendix F-1**.

A description of the proposed crossing method for each waterbody is provided in **Table 2.2-2**. FERC classifies surface waters based on size: major waterbodies are greater than 100 feet wide, intermediate

waterbodies are greater than 10 feet wide but less than or equal to 100 feet wide, and minor waterbodies are less than or equal to 10 feet wide.

TABLE 2.2-2
Waterbodies Crossed by the Franklin Farms Project

Project Component	Approximate Milepost	Feature Identification (ID)	Waterbody Name	Designated Use [a]	Flow Regime	Waterbody Width (feet)	Pipeline Crossing Length [b] (feet)	Proposed Crossing Method	Construction Impact (acres)	Operation Impact (Acres)
FF Lateral, TWS, FFL-PAR-002.000, FFL-PAR-002.100, FFL-TAR-001.000, FFL-TAR-003.000	0.6	D-T1F-01	Unnamed	None	AG DITCH (NON-RPW)	1	1	Conventional	0.01	0.00
FF Lateral, TWS	0	D-T1F-02	Unnamed	None	AG DITCH (NON-RPW)	1	1	Conventional	0.00	0.00
FF Lateral, TWS	0.7	D-T1F-03	Unnamed	None	AG DITCH (NON-RPW)	2	2	Conventional	0.00	0.00
FF Lateral, TWS	1	D-T1F-04	Unnamed	None	AG DITCH (NON-RPW)	1	1	Conventional	0.00	0.00
FF Lateral, TWS	1.1	D-T1F-05	Unnamed	None	AG DITCH (RPW)	7	7	Conventional	0.02	0.00
FF Lateral, TWS, FFL-TAR-004.000	1.6	D-T1F-06	Unnamed	None	AG DITCH (RPW)	4	7	Conventional	0.02	0.00
FF Lateral, FFL-TAR-012.000	5.1	D-T1F-07	Unnamed	None	ROADSIDE (RPW)	2	2	Bore	0.00	0.00
FF Lateral, TWS	5.3	D-T1F-08	Unnamed	None	AG DITCH (NON-RPW)	1	3	Conventional	0.01	0.00
FF Lateral, TWS	5.7	D-T1F-09	Unnamed	None	AG DITCH (RPW)	4	4	Conventional	0.01	0.00
FF Lateral, TWS	5.9	D-T1F-10	Unnamed	None	AG DITCH (NON-RPW)	2	2	Conventional	0.00	0.00
FF Lateral, TWS	6.7	D-T1F-11	Unnamed	None	AG DITCH (NON-RPW)	3	3	Conventional	0.01	0.00
TFL-PAR-001.000	8.5	D-T1F-12	Unnamed	None	AG DITCH (NON-RPW)	3	WS	Conventional	0.06	0.06
FF Lateral, TWS	8.1	D-T1F-13	Unnamed	None	AG DITCH (NON-RPW)	15	2	Conventional	0.00	0.00
FF Lateral, TWS	9	D-T1F-14	Unnamed	None	AG DITCH (RPW)	5	15	Conventional	0.03	0.00
FF Lateral, TWS	2.8	D-T2F-01	Unnamed	None	AG DITCH (NON-RPW)	2	2	Conventional	0.00	0.00
FF Lateral, TWS	3	D-T2F-02	Unnamed	None	AG DITCH (RPW)	7	7	Conventional	0.02	0.00
FF Lateral, TWS	2.5	D-T2F-03	Unnamed	None	AG DITCH (RPW)	29	30	Conventional	0.07	0.00
FFL-PAR-008.000, FFL-TAR-009.000	4.8	D-T2F-05a	Unnamed	None	AG DITCH (RPW)	7	WS	Conventional	0.00	0.00
FFL-TAR-010.000, FFL-TAR-011.000	5.1	D-T2F-05b	Unnamed	None	AG DITCH (NON-RPW)	2	2	Bore	0.01	0.00
Trio Farms Lateral, TWS	0.6	D-T2F-06	Unnamed	None	AG DITCH (RPW)	3	3	Conventional	0.00	0.00
Trio Farms Lateral, TWS	1	D-T2F-08	Unnamed	None	AG DITCH (RPW)	4	WS	Conventional	0.02	0.00
Trio Farms Lateral, TWS	1.2	D-T2F-09	Unnamed	A B C F	AG DITCH (RPW)	6	6	Conventional	0.01	0.00
TFL-PAR-002.000	1.5	D-T2F-12	Unnamed	None	AG DITCH (NON-RPW)	6	WS	Conventional	0.01	0.01
FF Lateral, TWS	3.2	D-T2F-14	Unnamed	None	AG DITCH (RPW)	16	16	Bore	0.03	0.00
FF Lateral, TWS	10.9	D-T3F-01	Bee Bayou	None	AG DITCH (RPW)	3	6	Conventional	0.01	0.00
FFL-TAR-016.000	10.5	D-T3F-02	Unnamed	None	AG DITCH (NON-RPW)	2	WS	Conventional	0.00	0.00
FFL-PAR-017.000	10.8	D-T3F-03	Unnamed	None	AG DITCH (NON-RPW)	3	WS	Conventional	0.00	0.00

TABLE 2.2-2
Waterbodies Crossed by the Franklin Farms Project

Project Component	Approximate Milepost	Feature Identification (ID)	Waterbody Name	Designated Use [a]	Flow Regime	Waterbody Width (feet)	Pipeline Crossing Length [b] (feet)	Proposed Crossing Method	Construction Impact (acres)	Operation Impact (Acres)
FFL-PAR-018.000, FFL-TAR-019.000, FFL-TAR-020.000	11.2	D-T3F-04	Unnamed	A B C	AG DITCH (NON-RPW)	2	WS	Conventional	0.00	0.00
FFL-TAR-021.000	11.3	D-T3F-05	Unnamed	None	ROADSIDE (NON-RPW)	2	WS	Conventional	0.00	0.00
FFL-TAR-028.000	0	D-T3F-8	Unnamed	None	ROADSIDE (NON-RPW)	2	WS	Conventional	0.00	0.00
FF Lateral, TWS, FFL-TAR-013.000	6	P-T1F-02	Unnamed	None	POND	75	70	Conventional	0.12	0.00
FF Lateral, TWS, FFL-TAR-013.000	6	S-T1F-01	Unnamed	None	PERENNIAL STREAM	60	60	Conventional	0.16	0.00
FF Lateral, TWS	6.9	S-T1F-02	Cow Bayou	None	PERENNIAL STREAM	26	26	Conventional	0.05	0.00
FF Lateral, TWS	7	S-T1F-04	Cow Bayou	None	PERENNIAL STREAM	42	25	Conventional	0.14	0.00
FF Lateral, TWS	10	S-T1F-05	Bee Bayou	None	PERENNIAL STREAM	6	7	Conventional	0.01	0.00
FF Lateral, TWS	4.4	S-T2F-02	Cow Bayou	None	PERENNIAL STREAM	70	70	Conventional	0.11	0.00
Trio Farms Lateral, TWS	1.4	S-T2F-04	Unnamed	None	PERENNIAL STREAM	25	25	Conventional	0.07	0.00
FF Lateral, TWS	11.6	S-T2F-05	Unnamed	None	EPHEMERAL STREAM	3	3	Conventional	0.01	0.00
FF Lateral, TWS	11.7	S-T2F-06	Unnamed	None	PERENNIAL STREAM	12	15	Conventional	0.03	0.00
FF Lateral, TWS	11.8	S-T2F-07	Unnamed	None	INTERMITTENT STREAM	5	5	Conventional	0.01	0.00
FF Lateral, TWS	12.6	S-T2F-08	Unnamed	None	INTERMITTENT STREAM	3	3	Conventional	0.01	0.00
FF Lateral, TWS	12.4	S-T2F-09	Boggy Bayou	None	PERENNIAL STREAM	28	44	Conventional	0.10	0.00
FF Lateral, TWS	12.2	S-T2F-10	Unnamed	None	INTERMITTENT STREAM	6	7	Conventional	0.01	0.00
FF Lateral, TWS	12.4	S-T2F-11	Unnamed	None	INTERMITTENT STREAM	2	WS	Conventional	0.00	0.00
Trio Farms Lateral, TWS	0.4	S-T3F-01	Bee Bayou	None	PERENNIAL STREAM	28	40	Conventional	0.07	0.00
FF Lateral, TWS	9.6	S-T3F-02	Unnamed	None	EPHEMERAL STREAM	2	WS	Conventional	0.00	0.00
FF Lateral, TWS	10.6	S-T3F-03	Bee Bayou	None	PERENNIAL STREAM	6	6	Conventional	0.01	0.00
FF Lateral, TWS	11.3	S-T3F-04	Cypress Creek	None	PERENNIAL STREAM	91	91	Conventional	0.24	0.00

Notes:

[a] A – Primary Contact Recreation; B – Secondary Contact Recreation; C – Fish and Wildlife Propagation; F - Agriculture

[b] WS= crossed by workspace, not by pipeline

Ag ditch – agricultural ditch; Non-RPW – non-relatively permanent water; Roadside – upland-excavated drainage conveyance constructed to convey surface drainage from adjacent roadways; Intermittent stream – surface water that flows continuously during certain times of year but may not flow during dry periods or extended droughts; Perennial stream – in the absence of abnormal, extended, or severe drought, continuously conveys surface water flow; Ephemeral stream – flows only briefly during and following a period of rainfall in the immediate locality; TWS – temporary workspace.

2.2.3 WATER QUALITY

2.2.3.1 Surface Water Intakes

None of the waters crossed by the Project are potable water supplies. The nearest surface water source is the Boeuf River, located approximately 0.85 miles southwest of the Project. The majority use of surface water from the Boeuf River and Boeuf subbasin is irrigation for agricultural use (USGS, 2019). There are no surface waters within the Project or within three downstream miles of the Project that are designated for drinking water supply.

2.2.3.2 Surface Water Classification

The State of Louisiana classifies surface waters based on seven water use designations: agriculture, drinking water supply, fish and wildlife propagation, outstanding natural resource waters, oyster propagation, primary contact recreation, and secondary contact recreation (Louisiana Administrative Code [LAC] 33:IX.1111 and 33:IX.1123). These designated uses are established based on chemical, physical, and biological conditions that preclude such uses.

Boggy Bayou and Cypress Creek, both crossed by the Franklin Farms lateral, have designated uses of primary contact recreation, secondary contact recreation, and fish and wildlife propagation. Additionally, Boggy Bayou is designated for agriculture (LAC 33:IX.1123.D).

2.2.3.3 Impaired Waters

Section 303(d) of the Clean Water Act (CWA) authorizes EPA to assist states in listing impaired waters and developing Total Maximum Daily Loads (TMDL) for the impaired waterbodies. A TMDL establishes the maximum amount of a pollutant allowed in a waterbody and serves as the starting point or planning tool for restoring water quality. EPA Region 6 reviews the State of Louisiana 303(d) list and approves or disapproves the list and related TMDLs. None of the waterbodies crossed are on the 303(d) list (EPA, 2025i). The nearest stream listed on the 2024 integrated 303(d)/305(b) list is the Boeuf River, located approximately 0.85-mile southwest of the Project (Louisiana Department of Environmental Quality [LDEQ], 2024). The Boeuf River from the Arkansas state line to the Ouachita River does not meet its designated use of fish and wildlife propagation (fishing) due to 4,4'- dichlorodiphenyltrichloroethane (DDT), carbofuran, dissolved oxygen, mercury, toxaphene, and turbidity. Mercury is the only impairment factor for the Boeuf River for which a TMDL has not yet been developed; therefore, it is the only factor contributing to the river's 303(d)-listed status.

2.2.4 CONTAMINATED SEDIMENTS

ETC Tiger searched the EPA's databases to identify any known or potential contamination sites within 1,000 feet of the proposed Project area (see Section 2.1.7). No contaminated sites were documented in or around the Project area.

2.2.5 FLOODPLAINS

Executive Order No. 11988 requires federal agencies to avoid (to the extent possible) long- and short-term adverse impacts associated with the occupancy and modification of floodplains. ETC Tiger reviewed the most recent published FEMA floodplain maps (FEMA, 2025). The assessed flood zone where the proposed Project facilities will be located is provided in **Table 2.2-4**, and FEMA mapping of the Project area is provided in **Appendix A-10**.

According to FEMA floodplain mapping, most of the Project area will be located outside FEMA-designated flood hazard areas. Portions of the Project area are located within Zone A, which are considered to have a one percent annual chance of flooding and are in the 100-year flood zone. Locations of designated FEMA flood hazard areas are provided in **Table 2.2-3**.

TABLE 2.2-3
FEMA Zones Crossed by the Franklin Farms Project

Facility	FEMA FIRM ID	MP		Flood Zones Crossed	Distance Crossed (miles)
		Begin	End		
Pipeline					
Franklin Farms Lateral	22083C	4.3	12.8	A	8.5
Aboveground Facilities					
FF Bee Bayou Delivery Meter Station	22083C	4.5		A	N/A
Other Work Areas					
PAR (FFL-PAR-008.000)	22083C	4.5		A	0.02
Source: FEMA, 2025					

2.2.6 CONSTRUCTION IMPACTS AND MITIGATION

Impacts to surface water resources are considered direct if the surface water is located within the proposed construction workspace and is temporarily or permanently disturbed. Indirect impacts include effects such as potential changes in downstream water quality or off-site migration of pollutants.

2.2.6.1 Surface Water Use

Estimated water use requirements for Project construction are identified in **Table 2.1-2**. Water required to construct the pipeline will be acquired from select groundwater wells. Therefore, water will not be withdrawn from surface waterbodies for construction.

2.2.6.2 Off-site Migration of Pollutants

Stormwater runoff and erosion may result in indirect impacts (e.g., sedimentation, turbidity) on surface waters adjacent to the Project area. However, these effects will be avoided and minimized by installation and maintenance of sediment and erosion control BMPs throughout construction in accordance with the FERC Plan and FERC Procedures. Sediment and erosion control BMPs will remain in place until revegetation is determined to be successful, thus limiting potential effects during Project construction.

The Project is not located within an area underlain by soluble rocks and sinkholes (USGS 2020), and karst topography is likely not present. If karst areas are encountered, stormwater will be diverted upland from the excavated karst areas utilizing approved erosion and control methods. If surface waters are present near the karst excavation, then water will be flumed to minimize the potential for stormwater entering the void. Sandbags, or similar materials, will be utilized to withhold water from entering the excavation, and water levels will be monitored to determine whether it is entering the excavation.

2.2.6.3 Waterbody Crossings

Waterbodies crossed by the Project are listed in **Table 2.2-3.** ETC Tiger will use the conventional crossing (trenching) method, flume crossing method, and horizontal bore method to construct across all waterbodies. A description of these methods is included in Resource Report No. 1 (*Project Description)*. During

construction, ETC Tiger will adhere to the BMPs contained in the FERC Procedures which are designed to avoid and minimize impacts to waterbodies. This includes restricting the duration of in-water construction to the shortest time possible to allow pipeline construction across a waterbody.

Horizontal Bore

Select waterbodies will be crossed via horizontal bore which is a trenchless pipeline construction method. Therefore, these waterbodies will not be impacted by the Project construction.

Conventional and Flume Crossing Method

In-water pipeline construction using the wet open-cut or flume method will result in short-term sedimentation and turbidity. The increase in total suspended solids (TSS) and turbidity concentrations will be temporary and limited to the period of active construction. TSS and turbidity levels are anticipated to rapidly return to baseline conditions following construction. Per the FERC Procedures, ETC Tiger will be required to complete instream construction activities (including trenching, pipe installation, backfill, and restoration of the streambed contours) within 24 hours for minor waterbodies and 48 hours for intermediate waterbodies. In addition, stabilizing the waterbody banks and installing temporary sediment barriers will be required within 24 hours of completing instream construction activities.

Removal of trees and other bank vegetation from the edges of waterbodies at the pipeline crossing locations may reduce shading of the waterbody and result in locally elevated water temperatures. Elevated water temperatures can, in turn, lead to reductions in levels of dissolved oxygen (DO). To minimize potential effects associated with loss of riparian shade and vegetation cover, the proposed Project has been designed to limit tree clearing, and clearing of trees and other vegetation will be restricted to only what is necessary to safely construct and operate the pipeline. Once construction is complete, streambeds and banks will be restored in accordance with the FERC Procedures, and post construction maintenance (or mowing) will be limited to when necessary to facilitate periodic corrosion/leak surveys or to protect the integrity of the pipeline coating. Overall, the impacts resulting from waterbody construction are anticipated to be minor, localized, and short-term.

ETC Tiger will minimize impacts on waterbodies crossed by the conventional and flume methods by implementing BMPs and protective measures in accordance with the FERC Plan, FERC Procedures, and ETC Tiger's SPAR Plan which include:

- Limiting clearing of vegetation between temporary extra workspaces and the edge of the waterbody to preserve riparian vegetation;
- Constructing the crossing as close to perpendicular to the waterbody as site conditions allow;
- Maintaining adequate flow rates throughout construction to protect aquatic life and prevent the interruption of existing downstream uses;
- Locating equipment parking areas, equipment refueling areas, concrete coating activities, and hazardous material storage at least 100 feet from surface waters, unless unfeasible;
- Requiring construction across waterbodies to be completed as quickly as possible and during the windows specified in the Procedures or required by applicable permits;
- Requiring temporary erosion and sediment control measures to be installed across the entire width of the construction ROW after clearing and before ground disturbance;
- Requiring maintenance of temporary erosion and sediment control measures throughout construction until streambanks and adjacent upland areas are stabilized;
- Requiring bank stabilization and re-establishment of bed and bank contours and riparian vegetation after construction;

- Maintaining a 25-foot-wide strip of riparian vegetation along the waterbody and restricting maintenance of vegetation to a 10-foot-wide strip centered over the pipeline to trees greater than 15 feet in height within 15 feet of the pipeline centerline;
- Implementing the SPAR Plan if a spill or leak occurs during construction;
- Limiting use of equipment operating in the waterbody to that needed to construct the crossing and using an equipment bridge for construction equipment; and
- Monitoring the success of vegetation restoration annually for a period of at least five years after construction, or until the waterbody banks are successfully revegetated.

Therefore, the proposed Project is not anticipated to have any significant impacts on waterbodies crossed by the Project.

2.2.6.4 Sensitive and Impaired Waterbody Crossings

Impaired waterbodies on the 303(d) list are discussed in Section 2.2.3.2. No 303(d) listed waterbodies will be crossed by the Project. The Project is not anticipated to further contribute to impairment of any 303(d)-listed waterbodies or alter the designated use of any waterbodies within the Project or in the vicinity. BMPs will be utilized to minimize sediment discharge into Project waterbodies. The Project will not involve mercury use or storage and will not exacerbate existing mercury levels in the Boeuf River. Contaminated Sediments

No contaminated sediments are expected to occur within or near the Project limits. In the event contaminated sediments are encountered during Project construction, ETC Tiger will implement procedures to identify and manage the contaminated areas in accordance with the *Unanticipated Discovery Plan* (**Appendix C-4**).

2.2.6.5 Hydrostatic Testing

Hydrostatic testing of the integrity of the completed Project and aboveground facilities will occur before final restoration and placing the Project into service. Water for hydrostatic testing will be withdrawn from select groundwater wells. Further, no chemicals will be added to the hydrostatic test water. Hydrostatic test water will be discharged in accordance with the FERC Plan, FERC Procedures, and applicable permit conditions. This includes discharging test water into a well vegetated upland area and filtering the water through a filter bag and/or erosion control barrier. Therefore, hydrostatic testing is not anticipated to impact surface water resources.

2.2.6.6 Floodplain

Avoidance of floodplain areas during construction of the Project will not be feasible. However, any alteration of current drainage patterns and surface water flow will be short-term and minor. Following pipe installation and backfilling, the construction ROW will be restored to preconstruction contours.

ETC Tiger does not propose constructing any aboveground facilities within floodplain areas apart from the FF Bee Bayou Delivery Meter Station. Although the station will be within a flood zone, the proposed meter station will be fenced and graveled using existing contours and elevations. Therefore, FF Bee Bayou Delivery Meter Station is not expected to significantly alter floodplain characteristics. In addition, a very small portion of the proposed permanent access road (FFL-PAR-008.000) crosses the floodplain; therefore, the construction of the access roads is not anticipated to change floodplain dynamics. Further, ETC Tiger will coordinate with the local floodplain administrator and obtain applicable floodplain permitting if it is required prior to construction. Therefore, ETC Tiger does not anticipate any adverse impacts on the floodplain.

2.2.6.7 Spills

Accidental spills of construction-related fluids (e.g., oil, gasoline, or hydraulic fluids) on the landscape or directly into waterbodies could result in surface water quality impacts. Impacts to surface waters would depend on the type and quantity of the spill, and the dispersal and attenuation characteristics of the waterbody. To reduce the potential for surface water contamination, ETC Tiger will implement its SPAR Plan, which includes BMPs to avoid and minimize the potential for accidental releases and contains measures that will be implemented to clean up any releases. In accordance with the FERC Procedures, construction equipment, vehicles, hazardous materials, chemicals, fuels, lubricating oils, and petroleum products would not be parked, stored, or serviced within 100 feet of any waterbody.

2.2.7 OPERATION IMPACTS AND MITIGATION

Project operations are not anticipated to adversely impact surface water resources. Routine maintenance activities will include clearing and mowing of vegetation along the permanent ROWs. These activities will adhere to the FERC Plan and FERC Procedures and are not expected to have any noticeable impact on surface waters within the Project area. The proposed Franklin Farms Lateral and aboveground facilities will not result in any surface water use.

During operations, impacts to surface water resources will be avoided or minimized with the use of BMPs and protective measures in accordance with the FERC Plan, FERC Procedures, and ETC Tiger SPAR Plan.

2.2.7.1 Floodplains

Project operations are not anticipated to adversely impact the floodplain or flood storage capacity. The proposed Project will be belowground during operations, and the proposed aboveground facilities will be located within areas either considered to have minimal flood potential or already developed. Coordination with the Richland Parish floodplain administrators will occur prior to construction for FF Bee Bayou Delivery Meter Station.

2.3 Wetlands

Wetlands are areas that are inundated or saturated by surface water or groundwater at a frequency and duration sufficient to support, and under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions (USACE, 1987). Wetlands can be a source of substantial biodiversity and serve a variety of functions that include providing wildlife habitat, recreational opportunities, flood control, and naturally improving water quality (EPA, 2016).

Section 404 of the CWA establishes standards to evaluate and reduce total and net impacts on wetlands under the regulatory jurisdiction of the United States Army Corps of Engineers (USACE). These standards require avoidance of wetlands where possible and minimization of disturbance where impacts are unavoidable, to the degree practicable. ETC Tiger must demonstrate that it has taken appropriate and practicable steps to minimize wetland impacts. The USACE Vicksburg District Office has authority under Section 404 of the CWA to review and issue permits for Project-related activities that will result in discharge of dredged or fill material into waters of the United States, including wetlands.

2.3.1 EXISTING WETLANDS IN THE PROJECT AREA

ETC Tiger conducted wetlands and waterbody field surveys of the Project area in February, March, and May 2026. The surveys were conducted in accordance with the *1987 Corps of Engineers Wetland Delineation Manual*; and the *Regional Supplement to the Corps of Engineers Wetland Delineation Manual:*

Atlantic and Gulf Coastal Plain Region (Version 2.0). Prior to the surveys, a desktop review was conducted using the databases listed in Section 2.2.2.

The survey included evaluating an area that includes the Project centerline. In addition, the entire footprint of the proposed workspace for the proposed stations, aboveground facilities, contractor yards, and access roads was surveyed. The field survey report is provided in **Appendix F-1**.

During the field surveys, wetlands were identified within the proposed Project area. No wetlands were identified within the construction workspace for the proposed aboveground facilities. Mapping of the wetland areas is provided in **Appendix A-4**.

The USFWS wetland classification system described by Cowardin et. al. (1979) was used to classify the wetlands that will be affected by the Project. All the wetlands identified within the Project area are classified as palustrine systems. Palustrine systems include non-tidal wetlands that are dominated by trees, shrubs, persistent emergents, and emergent mosses or lichens and all wetlands that occur in tidal areas where salinity due to ocean-derived salts is below 0.5 percent. The palustrine system was developed to group vegetated wetlands, commonly referred to as marshes, swamps, bogs, and prairies. This system also includes ponds and may be situated shoreward of lakes, river channels, estuaries, and river floodplains or in isolated catchments or on slopes (Cowardin et. al., 1979).

Wetlands in the Project area consist of Palustrine Emergent (PEM) and Palustrine Forested (PFO) wetlands (see **Table 2.3-1**). PEM wetlands are characterized by erect, rooted, herbaceous hydrophytes, excluding mosses and lichens. PFO wetlands are characterized by woody vegetation that is 19.7 feet (6 meters) tall, or taller, and often occur along riparian areas (Cowardin et. al. 1979). Details of the proposed Project-related impacts are provided in **Table 2.3-2**, and a summary of the impacts is provided in **Table 2.3-3**.

TABLE 2.3-1
Wetlands Crossed by the Franklin Farms Project

Project Facility	Approx. Milepost	Wetland ID	Linear Crossing Length (feet)[1]	Wetland Impact (acres)[2, 3, 4]								Crossing Method
				PEM		PSS		PFO		Total		
				Const.	Oper.	Const.	Oper.	Const.	Oper.	Const.	Oper.	
FF Permanent Easement, FF Temporary Workspace	5.3	PEM-T1F-03	95.30	0.30	0.00	0.00	0.00	0.00	0.00	0.30	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	9	PEM-T1F-05	84.39	0.14	0.00	0.00	0.00	0.00	0.00	0.14	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	9.4	PEM-T1F-06	234.64	0.38	0.00	0.00	0.00	0.00	0.00	0.38	0.00	Horizontal Bore
FF Permanent Easement, FF Temporary Workspace	10	PEM-T1F-07	50.05	0.08	0.00	0.00	0.00	0.00	0.00	0.08	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	3.2	PEM-T2F-01	33.80	0.06	0.00	0.00	0.00	0.00	0.00	0.06	0.00	Horizontal Bore
FF Temporary Workspace	12.5	PEM-T2F-06	N/A	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	Conventional
FF Temporary Workspace	12.4	PEM-T2F-08	N/A	0.02	0.00	0.00	0.00	0.00	0.00	0.02	0.00	Conventional
FFL-TAR-007.000, FFL-ATWS-0040, FFL-ATWS-0039	3.3	PEM-T2F-10	N/A	0.03	0.00	0.00	0.00	0.00	0.00	0.03	0.00	Conventional
TF Permanent Easement, TF Temporary Workspace	1.3	PEM-T2F-11	33.48	0.05	0.00	0.00	0.00	0.00	0.00	0.05	0.00	Conventional
TF Permanent Easement, TF Temporary Workspace	1.4	PEM-T2F-12	18.92	0.03	0.00	0.00	0.00	0.00	0.00	0.03	0.00	Conventional
TF Permanent Easement, TF Temporary Workspace	1.2	PEM-T2F-13	122.82	0.17	0.00	0.00	0.00	0.00	0.00	0.17	0.00	Conventional
TF Permanent Easement, TF Temporary Workspace	1.1	PEM-T2F-14	59.09	0.10	0.00	0.00	0.00	0.00	0.00	0.10	0.00	Conventional
TF Permanent Easement, TF Temporary Workspace	0.9	PEM-T2F-15	N/A	0.09	0.00	0.00	0.00	0.00	0.00	0.09	0.00	Conventional
TF Temporary Workspace	0.9	PEM-T2F-16	N/A	0.01	0.00	0.00	0.00	0.00	0.00	0.01	0.00	Conventional
TF Permanent Easement	1.4	PEM-T3F-03	44.11	0.04	0.00	0.00	0.00	0.00	0.00	0.04	0.00	Conventional
Yard	0	PEM-T3F-06	N/A	0.41	0.00	0.00	0.00	0.00	0.00	0.41	0.00	Conventional
TF Permanent Easement, TF Temporary Workspace	0.3	PEM-T4F-02	77.24	0.12	0.00	0.00	0.00	0.00	0.00	0.12	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace, FFL-ATWS-0067, FFL-ATWS-0068, FFL-ATWS-0069, FFL-ATWS-0070, FFL-ATWS-0071, FFL-ATWS-0072,	6.8	PFO-T1F-01	6796.09	0.00	0.00	0.00	0.00	12.65	7.76	12.65	0.00	Conventional
FFL-TAR-013.000, FF Permanent Easement, FF Temporary Workspace	6	PFO-T1F-02	15.75	0.00	0.00	0.00	0.00	0.03	0.02	0.03	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	5.4	PFO-T1F-03	448.91	0.00	0.00	0.00	0.00	0.70	0.50	0.70	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	9	PFO-T1F-05	1380.63	0.00	0.00	0.00	0.00	2.35	1.58	2.35	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace, FFL-ATWS-0079	9.4	PFO-T1F-06	625.60	0.00	0.00	0.00	0.00	1.28	0.72	1.28	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	4.1	PFO-T2F-03	489.43	0.00	0.00	0.00	0.00	0.87	0.56	0.87	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace, FFL-ATWS-0114	12.5	PFO-T2F-05	61.22	0.00	0.00	0.00	0.00	0.14	0.07	0.14	0.00	Horizontal Bore

TABLE 2.3-1
Wetlands Crossed by the Franklin Farms Project

Project Facility	Approx. Milepost	Wetland ID	Linear Crossing Length (feet)[1]	Wetland Impact (acres)[2, 3, 4]								Crossing Method
				PEM		PSS		PFO		Total		
				Const.	Oper.	Const.	Oper.	Const.	Oper.	Const.	Oper.	
FF Permanent Easement, FF Temporary Workspace	12.6	PFO-T2F-07	115.66	0.00	0.00	0.00	0.00	0.19	0.14	0.19	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	4.4	PFO-T2F-09	19.12	0.00	0.00	0.00	0.00	0.04	0.02	0.04	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	12.1	PFO-T2F-19	251.51	0.00	0.00	0.00	0.00	0.43	0.29	0.43	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	12.2	PFO-T2F-20	84.14	0.00	0.00	0.00	0.00	0.15	0.10	0.15	0.00	Conventional
FF Temporary Workspace	12.4	PFO-T2F-21	N/A	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	10.9	PFO-T3F-05	192.98	0.00	0.00	0.00	0.00	0.32	0.22	0.32	0.00	Conventional
FF Permanent Easement, FF Temporary Workspace	11.4	PFO-T3F-07	78.70	0.00	0.00	0.00	0.00	0.14	0.09	0.14	0.00	Conventional
TFL-PAR-001.000, TF Permanent Easement,TFL-ATWS-0002,TF Temporary Workspace	0.3	PFO-T4F-01	470.09	0.00	0.00	0.00	0.00	0.70	0.54	0.70	0.00	Conventional

[1] Represent crossing width at pipe center
[2]Palustrine Emergent (PEM), Palustrine Scrub Shrub (PSS), and Palustrine Forested (PFO)
[3] PEM wetlands within the permanent ROW will be allowed to return to pre-construction conditions, no permanent loss of wetlands. PFO wetlands within the permanent ROW will be converted to PEM wetlands; however, there will be no permanent loss of wetlands.
[4] Operational impacts are also included in construction impact total

TABLE 2.3-2
Summary of Wetland Impacts for the Franklin Farms Project

Facility	Temporary Construction Footprint (acres)			Operations Footprint (acres)	Temporary impact or permanent conversion [c] (acres)
	PEM [a]	PFO [a]	PSS [a]	PFO [b,c]	
Franklin Farms Lateral	1.00	19.29	0.00	12.06	32.35
Trio Farm Lateral	0.61	0.70	0.00	0.54	1.31
Contractor Yards	0.41	0.00	0.00	0.00	0.41
PAR	0.00	0.00	0.00	0.00	0.00
TAR	0.02	0.00	0.00	0.00	0.02
FF Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station	N/A	N/A	N/A	N/A	N/A
FF Bee Bayou Delivery Meter Station	N/A	N/A	N/A	N/A	N/A
FF Interconnect Site	N/A	N/A	N/A	N/A	N/A
FF Gulf Crossing Receipt Meter Station	N/A	N/A	N/A	N/A	N/A
MEP Tap Site	N/A	N/A	N/A	N/A	N/A
FF MEP Receipt Meter Station	N/A	N/A	N/A	N/A	N/A
Franklin Farms Delivery Meter Station	N/A	N/A	N/A	N/A	N/A
FF Trio Farm Delivery Meter Station	N/A	N/A	N/A	N/A	N/A

Notes:
[a] Palustrine Emergent (PEM), Palustrine Forested (PFO).
[b] In open-cut areas, the FERC Procedures require that 10 feet of the permanent right-of-way (ROW) (centered on the pipeline) be maintained in an herbaceous state and trees be restricted to less than 15 feet high within 15 feet of the pipeline. The maintained ROW in this table reflects the 10 feet area to be maintained within an herbaceous state.
[c] PEM wetlands within the permanent ROW will be allowed to return to pre-construction conditions, no permanent loss of wetlands. PFO wetlands within the permanent ROW will be converted to PEM wetlands; however, there will be no permanent loss of wetlands.

2.3.2 CONSTRUCTION IMPACTS AND MITIGATION

The proposed aboveground facilities will not impact any wetlands. Both short- and long-term impacts are expected from construction and operation of the proposed Franklin Farms Lateral, Trio Farm Lateral, and permanent access roads (PAR). The laydown yards and temporary access roads (TAR) will have short term impacts on wetlands. ETC Tiger will avoid and minimize potential impacts by following the BMPs described in the FERC Plan and FERC Procedures. Impacts to PEM wetlands are anticipated to be short-term and minor. Following construction, PEM wetlands will be returned to pre-construction conditions and

allowed to naturally revegetate from existing rootstocks and seedbanks. Revegetation of PEM wetlands to pre-construction conditions is anticipated to occur within one to two growing seasons.

Impacts to PFO wetlands are anticipated to be long-term. In conventional construction areas (trenched areas), the FERC Procedures require that ten feet of the permanent ROW (centered on the pipeline) be maintained in an herbaceous state during operations and trees be restricted to less than 15 feet high within 15 feet of the pipeline, herein referred to as the maintenance corridor. The remaining permanent ROW that lies outside of the maintenance corridor and the temporary construction ROW will be allowed to revegetate from existing rootstocks and seedbanks to pre-construction conditions. PFO wetlands within the permanent ROW and outside of the maintenance corridor, the construction ROW, and Additional Temporary Workspace (ATWS) are expected to return to pre-construction conditions within 10 to 30 years. PFO wetlands located within the maintenance corridor (10 feet centered on the pipeline) will be converted to PEM or PSS wetlands.

Further, to minimize potential impacts to wetlands from soil compaction and rutting, ETC Tiger will use timber mats (or equivalent) and low ground pressure equipment, when possible, during Project construction based on wetland saturation.

Following construction, ETC Tiger will perform annual monitoring and maintenance within temporarily impacted wetland areas per the FERC Procedures, as well as any USACE permit conditions, until restoration requirements have been met and the wetland areas have been successfully restored. Revegetation will be considered successful when the vegetative cover returns to at least 80 percent of the type, density, and distribution of the vegetation in adjacent undisturbed portions of the wetland. For any wetland where revegetation via natural recruitment is not successful at the end of three years after construction, a remedial revegetation plan will be developed and implemented to actively revegetate these wetlands.

2.3.2.1 Spills

Accidental spills of construction-related fluids (e.g., oil, gasoline, or hydraulic fluids) on the landscape or directly into wetlands could result in water quality impacts. Impacts to wetlands would depend on the type and quantity of the spill, and the dispersal and attenuation characteristics of the wetland. To reduce the potential for wetland contamination, ETC Tiger will implement its SPAR Plan, which includes BMPs to avoid and minimize the potential for accidental releases and contains measures that will be implemented to clean up any releases.

2.3.3 OPERATION IMPACTS AND MITIGATION

Although the proposed aboveground facilities will not impact wetlands, the proposed pipeline ROW will result in a permanent conversion of PFO wetlands. and routine maintenance (e.g., mowing) of the pipeline ROW and aboveground facilities will adhere to the FERC Plan, FERC Procedures, and ETC Tiger's SPAR Plan and are not expected to have any noticeable impact on wetlands within the Project area.

2.3.4 COMPENSATORY WETLAND MITIGATION

The permanent conversion of PFO wetlands (i.e., the permanent removal of trees within forested wetlands that will be maintained in an herbaceous state) is detailed in **Tables 2.3-1** and **2.3-2**. Under Sections 404 and 401 of the CWA, ETC Tiger will be required to obtain a permit from the USACE Vicksburg District (Section 404) for the Project's proposed wetland impacts. ETC Tiger will adhere to the USACE permit requirements regarding any compensatory mitigation for unavoidable impacts to wetlands. ETC Tiger plans to purchase compensatory wetland mitigation credits from a mitigation bank, as required. A copy of the final mitigation plan will be provided to the FERC Docket when available.

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ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 3
Fish, Vegetation, and Wildlife

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 3 FILING REQUIREMENTS	
Information	**Section Reference**
1. Classify the fishery type of each surface waterbody that would be crossed, including fisheries of special concern. (§ 380.12(e) (1))	Section 3.1.1
2. Describe terrestrial and wetland wildlife and habitats that would be affected by the project. (§ 380.12(e) (2))	Section 3.2.1 Section 3.3.1
3. Describe the major vegetative cover types that would be crossed and provide the acreage of each vegetative cover type that would be affected by construction. (§ 380.12(e) (3))	Section 3.2.1 Table 3.2-2
4. Describe the effects of construction and operation procedure on the fishery resources and proposed mitigation measures.	Section 3.1.2 Section 3.1.3
5. Evaluate the potential for short-term, long-term, and permanent impact on the wildlife resources and state-listed endangered or threatened species caused by construction and operation of the project and proposed mitigation measures. (§ 380.12(e) (4))	Section 3.3.3 Section 3.3.4 Section 3.4.2 Section 3.4.3
6. Identify all federally listed or proposed endangered or threatened species that potentially occur in the vicinity of the project and discuss the results of the consultations with other agencies. Include survey reports as specified in (§ 380.12(e) (5)).	Table 3.4-1 Appendix D-1
7. Identify all federally listed essential fish habitat (EFH) that potentially occurs in the vicinity of the project and the results of abbreviated consultations with National Marine and Fisheries Service (NMFS), and any resulting EFH assessment. (§ 380.12(e) (6))	Section 3.1.1.1
8. Describe any significant biological resources that would be affected. Describe impact and any mitigation proposed to avoid or minimize that impact. (§ 380.12(e) (4&7))	Section 3.1.1.1 Section 3.2.1.1 Section 3.3.2
Additional Information Often Missing and Resulting in Data Requests	
Provide copies of correspondence from federal and state fish and wildlife agencies along with responses to their recommendations to avoid or minimize impacts on fisheries, wildlife, and vegetation	Appendix D-1
Provide a list of significant wildlife habitats crossed by the project. Specify locations by milepost and include length and width of crossing at each significant wildlife habitat.	Section 3.3.2
Provide a description of project-specific measures that would be implemented during construction and operation of the project to avoid or minimize impacts on migratory birds. Include comments from the U.S. Fish and Wildlife Service on the proposed measures.	Section 3.3.3 Section 3.3.4 Section 3.5.1 Section 3.5.2
For aquatic and marine species, be sure to include effects of sedimentation, changes to substrate, effects of blasting, etc. This information may be needed on a location-specific (i.e., milepost) basis and may require geophysical and other surveys. Results of such surveys and analyses should be included in the application.	Section 3.1.2.1

RESOURCE REPORT NO. 3
FISH, WILDLIFE AND VEGETATION

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ABBREVIATIONS AND ACRONYMS

ATWS	additional temporary workspace
BCR	bird conservation region
BMP	best management practices
Certificate	Certificate of Public Convenience and Necessity
CFR	Code of Federal Regulations
EFH	essential fish habitat
EI	environmental inspector
EPA	Environmental Protection Agency
ESA	Endangered Species Act
ETC Tiger	ETC Tiger Pipeline, LLC
FERC or Commission	Federal Energy Regulatory Commission
FERC Plan	FERC Upland Erosion Control, Revegetation and Maintenance Plan
FERC Procedures	FERC Wetland and Waterbody Construction and Mitigation Procedures
FF	Franklin Farms
IPaC	Information for Planning and Consultation
LDWF	Louisiana Department of Wildlife and Fisheries
MBTA	Migratory Bird Treaty Act
MEP	Midcontinent Express Pipeline LLC
NABCI	North American Bird Conservation Initiative
NGA	Natural Gas Act
NMFS	Natural Marine Fishery Service
NWSRS	National Wild and Scenic Rivers System
PEM	palustrine emergent
PFO	palustrine forested
Project	Franklin Farms Project
PSS	palustrine scrub shrub
ROW	right-of-way
SPAR Plan	Spill Prevention and Response Plan
WRE	Wetland Reserve Easement
U.S.	United States
USACE	United States Army Corps of Engineers
USDA	United States Department of Agriculture
USFWS	United States Fish & Wildlife Service
USGS	United States Geological Survey

3.0 FISH, WILDLIFE AND VEGETATION

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, an interconnect site, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project in June 2027, and following the receipt of all applicable approvals, anticipating construction activities to begin in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

Resource Report No. 3 describes the existing fish, wildlife, and vegetation resources within the Project area, the potential impacts of construction and operation on these resources, and measures that will be implemented to reduce and mitigate potential Project-related impacts. Information on fisheries, wildlife, vegetation, and federally and state-listed threatened and endangered species was obtained from review of existing literature, consultation with various federal and state natural resource agencies, and biological field reconnaissance surveys.

To avoid and minimize impacts to fisheries, wildlife, and vegetation resources during construction and operation of the Project, ETC Tiger will implement the FERC *Upland Erosion Control, Revegetation and Maintenance Plan*, May 2013 (FERC Plan, FERC 2013a; **Appendix C-1**), the FERC *Wetland and Waterbody Construction and Mitigation Procedures*, May 2013 (FERC Procedures, FERC 2013b; **Appendix C-2**), with proposed site-specific exceptions, and ETC Tiger's *Spill Prevention and Response Plan* (SPAR Plan) (**Appendix C-3**).

3.1 FISHERIES RESOURCES

Fisheries resources are surface water areas that provide habitat for fish and are typically characterized according to water temperature (warmwater or cold water), salinity (freshwater, marine, or estuarine), types of fishing use (commercial or recreational), and utilization by open water marine fishes that require freshwater areas to spawn (i.e., anadromous).

Significant fisheries resources are defined by the FERC as waterbodies that (1) provide important habitat for foraging, rearing, or spawning of fish species; (2) represent important commercial or recreational fishing areas; and/or (3) support large populations of commercially or recreationally valuable fish species or species listed for protection at the federal, state, or local level. Resource Report No. 2 (*Water Use and Quality)* contains a detailed characterization of the waterbodies crossed by the proposed Project.

3.1.1 Existing Resources

Project workspace will cross perennial and ephemeral streams, ditches, and ponds (see Resource Report No. 2, Table 2.2-2). All waterbodies crossed by the Project are freshwater. Common freshwater fish species in Louisiana include largemouth bass (*Micropterus nigricans*), bluegill (*Lepomis macrochirus*), longear sunfish (*Lepomis megalotis*), redear sunfish (*Lepomis microlophus*), white crappie (*Pomoxis annulari*), black crappie (*Pomoxis nigromaculatus*), warmouth (*Lepomis gulosus*), striped bass (*Morone saxatilis*), white bass (*Morone chrysops*), hybrid striped bass (*Morone chrysops* × *M. saxatilis*), flathead catfish (*Pylodictis olivaris*), channel catfish (*Ictalurus punctatus*), chain pickerel (*Esox niger*), alligator gar (*Atractosteus spatula*), and bowfin (*Amia calva*) (Louisiana Department of Wildlife and Fisheries [LDWF], 2026a).

Although the LDWF does not have state designated fisheries, they enhance freshwater and saltwater fish populations and provide the public with recreational fishing opportunities through initiatives such as the operation of four freshwater fish hatcheries, the stocking of fish in ponds throughout the state, and their artificial reef program. The Project is not within close proximity to any LDWF hatcheries or coastal artificial reefs, and there are no LDWF stocked ponds within Richland Parish (LDWF, 2026b). All waterbodies crossed by the Project could support warmwater freshwater fish communities. The perennial streams within the Project area have a continuous flow of water and are likely to provide stable aquatic habitat for a variety of fish. Perennial streams with high water quality and suitable habitat support diverse fisheries. However, the ephemeral streams and ditches flow only in response to precipitation events and are unlikely to contain sufficient water to support fisheries.

3.1.1.1 Fisheries of Special Concern

Fisheries of special concern include waterbodies containing exceptional recreational or commercial fisheries, specially designated streams or rivers, and/or waterbodies supporting threatened or endangered aquatic species. The aquatic features crossed by the proposed Project do not support any commercial fisheries. Additionally, the aquatic features are not known to provide potentially suitable habitat for any Endangered Species Act (ESA)-listed fish species (United States [U.S.] Fish and Wildlife [USFWS], 2026a) or lie within USFWS designated critical habitat (USFWS, 2026b). Therefore, fisheries of special concern and ESA-listed fish species are not anticipated to be impacted as a result of Project construction or operation activities.

The Magnuson-Stevens Fishery and Conservation Management Act mandates the identification of habitats essential to managed species and execution of plans to conserve and enhance these habitats. Essential fish habitat (EFH) is defined by Congress as "*those waters necessary to fish for spawning, breeding, feeding, or growth to maturity*". The waterbodies crossed by the Project area do not contain species managed by the National Marine Fisheries Service (NMFS), nor does it contain designated EFH (NMFS, 2026).

3.1.2 Construction Impacts and Mitigation

Potential impacts to waterbodies from proposed Project construction are discussed in Resource Report No. 2. None of the waterbodies affected by the Project contain or have the potential to contain EFH (NMFS, 2026), and there are no state designated warmwater fisheries with the Project area vicinity (See Section 3.1.1); therefore, no impacts on EFH will occur as a result of the Project.

Waterbodies will be crossed according to the methods outlined in Table 2.2-2 of Resource Report No. 2. Refer to Section 1.3.1.1 in Resource Report No. 1 (*General Project Description)* for further details regarding the conventional and flume crossing methods. ETC Tiger will conduct waterbody crossings according to measures and Best Management Practices (BMP) contained in the FERC Procedures, which are designed to minimize impacts to waterbodies and aquatic resources and restrict the duration of impacts to the shortest time possible to allow pipeline construction across a waterbody.

Construction of the proposed Project could result in temporary impacts to fish and aquatic resources. Sedimentation and turbidity, alteration or removal of instream and stream bank cover, stream bank erosion, or introduction of water pollutants resulting from Project activities could increase stress, injury, and mortality of aquatic biota. The degree of impact would depend on the waterbody crossing method, the existing conditions at each crossing location, the restoration procedures and mitigation measures employed, and the timing of construction. The discussions below further describe construction impacts on fisheries and aquatic resources and the measures that would be implemented to minimize impacts.

3.1.2.1 Sedimentation and Turbidity

ETC Tiger will use the horizontal bore construction method to install the pipeline across some waterbodies along the proposed Pipeline. This construction method is a trenchless construction method and will avoid impacts to these waterbodies. Therefore, waterbodies crossed via horizontal bore are not anticipated to be impacted by the Project. Refer to Table 2.2-2 in Resource Report No. 2 for details regarding which waterbodies will be crossed via horizontal bore.

Pipeline construction using conventional and flume methods may result in a temporary increase in sedimentation and turbidity from in-stream and adjacent construction activities. Sedimentation could impact the health of fish eggs and other benthic biota and alter stream bottom characteristics, such as converting sand, gravel, or rock substrate to silt or mud substrate. These habitat alterations could reduce juvenile fish survival, spawning habitat, and benthic community diversity and health. Fish and other stream biota would be displaced to similar habitat upstream or downstream of the pipeline crossing, which could lead to increased competition for habitat and food sources and affect fish survival and health.

Increased turbidity could temporarily reduce dissolved oxygen levels in the water column and reduce respiratory functions in stream biota, which could temporarily displace fish to unaffected stream segments, reduce fish health, or increase fish mortality. Turbid conditions could also reduce the ability for biota to find food sources or avoid prey. The extent of impacts from sedimentation and turbidity would depend on sediment loads, stream flows, stream bank and stream bed composition, sediment particle size, and the duration of the disturbances.

Although the standard conventional and flume methods will generate in-stream sediment and turbidity, the elevated levels will be short-term, occurring only during active construction. The effects will be minimized by installing and maintaining sediment and erosion controls throughout construction until revegetation is successful. In accordance with the FERC Procedures, ETC Tiger will complete all in-stream work in less than 24 hours for minor streams (less than 10 feet across) and less than 48 hours for intermediate streams (between 10 and 100 feet across). Trench spoil will be stored above the banks of waterbodies and will be protected with erosion control devices that prevent, or significantly reduce, sediment runoff from entering the waterbody. Therefore, impacts to water quality are anticipated to be minor, localized, and short-term, limited to the time of active construction.

3.1.2.2 Loss of Stream Bank Cover

Removal of trees and other bank vegetation from the edges of waterbodies at conventional pipeline crossing locations may reduce shading of the waterbody, diminish escape cover, and result in locally elevated water temperatures. Elevated water temperatures can, in turn, lead to a reduction in dissolved oxygen levels. This could reduce the availability of habitat for certain fish species. The proposed pipeline routes were designed to minimize impacts on forested land to the extent feasible. To minimize potential effects associated with the loss of riparian shade and vegetation cover, clearing of trees and other vegetation will be restricted to only what is necessary to safely construct and operate the pipeline. Once construction is complete, streambeds and banks will be restored in accordance with the FERC Procedures, and post construction maintenance (or mowing) will be limited to when necessary to facilitate periodic corrosion/leak surveys or to protect the integrity of the pipeline coating. Overall, the potential impacts to aquatic habitats and species from construction of the proposed Project are anticipated to be minor, localized, and short-term.

3.1.2.3 Fluid and Chemical Spills

An inadvertent spill of fuel or equipment related fluids could result in water quality impacts affecting fish and other organisms. Impacts to aquatic resources would depend on the type and quantity of the spill, and the dispersal and attenuation characteristics of the waterbody. The chemicals released during spills could

have acute fish impacts, such as altered behavior, changes in physiological processes, or changes in food sources. Fish could also experience mortality if a hazardous liquid is spilled into a waterbody. Furthermore, ingestion of large numbers of contaminated prey could impact fish predators in the food chain.

Potential spill-related impacts from construction are mainly associated with fuel storage, equipment refueling, and equipment maintenance. These potential impacts will be avoided or greatly reduced by regulating fuel storage and refueling activities, and by requiring immediate cleanup should a spill or leak occur. ETC Tiger's SPAR Plan (see **Appendix C-3**) describes in detail the preventative measures to be used to avoid spills and leaks, as well as the mitigative measures to be utilized to minimize potential impacts should a spill or leak occur. In accordance with FERC's Procedures, construction equipment, vehicles, hazardous materials, chemicals, fuels, lubricating oils, and petroleum products will not be parked, stored, or serviced within 100 feet of any waterbody.

3.1.2.4 Hydrostatic Testing

Hydrostatic testing of the integrity of the completed pipeline and aboveground facilities will occur before final restoration and placing the Project into service. As discussed in Resource Report No. 2, water from select existing water wells will be used for hydrostatic testing and other construction water uses. Therefore, water will not be withdrawn from any waterbodies during construction. Further, no chemicals will be added to the hydrostatic test water. Hydrostatic test water will be discharged in accordance with the FERC Plan and FERC Procedures and state permitting. ETC Tiger will discharge test water into a well vegetated upland area and filter the water through a filter bag and/or erosion control barrier to prevent scour and sedimentation, flooding, or the introduction of foreign or toxic substances into the aquatic system. Therefore, hydrostatic testing is not anticipated to adversely impact aquatic resources.

3.1.3 Operation Impacts and Mitigation

Operation and routine maintenance of the pipeline right-of-way (ROW) (e.g., mowing) and aboveground facilities are not expected to have a noticeable impact on waterbodies and aquatic resources within the proposed Project area. During operations, impacts to waterbodies and aquatic resources will be avoided or minimized with the use of BMPs and protective measures in accordance with the FERC Plan and Procedures and ETC Tiger's SPAR Plan.

3.2 VEGETATION

This section provides a detailed description of the vegetation cover types that are found within the Project area. Vegetative cover types, or habitats, have been classified by the presence of the dominant vegetation species. Defining vegetative habitats is necessary to assess the potential presence of wildlife, threatened and endangered species, and other ecologically sensitive areas.

3.2.1 Existing Resources

The proposed Project falls within the Macon Ridge (United States Environmental Protection Agency [EPA] level IV ecoregion: 73j) of the Mississippi Alluvial Plain (Daigle et al. 2006). The Mississippi Alluvial Plain is primarily a broad, flat, alluvial plain with river terraces, swales, and levees providing the main elements of relief. Historically, this region contained one of the largest continuous wetland systems in North America, with bottomland deciduous forest covering the region prior to much of it being cleared for cultivation. The widespread loss of forest and wetland habitat has impacted wildlife and reduced bird populations, although the region remains a major bird migration corridor. The Macon Ridge region contains higher elevations, is underlain by soils with higher drainage, and supports drier plant communities than surrounding ecoregions. Native forests within the ecoregion range from well drained bottomlands dominated by willow oak (*Quercus phellos*), water oak (*Quercus nigra*), and swamp chestnut oak (*Quercus*

michauxii) with upland hardwood forests dominated by white oak (*Quercus alba*), southern red oak (*Quercus falcata*), and in drier areas, post oak (*Quercus stellata*). Historically, prairies and loblolly pine (*Pinus taeda*) dominated areas may have also occurred. In present day, the Macon Ridge region is a mosaic of cropland and pasture with some woodland and forest. Major crops produced in the ecoregion include cotton, corn, soybeans and wheat (Daigle et al. 2006).

Characterization of vegetation cover types within the Project area was completed using information gathered from field surveys conducted in February, March, and May 2026, as well as review of recent aerial photography (2026). The following classifications represent the major vegetation cover types present in the Project area:

- Agricultural land – Agricultural lands predominantly include cropland, hay fields, and pasturelands.
- Developed – Land used for residential, commercial, or industrial purposes. These areas are typically unvegetated and may contain areas containing grass species or landscaped land.
- Open Land– Open land includes non-forested, vegetated areas that are not used for agricultural production or landscaped. This vegetation category also includes rangeland, herbaceous dry prairie, shrub and brushlands, and maintained ROWs (e.g., road corridors).
- Forest/Woodland – Forest/Woodland areas consist of various ages of upland deciduous and coniferous tree species that are not used for specific commercial purposes. The canopy closure in areas considered to be forested must be 25 percent or more. Forests/Woodlands may support an understory of shrubs and herbaceous plants including grasses.
- Wetlands – Wetlands in the Project area include palustrine emergent (PEM) and palustrine forested (PFO) wetlands. PEM wetlands are typically dominated by perennial emergent and rooted herbaceous hydrophytes, excluding mosses and lichens that are present for most of the growing season in most years. PFO wetlands are dominated by woody vegetation that is at least 19.7 feet tall (Cowardin et al., 1979). Resource Report No. 2 includes a discussion of wetlands crossed in the Project area (see **Appendix A-4** for mapping).
- Open water – Open water consists of waterbodies that typically have less than 25 percent cover of vegetation and land, including streams and ponds (see Resource Report No. 2).
- Silviculture – wooded lands being managed for forest products (e.g., pine plantations, sugar maple stands, or tree nurseries).

The predominant vegetation cover type within the Project area is agricultural land. A breakdown of impacts by vegetation cover types is provided in Section 3.2.2. Habitats within the Project area are depicted in **Appendix A-6**.

3.2.1.1 Vegetation Communities of Special Concern

Vegetative communities of special concern include ecologically important natural communities, unique or critical plant communities, threatened or endangered plant species, or other rare and imperiled plants in need of special protection or minimal disturbance. ETC Tiger reviewed information from USFWS and LDWF to determine if vegetation communities of special concern occur within the Project area. No federally protected plant species were identified to have the potential to exist within the Project area (USFWS, 2026a).

The LDWF database identified two imperiled or vulnerable natural plant communities and three imperiled or vulnerable plant species that have the potential to occur within the Project area. The identified natural plant communities include mesic hardwood flatwoods and mixed hardwood-loblolly forest, and the

identified plant species include northern red oak (*Quercus rubra*), purple coneflower (*Echinacea purpurea*), and yellowleaf Tinker's-weed (*Triosteum angustifolium*) (LDWF, 2026c). The plant species which were identified to have the potential to occur within the Project area were not observed during the 2026 field investigation. ETC Tiger sent a consultation letter to the LDWF to request information regarding the natural communities within the Project area vicinity as detailed in **Table 1.7-1** of **Resource Report 1**. A copy of the correspondence is included in **Appendix D-1.**

3.2.1.2 Invasive Species

Invasive species are species that are non-native to the ecosystem and whose introduction causes or is likely to cause economic or environmental harm. They are spread primarily by human activities, often unintended. When introduced they can pose a significant threat to native communities, potentially leading to the extinction of native species, a change in biodiversity, and/or permanently alter habitats (U.S. Department of Agriculture [USDA], 2026a). Invasive plant species which occur in Louisiana are listed in **Table 3.2-1**. During field surveys the following non-native invasive species were observed: Chinese privet (*Ligustrum sinense*), Chinese tallow (*Triadica sebifera*), alligator weed (*Alternanthera philoxeroides*), bermuda grass (*Cynodon dactylon*), Japanese honeysuckle (*Lonicera japonica*), Asian spiderwort (*Murdannia keisak*), golden crown grass (*Paspalum dilatatum*), bahia grass (*Paspalum notatum*), Vasey's grass (*Paspalum urvillei*), curly dock (*Rumex crispus*), Brazilian vervain (*Verbena brasiliensis*), and muscadine (*Vitis rotundifolia*). See **Section 3.2.2** for further details regarding invasive and exotic species mitigation.

TABLE 3.2-1
Invasive Plant Species in Louisiana

Teir 1	
Coral Ardisia (*Ardisia crenata*)	Camphor Tree (*Cinnamomum camphora*)
Elephant Ear (*Colocasia esculenta*)	Bermuda Grass (*Cynodon dactylon*)
Japanese Twin-Sorus Fern (*Deparia petersenii*)	Air Yam (*Dioscorea alata & D. bulbifera*)
Brazilian Waterweed (*Egeria densa*)	Water Hyacinth (*Eichhornia crassipes*)
Chinese Parasol Tree (*Firmiana simplex*)	Hydrilla or Waterthyme (*Hydrilla verticillata*)
Cogon Grass (*Imperata cylindrica*)	Yellow Flag Iris (*Iris pseudacorus*)
Chinese Privet (*Ligustrum sinense*)	Japanese Climbing Fern (*Lygodium japonicum*)
Torpedo Grass (*Panicum repens*)	Water Paspalum (*Paspalum modestum (=P. hydrophyllum*))
Vasey Grass (*Paspalum urvillei*)	Trifoliate Orange (*Poncirus trifoliata*)
Kudzu (*Pueraria montana*)	McCartney Rose (*Rosa bracteata*)
Cherokee Rose (*Rosa laevigata*)	Common Salvinia (Water Spangles, *Salvinia minima*)
Giant Salvinia (*Salvinia molesta*)	Smut Grass (*Sporobolus indicus*)
Chinese Tallow Tree (*Triadica sebifera*)	Tungoil Tree (*Vernicia fordii*)
Teir 1I	
Yellow Floating Heart (*Nymphoides peltata*)	Tree-of-Heaven (*Ailanthus altissima*)
Mimosa (*Albizia julibrissin*)	Alligatorweed (*Alternanthera philoxeroides*)
Sessile Joyweed (*Alternanthera sessilis*)	Giant Reed (*Arundo donax*)
Mosquito Fern (*Azolla pinatta*)	Australian Bluestem (*Bothriochloa bladhii*)

TABLE 3.2-1 **Invasive Plant Species in Louisiana**	
King Ranch Bluestem (*Bothriochloa ischaemum var. songarica*)	Little Quaking Grass (*Briza minor*)
Paper Mulberry (*Broussonetia papyrifera*)	Balloon Vine (*Cardiospermum halicacabum*)
Nodding Thistle (*Carduus nutans*)	Bushkiller (*Cayratia japonica*)
Water Sprite (*Ceratopteris thalictroides*)	Paraguayan Windmill Grass (*Chloris canterai*)
Bull Thistle (*Cirsium vulgare*)	Large-Head Horseweed (*Conyza bonariensis*)
Deep-rooted Sedge (*Cyperus entrerianus*)	Ricefield Flatsedge (*Cyperus iria*)
Fuzzy Flatsedge (*Cyperus pilosus*)	Purple Nutsedge (*Cyperus rotundus*)
Asian Net-veined Holly Fern (*Cyrtomium fortunei*)	Kleberg Bluestem (*Dichanthium annulatum*)
Smooth Crabgrass (*Digitaria ischaemum*)	Hairy Crabgrass (*Digitaria sanguinalis*)
Dopatrium (*Dopatrium junceum*)	Junglerice (*Echinochloa colona*)
Barnyardgrass (*Echinochloa crus-galli*)	Thorny Olive (*Elaeagnus pungens*)
Autumn Olive (*Elaeagnus umbellata*)	Elodea (*Elodea canadensis*)
Centipedegrass (*Eremochloa ophiuroides*)	Reed Fescue (*Festuca arundinacea*)
Groundivy (*Glechoma hederacea*)	English Ivy (*Hedera helix*)
West Indian Marshgrass (*Hymenachne amplexicaulis*)	Tropical White Morning-Glory (*Ipomoea alba*)
Mile-a-Minute Vine (*Ipomoea cairica*)	Cypress Vine (*Ipomoea quamoclit*)
Tie Vine (*Jacquemontia tamnifolia*)	Japanese Clover (*Kummerowia striata*)
West India Camara (*Lantana camara*)	Trailing Shrub Verbena (*Lantana montevidensis*)
Shrubby Lespedeza (*Lespedeza bicolor*)	Japanese Privet (*Ligustrum japonicum*)
Glossy Privet (*Ligustrum lucdium*)	Common Privet (*Ligustrum vulgare*)
Indian Marshweed (*Limnophila indica*)	Marshweed (*Limnophila x ludoviciana*)
Asian Marshweed (*Limnophila sessiliflora*)	Monkeygrass (*Liriope muscari*)
Perennial Ryegrass (*Lolium perenne*)	Japanese Honeysuckle (*Lonicera japonica*)
Uruguay Seedbox (*Ludwigia hexapetala*)	Peruvian Water Grass (*Luziola peruviana*)
Purple Loosestrife (*Lythrum salicaria*)	Catclaw Vine (*Macfadyena unguis-cati*)
Big-foot Water Clover (*Marsilea macropoda*)	Chinaberry (*Melia azedarach*)
Parrotfeather (*Myriophyllum aquaticum*)	Eurasian Watermilfoil (Spike Milfoil) (*Myriophyllum spicatum*)
Brittle Naiad (Brittle Waternymph) (*Najas minor*)	Nandina (*Nandina domestica*)
Watercress (*Nasturtium officinale*)	White Egyptian Lotus (*Nymphaea lotus*)
Sacred Lotus (*Nelumbo nucifera*)	Crested Floating Hearts (*Nymphoides cristata*)
Duck Lettuce (*Ottelia alismoides*)	Cuban Bulrush (*Oxycaryum cubense*)
Stinkvine (*Paederia foetida*)	Dallis Grass (*Paspalum dilatatum*)

TABLE 3.2-1 **Invasive Plant Species in Louisiana**	
Common Bahia Grass (*Paspalum notatum*)	Beefsteak Plant (*Perilla frutescens*)
Timothy Grass (*Phleum pratense*)	Golden Bamboo (*Phyllostachys aurea*)
Water Lettuce (*Pistia stratiotes*)	Japanese Knotweed (*Polygonum cuspidatum*)
Curly Pondweed (*Potamogeton crispus*)	Spider Brake Fern (*Pteris multifida*)
Callery Pear (*Pyrus calleryana*)	Sawtooth Oak (*Quercus acutissima*)
Castorbean (*Ricinus communis*)	Multiflora Rose (*Rosa multiflora*)
Indian Toothcup (*Rotala indica*)	Itch Grass (*Rottboellia cochinchinensis*)
Linear Leaf Wild Petunia (*Ruellia simplex*)	Curly Dock (*Rumex crispus*)
Indian Cupscale (*Sacciolepis indica*)	Guyana Arrowhead (*Sagittaria guyanensis*)
Brazilian Rattlebox (*Sesbania punicea*)	Thin-Spike Bristlegrass (*Setaria pumila ssp. pallidefusca*)
Jerusalem Cherry (*Solanum pseudocapsicum*)	Tropical Soda Apple (*Solanum viarum*)
Johnson Grass (*Sorghum halepense*)	African Saltcedar (*Tamarix Africana*)
Canary Island Saltcedar (*Tamarix canariensis*)	French Tamarisk (*Tamarix gallica*)
Saltcedar (*Tamarix ramosissima*)	False Maiden Fern (*Macrothelypteris torresiana*)
Guinea Grass (*Urochloa maxima*)	Para Grass (*Urochloa mutica*)
Brazilian Vervain (*Verbena brasiliensis*)	Winter Vetch (*Vicia villosa*)
Chinese Wisteria (*Wisteria sinensis*)	Japanese Hawksbeard (*Youngia japonica*)
Teir 1II	
Brazilian Water-hyssop (*Bacopa egensis*)	Blyxa (*Blyxa aubertii*)
Ethiopian Rattlebox (*Crotalaria brevidens var. intermedia*)	Lanceleaf Rattlebox (*Crotalaria lanceolata*)
Rattleweed (*Crotalaria retusa*)	Showy Rattle (*Crotalaria spectabilis*)
Eucalyptus (*Eucalyptus spp*.)	Asian Spiderwort (*Murdannia keisak*)
Crownvetch (*Securigera varia*)	
Teir IV	
Australian Pine (*Casuarina spp*.)	"Cylindro" Blue Green Algae (*Cylindrospermopsis raciborskii*)
Rooting Water Hyacinth (*Eichhornia azurea*)	Indian Swampweed (*Hygrophila polysperma*)
Water Spinach (*Ipomoea aquatica*)	African Elodea (*Lagarosiphon major & L. muscoides*)
Old World Climbing Fern (*Lygodium microphyllum*)	Water Clovers (*Marsilea minuta & M. mutica*)
Punktree Melaleuca (*Melaleuca quinquenervia*)	False Pickerelweeds (*Monochoria hastata & M. vaginalis*)
Marine Naiad (*Najas marina*)	Water Floatinghearts (*Nymphoides indica*)
Roundleaf Toothcup (*Rotala rotundifolia*)	Brazilian Peppertree (*Schinus terebinthifolius*)

TABLE 3.2-1 **Invasive Plant Species in Louisiana**	
Aquatic Soda Apple (*Solanum tampicense*)	Water Chestnut (*Trapa natans*)
Source: LDWF, 2019	

3.2.2 Construction Impacts and Mitigation

Construction impacts to vegetation will include cutting, clearing, and/or removal of existing vegetation within the construction ROW and Additional Temporary Workspace (ATWS) areas. The vegetation impacts associated with the Project will vary by vegetation type and the rate at which vegetation regenerates after construction. A summary of impacts to vegetation cover is presented in **Table 3.2-2**.

TABLE 3.2-2
Vegetation and Habitat Types Affected by Construction and Operation of the Project (acres)

Project Facility	Forest/ Woodland		Silviculture (Pine Plantation)		Agricultural Land		Open Land		Open Water		Wetlands[3]						Developed[4]		Total	
											PEM		PSS		PFO					
	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation
Pipeline Facilities [5]																				
Franklin Farms Lateral	15.6	6.8	2.3	0.9	114.0	43.8	27.8	11.9	1.2	0.7	1.0	0.6	0.0	0.0	19.3	12.1	4.2	1.8	**185.4**	**78.5**
Franklin Farms (FF) Trio Farm Lateral	0.4	0.4	0.0	0.0	7.1	3.8	9.3	5.1	0.2	0.1	0.6	0.4	0.0	0.0	0.7	0.5	0.1	0.0	**18.3**	**10.5**
Aboveground Facilities																				
Meter Stations																				
FF Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station	0.0	0.0	0.0	0.0	6.0	3.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**6.0**	**3.6**
FF Bee Bayou Delivery Meter Station [6]	0.0	0.0	0.0	0.0	1.5	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**1.5**	**1.0**
FF Interconnect Site	0.0	0.0	0.0	0.0	1.0	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**1.0**	**0.9**
FF Gulf Crossing Receipt Meter Station	0.0	0.0	0.0	0.0	1.7	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**1.7**	**1.4**
MEP Tap Site	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.1**	**0.1**
FF MEP Receipt Meter Station	0.0	0.0	0.0	0.0	1.4	1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**1.4**	**1.2**
Franklin Farms Delivery Meter Station [6]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.6	1.0	**1.6**	**1.0**
FF Trio Farm Delivery Meter Station [6]	0.0	0.0	0.0	0.0	2.0	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**2.0**	**1.4**

TABLE 3.2-2
Vegetation and Habitat Types Affected by Construction and Operation of the Project (acres)

Project Facility	Forest/ Woodland		Silviculture (Pine Plantation)		Agricultural Land		Open Land		Open Water		Wetlands[3]						Developed[4]		Total	
											PEM		PSS		PFO					
	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation
Other Work Areas																				
Contractor Yards	0.0	0.0	0.0	0.0	0.0	0.0	4.7	0.0	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	9.7	0.0	**14.8**	**0.0**
Access Roads	0.1	0.0	0.0	0.0	6.4	2.2	1.5	0.5	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	6.7	4.6	**14.9**	**7.3**
Project Total [1]	16.1	7.2	2.3	0.9	141.3	59.4	43.3	17.5	1.6	0.9	2.0	1.0	0.0	0.0	20.0	12.6	22.2	7.4	**248.9**	**107.0**

Notes:

[1] Numbers may vary slightly due to rounding.

[2] Construction area includes the total acres of workspace required for construction of the Project, including the operation area.

[3] PEM = palustrine emergent, PSS = palustrine scrub-shrub, PFO = palustrine forested.

[4] Developed Land is land used for residential, commercial, or industrial purposes.

[5] Operational pipeline impacts presented in this table are based on the permanent easement; however, there will be no restrictions that prevent agricultural, open land, PEM wetlands, and open water from reverting to their prior use.

[6] As discussed in Section 1.1.4.1 of Resource Report 1, these meter stations will be previously disturbed by Entergy Louisiana, LLC prior to the construction of the Project.

Impacts on non-forested areas (i.e., agricultural land, open land, PEM wetlands, and developed land) will result in short-term, temporary vegetation disturbance because herbaceous and scrub shrub communities typically will revert to pre-construction conditions within one to three years. Forested communities, both upland and wetland, will be affected to a greater extent, due to the long-term conversion of these wooded habitats to earlier successional stages within the temporary ROW and ATWS areas (i.e., long-term, temporary disturbance) and permanent conversion to non-woody herbaceous species in the permanent, maintained ROW (i.e., permanent disturbance). Depending on the age of cleared trees, temporarily disturbed forested areas are expected to achieve preconstruction conditions within one to three decades. Permanent vegetation impacts will also occur at the new station sites which require converting small herbaceous areas to graveled, fenced areas.

To minimize vegetation impacts, ETC Tiger will implement BMP measures in the FERC Plan and FERC Procedures, which require the use of temporary and permanent erosion control measures, topsoil segregation in select areas, testing and mitigation for soil compaction, and limited routine vegetation maintenance. Furthermore, all temporarily disturbed upland areas will be reseeded in accordance with the FERC Plan, FERC Procedures, and recommendations from state agencies (i.e., the Natural Resources Conservation Service Field Service Center in Louisiana) and/or applicable permit requirements. PEM wetland areas will be returned to pre-construction conditions and allowed to naturally revegetate from existing rootstocks and seedbanks. PFO wetlands outside of the permanent ROW will also be allowed to naturally revegetate. Disturbed areas will be routinely monitored until restoration and revegetation are successful in accordance with the FERC Plan and FERC Procedures.

Invasive or exotic plant species may alter natural systems by out-competing native plants for resources and replacing native plants in the composition of native communities. Ground disturbance from construction activities provide ideal conditions for the establishment of invasive plant species. ETC Tiger will implement the following measures to minimize the potential to introduce or spread noxious and invasive vegetation species including:

- ETC Tiger will inform and train construction personnel regarding noxious weed and invasive species identification and the protocols to prevent or control the spread of invasive species. Environmental Inspectors (EI) will be employed during construction to monitor and provide construction oversight.
- Prior to construction, ETC Tiger will mark the entry and exit of any identified areas of noxious weed infestation with signage along the construction ROW.
- Vehicles and equipment will be inspected for remnant soils, vegetation, and debris, and will be cleaned of these materials before they are brought to the Project area.
- To prevent the spread of seeds, roots, or other viable plant materials, equipment used in areas containing invasive plant species will be cleaned before moving to an un-infested area.
- ETC Tiger will ensure that seeds for revegetation and straw or hay bales used for sediment barrier installations or mulch distribution, where appropriate, are weed-free.
- Post-construction monitoring of invasive plant species populations and colonization of the ROW will be conducted for a minimum of two years. Monitoring reports detailing the success of ROW restoration and revegetation measures will identify invasive plant species' colonization locations and densities and the management measures that will be implemented to control the identified populations.

- ETC Tiger will utilize mechanical treatment or herbicide application to control the spread of invasive species during and after construction. Herbicides will be applied according to manufacturer's printed recommendations and in accordance with applicable agency regulations governing herbicide application. No herbicides or pesticides will be used within 100 feet of waterbodies or wetlands. A qualified contractor will be utilized to determine the appropriate herbicide application method.

An accidental spill or release of hazardous materials (e.g., fuels, lubricants, solvents) during construction could also affect vegetation. However, potential vegetation impacts from accidental spills and releases will be avoided or minimized through implementation of ETC Tiger's SPAR Plan.

In summary, construction of the Project is not anticipated to have a significant impact on vegetation as most of the proposed construction activities will occur in areas of herbaceous vegetation which will be allowed to naturally revegetate from existing rootstocks and seedbanks following construction. Furthermore, ETC Tiger will implement the erosion and sediment control BMPs contained in the FERC Plan and FERC Procedures and the spill prevention and response procedures contained in ETC Tiger's SPAR Plan. Implementation of the measures outlined in these plans will promote revegetation in the Project area following construction.

3.2.3 Operation Impacts and Mitigation Measures

Operation and maintenance of the facilities will have little additional impact to vegetation after construction is complete. A summary of the vegetation cover operational impacts is presented in **Table 3.2-2**. To ensure access to its facilities and comply with requirements listed under 49 Code of Federal Regulations (CFR) Part 192, ETC Tiger's maintenance of the permanent ROW will include routine mowing and clearing activities using mechanical equipment. ETC Tiger will minimize impacts to vegetation resulting from operation and maintenance of the Project through the implementation of the FERC Plan, FERC Procedures, and ETC Tiger's SPAR Plan. Specifically, routine vegetation mowing or clearing over the full width of the permanent ROW in uplands will not be conducted more frequently than every three years. To facilitate periodic corrosion/leak surveys, a corridor not exceeding 10 feet in width in wetlands centered on the pipeline may be cleared at a frequency necessary to maintain the 10-foot corridor in an herbaceous state consistent with the FERC Procedures.

3.3 WILDLIFE

This section identifies and discusses the various wildlife species associated with the upland and wetland vegetation cover types to be traversed by the Project. Wildlife species occurring within the vicinity of the Project are characteristic of the habitats provided by the plant communities that occur in these areas. Protected wildlife species are discussed in Section 3.4.

3.3.1 Existing Resources

The composition, structure, and distribution of the plant community in an area constitute a large part of the cover and food components of wildlife habitat. As a result, areas with similar vegetative characteristics tend to have similar assemblages of wildlife species. Some wildlife species have very specific habitat requirements and are found in only a few habitats while other species have broader habitat requirements and are ubiquitous in nature. The proposed Project facilities will cross a variety of terrestrial, wetland, and stream/open water habitats that support wildlife species.

As noted above, the proposed Project will cross the Macon Region of the Mississippi Alluvial Plain. The Macon Region is characterized by cropland and pasture with some woodland and forest habitat (Daigle et al. 2006). Wildlife typical of the Macon Region is provided in **Table 3.3-1**.

TABLE 3.3-1
Wildlife Species Typical of Major Vegetation Cover Types Present in the Project Area

Common Wildlife Species	
Game Species	Mourning Dove (*Zenaida macroura*), White-winged Dove (*Zenaida asiatica*), Collared Dove (*Streptopelia decaocto*), King Rail (*Rallus elegans*), Clapper Rail (*Rallus crepitans*), Sora (*Porzana carolina*), Virginia Rail (*Rallus limicola*), Common Gallinule (*Gallinula galeata*), Wilson's Snipe (*Gallinago delicata*), American Woodcock (*Scolopax minor*), Blue-winged Teal (*Spatula discors*), Green-winged Teal (*Anas crecca*), Cinnamon Teal (*Spatula cyanoptera*), Wood Duck (*Aix sponsa*), Mallard (*Anas platyrhynchos*), American coot (*Fulica americana*), Snow Goose (*Anser caerulescens*), Greater White-fronted Goose (*Anser albifrons*), Canada Goose (*Branta canadensis*), Eastern Wild Turkey (*Meleagris gallopavo silvestris*), Northern Bobwhite (*Colinus virginianus*), swamp rabbit (*Sylvilagus aquaticus*), eastern cottontail (*Sylvilagus floridanus*), eastern grey squirrel (*Sciurus carolinensis*), white-tailed deer (*Odocoileus virginianus*), bobcat (*Lynx rufus*), common raccoon (*Procyon lotor*), Virginia opossum (*Didelphis virginiana*), American Crow (*Corvus brachyrhynchos*)
Non-Game Species	Forest/Woodland: Coal skink (*Plestiodon anthracinus*), timber rattlesnake (*Crotalus horridus*), Eastern hog-nosed snake (*Heterodon platirhinos*), lace-winged roadside-skipper (*Amblyscirtes aesculapius*), Louisiana slimy salamander (*Plethodon kisatchie*), Worm-eating Warbler (*Helmitheros vermivorum*)
	Wetlands: Louisiana Waterthrush (*Parkesia motacilla*)
	Open Land: Henslow's Sparrow (*Centronyx henslowii*), slender glass lizard (*Ophisaurus attenuatus*), yucca giant-skipper (*Megathymus yuccae*)
	Open Water: bigeye shiner (*Notropis boops*), blue sucker (*Cycleptus elongatus*), crystal darter (*Crystallaria asprella*), ebonyshell (*Reginaia ebenus*), fawnsfoot (*Truncilla donaciformis*), ironcolor shiner (Notropis chalybaeus), Ouachita map turtle (*Graptemys ouachitensis*), razor-backed musk turtle (Sternotherus carinatus), Red River mudpuppy (*Necturus louisianensis*), smooth softshell (*Apalone mutica*), spike (*Eurynia dilatata*), chub shiner (*Notropis potteri*), Quachita fencing crawfish (*Faxonella creaseri*), redspot darter (*Etheostoma artesiae*), saddleback darter (*Percina vigil*)
Critical Species	Forest/Woodland: Bells Vireo (*Vireo bellii*), northern long-eared bat (*Myotis septentrionalis*), Red-cockaded Woodpecker (*Dryobates borealis*), Western wormsnake (*Carphophis vermis*), ringtail (*Bassariscus astutus*), silver-haired bat (*Lasionycteris noctivagans*), southern dusky salamander (*Desmognathus auriculatus*), southern red-backed salamander (*Plethodon serratus*), tricolored bat (*Perimyotis subflavus*), Warbling Vireo (*Vireo gilvus*)
	Wetlands: pygmy rattlesnake (*Sistrurus miliarius*), Western chicken turtle (*Deirochelys reticularia miaria*)
	Open Land: Sandhill Crane (*Antigone canadensis*), southern prairie skink (*Plestiodon septentrionalis obtusirostris*), Short-eared Owl (*Asio flammeus*), southern crawfish frog (*Lithobates areolatus areolatus*)
	Open Water: alligator snapping turtle (*Macrochelys temminckii*), black sandshell (*Ligumia recta*), Blair's fencing crawfish (*Faxonella blairi*), bluehead shiner (*Pteronotropis hubbsi*), butterfly mussel (*Ellipsaria lineolata*), channel darter (*Percina copelandi*), elegant creek crawfish (*Procambarus elegans*), Fatmucket (*Lampsilis siliquoidea*), hickorynut (*Obovaria olivaria*), monkeyface (*Theliderma metanevra*), ouachita kidneyshell (*Ptychobranchus occidentalis*), pink mucket

TABLE 3.3-1 **Wildlife Species Typical of Major Vegetation Cover Types Present in the Project Area**	
Common Wildlife Species	
	(*Lampsilis abrupta*), plain pocketbook (*Lampsilis cardium*), pyramid pigtoe (*Pleurobema rubrum*), rabbitsfoot (*Theliderma cylindrica*), Silty Hornsnail (*Pleurocera canaliculata*), steelcolor shiner (*Cyprinella whipplei*), vernal crawfish (*Procambarus viaeviridis*), western sand darter (*Ammocrypta clara*), central stoneroller (*Campostoma anomalum*)
Source: LDWF, 2026c, LDWF, 2026d	

3.3.2 Designated Critical Wildlife Habitat

According to the USFWS, there are no designated unique, sensitive, or critical wildlife habitats within the Project area (USFWS, 2026a). No national parks, state forests, wildlife management areas, or local/private preserves are crossed by the Project (Louisiana State Parks, 2026; LDWF, 2026e, National Park Service, 2026a, 2026b; USDA Forest Service, 2026a, 2026b; USFWS, 2026c, National Wild and Scenic Rivers System [NWSRS], 2026). Two USDA Wetland Reserve Easements (WRE) and two wetland mitigation banks are located within close proximity to the Project workspace (U.S. Geological Service [USGS], 2026, U.S. Army Corps of Engineers [USACE], 2026). The Project workspace will not cross the WREs or wetland mitigation banks parcels; therefore, there will be no impacts to these designated areas as a result of the project. A description of the WRE areas is provided in Section 8.3.2.1 of Resource Report No. 8 (*Land Use, Recreation, and Aesthetics)*.

3.3.3 Construction Impacts and Mitigation

Construction of the Project will result in both short-term (or temporary) and long-term impacts to wildlife and wildlife habitat. Short-term impacts will be those associated with temporary disturbance to wildlife and wildlife habitat during construction, while long-term impacts will be those associated with conversion of forested habitats to scrub-shrub and herbaceous habitats, resulting from periodic maintenance of the permanent ROW, or the minimal loss of grassland/open land habitat within the new aboveground facility sites.

Direct impacts to wildlife could occur during clearing and grading operations resulting in the displacement of mobile wildlife species from the Project area. Indirect wildlife impacts associated with construction noise and increased human activity could also result in abandoned reproductive efforts, displacement, and avoidance of work areas. However, these impacts are not anticipated to have a significant impact on any local species populations. Impacts within any one area will be temporary and localized as construction progresses along the ROW.

During construction there is potential for wildlife (or livestock) to be injured or trapped by falling into the open trench. To protect wildlife from injury from the open trench, ETC Tiger's EI will inspect the trench daily prior to construction for wildlife. In addition, ETC Tiger will protect livestock from falling into the open trench through installation of temporary fencing, as negotiated with the landowner.

The extent and duration of impacts to wildlife from Project construction will vary depending on the species present in each affected habitat type and their individual life history. To minimize impacts to wildlife and promote the rapid stabilization and revegetation of disturbed areas, ETC Tiger will comply with the FERC Plan and FERC Procedures thereby minimizing disturbance to vegetation and providing for stabilization of affected areas to mitigate direct and indirect effects to wildlife. Following construction, temporarily disturbed areas will be allowed to revegetate via natural succession. Non-forested habitats (i.e., agricultural land, open land, ROW, PEM wetlands) are expected to recover quickly (i.e., one to three years) after

construction is completed. Wildlife would only temporarily be displaced with similar habitat types available in the immediate Project area. Immediately following restoration any displaced wildlife could return. Therefore, the temporary disturbance of these habitats is anticipated to have a negligible impact on individual wildlife species or wildlife populations.

Forested communities, both upland and wetland, will be affected to a greater extent because of the long-term conversion of these wooded habitats to earlier successional stages in the temporary ROW and the permanent conversion to scrub-shrub and/or non-woody herbaceous species in the permanent, maintained ROW. Temporary workspaces will be allowed to revegetate via natural succession. This natural revegetation process will gradually develop a stratified vegetative cover between the ROW and adjacent habitats. The return of forested habitat within the temporary construction areas could take approximately 10 to 30 years. Thus, species that depend upon trees for food, refuge, or nesting will be displaced to nearby forested habitat. The clearing activities may also result in a change from species favoring forest habitats to those using edge habitats and more open areas within these limited areas.

The forested areas along the proposed Franklin Farms Lateral and FF Trio Farm Lateral are currently primarily existing as edge habitat, not interior forested habitat. The permanent, maintained ROW may function as a travel corridor for some wildlife species and may provide food, cover, and breeding habitat for those species that use open and emergent habitats. Therefore, although some conversion of forested vegetation will occur, effects on habitat for forest-dwelling wildlife are anticipated to be minimal.

An accidental spill or release of hazardous materials (e.g., fuels, lubricants, solvents) during construction could affect wildlife or wildlife habitat. However, potential impacts from accidental spills and releases will be avoided or minimized through implementation of ETC Tiger's SPAR Plan. ETC Tiger's SPAR Plan includes BMPs to avoid and minimize the potential for accidental releases and contains measures that will be implemented to clean up any releases.

Overall, construction of the pipeline facilities is not expected to adversely affect the distribution or regional abundance of wildlife species given the short duration of the disturbance and the amount and distribution of similar habitat types available in the immediate Project area. Further, impacts to wildlife will be minimized by implementing the restoration methods outlined in the FERC Plan and FERC Procedures. Effects to wildlife at the aboveground facility sites will be negligible given that they are either in open land, agricultural land, or on a previously developed site.

3.3.4 Operation Impacts and Mitigation

Operation and maintenance of the facilities will have negligible additional impact to wildlife or wildlife habitat after construction is complete. The Project does not include any new aboveground facilities which will generate significant operational noise that could disturb wildlife. ETC Tiger's maintenance of the permanent ROW includes routine mowing and clearing activities using mechanical equipment. ETC Tiger will minimize impacts to wildlife species and habitats resulting from operation and maintenance of the Project through implementation of the FERC Plan, FERC Procedures, and ETC Tiger's SPAR Plan.

3.4 PROTECTED SPECIES

Game and non-game wildlife species are regulated and protected under state and federal agencies such as USFWS, and LDWF. Several regulations such as the ESA of 1973, the United States Fish and Wildlife Conservation Act of 1980, and the United States Fish and Wildlife Coordination Act of 1958 also regulate protected plant and animal species of concern.

The ESA declares that threatened and endangered plant and animal species are of aesthetic, ecological, educational, historic, and scientific value to the nation and requires protection of these species and their

habitat. The federal definition of "*endangered species*" is any species which is in danger of extinction throughout all or a significant portion of its range. A "*threatened*" species is any species that is likely to become an endangered species within the foreseeable future throughout all or a significant portion of its range. Endangered and threatened species may be in jeopardy due to destruction, modification, or curtailment of habitat, over utilization, effects of disease, pollution, or predation. Species likely to become threatened in the foreseeable future may be listed as "*rare*," "*protected*," "*candidate*," or "*of special concern*." In addition to the required protection of individual species, some rare, natural vegetative communities may also be protected.

Protection is also afforded under the ESA to "*Critical Habitat*," which the USFWS defines as specific areas both within and outside the geographic area occupied by a species on which are found those physical and biological features essential to its conservation.

3.4.1 Federally Protected Species

3.4.1.1 Field Surveys

Prior to initiating field surveys for listed species, available literature and public databases were reviewed to assess previously recorded on-site or nearby occurrences of regionally listed species. **Table 3.4-1** provides a list of the federally listed species that have the potential to occur in the Project area. This list was compiled using the USFWS Environmental Conservation Online System Information for Planning and Consultation (IPaC) System consultation tool (USFWS, 2026a).

TABLE 3.4-1
Federally Listed Species with Potential to Occur In or Near the Project Area

Species	Federal Status [a]	Potential to Occur in Project Area	Critical Habitat Present [b]	Effect Determination [c]
Mammals				
Tricolored Bat (*Perimyotis subflavus*)	PE [d]	Possible. The Project area contains potentially suitable habitat for this species; however, clearing windows will be coordinated with the agencies to minimize impacts during the winter torpor (December 15 – February 15) and roosting season/pup season (March 15 – July 15).	No	NLJ
Northern Long-eared Bat (*Myotis septentrionalis*)	T [e]	Possible. The Project area contains potentially suitable habitat for this species; however, the USFWS official IPaC did not list this species as one to occur within the Project area. Nonetheless, clearing windows will be consulted with agencies to minimize impacts during the winter torpor (December 15 – February 15) and roosting season/pup season (March 15 – July 15).	No	NLAA
Reptiles				
Alligator snapping turtle (*Macrochelys temminckii*)	PT [d]	Possible. The Project area contains limited suitable habitat for this species within Cow Bayou and a tributary between Cow Bayou and Bee Bayou; therefore, coordination with USFWS will occur to minimize potential impacts to the alligator snapping turtle. The Project area does contain suitable habitat that could support the preferred habitat of this species; however, to minimize impacts disturbance/alterations of nesting habitat will be minimized during nesting season (April-June). Additionally, in-stream work will avoid permanent impacts to streams to avoid alterations to habitat.	No	NLJ
Insects				

TABLE 3.4-1
Federally Listed Species with Potential to Occur In or Near the Project Area

Species	Federal Status [a]	Potential to Occur in Project Area	Critical Habitat Present [b]	Effect Determination [c]
Monarch butterfly (*Danaus plexippus*)	PT [d]	Possible. While the Project area does not contain obvious potentially suitable habitat for the monarch butterfly since the Project Area primarily consists of cropland, developed areas, and woody wetlands, and no milkweed species or monarch butterflies were observed during field surveys; there is still potential this species could be present. However, monarch butterflies are anticipated to be minimal and negligible.	No	NLJ

Notes:

[a] PE – Proposed Endangered; T – Threatened; PT – Proposed Threatened

[b] Designated Critical Habitat in the Project area

[c] NLJ – Not Likely to Jeopardize; NLAA - May Affect, Not Likely to Adversely Affect

[d] Proposed species do not receive statutory protection from "take" under the ESA; however, potential impacts to proposed species must be assessed for federal actions (e.g., funding, permits, etc.). Federal actions must not jeopardize the continued existence of proposed species

[e] This species was not identified on the USFWS IPaC report, but it was identified on the LDWF's Rare Species and Natural Communities by Parish List for Richland Parish

Source: USFWS, 2026a

Field surveys were conducted for the entire Project area for wildlife and potential habitat for protected species. Along the proposed Franklin Farms Lateral and FF Trio Farm Lateral, a 300-foot-wide corridor centered along the pipeline route (150 feet from each side of the centerline and around aboveground facilities) was surveyed in February, March, and May 2026. No ESA-listed species were observed during the field surveys; however, potential habitat was identified in the Project area. Additional details are provided in the Listed Species Report prepared for the Project (see **Appendix F-2**).

3.4.1.2 ESA-Listed Species Assessment

Based on the agency database information and field survey results, an assessment as to whether the proposed Project may impact federally listed species was made using the following USFWS effect determinations (USFWS and NMFS, 1998):

- *No effect* – This determination is appropriate when the proposed action will not directly or indirectly affect (neither negatively nor beneficially) individuals of listed, proposed, or candidate species or designated/proposed critical habitat of such species.
- *May affect, not likely to adversely affect* – This determination is appropriate when the proposed action is likely to cause insignificant, discountable, or wholly beneficial effects to individuals and designated critical habitat. Certain avoidance and minimization measures may need to be implemented to reach this level of effect.
- *May affect, likely to adversely affect* – Adverse effects to listed species may occur as a direct or indirect result of the proposed action or its interrelated or interdependent actions, and the effect is not discountable, insignificant, or beneficial.

In addition to the "*no effect*" determination, the following USFWS effect determinations are also applicable to proposed and candidate species or designated/proposed critical habitat:

- *Not likely to jeopardize* – This determination is appropriate when the proposed action is not expected to jeopardize the continued existence of a species proposed for listing or a candidate species, or adversely modify an area proposed for designation as critical habitat.
- *Likely to jeopardize* – This determination is appropriate when the proposed action is reasonably expected to jeopardize the continued existence of a species proposed for listing or a candidate species, or adversely modify an area proposed for designation as critical habitat.

It was determined that the proposed Project may contain potentially suitable habitat for each of the species listed as federally threatened, proposed endangered, or proposed threatened in **Table 3.4-1**, this includes the tricolored bat (*Perimyotis subflavus*), northern long-eared bat (*Myotis septentrionalis*), alligator snapping turtle (*Macrochelys temminckii*), and monarch butterfly (*Danaus plexippus*). The species with potential suitable habitat in the Project area are discussed in further detail below.

Tricolored Bat

Federally Listed – Proposed Endangered

Louisiana State Listed – Not Listed

The tricolored bat is a year-round resident within the area where the Project area lies (USFWS, 2024). In the spring, summer, and fall they roost among live and dead leaf clusters of live and dead hardwood trees. They have also been observed roosting in Spanish moss (*Tillandsia usneoides*), lichen (*Usnea trichodea*), among pine needles, and within man-made structures, such as barns and sheds. In the winter they hibernate in caves and mines, and in the southern U.S. they are often observed hibernating and roosting in road-associated culverts and trees (USFWS, 2026d).

Potentially suitable tricolored bat summer roosting habitat and winter hibernating habitat was identified within the Project area in the form of live and dead hardwood trees within the forested areas, dilapidated man-made structures, and a large culvert. ETC Tiger will conduct tree clearing activities outside of tricolored bat winter torpor (December 15 – February 15) and roosting season/pup season (March 15 – July 15) to minimize impacts to tricolored bats. Therefore, it is anticipated that the Project is "*May Affect"* the tricolored bat. See Section 3.4.1.3 for details regarding consultation with the USFWS for tricolored bats.

Northern Long-eared Bat

Federally Listed – Threatened

Louisiana State Listed – Threatened

The northern long-eared bat is primarily associated with hardwood and mixed forested landscapes during the spring, summer, and fall, where they forage near trees with a preference for low understory cover. They are often observed roosting singly or in colonies underneath bark, in cavities or crevices of both live and dead trees. They have also been found less commonly in roosting structures such as man-made barns and sheds. This species typically overwinters in caves or mines with constant temperature, high humidity, and no air currents (USFWS, 2026e).

Potentially suitable northern long-eared bat summer roosting habitat was identified within the Project area in the form of live and dead hardwood trees within the forested areas and dilapidated man-made structures. ETC Tiger will conduct tree clearing activities during tricolored bat winter torpor (December 15 – February 15) and roosting season/pup season (March 15 – July 15) to minimize impacts to tricolored bats. Therefore, it is anticipated that the Project "*May Affect, but is not Likely to Adversely Affect"* the northern long-eared bat.

Alligator Snapping Turtle

Federally Listed – Proposed Threatened

Louisiana State Listed – State Restricted

The alligator snapping turtle prefers to inhabit large rivers, major tributaries, bayous, canals, swamps, lakes, ponds, and oxbows. They are associated with deeper water, although they are known to occupy shallower water in early summer. Favored features for these habitats include high canopy forest areas and structures like tree root masses, stumps, and submerged trees. Hatchlings prefer shallow water and abundant canopy and vegetation (USFWS, 2021).

The Project Area contains limited suitable habitat for this species within Cow Bayou and a tributary between Cow Bayou and Bee Bayou. Disturbance to the bottom of streams within the Project Area where potential alligator snapping turtle habitat is present will occur when streams are crossed via the conventional crossing method and flume methods. Water within the streams will continue to flow through the trenching area during water crossing activities. If any alligator snapping turtle individuals happen to be within the vicinity during water crossing activities, they are likely to temporarily relocate to adjacent aquatic habitat. Following construction activities related to stream crossing, the trench will be backfilled, and the streambed profile will be restored to pre-construction contours and grade conditions. Although there will be impacts to stream beds during water crossing construction activities, BMPs will be in place to avoid permanent impacts to alligator snapping turtle habitat. Additionally, disturbance to potential nesting habitat will be minimized during nesting season (April-June). Therefore, it is anticipated that the Project is "*Not Likely to Jeopardize"* the alligator snapping turtle.

Monarch Butterfly

Federally Listed – Proposed Threatened

Louisiana State Listed – Not Listed

The monarch butterfly is a migratory species that utilizes a variety of habitat types. Except for a non-migratory population in Florida, monarch butterflies in eastern North America (i.e., east of the Rocky Mountains) travel north in the spring from overwintering areas in Mexico to summer breeding areas as far north as Canada, typically over two to three successive generations and breed along the way. In the fall, monarch butterflies migrate back to southern overwintering grounds. Depending upon environmental conditions, eastern North American monarch butterflies may require up to five generations to complete their annual migration cycle. During migration and breeding, adults depend on a diversity of blooming nectar resources (i.e., wildflowers) for feeding. Native milkweeds (primarily *Asclepias* spp.) are obligatory host plants for egg-laying and feeding during the larval life stage. In drier, more arid regions such as the southwestern U.S., migrating monarch butterflies tend to occur more frequently near water sources such as rivers, creeks, roadside ditches, and irrigated canals (USFWS, 2020).

The majority of the Project area is active farmland utilized for crop production. This routinely disturbed land would not provide optimal habitat for the monarch butterfly. Therefore, it is anticipated that the Project is "*Not Likely to Jeopardize"* the monarch butterfly.

3.4.1.3 USFWS Consultation

A regulatory review of the Project Area was conducted through the USFWS IPaC system using the range-wide Determination Key for the tricolored bat. The regulatory review resulted in the USFWS providing one consistency letter, dated May 20, 2026, verifying that the Project ***May Affect*** the tricolored bat, and that further consultation with USFWS is necessary. The consistency letter is included in **Appendix D-1**. ETC Tiger sent a consultation letter to USFWS that discusses the results of the field surveys for the Project (see

Section 3.4.1.2) as detailed in **Table 1.7-1** of **Resource Report 1.** A copy of the correspondence is included in **Appendix D-1**.

3.4.1.4 State-Listed Species Assessment

In addition to the federally protected species listed in **Table 3.4-1**, some of which are also state protected, there are species that are not federally protected but are considered state protected species in Louisiana. ETC Tiger utilized LDWF's Rare Species and Natural Communities by Parish Lists to evaluate the potential presence of protected state listed species within the Project area. It was found that only one state listed species occurs on the LDWF Richland Parish species list, the state threatened northern long-eared bat. This species is also a federally listed threatened species and is included in the federal listed species evaluation above. It was determined that the Project contains potentially suitable summer roosting habitat for northern long-eared bat and the Project "*May Affect, but is Not Likely to Adversely Affect"* this species, see Sections 3.4.1.1 and 3.4.1.2 for further detail.

3.4.2 Construction Impacts and Mitigation

Potential impacts to listed species habitat from Project construction will generally be similar to those described for wildlife habitat in Section 3.3.3. The Project will clear trees along the construction ROW during bat winter torpor (December 15 – February 15) to avoid and minimize the potential for affecting bats during the pup season. With adherence to this construction window and implementation of the BMPs in the FERC Plan, FERC Procedures, and ETC Tiger's SPAR Plan, it is anticipated that the Project construction will either have "*No Effect*" or "*May Affect, but is not Likely to Adversely Affect*" federal and state protected species with potential to occur in the Project area (**Table 3.4-1**).

ETC Tiger will provide environmental training for construction personnel including information on the identification and protection of listed species. In addition, construction personnel will be informed of the penalty for impacts to protected species. Further, protective measures will be incorporated within construction planning. ETC Tiger's EI will monitor construction activities and work areas so no protected species are harmed. Because the Project will have no significant long-term or permanent effects on threatened or endangered species and preferred habitat, no additional mitigation is proposed.

3.4.3 Operation Impacts and Mitigation

Operation and maintenance of the facilities is anticipated to have no additional impact to federal and state protected species after construction is complete. The pipeline will be buried during operations, and the Project does not include any new aboveground facilities which will generate significant operational noise or lighting that could disturb wildlife. ETC Tiger's maintenance of the permanent ROW will include routine mowing and clearing activities using mechanical equipment. Vegetative maintenance activities will comply with FERC Plan and FERC Procedures.

3.5 MIGRATORY BIRDS

Migratory birds are protected under the Migratory Bird Treaty Act (MBTA). The MBTA governs and prohibits the take, possession, import, export, transport, sale, purchase, barter, or offer for sale, purchase, or barter, of any migratory bird, their eggs, parts, and nests, except as authorized under a valid permit. The Project crosses the Mississippi Alluvial Valley Bird Conservation Region (BCR) (North American Bird Conservation Initiative [NCABI], 2021). Historically, this BCR was the greatest bottomland hardwood forest on earth and was subject to massive annual flood events of the Mississippi River and its tributaries. The pre-existing forested wetlands served as the primary wintering area for mid-continent Mallards, Wood Ducks, and other waterfowl species. The modern landscape has been significantly altered due to flood

control and deforestation for agriculture, with less than 25 percent of the region remaining forested and flooding events having reduced by about 90 percent. Despite the drastic alteration to the landscape within the region, large numbers of waterfowl still winter within the Mississippi Alluvial Valley BCR. Many shorebirds utilize managed wetlands for migration stopover sites and remnant forests provide habitat for populations of Swainson's Warblers (*Limnothlypis swainsonii*), Prothonotary Warblers (*Protonotaria citrea*), and Swallow-tailed Kites (*Elanoides forficatus*). The region provides excellent colonial waterbird habitat for White Ibis (*Eudocimus albus*), Yellow-crowned Night Heron (*Nyctanassa violacea*), and a variety of other heron and egret species (NCABI, 2021). A list of neotropical migratory species that breed in Louisiana and have been identified as focal migratory species is provided in **Table 3.5-1**.

TABLE 3.5-1
Focal Migratory Bird Species of Louisiana

Neotropical Migratory Bird Species	
Wood Thrush (*Hylocichla mustelina*)	Sprague's Pipit (*Anthus spragueii*)
LeConte's Sparrow (*Ammospiza leconteii*)	Henslow's Sparrow (*Centronyx henslowii*)
Golden-winged Warbler (*Vermivora chrysoptera*)	Swainson's Warbler (*Limnothlypis swainsonii*)
Kentucky Warbler (*Geothlypis formosa*)	Hooded Warbler (*Setophaga citrina*)
Cerulean Warbler (*Setophaga cerulea*)	Prairie Warbler (*Setophaga discolor*)
Painted Bunting (*Passerina ciris*)	Chimney Swift (*Chaetura pelagica*)
Northern Pintail (*Anas acuta*)	Yellow Rail (*Coturnicops noveboracensis*)
Whooping Crane (*Grus americana*)	Black-bellied Plover (*Pluvialis squatarola*)
Piping Plover (*Charadrius melodus*)	Hudsonian Godwit (*Limosa haemastica*)
Short-billed Dowitcher (*Limnodromus griseus*)	Ruddy Turnstone (*Arenaria interpres*)
Red Knot (*Calidris canutus*)	Sanderling (*Calidris alba*)
Least Tern (*Sternula antillarum*)	Least Bittern (*Ixobrychus exilis*)
Source: Southern Wings, 20256a	

Bald Eagles were removed from the USFWS endangered species list in 2007; however, Bald Eagles are protected under the Bald and Golden Eagle Protection Act and MBTA. No Bald or Golden Eagle nests were observed during field surveys for the Project.

3.5.1 Construction Impacts and Mitigation

Potential impacts on migratory birds include habitat loss, disruption of foraging adults, and abandonment or destruction of active nests. Increased human activity and noise from pipeline construction could result in short-term disturbance of migratory bird habitat, causing birds in the Project area to avoid the area and/or relocate during periods of active construction. Impacts to migratory bird habitat are anticipated to be short term and minimal given that the majority of the Project area is active agricultural lands. ETC Tiger will provide environmental training for construction personnel including specific instructions that no wildlife, including migratory birds, are to be harmed during construction activities. If nesting bald eagles are observed in the Project area during construction, ETC Tiger will adhere to measures in the 2007 USFWS *National Bald Eagle Management Guidelines* (USFWS, 2007).

3.5.2 Operation Impacts and Mitigation

Operation and maintenance of the facilities is not anticipated to have additional impacts to migratory bird species after construction is complete. ETC Tiger's maintenance of the permanent ROW will include routine mowing and clearing activities using mechanical equipment. Vegetative maintenance activities will comply with FERC Plan and FERC Procedures. In accordance with the FERC Plan, ETC Tiger will not conduct routine vegetation mowing or clearing during the migratory bird nesting season between April 15 and August 1 of any year unless specifically approved in writing by the responsible land management agency or the USFWS.

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ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 4
Cultural Resources

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 4 FILING REQUIREMENTS	
INFORMATION	**Section Reference**
1. Initial cultural resources consultation and documentation, and documentation of consultation with Native Americans (§ 380.12(f) (1) (I) & (2).	Section 4.3.2
2. Overview / Survey Reports (§ 380.12(f) (1) (ii) & (2).	Section 4.1.6; Appendix G
3. Identify the project APE in terms of direct or indirect effects to known cultural resources.	Section 4.1
4. Provide a project map with MPs, clearly showing boundaries of all areas surveyed (ROW, extra work areas, access roads, etc.) and to be surveyed with corridor widths clearly specified.	Appendix A-7
5. Provide documentation of consultation with SHPOs, THPOs, and applicable land-managing agencies regarding the need for and required extent of cultural resource surveys.	Section 4.3
6. Provide a narrative summary of overview results, cultural resource surveys completed, identified cultural resources, and any cultural resource issues.	Section 4.1
7. Provide a project specific Ethnographic Analysis (can be part of Overview / Survey Report).	Section 4.1; Appendix G
8. Identify by MPs any areas requiring survey for which the landowner denied access.	N/A
9. Provide written comments on the Overview and Survey Reports, if available, from the SHPOs or THPOs, as appropriate, and applicable land-managing agencies.	Section 4.3
10. Provide a Summary Table of completion status of cultural resource surveys and SHPO or THPO and land-managing agency comments on the reports.	Section 4.3
11. Provide a brief summary of the status of Native American consultation, including copies of all related correspondence and records of verbal communications.	Section 4.3.2
12. Provide an Unanticipated Discoveries Plan for the project area, referencing appropriate state statutes.	Section 4.2

RESOURCE REPORT NO. 4
CULTURAL RESOURCES

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C Project Construction and Mitigation Plans

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D Agency and Stakeholder Correspondence

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Appendix D-2 – SHPO Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

Appendix D-3 – Native American Organizations and Tribes Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

E Landowner Information

Appendix E-1 – Landowner/Stakeholder List (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE"*)

F Wetland and Waterbody Delineation Report and Listed Species Report

Appendix F-1 – Wetland and Waterbody Delineation Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

Appendix F-2 – Listed Species Report(Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

G Cultural Resource Assessment Survey Report (*Submitted as* "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

H Air Quality Calculations (Construction Emissions)

I Sound Study Report

ABBREVIATIONS AND ACRONYMS

APE	Area of Potential Effect
ATWS	additional temporary workspace
Certificate	Certificate of Public Convenience and Necessity
CFR	Code of Federal Regulations
ESC	environmental survey corridor
ETC Tiger	ETC Tiger Pipeline, LLC
FERC or Commission	Federal Energy Regulatory Commission
LA	Louisiana
LDHP	Louisiana Department of Historic Preservation
LDOA	Louisiana Divisions of Archaeology
LHRI	Louisiana Historic Resource Inventory
NGA	Natural Gas Act
NHPA	National Historic Preservation Act of 1966
NRHP	National Register of Historic Places
Project	Franklin Farms Project
ROW	right-of-way
SHPO	State Historical Preservation Office
Tribes	Native American Tribes
U.S.	United States

4.0 CULTURAL RESOURCES

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, an interconnect site, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project in June 2027, and following the receipt of all applicable approvals, anticipating construction activities to begin in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

(Project) Resource Report No. 4 describes the cultural resources investigation that has been conducted to date for the Project to obtain information that supports the Project's compliance with Section 106 of the National Historic Preservation Act of 1966 (NHPA), as amended, and its implementing regulations at 36 Code of Federal Regulations (CFR) Part 800. All testing has been conducted in accordance with the Louisiana Division of Archaeology's standards and guidelines for cultural resources surveys, and with FERC Guidelines for *Reporting on Cultural Resources Investigations for Pipeline Projects* and regulations at 18 CFR Section 380.12, 380.14, and Appendix A to Part 380.

4.1 CULTURAL RESOURCE SURVEYS AND RESULTS

The cultural resources survey status of the Project is listed in **Table 4.1-1**. Fieldwork occurred during the months of February through May 2026 and consisted of a Phase I cultural resources survey. That survey included a comprehensive pedestrian archaeological survey and shovel testing of mainline and lateral environmental survey corridor (ESC) (24.17 kilometers [km] or 15.02 miles [mi]), seven access roads (2.66 km or 1.65 mi), seven meter stations (3.56 hectares [ha] or 8.82 acres [ac]), and three contractor yards (7.42 ha or 18.33 ac). A total of three previously recorded archaeological sites, one newly recorded archaeological site, and four historic standing structures were identified during the survey.

TABLE 4.1-1
Survey Status of the Franklin Farms Project Facilities

Component	County / State	Milepost		Access Permission	Survey Status	Date of Report Submission to SHPO	Comments from SHPO or other Agencies
		Start	End				
Franklin Farms Lateral	Richland Parish, LA	0.0	13.2	Granted	Phase I Survey Completed	April 24, 2026 and June 10, 2026	Concurrence Received, Addendum Pending
FF Trio Farm Lateral	Richland Parish, LA	0.0	1.7	Granted	Phase I Survey Completed	April 24, 2026 and June 10, 2026	Concurrence Received, Addendum Pending
FF Gulf Run Bi Di Meter Station	Richland Parish, LA	0.0	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
FF Gulf South Receipt Meter Station	Richland Parish, LA	0.0	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received

TABLE 4.1-1
Survey Status of the Franklin Farms Project Facilities

Component	County / State	Milepost		Access Permission	Survey Status	Date of Report Submission to SHPO	Comments from SHPO or other Agencies
		Start	End				
FF Bee Bayou Delivery Meter Station	Richland Parish, LA	4.6	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
FF Gulf Crossing Receipt Meter Station	Richland Parish, LA	9.7	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
FF MEP Receipt Meter Station	Richland Parish, LA	10.8	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
Franklin Farms Delivery Meter Station	Richland Parish, LA	13.2	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
FF Interconnect	Richland Parish, LA	0.0	N/A	Granted	Phase I Survey Completed	June 10, 2026	Pending
FF Trio Farm Delivery Meter Station	Richland Parish, LA	1.75	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
LDY-Center Point	Richland Parish, LA	N/A	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
LDY-Construction Yard	Richland Parish, LA	N/A	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
LDY-Pump Station Construction Yard	Richland Parish, LA	N/A	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
AR-EC030326-002	Richland Parish, LA	N/A	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
AR-IH030326-002	Richland Parish, LA	N/A	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
AR-JRB022626A	Richland Parish, LA	N/A	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
AR-AB022826B	Richland Parish, LA	N/A	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received

TABLE 4.1-1
Survey Status of the Franklin Farms Project Facilities

Component	County / State	Milepost		Access Permission	Survey Status	Date of Report Submission to SHPO	Comments from SHPO or other Agencies
		Start	End				
AR-EC030326-001	Richland Parish, LA	N/A	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
AR-JRB022526C	Richland Parish, LA	N/A	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received
AR-IH030326-001	Richland Parish, LA	N/A	N/A	Granted	Phase I Survey Completed	April 24, 2026	Concurrence Received

Notes:
[a] Includes Additional Temporary Workspace (ATWS).

The recommended direct impact area varied based on Project component. Most of the proposed Franklin Farms pipeline survey corridor was 300 feet in width, centered on the proposed centerline; however, any workspaces that projected beyond the initial corridor were encompassed and included within the final direct impact area. Areas where engineering limitations or topography constrained the dimensions of the construction right-of-way (ROW) the impact area was similarly narrowed. The area of indirect effects extended 50 ft outside of the ESC, per consultation with the Louisiana Divisions of Archaeology (LDOA) and Historic Preservation (LDHP).

Site file research was initiated by reviewing records maintained by the LDOA and LDHP. Site file research resulted in a listing of all archaeological sites, historic built resources, and previously completed cultural resources surveys within 1-mile of the Project area. Documentary research, including review of historic maps, U.S. Geological Survey topographic maps, historic aerials, land grants, and county and regional histories, was conducted in order to provide an understanding of the development and history of the Project area, the surrounding area, and regional cultural history in general. Desktop assessment and survey area mapping was performed prior to initiating field investigations to better understand cultural, environmental, and geological settings. Results of the desktop assessment were then used to develop the field methodology, which was provided to Louisiana State Historic Preservation Office (SHPO) for review and approval. A copy of the field methodology and SHPO approval is provided in **Appendix D-2** (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*").

Preliminary research indicated that 64 known cultural resources were within 1.6 km (1.0 mi) of the proposed Project area. Of those, only three intersect the Franklin Farms Lateral (Sites 16RI300, 16RI407, and 16RI418) and no sites intersect the proposed FF Trio Farm Lateral, access roads, meters stations, or contractor yards. A summary of the resources identified is provided in the subsections below. A copy of the survey results is provided in **Appendix G** (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*").

4.1.1 Franklin Farms Lateral

The Franklin Farms Lateral was approximately 21.29 km (13.23 mi) and consisted of 13 survey segments which resulted in the revisit of three known cultural resources (Sites 16RI300, 16RI407, and 16RI411) and the identification of two newly identified cultural resources (Site 16RI439 and historic standing structure

Louisiana Historic Resource Inventory [LHRI] 42-00761) (**Table 4.1-2**). Sites revisited or newly identified during survey consisted of historic artifact scatters related to former homesteads. Sites 16RI300 and 16RI407 are late nineteenth to early twentieth century artifact scatters, Site 16RI411 is a middle nineteenth to early twentieth century artifact scatter, and Site 16RI439 is a late nineteenth to early twentieth century artifact scatter. LHRI 42-00761 is a two-story storage shed.

TABLE 4.1-2
Cultural Resources Examined During the Current Investigation.

Trinomial/Resource Type	Previous NRHP Status / Recommendation	Current Recommendations	SHPO Comment
Site 16RI300	Ineligible	Ineligible	Concurrence Received
Site 16407	Ineligible	Ineligible	Concurrence Received
Site 16RI411	Ineligible	Ineligible	Concurrence Received
Site 16RI439	N/A	Unevaluated; no avoidance necessary within ESC	Concurrence Received
Site 16RI1440	N/A	Ineligible	Pending
LHRI 42-00758	N/A	Ineligible	Concurrence Received
LHRI 42-00759	N/A	Ineligible	Concurrence Received
LHRI 42-00760	N/A	Ineligible	Concurrence Received
LHRI 42-00761	N/A	Ineligible	Concurrence Received

4.1.2 FF Trio Farm Lateral

The FF Trio Farm Lateral was approximately 2.88 km (1.79 mi) and consisted of two survey segments. One newly recorded archaeological resources, Site 16RI1440 was identified during additional survey work in May 2026. The status of this resource is pending consultation with the LDOA

4.1.3 Meter Stations

The Project consisted of seven meter stations located throughout the Franklin Farms Lateral and FF Trio Farm Lateral (totaling 3.56 ha or 8.82 ac). No cultural resources were identified during the cultural resources survey of the seven meter stations.

4.1.4 Contractor Yards

Three proposed contractor yards (totaling 7.42 ha or 18.33 ac) were surveyed within the Project Area. No cultural resources were identified during the cultural resources survey of the three contractor yards.

4.1.5 Access Roads

A total of 31 proposed access roads (totaling 6.46 km [4.02 mi]) were examined during the current investigation. No cultural resources were identified during the cultural resources survey of the seven access roads.

4.1.6 Results Summary

A total of 2,206 shovel tests were planned during this survey effort. Of those, TRC excavated 1,941 shovel tests, of which all but one were negative; the single positive test contained limited historic artifacts within the uppermost soil stratum (e.g., Stratum I). The remaining 265 shovel tests were unable to be excavated due to the presence of marked underground utilities, other man-made impediments (e.g., paved roads), and/or areas of standing water. TRC revisited and visually inspected three previously recorded historic period archeological resources (Sites 16RI300, 16RI407, and 16RI411). A single new archaeological resource (Site 16RI439) was also recorded. Additionally, four Historic Standing Structures (LHRI Resources 42-00758, 42-00759, 42-00760, and 42-00761), all domestic residences, were identified and recorded.

Previous investigations of Sites 16RI300, 16RI407, and 16RI411 (see Jones et al. 2026) resulted in a recommendation of no further investigations for these three resources. During the current investigation, the data TRC collected confirmed that these archaeological resources do not bear those qualities of significance and integrity as defined by the National Register of Historic Places (NRHP) Criteria for Evaluation (36 CFR 60.4 [a-d]). The NRHP status of the newly recorded archaeological site remains unevaluated as it could not be fully delineated as it extended outside the ESC boundaries. However, the examined portion of Site 16RI439 lacked artifact recoveries from subsurface contexts, indicating that the resource lacked integrity and research potential. Thus, the resource also does not possess those qualities of significance and integrity as defined by the NRHP Criteria for Evaluation (36 CFR 60.4[a-d]), and TRC does not recommend further investigations of that portion of the resource that lies within the current ESC. The newly recorded historic structures, LHRI Resources 42-00758, 42-00759, 42-00760, and 42-00761, are recommended as ineligible for listing in the NRHP as they are heavily altered from their period of construction, and significance, do not bear any distinct architectural characteristics, are not associated with significant individual or local history, and do not bear those qualities of significance and integrity as defined by the NRHP Criteria for Evaluation (36 CRR 60.4 [a-d]). As a result of these investigations, TRC recommended a finding of no historic properties affected, and that no further cultural resources investigations are necessary. A copy of the survey results is provided in **Appendix G** ("*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*").

Additional survey work from May 2026 will be addressed in a forthcoming addendum, which will include a discussion of newly recorded resource, Site 16RI1440, which TRC will be recommending as ineligible for NRHP inclusion.

4.2 UNEXPECTED DISCOVERIES AND EMERGENCY PROCEDURES PLAN

In the event unanticipated finds are revealed during Project construction, ETC Tiger and their contractors will follow the procedures outlined in the unexpected discoveries of cultural resources, burials and/or human remains plan included in **Appendix C-4**, *Unanticipated Discovery Plan*. This plan will be submitted to any Native American tribes (Tribes) that express interest in the Project area for review and comment. Results of this review will be filed to the FERC Docket when received. Copies will be on-site during construction, and construction field management and Environmental Inspectors will be trained on its contents.

4.3 AGENCY CONSULTATIONS

4.3.1 State Historic Preservation Office (SHPO)

An overview of the correspondence between ETC Tiger and the Louisiana SHPO is provided in **Table 4.3-1**. A copy of the correspondence is included in **Appendix D-2** (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*"). The survey results were submitted to the SHPO on April 24, 2026. SHPO concurrence of the survey results is pending. A copy of the survey results is provided in **Appendix G** ("*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*").

TABLE 4.3-1
SHPO Correspondence in Reference to the Project

Agency/State	Date Contacted	Contact Method	Summary of Contact	Comments Received
Louisiana (LA) SHPO	February 2, 2026	Email	TRC (Steven Filoromo) submitted a Scope of Work for SHPO review, and requested Teams call with Rachel Watson, LA State Archaeologist, and Sadie Whitehurst, Section 106 Reviewer to discuss Project, methodology.	Meeting was held February 4, 2026.
LA SHPO	February 4, 2026	Meeting/ Email	Meeting held in morning with Justin Minter (ET), Steven Filoromo (TRC), Ashley Thompson (TRC), Rachel Watson (LA SHPO), and Sadie Whitehurst (LA SHPO).	LA SHPO agreed with the approach via email.
LA SHPO	April 24, 2026	Email	TRC (Steven Filoromo) submitted a copy of the Phase I Cultural Resources Report to the LA SHPO	Automated response confirming receipt received.
LA SHPO	June 10, 2026	Email	TRC (Steven Filoromo) submitted a copy of the Phase I Cultural Resources Addendum Report to the LA SHPO	Automated response confirming receipt received.

4.3.2 Native American Tribes or Groups

Coordination letters were prepared to inform the Tribes about the Project and to solicit their interest in the Project and participation in identifying traditional cultural properties and historic properties within the Area of Potential Effect (APE) (**Table 4.3-2**). Letters were submitted to the Tribes on June 11, 2026. Any additional written communication, telephone conversation notes, or meeting notes will be filed to the FERC Docket when received. Copies of coordination letters are provided in **Appendix D-3** (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*").

TABLE 4.3-2
Native American Tribes Contacted in Reference to the Project

Native American Tribe	Date Contacted	Contact Method	Summary of Contact	Comments Received
Apache Tribe of Oklahoma	June 11, 2026	Email	Project introduction letter sent	TBD
Choctaw Nation of Oklahoma	June 11, 2026	Email	Project introduction letter sent	TBD

TABLE 4.3-2
Native American Tribes Contacted in Reference to the Project

Native American Tribe	Date Contacted	Contact Method	Summary of Contact	Comments Received
Coushatta Tribe of Louisiana	June 11, 2026	Email	Project introduction letter sent	TBD
Jena Band of Choctaw Indians	June 11, 2026	Email	Project introduction letter sent	TBD
Mississippi Band of Choctaw Indians	June 11, 2026	Email	Project introduction letter sent	TBD
Quapaw Nation	June 11, 2026	Email	Project introduction letter sent	TBD
Seminole Tribe of Florida	June 11, 2026	Email	Project introduction letter sent	TBD

4.4 REFERENCES

Jones, Travis; Steven Filoromo; and Austin Tranberg. 2026. *Phase I Cultural Resources Survey for the Franklin Farm Project, Richland Parish, Louisiana*. Report prepared by Energy Transfer, LP, Houston, Texas. LDOA Report 22-8040.

ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 5
Socioeconomics

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 5 FILING REQUIREMENTS	
Information	**Section Reference**
1. For major aboveground facilities and major pipeline projects that require an EIS, describe existing socioeconomic conditions within the project area. (§ 380.12(g)(1))	Section 5.1
2. For major aboveground facilities, quantify impact on employment, housing, local government services, local tax revenues, transportation, and other relevant factors within the project area. (§ 380.12(g)(2-6))	Section 5.2
3. Evaluate the impact of any substantial immigration of people on governmental facilities and services and describe plans to reduce the impact on local infrastructure.	Section 5.2.1, 5.2.2, 5.2.5
4. Describe on-site manpower requirements, including the number of construction personnel who currently reside within the impact area, would commute daily to the site from outside the impact area, or would relocate temporarily within the impact area.	Section 5.2.2
5. Estimate total worker payroll and material purchases during construction and operation.	Section 5.2.2
6. Determine whether existing housing within the impact area is sufficient to meet the needs of the additional population.	Section 5.2.2
7. Describe the number and types of residences and businesses that would be displaced by the project, procedures to be used to acquire these properties, and types and amounts of relocation assistance payments.	Section 5.2.3
8. Conduct a fiscal impact analysis evaluating incremental local government expenditures in relation to incremental local government revenues that would result from construction of the project. Incremental expenditures include, but are not limited to, school operating costs, road maintenance and repair, public safety, and public utility costs.	Section 5.2.2

RESOURCE REPORT NO. 5
SOCIOECONOMICS

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C Project Construction and Mitigation Plans

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D Agency and Stakeholder Correspondence

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Appendix D-2 – SHPO Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

Appendix D-3 – Native American Organizations and Tribes Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

E Landowner Information

Appendix E-1 – Landowner/Stakeholder List (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE"*)

F Wetland and Waterbody Delineation Report and Listed Species Report

Appendix F-1 – Wetland and Waterbody Delineation Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

Appendix F-2 – Listed Species Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

G Cultural Resource Assessment Survey Report (*Submitted as* "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

H Air Quality Calculations (Construction Emissions)

I Sound Study Report

ABBREVIATIONS AND ACRONYMS

$	United States dollar
%	percent/percentage
ACS	American Community Survey
BG	Census Block Group
Certificate	Certificate of Public Convenience and Necessity
CFR	Code of Federal Regulations
EMS	emergency medical services
EPAETC Tiger	United States Environmental Protection Agency
ETC Tiger	ETC Tiger Pipeline, LLC
LEP	limited English proficiency
NGA	Natural Gas Act
PCB	polychlorinated biphenyls
Project	Franklin Farms Project
RV	recreational vehicle
U.S.	United States

5.0 SOCIOECONOMICS

5.1 EXISTING SOCIOECONOMIC CONDITIONS

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, an interconnect site, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project in June 2027, and following the receipt of all applicable approvals, anticipating construction activities to begin in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

In accordance with Section 380.12(g) of the FERC regulations, 18 Code of Federal Regulations (CFR) § 380.12(g) (2025), Resource Report No. 5 is required for projects involving the construction of significant aboveground facilities, including, among others, conditioning or liquefied natural gas plants. ETC Tiger acknowledges that Resource Report No. 5 is not required for the proposed Project, as it does not include significant aboveground facilities, but has included the following information considering seven meter stations are proposed. Resource Report No. 5 discusses the existing socioeconomic conditions that will be affected by construction and operation of the proposed Project, as well as the expected impact on those conditions. As described below, the Project is expected to have a temporary impact on population, housing, public services, and transportation needs.

5.1.1 Population

Pipeline laterals and meter stations are in Richland Parish, Louisiana. Collectively, the Census block groups (BG) crossed by pipeline laterals and within 0.5 miles of meter stations are in the Study Area.

Population data and demographic data from the U.S. Census Bureau's American Community Survey (ACS) estimates for the Study Area are provided in **Table 5.1-1**. Percentage of minority individuals,[1] households below poverty,[2] and limited English proficiency (LEP) households[3] are included.

TABLE 5.1-1
Population and Demographics by Census Block Group, Study Area

Geographic Area Name	2024 Population	2024 Population Density (persons per square mile)	Minority Population (percent [%])	Households Below Poverty	LEP Households
Louisiana	**4,611,961**	**106.7**	**44.3%**	**19.0%**	**N/A**
Richland Parish	19,822	35.7	39.8%	23.5%	N/A

[1] Minority individuals self-identify as Hispanic (any race) and/or Black or African American, American Indian or Alaska Native, Asian, and Native Hawaiian and Other Pacific Islander.

[2] The annual thresholds for the federal poverty level vary with household size and composition. In 2025, the most current year for which data are available, the poverty level for a family of four related persons with two children under age 18 was $32,649. For more information, see https://www.census.gov/data/tables/time-series/demo/income-poverty/historical-poverty-thresholds.html.

[3] LEP households are those which no member 14 years old and over (1) speaks only English or (2) speaks a non-English language and speaks English "very well" or better. That is, all household members 14 years old and over have at least some difficulties with English.

TABLE 5.1-1
Population and Demographics by Census Block Group, Study Area

Geographic Area Name	2024 Population	2024 Population Density (persons per square mile)	Minority Population (percent [%])	Households Below Poverty	LEP Households
BG 2, Tract 9703	1,226	1,194.7	4.2%	29.7%	0.0%
BG 3, Tract 9703	917	539.6	7.2%	12.6%	0.0%
BG 1, Tract 9705	541	198.3	64.1%	24.9%	0.0%

Source: U.S. Census Bureau, 2025

5.1.2 Employment and the Economy

Table 5.1-2 presents data on the income, employment, and major industries within the Study Area.

TABLE 5.1-2
Existing Socioeconomic Conditions

Parish/State	Richland Parish	Louisiana
Per Capita Income (U.S. Dollars [$] 2024)[1]	$27,433	$35,038
Median Household Income (U.S. Dollars 2024)[1]	$53,544	$60,756
Civilian Labor Force (persons)[2]	7,937	2,110,788
Unemployment Rate[2]	5.2%	4.5%
Top Five Major Industries[1]	E, R, M, C, A	N/A

Labor force and unemployment rate data for Richland Parish are preliminary.

Industries: Full-Time, Year-Round Civilian Employed Population 16 Years and Over, in descending order of number of employees.

A = Arts, entertainment, recreation, and accommodation and food services; C = Construction; E = Educational services, and health care and social assistance; M = Manufacturing; R = Retail trade.

Sources:
[1] U.S. Census Bureau, 2025
[2] U.S. Bureau of Labor Statistics, 2026 (Labor Force & Unemployment Rate, February 2026)

5.1.3 Housing

Table 5.1-3 outlines housing data for the Project within Richland Parish based on 2020-2024 ACS 5-year estimates. The total number of housing units reflects the population size and housing availability in the Parish.

TABLE 5.1-3 Housing Characteristics		
Parish/State	**Richland Parish**	**Louisiana**
Total Housing Units	8,734	2,108,902
Owner-Occupied Housing Units	4,836	1,216,127
Percent Owner Occupied	55.4%	57.7%
Percent Renter Occupied	27.5%	27.9%
Median Gross Rent	$903	$1,064
Source: U.S. Census Bureau, 2025		

The ACS data were analyzed to evaluate vacancy trends and identify the availability of short-term housing options in the Study Area. Additional information regarding hotels, motels, campgrounds, and recreational vehicle (RV) parks was sourced from online platforms to provide a comprehensive overview. **Table 5.1-4** highlights the variety of rental units available in Richland Parish. Additional opportunities for short-term housing are available in nearby Monroe.

TABLE 5.1-4 Available Housing		
Parish/State	**Richland Parish**	**Louisiana**
Total Vacant Housing Units[1]	1,494	304,122
Vacant Units for Rent[1]	16	52,844
Percent Owner Occupied Vacancy[1]	1.0%	1.3%
Percent Rental Vacancy[1]	0.7%	7.8%
Number of Hotels and Motels[2]	29	1,017
Number of Campgrounds/RV Parks3,[4]	19	363
Sources: [1] U.S. Census Bureau, 2025 [2] Hotels.com, 2025 [3] RVLife.com, 2025 [4] RVPark.com, 2025		

5.1.4 Public Services and Infrastructure

Communities near the Project in Richland Parish are supported by a range of local government public services, including fire protection and emergency medical services (EMS), police departments and sheriffs' offices, medical facilities, and educational institutions. **Table 5.1-5** provides a list of facilities and their addresses within Richland Parish.

TABLE 5.1-5
Existing Public Services and Facilities in Richland Parish

Name	Address
Fire Departments	
Rayville Fire Department	902 Harrison Street, Rayville, LA 71269
Archibald - Alto Fire Station	Bess Lake Road, Rayville, LA 71269
Archibald - Alto Fire Station	1132 State Highway 856, Archibald, LA 71269
Ward One Fire Department Cook Station	293 Cook Road, Delhi, LA 71232
Archibald - Alto Fire Station	2737 State Highway 135, Rayville, LA 71269
Archibald Alto Fire Station 1	1526 State Highway 584, Rayville, LA 71269
Mangham Volunteer Fire Department	511 Horace Street, Mangham, LA 71259
Start - Girard Fire Protection District	26 Owl Loop Road, Start, LA 71279
Holly Ridge Fire District Station 1	597 Highway 134, Rayville, LA 71269
Delhi Fire Department	209 Broadway Street, Delhi, LA 71232
EMS/Hospitals	
Richardson Medical Center	254 State Highway 3048, Rayville, LA 71269
Delhi Hospital	407 Cincinnati Street, Delhi, LA 71232
Law Enforcement	
Rayville City Police Department	900 Harrison Street, Rayville, LA 71269
Mangham Police Department	306 Main Street, Mangham, LA 71259
Delhi Town Police Department	304 East Tennessee Street, Delhi, LA 71232
Richland Parish Sheriff's Office	708 Julia Street Room 113, Rayville, LA 71269
Source**:** U.S. Geological Survey, National Structures Dataset, 2026	

5.1.5 Traffic and Transportation

The Project will mainly be accessed by use of existing highways, roadways, and access roads. The following major routes in Richland Parish are anticipated to provide primary access to the Project:[4]

- State Routes 183, 582, and 853
- Interstate 20
- U.S. Highway 80

5.1.6 Tax Revenue

The general sales tax rates applicable to Richland Parish are listed in **Table 5.1-6**. Lodging is taxed at the local sales tax rate.

TABLE 5.1-6
Sales Tax Rates

Tax Jurisdiction	Rate
State of Louisiana	5%
Rayville, includes State sales tax	10.75%
Delhi, includes State sales tax	11.25%
Mangham, includes State sales tax	10.25%
Richland Parish—Unincorporated, includes State sales tax	9.25%
Source: SalesTaxHandbook, 2026	

5.2 CONSTRUCTION AND OPERATION IMPACTS

During construction, the Project is estimated to require roughly 300 full-time equivalent workers. Qualified local workers will be employed when available. Non-local workers will temporarily relocate to the area and are expected to secure temporary lodging within Richland Parish where construction will occur, or in larger nearby communities. Some may return to their primary residences on weekends, depending on distance, but few workers, if any, are expected to bring family members or relocate permanently.

Overall, any population effects within the Study Area resulting from non-local construction employment would be minor and short-lived. Once specialized tasks are completed, these workers generally depart the area, resulting in no lasting population change.

5.2.1 Economy and Tax Revenue

Although the Project does not involve the level of activity associated with a major pipeline project, the construction of the laterals and meter stations will stimulate economic activity within Richland Parish and the surrounding region. Non-local workers will make expenditures on lodging, food, and daily needs while

[4] U.S. Census Bureau, Geography Division, 2026

temporarily residing in the area. Local workers will benefit from enhanced job opportunities and from economic activity associated with spending by non-local workers. This spending generates secondary economic effects through both business-to-business transactions (indirect impacts) and household spending from worker income (induced impacts), creating a broader multiplier effect.

Richland Parish will also benefit from increased property tax revenues associated with the Project's facilities. These locally generated revenues provide funding for schools, emergency services, and parish operations and support a range of public programs identified in the Parish's approved fiscal budget.

5.2.2 Housing

During Project construction, the presence of construction workers in the Study Area will increase the demand for temporary short-term housing. Based on local labor availability, the majority of construction workers are anticipated to be non-local and will relocate temporarily to the vicinity of the Study Area for the duration of their employment, possibly commuting home on weekends, depending on the location of their primary residence. Non-local construction workers typically use available temporary housing such as area campgrounds/RV parks and hotel/motels in the Study Area and possibly adjacent towns or counties that are within a reasonable daily commuting distance of the Project. Non-local construction workers frequently use travel trailers or RV campers. Available lodging options in Richland Parish are summarized in **Table 5.1-4** in **Section 5.1.3**.

Given the size of the workforce needed and the availability of hotels/motels, RV parks, and campgrounds, the Project does not expect a shortage of temporary housing in the broader northeast Louisiana region, which includes the City of Monroe. The anticipated migration of non-local construction workers to the Study Area represents a small share of the total temporary housing (hotel and motel rooms and campsites); and, therefore, the temporary demand for these facilities is unlikely to displace permanent residents, adversely affect housing prices, or cause conflicts with tourism. Project impacts on local housing are expected to be negligible and short-term. No new operations staff are proposed; therefore, no long-term traffic impacts to housing are anticipated.

5.2.3 Displacement of Residences or Businesses

Aboveground facilities and routes are located on agricultural land. New construction will not require the displacement of any residences or businesses.

5.2.4 Public Services and Infrastructure

The Study Area contains a range of medical providers and emergency response agencies capable of supporting the temporary influx of construction personnel. A summary of available services is provided in **Table 5.1-5**. No measurable strain on existing fire protection, law enforcement, or healthcare facilities is anticipated from the temporary increase in workers. The Project will coordinate with local law enforcement, fire department, and EMS providers to ensure effective emergency response throughout the construction period.

Typically, temporary, non-local workers do not bring family members; therefore, the number of school-age children temporarily relocating to the area is anticipated to be minimal of any. As a result, no effects on enrollment within the local school district are expected.

5.2.5 Transportation and Traffic

Major state and federal transportation corridors in the Study Area are listed in **Section 5.1.5**. Existing public access roads will be the primary means of accessing the Project Facilities and contractor yards. Construction

activity will generate short-term, limited effects on the transportation network in the Study Area. Work planned within or adjacent to roadways will be scheduled to avoid local commuter periods and the operating hours of school transportation and other public transit services, where practicable.

Traffic management measures will be developed in coordination with parish transportation officials. Materials and equipment will be delivered directly to the designated construction areas. Most heavy equipment will remain onsite throughout the construction period. Construction is anticipated to occur Monday through Saturday from 6 AM through 10 PM. Construction will only occur 24 hours per day and/or on Sundays or holidays as necessary.

Construction personnel will travel to their assigned work areas in personal company vehicles. Workers will be distributed across multiple locations, reducing congestion at any one point along the roadway network. Typical start and end times are planned to align most worker commutes with off-peak traffic periods. Limited activities may extend outside these hours. Any Project-related traffic impacts will be temporary.

The Project does not anticipate substantial or long-term impacts on transportation infrastructure or traffic patterns during either construction or operation of the Project.

No new operations staff are proposed; therefore, no long-term traffic impacts to housing are anticipated.

5.2.6 Summary of Local Effects

Additional potential effects from the Project's construction and operation are discussed in **Table 5.2-1**.

TABLE 5.2-1
Anticipated Effects on Local Communities

Resources	Effects
Socioeconomics: Construction (Resource Report 5)	• Non-local workers will temporarily relocate to the Project vicinity. • Increases in demand for health care, municipal services, police, and fire protection services are expected to be temporary and are not expected to measurably affect the quality of services currently received by local communities and residents. • Revenues from construction employment, local expenditures by the construction companies for construction materials and equipment rentals, and non-local construction workers' expenditures for temporary housing, food, and entertainment will benefit the local economy. Taxing jurisdictions will also benefit from increased economic activity.
Socioeconomics: Operations (Resource Report 5)	• Permanent personnel are not anticipated to be added as a result of the project. • Long-term positive impacts from the Project's operation will arise from the increased property tax base. • Local communities will benefit from ad valorem taxes paid annually.

TABLE 5.2-1
Anticipated Effects on Local Communities

Resources	Effects
Water Use and Quality (Resource Report 2)	• Stormwater runoff and erosion may result in indirect impacts (e.g., sedimentation, turbidity) to surface waters adjacent to the Project area. However, these effects will be avoided and minimized by installation and maintenance of sediment and erosion control best management practices throughout construction in accordance with the *FERC Upland Erosion Control, Revegetation, and Maintenance Plan* (Appendix C-1) and *FERC Wetland and Waterbody Construction and Mitigation Procedures* (Appendix C-2).Therefore, the Project is not expected to adversely impact surface water quality or supplies. • An inadvertent spill of fuels, lubricants, or solvents could result in groundwater contamination. Spill-related impacts will be avoided or greatly reduced by prohibiting fuel storage and refueling activities within 200 feet of private wells and 500 feet of community or municipal wells; implementing proper storage, containment, and handling procedures; and by requiring immediate cleanup should a spill or leak occur. In the event of a spill, ETC Tiger will implement its Spill Prevention, Countermeasures, and Control Plan.
Wildlife, Fish, and Vegetation (Resource Report 3)	• Potential impacts to waterbodies from proposed Project construction are discussed in Resource Report No. 2. None of the waterbodies affected by the Project contain or have the potential to contain Essential Fish Habitat, therefore no impacts to Essential Fish Habitat will occur as a result of the Project. • According to the USFWS, there is no designated unique, sensitive, or critical wildlife habitat within the Project area. No national parks, state forests, wildlife management areas, or local / private preserves are crossed by the Project. • To minimize vegetation impacts during construction, ETC Tiger will use erosion control measures, topsoil segregation in select areas, testing and mitigation for soil compaction, and limited routine vegetation maintenance. Furthermore, all temporarily disturbed upland areas will be reseeded in accordance with the FERC Plan, FERC Procedures, and recommendations from state agencies. • ETC Tiger will minimize impacts to vegetation resulting from operation and maintenance of the Project through infrequent mowing and clearing. Specifically, routine vegetation mowing or clearing over the full width of the permanent rights-of-way in uplands will not be conducted more frequently than every 3 years.

TABLE 5.2-1
Anticipated Effects on Local Communities

Resources	Effects
Cultural Resources (Resource Report 4)	• ETC Tiger submitted coordination letters to representatives of Apache Tribe of Oklahoma, Choctaw Nation of Oklahoma, Coushatta Tribe of Louisiana, Jena Band of Choctaw Indians, Mississippi Band of Choctaw Indians, Quapaw Nation, and Seminole Tribe of Florida. • ETC Tiger has consulted with the Louisiana State Historic Preservation Office to identify resources listed or potentially eligible for listing on the National Register of Historic Places. Consultation is ongoing and ETC Tiger will avoid these resources to the extent practicable and implement mitigation measures where resources are unavoidable, if applicable. • ETC Tiger will implement the appropriate Unanticipated Discoveries Plan if previously unidentified cultural resources are identified during construction.
Land Use, Recreation, and Aesthetics: Construction (Resource Report 8)	• The majority of the land crossed is agricultural land. Following construction, agricultural lands will be returned to pre-construction conditions and will be available for use during pipeline operations. Therefore, impacts to agricultural lands are anticipated to be short-term and minor. • Project operations are not anticipated to affect any natural, recreational, or scenic areas. The pipelines will not have an impact on the viewshed in the vicinity upon the completion of construction activities and commencement of operation, as the pipelines will be buried underground. • No new residential or commercial development areas were identified within 0.25 miles of the Project area.
Air Quality (Resource Report 9)	Construction emissions will be temporary and of limited duration and will not significantly increase ambient air pollutant concentrations. Exhaust emissions from diesel- and gasoline-fueled construction equipment and vehicle engines will be minimized by federal design standards and will comply with U.S. Environmental Protection Agency (EPA) mobile and non-road emission regulations. Emissions also will be controlled by purchasing gasoline and fuel whose specifications are controlled by federal and state air pollution control regulations. Contractors and employees will also be encouraged to minimize vehicle and equipment idling time to the extent practical during construction activities.
Noise Quality (Resource Report 9)	Short-term increases in sound levels could occur during construction. Only standard construction equipment will be used in the construction of the meter stations, with no dynamic compaction expected. Temporary noise mitigation measures include utilizing equipment with muffled exhausts, minimizing equipment idling, minimizing the use of backup beepers, scheduling construction to end by 10PM, and utilizing temporary sound barriers.

TABLE 5.2-1
Anticipated Effects on Local Communities

Resources	Effects
Safety (Resource Report 11)	• Project Facilities will be designed, constructed, operated, and maintained in accordance with or to exceed the U.S. Department of Transportation Pipeline and Hazardous Materials Safety Administration minimum federal safety standards in 49 Code of Federal Regulations 192. • ETC Tiger maintains operating policies and procedures that are periodically reviewed by U.S. Department of Transportation. The policies provide specific directions in preventive maintenance and vigilant patrols of facilities, as well as procedures to be followed in the event of accident or natural catastrophe. All operating personnel are thoroughly trained to perform their activities in accordance with these policies and procedures.
Polychlorinated biphenyls (PCB) (Resource Report 12)	ETC Tiger will fully comply with the Toxic Substances Control Act. No PCBs (polychlorinated biphenyls) more than 50 parts per million in pipeline liquids have been identified within ETC Tiger's pipeline system and, therefore, the sampling protocols for characterization and abandonment of PCB-contaminated pipelines are not required under the Act.

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ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 6
Geological Resources

FERC Docket No. CP26-___-000

June 2026

RESOURCE REPORT NO. 6 FILING REQUIREMENTS	
Information	**Section Reference**
1. Identify the location (by milepost) of mineral resources and any planned or active surface mines crossed by the proposed facilities. (§ 380.12(h)(1&2)) • Describe hazards to the facilities from mining activities, including subsidence, blasting, slumping or landsliding or other ground failure.	Section 6.5 Section 6.3
2. Identify any geologic hazards to the proposed facilities. (§ 380.12(h)(2)) • For the offshore area this information is needed on a mile-by-mile basis and will require completion of geophysical and other surveys before filing.	Section 6.3
3. Discuss the need for and locations where blasting may be necessary in order to construct the proposed facilities. (§ 380.12(h)(3))	N/A
4. For LNG projects in seismic areas, the materials required by "Data Requirements for the Seismic Review of LNG Facilities," NBSIR84-2833. (§ 380.12(h)(5))	N/A
5. For underground storage facilities, how drilling activity by others within or adjacent to the facilities would be monitored, and how old wells would be located and monitored within the facility boundaries. (§ 380.12(h)(6))	N/A
Information Often Missing and Resulting in Data Requests	
Identify any sensitive paleontological resource areas crossed by the proposed facilities. (Usually only if raised in scoping or if the project affects federal lands.)	Section 6.4
Briefly summarize the physiography and bedrock geology of the project.	Section 6.1.2
If proposed pipeline crosses active drilling areas, describe plan for coordinating with drillers to ensure early identification of other companies' planned new wells, gathering lines, and aboveground facilities.	Section 6.2 and 6.5.2
If the application is for underground storage facilities: Describe monitoring of potential effects of the operation of adjacent storage or production facilities on the proposed facility, and vice versa; Describe measures taken to locate and determine the condition of old wells within the field and buffer zone and how the applicant would reduce risk from failure of known and undiscovered wells; and Identify and discuss safety and environmental safeguards required by state and federal drilling regulations	N/A

RESOURCE REPORT NO. 6
GEOLOGICAL RESOURCES

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D Agency and Stakeholder Correspondence

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Appendix D-2 – SHPO Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

Appendix D-3 – Native American Organizations and Tribes Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

E Landowner Information

Appendix E-1 – Landowner/Stakeholder List (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE"*)

F Wetland and Waterbody Delineation Report and Listed Species Report

Appendix F-1 – Wetland and Waterbody Delineation Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

Appendix F-2 – Listed Species Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

G Cultural Resource Assessment Survey Report (*Submitted as* "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

H Air Quality Calculations (Construction Emissions)

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ABBREVIATIONS AND ACRONYMS

BMP	best management practice
Certificate	Certificate of Public Convenience and Necessity
CFR	Code of Federal Regulations
CLECO	Central Louisiana Electric Company
EDMS	Electronic Document Management System
EI	environmental inspector
ETC Tiger	ETC Tiger Pipeline, LLC
FEMA	Federal Emergency Management Agency
FERC or Commission	Federal Energy Regulatory Commission
FERC Plan	Upland Erosion Control Revegetation and Maintenance Plan (2013)
FERC Procedures	Wetland and Waterbody Construction and Mitigation Procedures (2013)
KMZ	keyhole markup language zipped (Google Earth Pro)
Lateral	Franklin Farms Lateral and/or FF Trio Farms Lateral
LDCE	Louisiana Department of Conservation and Energy
LDEQ	Louisiana Department of Environmental Quality
LGS	Louisiana Geological Survey
LSU	Louisiana State University
MMI	modified Mercalli intensity
MP	milepost
NGA	Natural Gas Act
NGMDB	National Geologic Map Database
NPS	National Park Service
NRCS	National Resources Conservation Service
NSHM	National Seismic Hazard Model
O&G	oil and gas
OD	outer diameter
PAR	permanent access road
PGA	peak ground acceleration
Project	Franklin Farms Project
SONRIS	LDCE Data Portal
SPAR Plan	Spill Prevention and Response Plan
SVP	Society of Vertebrate Paleontology
SWEPCO	Southwestern Electric Power Company
TAR	temporary access road
U.S.	United States
USACE	United States Army Corps of Engineers
USDA	United States Department of Agriculture
USGS	United States Geological Survey

6.0 GEOLOGICAL RESOURCES

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, an interconnect site, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project in June 2027, and following the receipt of all applicable approvals, anticipating construction activities to begin in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

Resource Report No. 6 provides geologic information for the proposed Project. Information within this report includes details of the physiological setting, bedrock geology, mineral resources, and geologic hazards along the proposed pipeline route and at the aboveground facility locations. Also addressed are potential impacts to existing geological, mineral, and paleontological resources within the Project area due to construction, operations, and maintenance of the Project.

To avoid and minimize impacts to geological resources during construction and operation of the Project, ETC Tiger will implement the FERC *Upland Erosion Control, Revegetation and Maintenance Plan*, May 2013 (FERC Plan, **Appendix C-1**), the FERC *Wetland and Waterbody Construction and Mitigation Procedures*, May 2013 (FERC Procedures, **Appendix C-2**), with proposed site-specific exceptions, and ETC Tiger's *Spill Prevention and Response Plan* (SPAR Plan) (**Appendix C-3**).

6.1 GEOLOGIC SETTING

6.1.1 Geological Province

The contiguous United States (U.S.) is divided into physiographic provinces according to geomorphology. The Project area is located within the Coastal Plain province, the flattest physiographic province in the continental U.S., which extends for over 2,200 miles from Cape Cod to the Mexican border and into Central America. The Coastal Plain province is generally characterized by a thickening wedge of Cretaceous to Holocene unconsolidated to semi-consolidated sedimentary deposits that dip gently toward the coast and were deposited in marine, deltaic, and fluvial environments (NPS 2018, USGS 2008a).

Within this province, the Project area lies within the Mississippi Alluvial Plain, a major sub-province formed by the Mississippi River system. The Mississippi Alluvial Plain is a river-dominated landscape defined by Quaternary fluvial deposition including extensive Holocene floodplains, meander belts, and backswamps, as well as Pleistocene terrace features formed during glacial-interglacial cycles that controlled sedimentation within the Lower Mississippi Valley. These cycles produced braided-stream deposits composed primarily of sand and gravel (NPS 2018, Saucier 1984).

The site is positioned on a Pleistocene terrace feature known as Macon Ridge, which represents a relict upland within the alluvial valley forming a slightly elevated fluvial ridge of valley-train deposits in contrast to surrounding areas consisting of younger Holocene deposits (Snead, 1989). The Macon Ridge is divided into the Upper and Lower Macon Ridge allostratigraphic units, composed primarily of sand and gravel bar deposits formed during periods of high-energy glacial outwash transport. The Project area is located specifically within the Upper Macon Ridge alloformation, which is a stratigraphically higher and slightly younger braided-belt unit compared with the Lower Macon Ridge alloformation, and is characterized by a well-preserved sandy fluvial surface commonly mantled by Peoria Loess (windblown silt) or loess-derived silty sediments (LGS 2015a, LGS 2015b). Elevations on the Upper Macon Ridge are typically 80 to 120

feet above mean sea level (amsl), standing approximately 20 to 30 feet above the surrounding Holocene floodplain (Saucier 1984, LGS 2015a, USGS Topos).

6.1.2 Geology

In general, the Project area consists of braided-stream terrace deposits composed of glacial outwash of the ancestral Arkansas River, composed primarily of light gray, tan, and brown fine- to coarse-grained sand with some clay, silt, and gravel. In addition, portions of the proposed contractor yards are blanketed by thin eolian loess deposits described as tan to reddish-brown massive silt containing minor clay and very fine sand (NGMDB 1984, Geologic Map of Louisiana).

Late Pleistocene glaciation known as the Wisconsin Episode was a major control on regional sedimentation, and although ice sheets did not extend into Louisiana, glacial advance and retreat produced large volumes of fluvial sediment and braided-stream systems with subsequent valley incision and terrace formation. As a result, landforms in the Project area consist of slightly elevated ridges associated with Macon Ridge, surrounded by low-relief Holocene meander belts, natural levees, and backswamp floodplain deposits (Saucier 1984, USACE 1994). In Richland Parish, the thickness of Mississippi River alluvial deposits ranges from less than approximately 80 feet to greater than 160 feet (White, 2019). Upland areas such as Macon Ridge are mantled by eolian loess deposits derived from glacial outwash sediments of the Lower Mississippi Valley (Saucier, 1984); however, there is an estimated less than three feet of loess at the site location (LGS 2008). Beneath the Quaternary deposits, the suballuvial surface is developed on older Tertiary sedimentary units of the Coastal Plain, including the Cockfield Formation and Sparta Sand aquifer system (Covay 1985; White 2019).

The geologic units that underlie the proposed Project area were determined using geologic maps published by the Louisiana Geological Survey (NGMDB 1984) are listed in **Table 6.1-1** and depicted in **Appendix A-11**.

TABLE 6.1-1
Geologic Units Underlying the Franklin Farms Project

Facility	Map Symbol	Surficial Geology	Milepost		Distance (Miles)	Age	Description
			Enter	Exit			
Pipeline							
Franklin Farms Lateral	Qbs	Braided Stream Terraces	0.0-13.2		Entire Site	Pleistocene	Light gray, tan and brown fine to coarse sand; some clay, silt, and gravel.
Franklin Farm (FF) Trio Farm Lateral	Qbs	Braided Stream Terraces	0.0-1.7		Entire Site	Pleistocene	Light gray, tan and brown fine to coarse sand; some clay, silt, and gravel.
Contractor Yards							
Contractor Yards	Qbs-l	Braided Stream Terraces-Loess	N/A		Entire Yard	Pleistocene	Tan to reddish brown massive silt with some clay and minor amounts of very fine sand. 1-9 meters of Loess.

TABLE 6.1-1
Geologic Units Underlying the Franklin Farms Project

Facility	Map Symbol	Surficial Geology	Milepost		Distance (Miles)	Age	Description
			Enter	Exit			

Sources: NGMDB 1984

6.1.2.1 Bedrock

Bedrock is not anticipated to occur at shallow depth within the Project area. Soil units described as C-horizons represent relatively unaltered parent material that has been minimally affected by pedogenic processes (USDA 2022). A few areas along the Project area indicate soils with C-horizon properties extending to depths of approximately 72 inches (Draft Cultural Resources Survey, TRC). A review of the U.S. Department of Agriculture (USDA) Web Soil Survey indicates that bedrock within the Project area is generally mapped at depths greater than approximately 200 centimeters (80 inches) below ground surface (NRCS 2026).

Bedrock at the site in the Rayville-Delhi area generally lies beneath approximately 50 to 125 feet of Quaternary deposits. Bedrock is composed of the Cockfield Formation, which is described as Eocene (Tertiary) age and consisting of approximately 20 to 600 feet of interbedded sand, lignite, carbonaceous clay, and sandy glauconitic clay. The Sparta Formation below the Cockfield serves as a regionally extensive aquifer composed of fine to medium-grained sand with clay interbeds, ranging in thickness from approximately 500 to 700 feet (Covay 1985).

Blasting will not be required for the Project.

6.2 MINERAL RESOURCES

A review of available state and publicly accessible data sources indicates that no active or proposed surface mining operations are present within or adjacent to the Project area (LDEQ 2026, EDMS 2026). This review included agency databases and available aerial imagery. In Louisiana, non-coal mineral extraction activities such as sand and gravel operations are typically regulated through environmental permitting rather than formal mining permits, and no evidence of major sand, gravel, or borrow operations was identified within the immediate vicinity of the Project area (LDEQ 2026, Google Earth Pro 2026).

Lignite, a low-rank coal, is mined in Louisiana, including the Oxbow Lignite Mine (CLECO/AEP-SWEPCO) in Red River and Natchitoches Parishes, and the Dolet Hills Mine (Five Forks Mine) in DeSoto Parish. However, these operations are not located within the Project Area or Richland Parish (LDEQ 1993, LGS 2000, SONRIS LDCE 2026a).

Oil and gas production is a primary mineral resource in northeastern Louisiana, and several active and historic fields are present near the Project Area, including the Rayville, Sacksonia, Big Creek, and Delhi West fields (SONRIS 2026b).

Oil and gas wells were identified in the proposed Project vicinity as depicted in **Appendix A-12**. The wells located within 0.25-miles of the Project area are listed in **Table 6.2-1** (SONRIS 2026b). The closest well is approximately 695 feet from the Franklin Farms Lateral near milepost (MP) 0.1.

TABLE 6.2-1
Oil and Gas Wells within 0.25-miles of the Project

Facility	MP	Distance (feet)	Direction	Status
Pipeline				
Franklin Farms Lateral	0	1,028	South	Dry and Plugged
	0.1	695	South	Plugged and Abandoned
Contractor Yard				
Contractor Yard 1	-	1,282	South-West	Dry and Plugged
Contractor Yard 2	-	798	North-East	Permit Expired
Access Roads				
Temporary Access Road (TAR)				
TAR-009.000	5	860	North	Dry and Plugged
Permanent Access Road (PAR)				
PAR-024.000	13.2	1,026	North-East	Dry and Plugged

Source:
SONRIS LDCE 2026b

6.2.1 Planned Surface Mine Development

No planned surface mine development was identified in proximity to the Project area. A 21-acre mineral tract identified as "Minerals from Land Sales" is located approximately 1.3 miles west of the Project area along the east bank of the Boeuf River; however, this designation reflects mineral ownership and does not indicate active or proposed surface mining operations (SONRIS 2026c LDCE).

6.3 GEOLOGIC HAZARDS

Geologic hazards are conditions or phenomena, either naturally occurring or human-made, that present a risk to life and/or property. Geologic hazards in relation to the proposed Project area are discussed below and include earthquakes, soil liquefaction, landslide susceptibility, karst topography, land subsidence, and flooding.

6.3.1 Earthquakes and Seismicity

Based on a review of regional geology and seismic literature, northeastern Louisiana, including the Project area located east of Rayville, appears to be tectonically stable and lacks any confirmed growth faults or Quaternary-active faults (USGS 2026d, LGS 2001). Regional mapping from the Louisiana Geological Survey (LGS) identifies the area as part of the Mississippi Embayment, characterized by gently dipping sedimentary units and an absence of mapped surface faulting (LGS 2001). Regionally, the Monroe Uplift is recognized as a major structural feature underlying northeastern Louisiana and adjacent states, and the Project area is situated along the southern flank of this uplift. The Monroe Uplift itself is a broad, basement-

controlled structural high with no clearly defined surface fault expression, and available data suggest that any deformation is expressed as regional uplift rather than discrete fault movement (Cushing 1964).

Additional work evaluating "wrench faults" in Louisiana indicates that these features are best interpreted as deep-seated structural lineaments identified from subsurface data, with no clear evidence of Quaternary movement or significant contribution to seismic hazard (Zimmerman 1997). Consistent with this interpretation, the U.S. Geological Survey (USGS) classifies features in this region as either Class B or Class C, meaning they show either limited or uncertain evidence of Quaternary deformation (Class B) or lack sufficient geologic evidence to demonstrate tectonic faulting or Quaternary activity (Class C). All together these sources indicate that the Richland Parish area does not contain active faults and is not associated with significant tectonic seismic risk. (USGS 2021a, Wheeler 2000)

A review of the USGS Earthquake Catalog indicates that no recorded earthquakes have occurred within a 65-mile radius of the Project area since 1900. The nearest cataloged events during this period are located outside the search radius, primarily in southern Arkansas and west-central Mississippi (USGS 2026f, Google Earth Pro KMZ). This lack of nearby recorded seismicity further supports the interpretation that the Project area is characterized by very low historical seismic activity.

With no active faults identified at or near the Project area, the potential for seismic impacts is further assessed using regional ground motion estimates provided by the USGS (USGS 2026e). According to the latest USGS National Seismic Hazard Model, the Project area is in an area of very low seismic hazard, as reflected by low predicted peak ground acceleration (PGA) values as depicted in **Appendix A-13 and A-14**. The USGS estimates peak ground accelerations within the Project area between two and three percent gravity for a 500-year earthquake, or an earthquake with a 10 percent probability of exceedance within a 50-year interval (NSHM 2026). The estimated peak ground accelerations within the Project area (i.e., two to three percent) are associated with weak ground motions and correspond approximately to Modified Mercalli Intensity levels of IV to V based on empirical relationships between ground motion and felt intensity (USGS 2016, Worden 2012). If an earthquake were to occur within the Project area, the likelihood of significant damage would be low.

6.3.2 Soil Liquefaction

Soil liquefaction is the process by which soil loses a significant portion of its shear strength when subjected to rapid loading, causing the soil to behave in a manner similar to a liquid. During a seismic event, this condition can occur in saturated, cohesionless soils where increased pore water pressure reduces grain-to-grain contact. Liquefaction can result in ground settlement and loss of bearing capacity, potentially impacting structures that rely on soil for support. However, liquefaction potential at the Project area is considered low because the region experiences very little seismic shaking, as reflected in USGS hazard models, and there are no significant nearby earthquake sources (NSHM 2026, USGS 2008b). While the subsurface may include unconsolidated alluvial sediments that could be susceptible under stronger shaking, the overall likelihood of liquefaction is considered minimal given the low seismic hazard.

6.3.3 Flooding

Flash floods result from rapid increases in water volume and flow rate within waterbodies and adjacent floodplains, typically following intense rainfall over a short period, generally less than six hours. They can occur within minutes to a few hours depending on storm intensity and watershed characteristics.

As discussed in Resource Report No. 2 *Water Use and Quality*, portions of the Project pipeline and access road workspace are located within the 100-year floodplain (Zone A) as mapped by the Federal Emergency Management Agency (FEMA) (FEMA 2026). These zones represent areas subject to inundation by the one percent annual chance flood event. USGS topographic mapping indicates that portions of the Project

workspace are located within areas classified as wooded marsh or swamp, which are indicative of low-lying, poorly drained conditions where standing water may be present. Intermittent drainage features, including tributaries, mapped within the Project area indicate that surface water flow is primarily rainfall-driven and may contribute to localized surface runoff during storm events (USGS 2024a & 2024b-Topographic Maps).

Because the pipeline will be installed below ground surface, flooding is not anticipated to adversely affect pipeline integrity. In addition, access roads are not anticipated to significantly alter existing ground contours, and flooding conditions are not expected to change appreciably. Construction and mitigation measures for aboveground facilities related to flooding are described in Section 6.6. The Project will be designed in accordance with 49 Code of Federal Regulations (CFR) Part 192.

6.3.4 Landslide Susceptibility

Landslides are the downslope movement of soil, rock, or a combination of materials under the influence of gravity. These movements can range from slow displacement to rapid failure and are commonly classified by both the type of material and the mode of movement, including falls, slides, spreads, and flows (USGS 2008b). Slides typically occur along a defined surface of rupture, whereas flows behave more like a viscous fluid and are often associated with elevated water content. Landslides are generally triggered by factors such as heavy precipitation, stream or river erosion, seismic activity, or human activities including excavation and grading, with water playing a critical role by increasing pore pressure and reducing soil strength. They are most common in areas with steep slopes, weak or unconsolidated materials, and elevated groundwater conditions.

The Project area is in a relatively low-relief area characterized by generally gentle slopes and an absence of significant topographic features associated with landslide susceptibility. While minor, localized slope instability may occur in areas of erosion or where construction activities modify slope geometry or surface conditions, regional conditions do not support widespread landsliding. This is consistent with the USGS Landslide Inventory and Susceptibility Map, which classifies the Project area within an area of negligible to low landslide susceptibility (USGS 2026g). Based on these considerations, landslides are not anticipated to pose a significant risk to the Project area.

6.3.5 Land Subsidence and Karst Topography

Subsidence is the gradual settling or localized sinking of the ground surface resulting from subsurface movement of earth materials. The primary causes of subsidence include compaction of aquifer systems associated with groundwater withdrawal, drainage and oxidation of organic soils, and dissolution or collapse of susceptible subsurface materials. Subsidence related to aquifer-system compaction is the most widespread mechanism and can occur where unconsolidated or semi-consolidated sediments undergo compression in response to declining groundwater levels (USGS 2000, Galloway1999).

In northeastern Louisiana, subsidence potential is primarily associated with natural consolidation of unconsolidated alluvial sediments and, to a lesser extent, localized groundwater withdrawal. Although Louisiana and Mississippi are part of a broader Coastal Plain region underlain by evaporite deposits that have formed salt domes, these features are regionally concentrated and are not characteristic of northeastern Louisiana. The Project area is not associated with underground mining or solution mining of subsurface materials (e.g., salt caverns), which are commonly associated with collapse-related subsidence (USGS 1990). In addition, the Project area does not involve activities such as large-scale groundwater extraction or subsurface resource removal that are commonly associated with induced subsidence. While minor, localized subsidence may occur over time due to natural soil consolidation or changes in groundwater conditions, subsidence is not anticipated to represent a significant geologic hazard to the Project area.

Karst is a type of terrain formed by the dissolution of soluble rocks such as limestone, dolomite, gypsum, or salt, and is characterized by features including sinkholes, caves, and underground drainage systems. These features develop where slightly acidic groundwater dissolves susceptible bedrock, potentially creating subsurface voids that may lead to localized subsidence or collapse. The USGS has developed mapping that identifies areas of karst and the potential for karst development based on the distribution of soluble rocks and associated geologic conditions (USGS 2021b, Weary 1984).

Based on a review of USGS karst mapping, the Project area is not located within an area characterized by karst or significant potential for karst development. Northeastern Louisiana is predominantly underlain by unconsolidated to semi-consolidated clastic sediments of the Coastal Plain rather than shallow carbonate or evaporite rocks typically associated with karst terrain. Although soluble materials may be present regionally at depth, the absence of near-surface soluble bedrock and characteristic karst features indicates that karst-related hazards are not anticipated to represent a significant geologic concern for the Project area. There are no known sinkholes within 1,500 feet of the Project area (USGS 2014, USGS 2020).

6.4 PALEONTOLOGICAL RESOURCES

Paleontological resources consist of fossilized remains, traces, or imprints of organisms preserved in geologic materials, most commonly within sedimentary deposits. In northeastern Louisiana, mapped geologic units within the Project area consist of Quaternary alluvial and terrace deposits of the Mississippi Alluvial Plain, including Pleistocene braided-stream sediments associated with the Macon Ridge (LGS 2015a, 2015b, Snead 1989). These deposits may locally contain fossil material; however, repeated fluvial reworking tends to fragment, transport, and disperse fossil material rather than preserve it in place.

Although fossil-bearing units may be present regionally, surface exposures within the Project area are limited, and the likelihood of encountering significant or scientifically important paleontological resources during construction is considered low. While significant paleontological resources are not anticipated, appropriate precautions would be taken during construction. If paleontological resources are encountered, work would be halted in the immediate area, the resource would be protected, and FERC and the landowner would be notified to determine appropriate next steps (SVP 2010).

ETC Tiger has prepared an *Unanticipated Discoveries Plan* which includes paleontological resources for the Project. As part of this plan, the Louisiana State University (LSU) Museum of Natural Science will be contacted for guidance should a vertebrate fossil, or suspected vertebrate fossil, be uncovered. A copy of the plan is provided in **Appendix C-4**.

6.5 CONSTRUCTION IMPACTS AND MITIGATION

To minimize impacts to geology from construction activities on slopes and to prevent soil erosion, ETC Tiger will implement Best Management Practices (BMP) in the FERC Plan and FERC Procedures. These BMPs include installation of erosion control devices, management of trench dewatering activities, and revegetation. Restoration measures will be monitored to ensure their effectiveness. If monitoring identifies areas of erosion or ineffective revegetation, they will be addressed in a timely manner.

6.5.1 Blasting

Blasting is not anticipated for the Project area due to the nature of subsurface conditions in the vicinity. The site is underlain by alluvial and terrace deposits of the Mississippi Alluvial Plain, consisting primarily of sand, silt, clay, and gravel and no shallow competent bedrock is anticipated (NGMDB 1984). As a result, conventional excavation methods are anticipated to be sufficient for construction activities.

6.5.2 Mineral Resources

The Project area is located within a region of established oil and gas production, with multiple active and historic oil and gas fields identified in the vicinity, including the Rayville, Sacksonia, Big Creek, and Delhi West fields (SONRIS 2026b, LGS 2018 O&G). These resources are developed through subsurface wells and associated infrastructure and are not expected to influence near-surface construction activities. No significant mineral resource extraction operations (e.g., commercial sand and gravel pits, quarries, or surface mining operations) are present within the Project area.

6.5.3 Geologic Hazards

Given the low relief of the Project area and the limited potential for geologic hazards, significant impacts related to earthquakes, soil liquefaction, landslides, subsidence, or karst are not anticipated. Standard construction practices will be implemented to manage site conditions, including erosion and sediment control measures, routine inspection of work areas, and maintenance of existing drainage patterns during construction. In the event unexpected ground conditions are encountered, appropriate measures would be taken to stabilize affected areas and maintain Project integrity. Following construction, disturbed areas will be restored in accordance with applicable FERC Plans and Procedures.

Potential impacts related to construction in flood-prone areas are primarily associated with erosion. Other potential impacts may include a reduction in flood storage capacity resulting from displacement of volume within the flood zone by aboveground structures. Potential erosion impacts from Project construction will be minimized through implementation of the FERC Plan and FERC Procedures. Temporary erosion controls specified in the FERC Plan include sediment retention devices, sediment barriers, and mulching.

6.5.3.1 Karst

No known sinkholes are present within the vicinity of the Project. Historically, karst has not deterred the installation (and operation) of pipelines and other utilities, nor has karst deterred the construction of above ground structures including commercial and residential developments.

During Project planning, ETC Tiger assessed routing and design parameters of the Franklin Farms and Franklin Farms (FF) Trio Farms Lateral to avoid topographic and/or geologic features to the extent practicable. As part of the assessment, ETC Tiger reviewed topographic and geologic maps/data and performed site reconnaissance to evaluate sinkhole potential.

During construction, ETC Tiger will adhere to BMPs in the FERC Plan and FERC Procedures to control surface water flow and drainage within the construction work areas. ETC Tiger will maintain surface water flow(s) and drainage across the construction right-of-way (ROW) to prevent ponding and impoundment of surface water to the extent practicable during construction and restoration. Existing topography and drainage ways (e.g., drainage ditches or other surface water conveyances) will be maintained and restored unless otherwise specified by permit conditions, construction specifications, or landowner agreements. In addition, water discharges (e.g., trench dewatering, hydrostatic test-water discharge) will be directed to stable, well-vegetated upland areas in accordance with the FERC Plan and FERC Procedures. ETC Tiger will monitor dewatering activities for potential development of sinkholes or other problems. In the event sinkhole development is observed or suspected during construction, ETC Tiger will cease dewatering and evaluate the site in accordance with the measures described above.

6.5.4 Paleontological Resources

The potential for encountering significant paleontological resources within the Project area is low. If fossil material is encountered during construction, work will be halted in the immediate area and notifications

will be made to FERC and the landowner to determine appropriate next steps. With these measures, any resources encountered would be protected.

The potential for encountering significant paleontological resources (e.g., dinosaur) within the Project area is low. If one were found during construction, an *Unanticipated Discovery Plan* (**Appendix C-4**) has been developed that identifies the procedures for recognizing and handling significant fossils. This includes contacting the LSU Museum of Natural Science if significant fossils are discovered. With adherence to this plan, significant paleontological resources will be protected, if encountered.

6.6 OPERATION IMPACTS AND MITIGATION

Routine maintenance activities will be conducted in accordance with the FERC Plan, FERC Procedures, and ETC Tiger's standard operating procedures. If routine facility inspections identify areas of erosion or ineffective revegetation, they will be properly addressed in a timely manner. Based on the scope of the Project and ETC Tiger's proposed implementation of impact avoidance and minimization measures, including adherence to the FERC Plan and FERC Procedures, operation of the proposed facilities is not anticipated to significantly affect geology.

6.6.1 Geologic Hazards

Restoration to original contours, reestablishing vegetation and existing drainage patterns, and implementing measures in the FERC Plan and Procedures will minimize the potential for human-induced landslides and erosion in the Project area. If routine facility inspections identify any areas of erosion or ineffective revegetation, they will be properly addressed in a timely manner.

In general, modern welded steel pipelines have not sustained damage during seismic events except where permanent ground deformation or traveling ground-wave propagation associated with Modified Mercalli Intensity VIII or greater has occurred (O'Rourke & Palmer 1996). The primary seismic risk to pipelines and aboveground facilities would be fault displacement during an earthquake. However, there are no known active faults crossed by the Project. ETC Tiger will design and construct the pipeline and associated facilities to withstand seismic events and ground shaking should they occur. The pipeline will be constructed in accordance with 49 CFR Part 192 standards, which require that all practicable steps be taken to protect the transmission pipeline from unstable soil, landslides, or other hazards that may cause the pipeline to move or sustain abnormal loads.

ETC Tiger will design, construct, and operate aboveground facilities in accordance with applicable federal, state, and local requirements for development within a flood zone. All Project meter stations and the interconnect site will incorporate site-specific grading and drainage plans designed to minimize erosion and flooding impacts. Temporary workspaces within mapped floodplains will be restored to pre-construction conditions, therefore, long-term impacts are not anticipated. ETC Tiger does not anticipate adverse impacts to floodplain functions resulting from the Project.

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ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 7
Soils

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 7 FILING REQUIREMENTS	
Information	**Section Reference**
1. Identify, describe, and group the soils affected by the proposed pipeline and aboveground facilities. (§380.12 (I) (1))	Section 7.1; Table 7.1-1; Table 7.1-2
2. For aboveground facilities that would occupy sites over 5 acres, determine the acreage of prime farmland soils that would be affected by construction and operation. (§380.12 (i) (2))	Not Applicable
3. Describe potential impacts on soils. (§380.12 (i) (3,4)	Table 7.1-1; Table 7.1-2, Section 7.2; Section 7.3
4. Identify proposed mitigation to minimize impact on soils, and compare with the staff's Upland Erosion Control, Revegetation, and Maintenance Plan. (§380.12 (i) (5))	Section 7.3, Section 7.4
Information Often Missing and Resulting in Data Requests	
If the applicant generally proposes to adopt the Federal Energy Regulatory Commission staff's *Upland Erosion Control, Revegetation, and Maintenance Plan* except at certain locations, identify on a site-specific basis locations where alternative measures are proposed, and describe the alternative measures that will ensure an equal or greater level of protection.	Not Applicable, No Alternative Measures Proposed
Identify invasive species and/or noxious weeds that occur in the area and measure to prevent the introduction and/or spread of these species (if not addressed in Resource Report 3).	Resource Report No. 3 - Fish, Vegetation and Wildlife (see Sections 3.2.1.2 and 3.2.2)
Provide documentation of consultation with the U.S. Department of Agriculture's Natural Resources Conservation Service or other applicable agencies regarding seed mixes, erosion control, and invasive species/noxious weeds.	Section 7.2.4.1

RESOURCE REPORT NO. 7
SOILS

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Appendix A-14 – Peak Horizontal Ground Acceleration with 10 Percent Probability of Exceedance in 50 Years
Appendix A-15 – Major Route Alternative Map
Appendix A-16 – Route Variation Map

B Typical and Site-Specific Detail Drawings

Appendix B-1 – Project Typicals for Construction

Appendix B-2 – Plot Plans (Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

C Project Construction and Mitigation Plans

Appendix C-1 – FERC *Upland Erosion Control, Revegetation, and Maintenance Plan* (2013)
Appendix C-2 – FERC *Wetland and Waterbody Construction and Mitigation Procedures* (2013)
Appendix C-3 – *Spill Prevention and Response Plan*
Appendix C-4 – *Unanticipated Discovery Plan*
Appendix C-5 – *Fugitive Dust Control Plan*
Appendix C-6 – *Noxious and Invasive Plant Management Plan*
Appendix C-7 – *Environmental Complaint Resolution Procedures*

D Agency and Stakeholder Correspondence

Appendix D-1 – Agency Correspondence (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

Appendix D-2 – SHPO Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

Appendix D-3 – Native American Organizations and Tribes Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

E Landowner Information

Appendix E-1 – Landowner/Stakeholder List (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE"*)

F Wetland and Waterbody Delineation Report and Listed Species Report

Appendix F-1 – Wetland and Waterbody Delineation Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

Appendix F-2 – Listed Species Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

G Cultural Resource Assessment Survey Report (*Submitted as* "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

H Air Quality Calculations (Construction Emissions)

I Sound Study Report

ABBREVIATIONS AND ACRONYMS

Certificate	Certificate of Public Convenience and Necessity
CFR	Code of Federal Regulations
ETC Tiger	ETC Tiger Pipeline, LLC
FERC or Commission	Federal Energy Regulatory Commission
FERC Plan	Upland Erosion, Control, Revegetation, and Maintenance Plan (2013)
FERC Procedures	Wetland and Waterbody Construction and Mitigation Procedures (2013)
FF	Franklin Farms
Laterals	Franklin Farms Lateral and FF Trio Farm Lateral
MP	Milepost
NGA	Natural Gas Act
NRCS	Natural Resource Conservation Service
PAR	permanent access road
Project	Franklin Farms Project
ROW	right-of-way
SPAR Plan	Spill Prevention and Response Plan
TAR	temporary access road
USACE	United States Army Corps of Engineers
USDA	United States Department of Agriculture
WEG	Wind Erodibility Group

7.0 SOILS

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, an interconnect site, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project in June 2027, and following the receipt of all applicable approvals, anticipate beginning construction activities to begin in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

Resource Report No. 7 addresses the soil resources present within the Project footprint. The distribution and characteristics of soils were inventoried using information available from the United States Department of Agriculture's (USDA) Natural Resources Conservation Service (NRCS). This report also assesses potential impacts from the Project on soil resources and identifies proposed mitigation measures.

To avoid and minimize impacts to soil resources during construction and operation of the Project, ETC Tiger will implement the FERC *Upland Erosion Control, Revegetation and Maintenance Plan*, May 2013 (FERC Plan, FERC 2013a; **Appendix C-1**) and the FERC *Wetland and Waterbody Construction and Mitigation Procedures*, May 2013 (FERC Procedures, FERC 2013b; **Appendix C-2**), with proposed site-specific exceptions, ETC Tiger's *Spill Prevention and Response Plan* (SPAR Plan) (**Appendix C-3**).

7.1 SOIL CONDITIONS AT PROPOSED FACILITIES

Soils within the Project area were identified and characterized using the NRCS Soil Survey for Richland Parish, Louisiana, as well as data from the NRCS Web Soil Survey. The NRCS has assigned soils to map units (taxonomic classes). These map units were developed to separate the landscape into landforms or landform segments that have similar use and management requirements. Each map unit is defined by a unique combination of soil components and/or miscellaneous areas in predictable proportions and is named according to the taxonomic classification of the dominant soils present (USDA NRCS, 2025). Soils maps were generated for the Project area using the NRCS Soil Survey Geographic Database and they are provided in **Appendix A-5.** A description of the different soil map units crossed by the Project as well as information on the location (milepost [MP]) of each unit crossed, associated soil limitations, and the presence of hydric and prime or unique farmland soils is provided in **Table 7.1-1**. A summary of the soil limitations associated with soils crossed by the Project is provided in **Table 7.1-2**.

TABLE 7.1-1
Soil Descriptions and Physical Characteristics of the ETC Tiger Franklin Farms Project

Map Unit Name	Map Unit	Hydric Soils [a]	Drainage Characteristics [a]	Frequency of Flooding/ Ponding	Surface Texture	Prime Farmland	Slope (percent)	Shallow Depth to Bedrock	Revegetation Potential [b]	High Compaction Potential	Erodibility Factor		Acres of Soil Map Unit (temp)	Acres of Soil Map Unit (perm)
											Wind [c]	Water [d]		
Franklin Farms Lateral														
Deerford silt loam	Da	No	Somewhat Poorly Drained	None/None	Silt loam	Not Prime Farmland	1.0	No	High	Medium	5	0.49	0.7	0.7
Dexter silt loam, 1 to 3 percent slopes	De	No	Well Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.37	1.3	1.0
Dexter silt loam, 3 to 5 percent slopes	Df	No	Well Drained	None/None	Silt loam	All Areas Prime Farmland	4.0	No	High	Medium	5	0.37	<0.1	0.0
Egypt silt loam, 1 to 3 percent slopes	Eg	No	Somewhat Poorly Drained	Rare/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.49	0.4	0.3
Forestdale silty clay loam, 0 to 1 percent slopes, rarely flooded, upland phase	Fr	Yes	Poorly Drained	None/None	Silty Clay Loam	All Areas Prime Farmland	0.5	No	High	Medium	4	0.32	26.8	23.8
Gigger silt loam, 1 to 3 percent slopes	Ge	No	Moderately Well Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.43	1.6	1.0
Gigger-Gilbert silt loams, gently undulating	Gg	No	Moderately Well Drained	Rare/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.43	15.3	10.6
Gilbert silt loam	Gk	Yes	Poorly Drained	Rare/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	16.4	8.7
Gilbert-Egypt silt loams, gently undulating	Gm	Yes	Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	25.9	19.2
Necessity silt loam, 1 to 3 percent slopes	Ne	No	Somewhat Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.55	3.6	2.7
Necessity-Gilbert silt loams, gently undulating	Ng	No	Somewhat Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.55	11.3	8.2
Perry clay	Pd	Yes	Poorly Drained	Rare/None	Clay	All Areas Prime Farmland	0.5	No	High	Medium	4	0.24	1.6	1.0
Rilla silt loam, 0 to 1 percent slopes	Ra	No	Well Drained	None/None	Silt Loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.49	1.9	1.2
												Subtotal	**106.8**	**78.4**
Franklin Farms (FF) Trio Farm Lateral														
Dexter silt loam, 1 to 3 percent slopes	De	No	Well Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.37	0.2	0.2
Forestdale silty clay loam, 0 to 1 percent slopes, rarely flooded, upland phase	Fr	Yes	Poorly Drained	None/None	Silty Clay Loam	All Areas Prime Farmland	0.5	No	High	Medium	4	0.32	2.2	2.8
Gilbert-Egypt silt loams, gently undulating	Gm	Yes	Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	1.8	2.1
Necessity-Gilbert silt loams, gently undulating	Ng	No	Somewhat Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.55	2.6	3.3
Perry silty clay loam	Pc	No	Poorly Drained	None/None	Silty Clay Loam	All Areas Prime Farmland	0.5	No	High	Low	4	0.32	1.1	2.0

TABLE 7.1-1
Soil Descriptions and Physical Characteristics of the ETC Tiger Franklin Farms Project

Map Unit Name	Map Unit	Hydric Soils [a]	Drainage Characteristics [a]	Frequency of Flooding/ Ponding	Surface Texture	Prime Farmland	Slope (percent)	Shallow Depth to Bedrock	Revegetation Potential [b]	High Compaction Potential	Erodibility Factor		Acres of Soil Map Unit (temp)	Acres of Soil Map Unit (perm)
											Wind [c]	Water [d]		
												Subtotal	**7.9**	**10.4**
FF Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station														
Gilbert silt loam	Gk	Yes	Poorly Drained	Rare/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	0.2	0.8
Gilbert-Egypt silt loams, gently undulating	Gm	Yes	Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	2.1	2.9
												Subtotal	**2.3**	**3.7**
FF Bee Bayou Delivery Meter Station														
Gilbert silt loam	Gk	Yes	Poorly Drained	Rare/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	0.3	0.9
Gilbert-Egypt silt loams, gently undulating	Gm	Yes	Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	0.2	0.2
												Subtotal	**0.5**	**1.1**
FF Interconnect Site														
Forestdale silty clay loam, 0 to 1 percent slopes, rarely flooded, upland phase	Fr	Yes	Poorly Drained	None/None	Silty Clay Loam	All Areas Prime Farmland	0.5	No	High	Medium	4	0.32	0.1	0.4
Gilbert-Egypt silt loams, gently undulating	Gm	Yes	Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	<0.1	0.5
												Subtotal	**0.1**	**0.9**
FF Gulf Crossing Receipt Meter Station														
Forestdale silty clay loam, 0 to 1 percent slopes, rarely flooded, upland phase	Fr	Yes	Poorly Drained	None/None	Silty Clay Loam	All Areas Prime Farmland	0.5	No	High	Medium	4	0.32	0.0	0.5
Necessity-Gilbert silt loams, gently undulating	Ng	No	Somewhat Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.55	0.3	1.0
												Subtotal	**0.3**	**1.5**
MEP Tap Site														
Forestdale silty clay loam, 0 to 1 percent slopes, rarely flooded, upland phase	Fr	Yes	Poorly Drained	None/None	Silty Clay Loam	All Areas Prime Farmland	0.5	No	High	Medium	4	0.32	0.00	0.09
												Subtotal	**0.00**	**0.09**
FF MEP Receipt Meter Station														
Forestdale silty clay loam, 0 to 1 percent slopes, rarely flooded, upland phase	Fr	Yes	Poorly Drained	None/None	Silty Clay Loam	All Areas Prime Farmland	0.5	No	High	Medium	4	0.32	0.2	1.2
												Subtotal	**0.2**	**1.2**
Franklin Farms Delivery Meter Station														
Dexter silt loam, 1 to 3 percent slopes	De	No	Well Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.37	0.1	0.5

TABLE 7.1-1
Soil Descriptions and Physical Characteristics of the ETC Tiger Franklin Farms Project

Map Unit Name	Map Unit	Hydric Soils [a]	Drainage Characteristics [a]	Frequency of Flooding/ Ponding	Surface Texture	Prime Farmland	Slope (percent)	Shallow Depth to Bedrock	Revegetation Potential [b]	High Compaction Potential	Erodibility Factor		Acres of Soil Map Unit (temp)	Acres of Soil Map Unit (perm)
											Wind [c]	Water [d]		
Gilbert-Egypt silt loams, gently undulating	Gm	Yes	Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	0.5	0.5
												Subtotal	**0.6**	**1.0**
FF Trio Farm Delivery Meter Station														
Necessity-Gilbert silt loams, gently undulating	Ng	No	Somewhat Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.55	0.6	1.4
												Subtotal	**0.6**	**1.4**
Contractor Yards														
Calhoun-Calloway silt loams, gently undulating	Cc	Yes	Poorly Drained	Rare/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.49	3.4	0.0
Gigger-Gilbert silt loams, gently undulating	Gg	No	Moderately Well Drained	Rare/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.43	8.5	0.0
Grenada silt loam, 1 to 3 percent slopes	Gr	No	Moderately Well Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	High	5	0.49	2.9	0.0
												Subtotal	**14.8**	**0.0**
Access Roads														
Calhoun-Calloway silt loams, gently undulating	Cc	Yes	Poorly Drained	Rare/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.49	0.1	0.0
Dexter silt loam, 1 to 3 percent slopes	De	No	Well Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.37	0.9	0.5
Dexter silt loam, 3 to 5 percent slopes	Df	No	Well Drained	None/None	Silt loam	All Areas Prime Farmland	4.0	No	High	Medium	5	0.37	0.9	0.0
Forestdale silty clay loam, 0 to 1 percent slopes, rarely flooded, upland phase	Fr	Yes	Poorly Drained	None/None	Silty Clay Loam	All Areas Prime Farmland	0.5	No	High	Medium	4	0.32	0.9	1.9
Gigger-Gilbert silt loams, gently undulating	Gg	No	Moderately Well Drained	Rare/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.43	0.1	1.5
Gilbert silt loam	Gk	Yes	Poorly Drained	Rare/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	1.1	0.1
Gilbert-Egypt silt loams, gently undulating	Gm	Yes	Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.55	3.6	1.6
Hebert silt loam, 0 to 1 percent slopes	Hb	No	Somewhat Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.49	0.0	0.8
Necessity silt loam, 1 to 3 percent slopes	Ne	No	Somewhat Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.55	0.1	0.0
Necessity-Gilbert silt loams, gently undulating	Ng	No	Somewhat Poorly Drained	None/None	Silt loam	All Areas Prime Farmland	2.0	No	High	Medium	5	0.55	<0.0	1.0
Rilla silt loam, 0 to 1 percent slopes	Ra	No	Well Drained	None/None	Silt Loam	All Areas Prime Farmland	0.5	No	High	Medium	5	0.49	<0.0	0.0

TABLE 7.1-1
Soil Descriptions and Physical Characteristics of the ETC Tiger Franklin Farms Project

Map Unit Name	Map Unit	Hydric Soils [a]	Drainage Characteristics [a]	Frequency of Flooding/ Ponding	Surface Texture	Prime Farmland	Slope (percent)	Shallow Depth to Bedrock	Revegetation Potential [b]	High Compaction Potential	Erodibility Factor		Acres of Soil Map Unit (temp)	Acres of Soil Map Unit (perm)
											Wind [c]	Water [d]		
												Subtotal	**7.7**	**7.4**
												Project Total	**141.8**	**107.1**

Notes:

[a] Source: USDA NRCS 2025, 2026a, 2026b

[b] Soils with high revegetation potential are those classified as prime farmland, little to no slope, and low to medium compaction potential

[c] WEG - Based on wind erodibility group classification: High (1-2), Moderate (3-4), Low (≥ 5)

[d] The K Factor is a measure of the susceptibility of soils to water erosion. K Factor values range from 0.02 to 0.69 with soils of 0.69 having the highest susceptibility to water erosion. Soils with a K factor value of 0.02 to 0.24 are considered to have “Low” susceptibility to water erosion; K Factor values of 0.25 to 0.44 are considered to have “Moderate” susceptibility to water erosion; K Factors of 0.45 to 0.69 a “High” susceptibility to water erosion

TABLE 7.1-2
Soil Characteristics Affected by the ETC Tiger Franklin Farms Project (Acres)[a, b]

Facility	Prime Farmland [c]	Hydric Soils [c]	High Compaction Potential [d]	Highly Erodible		Poor Revegetation Potential [g]	Shallow Bedrock
				Water [e]	Wind [f]		
Pipeline							
Franklin Farms Lateral	184.0	123.5	0.0	103.7	0.0	0.0	0.0
FF Trio Farm Lateral	18.3	8.9	0.0	10.2	0.0	0.0	0.0
Aboveground Facilities							
FF Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station	6.0	6.0	0.0	6.0	0.0	0.0	0.0
FF Bee Bayou Delivery Meter Station	1.5	1.5	0.0	1.5	0.0	0.0	0.0
FF Interconnect Site	1.0	1.0	0.0	0.5	0.0	0.0	0.0
FF Gulf Crossing Receipt Meter Station	1.7	0.5	0.0	1.2	0.0	0.0	0.0
MEP Tap Site	0.1	0.1	0.0	0.0	0.0	0.0	0.0
FF MEP Receipt Meter Station	1.4	1.4	0.0	0.0	0.0	0.0	0.0
Franklin Farms Delivery Meter Station	1.6	1.0	0.0	1.6	0.0	0.0	0.0
FF Trio Farm Delivery Meter Station	2.0	0.0	0.0	2.0	0.0	0.0	0.0
Contractor Yards	14.8	3.4	2.9	6.3	0.0	0.0	0.0
Access Roads	14.9	9.2	0.0	10.6	0.0	0.0	0.0
Total Acreage	**247.3**	**156.5**	**2.9**	**143.6**	**0.00**	**0.00**	**0.00**

Notes:
[a] The area affected includes the permanent pipeline right-of- way, temporary pipeline right- of-way, and additional temporary workspace. The soils data in the table does not include areas of open water
[b] The numbers in this table have been rounded for presentation purposes
[c] As designated by the NRCS. Prime farmland does not include those soils that are considered prime if artificial drainage is implemented due to the lack of drain tile use in the Project area
[d] Includes soils in somewhat poor to very poor drainage classes with surface textures of sandy clay loam and finer
[e] See footnote "d" in Table 7.1-1 above
[f] WEG classifications 1 and 2
[g] See footnote "b" in Table 7.1-1 above
[h] Soils identified as containing bedrock at a depth of 5 feet ("ft") or less from the surface, all of which is paralithic and rippable with standard construction equipment

7.2 CONSTRUCTION IMPACTS AND MITIGATION

Project construction will result in minor, short-term and long-term impacts on soil resources. Specifically, installation of the new aboveground facilities and permanent access roads (PAR) will result in the permanent loss of soil use within the newly graveled areas. Construction of the Franklin Farms Lateral and FF Trio Farm Lateral (Laterals), use of the contractor yards, and use of temporary access roads (TAR) will only temporarily affect soils, pre-construction uses for the soils will be able to resume following construction.

During Project construction, clearing and grading will be required, and potential impacts to soils may include erosion by wind and water, compaction, rutting, loss of productivity associated with mixing of topsoil, and poor revegetation post-construction. Further, installing the Laterals (i.e., trenching) could affect soil structure and other characteristics including the soil's ability to retain moisture. Potential impacts will be minimized with adherence to erosion control and other protective measures in the FERC Plan and FERC Procedures.

Upon completion of the proposed construction activities, temporarily disturbed areas will be restored to pre-construction conditions through grading and seeding per the FERC Plan, FERC Procedures, and NRCS recommendations (see **Section 7.2.4.1**), or by landowner specifications. Revegetation success will be monitored, and if necessary, corrective actions will be completed.

Based on the scope of the Project, characteristics of the soils underlying the facilities, and ETC Tiger's implementation of impact minimization measures, including adherence to the FERC Plan and FERC Procedures, installation of the proposed facilities is not anticipated to significantly affect soils.

7.2.1 Shallow Bedrock

Typical profiles of the soils crossed by the Project indicate that they do not have a shallow depth to bedrock (see **Table 7.1-1**). If bedrock is encountered during trenching, ETC Tiger will first attempt to remove it using mechanical means. Blasting techniques are not anticipated to be utilized during Project construction.

7.2.2 High Compaction Potential

The majority of the soils crossed by the Project consist of silt loam or finer. Very few of the soil types crossed have a high compaction potential (see **Tables 7.1-1** and **7.1-2**). In areas of high compaction potential, ETC Tiger will minimize compaction by implementing measures in the FERC Plan, as applicable.

7.2.3 High Erosion Potential

Construction of the Project will remove existing ground cover and temporarily disturb soils. Water erosion is strongly related to the permeability of the soil type and to the cohesion of the soil particles that comprise it. Water erosion is also closely associated with other soil properties, such as texture, percent organic matter, structure, and infiltration capacity. Soils containing high portions of silt and very fine sand are the most susceptible to erosion. Well-drained and well-graded gravels and gravel-sand mixtures with little or no silt are the least erodible soils. Other soil properties that influence erosion can include slope length and gradient; the frequency, intensity, and duration of rainfall; and the amount of time bare soils are exposed. The soil types crossed by the Project have a moderate or high erosion potential from water, none were found to have a low erosion potential from water (see **Tables 7.1-1** and **7.1-2**). The erosion potential from water can largely be attributed to the sloping terrain encountered within the Project area.

Of the soil types crossed by the Project, none have the potential for severe wind erosion. The soil properties that are most important with respect to soil blowing include soil moisture, soil texture, organic matter

content, the presence of carbonates in the fine-earth fraction, rock and pararock fragment content, and mineralogy. A wind erodibility group (WEG) is a grouping of soils that have similar properties affecting their resistance to soil blowing in cultivated areas (USDA NRCS, 2005). The Project area soils all have a WEG classification of four or higher (see **Table 7.1-1**), which indicates they are stable against breakdown by tillage and abrasion from wind. ETC Tiger will implement measures provided in its *Fugitive Dust Control Plan* (**Appendix C-5**) to ensure that dust-control measures are utilized, including routine wetting of the construction workspace as necessary where soils are exposed. ETC Tiger will also minimize construction impacts by adherence to the FERC Plan.

7.2.4 Revegetation Potential

Construction will result in the removal of existing vegetation. Revegetation within soil types that have a low revegetation potential may be difficult. Soil properties that affect the growth of grasses and legumes include slope steepness, compaction potential, depth to bedrock or other restrictive layers, and fertility. Soils with high revegetation potential are those that are prime farmland, little to no slope, and low to medium compaction potential. Soils with medium to low revegetation potential include those with steep slopes, shallow restrictive layers, or areas otherwise not suited to growth of agricultural crops No soils within the Project area have a low revegetation potential (see **Table 7.1-1**).

Following construction within temporarily disturbed areas, ETC Tiger will prepare the seedbed and utilize seed mix and soil supplements as specified by the FERC Plan and NRCS recommendations, or by landowner specifications. Monitoring and necessary maintenance activities will be conducted per the FERC Plan.

7.2.4.1 Revegetation Recommendations

ETC Tiger received recommendations from the NRCS on seed mixes and seeding rates on June 9, 2026 (see correspondence in **Appendix D-1**). The NRCS provided reseeding and planting recommendations based on the pre-existing land use type. Within croplands, an annual grass mix including wheat (*Triticum* sp.), cereal rye (*Secale cereale*), and oat (*Avena sativa*) species was recommended and, within improved pastures, the reestablishment of existing pasture vegetation, including common Bermuda grass (*Cynodon dactylon*) and bahia grass (*Paspalum notatum*), was recommended. A cool season grass mix designed to benefit pollinators which includes flowering species such as butterfly milkweed (*Asclepias tuberosa*), purple coneflower (*Echinacea purpurea*), and rough blazingstar (*Liatris aspera*) was recommended for forested areas. ETC Tiger plans to follow this guidance in non-agricultural areas or other areas not exclusively covered by landowner agreements.

7.2.5 Hydric Soils

Hydric soils are defined as soils that formed under conditions of saturation, flooding, or ponding of sufficient duration during the growing season to develop anaerobic conditions in the upper soil surface. Under natural conditions, these soils are either saturated or inundated long enough during the growing season to support the growth and regeneration of hydrophytic vegetation (USDA NRCS, 2026c). A combination of hydric soils, hydrophytic vegetation, and hydrology properties are used to define wetlands as described by *Corps of Engineers Wetlands Delineation Manual* (United States Army Corps of Engineers [USACE], 1987).

As noted in **Tables 7.1-1** and **7.1-2**, several areas of the Project have the potential for hydric soils. Hydric soils can be more susceptible to compaction and rutting than non-hydric soils. ETC Tiger will minimize rutting and compaction of hydric soils by implementing measures in the FERC Procedures. A discussion concerning the presence of wetland habitat within the Project area and wetland protection measures is provided in Resource Report No. 2 (*Water Use and Quality*).

7.2.6 Prime or Unique Farmland Soils

As noted in **Tables 7.1-1** and **7.1-2**, almost the entirety of the Project workspace is located on prime farmland soils as classified by the NRCS. Prime farmland has the best combination of physical and chemical characteristics for producing food, feed, forage, fiber, and oilseed crops (United States Code of Federal Regulations [CFR], 2026). Pre-construction uses for the soils can resume following construction in areas with temporary impacts. Further, following construction, there will be no restrictions that prevent agricultural land from reverting to its prior use within the permanent easement for the Laterals. Tiger

7.2.7 Residential Soils

The Laterals, aboveground facilities, and access roads are located within rural areas. The Project is not located directly abutting any high-density residential tracts; therefore, impacts to residential yards will not occur.

7.2.8 Spills

Accidental spills of construction-related fluids (e.g., oil, gasoline, or hydraulic fluids) could impact soils. Impacts on soils would depend on the type and quantity of the spill. To reduce the potential for soil contamination, ETC Tiger will implement its SPAR Plan, which includes best management practices to avoid and minimize the potential for accidental releases and contains measures that will be implemented to clean up any releases.

7.3 OPERATION IMPACTS AND MITIGATION

Pre-construction uses for soils can resume during operation of the Lateral. However, the permanent right-of-way (i.e., operational ROW) will be maintained in a non-woody state and mowed periodically for safety and maintenance reasons. Installation of the new aboveground facilities and PARs will result in a minor, permanent loss of soil use within the newly graveled areas.

Based on the scope of the Project, the characteristics of the soils underlying the facilities, and ETC Tiger's adherence to the FERC Plan, FERC Procedures, and ETC Tiger's SPAR Plan, operation of the proposed facilities is not anticipated to significantly affect soils.

7.4 REQUESTED DEVIATIONS FROM THE FERC PLAN

ETC Tiger is adhering to the FERC Plan; however, there are places where the topography, ROW, and natural conditions make it impractical to implement some of the measures specified in the FERC Procedures. See **Table 1.3-2** in Resource Report No. 1 (*General Project Description*) for further details regarding site-specific deviations to the FERC Procedures.

7.5 REFERENCES

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ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 8
Land Use, Recreation, and Aesthetics

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 8 FILING REQUIREMENTS	
Information	**Section Reference**
1. Classify and quantify land use affected by: (§ 380.12(j) (1))	Table 8.1-1
a) Pipeline construction and permanent rights-of-way;	Section 8.1.1 Table 8.1-1
b) Extra work/staging areas;	Section 8.1.1.4 Table 8.1-2
c) Access roads;	Section 8.1.1.5 Section 8.1.2.10 Table 8.1-1
d) Pipe and contractor yards; and	Section 8.1.2.9 Table 8.1-1
e) Aboveground facilities.	Section 8.1.2 Table 8.1-1
2. Identify by milepost (MP) all locations where the pipeline right-of-way would at least partially coincide with existing right-of-way, where it would be adjacent to existing rights-of-way, and where it would be outside of existing right-of-way. (§ 380.12(j) (1))	Appendix A-7
3. Provide detailed typical construction right-of-way cross-section diagrams showing information such as widths and relative locations of existing rights-of-way, new permanent right of way and temporary construction right-of-way. (§ 380.12(j) (1))	Appendix B
4. Summarize the total acreage of land affected by construction and operation of the project. (§ 380.12(j) (1))	Table 8.1-1
5. Identify by milepost all planned residential or commercial/business development and the time frame for construction. (§ 380.12(j) (3))	Section 8.2.6
6. Identify by milepost special land uses (e.g., maple sugar stands, specialty crops, natural areas, national and state forests, conservation land, etc. (§ 380.12(j) (4))	Section 8.3
7. Identify by beginning milepost and length of crossing all land administered by Federal, state, or local agencies, or private conservation organizations. (§ 380.12(j) (4))	Section 8.3.1 Section 8.3.2
8. Identify by milepost all natural, recreational, or scenic areas and all registered natural landmarks crossed by the project. (§ 380.12(j) (4&6))	Section 8.3.2
9. Identify all facilities that would be within designated coastal zone management areas. Provide a consistency determination or evidence that a request for consistency determination has been filed with the appropriate state agency. (§ 380.12(j) (4&7))	Section 8.3.3
10. Identify by milepost all residence that would be within 50 feet of the construction right-of-way or extra work area. (§ 380.12(j) (5))	Table 8.2-1
11. Identify all National or State Wild and Scenic Rivers crossed by the project. (§ 380.12(j) (6))	Section 8.3.2

RESOURCE REPORT NO. 8 FILING REQUIREMENTS	
Information	**Section Reference**
12. Describe any measures to visually screen aboveground facilities, such as compressor stations. (§ 380.12(j) (11))	Section 8.3
13. Demonstrate that applications for rights-of-way or other proposed land use have been or soon will be filed with Federal land-managing agencies with jurisdiction over land that would be affected by the project. (§ 380.12(j) (12))	Section 8.5.3.1
Additional Information Often Missing and Resulting in Data Requests	
• Identify all buildings with 50 feet of the construction right-of-way and extra workspace.	Table 8.2-1
• Describe the management and use of all public lands that would be crossed.	Section 8.3.2.1
• Provide a list of landowners by milepost or tract number that corresponds to information on alignment sheets.	(Submitted as *CUI/Privileged Information*)
• Provide a site-specific construction plan for residences within 50 feet of construction.	Appendix B-6

RESOURCE REPORT NO. 8
LAND USE, RECREATION, AND AESTHETICS

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F Wetland and Waterbody Delineation Report and Listed Species Report

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G Cultural Resource Assessment Survey Report (*Submitted as* "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

H Air Quality Calculations (Construction Emissions)

I Sound Study Report

ABBREVIATIONS AND ACRONYMS

ACEP	Agricultural Conservation Easement Program
ATWS	additional temporary workspace
Certificate	Certificate of Public Convenience and Necessity
ECD	environmental control devices
EPA	United States Environmental Protection Agency
ETC Tiger	ETC Tiger Pipeline, LLC
FERC or Commission	Federal Energy Regulatory Commission
FERC Plan	Upland Erosion, Control, Revegetation, and Maintenance Plan (2013)
FERC Procedures	Wetland and Waterbody Construction and Mitigation Procedures (2013)
FF	Franklin Farms
FHWA	Federal Highway Administration
HUC	hydrologic unit code
ID	identification
Laterals	Franklin Farms Lateral and FF Trio Farm Lateral
LDWF	Louisiana Department of Wildlife and Fisheries
MP	milepost
MRLC	Multi-Resolution Land Characteristics
NGA	Natural Gas Act
NLCD	National Land Cover Database
NPS	National Park Service
NRCS	Natural Resources Conservation Service
NWSRS	National Wild and Scenic Rivers System
PAR	Permanent Access Road
PEM	Palustrine emergent wetlands
PFO	Palustrine forested wetlands
Project	Franklin Farms Project
RCRA	Resource Conservation and Recovery Act
ROW	right-of-way
SPAR Plan	Spill Prevention and Response Plan
TAR	Temporary Access Road
U.S.	United States
USACE	United States Army Corps of Engineers
USDA	United States Department of Agriculture
USDOT	United States Department of Transportation
USFWS	Unites States Fish and Wildlife Service
USGS	United States Geological Survey
UST	underground storage tank
WRE	Wetlands Reserve Easement
WRP	Wetland Reserve Program

8.0 LAND USE, RECREATION, AND AESTHETICS

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, an interconnect site, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project in June 2027, and following the receipt of all applicable approvals, anticipating construction activities to begin in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

Resource Report No. 8 assesses existing land use within the Project area. This report also identifies land ownership (public and private lands), the presence of any sensitive or special-use areas, and the presence of visual resources. Further, the report provides a description of potential impacts to land use, recreation, and aesthetics from Project construction and operations and proposed mitigation measures.

To avoid and minimize impacts to land use, recreation, and aesthetics during construction and operation of the Project, ETC Tiger will implement the FERC *Upland Erosion Control, Revegetation and Maintenance Plan*, May 2013 (FERC Plan, FERC 2013a; **Appendix C-1**), the FERC *Wetland and Waterbody Construction and Mitigation Procedures*, May 2013 (FERC Procedures, FERC 2013b; **Appendix C-2**), with proposed site-specific exceptions, and ETC Tiger's *Spill Prevention and Response Plan* (SPAR Plan) (**Appendix C-3**).

8.1 LAND USE

Land use categories impacted by the Project were classified and defined categories based on the United States (U.S.) Geological Survey's (USGS) 2024 National Land Cover Database (NLCD) Multi-Resolution Land Characteristics (MRLC) Consortium and field observations completed over the majority of the pipeline routes and locations of aboveground Project facilities, contractor yards, and access roads. The predominant land use categories affected by the Project are defined as follows:

- Agricultural Land – Cultivated or rotated cropland, orchards, vineyards, or hay fields.
- Forest/Woodland – Wooded lands not being used for specific commercial purposes, consisting of deciduous and coniferous types, including but not limited to state forest lands.
- Silviculture – wooded lands being managed for forest products (e.g., pine plantations, sugar maple stands, or tree nurseries).
- Open Land – Non-forested lands used for open space or pasture.
- Open Water – Open water consists of waterbodies that typically have less than 25 percent cover of vegetation and land, including streams and ponds (see Resource Report No. 2 [*Water Use and Quality]*).
- Wetlands – Wetlands in the Project area include palustrine emergent (PEM) and palustrine forested (PFO) wetlands. PEM wetlands are typically dominated by perennial emergent and rooted herbaceous hydrophytes, excluding mosses and lichens that are present for most of the growing season in most years. PFO wetlands are dominated by woody vegetation that is at least 19.7 feet tall (Cowardin et al., 1979). Resource Report No. 2 includes a discussion of wetlands crossed in the Project area (see **Appendix A-4** for mapping).
- Residential Land – Residential yards, residential subdivisions, and planned.

- Industrial/Commercial Land – Electric power or gas utility stations, manufacturing or industrial plants, landfills, mines, quarries, commercial or retail facilities, and roads.

Mapping of habitats in the Project area is provided in **Appendix A-6**. Additional details of resources in the Project area are provided for wetlands and waterbodies in Resource Report No. 2; vegetation in Resource Report No. 3 (*Fish, Wildlife, and Vegetation)*; and soils in Resource Report No. 7 (*Soils)*.

TABLE 8.1-1
Land Use Affected by Construction and Operation of the Project (acres)

Project Facility	Forest/ Woodland		Silviculture (Pine Plantation)		Agricultural		Open Land		Open Water		Wetlands		Residential		Industrial/ Commercial		Total [1]	
	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation	Construction[2]	Operation
Pipeline Facilities [3]																		
Franklin Farms Lateral	15.6	6.8	2.3	0.9	114.0	43.8	27.8	11.9	1.2	0.7	20.3	12.7	0.0	0.0	4.2	1.8	**185.4**	**78.5**
Franklin Farms (FF) Trio Farm Lateral	0.4	0.4	0.0	0.0	7.1	3.8	9.3	5.1	0.2	0.1	1.3	1.0	0.0	0.0	0.1	0.0	**18.3**	**10.5**
Aboveground Facilities																		
Meter Stations																		
FF Gulf Run Bi Di Meter Station & FF Gulf South Receipt Meter Station [4]	0.0	0.0	0.0	0.0	6.0	3.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**6.0**	**3.6**
FF Gulf Crossing Receipt Meter Station	0.0	0.0	0.0	0.0	1.5	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**1.5**	**1.0**
FF Bee Bayou Delivery Meter Station [5]	0.0	0.0	0.0	0.0	1.0	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**1.0**	**0.9**
FF Interconnect Site	0.0	0.0	0.0	0.0	1.7	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**1.7**	**1.4**
MEP Tap Site	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.1**	**0.1**
FF MEP Receipt Meter Station	0.0	0.0	0.0	0.0	1.4	1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**1.4**	**1.2**
Franklin Farms Delivery Meter Station [5]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.6	1.0	**1.6**	**1.0**
FF Trio Farm Delivery Meter Station [5]	0.0	0.0	0.0	0.0	2.0	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**2.0**	**1.4**
Other Work Areas																		
Contractor Yards	0.0	0.0	0.0	0.0	0.0	0.0	4.7	0.0	0.0	0.0	0.4	0.0	0.0	0.0	9.7	0.0	**14.8**	**0.0**
Access Roads	0.1	0.0	0.0	0.0	6.4	2.2	1.5	0.5	0.2	0.1	0.0	0.0	0.0	0.0	6.7	4.5	**14.9**	**7.3**
Project Total [1]	**16.1**	**7.2**	**2.3**	**0.9**	**141.3**	**59.4**	**43.3**	**17.5**	**1.6**	**0.9**	**22.0**	**13.6**	**0.0**	**0.0**	**22.2**	**7.4**	**248.9**	**107.0**

Notes:
[1] Numbers may vary slightly due to rounding.
[2] Construction area includes the total acres of workspace required for construction of the Project, including the operation area.
[3] Operational pipeline impacts presented in this table are based on the permanent easement; however, there will be no restrictions that prevent agricultural, open land, PEM wetlands, and open water from reverting to their prior use.
[4] The FF Gulf Run Bi Di Meter Station and the FF Gulf South Receipt Meter Station share temporary and permanent workspace.
[5] As discussed in Section 1.1.4.1 of Resource Report 1, these meter stations will be previously disturbed by Entergy Louisiana, LLC prior to the construction of the Project.

8.1.1 Pipeline Facilities

The following sections detail the land use requirements for construction and operation of the Franklin Farms Lateral and Franklin Farms (FF) Trio Farm Lateral (Laterals). Details on the aboveground facilities are provided under Section 8.1.2. Alignment sheets of the proposed pipeline facilities are provided in **Appendix A-7**.

8.1.1.1 Permanent (Operational) Right-of-Way

The permanent right-of-way (ROW) for the Laterals is 50 feet wide. The permanent ROW will be maintained in accordance with the FERC Plan, FERC Procedures, and applicable permit requirements. The predominant land use which will be crossed by the new permanent ROW comprises agricultural land (see **Table 8.1-1**). Typical drawings which depict the permanent ROW are provided in **Appendix B-1** and **Appendix B-2**.

8.1.1.2 Temporary (Construction) Right-of-Way

In addition to the new permanent ROW, construction of the Laterals will require temporary (construction) workspace to facilitate construction equipment, referred to as the construction ROW. In general, the construction ROW will extend beyond the permanent ROW to provide a 100-foot total corridor along the Franklin Farms Lateral and a 75-foot total corridor along the FF Trio Farm Lateral. In some areas (*e.g.*, wetland crossings), this width will be reduced in accordance with the FERC Procedures. Typical drawings which depict the construction ROW are provided in **Appendix B-1** and **Appendix B-2**.

The predominant land use which will be temporarily affected during construction comprises agricultural land (see **Table 8.1-1**). Following construction, all temporary workspaces will be allowed to revert to their previous land use.

8.1.1.3 Additional Temporary Work Space

In addition to the temporary (construction) ROW, ETC Tiger will require additional temporary workspace (ATWS) to facilitate construction at:

- Road, wetland, waterbody, railroad, and foreign utility crossings;
- Pipeline fitting and installation;
- Points of inflection;
- Aboveground facilities;
- Topsoil segregation in active agricultural lands;
- Equipment and material storage locations; and
- Equipment turnarounds.

Unless otherwise specified within the requested deviations, ATWS will adhere to wetland and waterbody setbacks in accordance with the FERC Procedures. The location, purpose, size, and existing land use of the proposed ATWS is summarized in **Table 8.1-2**. The current land use within the ATWS predominately comprises agricultural land. ATWS will be allowed to revert to previous land uses following construction and per landowner agreements.

TABLE 8.1-2
ATWS Required for the Franklin Farms Lateral and FF Trio Farm Lateral Construction

ATWS Identification (ID)	Milepost (MP)	Parish	Justification	Average Width (feet)	Average Length (feet)	Total Area (acres)	Predominant Existing Land Use [a]	Located within 50 feet of a Wetland or Waterbody
Franklin Farms Lateral								
FFL-ATWS-0001	0.00	Richland	Meter Station - Installation of Environmental Control Devices (ECD), Topsoil Segregation, Site Grading, Material and Equipment offloading and staging, Spoil Storage, Existing Pipeline Tie-ins, Permanent Fence Installation	Varies	Varies	2.26	Agricultural Land	No
FFL-ATWS-0002	0.00	Richland	Meter Stations - Installation of ECDs, Permanent Fence Installation	10	89	0.02	Agricultural Land	No
FFL-ATWS-0003	0.01	Richland	Meter Stations - Installation of ECDs, Permanent Fence Installation	10	167	0.04	Agricultural Land	No
FFL-ATWS-0004	0.04	Richland	Stream/Ditch Crossing – Material and Equipment Staging, Spoil Storage	50	100	0.11	Agricultural Land	No
FFL-ATWS-0005	0.30	Richland	Active Agricultural Land - Topsoil Segregation	25	2,691	1.54	Agricultural Land	No
FFL-ATWS-0006	0.57	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	50	150	0.17	Agricultural Land	No
FFL-ATWS-0007	0.63	Richland	Public Road Crossing/Horizontal Bore & Pipeline Fitting Installation - Material and Equipment offloading and staging, Spoil Storage, Parking	50	150	0.17	Agricultural Land	No
FFL-ATWS-0008	0.64	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	50	188	0.22	Agricultural Land	No
FFL-ATWS-0009	0.68	Richland	Active Agricultural Land - Topsoil Segregation	25	430	0.25	Agricultural Land	No
FFL-ATWS-0010	0.71	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	103	0.12	Agricultural Land	No
FFL-ATWS-0011	0.76	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	104	0.12	Agricultural Land	No
FFL-ATWS-0012	0.87	Richland	Active Agricultural Land - Topsoil Segregation	25	1,045	0.60	Agricultural Land	No
FFL-ATWS-0013	0.95	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	104	0.12	Agricultural Land	No
FFL-ATWS-0014	0.99	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Agricultural Land	No
FFL-ATWS-0015	1.05	Richland	Active Agricultural Land - Topsoil Segregation	25	647	0.37	Agricultural Land	No
FFL-ATWS-0016	1.10	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	106	0.12	Agricultural Land	No
FFL-ATWS-0017	1.14	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Agricultural Land	No
FFL-ATWS-0018	1.33	Richland	Active Agricultural Land - Topsoil Segregation	25	1,971	1.13	Agricultural Land	No
FFL-ATWS-0019	1.54	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Open Land	No
FFL-ATWS-0020	1.58	Richland	Stream/Ditch Crossing & Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	50	328	0.38	Agricultural Land	No
FFL-ATWS-0021	1.58	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	Varies	Varies	0.03	Agricultural Land	No
FFL-ATWS-0022	1.64	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Parking	50	145	0.17	Agricultural Land	No
FFL-ATWS-0023	1.65	Richland	Public Road Crossing/Horizontal Bore - Spoil Storage	25	188	0.11	Agricultural Land	No
FFL-ATWS-0024	1.93	Richland	Active Agricultural Land - Topsoil Segregation	25	3,083	1.77	Agricultural Land	No
FFL-ATWS-0025	2.25	Richland	Pipeline Fitting Installation & Open Cut Private Road Crossing - Material Staging, Spoil Storage	50	230	0.26	Agricultural Land	No
FFL-ATWS-0026	2.28	Richland	Open Cut Private Road Crossing - Material Staging, Spoil Storage	50	100	0.11	Agricultural Land	No
FFL-ATWS-0027	2.40	Richland	Active Agricultural Land - Topsoil Segregation	25	1,285	0.74	Agricultural Land	No
FFL-ATWS-0028	2.53	Richland	Pipeline Fitting Installation & Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	217	0.25	Agricultural Land	No
FFL-ATWS-0029	2.57	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	150	0.17	Agricultural Land	No
FFL-ATWS-0030	2.68	Richland	Active Agricultural Land - Topsoil Segregation	25	939	0.54	Agricultural Land	No

TABLE 8.1-2
ATWS Required for the Franklin Farms Lateral and FF Trio Farm Lateral Construction

ATWS Identification (ID)	Milepost (MP)	Parish	Justification	Average Width (feet)	Average Length (feet)	Total Area (acres)	Predominant Existing Land Use [a]	Located within 50 feet of a Wetland or Waterbody
FFL-ATWS-0031	2.78	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	50	150	0.17	Agricultural Land	No
FFL-ATWS-0032	2.82	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	50	150	0.17	Agricultural Land	No
FFL-ATWS-0033	2.92	Richland	Active Agricultural Land - Topsoil Segregation	25	936	0.54	Agricultural Land	No
FFL-ATWS-0034	3.02	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	150	0.17	Agricultural Land	No
FFL-ATWS-0035	3.07	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	150	0.17	Agricultural Land	No
FFL-ATWS-0036	3.15	Richland	Active Agricultural Land - Topsoil Segregation	25	708	0.41	Agricultural Land	No
FFL-ATWS-0037	3.21	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	75	200	0.34	Agricultural Land	No
FFL-ATWS-0038	3.31	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	75	361	0.62	Open Land	No
FFL-ATWS-0039	3.31	Richland	Construction Move Around - Material and Equipment offloading and staging	Varies	Varies	0.12	Open Land	Yes (wetland)
FFL-ATWS-0040	3.31	Richland	Construction Move Around - Material and Equipment offloading and staging	100	385	0.88	Open Land	Yes (wetland)
FFL-ATWS-0041	3.78	Richland	Pipeline Fitting Installation - Material Staging, Spoil Storage	50	100	0.11	Open Land	No
FFL-ATWS-0042	3.97	Richland	Wetland Crossing - Material and Equipment Staging	50	146	0.17	Open Land	No
FFL-ATWS-0043	4.13	Richland	Wetland Crossing, Public Road Crossing & Pipeline Fitting Installation - Material and Equipment offloading and staging, Spoil Storage, Parking	50	222	0.26	Silviculture	No
FFL-ATWS-0044	4.16	Richland	Wetland Crossing, Public Road Crossing & Pipeline Fitting Installation - Material and Equipment offloading and staging, Spoil Storage, Parking	60	172	0.24	Silviculture	No
FFL-ATWS-0045	4.34	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	75	152	0.26	Forested/Woodland	No
FFL-ATWS-0046	4.46	Richland	Stream/Ditch Crossing & Pipeline Fitting Installation - Material and Equipment Staging, Spoil Storage	100	146	0.33	Agricultural Land	No
FFL-ATWS-0047	4.55	Richland	Meter Station - Installation of ECDs, Site Grading, Permanent Fence Installation	25	800	0.46	Agricultural Land	No
FFL-ATWS-0048	4.64	Richland	Pipeline Fitting Installation - Material Staging, Spoil Storage	50	250	0.29	Agricultural Land	No
FFL-ATWS-0049	5.02	Richland	Pipeline Fitting Installation & Construction Move Around - Material Staging, Spoil Storage, Material and Equipment offloading and staging	75	367	0.63	Agricultural Land	No
FFL-ATWS-0050	5.10	Richland	Public Road Crossing/Horizontal Bore & Pipeline Fitting Installation - Material and Equipment offloading and staging, Spoil Storage, Parking	75	321	0.55	Agricultural Land	No
FFL-ATWS-0051	5.17	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	75	148	0.25	Open Land	No
FFL-ATWS-0052	5.29	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Open Land	No
FFL-ATWS-0053	5.45	Richland	Construction Move Around & Wetland Crossing - Material and Equipment offloading and staging	120	299	0.82	Agricultural Land	No
FFL-ATWS-0054	5.51	Richland	Active Agricultural Land - Topsoil Segregation	25	317	0.18	Agricultural Land	No
FFL-ATWS-0055	5.55	Richland	Pipeline Fitting Installation - Material Staging, Spoil Storage	50	156	0.18	Agricultural Land	No
FFL-ATWS-0056	5.62	Richland	Active Agricultural Land - Topsoil Segregation	25	257	0.15	Agricultural Land	No
FFL-ATWS-0057	5.66	Richland	Pipeline Fitting Installation & Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	223	0.26	Agricultural Land	No
FFL-ATWS-0058	5.69	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	101	0.12	Agricultural Land	No
FFL-ATWS-0059	5.80	Richland	Active Agricultural Land - Topsoil Segregation	25	1,047	0.60	Agricultural Land	No

TABLE 8.1-2
ATWS Required for the Franklin Farms Lateral and FF Trio Farm Lateral Construction

ATWS Identification (ID)	Milepost (MP)	Parish	Justification	Average Width (feet)	Average Length (feet)	Total Area (acres)	Predominant Existing Land Use [a]	Located within 50 feet of a Wetland or Waterbody
FFL-ATWS-0060	5.91	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Agricultural Land	No
FFL-ATWS-0061	5.97	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	298	0.34	Agricultural Land	No
FFL-ATWS-0062	6.07	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	201	0.23	Agricultural Land	No
FFL-ATWS-0063	6.37	Richland	Active Agricultural Land - Topsoil Segregation	25	3,040	1.74	Agricultural Land	No
FFL-ATWS-0064	6.67	Richland	Pipeline Fitting Installation & Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	242	0.28	Agricultural Land	No
FFL-ATWS-0065	6.71	Richland	Stream/Ditch Crossing & Pipeline Fitting Installation – Material and Equipment Staging, Spoil Storage	50	198	0.23	Forested/Woodland	No
FFL-ATWS-0066	6.72	Richland	Stream/Ditch Crossing & Pipeline Fitting Installation – Material and Equipment Staging, Spoil Storage	25	128	0.07	Forested/Woodland	No
FFL-ATWS-0067	6.91	Richland	Stream/Ditch Crossing – Material and Equipment Staging, Spoil Storage	50	155	0.18	Wetlands	Yes (wetland)
FFL-ATWS-0068	6.96	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	89	0.10	Wetlands	Yes (wetland and waterbody)
FFL-ATWS-0069	7.02	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	168	0.19	Wetlands	Yes (wetland)
FFL-ATWS-0070	7.09	Richland	Pipeline Fitting Installation - Material Staging, Spoil Storage	25	202	0.12	Wetlands	Yes (wetland)
FFL-ATWS-0071	7.13	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	164	0.19	Wetlands	Yes (wetland)
FFL-ATWS-0072	7.19	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	165	0.19	Wetlands	Yes (wetland)
FFL-ATWS-0073	8.11	Richland	Stream/Ditch Crossing, Pipeline Fitting Installation & Wetland Crossing - Material and Equipment Staging, Spoil Storage	50	250	0.29	Agricultural Land	No
FFL-ATWS-0074	8.29	Richland	Active Agricultural Land - Topsoil Segregation	25	1,748	1.00	Agricultural Land	No
FFL-ATWS-0075	8.52	Richland	Pipeline Fitting Installation - Material Staging, Spoil Storage	50	249	0.29	Agricultural Land	No
FFL-ATWS-0076	8.73	Richland	Active Agricultural Land - Topsoil Segregation	25	2,139	1.23	Agricultural Land	No
FFL-ATWS-0077	8.95	Richland	Stream/Ditch Crossing & Wetland Crossing - Material and Equipment Staging, Spoil Storage	50	152	0.17	Agricultural Land	No
FFL-ATWS-0078	9.27	Richland	Wetland Crossing - Material and Equipment Staging	50	123	0.14	Open Land	No
FFL-ATWS-0079	9.41	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage	50	150	0.17	Wetlands	Yes (wetland)
FFL-ATWS-0080	9.50	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	50	181	0.21	Open Land	No
FFL-ATWS-0081	9.62	Richland	Pipeline Crossing - Material and Equipment Staging, Spoil Storage	75	138	0.24	Agricultural Land	No
FFL-ATWS-0082	9.66	Richland	Meter Station - Installation of ECDs, Site Grading, Permanent Fence Installation	25	510	0.29	Agricultural Land	No
FFL-ATWS-0083	9.78	Richland	Active Agricultural Land - Topsoil Segregation	25	923	0.53	Agricultural Land	No
FFL-ATWS-0084	9.88	Richland	Pipeline Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Agricultural Land	No
FFL-ATWS-0085	9.93	Richland	Stream/Ditch Crossing & Pipeline Crossing - Material and Equipment Staging, Spoil Storage, Parking	50	345	0.40	Agricultural Land	No
FFL-ATWS-0086	10.02	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	184	0.21	Agricultural Land	No
FFL-ATWS-0087	10.23	Richland	Active Agricultural Land - Topsoil Segregation	25	2,036	1.17	Agricultural Land	No
FFL-ATWS-0088	10.48	Richland	Pipeline Fitting Installation - Material Staging, Spoil Storage	50	247	0.28	Agricultural Land	No
FFL-ATWS-0089	10.54	Richland	Active Agricultural Land - Topsoil Segregation	25	405	0.23	Agricultural Land	No
FFL-ATWS-0090	10.59	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	150	0.17	Agricultural Land	No
FFL-ATWS-0091	10.64	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	150	0.17	Agricultural Land	No

TABLE 8.1-2
ATWS Required for the Franklin Farms Lateral and FF Trio Farm Lateral Construction

ATWS Identification (ID)	Milepost (MP)	Parish	Justification	Average Width (feet)	Average Length (feet)	Total Area (acres)	Predominant Existing Land Use [a]	Located within 50 feet of a Wetland or Waterbody
FFL-ATWS-0092	10.71	Richland	Active Agricultural Land - Topsoil Segregation	25	538	0.31	Agricultural Land	No
FFL-ATWS-0093	10.76	Richland	Pipeline Crossing - Material and Equipment Staging, Spoil Storage	70	189	0.30	Agricultural Land	No
FFL-ATWS-0094	10.81	Richland	Pipeline Crossing - Material and Equipment Staging, Spoil Storage	50	558	0.64	Agricultural Land	No
FFL-ATWS-0095	10.83	Richland	Meter Station - Installation of ECDs, Site Grading, Permanent Fence Installation	70	55	0.09	Agricultural Land	No
FFL-ATWS-0096	10.89	Richland	Meter Station - Installation of ECDs, Site Grading, Permanent Fence Installation	25	253	0.15	Agricultural Land	No
FFL-ATWS-0097	10.96	Richland	Stream/Ditch Crossing & Wetland Crossing - Material and Equipment Staging, Spoil Storage	50	150	0.17	Agricultural Land	No
FFL-ATWS-0098	11.08	Richland	Active Agricultural Land - Topsoil Segregation	25	1,123	0.64	Agricultural Land	No
FFL-ATWS-0099	11.21	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	210	0.24	Forested/Woodland	No
FFL-ATWS-0100	11.30	Richland	Stream/Ditch Crossing, Pipeline Crossing & Pipeline Fitting Installation - Material and Equipment Staging, Spoil Storage, Parking	50	324	0.37	Forested/Woodland	No
FFL-ATWS-0101	11.37	Richland	Pipeline Crossing - Material and Equipment Staging, Spoil Storage	50	150	0.17	Open Land	No
FFL-ATWS-0102	11.39	Richland	Pipeline Fitting Installation - Material Staging, Spoil Storage	25	120	0.07	Open Land	No
FFL-ATWS-0103	11.47	Richland	Wetland Crossing - Material and Equipment Staging	50	100	0.11	Open Land	No
FFL-ATWS-0104	11.54	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Open Land	No
FFL-ATWS-0105	11.60	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	Varies	Varies	0.18	Forested/Woodland	No
FFL-ATWS-0106	11.73	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Open Land	No
FFL-ATWS-0107	11.79	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	162	0.19	Forested/Woodland	No
FFL-ATWS-0108	11.85	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Forested/Woodland	No
FFL-ATWS-0109	12.01	Richland	Wetland Crossing - Material and Equipment Staging	50	109	0.13	Open Land	No
FFL-ATWS-0110	12.11	Richland	Wetland Crossing & Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	192	0.22	Open Land	No
FFL-ATWS-0111	12.19	Richland	Wetland Crossing & Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Forested/Woodland	No
FFL-ATWS-0112	12.38	Richland	Wetland Crossing & Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	176	0.20	Forested/Woodland	No
FFL-ATWS-0113	12.46	Richland	Stream/Ditch Crossing & Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	50	167	0.19	Forested/Woodland	No
FFL-ATWS-0114	12.50	Richland	Public Road Crossing/Horizontal Bore & Pipeline Fitting Installation - Material and Equipment offloading and staging, Spoil Storage, Parking	Varies	Varies	0.13	Forested/Woodland	Yes (wetland)
FFL-ATWS-0115	12.55	Richland	Public Road Crossing/Horizontal Bore, Pipeline Fitting Installation & Stream/Ditch Crossing - Material and Equipment offloading and staging, Spoil Storage	25	207	0.12	Forested/Woodland	Yes (wetland and waterbody)
FFL-ATWS-0116	12.62	Richland	Stream/Ditch Crossing & Pipeline Fitting Installation - Material and Equipment Staging, Spoil Storage	50	173	0.20	Open Land	No
FFL-ATWS-0117	13.09	Richland	Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	75	200	0.34	Industrial/Commercial	No
FFL-ATWS-0118	13.17	Richland	Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	75	200	0.34	Industrial/Commercial	No
FFL-ATWS-0119	13.22	Richland	Meter Station - Installation of ECDs, Site Grading, Permanent Fence Installation	90	89	0.18	Industrial/Commercial	No
FFL-ATWS-0120	13.24	Richland	Meter Station - Installation of ECDs, Site Grading, Permanent Fence Installation	25	740	0.42	Industrial/Commercial	No

TABLE 8.1-2
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ATWS Identification (ID)	Milepost (MP)	Parish	Justification	Average Width (feet)	Average Length (feet)	Total Area (acres)	Predominant Existing Land Use [a]	Located within 50 feet of a Wetland or Waterbody
FF Trio Farm Lateral								
FTL-ATWS-0001	0.00	Richland	Meter Station - Installation of ECDs, Site Grading, Permanent Fence Installation	25	290	0.17	Agricultural Land	No
TFL-ATWS-0002	0.12	Richland	Active Agricultural Land - Topsoil Segregation	25	1,134	0.65	Agricultural Land	No
TFL-ATWS-0003	0.21	Richland	Pipeline Fitting Installation & Wetland Crossing - Material and Equipment Staging, Spoil Storage	50	150	0.17	Forested/Woodland	No
TFL-ATWS-0004	0.38	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	150	0.17	Agricultural Land	No
TFL-ATWS-0005	0.46	Richland	Pipeline Crossing & Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	218	0.25	Open Land	No
TFL-ATWS-0006	0.53	Richland	Pipeline Crossing - Material and Equipment Staging, Spoil Storage	50	156	0.18	Open Land	No
TFL-ATWS-0007	0.62	Richland	Pipeline Crossing & Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	25	188	0.11	Open Land	No
TFL-ATWS-0008	0.67	Richland	Pipeline Crossing & Pipeline Fitting Installation - Material and Equipment Staging, Spoil Storage	25	307	0.18	Open Land	No
TFL-ATWS-0009	1.05	Richland	Confined Construction ROW - Material and Equipment Staging, Spoil Storage	25	150	0.09	Open Land	No
TFL-ATWS-0010	1.12	Richland	Confined Construction ROW & Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	25	144	0.08	Open Land	No
TFL-ATWS-0011	1.18	Richland	Public Road Crossing/Horizontal Bore & Stream/Ditch Crossing - Material and Equipment offloading and staging, Spoil Storage, Parking	50	153	0.18	Open Land	No
TFL-ATWS-0012	1.25	Richland	Public Road Crossing/Horizontal Bore - Material and Equipment offloading and staging, Spoil Storage, Parking	50	150	0.17	Open Land	No
TFL-ATWS-0013	1.25	Richland	Public Road Crossing/Horizontal Bore - Spoil Storage	25	150	0.09	Open Land	No
TFL-ATWS-0014	1.32	Richland	Stream/Ditch Crossing & Wetland Crossing - Material and Equipment Staging, Spoil Storage	50	111	0.13	Open Land	No
TFL-ATWS-0015	1.39	Richland	Stream/Ditch Crossing & Wetland Crossing - Material and Equipment Staging, Spoil Storage	50	100	0.11	Agricultural Land	No
TFL-ATWS-0016	1.42	Richland	Stream/Ditch Crossing - Material and Equipment Staging, Spoil Storage	50	101	0.12	Agricultural Land	No
TFL-ATWS-0017	1.45	Richland	Stream/Ditch Crossing & Pipeline Fitting Installation - Material and Equipment Staging, Spoil Storage	50	100	0.11	Agricultural Land	No
TFL-ATWS-0018	1.71	Richland	Pipeline Fitting Installation - Material Staging, Spoil Storage	25	198	0.11	Agricultural Land	No
TFL-ATWS-0019	1.75	Richland	Meter Station - Installation of ECDs, Site Grading, Permanent Fence Installation	25	185	0.11	Agricultural Land	No
TFL-ATWS-0020	1.75	Richland	Meter Station - Installation of ECDs, Site Grading, Permanent Fence Installation	25	810	0.46	Agricultural Land	No

Notes:
[a] Source: MRLC Consortium, 2024

8.1.1.4 Access Roads

Access by construction crews and the delivery of materials and equipment for the Project will primarily be via existing permanent public and private roads or along the construction ROW. The Project will use temporary access roads (TAR) and permanent access roads (PAR) to access Project facilities. TARs will be returned to pre-construction conditions or per landowner agreement following construction. Details of the planned permanent and temporary access roads are provided in Table 1.2-2 of Resource Report No. 1 (*General Project Description).*

Table 8.1-1 identifies the land uses associated with the Project access roads and summarizes the land requirements for construction (temporary) and operation (permanent). The predominant land uses impacted by access roads are industrial/commercial and agricultural land. Alignment sheets depicting the access roads are provided in **Appendix A-7**.

8.1.2 Aboveground Facilities

Table 8.1-1 identifies the land uses associated with the proposed aboveground facility locations and summarizes the land requirements for facility construction (temporary) and operation (permanent). The predominant land use affected by the proposed aboveground facilities (all meter stations and interconnect site) is agricultural land. Alignment sheets depicting the aboveground facilities are provided in **Appendix A-7**.

8.1.2.1 Contractor Yards

ETC Tiger intends to use three construction pipe and contractor yards in Richland Parish, Louisiana. The predominant land use for the contractor yards is industrial/commercial land. The contractor yards will only be used during construction, and the yards will be allowed to revert to previous land uses following construction and per landowner agreements.

8.1.3 Land Use Impacts and Mitigation

Land requirements for the Project and the land use types which will be affected by construction (temporary impacts) are provided in **Table 8.1-1**. ETC Tiger does not anticipate that installing the proposed Project facilities will significantly affect land use based on (1) the minor amount of land necessary to install the facilities, (2) the current use of these lands, and (3) except for a very limited amount of area, current land uses will be allowed to resume following construction.

8.1.3.1 Agricultural Lands

The majority of the land crossed by the Project is agricultural land. Following construction, agricultural land will be returned to pre-construction conditions and will be available for use during pipeline operations. Given the availability of agricultural land in the surrounding area, construction of all seven Project meter stations will result in a negligible loss of agricultural land. Therefore, overall impacts to agricultural lands are anticipated to be short-term and minor.

8.1.3.2 Forest/Woodland

Both short- and long-term impacts are expected to upland forests as a result of Project construction. In open-cut areas (trenched areas), trees will be cleared from the new permanent ROW, construction ROW, and ATWS, as necessary, to facilitate construction. The FERC Plan requires that 10 feet of the permanent ROW (centered on the pipeline) be maintained in an herbaceous state during operations and trees be restricted to less than 15 feet high within 15 feet of the pipeline, herein referred to as the maintenance

corridor[1]. The remaining permanent ROW that lies outside of the maintenance corridor, construction ROW, and ATWS will all be allowed to revert to pre-construction conditions. ETC Tiger anticipates that upland forest will regenerate to pre-construction conditions within one to three decades, depending on the woodland species and management practices.

8.1.3.3 Open Lands

The majority of open land will be temporarily impacted by construction of the pipeline laterals, contractor yards, and access roads. Within the operational pipeline workspace, nothing will restrict impacted open land from reverting to its prior use after construction. Temporarily impact areas will be restored to pre-construction conditions and contours. A small amount of open land will be permanently impacted due to construction of a permanent access road. Therefore, impacts to open lands are anticipated to be minor and primarily short-term.

8.1.3.4 Residential Lands

Impacts to residential lands (i.e., developed lands) adjacent to the Project area are anticipated to be short-term and minor. Potential impacts from construction will include temporary increases in noise and dust levels (see Resource Report No. 9 [*Air Quality and Noise]*). Effects in any one location will be short-term in nature as construction progresses along the ROW.

8.1.3.5 Industrial/Commercial Lands

Industrial/Commercial land that will be impacted as a result of the Project is currently under construction for new developments. Impacts to other industrial/commercial lands are not anticipated from construction of the Project.

8.1.3.6 Open Water

ETC Tiger will follow the methods described in the FERC Plan and FERC Procedures to minimize the potential for indirect impacts to streams. Impacts to open water areas which are crossed by the open-cut method are anticipated to be short-term and minor. See Resource Report No. 2 for additional discussion on crossing methods and potential impacts to waterbodies.

8.1.3.7 Wetlands

Both short- and long-term impacts are expected to wetlands from construction and operation of the proposed Project. ETC Tiger will minimize the potential impacts by following the methods described in the FERC Plan and FERC Procedures. In addition, no aboveground facilities have been sited within wetland areas.

Impacts to PEM wetlands are anticipated to be short-term and minor. Following construction, the wetland areas will be returned to pre-construction conditions and allowed to revegetate from existing rootstocks and seedbanks to pre-construction conditions.

Impacts to PFO wetlands are anticipated to be long-term to permanent. ETC Tiger will implement the FERC Procedures which require that the maintenance corridor (10 feet in width centered on the pipeline) be maintained in an herbaceous state during operations and that trees be restricted to less than 15 feet high within 15 feet of the pipeline. The remaining permanent ROW that lies outside of the maintenance corridor,

[1] To facilitate periodic corrosion/leak surveys, a corridor not exceeding 10 feet in width centered on the pipeline will be cleared at a frequency necessary to maintain the corridor in an herbaceous state.

construction ROW, and ATWS will all be allowed to revegetate from existing rootstocks and seedbanks to pre-construction conditions. Therefore, PFO wetlands outside of the maintenance corridor are expected to return to pre-construction conditions within 10 to 30 years. PFO wetlands located within the maintenance corridor of the new permanent ROW will be permanently converted to palustrine scrub shrub or PEM wetlands. These modifications will result in a change in wetland habitat type but will not result in a net loss of wetlands.

See Resource Report No. 2 for additional details on the proposed wetland crossings and potential impacts.

8.1.3.8 Roadways

In locations where the Laterals will traverse public roads, the pipelines will be installed using the boring method, a trenchless construction method. No paved roads will be open-cut. A typical drawing of a road crossing is provided in **Appendix B**. In the unlikely event a public road is open-cut (due to failed bore or other complications), at least one traffic lane will be maintained, except for brief periods essential to laying the new pipeline. If Project construction crosses roads that access private residences or businesses and no alternate entrance exists, measures will be taken to maintain passage for landowners during construction. Attempts will be made to avoid peak traffic time periods during construction that temporarily closes roads.

To maintain safe conditions, ETC Tiger will keep roads free of mud and debris that may be left by crossing construction equipment. ETC Tiger will also minimize the potential for road damage by enforcing local weight limitations and restrictions. Roadways damaged during construction will be repaired to pre-construction conditions. No long-term effects are expected to roadways.

8.1.3.9 Railroads

The Franklin Farms Lateral will cross under the Canadian Pacific KC Railroad. There will be no impacts to normal operation of the railroad. Where the Franklin Farms Lateral will cross the Canadian Pacific KC Railroad, it will be installed using the horizontal boring method. No modifications to the railroad are proposed and there will be no impacts to normal operation of the railroad. For safety purposes, ETC Tiger will consider the specific requirements of the railroad company when designing and constructing the railroad crossing.

8.1.4 Existing Residences and Buildings

The structures identified within 50 feet of the proposed Project area are identified in **Table 8.1-3**. In addition, **Table 8.1-3** identifies if each identified structure is a habitable structure.

TABLE 8.1-3
Residences or Buildings Within 50 Feet of the Franklin Farms Project Construction Work Area

Facility	MP	Parish	Description	Habitable	Distance Away Construction Work Area (feet)
STR - #1	N/A	Richland	Single Family Home	Yes	27
STR - #2	N/A	Richland	Shed	No	33
STR - #3	N/A	Richland	Shed	No	47
STR - #4	11.6	Richland	Abandoned Barn	No	Within Construction ROW

No residences are less than 25 feet from the construction workspace. ETC Tiger will consult with affected landowners regarding all non-residential buildings (such as barns, outbuildings, sheds, and/or garages) within the proposed Project workspaces. ETC Tiger will work with the affected landowners to determine the best path forward for these impacted structures such as relocating the buildings to a location outside of the Project construction workspace or compensating the landowner for removal of the buildings.

ETC Tiger will implement the following general measures to minimize construction-related impacts on all residences and other structures located within 50 feet of the construction ROW:

- Attempt to avoid removing mature trees and landscaping from within the edge of the construction work area unless necessary to safely operate construction equipment or as specified in landowner agreements.
- Restore all lawn areas and landscaping within the construction work area immediately after cleanup operations, or as specified in landowner agreements, consistent with the requirements of the FERC Plan.
- Install safety fence along the edge of the construction work area adjacent to the residence for a distance of 100 feet on either side of the residence to ensure that construction equipment and materials, including the spoil pile, remain within the construction work area.
- Maintain fencing throughout active construction in the area.
- Maintain a minimum of 25 feet between the residence and the construction work area for a distance of 100 feet on either side of the residence (i.e., reduce the construction work area as necessary to maintain the minimum distance).
- Complete final cleanup, grading, and installation of permanent erosion control devices within 10 days after backfilling the trench.

To minimize impacts on residences, ETC Tiger will stake the limits of disturbance and the centerline of the pipeline and will coordinate with property owners before and throughout the construction process. Owners of residential property crossed by the pipelines will be contacted to notify them of the approximate time that construction will take place on their property and to keep them informed about any special construction activities that may concern them. ETC Tiger has developed *Environmental Complaint Resolution Procedures* to address concerns or complaints received from affected landowners during construction of the Project (**Appendix C-7**). Affected landowners will be provided a copy of the *Environmental Complaint Resolution Procedures* prior to the commencement of construction. ETC Tiger is committed to work closely with affected landowners to ensure that any concerns or complaints are resolved satisfactorily and in a timely manner.

ETC Tiger will attempt to maintain residential access at all times; however, interruption of access necessary to complete a driveway crossing will be coordinated in advance with the landowner. ETC Tiger will generally complete construction across driveways within one day and install a temporary surface patch. Final surfacing of driveways will occur separately in conjunction with other driveway final restorations within the area. If a road needs to be open cut (e.g., failed bore), traffic flow will be maintained except for temporary periods essential for pipeline installation.

Following construction, all residential areas will be restored to preconstruction conditions or as specified in written landowner agreements.

8.1.5 Planned Residential and Commercial Areas

No new residential or commercial development areas were identified within 0.25 miles of the Project area.

8.2 PUBLIC LAND, RECREATION, AND OTHER DESIGNATED AREAS

8.2.1 Public or Conservation Land

The proposed Project facilities will not cross any lands or waters administered by federal, state, or local agencies, or private conservation organizations. All lands which will be crossed by the Project are privately owned. A list of landowners is provided in **Appendix E-1** and is submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE"*.

8.2.2 Natural, Recreational, or Scenic Areas

The proposed Project facilities will not be located within 0.25 miles of any:

- Designated or proposed candidate National Wild and Scenic Rivers or Louisiana Natural and Scenic Rivers (National Wild and Scenic Rivers System [NWSRS], 2026; Louisiana Department of Wildlife and Fisheries [LDWF] 2026a),
- National battlefields or military parks (National Park Service [NPS], 2026a, 2026b, 2026c, 2026d),
- Registered National Landmark (NPS, 2026e),
- State or National Forests and Parks (United States Department of Agriculture [USDA] Forest Service, 2026a, 2026b; NPS 2026c; Louisiana State Parks 2026),
- Wildlife Management Areas (LDWF, 2026b),
- National Wildlife Refuges (United States Fish and Wildlife Service [USFWS], 2026a),
- U.S. Department of Transportation Federal Highway Administration (USDOT FHWA) designated American Byways which include National Scenic Byways and All-American Roads (USDOT FHWA, 2026).

Natural, recreational, or scenic areas which were identified within 0.25 miles of the proposed Project facilities are described below.

8.2.3 Special Use Areas

Table 8.2-1 details Project facilities within 0.25-mile of special use areas.

TABLE 8.2-1
Public Land, Recreation, and Special Use Areas within 0.25 mile of the Project Workspaces

Project Facility	Special Use Area	Approximate MP	Distance from Project (miles)	Approximate Crossing length (miles)
FF Trio Farms Lateral	USDA Agricultural Conservation Easement Program (ACEP) -Wetland Reserve Easement, Richland, Louisiana	0.6, 9.0	0.0 (Adjacent)	0.0 (Project does not cross)
Franklin Farms Lateral	USDA ACEP - Wetland Reserve Easement, Richland, Louisiana	7.3, 8.1	0.1	0.0 (Project does not cross)
Franklin Farms Lateral	Bee Bayou Mitigation Bank	6.7	0.5	0.0 (Project does not cross)

TABLE 8.2-1
Public Land, Recreation, and Special Use Areas within 0.25 mile of the Project Workspaces

Project Facility	Special Use Area	Approximate MP	Distance from Project (miles)	Approximate Crossing length (miles)
Franklin Farms Lateral	Natural Resources and Conservation Service (NRCS) Private Easement (Mitigation Bank)	2.7	0.2	0.0 (Project does not cross)
Franklin Farms Lateral	NRCS Private Easement (Mitigation Bank)	3.4	0.1	0.0 (Project does not cross)
Franklin Farms Lateral	NRCS Private Easement (Mitigation Bank)	3.3	0.1	0.0 (Project does not cross)

The Project is within 0.25 miles of land parcels identified as being a part of USDA Natural Resources Conservation Service (NRCS) Agricultural Conservation Easement Program - Wetlands Reserve Easement (WRE), and wetland mitigation banks (USGS, 2026; U.S. Army Corps of Engineers [USACE], 2026). Land eligible for WRE easements includes privately held farmed or converted wetlands that were previously degraded due to agricultural uses and can be successfully and cost-effectively restored. The USDA NRCS enters into purchase agreements with eligible private landowners or Indian tribes that include the right for NRCS to develop and implement a plan to help restore, protect, and enhance the wetlands functions and values (NRCS, 2026). A mitigation bank is a wetland, stream, or other aquatic resource area that has been restored, established, enhanced, or preserved for the purpose of providing compensation for unavoidable impacts to aquatic resources permitted under Section 404 or a similar state or local wetland regulation (EPA, 2026d).

The Project workspace will not cross any of the identified parcels within the WRE program or wetland mitigation banks; therefore, the Project does not anticipate any impacts on these special use areas.

8.2.4 Coastal Zone Management Areas

In 1972, the United States Congress passed the federal Coastal Zone Management Act to address the increasing conflict between protection and use of our nation's coastal zone. The Project is not located within the coastal zone.

8.2.5 Historical Resources

Details concerning cultural resources and places/areas identified on the National Register of Historic Places are provided in Resource Report No. 4 (*Cultural Resources*).

8.2.6 Landfills and Hazardous Waste Sites

ETC Tiger reviewed Louisiana Department of Environmental Quality and United States Environmental Protection Agency (EPA) enforcement and compliance websites to identify landfills or hazardous waste and contamination clean-up sites within the Project vicinity. As detailed in **Table 8.2-2** and depicted in **Appendix A-9**, several sites that have the potential to be sources of contamination were identified within 0.25 miles of the Project area (EPA, 2026a, 2026b, 2026c). While the facilities in **Table 8.2-2** are part of hazardous waste reporting programs, the identified sites are not confirmed sites of hazardous waste spills but are potential sources of hazardous waste because the sites handle hazardous waste within their normal operation. ETC Tiger has prepared an *Unanticipated Discovery Plan* (**Appendix C-4**) should contamination or possible contamination be identified during construction.

TABLE 8.2-2
EPA Potential Hazardous Waste Facilities within 0.25 miles of the Franklin Farms Project

EPA Facility Name	12-digit Hydrologic Unit Code (HUC)	Approximate Distance from Construction Workspace (ft)	Program [a]
Closed Underground Storage Tank (UST)	080500020505	30	RCRA
Gulf South Pipeline Co, Delhi	080500020505	407	RCRA
Columbia Gulf Transmission Co.	080500020505	689	RCRA
Southeast Supply Header LLC (SESH) - Delhi	080500020505	689	RCRA
SESH LLC - Warehouse	080500020505	1,034	RCRA
Franklin Farms Power Station	080500011003	1,200	RCRA

Source: EPA, 2026b, 2026c
[a] RCRA = Resource Conservation Recovery Act

8.2.7 Roadways, Railroads, and Utilities

A list of the roadways, railroads, and utilities that will be crossed by the Project is provided in **Table 8.2-3**.

TABLE 8.2-3
Roadways and Utilities Crossed by the Franklin Farms and FF Trio Farms Laterals

MP	Roadway, Railroad, or Utility	ROW Width [a] (feet)	Crossing Method
Franklin Farms Lateral			
0 [b]	Transmission Pipeline: Gulf Run Transmission, LLC	75	Open-Cut
0 [b]	Transmission Pipeline: ETC Tiger Pipeline, LLC	75	Open-Cut
0.6	Distribution Powerline: Northeast Louisiana Electric Power Cooperative, Inc.	20	Open-Cut
0.6	Local Road: Mengle Road	40	Horizontal Bore
1.6	Distribution Powerline: Unknown	20	Open-Cut
1.6	Local Road: Pony Greer Road	40	Horizontal Bore
2.8	Distribution Powerline: Unknown	20	Open-Cut
2.8	Local Road: Ollie Morris Road	40	Horizontal Bore
3.2	Local Road: Interstate 20	200	Horizontal Bore
4.1	Distribution Powerline: Unknown	25	Open-Cut
5.1	Railroad and Local Road: Highway 80	175	Horizontal Bore
5.2	Distribution Powerline: Northeast Louisiana Electric Power Cooperative, Inc.	20	Open-Cut
6.7	Distribution Powerline: Entergy Corporation	20	Open-Cut

TABLE 8.2-3
Roadways and Utilities Crossed by the Franklin Farms and FF Trio Farms Laterals

MP	Roadway, Railroad, or Utility	ROW Width [a] (feet)	Crossing Method
9.0	Transmission Powerline: Entergy Corporation	120	Open-Cut
9.4	Transmission Pipeline: DAPL-ETCO Operations Management, LLC	30	Open-Cut
9.4	Transmission Pipeline: MPLX, LP	100	Open-Cut
9.4	Pipeline Corridor and Smalling Road	50	Horizontal Bore
9.5	Transmission Pipeline: Trunkline Gas Company, LLC	50	Open-Cut
9.5	Distribution Powerline: Unknown	20	Open-Cut
9.6	Transmission Pipeline: Gulf Crossing	75	Open-Cut
9.9	Transmission Pipeline: Gulf South Pipeline Company, LLC	75	Open-Cut
10.8	Transmission Pipeline: Gulf South Pipeline Company, LLC	30	Open-Cut
10.8	Transmission Pipeline: Midcontinent Express Pipeline, LLC	30	Open-Cut
10.8	Transmission Pipeline: Gulf South Pipeline Company, LLC	30	Open-Cut
11.3	Transmission Pipeline: Trunkline Gas Company, LLC	30	Open-Cut
11.3	Transmission Pipeline: MPLX, LP	100	Open-Cut
11.4	Transmission Pipeline: DAPL-ETCO Operations Management, LLC	30	Open-Cut
12.5	Distribution Powerline: Entergy Louisiana, LLC	20	Open-Cut
12.5	Local Road: Jaggers Lane	40	Horizontal Bore
13.0	Distribution Powerline: Entergy Louisiana, LLC	20	Open-Cut
13.1	Entergy Laydown Yard	530	Horizontal Bore
FF Trio Farm Lateral			
0.5	Transmission Pipeline: DAPL-ETCO Operations Management, LLC	30	Open-Cut
0.5	Transmission Pipeline: MPLX, LP	100	Open-Cut
0.5	Transmission Pipeline: Trunkline Gas Company, LLC	30	Open-Cut
0.7	Transmission Pipeline: Boardwalk Pipeline Partners, LP	30	Open-Cut
1.2	Distribution Powerline: Boardwalk Pipeline Partners, LP	20	Open-Cut
1.2	Distribution Pipeline: Unknown	20	Open-Cut
1.2	Local Road: Highway 583	40	Horizontal Bore

Notes:
[a] ROW width is based on aerial interpretation
[b] MP not applicable, within FF Gulf Run Bi Di Meter Station

8.3 VISUAL RESOURCES

The potential for visual resource impacts from a project is typically determined by considering the general character of the existing landscape and construction of any visually prominent features. The subsections below outline the existing viewshed of the proposed Project facilities. The Project is not located within the viewshed of any federal, state, or locally designated scenic areas (as discussed in **Section 8.3**).

8.3.1 Proposed Franklin Farms Lateral and FF Trio Farm Lateral

There are visually sensitive areas, such as residences and churches, within the viewshed of the Laterals where the pipeline will be installed using the open-cut method. The pipelines will not have an impact on the viewshed in the vicinity upon the completion of construction activities and commencement of operation, as the pipelines will be buried underground. Therefore, the proposed lateral pipelines will only have a temporary impact on the viewshed within the vicinity and no long-term visual impacts are anticipated.

8.3.2 Proposed Aboveground Facilities

There are visually sensitive areas within the viewshed of the aboveground facilities. The churches within the vicinity of meter stations are screened from the proposed facilities by vegetative buffers and are a significant distance (greater than 500 feet) from the proposed meter stations; therefore, the construction and operation of the Project facilities would not have an impact on the viewshed of any churches. Many of the residences in the Project vicinity are a significant distance from the proposed meter stations and/or screened by vegetative buffers.

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ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 9
Air and Noise Quality

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 9 FILING REQUIREMENTS	
Information	**Section Reference**
1. Describe existing air quality in the vicinity of the project. (§ 380.12(k)(1)) • Identify criteria pollutants that may be emitted above EPA-identified significance levels.	9.1.1
2. Quantify the existing noise levels (day-night sound level (L_{dn}) and other applicable noise parameters) at noise-sensitive areas and at other areas covered by relevant state and local noise ordinances. (§380.12 (k) (2))	Table 9.2-2
3. Quantify existing and proposed emissions of compressor equipment, plus construction emissions, including nitrogen oxides (NO_x) and carbon monoxide (CO), and the basis for these calculations. Summarize anticipated air quality impacts for the project. (§380.12 (k) (3))	9.1.2 (construction)
4. Describe the existing and proposed compressor units at each station where new, additional, or modified compression units are proposed, including the manufacturer, model number, and horsepower (hp) of the compressor units. (§380.12 (k) (4))	N/A (no compression equipment)
5. Identify any nearby noise-sensitive area by distance and direction from the proposed compressor unit building/enclosure. (§380.12 (k) (4))	Table 9.2-1
6. Identify any applicable state or local noise regulations. (§380.12 (k) (4))	9.2.1
7. Calculate the noise impact at noise-sensitive areas of the proposed compressor unit modifications or additions, specifying how the impact was calculated, including manufacturer's data and proposed noise control equipment. (§380.12 (k) (4))	Table 9.2-3 (construction)
Information Often Missing and Resulting in Data Requests	
Air Quality Information	
Include climate information as part of the air quality information provided for the project area.	9.1.1
Identify potentially applicable federal and state air quality regulations.	9.1.1.2
Provide construction emissions (criteria pollutants, hazardous air pollutants, greenhouse gases) for proposed pipelines and aboveground facilities.	9.1.2
Provide copies of state and federal applications for air permits.	N/A
Provide operation and fugitive emissions (criteria pollutants, hazardous air pollutants, greenhouse gases) for pipelines and aboveground facilities.	9.1.3
Provide air quality modeling for entire compressor stations.	N/A
Identify temporary and permanent emissions sources that may have cumulative air quality effects in addition to those resulting from the project.	Table 1.10-2 of Resource Report 1
Noise and Vibration	
Describe the existing noise environment and ambient noise surveys for compressor stations, liquefied natural gas facilities, meter and regulation facilities, and drilling locations.	Table 9.2-2
Identify any state or local noise regulations applicable to construction and operation of the project.	9.2.1
Indicate whether construction activities would occur over 24-hour periods.	9.2.4
Discuss construction noise impacts and quantify construction noise impacts from drilling, pile driving, dredging, etc.	9.2.4

RESOURCE REPORT NO. 9 FILING REQUIREMENTS	
Quantify operation noise from aboveground facilities, including blowdowns.	--
Describe the potential for the operation of the proposed facilities to result in an increase in perceptible vibration and how this would be prevented.	9.2.3.4
Identify temporary and permanent noise sources that may have cumulative noise effects in addition to those resulting from the project.	9.2.4

RESOURCE REPORT NO. 9
AIR AND NOISE QUALITY

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Appendix C-6 – *Noxious and Invasive Plant Management Plan*
Appendix C-7 – *Environmental Complaint Resolution Procedures*

D Agency and Stakeholder Correspondence

Appendix D-1 – Agency Correspondence (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")
Appendix D-2 – SHPO Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")
Appendix D-3 – Native American Organizations and Tribes Correspondence (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

E Landowner Information

Appendix E-1 – Landowner/Stakeholder List (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE"*)

F Wetland and Waterbody Delineation Report and Listed Species Report

Appendix F-1 – Wetland and Waterbody Delineation Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")
Appendix F-2 – Listed Species Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

G Cultural Resource Assessment Survey Report (*Submitted as* "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

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ABBREVIATIONS AND ACRONYMS

°	degrees
µg/m^3	microgram per cubic meter
ANSI	American National Standards Institute
AP-42	EPA's AP-42, Fifth Edition Compilation of Air Pollutant Emission Factors
AQCR	Air Quality Control Region
CAA	Clean Air Act
Certificate	Certificate of Public Convenience and Necessity
CFR	Code of Federal Regulations
CH_4	methane
CO	carbon monoxide
CO_2	carbon dioxide
CO_2e	carbon dioxide equivalent
dBA	A-weighted decibels
EPA	United States Environmental Protection Agency
ETC Tiger	ETC Tiger Pipeline, LLC
F	Fahrenheit
FERC/Commission	Federal Energy Regulatory Commission
GHG	greenhouse gases
g/hp-hr	grams per horsepower hour
g/VMT	grams per vehicle mile traveled
GWP	global warming potential
HAP	hazardous air pollutant
HC	hydrocarbon
LAC	Louisiana Administrative Code
LDEQ	Louisiana Department of Environmental Quality
L_{dn}	day-night sound level in decibels (dBA)
L_{eq}	equivalent continuous sound level in dBA
MOVES	Mobile Vehicle Emissions Simulator
N_2O	nitrous oxide
NAAQS	National Ambient Air Quality Standards
NGA	Natural Gas Act
NNSR	Nonattainment New Source Review
NO_2	nitrogen dioxide
NO_x	nitrogen oxides
NSA	Noise Sensitive Area
NSR	New Source Review
O_2	oxygen
O_3	ozone
Pb	lead
PM	Particulate matter
PM_{10}	Particulate matter, aerodynamic diameter ≤ 10 microns

$PM_{2.5}$	Particulate matter, aerodynamic diameter ≤ 2.5 microns
Project	Franklin Farms Project
ppb	parts per billion
ppm	parts per million
PSD	Prevention of Significant Deterioration
PTE	potential to emit
SO_2	sulfur dioxide
THC	Total Hydrocarbon
tpy	tons per year
VOC	Volatile Organic Compound

9.0 AIR AND NOISE QUALITY

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project on June 2027, and following the receipt of all applicable approvals, anticipating construction activities to begin in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

Resource Report No. 9 assesses existing air and noise quality within the Project area. This report identifies relative meteorological, air quality, and noise level data, and the presence of any sensitive or special-use areas, including Noise Sensitive Areas (NSA). Further, the report provides a description of potential impacts to air and noise quality from Project construction and operations.

9.1 AIR QUALITY

This section describes the climate and existing ambient air quality in the proposed Project area. Relevant Federal, and Louisiana air permitting requirements and air regulatory compliance programs are summarized in this section. Additionally, potential air quality effects from construction and operation of the Project are provided.

9.1.1 Existing Conditions

9.1.1.1 Regional Climate and Local Meteorology

The regional climate of Richland Parish, Louisiana, Where the proposed Project sites are located, is generally hot during summer and cooler during winter. Richland Parish, LA is approximately seven miles to the North of the Rayville, Louisiana station and has climatological data (1991 through 2020 monthly normal) tabulated by the National Centers for Environmental Information (National Centers for Environmental Information 2026). The mean daily temperature ranges from 47.7 degrees Fahrenheit (°F) in January to 83.6°F in August. Precipitation is generally well distributed throughout the year, with slight decrease over the summer months, apart from snowfall during winter. The monthly variability of temperature and precipitation in the Project area is shown in **Table 9.1-1** for Rayville, LA.

TABLE 9.1-1
Monthly Mean Climate Parameters for the Rayville, LA Weather Station (1991- 2020)

Month	Precipitation (inches)		Temperature (Degrees Fahrenheit)		
	Rain	Snow	Daily Minimum	Average	Daily Maximum
January	5.54	0.30	38.0	47.7	57.4
February	5.11	0.40	41.0	51.3	61.6
March	5.65	0.00	48.2	59.0	69.8
April	6.70	0.00	55.6	66.6	77.6
May	5.25	0.00	64.1	74.0	84.0
June	4.75	0.00	71.4	80.7	90.1

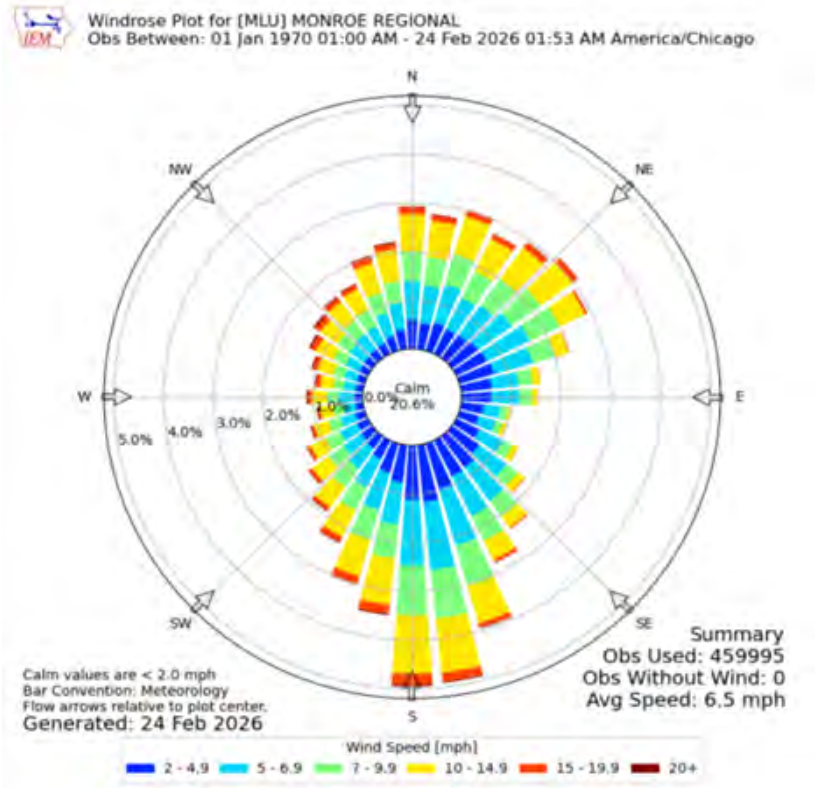
TABLE 9.1-1
Monthly Mean Climate Parameters for the Rayville, LA Weather Station (1991- 2020)

Month	Precipitation (inches)		Temperature (Degrees Fahrenheit)		
	Rain	Snow	Daily Minimum	Average	Daily Maximum
July	4.16	0.00	74.3	83.5	92.7
August	3.88	0.00	73.6	83.6	93.6
September	3.82	0.00	67.2	77.9	88.5
October	4.63	0.00	56.0	68.0	80.1
November	4.73	0.00	45.7	57.0	68.3
December	5.78	0.00	40.0	49.8	59.7
Yearly	60.00	0.70	56.3	66.6	77.0

Source: National Centers for Environmental Information, 1991-2020 US Climate Normals, Rayville, LA, US

A windrose for Monroe Regional Airport, which is approximately 30 kilometers west from the Project site, between January 1970 and February 2026 is depicted in Figure 9.1-1. Winds in the area are predominantly from the south, with secondary maximum frequency from the northeast. Calm winds account for 20.6 percent of the records during that period.

Figure 9.1-1
Monroe Regional Airport Windrose



Source: Iowa State University, Iowa Environmental Mesonet ()

9.1.1.2 Federal and State of Louisiana Air Quality Standards

Federal and state air quality standards have been designed to protect human health and the environment from airborne pollutants. The United States Environmental Protection Agency (EPA) has developed National Ambient Air Quality Standards (NAAQS) for criteria air pollutants such as nitrogen oxides (NO_x), carbon monoxide (CO), ozone (O_3), sulfur dioxide (SO_2), and inhalable particulate matter (PM), including PM less than 10 microns in aerodynamic diameter (PM_{10}) and PM less than 2.5 microns in aerodynamic diameter ($PM_{2.5}$). The NAAQS are set at levels the EPA has determined are necessary to protect human health and the general public's welfare as well as sensitive populations. The State of Louisiana has adopted the federal NAAQS, though current state regulations still maintain some older NAAQS standards that have recently changed (such as 8-hour O_3 and annual $PM_{2.5}$). **Table 9.1-2** lists the current NAAQS which apply to the Project area.

TABLE 9.1-2
NAAQS for Criteria Pollutants

Pollutant	Primary or Secondary	Averaging Time	Level	Form
Carbon Monoxide	Primary	8-hour	9 parts per million (ppm)	Not to be exceeded more than once per year
		1 -hour	35 ppm	
Lead	Primary and Secondary	Rolling 3-month average	0.15 micrograms per cubic meter ($\mu g/m^3$) [a]	Not to be exceeded
NO_2	Primary	1 -hour	100 parts per billion (ppb)	98th percentile of 1-hour daily maximum concentrations in a calendar year, averaged over 3 years
	Primary and Secondary	Annual	0.053 ppm[b]	Annual mean
Ozone	Primary and Secondary	8-hour	0.070 ppm[c]	Annual fourth-highest daily maximum 8-hr concentration, averaged over 3 years
$PM_{2.5}$	Primary	Annual	9.0 $\mu g/m^3$	Annual mean, averaged over 3 years
	Secondary	Annual	15.0 $\mu g/m^3$	
	Primary and Secondary	24-hour	35 $\mu g/m^3$	98th percentile, averaged over 3 years
PM_{10}	Primary and Secondary	24-hour	150 $\mu g/m^3$	Not to be exceeded more than once per year on average over 3 years
Sulfur Dioxide	Primary	1 -hour	75 ppb [d]	99th percentile of 1-hour daily maximum concentrations in a calendar year, averaged over 3 years
	Secondary	1 -year	10 ppb [e]	Annual mean, averaged over 3 years
Notes:				

TABLE 9.1-2
NAAQS for Criteria Pollutants

Pollutant	Primary or Secondary	Averaging Time	Level	Form

[a] Final rule signed October 15, 2008. The 1978 lead standard (1.5 micrograms per cubic meter [µg/m3] as a quarterly average) remains in effect until one year after an area is designated for the 2008 standard, except that In areas designated nonattainment for the 1978 standard, the 1978 standard remains in effect until implementation plans to attain or maintain the 2008 standard are approved.

[b] The official level of the annual NO_2 standard is 0.053 parts per million (ppm), equal to 53 parts per billion (ppb), which is shown here for the purpose of clearer comparison to the 1-hour standard.

[c] Final rule signed October 1, 2015, and effective December 28, 2015. The previous (2008) ozone standards additionally remain in effect in some areas. Revocation of The previous (2008) ozone standards and transitioning to the current (2015) standards will be addressed in the implementation rule for the current standards.

[d] The previous SO2 standards (0.14 ppm 24-hour and 0.03 ppm annual) will additionally remain in effect in certain areas: (1) any area for which it is not yet 1 year since the effective date of designation under the current (2010) standards, and (2) any area for which an implementation plan providing for attainment of the current (2010) standard has not been submitted and approved and which is designated nonattainment under the previous SO_2 standards or is not meeting the requirements of a SIP call under the previous SO2 standards (40 CFR 50.4(3)). A SIP call is an EPA action requiring a state to resubmit all or part of its State Implementation Plan to demonstrate attainment of the required NAAQS.

[e] The 2024 rulemaking added a new annual secondary NAAQS in 40 CFR Part 50 without altering the 1971 secondary 3-hour SO2 NAAQS. Note that some areas may have certain continuing implementation obligations.

(Source: 40 Code of Federal Regulations (CFR) Part 50 and Federal Register citations provided in table).

9.1.1.3 Existing Ambient Air Quality - Attainment Status

To understand existing ambient air quality, attainment status of NAAQS within Project area was conducted. Air Quality Control Regions (AQCR) are areas for which implementation plans describe how ambient air quality standards will be achieved and maintained. AQCRs are defined by the EPA and state agencies in accordance with the Clean Air Act (CAA) of 1970. The 1977 CAA Amendments in Section 107 require the EPA and states to identify by category those AQCRs meeting and not meeting the NAAQS. Areas meeting the NAAQS are designated as "attainment areas". The attainment status designations for the AQCRs appear in 40 Code of Federal Regulations (CFR) Part 81 (Office of Federal Register 2026). Richland Parish is part of the Monroe-El Dorado Interstate AQCR. The attainment status of the region, in addition to a Project's emission rates, determines the regulatory review process for each proposed project. **Table 9.1-3** summarizes the attainment status for specific pollutants within Richland Parish where emissions from the Project will occur.

TABLE 9.1-3
Attainment Status of Richland Parish

Air Pollutant	Richland Parish, Louisiana
Sulfur Dioxide (SO_2)	Unclassified/Attainment
Carbon Monoxide (CO)	Unclassified/Attainment
Nitrogen Dioxide (NO_2)	Unclassified/Attainment
Ozone (1997 8-hr standard)	Unclassified/Attainment
Ozone (2008 8-hr standard)	Unclassified/Attainment
Ozone (20015 8-hr standard)	Unclassified/Attainment
PM_{10}	Unclassified/Attainment
$PM_{2.5}$ (1997 annual & 24-hr standard)	Unclassified/Attainment

TABLE 9.1-3 Attainment Status of Richland Parish	
Air Pollutant	**Richland Parish, Louisiana**
$PM_{2.5}$ (2012 annual standard)	Unclassified/Attainment
$PM_{2.5}$ (2006 24-hr standard)	Unclassified/Attainment
Lead (Pb)	Unclassified/ Attainment

9.1.1.4 Regulatory Applicability

Authorization to begin construction activities and initially operate a new or modified source must be obtained by complying with key regulatory elements. This section of the report addresses the various, potentially applicable programs and requirements.

Relevant Applicable Federal Standards

The CAA is the basic federal statute governing air pollution in the United States. The provisions of the Act that are potentially relevant to the Project include:

- New Source Review (NSR) program – The NSR program is a construction permitting program that applies to both major and minor stationary sources. The NSR permitting regulations for new major sources and major modifications to existing major sources are comprised of two programs: 1) Prevention of Significant Deterioration (PSD); and 2) Nonattainment NSR (NNSR).
- Title V Operating Permit program – The Tite V program consolidates state and federal requirements applicable to major sources into a single comprehensive operating permit for the purposes of facilitating ongoing compliance. Sources with a potential to emit (PTE) of 100 tons per year (tpy) or more for criteria pollutants, 25 tpy or more for total hazardous air pollutants (HAPs), or 10 tpy or more for individual HAP are considered major sources for which applicants must obtain a Title V Operating Permit.

Emissions associated with the Project will be primarily from construction activities. The additional sources that will be added on a continuous operational basis have emissions that are below the de minimis thresholds. Therefore, neither of these federal programs apply to the Project.

- Standards of Performance for New Stationary Sources (40 CRP Part 60):
    - Subpart OOOOb – Standards for Performance for Crude Oil and Natural Gas Facilities for Which Construction, Modification or Reconstruction Commences After December 6, 2022.

Subpart OOOOb is applicable to well facilities, centrifugal compressor facilities, reciprocating compressor facilities, process controlled facilities, storage vessel facilities, sweetening unit facilities, pump facilities, and fugitive emission component facilities. The meter stations are not well, centrifugal compressor, reciprocating compressor, process controlled, sweetening unit or pump facilities. A storage vessel affected facility is a tank battery that has potential emissions greater than 6 tpy of volatile organic compounds (VOC) and 20 tpy of methane (CH_4) (cumulative emissions from all storage tanks at a site). Emissions from the storage tanks at the meter stations are below these thresholds. Fugitive components affected facility is the collection of fugitive components at a well site, centralized production facility or compressor station. The meter stations do not meet the definition of a well site, centralized production facility or compressor station; therefore, the meter stations are not subject to Subpart OOOOb.

Relevant Applicable State Standards

Louisiana Department of Environmental Quality (LDEQ) states in Louisiana Administrative Code (LAC) 33:III.501.B.5 (Table 1) that construction activity does not require permitting. In addition, Table 1 outlines that emergency generators used only during power outages and that operate no more than 500 hours per year do not require permitting. Table 1 also notes that in general facilities with potential air emissions less than five tpy of any regulated air pollutant as defined by the CAA, less than 15 tpy of all such defined pollutants combined, and less than the minimum emission rate for each toxic air pollutant do not require an air permit in order to operate. Since Project's operational emissions per meter station fall below the specified thresholds (as noted in **Section 9.1.3** and in **Appendix H**), no permit is required for each separate operating facilities (meter stations).

9.1.1.5 Class I Areas

Federal Class I areas are established by Congress and afforded special protections under the CAA (e.g., wilderness areas, national parks). Class I areas are allowed the smallest degree of air quality deterioration through NSR/PSD permitting, and special considerations must be made in the NSR permitting process when a Class I area is located close to a proposed site. There are no Class I area within 100 kilometers (~62 miles) of the proposed Project (EPA 2026c). The closest Class I area is Carney Creek Wilderness in Arkansas and is approximately 290 kilometers to the northwest of the Project.

9.1.1.6 General Conformity

The proposed Project is also subject to the General Conformity Rule in the CAA. The CAA mandates the General Conformity Rule to ensure that actions taken by federal agencies in non-attainment areas and maintenance areas do not interfere with a state's plans to meet attainment of the NAAQS (EPA 2026a). The General Conformity Rule divides the air conformity process into two parts, including applicability analysis and determination. The applicability analysis process (40 CFR §93.153) requires federal agencies to determine if proposed actions within NAAs and maintenance areas will increase emissions of criteria pollutants above preset threshold levels (Office of Federal Register 2023). The applicability thresholds vary based on the severity of the non-attainment area. De minimis levels are the minimum thresholds for the rates of total direct and indirect emissions of a criteria pollutant by a federal action in non-attainment or maintenance areas which require a conformity determination. These thresholds are presented in **Table 9.1-4**. Since Richland Parish is in an attainment area for all pollutants, comparison to the General Conformity thresholds is not required.

TABLE 9.1-4
***De Minimis* General Conformity Thresholds**

Pollutant/Area Type	Tons/Year
Ozone (VOC or NO_x):	
Serious NAAs	50
Severe NAAs	25
Extreme NAAs	10
Other Ozone NAA's areas outside an ozone transport region (OTR)	100
Ozone (NO_x):	
Marginal and moderate NAAs inside an OTR	100
Maintenance	100
Ozone (VOC):	
Marginal and moderate NAAs inside an OTR	50

TABLE 9.1-4 ***De Minimis* General Conformity Thresholds**	
Pollutant/Area Type	**Tons/Year**
Maintenance within an OTR	50
Maintenance outside an OTR	100
CO, SO_2 and NO_2:	
All NAAs and maintenance	100
PM_{10}:	
Serious NAAs	70
Moderate NAAs and maintenance	100
$PM_{2.5}$ (direct emissions, SO_2 as precursor or NO_x as precursor):	
Serious NAAs	70
Moderate NAAs and maintenance	100
Lead (Pb):	
All NAAs and maintenance	25

Source: 40 CFR § 93.153 (Federal Register 2023)

9.1.2 Construction Impacts

Air quality impacts associated with construction of the proposed Project will include emissions from fossil fuel-fired construction equipment, fugitive dust from earth disturbance and vehicles traveling on unpaved roads. All air quality impacts during construction will generally be temporary and localized. Large earth-moving equipment and other vehicles that are powered by diesel or gasoline engines are sources of combustion-related emissions including criteria pollutants, greenhouse gases (GHG), and small amounts of hazardous air pollutants (HAP). ETC Tiger will not use open burning to dispose of cleared vegetation. To accommodate unanticipated construction schedule delays, conservative assumptions were used to estimate the Project's anticipated construction emissions, which are described below.

Exhaust emissions of NOx, CO, VOCs, SO_2, PM_{10}, $PM_{2.5}$, GHGs, and HAPs from construction equipment and vehicle engines used during Project construction have been estimated based on the anticipated types of equipment and the levels of use. Annual average emission factors [grams per horsepower hour (g/hp-hr)] for NOx, CO, VOC, PM_{10}, $PM_{2.5}$, SO_2, and carbon dioxide (CO_2) for non-road equipment engines were obtained using EPA's Mobile Vehicle Emissions Simulator (MOVES) version MOVES5. Emission factors for CH_4 were obtained by using hydrocarbon (HC) emission factor from MOVES5 and subtracting ratio of 0.984 and Total HC (THC) from THC for diesel fuel, while using ratio of 0.9 for gasoline fuel. This method is from "Conversion Factors for Hydrocarbon Emission Components", July 2010, EPA-420-R-10-015. Emission factors for nitrous oxide (N_2O) were obtained from Table C-2 of 40 CFR Part 98, Subpart C for General Stationary Fuel Combustion Sources. Total GHG emission factors on a carbon dioxide equivalent (CO_2e) basis were determined using the global warming potential (GWP) obtained from 40 CFR Part 98, Subpart A.

Emission factors (grams per vehicle mile traveled [g/VMT]) for on-road vehicles were obtained from MOVES5. Emission factors were obtained for NOx, CO, PM_{10}, $PM_{2.5}$, VOC (as total hydrocarbon), CO_2, CH_4, N_2O, total GHG on a CO_2e basis, and HAPs.

Construction activities will emit particulate matter (total suspended particulates, PM_{10}, and $PM_{2.5}$) in the form of fugitive dust. Fugitive dust will result from land clearing, grading, excavation, concrete work, and vehicle traffic on paved and unpaved roads. The amount of dust generated will be a function of construction

activities, soil type, moisture content, wind speed, frequency of precipitation, vehicle traffic, vehicle types, and roadway characteristics. Emissions will typically be greater during dry periods and in areas of fine-textured soils subject to surface activity. The magnitude of fugitive dust emissions is a function of construction vehicle weight and speed.

Worst-case fugitive dust emissions for PM_{10} and $PM_{2.5}$ were calculated based on the EPA's AP-42, Fifth Edition Compilation of Air Pollutant Emission Factors (AP-42) (EPA 2026b). The methodologies in Sections 13.2.1 and 13.2.2 of the AP-42 were used to estimate fugitive dust emissions from vehicular travel on paved roads and unpaved roads respectively. Additionally, emissions were calculated using a EPA report on estimating particulate matter from construction operations (EPA 1999). This calculation is meant to be general and is used to cover a wide range of construction activities over the area of land that is disturbed from the Project. Overall, these emission factors may over-estimate potential fugitive dust generated by construction of the Project.

Although dust impacts will be temporary, dust suppression techniques will be employed to minimize these emissions during construction. ETC Tiger will implement the following mitigation measures on an as-needed basis, as detailed in the *Fugitive Dust Control Plan* (**Appendix C**, Attachment C-5):

- Apply water (or other approved dust suppressant), as necessary, to construction and demolition sites which may create significant airborne dust;
- Reduce vehicle speed on unpaved roads;
- Construct and maintain construction entrances to prevent tracking mud and soil onto paved roads; and
- Pave/gravel roadways, where possible, and maintain them in a clean condition.

The conservative assumptions, data, and emission factors used to estimate emissions from construction equipment engines and vehicles are provided in **Appendix H**, along with more detailed listings of calculated emissions estimates.

Pending receipt of applicable federal permit authorizations and a Notice to Proceed from FERC, a construction start date in late July 2027 is anticipated, which is based on the need to meet an in-service date in January 2028.

Total construction emissions were calculated for the Project area to determine the total estimated emissions for the AQCR and nonattainment/maintenance areas. A summary of emission estimates for construction of the Project is presented in **Table 9.1-5**.

TABLE 9.1-5
Summary of Potential Construction Emissions from the Project (tons)[a]

Source / Area	NO_x	CO	VOC	PM_{10}	$PM_{2.5}$	SO_2	Total HAPs	GHGs[b]
Non-Road Equipment - Franklin Farms Lateral	6.82	2.21	0.61	0.43	0.42	0.02	0.23	9,094.66
Non-Road Equipment – FF Gulf Run Bi Di MS	1.31	0.48	0.13	0.09	0.08	0.004	0.05	1,531.87
Non-Road Equipment – FF Gulf South Receipt MS	1.31	0.48	0.13	0.09	0.08	0.004	0.05	1,531.87
Non-Road Equipment – FF Gulf Crossing Receipt MS	1.31	0.48	0.13	0.09	0.08	0.004	0.05	1,531.87
Non-Road Equipment – FF MEP Receipt MS	1.31	0.48	0.13	0.09	0.08	0.004	0.05	1,531.87
Non-Road Equipment - Franklin Farms Delivery MS	1.31	0.48	0.13	0.09	0.08	0.004	0.05	1,531.87
Non-Road Equipment - FF Bee Bayou Delivery MS	1.31	0.48	0.13	0.09	0.08	0.004	0.05	1,531.87
Non-Road Equipment – FF Trio Farm Delivery MS	1.31	0.48	0.13	0.09	0.08	0.004	0.05	1,531.87
Construction Activity Fugitive Dust	N/A	N/A	N/A	451.34	67.70	N/A	N/A	N/A
Roadway Fugitive Dust	N/A	N/A	N/A	171.70	37.34	N/A	N/A	N/A
On-Road Equipment	53.83	183.62	10.24	4.02	2.78	0.13	0.70	21,641.29
Total Project Construction Activities	**69.81**	**189.2**	**11.77**	**628.11**	**108**.84	**0.18**	**1.27**	**41,459.05**

a. All construction emissions are expected to occur for a 7-month period.

TABLE 9.1-5 Summary of Potential Construction Emissions from the Project (tons)[a]								
Source / Area	**NO_x**	**CO**	**VOC**	**PM_{10}**	**$PM_{2.5}$**	**SO_2**	**Total HAPs**	**GHGs[b]**
b. As CO_2e (includes CO_2, CH_4 and N_2O).								

The air emissions and fugitive dust that will be generated during construction of the Project will primarily be limited to the immediate vicinity of the construction activities. Construction emissions will be temporary, intermittent, and minor.

9.1.3 Operational Impacts

Fugitive emissions from operation of the Franklin Farms Lateral pipeline, and emissions associated with emergency generators and condensate tanks at meter stations may result in small amounts of pollutants. Detailed emissions and calculation methodology are provided in **Appendix H**.

As discussed in the previous sections, operational emissions per meter station fall below the de-minimis thresholds for both federal and state level reporting requirements and do not require an air permit. Emissions from the operation of the Project listed in **Table 9.1-6** are not anticipated to be significant nor result in impacts on local or regional air quality.

TABLE 9.1-6
Summary of Potential Operational Emissions from the Project (tons)

Source / Area	NO_x	CO	VOC	PM_{10}	$PM_{2.5}$	SO_2
Franklin Farms Delivery MS	0.03	1.28	0.99	1.65E-04	1.65E-04	2.34E-04
FF MEP Receipt MS	0.03	1.28	0.93	1.65E-04	1.65E-04	2.34E-04
FF Gulf Crossing Receipt MS	0.03	1.28	0.93	1.65E-04	1.65E-04	2.34E-04
FF Trio Farm Delivery MS	0.03	1.28	0.99	1.65E-04	1.65E-04	2.34E-04
FF Bee Bayou Delivery MS	0.03	1.28	0.99	1.65E-04	1.65E-04	2.34E-04
FF Gulf South Receipt MS	0.03	1.28	0.93	1.65E-04	1.65E-04	2.34E-04
FF Gulf Run Bi Di MS	0.03	1.28	0.98	1.65E-04	1.65E-04	2.34E-04
Total Operational Activities	**0.19**	**8.96**	**6.72**	**0.001**	**0.001**	**0.002**

9.1.4 Air Quality Mitigation Measures – Construction Activities

As described above, construction will generate potential air emissions of NOx, CO, VOC, $PM_{10}/PM_{2.5}$, SO_2, HAPs, and GHGs. Construction emissions will be temporary and of limited duration and will not significantly increase ambient air pollutant concentrations. Potential impacts will be mitigated and minimized as described below.

Exhaust emissions from diesel- and gasoline-fueled construction equipment and vehicle engines will be minimized by federal design standards imposed at the time of manufacture of the vehicles and will comply with EPA mobile and non-road emission regulations (40 CFR Parts 85, 86, and 89). Emissions also will be controlled by purchasing commercial gasoline and diesel fuel products whose specifications are controlled by federal and state air pollution control regulations applicable to fuel suppliers and distributors. Contractors and employees will also be encouraged to minimize vehicle and equipment idling time to the extent practical during construction activities.

Construction activities, coupled with certain weather conditions, can create situations that promote the production and dispersal of dust. The amount of dust generated is a function of construction activities, soil type, moisture content, wind speed, frequency of precipitation, vehicle traffic, vehicle types, and roadway characteristics. Although dust impacts will be temporary and minor, dust suppression techniques will be employed to control these emissions during construction.

9.2 NOISE QUALITY

The following sections describe potential sound pressure level (noise) effects due to construction and operation of the Project. This includes potential effects to nearby NSAs. Examples of NSAs include schools, hospitals, residences, etc. The Project does not include any new compressor stations but will include the installation of new meter stations. A discussion of potential noise effects from typical pipeline construction and the proposed meter stations is included below, as well as Lateral operations.

9.2.1 Noise Regulations

ETC Tiger will adhere to all applicable federal, state, county, or local noise ordinance limits applicable to the Project. At this point, no applicable state, county, or local noise ordinances were identified.

9.2.2 Existing Noise Sensitive Areas

Table 9.2-1 list the existing NSAs located within a half mile of the meter station and provide their approximate distance and direction from each station.

TABLE 9.2-1
Nearby Noise Sensitive Areas

NSA	Description	Distance (feet)	Direction
1	Residence	250	SW
2	Residence	550	SE
3	Residence	1,500	SE
4	Residence	1,700	S
5	Residence	1,600	NW
6	Residence	1,300	S
7	Residence	2,200	SW
8	Residence	1,700	S
9	Residence	1,200	S
10	Residence	2,000	SE
11	Residence	2,200	E
12	Residence	2,300	E
13	Residence	1,900	SE
14	Residence	1,600	E
15	Residence	650	NW
16	Residence	1,200	N

17	Residence	2,300	N
18	Residence	2,200	N
19	Residence	2,200	N

9.2.3 Existing Noise Levels

A sound level survey was conducted at the site property lines and at the nearby NSAs to measure the existing sound levels near each Project meter station (**Appendix I**). The sound level measurements were taken for approximately ten minutes at each location during both daytime and nighttime periods.

The measurements were taken using a Larson Davis Model 831C sound level meter that meets the requirements of the American National Standards Institute (ANSI) Standards for Type I instruments. The sound level meter was calibrated before and after each monitoring period using a CAL200 acoustic calibrator. The microphone was positioned according to the ANSI Standard on a tripod 1.5 meters above ground. 7.5 meters from large reflecting surfaces, and at least 1.5 meters from tall trees.

The measured sound levels at each location are summarized in the following tables.

TABLE 9.2-2
Existing NSA Sound Levels

MP	Daytime Leq	Daytime L90	Nighttime Leq	Nighttime L90	Day-Night (Ldn)
1	51.3	45.0	45.2	41.2	53.2
2	50.1	45.0	49.3	48.7	55.8
3	52.2	48.5	55.7	55.0	61.8
4	57.9	53.5	60.7	52.0	66.8
5	48.8	42.8	53.6	53.0	59.6
6	57.6	42.5	61.2	49.4	67.2
7	60.6	43.4	70.2	52.5	76.0
8	64.6	42.3	63.0	37.9	69.7
9	60.3	40.3	61.2	37.3	67.5
10	61.0	41.0	71.0	49.3	76.8
11	45.4	34.3	61.4	37.2	67.2
12	52.7	30.6	57.6	38.1	63.6
13	47.0	40.3	59.0	38.2	64.8
14	59.5	49.5	58.8	48.8	65.3
15	59.8	48.0	52.4	42.5	61.0
16	64.0	50.3	56.2	43.4	65.0
17	66.8	46.9	64.4	40.3	71.2
18	65.1	41.7	66.0	53.0	72.3
19	49.9	42.1	50.0	45.2	56.4

9.2.3.1 Vibration

No blasting is anticipated for the construction of this Project.

9.2.4 Construction Noise Impacts

The level of construction noise at any one time will vary over the course of the entire construction period and will be highly dependent on the type of equipment being used. Concentrated construction activities will only occur at periodic intervals.

Short-term increases in sound levels could occur during construction. Only standard construction equipment will be used in the construction of the meter stations, with no dynamic compaction expected. Construction will take place between the hours of 7 AM and 7 PM However, weather conditions, site conditions, specialized construction techniques, emergencies, or other atypical circumstances may necessitate nighttime work or extended work on Sundays. To be conservative in the construction noise calculations, all Project construction equipment was included in the noise model as operating at the same time.

Based on the Project equipment usage predictions, a sound level calculation was performed using the Federal Highway Administration's Roadway Construction Noise Model version 1.1 (Federal Highway Administration, 2006). **Tables 9.2-3** summarize the results of the construction noise modeling, which was completed using what is assumed to be the loudest 20 pieces of construction equipment.

TABLE 9.2-3
Construction Noise Modeling Results

NSA	Existing Leq	Modeled Leq	Total Leq	Temporary Increase Leq
1	55.8	75.5	75.5	19.7
2	53.2	68.7	68.8	15.6
3	53.2	60	60.8	7.6
4	59.6	58.9	62.3	2.7
5	67.2	59.4	67.9	0.7
6	67.2	61.2	68.2	1.0
7	76.8	56.6	76.8	0.0
8	67.5	58.9	68.1	0.6
9	67.5	61.9	68.6	1.1
10	72.3	57.5	72.4	0.1
11	69.7	56.6	69.9	0.2
12	69.7	56.2	69.9	0.2
13	67.2	57.9	67.7	0.5
14	64.8	59.4	65.9	1.1
15	65.0	67.2	69.2	4.2
16	65.0	61.9	66.7	1.7
17	61.8	56.2	62.9	1.1
18	61.8	56.6	62.9	1.1
19	61.8	56.6	62.9	1.1

As shown in the above tables, there could be significant temporary increases in sound levels at nearby NSAs during Project construction activities. However, the modelled results are extremely conservative, in that the model assumes that all proposed construction equipment is operating simultaneously and that there will be no sound mitigation from intervening structures and geographic features such as trees and elevation

changes. The actual temporary sound level increases will be significantly lower than those predicted by the Roadway Construction Noise Model.

9.2.5 Noise Mitigation Measures

The construction activities related to the Project will have short-term noise impacts to nearby noise sensitive areas. The following temporary noise mitigation measures could be adopted as needed to reduce temporary Project noise impacts during construction.

- Operate equipment according to manufacturer's recommendations to reduce excess sound;
- Utilize equipment with muffled exhausts;
- Minimize equipment idling;
- Minimize the use of backup beepers;
- Schedule construction activities for periods when nearby residents are less likely to be home; and
- Utilize temporary sound barriers

Predicted sound levels from Project construction exceed the recommended FERC guideline of 55 A-weighted decibels (dBA); however, measured existing background sound levels already exceed this guideline at every NSA except for NSAs 2 and 3, which were just below 55 dBA.

9.2.6 Conclusions

The Project site is in a residential area near numerous residences. The results of the ambient sound survey have indicated that the existing sound levels at and near the Project site currently exceed the sound levels recommended by FERC at several locations. As a result, it is likely that the Project will be required to demonstrate using noise mitigation techniques during construction. The implementation of noise mitigation measures, such as sound walls, may be required to make such a demonstration.

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ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 10
Alternatives

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 10 FILING REQUIREMENTS	
Information	**Section Reference**
1. Address the "no action" alternative. (§ 380.12 (*l*) (1))	Section 10.1
2. For large Projects, address the effect of energy conservation or energy alternatives to the Project. (§ 380.12 (l) (1))	Sections 10.2 and 10.3
3. Identify system alternatives considered during the identification of the Project and provide the rationale for rejecting each alternative. (§ 380.12 (*l*) (1))	Section 10.4
4. Identify major and minor route alternatives considered to avoid impact on sensitive environmental areas (e.g., wetlands, parks, or residences) and provide sufficient comparative data to justify the selection of the proposed route. (§ 380.12 (*l*) (3))	Section 10.6
5. Identify alternative sites considered for the location of major new aboveground facilities and provide sufficient comparative data to justify the selection of the proposed site. (§ 380.12 (*l*) (3))	Section 10.7
Information Often Missing and Resulting in Data Requests	
Ensure that project objectives that serve as the basis for evaluating alternatives are consistent with the purpose and need discussion in Resource Report 1.	Section 10.0
Identify and evaluate alternatives identified by stakeholders.	Section 10.6.2
Clearly identify and compare the corresponding segments of route alternatives and route variations with the segments of the proposed route that they would replace if adopted.	Section 10.6

RESOURCE REPORT NO. 10
ALTERNATIVES

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Appendix E-1 – Landowner/Stakeholder List (Submitted as "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE"*)

F Wetland and Waterbody Delineation Report and Listed Species Report

Appendix F-1 – Wetland and Waterbody Delineation Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

Appendix F-2 – Listed Species Report (Portions Submitted as "*CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE*")

G Cultural Resource Assessment Survey Report (*Submitted as* "*CUI/PRIVILEGED INFORMATION – DO NOT RELEASE*")

H Air Quality Calculations (Construction Emissions)

I Sound Study Report

ABBREVIATIONS AND ACRONYMS

Certificate	Certificate of Public Convenience and Necessity
Dth/day	dekatherms per day
ETC Tiger	ETC Tiger Pipeline, LLC
FERC or Commission	Federal Energy Regulatory Commission
FF	Franklin Farms
MEP	Midcontinent Express Pipeline LLC
NGA	Natural Gas Act
NLCD	National Land Cover Database
NWI	National Wetland Inventory
Project	Franklin Farms Project

10.0 ALTERNATIVES

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, an interconnect site, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project in June 2027, and following the receipt of all applicable approvals, anticipating construction activities to begin in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

Resource Report No. 10 addresses the alternatives analysis that was undertaken for the Project. Possible alternatives to the Project include the No-Action alternative, energy conservation, energy alternatives, system alternatives, design alternatives, and route or site alternatives. The proposed Project will supply natural gas capacity to Entergy Louisiana, LLC's new generating facilities in Richland Parish, Louisiana. Any viable alternative would need to meet the Project's purpose and need. To be a preferred alternative, it would also need to provide a significant environmental advantage over the proposed Project with consideration of the landowner(s), technical feasibility, schedule, and cost.

10.1 PURPOSE AND NEED

The proposed Project is designed to provide 250,000 dekatherms per day (Dth/day) of firm transportation service mainline capacity and 1,000,000 Dth/day of capacity on a newly constructed lateral to meet the increasing market demand for industrial customers in Richland Parish, Louisiana. ETC Tiger has executed a Precedent Agreement with Entergy Louisiana, LLC for the full capacity of the Project. The Project will supply natural gas to Entergy Louisiana, LLC's new Combined Cycle Combustion Turbine generating facilities which will be utilized for power generation to support a data center campus in Richland Parish, Louisiana. In support of powering artificial intelligence data centers consistent with the current Administration and FERC priorities, the Project expands natural gas infrastructure which will allow greater amounts of natural gas to be delivered to new electric generator facilities, such as the natural gas-fired turbines that will generate electricity for the new data center in Richland Parish, Louisiana. The proposed Project will enable ETC Tiger to meet its contractual obligations to Entergy Louisiana, LLC, and to continue fulfilling existing contractual obligations thereby ensuring that the Project has no impact on service to ETC Tiger's existing customers.

10.2 NO-ACTION ALTERNATIVE

Under the No Action alternative, the proposed Project would not be constructed and all impacts associated with construction and operation of the Project would not occur. However, the purpose and need of the Project would not be fulfilled. Under the No-Action alternative, ETC Tiger would be unable to meet Entergy Louisiana, LLC's requirements for natural gas for their new generating facilities. The Project will leverage nearby, existing pipeline systems to deliver natural gas to Entergy Louisiana, LLC's sites; thereby, minimizing the Project footprint and associated environmental impacts. Any other means of meeting the Project's purpose and need would not provide a significant environmental advantage.

10.3 ENERGY CONSERVATION

Energy conservation would not provide a viable alternative to the proposed Project. The goal of energy conservation is to use fewer non-renewable natural resources and increase efficiencies to reduce energy consumption and minimize environmental pollution while still meeting energy needs. However, Entergy

Louisiana, LLC's customers have an energy demand from Entergy Louisiana, LLC's new generating facilities which current energy conservation cannot meet. Therefore, providing natural gas supply to Entergy Louisiana, LLC's new generating facilities is needed, making energy conservation an unviable alternative.

10.4 ENERGY ALTERNATIVES

ETC Tiger evaluated the potential for other energy sources to meet the objectives of the Project. A viable alternative energy source would need to be capable of providing the equivalent energy to Entergy Louisiana, LLC's new generating facilities as the proposed Project. Potential alternative energy sources to natural gas include traditional fuels (e.g., coal and oil), nuclear energy, and renewable energy sources (e.g., wind, solar, hydroelectric, biomass, and tidal and wave). Depending on the location of the identified alternative energy source, new infrastructure would be required, including transmission facilities, which would result in their own set of impacts that could be less than, equal to, or greater than those associated with the proposed Project. Each alternative energy source is associated with its own set of advantages and drawbacks and no viable energy alternatives to the proposed Project were identified.

10.4.1 TRADITIONAL ENERGY SOURCES

Oil and coal are abundant and readily available and, when combusted, have higher air pollution emissions than other energy sources. Combustion of natural gas emits approximately half as much carbon dioxide per unit energy than coal and approximately 30 percent less than oil. Further, electricity generated from liquid petroleum products and coal tends to involve higher costs. Moreover, increased use of oil and coal would result in additional environmental impacts associated with their extraction (e.g., mining, drilling) and transportation.

10.4.2 NUCLEAR ENERGY

Nuclear power can be produced in massive quantities with little or no air emissions. However, a historically unfavorable regulatory climate exists for nuclear projects due to operational safety and environmental concerns associated with the disposal of radioactive waste. Additionally, the time required to design, permit, and construct a nuclear generation facility would be extensive and significantly greater than the amount of time required to design, permit, and construct the proposed Project.

10.4.3 RENEWABLE ENERGY SOURCES

While renewable energy sources are expected to continue to meet part of increased energy consumption demands overall, renewable energy capacity is not expected to be able to support all the specific requirements for the demand proposed to be met by this Project. Additionally, due to infrastructure constraints, renewable energy sources cannot easily replace natural gas nor meet the necessary timeframe. Substantial new renewable infrastructure (power generation and transmission) would be needed to supply the equivalent energy value of natural gas.

10.5 SYSTEM ALTERNATIVES

System alternatives would involve the use of existing or other proposed natural gas transmission systems in lieu of construction of some or the entire Project. Although existing pipeline systems are in place in the region to provide natural gas services to customers, they do not have the current capacity to deliver the natural gas required to meet the Project purpose and need. Therefore, modifications to and expansions of these existing facilities would be required to deliver the proposed Project gas volume. Such modifications or additions would result in environmental impacts that could be less than, similar to, or greater than those associated with the proposed Project.

10.5.1 GULF RUN

Gulf Run's existing pipeline system connects production basins in the Haynesville supply basin with market areas in the Perryville Hub and south into the Gulf market area. Although the Gulf Run system serves a portion of the Project's proposed service area in north Louisiana, it does not have sufficient capacity to deliver the natural gas required to meet the Project need without significant looping, pipeline expansion, and additional compression requirements. Because the Gulf Run system does not have sufficient capacity to meet the Project need and Gulf Run has not entered into the long-term binding precedent agreements executed by ETC Tiger, the Gulf Run system is not a reasonable alternative.

10.5.2 GULF SOUTH

Gulf South/Gulf Crossing's existing Gulf South pipeline system transports natural gas from Texas and the Haynesville supply basin to market areas throughout Louisiana, Alabama, Mississippi, and LNG exports in the Gulf of Mexico. The Gulf South system does not have sufficient capacity to deliver the natural gas required to meet the Project need without significant looping, pipeline expansion, and/or additional compression. Because the Gulf South system does not have sufficient capacity to meet the Project need and Gulf South has not entered into the long-term binding precedent agreements executed by ETC Tiger, the Gulf South system is not a reasonable alternative.

10.5.3 MIDCONTINENT EXPRESS PIPELINE

Midcontinent Express Pipeline LLC (MEP) is jointly owned by Energy Transfer and Kinder Morgan. The MEP system transports natural gas from Oklahoma to market areas in Alabama. The MEP system does not have sufficient capacity to deliver the natural gas required to meet the Project need without significant looping, pipeline expansion, and/or additional compression. Because the MEP system does not have sufficient capacity to meet the Project need and MEP has not entered into the long-term binding precedent agreements executed by ETC Tiger, the MEP system is not a reasonable alternative.

10.6 DESIGN ALTERNATIVES

Alternative designs for the Franklin Farms Lateral and FF Trio Farm Lateral were evaluated during the planning and design phase of the Project to achieve the requested gas supply volume at the minimum delivery point pressure. This included conducting a detailed hydraulic analysis to determine the necessary pipeline location and diameter to meet the proposed Project's purpose and need. The proposed Project as described in Resource Report No. 1 *Project Description* provides the best combination of additional pipeline mileage and overall construction and operations/maintenance costs.

10.7 PIPELINE ROUTE ALTERNATIVES

ETC Tiger evaluated major and minor route alternatives for the Franklin Farms Lateral and FF Trio Farm Lateral. Major route alternatives are alignments that may have different starting and ending points and/or deviate for an extended distance (i.e., miles). Minor route alternatives are alignment deviations over a relatively short distance in consideration of a specific resource, constructability issue (e.g., topographic feature), or landowner concern.

10.7.1 MAJOR ROUTE ALTERNATIVES

The intent of initial routing efforts was to identify the most environmentally sound, technically efficient, and cost-effective route for transportation of natural gas from the existing Gulf Run Pipeline and Tiger Pipeline to Entergy Louisiana, LLC's generating facilities. ETC Tiger evaluated one major route alternative

during the development of the proposed Project route. The proposed and alternative routes are discussed below and shown in **Appendix A-15**. A comparison of the routes is provided in **Table 10.6-1**.

TABLE 10.7-1
Summary of Major Route Alternatives

Criteria		Preferred Route [a]	Alternative No. 1
Length (miles)	Total	2.6	2.7
	Collocation [b] (Percent)	17.0%	5.0%
Public Lands Crossed (miles)		0.0	0.0
National Land Cover Database (NLCD) (miles)	Agricultural	1.2	0.7
	Developed	0.2	0.1
	Open Land / Barren Land	0.0	0.0
	Forested	<0.05	<0.05
	Shrub / Scrub / Herbaceous	0.0	0.0
	Waterbody	0.0	0.0
	Wetland	1.2	1.9
	Total	2.6	2.7
Delineated Wetland Impacts [c] (miles)	Palustrine Emergent (PEM)	<0.05	0.1
	Palustrine Scrub Shrub (PSS)	0.0	0.0
	Palustrine Forested (PFO)	1.2	1.8
	Lake / Pond	0.0	0.0
	Total	1.2	1.8
Delineated Waterbody Crossings [c] (number)	Perennial Stream	2	2
	Intermittent Stream	0	0
	Canal / Ditch	2	5
Wetland Reserve Easement impacts (acres)		0.0	2.5
USFWS Designated Critical Habitat (miles)		0.0	0.0
Buildings within 50 feet of the centerline [d] (number)		0	0
Prime Farmland Soils (includes farmland of statewide importance, "if drained," and "if protected," and prime farmland classification) (miles)		2.6	2.7
Pipeline Crossings (number)		3	3
Electric Transmission Line Crossings (number)		2	4
Road Crossings (number)		2	0
National Register of Historic Places sites within 0.25-mile (number of crossings)		0	0
Constructability Assessment		Least environmental impacts	More environmental impacts

Notes:

[a] In order to provide an accurate comparison, this column includes impacts from the portion of the Project specific to the Alternative route, not the anticipated impacts from the complete proposed Project.
[b] Collocation includes linear infrastructure within 200 feet of the route based on centerline-to-centerline measurement for a minimum length of 100 feet.
[c] Based on survey data.
[d] Based on aerial interpretation.

10.7.1.1 Franklin Farms Lateral and FF Trio Farm Lateral Interconnection

ETC Tiger evaluated one route alternative for the interconnection between the Franklin Farms Lateral and the FF Trio Farm Lateral. Significant consideration was given to minimizing environmental impacts and impacts to landowners. The alternative route is located just to the east and north of the proposed route. While feasible, ETC Tiger determined that the alternative route was abutting parcels enrolled in permanent Wetland Reserve Easements under the Agricultural Conservation Easement Program administered by the Natural Resources Conservation Service. The proposed route is sited further from and does not impact the Wetland Reserve Easement lands. Additionally, the proposed route is collocated more with existing infrastructure than the alternative which minimizes environmental impacts. Thus, ETC Tiger elected not to proceed with the alternative route since it has greater environmental impacts than the proposed route.

10.7.2 MINOR ROUTE VARIATIONS

In addition to the above analysis of major pipeline route alternatives, ETC Tiger also identified three minor route variations during design of the proposed Franklin Farms Lateral and one minor route variation during design of the proposed FF Trio Farm Lateral, all of which have been incorporated into the proposed routes. Route variations are small shifts identified during the design of a route to address constructability issues, such as topographic features avoid sensitive environmental resources (e.g., wetlands), and/or accommodate landowner concerns. The four minor route variations which were considered in the refinement of the proposed Project route are summarized in **Table 10.6-2** and shown in **Appendix A-16**.

TABLE 10.6-2
Minor Route Variations for the Project

Minor Route Variation		Approximate Milepost [a]		At Landowner Request	Rationale
		Begin	End		
Franklin Farms Lateral	Variation No. 1	3.8	4.1	No	Adjustment to avoid two streams parallel within the construction right-of-way discovered during field surveys (S-T2F-03 and D-T2F-04). Adjustment also eliminates impacts to two wetlands (PFO-T2F-04 and PEM-T2F-04).
	Variation No. 2	10.8	11.4	Yes	Adjustment at landowners request to continue to parallel Fortenberry Road as much as possible to reduce impact to farming operations.
	Variation No. 3	12.5	13.2	Yes	Adjustment in collaboration with landowner on development plans associated with major transmission powerline corridor and location of Franklin Farms Delivery Meter Station.
FF Trio Farm Lateral	Variation No. 4	1.4	1.7	Yes	Adjustment in collaboration with landowner on future development plans for the tract.

TABLE 10.6-2
Minor Route Variations for the Project

Minor Route Variation	Approximate Milepost [a]		At Landowner Request	Rationale
	Begin	End		

Notes:

[a] Milepost of the proposed Franklin Farms Lateral or FF Trio Farm Lateral.

10.7.2.1 Frankin Farms Lateral

Variation No. 1

Variation No. 1 was incorporated into the proposed route to avoid impacts to two streams, a palustrine emergent wetland, and a palustrine forested wetland. This minor route variation deviates to the east for approximately 0.3-mile to avoid the wetland and water features. Without Variation No. 1, the Project would have additional temporary and permanent impacts to wetlands and waters. Therefore, Variation No. 1 was incorporated into the proposed route to avoid and minimize impacts to wetland and water features.

Variation No. 2

Variation No. 2 was incorporated into the proposed route to minimize landowner impacts. This minor route variation deviates to the north along Fortenberry Road for approximately 0.6-mile to reduce impacts to active farming operations. Without Variation No. 2, the Project would have encroached further into the landowner's property where active farming occurs. Therefore, Variation No. 2 was incorporated into the proposed route at the landowner's request to minimize impacts to farming operations.

Variation No. 3

Variation No. 3 was incorporated into the proposed route based on feedback from the landowner regarding new development plans in the area. Variation No. 3 was routed to accommodate a major transmission line corridor currently under development as well as locating the proposed Franklin Farms Delivery Meter Station at the landowner's preferred location. This minor route variation deviates to the north along Fortenberry Road for approximately 0.7-mile to accommodate these siting requirements. Therefore, Variation No. 3 was incorporated into the proposed route at the landowner's request to accommodate development plans.

10.7.2.2 FF Trio Farm Lateral

Variation No. 4

Variation No. 4 was incorporated into the proposed route based on feedback from the landowner regarding future development plans for the tract. This minor route variation deviates to the east and north for approximately 0.3-mile to stay closer to the edge of the tract. Without Variation No. 4, the Project would have encroached further into the landowner's property where active farming occurs. Therefore, Variation No. 4 was incorporated into the proposed route at the landowner's request to allow for the landowner's future development plans.

10.8 ABOVEGROUND FACILITY ALTERNATIVES

ETC Tiger considered various factors when selecting the sites for the proposed meter stations including facility footprint, site use, land availability, access road and/or pipeline lateral lengths required to reach the facility, engineering constraints, costs, and environmental impacts. The seven proposed meter stations were

carefully sited to be collocated with either the Franklin Farms Lateral or the FF Trio Farm Lateral with minimal environmental impacts. Three of the proposed meter stations (FF Bee Bayou Delivery Meter Station, Franklin Farms Delivery Meter Station, and FF Trio Farm Delivery Meter Station) and one interconnect site (FF Interconnect Site) are adjacent to the proposed the Franklin Farms Lateral or the FF Trio Farm Lateral at delivery points while four of the proposed meter stations (FF Gulf Run Bi Di Meter Station, FF Gulf South Receipt Meter Station, FF Gulf Crossing Receipt Meter Station, and FF MEP Receipt Meter Sation) are also adjacent to existing pipeline infrastructure at receipt points. Alternative meter station sites not collocated with the proposed Project pipelines and existing pipeline infrastructure would require additional installation of pipeline laterals to reach the interconnect points. As a result, construction of the laterals would have additional environmental and landowner impacts. Since the proposed meter station sites are located at the intersection of the Project pipelines and the delivery or receipt points, no additional lateral is required for interconnection which minimizes the environmental impact of these facilities. Further, all seven proposed meter station sites are in previously disturbed agricultural land, no tree clearing is required, and no wetlands or waterbodies will be impacted. Based on these considerations, no feasible alternative meter station sites were identified that have fewer environmental impacts than the proposed sites. Thus, no alternative meter station sites were further evaluated.

ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 11
Reliability and Safety

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 11 FILING REQUIREMENTS	
Information	**Section Reference**
1. Describe how the Project facilities would be designed, constructed, operated, and maintained to minimize potential hazard to the public from the failure of Project components as a result of accidents or natural catastrophes. (§ 380.12 (m))	Sections 11.0 and 11.1
Information Often Missing and Resulting in Data Requests	
Identify by milepost and in table form, all U.S. Department of Transportation class locations and areas of concern (for example, high consequence areas) as defined in Title 49 of the Code of Federal Regulations, Section 192.903 for the proposed route, alternative routes, and compressor stations and explain the basis for high consequence area identification.	Table 11.3-1
Discuss the outcome of the consultations with local fire departments and emergency response agencies regarding whether additional equipment, training, and support are needed in the project area.	Section 11.2.3

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RELIABILITY AND SAFETY

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ABBREVIATIONS AND ACRONYMS

Certificate	Certificate of Public Convenience and Necessity
CFR	Code of Federal Regulations
ETC Tiger	ETC Tiger Pipeline, LLC
FERC or Commission	Federal Energy Regulatory Commission
FF	Franklin Farms
HCA	High Consequence Area
Laterals	Franklin Farms Lateral and/or FF Trio Farms Lateral
MAOP	maximum allowable operating pressure
NGA	Natural Gas Act
NSC	National Safety Council
OPS	Office of Pipeline Safety
OQ	Operator Qualification
PHMSA	Pipeline and Hazardous Materials Safety Administration
psig	pounds per square inch gauge
Project	Franklin Farms Project
USDOT	U.S. Department of Transportation

11.0 RELIABILITY AND SAFETY

ETC Tiger Pipeline, LLC (ETC Tiger) is seeking a Certificate of Public Convenience and Necessity (Certificate) from the Federal Energy Regulatory Commission (FERC or Commission) pursuant to Section 7(c) of the Natural Gas Act (NGA) for authorization to construct, own, operate, and maintain certain natural gas pipeline, aboveground, and appurtenant facilities, herein referred to as the Franklin Farms Project (Project), located in Richland Parish, Louisiana. The proposed Project consists of two new natural gas pipeline laterals, seven new meter stations, an interconnect site, and other appurtenant facilities within Richland Parish, Louisiana. ETC Tiger is requesting issuance of a Certificate for the Project in June 2027, and following the receipt of all applicable approvals, anticipating construction activities to begin in July 2027. ETC Tiger plans to place the Project facilities into service on February 1, 2028.

The proposed Project facilities are designed for a Maximum Allowable Operating Pressure (MAOP) of 1440 pounds per square inch gauge (psig) and will enable ETC Tiger to deliver new incremental volume to fulfill the Project's purpose and need. The proposed Project will not change ETC Tiger's system daily capacity.

Resource Report No. 11 describes the reliability and safety aspects of the proposed Project. The purpose of this report is to gauge the potential risk to the public associated with an accident to the existing natural gas system, as modified by the addition of the proposed facilities. Section 11.2 of this resource report identifies the potential for accidents and related hazards based on the actual operating experience of ETC Tiger. The procedures and design features to ensure operational reliability and safety are detailed in Section 11.3. A discussion of the key safety considerations associated with the natural gas pipeline industry is presented in Section 11.4.

ETC Tiger will own, operate, and maintain the new facilities under the procedures and policies that ETC Tiger utilizes for its existing natural gas facilities to avoid undue hazards and effects.

11.1 SYSTEM RELIABILITY AND SAFETY

The Project will fully adhere to United States Department of Transportation (USDOT) regulatory requirements pertaining to safety. All facilities will be designed, constructed, tested, operated, and maintained to conform with or exceed the requirements of Title 49 of the Code of Federal Regulations (CFR) Part 192, Transportation of Natural and Other Gas by Pipeline: Minimum Safety Standards; 18 CFR Section 380.15, Siting and Maintenance Requirements; and all other applicable federal and state regulations. These safety regulations will be reinforced by ETC Tiger's comprehensive and strictly enforced corporate practices. The empirical information presented below illustrates that with adherence to these regulations and ETC Tiger corporate practices, the potential for public hazard from accidents associated with operation of ETC Tiger's existing system and the proposed new Project facilities is low.

ETC Tiger and the natural gas transmission industry have an excellent record of public safety. Pipelines and related facilities are designed and maintained with strict adherence to USDOT standards to ensure public safety, reliability, and to minimize the opportunity for system failure. ETC Tiger will continue to employ similar system design, construction, operation, and maintenance practices to ensure this excellent record is maintained.

11.2 MEASURES TO PROTECT THE PUBLIC

ETC Tiger maintains operating policies and procedures that are periodically reviewed by USDOT. The policies provide specific directions in preventive maintenance and vigilant patrols of facilities, as well as procedures to be followed in the event of accident or natural catastrophe. All operating personnel are thoroughly trained to perform their activities in accordance with these policies and procedures.

Operations employees undergo periodic training sessions and review of operating and emergency procedures, as applicable to their job duties. The training includes safe operation of pipeline valves and equipment; facilities, including meter stations and compressor stations; hazardous material handling procedures; public liaison programs; and general operating procedures. The proposed Project facilities will be operated and maintained in accordance with these procedures.

ETC Tiger has implemented the federally mandated Operator Qualification (OQ) Plan as referenced in 49 CFR Part 192. All personnel who operate and maintain pipeline and station facilities and construct certain facilities are trained, evaluated, and qualified to perform company identified "OQ Tasks" and have the ability to recognize and react appropriately to any Abnormal Operating Conditions encountered while performing work. The OQ Plan applies to company, personnel, contractor personnel, subcontractor personnel, and personnel of all other entities who perform OQ Task requirements.

11.2.1 Surveys

Aerial and vehicle patrols of pipeline facilities are performed along with scheduled preventive maintenance, including smart-pigging and hydrostatic testing of pipelines. Any unusual situation or condition is reported and investigated in a timely manner. ETC Tiger performs annual leak detection surveys of its pipeline facilities, which are instrumental for the early detection of leaks and can reduce the likelihood of pipeline failure. The facilities to be constructed by ETC Tiger for this Project will follow all company safety and preventative measures and protocols.

ETC Tiger is also a member of "One Call" and related pre-excavation notification organizations in the states in which it operates. Through "One Call," contractors provide notification to a central agency of proposed excavation that in turn notifies ETC Tiger of the excavation locations. If ETC Tiger facilities are in an area of proposed contractor activity, the ETC Tiger facilities will be marked in the field and an ETC Tiger representative will be present during excavation to ensure that the facility is not compromised.

11.2.2 Equipment

The ETC Tiger pipeline system includes many equipment features that are designed to increase the overall safety of the system and protect the public from a potential failure of the system due to accidents or natural catastrophes. These include cathodic protection systems, a gas control center, data acquisition systems, and competent emergency response construction crew.

Cathodic protection systems are installed at various points along the ETC Tiger system pipelines and within compressor stations to mitigate corrosion of the pipeline facilities. The cathodic protection system impresses a low voltage current to the pipelines to off-set natural soil and groundwater corrosion potential. The functional capability of cathodic protection systems is inspected frequently to ensure proper operating conditions for corrosion mitigation.

Data acquisition systems are present at all meter stations along the system. If system pressures fall outside a predetermined range, an alarm becomes activated, and a notice is transmitted to the Houston Gas Control Center. The alarm provides notice that pressures at the station are not within an acceptable range. ETC Tiger also operates area and sub-area offices whose personnel can provide the appropriate response to emergency situations and direct safety operations as necessary.

ETC Tiger has a pipeline construction crew who are available to respond to ETC Tiger needs in the event of an emergency. ETC Tiger employs qualified and licensed field personnel who can be immediately dispatched to the scene of an emergency if the need should arise.

11.2.3 Liaison Procedures with Local Authorities

Liaison with local emergency officials, including fire, police and other public officials, is established and maintained in the communities in which ETC Tiger operates. Key components of the liaison program consist of:

1. Periodic mock firefighting drills;
2. Special informational meetings and training at the request of the municipality; and
3. Periodic literature distribution listing emergency telephone numbers and other pertinent data.

Excerpts from the ETC Tiger Liaison Program which will be used for the new Project facilities are as follows:

1. Contact is made with the police and fire departments and/or public officials of all communities that contain ETC Tiger facilities to accomplish the following:
    a. Ascertain how the officials may be able to assist ETC Tiger during an emergency, including the determination of the jurisdiction and/or responsibility with resources that may be involved in a response to an emergency;
    b. Acquaint the officials with how ETC Tiger responds to an emergency on its pipeline system;
    c. Notify the officials of the types of pipeline emergencies for which they may be contacted; and
    d. Inform them how ETC Tiger, in working with their departments, will cooperate in mutually assisting in protecting life or property during an emergency.
2. For ETC Tiger to quickly establish contact with police or fire departments and public officials in the event of an emergency at any location on the pipeline system, a current listing of their telephone numbers is maintained. This listing is reviewed on a periodic basis and necessary revisions are made.
3. ETC Tiger invites fire companies to participate in its periodic fire demonstrations where emphasis is placed on the following areas:
    a. When and when not to extinguish a natural gas fire during an emergency;
    b. How to extinguish different types of natural gas fires; and
    c. Periodic emergency simulation exercises.
4. ETC Tiger will continue to participate, on an invitational basis, in meetings with fire departments in communities in which the ETC Tiger facilities are located. The following subjects will be emphasized at these meetings:
    a. ETC Tiger’s role in emergencies on the pipeline system;
    b. The properties of natural gas and precautionary measures around an emergency; and
    c. The local fire company's participation during an emergency on the ETC Tiger system.

11.3 NATURAL GAS PIPELINE INDUSTRY SAFETY OVERVIEW

The following information provides a perspective on ETC Tiger's experience with respect to safety and reliability compared to industry-wide operational data. The information presented also assists in defining key industry-related safety issues.

11.3.1 Pipeline Safety

11.3.1.1 Hazards

The transportation of natural gas by pipeline involves some degree of risk to the public in the event of an accident and subsequent release of gas. The greatest danger is a fire or explosion following a major pipeline rupture. Methane, the primary component of natural gas, is colorless, odorless, and tasteless. Methane has an ignition temperature of 1,000 degrees Fahrenheit and is flammable at concentrations between 5.0 percent and 15.0 percent in air. Unconfined mixtures of methane in the air are not explosive. However, a flammable concentration within an enclosed space and in the presence of an ignition source can explode.

The specific gravity of methane is 0.55; therefore, it is buoyant at atmospheric temperatures. It is not toxic, but is classified as a simple asphyxiant, possessing only a slight inhalation hazard. If methane is breathed in high concentration, oxygen deficiency can result in serious injury or death.

No hazards to aquatic life have been identified should a leak occur under a waterbody or wetland system. Methane gas is a natural by-product of the decomposition of organic material in these systems.

11.3.1.2 Safety Standards

The proposed Project facilities will be designed, constructed, operated, and maintained in accordance with the USDOT Minimum Federal Safety Standards in 49 CFR Part 192. These standards are intended to ensure adequate protection for the public from natural gas pipeline failures. Part 192 specifies material selection and qualification, minimum design requirements, and protection from internal, external, and atmospheric corrosion. Part 192 also defines area classifications based on population density in the vicinity of the pipeline, which determine more rigorous safety requirements for populated areas. The class location unit is an area that extends 220 yards on either side of the centerline of any continuous 1-mile length of pipeline. The four area classifications are defined as follows:

- Class 1 – Location with 10 or fewer buildings intended for human occupancy;
- Class 2 – Location with more than 10 but less than 46 buildings intended for human occupancy;
- Class 3 – Location with 46 or more buildings intended for human occupancy or where pipeline lies within 100 yards of any building, or small, well-defined outside area occupied by 20 or more people during normal use; and
- Class 4 – Location where buildings with four or more stories aboveground are prevalent.

Class locations representing more populated areas require higher safety factors in pipeline design, testing, and operation. Pipelines constructed in Class 1 locations must be installed with a minimum depth of cover of 30 inches in normal soil, and 18 inches in consolidated rock. Class 2, 3, and 4 locations, as well as drainage ditches of public roads and railroad crossings, require a minimum of 36 inches of cover in normal soil and 24 inches in consolidated rock. Class locations also specify the maximum distance to a sectionalizing block valve: 10 miles in Class 1; 7.5 miles in Class 2; 4 miles in Class 3; and 2.5 miles in Class 4. Pipeline design pressures, hydrostatic test pressures, maximum allowable operating pressure, inspection and testing of welds, and frequency of pipeline patrols and leak surveys must also conform to

higher standards in more populated areas. The class specifications for the proposed Laterals are provided in **Table 11.3-1**.

TABLE 11.3-1
Pipeline Classification Specifications Along the Franklin Farms and Franklin Farms (FF) Trio Farm Laterals

Class	Begin Milepost	End Milepost	Length (miles)
Franklin Farms Lateral			
1	0.0	0.4	0.4
2	0.4	1.8	1.4
1	1.8	8.1	6.3
2	8.1	8.7	0.6
1	8.7	11.1	2.4
2	11.1	11.5	0.5
1	11.5	12.9	1.3
3	12.9	13.2	0.4
FF Trio Farm Lateral			
1	0.0	1.8	1.8

Part 192 prescribes the minimum standards for operating and maintaining pipeline facilities, including the requirement to establish a written plan governing these activities. Under Section 192.615, each pipeline operator must establish an emergency plan that provides written procedures to minimize the hazards from a gas pipeline emergency. Key elements of the plan include procedures for:

1. Receiving, identifying, and classifying emergency events, including gas leakage, fires, explosions, and natural disasters;
2. Establishing and maintaining communications with local fire, police, and public officials, and coordinating emergency response;
3. Making personnel, equipment, tools, and materials available at the scene of an emergency;
4. Protecting people first and then property, making them safe from actual or potential hazards; and
5. Emergency shutdown of the system and safely restoring service.

Each operator must establish and maintain liaison with appropriate fire, police, and public officials to learn the resources and responsibilities of each organization that may respond to a gas pipeline emergency and coordinate mutual assistance in responding to emergencies. The operator must also establish a continuing education program to ETC Tiger customers, the public, government officials, and those engaged in excavation activities to recognize a gas pipeline emergency and report it to appropriate public officials.

11.3.1.3 High Consequence Areas

High Consequence Areas (HCA) (as defined in 49 CFR Section 192.903) were evaluated along the Franklin Farms and FF Trio Farm Laterals. To identify HCAs along the route, a potential impact radius is calculated to depict potential impact circles. The population within the impact radius is then evaluated. Potential impact circles that contain 20 or more structures intended for human occupancy; buildings housing

populations of limited mobility; buildings that would be hard to evacuate (e.g., nursing homes, schools); or buildings and outside areas occupied by more than 20 persons on a specified minimum number of days each year, are defined as HCAs (USDOT PHMSA, 2011). HCAs along the Franklin Farms Lateral are shown in **Table 11.3-2**. No HCAs were identified along the FF Trio Farm Lateral.

TABLE 11.3-2
High Consequence Areas along the Franklin Farms Lateral

HCA Area	Milepost		Length (miles)
	Begin	End	
Area #1	12.8	13.2	0.5

11.3.1.4 Pipeline Accident Data

Since 1970, all operators of transmission and gathering systems under the jurisdiction of the USDOT have been required to report pipeline incidents to the Pipeline and Hazardous Materials Safety Administration (PHMSA) Office of Pipeline Safety (OPS). The OPS is the federal safety authority for the nation's 2.6 million miles of natural gas and hazardous liquid pipelines. PHMSA gathers information about pipeline incidents, including property damage more than $50,000, injury resulting in hospitalization, death, or other incidents that are otherwise considered significant by the operator resulting from a release of gas. This comprehensive data have been used by PHMSA to review all pipeline incidents for the 20-year period between 2006 and 2025 (USDOT PHMSA, 2025a).

Table 11.3-3 provides a summary of natural gas transmission pipeline significant incidents by category in the State of Louisiana from 2006 to 2025 (USDOT PHMSA, 2025a). Material, weld, and equipment failure and excavation damage (primarily third-party excavation damage) are the leading causes of recent incidents. Potential geological hazards specific to the Project area are discussed in Resource Report No. 6 (*Geological Resources*).

TABLE 11.3-3
Reported Natural Gas Transmission Pipeline Significant Incidents in Louisiana

Time Period	Number of Pipeline Incidents	Cause of Incident							Fatalities / Injuries
		Material/Weld/ Equipment Failure	Corrosion	Excavation Damage	Incorrect Operation	Natural Force Damage	Other Outside Force Damage	All Other Causes	
State of Louisiana									
2006 - 2025	170	16	62	13	8	16	19	18	14

Source: USDOT PHMSA (2025a).

11.3.1.5 Impact on Public Safety

The nationwide totals for accidental fatalities due to various manmade and natural hazards are provided in **Table 11.3-4**. These totals provide a relative measure of the industry-wide safety of natural gas pipelines; however, direct comparisons between accident categories should be made cautiously since individual exposures to hazards are not uniform among all categories. Overall, the average number of fatalities resulting from natural gas transmission and gathering pipelines is relatively small considering the more than

413,022 miles of transmission and gathering lines (405,583 miles onshore) in service nationwide as of 2025 (USDOT PHMSA, 2025b).

TABLE 11.3-4
Preventable-Injury-Related Deaths by Principal Cause in the United States (2025)

Type of Accident	Fatalities [a]
All accidents	197,449
Motor vehicles	41,064
Falls	47,026
Poisoning	75,761
Drowning	4,201
Choking	5,923
Fire, Flames, or Smoke	3,155
Mechanical Suffocation	1,719
Firearms	450
Other	15,143
Natural Gas Gathering and Transmission Pipelines [b]	11

Notes:
[a] NSC, 2025
[b] USDOT PHMSA, 2025c

11.3.2 REFERENCES

National Safety Council (NSC). 2026. Historical Preventable Fatality Trends - Preventable-injury-related deaths by cause, United States, 1903-2023. Available online at: https://injuryfacts.nsc.org/all-injuries/historical-preventable-fatality-trends/deaths-by-cause/. Accessed May 2026.

U.S. Department of Transportation (USDOT) Pipeline and Hazardous Materials Safety Administration (PHMSA). 2011. Fact Sheet: High Consequence Areas (HCA). Available online at: https://primis.phmsa.dot.gov/comm/factsheets/fshca.htm. Accessed May 2026.

USDOT PHMSA. 2026a. Significant Incidents. Available online at: GT Incident Trend Drill to Cause w subtotal 20 yr totals first drill. Accessed May 2026.

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USDOT PHMSA. 2026c. Pipeline Facility Incident Report – All Reported Incidents 20 Year Trend. Available online at: https://portal.phmsa.dot.gov/analytics/saw.dll?Portalpages&PortalPath=%2Fshared%2FPDM%20Public%20Website%2F_portal%2FSC%20Incident%20Trend&Page=All%20Reported. Accessed May 2026.

ETC Tiger Pipeline, LLC

Franklin Farms Project

RESOURCE REPORT NO. 12
PCB Contamination

FERC Docket No. CP26-___-000

June 2026

ENERGY TRANSFER

RESOURCE REPORT NO. 12 FILING REQUIREMENTS	
Information	**Section Reference**
1. For Projects involving the replacement or abandonment of facilities determined to have PCBs, provide a statement that activities would comply with an approved EPA disposal permit or with the requirements of the TSCA. (§ 380.12 (n)(1))	Not Applicable
2. For compressor station modifications on sites that have been determined to have soils contaminated with PCBs, describe the status of remediation efforts completed to date. (§ 380.12 (n) (2))	Not Applicable

RESOURCE REPORT NO. 12
PCB CONTAMINATION

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12.0 PCB CONTAMINATION 12-1

ABBREVIATIONS AND ACRONYMS

ETC Tiger	ETC Tiger Pipeline, LLC
EPA	United States Environmental Protection Agency
PCB	Polychlorinated biphenyls
TSCA	Toxic Substances Control Act

12.0 PCB CONTAMINATION

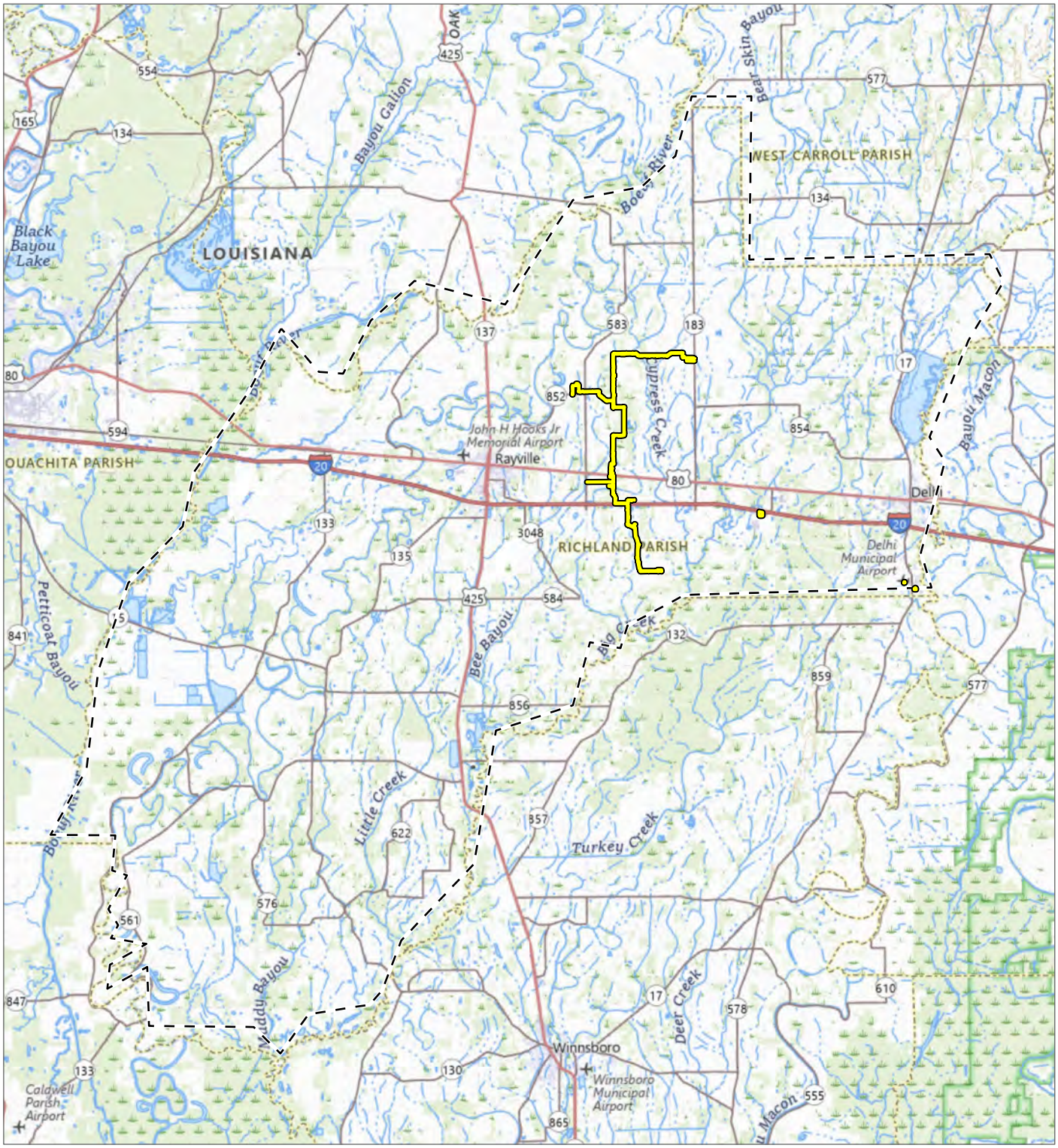
ETC Tiger Pipeline, LLC (ETC Tiger) will fully comply with the Toxic Substances Control Act (TSCA). Under the TSCA, there is no presumption of contamination of pipelines with polychlorinated biphenyls (PCB) regardless of age of the pipeline. ETC Tiger was not one of the thirteen pipelines that the United States Environmental Protection Agency (EPA) considered to be presumed PCB-contaminated prior to the issuance of the PCB Mega-Rule. Under TSCA, pipelines are characterized based upon PCBs within the pipeline system. No PCBs more than 50 parts per million in pipeline liquids have been identified within ETC Tiger's pipeline system and, therefore, the sampling protocols for characterization and abandonment of PCB-contaminated pipelines are not required under TSCA.

APPENDIX A

Project Mapping

APPENDIX A-1

Project Overview Map



PROJECT BOUNDARY

RICHLAND PARISH

0 14,000 28,000

FEET

1:336,000 1" = 28,000'

BASE MAP: USGS "THE NATIONAL MAP" ONLINE SERVICE LAYER;
DATA SOURCES: TRC, USGS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

GENERAL PROJECT LOCATION MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 1.B.1

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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APPENDIX A-2

USGS Topographic Maps

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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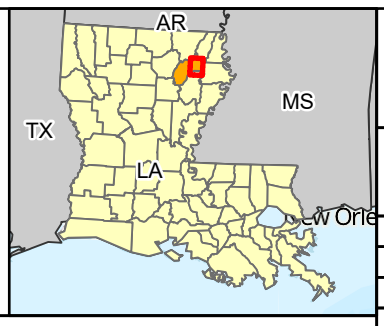
PROJECT BOUNDARY



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 3,500 7,000 FEET
1:84,000 1" = 7,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP (OVERVIEW)

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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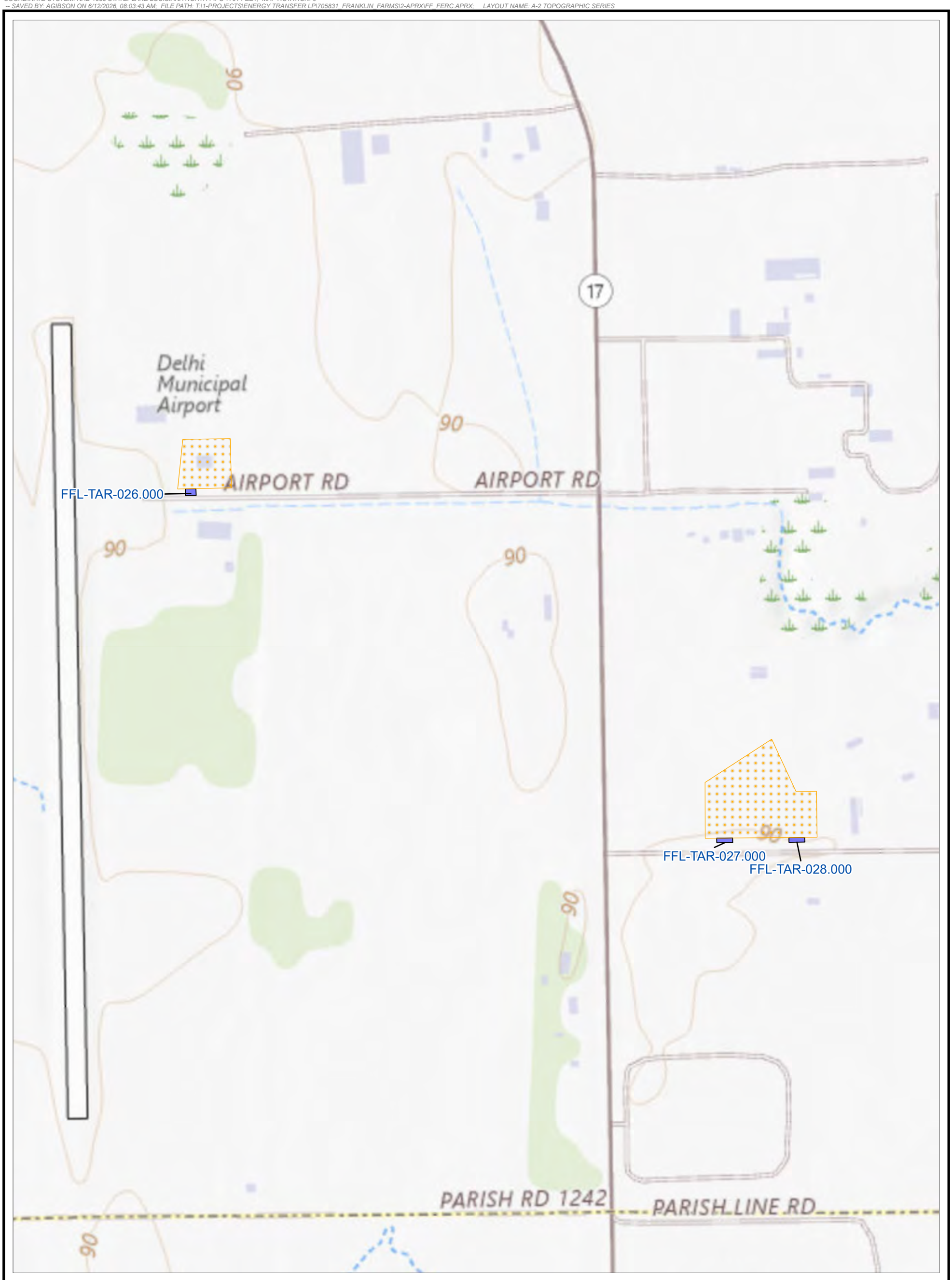
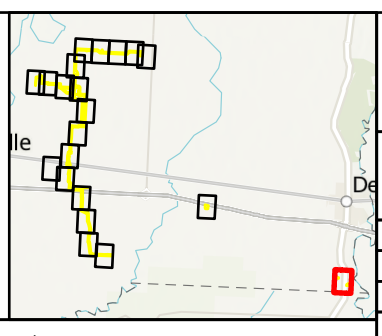


TEMPORARY ACCESS ROAD
CONTRACTOR YARD
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 1 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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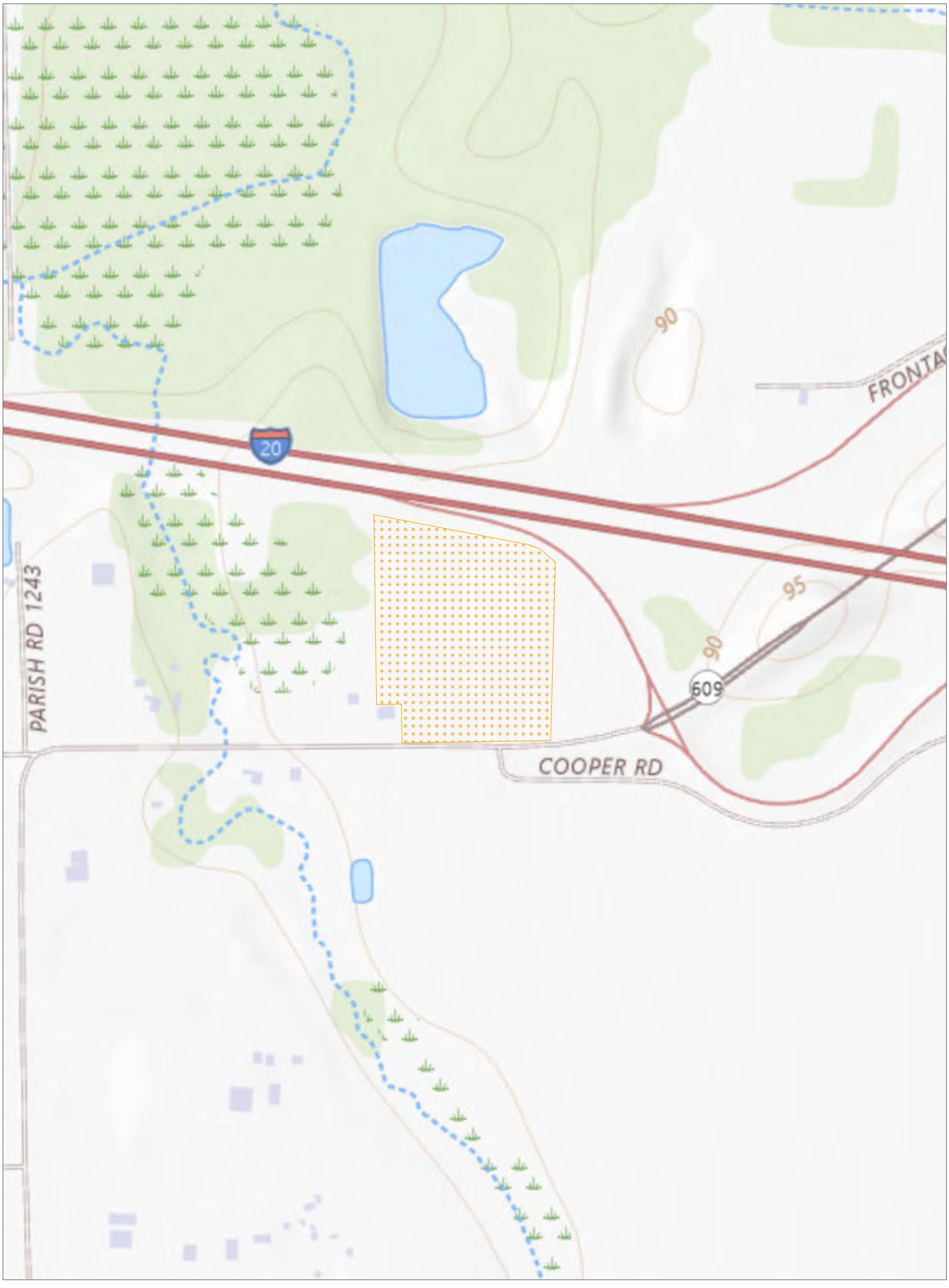
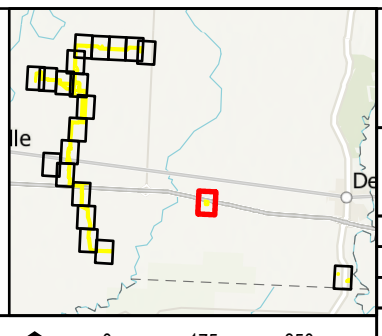


CONTRACTOR YARD

MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 2 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES

FF Gulf Run Bi Di Meter Station
& FF Gulf South Receipt Meter Station

0.4 0.3 0.2 0.1 0.0

FFL-TAR-003.000

FFL-PAR-002.100

FFL-PAR-002.000

FFL-TAR-001.000

LITTLE RD

80

- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION
- MILE POST

BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

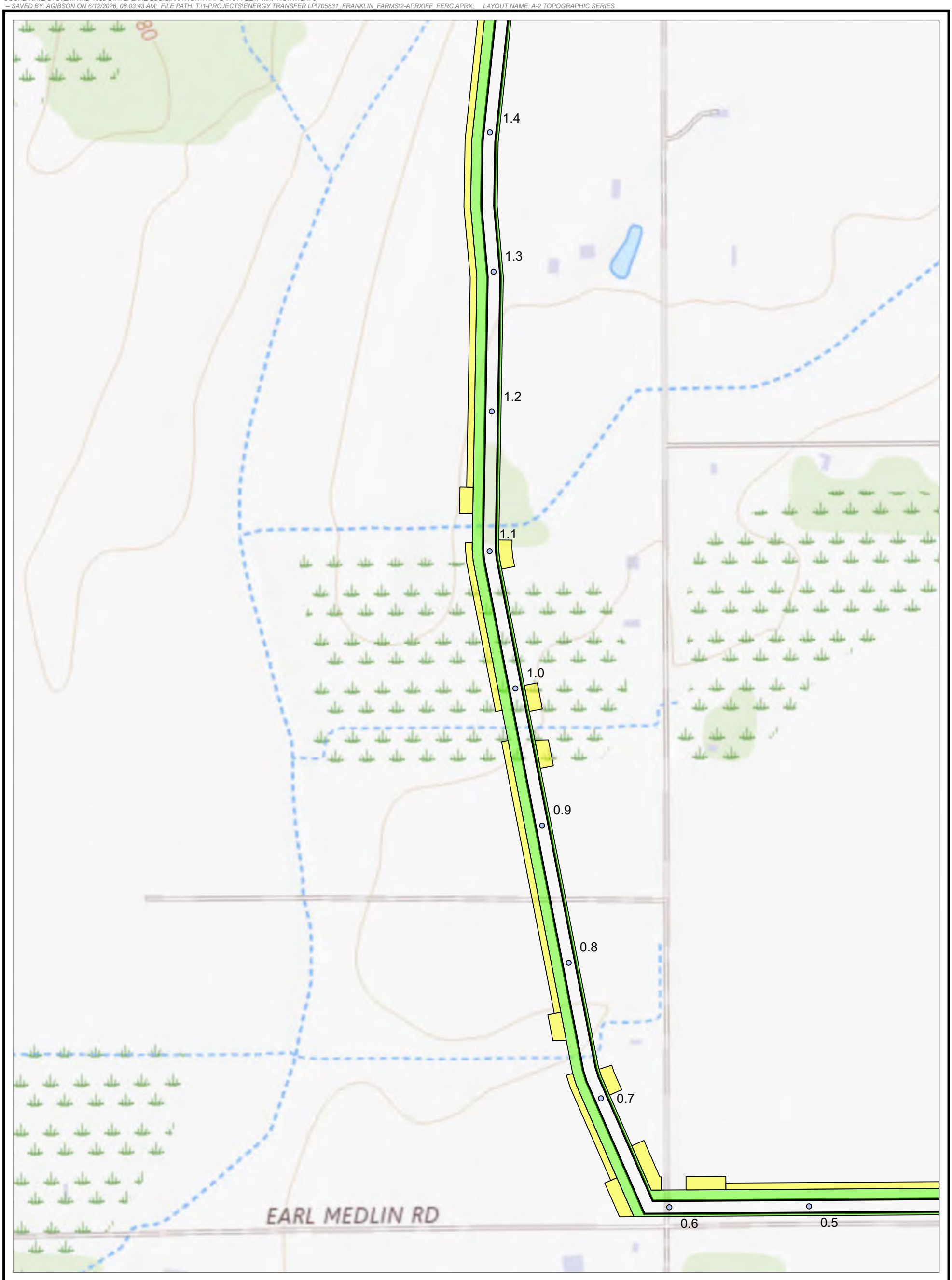
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

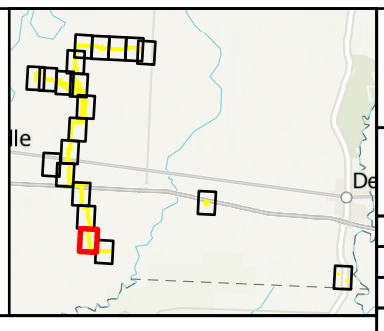
FIGURE A-2
PAGE 3 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 4 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES

FFL-TAR-006.000
2.5
2.4
2.3
2.2
2.1
2.0
1.9
1.8
1.7
1.6
1.5
FFL-TAR-005.000
FFL-TAR-004.000
PARISH RD 2263
TURN ROW
PONY GREER RD

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST

BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 5 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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FFL-TAR-007.000

3.6 3.5 3.4 3.3 3.2 3.1 3.0 2.9 2.8 2.7 2.6

OLLIE MORRIS RD

OLLIE MORRIS RD

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST

BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 6 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES

FFL-TAR-010.000

4.8

4.7

4.6

FFL-PAR-008.000

4.5

FF Bee Bayou
Delivery
Meter Station

4.4

Cow Bayou

4.3

4.2

4.1

4.0

3.9

3.8

3.7

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
MILE POST

BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

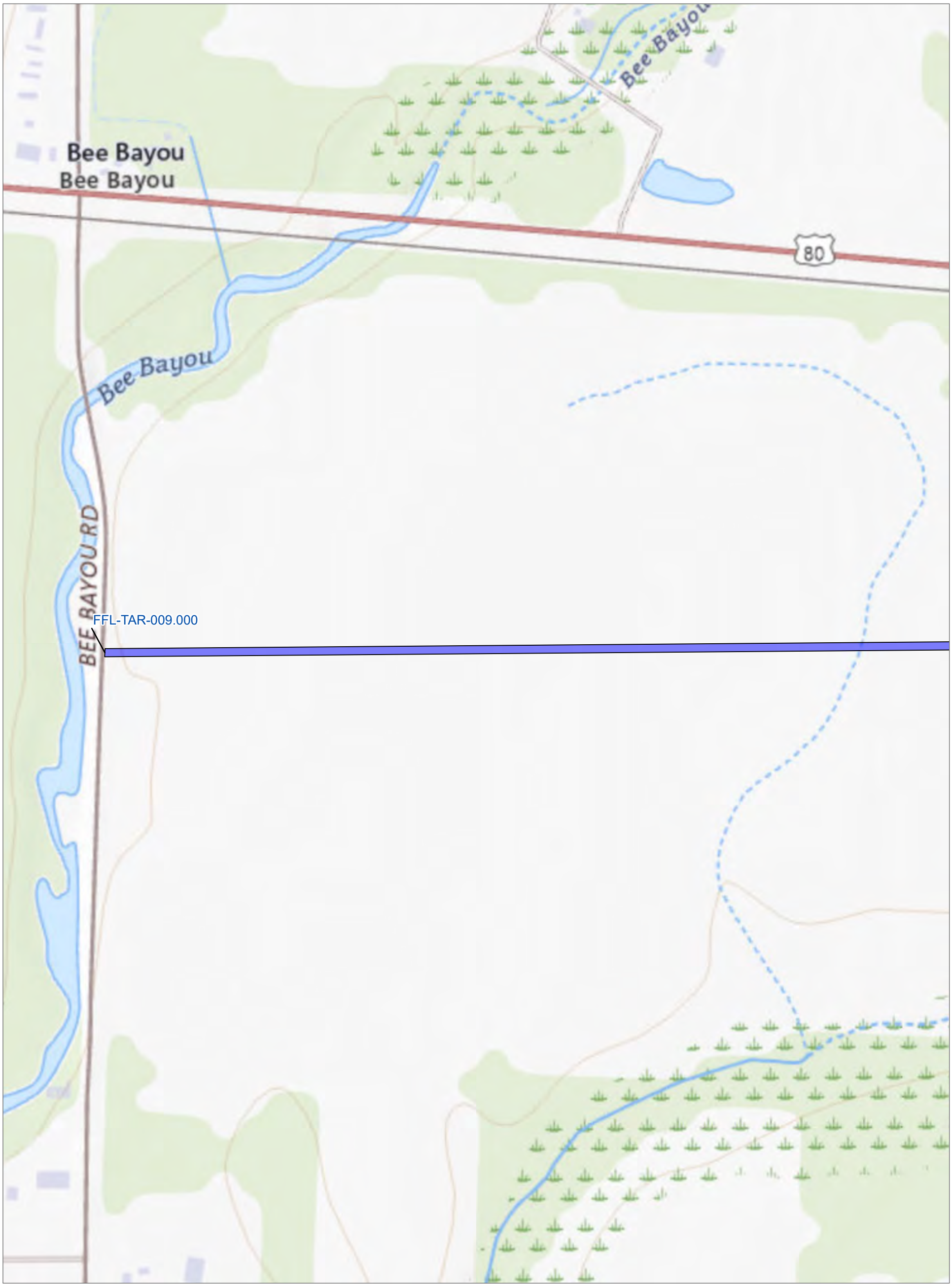
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 7 OF 21

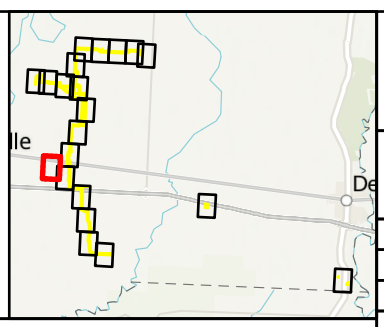
TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



TEMPORARY ACCESS ROAD

MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET

1:4,200 1" = 350'

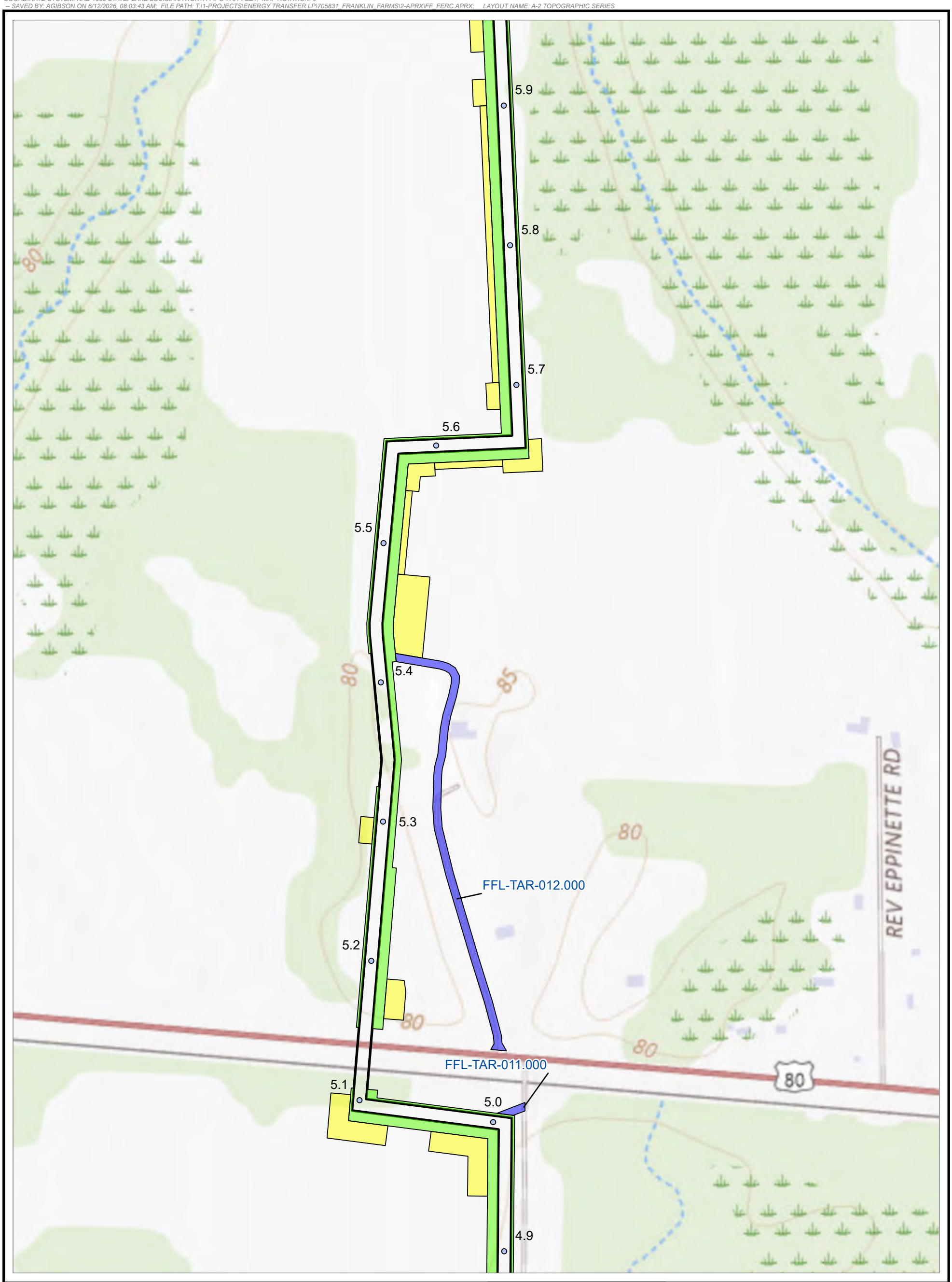
ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP

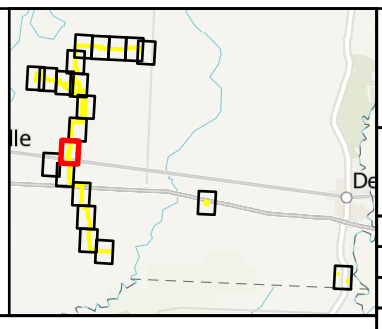
DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 8 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP

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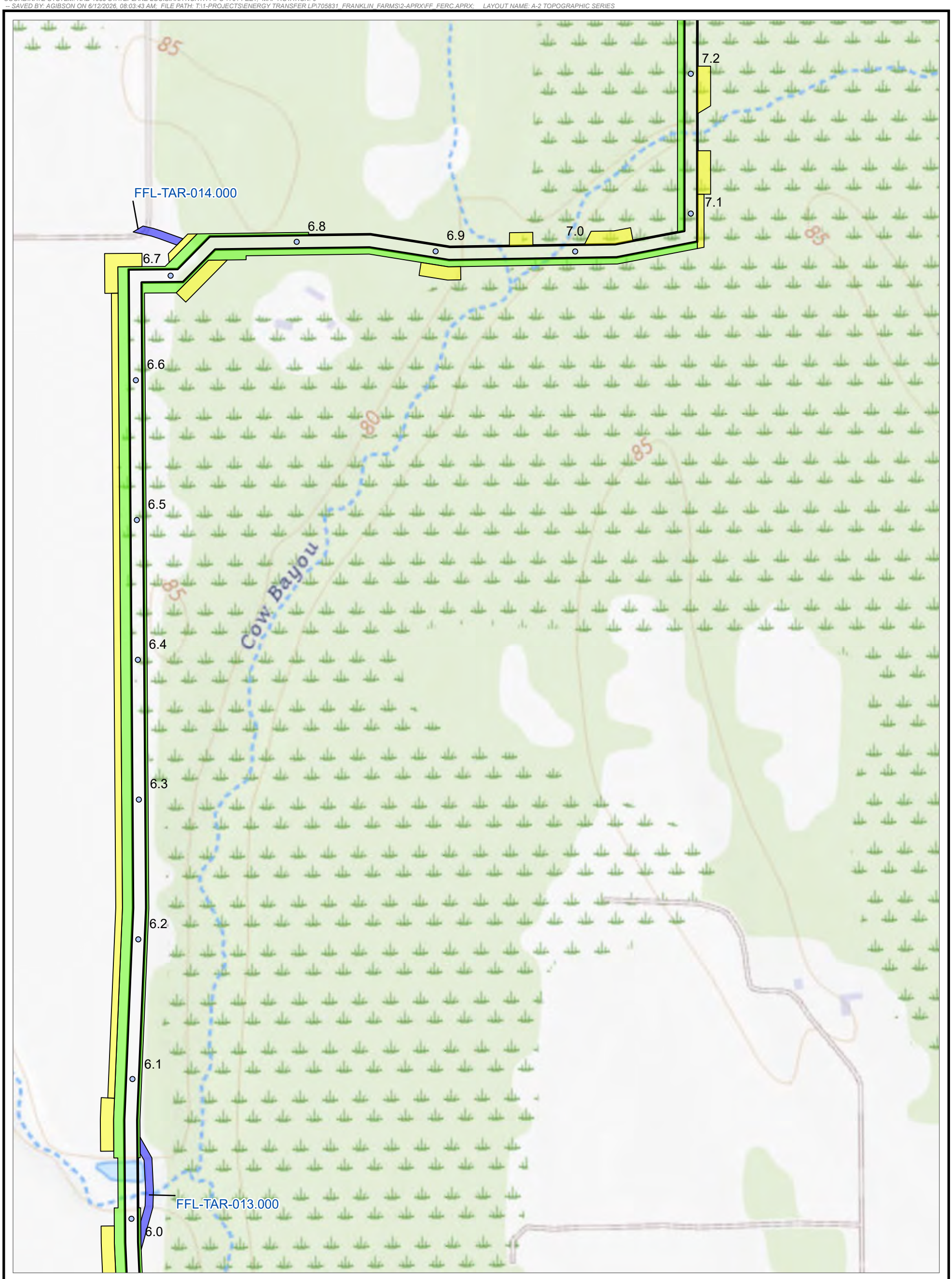
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 9 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST

BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 10 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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8.3
8.2
8.1
TEMPLE RD
8.0
7.9
7.8
7.7
7.6
7.5
7.4
7.3

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
MILE POST

BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

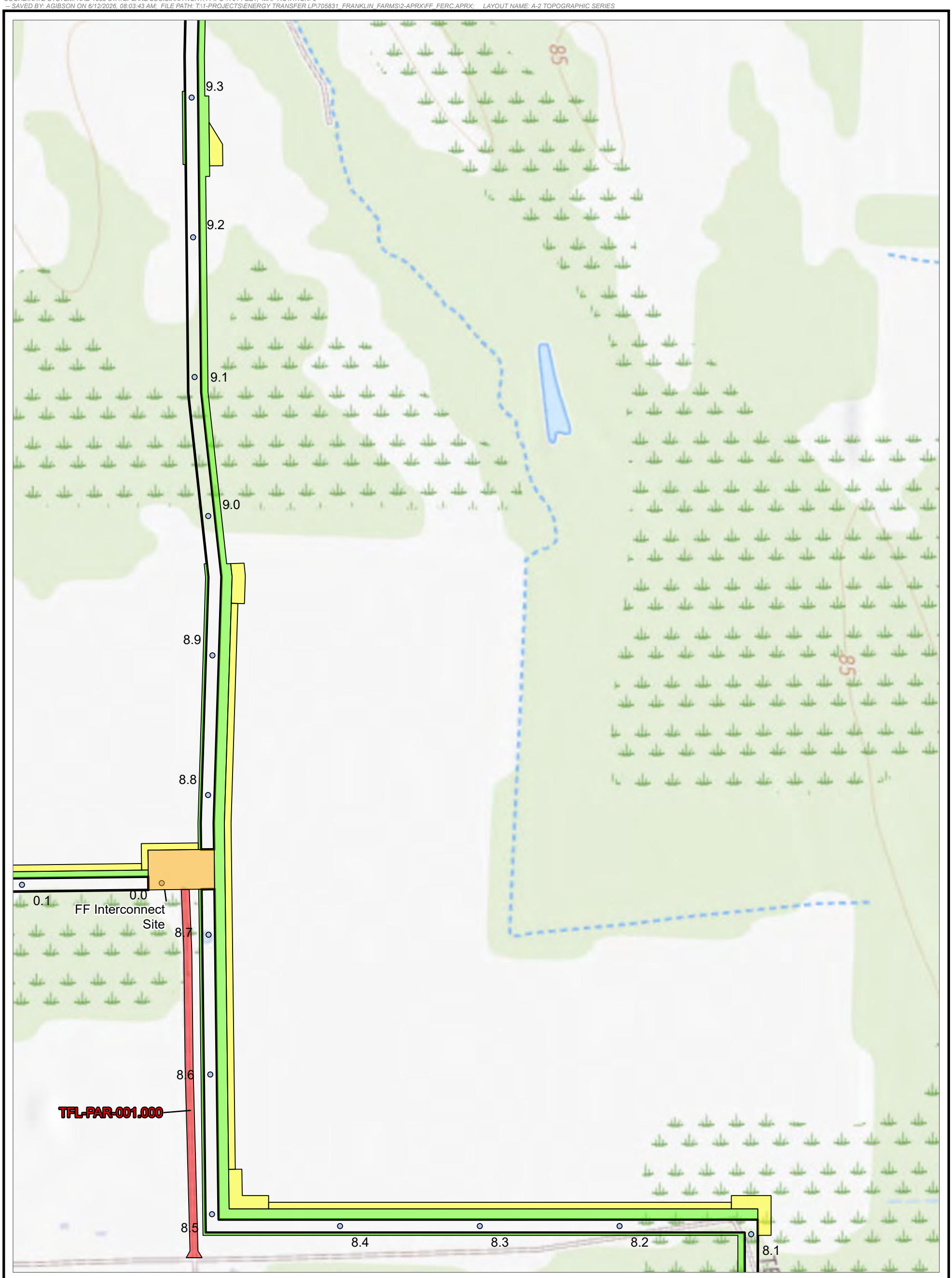
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

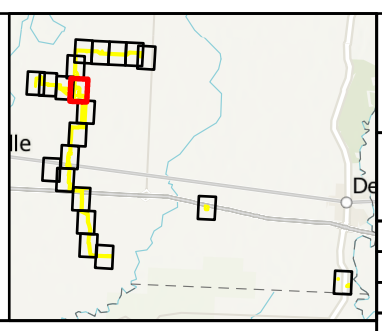
FIGURE A-2
PAGE 11 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
INTERCONNECT SITE
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

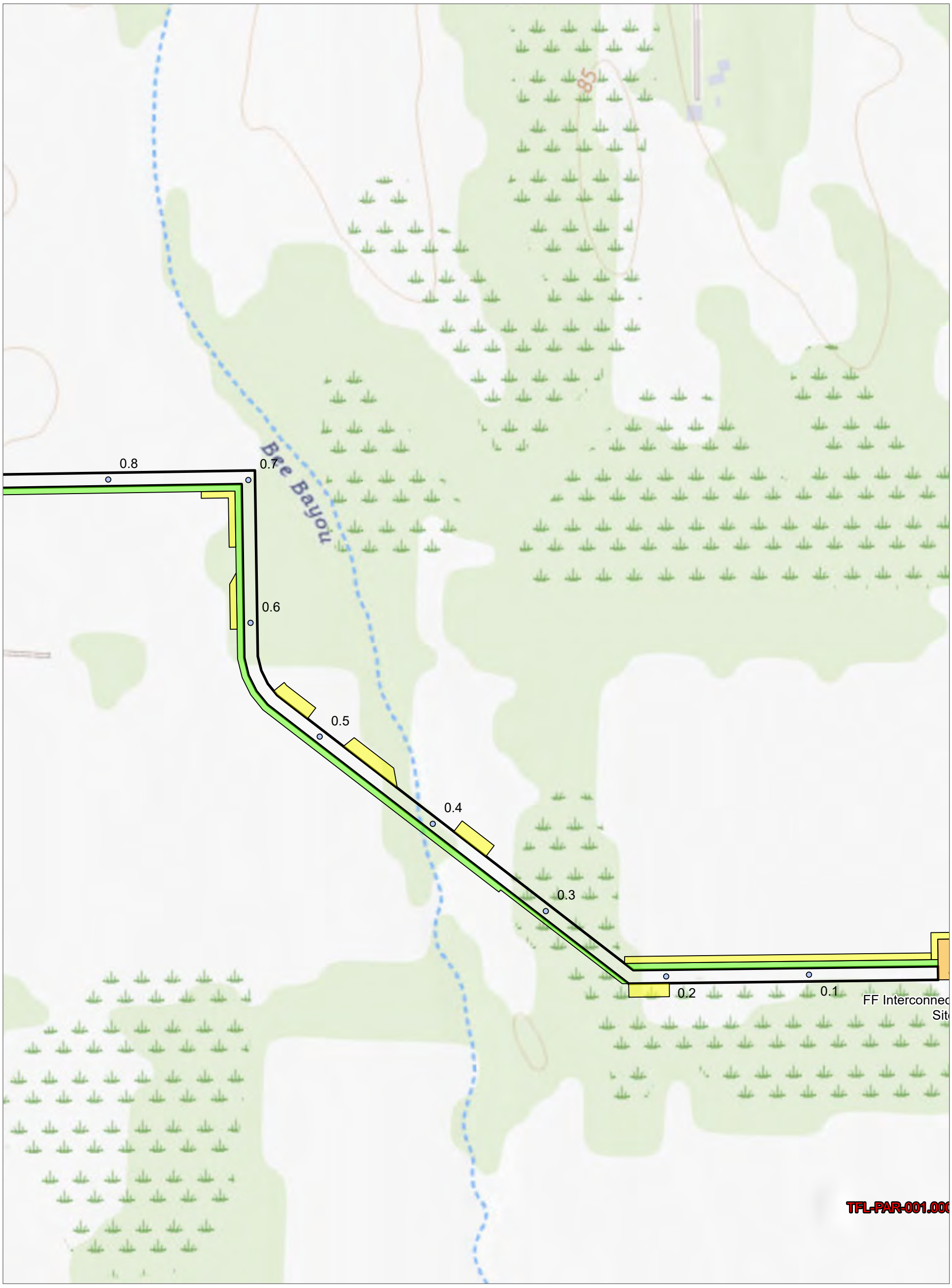
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

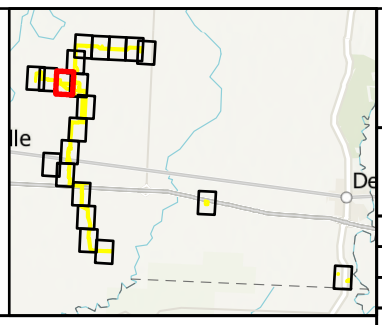
FIGURE A-2
PAGE 12 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
INTERCONNECT SITE
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

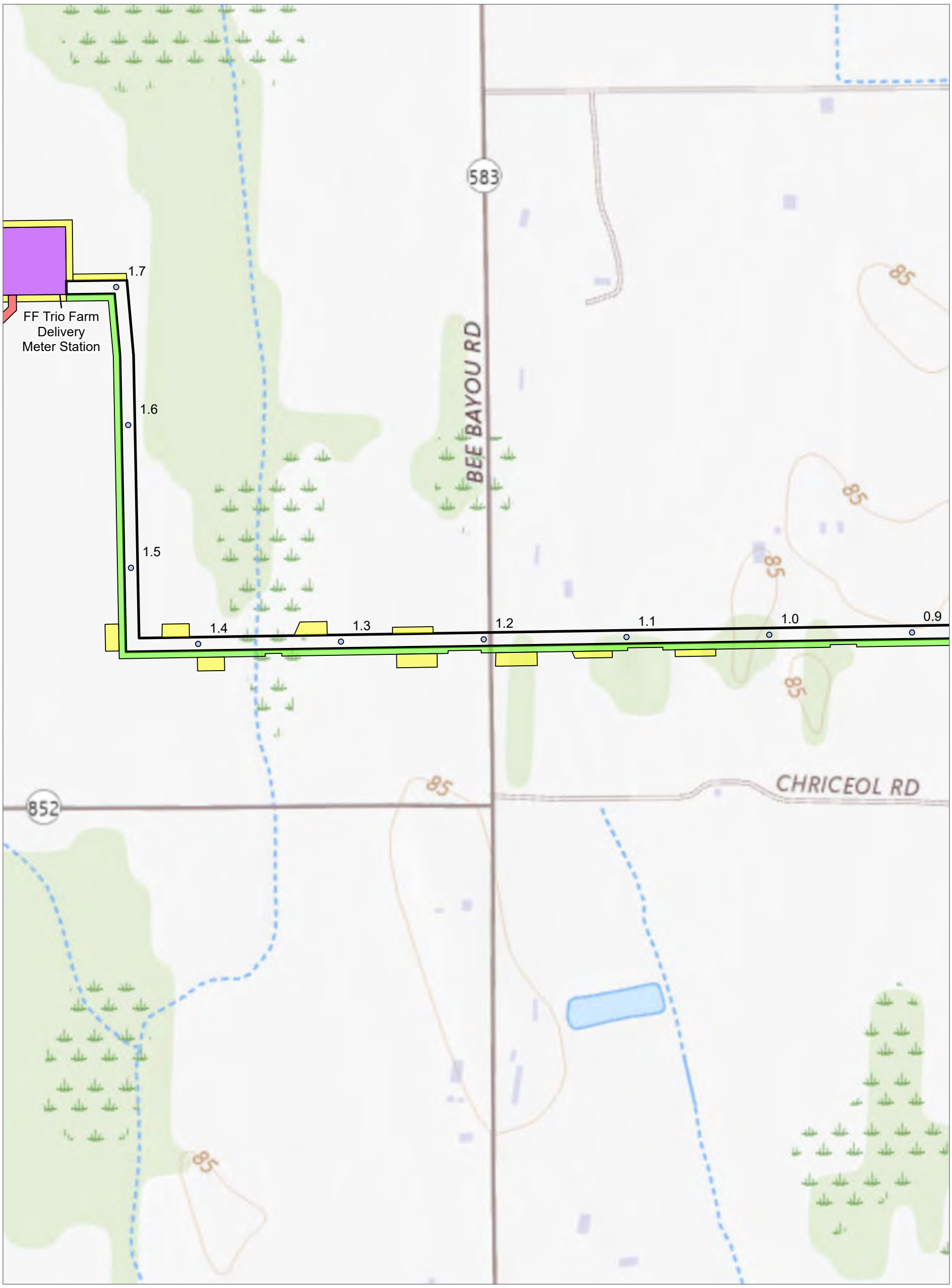
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

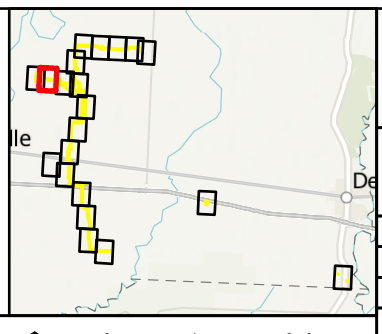
FIGURE A-2
PAGE 13 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
METER STATION
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

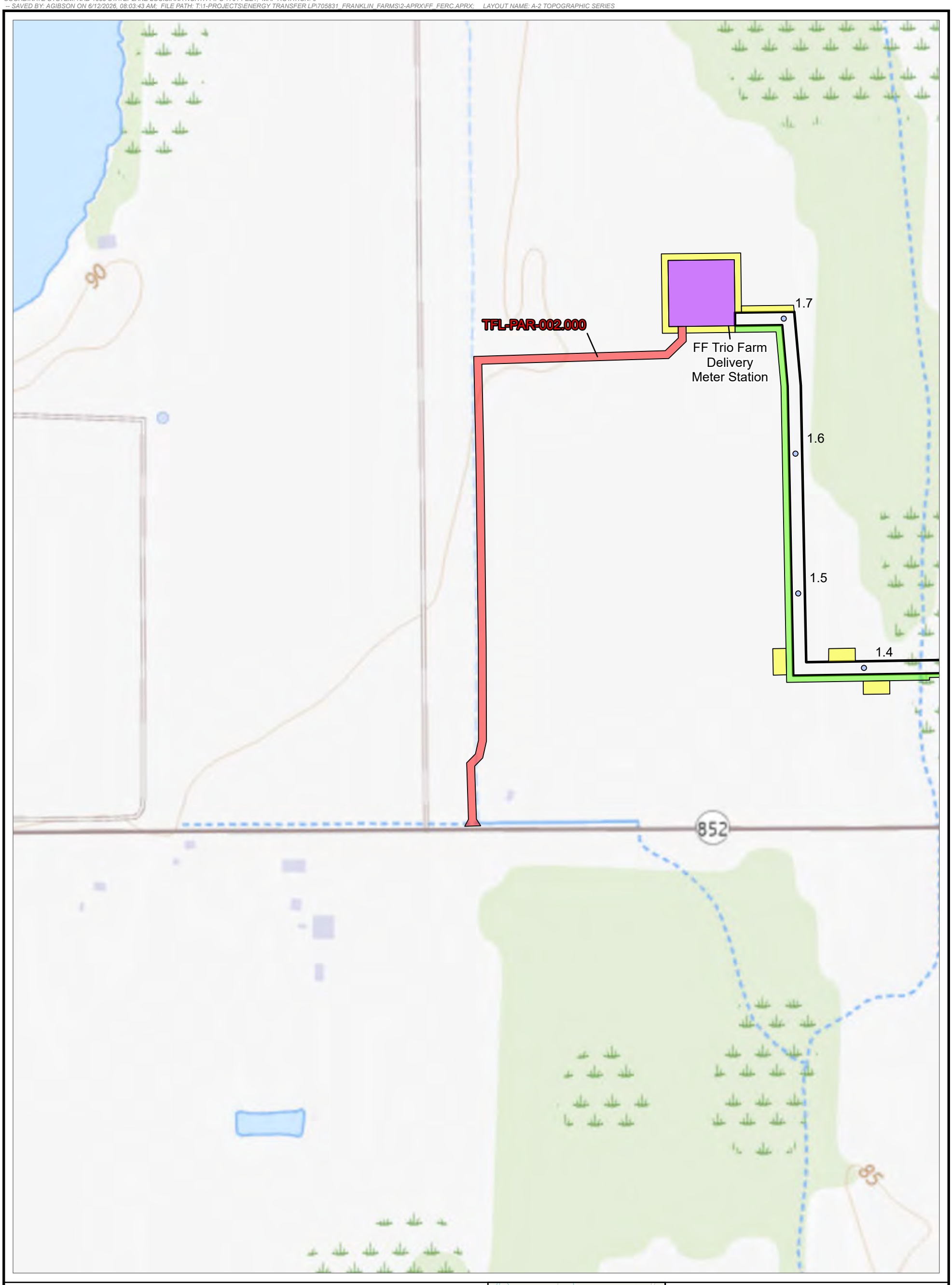
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

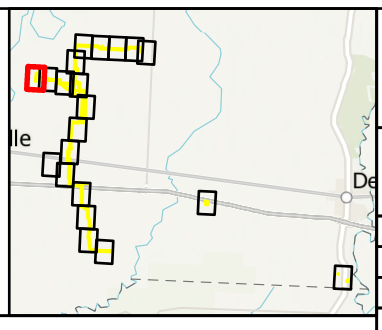
FIGURE A-2
PAGE 14 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
METER STATION
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

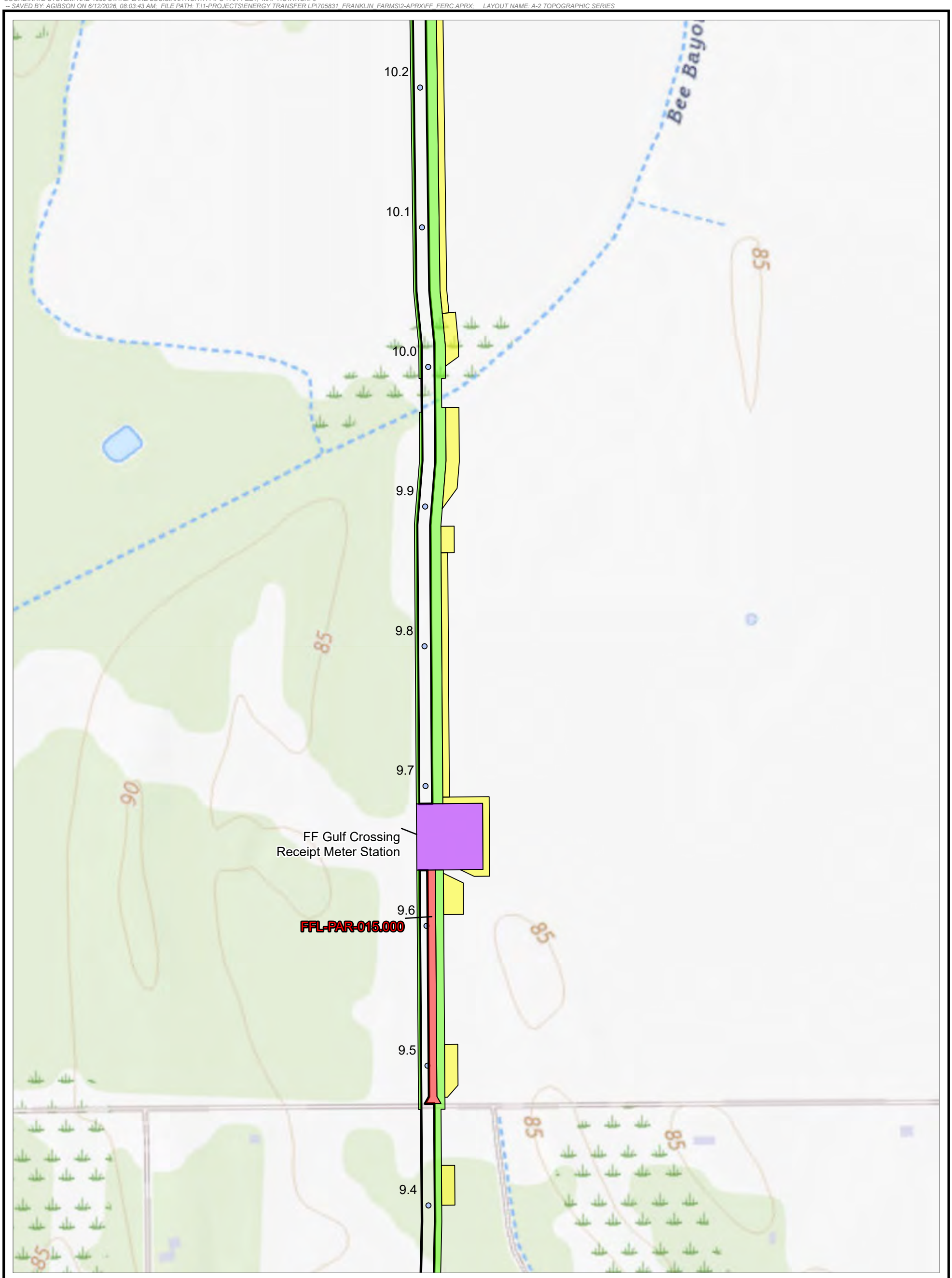
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

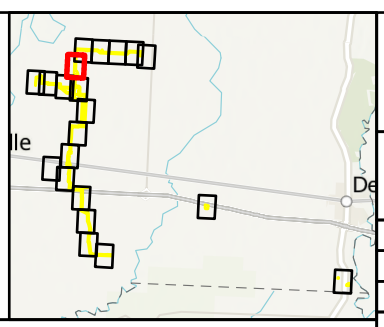
FIGURE A-2
PAGE 15 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
METER STATION
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

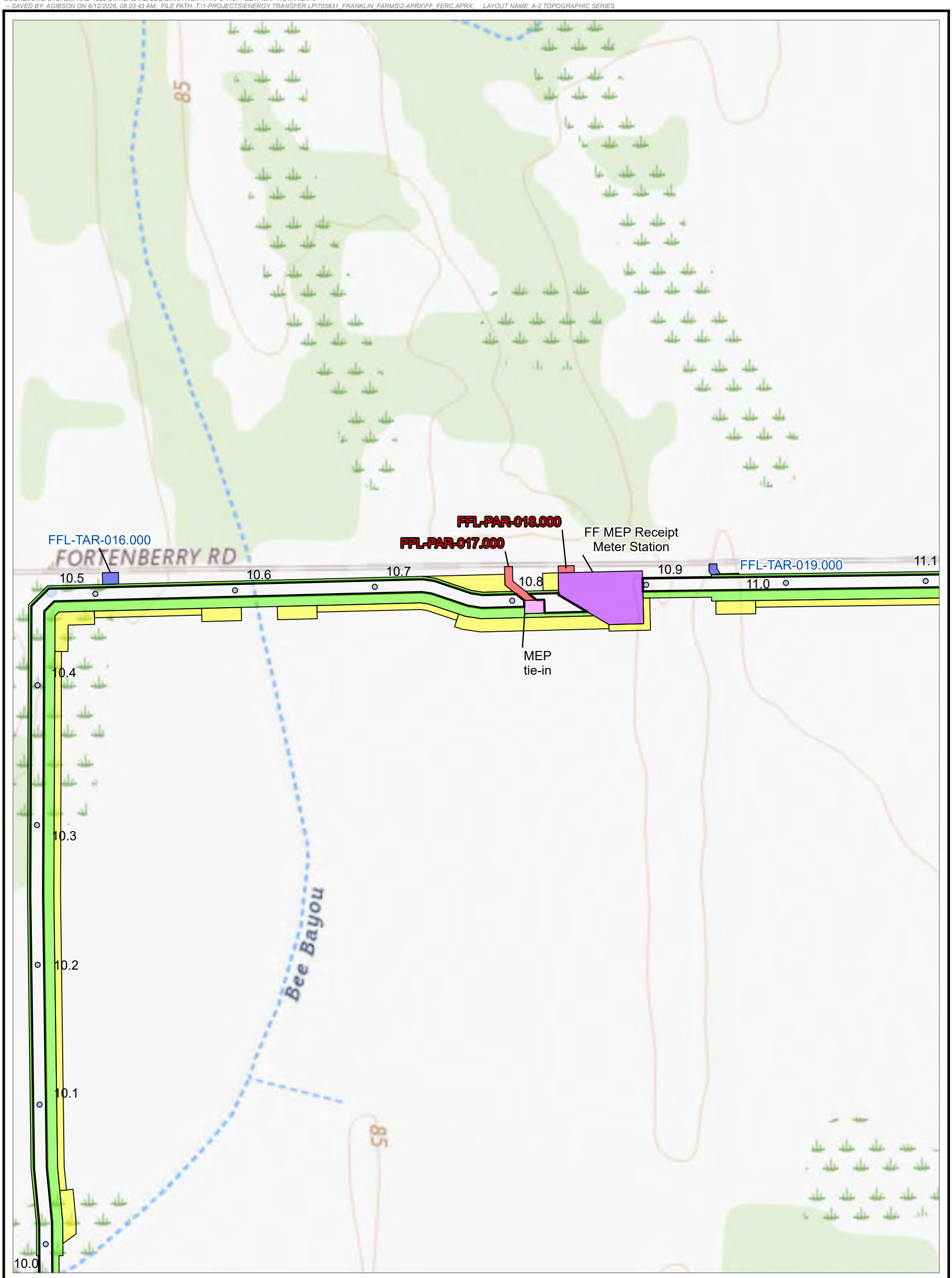
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 16 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
TAP SITE
MILE POST

BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

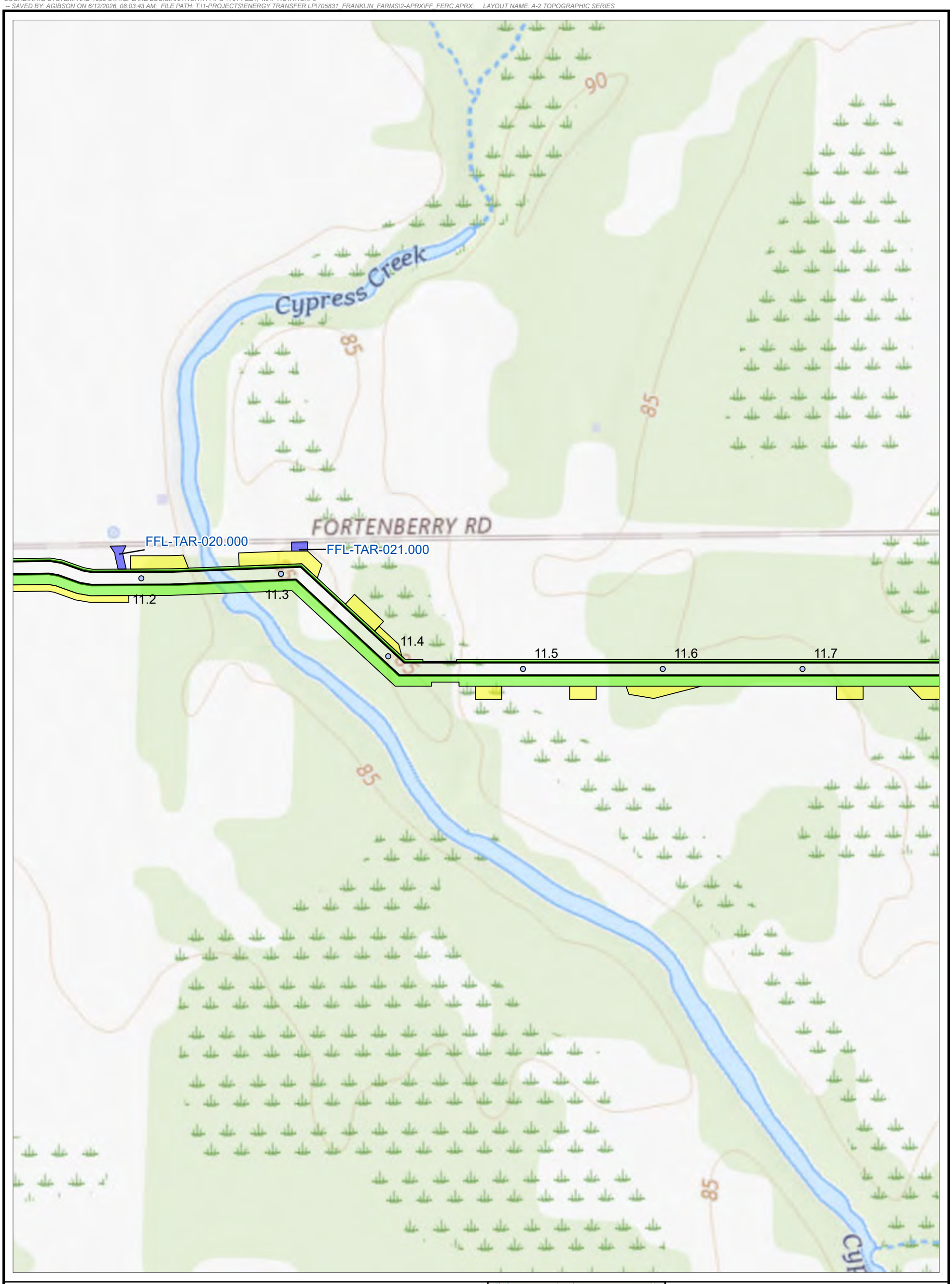
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

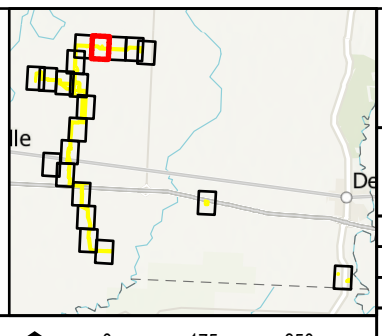
FIGURE A-2
PAGE 17 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

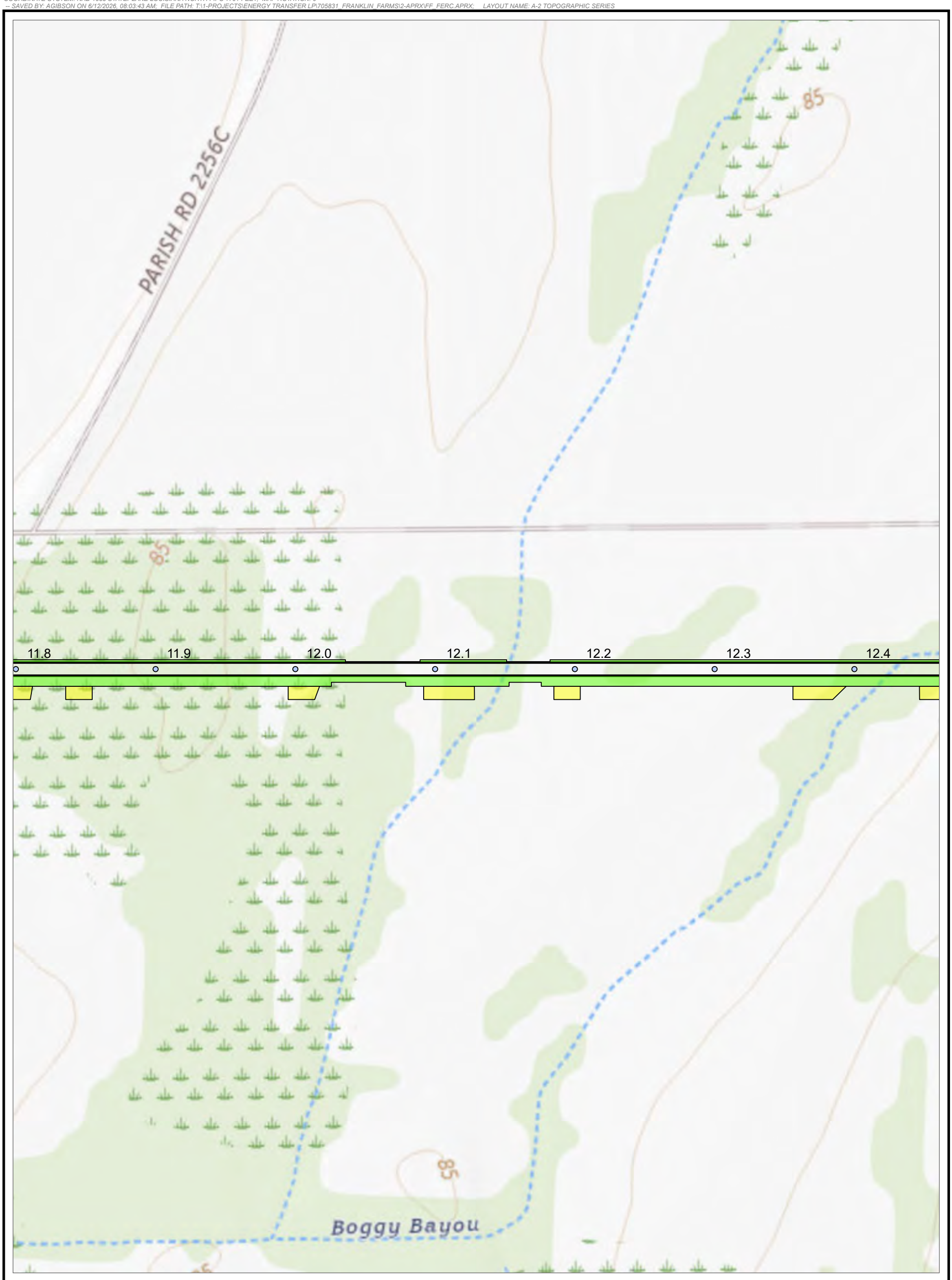
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

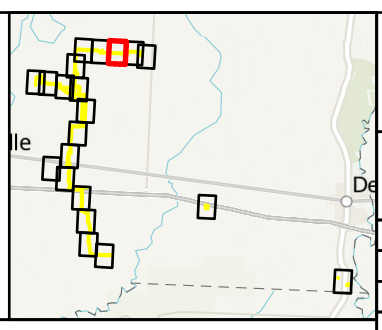
FIGURE A-2
PAGE 18 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

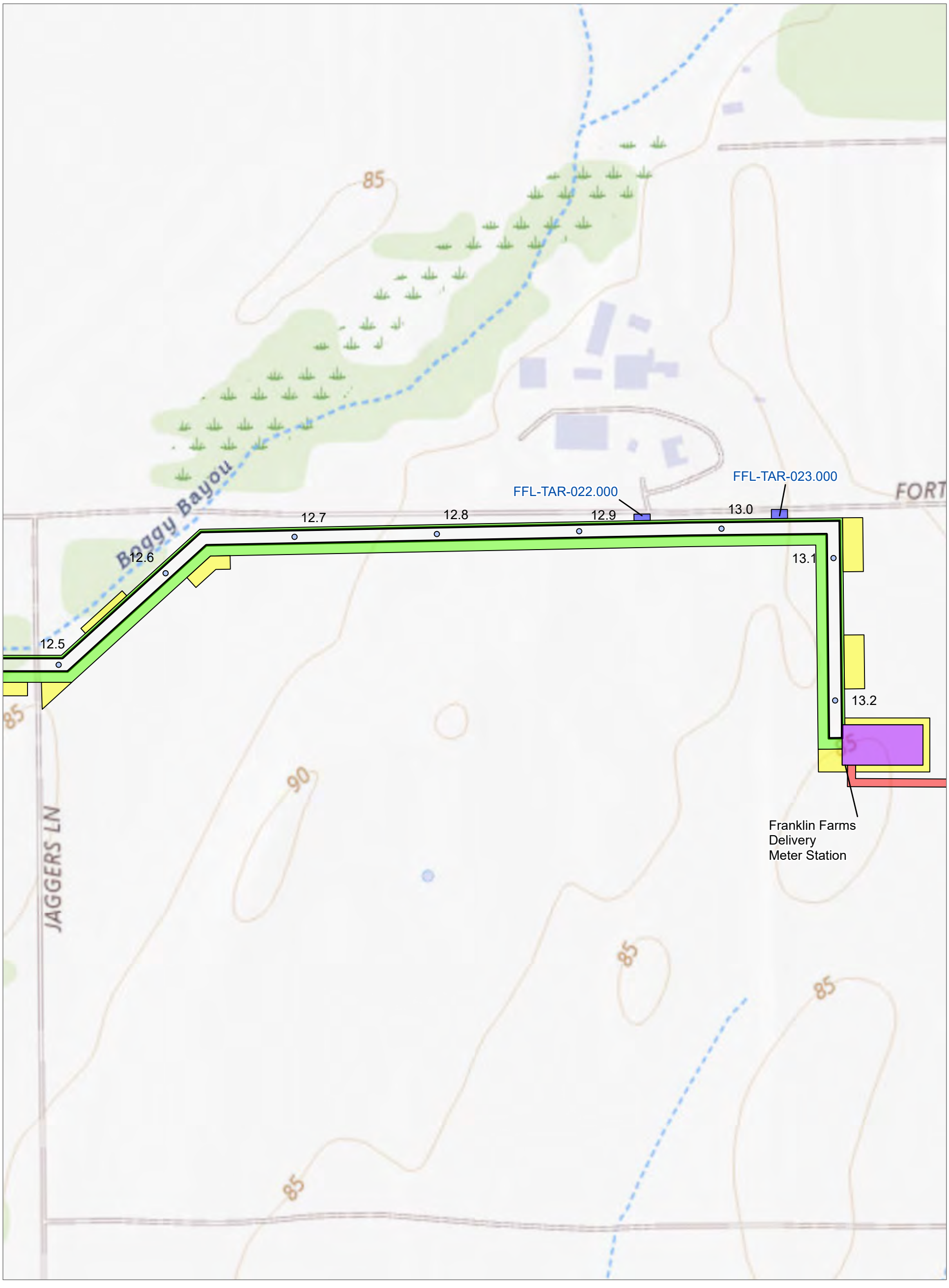
TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

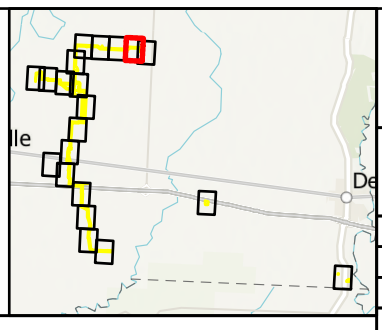
FIGURE A-2
PAGE 19 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
MILE POST



BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-2
PAGE 20 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-2 TOPOGRAPHIC SERIES

FFL-TAR-023.000

FORTENBERRY RD

13.0

13.1

13.2

FFL-PAR-024.000

Franklin Farms
Delivery
Meter Station

FFL-PAR-025.000

183

PARISH RD

85

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
MILE POST

BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

TOPOGRAPHIC MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

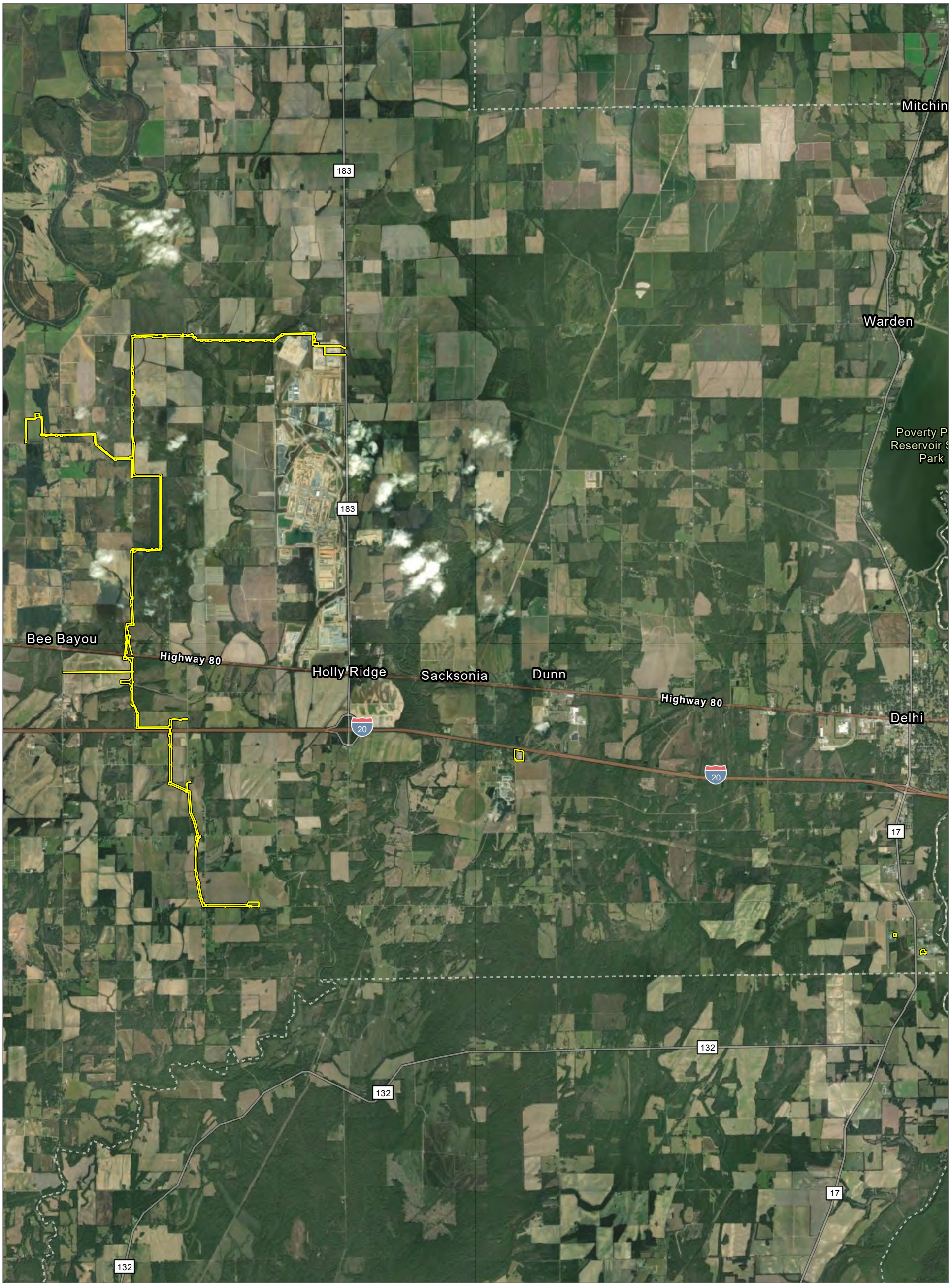
FIGURE A-2
PAGE 21 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

APPENDIX A-3

Aerial Mapping

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/9/2026, 12:53:29 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL OVERVIEW



PROJECT BOUNDARY

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 3,500 7,000 FEET

1:84,000 1" = 7,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP
(OVERVIEW)

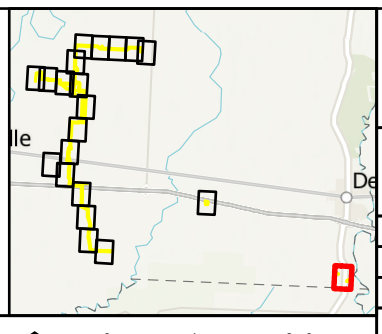
DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

TEMPORARY ACCESS ROAD
CONTRACTOR YARD
MILE POST



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

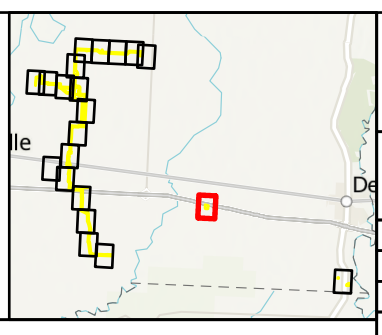
FIGURE A-3
PAGE 1 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

CONTRACTOR YARD

MILE POST



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 2 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

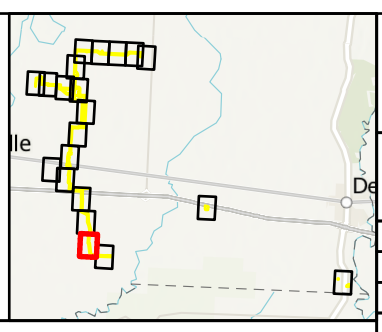
DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 3 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
MILE POST



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 4 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 5 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

FFL-TAR-007.000

Mengle Rd

3.6 3.5 3.4 3.3 3.2 3.1 3.0 2.9 2.8 2.7 2.6

20

Ollie Morris Rd

O'Quin

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

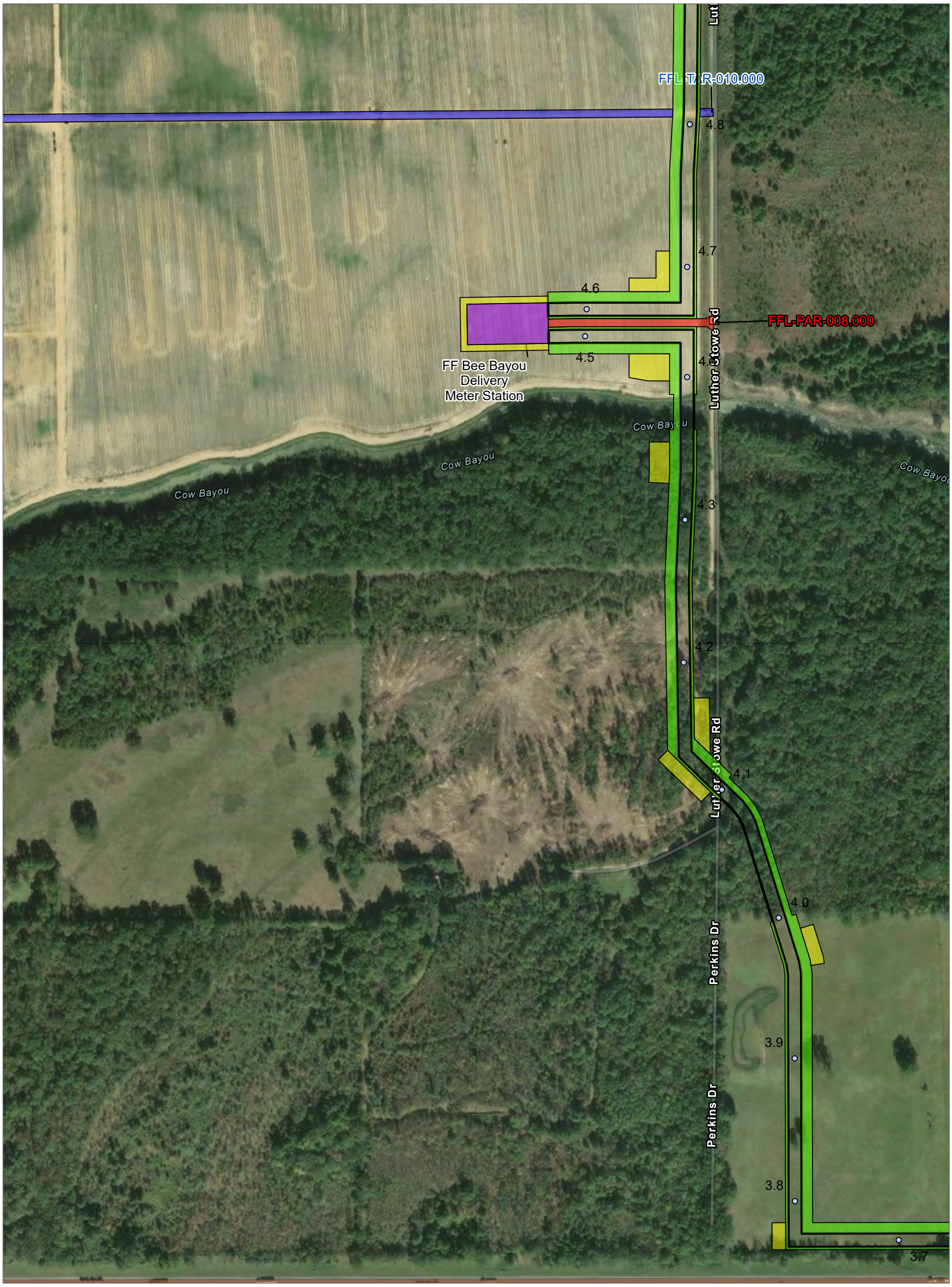
ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 6 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

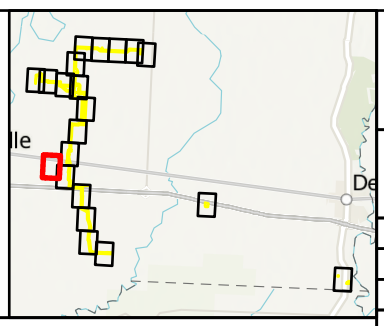
FIGURE A-3
PAGE 7 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

TEMPORARY ACCESS ROAD

MILE POST



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

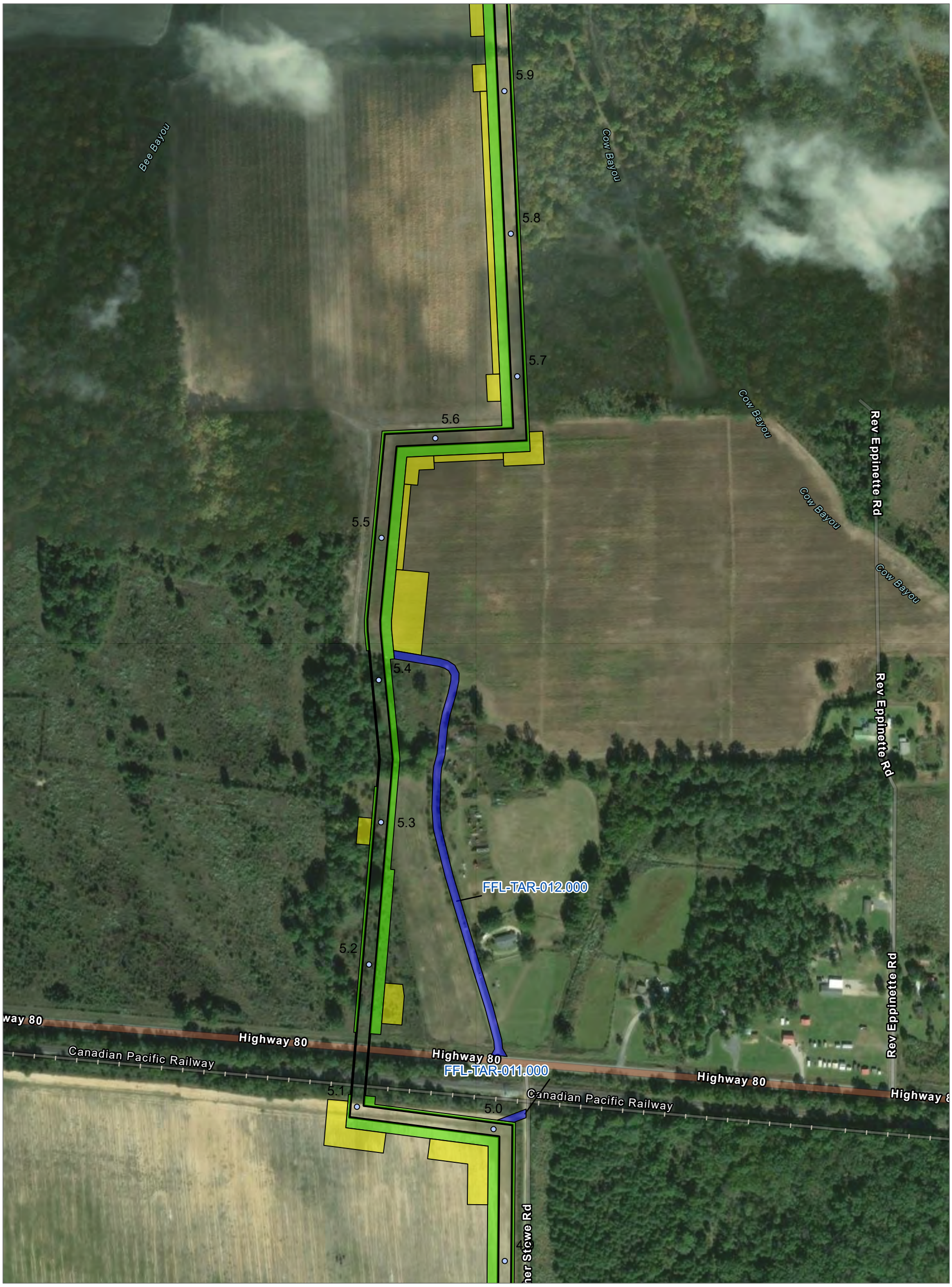
AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 8 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

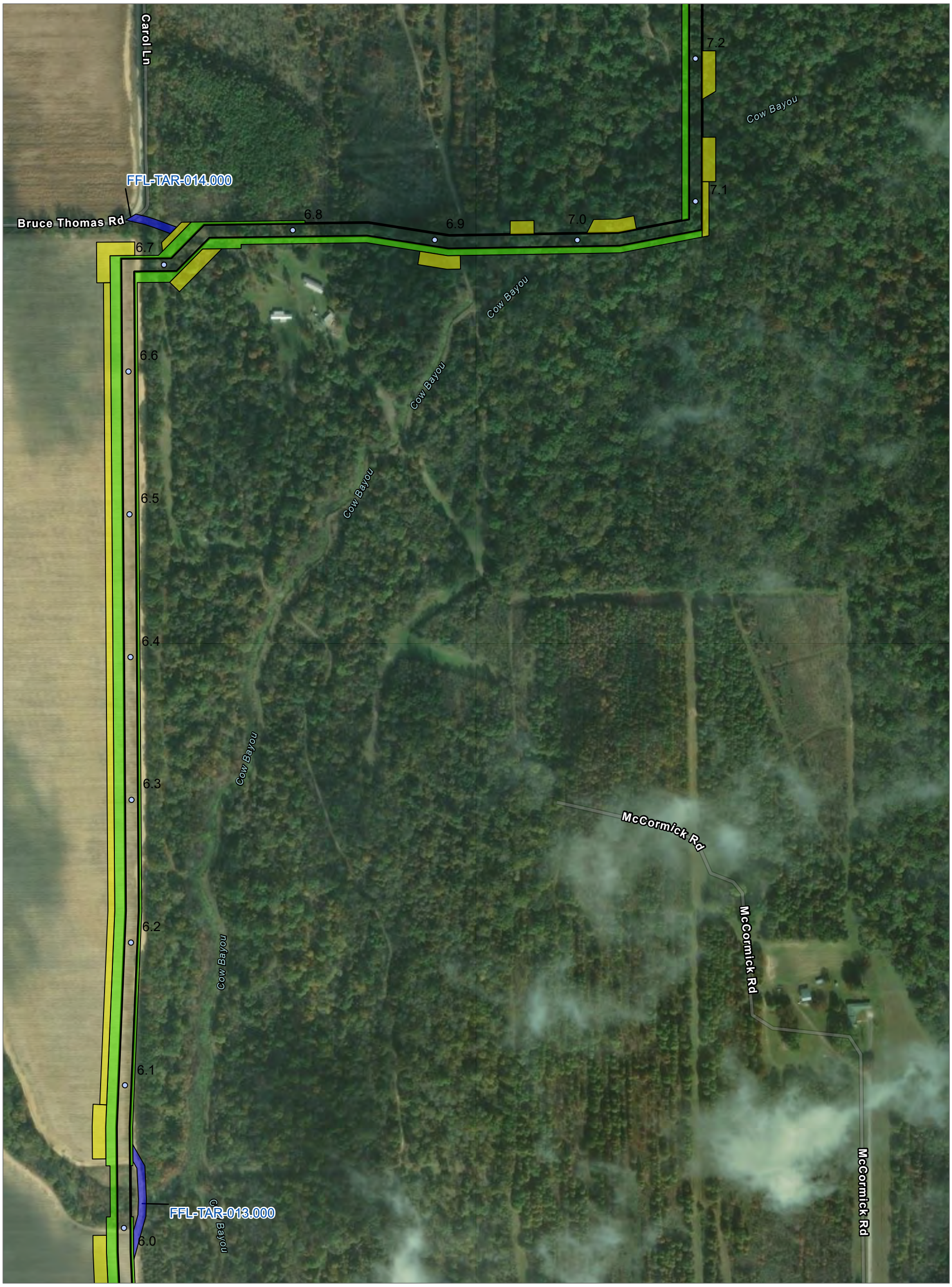
AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 9 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

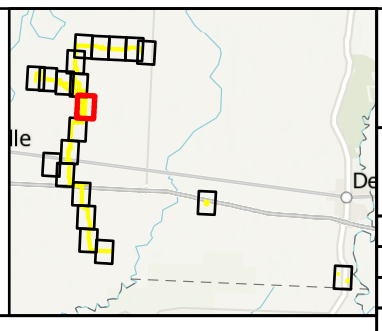
DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 10 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
MILE POST



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

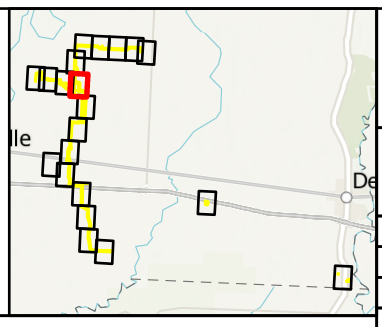
DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 11 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
INTERCONNECT SITE
MILE POST



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 12 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
INTERCONNECT SITE
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

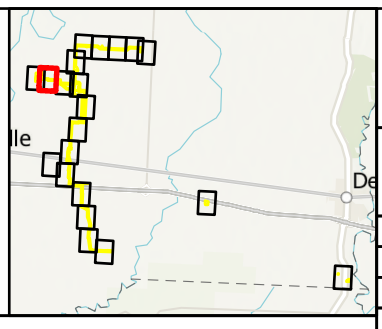
FIGURE A-3
PAGE 13 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
METER STATION
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,



0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

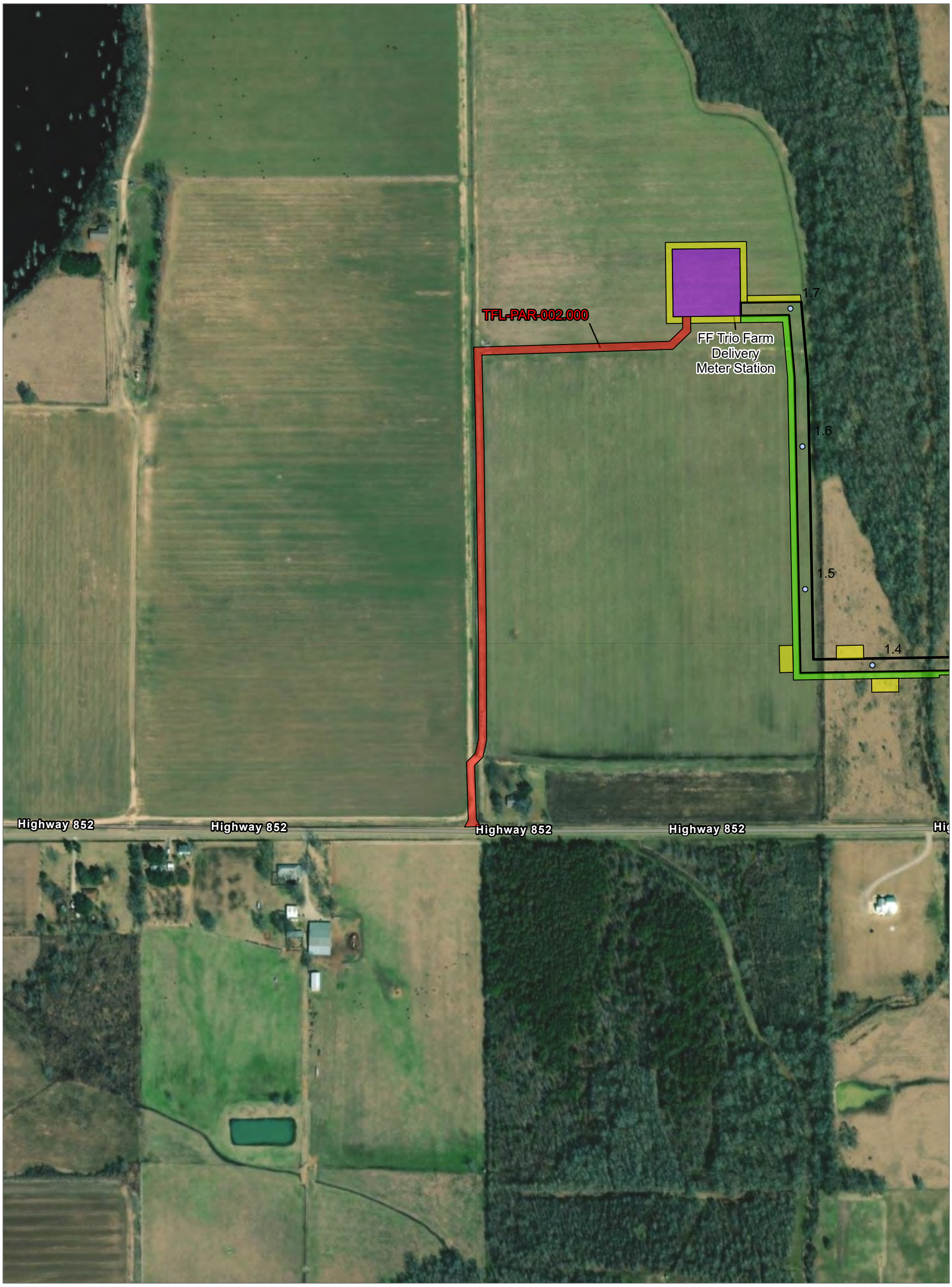
AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

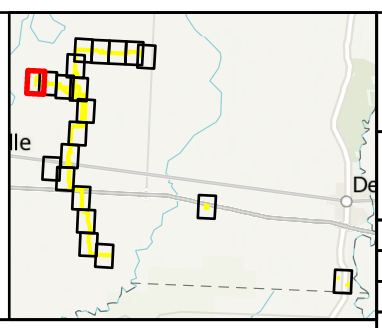
FIGURE A-3
PAGE 14 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
METER STATION
MILE POST



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 15 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
METER STATION
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

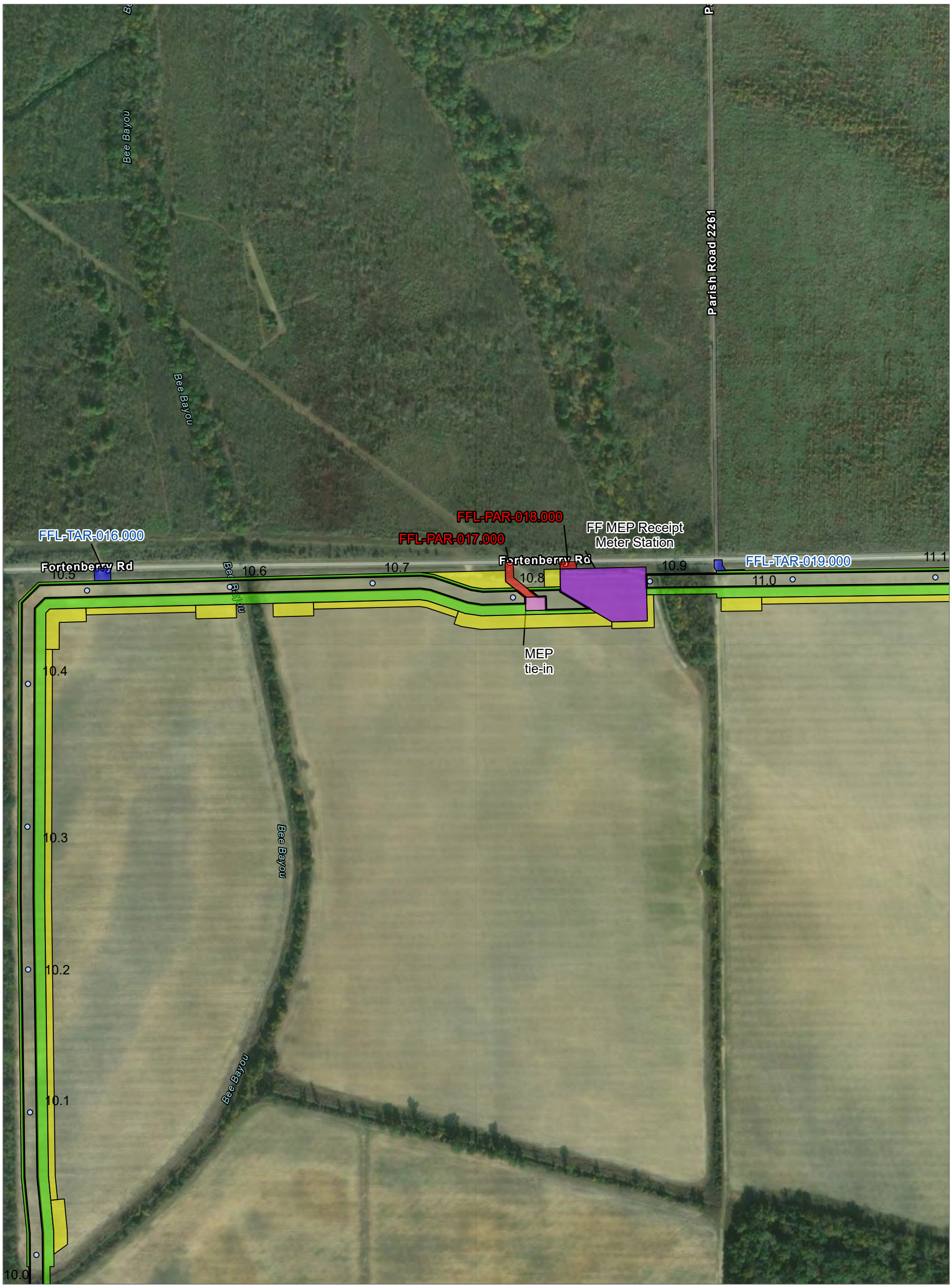
AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 16 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
TAP SITE
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

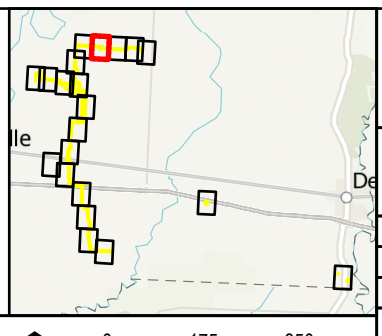
DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 17 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

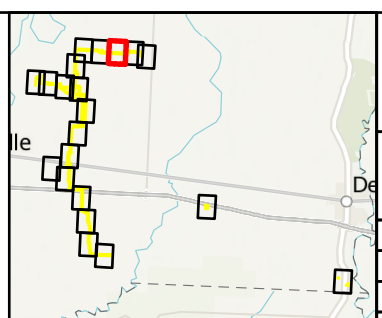
DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 18 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
MILE POST



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 19 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 20 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-3 AERIAL SERIES

FFL-TAR-023.000
13.0
13.1
13.2
Fortenberry Rd
Fortenberry Rd
New Ground Rd
Highway 183
Highway 183
FFL-PAR-024.000
Franklin Farms
Delivery
Meter Station
FFL-PAR-025.000
Highway 183
Parish Road 2304
Highway 183
Parish Road 2303

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
MILE POST

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC,

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

AERIAL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-3
PAGE 21 OF 21

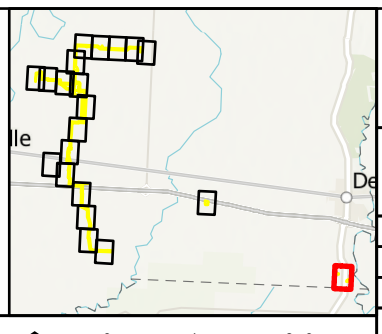
TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

APPENDIX A-4

National Wetland Inventory Map

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:42:53 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-4 NWI

TEMPORARY ACCESS ROAD
CONTRACTOR YARD
NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND
RIVERINE



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 1 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:42:53 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-4 NWI

CONTRACTOR YARD

NWI WETLAND TYPE

FRESHWATER FORESTED/SHRUB WETLAND

FRESHWATER POND

RIVERINE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

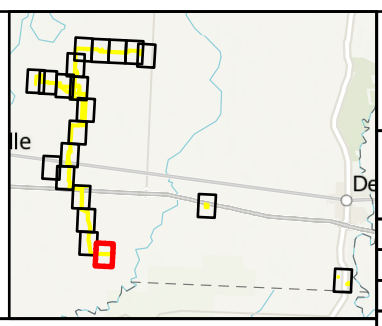
FIGURE A-4
PAGE 2 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:42:53 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-4 NWI

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
MILE POST

NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND
RIVERINE



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

N
0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

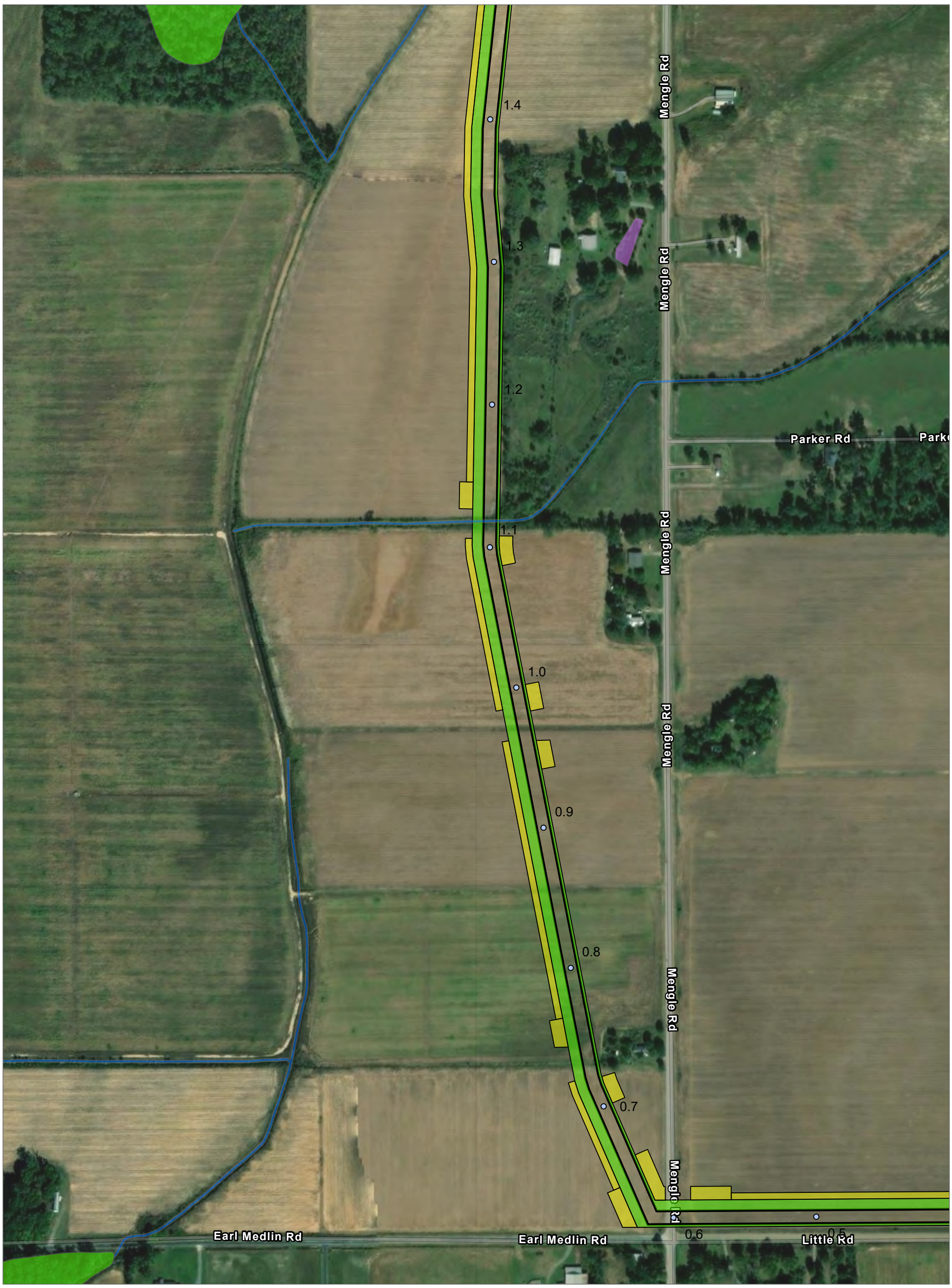
NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 3 OF 21

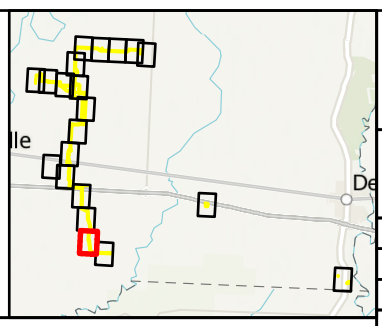
TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:42:53 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-4 NWI



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
MILE POST

NWI WETLAND TYPE
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND
RIVERINE



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE	**FIGURE A-4**	
APPROVED BY:	A. THOMPSON	**PAGE 4 OF 21**	
DATE:	JUNE 2026		

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:42:53 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-4 NWI

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST
NWI WETLAND TYPE
FRESHWATER FORESTED/SHRUB WETLAND
RIVERINE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 5 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST

NWI WETLAND TYPE
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND
RIVERINE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

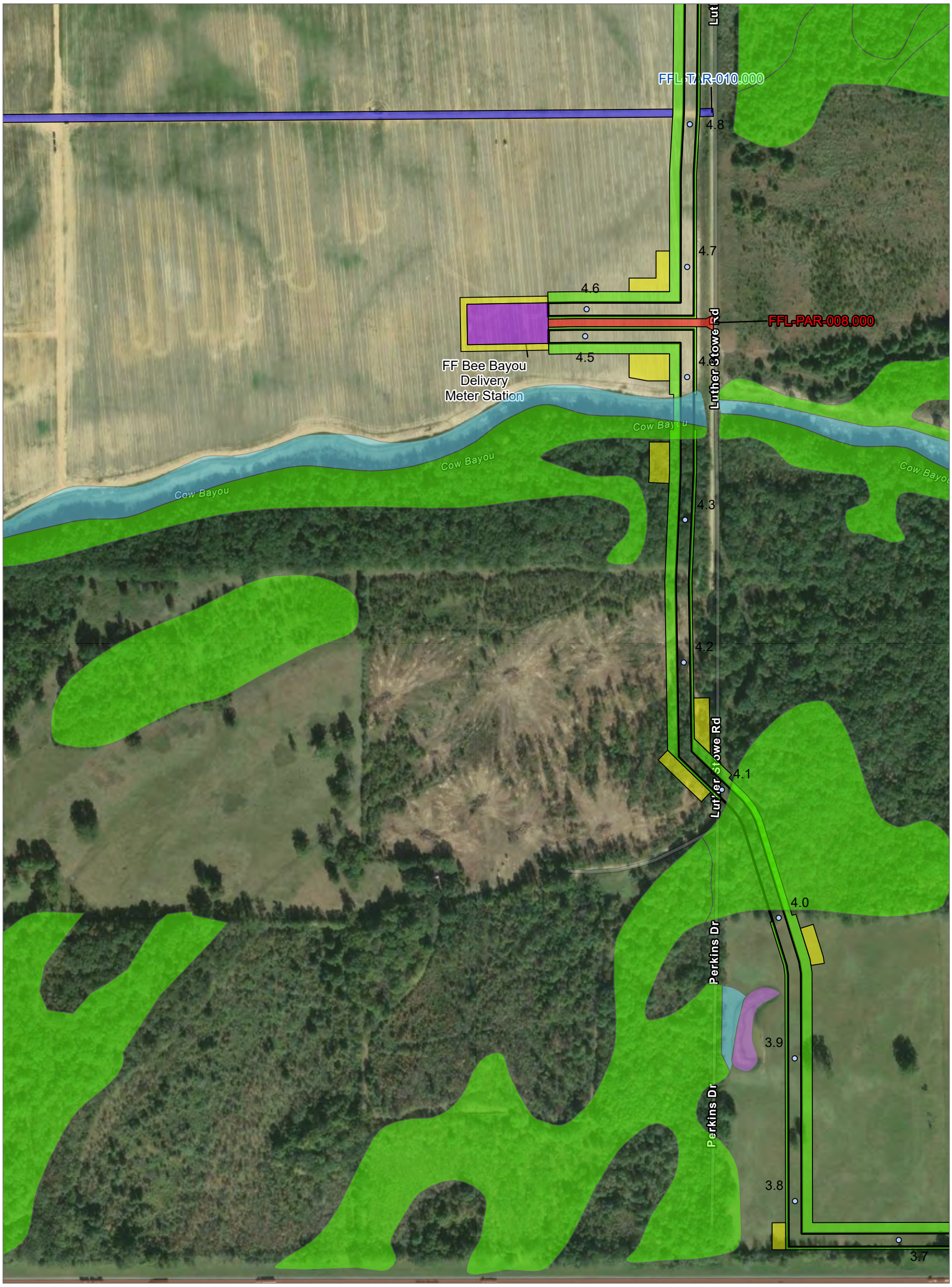
NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 6 OF 21

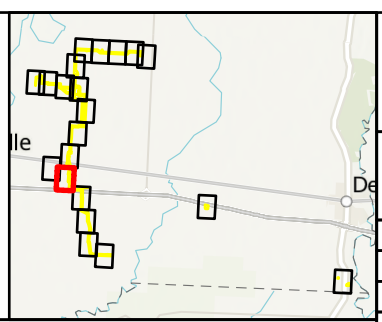
TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:42:53 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-4 NWI



PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
MILE POST

NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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DATE:	JUNE 2026		

FIGURE A-4
PAGE 7 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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Bee Bayou
Highway 583
Franklin Dr
Highway 80
Bee Bayou
Highway 80
Canadian Pacific Railway
Highway 80
Highway 80
Canadian Pacific Railway
Highway 583
Bee Bayou
Highway 583
FFL-TAR-009.000
Bee Bayou
Highway 583
Cow Bayou
Saint Peter Church
Cow Bayou
583

TEMPORARY ACCESS ROAD

NWI WETLAND TYPE

FRESHWATER EMERGENT WETLAND

FRESHWATER FORESTED/SHRUB WETLAND

FRESHWATER POND

RIVERINE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

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CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 8 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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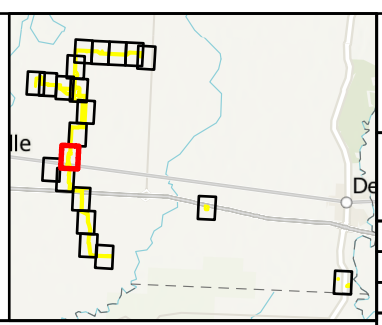
PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST
NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 9 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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Carol Ln
FFL-TAR-014.000
Bruce Thomas Rd
6.8
6.9
7.0
7.1
7.2
6.7
6.6
6.5
6.4
6.3
6.2
6.1
6.0
Cow Bayou
McCormick Rd
FFL-TAR-013.000

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD
MILE POST
NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

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CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 10 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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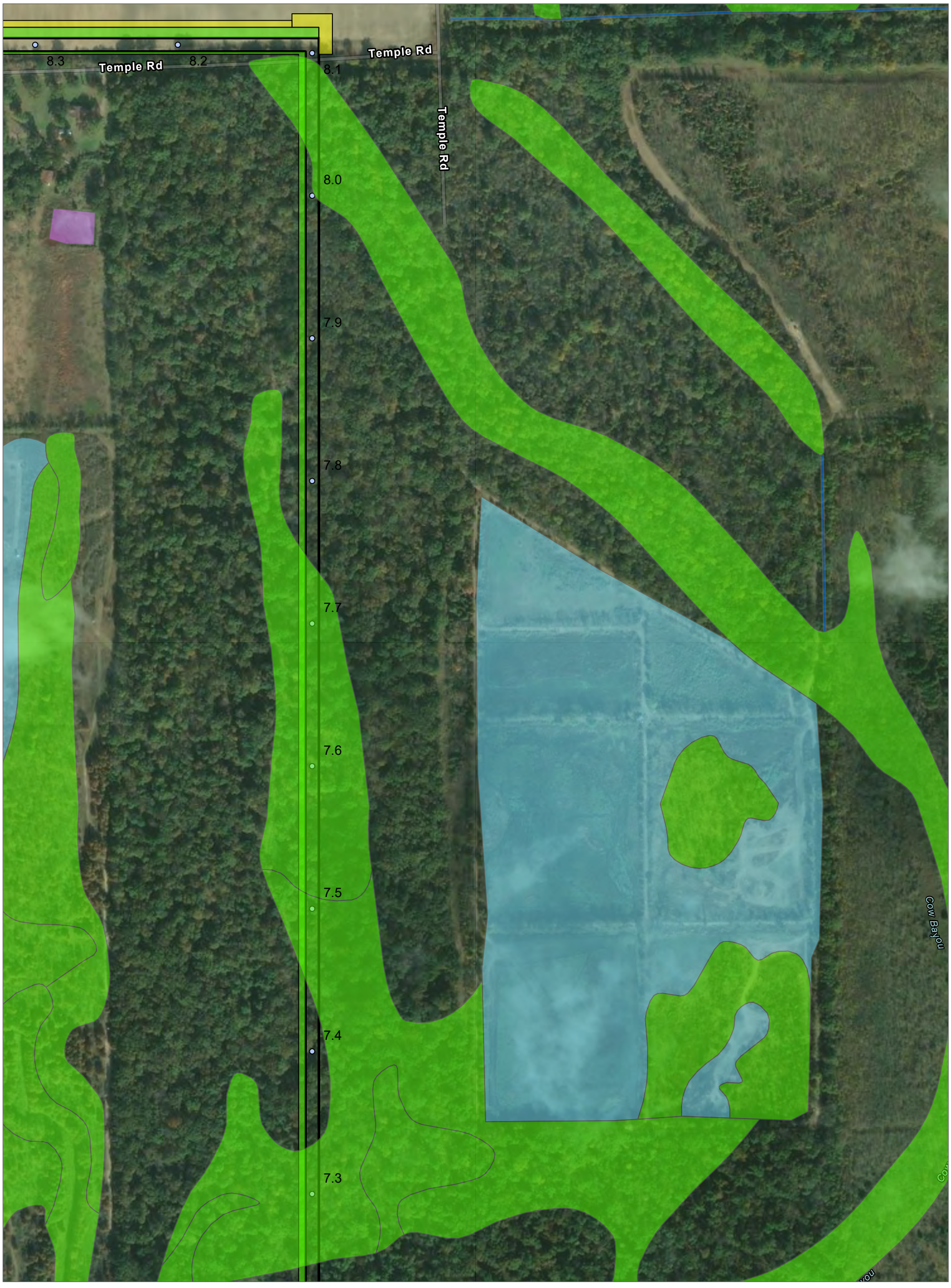
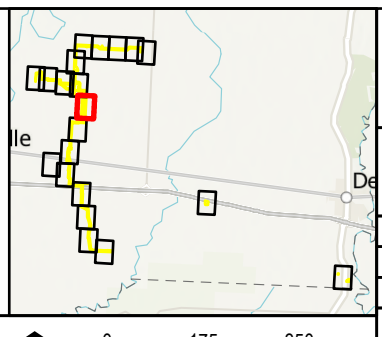


PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
MILE POST

NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND
RIVERINE



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 11 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
INTERCONNECT SITE
MILE POST

NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND
RIVERINE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 12 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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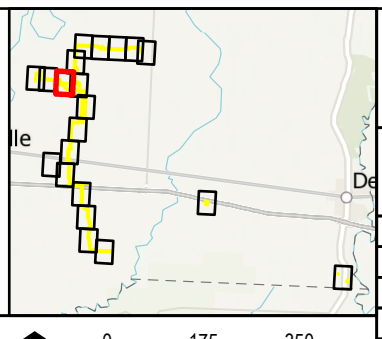
PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
INTERCONNECT SITE
MILE POST

NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND
RIVERINE



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

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APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 13 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
METER STATION
MILE POST

NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND
RIVERINE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 14 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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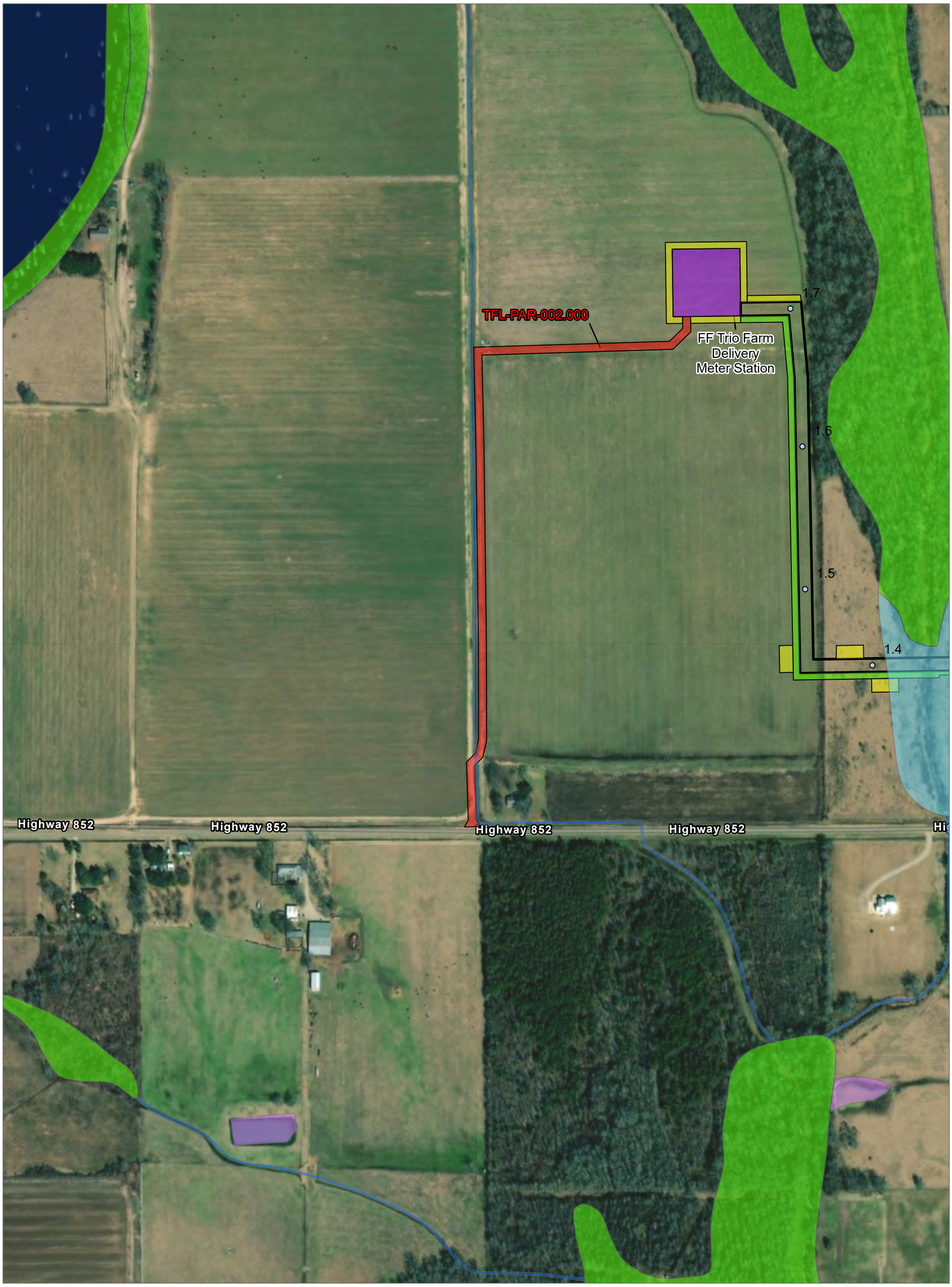
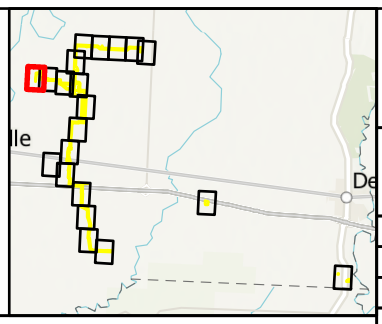


PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
METER STATION
MILE POST

NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND
LAKE
RIVERINE



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 15 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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10.2
10.1
10.0
9.9
9.8
9.7
9.6
9.5
9.4
Bee Bayou
Bee Bayou
Bee Bayou
Bee Bayou
FF Gulf Crossing
Receipt Meter Station
FFL-PAR-015.000
Smalling Rd
Smalling Rd
Smalling Rd
Smalling Rd

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
METER STATION
MILE POST

NWI WETLAND TYPE
FRESHWATER EMERGENT WETLAND
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND
RIVERINE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 16 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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Bee Bayou
Bee Bayou
Parish Road 2261
FFL-TAR-016.000
Fortenberry Rd
FFL-PAR-018.000
FFL-PAR-017.000
FF MEP Receipt Meter Station
Fortenberry Rd
FFL-TAR-019.000
10.5
10.6
10.7
10.8
10.9
11.0
11.1
MEP tie-in
10.4
10.3
10.2
10.1
10.0
Bee Bayou
Bee Bayou
Bee Bayou

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
TAP SITE
MILE POST
NWI WETLAND TYPE
FRESHWATER FORESTED/SHRUB WETLAND
RIVERINE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 17 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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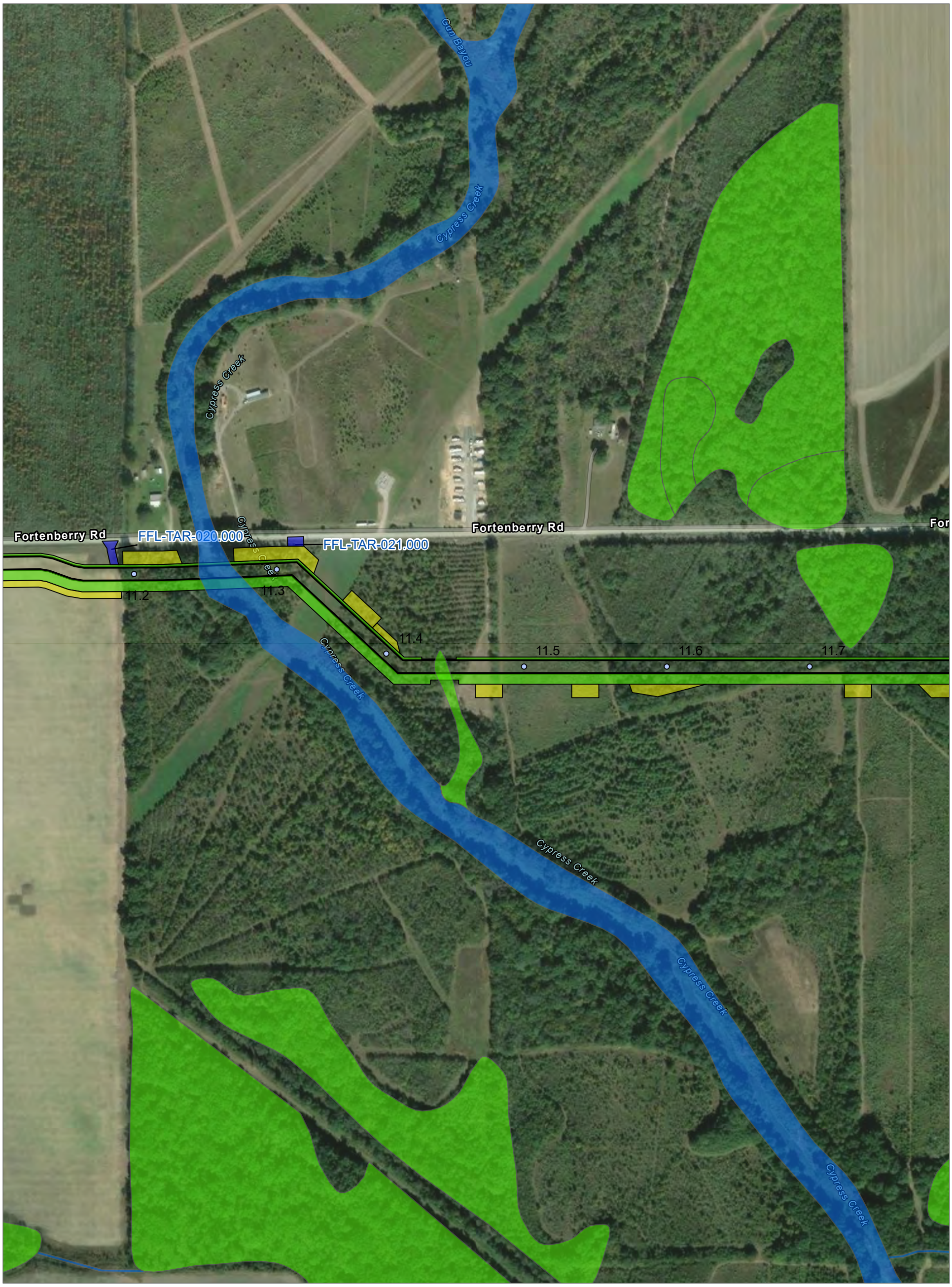
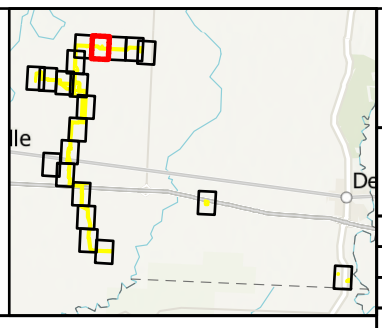


PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
TEMPORARY ACCESS ROAD

MILE POST

NWI WETLAND TYPE
FRESHWATER FORESTED/SHRUB WETLAND
RIVERINE



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
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DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 18 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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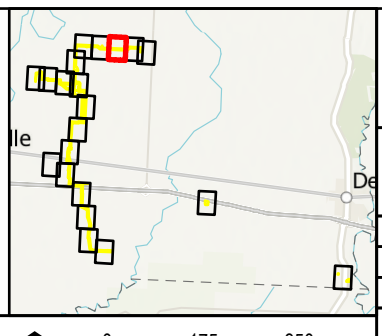
PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
MILE POST

NWI WETLAND TYPE
FRESHWATER FORESTED/SHRUB WETLAND
RIVERINE



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IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 19 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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Boggy Bayou
Boggy Bayou
Boggy Bayou
Boggy Bayou
Boggy Bayou
FFL-TAR-022.000
FFL-TAR-023.000
Fortenberry Rd
Fortenberry Rd
Fortenberry R
12.5
12.6
12.7
12.9
13.0
13.1
13.2
Jaggers Ln
Jaggers Ln
Franklin Farms
Delivery
Meter Station

- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION
- MILE POST

NWI WETLAND TYPE
- FRESHWATER EMERGENT WETLAND
- FRESHWATER FORESTED/SHRUB WETLAND
- RIVERINE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE A-4
PAGE 20 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:42:53 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: A-4 NWI

FFL-TAR-023.000
13.0
13.1
13.2
Fortenberry Rd
Fortenberry Rd
New Ground Rd
Highway 183
Highway 183
Highway 183
Highway 183
Highway 183
FFL-PAR-024.000
FFL-PAR-025.000
Franklin Farms
Delivery
Meter Station
Parish Road 2304
Parish Road 2303

PERMANENT EASEMENT
TEMPORARY WORKSPACE
ADDITIONAL TEMPORARY WORKSPACE
PERMANENT ACCESS ROAD
TEMPORARY ACCESS ROAD
METER STATION
MILE POST
NWI WETLAND TYPE
FRESHWATER FORESTED/SHRUB WETLAND
FRESHWATER POND
RIVERINE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NWI

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NATIONAL WETLAND INVENTORY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

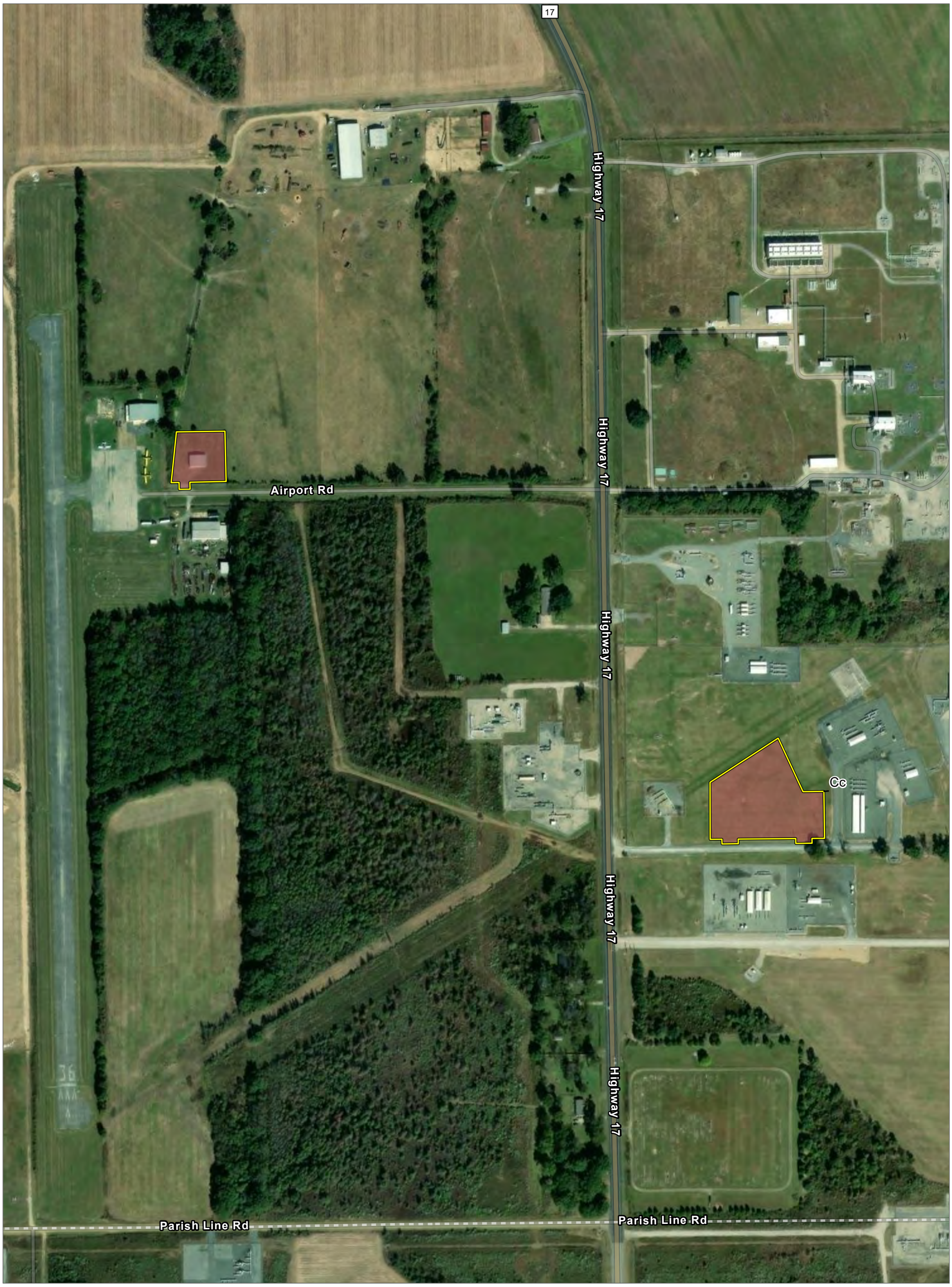
FIGURE A-4
PAGE 21 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

APPENDIX A-5

Soils Map

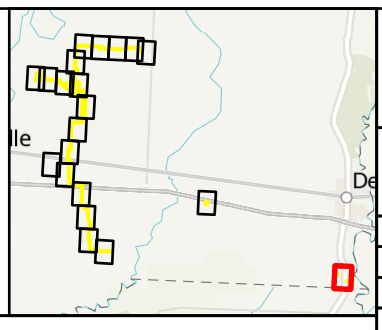
COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS



PROJECT BOUNDARY

MAP UNIT

CC, CALHOUN-CALLOWAY SILT LOAMS, GENTLY UNDULATING



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 1 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

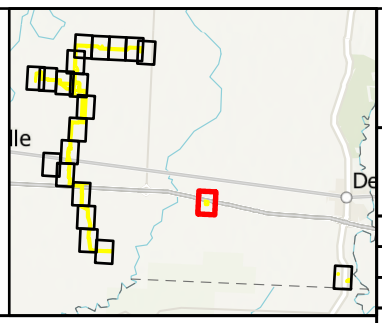
COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

PROJECT BOUNDARY

MAP UNIT

GG, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING

GR, GRENADA SILT LOAM, 1 TO 3 PERCENT SLOPES



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

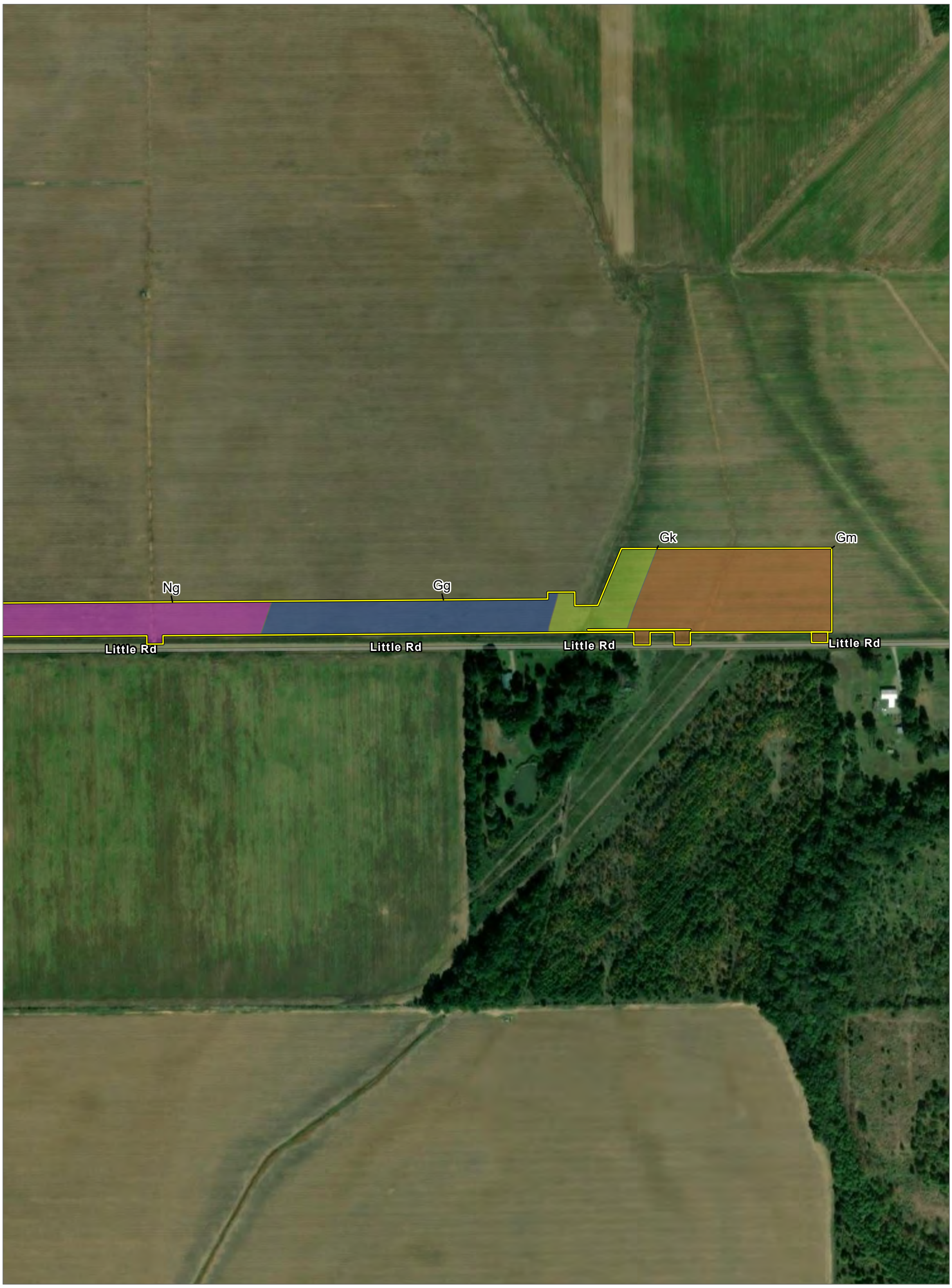
NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 2 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS



PROJECT BOUNDARY

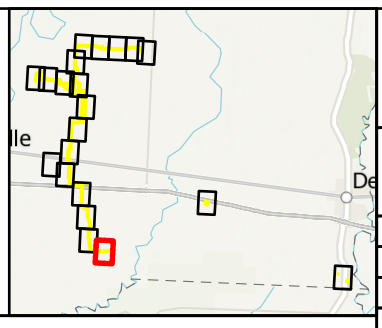
MAP UNIT

GG, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING

GK, GILBERT SILT LOAM

GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING

NG, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

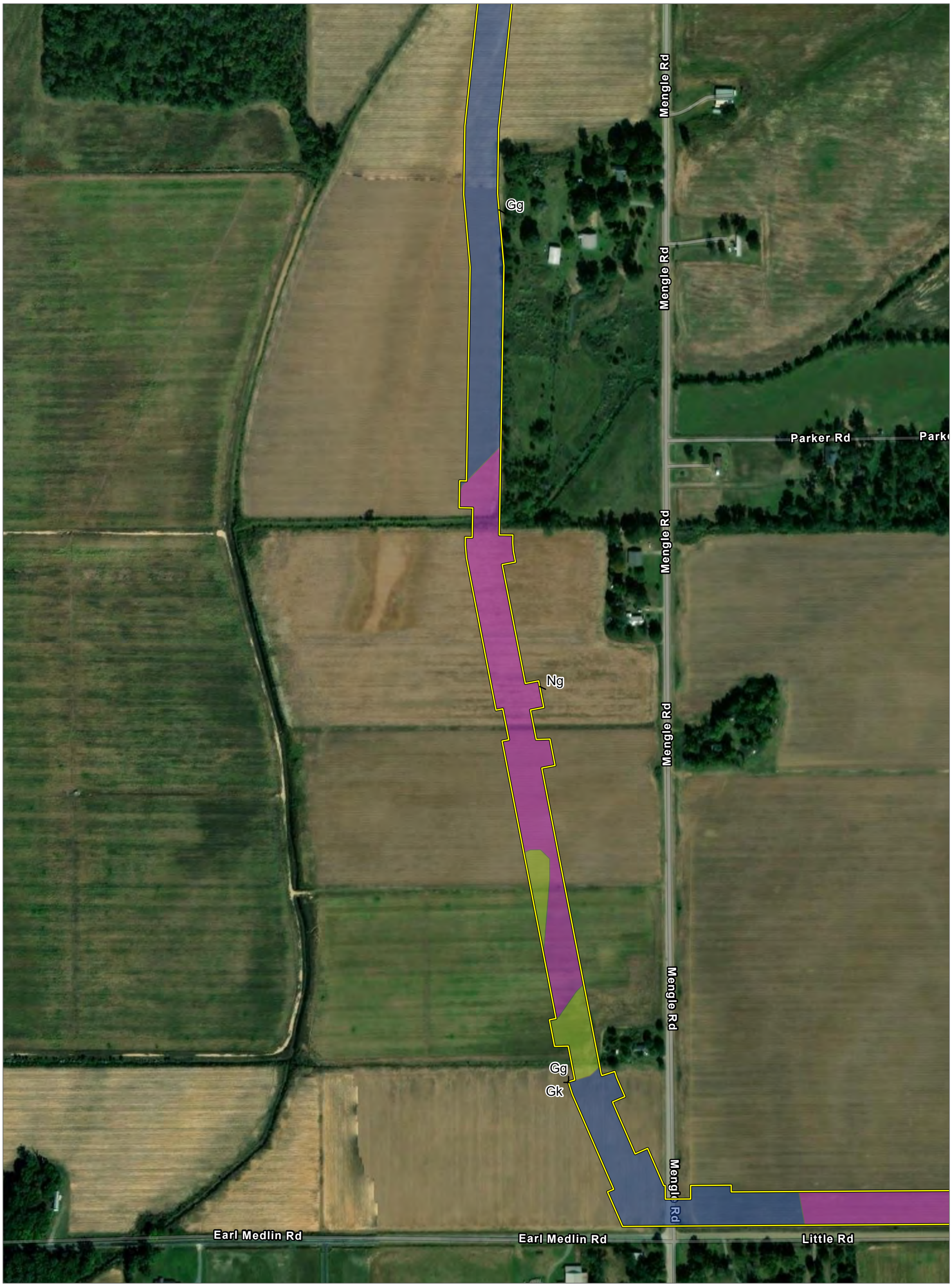
NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 3 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS



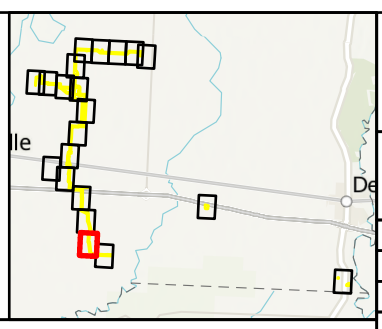
PROJECT BOUNDARY

MAP UNIT

GG, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING

GK, GILBERT SILT LOAM

NG, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 4 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

PROJECT BOUNDARY

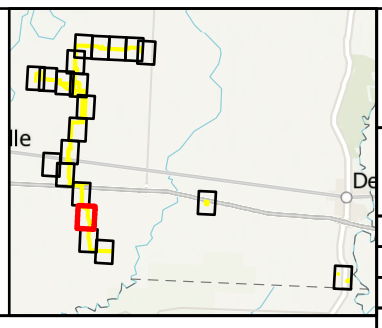
MAP UNIT

DE, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES

GG, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING

GK, GILBERT SILT LOAM

GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

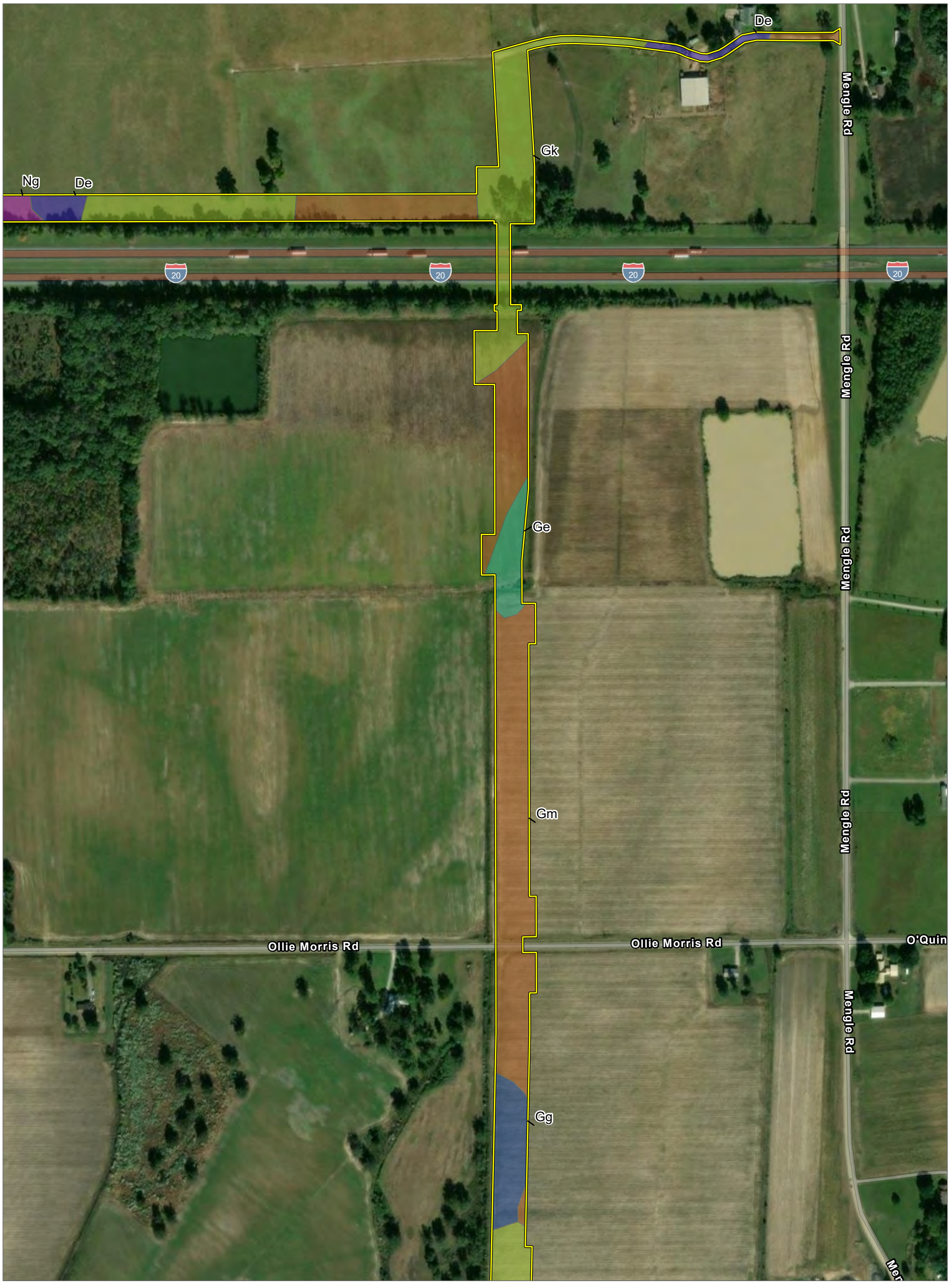
NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 5 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

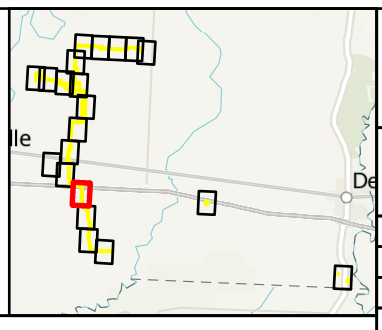
COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS



PROJECT BOUNDARY

MAP UNIT

- DE, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- GE, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- GG, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING
- GK, GILBERT SILT LOAM
- GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- NG, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 6 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

PROJECT BOUNDARY

MAP UNIT

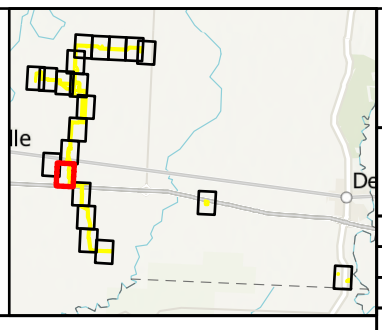
DE, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

GK, GILBERT SILT LOAM

GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING

NG, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 7 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

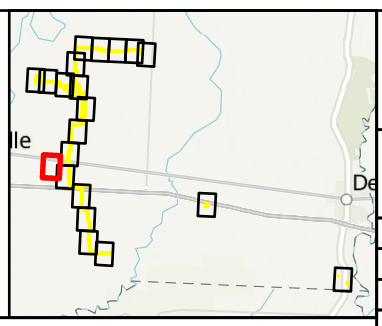
PROJECT BOUNDARY

MAP UNIT

DE, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 8 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

Bee Bayou
Gg
Cow Bayou
Cow Bayou
Rev Eppinette Rd
Cow Bayou
Cow Bayou
Fr
Gk
Df
De
Highway 80
Highway 80
Highway 80
Canadian Pacific Railway
Canadian Pacific Railway
Gm
Rev Eppinette Rd
Rev Eppinette Rd
ner Stowe Rd

PROJECT BOUNDARY

MAP UNIT

DE, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES

DF, DEXTER SILT LOAM, 3 TO 5 PERCENT SLOPES

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

GG, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING

GK, GILBERT SILT LOAM

GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

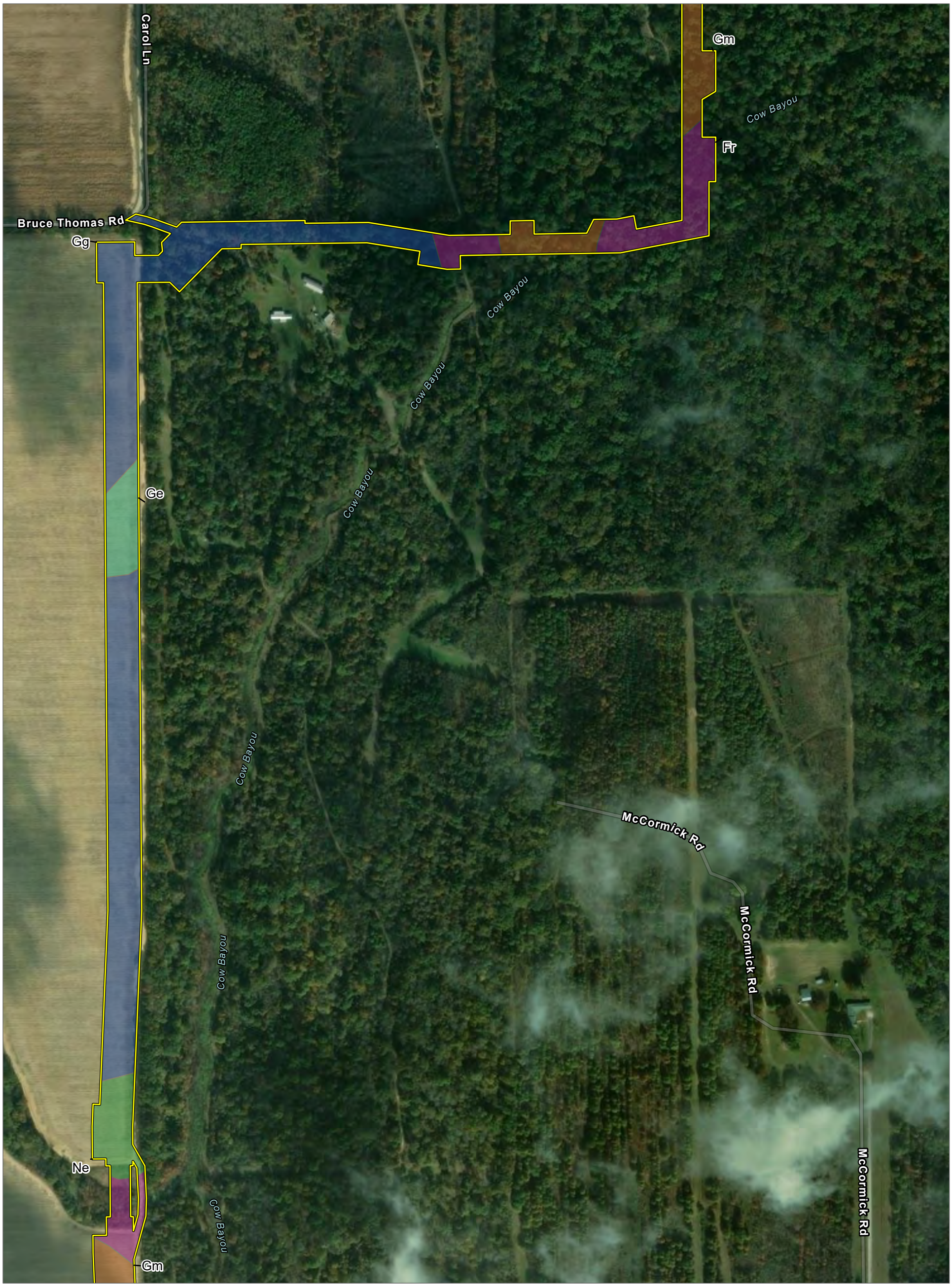
NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 9 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

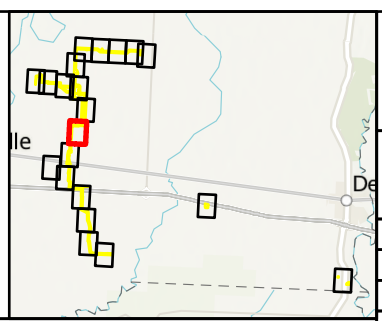
COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS



PROJECT BOUNDARY

MAP UNIT

- FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- GE, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- GG, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING
- GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- NE, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 10 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

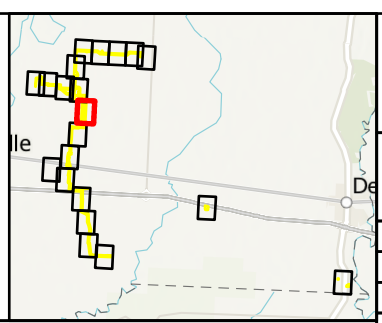
COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

PROJECT BOUNDARY

MAP UNIT

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 11 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

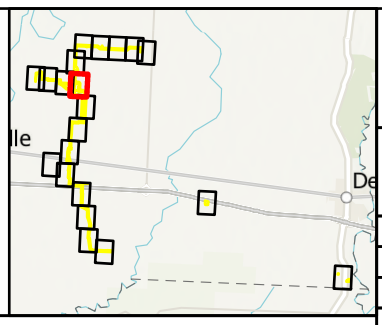
PROJECT BOUNDARY

MAP UNIT

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING

NG, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 12 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

PROJECT BOUNDARY

MAP UNIT

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING

NG, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 13 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

PROJECT BOUNDARY

MAP UNIT

DE, DEXTER SILTY LOAM, 1 TO 3 PERCENT SLOPES

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

NG, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING

PC, PERRY SILTY CLAY LOAM

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 14 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

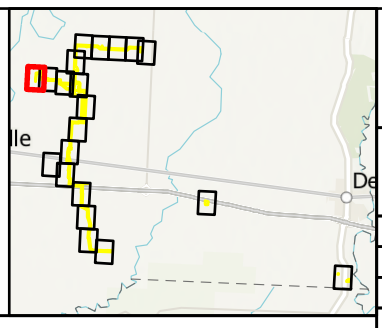
PROJECT BOUNDARY

MAP UNIT

HB, HEBERT SILT LOAM, 0 TO 1 PERCENT SLOPES

NG, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING

PC, PERRY SILTY CLAY LOAM



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 15 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

PROJECT BOUNDARY

MAP UNIT

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

NE, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES

NG, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING

PD, PERRY CLAY

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 16 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

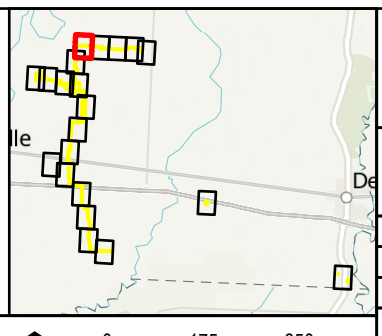
PROJECT BOUNDARY

MAP UNIT

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

NE, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES

PD, PERRY CLAY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

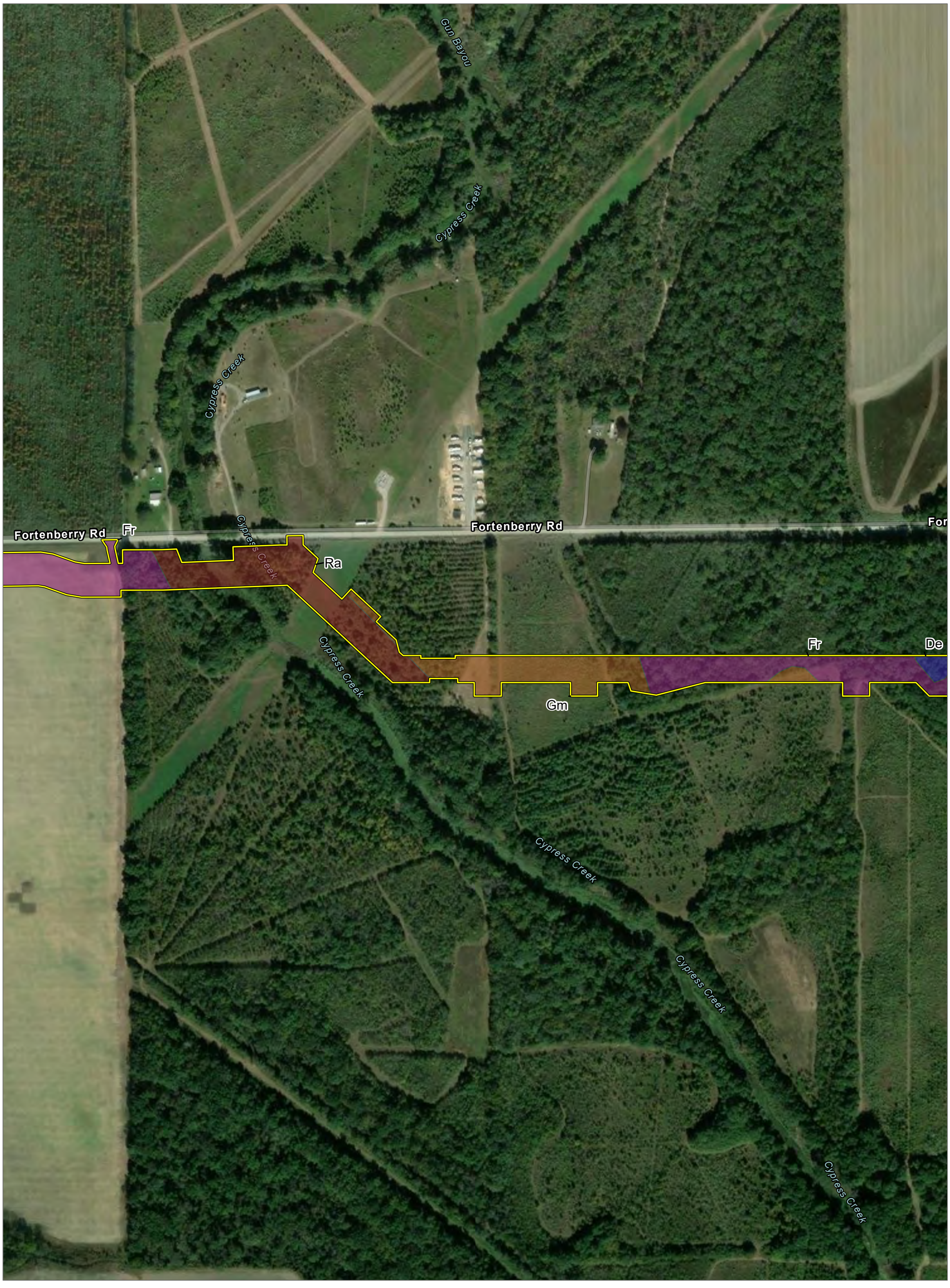
NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 17 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS



PROJECT BOUNDARY

MAP UNIT

DE, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING

RA, RILLA SILT LOAM, 0 TO 1 PERCENT SLOPES

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 18 OF 21

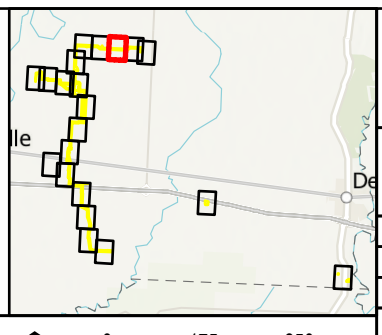
TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

PROJECT BOUNDARY

MAP UNIT

- DA, DEERFORD SILT LOAM
- DE, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- EG, EGYPT SILT LOAM, 1 TO 3 PERCENT SLOPES
- FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- NE, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

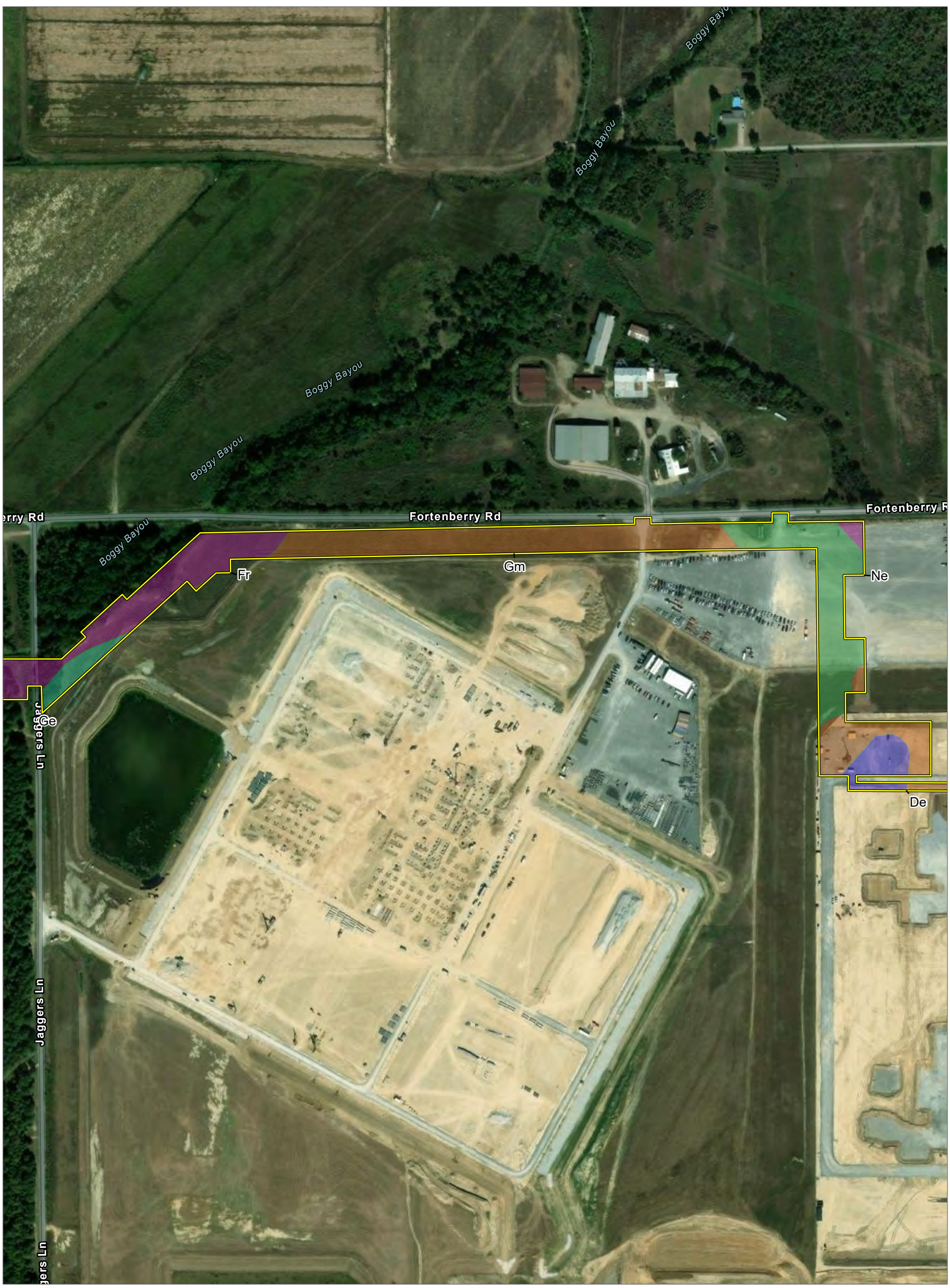
NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 19 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

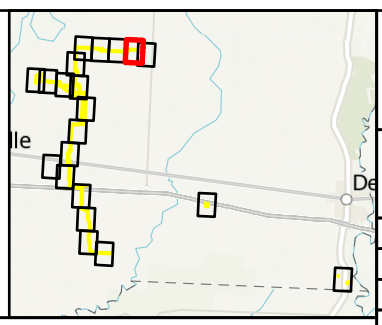
COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS



PROJECT BOUNDARY

MAP UNIT

- DE, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- GE, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- NE, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 20 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 7.A-1 NRCS SOILS

PROJECT BOUNDARY

MAP UNIT

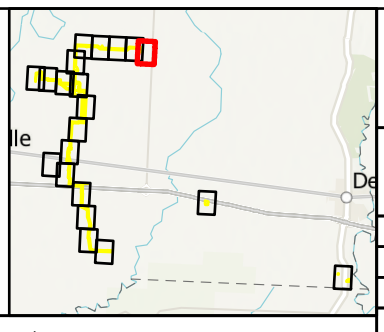
DE, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES

FR, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE

GG, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING

GM, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING

NE, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS SSURGO

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

NRCS SSURGO SOIL MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7.A-1
PAGE 21 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

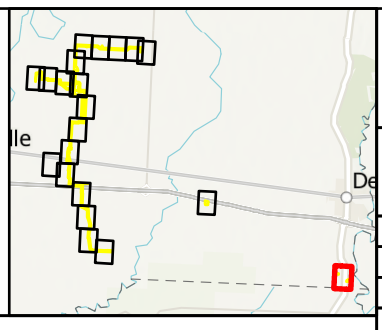
APPENDIX A-6

Habitats Crossed by the Project

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 3.A-1 HABITATS

WORKSPACE TYPE
- TEMPORARY ACCESS ROAD
- CONTRACTOR YARD

LANDCOVER
- DEVELOPED
- OPEN LAND
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 1 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 3.A-1 HABITATS

WORKSPACE TYPE
- CONTRACTOR YARD

LANDCOVER
- DEVELOPED
- OPEN LAND
- PEM WETLAND

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

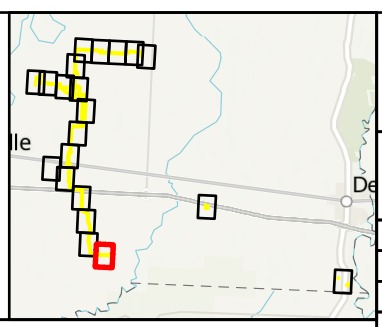
FIGURE 3.A-1
PAGE 2 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 3.A-1 HABITATS

WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

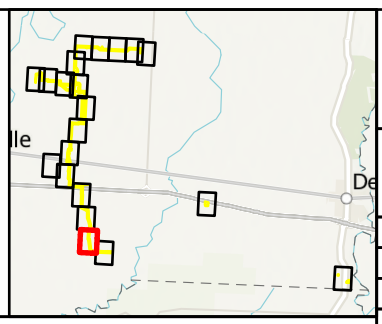
FIGURE 3.A-1
PAGE 3 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 3.A-1 HABITATS

WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- OPEN LAND
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

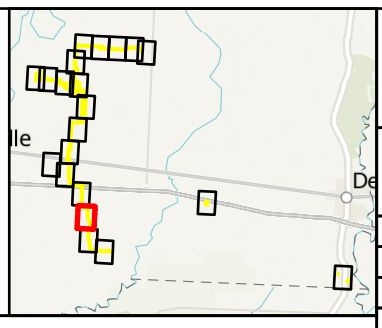
FIGURE 3.A-1
PAGE 4 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 3.A-1 HABITATS

WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- TEMPORARY ACCESS ROAD

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- OPEN LAND
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 5 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 3.A-1 HABITATS

FFL-TAR-007.000

Mengle Rd

20

Ollie Morris Rd

O'Quin

WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- TEMPORARY ACCESS ROAD

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- OPEN LAND
- PEM WETLAND
- SURFACE WATER

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

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FIGURE 3.A-1
PAGE 6 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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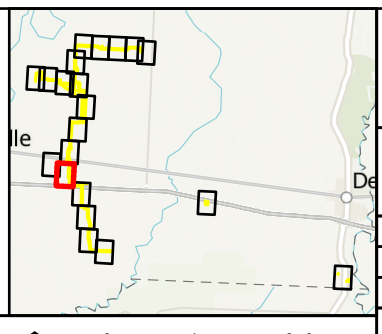
WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- FORESTED/WOODLAND
- OPEN LAND
- PFO WETLAND
- SILVICULTURE
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
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HABITATS CROSSED BY PROJECT FACILITIES

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FIGURE 3.A-1
PAGE 7 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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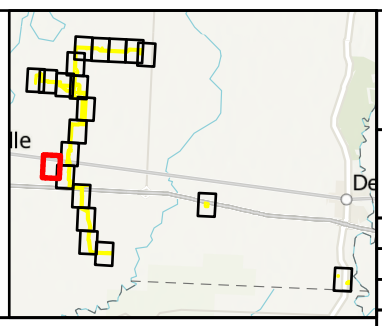
WORKSPACE TYPE

TEMPORARY ACCESS ROAD

LANDCOVER

AGRICULTURAL LAND



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

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FIGURE 3.A-1
PAGE 8 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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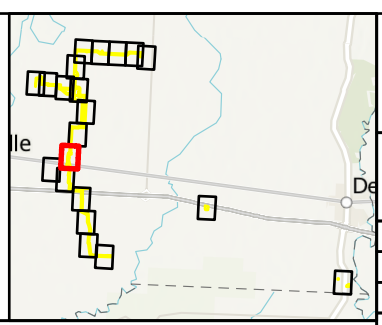
WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- TEMPORARY ACCESS ROAD

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- FORESTED/WOODLAND
- OPEN LAND
- PEM WETLAND
- PFO WETLAND
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

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FIGURE 3.A-1
PAGE 9 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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Carol Ln
Cow Bayou
FFL-TAR-014.000
Bruce Thomas Rd
Cow Bayou
Cow Bayou
Cow Bayou
Cow Bayou
Cow Bayou
McCormick Rd
McCormick Rd
McCormick Rd
Cow Bayou
FFL-TAR-013.000

WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- TEMPORARY ACCESS ROAD

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- FORESTED/WOODLAND
- OPEN LAND
- PFO WETLAND
- SURFACE WATER

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

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1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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DATE:	JUNE 2026		

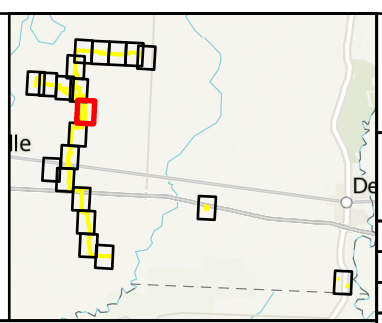
FIGURE 3.A-1
PAGE 10 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 3.A-1 HABITATS

WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE

LANDCOVER
- AGRICULTURAL LAND
- PFO WETLAND
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

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DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 11 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- INTERCONNECT SITE

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- FORESTED/WOODLAND
- OPEN LAND
- PEM WETLAND
- PFO WETLAND
- SURFACE WATER

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

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APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 12 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- INTERCONNECT SITE

LANDCOVER
- AGRICULTURAL LAND
- FORESTED/WOODLAND
- OPEN LAND
- PEM WETLAND
- PFO WETLAND
- SURFACE WATER

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

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APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 13 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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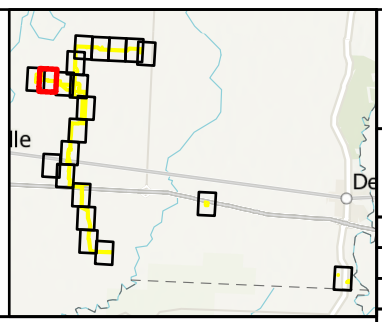
WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- METER STATION

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- FORESTED/WOODLAND
- OPEN LAND
- PEM WETLAND
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

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CHECKED BY:	A. CLINE		
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DATE:	JUNE 2026		

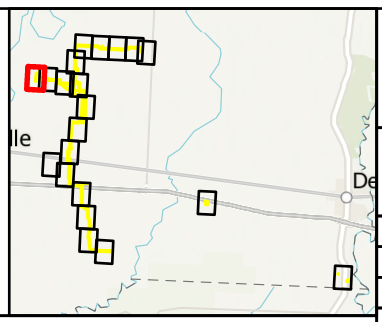
FIGURE 3.A-1
PAGE 14 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 3.A-1 HABITATS

WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- METER STATION

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- FORESTED/WOODLAND
- OPEN LAND
- PEM WETLAND
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

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DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 15 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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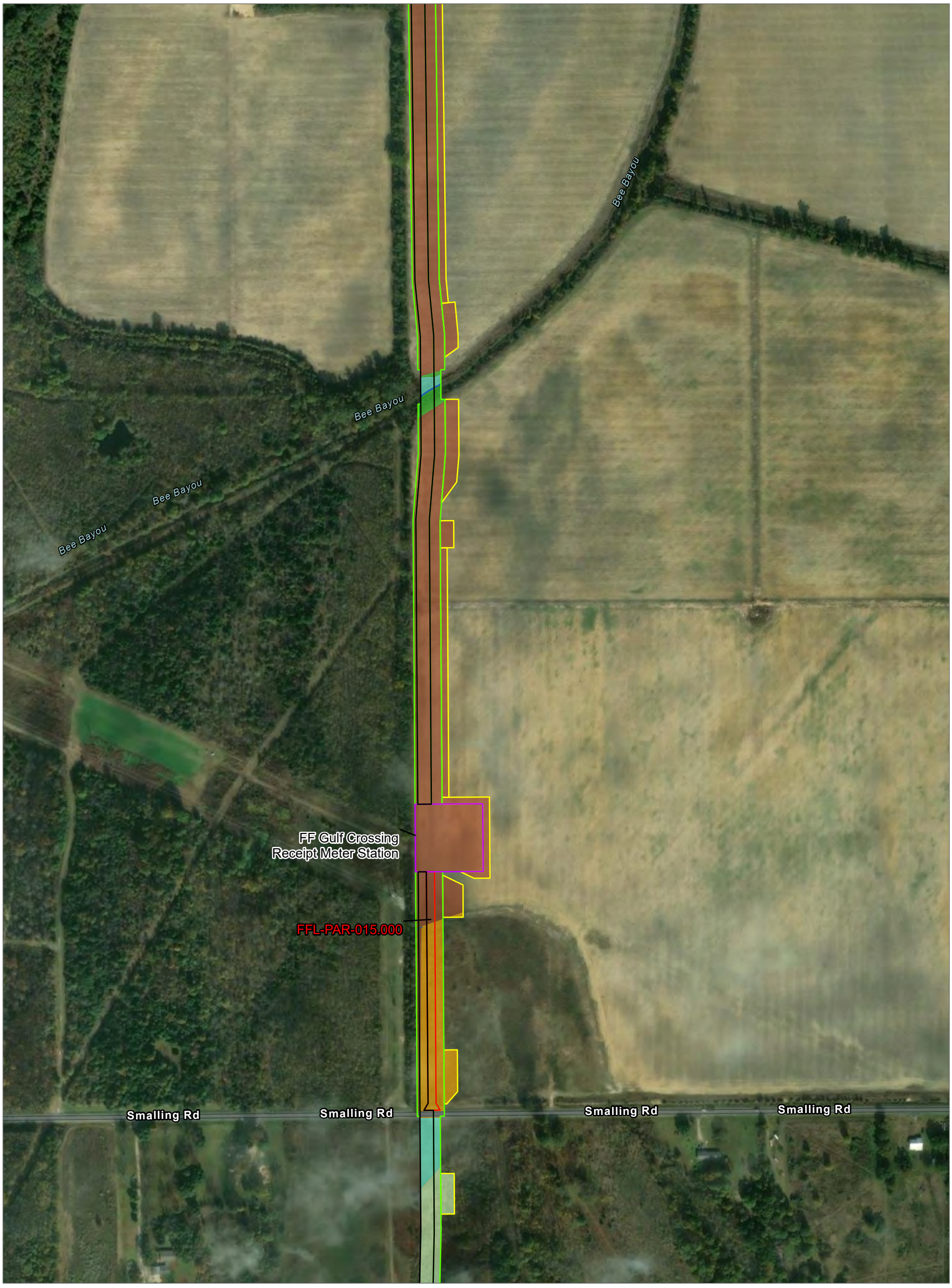

WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- METER STATION

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- FORESTED/WOODLAND
- OPEN LAND
- PEM WETLAND
- PFO WETLAND
- SURFACE WATER

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

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DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 16 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION
- TAP SITE

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- FORESTED/WOODLAND
- OPEN LAND
- PEM WETLAND
- PFO WETLAND
- SURFACE WATER

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

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CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 17 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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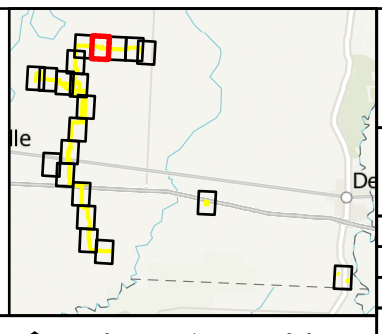
WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- TEMPORARY ACCESS ROAD

LANDCOVER
- AGRICULTURAL LAND
- DEVELOPED
- FORESTED/WOODLAND
- OPEN LAND
- PFO WETLAND
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

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ETC TIGER PIPELINE, LLC
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RICHLAND PARISH, LA

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APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 18 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE

LANDCOVER
- FORESTED/WOODLAND
- OPEN LAND
- PEM WETLAND
- PFO WETLAND
- SURFACE WATER

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

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CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 19 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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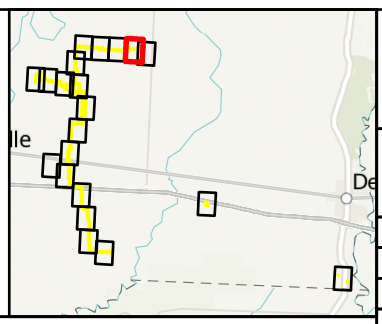
WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION

LANDCOVER
- DEVELOPED
- FORESTED/WOODLAND
- OPEN LAND
- PEM WETLAND
- PFO WETLAND
- SURFACE WATER



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

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1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 20 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 3.A-1 HABITATS

FFL-TAR-023.000

Fortenberry Rd

Fortenberry Rd

New Ground Rd

Highway 183

Highway 183

FFL-PAR-024.000

Franklin Farms
Delivery
Meter Station

FFL-PAR-025.000

Highway 183

Parish Road 2304

Highway 183

Parish Road 2303

WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIONAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION

LANDCOVER
- DEVELOPED
- OPEN LAND

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HABITATS CROSSED BY PROJECT FACILITIES

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 3.A-1
PAGE 21 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

APPENDIX A-7

Alignment Sheets

ENERGY TRANSFER

PROPOSED FRANKLIN FARMS LATERAL ALIGNMENT PACKAGE

WBS: C-24328-GL-2610001

ISSUE FOR PERMIT

PREPARED BY: FIRM No:23055
8521 McHARD RD, HOUSTON, TX 77053

DATE 05/29/2026

DRAWING NO. P0-01

CONSTRUCTION DRAWINGS	
DRAWING NO.	TITLE
P0-01	COVER SHEET
P1-01	INDEX SHEET
P3-001	ALIGNMENT SHEET (MP 0.00 TO MP 0.68)
P3-002	ALIGNMENT SHEET (MP 0.68 TO MP 1.72)
P3-003	ALIGNMENT SHEET (MP 1.72 TO MP 2.77)
P3-004	ALIGNMENT SHEET (MP 2.77 TO MP 3.37)
P3-005	ALIGNMENT SHEET (MP 3.37 TO MP 3.90)
P3-006	ALIGNMENT SHEET (MP 3.90 TO MP 5.06)
P3-006A	ACCESS ROAD DETAIL (MP 3.90 TO MP 5.06)
P3-007	ALIGNMENT SHEET (MP 5.06 TO MP 6.25)
P3-008	ALIGNMENT SHEET (MP 6.25 TO MP 6.76)
P3-009	ALIGNMENT SHEET (MP 6.76 TO MP 7.22)
P3-010	ALIGNMENT SHEET (MP 7.22 TO MP 8.07)
P3-011	ALIGNMENT SHEET (MP 8.07 TO MP 8.62)
P3-012	ALIGNMENT SHEET (MP 8.62 TO MP 9.66)
P3-013	ALIGNMENT SHEET (MP 9.66 TO MP 10.53)
P3-014	ALIGNMENT SHEET (MP 10.53 TO MP 11.57)
P3-015	ALIGNMENT SHEET (MP 11.57 TO MP 12.48)
P3-016	ALIGNMENT SHEET (MP 12.48 TO MP 13.24)
P3-016A	ACCESS ROAD DETAIL (MP 12.48 TO MP 13.24)

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						

CADDS: PLOT DATE: 05/29/26
FILE NAME: P1-01

P.L./STA. NO.
ACCOUNT NO.
CONSTRUCTION YEAR

	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.
SCALE: AS NOTED

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
INDEX SHEET
STA. 0+00 TO STA. 699+12
MP 0.00 TO MP 13.24
RICHLAND PARISH, LOUISIANA

PROJECT NO.
C-24328-GL-26100001
PREVIOUS DWG. NO.
SHEET 1 OF 1
DWG. NO.
P1-01
A
SHEET 1 OF 1

OWNERSHIP & TRACT NO.	0+00	LA-RI-01.00	31+76	LA-RI-02.00 / 36+00
COUNTY & STATE	RICHLAND PARISH, LOUISIANA			

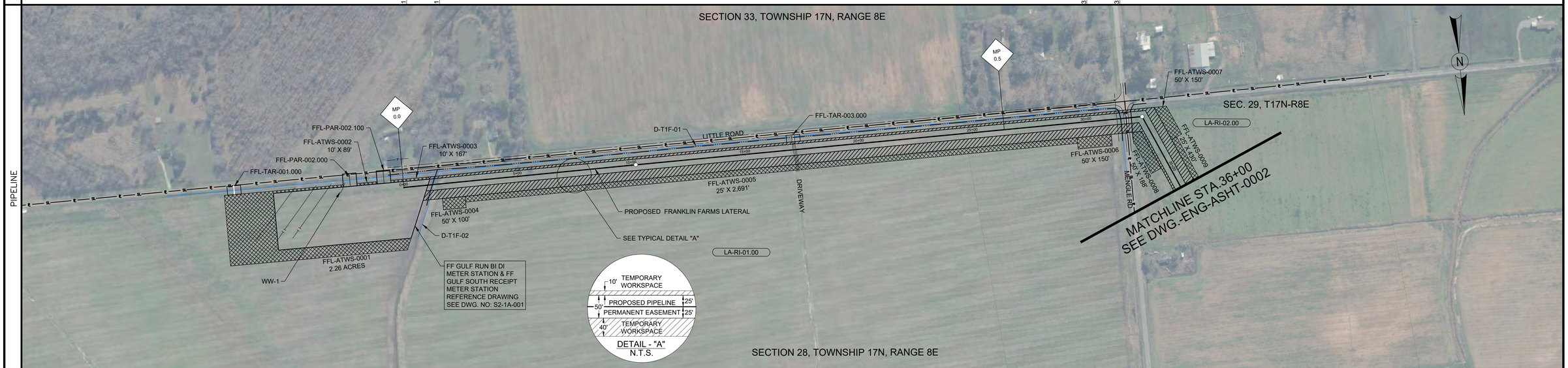
STATIONING

0+00 BEGIN STA.
1+29 ENTER AG-DITCH (D-T1F-02)
1+29 CL OF DRAINAGE DITCH
1+30 EXIT AG-DITCH (D-T1F-02)
10+20 P.I. < 000°36" RT.
17+12 EDGE OF DIRT ROAD
17+24 EDGE OF DIRT ROAD
31+35 OVERHEAD POWER LINE
31+51 ENTER AG-DITCH (D-T1F-01)
31+52 EXIT AG-DITCH (D-T1F-01)
31+52 CL OF DRAINAGE DITCH
31+66 EDGE OF ASPHALT ROAD
31+76 CL OF ASPHALT ROAD
31+78 PROPERTY LINE SECTION LINE
31+85 EDGE OF ASPHALT ROAD
31+99 CL OF DRAINAGE DITCH
32+49 P.I. < 090°57'32" RT.
36+00 MATCHLINE

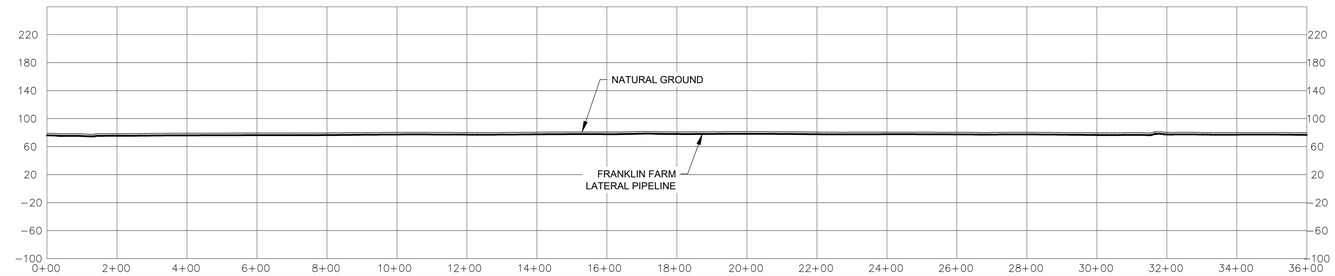
ENVIROMENTAL

D-T1F-02 AG-DITCH 1+29 1+30
D-T1F-01 AG-DITCH 31+51 31+52

PIPELINE



PROFILE



PIPE DETAIL

0+00	2080' (2)	20+80	1039' (3)	31+19	100' (5)	32+19	381' (3)	36+00

DESIGN FACTOR	0.72	0.60	0.50 / 0.60
CLASS LOCATION	CLASS 1	CLASS 2	
HCA			

MATERIAL SUMMARY

NO.	DESCRIPTION	QUANTITY
2	36.00" O.D. X 0.515" WT., GRADE X-70, 14-16 MILS FBE	2080 LF
3	36.00" O.D. X 0.618" WT., GRADE X-70, 14-16 MILS FBE	1420 LF
5	36.00" O.D. X 0.750" WT., GRADE X-70, 14-16 MILS FBE, 30-40 MILS ARO	100 LF

DWG. NO.	REFERENCE DRAWING TITLE
S2-1A-001	FF GULF RUN BI-DI MS & GULF SOUTH RECEIPT MS

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE: UPD FILE NAME: GM00001					

P.L./STA. NO. PL/STA

CONSTRUCTION YEAR 2026

	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.

SCALE: 1" = 200'

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ALIGNMENT SHEET
STA. 0+00 TO STA.36+00
MP 0.00 TO MP 0.68
RICHLAND PARISH, LOUISIANA

PROJECT NO. C-24328-GL-26100001
PREVIOUS DWG. NO. N/A
SHEET N/A OF 18
DWG. NO. P3-001 (A)
SHEET 1 OF 18

| | | |

OWNERSHIP & TRACT NO.	91+00	LA-RI-07.00	98+78	LA-RI-08.00	112+15	LA-RI-09.00	119+80	LA-RI-09.00	134+46	LA-RI-10.00	146+00
COUNTY & STATE	RICHLAND PARISH, LOUISIANA										

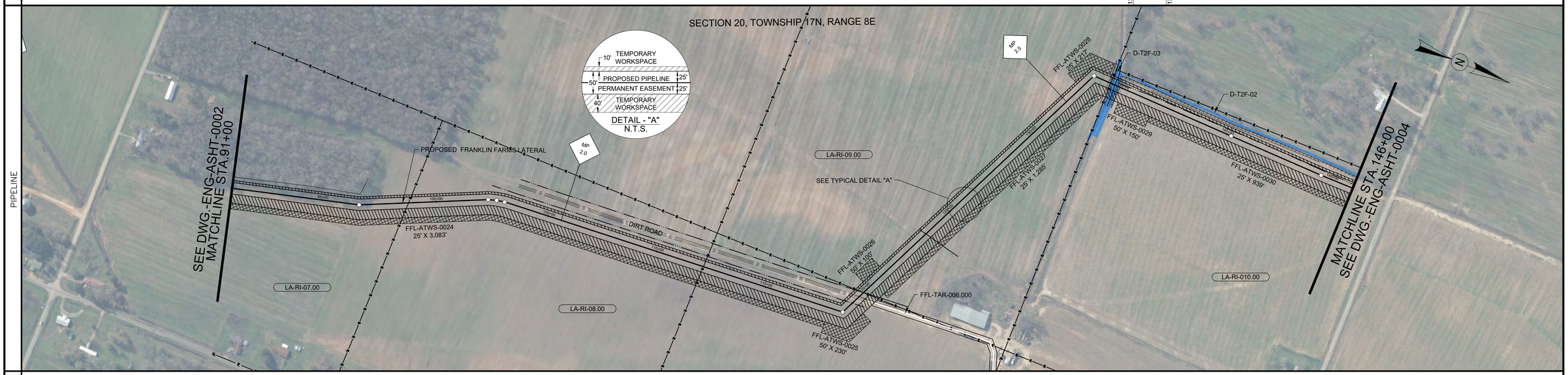
STATIONING

91+00 MATCHLINE
96+61 P.I. < 9°43'16" LT.
98+78 PROPERTY LINE
102+26 P.I. < 6°57'38" RT.
102+63 P.I. < 6°42'00" RT.
103+03 P.I. < 6°42'00" RT.
112+15 PROPERTY LINE
118+56 P.I. < 61°16'08" LT.
119+13 CL OF DRAINAGE DITCH
119+42 EDGE OF DIRT ROAD
119+49 CL OF DIRT ROAD
119+56 EDGE OF DIRT ROAD
119+80 PROPERTY LINE
119+81 CL OF DRANAGE DITCH
123+55 CL OF DRAINAGE DITCH
133+65 P.I. < 66°53'16" RT.
134+35 EDGE OF CROP
134+42 TOP OF BANK
134+46 PROPERTY LINE
134+54 TOE OF SLOPE (D-T2F-03)
134+65 CL OF DRAINAGE DITCH
134+75 TOE OF SLOPE (D-T2F-03)
134+79 EDGE AG-DITCH
134+81 TOP OF BANK
134+88 EDGE OF CROP
140+21 P.I. < 0°07'36" LT.
144+50 P.I. < 1°15'38" LT.
146+00 MATCHLINE

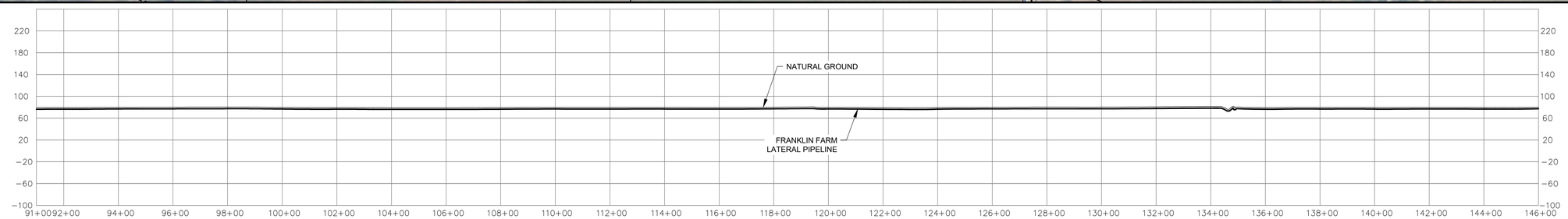
ENVIROMENTAL

D-T2F-03
AG-DITCH
134+49
134+79

PIPELINE



PROFILE



PIPE DETAIL

590' ③ — 91+00 to 96+90
4910' ② — 96+90 to 146+00

DESIGN FACTOR	0.60	0.72	
CLASS LOCATION	CLASS 2	CLASS 1	
HCA			

MATERIAL SUMMARY

NO.	DESCRIPTION	QUANTITY
2	36.00" O.D. X 0.515" WT., GRADE X-70, 14-16 MILS FBE	4190 LF
3	36.00" O.D. X 0.618" WT., GRADE X-70, 14-16 MILS FBE	590 LF

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	BY	DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:		FILE NAME: SHT0003			

P.L./STA. NO.

CONSTRUCTION YEAR	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.
SCALE: 1" = 200'

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ALIGNMENT SHEET
STA. 91+00 TO STA. 146+00
MP 1.72 TO MP 2.77
RICHLAND PARISH, LOUISIANA

PROJECT NO. C-24328-GL-26100001
PREVIOUS DWG. NO. P3-002
SHEET 2 OF 18
DWG. NO. P3-003
SHEET 3 OF 18
A

| | | |

OWNERSHIP & TRACT NO.	178+00	LA-RI-013.00	206+00
COUNTY & STATE		RICHLAND PARISH, LOUISIANA	

STATIONING

178+00 MATCHLINE

184+51 5' BARBED WIRE FENCE

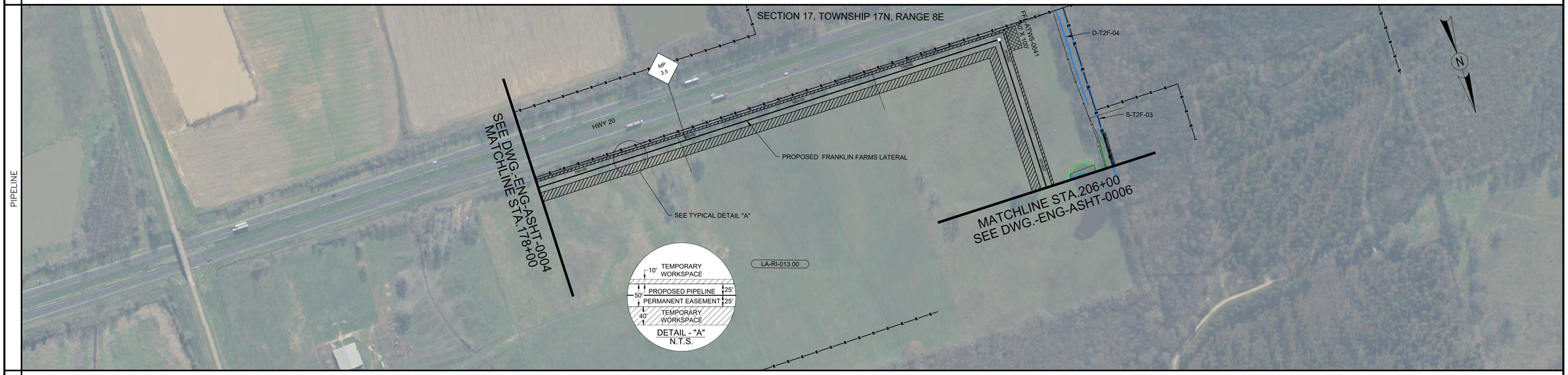
193+45 5' BARBED WIRE FENCE

199+19 P.I. < 90°00'00" RT.

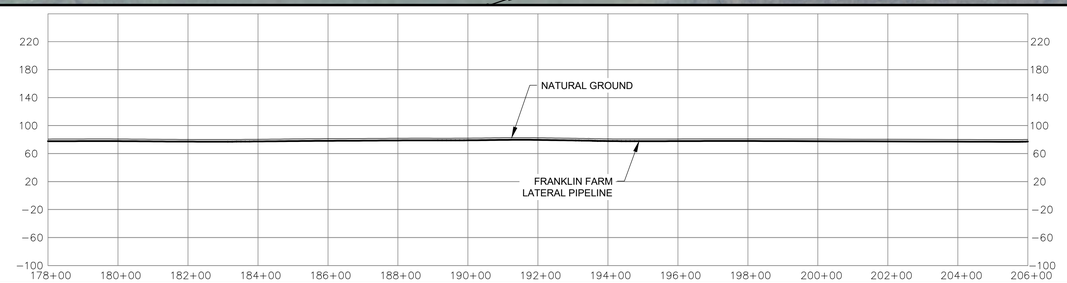
206+00 MATCHLINE

ENVIROMENTAL

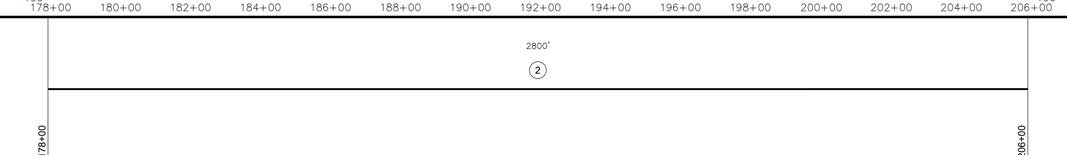
PIPELINE



PROFILE



PIPE DETAIL



DESIGN FACTOR	0.72
CLASS LOCATION	CLASS 1
HCA	

NO.	MATERIAL SUMMARY DESCRIPTION	QUANTITY
2	36.00" O.D. X 0.515" WT., GRADE X-70, 14-16 MILS FBE	2800 LF

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME: SHT0005					

P.I./STA. NO.

CONSTRUCTION YEAR

	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.

SCALE: 1" = 200'

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ALIGNMENT SHEET
STA. 178+00 TO STA. 206+00
MP 3.37 TO MP 3.90
RICHLAND PARISH, LOUISIANA

PROJECT NO. C-24328-GL-26100001

PREVIOUS DWG. NO. P3-004 SHEET 4 OF 18

DWG. NO. P3-005 (A) SHEET 5 OF 18

| | | |

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	BY	DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME:					

P.L./STA. NO.			
ACCOUNT NO.			
CONSTRUCTION YEAR			
		BY	DATE
DESIGN		TROY	05/29/26
DRAWN		TROY	05/29/26
ASBUILT			
FILE NO.			
SCALE: 1" = 200'			

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ACCESS ROAD DETAIL
STA. 206+00 TO STA. 267+00
MP 3.90 TO MP 5.06
RICHLAND PARISH, LOUISIANA

PROJECT NO.	C-24328-GL-26100001
PREVIOUS DWG. NO.	P3-006
SHEET 6 OF 18	
DWG. NO.	P3-006A (A)
SHEET 7 OF 18	

OWNERSHIP & TRACT NO.	267+00	LA-RI-016.00	269+71	LA-RI-016.50RR	270+97	LA-RI-017.00	313+09	LA-RI-018.00	330+00

COUNTY & STATE: RICHLAND PARISH, LOUISIANA

STATIONING

267+00 MATCHLINE

269+11 P.I. < 87°28'28" RT.
269+52 ENTER DITCH (D-T2F-05b)
269+54 EXIT DITCH (D-T2F-05b)
269+55 CL OF DRAINAGE DITCH
269+71 ENTER RIGHT-OF-WAY
270+20 TOE OF SLOPE
270+42 TOP OF BANK
270+45 TOP OF RAILROAD
270+50 TOP OF BANK
270+78 TOE OF SLOPE
271+13 ENTER ASPHALT
271+33 EDGE OF ASPHALT
271+46 CL OF ST. HWY. 80
271+60 EDGE OF ASPHALT
271+74 TOE OF SLOPE
271+86 ENTER ROAD DITCH (D-T1F-07)
271+88 EXIT ROAD DITCH (D-T1F-07)
272+98 EDGE OF TREE LINE
273+98 EDGE OF TREE LINE (TRUNK)

280+53 ENTER AG-DITCH (D-T1F-08)
280+56 EXIT AG-DITCH (D-T1F-08)
281+52 ENTER WETLAND (PEM-T1F-03)
281+66 EXIT WETLAND (PEM-T1F-03)
281+81 EDGE OF TREE LINE
282+18 P.I. < 10°16'34" LT.
5' BARBED WIRE FENCE

285+91 5' BARBED WIRE FENCE
286+00 EDGE OF TREE LINE (TRUNK)
286+01 EXIT WETLAND (PFO-T1F-03)
286+08 EDGE OF GRAVEL ROAD
286+08 CL OF DIRT ROAD
286+14 EDGE OF DIRT ROAD
286+15 EDGE OF VEGETATION - (CORN)
286+96 P.I. < 41°17'42" RT.
287+36 P.I. < 6°30'00" RT.

294+03 P.I. < 82°03'51" RT.

298+81 P.I. < 90°00'00" LT.
299+16 EDGE OF GRAVEL ROAD
299+26 CL OF GRAVEL ROAD
299+32 EDGE OF GRAVEL ROAD
299+34 ENTER AG-DITCH (D-T1F-09)
299+45 CL OF DRAINAGE DITCH
299+48 EXIT AG-DITCH (D-T1F-09)
299+48 EDGE OF TREE LINE
299+53 EDGE OF VEGETATION - (CORN)

312+51 EDGE OF VEGETATION - (CORN)
312+74 CL OF DRAINAGE DITCH (DRIP LINE)
312+98 CL OF DRAINAGE DITCH
312+99 ENTER STREAM (S-T1F-01)
313+01 EXIT STREAM (S-T1F-01)
313+09 PROPERTY LINE
313+12 EDGE OF TREE LINE-(DRIP LINE)
313+18 EDGE OF VEGETATION - (CORN)

315+91 P.I. < 2°10'44" RT.

316+64 EDGE OF VEGETATION - (CORN)
317+38 EDGE OF WATER
317+43 ENTER STREAM (S-T1F-01)
318+04 EDGE OF WATER
318+04 ENTER WETLAND (PFO-T1F-02)
318+19 EXIT STREAM (S-T1F-01)
318+31 EXIT WETLAND (PFO-T1F-02)
319+01 EXIT POND (P-T1F-02)
319+39 EDGE OF VEGETATION

320+53 P.I. < 2°48'39" RT.

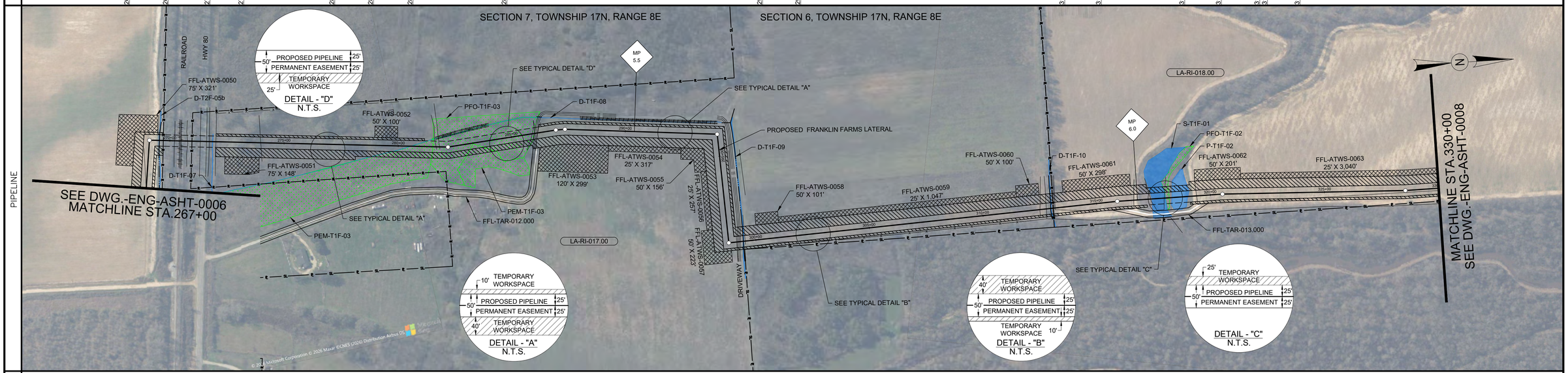
328+54 P.I. < 2°48'39" LT.

330+00 MATCHLINE

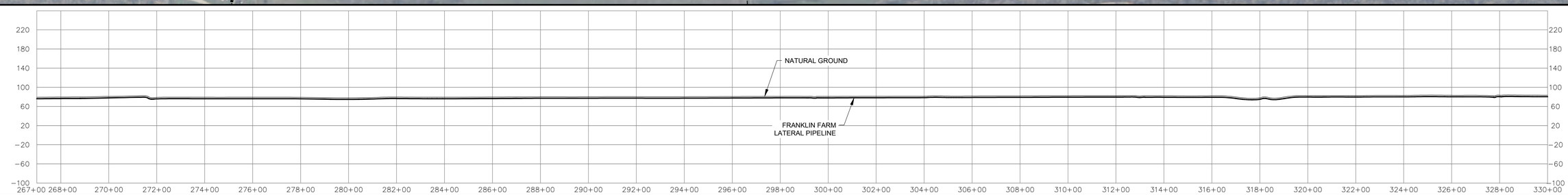
ENVIROMENTAL

289+52	289+54	271+86	271+88	280+53	280+56	281+52	286+01	299+44	299+48	312+99	313+01	317+43	318+04	318+19	318+31	319+01
D-T2F-05b	AG DITCH	D-T1F-07	ROAD DITCH	D-T1F-08	AG DITCH	PEM-T1F-03	PFO-T1F-03	D-T1F-09	AG DITCH	S-T1F-01	STREAM	S-T1F-01	STREAM	PFO-T1F-02	PFO	P-T1F-02	POND

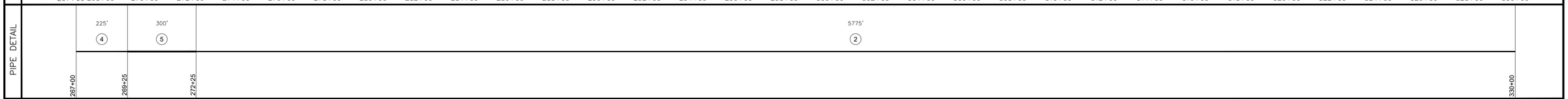
PIPELINE



PROFILE



PIPE DETAIL



DESIGN FACTOR	0.50	0.72	
CLASS LOCATION	CLASS 1		
HCA			

MATERIAL SUMMARY

NO.	DESCRIPTION	QUANTITY
2	36.00" O.D. X 0.515" WT., GRADE X-70, 14-16 MILS FBE	5775 LF
4	36.00" O.D. X 0.750" WT., GRADE X-70, 14-16 MILS FBE	225 LF
5	36.00" O.D. X 0.750" WT., GRADE X-70, 14-16 MILS FBE, 30-40 MILS ARO	300 LF

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	BY	DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME: SHT0007					

P.I./STA. NO.

CONSTRUCTION YEAR

	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.

SCALE: 1" = 200'

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ALIGNMENT SHEET
STA. 267+00 TO STA. 330+00
MP 5.06 TO MP 6.25
RICHLAND PARISH, LOUISIANA

PROJECT NO. C-24328-GL-26100001
PREVIOUS DWG. NO. P3-006A
SHEET 7 OF 18
DWG. NO. P3-007 (A)
SHEET 8 OF 18

OWNERSHIP & TRACT NO.	330+00	LA-RI-018.00	352+97	LA-RI-019.00	357+00
COUNTY & STATE	RICHLAND PARISH, LOUISIANA			

STATIONING

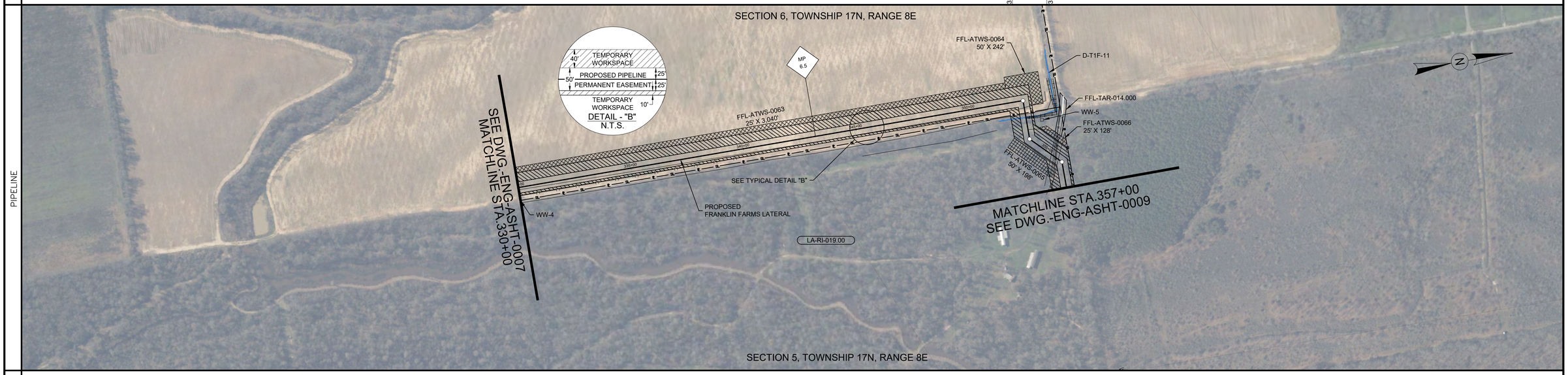
330+00 MATCHLINE

352+42 P.I. <- 90°00'00" RT.
352+73 EDGE OF VEGETATION
352+92 EDGE OF TREE LINE (DRIP LINE)
353+02 TOP OF BANK
353+14 ENTER AG-DITCH (D-T1F-11)
353+15 TOE OF SLOPE
353+18 EXIT AG-DITCH (D-T1F-11)
353+19 TOE OF SLOPE
353+22 TOP OF BANK
353+70 EDGE OF TREE LINE (DRIP LINE)
354+13.37 P.I. <- 90°00'00" RT.
355+29 EDGE OF TREE LINE (DRIP LINE)
355+41 EDGE OF GRAVEL ROAD
355+48 CL OF DRAINAGE DITCH
355+51 CL OF ROAD DITCH LINE
355+55 EDGE OF GRAVEL ROAD
355+66 EDGE OF TREE LINE (DRIP LINE)
356+48 P.I. <- 45°00'47" RT.
357+00 MATCHLINE

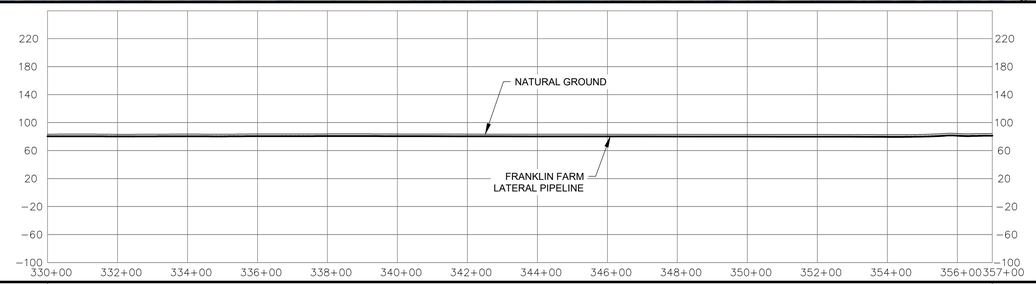
ENVIROMENTAL

D-T1F-11
353+15 AG DITCH 353+18

PIPELINE



PROFILE



PIPE DETAIL

DESIGN FACTOR	0.72
CLASS LOCATION	CLASS 1
HCA	

MATERIAL SUMMARY		
NO.	DESCRIPTION	QUANTITY
2	36.00" O.D. X 0.515" WT., GRADE X-70, 14-16 MILS FBE	2700 LF

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	BY	DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME: SHT0008					

P.L./STA. NO.

CONSTRUCTION YEAR	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.
SCALE: 1" = 200'

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ALIGNMENT SHEET
STA. 330+00 TO STA. 357+00
MP 6.25 TO MP 6.76
RICHLAND PARISH, LOUISIANA

PROJECT NO. C-24328-GL-26100001
PREVIOUS DWG. NO. P3-007
SHEET 8 OF 18
DWG. NO. P3-008
SHEET 9 OF 18
A

| | | |

OWNERSHIP & TRACT NO.	381+00	LA-RI-021.00	426+00
COUNTY & STATE		RICHLAND PARISH, LOUISIANA	

STATIONING

381+00 MATCHLINE

391+31 EDGE OF WATER
391+47 EDGE OF WATER

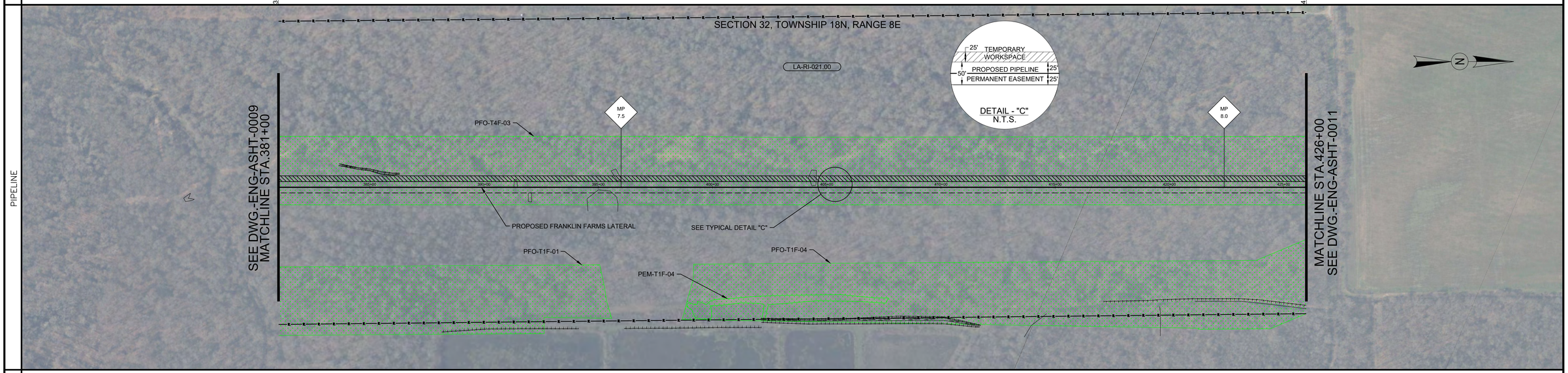
426+00 MATCHLINE

ENVIROMENTAL

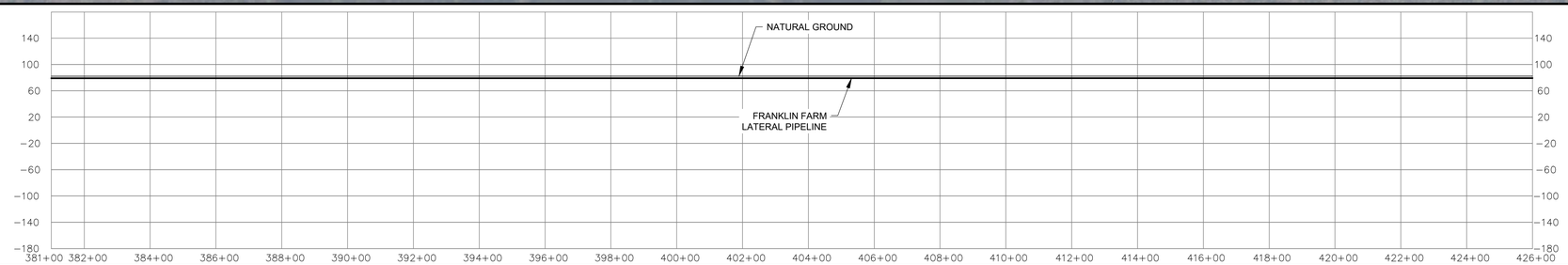
381+00: PFO-T4F-03

426+00: PFO

PIPELINE



PROFILE



PIPE DETAIL

DESIGN FACTOR	0.72
CLASS LOCATION	CLASS 1
HCA	

MATERIAL SUMMARY		
NO.	DESCRIPTION	QUANTITY
2	36.00" O.D. X 0.515" WT., GRADE X-70, 14-16 MILS FBE	4500 LF

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME: SHT0010					

P.L./STA. NO.

CONSTRUCTION YEAR	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.

SCALE: 1" = 200'

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ALIGNMENT SHEET
STA. 381+00 TO STA. 426+00
MP 7.22 TO MP 8.07
RICHLAND PARISH, LOUISIANA

PROJECT NO.	C-24328-GL-26100001
PREVIOUS DWG. NO.	P3-009
	SHEET 10 OF 18
DWG. NO.	P3-010 A
	SHEET 11 OF 18

OWNERSHIP & TRACT NO.	426+00	LA-RI-021.00	427+02	LA-RI-022.00	455+00
COUNTY & STATE	RICHLAND PARISH, LOUISIANA				

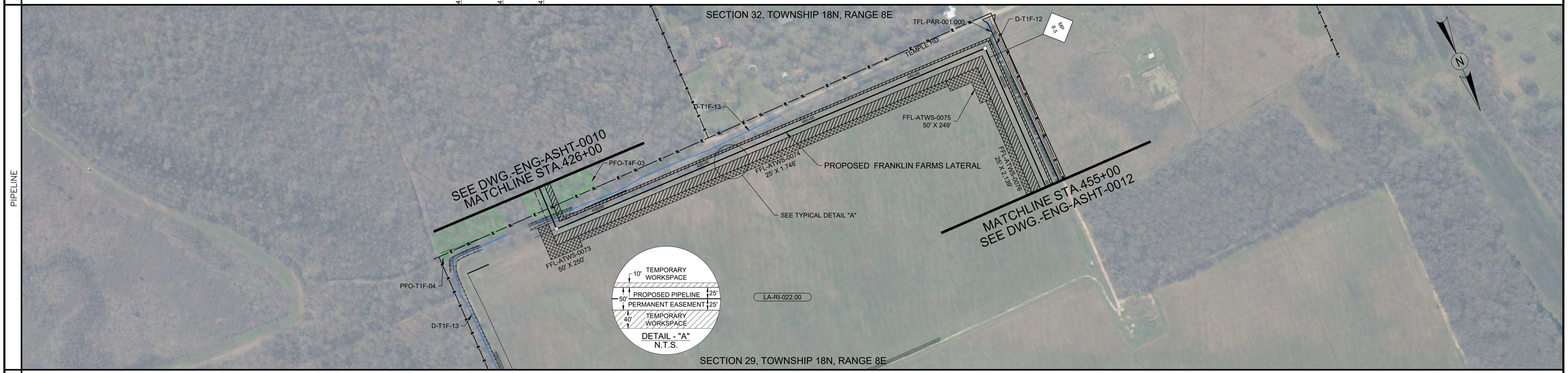
STATIONING

STA	
426+00	MATCHLINE
427+02	PROPERTY LINE, SECTION LINE
427+33	TOE OF BANK
427+36	EXIT WETLAND (PFO-T4F-03)
427+41	ENTER AG-DITCH (D-T1F-13)
427+43	EXIT AG-DITCH (D-T1F-13)
427+47	TOE OF SLOPE
427+54	TOE OF BANK
427+65	EDGE OF TREE LINE-DRIP LINE
427+99	EDGE OF CROP LINE-UNK. CROP
427+99	P.I. < 90°00'00" LT.
428+02	EDGE OF CROP LINE-UNK. CROP
448+35	P.I. < 89°17'57" RT.
455+00	MATCHLINE

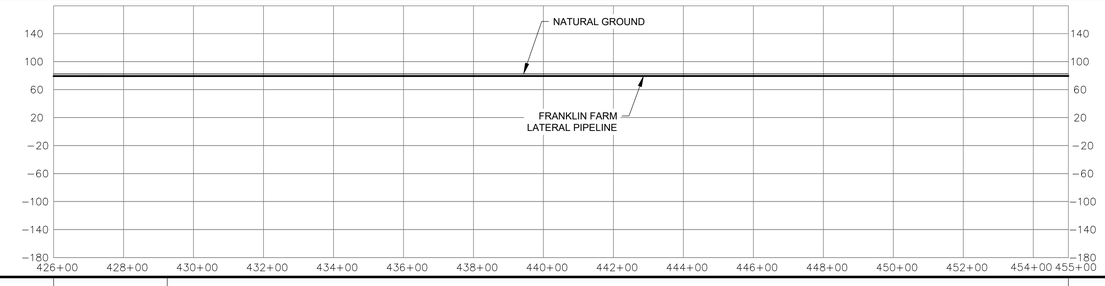
ENVIROMENTAL

STA	
426+00	PFO-T4F-03
427+41	PFO
427+43	D-T1F-13
	AG DITCH

PIPELINE



PROFILE



PIPE DETAIL

	426+00	429+25	455+00
	325' ②	2575' ③	

DESIGN FACTOR	0.72	0.60	
CLASS LOCATION	CLASS 1	CLASS 2	
HCA			

MATERIAL SUMMARY

NO.	DESCRIPTION	QUANTITY
2	36.00" O.D. X 0.515" WT., GRADE X-70, 14-16 MILS FBE	325 LF
3	36.00" O.D. X 0.618" WT., GRADE X-70, 14-16 MILS FBE	2575 LF

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME: SHT0011					

P.L./STA. NO.

CONSTRUCTION YEAR	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.

SCALE: 1" = 200'

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ALIGNMENT SHEET
STA. 426+00 TO STA. 455+00
MP 8.07 TO MP 8.62
RICHLAND PARISH, LOUISIANA

PROJECT NO. C-24328-GL-26100001
PREVIOUS DWG. NO. P3-010
SHEET 11 OF 18
DWG. NO. P3-011 (A)
SHEET 12 OF 18

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OWNERSHIP & TRACT NO.	510+00	LA-RI-025.00	556+00
COUNTY & STATE		RICHLAND PARISH, LOUISIANA	

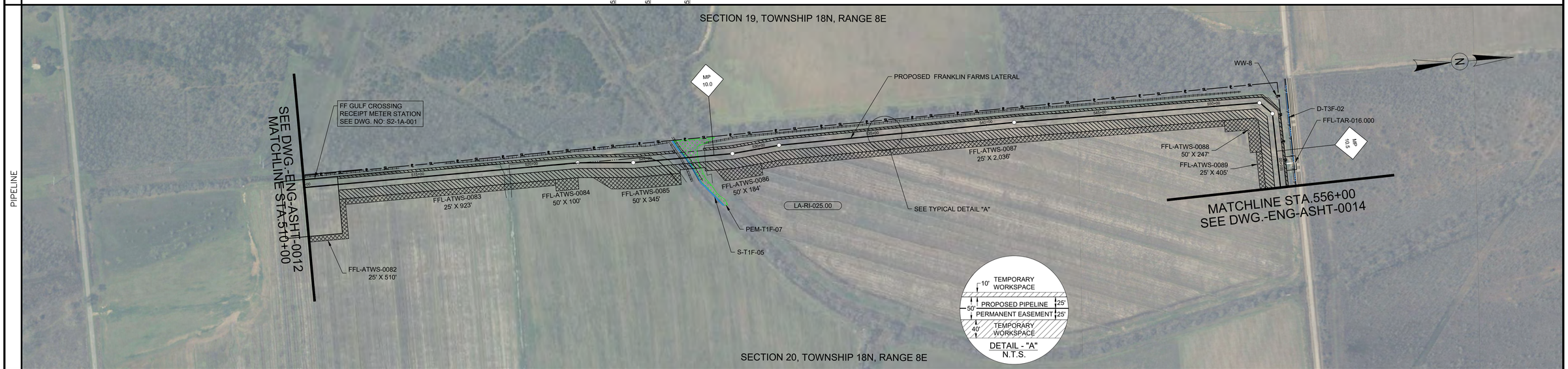
STATIONING

- 510+00 MATCHLINE
- 518+71 EDGE OF VEGETATION – COTTON
- 519+01 EDGE OF VEGETATION – COTTON
- 522+02 P.I. < 5°00'00" RT.
- 522+26 CL OF WATER WAY
- 524+45 P.I. < 5°00'00" LT.
- 525+46 CL OF WATER WAY
- 525+95 EDGE OF VEGETATION – COTTON
- 526+50 TOP OF BANK
- 526+57 TOE
- 526+72 ENTER STREAM (S–T1F–05)
- 526+74 TOE
- 526+79 EXIT STREAM (S–T1F–05)
- 526+79 ENTER WETLAND (PEM–T1F–07)
- 526+90 TOP OF BANK
- 527+29 EXIT WETLAND (PEM–T1F–07)
- 527+60 EDGE OF VEGETATION – COTTON
- 528+81 P.I. < 5°00'00" LT.
- 530+88 P.I. < 4°37'40" RT.
- 541+23 P.I. < 0°59'37" RT.
- 552+02 P.I. < 45°00'00" RT.
- 552+74 P.I. < 43°19'36" RT.
- 556+00 MATCHLINE

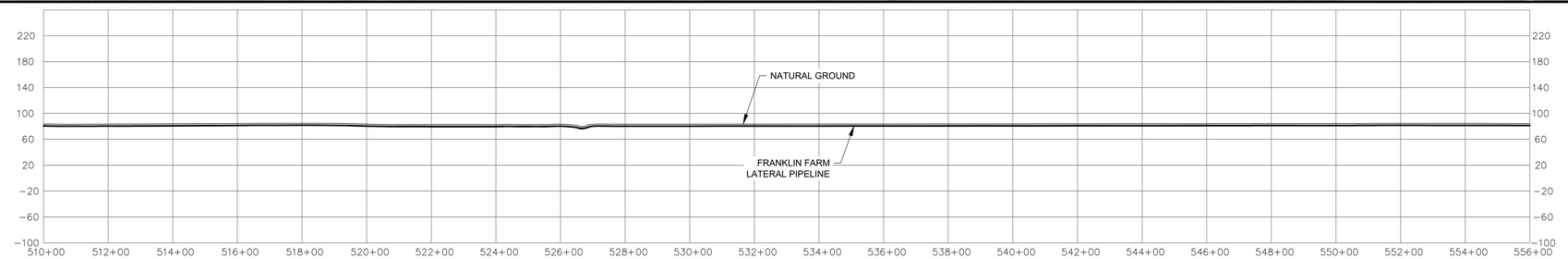
ENVIROMENTAL

- S-T1F-05 — STREAM — 526+72 / 526+79
- PEM-T1F-07 — PEM — 527+29

PIPELINE



PROFILE



PIPE DETAIL

DESIGN FACTOR	0.72
CLASS LOCATION	CLASS 1
HCA	

MATERIAL SUMMARY		
NO.	DESCRIPTION	QUANTITY
2	36.00" O.D. X 0.515" WT., GRADE X-70, 14-16 MILS FBE	4600 LF

DWG. NO.	REFERENCE DRAWING TITLE
S2–1A–001	FF GULF CROSSING RECEIPT METER STATION

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME: SHT0013					

P.L./STA. NO.

CONSTRUCTION YEAR

	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.

SCALE: 1" = 200'

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ALIGNMENT SHEET
STA. 510+00 TO STA. 556+00
MP 9.66 TO MP 10.53
RICHLAND PARISH, LOUISIANA

PROJECT NO. C–24328–GL–26100001

PREVIOUS DWG. NO. P3–012
SHEET 13 OF 18

DWG. NO. P3–013
SHEET 14 OF 18

A

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OWNERSHIP & TRACT NO.	611+00	LA-RI-027.00	619+31	LA-RI-028.00	632+56	LA-RI-029.00	659+00
COUNTY & STATE	RICHLAND PARISH, LOUISIANA						

STATIONING

611+00	MATCHLINE
618+05	3' HOG WIRE FENCE
619+31	PROPERTY LINE
620+19	TOP OF BANK
620+21	ENTER STREAM (S-T2F-06)
620+25	TOE OF SLOPE
620+35	TOE OF SLOPE
620+35	EXIT STREAM (S-T2F-06)
620+39	TOP OF BANK
624+27	ENTER STREAM (S-T2F-07)
624+32	CL OF DRAINAGE DITCH
624+32	EXIT STREAM (S-T2F-07)
632+56	PROPERTY LINE
633+36	EDGE OF TREE LINE (DRIP LINE)
634+48	EDGE OF TREE LINE (DRIP LINE)
635+45	ENTER WETLAND (PFO-T2F-19)
637+97	EXIT WETLAND (PFO-T2F-19)
641+54	TOP OF BANK
641+57	ENTER STREAM (S-T2F-10)
641+58	TOE OF SLOPE
641+64	EXIT STREAM (S-T2F-10)
641+65	TOE OF SLOPE
642+15	TOP OF BANK
642+16	ENTER WETLAND (PFO-T2F-20)
643+00	EXIT WETLAND (PFO-T2F-20)
649+23	EDGE OF TREE LINE (DRIP LINE)
650+85	EDGE OF TREE LINE (DRIP LINE)
655+99	ENTER STREAM (S-T2F-09)
656+08	TOP OF BANK
656+17	TOE OF SLOPE
656+35	TOE OF SLOPE
656+43	EXIT STREAM (S-T2F-09)
656+45	TOP OF BANK
658+68	OVERHEAD POWER LINE
658+82	TOP OF BANK
658+88	TOE OF SLOPE
658+95	TOE OF SLOPE
659+00	MATCHLINE

ENVIROMENTAL

| | | | | | |

OWNERSHIP & TRACT NO.	659+00 LA-RI-029.00 659+16	LA-RI-030.00	699+12
COUNTY & STATE	RICHLAND PARISH, LOUISIANA		

STATIONING

659+00 TEMPLATE ML
659+03 TOP OF BANK
659+04 EDGE OF ASPHALT ROAD
659+13 CL OF ASPHALT ROAD
659+16 EXISTING LATEST SECTION LINE
659+23 EDGE OF ASPHALT ROAD
659+46 ENTER WETLAND (PFO-T2F-05)
660+07 EXIT WETLAND (PFO-T2F-05)
660+24 P.I. < 42°18'18" LT.

662+88 EDGE OF DIRT ROAD
662+94 CL OF DIRT ROAD
663+00 EDGE OF DIRT ROAD

664+33 EDGE OF TREE LINE (DRIP LINE)
664+51 ENTER WETLAND (PFO-T2F-07)
665+07 EXIT WETLAND (PFO-T2F-07), ENTER STREAM (S-T2F-08)
665+10 EXIT STREAM (S-T2F-08), ENTER WETLAND (PFO-T2F-07)
665+70 EXIT WETLAND (PFO-T2F-07), EDGE OF TREE LINE (DRIP LINE)
667+19 P.I. < 41°10'18" RT.

683+39 EDGE OF ROAD
683+46 CL OF ROAD
683+54 EDGE OF ROAD
684+22 OVERHEAD POWER LINE
685+21 P.I. < 0°28'35" RT.

686+26 8' CHAINLINK FENCE

687+35 8' CHAINLINK FENCE

688+17 8' CHAINLINK FENCE
688+40 EDGE OF GRAVEL ROAD
688+57 CL OF GRAVEL ROAD
688+64 EDGE OF GRAVEL ROAD
689+43 EXISTING BUILDING
690+00 EXISTING BUILDING
690+54 P.I. < 90°00'00" RT.

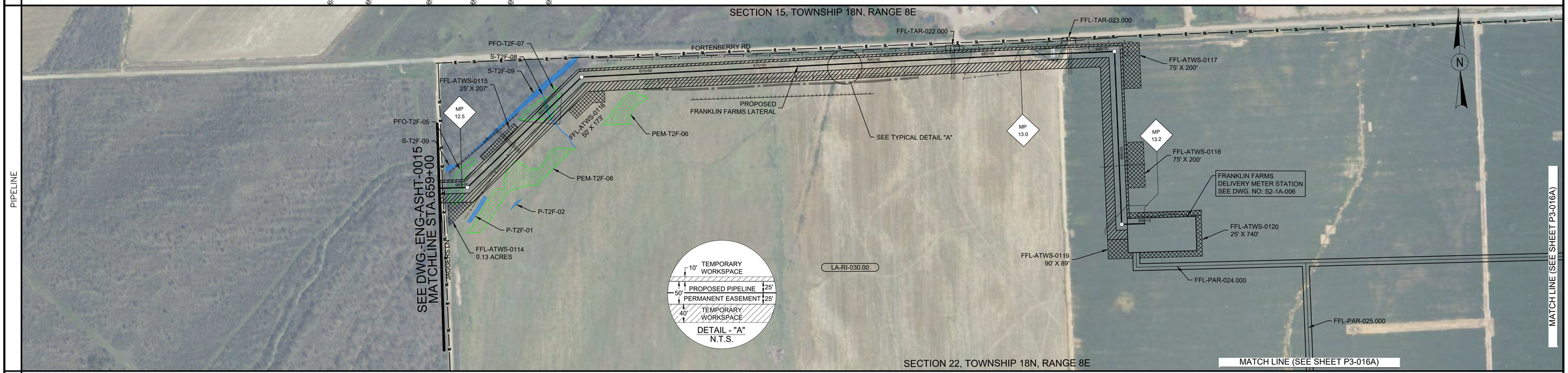
695+55 8' CHAINLINK FENCE

697+56 8' CHAINLINK FENCE
698+11 P.I. < 89°20'35" LT.
699+12 END STA.

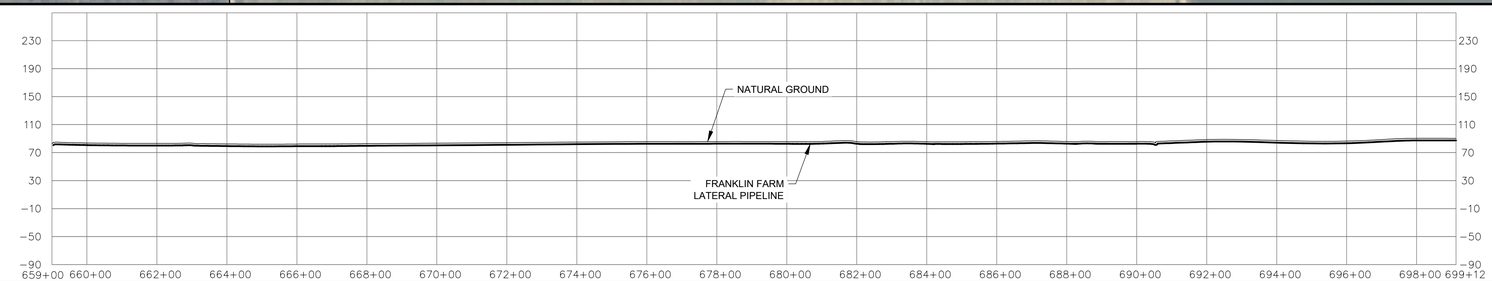
ENVIROMENTAL

659+46 PFO-T2F-05 660+07 PFO
664+51 PFO-T2F-07 665+07 PFO
665+07 S-T2F-08 665+10 STREAM
665+10 PFO-T2F-07 665+70 PFO

PIPELINE



PROFILE



PIPE DETAIL

	659+00	659+38	660+14	691+90	695+70	699+12
	38' (5)	2076' (2)	1176' (4)	380' (5)	342' (4)	
DESIGN FACTOR	0.50	0.72		0.50		
CLASS LOCATION	CLASS 1			CLASS 3		
HCA				HCA		

MATERIAL SUMMARY		
NO.	DESCRIPTION	QUANTITY
2	36.00" O.D. X 0.515" WT., GRADE X-70, 14-16 MILS FBE	2076 LF
4	36.00" O.D. X 0.750" WT., GRADE X-70, 14-16 MILS FBE	1518 LF
5	36.00" O.D. X 0.750" WT., GRADE X-70, 14-16 MILS FBE, 30-40 MILS ARO	418 LF

DWG. NO.	REFERENCE DRAWING TITLE
S2-1A-006	FRANKLIN FARMS DELIVERY METER STATION

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME: SHT0016					

P.L./STA. NO.		
CONSTRUCTION YEAR		
	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		
FILE NO.		
SCALE: 1" = 200'		

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ALIGNMENT SHEET
STA. 659+00 TO STA. 699+12
MP 12.48 TO MP 13.24
RICHLAND PARISH, LOUISIANA

PROJECT NO.	C-24328-GL-26100001
PREVIOUS DWG. NO.	P3-015
SHEET	16 OF 18
DWG. NO.	P3-016 (A)
SHEET	17 OF 18

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME:					

P.L./STA. NO.			
ACCOUNT NO.			
CONSTRUCTION YEAR			
		BY	DATE
DESIGN		TROY	05/29/26
DRAWN		TROY	05/29/26
ASBUILT			
FILE NO.			
SCALE: 1" = 200'			

ENERGY TRANSFER

FRANKLIN FARMS LATERAL
ACCESS ROAD DETAIL
STA. 659+00 TO STA. 699+00
MP 12.48 TO MP 13.24
RICHLAND PARISH, LOUISIANA

PROJECT NO.	C-24328-GL-26100001
PREVIOUS DWG. NO.	P3-016
SHEET 17 OF 18	
DWG. NO.	P3-016A (A)
SHEET 18 OF 18	

ENERGY TRANSFER

PROPOSED FF TRIO FARM LATERAL ALIGNMENT PACKAGE

WBS: TBD

ISSUE FOR PERMIT

PREPARED BY: FIRM No:23055

8521 McHARD RD, HOUSTON, TX 77053

DATE 05/29/2026

DRAWING NO. P0-01

CONSTRUCTION DRAWINGS	
DRAWING NO.	TITLE
P0-01	COVER SHEET
P1-01	INDEX SHEET
P3-01	ALIGNMENT SHEET (MP 0.00 TO MP 0.91)
P3-02	ALIGNMENT SHEET (MP 0.91 TO MP 1.40)
P3-03	ALIGNMENT SHEET (MP 1.40 TO MP 1.75)

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE: 05/29/26					
	FILE NAME: P1-01					

P.I./STA. NO.
ACCOUNT NO.
CONSTRUCTION YEAR

	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.
SCALE: AS NOTED

ENERGY TRANSFER

FF TRIO FARM LATERAL
INDEX SHEET
STA. 0+00 TO STA. 92+51
MP 0.00 TO MP 1.75
RICHLAND PARISH, LOUISIANA

PROJECT NO.
C-24328-GL-26100001
PREVIOUS DWG. NO.
SHEET 1 OF 1
DWG. NO.
P1-01
A
SHEET 1 OF 1

| | | |

OWNERSHIP & TRACT NO.	48+00	TF-LA-RI-03.00	63+15 TF-LA-RI-04.00 74+00
COUNTY & STATE			

STATIONING

48+00 MATCHLINE
48+48 EDGE OF TREE LINE (DRIP LINE)

56+77 EDGE OF WATER
57+13 ENTER WETLAND (PEM-T2F-14)
57+72 EXIT WETLAND (PEM-T2F-14)
58+05 EDGE OF WATER

60+85 TOP OF BANK
60+88 TOE OF SLOPE
60+89 ENTER AG DITCH (D-T2F-09)
60+95 EXIT AG DITCH (D-T2F-09)
60+98 TOE OF SLOPE
60+99 TOP OF BANK

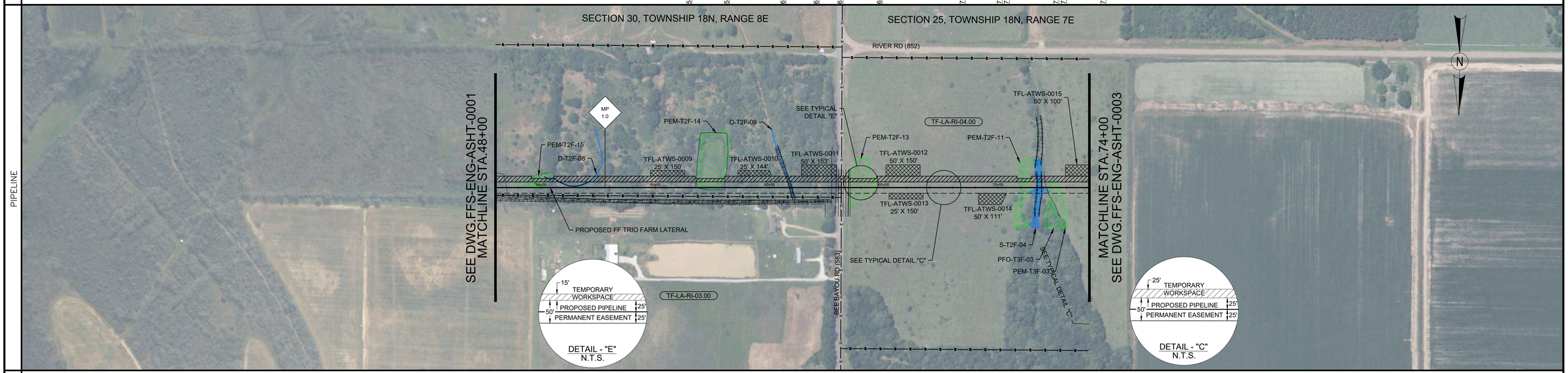
62+72 4' HOG WIRE FENCE
62+91 TOE OF SLOPE
62+92 EDGE OF ASPHALT ROAD
63+12 C/L OF ASPHALT ROAD
63+15 PROPERTY LINE, SECTION LINE
63+22 EDGE OF ASPHALT ROAD
63+34 TOE OF SLOPE
63+45 ENTERGY LA, LLC - OVH PWR LN
63+46 ENTER WETLAND (PEM-T2F-15)
63+47 3\ BARBED WIRE FENCE
63+53 UNK. EXISTING 4" P/L (4" COV.)
64+69 EXIT WETLAND (PEM-T2F-15)

71+01 ENTER WETLAND (PERM-T2F-11)
71+34 EXIT WETLAND (PERM-T2F-11)
71+69 TOP OF BANK
71+70 ENTER STREAM (S-T2F-04)
71+72 TOE OF BANK
71+90 TOE OF BANK
71+92 TOP OF BANK
71+96 EXIT STREAM (S-T2F-04)
72+13 CROP CHANGE
72+23 ENTER WETLAND (PERM-T3F-03)
72+67 EXIT WETLAND (PEM-T3F-03)
74+00 MATCHLINE

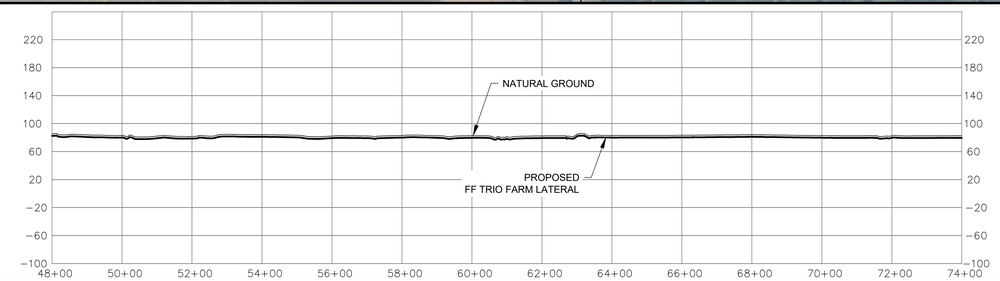
ENVIROMENTAL

57+13	PEM-T2F-14	57+72	60+89	D-T2F-09	60+96	63+46	PEM-T2F-13	64+69	71+01	PEM-T2F-11	71+34	71+70	S-T2F-04	71+96	72+23	PEM-T3F-03	72+67
	PEM			AG DITCH			PEM			PEM			STREAM			PEM	

PIPELINE



PROFILE



PIPE DETAIL

48+00	1462' (2)	62+62	120' (5)	63+82	1018' (2)	74+00

DESIGN FACTOR	48+00	0.72 62+62	0.50 63+82 0.72 74+00
CLASS LOCATION		CLASS 1	
HCA			

MATERIAL SUMMARY

NO.	DESCRIPTION	QUANTITY
2	20.00" O.D. X 0.375" WT., GRADE X-65, 14-16 MILS FBE	2480 LF
5	20.00" O.D. X 0.500" WT., GRADE X-65, 14-16 MILS FBE, 30-40 MILS ARO	120 LF

DWG. NO.	REFERENCE DRAWING TITLE

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME: SHT0002					

P.L./STA. NO.		
CONSTRUCTION YEAR		
	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		
FILE NO.		
SCALE: 1 = 200'		

ENERGY TRANSFER

FF TRIO FARM LATERAL
ALIGNMENT SHEET
STA. 48+00 TO STA. 74+00
MP 0.91 TO MP 1.40
RICHLAND PARISH, LOUISIANA

PROJECT NO.	C-24328-GL-26100001
PREVIOUS DWG. NO.	P3-001
SHEET 1 OF 3	
DWG. NO.	P3-002 A
SHEET 2 OF 3	

OWNERSHIP & TRACT NO.

74+00 | TF-LA-RI-04.00 | 75+81 | TF-LA-RI-05.00 | 83+50 | TF-LA-RI-06.00 | 92+51

COUNTY & STATE

STATIONING

74+00 MATCHLINE

75+67 3' BARBED WIRE FENCE
75+75 ENTER WETLAND (PERM-T2F-12)
75+81 PROPERTY LINE
75+84 C/L OF DRAINAGE DITCH
75+84 EXIT WETLAND (PERM-T2F-12)
75+91 EDGE OF DIRT ROAD
76+35 P.I. < 89°47'47" RT.

83+50 PROPERTY LINE

87+04 P.I. < 37°57'11" LT.

88+15 EDGE OF DIRT ROAD
88+19 C/L OF DIRT ROAD
88+23 EDGE OF DIRT ROAD

89+59 P.I. < 85°37'36" LT.

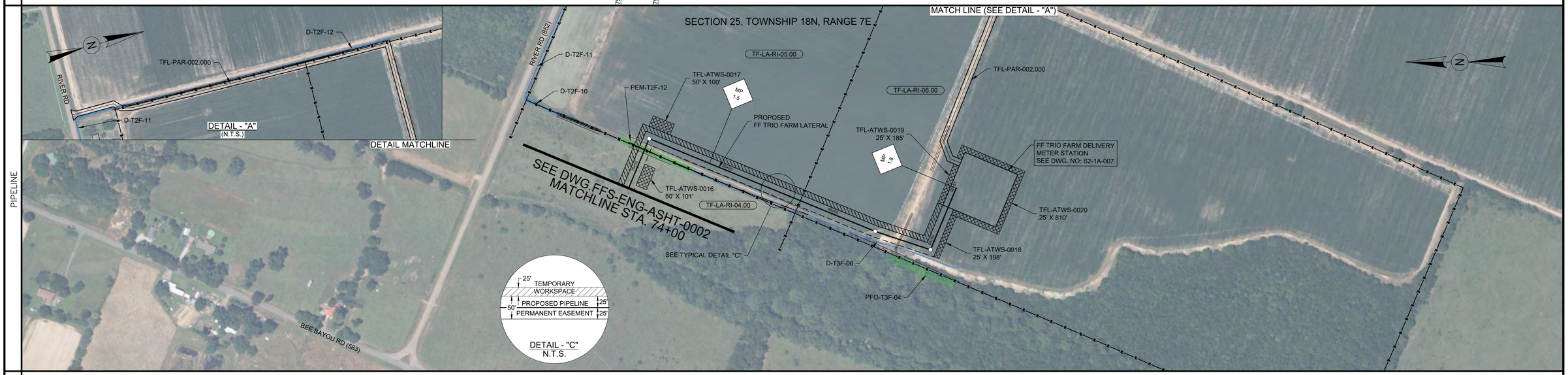
92+51 END STA.

ENVIROMENTAL

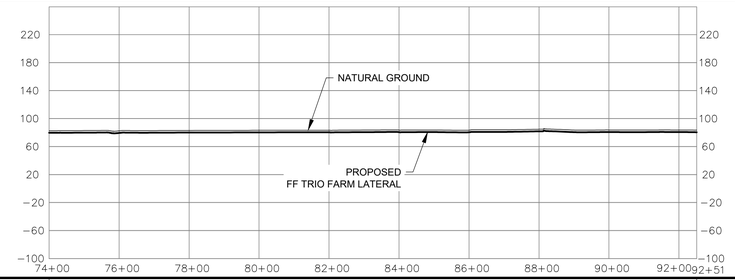
75+75 PEM-T2F-12

75+94 PEM

PIPELINE



PROFILE



PIPE DETAIL

DESIGN FACTOR: 0.72

CLASS LOCATION: CLASS 1

HCA

MATERIAL SUMMARY		
NO.	DESCRIPTION	QUANTITY
2	20.00" O.D. X 0.375" WT., GRADE X-65, 14-16 MILS FBE	1851 LF

DWG. NO.	REFERENCE DRAWING TITLE
S2-1A-007	FF TRIO FARM DELIVERY METER STATION

NO.	REVISION – DESCRIPTION	BY	DATE	CHK'D	APP'D
A	ISSUED FOR PERMIT	JBC	05/29/26	JSD	CF

DWG. STATUS	CHECKED BY	CHECKED DATE	APPROVED BY	APPROVED DATE	APPROVED BY	APPROVED DATE
PREL'Y						
BID						
CONSTR.						
CADDS	PLOT DATE:					
	FILE NAME: SHT0003					

P.I./STA. NO.

CONSTRUCTION YEAR

	BY	DATE
DESIGN	TROY	05/29/26
DRAWN	TROY	05/29/26
ASBUILT		

FILE NO.

SCALE: 1 = 200'

ENERGY TRANSFER

FF TRIO FARM LATERAL
ALIGNMENT SHEET
STA. 74+00 TO STA. 92+51
MP 1.40 TO MP 1.75
RICHLAND PARISH, LOUISIANA

PROJECT NO. C-24328-GL-26100001

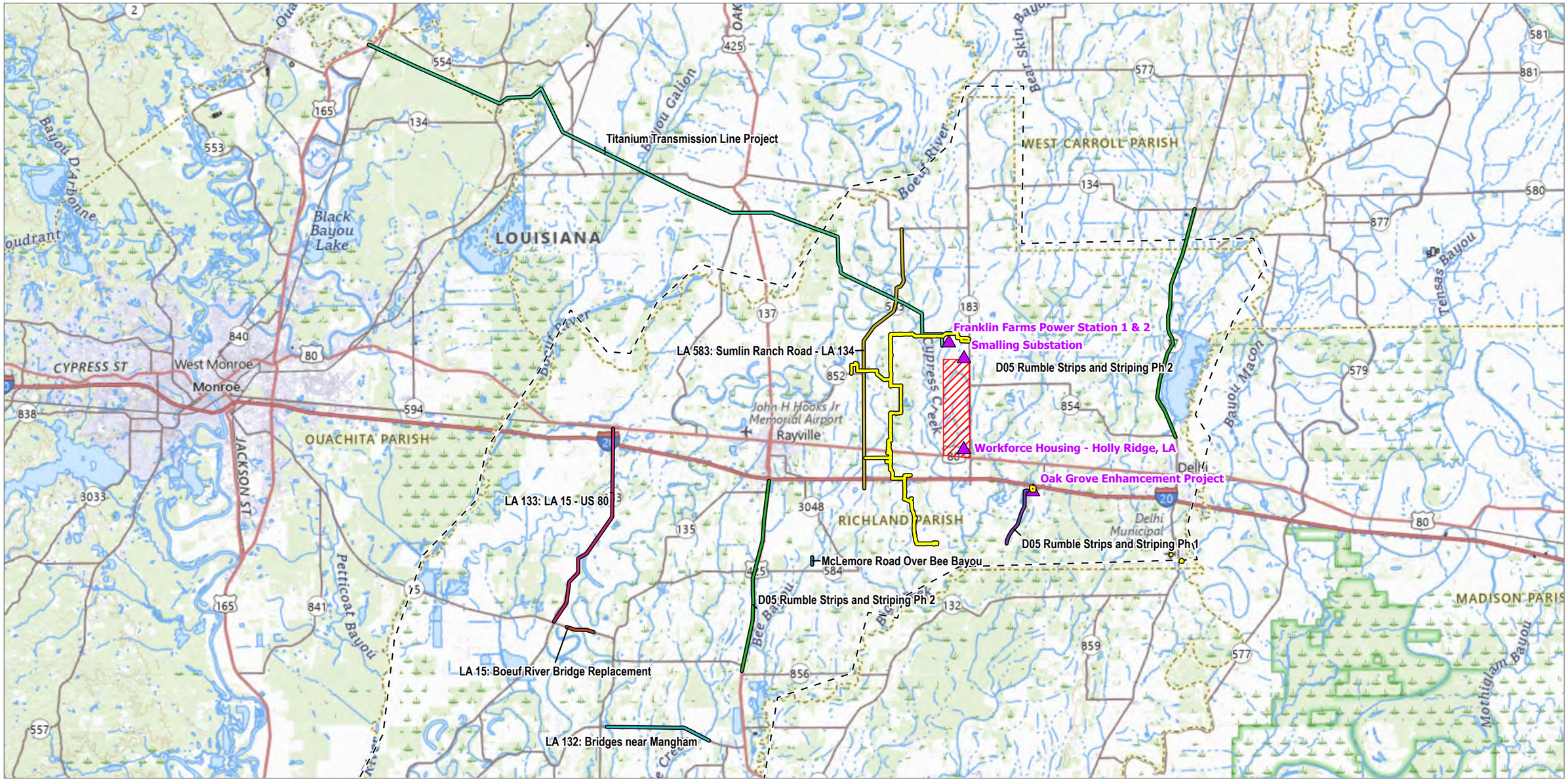
PREVIOUS DWG. NO. P3-002
SHEET 2 OF 3

DWG. NO. P3-003
SHEET 3 OF 3

A

APPENDIX A-8

Cumulative Map



PROJECT BOUNDARY
CUMULATIVE IMPACT LOCATION
D05 RUMBLE STRIPS AND STRIPING PH 1
D05 RUMBLE STRIPS AND STRIPING PH 2
LA 15: BOEUF RIVER BRIDGE REPLACEMENT
LA 132: BRIDGES NEAR MANGHAM
LA 133: LA 15 - US 80
LA 583: SUMLIN RANCH ROAD - LA 134
MCLEMORE ROAD OVER BEE BAYOU
TITANIUM TRANSMISSION LINE PROJECT
OUTLINES
HYPERION DATA CENTER
RICHLAND PARISH

BASE MAP: USGS "USA TOPO MAPS" ONLINE SERVICE LAYER
DATA SOURCES: TRC

N

0 10,000 20,000 FEET
1:240,000 1" = 20,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

CUMULATIVE IMPACT ASSESSMENT AREA MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

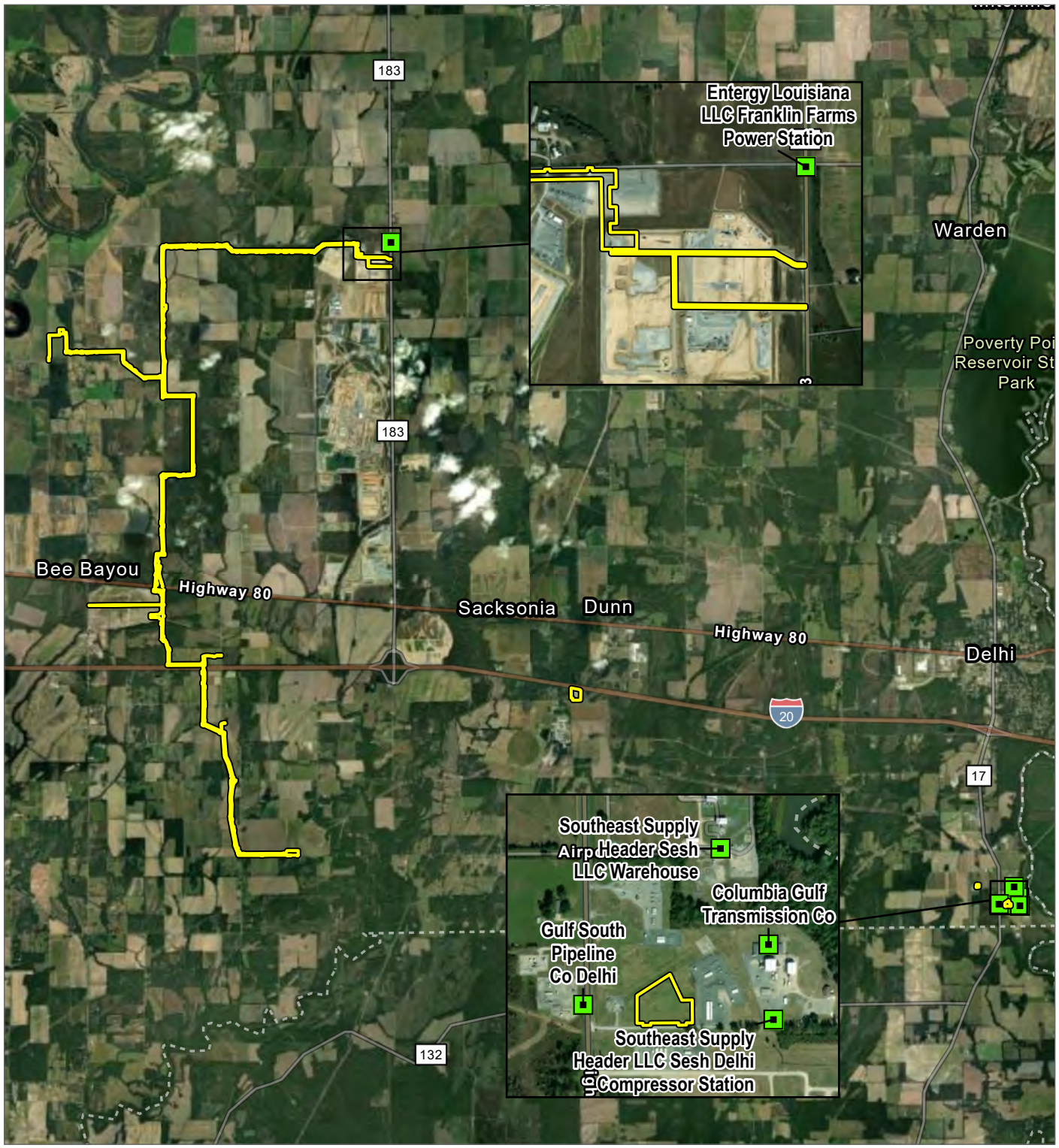
FIGURE 1.B.3

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

Coordinate System:NAD 1983 StatePlane Louisiana North FIPS 1701 Feet; Map Rotation: 0
Saved By: AGIBSON on 5/22/2026, 8:59:09 AM; File Path: T:\PROJECTS\Energy Transfer LP\705831_Franklin_Farms\2-APRX\FE_FERC.aprx; Layout Name: 1.B.3 Cumulative Analysis

APPENDIX A-9

Hazardous Material Sites in the Vicinity of the Project



PROJECT BOUNDARY

HAZARDOUS WASTE SITE

0 5,000 10,000

FEET

1:120,000 1" = 10,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, USEPA

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

HAZARDOUS MATERIAL SITES

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 2.C-2

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; -- FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FF_FERC_APRX; -- LAYOUT NAME: 2.C-2 HAZARDOUS MATERIALS

APPENDIX A-10

FEMA Floodplains Crossed by the Project

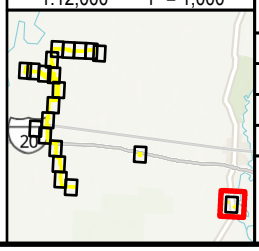
WORKSPACE TYPE

- TEMPORARY ACCESS ROAD
- CONTRACTOR YARD
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

N

0 500 1,000
FEET
1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

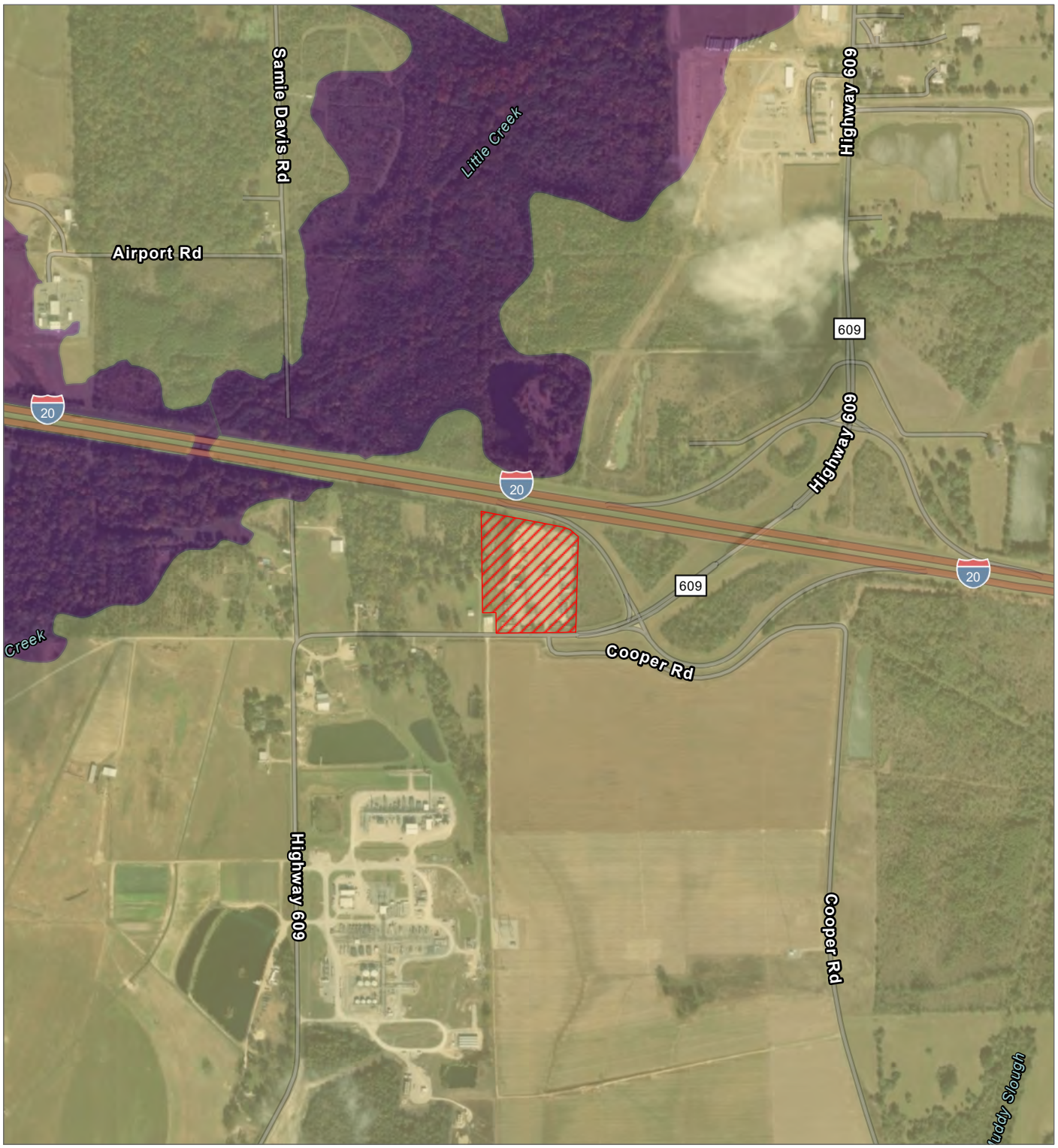
FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 1 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FF_FERC_APPX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN



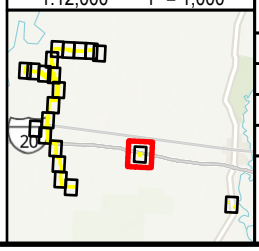
WORKSPACE TYPE

- CONTRACTOR YARD
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

N

0 500 1,000
FEET
1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

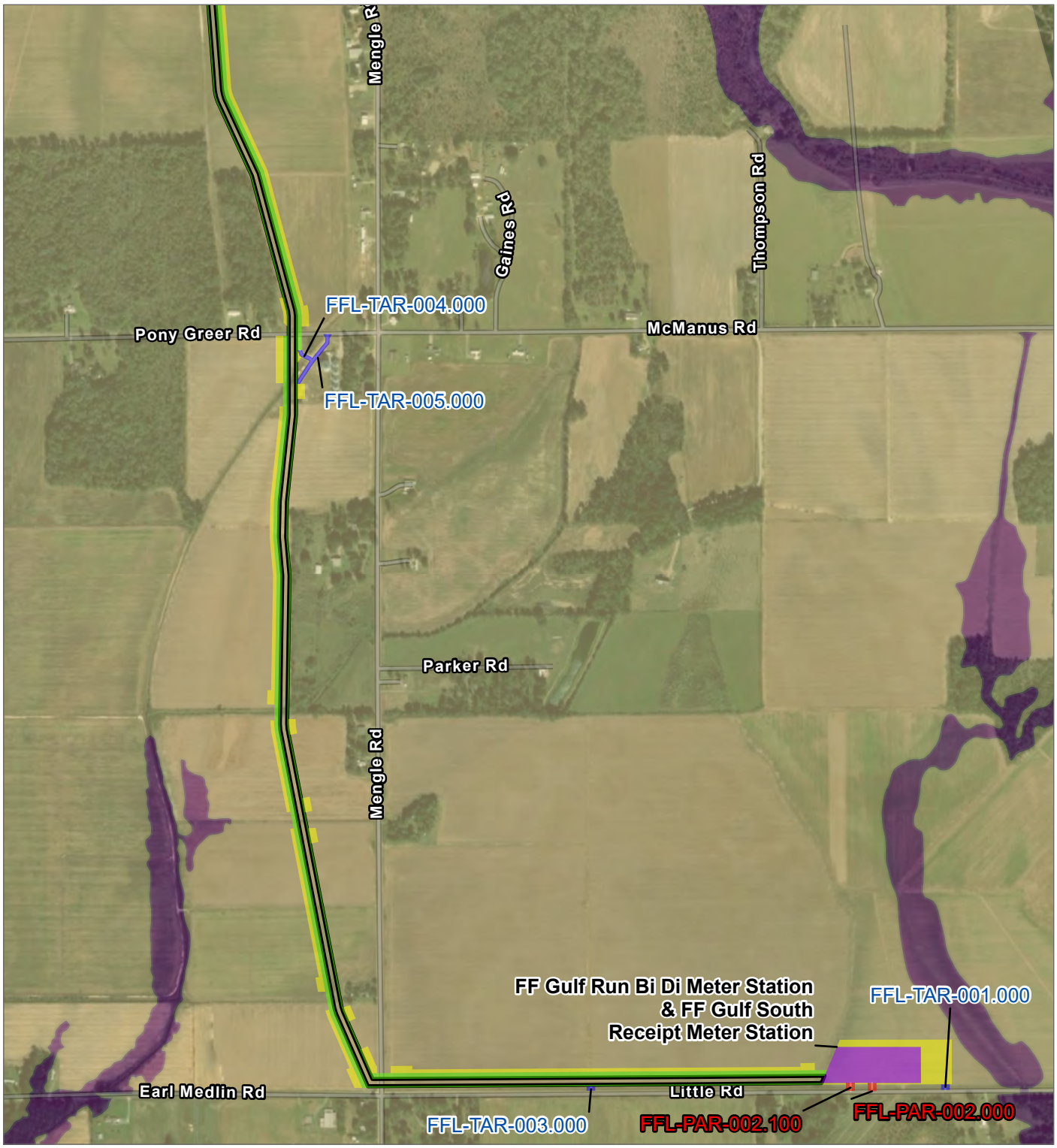
FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 2 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC_APRX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN



WORKSPACE TYPE

- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIOINAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

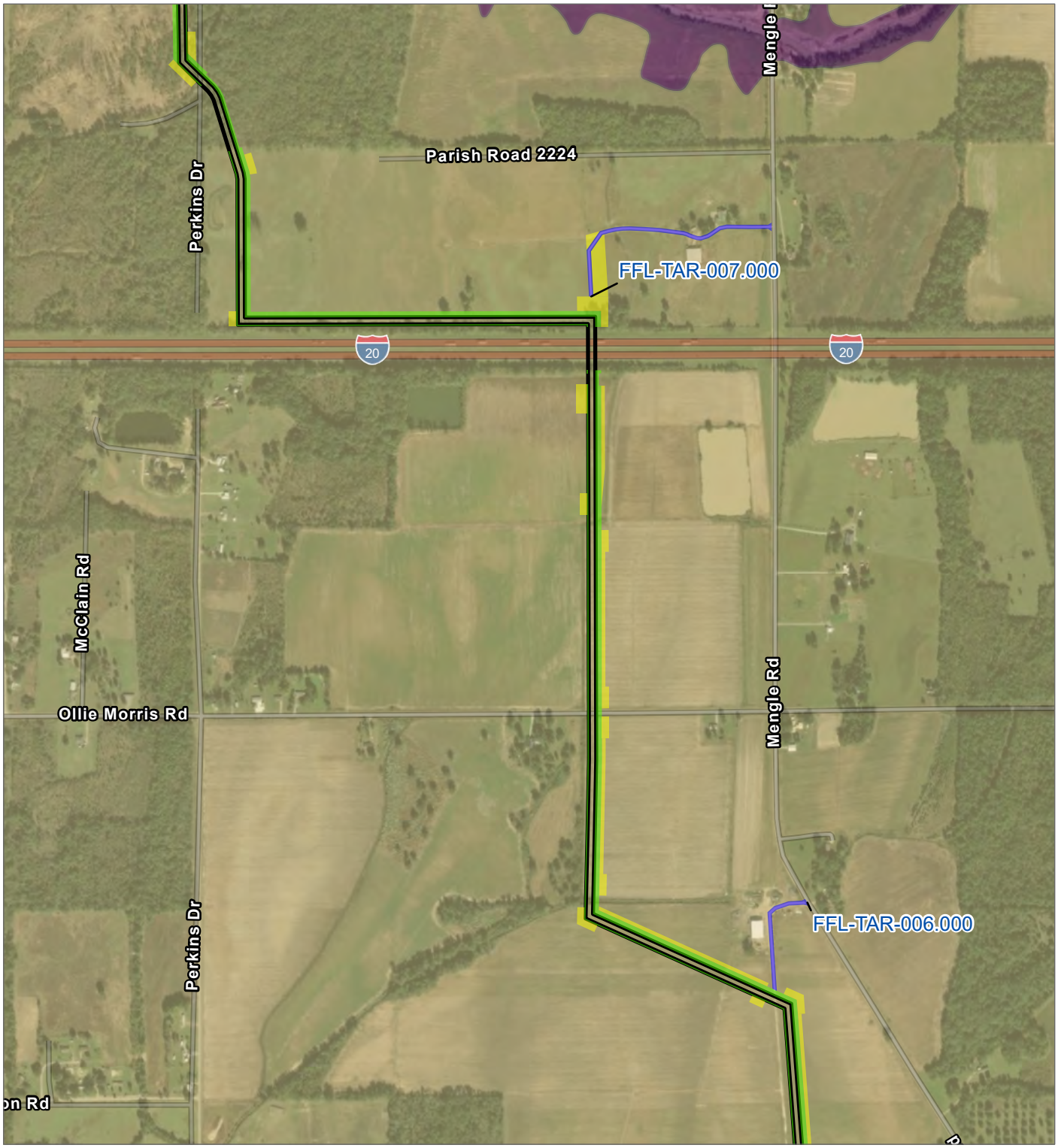
FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 3 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FF_FERC_APPX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN



WORKSPACE TYPE

- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIOINAL TEMPORARY WORKSPACE
- TEMPORARY ACCESS ROAD
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

0 500 1,000
FEET
1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

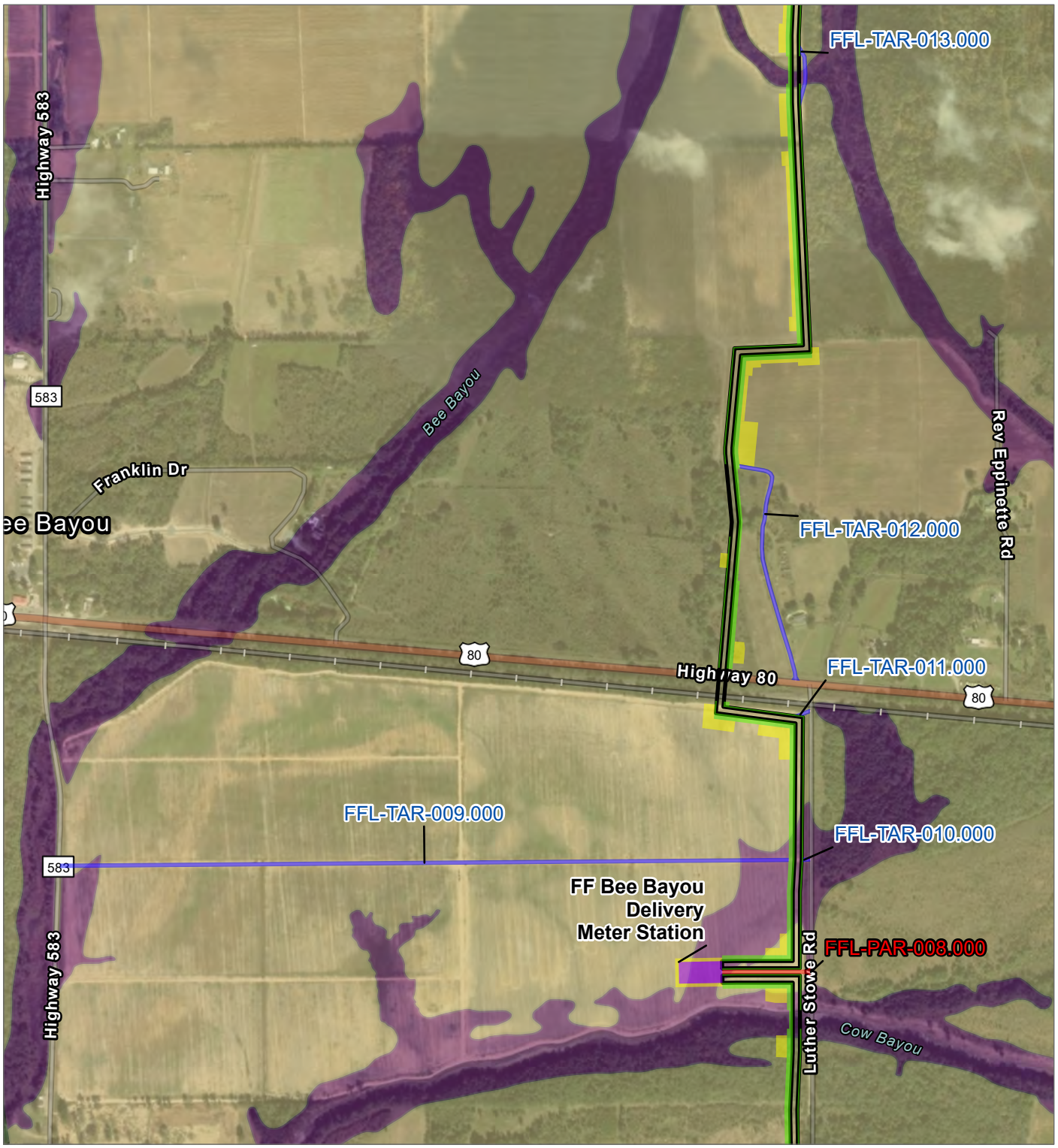
FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 4 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC_APRX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN



WORKSPACE TYPE

- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIOINAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

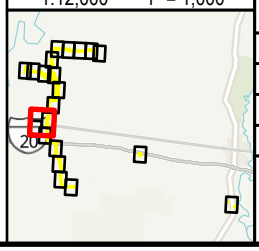
BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

N

0 500 1,000

FEET

1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

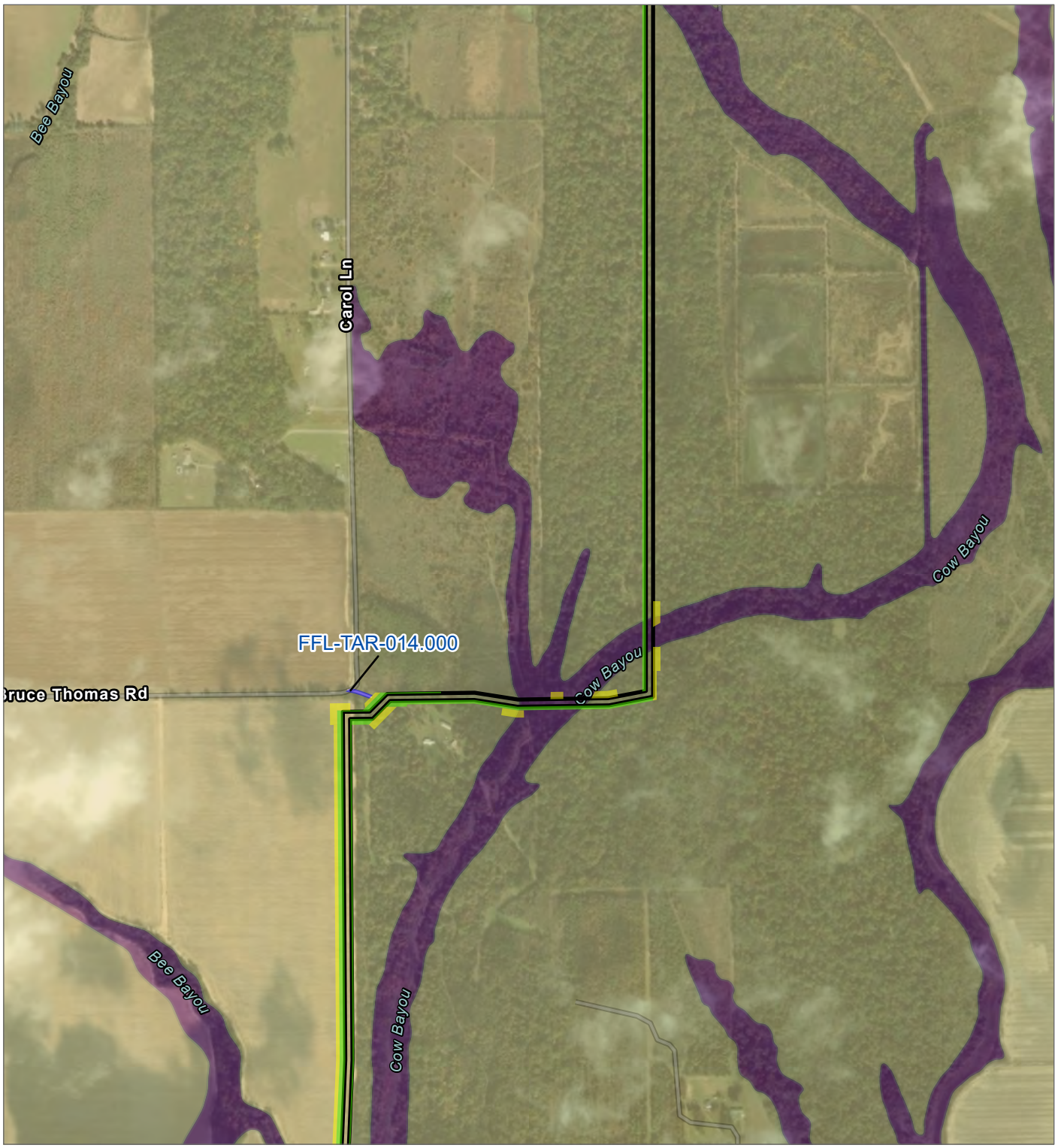
FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 5 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC_APRX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN



WORKSPACE TYPE

- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIOINAL TEMPORARY WORKSPACE
- TEMPORARY ACCESS ROAD
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

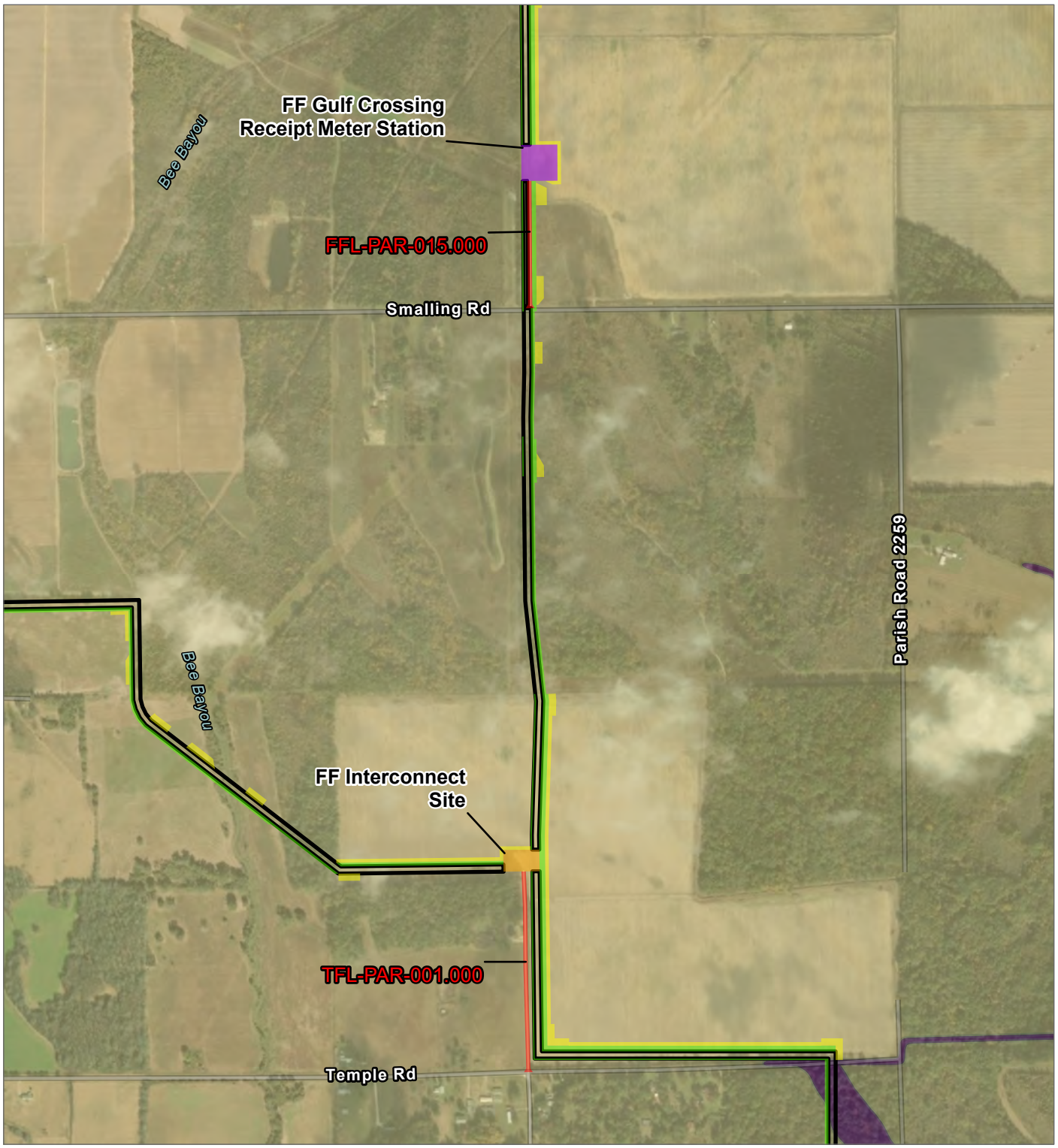
FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 6 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPK\FF_FERC_APPX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN



WORKSPACE TYPE

- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIOINAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- INTERCONNECT SITE
- METER STATION
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

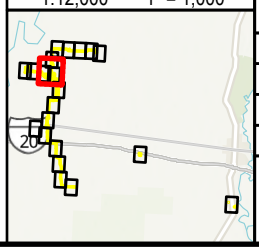
BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

N

0 500 1,000

FEET

1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 7 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FF_FERC_APPX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN

WORKSPACE TYPE

- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIOINAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- METER STATION
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

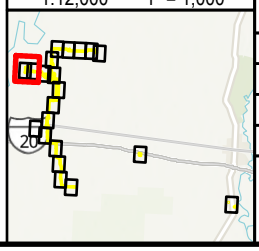
BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

N

0 500 1,000

FEET

1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

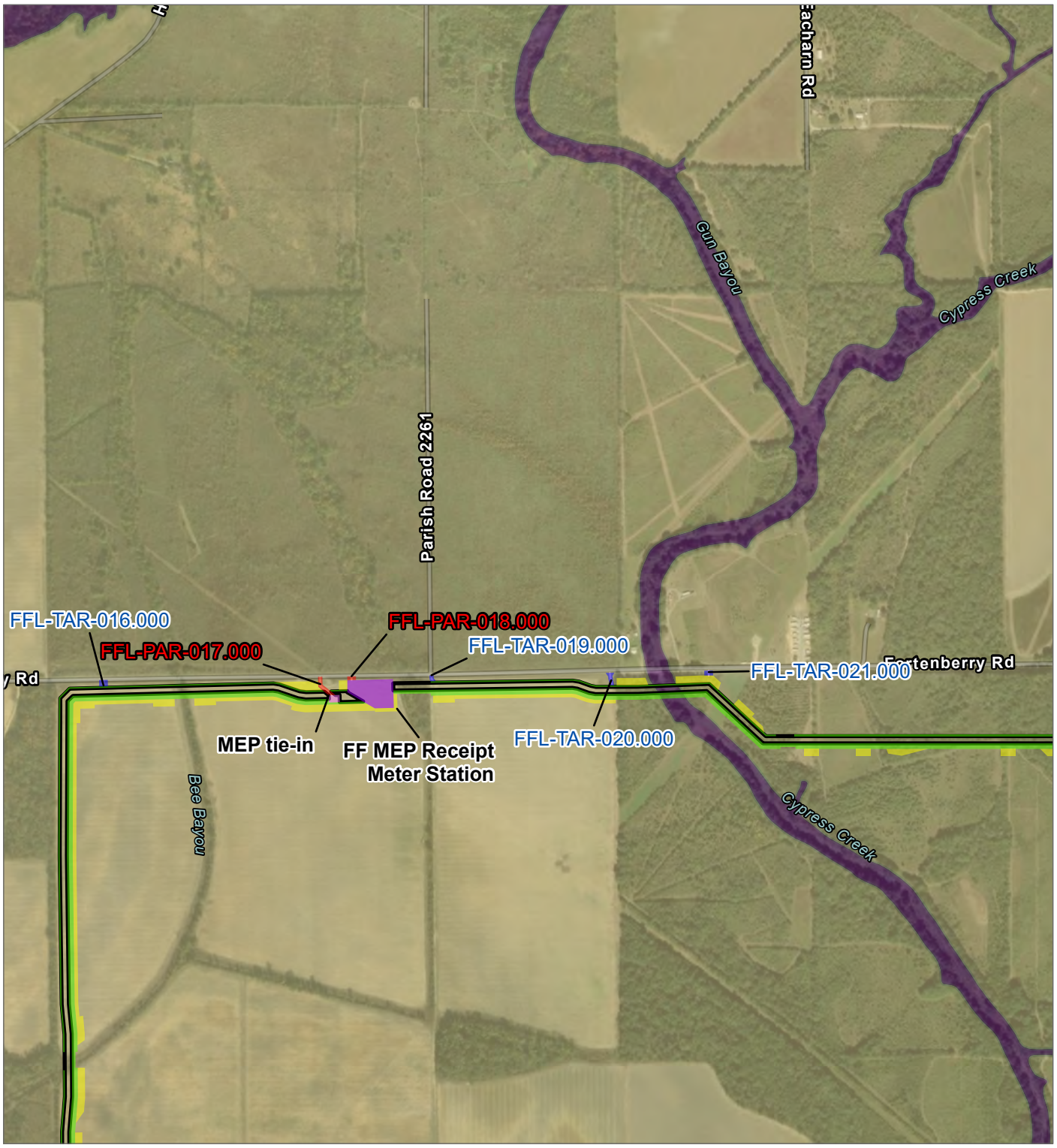
FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 8 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC_APRX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN



WORKSPACE TYPE
- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIOINAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION
- TAP SITE
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

0 500 1,000
FEET
1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

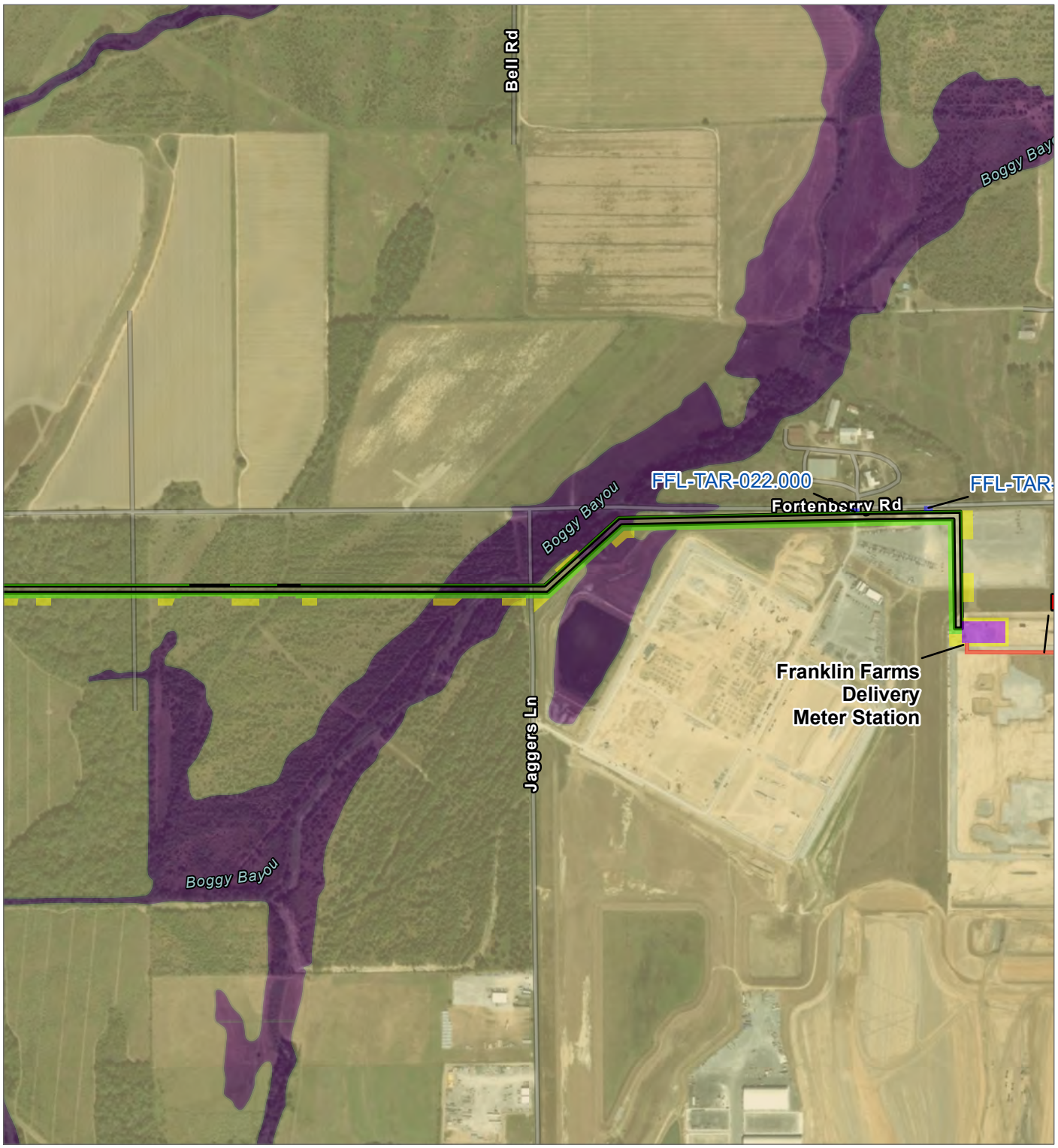
FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 9 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC_APRX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN



WORKSPACE TYPE

- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIOINAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

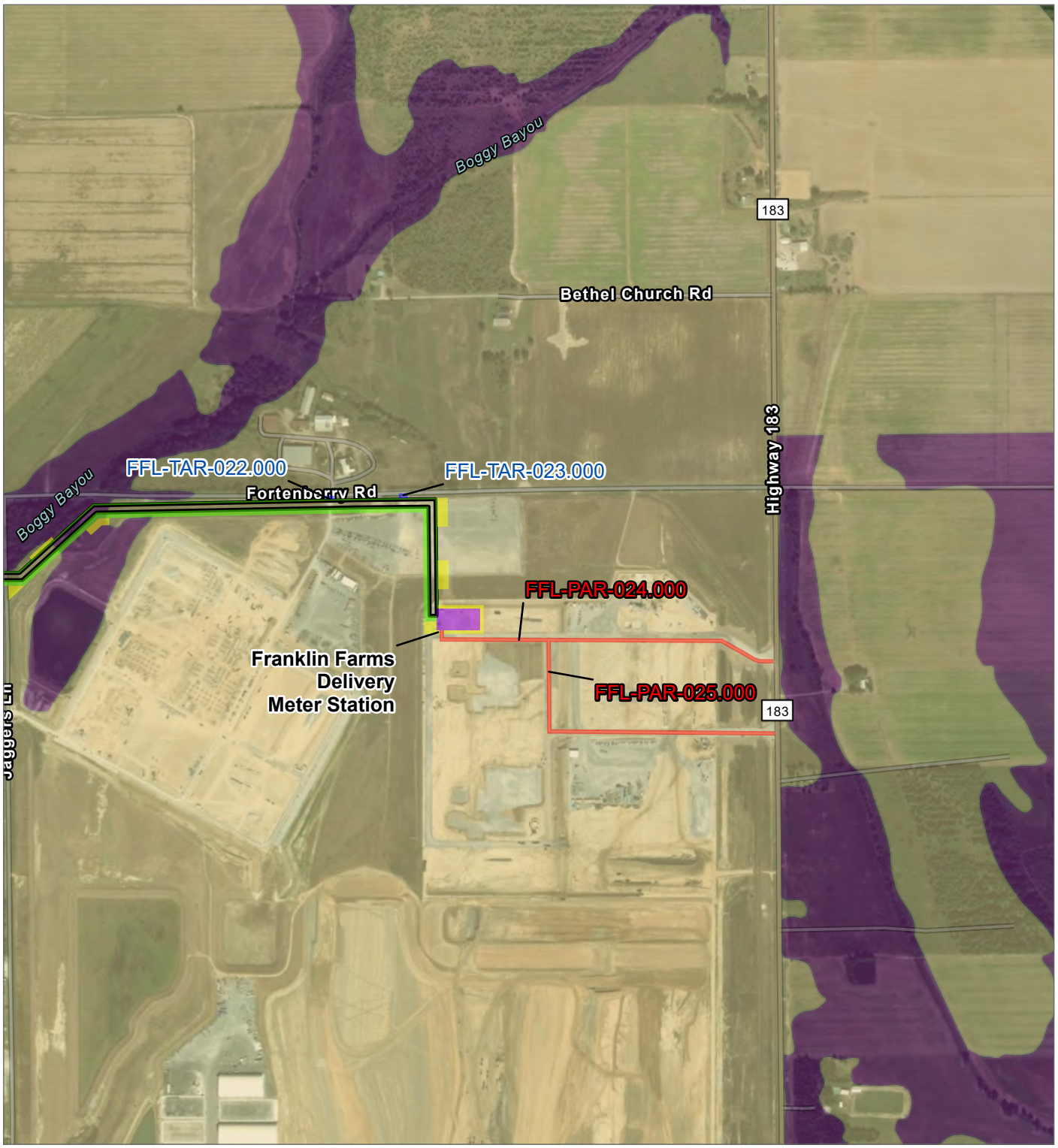
FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 10 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC_APRX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN



WORKSPACE TYPE

- PERMANENT EASEMENT
- TEMPORARY WORKSPACE
- ADDITIOINAL TEMPORARY WORKSPACE
- PERMANENT ACCESS ROAD
- TEMPORARY ACCESS ROAD
- METER STATION
- 1% ANNUAL CHANCE FLOOD HAZARD
- AREA WITH REDUCED RISK DUE TO LEVEE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, FEMA

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

FEMA FLOODPLAIN MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	JUNE 2026		

FIGURE 2.C-1
PAGE 11 OF 11

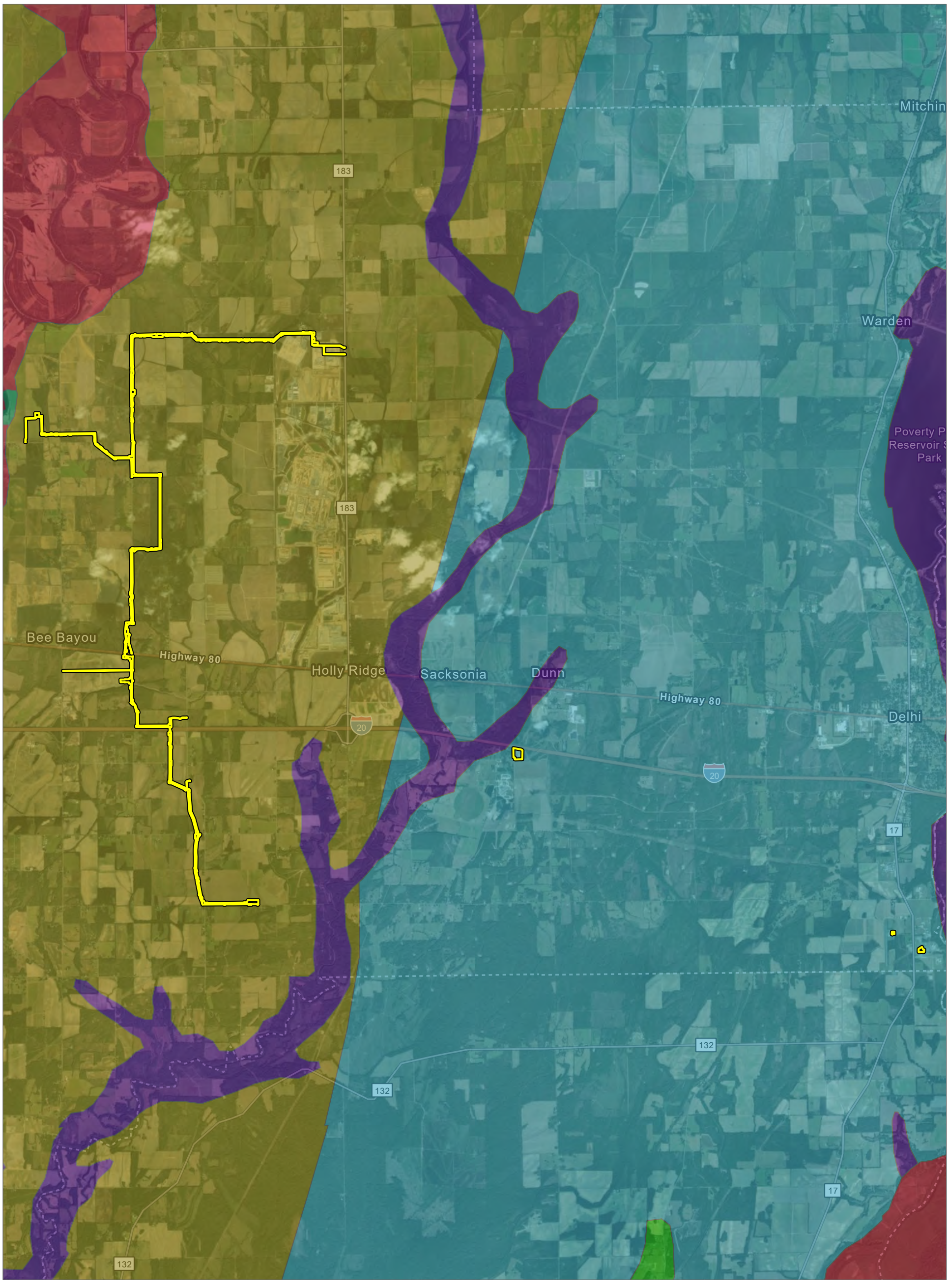
TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 6/12/2026, 08:03:43 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FF_FERC_APPX; LAYOUT NAME: 2.C-1 FEMA FLOODPLAIN

APPENDIX A-11

Surficial Geology Map

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 6.A-1 SURFICIAL GEOLOGY



PROJECT BOUNDARY

GEOLOGICAL UNIT

- QAL, ALLUVIUM
- QBS, BRAIDED STREAM TERRACES
- QBS-L, BRAIDED STREAM TERRACES
- QNL, NATURAL LEVEES
- QTP, PRAIRIE TERRACES
- WATER

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, USGS

0 3,500 7,000 FEET

1:84,000 1" = 7,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

SURFICIAL GEOLOGY

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

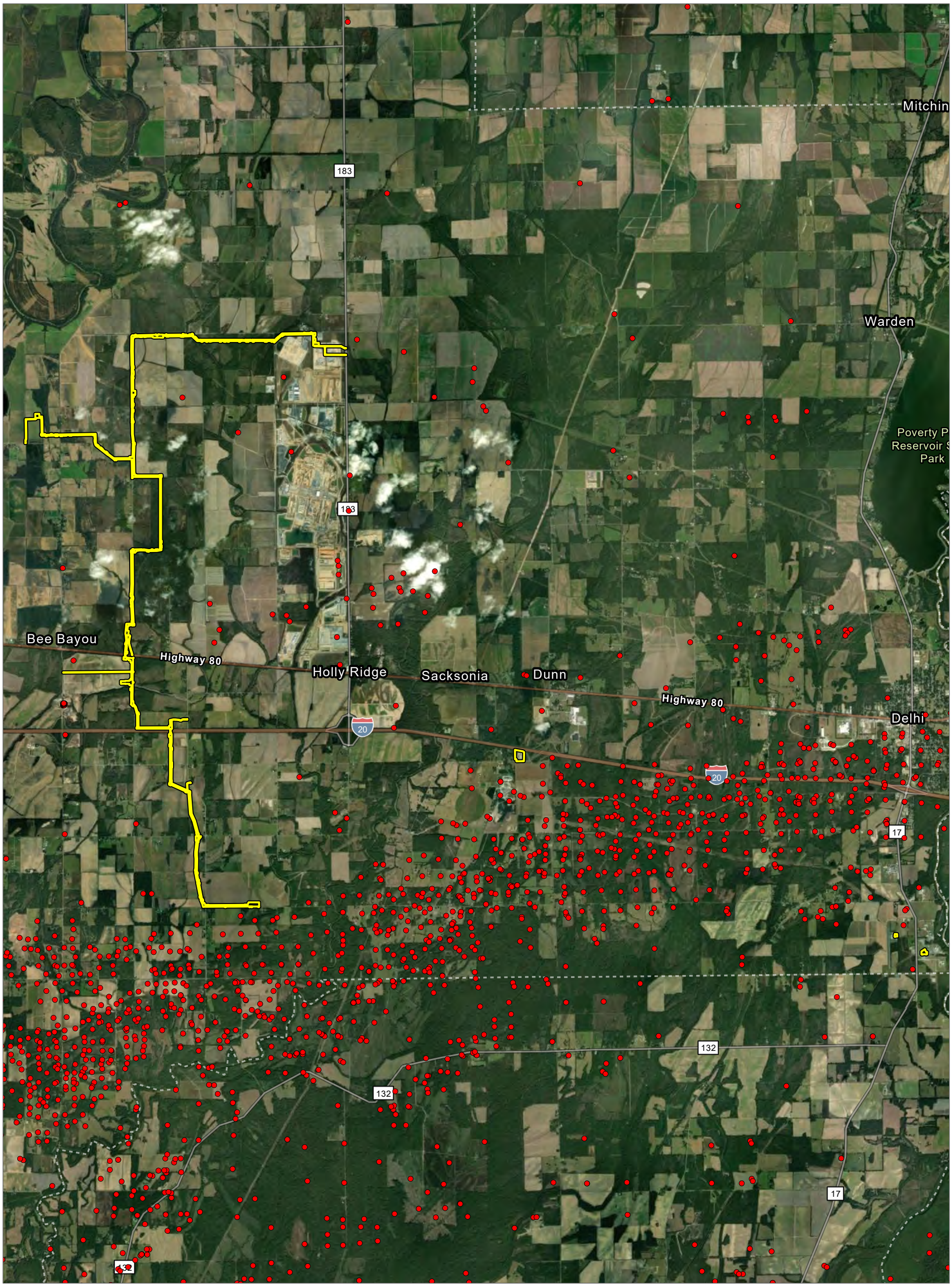
FIGURE 6.A-1

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

APPENDIX A-12

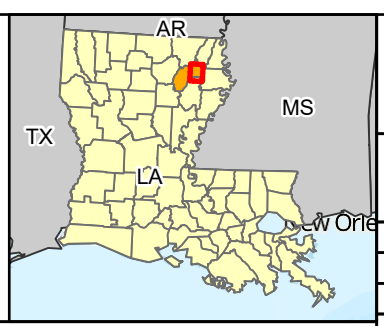
Oil and Gas Wells in the Vicinity of the Project

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 6.A-2 OIL AND GAS WELLS



PROJECT BOUNDARY

OIL AND GAS WELLS



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, LADNR, SONRIS

0 3,500 7,000 FEET

1:84,000 1" = 7,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

OIL AND GAS WELLS

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

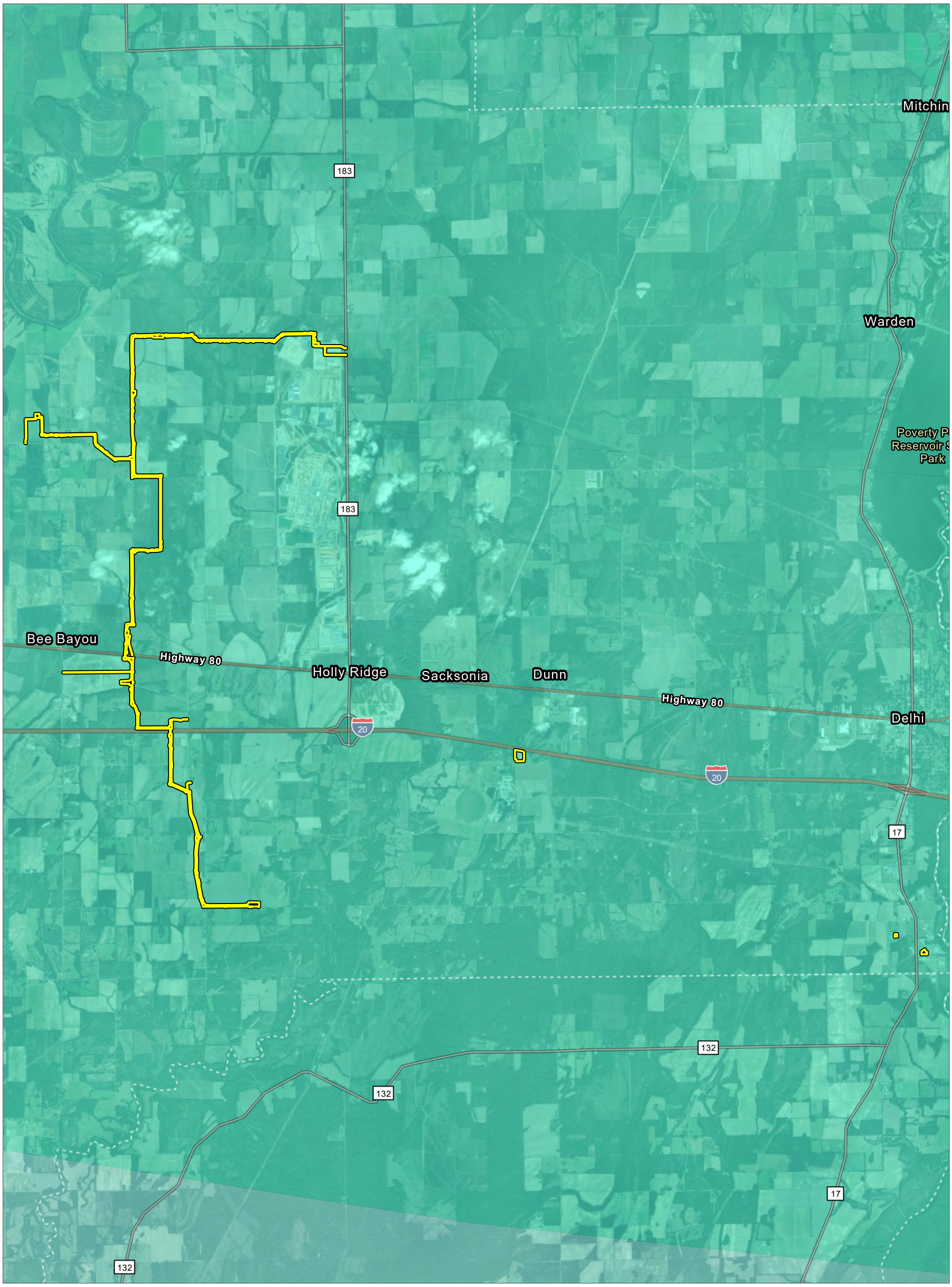
FIGURE 6.A-2

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

APPENDIX A-13

Peak Horizontal Ground Acceleration with 2 Percent Probability of Exceedance in 50 Years

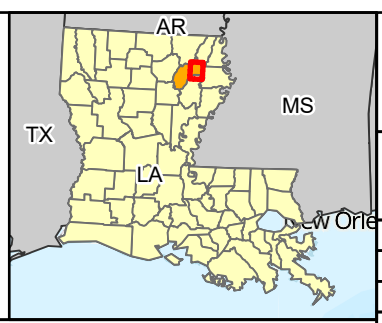
COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 6.A-3 PGA 2%



PROJECT BOUNDARY

PGA WITH 2% PROBABILITY OF EXCEEDANCE

- 0.1 – 0.12
- 0.12 – 0.14



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, USGS

0 3,500 7,000 FEET

1:84,000 1" = 7,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

PEAK HORIZONTAL GROUND ACCELERATION (PERCENT OF GRAVITY) WITH 2 PERCENT PROBABILITY OF EXCEEDANCE IN 50 YEARS

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

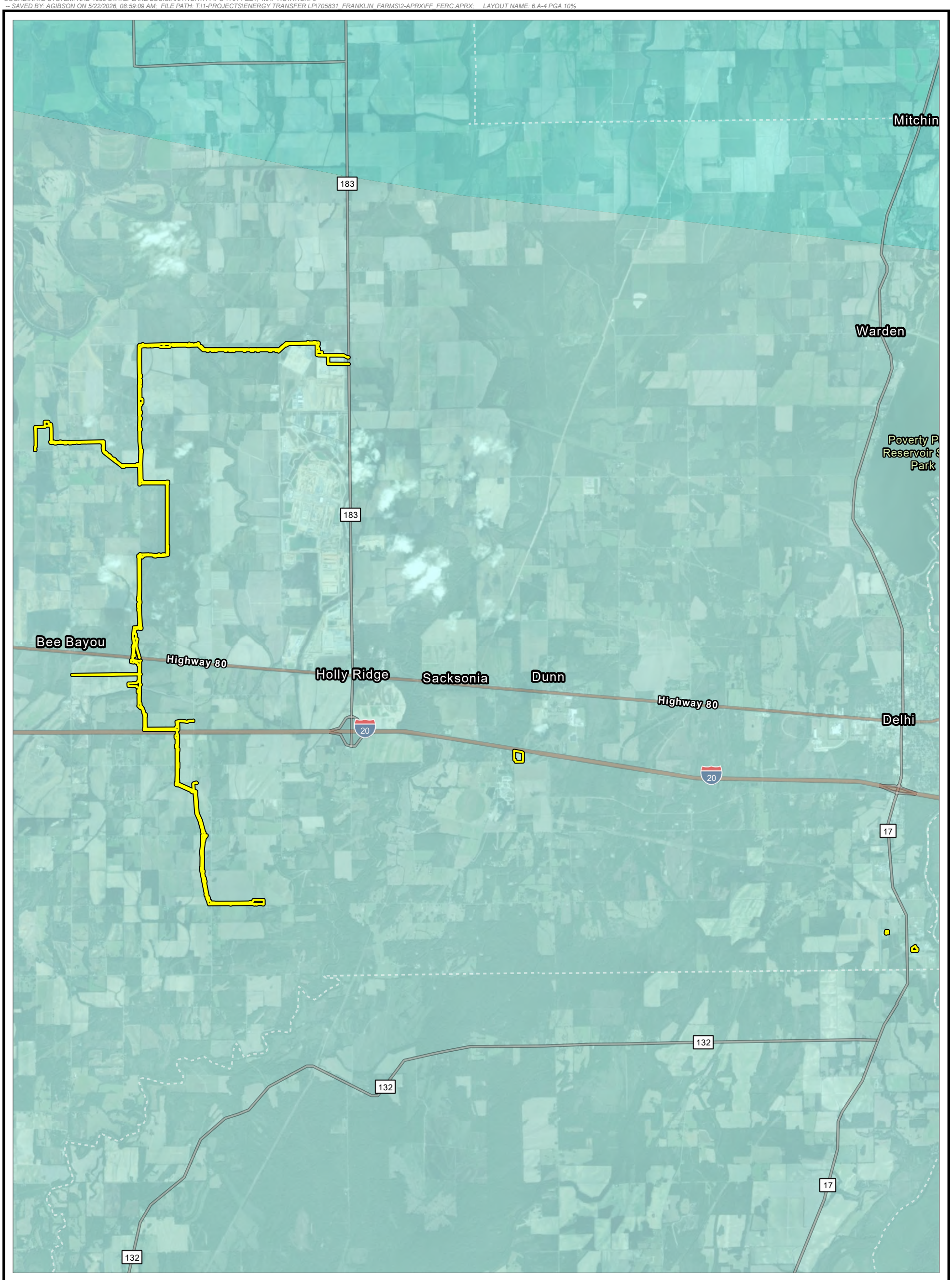
FIGURE 6.A-3

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

APPENDIX A-14

Peak Horizontal Ground Acceleration with 10 Percent Probability of Exceedance in 50 Years

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 6.A-4 PGA 10%



PROJECT BOUNDARY

PGA WITH 10% PROBABILITY OF EXCEEDANCE

- 0.04 – 0.05
- 0.05 – 0.06

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, USGS

0 3,500 7,000 FEET

1:84,000 1" = 7,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

PEAK HORIZONTAL GROUND ACCELERATION (PERCENT OF GRAVITY) WITH 10 PERCENT PROBABILITY OF EXCEEDANCE IN 50 YEARS

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

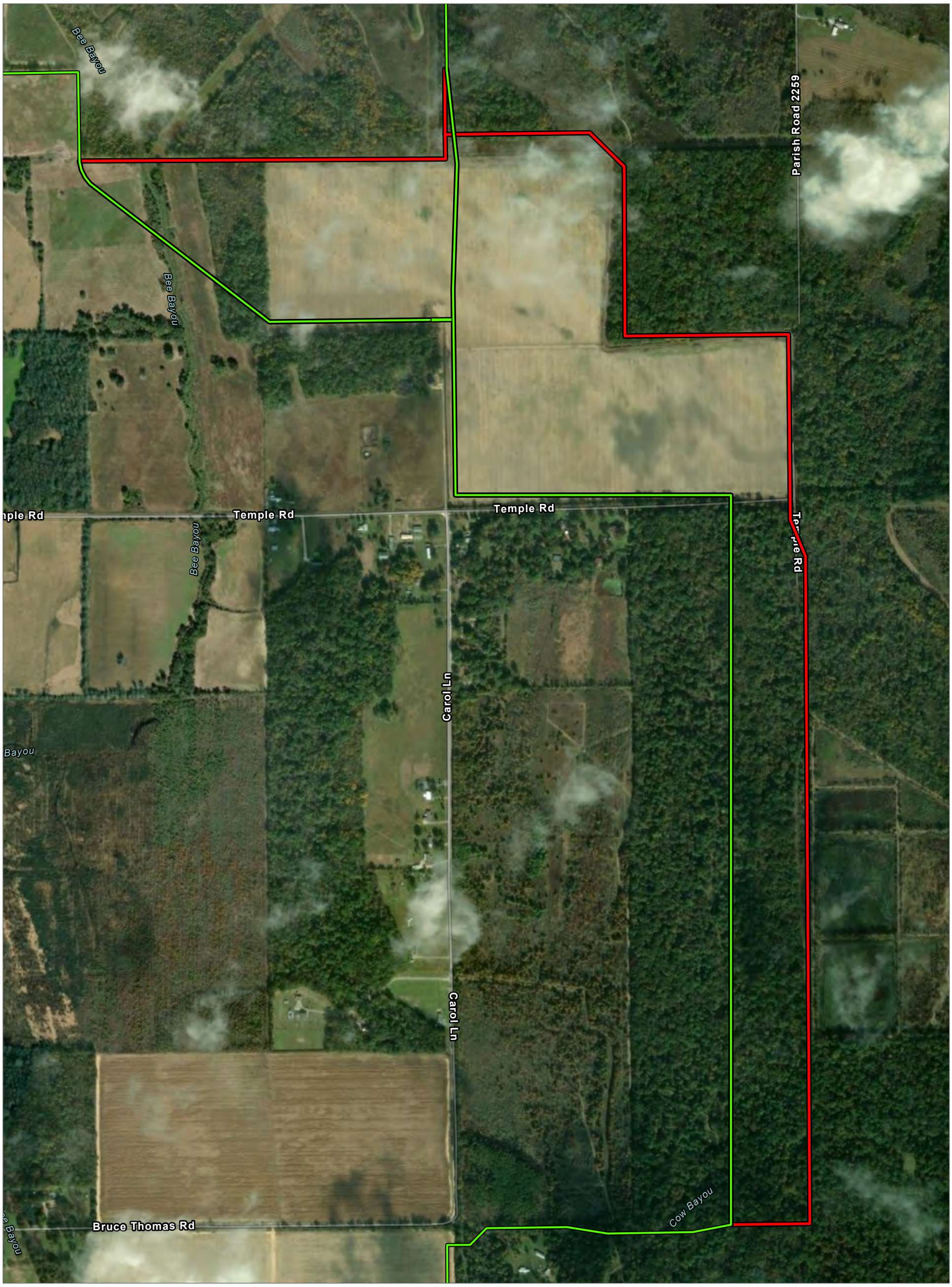
FIGURE 6.A-4

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

APPENDIX A-15

Major Route Alternative Map

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 10.B MAJOR ROUTE ALTERNATIVES



ALTERNATE ROUTE

PREFERRED ROUTE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 350 700 FEET

1:8,400 1" = 700'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

MAJOR ROUTE ALTERNATIVES

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 10.B

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

APPENDIX A-16

Route Variation Map

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 10.C ROUTE VARIATION OVERVIEW

Page 2
Page 3
Page 4
Page 1
Jordan Lake
Bee Bayou
Highway 80
Holly Ridge
Highway 183
Good Rd
583
852
183
80
20
3048
584

PROPOSED PIPELINE ROUTE
ORIGINAL PIPELINE ROUTE
PAGE EXTENT

AR
MS
TX
LA

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 2,000 4,000 FEET
1:48,000 1" = 4,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

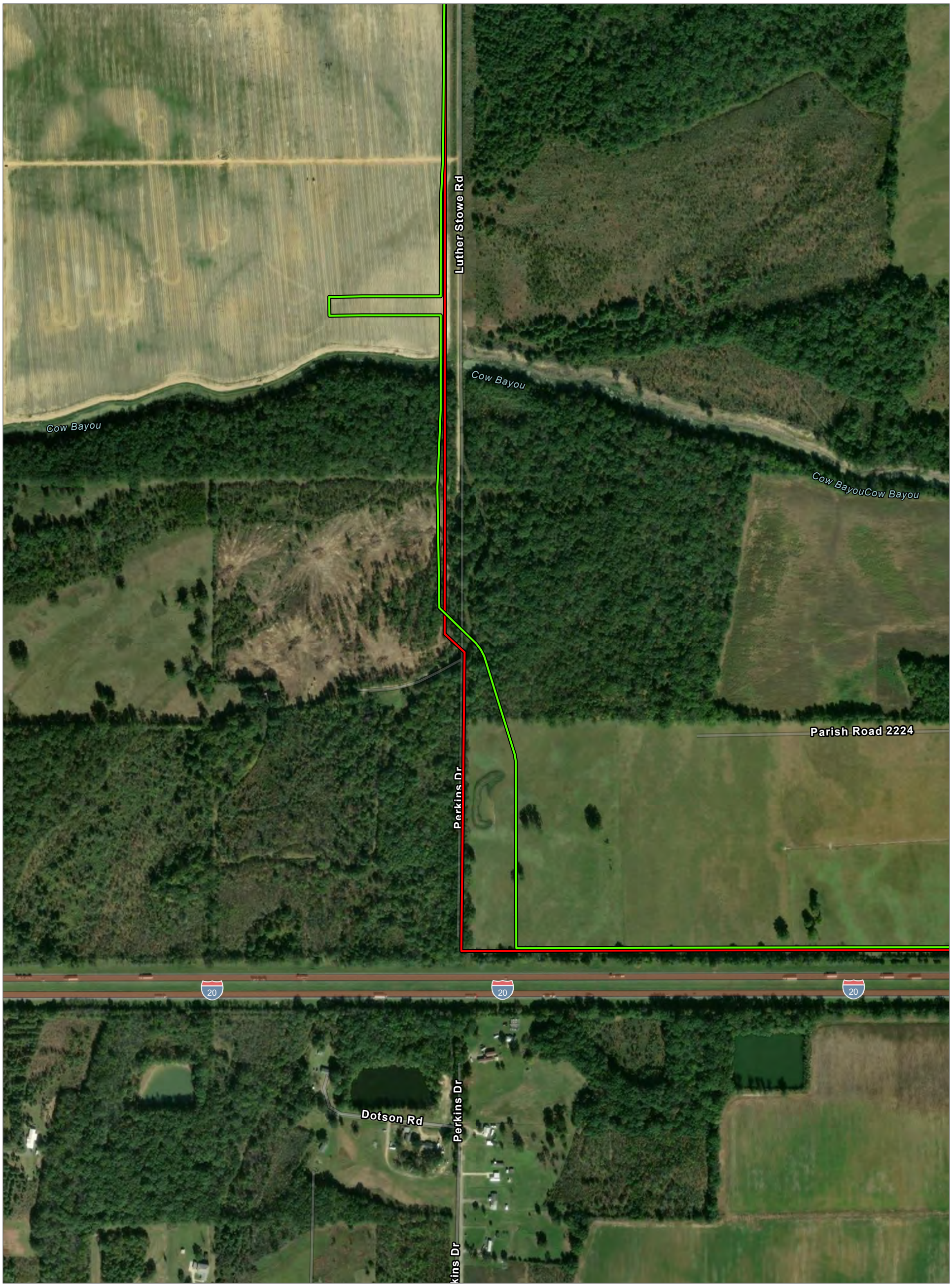
ROUTE VARIATION

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 10.C

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 10.C ROUTE VARIATION SERIES



PROPOSED PIPELINE ROUTE

ORIGINAL PIPELINE ROUTE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N

0 250 500 FEET

1:6,000 1" = 500'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

ROUTE VARIATION

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 10.C
PAGE 1 OF 4

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 10.C ROUTE VARIATION SERIES

PROPOSED PIPELINE ROUTE

ORIGINAL PIPELINE ROUTE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N

0 250 500 FEET

1:6,000 1" = 500'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

ROUTE VARIATION

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 10.C
PAGE 2 OF 4

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 10.C ROUTE VARIATION SERIES

PROPOSED PIPELINE ROUTE

ORIGINAL PIPELINE ROUTE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 250 500 FEET

1:6,000 1" = 500'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

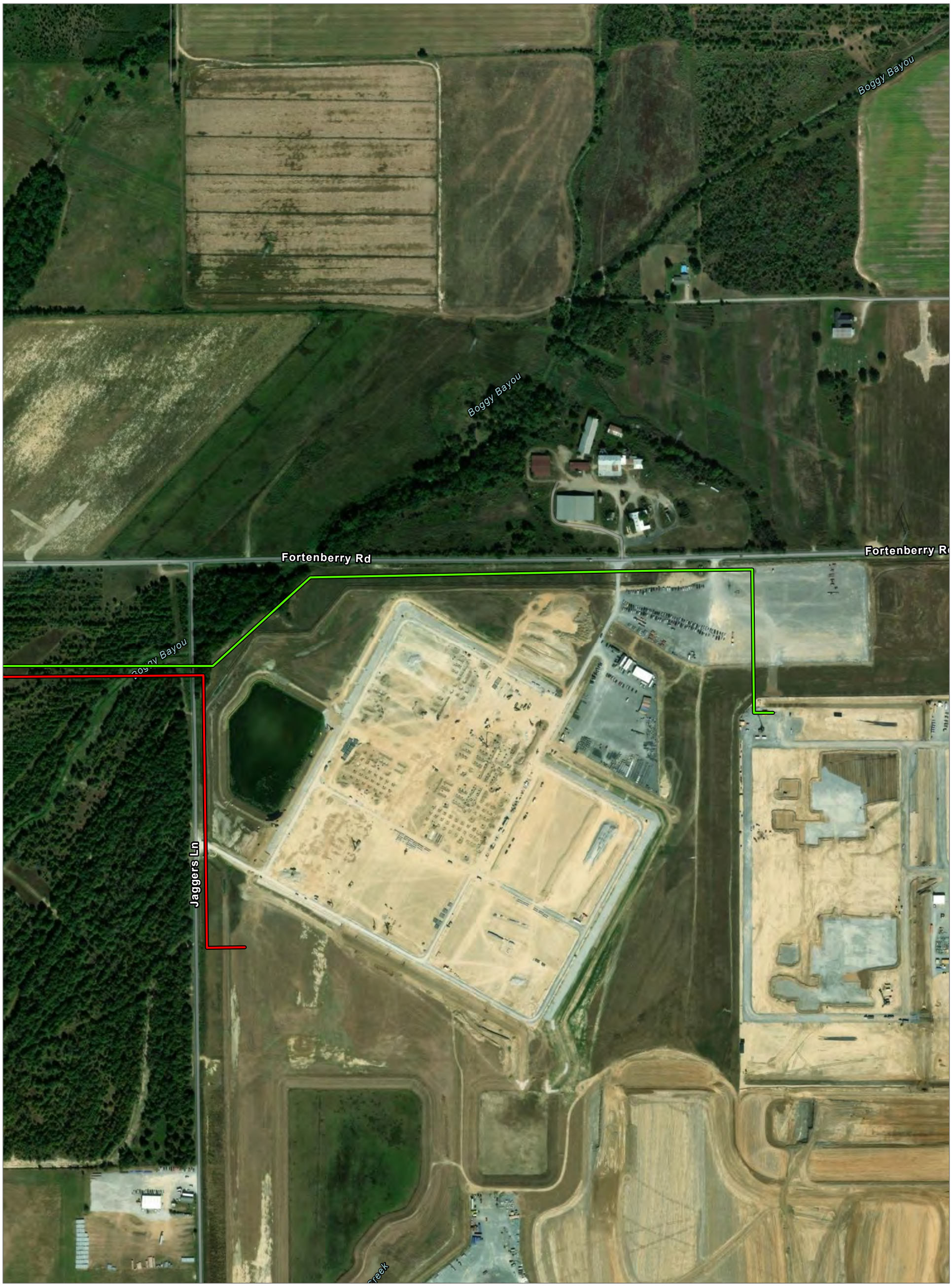
ROUTE VARIATION

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 10.C
PAGE 3 OF 4

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/22/2026, 08:59:09 AM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FF_FERC.APRX; LAYOUT NAME: 10.C ROUTE VARIATION SERIES



PROPOSED PIPELINE ROUTE

ORIGINAL PIPELINE ROUTE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 250 500 FEET

1:6,000 1" = 500'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT
RICHLAND PARISH, LA

ROUTE VARIATION

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 10.C
PAGE 4 OF 4

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

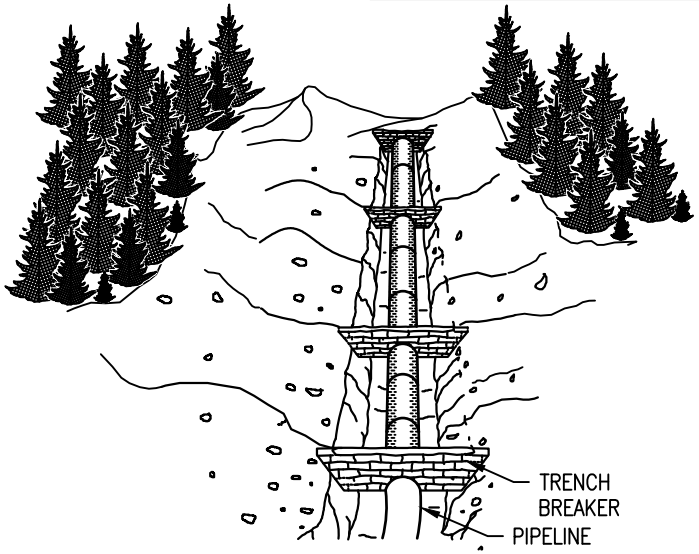
APPENDIX B

Typical and Site-Specific Detail Drawings

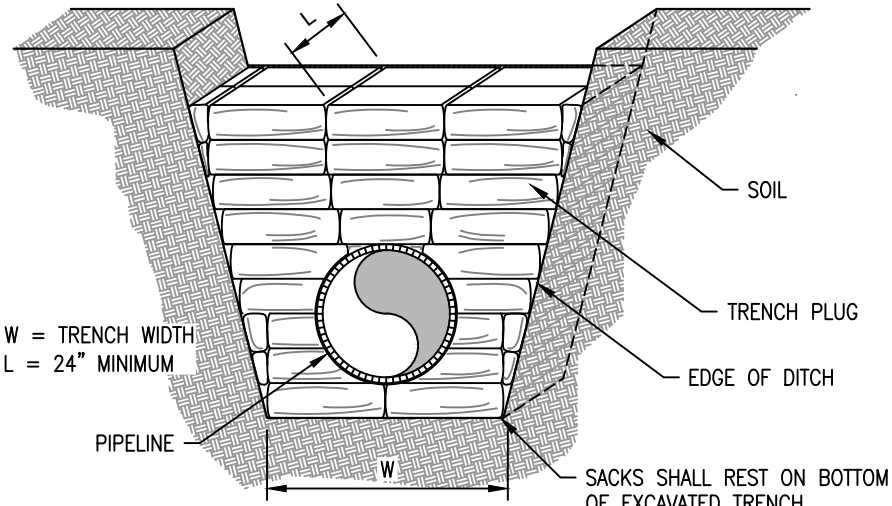
APPENDIX B-1

Project Typicals for Construction

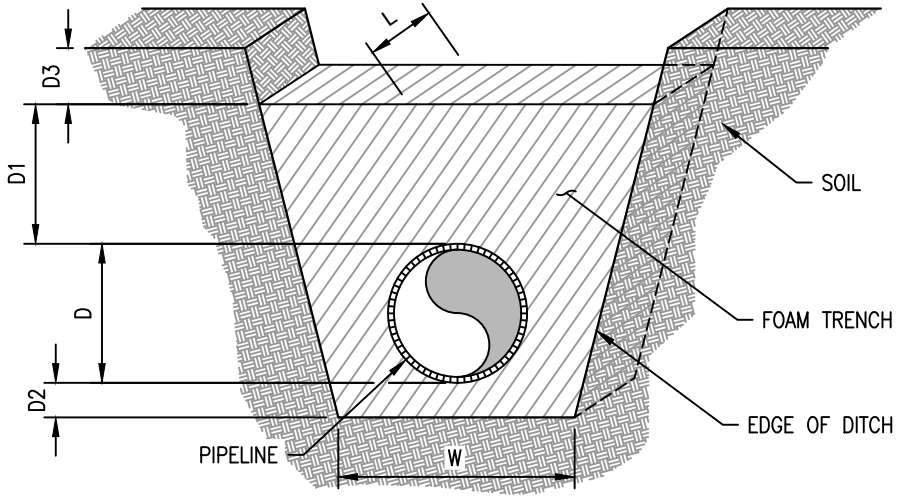
TRENCH BREAKERS INSTALLATION – ENVIRONMENTAL TYPICAL



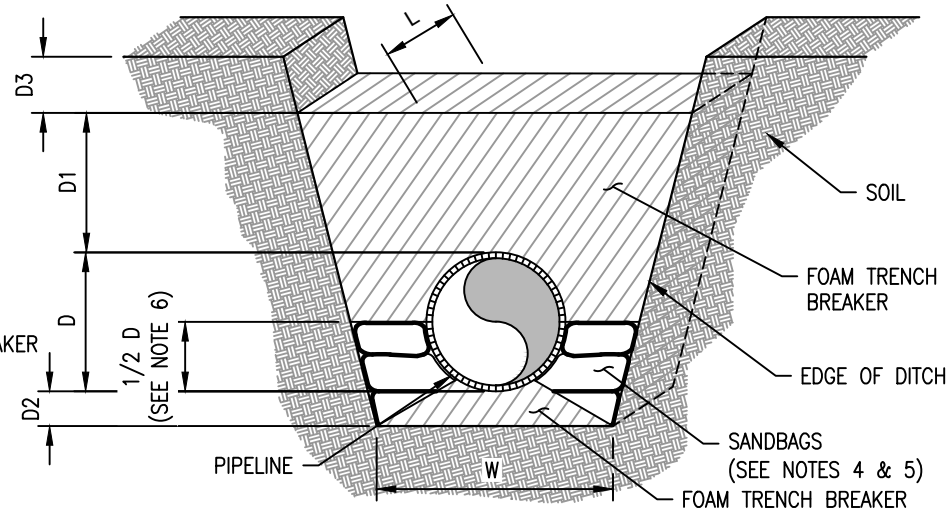
TRENCH BREAKERS



TRENCH PLUG DETAIL



FOAM TRENCH BREAKER



FOAM TRENCH BREAKER WITH SANDBAGS

D= PIPE DIAMETER
D1 = APPROXIMATELY 24"
D2 = APPROXIMATELY 6" (8" MIN. IN ROCK)
D3 = APPROXIMATELY 12"
W = D + 2 TO 4 FEET
L = 24" MINIMUM
D1 + D3 = 36" MINIMUM

SEE SHEET 2 OF 2: TYPICAL
TYP–0001
FOR NOTES.

REVISIONS

REV#	REV DATE	REV NAME	DRN	CHK	APP
A	05-06-2026	ISSUE FOR PERMIT	HW	JD	CF

PREPARED BY: FIRM No:EF.0007320
TRC
8521 McHARD RD, HOUSTON, TX 77053

ENERGY TRANSFER

FRANKLIN FARMS PROJECT
TRENCH BREAKERS INSTALLATION

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0001

REV
A

TRENCH BREAKERS INSTALLATION
ENVIRONMENTAL TYPICAL

NOTES:

1. TRENCH BREAKERS SHALL BE INSTALLED:
   - 1.a. ON SLOPES ALONG THE TRENCH LINE WHERE THE NATURAL DRAINAGE PATTERN, PROFILE AND TYPE OF BACKFILL MATERIAL MAY RESULT IN LOSS OF BACKFILL MATERIAL OR ALTERATION OF THE NATURAL PATTERN;
   - 1.b. AT THE BASE OF SLOPES ADJACENT TO WATERBODIES AND WETLANDS;
   - 1.c. WHERE NEEDED TO AVOID DRAINING A WETLAND; ON UPLAND SLOPES,
   - 1.d. AT THE SAME SPACING AS SLOPE BREAKERS AND UP SLOPE OF SLOPE BREAKERS; IN CULTIVATED LAND AND RESIDENTIAL AREAS WHERE PERMANENT SLOPE BREAKERS ARE NOT TYPICALLY INSTALLED,
   - 1.e. AT THE SAME SPACING AS IF PERMANENT SLOPE BREAKERS WHERE REQUIRED.
2. BREAKERS SHALL BE KEYED 6 INCHES INTO TRENCH WALLS AND BOTTOM.
3. IN ROCK OR BOULDER AREAS, BREAKERS SHALL BE KEYED 6 INCHES BELOW PADDING.
4. THE ADDITION OF SAND BAGS UNDER FOAM BREAKERS WILL BE PLACED AT THE DISCRETION OF THE COMPANY REPRESENTATIVE.
5. SAND BAGS WILL NOT BE INSTALLED WITH THE FOAM BREAKER AT THE EDGE OF WETLANDS AND WATERBODIES.
6. WHEN SAND BAGS ARE REQUIRED UNDER THE FOAM BREAKERS, A MINIMUM OF TWO LAYERS SHALL BE INSTALLED.
7. OPEN WEAVE HEMP OR JUTE SACKS SHALL BE FILLED WITH A MINIMUM OF 55lbs. OF MIXTURE OF 1 PART CEMENT TO 6 PARTS SAND OR SUBSOIL SO THAT NATURAL GROUND WATER WILL PERMIT MIXTURE TO EXUDE AND BOND SACKS TOGETHER.
8. BREAKER SPACING AND CONFIGURATION MAY BE CHANGED AS DIRECTED BY COMPANY. DEPTH OF DITCH MAY VARY WITH SITE CONDITIONS.
9. ALL MATERIALS SHALL BE SUPPLIED BY CONTRACTOR.

REVISIONS

REV#	REV DATE	REV NAME	DRN	CHK	APP
A	05-06-2026	ISSUE FOR PERMIT	HW	JD	CF

PREPARED BY: FIRM No:EF.0007320
TRC
8521 McHARD RD, HOUSTON, TX 77053

ENERGY TRANSFER

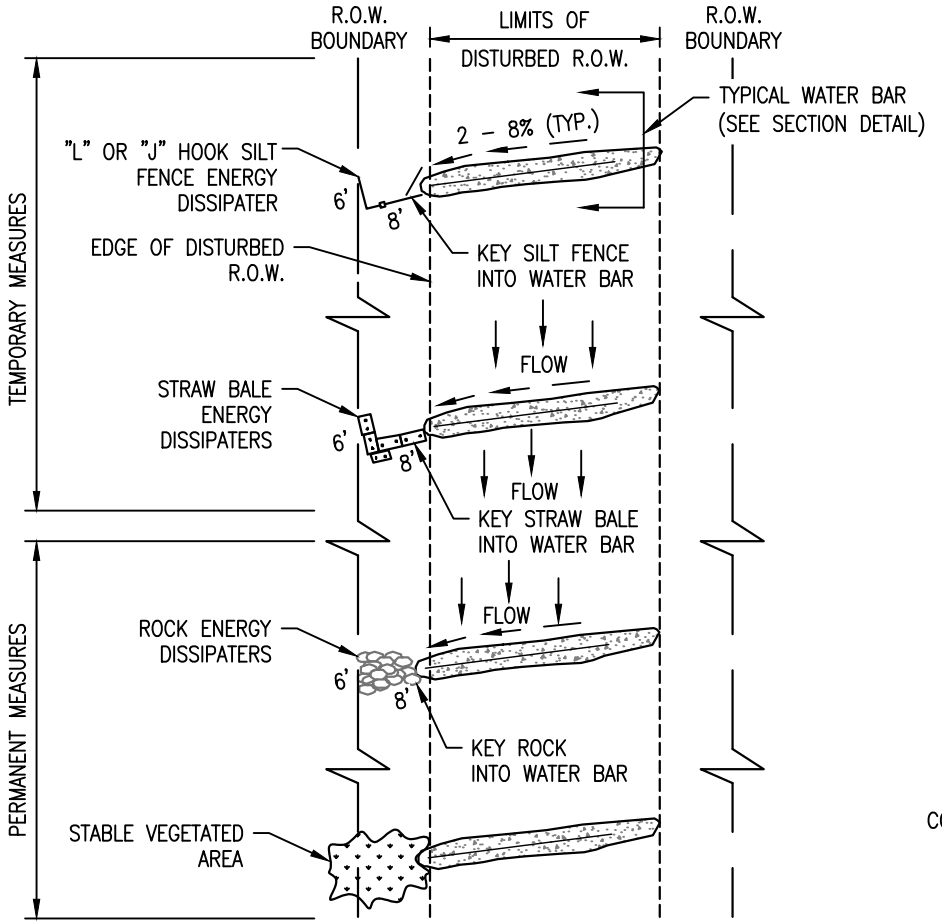
FRANKLIN FARMS PROJECT
TRENCH BREAKERS INSTALLATION

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

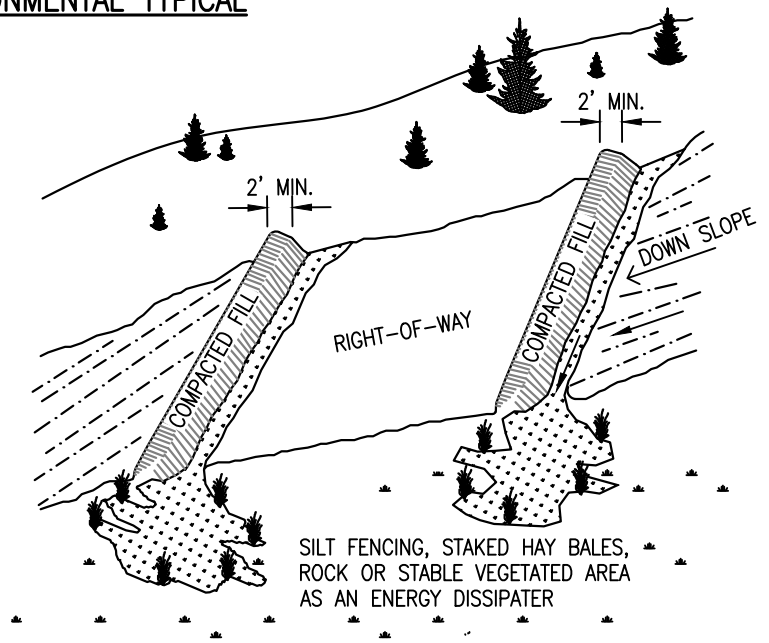
DWG NUMBER
TYP-0001

REV
A

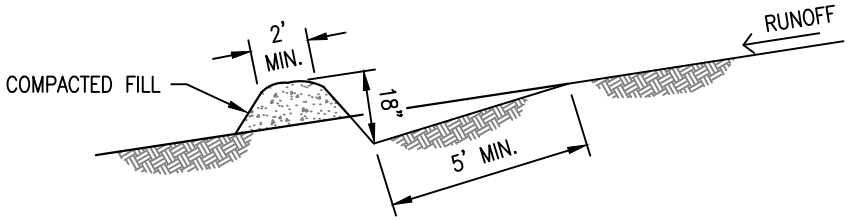
SLOPE BREAKER – ENVIRONMENTAL TYPICAL



PLAN VIEW
N.T.S.



ISO VIEW
N.T.S.



TYPICAL WATER BAR CROSS SECTION DETAIL
N.T.S.

SEE SHEET 2 OF 2: TYPICAL TYP-0002 FOR NOTES.

REVISIONS					
REV#	REV DATE	REV NAME	DRN	CHK	APP
A	05-06-2026	ISSUE FOR PERMIT	HW	JD	CF

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ENERGY TRANSFER

FRANKLIN FARMS PROJECT
SLOPE BREAKER

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0002

REV
A

SLOPE BREAKER – ENVIRONMENTAL TYPICAL

NOTES:

1. SLOPE BREAKERS SHALL BE CONSTRUCTED OF COMPACTED NATIVE SOIL AND INSTALLED AT LOCATIONS AS PER CONSTRUCTION SPECIFICATIONS, AS SHOWN ON THE CONSTRUCTION DRAWINGS OR AS DIRECTED BY THE COMPANY ENVIRONMENTAL INSPECTOR.
2. SLOPE BREAKERS SHALL BE ORIENTED AS SHOWN OR OTHER PATTERN AS DIRECTED BY THE COMPANY ENVIRONMENTAL INSPECTOR TO DIRECT THE WATER OFF THE RIGHT-OF-WAY.
3. SLOPE BREAKERS SHALL BE CONSTRUCTED AT A 2-8% GRADIENT ACROSS THE SLOPE.
4. THE SLOPE BREAKERS SHALL BE 18" DEEP (AS MEASURED FROM THE TROUGH TO THE TOP OF THE SLOPE BREAKER). THE TROUGH WILL BE A MINIMUM OF 5' WIDE ACROSS THE WIDTH OF THE RIGHT-OF-WAY.
5. THE OUTLET OF THE SLOPE BREAKER MUST FREELY DISCHARGE ALL RUNOFF OFF THE DISTURBED RIGHT-OF-WAY INTO A STABLE, WELL VEGETATED AREA OR INTO AN ENERGY DISSIPATER.
6. WHERE SLOPE BREAKERS EXTEND BEYOND THE EDGE OF THE CONSTRUCTION R.O.W. TO DIRECT RUNOFF INTO STABLE, WELL VEGETATED AREAS, THESE LOCATIONS MUST BE APPROVED BY THE COMPANY ENVIRONMENTAL INSPECTOR.
7. THE OUTLET SHALL CONTAIN AN ENERGY DISSIPATER IF THE COMPANY ENVIRONMENTAL INSPECTOR DETERMINES EXISTING VEGETATION IS NOT SUFFICIENTLY STABLE TO PREVENT EROSION. THE ENERGY DISSIPATER SHALL BE CONSTRUCTED AS FOLLOWS:
   - OUTFALL END OF DISSIPATER SHOULD BE LOWER THAN SLOPE BREAKER END.
   - SILT FENCE, STRAW BALE OR ROCK DISSIPATERS SHOULD BE KEYED INTO THE END OF THE SLOPE BREAKER.
   - PROVIDE ENOUGH AREA INSIDE "L" TO CAPTURE AND HOLD SEDIMENT.
8. TYPICAL SPACING FOR SLOPES 5-15% = 300 FEET, SLOPES 15-30% = 200 FEET, AND SLOPES >30% = 100 FEET. OTHERWISE, COMPANY ENVIRONMENTAL INSPECTOR SHOULD DETERMINE APPROPRIATE SPACING TO CONTROL EROSION AND RUNOFF PER SITE CONDITIONS.

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SLOPE BREAKER

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0002

REV
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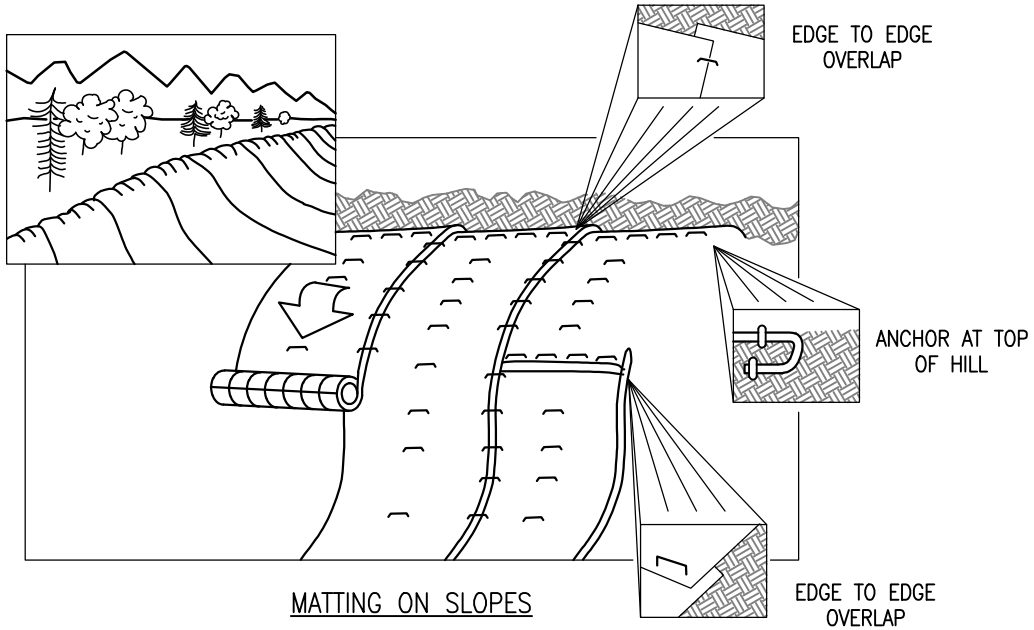
MATTING ON SLOPES – ENVIRONMENTAL TYPICAL



MATTING ON SLOPES

NOTES:

1. EROSION CONTROL MATTING (BLANKETS) SHALL BE USED AT LOCATIONS IDENTIFIED IN THE PLAN AND/OR AS DIRECTED BY THE ENVIRONMENTAL INSPECTOR.
2. EROSION CONTROL MATTING SHALL MEET THE REQUIREMENTS SPECIFIED IN THE PLAN AND/OR AS DIRECTED BY THE ENVIRONMENTAL INSPECTOR.
3. STAPLES SHALL BE MADE OF 11 GAUGE WIRE, U–SHAPED WITH 6" LEGS AND A 1" CROWN. STAPLES SHALL BE DRIVEN INTO THE GROUND FOR THE FULL LENGTH OF THE STAPLE LEGS.
4. MATTING SHALL BE INSTALLED ACCORDING TO MANUFACTURER SPECIFICATIONS OR AS STATED BELOW:
   - 4.a. EXTEND TOP OF BLANKET 3–FEET PAST THE UPPER EDGE OF THE SLOPE.
   - 4.b. ANCHOR ("KEY") THE UPPER EDGE OF THE BLANKET INTO THE SLOPE USING A 6" DEEP TRENCH AND ROLL THE BLANKET DOWN THE HILL. DOUBLE STAPLE EVERY 12" BEFORE BACKFILLING AND COMPACTING TRENCH.
   - 4.c. AVOID STRETCHING EROSION CONTROL MATTING DURING INSTALLATION.
   - 4.d. BRING MAT ROLL BACK OVER THE TOP OF THE TRENCH AND CONTINUE TO ROLL DOWN SLOPE. STAPLE EVERY 12" WHERE MAT EXITS THE TRENCH AT THE TOP OF THE SLOPE.
   - 4.e. WHEN BLANKETS ARE SPLICED DOWN–SLOPE TO ADJOINING MATS (SLOPE OR STREAM BANK MATS), THE UPPER BLANKET SHALL BE PLACED OVER THE LOWER MAT (SHINGLE STYLE) WITH APPROXIMATELY 6" OF OVERLAP. STAPLE THROUGH THE OVERLAPPED AREA EVERY 12".
   - 4.f. OVERLAP ADJACENT BLANKETS 6". STAPLE EDGES OF BLANKETS AND CENTER EVERY 36".
5. IN LIVESTOCK AREAS WHERE EROSION CONTROL MATTING IS APPLIED TO THE SLOPES, FENCING WILL BE USED IF NECESSARY TO EXCLUDE LIVESTOCK, WITH PERMISSION OF THE LANDOWNER.
6. MONITOR WASHOUTS, STAPLE INTEGRITY, OR MAT MOVEMENT, REPLACE OR REPAIR AS NECESSARY.
7. DO NOT USE SYNTHETIC MONOFILAMENT MESH/NETTED MATERIALS IN AREAS DESIGNATED AS SENSITIVE WILDLIFE HABITAT, UNLESS THE PRODUCT IS SPECIFICALLY DESIGNED TO MINIMIZE HARM TO WILDLIFE.

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ENERGY TRANSFER

FRANKLIN FARMS PROJECT
MATTING ON SLOPES

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0003

REV
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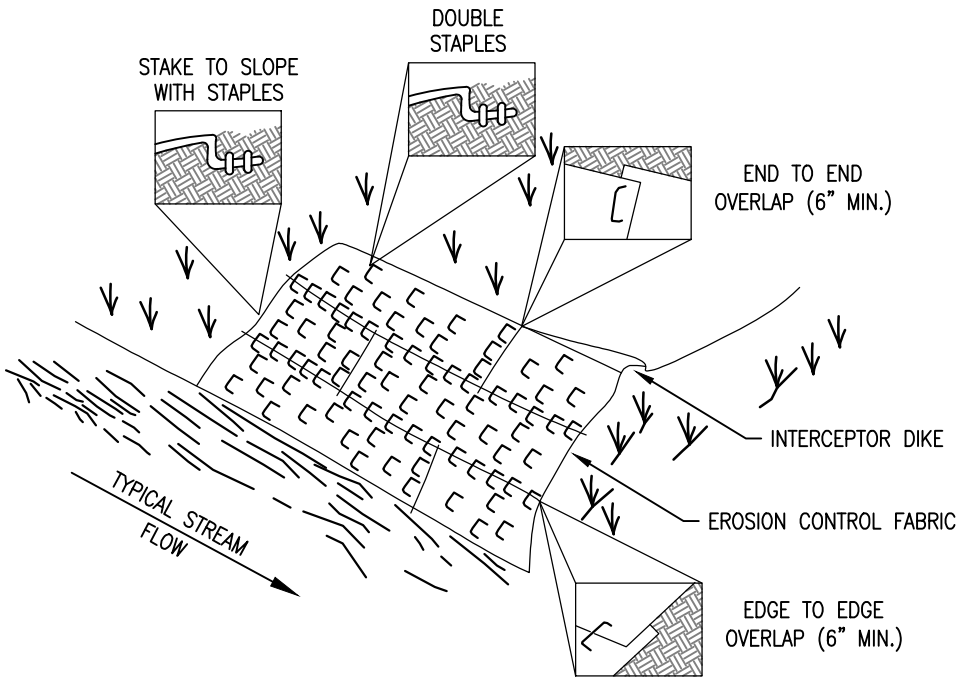
MATTING OF STREAM BANKS – ENVIRONMENTAL TYPICAL



MATTING OF STREAM BANKS

NOTES:

1. EROSION CONTROL MATTING SHALL BE IN PLACE ON THE BANKS OF FLOWING STREAMS WHERE VEGETATION HAS BEEN REMOVED OR AS DIRECTED BY THE ENVIRONMENTAL INSPECTOR.
2. EROSION CONTROL MATTING SHALL MEET THE REQUIREMENTS SPECIFIED IN THE PLAN AND/OR AS DIRECTED BY THE ENVIRONMENTAL INSPECTOR.
3. STAPLES SHALL BE MADE OF 11 GAUGE WIRE, U–SHAPED WITH 6" LEGS AND 1" CROWN. STAPLES SHALL BE DRIVEN INTO THE GROUND FOR THE FULL LENGTH OF THE STAPLE LEGS. ALTERNATELY 1" WOODEN PEGS 6" LONG AND BEVELED TO SECURE MATTING.
4. MATTING SHALL BE INSTALLED ACCORDING TO MANUFACTURER SPECIFICATIONS OR AS STATED BELOW:
   - 4.a. EXTEND TOP OF BLANKET 2–FEET PAST THE UPPER EDGE OF THE HIGH WATER MARK. IF AN INTERCEPTOR DIKE IS PRESENT ON THE APPROACH SLOPE, BEGIN THE BLANKET ON THE UPHILL SIDE OF THE INTERCEPTOR DIKE.
   - 4.b. INSTALL BLANKET(S) ACROSS THE SLOPE IN THE DIRECTION OF THE WATER FLOW.
   - 4.c. ANCHOR ("KEY") THE UPSTREAM EDGE OF THE BLANKET(S) INTO THE SLOPE USING A 6" DEEP TRENCH. DOUBLE STAPLE EVERY 12" BEFORE BACKFILLING AND COMPACTING TRENCH.
   - 4.d. ANCHOR ("KEY") THE UPPER EDGE OF THE BLANKET INTO THE SLOPE USING A 6" DEEP TRENCH. DOUBLE STAPLE EVERY 12" BEFORE BACKFILLING AND COMPACTING TRENCH.
   - 4.e. OVERLAP THE EDGES OF PARALLEL BLANKETS A MINIMUM OF 6". PLACE THE UPPER BLANKET OVER THE LOWER BLANKET (SHINGLE STYLE) AND STAPLE EVERY 12" ALONG THE LENGTH OF THE EDGE.
   - 4.f. WHEN BLANKET ENDS ARE ADJOINED, PLACE THE UPSTREAM BLANKET OVER THE DOWNSTREAM BLANKET (SHINGLE STYLE) WITH APPROXIMATELY 6" OF OVERLAP AND STAPLE THROUGH THE OVERLAPPED AREAS EVERY 12".
   - 4.g. STAPLE DOWN THE CENTER OF THE BLANKET(S). THREE STAPLES IN EVERY SQUARE YARD.
5. IN LIVESTOCK AREAS WHERE EROSION CONTROL MATTING IS APPLIED TO THE STREAMBANKS, FENCING WILL BE USED IF NECESSARY TO EXCLUDE LIVESTOCK, WITH PERMISSION OF THE LANDOWNER.
6. MONITOR WASHOUTS, STAPLE INTEGRITY OR MAT MOVEMENT. REPLACE OR REPAIR AS NECESSARY.
7. DO NOT USE SYNTHETIC MONOFILAMENT MESH/NETTED MATERIALS IN AREAS DESIGNATED AS SENSITIVE WILDLIFE HABITAT, UNLESS THE PRODUCT IS SPECIFICALLY DESIGNED TO MINIMIZE HARM TO WILDLIFE.

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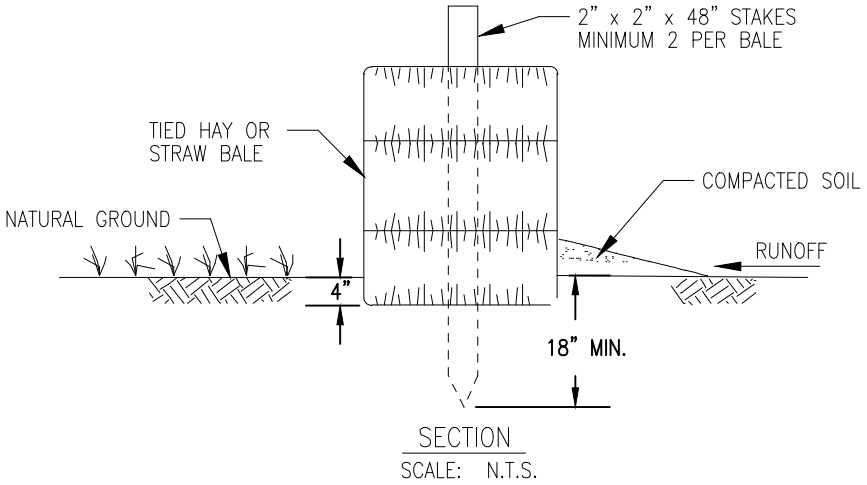
FRANKLIN FARMS PROJECT
MATTING ON STREAM BANKS

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0004

REV
A

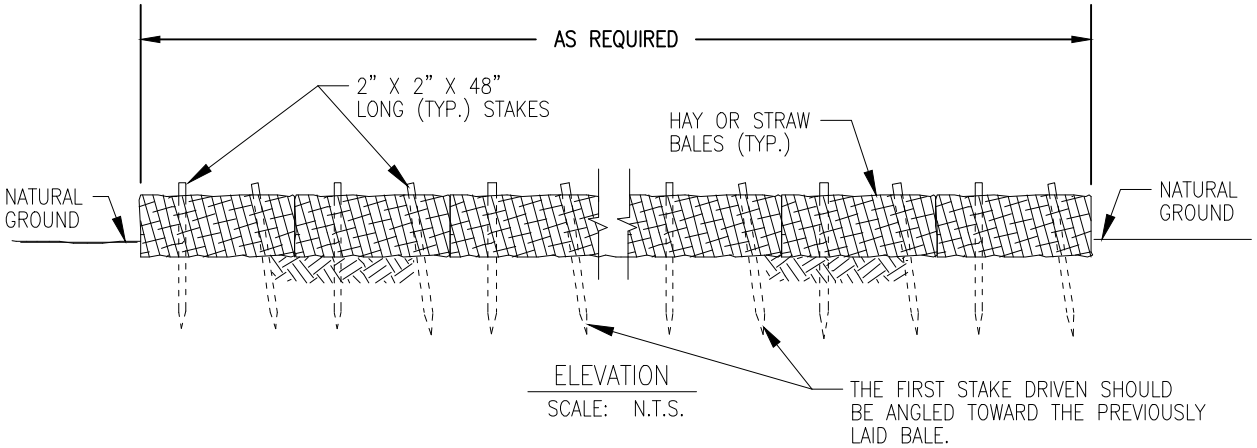
STRAW BALE BARRIER
ENVIRONMENTAL TYPICAL



SECTION
SCALE: N.T.S.

NOTES:

1. TO ELIMINATE POSSIBLE END FLOW, BOTH ENDS OF THE STRAW BALE BARRIER SHOULD BE TURNED AND EXTENDED UPSLOPE.
2. EACH BALE SHOULD BE SECURED BY AT LEAST TWO (2) STAKES. THE FIRST STAKE IN EACH BALE SHALL BE DRIVEN TOWARD THE PREVIOUSLY LAID BALE TO FORCE THE BALES TOGETHER. ANY GAPS CAN BE FILLED IN BY WEDGING LOOSE STRAW BETWEEN THE BALES. STAKES SHOULD BE DRIVEN A MINIMUM OF 18 INCHES INTO THE GROUND.
3. COMPACT EXCAVATED SOIL AS NECESSARY AGAINST THE UPHILL SIDE OF THE BARRIER TO PREVENT WATER TUNNELING UNDER THE BALES.
4. STRAW AND HAY BALE BARRIERS REQUIRE CONTINUAL MAINTENANCE TO REMOVE COLLECTED SEDIMENT AND REPLACE DAMAGED BALES. CLOSE ATTENTION SHOULD BE PAID TO THE REPAIR OF DAMAGED BALES, END RUNS AND UNDERCUTTING BENEATH BALES. BALE BARRIERS SHOULD NOT BE USED MORE THAN 3 MONTHS.
5. SEDIMENT DEPOSITS SHOULD BE REMOVED AFTER EACH RAINFALL. THEY MUST BE REMOVED WHEN THE LEVEL OF DEPOSITION REACHES APPROXIMATELY ONE-THIRD THE HEIGHT OF THE BARRIER.
6. IF USED IN CONJUNCTION WITH SILT FENCE, BALES ARE PLACED UP-SLOPE OF THE SILT FENCE AND DO NOT NEED TO BE TRENCHED IN.



ELEVATION
SCALE: N.T.S.

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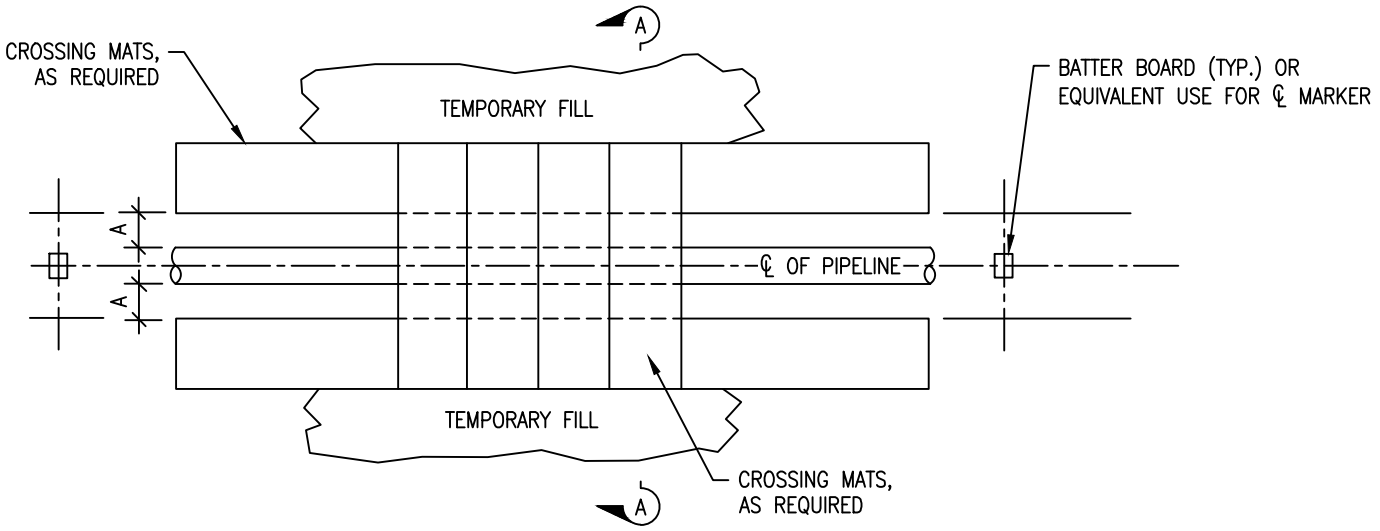
FRANKLIN FARMS PROJECT
STRAW BALE BARRIER

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

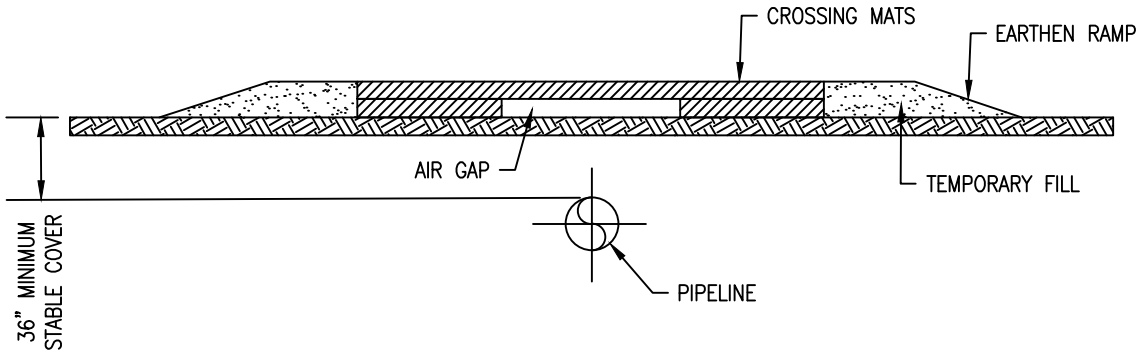
DWG NUMBER
TYP-0005

REV
A

TIMBER MAT BRIDGE INSTALLATION – CONSTRUCTION TYPICAL



PLAN



SECTION A–A

REVISIONS					
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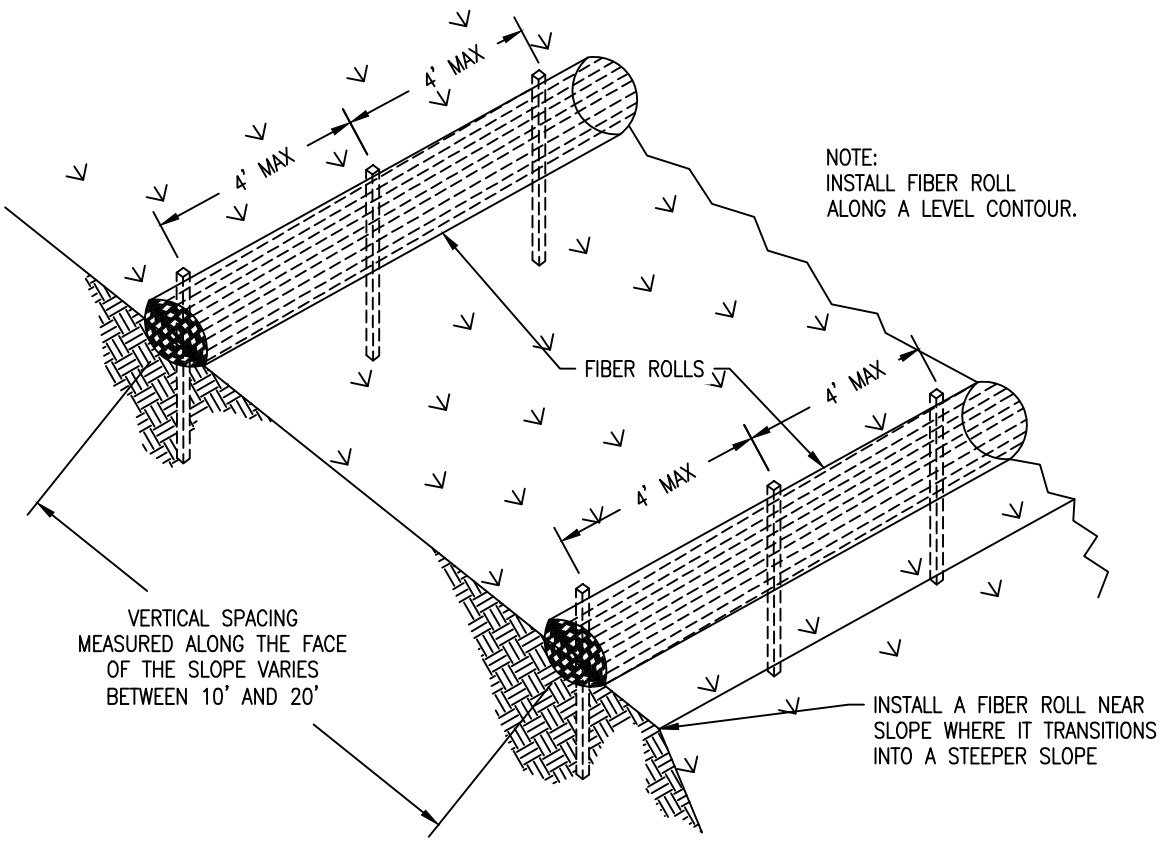
FRANKLIN FARMS PROJECT
TIMBER MAT BRIDGE INSTALLATION

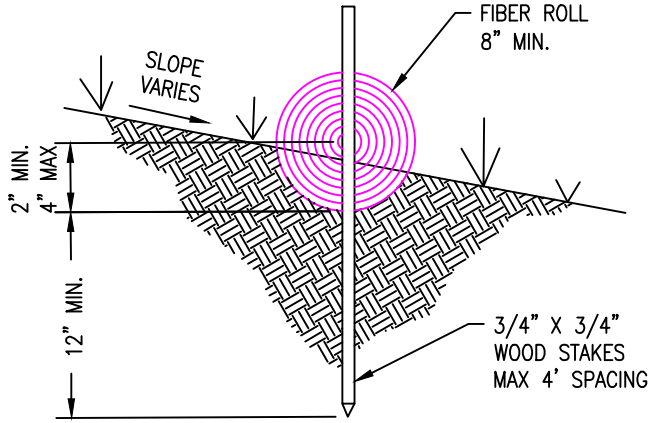
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0006

REV
A

BIOLOG – ENVIRONMENTAL TYPICALS





NOTES:

1. INSTALL ON DISTURBED AREAS THAT REQUIRE IMMEDIATE EROSION PROTECTION.
2. USE ON SLOPES REQUIRING STABILIZATION UNTIL PERMANENT VEGETATION CAN BE ESTABLISHED.
3. CAN BE USED ALONG THE PERIMETER OF A PROJECT AND AROUND TEMPORARY STOCKPILES.
4. BIOLOGS TO BE INSTALLED PERPENDICULAR TO THE FLOW DIRECTION AND PARALLEL TO THE SLOPE CONTOUR.

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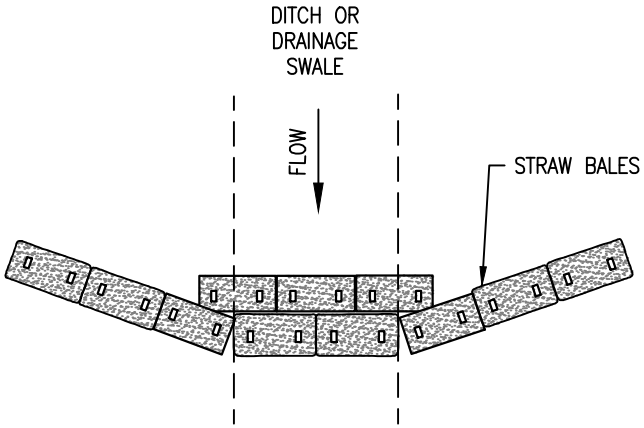
FRANKLIN FARMS PROJECT
BIOLOG

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

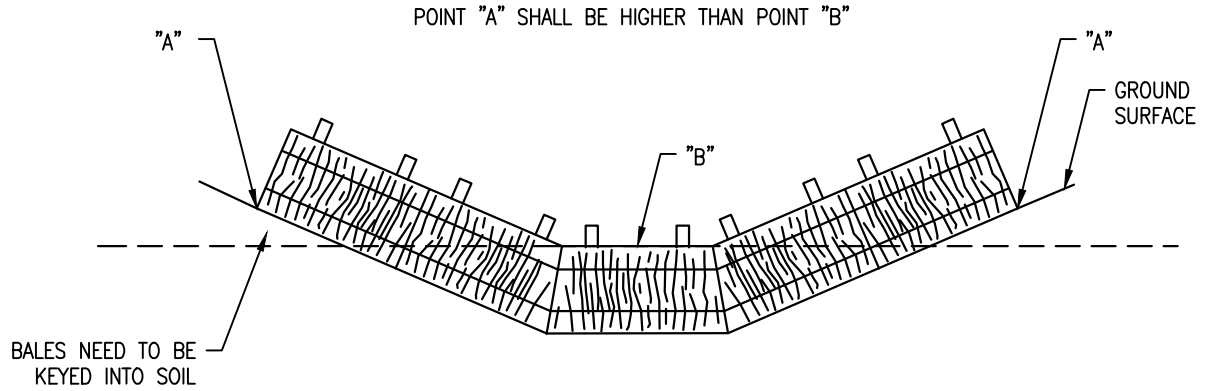
DWG NUMBER
TYP-0007

REV
A

CHECK DAM STRAW BALE INSTALLATION
ENVIRONMENTAL TYPICAL



PLAN VIEW



CROSS–SECTION

REVISIONS					
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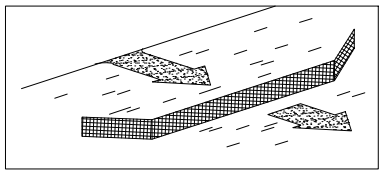
ENERGY TRANSFER

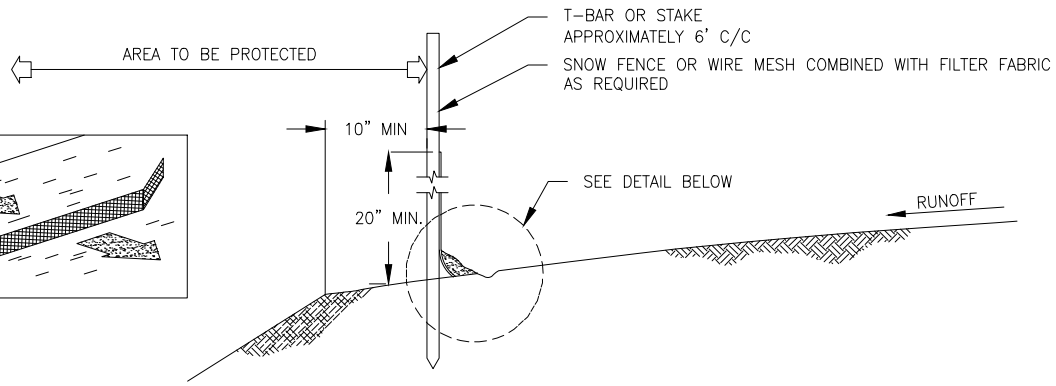
FRANKLIN FARMS PROJECT
CHECK DAM STRAW BALE INSTALLATION
ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

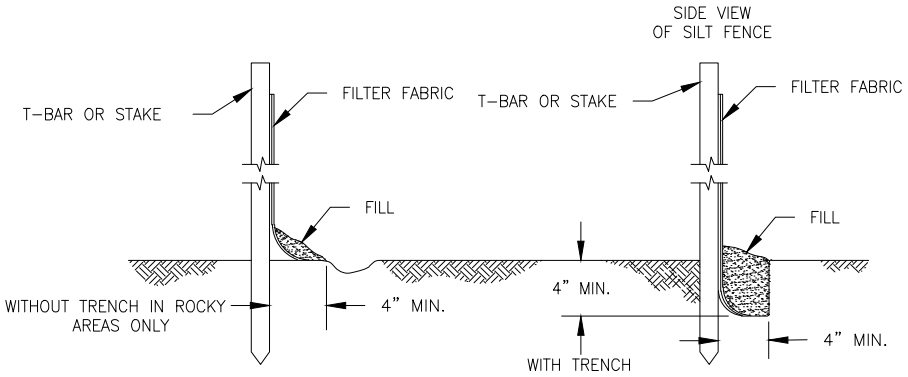
DWG NUMBER
TYP-0008

REV
A

SEDIMENT BARRIER – ENVIRONMENTAL TYPICALS







NOTES:

1. GENERALLY WHEN A LONG SEDIMENT BARRIER IS REQUIRED, SILT FENCE WILL BE UTILIZED RATHER THAN STRAW BALES AT:
   - THE BASE OF ALL SLOPES ABOVE ROADS, SPRINGS, WETLANDS, IMPOUNDMENTS, PERENNIAL AND INTERMITTENT STREAMS.
   - THE DOWN SLOPE RIGHT–OF–WAY EDGE WHERE ANY OF THE ABOVE MENTIONED LOCATIONS ARE
   - ADJACENT TO THE RIGHT–OF–WAY.
   - BETWEEN TOPSOIL/SPOIL STOCKPILES AND PERENNIAL OR INTERMITTENT STREAMS OR WETLANDS WHERE BUFFER ZONE REQUIREMENTS CANNOT BE MET.
   - ALONG RIGHT–OF–WAY BOUNDARIES OF WETLAND CONSTRUCTION.
   - AS SPECIFIED IN THE SPILL PREVENTION, CONTAINMENT AND COUNTERMEASURE PLAN.
2. THE SILT FENCE SHALL BE CONSTRUCTED AS FOLLOWS:
   - FABRIC USED FOR THE SILT FENCE SHALL BE A "STANDARD STRENGTH" GEOTEXTILE, SUCH AS MIRAFI
   - 100X OR AN APPROVED EQUIVALENT.
   - THE FABRIC SHALL BE CUT FROM A CONTINUOUS FABRIC ROLL.
   - THE HEIGHT OF THE FENCE SHALL NOT EXCEED 24".
   - SPLICES SHALL ONLY BE DONE AT POSTS AND SHALL CONSIST OF A MINIMUM OF 6" OF OVERLAP WITH BOTH ENDS SECURED TO THE POST.
   - FENCING OVERLAP TO BE ROLLED TO ENSURE CLOSURE.
   - POST SHALL BE POSITIONED A MAXIMUM OF 6' APART.
   - POST SHALL CONSIST OF 2"x2" WOODEN STAKES OF SUFFICIENT LENGTH TO EXTEND A MINIMUM OF 12" INTO THE GROUND.
   - FABRIC SHALL BE STAPLED OR WIRED TO POSTS A MAXIMUM OF EVERY 9".
3. THE SILT FENCE SHALL BE INSTALLED AS SPECIFIED BY THE MANUFACTURER OR AS FOLLOWS:
   - A TRENCH, 4" WIDE AND 4" DEEP SHALL BE EXCAVATED ALONG THE CONTOUR.
   - THE POST SHALL BE DRIVEN INTO THE BOTTOM OF THE TRENCH ON THE DOWNSTREAM SIDE OF THE FILTER FABRIC.
   - THE TRENCH SHALL BE BACKFILLED AND COMPACTED, ENSURING 4" OF FENCE BURIED WITHIN THE TRENCH.
   - IN AREAS WHERE THE TERRAIN IS TOO ROCKY FOR TRENCHING, A 4" GROUND FLAP WITH ROCK FILL TO HOLD IT IN PLACE SHALL BE USED.

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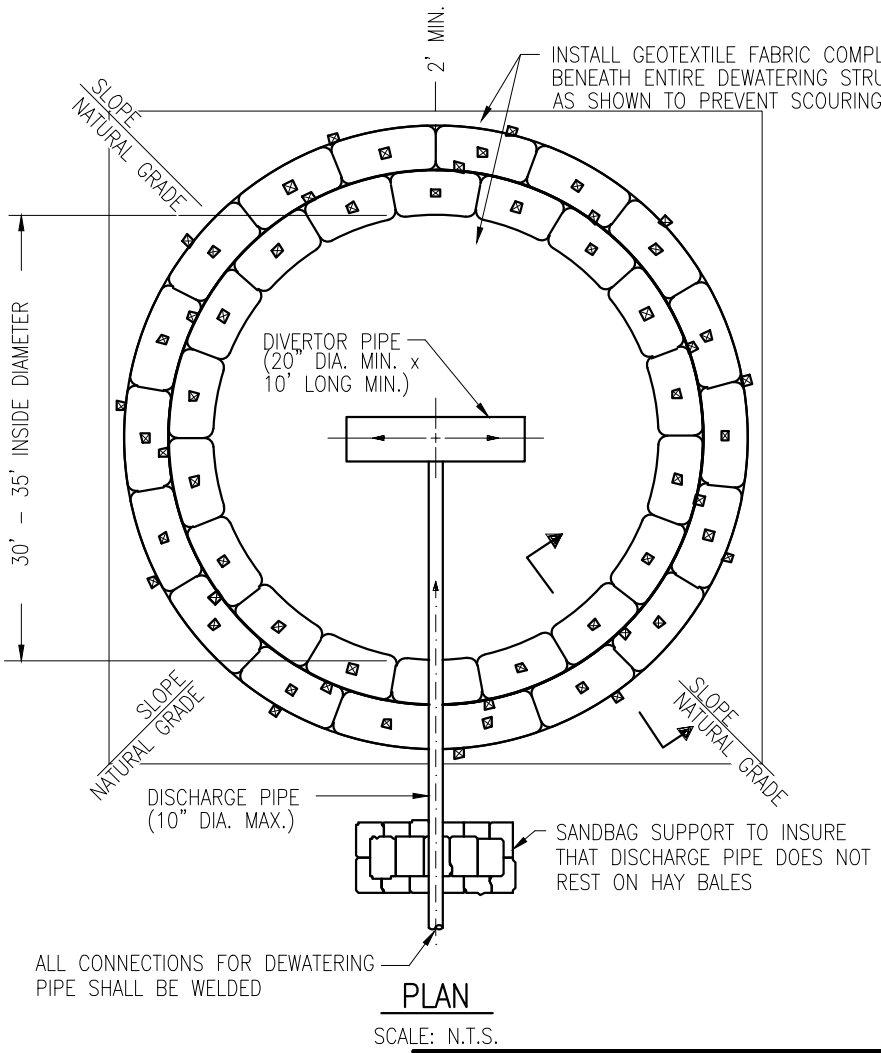
FRANKLIN FARMS PROJECT
SEDIMENT BARRIER

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0009

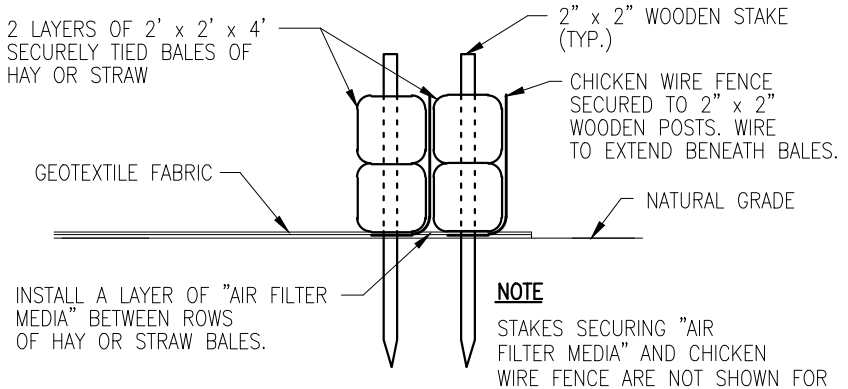
REV
A

HYDROSTATIC DEWATERING STRUCTURE
ENVIRONMENTAL TYPICAL



PLAN

SCALE: N.T.S.



SECTION

SCALE: N.T.S.

NOTES:

1. STRUCTURE SHALL BE PLACED ON A LEVEL WELL VEGETATED SITE SUCH THAT WATER WILL FLOW AWAY FROM STRUCTURE AND ANY WORK AREAS.
2. FLOW RATES THROUGH DISCHARGE AND DIVERTER PIPES SHALL BE SUCH THAT STRUCTURE WILL NOT OVERFLOW.
3. WHERE CONDITIONS WARRANT A 30' x 30' RECTANGULAR STRUCTURE MAY BE SUBSTITUTED FOR THE CIRCULAR CONFIGURATION SHOWN.
4. DIMENSIONS SHOWN ARE THE MINIMUM ACCEPTABLE VALUES AND MAY BE VARIED DEPENDING UPON SPECIFIC LOCATION.
5. CONTRACTOR SHALL PROPERLY REMOVE AND DISPOSE OF DEWATERING STRUCTURE IMMEDIATELY UPON COMPLETION OF DEWATERING OPERATIONS. UNDER NO CIRCUMSTANCES SHALL USED DEWATERING STRUCTURES BE LEFT IN PLACE FOR A PERIOD OF TIME GREATER THAN 48 HOURS AFTER DEWATERING OPERATIONS ARE COMPLETE.

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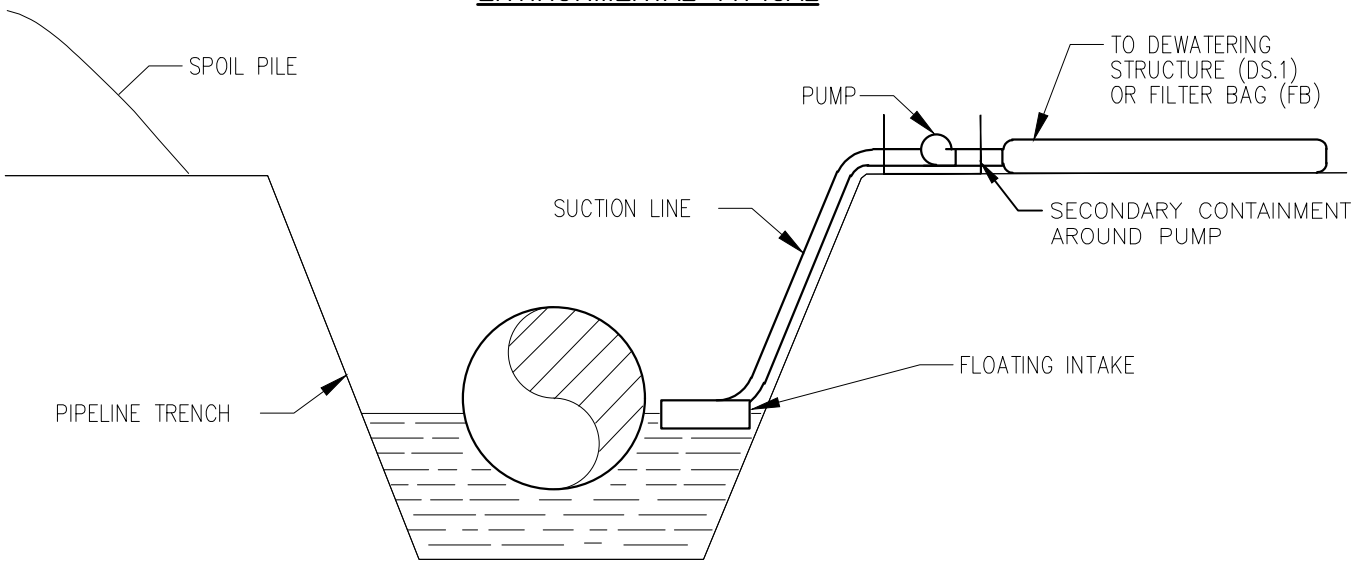
HYDROSTATIC DEWATERING STRUCTURE
ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0010

REV
A

TRENCH DEWATERING
ENVIRONMENTAL TYPICAL



NOTES:

1. WATER PUMPED OUT OF TRENCH SHALL NOT BE DISCHARGED INTO WATERWAYS. WATER SHALL BE DISCHARGED INTO A FILTER BAG OR DEWATERING STRUCTURE.

2. PUMP SHALL BE CONTROLLED SO THAT DISCHARGE DOES NOT OVERFLOW DEWATERING STRUCTURE.

3. PUMP SUCTION HOSE MUST NOT BE ALLOWED TO COME IN CONTACT WITH TRENCH BOTTOM. PROVISIONS MUST BE MADE TO ELEVATE THE SUCTION HOSE TO AT LEAST ONE FOOT ABOVE THE BOTTOM OF THE PIPE TRENCH UNTIL BOTTOM DEWATERING IS NECESSARY.

4. DEWATERING SHALL NOT OCCUR DURING TIMES OF HEAVY RAINFALL EXCEPT AS REQUIRED TO PREVENT FLOODING OF CONSTRUCTION EQUIPMENT LOCATED IN BORE PITS AND TRENCHES.

5. PUMPS UTILIZED DURING DEWATERING SHALL BE PLACED WITHIN SECONDARY CONTAINMENT IF POSITIONED WITHIN 100 FEET OF A WETLAND OR WATERBODY.

REVISIONS

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PREPARED BY: FIRM No:EF.0007320
TRC
8521 McHARD RD, HOUSTON, TX 77053

ENERGY TRANSFER

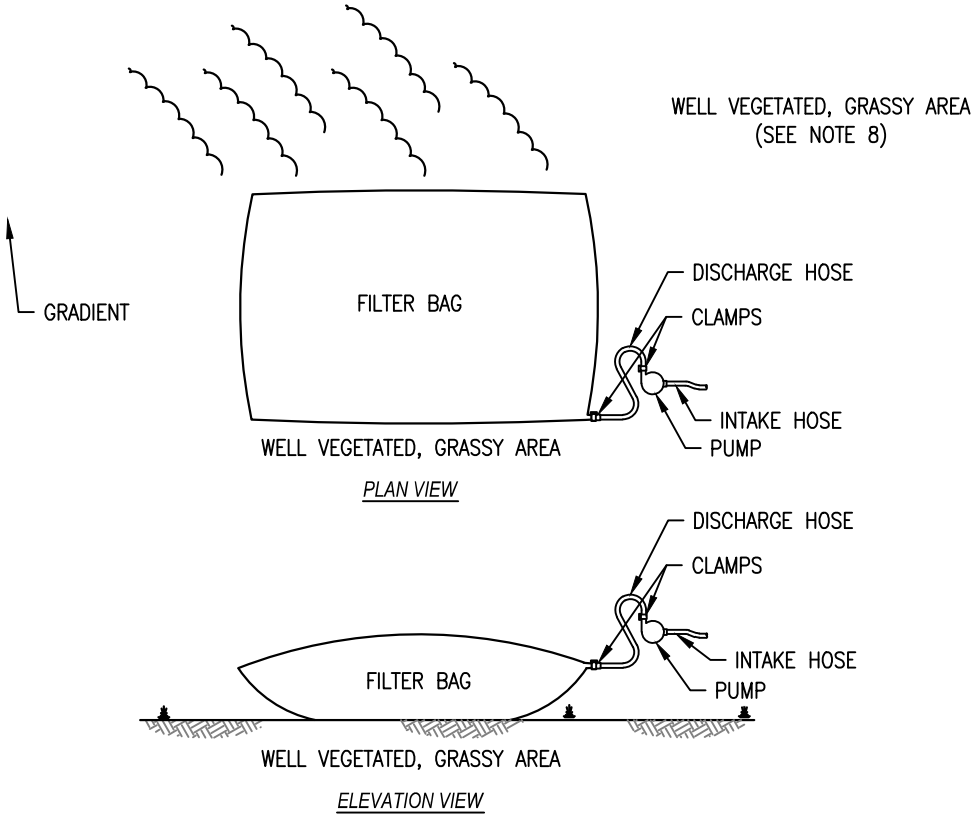
FRANKLIN FARMS PROJECT
TRENCH DEWATERING

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0011

REV
A

GEOTEXTILE FILTER BAG
ENVIRONMENTAL TYPICAL



NOTES:

1. FILTER BAGS SHALL BE MADE FROM NON-WOVEN GEO-TEXTILE MATERIAL WITH HIGH STRENGTH, DOUBLE STITCHED "J" TYPE SEAMS. FILTER BAGS SHALL BE CAPABLE OF TRAPPING PARTICLES LARGER THAN 150 MICRONS.
2. A SUITABLE MEANS OF ACCESSING THE FILTER BAG WITH MACHINERY REQUIRED FOR DISPOSAL PROVIDED. FILTER BAGS SHALL BE REPLACED WHEN THEY BECOME 1/2 FULL. SPARE BAGS SHALL BE KEPT AVAILABLE FOR REPLACEMENT OF THOSE THAT HAVE FAILED OR ARE FILLED.
3. BAGS SHALL BE LOCATED IN WELL-VEGETATED (GRASSY) AREA, AND DISCHARGE ONTO STABLE, EROSION RESISTANT AREAS. WHERE THIS IS NOT POSSIBLE, A GEO-TEXTILE FLOW PATH SHALL BE PROVIDED. BAGS SHALL NOT BE PLACED ON SLOPES GREATER THAN 5%.
4. THE PUMP DISCHARGE HOSE SHALL BE INSERTED INTO THE BAGS IN THE MANNER SPECIFIED BY THE MANUFACTURER AND SECURELY CLAMPED.
5. THE PUMPING RATE SHALL BE NO GREATER THAN 750 GPM OR 1/2 THE MAXIMUM SPECIFIED BY THE MANUFACTURER, WHICHEVER IS LESS. PUMP INTAKES SHOULD BE FLOATING AND SCREENED.
6. FILTER BAGS WILL BE INSPECTED PER PERMIT CONDITIONS. IF ANY PROBLEM IS DETECTED, PUMPING SHALL CEASE IMMEDIATELY AND NOT RESUME UNTIL THE PROBLEM IS CORRECTED.
7. CONTRACTOR SHALL PROPERLY REMOVE AND PROPERLY DISPOSE OF USED FILTER BAGS IMMEDIATELY UPON COMPLETION OF THE DE-WATERING OPERATION.
8. IF VEGETATIVE AREA IS NOT AVAILABLE, INSTALL EROSION RESISTANT GEO-TEXTILE FABRIC TO A WELL-VEGETATED ARES DOWN GRADIENT OF THE FILTER BAG.

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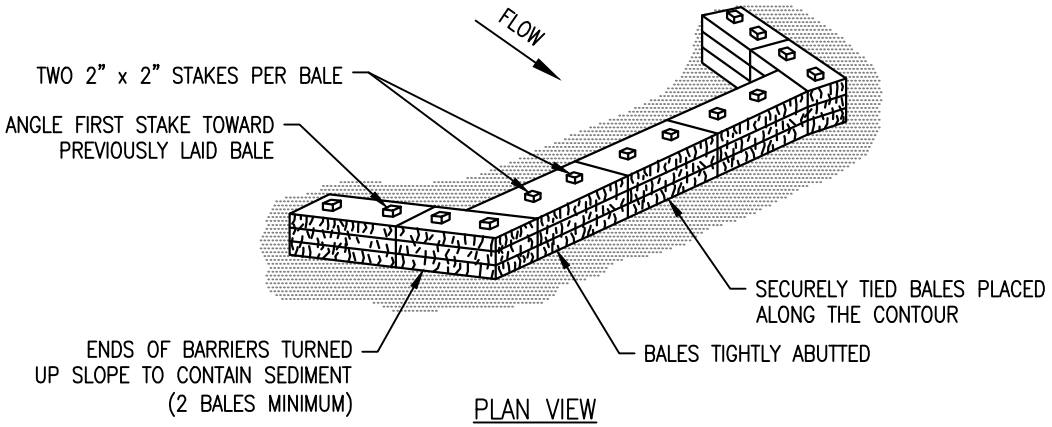
FRANKLIN FARMS PROJECT
GEOTEXTILE FILTER BAG

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

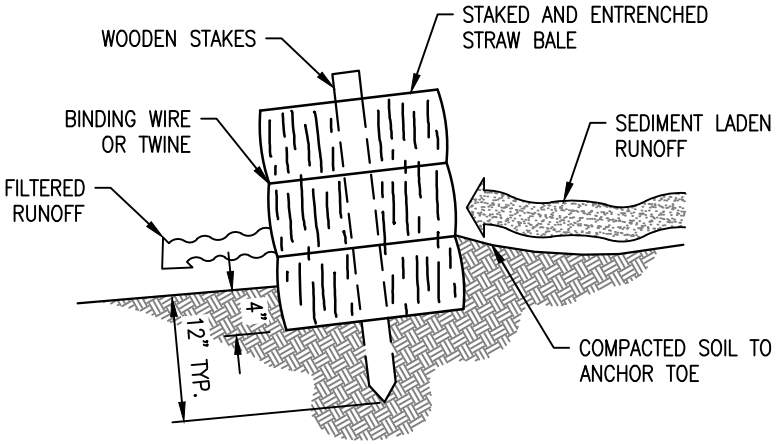
DWG NUMBER
TYP-0012

REV
A

STRAW BALE INSTALLATION
ENVIRONMENTAL TYPICAL



PLAN VIEW



CROSS–SECTION

STRAW BALE INSTALLATION

INSTALLATION REQUIREMENTS

- WHEN USING STRAW BALES, PLACE THEM:
  - WITH THEIR ENDS TIGHTLY ABUTTING AND EMBEDDED IN THE SOIL A TYPICAL OF 4".
  - BETWEEN DISTURBED AREAS AND DOWN–SLOPE ENVIRONMENTAL RESOURCE AREAS.
  - AT THE BASE OF ALL SLOPES NEXT TO WETLANDS, WATERBODIES AND ROAD CROSSINGS.
  - AT THE INLET AND OUTLET OF OPEN DRAINAGE STRUCTURES.
  - APPROXIMATELY 6–FEET BEYOND THE TOE OF THE SLOPE TO GIVE THE SEDIMENT ROOM TO COLLECT.
- KEY IN THE BOTTOM OF THE BALE. IN AREAS WHERE IT IS NOT FEASIBLE TO TRENCH IT IN (LEDGES, ROCKY SOIL, LARGE TREE ROOTS, ETC.), USE NATIVE SOIL AS BACKFILL UP–SLOPE OF THE BALE.
- IF USED IN CONJUNCTION WITH SILT FENCE, BALES ARE PLACED UP–SLOPE OF THE SILT FENCE AND DO NOT NEED TO BE TRENCHED IN.

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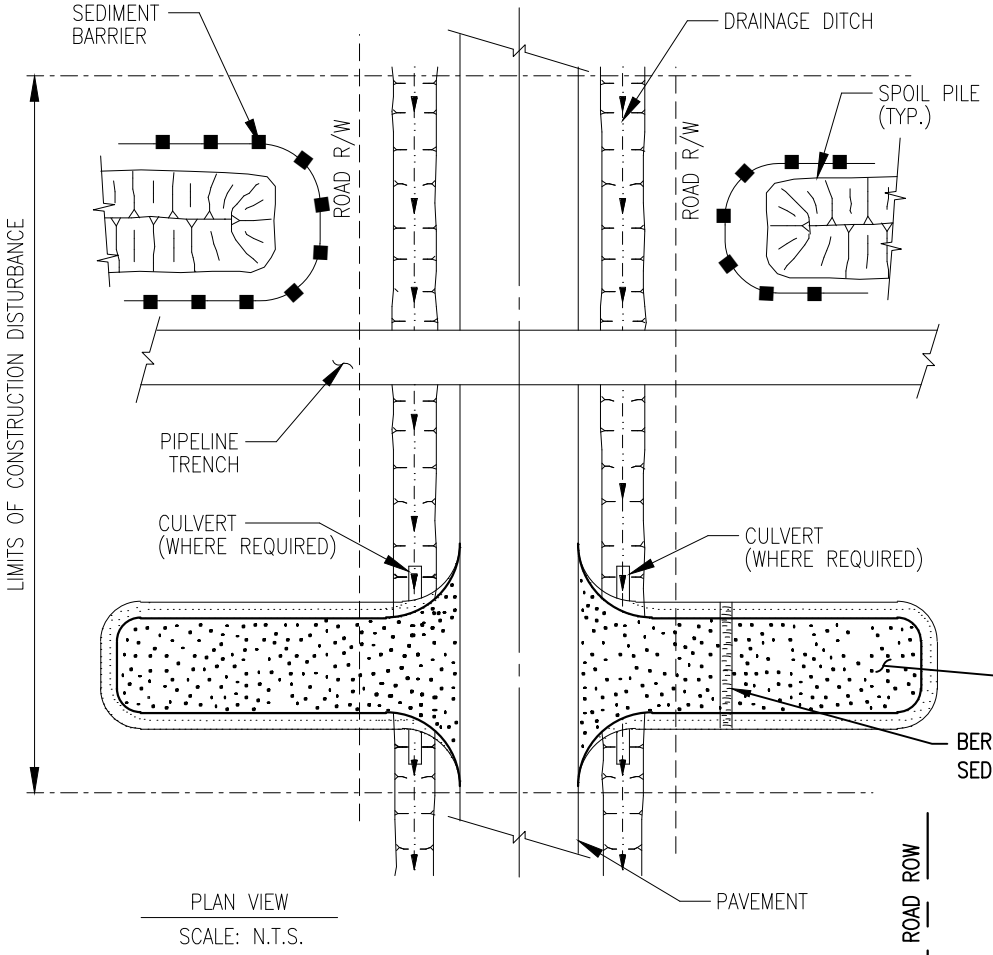
FRANKLIN FARMS PROJECT
STRAW BALE INSTALLATION

ENVIRONMENTAL TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0013

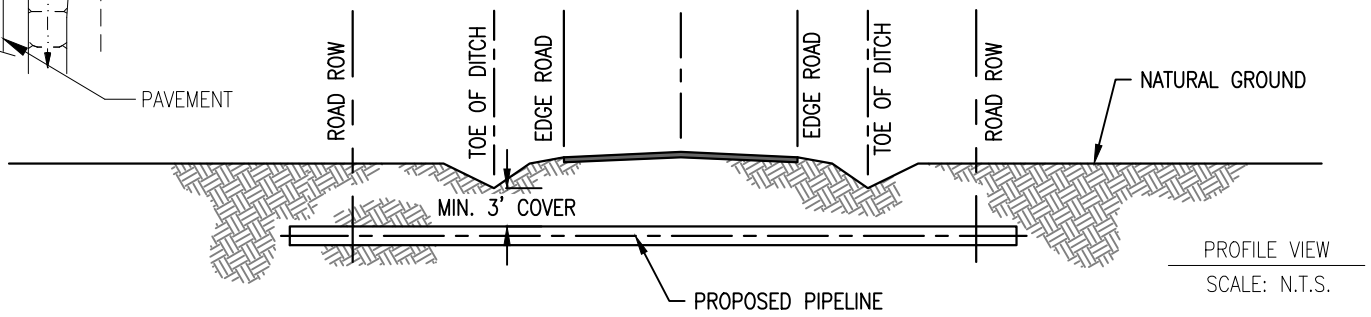
REV
A

ROAD CROSSING – OPEN CUT CONSTRUCTION TYPICAL



PLAN VIEW

SCALE: N.T.S.



PROFILE VIEW

SCALE: N.T.S.

NOTES:

1. SEDIMENT BARRIER SHALL BE INSTALLED AT THE BASE OF SLOPES ADJACENT TO ROAD CROSSINGS WHERE VEGETATION IS DISTURBED, TO INTERCEPT SURFACE RUNOFF, AS NEEDED.
2. PROTECTION FOR SPOIL PILES SHALL BE INSTALLED ONLY WHERE SEDIMENT BARRIERS ACROSS THE ENTIRE DISTURBED AREA ARE NOT REQUIRED.
3. CULVERTS TO BE SIZED AND PLACED WHERE REQUIRED TO MAINTAIN WATER FLOW. CULVERT SHALL BE REMOVED FOLLOWING CONSTRUCTION ACTIVITIES.
4. CONTRACTOR SHALL BE REQUIRED TO KEEP THE ROAD CLEAN OF DEBRIS AT ALL TIMES.
5. CONTRACTOR MAY ELECT TO UTILIZE SHEET PILING IN ORDER TO STABILIZE PIPE TRENCH.
6. CONTRACTOR MAY ELECT TO UTILIZE WELL–POINTS IN ORDER TO REDUCE THE WATER TABLE PRIOR TO COMMENCING EXCAVATION.

REVISIONS

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FRANKLIN FARMS PROJECT
ROAD CROSSING - OPEN CUT

CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0014

REV
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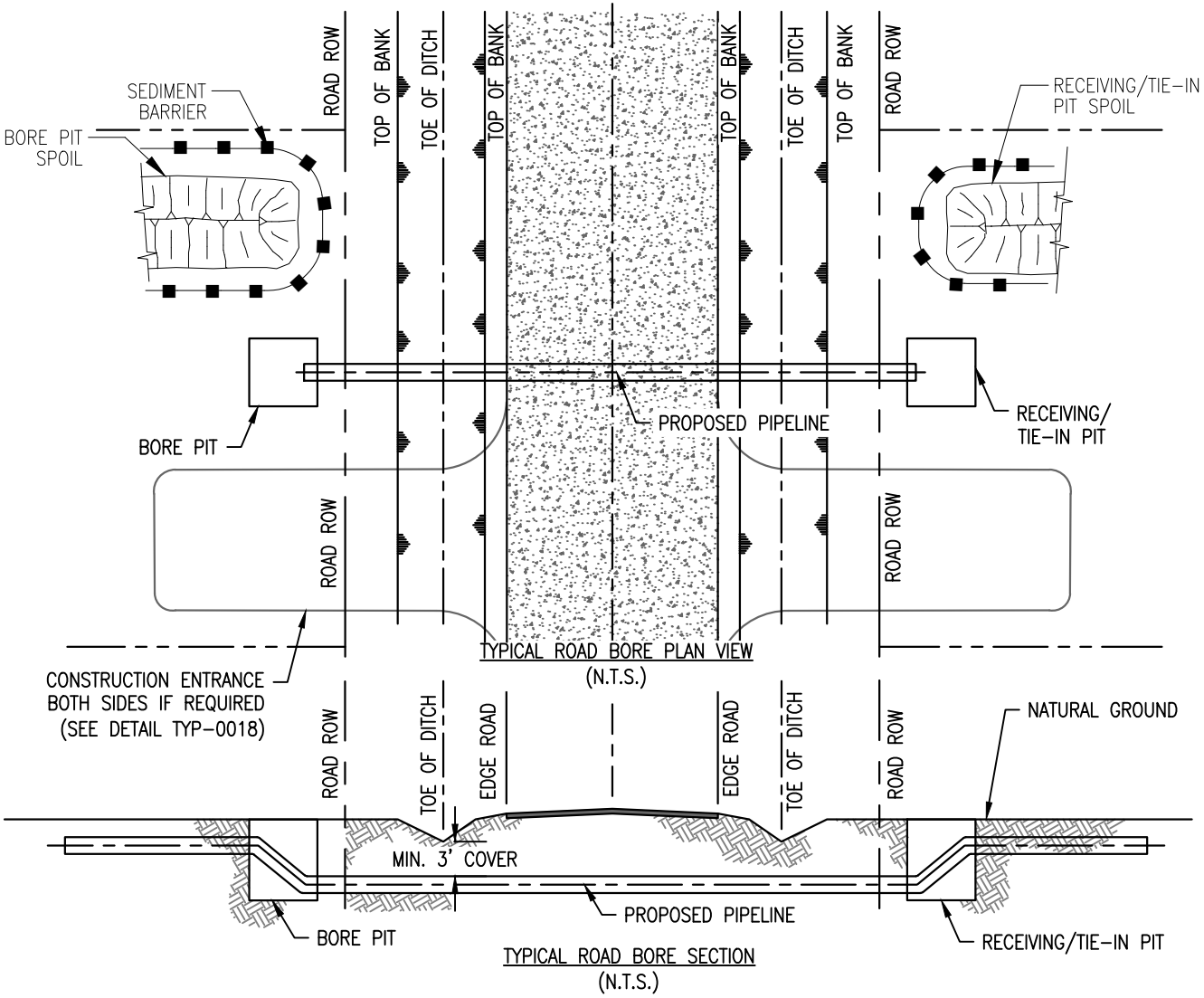
ROAD CROSSING – BORE CONSTRUCTION TYPICAL



NOTES:

1. SEDIMENT BARRIER SHALL BE INSTALLED AT THE BASE OF SLOPES ADJACENT TO ROAD WHERE VEGETATION IS DISTURBED, TO INTERCEPT SURFACE RUNOFF, AS NEEDED.
2. PROTECTION FOR SPOIL PILES SHALL BE INSTALLED ONLY WHERE SEDIMENT BARRIERS ACROSS THE ENTIRE DISTURBED AREA ARE NOT REQUIRED.
3. WATER REMOVED FROM BORE PIT AND RECEIVING PIT SHALL BE FILTERED THROUGH A DEWATERING STRUCTURE OR FILTER BAG.
4. IF WELL POINTING IS REQUIRED PRIOR TO EXCAVATING BORE PITS, CONTRACTOR SHALL CONSULT WITH COMPANY'S ENVIRONMENTAL INSPECTOR PRIOR TO COMMENCEMENT OF WORK IN ORDER TO DETERMINE PROPER DEWATERING LOCATION.
5. CONTRACTOR SHALL BE REQUIRED TO KEEP THE ROAD CLEAN OF DEBRIS AT ALL TIMES.
6. CONTRACTOR MAY ELECT TO UTILIZE SHEET PILING IN ORDER TO STABILIZE BORE PITS.

REVISIONS

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ENERGY TRANSFER

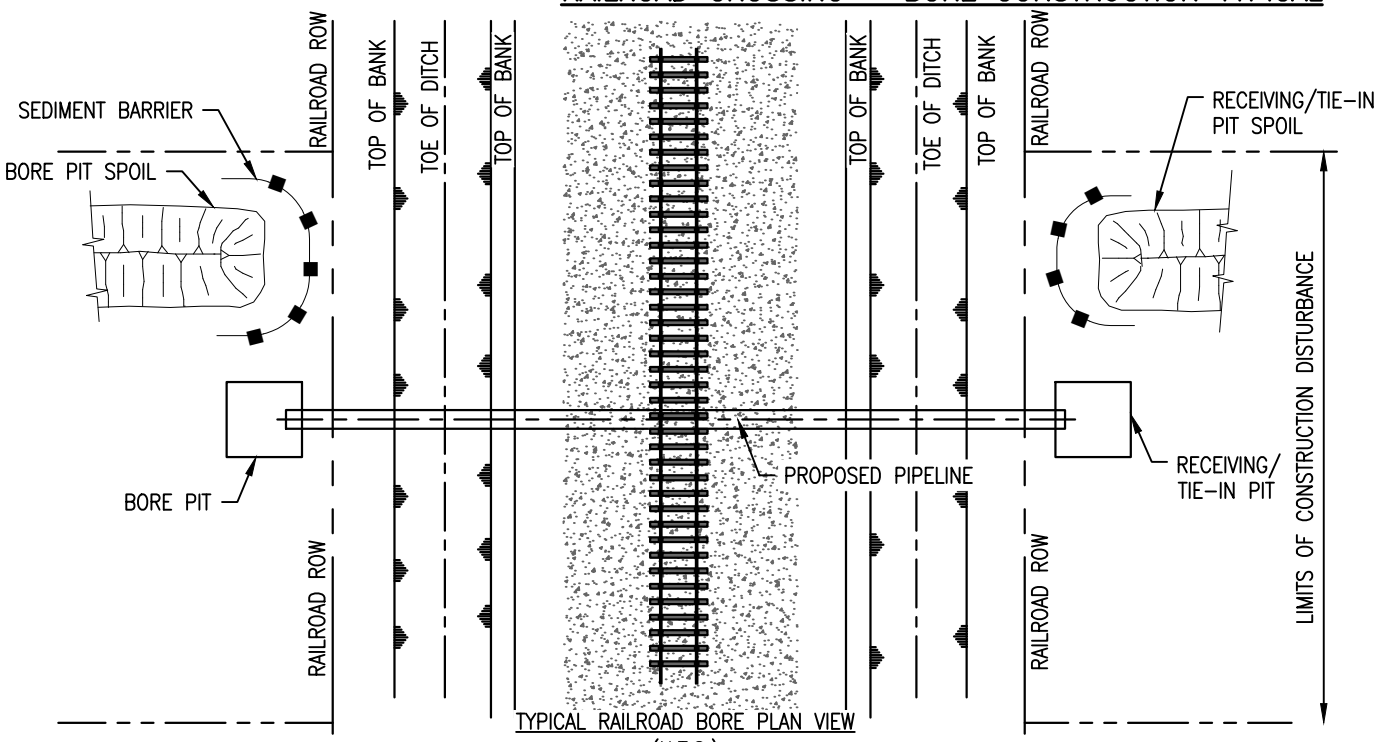
FRANKLIN FARMS PROJECT
ROAD CROSSING - BORE

CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0015

REV
A

RAILROAD CROSSING – BORE CONSTRUCTION TYPICAL



NOTES:

1. SEDIMENT BARRIER SHALL BE INSTALLED AT THE BASE OF SLOPES ADJACENT TO ROAD WHERE VEGETATION IS DISTURBED, TO INTERCEPT SURFACE RUNOFF, AS NEEDED.
2. PROTECTION FOR SPOIL PILES SHALL BE INSTALLED ONLY WHERE SEDIMENT BARRIERS ACROSS THE ENTIRE DISTURBED AREA ARE NOT REQUIRED.
3. WATER REMOVED FROM BORE PIT AND RECEIVING PIT SHALL BE FILTERED THROUGH A DEWATERING STRUCTURE OR FILTER BAG.
4. IF WELL POINTING IS REQUIRED PRIOR TO EXCAVATING BORE PITS, CONTRACTOR SHALL CONSULT WITH COMPANY'S ENVIRONMENTAL INSPECTOR PRIOR TO COMMENCEMENT OF WORK IN ORDER TO DETERMINE PROPER DEWATERING LOCATION.
5. CONTRACTOR MAY ELECT TO UTILIZE SHEET PILING IN ORDER TO STABILIZE BORE PITS.

REVISIONS

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ENERGY TRANSFER

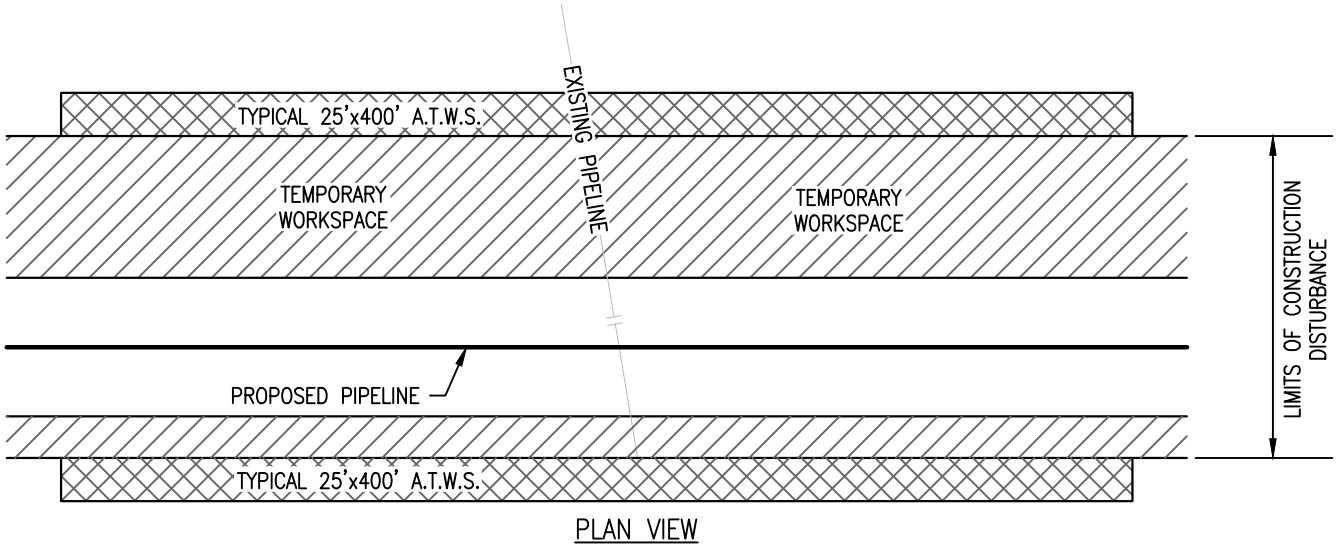
FRANKLIN FARMS PROJECT
RAILROAD CROSSING - BORE

CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

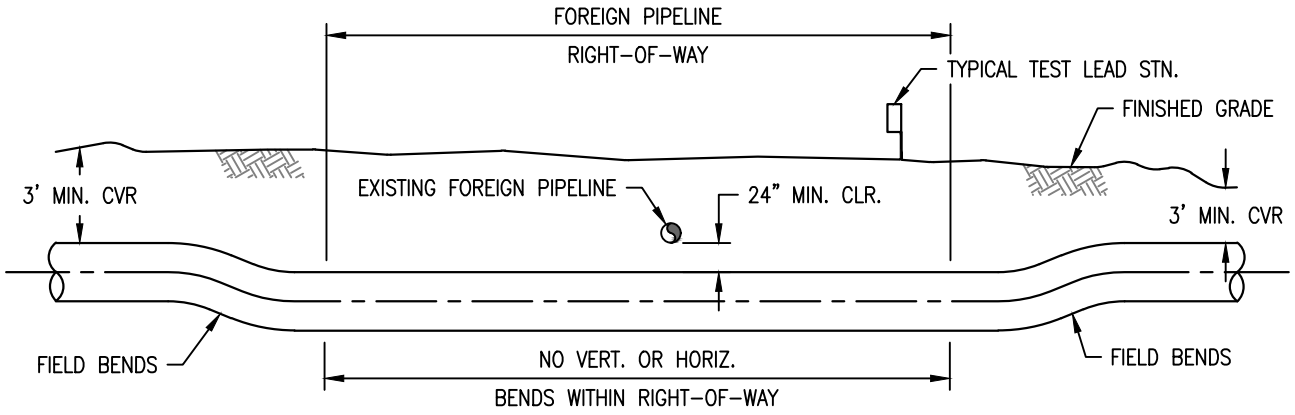
DWG NUMBER
TYP-0016

REV
A

FOREIGN PIPELINE & UTILITY CROSSINGS



PLAN VIEW



PROFILE VIEW

NOTES:

1. CONTRACTOR SHALL MAINTAIN A MINIMUM OF 24" CLEARANCE OR AS SPECIFIED OR AGREED TO IN THE CROSSING AGREEMENT.
2. ALL UNDERGROUND FACILITIES SUCH AS COMMUNICATION OR POWER CABLES, DRAIN TILES, IRRIGATION LINES, ETC. SHALL HAVE A 24" MINIMUM CLEARANCE. SAND BAGS OR OTHER SUPPORT STRUCTURE SHALL BE PLACED TO PREVENT THE PIPELINE WEIGHT FROM BEING PLACED ON THE CABLE. CROSSINGS SHOULD BE MADE AT THE ANGLE AGREED TO IN THE CROSSING AGREEMENT.
3. CONTRACTOR SHALL COMPLY WITH COMPANY'S FOREIGN LINE CROSSING CONSTRUCTION SPECIFICATIONS OR OTHER APPLICABLE REQUIREMENTS.
4. THE PROPOSED PIPELINE SHALL BE INSTALLED ABOVE THE FOREIGN LINE AT LOCATIONS WHERE THE FOREIGN LINE IS BURIED AT A DEPTH SUCH THAT 24" SEPERATION AND THE MINIMUM REQUIRED COVER CAN BE MAINTAINED.
5. CONTRACTOR SHALL SUBMIT ALL APPLICABLE LOUISIANA 811 ONE CALLS PRIOR TO EXCAVATION.

REVISIONS

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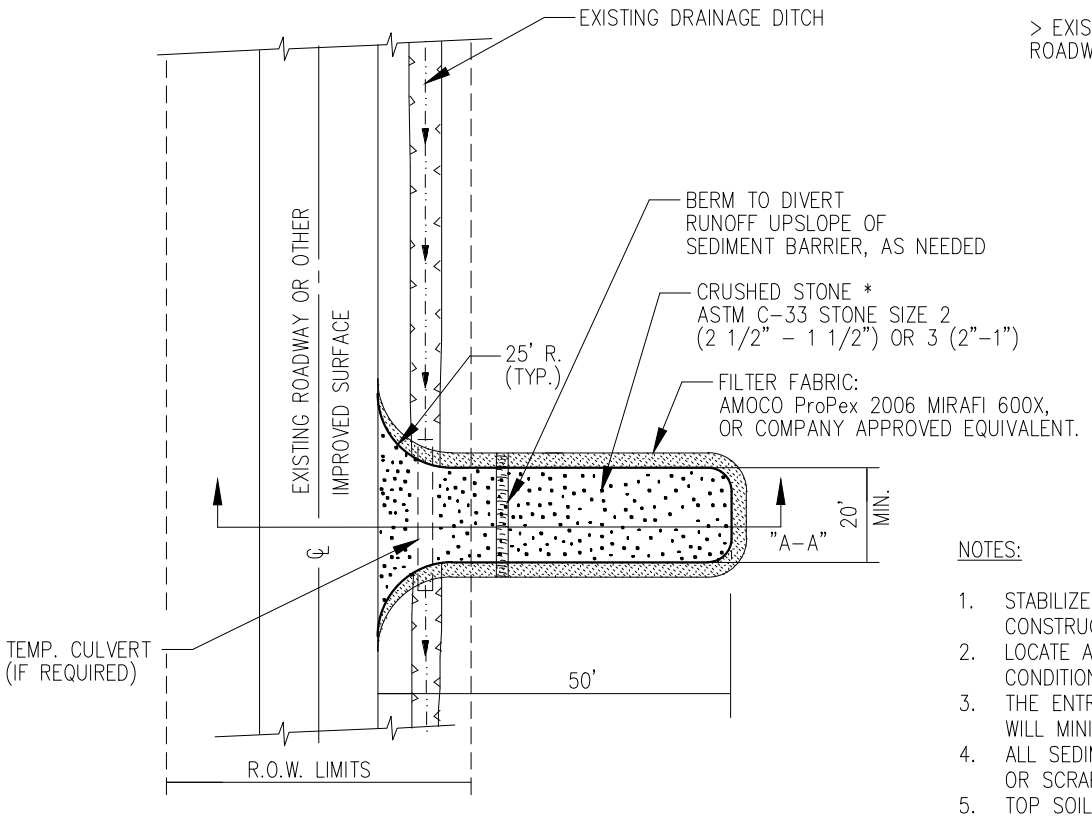
ENERGY TRANSFER

FRANKLIN FARMS PROJECT
FOREIGN PIPELINE & UTILITY CROSSINGS
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0017

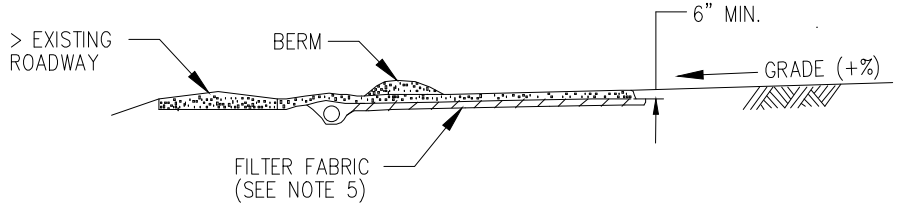
REV
A

CONSTRUCTION ENTRANCE
ENVIRONMENTAL TYPICALS



PLAN VIEW

SCALE: N.T.S.



SECTION "A-A"

SCALE: N.T.S.

NOTES:

1. STABILIZED ENTRANCE SHALL BE INSTALLED WHERE EQUIPMENT ENTERS AND EXITS CONSTRUCTION SITES ONTO PAVED ROADWAY OR OTHER IMPROVED SURFACE.
2. LOCATE ALL ROADWAY CROSSINGS AND ENTRANCES TO ENSURE SAFE AND ACCESSIBLE CONDITIONS THROUGHOUT THE CONSTRUCTION PHASE.
3. THE ENTRANCE SHALL BE MAINTAINED THROUGHOUT CONSTRUCTION IN A CONDITION WHICH WILL MINIMIZE TRACKING OF SEDIMENT ONTO A PUBLIC ROADWAY.
4. ALL SEDIMENT TRACKED ONTO PAVEMENT SHALL BE REMOVED IMMEDIATELY BY SWEEPING OR SCRAPING.
5. TOP SOIL SHALL BE REMOVED TO A MINIMUM DEPTH OF 6 INCHES PRIOR TO PLACEMENT OF GEOTEXTILE FABRIC AND STONE.
6. WHERE DRAINAGE DITCH EXISTS, CONTRACTOR SHALL PROVIDE A FLUME PIPE.
7. THE CONSTRUCTION ENTRANCE SHALL BE REMOVED AND AREA RESTORED AS PART OF FINAL GRADING AND CLEAN UP. REMOVAL IS NOT CONTINGENT UPON ESTABLISHMENT OF PERMANENT VEGETATION.
8. CULVERTS TO BE SIZED AND PLACED WHERE REQUIRED TO MAINTAIN WATER FLOW. CULVERT SHALL BE REMOVED FOLLOWING CONSTRUCTION ACTIVITIES.

REVISIONS

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ENERGY TRANSFER

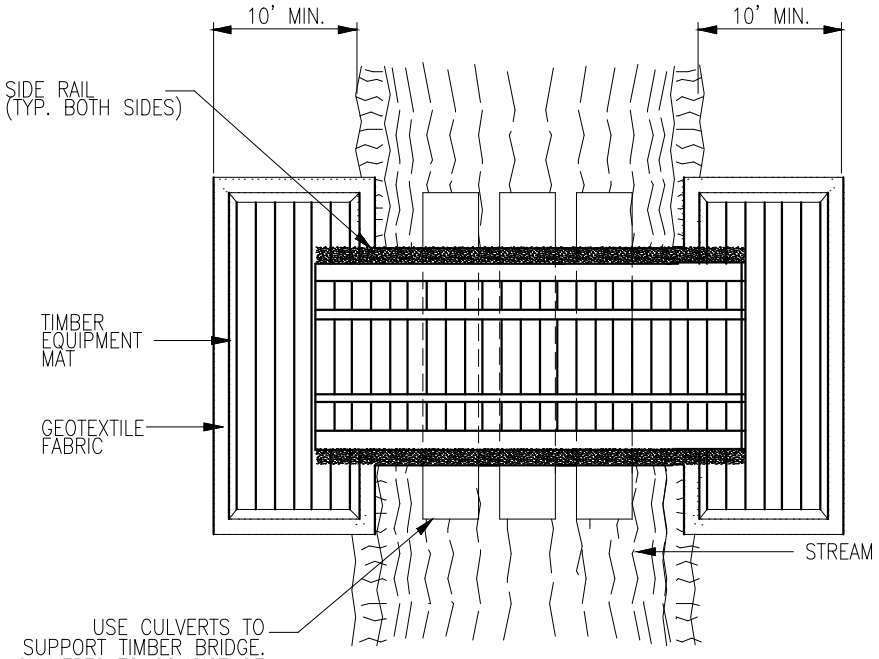
FRANKLIN FARMS PROJECT
CONSTRUCTION ENTRANCE

CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

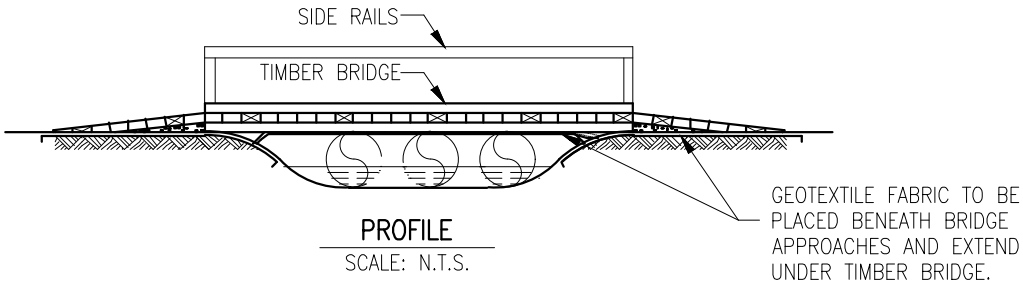
DWG NUMBER
TYP-0018

REV
A

TEMPORARY EQUIPMENT BRIDGE
CONSTRUCTION TYPICAL



PLAN VIEW
SCALE: N.T.S.



PROFILE
SCALE: N.T.S.

NOTES:

1. TIMBER BRIDGES SHALL BE ADEQUATELY ANCHORED AT BOTH ENDS.
2. PERIODICALLY CHECK BRIDGE INSTALLATION AND REMOVE BUILD-UP OF SEDIMENT OR DEBRIS ON BRIDGE.
3. BRIDGE APPROACHES SHALL BE TIMBER EQUIPMENT MATS.
4. MATERIALS PLACED ALONG STREAM CHANNEL SHALL BE COMPLETELY REMOVED DURING FINAL CLEAN-UP. REMOVAL OF THIS STRUCTURE IS NOT CONTINGENT UPON ESTABLISHMENT OF PERMANENT VEGETATION.
5. DEPENDING ON TOPOGRAPHY, SITE CONDITIONS AND/OR STATE REQUIREMENTS, CULVERTS MAY BE USED TO SUPPORT THE TIMBER BRIDGE TO PREVENT SETTLEMENT OF THE BRIDGE, IF THE GEOMETRY OF THE STREAM ALLOWS FOR SUCH INSTALLATION. THE TIMBER BRIDGE AND GEOTEXTILE FABRIC SHALL REMAIN ABOVE THE WATER SURFACE ELEVATION AT ALL TIMES.
6. SIDE RAILS SHALL BE INSTALLED ON BOTH SIDES OF THE BRIDGE EQUIPMENT CROSSING IN ORDER TO PREVENT SEDIMENT FROM ENTERING THE WATERBODY. SIDE RAILS TO BE CONSTRUCTED OF PLYWOOD NAILED TO THE OUTER EDGES OF THE TIMBER EQUIPMENT MATS.
7. TIMBER EQUIPMENT MATS SHALL EXTEND A MINIMUM OF 10 FEET OUTSIDE OF THE WATERBODY OR WETLAND BOUNDARIES.

REVISIONS

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ENERGY TRANSFER

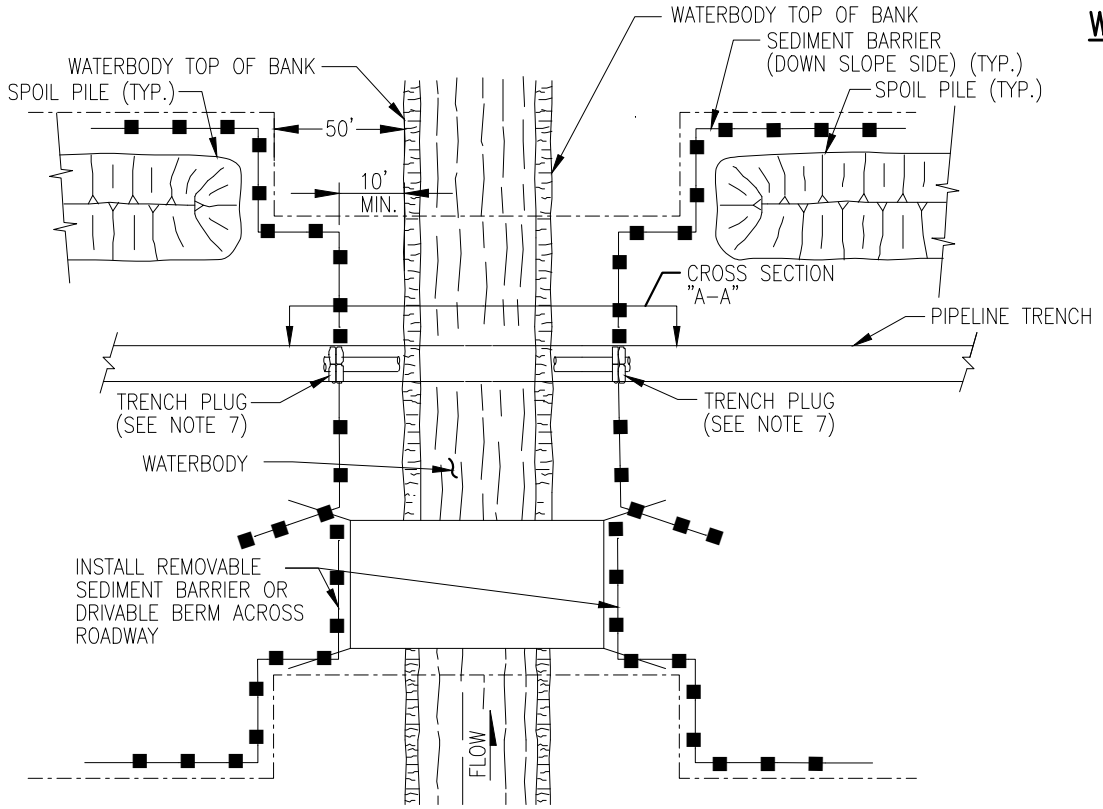
FRANKLIN FARMS PROJECT
TEMPORARY EQUIPMENT BRIDGE

CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0019

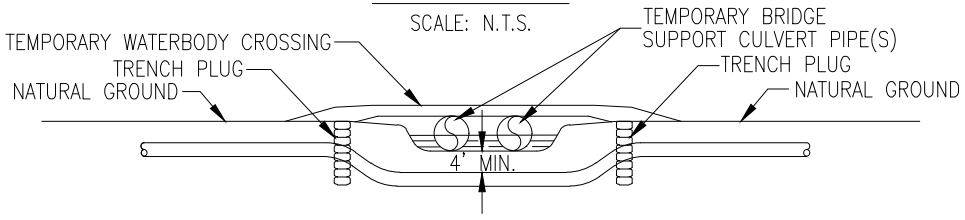
REV
A

WET WATERBODY CROSSING CONSTRUCTION TYPICAL



PLAN

SCALE: N.T.S.



CROSS SECTION "A-A"

SCALE: N.T.S.

NOTES:

1. SEDIMENT BARRIERS SHALL BE INSTALLED AS DEPICTED AND ALONG DOWN GRADIENT SIDES OF WORK AREAS AND STAGING AREAS SUCH THAT NO HEAVILY SILT LADEN WATER ENTERS THE WATERBODY OR LEAVES THE CONSTRUCTION RIGHT OF WAY.
2. HARD DITCH PLUGS MUST REMAIN IN PLACE AT CONVENIENT LOCATIONS TO SEPARATE MAINLINE DITCH FROM THE WATERBODY CROSSING UNTIL THE WATERBODY CROSSING IS INSTALLED AND BACK FILLED, AS NEEDED.
3. EQUIPMENT OPERATING IN THE WATERBODY SHALL BE LIMITED TO THAT NEEDED TO PERFORM CONSTRUCTION. IF OTHER TYPES OF EQUIPMENT MUST CROSS THE WATERBODY, CONTRACTOR SHALL PROVIDE AND USE TEMPORARY WATERBODY CROSSING.
4. STAGING AREA(S) FOR WATERBODY CROSSING(S), WHEN REQUIRED, SHALL BE LOCATED AT LEAST 10 FEET FROM WATER'S EDGE AND SHALL BE OF A MINIMUM SIZE NEEDED FOR CONVENIENT PREPARATION.
5. EROSION AND SEDIMENT CONTROL MEASURES SHALL BE INSPECTED PER PERMIT CONDITIONS AND REPAIRED IF NECESSARY.
6. CHEMICALS, FUELS AND LUBRICATING OILS SHALL NOT BE STORED AND EQUIPMENT SHALL NOT BE REFUELED WITHIN 100 FEET OF THE WATERBODY.
7. INSTALL TRENCH PLUGS ON BOTH SIDES OF THE WATERBODY TO PREVENT DIVERSION OF WATER INTO UPLAND PORTIONS OF THE PIPELINE TRENCH AND TO KEEP ANY ACCUMULATED TRENCH WATER OUT OF THE WATERBODY, AS NEEDED.
8. CONTRACTOR SHALL POSTPONE GRADING OF RIGHT-OF-WAY ADJACENT TO WATERBODY UNTIL STAGING AREA IS PREPARED AND WORK IN THE WATERBODY IS READY TO COMMENCE.
9. COMPLETE IN STREAM CONSTRUCTION ACTIVITIES (NOT INCLUDING BLASTING AND OTHER ROCK BREAKING MEASURES) WITHIN 24 HOURS, UNLESS SITE SPECIFIC CONDITIONS MAKE COMPLETION WITHIN 24 HOURS INFEASIBLE.

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ENERGY TRANSFER

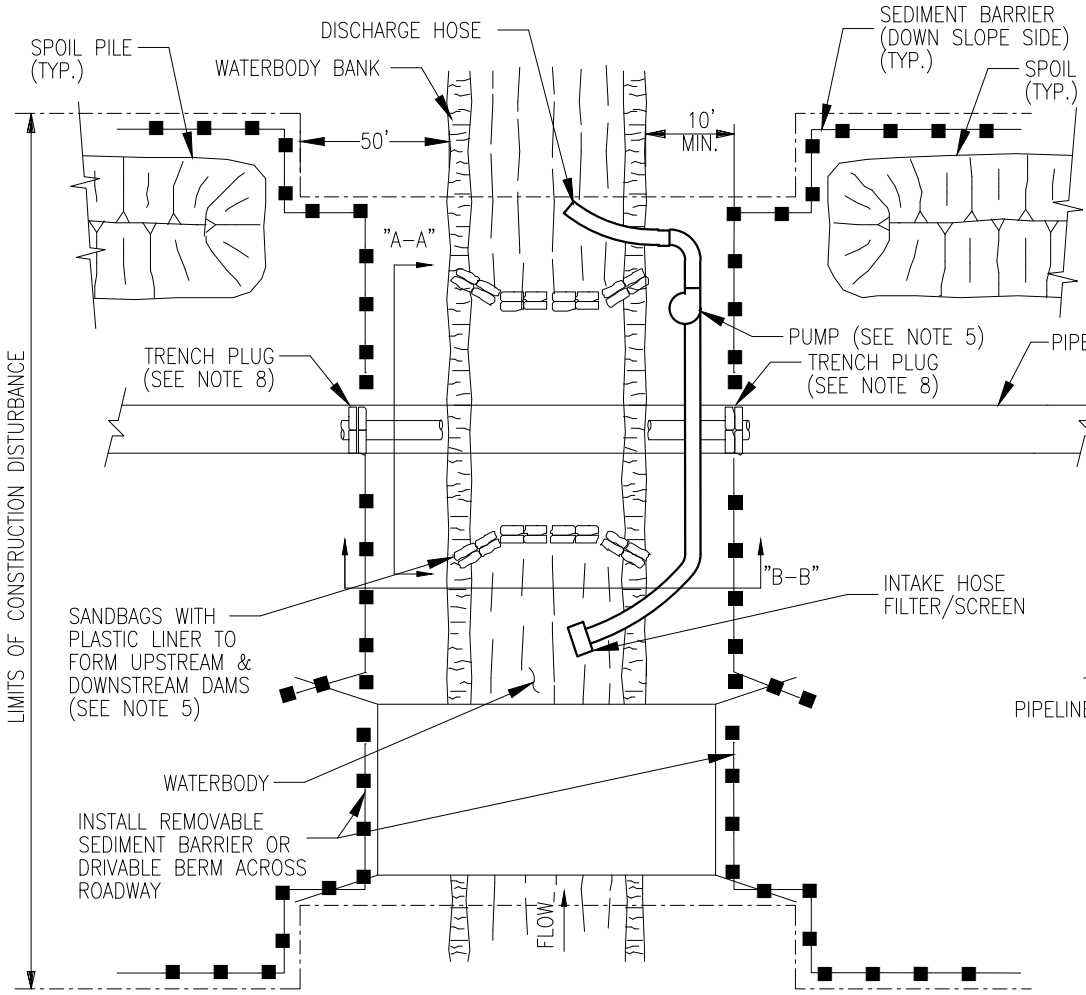
FRANKLIN FARMS PROJECT
WET WATERBODY CROSSING

CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

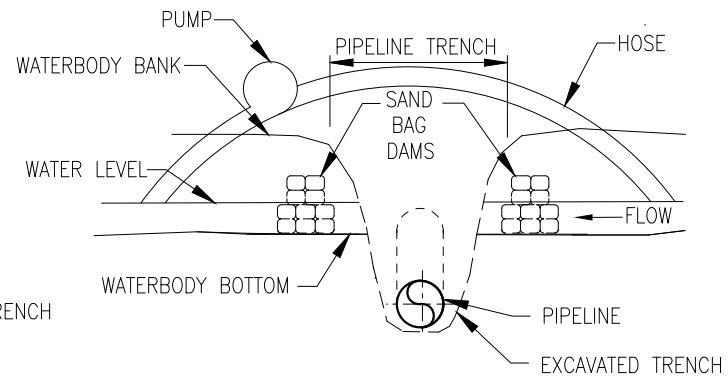
DWG NUMBER
TYP-0020

REV
A

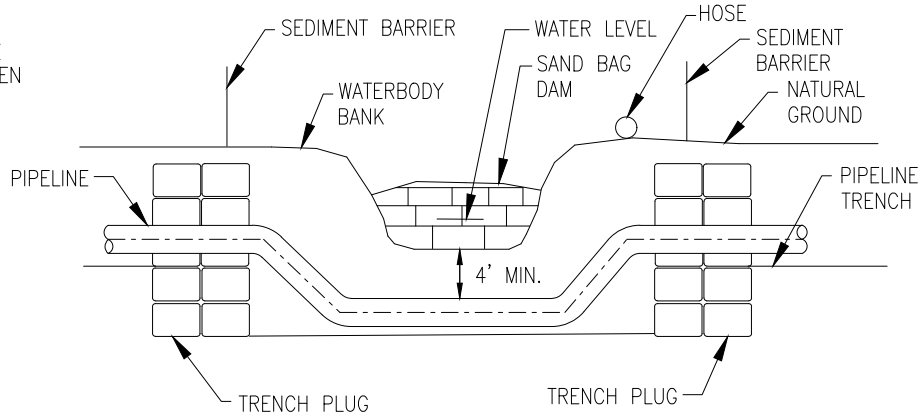
DAM AND PUMP WATERBODY CROSSING – CONSTRUCTION TYPICAL



PLAN
SCALE: N.T.S.



SECTION "A-A"
SCALE: N.T.S.



SECTION "B-B"
SCALE: N.T.S.

SEE SHEET 2 OF 2: TYPICAL TYP-0021 FOR NOTES.

REVISIONS					
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ENERGY TRANSFER

FRANKLIN FARMS PROJECT
DAM AND PUMP WATERBODY CROSSING
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER: TYP-0021
REV: A

DAM AND PUMP WATERBODY CROSSING – CONSTRUCTION TYPICAL

NOTES:

1. SEDIMENT BARRIERS SHALL BE INSTALLED AS DEPICTED AND ALONG DOWN GRADIENT SIDES OF WORK AREAS AND STAGING AREAS SUCH THAT NO HEAVY SILT LADEN WATER ENTERS THE WATERBODY OR LEAVES THE CONSTRUCTION RIGHT–OF–WAY.
2. HARD DITCH PLUGS MUST REMAIN IN PLACE AT CONVENIENT LOCATIONS TO SEPARATE MAINLINE DITCH FROM THE WATERBODY CROSSING UNTIL THE WATERBODY CROSSING IS INSTALLED AND BACKFILLED, AS NEEDED.
3. EQUIPMENT OPERATING IN THE WATERBODY SHALL BE LIMITED TO THAT NEEDED TO PERFORM CONSTRUCTION. IF OTHER TYPES OF EQUIPMENT MUST CROSS THE WATERBODY, CONTRACTOR SHALL PROVIDE AND USE A TEMPORARY STREAM CROSSING.
4. STAGING AREA(S) FOR WATERBODY CROSSING(S), WHEN REQUIRED, SHALL BE LOCATED AT LEAST 10 FEET FROM WATER'S EDGE AND SHALL BE OF A MINIMUM SIZE NEEDED FOR CONVENIENT PREPARATION.
5. IMPLEMENTATION OF THE DAM–AND–PUMP CROSSING METHOD MUST MEET THE FOLLOWING PERFORMANCE CRITERIA:
   - (A) USE SUFFICIENT PUMPS, INCLUDING ON–SITE BACKUP PUMPS, TO MAINTAIN DOWNSTREAM FLOWS.
   - (B) CONSTRUCT DAMS WITH MATERIALS THAT PREVENT SEDIMENT AND OTHER POLLUTANTS FROM ENTERING THE WATERBODY (E.G., SANDBAGS OR CLEAN GRAVEL WITH PLASTIC LINER).
   - (C) SCREEN PUMP INTAKES;
   - (D) PREVENT STREAM BED SCOUR AT PUMP DISCHARGE.
   - (E) MONITOR THE DAM AND PUMPS TO ENSURE PROPER OPERATION THROUGHOUT THE WATERBODY CROSSING.
6. EROSION AND SEDIMENT CONTROL MEASURES SHALL BE INSPECTED PER PERMIT CONDITIONS AND REPAIRED IF NECESSARY.
7. CHEMICALS, FUELS AND LUBRICATING OILS SHALL NOT BE STORED AND EQUIPMENT SHALL NOT BE REFUELED WITHIN 100 FEET OF THE WATERBODY.
8. INSTALL TRENCH PLUGS ON BOTH SIDES OF THE WATERBODY TO PREVENT DIVERSION OF WATER INTO UPLAND PORTIONS OF THE PIPELINE TRENCH AND TO KEEP ANY ACCUMULATED TRENCH WATER OUT OF THE WATERBODY, AS NEEDED.
9. CONTRACTOR SHALL POSTPONE GRADING OF RIGHT–OF–WAY ADJACENT TO WATERBODY UNTIL STAGING AREA IS PREPARED AND WORK IN THE WATERBODY IS READY TO COMMENCE.
10. REMOVE SEDIMENT DEPOSITS AS NECESSARY TO PROVIDE ADEQUATE STORAGE VOLUME.
11. REMOVE ALL SEDIMENT FENCE AND UNSTABLE SEDIMENT DEPOSITS AND BRING THE AREA TO ORIGINAL GRADE AND STABILIZE IT AFTER THE CONTRIBUTING DRAINAGE AREA HAS BEEN STABILIZED.

REVISIONS

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PREPARED BY: FIRM No:EF.0007320
TRC
8521 McHARD RD, HOUSTON, TX 77053

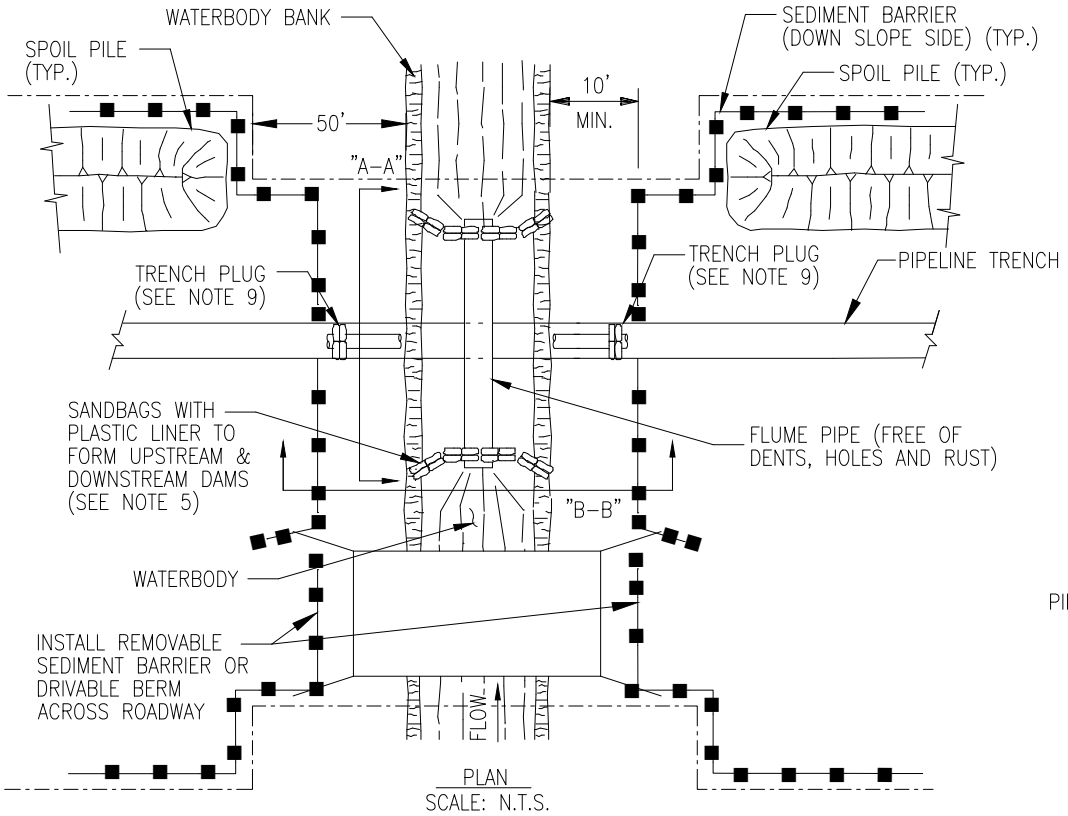
ENERGY TRANSFER

FRANKLIN FARMS PROJECT
DAM AND PUMP WATERBODY CROSSING
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

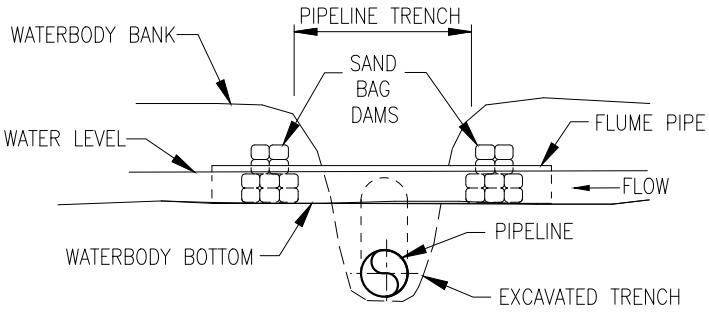
DWG NUMBER
TYP-0021

REV
A

FLUME WATERBODY CROSSING
CONSTRUCTION TYPICAL



PLAN
SCALE: N.T.S.



SECTION "A-A"
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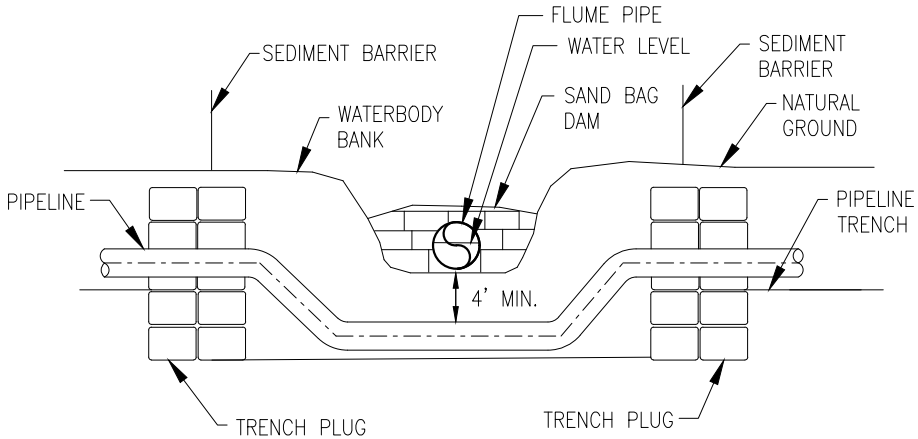

SECTION "B-B"
SCALE: N.T.S.

SEE SHEET 2 OF 2: TYPICAL TYP-0022 FOR NOTES.

REVISIONS					
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ENERGY TRANSFER

FRANKLIN FARMS PROJECT
FLUME WATERBODY CROSSING

CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0022

REV
A

DAM AND PUMP WATERBODY CROSSING – CONSTRUCTION TYPICAL

NOTES:

1. SEDIMENT BARRIERS SHALL BE INSTALLED AS DEPICTED AND ALONG DOWN GRADIENT SIDES OF WORK AREAS AND STAGING AREAS SUCH THAT NO HEAVY SILT LADEN WATER ENTERS THE WATERBODY OR LEAVES THE CONSTRUCTION RIGHT–OF–WAY.
2. HARD DITCH PLUGS MUST REMAIN IN PLACE AT CONVENIENT LOCATIONS TO SEPARATE MAINLINE DITCH FROM THE WATERBODY CROSSING UNTIL THE WATERBODY CROSSING IS INSTALLED AND BACKFILLED, AS NEEDED.
3. EQUIPMENT OPERATING IN THE WATERBODY SHALL BE LIMITED TO THAT NEEDED TO PERFORM CONSTRUCTION. IF OTHER TYPES OF EQUIPMENT MUST CROSS THE WATERBODY, CONTRACTOR SHALL PROVIDE AND USE A TEMPORARY STREAM CROSSING.
4. STAGING AREA(S) FOR WATERBODY CROSSING(S), WHEN REQUIRED, SHALL BE LOCATED AT LEAST 10 FEET FROM WATER'S EDGE AND SHALL BE OF A MINIMUM SIZE NEEDED FOR CONVENIENT PREPARATION.
5. FLUME CROSSING METHOD REQUIREMENTS INCLUDE:
   (A) INSTALL FLUME PIPE(S) AFTER BLASTING (IF NECESSARY), BUT BEFORE ANY TRENCHING.
   (B) USE SAND BAG OR SAND BAG AND PLASTIC SHEETING DIVERSION STRUCTURE OR EQUIVALENT TO DEVELOP AN EFFECTIVE SEAL AND TO DIVERT STREAM FLOW THROUGH THE FLUME PIPE (SOME MODIFICATIONS TO THE STREAM BOTTOM MAY BE REQUIRED TO ACHIEVE AN EFFECTIVE SEAL).
   (C) PROPERLY ALIGN FLUME PIPE(S) TO PREVENT BANK EROSION AND STREAM BED SCOUR.
   (D) DO NOT REMOVE FLUME PIPE DURING TRENCHING, PIPE LAYING, OR BACKFILLING ACTIVITIES, OR INITIAL STREAM BED RESTORATION EFFORTS.
   (E) REMOVE ALL FLUME PIPES AND DAMS THAT ARE NOT ALSO PART OF THE EQUIPMENT BRIDGE AS SOON AS FINAL CLEANUP OF THE STREAM BED AND BANK IS COMPLETE.
6. THE FLUME PIPE MUST BE SIZED TO PREVENT IMPEDIMENT OF THE UPSTREAM FLOW AND TO MAINTAIN ADEQUATE FLOW RATES TO PROTECT AQUATIC LIFE, AND PREVENT THE INTERRUPTION OF EXISTING DOWNSTREAM USES.
7. EROSION AND SEDIMENT CONTROL MEASURES SHALL BE INSPECTED PER PERMIT CONDITIONS AND REPAIRED IF NECESSARY.
8. CHEMICALS, FUELS AND LUBRICATING OILS SHALL NOT BE STORED AND EQUIPMENT SHALL NOT BE REFUELED WITHIN 100 FEET OF THE WATERBODY.
9. INSTALL TRENCH PLUGS ON BOTH SIDES OF THE WATERBODY TO PREVENT DIVERSION OF WATER INTO UPLAND PORTIONS OF THE PIPELINE TRENCH AND TO KEEP ANY ACCUMULATED TRENCH WATER OUT OF THE WATERBODY, AS NEEDED.
10. CONTRACTOR SHALL POSTPONE GRADING OF RIGHT–OF–WAY IMMEDIATELY ADJACENT TO WATERBODY UNTIL STAGING AREA IS PREPARED AND WORK IN THE WATERBODY IS READY TO COMMENCE.
11. REMOVE SEDIMENT DEPOSITS AS NECESSARY TO PROVIDE ADEQUATE STORAGE VOLUME.
12. REMOVE ALL SEDIMENT FENCE AND UNSTABLE SEDIMENT DEPOSITS AND BRING THE AREA TO ORIGINAL GRADE AND STABILIZE IT AFTER THE CONTRIBUTING DRAINAGE AREA HAS BEEN STABILIZED.

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ENERGY TRANSFER

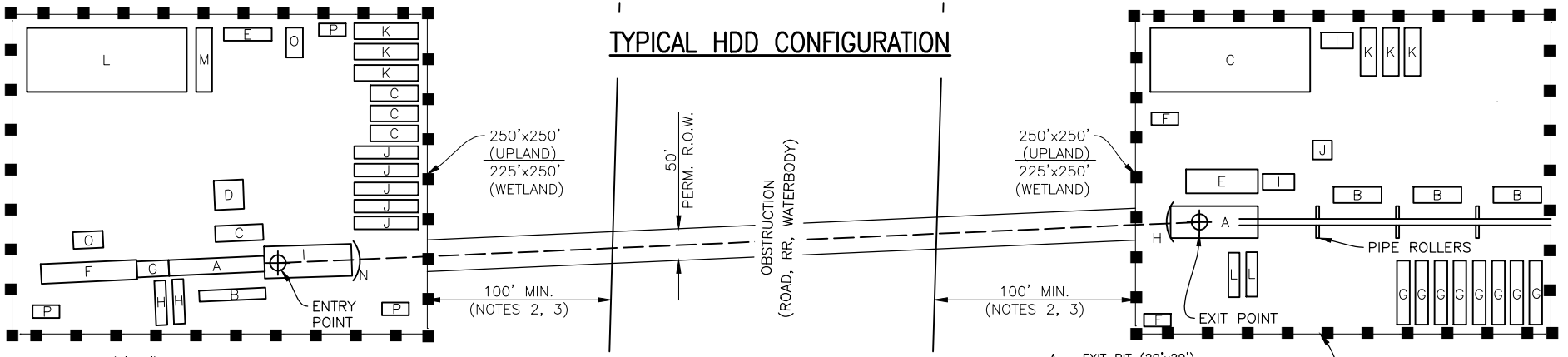
FRANKLIN FARMS PROJECT
FLUME WATERBODY CROSSING

CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0022

REV
A

TYPICAL HDD CONFIGURATION



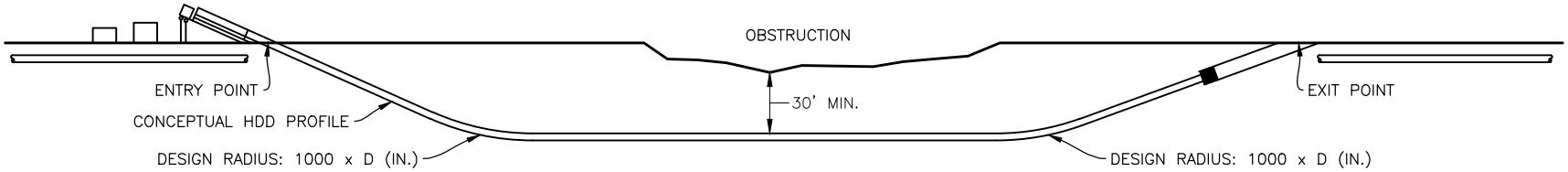
A – DRILL RIG (9'x51')
B – DRILLERS'S CONSOLE (8'x20')
GENERATOR (6'x15')
C – DRILL PIPE ON FLOAT (8'x40')
D – CRANE (17'x30')
E – BOX TRAILER (8'x53')
F – MUD CLEANING UNIT (9'x52')
G – MUD MIXING UNIT (5'x16')
H – MUD PUMPS (4'x9')
I – MUD PIT (20'x40')
J – FRAC TANKS (8'x45')

K – DRILLING MUD (8'x40')
L – PARKING (50'x80')
M – OFFICE CONEX (8'x40')
N – CONTAINMENT BERM
O – DUMPSTER (8'x13')
P – LIGHT PLANT (6'x15')
Q – EXCAVATOR (17'x23')

PLAN
N.T.S.

A – EXIT PIT (20'x20')
B – LIFT EQUIPMENT (17'x23')
C – PARKING AREA (50'x80')
D – MUD PIT (20'x40')
E – MUD CLEANING UNIT (9'x52')
F – LIGHT PLANT (6'x15')
G – FRAC TANKS (8'x45')
H – CONTAINMENT BERM
I – DUMPSTER (8'x13')
J – EXCAVATOR (17'x23')
K – DRILL PIPE ON FLOAT (8'x40')
L – MUD PUMPS (9'x4')

INSTALL SILT FENCE AROUND ENTIRE WORK AREA (TYP. BOTH SIDES)



PROFILE
N.T.S.

NOTES:
1. ANY DEVIATION FROM RECOMMENDED DESIGN IS SUBJECT TO OWNER ENGINEER'S APPROVAL.
2. SETBACK MAY BE ADJUSTED BASED ON GEOTECHNICAL ANALYSIS.
3. DO NOT CLEAR OR GRADE WITHIN THE SETBACK ZONE.
4. DESIGN CURVATURE RADIUS: 1000 X D.
5. ONLY HAND CLEARING WILL OCCUR IN THE HDD CONSTRUCTION R.O.W. WIDTH TO ALLOW FOR THE PLACEMENT OF THE TRU TRACKER GUIDANCE WIRES (APPROX. 5' WIDTH FOR EACH WIRE).
6. ONLY MINOR CLEARING WILL OCCUR IN THE HDD CONSTRUCTION R.O.W. WIDTH FOR THE PLACEMENT OF TEMPORARY WATER PIPELINES FOR HYDROSTATIC TESTING & PUMP AND FILL; AS DETAILED ON SITE SPECIFIC HDD DRAWINGS, ACCESS LOCATIONS.
7. THE EQUIPMENT LAYOUT SHOWN IS TYPICAL AND MAY VARY DUE TO SITE CONDITIONS AND WILL BE FIELD DETERMINED BY THE CONTRACTOR.

REVISIONS

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ENERGY TRANSFER

FRANKLIN FARMS PROJECT
TYPICAL HDD CONFIGURATION

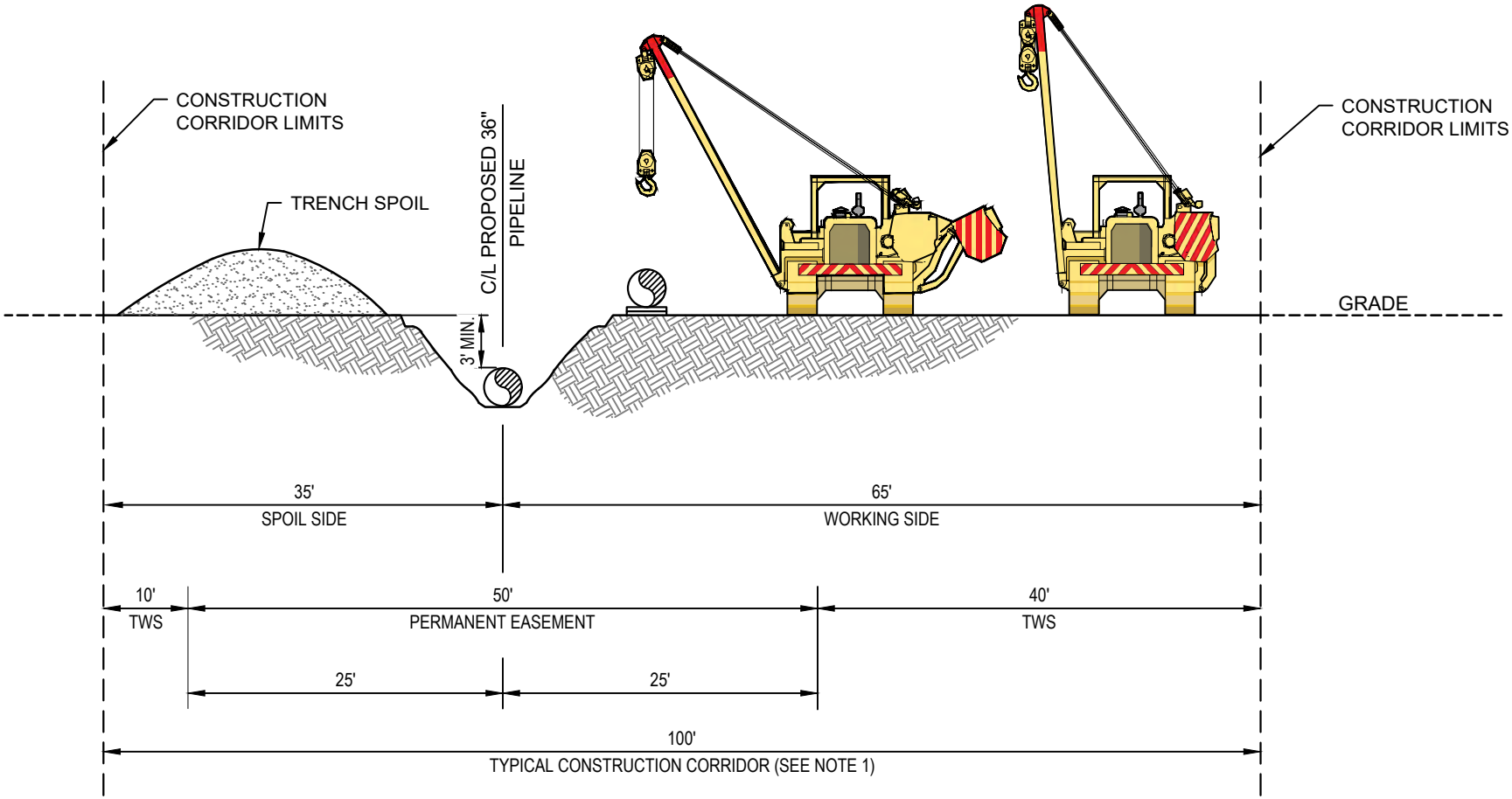
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
TYP-0023

REV
A

TYPICAL UPLAND WORKSPACE - 36"

N.T.S.



NOTES:

1. CONSTRUCTION RIGHT-OF-WAY (ROW) WILL TYPICALLY BE 100 FEET WIDE CONSISTING OF 50 FEET OF PERMANENT EASEMENT AND 50 FEET OF TEMPORARY WORKSPACE (TWS). ADDITIONAL TEMPORARY WORKSPACE (ATWS) WILL BE REQUIRED AT ROAD, RAILROAD AND WATERBODY CROSSINGS AND OTHER LOCATIONS AS APPLICABLE. CERTAIN SITUATIONS MAY REQUIRE A NARROWER WIDTH.
2. LEAVE GAPS IN SPOIL PILES FOR LIVESTOCK AND AT OBVIOUS DRAINAGE. DO NOT PUSH UPLAND SOILS INTO CANALS OR WETLANDS.
3. RESTORE GRADE TO NEAR PRE-CONSTRUCTION TOPOGRAPHY.

REVISIONS

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ENERGY TRANSFER

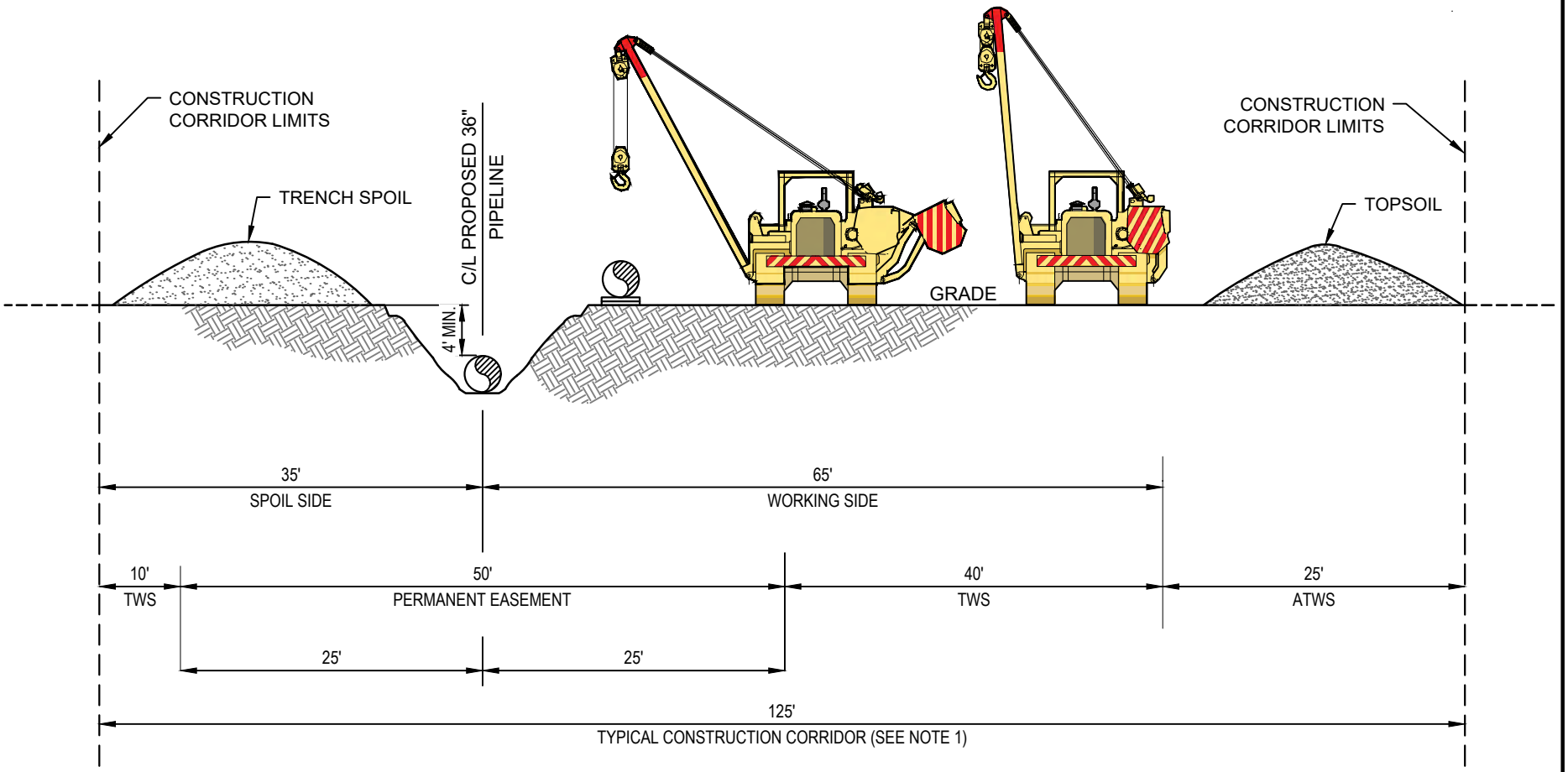
FRANKLIN FARMS LATERAL
TYPICAL UPLAND WORKSPACE
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
FIGURE-001

REV
A

TYPICAL AGRICULTURAL WORKSPACE - 36"

N.T.S.



NOTES:

1. CONSTRUCTION RIGHT-OF-WAY (ROW) WILL TYPICALLY BE 125 FEET WIDE CONSISTING OF 50 FEET OF PERMANENT EASEMENT, 50 FEET OF TEMPORARY WORKSPACE (TWS) AND 25 FEET OF ADDITIONAL TEMPORARY WORKSPACE (ATWS). ADDITIONAL TEMPORARY WORKSPACE (ATWS) WILL BE REQUIRED AT ROAD, RAILROAD AND WATERBODY CROSSINGS AND OTHER LOCATIONS AS APPLICABLE. CERTAIN SITUATIONS MAY REQUIRE A NARROWER WIDTH.
2. LEAVE GAPS IN SPOIL PILES FOR LIVESTOCK AND AT OBVIOUS DRAINAGE. DO NOT PUSH UPLAND SOILS INTO CANALS OR WETLANDS.
3. AT A MINIMUM, TOPSOIL SEGREGATION SHALL BE PERFORMED ALONG THE DITCHLINE.
4. VARIATIONS FOR STAGING AREAS/ADDITIONAL TEMPORARY WORKSPACE MAY BE NECESSARY DUE TO SITE-SPECIFIC TERRAIN FEATURES/CROSSINGS.
5. RESTORE GRADE TO NEAR PRE-CONSTRUCTION TOPOGRAPHY. REPLACE TOPSOIL AND INSTALL PERMANENT EROSION CONTROL IF APPLICABLE.

REVISIONS

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ENERGY TRANSFER

FRANKLIN FARMS LATERAL
TYPICAL AGRICULTURAL WORKSPACE
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER

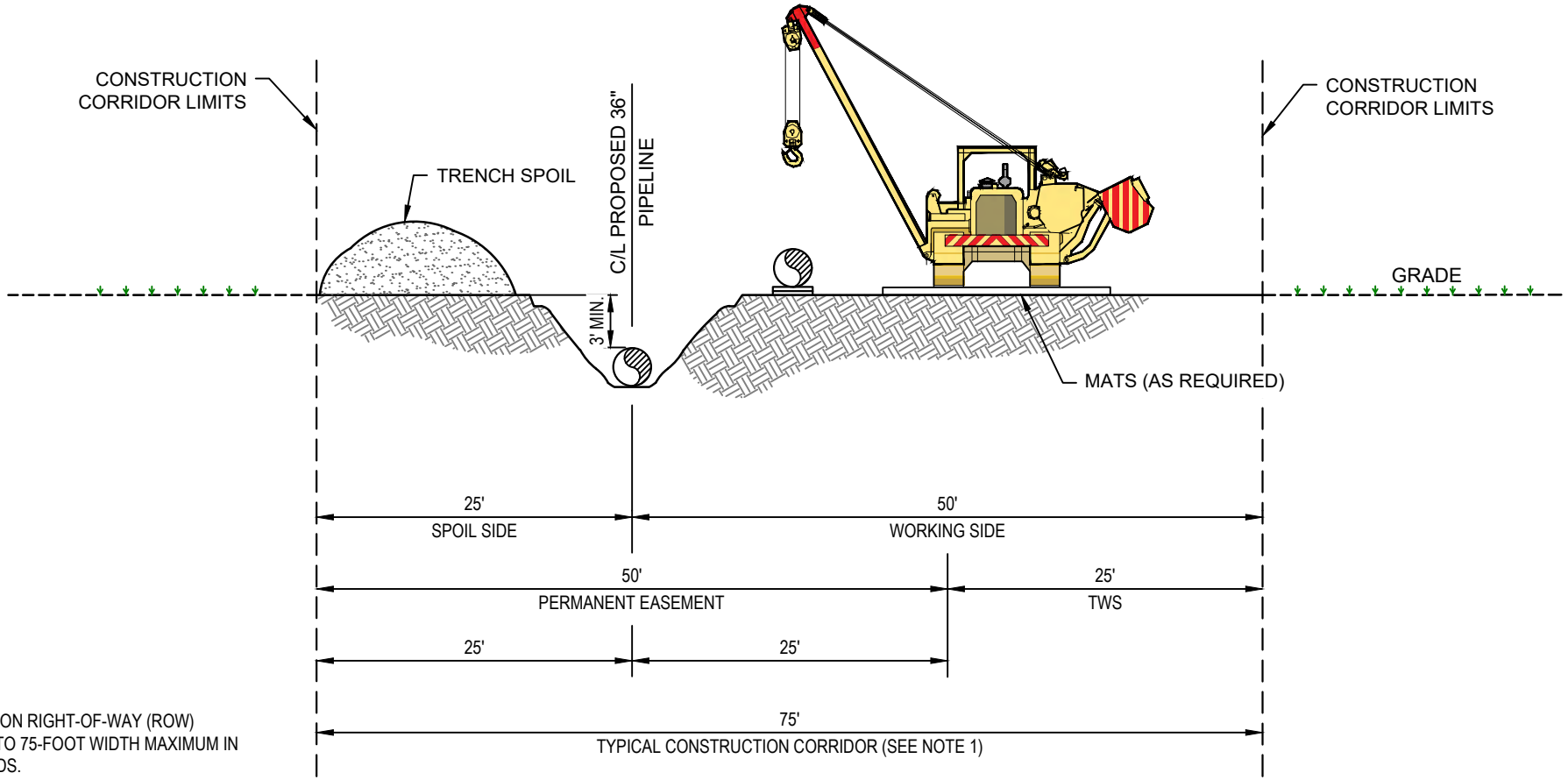
FIGURE-002

REV

A

TYPICAL UNSATURATED WETLAND WORKSPACE - 36"

N.T.S.



NOTES:

1. CONSTRUCTION RIGHT-OF-WAY (ROW) NARROWED TO 75-FOOT WIDTH MAXIMUM IN ALL WETLANDS.
2. LEAVE GAPS IN SPOIL PILES FOR LIVESTOCK AND AT OBVIOUS DRAINAGE. DO NOT PUSH UPLAND SOILS INTO CANALS OR WETLANDS.
3. TOPSOIL SEGREGATION SHALL BE PERFORMED FOR DITCHLINE ONLY.
4. VARIATIONS FOR STAGING AREAS/ADDITIONAL TEMPORARY WORKSPACE MAY BE NECESSARY DUE TO SITE-SPECIFIC TERRAIN FEATURES/CROSSINGS.
5. RESTORE GRADE TO NEAR PRE-CONSTRUCTION TOPOGRAPHY. REPLACE TOPSOIL AND INSTALL PERMANENT EROSION CONTROL IF APPLICABLE.

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ENERGY TRANSFER

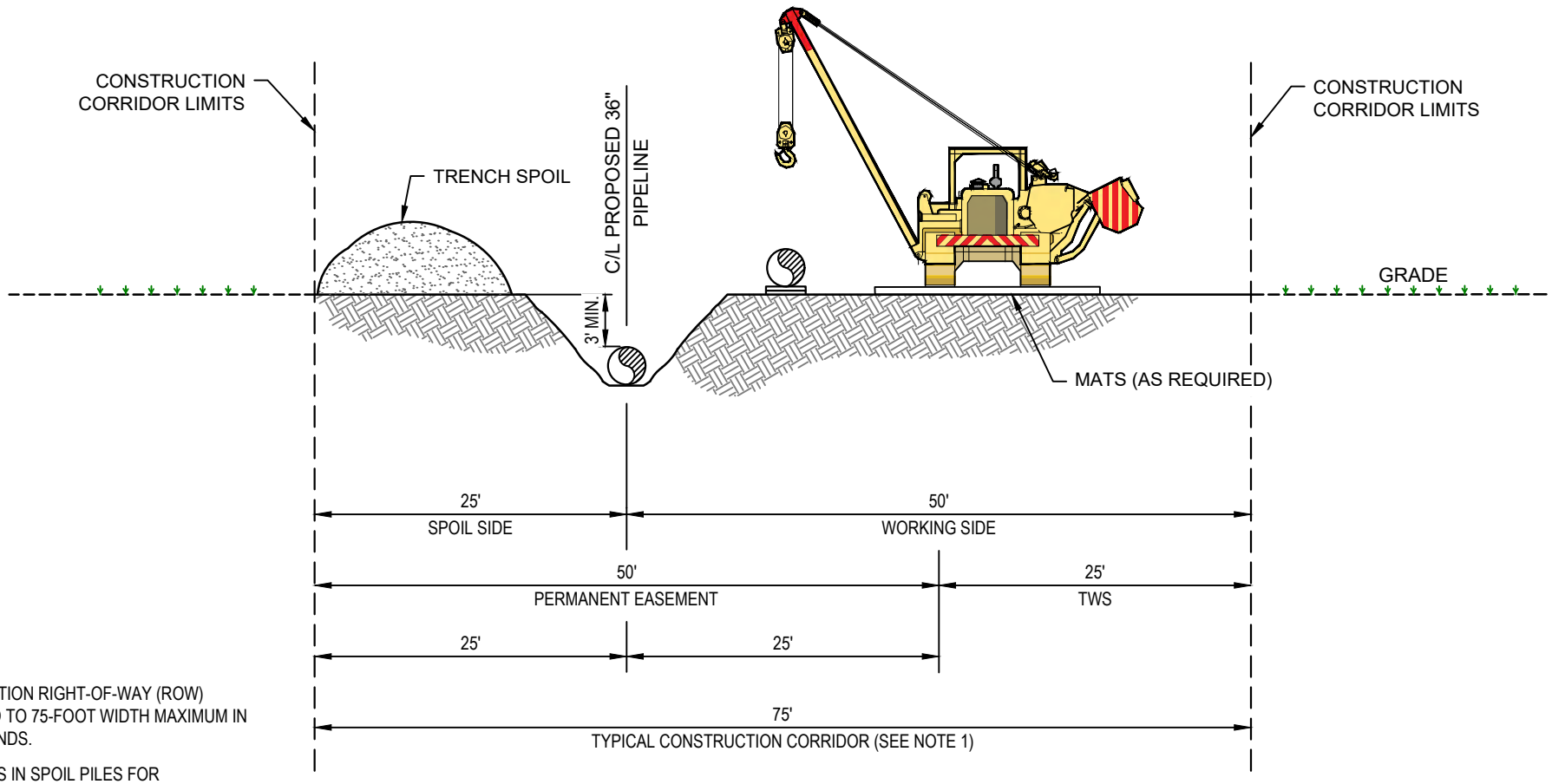
FRANKLIN FARMS LATERAL
TYPICAL UNSATURATED WETLAND WORKSPACE
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
FIGURE-003

REV
A

TYPICAL SATURATED WETLAND WORKSPACE - 36"

N.T.S.



NOTES:

1. CONSTRUCTION RIGHT-OF-WAY (ROW) NARROWED TO 75-FOOT WIDTH MAXIMUM IN ALL WETLANDS.
2. LEAVE GAPS IN SPOIL PILES FOR LIVESTOCK AND AT OBVIOUS DRAINAGE. DO NOT PUSH UPLAND SOILS INTO CANALS OR WETLANDS.
3. TOPSOIL SEGREGATION SHALL NOT BE REQUIRED IN SATURATED SOIL CONDITIONS.
4. VARIATIONS FOR STAGING AREAS/ADDITIONAL TEMPORARY WORKSPACE MAY BE NECESSARY DUE TO SITE-SPECIFIC TERRAIN FEATURES/CROSSINGS.
5. RESTORE GRADE TO NEAR PRE-CONSTRUCTION TOPOGRAPHY. REPLACE TOPSOIL AND INSTALL PERMANENT EROSION CONTROL IF APPLICABLE.

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ENERGY TRANSFER

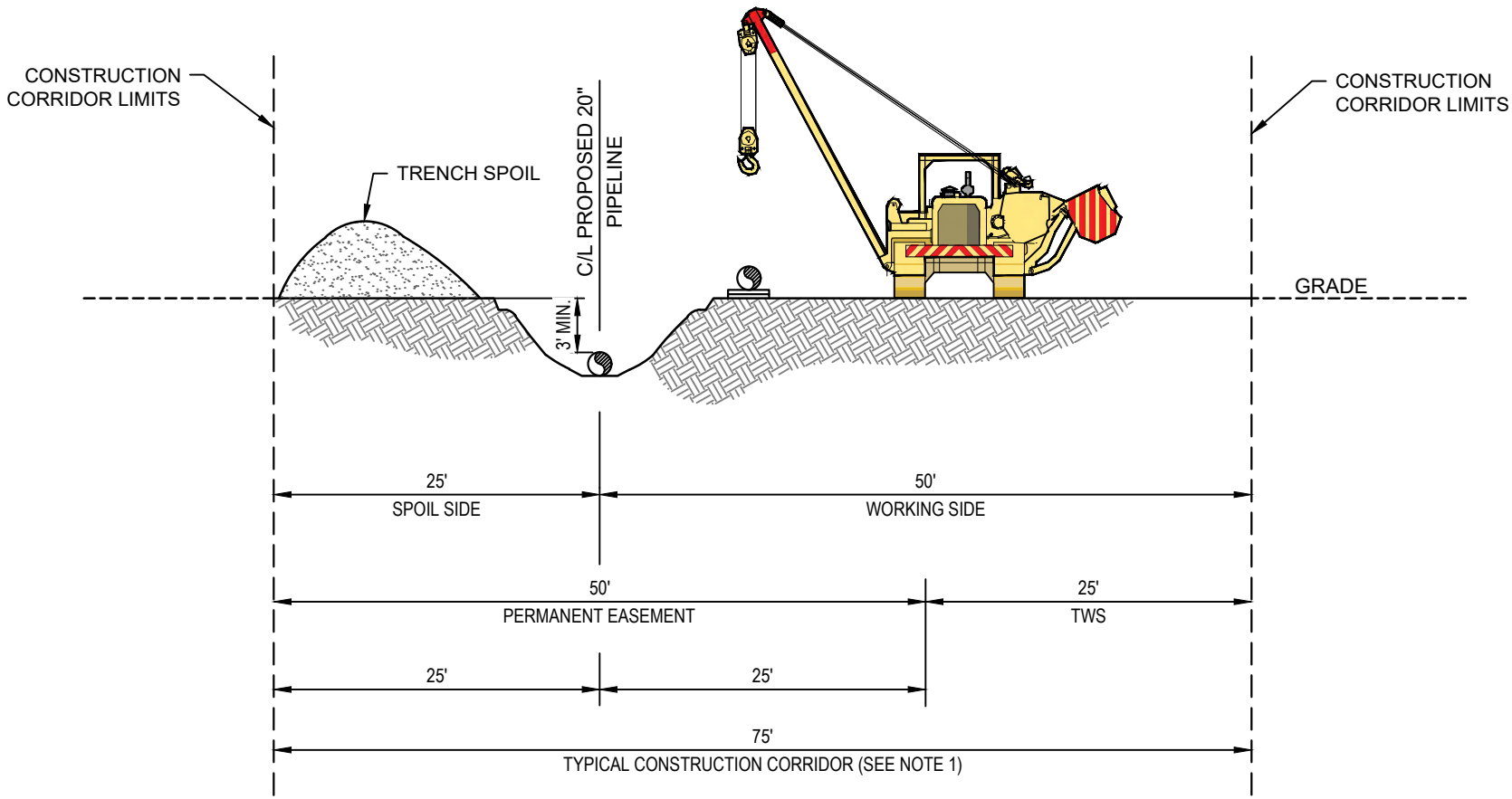
FRANKLIN FARMS LATERAL
TYPICAL SATURATED WETLAND WORKSPACE
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
FIGURE-004

REV
A

TYPICAL UPLAND WORKSPACE - 20"

N.T.S.



NOTES:

1. CONSTRUCTION RIGHT-OF-WAY (ROW) WILL TYPICALLY BE 75 FEET WIDE CONSISTING OF 50 FEET OF PERMANENT EASEMENT AND 25 FEET OF TEMPORARY WORKSPACE (TWS). ADDITIONAL TEMPORARY WORKSPACE (ATWS) WILL BE REQUIRED AT ROAD, RAILROAD AND WATERBODY CROSSINGS AND OTHER LOCATIONS AS APPLICABLE. CERTAIN SITUATIONS MAY REQUIRE A NARROWER WIDTH.
2. LEAVE GAPS IN SPOIL PILES FOR LIVESTOCK AND AT OBVIOUS DRAINAGE. DO NOT PUSH UPLAND SOILS INTO CANALS OR WETLANDS.
3. RESTORE GRADE TO NEAR PRE-CONSTRUCTION TOPOGRAPHY.

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ENERGY TRANSFER

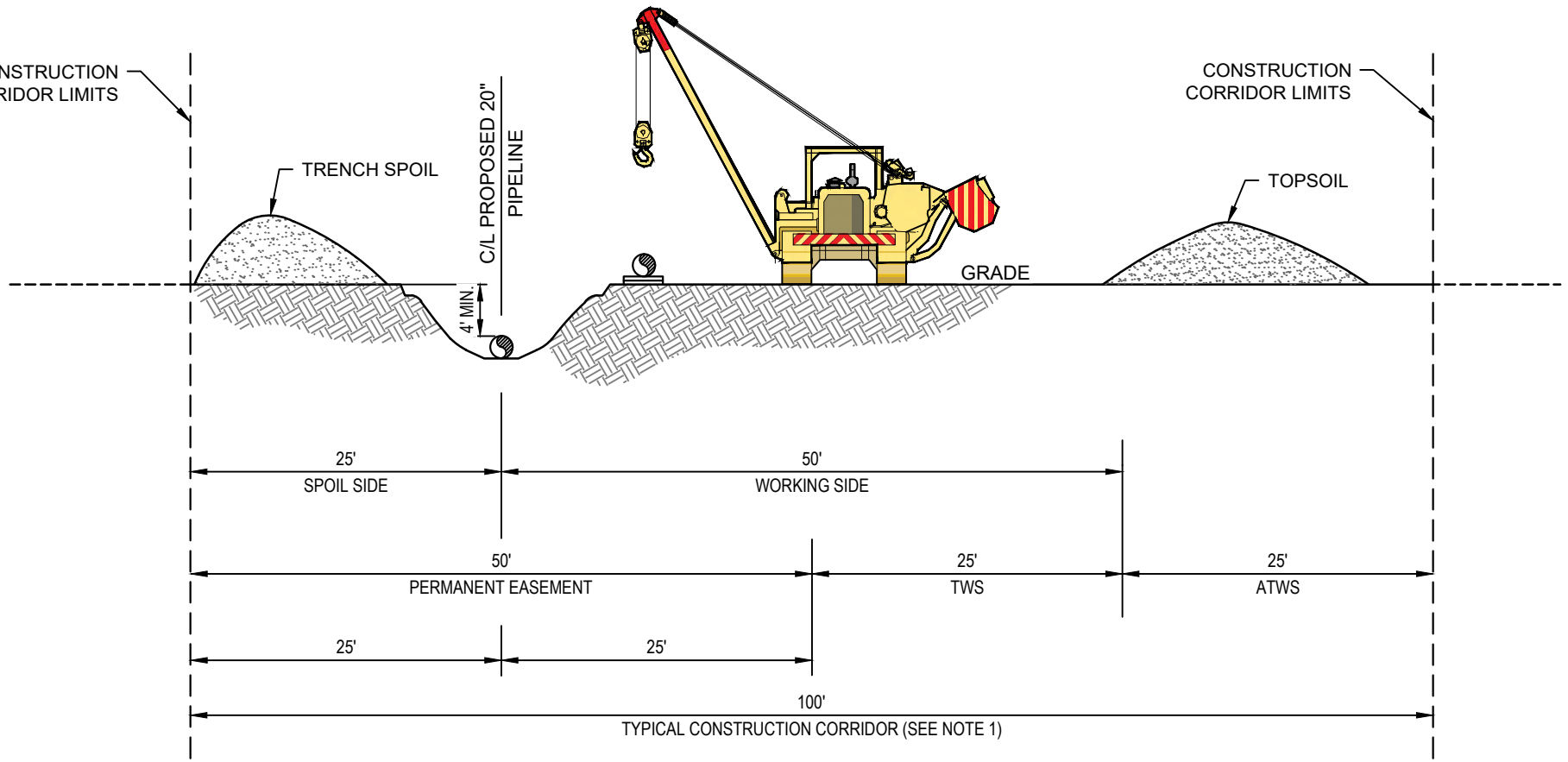
FF TRIO FARM LATERAL
TYPICAL UPLAND WORKSPACE
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
FIGURE-005

REV
A

TYPICAL AGRICULTURAL WORKSPACE - 20"

N.T.S.



NOTES:

1. CONSTRUCTION RIGHT-OF-WAY (ROW) WILL TYPICALLY BE 100 FEET WIDE CONSISTING OF 50 FEET OF PERMANENT EASEMENT, 25 FEET OF TEMPORARY WORKSPACE (TWS) AND 25 FEET OF ADDITIONAL TEMPORARY WORKSPACE (ATWS). ADDITIONAL TEMPORARY WORKSPACE (ATWS) WILL BE REQUIRED AT ROAD, RAILROAD AND WATERBODY CROSSINGS AND OTHER LOCATIONS AS APPLICABLE. CERTAIN SITUATIONS MAY REQUIRE A NARROWER WIDTH.
2. LEAVE GAPS IN SPOIL PILES FOR LIVESTOCK AND AT OBVIOUS DRAINAGE. DO NOT PUSH UPLAND SOILS INTO CANALS OR WETLANDS.
3. AT A MINIMUM, TOPSOIL SEGREGATION SHALL BE PERFORMED ALONG THE DITCHLINE.
4. VARIATIONS FOR STAGING AREAS/ADDITIONAL TEMPORARY WORKSPACE MAY BE NECESSARY DUE TO SITE-SPECIFIC TERRAIN FEATURES/CROSSINGS.
5. RESTORE GRADE TO NEAR PRE-CONSTRUCTION TOPOGRAPHY. REPLACE TOPSOIL AND INSTALL PERMANENT EROSION CONTROL IF APPLICABLE.

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ENERGY TRANSFER

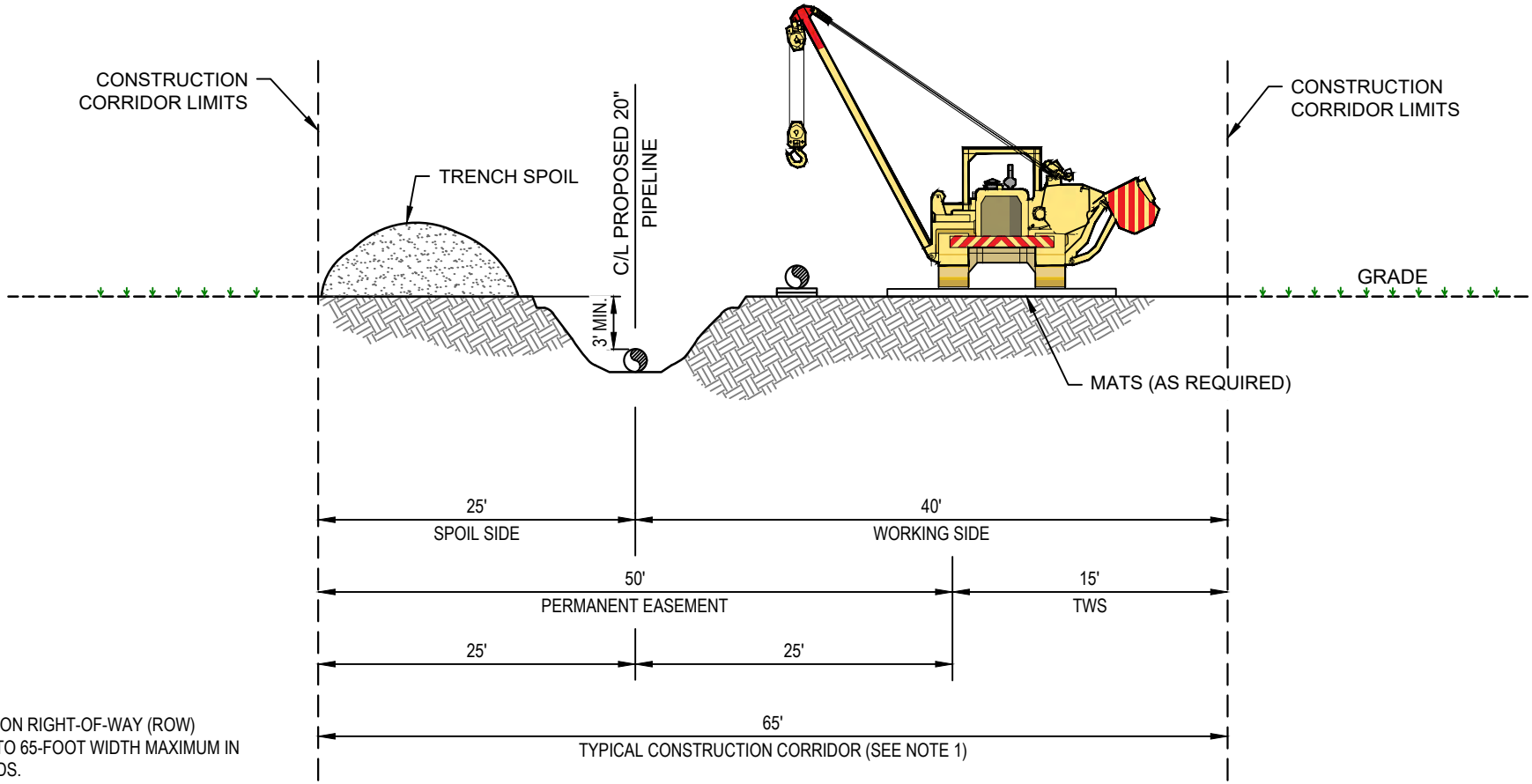
FF TRIO FARM LATERAL
TYPICAL AGRICULTURAL WORKSPACE
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
FIGURE-006

REV
A

TYPICAL UNSATURATED WETLAND WORKSPACE - 20"

N.T.S.



NOTES:

1. CONSTRUCTION RIGHT-OF-WAY (ROW) NARROWED TO 65-FOOT WIDTH MAXIMUM IN ALL WETLANDS.
2. LEAVE GAPS IN SPOIL PILES FOR LIVESTOCK AND AT OBVIOUS DRAINAGE. DO NOT PUSH UPLAND SOILS INTO CANALS OR WETLANDS.
3. TOPSOIL SEGREGATION SHALL BE PERFORMED FOR DITCHLINE ONLY.
4. VARIATIONS FOR STAGING AREAS/ADDITIONAL TEMPORARY WORKSPACE MAY BE NECESSARY DUE TO SITE-SPECIFIC TERRAIN FEATURES/CROSSINGS.
5. RESTORE GRADE TO NEAR PRE-CONSTRUCTION TOPOGRAPHY. REPLACE TOPSOIL AND INSTALL PERMANENT EROSION CONTROL IF APPLICABLE.

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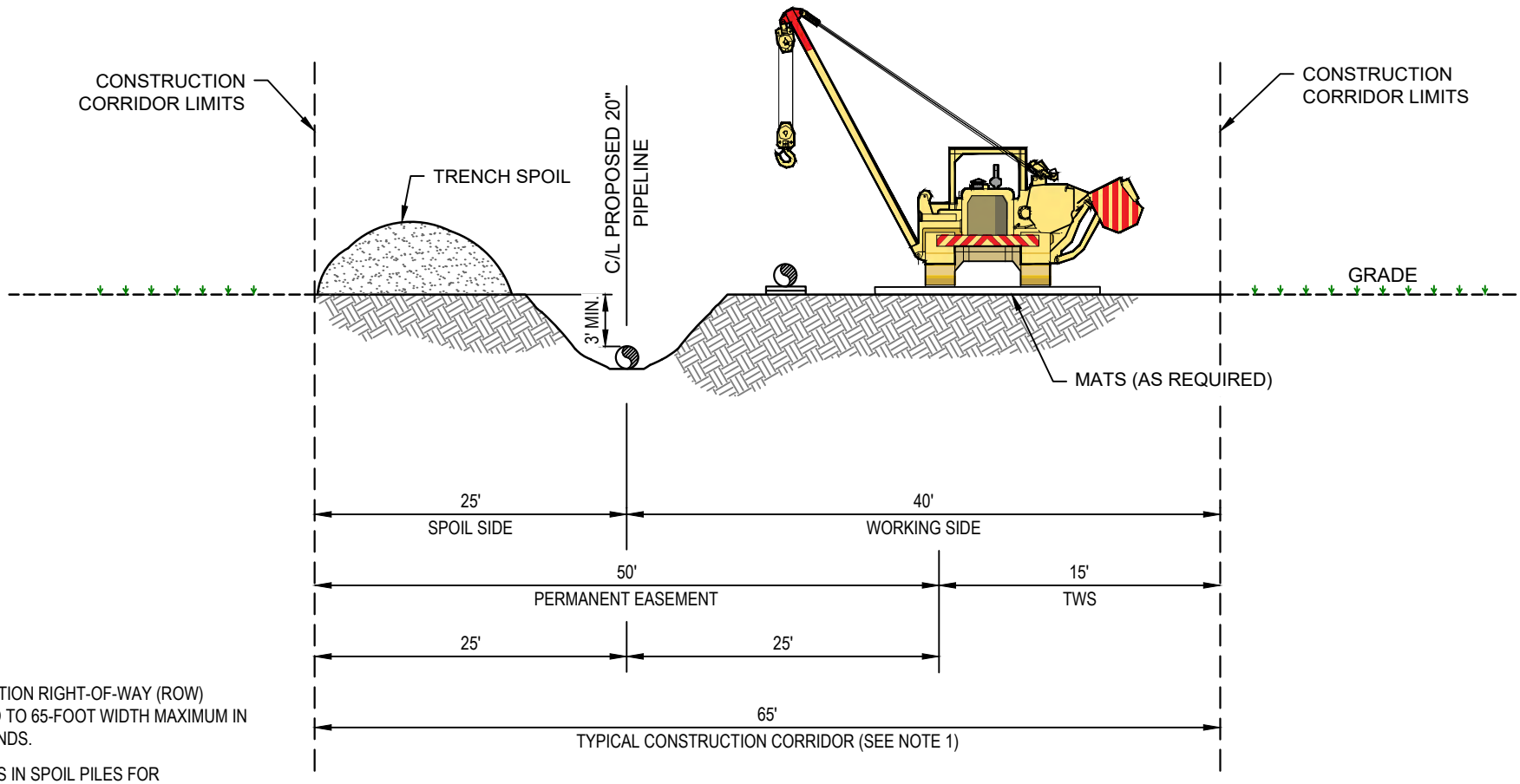
FF TRIO FARM LATERAL
TYPICAL UNSATURATED WETLAND WORKSPACE
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
FIGURE-007

REV
A

TYPICAL SATURATED WETLAND WORKSPACE - 20"

N.T.S.



NOTES:

1. CONSTRUCTION RIGHT-OF-WAY (ROW) NARROWED TO 65-FOOT WIDTH MAXIMUM IN ALL WETLANDS.
2. LEAVE GAPS IN SPOIL PILES FOR LIVESTOCK AND AT OBVIOUS DRAINAGE. DO NOT PUSH UPLAND SOILS INTO CANALS OR WETLANDS.
3. TOPSOIL SEGREGATION SHALL NOT BE REQUIRED IN SATURATED SOIL CONDITIONS.
4. VARIATIONS FOR STAGING AREAS/ADDITIONAL TEMPORARY WORKSPACE MAY BE NECESSARY DUE TO SITE-SPECIFIC TERRAIN FEATURES/CROSSINGS.
5. RESTORE GRADE TO NEAR PRE-CONSTRUCTION TOPOGRAPHY. REPLACE TOPSOIL AND INSTALL PERMANENT EROSION CONTROL IF APPLICABLE.

REVISIONS

REV#	REV DATE	REV NAME	DRN	CHK	APP
A	05-06-2026	ISSUE FOR PERMIT	HW	JD	CF

PREPARED BY: FIRM No:EF.0007320

TRC

8521 McHARD RD, HOUSTON, TX 77053

ENERGY TRANSFER

FF TRIO FARM LATERAL
TYPICAL SATURATED WETLAND WORKSPACE
CONSTRUCTION TYPICALS
RICHLAND PARISH, LOUISIANA

DWG NUMBER
FIGURE-008

REV
A

APPENDIX B

Typical and Site-Specific Detail Drawings

APPENDIX B-2

Plot Plans

Submitted as CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE

APPENDIX C

Project Construction and Mitigation Plans

APPENDIX C-1

FERC *Upland Erosion Control, Revegetation, and Maintenance Plan* (2013)

Federal Energy Regulatory Commission

Office of Energy Projects

May 2013

UPLAND EROSION CONTROL, REVEGETATION, AND MAINTENANCE PLAN

Washington, DC 20426

MAY 2013 VERSION

UPLAND EROSION CONTROL, REVEGETATION, AND MAINTENANCE PLAN

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UPLAND EROSION CONTROL, REVEGETATION, AND MAINTENANCE PLAN (PLAN)

I. APPLICABILITY

A. The intent of this Plan is to assist project sponsors by identifying baseline mitigation measures for minimizing erosion and enhancing revegetation. Project sponsors shall specify in their applications for a new FERC authorization and in prior notice and advance notice filings, any individual measures in this Plan they consider unnecessary, technically infeasible, or unsuitable due to local conditions and fully describe any alternative measures they would use. Project sponsors shall also explain how those alternative measures would achieve a comparable level of mitigation.

Once a project is authorized, project sponsors can request further changes as variances to the measures in this Plan (or the applicant's approved plan). The Director of the Office of Energy Projects (Director) will consider approval of variances upon the project sponsor's written request, if the Director agrees that a variance:

1. provides equal or better environmental protection;

2. is necessary because a portion of this Plan is infeasible or unworkable based on project-specific conditions; or

3. is specifically required in writing by another federal, state, or Native American land management agency for the portion of the project on its land or under its jurisdiction.

Sponsors of projects planned for construction under the automatic authorization provisions in the FERC's regulations must receive written approval for any variances in advance of construction.

Project-related impacts on wetland and waterbody systems are addressed in the staff's Wetland and Waterbody Construction and Mitigation Procedures (Procedures).

II. SUPERVISION AND INSPECTION

A. ENVIRONMENTAL INSPECTION

1. At least one Environmental Inspector is required for each construction spread during construction and restoration (as defined by section V). The number and experience of Environmental Inspectors assigned to each construction spread shall be appropriate for the length of the construction spread and the number/significance of resources affected.

2. Environmental Inspectors shall have peer status with all other activity inspectors.

3. Environmental Inspectors shall have the authority to stop activities that violate the environmental conditions of the FERC's Orders, stipulations of other environmental permits or approvals, or landowner easement agreements; and to order appropriate corrective action.

B. RESPONSIBILITIES OF ENVIRONMENTAL INSPECTORS

At a minimum, the Environmental Inspector(s) shall be responsible for:

1. Inspecting construction activities for compliance with the requirements of this Plan, the Procedures, the environmental conditions of the FERC's Orders, the mitigation measures proposed by the project sponsor (as approved and/or modified by the Order), other environmental permits and approvals, and environmental requirements in landowner easement agreements.

2. Identifying, documenting, and overseeing corrective actions, as necessary to bring an activity back into compliance;

3. Verifying that the limits of authorized construction work areas and locations of access roads are visibly marked before clearing, and maintained throughout construction;

4. Verifying the location of signs and highly visible flagging marking the boundaries of sensitive resource areas, waterbodies, wetlands, or areas with special requirements along the construction work area;

5. Identifying erosion/sediment control and soil stabilization needs in all areas;

6. Ensuring that the design of slope breakers will not cause erosion or direct water into sensitive environmental resource areas, including cultural resource sites, wetlands, waterbodies, and sensitive species habitats;

7. Verifying that dewatering activities are properly monitored and do not result in the deposition of sand, silt, and/or sediment into sensitive environmental resource areas, including wetlands, waterbodies, cultural resource sites, and sensitive species habitats; stopping dewatering activities if such deposition is occurring and ensuring the design of the discharge is changed to prevent reoccurrence; and verifying that dewatering structures are removed after completion of dewatering activities;

8. Ensuring that subsoil and topsoil are tested in agricultural and residential areas to measure compaction and determine the need for corrective action;

9. Advising the Chief Construction Inspector when environmental conditions (such as wet weather or frozen soils) make it advisable to restrict or delay construction activities to avoid topsoil mixing or excessive compaction;

10. Ensuring restoration of contours and topsoil;

11. Verifying that the soils imported for agricultural or residential use are certified as free of noxious weeds and soil pests, unless otherwise approved by the landowner;

12. Ensuring that erosion control devices are properly installed to prevent sediment flow into sensitive environmental resource areas (e.g., wetlands, waterbodies, cultural resource sites, and sensitive species habitats) and onto roads, and determining the need for additional erosion control devices;

13. Inspecting and ensuring the maintenance of temporary erosion control measures at least:

    a. on a daily basis in areas of active construction or equipment operation;

    b. on a weekly basis in areas with no construction or equipment operation; and

    c. within 24 hours of each 0.5 inch of rainfall;

14. Ensuring the repair of all ineffective temporary erosion control measures within 24 hours of identification, or as soon as conditions allow if compliance with this time frame would result in greater environmental impacts;

15. Keeping records of compliance with the environmental conditions of the FERC's Orders, and the mitigation measures proposed by the project sponsor in the application submitted to the FERC, and other federal or state environmental permits during active construction and restoration;

16. Identifying areas that should be given special attention to ensure stabilization and restoration after the construction phase; and

17. Verifying that locations for any disposal of excess construction materials for beneficial reuse comply with section III.E.

III. PRECONSTRUCTION PLANNING

The project sponsor shall do the following before construction:

A. CONSTRUCTION WORK AREAS

1. Identify all construction work areas (e.g., construction right-of-way, extra work space areas, pipe storage and contractor yards, borrow and disposal areas, access roads) that would be needed for safe construction. The project sponsor must ensure that appropriate cultural resources and biological surveys are conducted, as determined necessary by the appropriate federal and state agencies.

2. Project sponsors are encouraged to consider expanding any required cultural resources and endangered species surveys in anticipation of the need for activities outside of authorized work areas.

3. Plan construction sequencing to limit the amount and duration of open trench sections, as necessary, to prevent excessive erosion or sediment flow into sensitive environmental resource areas.

B. DRAIN TILE AND IRRIGATION SYSTEMS

1. Attempt to locate existing drain tiles and irrigation systems.

2. Contact landowners and local soil conservation authorities to determine the locations of future drain tiles that are likely to be installed within 3 years of the authorized construction.

3. Develop procedures for constructing through drain-tiled areas, maintaining irrigation systems during construction, and repairing drain tiles and irrigation systems after construction.

4. Engage qualified drain tile specialists, as needed to conduct or monitor repairs to drain tile systems affected by construction. Use drain tile specialists from the project area, if available.

C. GRAZING DEFERMENT

Develop grazing deferment plans with willing landowners, grazing permittees, and land management agencies to minimize grazing disturbance of revegetation efforts.

D. ROAD CROSSINGS AND ACCESS POINTS

Plan for safe and accessible conditions at all roadway crossings and access points during construction and restoration.

E. DISPOSAL PLANNING

Determine methods and locations for the regular collection, containment, and disposal of excess construction materials and debris (e.g., timber, slash, mats, garbage, drill cuttings and fluids, excess rock) throughout the construction process. Disposal of materials for beneficial reuse must not result in adverse environmental impact and is subject to compliance with all applicable survey, landowner or land management agency approval, and permit requirements.

F. AGENCY COORDINATION

The project sponsor must coordinate with the appropriate local, state, and federal agencies as outlined in this Plan and/or required by the FERC's Orders.

1. Obtain written recommendations from the local soil conservation authorities or land management agencies regarding permanent erosion control and revegetation specifications.

2. Develop specific procedures in coordination with the appropriate agencies to prevent the introduction or spread of invasive species, noxious weeds, and soil pests resulting from construction and restoration activities.

3. Develop specific procedures in coordination with the appropriate agencies and landowners, as necessary, to allow for livestock and wildlife movement and protection during construction.

4. Develop specific blasting procedures in coordination with the appropriate agencies that address pre- and post-blast inspections; advanced public notification; and mitigation measures for building foundations, groundwater wells, and springs. Use appropriate methods (e.g., blasting mats) to prevent damage to nearby structures and to prevent debris from entering sensitive environmental resource areas.

G. SPILL PREVENTION AND RESPONSE PROCEDURES

The project sponsor shall develop project-specific Spill Prevention and Response Procedures, as specified in section IV of the staff's Procedures. A copy must be filed with the Secretary of the FERC (Secretary) prior to construction and made available in the field on each construction spread. The filing requirement does not apply to projects constructed under the automatic authorization provisions in the FERC's regulations.

H. RESIDENTIAL CONSTRUCTION

For all properties with residences located within 50 feet of construction work areas, project sponsors shall: avoid removal of mature trees and landscaping within the construction work area unless necessary for safe operation of construction equipment, or as specified in landowner agreements; fence the edge of the construction work area for a distance of 100 feet on either side of the residence; and restore all lawn areas and landscaping immediately following clean up operations, or as specified in landowner agreements. If seasonal or other weather conditions prevent compliance with these time frames, maintain and monitor temporary erosion controls (sediment barriers and mulch) until conditions allow completion of restoration.

I. WINTER CONSTRUCTION PLANS

If construction is planned to occur during winter weather conditions, project sponsors shall develop and file a project-specific winter construction plan with the FERC application. This filing requirement does not apply to projects constructed under the automatic authorization provisions of the FERC's regulations.

The plan shall address:

1. winter construction procedures (e.g., snow handling and removal, access road construction and maintenance, soil handling under saturated or frozen conditions, topsoil stripping);

2. stabilization and monitoring procedures if ground conditions will delay restoration until the following spring (e.g., mulching and erosion controls, inspection and reporting, stormwater control during spring thaw conditions); and

3. final restoration procedures (e.g., subsidence and compaction repair, topsoil replacement, seeding).

IV. INSTALLATION

A. APPROVED AREAS OF DISTURBANCE

1. Project-related ground disturbance shall be limited to the construction right-of-way, extra work space areas, pipe storage yards, borrow and disposal areas, access roads, and other areas approved in the FERC's Orders. Any project-related ground disturbing activities outside these areas will require prior Director approval. This requirement does not apply to activities needed to comply with the Plan and Procedures (i.e., slope breakers, energy-dissipating devices, dewatering structures, drain tile system repairs) or minor field realignments and workspace shifts per landowner needs and requirements that do not affect other landowners or sensitive environmental resource areas. All construction or restoration activities outside of authorized areas are subject to all applicable survey and permit requirements, and landowner easement agreements.

2. The construction right-of-way width for a project shall not exceed 75 feet or that described in the FERC application unless otherwise modified by a FERC Order. However, in limited, non-wetland areas, this construction right-of-way width may be expanded by up to 25 feet without Director approval to accommodate full construction right-of-way topsoil segregation and to ensure safe construction where topographic conditions (e.g., side-slopes) or soil limitations require it. Twenty-five feet of extra construction right-of-way width may also be used in limited, non-wetland or non-forested areas for truck turn-arounds where no reasonable alternative access exists.

   Project use of these additional limited areas is subject to landowner or land management agency approval and compliance with all applicable survey and permit requirements. When additional areas are used, each one shall be identified and the need explained in the weekly or biweekly construction reports to the FERC, if required. The following material shall be included in the reports:

   a. the location of each additional area by station number and reference to previously filed alignment sheets, or updated alignment sheets showing the additional areas;

   b. identification of the filing at FERC containing evidence that the additional areas were previously surveyed; and

c. a statement that landowner approval has been obtained and is available in project files.

Prior written approval of the Director is required when the authorized construction right-of-way width would be expanded by more than 25 feet.

B. TOPSOIL SEGREGATION

1. Unless the landowner or land management agency specifically approves otherwise, prevent the mixing of topsoil with subsoil by stripping topsoil from either the full work area or from the trench and subsoil storage area (ditch plus spoil side method) in:

   a. cultivated or rotated croplands, and managed pastures;

   b. residential areas;

   c. hayfields; and

   d. other areas at the landowner's or land managing agency's request.

2. In residential areas, importation of topsoil is an acceptable alternative to topsoil segregation.

3. Where topsoil segregation is required, the project sponsor must:

   a. segregate at least 12 inches of topsoil in deep soils (more than 12 inches of topsoil); and

   b. make every effort to segregate the entire topsoil layer in soils with less than 12 inches of topsoil.

4. Maintain separation of salvaged topsoil and subsoil throughout all construction activities.

5. Segregated topsoil may not be used for padding the pipe, constructing temporary slope breakers or trench plugs, improving or maintaining roads, or as a fill material.

6. Stabilize topsoil piles and minimize loss due to wind and water erosion with use of sediment barriers, mulch, temporary seeding, tackifiers, or functional equivalents, where necessary.

C. DRAIN TILES

1. Mark locations of drain tiles damaged during construction.

2. Probe all drainage tile systems within the area of disturbance to check for damage.

3. Repair damaged drain tiles to their original or better condition. Do not use filter-covered drain tiles unless the local soil conservation authorities and the landowner agree. Use qualified specialists for testing and repairs.

4. For new pipelines in areas where drain tiles exist or are planned, ensure that the depth of cover over the pipeline is sufficient to avoid interference with drain tile systems. For adjacent pipeline loops in agricultural areas, install the new pipeline with at least the same depth of cover as the existing pipeline(s).

D. IRRIGATION

Maintain water flow in crop irrigation systems, unless shutoff is coordinated with affected parties.

E. ROAD CROSSINGS AND ACCESS POINTS

1. Maintain safe and accessible conditions at all road crossings and access points during construction.

2. If crushed stone access pads are used in residential or agricultural areas, place the stone on synthetic fabric to facilitate removal.

3. Minimize the use of tracked equipment on public roadways. Remove any soil or gravel spilled or tracked onto roadways daily or more frequent as necessary to maintain safe road conditions. Repair any damages to roadway surfaces, shoulders, and bar ditches.

F. TEMPORARY EROSION CONTROL

Install temporary erosion controls immediately after initial disturbance of the soil. Temporary erosion controls must be properly maintained throughout construction (on a daily basis) and reinstalled as necessary (such as after backfilling of the trench) until replaced by permanent erosion controls or restoration is complete.

1. Temporary Slope Breakers

   a. Temporary slope breakers are intended to reduce runoff velocity and divert water off the construction right-of-way. Temporary slope

breakers may be constructed of materials such as soil, silt fence, staked hay or straw bales, or sand bags.

b. Install temporary slope breakers on all disturbed areas, as necessary to avoid excessive erosion. Temporary slope breakers must be installed on slopes greater than 5 percent where the base of the slope is less than 50 feet from waterbody, wetland, and road crossings at the following spacing (closer spacing shall be used if necessary):

Slope (%)	Spacing (feet)
5 - 15	300
>15 - 30	200
>30	100

c. Direct the outfall of each temporary slope breaker to a stable, well vegetated area or construct an energy-dissipating device at the end of the slope breaker and off the construction right-of-way.

d. Position the outfall of each temporary slope breaker to prevent sediment discharge into wetlands, waterbodies, or other sensitive environmental resource areas.

2. Temporary Trench Plugs

Temporary trench plugs are intended to segment a continuous open trench prior to backfill.

a. Temporary trench plugs may consist of unexcavated portions of the trench, compacted subsoil, sandbags, or some functional equivalent.

b. Position temporary trench plugs, as necessary, to reduce trenchline erosion and minimize the volume and velocity of trench water flow at the base of slopes.

3. Sediment Barriers

Sediment barriers are intended to stop the flow of sediments and to prevent the deposition of sediments beyond approved workspaces or into sensitive resources.

a. Sediment barriers may be constructed of materials such as silt fence, staked hay or straw bales, compacted earth (e.g., driveable berms across travelways), sand bags, or other appropriate materials.

b. At a minimum, install and maintain temporary sediment barriers across the entire construction right-of-way at the base of slopes greater than 5 percent where the base of the slope is less than 50 feet from a waterbody, wetland, or road crossing until revegetation is successful as defined in this Plan. Leave adequate room between the base of the slope and the sediment barrier to accommodate ponding of water and sediment deposition.

c. Where wetlands or waterbodies are adjacent to and downslope of construction work areas, install sediment barriers along the edge of these areas, as necessary to prevent sediment flow into the wetland or waterbody.

4. Mulch

a. Apply mulch on all slopes (except in cultivated cropland) concurrent with or immediately after seeding, where necessary to stabilize the soil surface and to reduce wind and water erosion. Spread mulch uniformly over the area to cover at least 75 percent of the ground surface at a rate of 2 tons/acre of straw or its equivalent, unless the local soil conservation authority, landowner, or land managing agency approves otherwise in writing.

b. Mulch can consist of weed-free straw or hay, wood fiber hydromulch, erosion control fabric, or some functional equivalent.

c. Mulch all disturbed upland areas (except cultivated cropland) before seeding if:

(1) final grading and installation of permanent erosion control measures will not be completed in an area within 20 days after the trench in that area is backfilled (10 days in residential areas), as required in section V.A.1; or

(2) construction or restoration activity is interrupted for extended periods, such as when seeding cannot be completed due to seeding period restrictions.

d. If mulching before seeding, increase mulch application on all slopes within 100 feet of waterbodies and wetlands to a rate of 3 tons/acre of straw or equivalent.

e. If wood chips are used as mulch, do not use more than 1 ton/acre and add the equivalent of 11 lbs/acre available nitrogen (at least 50 percent of which is slow release).

f. Ensure that mulch is adequately anchored to minimize loss due to wind and water.

g. When anchoring with liquid mulch binders, use rates recommended by the manufacturer. Do not use liquid mulch binders within 100 feet of wetlands or waterbodies, except where the product is certified environmentally non-toxic by the appropriate state or federal agency or independent standards-setting organization.

h. Do not use synthetic monofilament mesh/netted erosion control materials in areas designated as sensitive wildlife habitat, unless the product is specifically designed to minimize harm to wildlife. Anchor erosion control fabric with staples or other appropriate devices.

V. RESTORATION

A. CLEANUP

1. Commence cleanup operations immediately following backfill operations. Complete final grading, topsoil replacement, and installation of permanent erosion control structures within 20 days after backfilling the trench (10 days in residential areas). If seasonal or other weather conditions prevent compliance with these time frames, maintain temporary erosion controls (i.e., temporary slope breakers, sediment barriers, and mulch) until conditions allow completion of cleanup.

   If construction or restoration unexpectedly continues into the winter season when conditions could delay successful decompaction, topsoil replacement, or seeding until the following spring, file with the Secretary for the review and written approval of the Director, a winter construction plan (as specified in section III.I). This filing requirement does not apply to projects constructed under the automatic authorization provisions of the FERC's regulations.

2. A travel lane may be left open temporarily to allow access by construction traffic if the temporary erosion control structures are installed as specified in section IV.F. and inspected and maintained as specified in sections II.B.12 through 14. When access is no longer required the travel lane must be removed and the right-of-way restored.

3. Rock excavated from the trench may be used to backfill the trench only to the top of the existing bedrock profile. Rock that is not returned to the trench shall be considered construction debris, unless approved for use as mulch or for some other use on the construction work areas by the landowner or land managing agency.

4. Remove excess rock from at least the top 12 inches of soil in all cultivated or rotated cropland, managed pastures, hayfields, and residential areas, as well as other areas at the landowner's request. The size, density, and distribution of rock on the construction work area shall be similar to adjacent areas not disturbed by construction. The landowner or land management agency may approve other provisions in writing.

5. Grade the construction right-of-way to restore pre-construction contours and leave the soil in the proper condition for planting.

6. Remove construction debris from all construction work areas unless the landowner or land managing agency approves leaving materials onsite for beneficial reuse, stabilization, or habitat restoration.

7. Remove temporary sediment barriers when replaced by permanent erosion control measures or when revegetation is successful.

B. PERMANENT EROSION CONTROL DEVICES

1. Trench Breakers

a. Trench breakers are intended to slow the flow of subsurface water along the trench. Trench breakers may be constructed of materials such as sand bags or polyurethane foam. Do not use topsoil in trench breakers.

b. An engineer or similarly qualified professional shall determine the need for and spacing of trench breakers. Otherwise, trench breakers shall be installed at the same spacing as and upslope of permanent slope breakers.

c. In agricultural fields and residential areas where slope breakers are not typically required, install trench breakers at the same spacing as if permanent slope breakers were required.

d. At a minimum, install a trench breaker at the base of slopes greater than 5 percent where the base of the slope is less than 50 feet from a waterbody or wetland and where needed to avoid draining a waterbody or wetland. Install trench breakers at wetland boundaries, as specified in the Procedures. Do not install trench breakers within a wetland.

2. Permanent Slope Breakers

a. Permanent slope breakers are intended to reduce runoff velocity, divert water off the construction right-of-way, and prevent sediment deposition into sensitive resources. Permanent slope breakers may be constructed of materials such as soil, stone, or some functional equivalent.

b. Construct and maintain permanent slope breakers in all areas, except cultivated areas and lawns, unless requested by the landowner, using spacing recommendations obtained from the local soil conservation authority or land managing agency.

In the absence of written recommendations, use the following spacing unless closer spacing is necessary to avoid excessive erosion on the construction right-of-way:

Slope (%)	Spacing (feet)
5 - 15	300
>15 - 30	200
>30	100

c. Construct slope breakers to divert surface flow to a stable area without causing water to pool or erode behind the breaker. In the absence of a stable area, construct appropriate energy-dissipating devices at the end of the breaker.

d. Slope breakers may extend slightly (about 4 feet) beyond the edge of the construction right-of-way to effectively drain water off the disturbed area. Where slope breakers extend beyond the edge of the construction right-of-way, they are subject to compliance with all applicable survey requirements.

C. SOIL COMPACTION MITIGATION

1. Test topsoil and subsoil for compaction at regular intervals in agricultural and residential areas disturbed by construction activities. Conduct tests on the same soil type under similar moisture conditions in undisturbed areas to approximate preconstruction conditions. Use penetrometers or other appropriate devices to conduct tests.

2. Plow severely compacted agricultural areas with a paraplow or other deep tillage implement. In areas where topsoil has been segregated, plow the subsoil before replacing the segregated topsoil.

If subsequent construction and cleanup activities result in further compaction, conduct additional tilling.

3. Perform appropriate soil compaction mitigation in severely compacted residential areas.

D. REVEGETATION

1. General

a. The project sponsor is responsible for ensuring successful revegetation of soils disturbed by project-related activities, except as noted in section V.D.1.b.

b. Restore all turf, ornamental shrubs, and specialized landscaping in accordance with the landowner's request, or compensate the landowner. Restoration work must be performed by personnel familiar with local horticultural and turf establishment practices.

2. Soil Additives

Fertilize and add soil pH modifiers in accordance with written recommendations obtained from the local soil conservation authority, land management agencies, or landowner. Incorporate recommended soil pH modifier and fertilizer into the top 2 inches of soil as soon as practicable after application.

3. Seeding Requirements

a. Prepare a seedbed in disturbed areas to a depth of 3 to 4 inches using appropriate equipment to provide a firm seedbed. When hydroseeding, scarify the seedbed to facilitate lodging and germination of seed.

b. Seed disturbed areas in accordance with written recommendations for seed mixes, rates, and dates obtained from the local soil conservation authority or the request of the landowner or land management agency. Seeding is not required in cultivated croplands unless requested by the landowner.

c. Perform seeding of permanent vegetation within the recommended seeding dates. If seeding cannot be done within those dates, use appropriate temporary erosion control measures discussed in section IV.F and perform seeding of permanent vegetation at the beginning of the next recommended seeding season. Dormant seeding or temporary

seeding of annual species may also be used, if necessary, to establish cover, as approved by the Environmental Inspector. Lawns may be seeded on a schedule established with the landowner.

d. In the absence of written recommendations from the local soil conservation authorities, seed all disturbed soils within 6 working days of final grading, weather and soil conditions permitting, subject to the specifications in section V.D.3.a through V.D.3.c.

e. Base seeding rates on Pure Live Seed. Use seed within 12 months of seed testing.

f. Treat legume seed with an inoculant specific to the species using the manufacturer's recommended rate of inoculant appropriate for the seeding method (broadcast, drill, or hydro).

g. In the absence of written recommendations from the local soil conservation authorities, landowner, or land managing agency to the contrary, a seed drill equipped with a cultipacker is preferred for seed application.

Broadcast or hydroseeding can be used in lieu of drilling at double the recommended seeding rates. Where seed is broadcast, firm the seedbed with a cultipacker or roller after seeding. In rocky soils or where site conditions may limit the effectiveness of this equipment, other alternatives may be appropriate (e.g., use of a chain drag) to lightly cover seed after application, as approved by the Environmental Inspector.

VI. OFF-ROAD VEHICLE CONTROL

To each owner or manager of forested lands, offer to install and maintain measures to control unauthorized vehicle access to the right-of-way. These measures may include:

A. signs;

B. fences with locking gates;

C. slash and timber barriers, pipe barriers, or a line of boulders across the right-of-way; and

D. conifers or other appropriate trees or shrubs across the right-of-way.

VII. POST-CONSTRUCTION ACTIVITIES AND REPORTING

A. MONITORING AND MAINTENANCE

1. Conduct follow-up inspections of all disturbed areas, as necessary, to determine the success of revegetation and address landowner concerns. At a minimum, conduct inspections after the first and second growing seasons.

2. Revegetation in non-agricultural areas shall be considered successful if upon visual survey the density and cover of non-nuisance vegetation are similar in density and cover to adjacent undisturbed lands. In agricultural areas, revegetation shall be considered successful when upon visual survey, crop growth and vigor are similar to adjacent undisturbed portions of the same field, unless the easement agreement specifies otherwise.

   Continue revegetation efforts until revegetation is successful.

3. Monitor and correct problems with drainage and irrigation systems resulting from pipeline construction in agricultural areas until restoration is successful.

4. Restoration shall be considered successful if the right-of-way surface condition is similar to adjacent undisturbed lands, construction debris is removed (unless otherwise approved by the landowner or land managing agency per section V.A.6), revegetation is successful, and proper drainage has been restored.

5. Routine vegetation mowing or clearing over the full width of the permanent right-of-way in uplands shall not be done more frequently than every 3 years. However, to facilitate periodic corrosion/leak surveys, a corridor not exceeding 10 feet in width centered on the pipeline may be cleared at a frequency necessary to maintain the 10-foot corridor in an herbaceous state. In no case shall routine vegetation mowing or clearing occur during the migratory bird nesting season between April 15 and August 1 of any year unless specifically approved in writing by the responsible land management agency or the U.S. Fish and Wildlife Service.

6. Efforts to control unauthorized off-road vehicle use, in cooperation with the landowner, shall continue throughout the life of the project. Maintain signs, gates, and permanent access roads as necessary.

B. REPORTING

1. The project sponsor shall maintain records that identify by milepost:

   a. method of application, application rate, and type of fertilizer, pH modifying agent, seed, and mulch used;

   b. acreage treated;

   c. dates of backfilling and seeding;

   d. names of landowners requesting special seeding treatment and a description of the follow-up actions;

   e. the location of any subsurface drainage repairs or improvements made during restoration; and

   f. any problem areas and how they were addressed.

2. The project sponsor shall file with the Secretary quarterly activity reports documenting the results of follow-up inspections required by section VII.A.1; any problem areas, including those identified by the landowner; and corrective actions taken for at least 2 years following construction.

   The requirement to file quarterly activity reports with the Secretary does not apply to projects constructed under the automatic authorization, prior notice, or advanced notice provisions in the FERC's regulations.

APPENDIX C-2

FERC *Wetland and Waterbody Construction and Mitigation Procedures* (2013)

Federal Energy Regulatory Commission

Office of Energy Projects

May 2013

WETLAND AND WATERBODY CONSTRUCTION AND MITIGATION PROCEDURES

Washington, DC 20426

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WETLAND AND WATERBODY CONSTRUCTION AND MITIGATION PROCEDURES

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WETLAND AND WATERBODY
CONSTRUCTION AND MITIGATION PROCEDURES (PROCEDURES)

I. APPLICABILITY

A. The intent of these Procedures is to assist project sponsors by identifying baseline mitigation measures for minimizing the extent and duration of project-related disturbance on wetlands and waterbodies. Project sponsors shall specify in their applications for a new FERC authorization, and in prior notice and advance notice filings, any individual measures in these Procedures they consider unnecessary, technically infeasible, or unsuitable due to local conditions and fully describe any alternative measures they would use. Project sponsors shall also explain how those alternative measures would achieve a comparable level of mitigation.

Once a project is authorized, project sponsors can request further changes as variances to the measures in these Procedures (or the applicant's approved procedures). The Director of the Office of Energy Projects (Director) will consider approval of variances upon the project sponsor's written request, if the Director agrees that a variance:

1. provides equal or better environmental protection;

2. is necessary because a portion of these Procedures is infeasible or unworkable based on project-specific conditions; or

3. is specifically required in writing by another federal, state, or Native American land management agency for the portion of the project on its land or under its jurisdiction.

Sponsors of projects planned for construction under the automatic authorization provisions in the FERC's regulations must receive written approval for any variances in advance of construction.

Project-related impacts on non-wetland areas are addressed in the staff's Upland Erosion Control, Revegetation, and Maintenance Plan (Plan).

B. DEFINITIONS

1. "Waterbody" includes any natural or artificial stream, river, or drainage with perceptible flow at the time of crossing, and other permanent waterbodies such as ponds and lakes:

   a. "minor waterbody" includes all waterbodies less than or equal to 10 feet wide at the water's edge at the time of crossing;

   b. "intermediate waterbody" includes all waterbodies greater than 10 feet wide but less than or equal to 100 feet wide at the water's edge at the time of crossing; and

   c. "major waterbody" includes all waterbodies greater than 100 feet wide at the water's edge at the time of crossing.

2. "Wetland" includes any area that is not in actively cultivated or rotated cropland and that satisfies the requirements of the current federal methodology for identifying and delineating wetlands.

II. PRECONSTRUCTION FILING

A. The following information must be filed with the Secretary of the FERC (Secretary) prior to the beginning of construction, for the review and written approval by the Director:

1. site-specific justifications for extra work areas that would be closer than 50 feet from a waterbody or wetland; and

2. site-specific justifications for the use of a construction right-of-way greater than 75-feet-wide in wetlands.

B. The following information must be filed with the Secretary prior to the beginning of construction. These filing requirements do not apply to projects constructed under the automatic authorization provisions in the FERC's regulations:

1. Spill Prevention and Response Procedures specified in section IV.A;

2. a schedule identifying when trenching or blasting will occur within each waterbody greater than 10 feet wide, within any designated coldwater fishery, and within any waterbody identified as habitat for federally-listed threatened or endangered species. The project sponsor will revise the schedule as necessary to provide FERC staff at least 14 days advance notice. Changes within this last 14-day period must provide for at least 48 hours advance notice;

3. plans for horizontal directional drills (HDD) under wetlands or waterbodies, specified in section V.B.6.d;

4. site-specific plans for major waterbody crossings, described in section V.B.9;

5. a wetland delineation report as described in section VI.A.1, if applicable; and

6. the hydrostatic testing information specified in section VII.B.3.

III. ENVIRONMENTAL INSPECTORS

A. At least one Environmental Inspector having knowledge of the wetland and waterbody conditions in the project area is required for each construction spread. The number and experience of Environmental Inspectors assigned to each construction spread shall be appropriate for the length of the construction spread and the number/significance of resources affected.

B. The Environmental Inspector's responsibilities are outlined in the Upland Erosion Control, Revegetation, and Maintenance Plan (Plan).

IV. PRECONSTRUCTION PLANNING

A. The project sponsor shall develop project-specific Spill Prevention and Response Procedures that meet applicable requirements of state and federal agencies. A copy must be filed with the Secretary prior to construction and made available in the field on each construction spread. This filing requirement does not apply to projects constructed under the automatic authorization provisions in the FERC's regulations.

1. It shall be the responsibility of the project sponsor and its contractors to structure their operations in a manner that reduces the risk of spills or the accidental exposure of fuels or hazardous materials to waterbodies or wetlands. The project sponsor and its contractors must, at a minimum, ensure that:

a. all employees handling fuels and other hazardous materials are properly trained;

b. all equipment is in good operating order and inspected on a regular basis;

c. fuel trucks transporting fuel to on-site equipment travel only on approved access roads;

d. all equipment is parked overnight and/or fueled at least 100 feet from a waterbody or in an upland area at least 100 feet from a wetland boundary. These activities can occur closer only if the Environmental Inspector determines that there is no reasonable alternative, and the

project sponsor and its contractors have taken appropriate steps (including secondary containment structures) to prevent spills and provide for prompt cleanup in the event of a spill;

e. hazardous materials, including chemicals, fuels, and lubricating oils, are not stored within 100 feet of a wetland, waterbody, or designated municipal watershed area, unless the location is designated for such use by an appropriate governmental authority. This applies to storage of these materials and does not apply to normal operation or use of equipment in these areas;

f. concrete coating activities are not performed within 100 feet of a wetland or waterbody boundary, unless the location is an existing industrial site designated for such use. These activities can occur closer only if the Environmental Inspector determines that there is no reasonable alternative, and the project sponsor and its contractors have taken appropriate steps (including secondary containment structures) to prevent spills and provide for prompt cleanup in the event of a spill;

g. pumps operating within 100 feet of a waterbody or wetland boundary utilize appropriate secondary containment systems to prevent spills; and

h. bulk storage of hazardous materials, including chemicals, fuels, and lubricating oils have appropriate secondary containment systems to prevent spills.

2. The project sponsor and its contractors must structure their operations in a manner that provides for the prompt and effective cleanup of spills of fuel and other hazardous materials. At a minimum, the project sponsor and its contractors must:

a. ensure that each construction crew (including cleanup crews) has on hand sufficient supplies of absorbent and barrier materials to allow the rapid containment and recovery of spilled materials and knows the procedure for reporting spills and unanticipated discoveries of contamination;

b. ensure that each construction crew has on hand sufficient tools and material to stop leaks;

c. know the contact names and telephone numbers for all local, state, and federal agencies (including, if necessary, the U. S. Coast Guard and the National Response Center) that must be notified of a spill; and

d. follow the requirements of those agencies in cleaning up the spill, in excavating and disposing of soils or other materials contaminated by a spill, and in collecting and disposing of waste generated during spill cleanup.

B. AGENCY COORDINATION

The project sponsor must coordinate with the appropriate local, state, and federal agencies as outlined in these Procedures and in the FERC's Orders.

V. WATERBODY CROSSINGS

A. NOTIFICATION PROCEDURES AND PERMITS

1. Apply to the U.S. Army Corps of Engineers (COE), or its delegated agency, for the appropriate wetland and waterbody crossing permits.

2. Provide written notification to authorities responsible for potable surface water supply intakes located within 3 miles downstream of the crossing at least 1 week before beginning work in the waterbody, or as otherwise specified by that authority.

3. Apply for state-issued waterbody crossing permits and obtain individual or generic section 401 water quality certification or waiver.

4. Notify appropriate federal and state authorities at least 48 hours before beginning trenching or blasting within the waterbody, or as specified in applicable permits.

B. INSTALLATION

1. Time Window for Construction

Unless expressly permitted or further restricted by the appropriate federal or state agency in writing on a site-specific basis, instream work, except that required to install or remove equipment bridges, must occur during the following time windows:

a. coldwater fisheries - June 1 through September 30; and

b. coolwater and warmwater fisheries - June 1 through November 30.

2. Extra Work Areas

a. Locate all extra work areas (such as staging areas and additional spoil storage areas) at least 50 feet away from water's edge, except where

the adjacent upland consists of cultivated or rotated cropland or other disturbed land.

b. The project sponsor shall file with the Secretary for review and written approval by the Director, site-specific justification for each extra work area with a less than 50-foot setback from the water's edge, except where the adjacent upland consists of cultivated or rotated cropland or other disturbed land. The justification must specify the conditions that will not permit a 50-foot setback and measures to ensure the waterbody is adequately protected.

c. Limit the size of extra work areas to the minimum needed to construct the waterbody crossing.

3. General Crossing Procedures

a. Comply with the COE, or its delegated agency, permit terms and conditions.

b. Construct crossings as close to perpendicular to the axis of the waterbody channel as engineering and routing conditions permit.

c. Where pipelines parallel a waterbody, maintain at least 15 feet of undisturbed vegetation between the waterbody (and any adjacent wetland) and the construction right-of-way, except where maintaining this offset will result in greater environmental impact.

d. Where waterbodies meander or have multiple channels, route the pipeline to minimize the number of waterbody crossings.

e. Maintain adequate waterbody flow rates to protect aquatic life, and prevent the interruption of existing downstream uses.

f. Waterbody buffers (e.g., extra work area setbacks, refueling restrictions) must be clearly marked in the field with signs and/or highly visible flagging until construction-related ground disturbing activities are complete.

g. Crossing of waterbodies when they are dry or frozen and not flowing may proceed using standard upland construction techniques in accordance with the Plan, provided that the Environmental Inspector verifies that water is unlikely to flow between initial disturbance and final stabilization of the feature. In the event of perceptible flow, the project sponsor must comply with all applicable Procedure requirements for "waterbodies" as defined in section I.B.1.

4. Spoil Pile Placement and Control

   a. All spoil from minor and intermediate waterbody crossings, and upland spoil from major waterbody crossings, must be placed in the construction right-of-way at least 10 feet from the water's edge or in additional extra work areas as described in section V.B.2.

   b. Use sediment barriers to prevent the flow of spoil or silt-laden water into any waterbody.

5. Equipment Bridges

   a. Only clearing equipment and equipment necessary for installation of equipment bridges may cross waterbodies prior to bridge installation. Limit the number of such crossings of each waterbody to one per piece of clearing equipment.

   b. Construct and maintain equipment bridges to allow unrestricted flow and to prevent soil from entering the waterbody. Examples of such bridges include:

      (1) equipment pads and culvert(s);
      (2) equipment pads or railroad car bridges without culverts;
      (3) clean rock fill and culvert(s); and
      (4) flexi-float or portable bridges.

      Additional options for equipment bridges may be utilized that achieve the performance objectives noted above. Do not use soil to construct or stabilize equipment bridges.

   c. Design and maintain each equipment bridge to withstand and pass the highest flow expected to occur while the bridge is in place. Align culverts to prevent bank erosion or streambed scour. If necessary, install energy dissipating devices downstream of the culverts.

   d. Design and maintain equipment bridges to prevent soil from entering the waterbody.

   e. Remove temporary equipment bridges as soon as practicable after permanent seeding.

   f. If there will be more than 1 month between final cleanup and the beginning of permanent seeding and reasonable alternative access to the right-of-way is available, remove temporary equipment bridges as soon as practicable after final cleanup.

g. Obtain any necessary approval from the COE, or the appropriate state agency for permanent bridges.

6. Dry-Ditch Crossing Methods

a. Unless approved otherwise by the appropriate federal or state agency, install the pipeline using one of the dry-ditch methods outlined below for crossings of waterbodies up to 30 feet wide (at the water's edge at the time of construction) that are state-designated as either coldwater or significant coolwater or warmwater fisheries, or federally-designated as critical habitat.

b. Dam and Pump

(1) The dam-and-pump method may be used without prior approval for crossings of waterbodies where pumps can adequately transfer streamflow volumes around the work area, and there are no concerns about sensitive species passage.

(2) Implementation of the dam-and-pump crossing method must meet the following performance criteria:

(i) use sufficient pumps, including on-site backup pumps, to maintain downstream flows;
(ii) construct dams with materials that prevent sediment and other pollutants from entering the waterbody (e.g., sandbags or clean gravel with plastic liner);
(iii) screen pump intakes to minimize entrainment of fish;
(iv) prevent streambed scour at pump discharge; and
(v) continuously monitor the dam and pumps to ensure proper operation throughout the waterbody crossing.

c. Flume Crossing

The flume crossing method requires implementation of the following steps:

(1) install flume pipe after blasting (if necessary), but before any trenching;

(2) use sand bag or sand bag and plastic sheeting diversion structure or equivalent to develop an effective seal and to divert stream flow through the flume pipe (some modifications to the stream bottom may be required to achieve an effective seal);

(3) properly align flume pipe(s) to prevent bank erosion and streambed scour;

(4) do not remove flume pipe during trenching, pipelaying, or backfilling activities, or initial streambed restoration efforts; and

(5) remove all flume pipes and dams that are not also part of the equipment bridge as soon as final cleanup of the stream bed and bank is complete.

d. Horizontal Directional Drill

For each waterbody or wetland that would be crossed using the HDD method, file with the Secretary for the review and written approval by the Director, a plan that includes:

(1) site-specific construction diagrams that show the location of mud pits, pipe assembly areas, and all areas to be disturbed or cleared for construction;

(2) justification that disturbed areas are limited to the minimum needed to construct the crossing;

(3) identification of any aboveground disturbance or clearing between the HDD entry and exit workspaces during construction;

(4) a description of how an inadvertent release of drilling mud would be contained and cleaned up; and

(5) a contingency plan for crossing the waterbody or wetland in the event the HDD is unsuccessful and how the abandoned drill hole would be sealed, if necessary.

The requirement to file HDD plans does not apply to projects constructed under the automatic authorization provisions in the FERC's regulations.

7. Crossings of Minor Waterbodies

Where a dry-ditch crossing is not required, minor waterbodies may be crossed using the open-cut crossing method, with the following restrictions:

a. except for blasting and other rock breaking measures, complete instream construction activities (including trenching, pipe installation, backfill, and restoration of the streambed contours) within 24 hours.

Streambanks and unconsolidated streambeds may require additional restoration after this period;

b. limit use of equipment operating in the waterbody to that needed to construct the crossing; and

c. equipment bridges are not required at minor waterbodies that do not have a state-designated fishery classification or protected status (e.g., agricultural or intermittent drainage ditches). However, if an equipment bridge is used it must be constructed as described in section V.B.5.

8. Crossings of Intermediate Waterbodies

Where a dry-ditch crossing is not required, intermediate waterbodies may be crossed using the open-cut crossing method, with the following restrictions:

a. complete instream construction activities (not including blasting and other rock breaking measures) within 48 hours, unless site-specific conditions make completion within 48 hours infeasible;

b. limit use of equipment operating in the waterbody to that needed to construct the crossing; and

c. all other construction equipment must cross on an equipment bridge as specified in section V.B.5.

9. Crossings of Major Waterbodies

Before construction, the project sponsor shall file with the Secretary for the review and written approval by the Director a detailed, site-specific construction plan and scaled drawings identifying all areas to be disturbed by construction for each major waterbody crossing (the scaled drawings are not required for any offshore portions of pipeline projects). This plan must be developed in consultation with the appropriate state and federal agencies and shall include extra work areas, spoil storage areas, sediment control structures, etc., as well as mitigation for navigational issues. The requirement to file major waterbody crossing plans does not apply to projects constructed under the automatic authorization provisions of the FERC's regulations.

The Environmental Inspector may adjust the final placement of the erosion and sediment control structures in the field to maximize effectiveness.

10. Temporary Erosion and Sediment Control

Install sediment barriers (as defined in section IV.F.3.a of the Plan) immediately after initial disturbance of the waterbody or adjacent upland.

Sediment barriers must be properly maintained throughout construction and reinstalled as necessary (such as after backfilling of the trench) until replaced by permanent erosion controls or restoration of adjacent upland areas is complete. Temporary erosion and sediment control measures are addressed in more detail in the Plan; however, the following specific measures must be implemented at stream crossings:

a. install sediment barriers across the entire construction right-of-way at all waterbody crossings, where necessary to prevent the flow of sediments into the waterbody. Removable sediment barriers (or driveable berms) must be installed across the travel lane. These removable sediment barriers can be removed during the construction day, but must be re-installed after construction has stopped for the day and/or when heavy precipitation is imminent;

b. where waterbodies are adjacent to the construction right-of-way and the right-of-way slopes toward the waterbody, install sediment barriers along the edge of the construction right-of-way as necessary to contain spoil within the construction right-of-way and prevent sediment flow into the waterbody; and

c. use temporary trench plugs at all waterbody crossings, as necessary, to prevent diversion of water into upland portions of the pipeline trench and to keep any accumulated trench water out of the waterbody.

11. Trench Dewatering

Dewater the trench (either on or off the construction right-of-way) in a manner that does not cause erosion and does not result in silt-laden water flowing into any waterbody. Remove the dewatering structures as soon as practicable after the completion of dewatering activities.

C. RESTORATION

1. Use clean gravel or native cobbles for the upper 1 foot of trench backfill in all waterbodies that contain coldwater fisheries.

2. For open-cut crossings, stabilize waterbody banks and install temporary sediment barriers within 24 hours of completing instream construction activities. For dry-ditch crossings, complete streambed and bank stabilization before returning flow to the waterbody channel.

3. Return all waterbody banks to preconstruction contours or to a stable angle of repose as approved by the Environmental Inspector.

4. Install erosion control fabric or a functional equivalent on waterbody banks at the time of final bank recontouring. Do not use synthetic monofilament

mesh/netted erosion control materials in areas designated as sensitive wildlife habitat unless the product is specifically designed to minimize harm to wildlife. Anchor erosion control fabric with staples or other appropriate devices.

5. Application of riprap for bank stabilization must comply with COE, or its delegated agency, permit terms and conditions.

6. Unless otherwise specified by state permit, limit the use of riprap to areas where flow conditions preclude effective vegetative stabilization techniques such as seeding and erosion control fabric.

7. Revegetate disturbed riparian areas with native species of conservation grasses, legumes, and woody species, similar in density to adjacent undisturbed lands.

8. Install a permanent slope breaker across the construction right-of-way at the base of slopes greater than 5 percent that are less than 50 feet from the waterbody, or as needed to prevent sediment transport into the waterbody. In addition, install sediment barriers as outlined in the Plan.

   In some areas, with the approval of the Environmental Inspector, an earthen berm may be suitable as a sediment barrier adjacent to the waterbody.

9. Sections V.C.3 through V.C.7 above also apply to those perennial or intermittent streams not flowing at the time of construction.

D. POST-CONSTRUCTION MAINTENANCE

1. Limit routine vegetation mowing or clearing adjacent to waterbodies to allow a riparian strip at least 25 feet wide, as measured from the waterbody's mean high water mark, to permanently revegetate with native plant species across the entire construction right-of-way. However, to facilitate periodic corrosion/leak surveys, a corridor centered on the pipeline and up to 10 feet wide may be cleared at a frequency necessary to maintain the 10-foot corridor in an herbaceous state. In addition, trees that are located within 15 feet of the pipeline that have roots that could compromise the integrity of the pipeline coating may be cut and removed from the permanent right-of-way. Do not conduct any routine vegetation mowing or clearing in riparian areas that are between HDD entry and exit points.

2. Do not use herbicides or pesticides in or within 100 feet of a waterbody except as allowed by the appropriate land management or state agency.

3. Time of year restrictions specified in section VII.A.5 of the Plan (April 15 – August 1 of any year) apply to routine mowing and clearing of riparian areas.

VI. WETLAND CROSSINGS

A. GENERAL

1. The project sponsor shall conduct a wetland delineation using the current federal methodology and file a wetland delineation report with the Secretary before construction. The requirement to file a wetland delineation report does not apply to projects constructed under the automatic authorization provisions in the FERC's regulations.

   This report shall identify:

   a. by milepost all wetlands that would be affected;

   b. the National Wetlands Inventory (NWI) classification for each wetland;

   c. the crossing length of each wetland in feet; and

   d. the area of permanent and temporary disturbance that would occur in each wetland by NWI classification type.

   The requirements outlined in this section do not apply to wetlands in actively cultivated or rotated cropland. Standard upland protective measures, including workspace and topsoiling requirements, apply to these agricultural wetlands.

2. Route the pipeline to avoid wetland areas to the maximum extent possible. If a wetland cannot be avoided or crossed by following an existing right-of-way, route the new pipeline in a manner that minimizes disturbance to wetlands. Where looping an existing pipeline, overlap the existing pipeline right-of-way with the new construction right-of-way. In addition, locate the loop line no more than 25 feet away from the existing pipeline unless site-specific constraints would adversely affect the stability of the existing pipeline.

3. Limit the width of the construction right-of-way to 75 feet or less. Prior written approval of the Director is required where topographic conditions or soil limitations require that the construction right-of-way width within the boundaries of a federally delineated wetland be expanded beyond 75 feet. Early in the planning process the project sponsor is encouraged to identify site-specific areas where excessively wide trenches could occur and/or where spoil piles could be difficult to maintain because existing soils lack adequate unconfined compressive strength.

4. Wetland boundaries and buffers must be clearly marked in the field with signs and/or highly visible flagging until construction-related ground disturbing activities are complete.

5. Implement the measures of sections V and VI in the event a waterbody crossing is located within or adjacent to a wetland crossing. If all measures of sections V and VI cannot be met, the project sponsor must file with the Secretary a site-specific crossing plan for review and written approval by the Director before construction. This crossing plan shall address at a minimum:

   a. spoil control;

   b. equipment bridges;

   c. restoration of waterbody banks and wetland hydrology;

   d. timing of the waterbody crossing;

   e. method of crossing; and

   f. size and location of all extra work areas.

6. Do not locate aboveground facilities in any wetland, except where the location of such facilities outside of wetlands would prohibit compliance with U.S. Department of Transportation regulations.

B. INSTALLATION

1. Extra Work Areas and Access Roads

   a. Locate all extra work areas (such as staging areas and additional spoil storage areas) at least 50 feet away from wetland boundaries, except where the adjacent upland consists of cultivated or rotated cropland or other disturbed land.

   b. The project sponsor shall file with the Secretary for review and written approval by the Director, site-specific justification for each extra work area with a less than 50-foot setback from wetland boundaries, except where adjacent upland consists of cultivated or rotated cropland or other disturbed land. The justification must specify the site-specific conditions that will not permit a 50-foot setback and measures to ensure the wetland is adequately protected.

   c. The construction right-of-way may be used for access when the wetland soil is firm enough to avoid rutting or the construction right-of-way has been appropriately stabilized to avoid rutting (e.g., with timber riprap, prefabricated equipment mats, or terra mats).

      In wetlands that cannot be appropriately stabilized, all construction equipment other than that needed to install the wetland crossing shall

use access roads located in upland areas. Where access roads in upland areas do not provide reasonable access, limit all other construction equipment to one pass through the wetland using the construction right-of-way.

d. The only access roads, other than the construction right-of-way, that can be used in wetlands are those existing roads that can be used with no modifications or improvements, other than routine repair, and no impact on the wetland.

2. Crossing Procedures

a. Comply with COE, or its delegated agency, permit terms and conditions.

b. Assemble the pipeline in an upland area unless the wetland is dry enough to adequately support skids and pipe.

c. Use "push-pull" or "float" techniques to place the pipe in the trench where water and other site conditions allow.

d. Minimize the length of time that topsoil is segregated and the trench is open. Do not trench the wetland until the pipeline is assembled and ready for lowering in.

e. Limit construction equipment operating in wetland areas to that needed to clear the construction right-of-way, dig the trench, fabricate and install the pipeline, backfill the trench, and restore the construction right-of-way.

f. Cut vegetation just above ground level, leaving existing root systems in place, and remove it from the wetland for disposal.

The project sponsor can burn woody debris in wetlands, if approved by the COE and in accordance with state and local regulations, ensuring that all remaining woody debris is removed for disposal.

g. Limit pulling of tree stumps and grading activities to directly over the trenchline. Do not grade or remove stumps or root systems from the rest of the construction right-of-way in wetlands unless the Chief Inspector and Environmental Inspector determine that safety-related construction constraints require grading or the removal of tree stumps from under the working side of the construction right-of-way.

h. Segregate the top 1 foot of topsoil from the area disturbed by trenching, except in areas where standing water is present or soils are

saturated. Immediately after backfilling is complete, restore the segregated topsoil to its original location.

i. Do not use rock, soil imported from outside the wetland, tree stumps, or brush riprap to support equipment on the construction right-of-way.

j. If standing water or saturated soils are present, or if construction equipment causes ruts or mixing of the topsoil and subsoil in wetlands, use low-ground-weight construction equipment, or operate normal equipment on timber riprap, prefabricated equipment mats, or terra mats.

k. Remove all project-related material used to support equipment on the construction right-of-way upon completion of construction.

3. Temporary Sediment Control

Install sediment barriers (as defined in section IV.F.3.a of the Plan) immediately after initial disturbance of the wetland or adjacent upland. Sediment barriers must be properly maintained throughout construction and reinstalled as necessary (such as after backfilling of the trench). Except as noted below in section VI.B.3.c, maintain sediment barriers until replaced by permanent erosion controls or restoration of adjacent upland areas is complete. Temporary erosion and sediment control measures are addressed in more detail in the Plan.

a. Install sediment barriers across the entire construction right-of-way immediately upslope of the wetland boundary at all wetland crossings where necessary to prevent sediment flow into the wetland.

b. Where wetlands are adjacent to the construction right-of-way and the right-of-way slopes toward the wetland, install sediment barriers along the edge of the construction right-of-way as necessary to contain spoil within the construction right-of-way and prevent sediment flow into the wetland.

c. Install sediment barriers along the edge of the construction right-of-way as necessary to contain spoil and sediment within the construction right-of-way through wetlands. Remove these sediment barriers during right-of-way cleanup.

4. Trench Dewatering

Dewater the trench (either on or off the construction right-of-way) in a manner that does not cause erosion and does not result in silt-laden water flowing into any wetland. Remove the dewatering structures as soon as practicable after the completion of dewatering activities.

C. RESTORATION

1. Where the pipeline trench may drain a wetland, construct trench breakers at the wetland boundaries and/or seal the trench bottom as necessary to maintain the original wetland hydrology.

2. Restore pre-construction wetland contours to maintain the original wetland hydrology.

3. For each wetland crossed, install a trench breaker at the base of slopes near the boundary between the wetland and adjacent upland areas. Install a permanent slope breaker across the construction right-of-way at the base of slopes greater than 5 percent where the base of the slope is less than 50 feet from the wetland, or as needed to prevent sediment transport into the wetland. In addition, install sediment barriers as outlined in the Plan. In some areas, with the approval of the Environmental Inspector, an earthen berm may be suitable as a sediment barrier adjacent to the wetland.

4. Do not use fertilizer, lime, or mulch unless required in writing by the appropriate federal or state agency.

5. Consult with the appropriate federal or state agencies to develop a project-specific wetland restoration plan. The restoration plan shall include measures for re-establishing herbaceous and/or woody species, controlling the invasion and spread of invasive species and noxious weeds (e.g., purple loosestrife and phragmites), and monitoring the success of the revegetation and weed control efforts. Provide this plan to the FERC staff upon request.

6. Until a project-specific wetland restoration plan is developed and/or implemented, temporarily revegetate the construction right-of-way with annual ryegrass at a rate of 40 pounds/acre (unless standing water is present).

7. Ensure that all disturbed areas successfully revegetate with wetland herbaceous and/or woody plant species.

8. Remove temporary sediment barriers located at the boundary between wetland and adjacent upland areas after revegetation and stabilization of adjacent upland areas are judged to be successful as specified in section VII.A.4 of the Plan.

D. POST-CONSTRUCTION MAINTENANCE AND REPORTING

1. Do not conduct routine vegetation mowing or clearing over the full width of the permanent right-of-way in wetlands. However, to facilitate periodic corrosion/leak surveys, a corridor centered on the pipeline and up to 10 feet wide may be cleared at a frequency necessary to maintain the 10-foot corridor in an herbaceous state. In addition, trees within 15 feet of the pipeline with roots that could compromise the integrity of pipeline coating may be selectively cut and removed from the permanent right-of-way. Do not conduct any routine vegetation mowing or clearing in wetlands that are between HDD entry and exit points.

2. Do not use herbicides or pesticides in or within 100 feet of a wetland, except as allowed by the appropriate federal or state agency.

3. Time of year restrictions specified in section VII.A.5 of the Plan (April 15 – August 1 of any year) apply to routine mowing and clearing of wetland areas.

4. Monitor and record the success of wetland revegetation annually until wetland revegetation is successful.

5. Wetland revegetation shall be considered successful if all of the following criteria are satisfied:

   a. the affected wetland satisfies the current federal definition for a wetland (i.e., soils, hydrology, and vegetation);

   b. vegetation is at least 80 percent of either the cover documented for the wetland prior to construction, or at least 80 percent of the cover in adjacent wetland areas that were not disturbed by construction;

   c. if natural rather than active revegetation was used, the plant species composition is consistent with early successional wetland plant communities in the affected ecoregion; and

   d. invasive species and noxious weeds are absent, unless they are abundant in adjacent areas that were not disturbed by construction.

6. Within 3 years after construction, file a report with the Secretary identifying the status of the wetland revegetation efforts and documenting success as defined in section VI.D.5, above. The requirement to file wetland restoration reports with the Secretary does not apply to projects constructed under the automatic authorization, prior notice, or advance notice provisions in the FERC's regulations.

   For any wetland where revegetation is not successful at the end of 3 years after construction, develop and implement (in consultation with a

professional wetland ecologist) a remedial revegetation plan to actively revegetate wetlands. Continue revegetation efforts and file a report annually documenting progress in these wetlands until wetland revegetation is successful.

VII. HYDROSTATIC TESTING

A. NOTIFICATION PROCEDURES AND PERMITS

1. Apply for state-issued water withdrawal permits, as required.

2. Apply for National Pollutant Discharge Elimination System (NPDES) or state-issued discharge permits, as required.

3. Notify appropriate state agencies of intent to use specific sources at least 48 hours before testing activities unless they waive this requirement in writing.

B. GENERAL

1. Perform 100 percent radiographic inspection of all pipeline section welds or hydrotest the pipeline sections, before installation under waterbodies or wetlands.

2. If pumps used for hydrostatic testing are within 100 feet of any waterbody or wetland, address secondary containment and refueling of these pumps in the project's Spill Prevention and Response Procedures.

3. The project sponsor shall file with the Secretary before construction a list identifying the location of all waterbodies proposed for use as a hydrostatic test water source or discharge location. This filing requirement does not apply to projects constructed under the automatic authorization provisions of the FERC's regulations.

C. INTAKE SOURCE AND RATE

1. Screen the intake hose to minimize the potential for entrainment of fish.

2. Do not use state-designated exceptional value waters, waterbodies which provide habitat for federally listed threatened or endangered species, or waterbodies designated as public water supplies, unless appropriate federal, state, and/or local permitting agencies grant written permission.

3. Maintain adequate flow rates to protect aquatic life, provide for all waterbody uses, and provide for downstream withdrawals of water by existing users.

4. Locate hydrostatic test manifolds outside wetlands and riparian areas to the maximum extent practicable.

D. DISCHARGE LOCATION, METHOD, AND RATE

1. Regulate discharge rate, use energy dissipation device(s), and install sediment barriers, as necessary, to prevent erosion, streambed scour, suspension of sediments, or excessive streamflow.

2. Do not discharge into state-designated exceptional value waters, waterbodies which provide habitat for federally listed threatened or endangered species, or waterbodies designated as public water supplies, unless appropriate federal, state, and local permitting agencies grant written permission.

APPENDIX C-3

Spill Prevention and Response Plan

ETC Tiger Pipeline, LLC

Franklin Farms Project

Spill Prevention and Response (SPAR) Plan

June 2026

ENERGY TRANSFER

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ABBREVIATIONS AND ACRONYMS

EI	Environmental Inspector
ETC Tiger	ETC Tiger Pipeline, LLC
ft	feet
LDEQ	Louisiana Department of Environmental Quality
Project	Franklin Farms Project
SPAR Plan or Plan	*Spill Prevention and Response Plan*

1.0 GENERAL DESCRIPTION OF SPILL PREVENTION AND RESPONSE PLAN

ETC Tiger Pipeline, LLC (ETC Tiger) has prepared a *Spill Prevention and Response* (SPAR) *Plan*, or Plan, for the Franklin Farms Project (Project) that is designed to minimize hazards to human health and/or the environment from any unplanned sudden or non-sudden releases of oils, toxic, hazardous, or other polluting materials to the air, soil, surface water or groundwater. ETC Tiger, through its Contractors and Inspectors, shall be responsible for the administration and implementation of the SPAR Plan. This Plan is intended to provide minimum requirements for spill prevention and response during construction activities. The Contractor may develop their own spill prevention and response plan or use an existing plan provided that the plan used contains, at a minimum, all of the provisions of ETC Tiger's SPAR Plan and 40 CFR 112, as applicable.

This plan identifies:

- Measures taken for spill preparedness and prevention.
- Emergency response procedures describing the actions that ETC Tiger and Contractor personnel will take in response to leaks, spills, or discharges of oil and hazardous substances/materials.
- Designated emergency coordinator(s) and his/her responsibilities.
- Spill incident reporting procedures.
- Contact numbers for the local police and fire departments, hospitals, and state and local emergency planning committees.

Prior to the start of construction in an area, the Contractor shall designate storage, refueling, loading, and unloading locations that minimize the environmental and safety impacts associated with releases of fuel, lubricants, or hazardous substances. These areas will be designated using the following guidelines:

- Refueling shall not occur within 100 feet (ft) of a waterbody or 100 ft from a wetland boundary without ETC Tiger Environmental Inspector (EI) coordination and approval.
- Hazardous materials, including chemicals, fuels, and lubricating oils, shall not be stored within 100 ft of a wetland, waterbody, or designated municipal watershed area without ETC Tiger EI coordination and approval.
- Refueling and storage of hazardous materials, including chemicals, fuels, and lubricating oils is prohibited within 200 ft of private wells and 500 ft of community and municipal wells.
- No potentially hazardous materials, other than essential equipment fuels (e.g., gasoline, diesel, etc.) or standard lubricants (e.g., engine oils, grease, etc.) shall be transported into the right-of-way or construction area without ETC Tiger EI coordination and approval.

The CONTRACTOR will be required to comply with all applicable requirements of 40 CFR 112, Oil Pollution Prevention, for any facility set up for the storage of fuel, oil, or other hydrocarbons, or refueling of vehicles and equipment, if the facility triggers compliance with the rule. This would include the development and implementation by CONTRACTOR of a Spill Prevention, Control, and Countermeasures (SPCC) Plan, if necessary.

2.0 SPILL AND LEAK PREVENTION AND PREPAREDNESS

2.1 PREVENTION AND PREPAREDNESS

The Contractor will take the following precautions to prevent a spill from occurring and to be prepared in the event that a spill does occur.

2.1.1 Containers

- All containers shall be stored on pallets and surrounded with temporary containment.
  - Small cans of gasoline, diesel, solvents, etc., should be stored within the temporary containment when not in use.
  - No incompatible materials shall be stored in the same containment area.
- Containment areas shall be capable of containing 110% of the volume of the largest container in the storage area plus sufficient freeboard for rainfall.
  - All container storage areas shall be inspected daily for leaks and deterioration.
- Leaking or deteriorated containers shall be replaced as soon as the condition is first detected.
- Container storage areas shall be secured during non-working hours.
- Any accumulated rainwater in the containment areas shall be inspected for sheens and indications of any other hazardous materials prior to release to the ground, and if a sheen or hazardous substance is detected, accumulated rainwater shall be properly disposed.

2.1.2 Tanks

- The Contractor shall operate only those tanks for fuel and material storage that meet the approval of ETC Tiger. Single wall tanks shall be provided with temporary containment as described in Section
  2.1.1 for Containers.
- Self-supporting tanks shall be constructed of carbon steel or other materials compatible with the contents of each tank.
- All tanks and storage areas shall be inspected daily for leaks and deterioration.
- Vehicle mounted tanks shall be equipped with flame/spark arrestors on all vents to ensure that self-ignition does not occur.
- Tanks will not be used to store incompatible materials in sequence unless first thoroughly decontaminated.

2.1.3 Loading/Unloading Areas

- Transferring of liquids and refueling shall only occur in pre-designated locations at least 100 ft from all waterbodies and wetlands, 200 ft from any private water well, and 500 ft from municipal or community water supply wells unless prior approval is obtained from the ETC Tiger EI.
- All loading/unloading areas will be inspected for spills prior to and immediately after each use and closely monitored during use to prevent leaks and spills and ensure immediate response in the event of a spill.
- All those connections shall be inspected for leaks. If leaks should occur, the operation shall cease until the leak is repaired or a containment pan is placed under the leaking connection.

2.1.4 Equipment

- The Contractor is to ensure that all equipment operating on the Project is in good working order and free of any leaks.
- The EI has the authority to request the contractor to remove a piece of equipment from the Project if it is found to be leaking hazardous fluids on the ground and is not repaired by the contractor.

2.1.5 Spill Response Kits

- Any service vehicle used to transport lubricants and fuel shall be equipped with an oil spill response kit adequately stocked to respond to a minor oil/fuel spill event.
- Chemical spill response kits, adequately stocked to respond to a minor chemical spill event, shall be available in areas where appropriate.
- Additionally, spill response kits shall be available on the right-of-way and on or near operating equipment as deemed appropriate by the ETC Tiger EI.
- Equipment such as hydraulic track hoes and hydraulic pumps that could fail and cause a reportable spill must be equipped with an oil spill response kit adequately stocked to respond to a minor oil/fuel spill event.

2.2 EMPLOYEE TRAINING

All personnel involved in the construction of the proposed facilities will be trained on the contents of the SPAR Plan. Training briefings will be conducted by the Contractor Superintendent or his designee and the ETC Tiger EI on the job site.

2.3 SPILL RESPONSE EQUIPMENT

Project construction shall have adequate manpower and equipment necessary to divert any spill from reaching water bodies and wetland areas. Emergency equipment may include, but is not limited to, shovels, backhoes, dozers, front-end loaders, oil absorbent booms, pillows, chemical absorbent pulp, pillows, socks, and/or mats.

3.0 INITIAL SPILL RESPONSE PROCEDURES

This section provides a description of spill response procedures to be performed in response to any spills that may occur during Project construction.

3.1 COMPANY AND CONTRACTOR RESPONSIBILITIES

The Contractor and ETC Tiger on-site personnel have responsibilities for spill prevention and response. **In addition to the oversight of initial spill response activities, ETC Tiger's EI and Environmental Project Manager will determine if state and/or Federal notifications are required and make notification accordingly.**

The Contractor will have a designated Environmental Coordinator for the site. The Contractor's Environmental Coordinator will be responsible for the Contractor's initial spill response activities. The responsibilities of the Contractor and ETC Tiger will be as follows:

3.1.1 Contractor Responsibilities

- The Contractor will be responsible for taking immediate action to safely control and contain any spills or releases of oil, petroleum products, and hazardous substances/materials.
- All spills or releases, including any sheen created on water or releases to the atmosphere, must be reported immediately to the ETC Tiger EI.
- The Contractor shall supply necessary manpower and equipment to control, contain, and clean up all spills and releases resulting from their operations.

3.1.2 ETC Tiger Environmental Inspector (EI) Responsibilities

- ETC Tiger 's EI or his/her designee will be responsible for making appropriate agency notifications spills and releases.
- ETC Tiger will be responsible for the oversight of the initial spill response activities.
- ETC Tiger will provide supporting personnel and equipment to address releases as required.
- In the event of a spill, the EI shall obtain as much information as possible regarding the cause of the event, the type and amount of material spilled or released, and corrective measures or response activities being taken.
- Consult the ETC Tiger Environmental Project Manager immediately and determine if the spill or release is a reportable event. The EI will also notify the ETC Tiger Field Construction Office for releases of:
    - one pound or more of a solid material
    - five gallons or more of a liquid material
    - any spill to water, including any sheen on water
- Obtain a copy of the Contractor's written spill report as soon as it is available and forward a copy to the Environmental Project Manager.

3.1.3 ETC Tiger Environmental Project Manager Responsibilities

- Upon receiving spill information from the EI, determine if the release requires reporting to any Federal, state, or local regulatory agencies. At a minimum, notification to the following agencies may be required immediately upon detection of a spill:
    - Federal Energy Regulatory Commission,
    - Louisiana Department of Environmental Quality (LDEQ)
- If reporting is required, direct the EI to notify the appropriate regulatory agencies. This includes both verbal and any follow-up written reports.
- Contact outside remediation services if necessary, in coordination with the ETC Tiger EI, to assist with incidents that require additional resources.

3.1.4 Reporting Requirements

The Federal Energy Regulatory Commission and LDEQ require notification for the following conditions:

- General Spills
  - For emergency spill incidents, contact the Louisiana State Police Hazardous Materials Hotline at 225-925-6595 as soon as possible within the first hour of the emergency. For spills that do not cause an emergency condition the Louisiana State Police Hazardous Materials Hotline must be called within 24 hours of the discharge.
  - The Richland Parish Local Emergency Planning Committee must be notified of spills at 318-728-0453 (Dawn Williams).
  - The U.S. Environmental Protection Agency National Response Center should be notified of spills at 800-424-8802.
  - A written notification report regarding spills muse be reported to the LDEQ within seven calendar days after the initial reporting to the Louisiana State Police.
- Petroleum Based Spills
  - Spills into or involving state waterways (any amount).
  - Spills greater than 25 gallons (or potential > 25 gallons).
  - Spills requiring any State/Federal notifications or assistance.
  - Chemical spills.
  - All SARA/EHS/CERCLA Releases.
  - All spills threatening population or the environment.
  - All spills requiring evacuation of the public.
- Weather
  - Any incident associated with weather phenomena involving possible or actual damage to property or persons (e.g., wind damage, tornadoes, lightning strikes, flooding).
- Transportation
  - Incidents involving major thoroughfare closures.
  - All aircraft incidents.
  - All railroad incidents.
  - Incidents involving mass casualties.
  - All major incidents involving commercial vehicles/vessels.
- Fire
  - Major forest fires.
  - Fires involving chemicals or significant amounts of petroleum products.
  - Large or multiple structure fires.
- General
  - Sinkholes.
  - Public water source contamination.
  - National security.
  - Medical waste.
  - Immigration issues.
  - Potential/actual dam failures.
  - Incidents with potential effects to adjacent countries/states.
  - Incidents requiring assistance from state/Federal agencies.
  - Incidents with a prolonged effect on public utilities.

- Incidents involving potential or actual evacuations.

3.2 UNPLANNED AND PLANNED NATURAL GAS RELEASES

3.2.1 Unplanned Natural Gas Releases

Unplanned natural gas releases are reportable events in some of the states that ETC Tiger operates in. In the event that an unplanned release of natural gas occurs during activities related to the Project the Contractor shall immediately notify the ETC Tiger EI of the event.

3.2.2 Planned Natural Gas Releases

Some of the states that ETC Tiger operates in require prior notification and/or approval for planned releases of natural gas to the atmosphere such as blowdowns. In the event that a planned release of natural gas is scheduled to occur during activities related to the Project, the Contractor shall contact the ETC Tiger EI a minimum of two weeks prior to the event and confirm that notifications have been made and/or approvals obtained if required.

3.3 SPILL CLEAN-UP AND WASTE DISPOSAL

Spill clean-up and subsequent waste disposal of contaminated media will be the responsibility of the Contractor subject to the approval of the ETC Tiger Environmental Project Manager.

4.0 KEY EMERGENCY CONTACTS

The key personnel who will be contacted in the event of an emergency or spill incident include the following: **(Information to be provided prior to construction.)**

4.1 ETC Tiger Emergency Contacts

a) ETC Tiger Emergency Coordinator: Construction Manager, TBD

b) Field Construction Office: TBD

c) Lead Environmental Inspector: TBD

d) Environmental Project Manager: Justin Minter, office: 409-749-3902, cell: 409-377-0054

e) Area Office (in case of pipeline liquid spills): TBD

4.2 Contractor Emergency Contact

f) Contractor Emergency Coordinator: Contractor Superintendent, TBD

4.3 Local Authorities

Department	Phone
Local Police	911
Non-Emergency Sheriff's Office	318-728-2071 (Richland Parish, LA)
Local Fire Department	911
Hospital	318-728-4181 (Richardson Medical Center, in Rayville, LA) 318-878-5171 (Richland Parish Hospital, in Delhi, LA)

Ambulance	911
Local Emergency Planning Committee (LEPC)	318-728-0453 (Richland Parish, LA)
Louisiana State Police Hazardous Materials Hotline	225-925-6595

APPENDIX C-4

Unanticipated Discovery Plan

ETC Tiger Pipeline, LLC

Franklin Farms Project

Unanticipated Discoveries Plan

Cultural Resources, Human Remains, Contaminated Media, and Paleontological Resources

June 2026

ENERGY TRANSFER

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ABBREVIATIONS AND ACRONYMS

ACHP	Advisory Council on Historic Preservation
CM	Construction Manager
EI	Environmental Inspector
ETC Tiger	ETC Tiger Pipeline, LLC
FERC	Federal Energy Regulatory Commission
NRHP	National Register of Historic Places
Plan	Unanticipated Discoveries Plan
Project	Franklin Farms
SHPO	State Historic Preservation Office

1.0 INTRODUCTION

ETC Tiger Pipeline, LLC (ETC Tiger) has developed this *Unanticipated Discoveries Plan* (Plan) to provide general procedures for the handling of unanticipated finds during construction and operation of the Franklin Farms Project (Project). If previously unidentified resources, or unanticipated discoveries are encountered, they could consist of:

- Archaeological or historical resources.
- Human remains.
- Contaminated media.
- Paleontological resources.

The procedures outlined in this Plan are intended to:

- Maintain compliance with applicable federal and state laws and regulations.
- Describe to regulatory and review agencies the actions the Project or its representative will follow to prepare for and handle unanticipated discoveries.
- Provide direction and guidance to Project personnel as to the proper procedures to be followed should an unanticipated discovery occur.

The Environmental Inspector (EI) will be responsible for advising Contractor personnel on the procedures to follow in the event that an unanticipated discovery is made. Training will occur as part of the pre-construction on-site training program for foremen, company inspectors, and construction supervisors. The EI will advise all operators of equipment involved in grading, stripping, or trenching activities to:

1. Stop work immediately if they observe any indications of the presence of cultural materials (artifacts or other man-made features), bone, or contamination.
2. Contact the EI as soon as possible.
3. Comply with the applicable unanticipated discovery procedures (see below).
4. Treat human remains with dignity and respect.

2.0 PROCEDURES FOR THE UNANTICIPATED DISCOVERY OF CULTURAL RESOURCES

In the event that any member of the construction workforce believes that a cultural resource is encountered, the following procedures will be implemented:

1. All work within 100 feet of the discovery will immediately stop and the EI and Construction Manager (CM) will be notified. The area of work stoppage will be adequate to provide for the security, protection, and integrity of the materials. A cultural resource can be prehistoric or historic and could consist of, but not be limited to:
    - An accumulation of shell, burned rocks, or other subsistence related materials.
    - An area of charcoal or very dark soil with artifacts.
    - Stone tools, arrowheads, or dense concentrations of stone artifacts.
    - A cluster of bones in association with shell, charcoal, burned rocks, or stone artifacts.
    - A historic structure or assemblage of historic materials older than 50 years.
2. If the EI and/or CM believes that the discovery is a cultural resource, the EI will take appropriate steps to protect the discovery site. This will include flagging the immediate area of discovery, stopping work, or setting up an exclusion zone, as well as notifying the Environmental Project Manager and/or Company Representative. Work in the exclusion zone will not resume until treatment of the discovery has been completed.
3. ETC Tiger or its representative will arrange for the discovery to be evaluated by a qualified archaeologist in accordance with applicable regulations. The archaeologist will evaluate the discovery and provide recommendations for how to manage the resource under the Louisiana Division of Archaeology (LDOA), or Native American Graves Protection and Repatriation Act (NAGPRA), as applicable.
4. If the discovery is determined to be a cultural resource with the potential for eligibility for the National Register of Historic Places (NRHP), the archaeologist and ETC Tiger will consult with the State Historic Preservation Office (SHPO), the Federal Energy Regulatory Commission (FERC), and, if necessary, the State Archaeologist on how best to avoid, minimize, treat, or otherwise mitigate further impacts. Consultation with the SHPO and applicable federal agency should occur to obtain technical advice and guidance for the evaluation of the discovered cultural resource(s). Treatment measures may include mapping, photography, sample collection, and/or excavation activity.
5. The archaeologist will implement the appropriate treatment/mitigation measure(s) as outlined in the mitigation plan, and provide a report on its methods and results as required. The investigation and technical report will be performed in compliance with the Secretary of the Interior's *Standards and Guidelines for Archaeological Documentation (*48 CFR 44734--44737); the Advisory Council on Historic Preservation (ACHP) publication *"Treatment of Archaeological Properties*" (ACHP 1980); and follow the guidelines set forth by the Louisiana SHPO.

3.0 PROCEDURES FOR THE UNANTICIPATED DISCOVERY OF HUMAN REMAINS

In the event that human remains are encountered during either construction or maintenance activities for the Project, the specific procedures to be followed are outlined below. These procedures meet or exceed the *Policy Statement Regarding Treatment of Burial Sites, Human Remains, and Funerary Objects set forth by the National Historic Preservation Act* (Public Law [PL] 89-665), its implementing regulations, "*Protection of Historic and Cultural Properties*" (36 CFR Part 800); the Native American Grave and Repatriation Act (43 CFR Part 10); Procedures for the Protection of Historic Properties (33 CFR 325 Appendix C); the Archaeological and Historic Preservation Act; and Consultation and Coordination with Indian Tribal Governments (EO 13175).

All activities that might disturb the remains shall cease and may not resume until authorized by appropriate law enforcement officials or the State Archaeologist. Any human remains, burial sites, or burial related materials that are discovered will at all times be treated with dignity and respect. If any member of the construction workforce believes that human remains are encountered, the following procedures will be implemented:

1. All work in the near vicinity of the human remains shall immediately cease and reasonable efforts shall be made to avoid and protect the remains from additional impact. In cases of inclement weather, the human remains shall be protected with tarpaulins.
2. The EI and CM shall be notified immediately and will take appropriate measures to protect the discovery site. This shall include flagging the immediate area of discovery as an exclusion zone, as well as notifying the Environmental Project Manager and Project Manager. Work in the immediate area will not resume until authorized by the applicable agencies.
3. ETC Tiger, or its representative, shall immediately contact a qualified Professional Archeologist to investigate the reported discovery; inventory the remains and any associated artifacts; and assist in coordinating with state and local officials.
4. The Parish Coroner shall be immediately notified as to the findings. If the remains are found to be other than human, any construction will be cleared to proceed. If the remains are human and are less than 50 years old, the Coroner and local law enforcement officials will assume jurisdiction. If the remains are found to be human and older than 50 years, the State Archaeologist shall be notified and may assume jurisdiction of the remains. The SHPO and the FERC will also be notified. As much information as possible concerning the human remains, such as location and size, as well as any information on its significance, should be provided to the SHPO and the FERC.
5. If jurisdiction is assumed by the State Archaeologist, they will (a) determine whether the human remains represent a significant archaeological resource, and (b) make a reasonable effort to identify and locate persons who can establish direct kinship, tribal community, or ethnic relationship with the remains. If such a relationship cannot be established, then the State Archaeologist may consult with a committee of four to determine the proper disposition of the remains. This committee shall consist of a human skeletal analyst, two Native American members of current state tribes recommended by the Governor's Council on Indian Affairs, and "*an individual who has special knowledge or expertise regarding the particular type of the unmarked human burial*."
6. A plan for the avoidance of any further impact to the human remains and/or mitigative excavation, reinternment, or a combination of these treatments shall be developed in consultation with the State Archaeologist, the SHPO, the FERC and, if applicable, appropriate Indian tribes or closest lineal descendants. All parties will be expected to respond with advice and guidance in an efficient time frame. Once the plan is agreed to by all parties, the plan will be implemented.

4.0 PROCEDURES FOR THE UNANTICIPATED DISCOVERY OF CONTAMINATED MEDIA

To date, no known areas of contamination have been identified in the proposed Project area. However, the potential for discovery remains. Indicators of possible contamination include, but are not limited to:

- Drums or containers which are rusted or in otherwise poor condition, including buried drums or containers.
- Stained or otherwise discolored soil in contrast to adjoining materials.
- Spoil material containing debris other than obvious construction material.
- Chemical or hydrocarbon odors emanating from excavations.
- Oily residues.
- Visible sheens or other discoloration on groundwater or surface water.
- Underground structures (e.g., pipelines, storage tanks).

The EI and appropriate contractor personnel shall be trained in hazard identification, including potential contamination and worker protection. These topics shall be discussed regularly in safety meetings. The EI's duties will include, but are not limited to, ensuring compliance with all environmental conditions.

A desktop assessment for contamination within the Project footprint indicated that contamination is not likely to be encountered during construction or maintenance of the Project. In the unlikely event that contamination is encountered, the following procedures shall be implemented:

1. All construction activities in the immediate area of the discovery shall cease and the EI shall be notified. Work in the immediate area will not resume until an assessment of the discovery has been completed and ETC Tiger has released the site.
2. If safe to do so, the EI shall take appropriate steps to flag the area as an exclusion zone and notify the Environmental Project Manager.
3. If potentially contaminated groundwater or soil reaches (or has the potential to reach) surface waters, booms and/or absorbent materials shall be immediately deployed to contain and reduce downstream migration of the material.
4. Upon notification, the Environmental Project Manager shall perform or direct a hazard assessment to determine appropriate control measures to be implemented at the specific site. Activities may include sampling vapors, soil, sediments, groundwater, and/or wipe samples of materials.
5. Upon evaluation of the sampling results, additional notifications may be made to coordinate a work plan for measures to be implemented in the contaminated area to resume activities in a safe, environmentally compliant, and effective manner. Measures may include additional personal protective equipment, segregation of contaminated media, treatment, or off-site disposal of contaminated media.
6. If warranted by the assessment, the Environmental Project Manager will notify appropriate federal, state and local agencies.
7. All identification /characterization, handling, labeling, storage, manifesting, transportation, record keeping, and disposal of potentially contaminated materials will be conducted in accordance with applicable federal, state, and local regulations and guidance.

5.0 PROCEDURES FOR THE UNANTICIPATED DISCOVERY OF PALEONTOLOGICAL RESOURCES

A paleontological resource would be expected to be in the form of fossils. *In-situ* fossils are usually found within layers of geologically old sediments and rocks where the creature lived, died, and became fossilized. However, through geologic, hydrologic, and marine activity, many fossils and parts of fossils have been carried into younger geologic areas. In the event that any member of the construction or maintenance workforce believes that an unanticipated discovery of paleontological resources has been made, the following procedures shall be implemented:

1. All work in the immediate area of the discovery shall immediately cease and the EI shall be notified. The area of work stoppage shall be adequate to provide for the security, protection, and integrity of the suspected discovery.
2. If the EI believes that the discovery is a paleontological resource, the EI shall take appropriate measures to protect the discovery site. This shall include flagging the immediate area of discovery as an exclusion zone, as well as notifying the Environmental Project Manager and/or Project Manager. Work in the immediate area will not resume until treatment of the discovery has been completed.
3. The Environmental Project Manager shall arrange for the discovery to be evaluated by a qualified geologist/paleontologist in accordance with applicable regulations. The geologist/paleontologist will evaluate the discovery and provide recommendations for how to manage the resource.

Specimens destined for the University of Louisiana at Lafayette Geology Museum will be sent to the School of Geosciences at:

School of Geosciences
University of Louisiana at Lafeyette
611 McKinley Street
Hamilton Hall #323
P.O. Box 43717
Lafayette, LA 70504
337-482-6647
geology@louisiana.edu

6.0 KEY CONTACTS

The table below identifies key contacts that may need to be notified in the event of an unanticipated discovery of archeological resources, human remains, contaminated media, or paleontological resources.

Contact	Phone Number
Federal Energy Regulatory Commission (FERC) – Cultural Resources and Paleontological Resources	
Allison King	(202) 502-8847
Environmental Inspector	
TBD Prior to Construction	TBD
Construction Manager	
TBD Prior to Construction	TBD
Environmental Project Manager	
Justin Minter, Sr. Manager – Environmental Projects justin.minter@energytransfer.com	O: (409) 749-3902 C: (409) 377-0054
Cultural Resources Specialist	
Steven J. Filoromo, RPA, Sr. Archaeologist/Cultural Resources Team Lead Sfiloromo@trccompanies.com	C: (205)641-0696
State Historic Preservation Office	
Dr. Rachel Watson, State Archaeologist Louisiana Division of Archaeology 1051 N 3rd St, Room 316 Baton Rouge, Louisiana 70802	(225)342-8165 rawtson@crt.la.gov
Law Enforcement	
Richland Parish, LA Sheriff's Office	(318) 728-2071 or 911
Parish Coroner	
Dr. Matthew Prine, M.D. 256 Hwy 3048 Rayville, Louisiana 71269	(318)728-2046 tracy_morris2003@yahoo.com
State Agency Spill Reporting Contacts	
Louisiana State Police Hazardous Materials Hotline Richland Parish Local Emergency Planning Committee	(225) 925-6595 (318) 728-0453

APPENDIX C-5

Fugitive Dust Control Plan

ETC Tiger Pipeline, LLC

Franklin Farms Project

Fugitive Dust Control Plan

June 2026

ENERGY TRANSFER

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ABBREVIATIONS AND ACRONYMS

EI	Environmental Inspector
ETC Tiger	ETC Tiger Pipeline, LLC
FERC Plan	*Upland Erosion Control, Revegetation, and Maintenance Plan* (2013)
FERC Procedures	*Wetland and Waterbody Construction and Mitigation Procedures* (2013)
Plan	*Fugitive Dust Control Plan*
Project	Franklin Farms Project

1.0 INTRODUCTION

ETC Tiger Pipeline, LLC (ETC Tiger) has developed this *Fugitive Dust Control Plan* (Plan) to minimize fugitive dust emissions during ground-disturbing activities associated with construction of the Franklin Farms Project (Project).

Land disturbance from clearing and excavation activities has the potential to generate fugitive dust particles. Fugitive dust is generated by the mechanical disturbance of granular material exposed to the atmosphere. Dust from open sources is termed "*fugitive*" because it is not discharged to the atmosphere in a confined flow stream. Dust control measures are practices that help reduce the surface and air movement of dust from disturbed soil surfaces. This Plan outlines fugitive dust control methods that will be used on the Project to minimize fugitive dust emissions.

2.0 FUGITIVE DUST EMISSION SOURCES

The following Project activities have the potential to generate fugitive dust emissions:

- Vegetation removal.
- Clearing and grading.
- Topsoil removal.
- Cutting and filling.
- Trenching.
- Backfilling.
- Track-out onto roads.
- Bulk material loading, hauling, and unloading.
- Vehicle and motorized equipment movement on unpaved access roads.
- Use of material storage piles.
- Use of parking, staging, and storage areas.

3.0 FUGITIVE DUST MITIGATION MEASURES

It is the responsibility of the Contractor(s) and Environmental Inspector (EI) to ensure that Contractor personnel are complying with all fugitive dust control measures. The EI has authority to enforce and require compliance with this Plan and to stop work activities if a Contractor does not comply with fugitive dust control measures.

Implementation of construction and restoration best management practices and operational controls will be used to mitigate fugitive dust emissions. Project specific construction permits will likely outline specific practices to control fugitive dust emissions, including a construction sequence, use of temporary rock entrances, and temporary soil stabilization measures. ETC Tiger will also implement operational controls, including the use of a reduced speed limit on unpaved access roads as well as sweeping/vacuuming paved roadways when Project-related soils are tracked out onto paved surfaces.

Wet suppression is the predominate method of suppressing fugitive dust emissions on unpaved roads and gravel pads. The application of water causes finer materials to adhere to larger particles, thereby reducing the possibility of particles becoming airborne. Increasing the moisture content of finer materials can be accomplished either naturally or mechanically. Moisture content of unpaved surfaces can be naturally increased through precipitation or mechanically through the application of water, mainly via water trucks.

The amount of water required to sufficiently control fugitive dust emissions is dependent on a number of variables including surface moisture content, ambient conditions (e.g., temperature, wind, humidity, precipitation), and construction activities occurring (e.g., vehicle/equipment traffic, vehicle speeds, vehicle weight). Disturbed and unpaved areas will be kept sufficiently damp during working hours in dry conditions to minimize wind-blown or traffic-generated fugitive dust emissions. Areas to be watered include, but are not limited to:

- The construction corridor, including temporary workspace.
- Access roads.
- Aboveground facility sites.
- Active grading areas.
- Un-stabilized areas.
- Soil stockpiles.
- Unpaved parking/staging areas.

The frequency of water applications will vary based on weather and site conditions. More frequent applications will be required in dry conditions and in areas with a high potential for fugitive dust generation. Water for fugitive dust control will be obtained from select water wells.

3.1 Pipeline and Aboveground Facility Construction Activities

Fugitive dust emissions from vegetation removal, clearing, grading, cutting, filling, topsoil removal, trenching, backfilling, and stockpile storage will be controlled, to a great extent, by disturbing limited areas at a time and following the construction sequencing. Should visible dust plumes occur during construction activities, the plumes will be abated by applying water along the travel lane and disturbed areas via water truck.

3.2 Unpaved Access Roads and Staging Areas

Fugitive dust emissions from vehicle and equipment travel on unpaved surfaces will be controlled, to a great extent, by reducing speed limits on unpaved surfaces. Fugitive dust emissions generated by vehicle and equipment traffic will be controlled by wet suppression, as necessary. Specifically, fugitive dust emissions occurring on active access roads or unpaved surfaces will be controlled by periodic wetting of unpaved surfaces via water truck. The wetting of unpaved surfaces will occur more frequently during periods of high vehicle or equipment traffic to minimize fugitive dust emissions. If weather conditions are adequate to suppress fugitive dust emissions, watering of unpaved surfaces will occur less frequently.

3.3 Track-out onto Paved Roads

The track-out of loose materials will be controlled by installing rock, timber mats, or paved construction entrances on access roads that begin at a junction with paved roads. Any loose material tracked beyond the construction entrances will be recovered via sweeper trucks and/or vacuum trucks.

4.0 INSPECTION, MONITORING, AND RECORDKEEPING

The construction Contractor is responsible for implementing the fugitive dust control measures specified in this Plan. All construction personnel will be informed of the measures outlined in this Plan. The EI will have primary responsibility for monitoring and enforcing the implementation of fugitive dust control measures by the construction Contractor. The EI will also be responsible for ensuring control measures are effective and proper documentation is maintained. Field inspections for fugitive dust control will occur daily. The construction Contractor and EI will be responsible for recording the following information on a daily basis:

- Weather conditions (e.g., temperature, precipitation, wind speed and direction, etc.).
- Number of water trucks in use.
- Locations where fugitive dust suppression measures were implemented.
- Presence of track-outs and when they were cleaned.
- Overall status of fugitive dust control compliance.

This information will be incorporated into the EI's daily report.

5.0 PERSONNEL TRAINING

Prior to commencing construction, ETC Tiger will conduct environmental and safety training for ETC Tiger and contractor personnel. The training will focus on the Federal Energy Regulatory Commission's *Upland Erosion Control, Revegetation, Maintenance Plan* (FERC Plan) and *Wetland and Waterbody Construction and Mitigation Procedures* (FERC Procedures), and other construction, restoration, and mitigation plans, including this *Fugitive Dust Control Plan*, and any applicable permit conditions.

APPENDIX C-6

Noxious and Invasive Plant Management Plan

ETC Tiger Pipeline, LLC

Franklin Farms Project

Noxious Weed and Invasive Plant Management Plan

June 2026

ENERGY TRANSFER

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ABBREVIATIONS AND ACRONYMS

BMP	Best Management Practice
EI	Environmental Inspector
ETC Tiger	ETC Tiger Pipeline, LLC
FERC	Federal Energy Regulatory Commission
FERC Plan	*Upland Erosion Control, Revegetation, and Maintenance Plan* (2013)
FERC Procedures	*Wetland and Waterbody Construction and Mitigation Procedures* (2013)
Plan	*Noxious Weed and Invasive Plant Management Plan*
Project	Franklin Farms Project
ROW	right-of-way
USFWS	United States Fish and Wildlife Service

1.0 INTRODUCTION

ETC Tiger Pipeline, LLC (ETC Tiger) has developed this *Noxious Weed and Invasive Plant Management Plan* (Plan) to minimize the establishment and spread of noxious weeds and invasive plants during ground-disturbing activities associated with Franklin Farms Project (Project) construction.

Noxious weeds are non-native, undesirable native, or introduced species that can outcompete desired native species and thereby decrease overall species diversity. Noxious weeds often invade and persist in areas after disturbance.

Invasive plant species are species that display rapid growth and spread, becoming established over large areas. Most commonly they are exotic species that have been introduced from another part of the United States, another region, or another continent, although native species that exhibit rapid growth and spread are sometimes considered invasive. The United States Fish and Wildlife Service (USFWS) defines invasive species as "*non-native plants, animals and other living organisms that thrive in areas where they don't naturally live and cause (or are likely to cause) economic or environmental harm, or harm to human, animal or plant health*" (USFWS, 2026). Invasive plant species can change or degrade natural vegetation communities, which can reduce the quality of habitat for wildlife and native plant species.

2.0 INVASIVE SPECIES MANAGEMENT

The management of noxious weeds and invasive plant species will be conducted throughout all stages of the Project including:

- Pre-construction identification of known areas of infestation.
- Construction personnel training regarding the areas that have been identified and the importance of preventative measures and treatment methods.
- Specific prevention measures to prevent the spread of noxious weeds and invasive plant species.
- Pre- and post-construction treatment methods to be applied to areas of noxious and invasive plant infestation.

3.0 INVASIVE PLANT SPECIES IN PROJECT AREA

Non-native invasive plant species identified within the Project Area during field investigations include Chinese privet (*Ligustrum sinense*), Chinese tallow (*Triadica sebifera*), alligator weed (*Alternanthera philoxeroides*), Bermuda grass (*Cynodon dactylon*), Japanese honeysuckle (*Lonicera japonica*), wart-removing herb (*Murdannia keisak*), golden crown grass (*Paspalum dilatatum*), Bahia grass (*Paspalum notatum*), Vasey's grass (*Paspalum urvillei*), curly dock (*Rumex crispus*), Brazilian vervain (*Verbena brasiliensis*), and muscadine (*Vitis rotundifolia*). ETC Tiger will follow the guidelines set forth in this Plan to control the further spread of invasive species within the Project area.

4.0 PERSONNEL TRAINING

Prior to commencing construction, ETC Tiger will conduct environmental and safety training for ETC Tiger and Contractor personnel. The training will focus on the Federal Energy Regulatory Commission's (FERC) 2013 *Upland Erosion Control, Revegetation, Maintenance Plan* (FERC Plan, FERC 2013a), 2013 *Wetland and Waterbody Construction and Mitigation Procedures* (FERC Procedures, FERC 2013b), and other construction, restoration, and mitigation plans, including this Plan, and applicable permit conditions.

5.0 PREVENTION AND CONTROL MEASURES

To minimize vegetation impacts, ETC Tiger will implement Best Management Practices (BMPs) in the FERC Plan and FERC Procedures, which require the use of temporary and permanent erosion control measures, topsoil segregation in select areas, testing and mitigation for soil compaction, and limited routine vegetation maintenance. Furthermore, all temporarily disturbed upland areas will be reseeded in accordance with the FERC Plan, FERC Procedures, and/or applicable permit requirements. Disturbed areas will be routinely monitored until restoration and revegetation are successful in accordance with the FERC Plan and FERC Procedures.

Ground disturbance from construction activities provides ideal conditions for the establishment of noxious and invasive plant species. ETC Tiger will implement the following measures to minimize the potential to introduce or spread noxious and invasive plant species including:

- Prior to construction, noxious and invasive species locations will be marked or flagged and recorded using GPS equipment. This flagging will alert construction personnel that the appropriate mitigation measures identified in this plan must be followed.
- All movement of construction vehicles, other than on the right-of-way (ROW), will be restricted to pre-approved areas. All personnel will be informed that their activities must be confined to pre-approved areas.
- Vehicles and equipment will be inspected for remnant soils, vegetation, and debris, and will be cleaned of these materials before they are brought to the Project area.
- Prior to arrival at the work site, all Contractor vehicles and equipment will be cleaned. Any required power washing with water will be done before vehicles enter the Project ROW. Power washing will be done at a location where wash water and solids can be returned to an authorized disposal facility.
- Prior to the clearing of vegetation within Project areas, inspections for existing noxious weeds and invasive plant species will be conducted. Infestations will be flagged for reference in clearing the ROW and for post-construction monitoring.
- ETC Tiger will ensure that seeds for revegetation and straw or hay bales used for sediment barrier installations or mulch distribution, where appropriate, are free of noxious weeds and invasive plant species.
- ETC Tiger will utilize mechanical treatment or herbicide application to pre-treat or control the spread of noxious weeds and invasive species before, during, and/or after construction.
- Herbicides will be applied according to manufacturer's printed recommendations and in accordance with applicable agency regulations governing herbicide application. No herbicides or pesticides will be used within 100 feet of waterbodies or wetlands. The Contractor will ultimately determine the appropriate herbicide application method.
- Waterbodies with known or potential populations of aquatic noxious and invasive plant species will be identified to ensure the proper measures are implemented during in-stream construction activities. Additional BMPs specific to minimizing the potential transport of aquatic noxious and invasive plant species include restricting equipment and materials to that necessary for waterbody construction and reviewing all materials to make certain they are visibly free of vegetation and debris prior to entering and exiting a waterbody. Clean equipment will be used to install waterbody

crossings. Equipment condition and cleanliness will be verified by inspection staff prior to initiating in-stream activities.

6.0 TREATMENT METHODS

Herbicides will be applied according to the manufacturer's printed recommendations and in accordance with applicable agency regulations. If herbicides are not approved for use in an area by the applicable agencies, then mechanical methods may be used in lieu of herbicides. Reapplication of herbicides will occur as needed, based on the findings of monitoring activities described above. Additional mechanical methods may be warranted to remove future growth of noxious weed and invasive plant species.

In the event that noxious weed or invasive plant species become established in the ROW and are not influenced by adjacent weed-infested areas (off ROW), ETC Tiger will make good faith efforts to control weeds in the ROW and to work with adjacent landowners to prevent the spread of the species to adjacent lands. ETC Tiger ROW representatives will contact the landowner/land manager to inform them of any noxious weeds or invasive plant species found on the land in question and will coordinate with the landowner / land manager for the approval of recommended treatment methods. If the landowner requests treatment, ETC Tiger will treat and monitor the noxious weed or invasive plant population, as allowable by the landowner. Pre-existing infestations of noxious weeds or invasive plant species off the ROW will not be treated under this Plan.

7.0 COMPLIANCE

It is the responsibility of the Contractor and the designated Environmental Inspector (EI) to ensure that Contractor personnel are complying with all noxious weed and invasive plant species control measures. The EI has authority to enforce and require compliance with this Plan and to stop work activities if a Contractor does not comply with the appropriate control measures.

8.0 POST-CONSTRUCTION MONITORING

Post-construction monitoring of noxious weeds and invasive plant species populations and colonization of the ROW will be conducted in accordance to the FERC Plan by implementing the following measures:

- Conduct follow-up inspections of all disturbed areas, as necessary, to determine the success of revegetation and address landowner concerns. At a minimum, conduct inspections after the first and second growing seasons.
- Revegetation in non-agricultural areas shall be considered successful if upon visual survey the density and cover of non-nuisance vegetation are similar in density and cover to adjacent undisturbed lands. In agricultural areas, revegetation shall be considered successful when upon visual survey, crop growth and vigor are similar to adjacent undisturbed portions of the same field, unless the easement agreement specifies otherwise. Revegetation efforts will continue until revegetation is successful.
- Monitor and correct problems with drainage and irrigation systems resulting from pipeline construction in agricultural areas until restoration is successful.
- Restoration shall be considered successful if the ROW surface condition is similar to adjacent undisturbed lands, construction debris is removed (unless otherwise approved by the landowner or

land managing agency per section V.A.6 of the FERC Plan), revegetation is successful, and proper drainage has been restored.

- Efforts to control unauthorized off-road vehicle use, in cooperation with the landowner, shall continue throughout the life of the Project. Maintain signs, gates, and permanent access roads, as necessary.

Monitoring reports detailing the success of ROW restoration and revegetation measures will identify noxious weed and invasive plant species' colonization locations and densities, as well as the management measures that will be implemented to control the identified populations. If after two years of monitoring the densities of noxious weed and invasive plant species are documented below or consistent with off ROW densities, then ETC Tiger will cease monitoring activities upon approval from the appropriate regulatory agencies.

ETC Tiger will submit quarterly activity reports to the FERC for at least two years, following construction. These reports may include information such as:

- A summary of the general vegetative cover and density between the ROW and the comparison with off-ROW vegetation.
- Photographs.
- Identification of areas that require remedial action.
- Recommendations and schedule for remedial action(s).
- Monitoring forms.

Copies of these activity reports will be kept on file with ETC Tiger and will be provided to the appropriate agency personnel upon request.

9.0 REFERENCES

Federal Energy Regulatory Commission (FERC). 2013a. *Upland Erosion Control, Revegetation and Maintenance Plan.* Available at: https://www.ferc.gov/sites/default/files/2020-04/upland-pocket-guide.pdf. Accessed May 2026.

FERC. 2013b. *Wetland and Waterbody Construction and Mitigation Procedures.* Available at: https://www.ferc.gov/sites/default/files/2020-04/wetland-waterbody-construction-mitigation-procedures.pdf. Accessed May 2026.

U.S. Fish and Wildlife Service (USFWS). 2026. Invasive species. Available online at: http://www.fws.gov/invasives/. Accessed May 2026.

APPENDIX C-7

Environmental Complaint Resolution Procedures

ETC Tiger Pipeline, LLC

Franklin Farms Project

Environmental Complaint Resolution Procedures

June 2026

ENERGY TRANSFER

ETC TIGER PIPELINE, LLC
ENVIRONMENTAL COMPLAINT RESOLUTION PROCEDURES

ETC Tiger Pipeline, LLC (ETC Tiger) has developed its Environmental Complaint Resolution Procedures to address concerns or complaints received from affected landowners during construction of the Franklin Farms Project (Project). Affected landowners are being provided with a copy of the Environmental Complaint Resolution Procedures in this mailer, prior to the commencement of construction. ETC Tiger is committed to working closely with affected landowners to ensure that any concerns or complaints are resolved satisfactorily and in a timely manner.

Affected landowners are encouraged to contact ETC Tiger with any problems or concerns regarding construction of the Project. ETC Tiger will respond to any complaints or concerns within 48 hours of receiving a call from an affected party. The Project Right-of-Way Manager should be the first contact for affected landowners. The ETC Tiger Project Right-of-Way Manager can be contacted via telephone at (800) 777-6444 (Toll Free), and will direct them to an appropriate person, such as the Lead Environmental Inspector, Chief Inspector, or Right-of-Way Agent, to resolve any issues. If a complaint has been received, a site visit will be scheduled with the affected landowner to assess the problem and provide an appropriate response. The Right-of-Way Manager will also maintain a voice mail system for any calls received after business hours.

If an affected landowner has contacted ETC Tiger and still feels that an appropriate response has not been provided, the landowner may contact the Federal Energy Regulatory Commission (FERC or Commission) directly via the Commission's Dispute Resolution Service (DRS) Landowner Helpline at 1-877-337-2237 (Toll-Free) or by email to LandownerHelp@ferc.gov, or writing to the Dispute Resolution Service, Federal Energy Regulatory Commission, 888 First Street, N.E., Washington, DC 20426.

ETC Tiger will maintain a written record of all in-coming and out-going calls regarding landowner complaints related to its Project. A copy of this record will be maintained in ETC Tiger's Right-of-Way office. Each complaint received will be documented on a Landowner Complaint Resolution Form (example attached). Copies of this form will be distributed to the Chief Inspector, the Lead Environmental Inspector and ETC Tiger's Right-of-Way Agent.

Information regarding landowner complaints will be summarized in tabular form in a weekly report (example attached), and the table will be included in the Weekly Environmental Status Report submitted to the Commission.

ETC Tiger Pipeline, LLC
Franklin Farms Project
Landowner Complaint Resolution Form

Report Number: _____

Date: _____

Time: _____

Landowners Name: _____

Location of Concern: _____

Tract: _____

Facility: _____

Call taken by: _____

Description of Concern: _____

_____

_____

_____

Inspected By: _____

Date and Time: _____

Description: _____

_____

_____

Corrective Action(s): _____

_____

Status: _____

ROW Agent Notified: _____

Date & Time: _____

Environmental Complaint Resolution Procedures

Franklin Farms Project

Environmental Complaint Resolution Program
Weekly Environmental Status Report

Date Prepared: _____

Date of Call	Facility Name	Description of Concern	Status of Concern

APPENDIX D-1

Agency Correspondence

(Portions Submitted as "CUI/CRITICAL ENERGY INFRASTRUCTURE INFORMATION – DO NOT RELEASE")

Outlook

Re: [EXTERNAL] RE: Franklin Farms Project Review Request- Part 3

From Thompson, Ashley <TAThompson@trccompanies.com>
Date Wed 6/10/2026 6:36 PM
To cmichon <cmichon@wlf.la.gov>
Cc Minter, Justin D <justin.minter@energytransfer.com>; dbutler <dbutler@wlf.la.gov>

1 attachment (410 KB)
Franklin Farms Project_20260513.kml;

Hi Carolyn,

Dave had also requested that I provide the kmz of the project area. Please see the attached kmz of the Project area.

Reach out if you have any questions.

Also, I realized I copied the incorrect Dave, my apologies!

Thank you,

Ashley Thompson
Southern Plains Area Manager- Planning, Permitting, & Licensing

TRC

11767 Katy Freeway, Suite 850, Houston, TX 77079
C 941.773.1848 | TAThompson@trccompanies.com
LinkedIn | Instagram | TRCcompanies.com

From: Thompson, Ashley <TAThompson@trccompanies.com>
Sent: Wednesday, June 10, 2026 7:06 AM
To: cmichon <cmichon@wlf.la.gov>
Cc: dave <dave@swgeophys.com>; Minter, Justin D <justin.minter@energytransfer.com>
Subject: Re: [EXTERNAL] RE: Franklin Farms Project Review Request- Part 3

Part 3 of the pdf package attached.

Thank you,
Ashley

From: Thompson, Ashley <TAThompson@trccompanies.com>
Sent: Wednesday, June 10, 2026 7:04 AM
To: cmichon@wlf.la.gov <cmichon@wlf.la.gov>

Cc: Dave Decker <dave@swgeophys.com>; Minter, Justin D <justin.minter@energytransfer.com>
Subject: Re: [EXTERNAL] RE: Franklin Farms Project Review Request- Part 2

Part 2 of the pdf package attached.

Thank you,
Ashley

From: Thompson, Ashley <TAThompson@trccompanies.com>
Sent: Wednesday, June 10, 2026 6:56 AM
To: cmichon@wlf.la.gov <cmichon@wlf.la.gov>
Cc: Dave Decker <dave@swgeophys.com>; Minter, Justin D <justin.minter@energytransfer.com>
Subject: Fw: [EXTERNAL] RE: Franklin Farms Project Review Request- Part 1

Good Morning Carolyn,

I spoke with Dave on Monday after sending in the attached. He requested that I send this over to you for review and comment.

There were access issues getting into the external SharePoint link I provided, so I broke apart the pdf due to size and have attached part 1 of 3 of the pdf request package.

The remaining pdf parts will come soon after this email.

Please feel free to call me with any questions, thank you!

Ashley Thompson
Southern Plains Area Manager- Planning, Permitting, & Licensing

TRC

11767 Katy Freeway, Suite 850, Houston, TX 77079
C 941.773.1848 | TAThompson@trccompanies.com
LinkedIn | Instagram | TRCcompanies.com

From: Dave Butler <dbutler@wlf.la.gov>
Sent: Monday, June 8, 2026 11:09 AM
To: Thompson, Ashley <TAThompson@trccompanies.com>
Subject: [EXTERNAL] RE: Franklin Farms Project Review Request

This is an **External** email. Do not click links or open attachments unless you validate the sender and know the content is safe.

ALWAYS hover over the link to preview the actual URL/site and confirm its legitimacy.

cmichon@wlf.la.gov

Carolyn Michon will be the contact for this. 337-735-8734

Dave Butler

Permits Coordinator
Louisiana Department of Wildlife and Fisheries

2000 Quail Drive

Baton Rouge, LA 70808

(504) 286-4173 New Orleans Office

(225) 763-3595 Baton Rouge Office

From: Thompson, Ashley <TAThompson@trccompanies.com>
Sent: Monday, June 8, 2026 8:15 AM
To: Dave Butler <dbutler@wlf.la.gov>
Cc: justin.minter <justin.minter@energytransfer.com>
Subject: Franklin Farms Project Review Request

You don't often get email from tathompson@trccompanies.com. Learn why this is important

EXTERNAL EMAIL: Please do not click on links or attachments unless you know the content is safe.

Good Morning Dave,

ETC Tiger Pipelines LLC (ETC), an affiliate of Energy Transfer Companies LP, is requesting your review and concurrence with the recommended biological determinations for state protected species with the potential to occur in the

Franklin Farms Project (Project). The proposed Project is located in Richland Parish, Louisiana and is a Federal Energy Regulatory Commission (FERC) regulated Project.

Due to the size of the file, I have uploaded and shared the document to you on our secure external SharePoint site here:
LDWF_FranklinFarms_6.8.26.pdf

If you have an issue downloading or questions on this request, feel free to give me a call on my cell.

Thank you!

Ashley Thompson
Southern Plains Area Manager- Planning, Permitting, & Licensing

11767 Katy Freeway, Suite 850, Houston, TX 77079

C 941.773.1848 | TAThompson@trccompanies.com

LinkedIn | Instagram | TRCcompanies.com

Seguy, Lauren

From: Thompson, Ashley <TAThompson@trccompanies.com>
Sent: Wednesday, June 10, 2026 8:06 AM
To: cmichon
Cc: dave; Minter, Justin D
Subject: Re: [EXTERNAL] RE: Franklin Farms Project Review Request- Part 3
Attachments: LDWF_FranklinFarms_6.8.26 Part 3_opt.pdf

Part 3 of the pdf package attached.

Thank you,
Ashley

From: Thompson, Ashley <TAThompson@trccompanies.com>
Sent: Wednesday, June 10, 2026 7:04 AM
To: cmichon@wlf.la.gov <cmichon@wlf.la.gov>
Cc: Dave Decker <dave@swgeophys.com>; Minter, Justin D <justin.minter@energytransfer.com>
Subject: Re: [EXTERNAL] RE: Franklin Farms Project Review Request- Part 2

Part 2 of the pdf package attached.

Thank you,
Ashley

From: Thompson, Ashley <TAThompson@trccompanies.com>
Sent: Wednesday, June 10, 2026 6:56 AM
To: cmichon@wlf.la.gov <cmichon@wlf.la.gov>
Cc: Dave Decker <dave@swgeophys.com>; Minter, Justin D <justin.minter@energytransfer.com>
Subject: Fw: [EXTERNAL] RE: Franklin Farms Project Review Request- Part 1

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Ashley Thompson
Southern Plains Area Manager- Planning, Permitting, & Licensing

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To: Thompson, Ashley <TAThompson@trccompanies.com>
Subject: [EXTERNAL] RE: Franklin Farms Project Review Request

This is an **External** email. Do not click links or open attachments unless you validate the sender and know the content is safe.

ALWAYS hover over the link to preview the actual URL/site and confirm its legitimacy.

cmichon@wlf.la.gov

Carolyn Michon will be the contact for this. 337-735-8734

Dave Butler

Permits Coordinator
Louisiana Department of Wildlife and Fisheries

2000 Quail Drive

Baton Rouge, LA 70808

(504) 286-4173 New Orleans Office

(225) 763-3595 Baton Rouge Office

From: Thompson, Ashley <TAThompson@trccompanies.com>
Sent: Monday, June 8, 2026 8:15 AM
To: Dave Butler <dbutler@wlf.la.gov>
Cc: justin.minter <justin.minter@energytransfer.com>
Subject: Franklin Farms Project Review Request

You don't often get email from tathompson@trccompanies.com. Learn why this is important

EXTERNAL EMAIL: Please do not click on links or attachments unless you know the content is safe.

Good Morning Dave,

ETC Tiger Pipelines LLC (ETC), an affiliate of Energy Transfer Companies LP, is requesting your review and concurrence with the recommended biological determinations for state protected species with the potential to occur in the

Franklin Farms Project (Project). The proposed Project is located in Richland Parish, Louisiana and is a Federal Energy Regulatory Commission (FERC) regulated Project.

Due to the size of the file, I have uploaded and shared the document to you on our secure external SharePoint site here: LDWF_FranklinFarms_6.8.26.pdf

If you have an issue downloading or questions on this request, feel free to give me a call on my cell.

Thank you!

Ashley Thompson
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Seguy, Lauren

From: Cox, Maxwell - FPAC-NRCS, LA <max.cox@usda.gov>
Sent: Tuesday, June 9, 2026 3:12 PM
To: Soileau, Joshua
Cc: Barrera, Andrea; Marsano, Kaylyn; Seguy, Lauren; Thompson, Ashley
Subject: [EXTERNAL] RE: [External Email]Seed Mix Recommendation
Attachments: FERC Seed Mix_ Franklin Farms Pipeline_Richland.docx

This is an **External** email. Do not click links or open attachments unless you validate the sender and know the content is safe.

ALWAYS hover over the link to preview the actual URL/site and confirm its legitimacy.

Josh,

Attached is my recommendations for re-establishing vegetation on disturbed areas post project. If you need anything else please don't hesitate to reach out.

V/R

Max

Max Cox
Area Biologist
USDA – NRCS - Louisiana | Area 1
Monroe Area Office
U.S. DEPARTMENT OF AGRICULTURE
Natural Resources Conservation Service
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Monroe, LA 71203
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From: Soileau, Joshua <JSoileau@trccompanies.com>
Sent: Tuesday, June 9, 2026 1:08 PM
To: Cox, Maxwell - FPAC-NRCS, LA <max.cox@usda.gov>
Cc: Barrera, Andrea <ABarrera@trccompanies.com>; Marsano, Kaylyn <KMarsano@trccompanies.com>; Seguy, Lauren <LSeguy@trccompanies.com>; Thompson, Ashley <TAThompson@trccompanies.com>
Subject: [External Email]Seed Mix Recommendation

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Max,

I am currently working on an environmental report for a proposed pipeline in Richland Parish. Per Federal Energy Regulatory Commission (FERC) guidance, we need a seed mix recommendation from NRCS for areas that will be revegetated following construction. I have attached a KMZ of the proposed route.

The route is composed of a mix of landuses, predominantly cropland and forestland, but some improved pasture will be crossed. Soils within non-agricultural areas (i.e., forestland) are primarily of the Necessity, Gilbert, Forestdale, and Egypt series. The proposed route is a mix of upland and wetlands.

Can you provide us with a seeding recommendation suitable for these conditions?

Joshua Soileau
Project Manager/Senior Biologist

4545 Sherwood Common Blvd., Building 3, Suite A, Baton Rouge, LA 70816
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USDA
United States Department of Agriculture

June 9, 2026

USDA-NRCS
141 Industrial Loop
Rayville, LA 71269

Subject: Seed Mixes for Franklin Farms Pipeline Project (Richland Parish)

Joshua Soileau – Project Manager,

After reviewing the footprint of the proposed pipeline in Richland Parish, I would suggest the following for reseeding/planting the areas that would potentially be disturbed by the project.

1. Cropland Areas:
    a. Reestablish existing elevation and topographic features to existing condition post project.
    b. Plant an annual grass (ex: Wheat, Cereal Rye, or Oats) to maintain soil integrity and reduce the potential for soil erosion post project.
2. Improved Pastures:
    a. Reestablish existing pasture vegetation (Common Bermuda, Bahia, etc…) Post project.
3. Forested Areas:
    a. Includes (#s/ac): Rice Hulls (filler) (10), Yarrow (0.015), Milkweed, Butterfly (Butterflyweed) (0.01), Oats, Jerry (7), Partridge Pea (0.2), River Oats (0.01), Coreopsis, bigflower (0.3), Coreopsis, Lanceleaf (sand) (0.5), Coreopsis, Plains (0.03), Prairie Clover, White (0.1), Prairie Clover, Purple (0.3), Coneflower, Pale Purple (0.005), Coneflower, Purple (0.5), Wildrye, Virginia (0.1), Rattlesnake Master (0.02), Indian blanket (0.2), Blazingstar, Rough (Tall, Button) (0.005), Cardinal Flower (0.01), Sensitive briar(Nuttall's) (0.005), Mint, Lemon (0.04), Mint, Dotted (Spotted Bee Balm) (0.015), Beardtongue, Foxglove (Smooth penstemon) (0.04), Mint, Narrowleaf Mountain (Slender) (0.023), Susan, Black-eyed (0.05), Bluestem, Little (Aldous) (0.5), Goldenrod, Gray (Field) (Old Field) (0.05), Spiderwort, Ohio (Common) (0.005).
    b. The above mix can be found at (https://www.pfhabitatstore.com/store/22081/LA/LA-SM01-Solar-ROW-Mix-With-Virginia-Wildrye-cool-season-grass-2026). This mix was created in cooperation between NRCS and Quail Forever.

If you have any questions, please feel free to contact me by email (Max.Cox@usda.gov) or by phone (318-488-2323).

Sincerely,

Max Cox

Max Cox
USDA-NRCS
Area Biologist
LA-Area 1 Wetland & HEL Compliance Lead

APPENDIX D-2

SHPO Correspondence

Submitted as CUI/PRIVILEGED AND CONFIDENTIAL – DO NOT RELEASE

APPENDIX D-3

Native American Organizations and Tribes Correspondence

Submitted as CUI/PRIVILEGED AND CONFIDENTIAL – DO NOT RELEASE

APPENDIX E

Landowner Information

APPENDIX E-1

Landowner/Stakeholder List

Submitted as CUI/PRIVILEGED AND CONFIDENTIAL – DO NOT RELEASE

APPENDIX F

Wetland and Waterbody Delineation Report and Listed Species Report

(Portions filed in Critical Energy Infrastructure Information)

APPENDIX F-1

Wetland and Waterbody Delineation Report

(Portions filed in Critical Energy Infrastructure Information)

Wetlands and Other Waters Delineation Report

Franklin Farms Project
Richland Parish, Louisiana

TRC Project No.: 000705831.0000.0000

May 2026

Prepared For:
ETC Tiger Pipeline, LLC
1300 Main Street
Houston, TX 77002

Prepared By:
TRC Companies, Inc.
11767 Katy Freeway, Suite 850
Houston, TX 77079

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Acronyms and Abbreviations

Notation	Definition
1987 Manual	*Corps of Engineers Wetlands Delineation Manual*
APT	Antecedent Precipitation Tool
CFR	Code of Federal Regulations
CWA	Clean Water Act
EPA	U.S. Environmental Protection Agency
FAC	Facultative
FACU	Facultative Upland
FACW	Facultative Wetland
FEMA	Federal Emergency Management Agency
FF	Franklin Farms
GIS	Geographic Information System
GPS	Global Positioning System
Guidebook	*USACE Jurisdictional Determination Form Instructional Guidebook*
MRLC	Multi-Resolution Land Characteristics Consortium
LDWF	Louisiana Department of Wildlife and Fisheries
NHD	National Hydrography Dataset
NLCD	National Land Cover Database
NRCS	Natural Resources Conservation Service
NRPW	Non-relatively permanent water
NWI	National Wetlands Inventory
NWPL	National Wetland Plant List
NWPR	Navigable Waters Protection Rule
OBL	Obligate Wetland
OHWM	Ordinary high-water mark
Project	Franklin Farms 7(c) Project
Regional Supplement	Regional Supplement to the Corps of Engineers Wetland Delineation Manual: Atlantic and Gulf Coastal Plain Region, Version 2.0
RHA	River and Harbors Act
ROW	Right-of-way
RPW	Relatively permanent water
Sackett	*Sackett v. Environmental Protection Agency* U.S. Supreme Court Case
TNW	Traditional navigable water
TRC	TRC Environmental Corporation
UPL	Obligate Upland
USACE	U.S. Army Corps of Engineers
USFWS	U.S. Fish and Wildlife Service
USGS	U.S. Geological Survey
WOTUS	Waters of the U.S.

1.0 Introduction

ETC Tiger Pipeline, LLC contracted TRC Environmental Corporation (TRC) to support their environmental due diligence review of the proposed Franklin Farms Project (Project) located in Richland Parish, Louisiana (Project). The Project includes the following components:

- Franklin Farms (FF) Lateral - approximately 13.2-mile 36” pipeline lateral
- FF Trio Farm Lateral - approximately 1.7-mile 20” pipeline lateral
- FF Gulf Run Bi Di Meter Station
- FF Gulf South Receipt Meter Station
- FF Bee Bayou Delivery Meter Station
- FF Interconnect Site
- FF Gulf Crossing Receipt Meter Station
- FF MEP Receipt Meter Station
- Franklin Farms Delivery Meter Station
- FF Trio Farm Delivery Meter Station
- MEP Tap Site

This report summarizes the results of a desktop data review, on-site delineation of wetlands and other waters for the Project and assesses the anticipated U.S. Army Corps of Engineers (USACE) jurisdictional status of identified aquatic features. TRC performed this delineation of wetlands and other waters to identify the spatial extent and location of wetlands, streams, and other aquatic resources within the Project Area. Aquatic resources that are considered Waters of the U.S. (WOTUS), as defined by the U.S. Environmental Protect Agency (EPA) and USACE, are subject to regulation under Section 404 of the Clean Water Act (CWA) and Section 10 of the Rivers and Harbors Act (RHA). The regulatory authority of jurisdictional aquatic resources within the Project Area is the USACE Vicksburg District. The wetland and other waters delineation and an opinion on the jurisdictional status of identified aquatic features within the Project Area may be used to determine the extent of potential impacts to identified waters as a result of the Project.

1.1 Regulatory Background

WOTUS are regulated under Section 404 of the CWA, and a subset of those waters are subject to regulation under Section 10 of the RHA. The EPA is responsible for administering the laws and regulations of the CWA; however, the USACE has the primary regulatory authority for enforcing Sections 404 and 10 requirements for WOTUS, including wetlands. Section 404 of the CWA governs the discharge of dredged or fill material into WOTUS and Section 10 of the RHA regulates any work or structures that potentially affect the navigable capacity of a waterbody. Any activity that involves the discharge of dredged or fill material into WOTUS is subject to regulation by the USACE. WOTUS are defined to encompass navigable waterways; interstate waters; all other waters where their use, degradation, or destruction could affect interstate or foreign commerce;

tributaries of any of these waters; and wetlands that meet any of these criteria or are adjacent to any of these waters or their tributaries.

The regulatory definition and rules defining which waters are considered jurisdictional WOTUS have been contested in recent years and are currently in transition. On April 21, 2020, the Navigable Waters Protection Rule (NWPR) revised the definition of WOTUS and effectively changed which waters are considered WOTUS. The regulations (e.g., 40 Code of Federal Regulations [CFR] 230.3[s]); court decisions (e.g., *Rapanos v. United States*; *Carabell v. United States*; *Solid Waste Agency of Northern Cook County v. United States*), subsequent guidance documents, agency practices, memoranda, and guidance in place prior to the NWPR can be referenced as the pre-2015 regulatory regime. The NWPR was vacated and remanded by the U.S. District Court for the District of Arizona on August 30, 2021. The USACE and EPA issued a "Revised Definition of 'Waters of the United States'" on January 18, 2023, that became effective on March 20, 2023. Due to litigation, this final rule was not operative in all states, including Louisiana.

On May 25, 2023, the U.S. Supreme Court issued a decision in the *Sackett v. Environmental Protection Agency* case (*Sackett*) calling for further revisions to the definition of WOTUS. The *Sackett* decision restricts the agencies (i.e., USACE and EPA) from regulating certain types of wetlands and streams – specifically, wetlands that do not have a continuous surface connection with a navigable water. While streams were not specifically addressed in the *Sackett* decision, ephemeral streams and other waterbodies that are not relatively permanent would not be considered jurisdictional and, thereby, would no longer be federally regulated.

On August 29, 2023, the EPA and USACE issued a final rule to amend the "Revised Definition of 'Waters of the United States'" rule to conform with the Supreme Court's *Sackett* decision and definition of WOTUS. The conforming rule, "Revised Definition of 'Waters of the United States'; Conforming," was published in the Federal Register and became effective on September 8, 2023 in all states without ongoing litigation regarding the "Revised Definition of 'Waters of the United States'." Louisiana, along with 26 other states, have ongoing litigation in respect to the "Revised Definition of 'Waters of the United States'. For the 27 states currently in litigation, the EPA and USACE are interpreting WOTUS consistent with the pre-2015 regulatory regime and the Supreme Court's decision in *Sackett*. On March 12, 2025, the EPA and USACE issued regulatory guidance pertaining to the jurisdiction of wetlands for all states (USACE and EPA, 2025); however, the guidance did not address the jurisdiction of streams with respect to the *Sackett* decision. On March 24, 2025, the EPA and USACE announced a FR notice publicizing a series of six listening sessions and a 30-day recommendations docket to solicit feedback on key aspects of the definition of WOTUS before issuing further changes to the definition of WOTUS in response to the *Sackett* ruling.

Recent training and discussions with the USACE have resulted in the determination that most ephemeral streams and watercourses (e.g., swales, ditches, erosional areas, etc.) would be considered non-jurisdictional with consideration of the *Sackett* ruling. Some ephemeral streams may still be considered jurisdictional if more than half of the entire stream order in question meets

the definition of a relatively permanent water (RPW). A formal guidance document on the jurisdiction of ephemeral streams with respect to the *Sackett* decision has not been issued by the EPA or USACE.

2.0 Methods

2.1 Project Location

The Project encompasses approximately 691 acres and is located in Richland Parish, Louisiana approximately 4 to 5 miles east of Rayville, Louisiana (Project Area). A map illustrating the general vicinity of the Project Area is shown in **Figure 1** of **Appendix A**. Maps illustrating aerial imagery and topography of the Project Area are shown in **Figures 2 and 3** of **Appendix A**. The Project Area primarily consists of the forested areas, agricultural land, and agricultural ditches.

2.2 Desktop Data Review

Prior to conducting the on-site wetland and other waters delineation, TRC conducted a desktop-level review of readily and publicly available site information including, but not limited to, the following:

- Recent and historical aerial imagery (Google Earth Pro, 2026);
- U.S. Geological Survey (USGS) 7.5-minute Series Topographic Maps – Whelen Springs, Arkansas (USGS, 2026a);
- USGS National Land Cover Database ([NLCD]; Multi-Resolution Land Characteristics Consortium [MRLC], 2023);
- USGS National Hydrography Dataset ([NHD]; USGS, 2026b);
- Natural Resource Conservation Service (NRCS) Web Soil Survey (NRCS, 2026a) and;
- U.S. Fish and Wildlife Service (USFWS) National Wetlands Inventory (NWI) data (USFWS, 2026); and
- Federal Emergency Management Agency (FEMA) National Floor Hazard Layer data and Flood Insurance Rate Maps (FEMA, 2026).

Recent and historical aerial imagery were reviewed for signatures of hydrologic activity and saturation that could indicate presence of unmapped aquatic features. USGS topographic maps were reviewed for mapped aquatic features and other topographic features that may favor the formation of wetlands and streams such as valleys, depressions, and low-lying areas. Soil survey data was reviewed to assess the Project Area for the presence of hydric soils. The Project Area was screened for aquatic features included in the USFWS NWI dataset and the USGS NHD. Available FEMA flood maps and spatial data were reviewed to identify areas within mapped floodplains that could be subject to flooding. The results from the desktop analysis are presented in **Section 3.1**.

2.3 Field Investigation

TRC Biologists, Jeff Bradstreet, Emme Elliott, Ben Parrish, Elisabeth Offer, and Morgan Thigpen, conducted field surveys for aquatic and biological resources on February 23 to February 28, 2026, March 2 to March 4, 2026, and May 5 to May 8, 2026. The Project Area was examined for the presence of atypical situations to identify if any recent natural or human-induced alterations may have significantly changed the area's vegetation, soils, and/or hydrology.

The objectives of the wetlands and other waters delineation were to: 1) characterize on-site vegetation; 2) determine if on-site soils were hydric or non-hydric; 3) inspect on-site hydrology; and 4) assess, identify, and delineate wetlands and other waters present within the Project Area. The delineation was performed in accordance with the *Corps of Engineers Wetlands Delineation Manual* (*1987 Manual;* Environmental Laboratory, 1987) and the *Regional Supplement to the Corps of Engineers Wetland Delineation Manual: Atlantic and Gulf Coastal Plain, Version 2.0* (*Regional Supplement*; USACE, 2012). During the field survey, TRC identified aquatic features and delineated the boundaries while collecting sampling points using a global positioning system (GPS) with submeter accuracy. Results of the field investigation were mapped with geographic information system (GIS) software and shown in **Figures 7 and 8** of **Appendix A** and further detailed in **Section 3.2**. Field data was recorded on datasheets and representative photographs were taken of the Project Area and are included in **Appendix B**.

2.3.1 Wetlands

On-site wetland determinations were made in accordance with the technical approach outlined in the *1987 Manual* and *Regional Supplement*. The EPA and USACE define wetlands as "areas that are inundated or saturated by surface or ground water at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for line in saturated soil conditions." To be classified as a wetland, a feature must meet specific diagnostic characteristics for hydrophytic vegetation, hydric soils, and wetland hydrology. For identified features determined to be a wetland, a minimum of one USACE Wetland Determination Data Sheet was completed for each identified wetland as well as an associated upland data point. The boundary of the wetland and all Wetland Determination Data Sheet sample point locations were recorded using a submeter GPS.

Under the pre-2015 regulatory regime, the USACE criteria used in determining jurisdictional status of a wetland includes the wetland's physical proximity to a jurisdictional stream or a significant nexus (e.g., a physical, biological, or chemical connection) to a jurisdictional stream. The 2023 *Sackett* decision and subsequent changes to the definition of WOTUS have removed the significant nexus criteria when defining a wetland and determining its jurisdictional status. The March 12, 2025, regulatory guidance issued by the EPA and USACE pertaining to the jurisdiction of wetlands states a jurisdictional wetland must: 1) Be adjacent to (i.e., directly abutting; defined as "having a continuous surface connection" [40 CFR 120 and 33 CFR 328.3]) a WOTUS, and 2) have a continuous surface connection such that it is difficult to determine where the covered water ends and the wetland begins (USACE and EPA 2025). Wetlands must be truly physically abutting

jurisdictional water and cannot be separated by berms, dikes, or uplands. Additionally, wetlands that rely on intermittent flows via channels, ditches, swales, or culverts are no longer considered "adjacent" and are not considered jurisdictional.

The following is a description of the evaluation of hydrophytic vegetation, hydric soils, and wetland hydrology used when determining the presence of a wetland:

Hydrophytic Vegetation

The *1987 Manual* defines hydrophytic vegetation as, "the sum total of macrophytic plant life that occurs in areas where the frequency and duration of inundation or soil saturation produce permanently or periodically saturated soils of sufficient duration to exert a controlling influence on the plant species present" (Environmental Laboratory, 1987). Plant species are categorized according to their probability of occurrence in wetlands and have been assigned an "Indicator Status" that ranges from Obligate Wetland (i.e., greater than 99 percent occurrence in wetlands) to Obligate Upland (i.e., almost always does not occur in wetlands). The indicator status of plant species is maintained by The USACE National Wetland Plant List ([NWPL]; USACE, 2023). Indicator status categories are defined as follows:

- Obligate Wetland (OBL): A species that almost always (under natural conditions) occurs in wetlands (estimated probability greater than 99 percent).
- Facultative Wetland (FACW): A species that usually occurs in wetlands (estimated probability 67 to 99 percent) but occasionally is found in non-wetlands.
- Facultative (FAC): A species that is equally likely to occur in wetlands and non-wetlands (estimated probability in wetlands 34 to 66 percent).
- Facultative Upland (FACU): A species that usually occurs in non-wetlands (estimated probability 67 to 99 percent) but is occasionally found in wetlands.
- Obligate Upland (UPL): A species that almost always (under natural conditions) occurs in non-wetlands (estimated probability greater than 99 percent).

When assessing the presence of hydrophytic vegetation, the plant community at a sample location is characterized on Wetland Determination Data Sheets by identifying dominant plant species using the Dominance Test. For each stratum in the plant community (e.g., tree, sapling and shrub, herbaceous, and woody vine), a list of plant species and their respective percent cover was recorded. Species naming conventions and indicator status were obtained from the USACE NWPL (USACE, 2023). Absolute percent cover for each plant species was recorded within a 5-foot radius around a central observation point for the herbaceous stratum, a 15-foot radius for the saplings/shrub stratum, and 30-foot radius for the tree and woody vine strata. The total absolute cover for each stratum may range from zero to over 100 percent, depending on the density and amount of overlapping of vegetation.

"Dominant" plants were classified using the 50/20 rule, in which any plant species that equaled or exceeded 50 percent of the total percent aerial coverage for each stratum, and any additional species comprising 20 percent or more of the same stratum, was classified as a dominant plant.

Vegetation was reevaluated using the Prevalence Index in cases where indicators of hydric soil and wetland hydrology were present, but the percentage of dominant species did not exceed 50 percent utilizing the Dominance Test. The Prevalence Index is a weighted-average wetland indicator status of all plant species in the sample plot, where each indicator status category is given a numeric code and the abundance as evaluated by percent cover is weighted. A site scoring less than or equal to 3.0 on the Prevalence Index meets the wetland hydrophytic vegetation criterion.

Hydric Soils

The *1987 Manual* defines hydric soil as "a soil that is saturated, flooded or ponded long enough during the growing season to develop anaerobic conditions that favor the growth and regeneration of hydrophytic vegetation" (Environmental Laboratory, 1987). Hydric soil indicators within the Atlantic and Gulf Coastal Plain Region were utilized when assessing for the presence of hydric soils at a sample location. The presence or absence of hydric soils was determined by pit sampling to a depth of 16 in or more (when not limited by a restrictive layer) and characterizing the soil profile layers using Munsell soil color charts (X-Rite Incorporated, 2009). For each sample location, the soil profile description and any other applicable hydric soil indicators were recorded on the Wetland Determination Data Sheets. Complete descriptions of the hydric soil indicators within the Atlantic and Gulf Coastal Plain Region are included in the *Regional Supplement* (USACE, 2010).

Wetland Hydrology

The 1987 Manual states that "'wetland hydrology' encompasses all hydrologic characteristics of areas that are periodically inundated or have soils saturated to the surface at some time during the growing season. Areas with evident characteristics of wetland hydrology are those where the presence of water has an overriding influence on characteristics of vegetation and soils due to anaerobic and reducing conditions, respectively" (Environmental Laboratory, 1987). The frequency of soil inundation or saturation is dependent on a variety of factors, including topography, soil stratigraphy, and soil permeability, in conjunction with the water source(s) of precipitation, runoff, stormwater, and groundwater discharge. Wetland hydrology is classified according to the extent of soil saturation or inundation and ranges from permanently inundated to irregularly inundated or saturated. Those areas which are either intermittently or never inundated or saturated are not considered to have wetland hydrology. Complete descriptions of the wetland hydrology indicators within the Atlantic and Gulf Coastal Plain Region are included in the *Regional Supplement* (USACE, 2010). Observed wetland hydrology indicators were recorded on Wetland Determination Data Sheets for each sample location.

2.3.2 Special Aquatic Sites

Special Aquatic Sites are a subset of WOTUS that are defined as "geographic areas, large or small, possessing special ecological characteristics of productivity, habitat, wildlife protection, or other important and easily disrupted ecological values" (40 CFR 230.3) and include (in addition

to wetlands) sanctuaries and refuges, mud flats, vegetated shallows, coral reefs, and riffle and pool complexes (40 CFR 230.40–230.45). The Project Area was evaluated for the presence of Special Aquatic Sites using aerial imagery, topographic maps, and field observations. A Wetland Reserve Easement was identified on the eastern boundary of PFO-T1F-01 and PFO-T1F-04 (shown on Page 10 of Figure 7) and continues east beyond the boundary.

2.3.3 Other Waters

Other waters such as territorial seas, streams, impoundments, ponds, and drainage features observed within the Project Area were identified by the presence of ordinary high-water mark (OHWM) indicators, including but not limited to:

- Presence of a defined bed and bank;
- Open water;
- Evidence of a clear, natural line visible on the bank;
- Distinct channel or depressions;
- Shelving;
- Changes in soil characteristics,
- Observed flow or indicators of recent flow of water;
- Presence of litter and debris; and/or
- Inundation or watermarks indicative of inundation during high water conditions.

When identified, the boundary or OHWM of the features were delineated using a GPS. Data collected for each identified waterbody includes flow type (e.g., tidal, perennial, intermittent, or ephemeral), substrate type (e.g., mud/silt, sand, gravel, large rock, boulder, and/or bedrock), and OHWM width and depth.

Under the pre-2015 regulatory regime, the USACE criteria used in determining jurisdictional status of a waterbody includes the continuous presence of OHWM indicators and downstream connectivity to jurisdictional waterbodies. Downstream connectivity for delineated waterbodies in the field was determined based on the continuous presence of an OHWM and connection to downstream waterbodies. Waterbodies lacking a continuous bed and bank or continuous evidence of an OHWM do not have downstream connectivity and therefore are not considered jurisdictional WOTUS. The pre-2015 regulatory regime included ephemeral streams as jurisdictional features; however, in consideration of the 2023 *Sackett* ruling most ephemeral streams are now considered non-jurisdictional because they do not meet the definition of an RPW. Some ephemeral streams may still be considered jurisdictional if more than half of the entire stream order in question meets the definition of an RPW that intersected the Project Area (**Appendix A**).

3.0 Results

3.1 Desktop Data Review

The desktop analysis provided a preliminary indication of the potential presence of wetlands and other waters, based on a review of available aerial imagery, soils data, floodplain information, and other resources discussed in **Section 2.2**. Review of USGS topographic maps (USGS, 2026a), NHD data (USGS, 2026b), and USFWS NWI data (USFWS, 2026) identified flowlines, waterbodies, and wetlands associated with some waterbodies throughout the Project Area.

3.2 Field Investigation

Results of the February, March, and May 2026 field surveys are illustrated as identified features in **Figures 7 and 8** of **Appendix A**. Datasheets and representative photographs of the Project Area and identified features are provided in **Appendix B**.

3.2.1 Ecoregion

The Project Area is located within the Macon Ridge (Level IV ecoregion 73j) ecoregion of the Mississippi Alluvial Plain ecoregion (Level III ecoregion 73; Daigle et al., 2006). The Macon Ridge ecoregion is a wide, flat to irregular alluvial terrace with relict patterns of branching channels, irregular braided bars, and interfluves that are overlaid by Holocene alluvium. This area contains low-gradient, channelized streams and canals with silty substrates. The Macon Ridge is characterized mainly by forested wetlands, deciduous forest, cropland with sugarcane, cotton, soybeans and corn, pasture, crawfish aquaculture, and some urban and industrial developments. Wetness and flooding are the main limitations to farming.

Review of the NLCD (MRLC, 2023) found that the Project Area contains ten mapped land cover classifications summarized in **Table 1**. A map illustrating the land cover classifications within the Project Area is provided in **Figure 4** of **Appendix A**.

Table 1. Land Cover Classifications Mapped within the Project Area

Land Cover Classification	Acres within Project Area	Percentage within Project Area
Cultivated Crops	455.42	65.92%
Developed, High Intensity	0.24	0.03%
Developed, Low Intensity	26.70	3.87%
Developed, Medium Intensity	3.39	0.49%
Developed, Open Space	41.69	6.03%
Evergreen Forest	1.30	0.19%
Mixed Forest	3.70	0.54%
Pasture/Hay	11.00	1.59%
Shrub/Scrub	0.22	0.03%

Land Cover Classification	Acres within Project Area	Percentage within Project Area
Woody Wetlands	147.17	21.30%
Total	**690.82**	**100.00%**

[1] Land cover classifications from NLCD (MRLC, 2023).

During the on-site field survey, similar landcover types were observed that are consistent with the NLCD landcover classifications described in **Table 1**. Landcover types observed during the field survey include cultivated crops, woody wetlands, and developed space.

3.2.2 Plant Communities

Plant species diversity and vegetation density in the Project Area is consistent with similar habitats of the surrounding region. Plant species observed within the Project Area during the February to March 2026 on-site field survey, along with its wetland indicator status (USACE, 2023) and non-native/invasive status (LDWF, 2019) are presented in **Table 2.**

Table 2. Plant Species Observed within the Project Area

Common Name	Scientific Name	Wetland Indicator Status[1]	Non-native/Invasive Status[2]
Tree Stratum			
Red Maple	*Acer rubrum*	FAC	No
Pignut Hickory	*Carya glabra*	FACU	No
Sugarberry	*Celtis laevigata*	FACW	No
Common Persimmon	*Diospyros virginiana*	FAC	No
Honey-Locust	*Gleditsia triacanthos*	FAC	No
Eastern Red-Cedar	*Juniperus virginiana*	FACU	No
Sweetgum	*Liquidambar styraciflua*	FAC	No
Osage-orange	*Maclura pomifera*	FACU	No
Water Tupelo	*Nyssa aquatica*	OBL	No
Swamp Tupelo	*Nyssa biflora*	OBL	No
Eastern Hop-Hornbeam	*Ostrya virginiana*	FACU	No
Loblolly Pine	*Pinus taeda*	FAC	No
Northern White Oak	*Quercus alba*	FACU	No
Southern Red Oak	*Quercus falcata*	FACU	No
Overcup Oak	*Quercus lyrata*	OBL	No
Blackjack oak	*Quercus marilandica*	UPL	No
Water Oak	*Quercus nigra*	FAC	No
Cherrybark Oak	*Quercus pagoda*	FAC	No
Willow Oak	*Quercus phellos*	FACW	No
Post Oak	*Quercus stellata*	UPL	No

Common Name	Scientific Name	Wetland Indicator Status[1]	Non-native/Invasive Status[2]
Southern Dewberry	*Rubus trivialis*	FACU	No
Bagpod	*Sesbania vesicaria*	FAC	No
Winged Elm	*Ulmus alata*	FACU	No
American Elm	*Ulmus americana*	FAC	No
Cedar Elm	*Ulmus crassifolia*	FAC	No
Sapling/Shrub Stratum			
Red Maple	*Acer rubrum*	FAC	No
Groundseltree	*Baccharis halimifolia*	FAC	No
Sugarberry	*Celtis laevigata*	FACW	No
Common Persimmon	*Diospyros virginiana*	FAC	No
Honey-Locust	*Gleditsia triacanthos*	FAC	No
Yaupon	*Ilex vomitoria*	FAC	No
Eastern Red-Cedar	*Juniperus virginiana*	FACU	No
Chinese Privet	*Ligustrum sinense*	FAC	Yes
Sweetgum	*Liquidambar styraciflua*	FAC	No
Osage-Orange	*Maclura pomifera*	FACU	No
Water Tupelo	*Nyssa aquatica*	OBL	No
Northern White Oak	*Quercus alba*	FACU	No
Water Oak	*Quercus nigra*	FAC	No
Cherrybark Oak	*Quercus pagoda*	FAC	No
Black Oak	*Quercus velutina*	NI	No
Saw-Tooth Blackberry	*Rubus argutus*	FAC	No
Dwarf Palmetto	*Sabal minor*	FACW	No
Chinese Tallow	*Triadica sebifera*	FAC	Yes
Winged Elm	*Ulmus alata*	FACU	No
American Elm	*Ulmus americana*	FAC	No
Cedar Elm	*Ulmus crassifolia*	FAC	No
Herbaceous Stratum (Groundcover)			
Alligator weed	*Alternanthera philoxeroides*	OBL	Yes
Broom-Sedge	*Andropogon virginicus*	FAC	No
Narrow-Leaf Carpet Grass	*Axonopus fissifolius*	FACW	No
Shepherd's-Purse	*Capsella bursa-pastoris*	FACU	No
Hairy Bittercress	*Cardamine hirsuta*	FACU	No
Sand Bittercress	*Cardamine parviflora*	FACU	No
Eastern Woodland Sedge	*Carex blanda*	FAC	No
Cherokee Sedge	*Carex cherokeensis*	FACW	No
Cypress-swamp Sedge	*Carex joorii*	OBL	No
Louisiana Sedge	*Carex louisianica*	OBL	No

Common Name	Scientific Name	Wetland Indicator Status[1]	Non-native/Invasive Status[2]
Shallow Sedge	*Carex lurida*	OBL	No
Hairy-Fruit Chervil	*Chaerophyllum tainturieri*	FAC	No
Indian Wood-Oats	*Chasmanthium latifolium*	FAC	No
Slender Wood-Oats	*Chasmanthium laxum*	FACW	No
Long-Leaf Wood-Oats	*Chasmanthium sessiliflorum*	FAC	No
Bermuda Grass	*Cynodon dactylon*	FACU	Yes
Tall Flat Sedge	*Cyperus eragrostis*	FACW	No
Rusty Flat Sedge	*Cyperus odoratus*	FACW	No
Marsh Flat Sedge	*Cyperus pseudovegetus*	FACW	No
Straw-Color Flat Sedge	*Cyperus strigosus*	FACW	No
Silky Wild Oat Grass	*Danthonia sericea*	FACU	No
Heller's Rosette Grass	*Dichanthelium oligosanthes*	FACU	No
Poorjoe	*Diodia teres*	FACU	No
Virginia Buttonweed	*Diodia virginiana*	FACW	No
Blunt Spike-Rush	*Eleocharis obtusa*	OBL	No
Eastern Daisy Fleabane	*Erigeron annuus*	FACU	No
Asthmaweed	*Erigeron bonariensis*	FACU	No
Dogfennel	*Eupatorium capillifolium*	FACU	No
Evening Trumpet-Flower	*Gelsemium sempervirens*	FAC	No
Blue Mud-Plantain	*Heteranthera limosa*	OBL	No
St. Andrew's-Cross	*Hypericum hypericoides*	FAC	No
Lamp Rush	*Juncus effusus*	OBL	No
Torrey's Rush	*Juncus torreyi*	FACW	No
Eastern Red-Cedar	*Juniperus virginiana*	FACU	No
Henbit Deadnettle	*Lamium amplexicaule*	UPL	No
Common Duckweed	*Lemna minor*	OBL	No
Poorman's-Pepperwort	*Lepidium virginicum*	UPL	No
Chinese Privet	*Ligustrum sinense*	FAC	Yes
Japanese Honeysuckle	*Lonicera japonica*	FACU	Yes
seedbox	*Ludwigia alternifolia*	OBL	No
Marsh Primrose-Willow	*Ludwigia palustris*	OBL	No
Creeping Primrose-Willow	*Ludwigia repens*	OBL	No
Trailing Yellow-Loosestrife	*Lysimachia radicans*	OBL	No
Wart-Removing-Herb	*Murdannia keisak*	OBL	Yes
Tiny Mousetail	*Myosurus minimus*	FACW	No
Wand Panic Grass	*Panicum virgatum*	FAC	No
Golden Crown Grass	*Paspalum dilatatum*	FAC	Yes
Bahia Grass	*Paspalum notatum*	FACU	Yes

Common Name	Scientific Name	Wetland Indicator Status[1]	Non-native/Invasive Status[2]
Vasey's Grass	*Paspalum urvillei*	FAC	Yes
Swamp Smartweed	*Persicaria hydropiperoides*	OBL	No
Savannah-Panic Grass	*Phanopyrum gymnocarpon*	OBL	No
English Plantain	*Plantago lanceolata*	FACU	No
Annual Blue Grass	Poa *annua*	FACU	No
Spiny-Fruit Buttercup	*Ranunculus muricatus*	FACW	No
Hairy Buttercup	*Ranunculus sardous*	FAC	No
saw-Tooth Blackberry	*Rubus argutus*	FAC	No
curly Dock	*Rumex crispus*	FAC	Yes
Dwarf Palmetto	*Sabal minor*	FACW	No
Giant Plume Grass	*Saccharum giganteum*	FACW	No
Canadian Black-Snakeroot	*Sanicula canadensis*	FACU	No
Sawbrier	*Smilax glauca*	FAC	No
Horsebrier	*Smilax rotundifolia*	FAC	No
Spiny-Leaf Sow-Thistle	*Sonchus asper*	FACU	No
Bluejacket	*Tradescantia ohiensis*	FAC	No
Virginia Spiderwort	*Tradescantia virginiana*	FAC	No
Long-Spike Fluff Grass	*Tridens strictus*	FACW	No
White Clover	*Trifolium repens*	FACU	No
Purple-Top Vervain	*Verbena bonariensis*	FAC	No
Brazilian vervain	*Verbena brasiliensis*	FACU	Yes
Brazilian Vervain	*Verbena incompta*	FACU	No
White Vervain	*Verbena urticifolia*	FAC	No
Yellow Crownbeard	*Verbesina occidentalis*	FACU	No
Neckweed	*Veronica peregrina*	FAC	No
Louisiana Vetch	*Vicia ludoviciana*	FACU	No
Woody Vine Stratum			
Japanese Honeysuckle	*Lonicera japonica*	FACU	Yes
Saw-Tooth Blackberry	*Rubus argutus*	FAC	No
Southern Dewberry	*Rubus trivialis*	FACU	No
Horsebrier	*Smilax rotundifolia*	FAC	No
Muscadine	*Vitis rotundifolia*	FAC	Yes

[1] UPL = Upland; FACU = Facultative Upland; FAC = Facultative; FACW = Facultative Wetland; OBL = Obligate. Wetland indicator status from USACE National Wetland Plant List (USACE, 2023).

[2] Non-native/invasive status from LDWF Wildlife Action Plan Plants Database (LDWF, 2019).

3.2.3 Soils

Soils within the Project Area were identified using the NRCS Web Soil Survey database (NRCS,

2026). Review of the soil survey data found a total of 17 soil map units that are mapped within the Project Area. A map of the soils within the Project Area is illustrated in **Figure 5** of **Appendix A**. A summary and abbreviated description of the soils mapped within the Project Area are provided in **Table 3**.

Table 3. Soils within the Project Area

Map Unit Name (Symbol ID)	Acreage of Project Area (percentage)	Hydric Rating (Yes/No)[1]	Description[1]
Calhoun-Calloway silt loams, gently undulating (Cc)	6.31 (0.91%)	Yes	All areas are considered prime farmland. Parent material consists of loess or loess-like material. Minor components are non-hydric.
Deerford silt loam (Da)	4.30 (0.62%)	No	Not prime farmland. Parent material consists of loess. Minor components are a mix of hydric and non-hydric.
Dexter silt loam, 1 to 3 percent slopes (De)	13.23 (1.92%)	No	All areas are considered prime farmland. Parent material consists of thin loess and underlying loamy and sandy sediments on terraces. Minor components are a mix of hydric and non-hydric.
Dexter silt loam, 3 to 5 percent slopes (Df)	3.68 (0.53%)	No	All areas are considered prime farmland. Parent material consists of thin loess and underlying loamy and sandy sediments on terraces. Minor components are a mix of hydric and non-hydric.
Egypt silt loam, 1 to 3 percent slopes (Eg)	4.17 (0.60%)	No	All areas are considered prime farmland. Parent material consists of mixed loess terrace sediments. Minor components are a mix of hydric and non-hydric.
Forestdale silty clay loam, 0 to 1 percent slopes, rarely flooded, upland phase (Fr)	224.53 (32.50%)	Yes	All areas are considered prime farmland. Parent material consists of loess. Minor components are a mix of hydric and non-hydric.
Gigger silt loam, 1 to 3 percent slopes (Ge)	6.35 (0.92%)	No	All areas are considered prime farmland. Parent material consists of a thin loess over terrace sediments. Minor components are a mix of hydric and non-hydric.
Gigger-Gilbert silt loams, gently undulating (Gg)	69.26 (10.03%)	No	All areas are considered prime farmland. Parent material consists of thin and mixed loess over terrace sediments. Minor components are a mix of hydric and non-hydric.
Gilbert silt loam (Gk)	69.31 (10.03%)	Yes	All areas are considered prime farmland. Parent material consists of mixed loess and loamy sediments of late Pleistocene age. Minor components are a mix of hydric and non-hydric.
Gilbert-Egypt silt loams, gently undulating (Gm)	166.51 (24.10%)	Yes	All areas are considered prime farmland. Parent material consists of mixed loess and terrace sediments. Minor components are a mix of hydric and non-hydric.
Grenada silt loam, 1 to 3 percent slopes (Gr)	3.35 (0.49%)	No	All areas are considered prime farmland. Parent material consists of loess. Minor components are a mix of hydric and non-hydric.

Map Unit Name (Symbol ID)	Acreage of Project Area (percentage)	Hydric Rating (Yes/No)[1]	Description[1]
Hebert silt loam, 0 to 1 percent slopes (Hb)	1.28 (0.19%)	No	All areas are considered prime farmland. Parent material consists of Holocene silty alluvium. Minor components are a mix of hydric and non-hydric.
Necessity silt loam, 1 to 3 percent slopes (Ne)	13.19 (1.91%)	No	All areas are considered prime farmland. Parent material consists of loamy sediments. Minor components are a mix of hydric and non-hydric.
Necessity-Gilbert silt loams, gently undulating (Ng)	83.48 (12.08%)	No	All areas are considered prime farmland. Parent material consists of loamy sediments and mixed loess over terrace sediments. Minor components are a mix of hydric and non-hydric.
Perry silty clay loam (Pc)	8.17 (1.18%)	Yes	All areas are considered prime farmland. Parent material consists of clayey alluvium. Minor components are a mix of hydric and non-hydric.
Perry clay (Pd)	7.03 (1.02%)	Yes	All areas are considered prime farmland. Parent material consists of clayey alluvium. Minor components are a mix of hydric and non-hydric.
Rilla silt loam, 0 to 1 percent slopes (Ra)	6.65 (0.96%)	No	All areas are considered prime farmland. Parent material consists of silty alluvium. Minor components are a mix of hydric and non-hydric.

[1] Hydric soil rating and soil descriptions from the NRCS Web Soil Survey (NRCS, 2026).

3.2.4 Hydrology

The Project Area is located within the Big Creek watershed (Hydrologic Unit Code 10: 0805000110). The hydrology of the Project Area generally flows into agricultural ditches, intermittent streams, and perennial streams that flow into Cypress Creek, Bee Bayou, Boggy Bayou, and Cow Bayou.

The FEMA National Flood Hazard Layer (FEMA, 2026) depicts portions of the Project Area within Zone A 100-year floodplain (1% annual chance of flooding) and Zone X (Area with reduced risk due to levee). A summary of the FEMA floodplains within the Project Area are in **Table 4** below.

Table 4. Identified FEMA Floodplains within the Project Area

FIRM Panel	Zone	Acreage of Project Area	Acreage of Project Area	Percentage of Project Area
22083C	A	1% Annual Chance Flood Hazard	35.19	5%
22083C	X	Area with Reduced Risk Due to Levee	655.64	95%

The USACE developed the Antecedent Precipitation Tool ([APT] USACE, 2026) to aid in determining if site-specific observations occurred under normal climatic conditions. The APT was run for the Project Area to assess the climatic conditions at the site for February 23 to 28, March 2 to 4, and May 5 to 8, 2026 and found that hydrological conditions were drier than normal with

the drought index indicating moderate to severe drought during the wet season. Results from the APT analysis are included in **Appendix C**.

3.2.5 Rivers, Streams, and Drainages

Rivers, streams, and drainages can be classified into three categories: traditional navigable waters (TNWs), relatively permanent waters (RPWs), and non-relatively permanent waters (NRPWs).

Traditional Navigable Waters

The USACE RHA regulations (33 CFR 329.4) define navigable WOTUS, or TNWs, as "those waters that are subject to the ebb and flow of the tide, and/or are presently used, or have been used in the past, or may be susceptible for use to transport interstate or foreign commerce."

No TNWs were identified within the Project Area during the February, March, and May 2026 field surveys.

Relatively Permanent Waters

The *USACE Jurisdictional Determination Form Instructional Guidebook* (*Guidebook*; USACE 2007) defines RPWs as a waterbody "where flow is year-round or at during the wet season." Seasonal flow is typically defined as continuous flow for a consecutive period of at least three months. Perennial streams would be considered to have year-round flow and intermittent streams would be considered as possessing seasonal flow. RPWs must be connected to a TNW or have a continuous surface connection with another RPW or TNW to be considered jurisdictional.

RPWs are noted within **Table 5** below.

Non-Relatively Permanent Waters

The *USACE Guidebook* defines NRPWs as "non-navigable tributaries that are not relatively permanent." Ephemeral streams meet this definition and would be considered NRPWs. In some cases, erosion gullies, ditches, and other types of water conveyances that allow for the non-continuous flow of water also meet the USACE definition for NRPWs.

NRPWs are noted in **Table 5** below.

4.0 Identified Aquatic Features

A total of one hundred and eight aquatic features were identified within the Project Area during the February, March, and May 2026 field surveys and are depicted in **Figure 7** of **Appendix A** and a summary of each aquatic feature is provided in **Table 5** below. Detailed datasheets and representative photographs of identified features are provided in **Appendix B**.

Table 5. Identified Aquatic Features within the Project Area

Feature Name	Stream / Wetland Type	Linear Feet	Acres	Jurisdictional Opinion
D-T1F-01	Agriculture Ditch (NRPW)	4456.52	0.10	No
D-T1F-02	Agriculture Ditch (NRPW)	299.37	0.01	No
D-T1F-03	Agriculture Ditch (NRPW)	377.91	0.02	No
D-T1F-04	Agriculture Ditch (NRPW)	304.42	0.01	No
D-T1F-05	Agriculture Ditch (RPW)	300.99	0.05	Yes
D-T1F-06	Agriculture Ditch (RPW)	445.25	0.04	Yes
D-T1F-07	Roadside Ditch (RPW)	565.32	0.03	No
D-T1F-08	Agriculture Ditch (NRPW)	2272.66	0.05	No
D-T1F-09	Agriculture Ditch (RPW)	707.45	0.06	Yes
D-T1F-10	Agriculture Ditch (NRPW)	300.26	0.01	No
D-T1F-11	Agriculture Ditch (NRPW)	534.07	0.04	No
D-T1F-12	Agriculture Ditch (NRPW)	1192.13	0.08	No
D-T1F-13	Agriculture Ditch (RPW)	5452.89	0.25	Yes
D-T1F-14	Agriculture Ditch (RPW)	2058.67	0.63	Yes
D-T2F-01	Agriculture Ditch (NRPW)	265.08	0.01	No
D-T2F-02	Agriculture Ditch (RPW)	3521.84	0.59	No
D-T2F-03	Agriculture Ditch (RPW)	679.90	0.22	Yes
D-T2F-04	Agriculture Ditch (RPW)	438.85	0.06	No
D-T2F-05a	Agriculture Ditch (NRPW)	1736.00	0.17	No
D-T2F-05b	Agriculture Ditch (RPW)	1040.68	0.08	Yes
D-T2F-06	Agriculture Ditch (RPW)	300.78	0.02	No
D-T2F-07	Agriculture Ditch (NRPW)	174.36	0.01	No
D-T2F-08	Agriculture Ditch (RPW)	440.02	0.04	No
D-T2F-09	Agriculture Ditch (RPW)	316.88	0.04	Yes
D-T2F-10	Agriculture Ditch (RPW)	273.10	0.04	Yes
D-T2F-11	Agriculture Ditch (RPW)	1246.15	0.06	Yes

Feature Name	Stream / Wetland Type	Linear Feet	Acres	Jurisdictional Opinion
D-T2F-12	Agriculture Ditch (NRPW)	565.24	0.11	No
D-T2F-13	Agriculture Ditch (NRPW)	28.55	<0.01	No
D-T2F-14	Agriculture Ditch (RPW)	300.27	0.11	No
D-T3F-01	Agriculture Ditch (RPW)	317.48	0.04	Yes
D-T3F-02	Agriculture Ditch (NRPW)	883.07	0.04	No
D-T3F-03	Agriculture Ditch (NRPW)	983.83	0.07	No
D-T3F-04	Agriculture Ditch (NRPW)	1923.84	0.09	No
D-T3F-05	Roadside Ditch (NRPW)	462.57	0.02	No
D-T3F-06	Agriculture Ditch (NRPW)	1244.68	0.06	No
D-T3F-07	Roadside Ditch (NRPW)	270.84	0.01	No
D-T3F-08	Roadside Ditch (NRPW)	826.03	0.04	No
S-T1F-01	Perennial	340.90	0.56	Yes
S-T1F-02	Perennial	278.54	0.18	Yes
S-T1F-03	Perennial	44.66	0.01	Yes
S-T1F-04	Perennial	1246.10	0.88	Yes
S-T1F-05	Perennial	360.39	0.05	Yes
S-T2F-01	Intermittent	94.87	0.01	Yes
S-T2F-02	Perennial	247.64	0.36	Yes
S-T2F-03	Intermittent	790.59	0.11	Yes
S-T2F-04	Perennial	301.36	0.23	Yes
S-T2F-05	Ephemeral	804.53	0.06	No
S-T2F-06	Perennial	403.12	0.10	Yes
S-T2F-07	Intermittent	305.86	0.04	Yes
S-T2F-08	Intermittent	287.66	0.02	Yes
S-T2F-09	Perennial	1387.99	0.61	Yes
S-T2F-10	Intermittent	350.87	0.05	Yes
S-T2F-11	Intermittent	23.98	<0.01	Yes
S-T3F-01	Perennial	309.81	0.50	Yes
S-T3F-02	Ephemeral	52.44	<0.01	No
S-T3F-03	Perennial	301.22	0.04	Yes

Feature Name	Stream / Wetland Type	Linear Feet	Acres	Jurisdictional Opinion
S-T3F-04	Perennial	385.44	0.73	Yes
S-T3F-05	Intermittent	100.99	0.01	Yes
S-T3F-06	Intermittent	209.16	0.02	Yes
P-T1F-01	Pond	--	0.18	Yes
P-T1F-02	Pond	--	0.23	No
P-T2F-01	Pond	--	0.05	No
P-T2F-02	Pond	--	0.01	No
P-T2F-03	Pond	--	0.08	No
P-T2F-04	Pond	--	0.28	No
PFO-T1F-01	PFO	--	61.91	Yes
PFO-T1F-02	PFO	--	0.15	Yes
PFO-T1F-03	PFO	--	2.00	Yes
PEM-T1F-03a	PEM	--	0.59	Yes
PEM-T1F-03b	PEM	--	3.72	Yes
PFO-T1F-04	PFO	--	17.28	Yes
PEM-T1F-04	PEM	--	0.65	Yes
PFO-T1F-05a	PFO	--	11.61	Yes
PFO-T1F-05b	PFO	--	5.54	Yes
PEM-T1F-05	PEM	--	2.36	Yes
PFO-T1F-06	PFO	--	5.62	Yes
PEM-T1F-06	PEM	--	0.87	Yes
PEM-T1F-07	PEM	--	0.31	Yes
PEM-T2F-01	PEM	--	0.25	No
PFO-T2F-03	PFO	--	2.77	Yes
PFO-T2F-04	PFO	--	0.31	No
PEM-T2F-04	PEM	--	0.91	No
PFO-T2F-05	PFO	--	0.22	No
PEM-T2F-06	PEM	--	1.08	No
PFO-T2F-07	PFO	--	0.27	Yes
PEM-T2F-08	PEM	--	0.18	No
PFO-T2F-09	PFO	--	0.12	Yes
PEM-T2F-10	PEM	--	0.08	No

Feature Name	Stream / Wetland Type	Linear Feet	Acres	Jurisdictional Opinion
PEM-T2F-11	PEM	--	0.25	Yes
PEM-T2F-12	PEM	--	0.16	Yes
PEM-T2F-13	PEM	--	0.41	No
PEM-T2F-14	PEM	--	0.63	No
PEM-T2F-15	PEM	--	0.11	No
PEM-T2F-16	PEM	--	0.01	No
PEM-T2F-17	PEM	--	0.05	No
PEM-T2F-18	PEM	--	0.03	No
PSS-T2F-18	PSS	--	0.03	No
PFO-T2F-19	PFO	--	1.82	Yes
PFO-T2F-20	PFO	--	0.68	Yes
PEM-T2F-20	PEM	--	0.08	Yes
PFO-T2F-21	PFO	--	0.08	Yes
PFO-T3F-01	PFO	--	7.33	Yes
PSS-T3F-02	PSS	--	0.28	No
PFO-T3F-03	PFO	--	0.12	Yes
PEM-T3F-03	PEM	--	0.21	Yes
PFO-T3F-04	PFO	--	0.17	Yes
PFO-T3F-05	PFO	--	1.00	Yes
PEM-T3F-06	PEM	--	0.41	No
PFO-T3F-07	PFO	--	0.84	Yes
PEM-T4F-02	PEM	--	0.39	No

D = ditch; S = stream; Wetlands: PEM = palustrine emergent wetland; PSS = palustrine scrub-shrub; PFO = palustrine forested wetland.

Potentially Jurisdictional Features

Fifty-seven features are considered potentially jurisdictional as they would classify as RPWs. This includes 26 wetlands that are considered potentially jurisdictional as they are connected to RPWs that eventually lead to traditionally navigable waterways.

The following OHWM indicators were recorded for streams within the Project Area bed and bank present; matted, bent, or absent vegetation; disturbed/washed-away leaf litter; natural line impressed on bank; plant community change; terrestrial vegetation destroyed; water staining; scour; shelving; litter and debris; soil character change; deposition; multiple observed flow events; and wrack. Aquatic vegetation was also observed.

Potentially Non-Jurisdictional Features

Fifty-three features are considered to be non-jurisdictional as they are considered NRPWs. These features do not have connectivity to RPWs or TNWs or do not present the flow regime necessary to be considered jurisdictional.

4.1 Additional Observations

In addition to the delineated aquatic features previously discussed, multiple observation points were recorded during the February, March, and May 2026 field surveys to further document on-site conditions and support the findings presented in this report. Observation points documented within the Project Area are illustrated in **Figure 8** of **Appendix A** and described further in **Appendix D**.

5.0 Summary and Conclusions

TRC was contracted to conduct a delineation of wetlands and other waters within the Project Area. The delineation was performed to identify the presence of aquatic features as well as delineate the boundaries of identified aquatic features within the Project Area. Based on a desktop analysis of background data and a field investigation conducted by TRC on February 23 to February 28, 2026, March 2 to March 4, 2026, and May 5 to May 8, 2026, TRC determined that one-hundred and ten aquatic features were identified within the Project Area. Datasheets from the field delineation and representative photographs of the Project Area and the identified aquatic feature are provided in **Appendix B**.

It is TRC's professional opinion that fifty-seven of these features would be considered jurisdictional WOTUS. These aquatic features are either named waterbodies such as Cypress Creek, Bee Bayou, Boggy Bayou, and Cow Bayou, or have several OHWM indicators and downstream offsite connections to named waterbodies. TRC's opinion on the jurisdictional status of identified aquatic features within the Project Area is based on the pre-2015 regulatory regime and definition of WOTUS and consideration of the 2023 Sackett decision. However, it should be noted that the rules and regulations defining WOTUS are currently in a state of change. Nevertheless, a degree of change to the definition of WOTUS resulting from the Sackett decision is anticipated which could potentially change TRC's opinion of the jurisdictional status of identified aquatic features within the Project Area.

The delineation findings contained within this report represent the professional opinion of TRC and are not a verification or jurisdictional determination of WOTUS. No other warranty, expressed or implied, is made. The USACE Vicksburg District has the authority to make final decisions on all potential jurisdictional WOTUS through the jurisdictional determination process. It is anticipated that the Project will have impacts to WOTUS and a CWA Section 404 permit (Nationwide Permit 12 – Oil or Natural Gas Pipeline Activities) will be necessary for the Project. A pre-construction notification is anticipated to be required.

6.0 References

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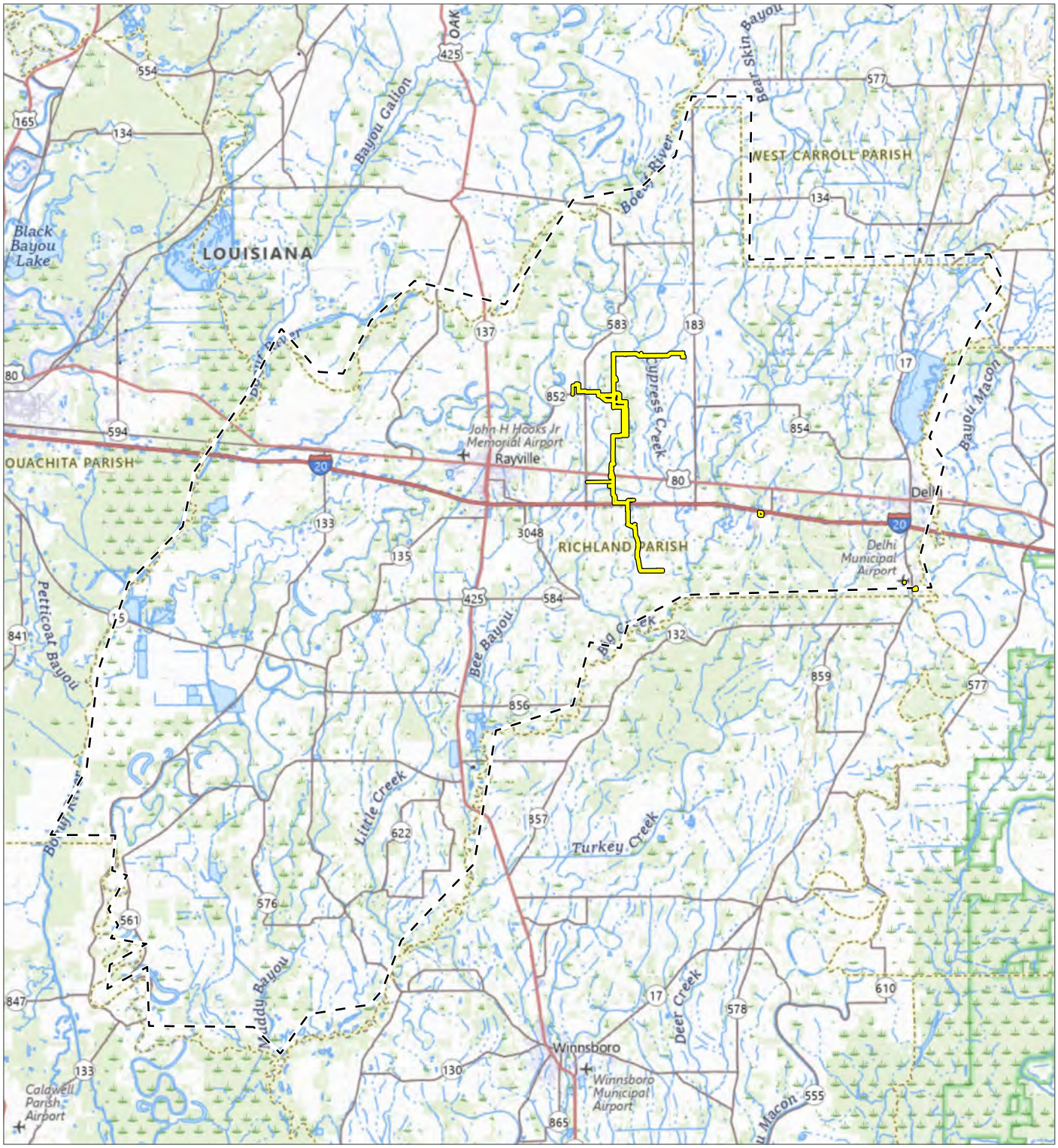
Appendix A
Figures



PROJECT BOUNDARY

RICHLAND PARISH

0 14,000 28,000

FEET

1:336,000 1" = 28,000'

BASE MAP: USGS "THE NATIONAL MAP" ONLINE SERVICE LAYER;
DATA SOURCES: TRC, USGS

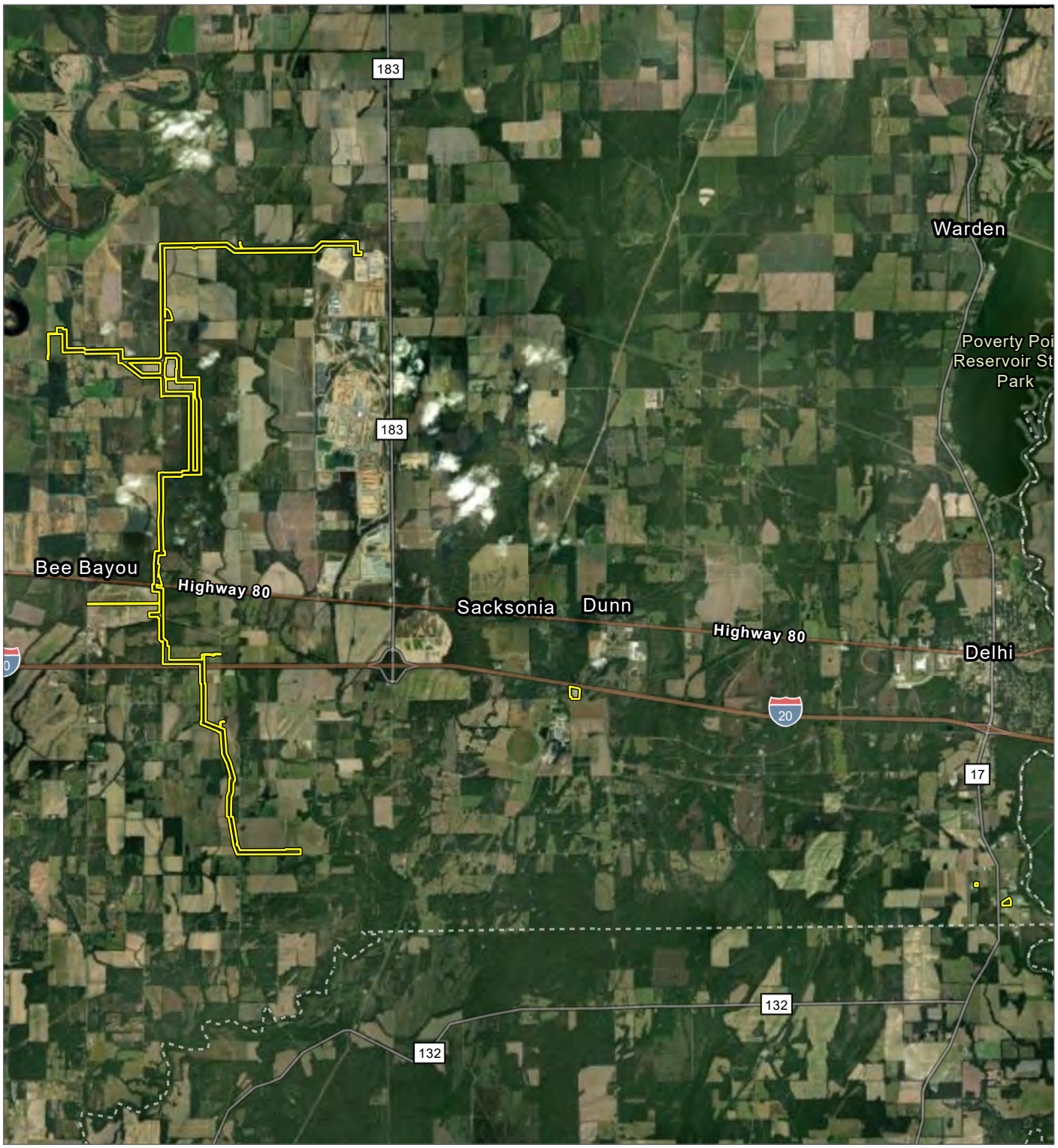
ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

PROJECT VICINITY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON	**FIGURE 1**	
DATE:	MAY 2026		

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM -- FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX -- LAYOUT NAME: 1_PROJECT VICINITY



PROJECT BOUNDARY

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, ESRI

N

0 5,000 10,000

FEET
1:120,000 1" = 10,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

PROJECT LOCATION MAP (AERIAL)

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 2

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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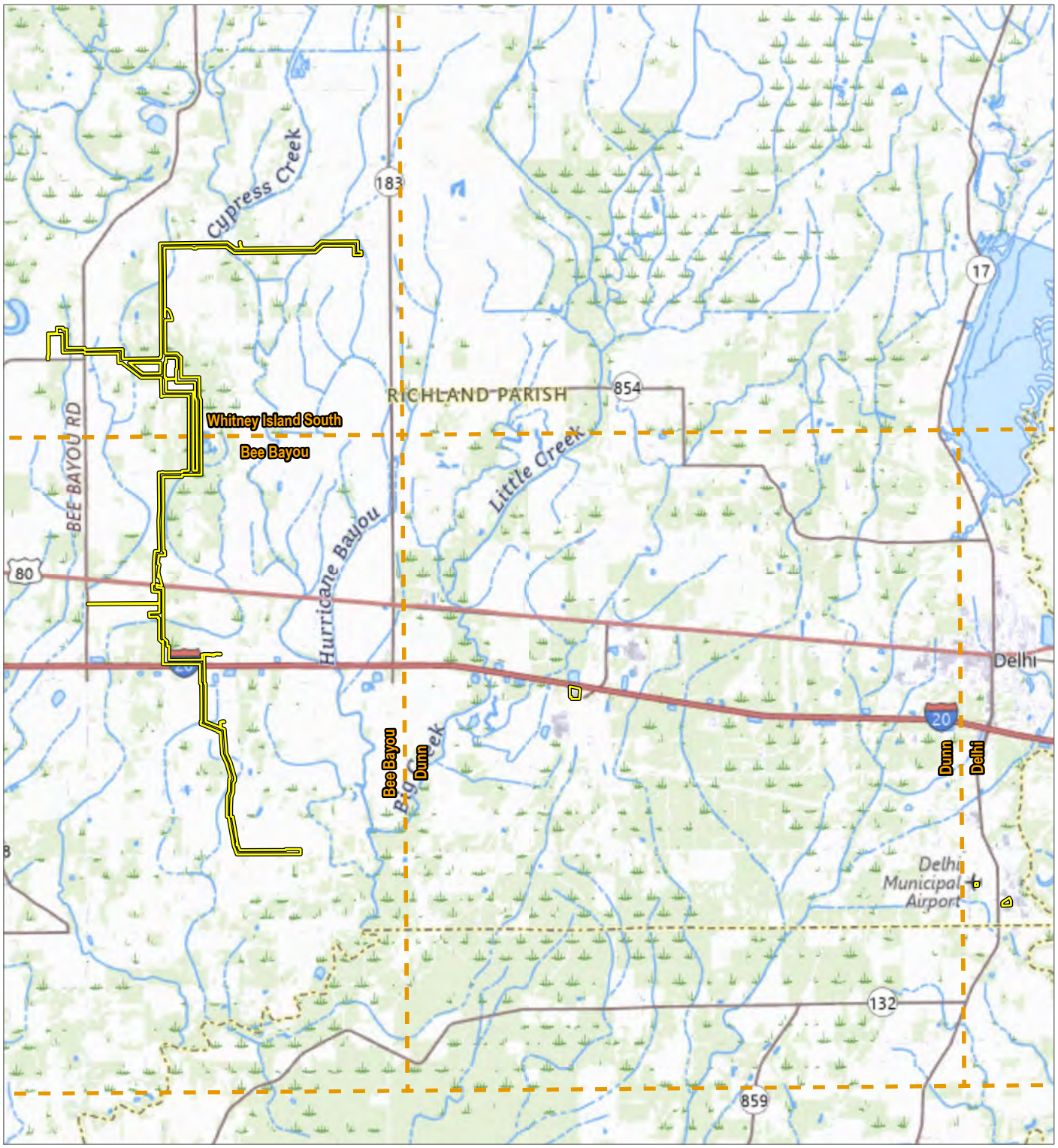

PROJECT BOUNDARY

USGS QUAD BOUNDARIES

0 5,000 10,000

FEET

1:120,000 1" = 10,000'

BASE MAP: USGS "THE NATIONAL MAP" ONLINE SERVICE LAYER; 7.5' QUADS: WHITNEY ISLAND SOUTH, LA (1986), BEE BAYOU, LA (1987), DUNN, LA (1987), DELHI, LA (1986)
DATA SOURCES: TRC, USGS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

PROJECT LOCATION MAP (TOPOGRAPHIC)

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		**FIGURE 3**
DATE:	MAY 2026		

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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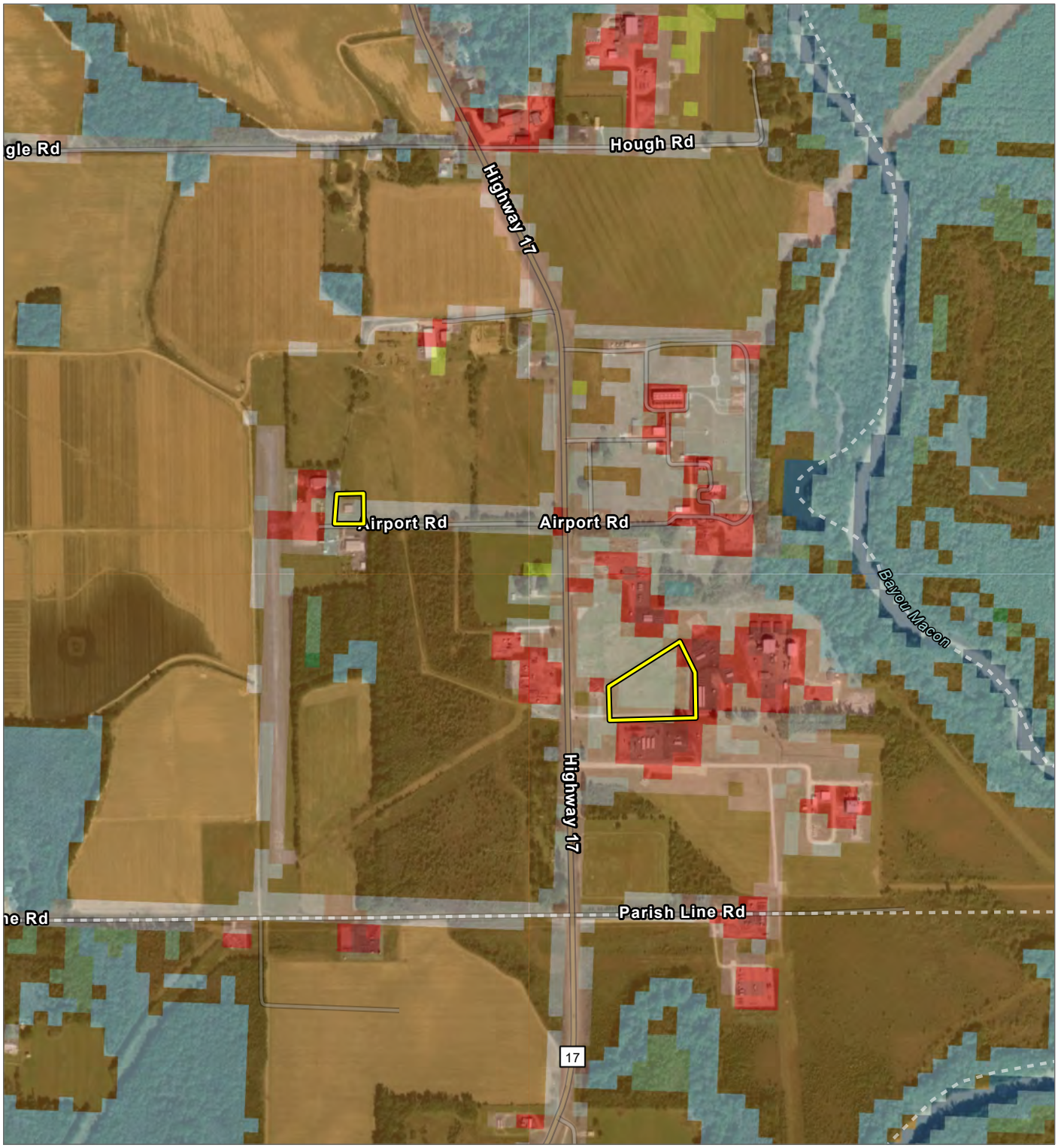

- PROJECT BOUNDARY
- OPEN WATER
- DEVELOPED, OPEN SPACE
- DEVELOPED, LOW INTENSITY
- DEVELOPED, MEDIUM INTENSITY
- DEVELOPED, HIGH INTENSITY
- DECIDUOUS FOREST
- MIXED FOREST
- PASTURE/HAY
- CULTIVATED CROPS
- WOODY WETLANDS
- EMERGENT HERBACEOUS WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000
FEET
1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 1 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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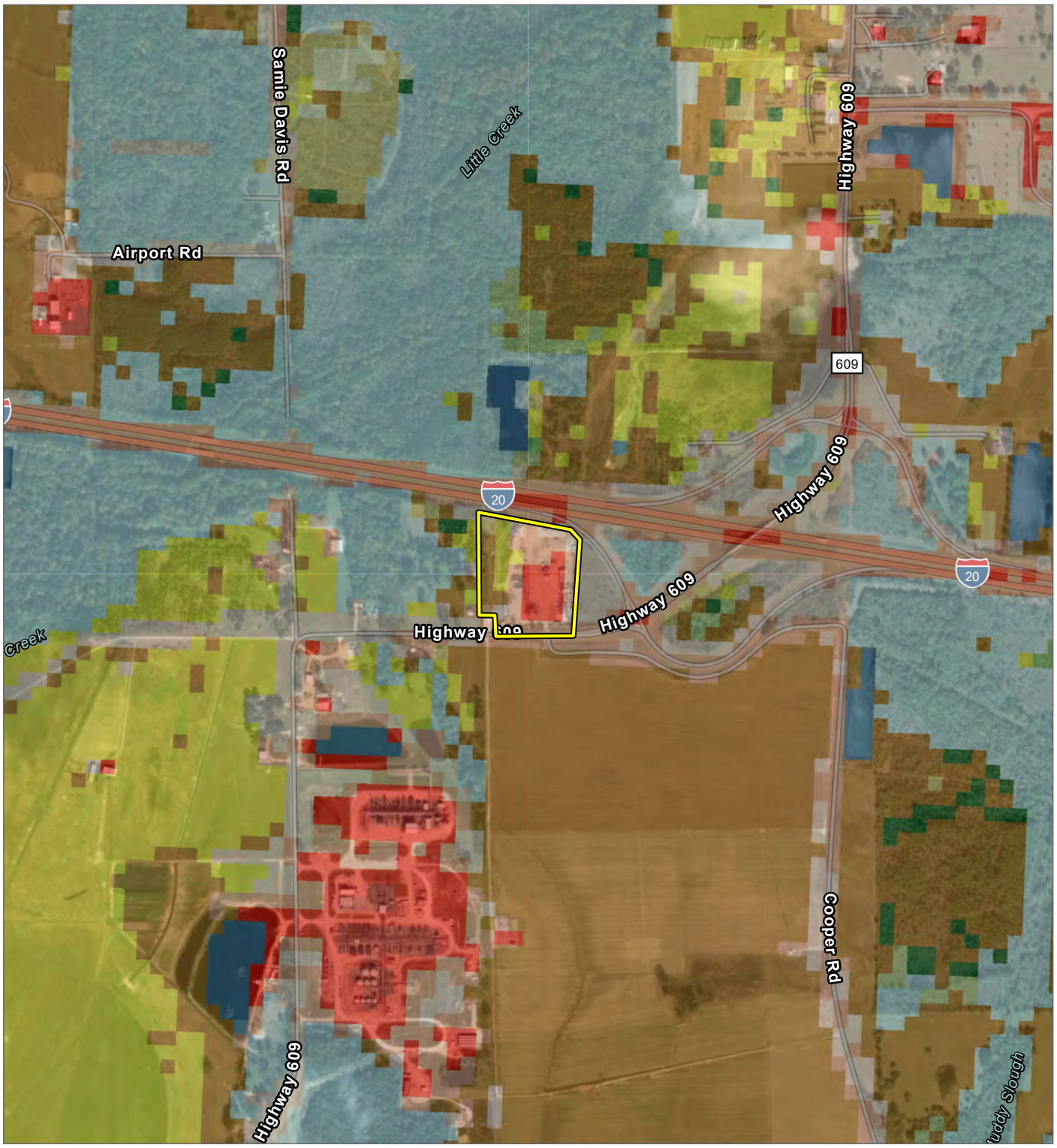

PROJECT BOUNDARY
OPEN WATER
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
DEVELOPED, MEDIUM INTENSITY
DEVELOPED, HIGH INTENSITY
DECIDUOUS FOREST
EVERGREEN FOREST
MIXED FOREST
SHRUB/SCRUB
GRASSLAND/HERBACEOUS
PASTURE/HAY
CULTIVATED CROPS
WOODY WETLANDS
EMERGENT HERBACEOUS

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1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 2 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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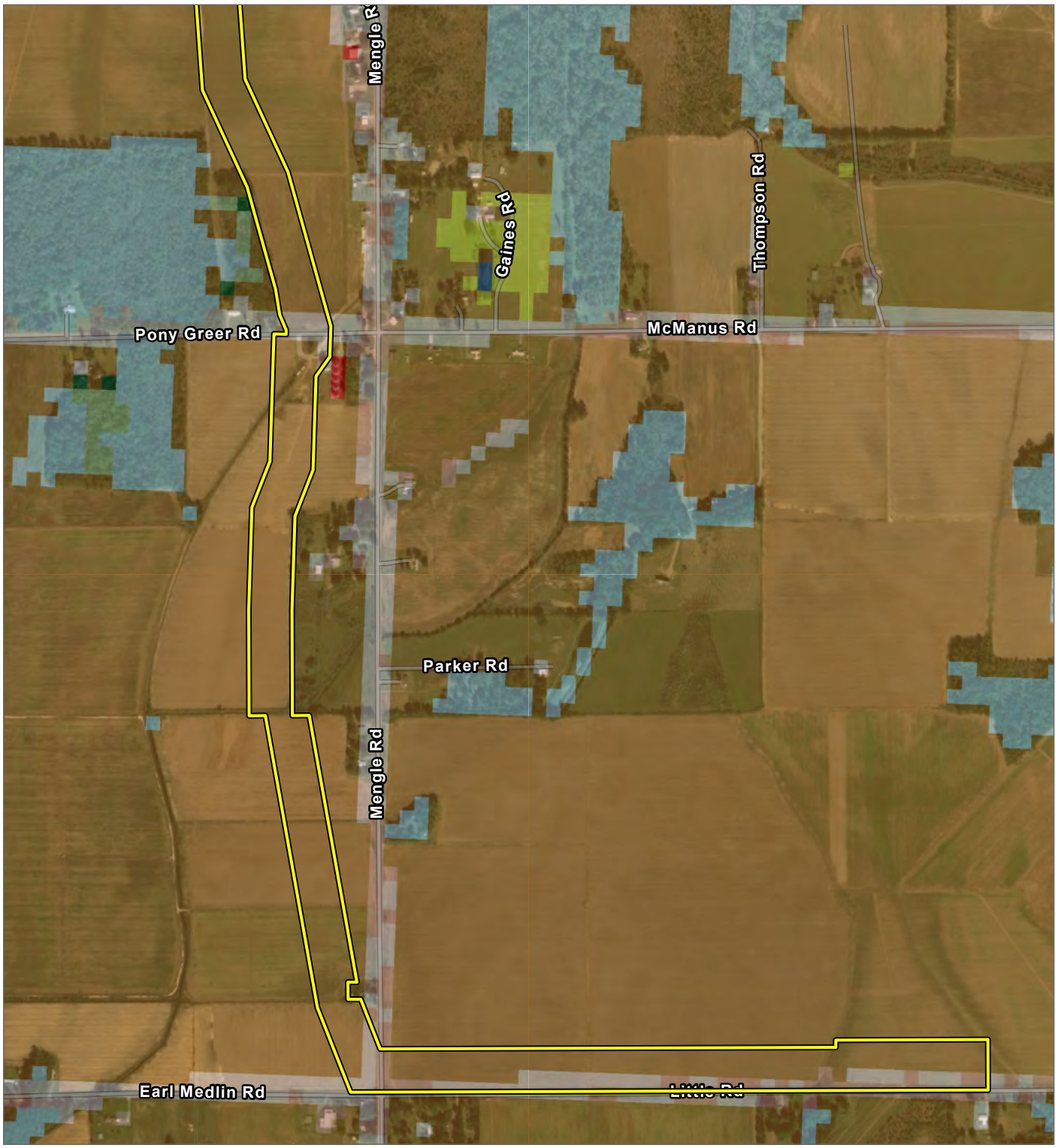

PROJECT BOUNDARY
OPEN WATER
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
DEVELOPED, MEDIUM INTENSITY
EVERGREEN FOREST
MIXED FOREST
PASTURE/HAY
CULTIVATED CROPS
WOODY WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

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1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 3 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET: MAP ROTATION: 0
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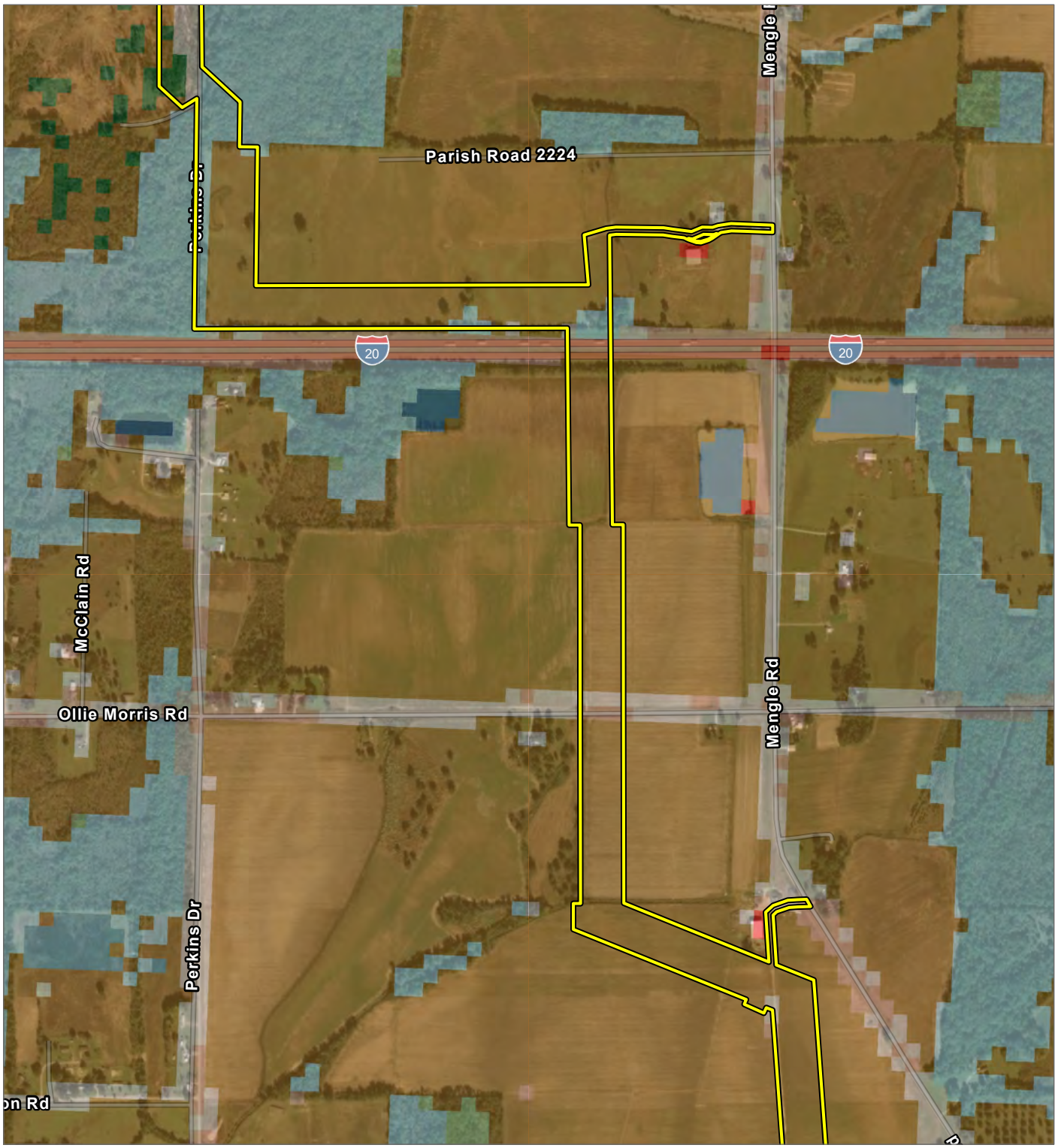

- PROJECT BOUNDARY
- OPEN WATER
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- DEVELOPED, LOW INTENSITY
- DEVELOPED, MEDIUM INTENSITY
- EVERGREEN FOREST
- MIXED FOREST
- GRASSLAND/HERBACEOUS
- CULTIVATED CROPS
- WOODY WETLANDS
- EMERGENT HERBACEOUS WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

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1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 4 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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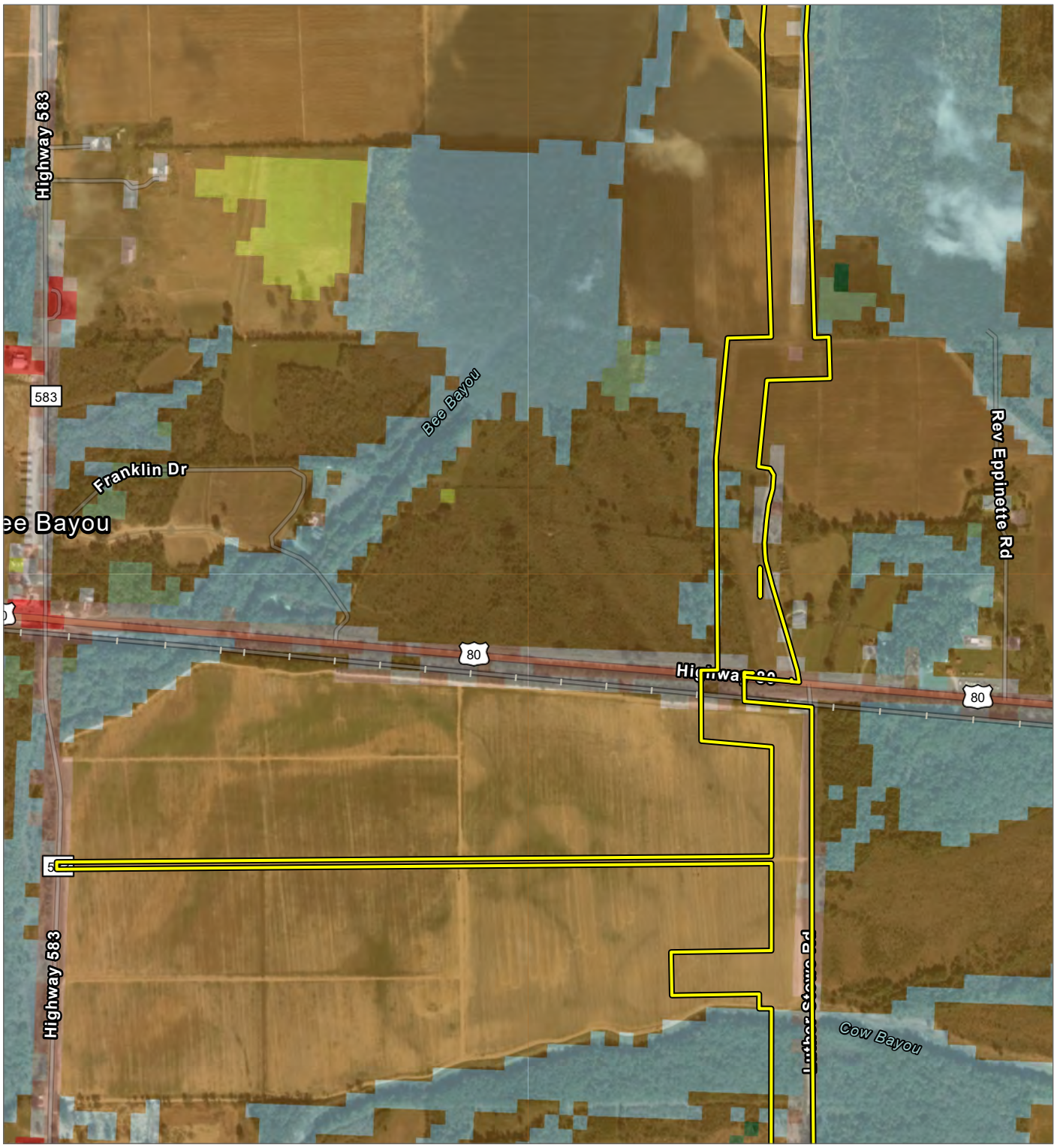

- PROJECT BOUNDARY
- DEVELOPED, OPEN SPACE
- DEVELOPED, LOW INTENSITY
- DEVELOPED, MEDIUM INTENSITY
- EVERGREEN FOREST
- MIXED FOREST
- PASTURE/HAY
- CULTIVATED CROPS
- WOODY WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000
FEET
1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 5 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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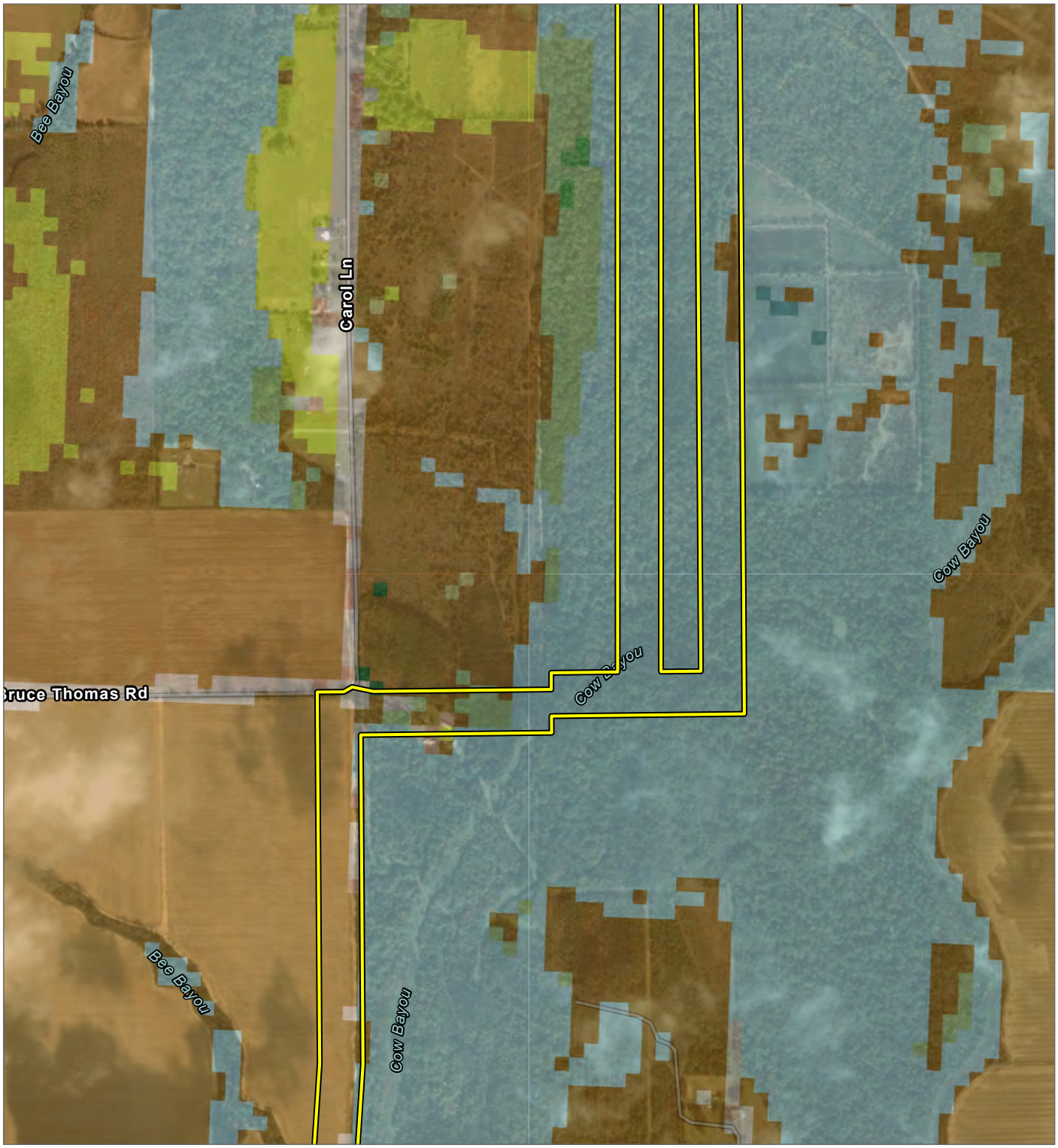


- PROJECT BOUNDARY
- DEVELOPED, OPEN SPACE
- DEVELOPED, LOW INTENSITY
- DECIDUOUS FOREST
- EVERGREEN FOREST
- MIXED FOREST
- SHRUB/SCRUB
- PASTURE/HAY
- CULTIVATED CROPS
- WOODY WETLANDS
- EMERGENT HERBACEOUS WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

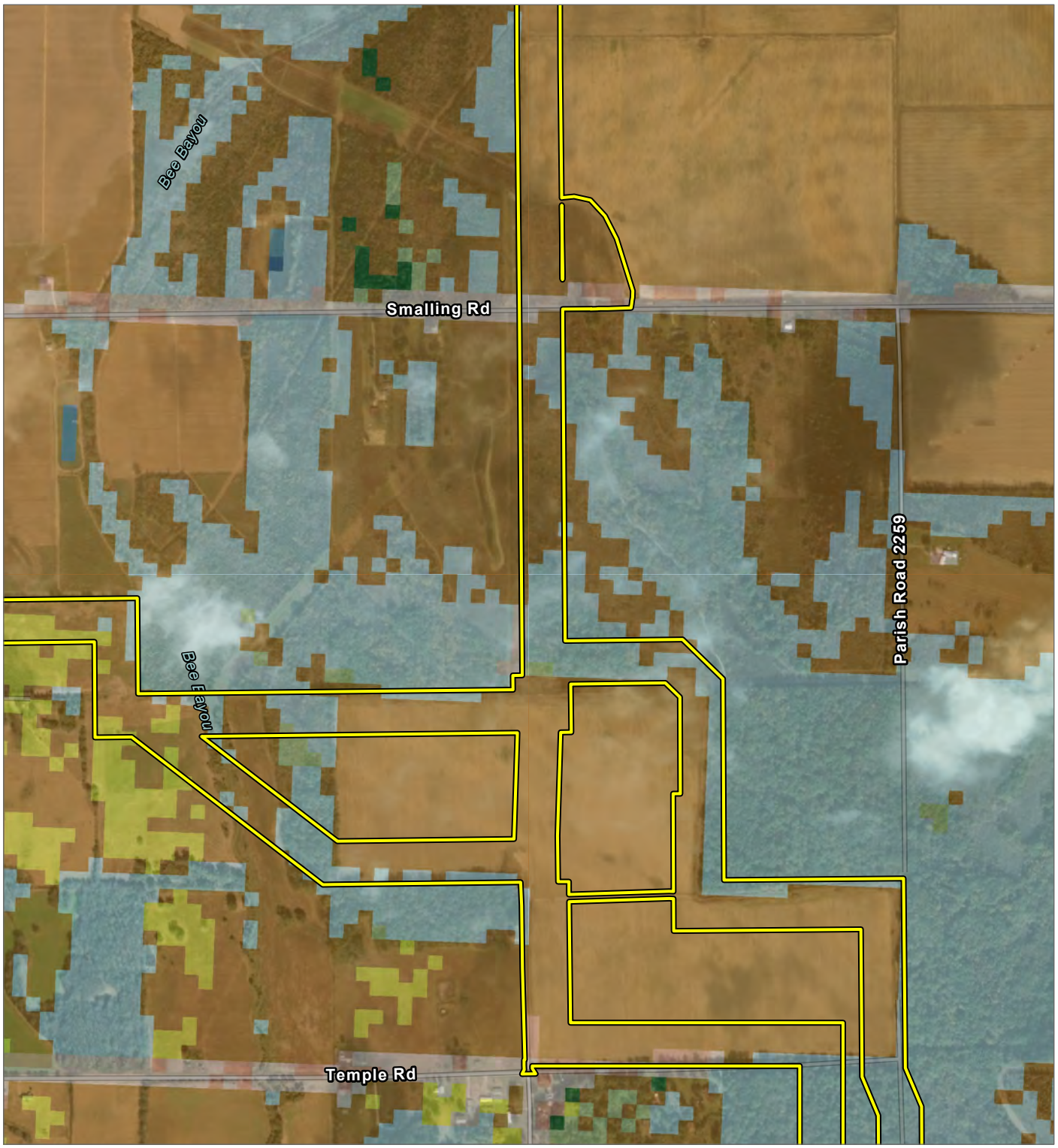
NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 6 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET: MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026 16:43:59 PM -- FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX -- LAYOUT NAME: 4_NLCD



- PROJECT BOUNDARY
- OPEN WATER
- DEVELOPED, OPEN SPACE
- DEVELOPED, LOW INTENSITY
- EVERGREEN FOREST
- MIXED FOREST
- PASTURE/HAY
- CULTIVATED CROPS
- WOODY WETLANDS
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BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

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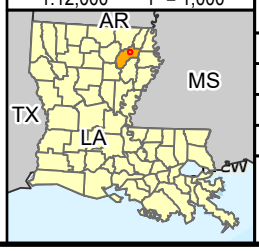
FEET
1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 7 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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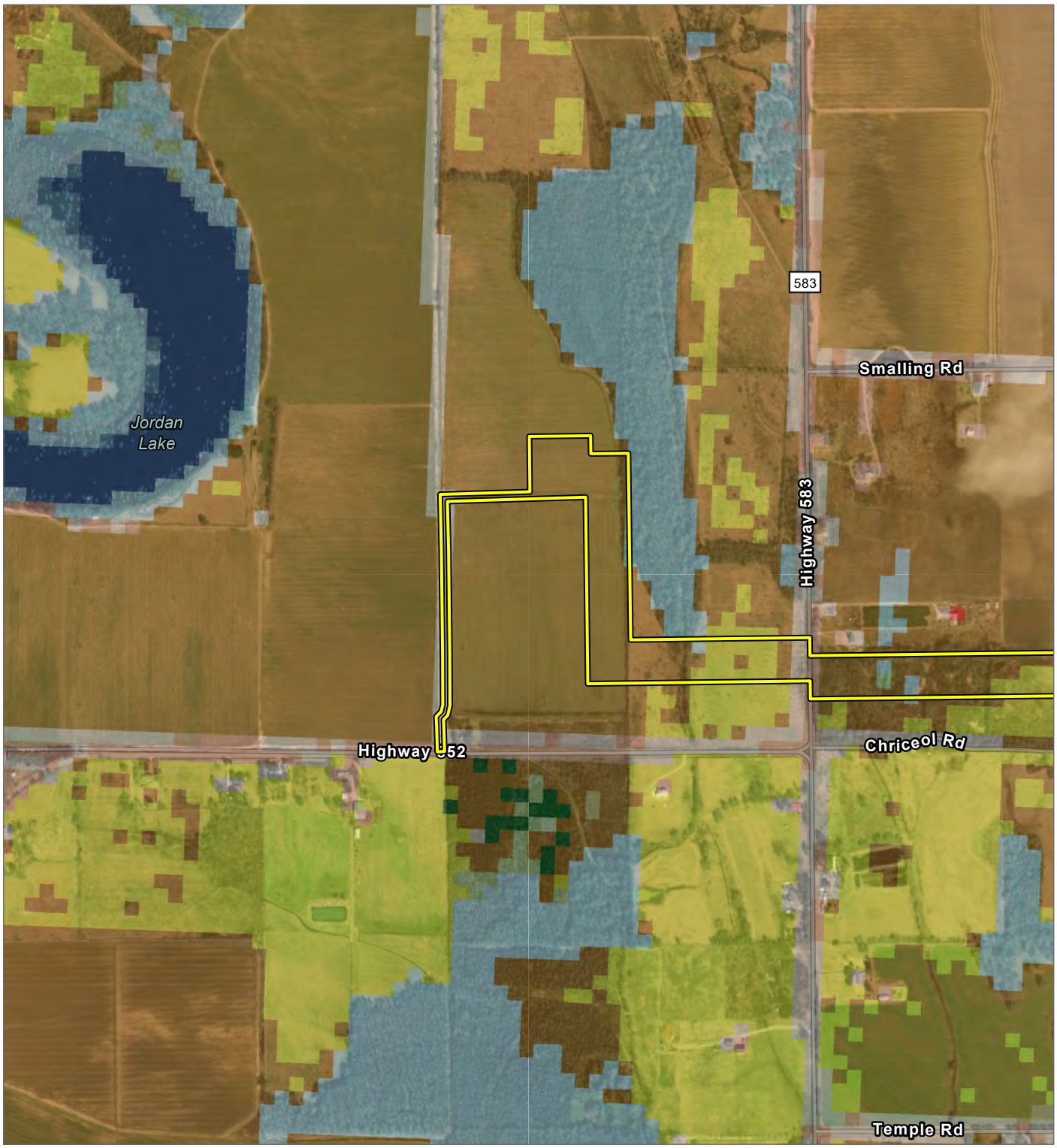
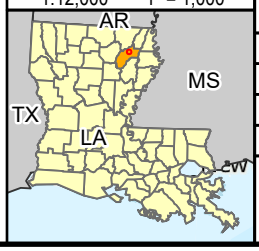


PROJECT BOUNDARY
OPEN WATER
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
DEVELOPED, MEDIUM INTENSITY
EVERGREEN FOREST
MIXED FOREST
SHRUB/SCRUB
GRASSLAND/HERBACEOUS
PASTURE/HAY
CULTIVATED CROPS
WOODY WETLANDS
EMERGENT HERBACEOUS

N

0 500 1,000
FEET
1:12,000 1" = 1,000'



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

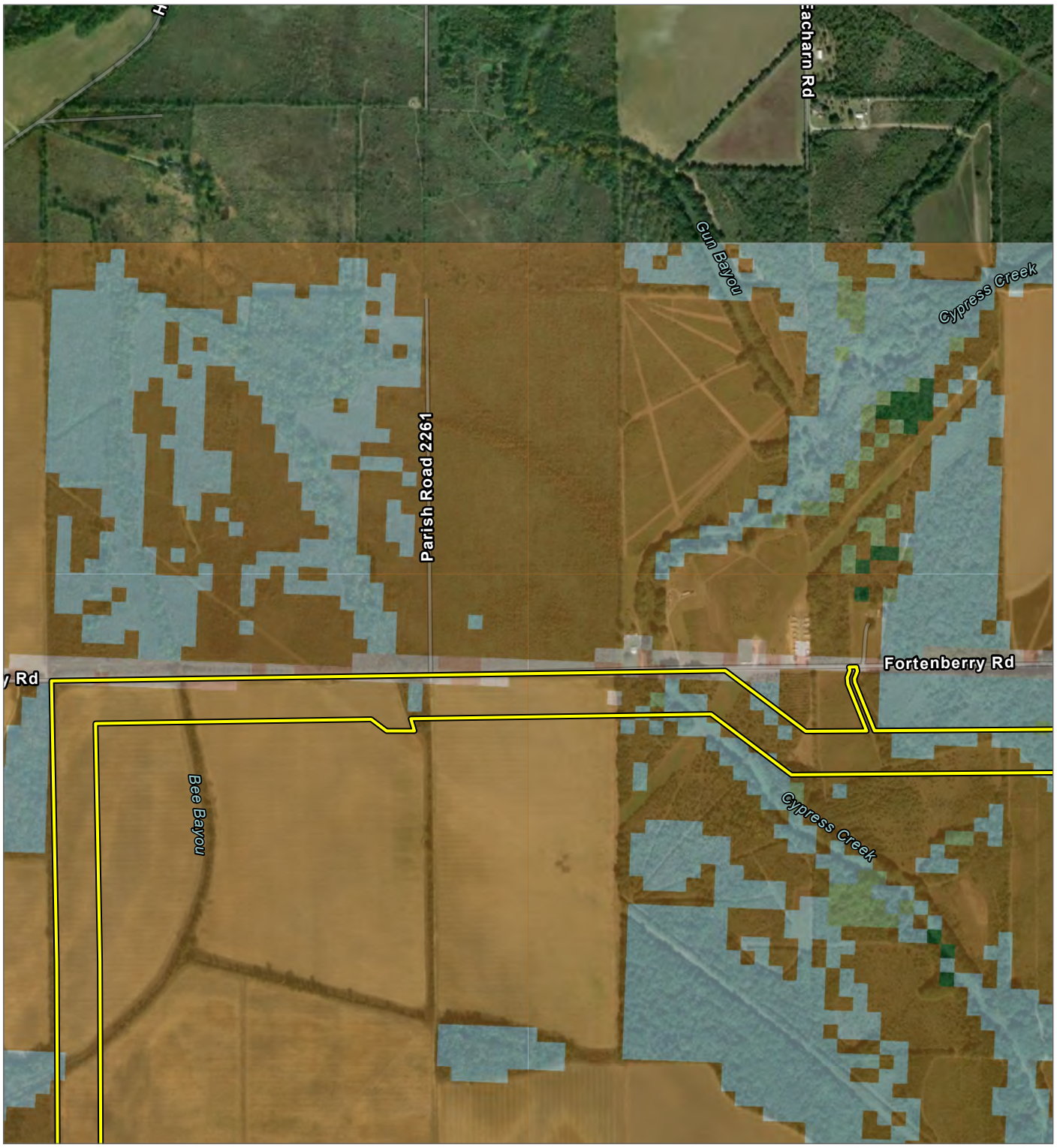
NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 8 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FRANKLIN_FARMS_BIO_SURVEY_CLONE APPX; LAYOUT NAME: 4_NLCD



PROJECT BOUNDARY
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
EVERGREEN FOREST
MIXED FOREST
CULTIVATED CROPS
WOODY WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000
FEET
1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 9 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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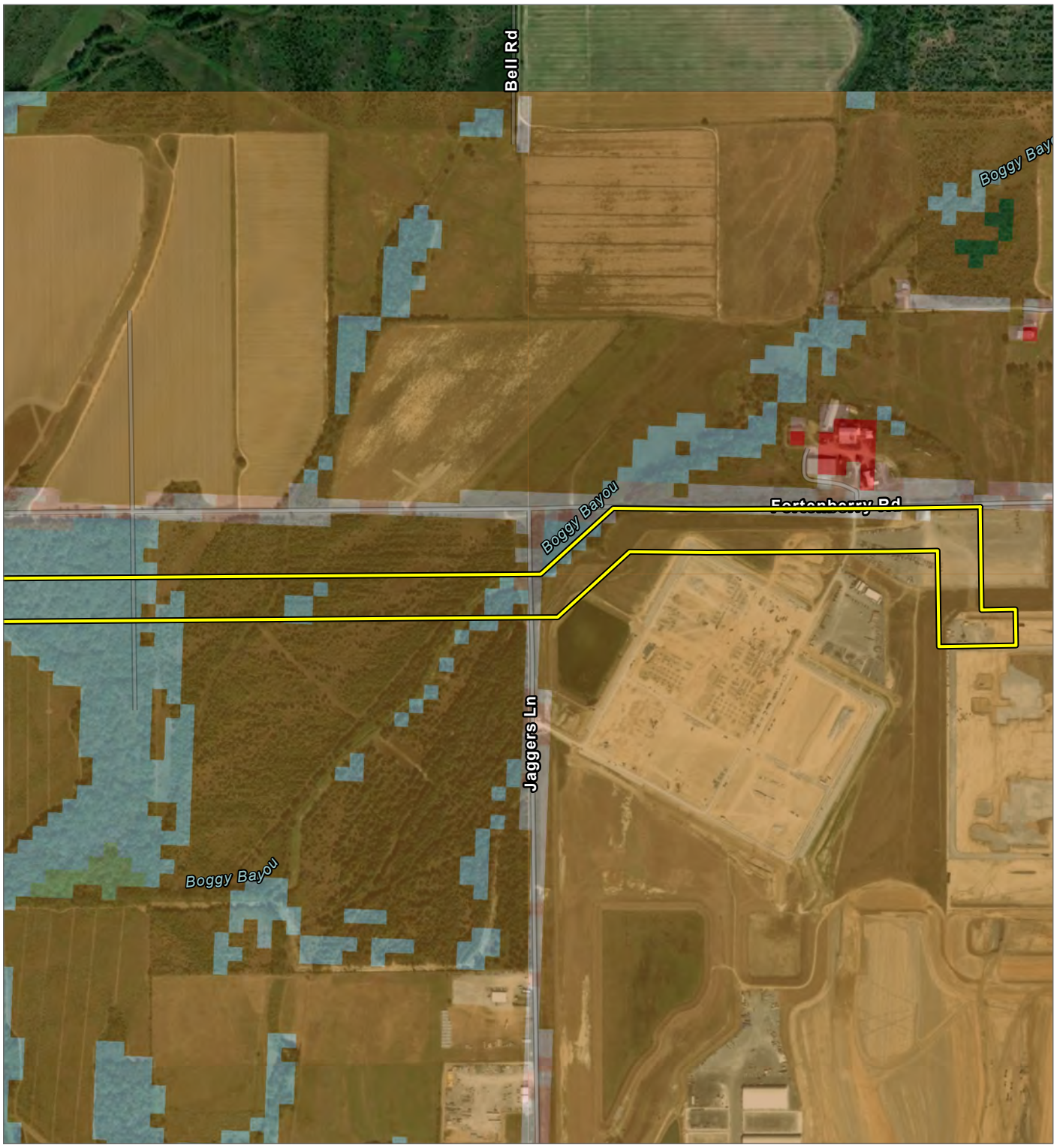
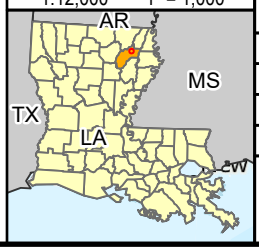


PROJECT BOUNDARY
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
DEVELOPED, MEDIUM INTENSITY
EVERGREEN FOREST
MIXED FOREST
CULTIVATED CROPS
WOODY WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000
FEET
1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

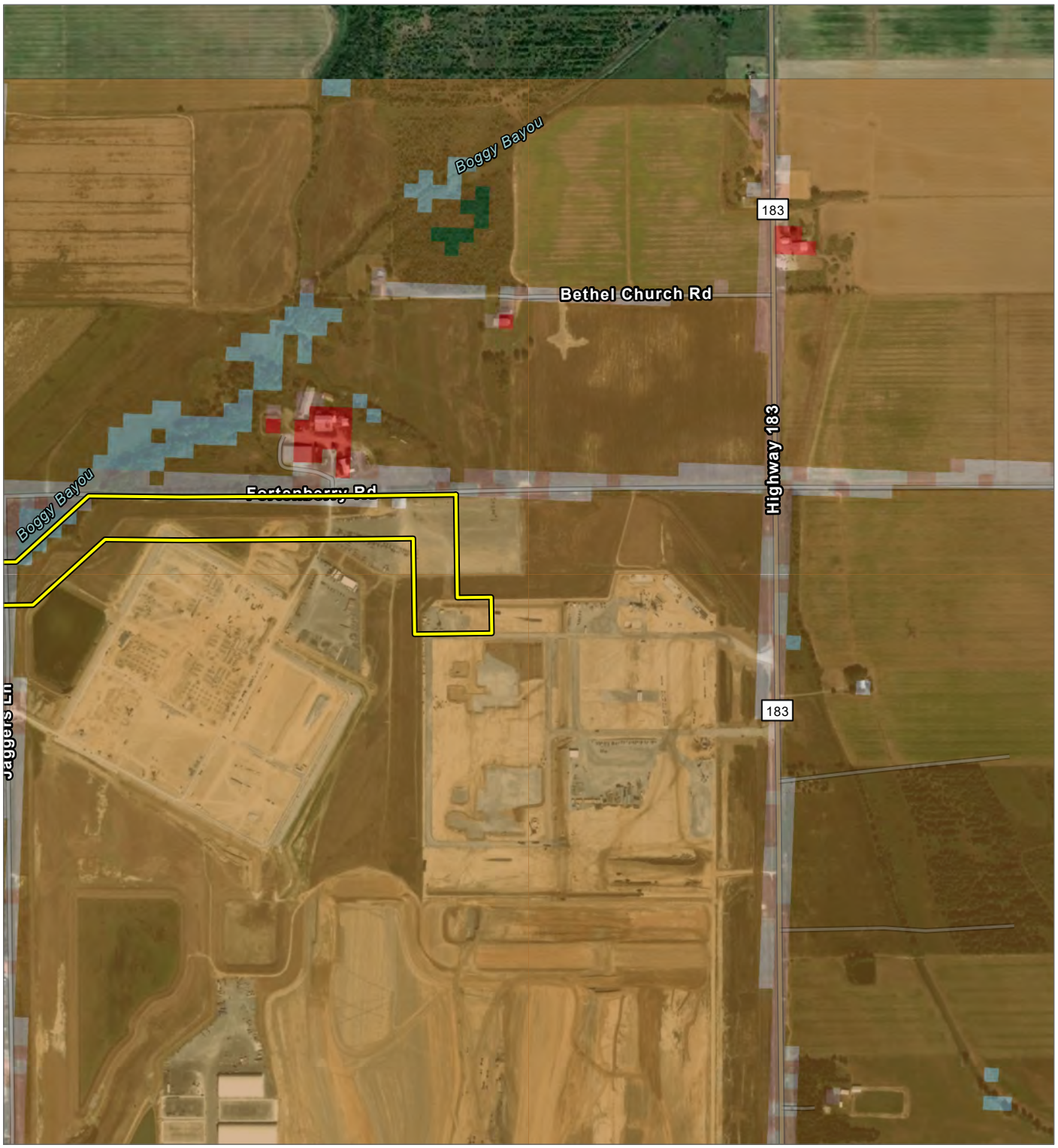
NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 10 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET: MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM -- FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FRANKLIN_FARMS_BIO_SURVEY_CLONE APPX- LAYOUT NAME: 4_NLCD

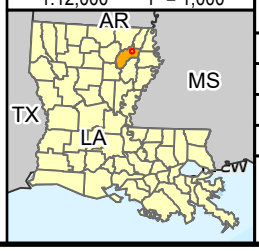


PROJECT BOUNDARY
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
DEVELOPED, MEDIUM INTENSITY
EVERGREEN FOREST
CULTIVATED CROPS
WOODY WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000
FEET
1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 11 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FRANKLIN_FARMS_BIO_SURVEY_CLONE APPX; -- LAYOUT NAME: 4_NLCD

PROJECT BOUNDARY

SOIL MAP UNIT

Cc, CALHOUN-CALLOWAY SILT LOAMS, GENTLY UNDULATING

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 1 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PROJECT BOUNDARY

SOIL MAP UNIT

Gg, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING

Gr, GRENADA SILT LOAM, 1 TO 3 PERCENT SLOPES

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 2 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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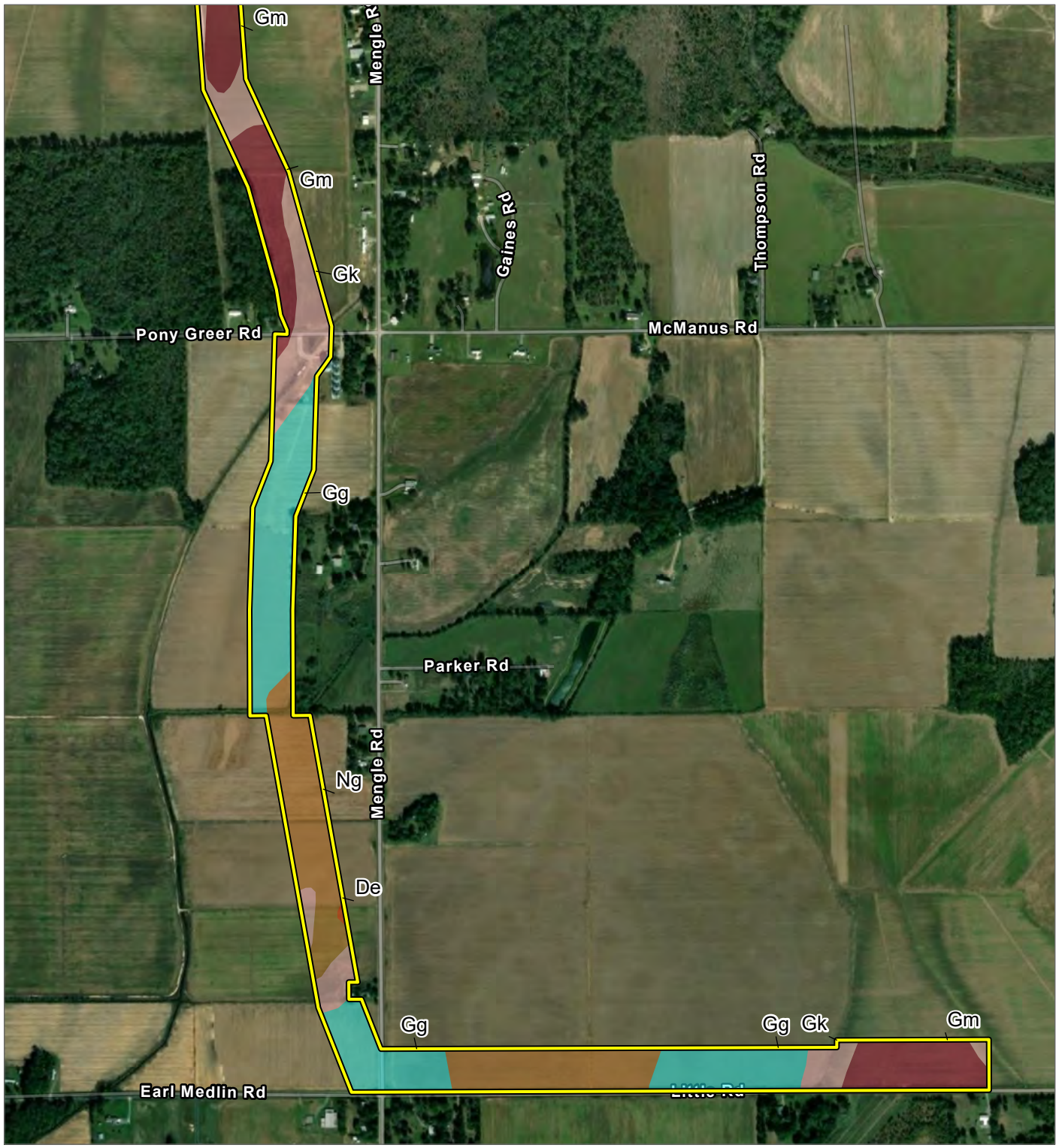


PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gg, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING
- Gk, GILBERT SILT LOAM
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ng, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

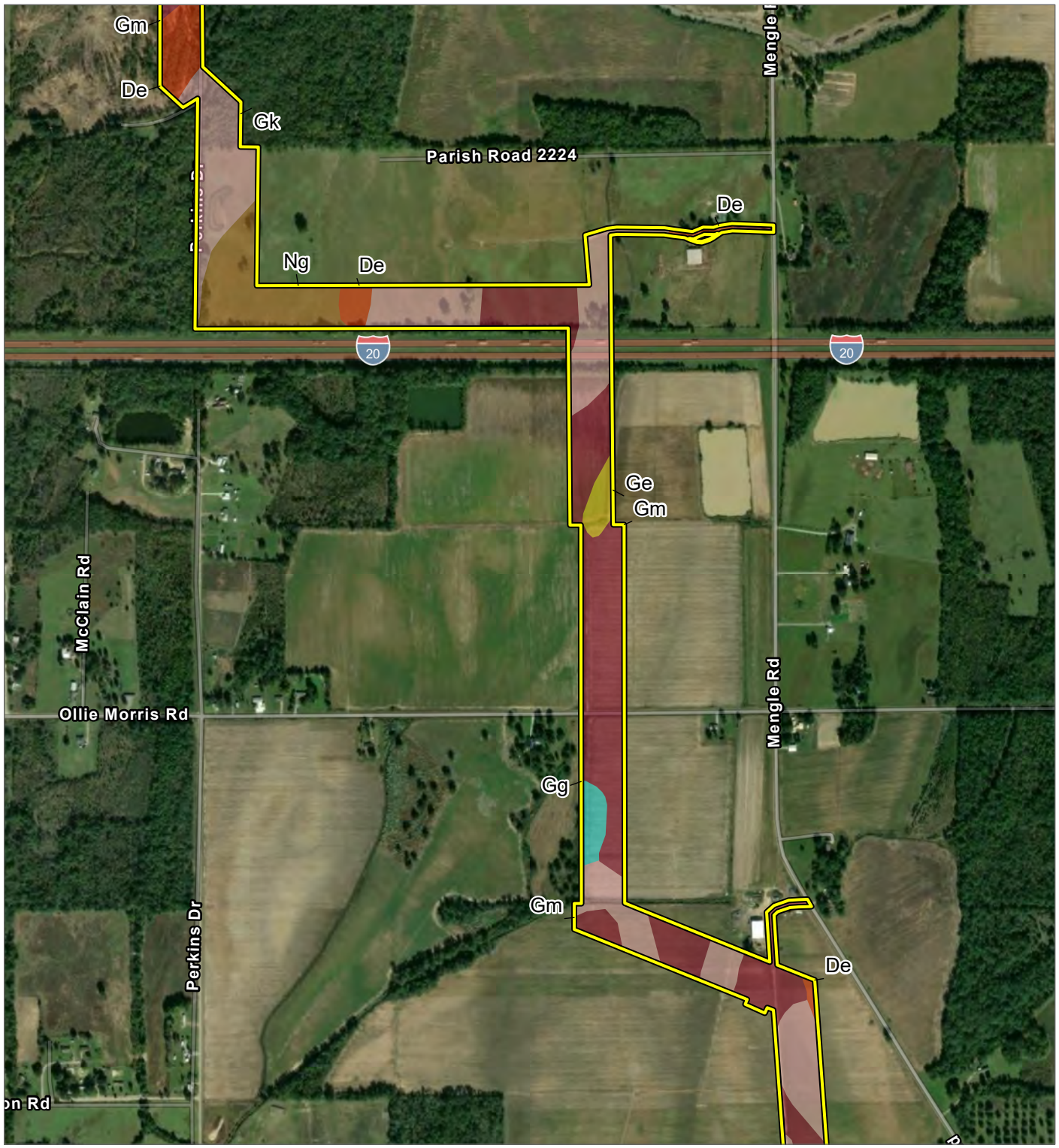
NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 3 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET: MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM: FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FRANKLIN_FARMS_BIO_SURVEY_CLONE APPX: - LAYOUT NAME: 5_NRCS SOILS



PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Ge, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gg, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING
- Gk, GILBERT SILT LOAM
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ng, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 4 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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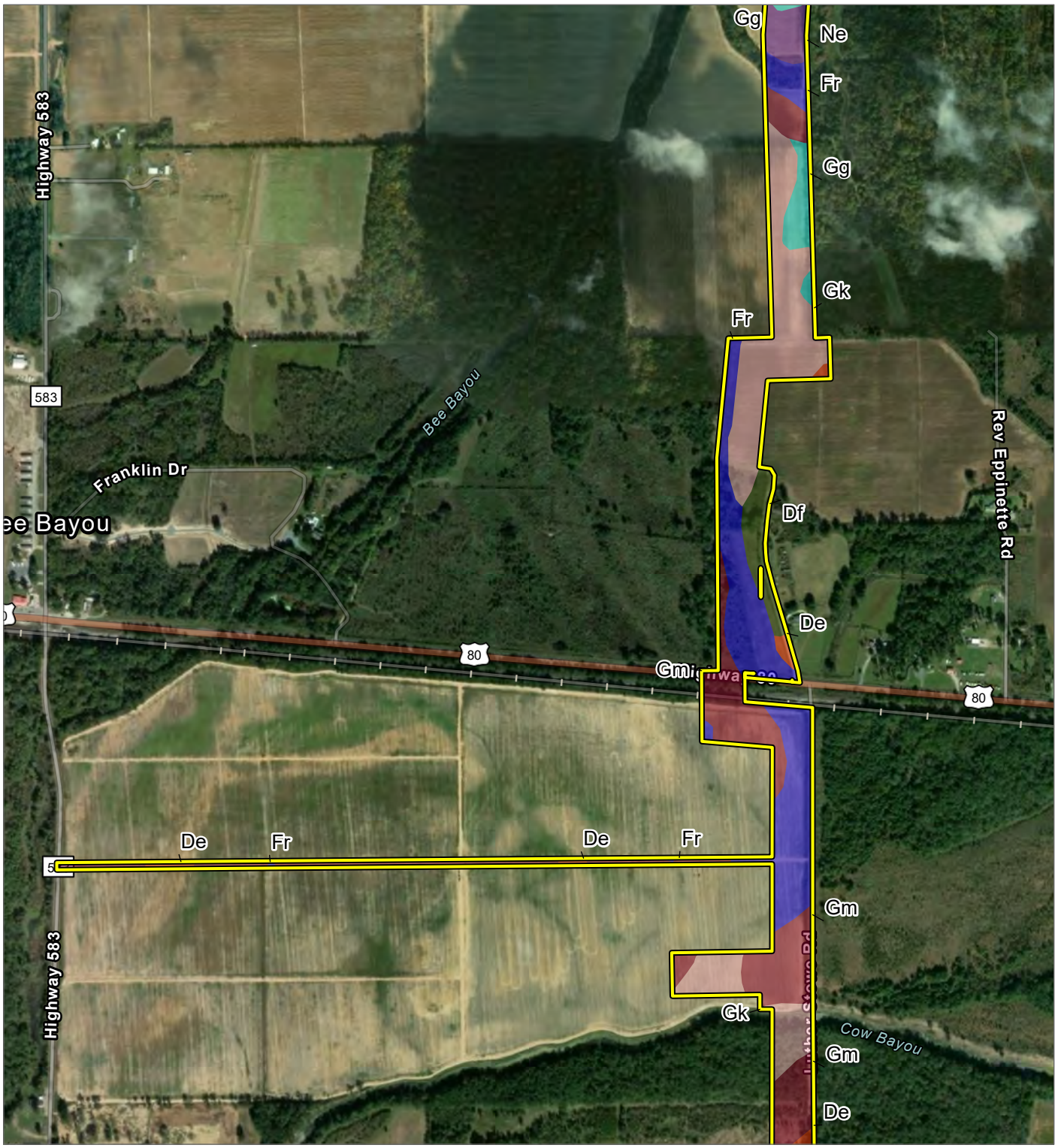


PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Df, DEXTER SILT LOAM, 3 TO 5 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Gg, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING
- Gk, GILBERT SILT LOAM
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ne, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

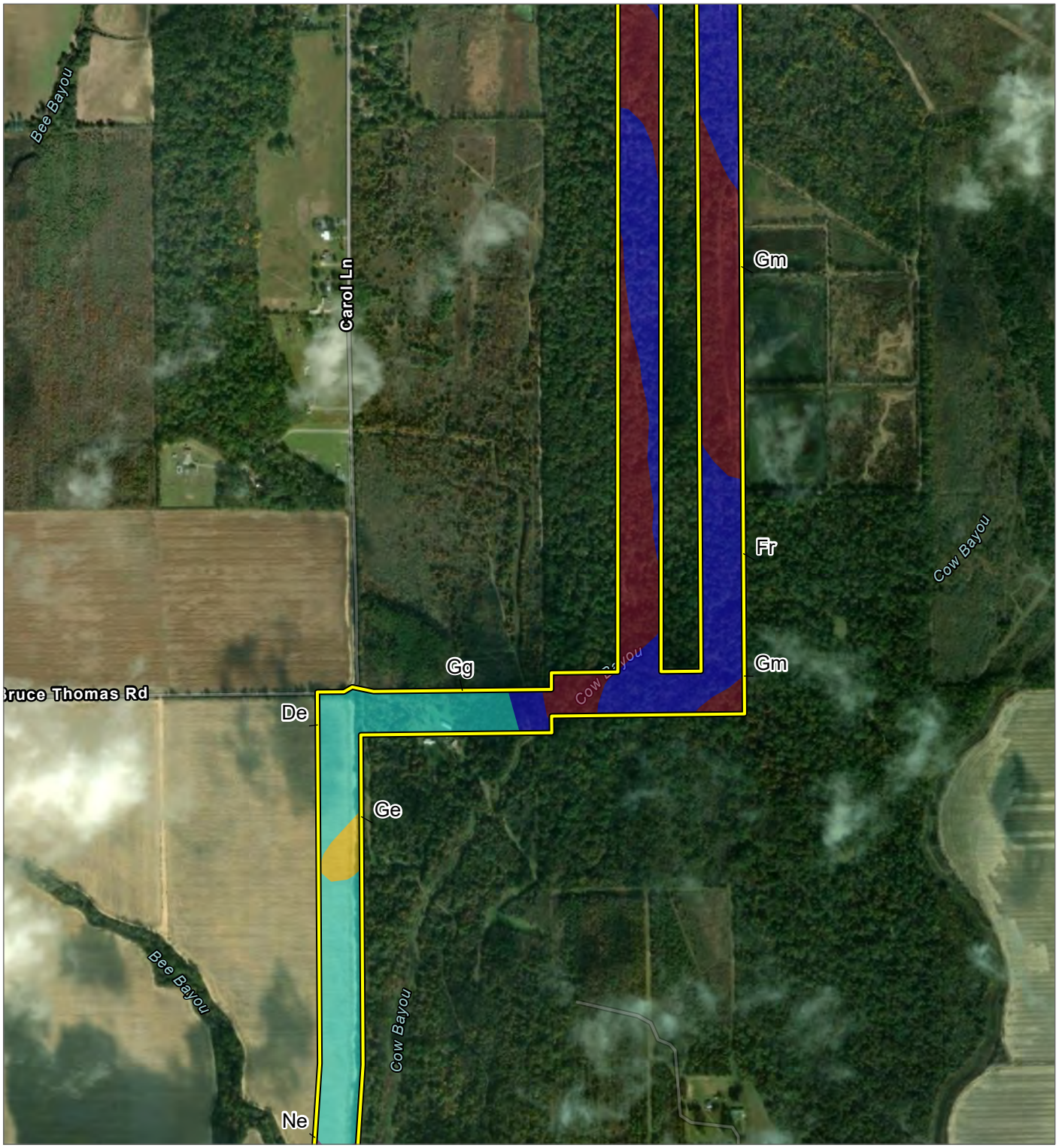
NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 5 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 1:16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FRANKLIN_FARMS_BIO_SURVEY_CLONE APPX; LAYOUT NAME: 5_NRCS SOILS



PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Ge, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gg, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ne, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

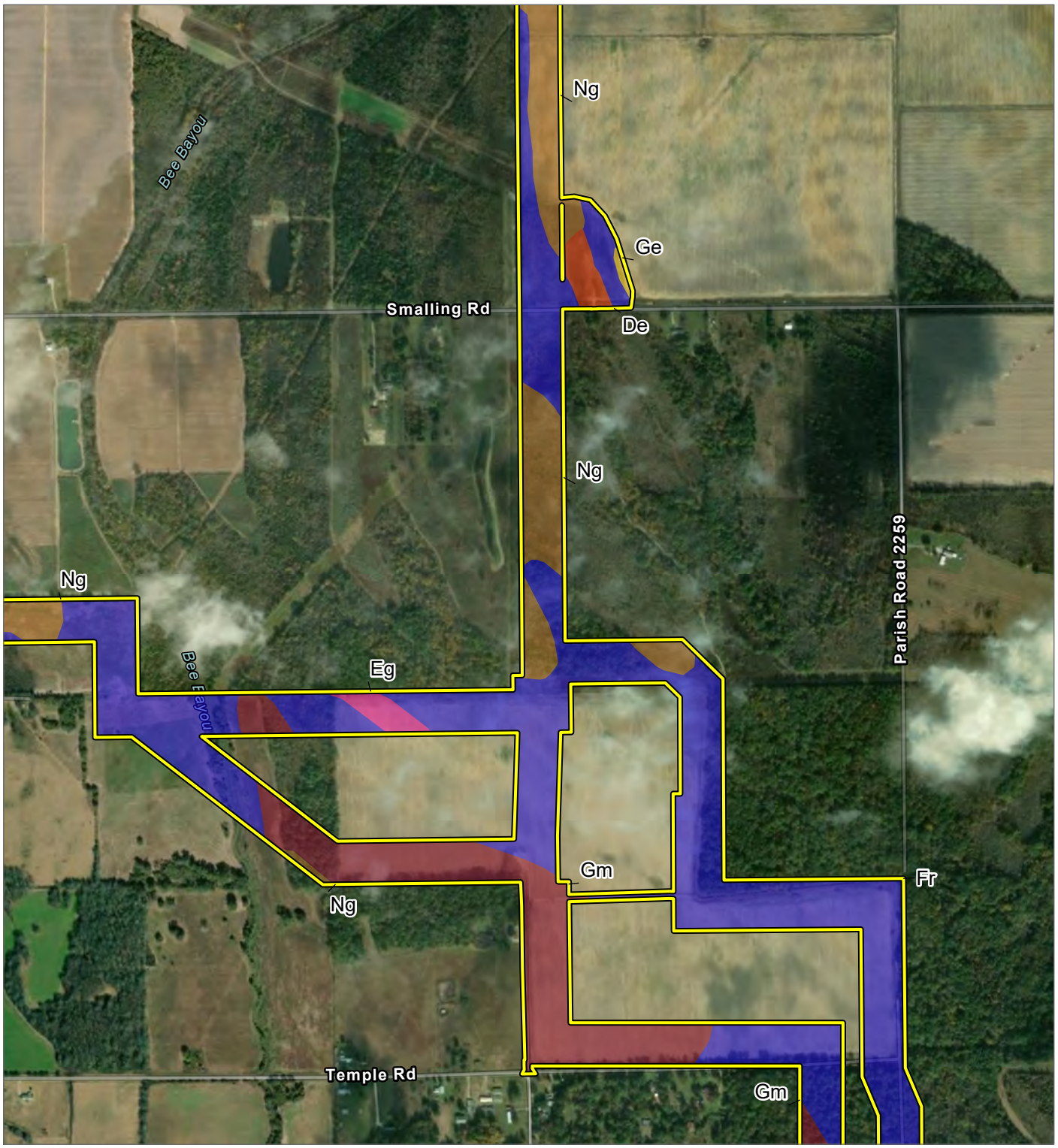
NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 6 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 5_NRCS SOILS



PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Eg, EGYPT SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Ge, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ng, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

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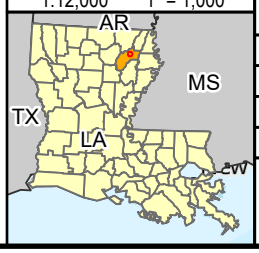
FEET

1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

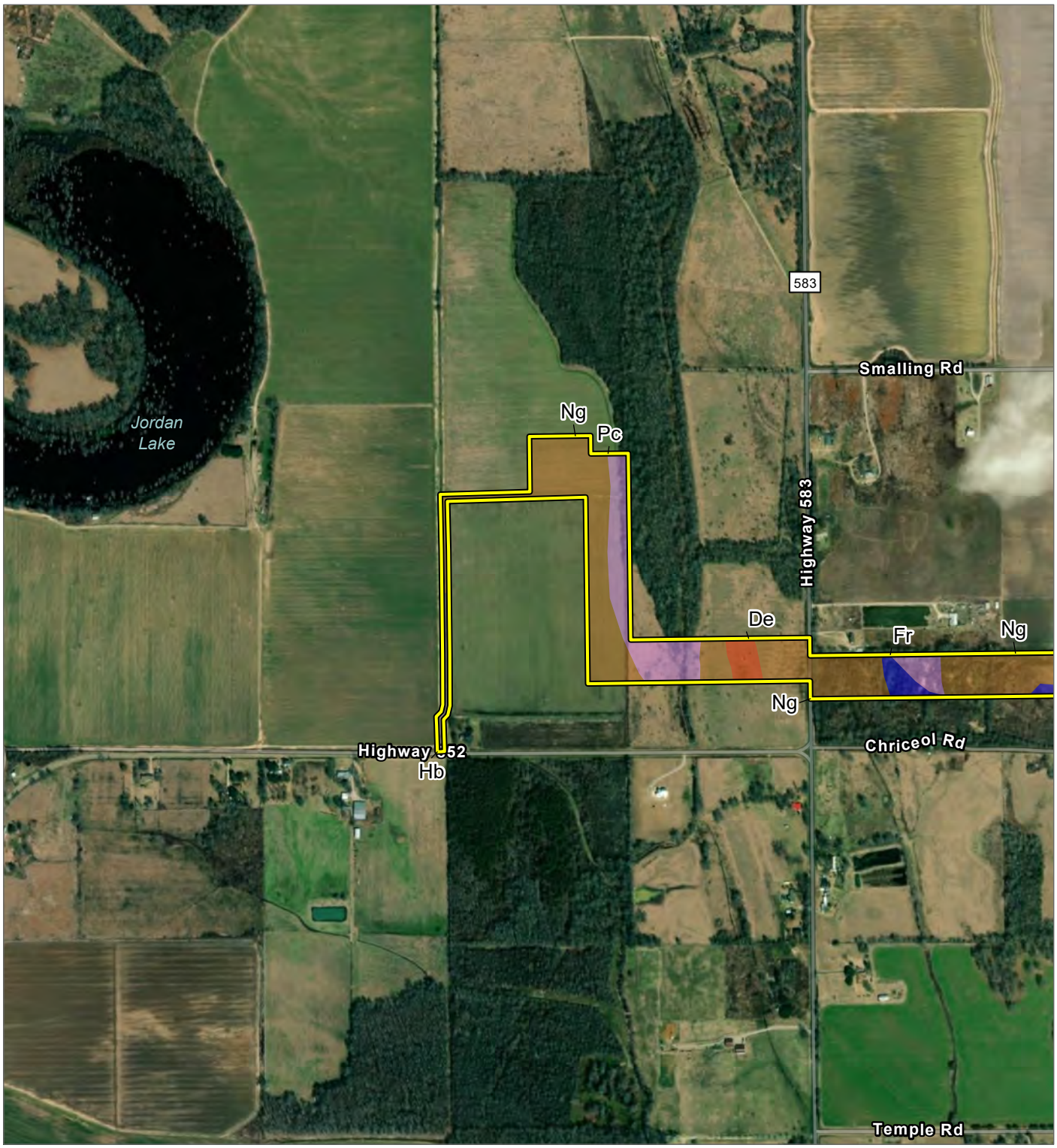
NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 7 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 1:16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FRANKLIN_FARMS_BIO_SURVEY_CLONE APPX; LAYOUT NAME: 5_NRCS SOILS



PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Hb, HEBERT SILT LOAM, 0 TO 1 PERCENT SLOPES
- Ng, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING
- Pc, PERRY SILTY CLAY LOAM

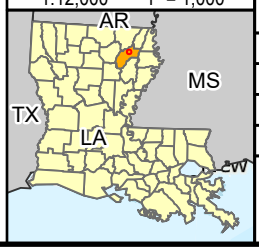
BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

0 500 1,000

FEET

1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

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FIGURE 5
PAGE 8 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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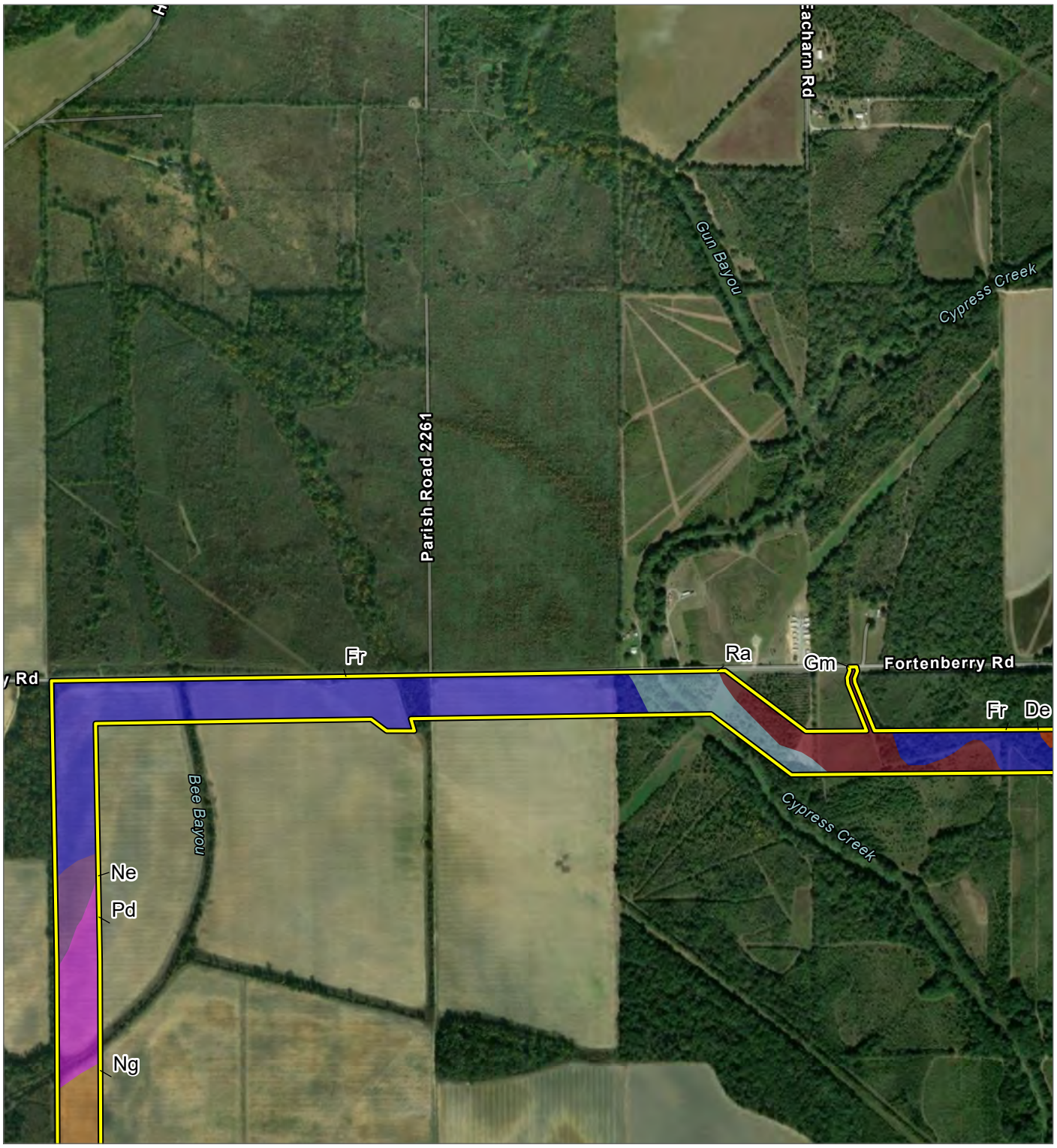

PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ne, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES
- Ng, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING
- Pd, PERRY CLAY
- Ra, RILLA SILT LOAM, 0 TO 1 PERCENT SLOPES

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 9 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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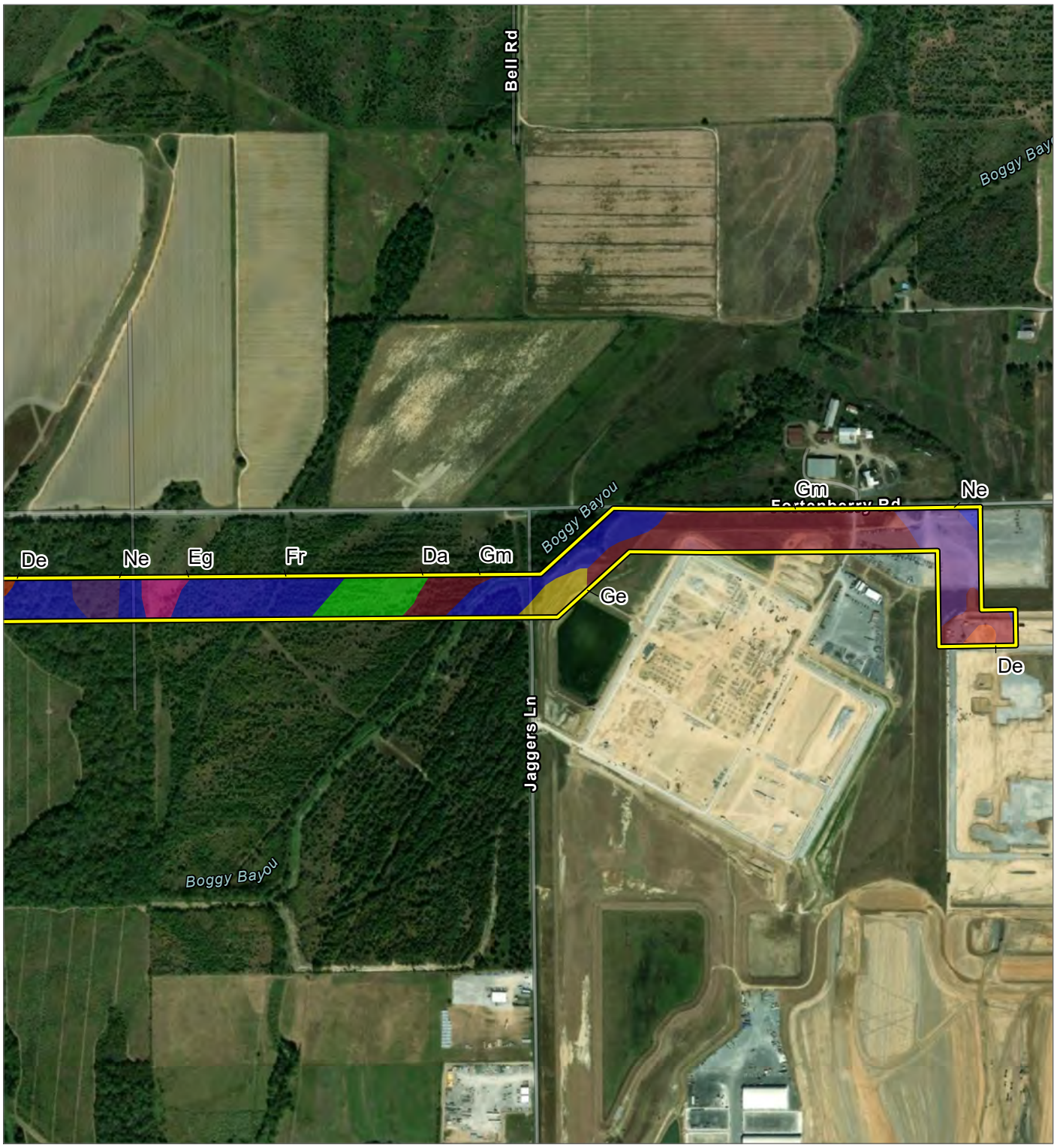


PROJECT BOUNDARY

SOIL MAP UNIT

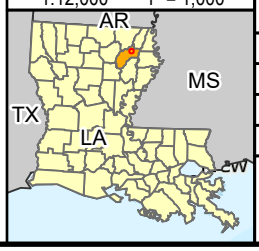
- Da, DEERFORD SILT LOAM
- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Eg, EGYPT SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Ge, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ne, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES

N

0 500 1,000

FEET

1:12,000 1" = 1,000'



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

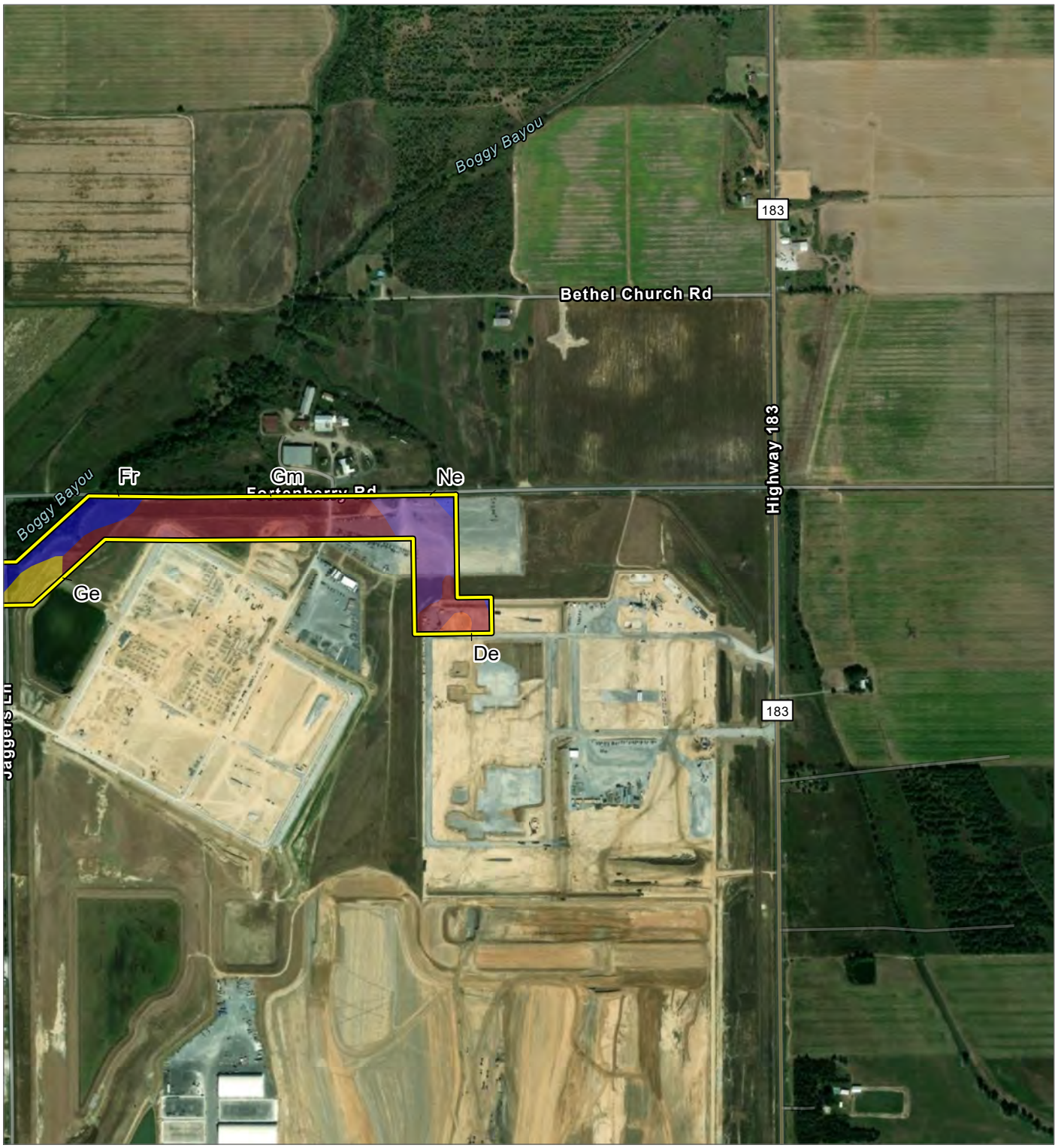
NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 10 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 1:16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; -- LAYOUT NAME: 5_NRCS SOILS



PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Ge, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ne, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES

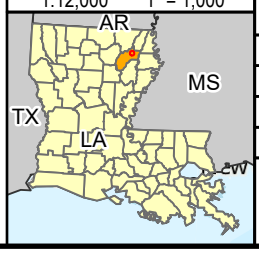
BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

0 500 1,000

FEET

1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 11 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FRANKLIN_FARMS_BIO_SURVEY_CLONE APPX; -- LAYOUT NAME: 5_NRCS SOILS

PROJECT BOUNDARY
3DHP FLOWLINE
3DHP WATERBODY
NWI WETLAND
1% ANNUAL CHANCE FLOOD HAZARD
AREA WITH REDUCED RISK DUE TO LEVEE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, 3DHP, NWI, FEMA

N

0 500 1,000
FEET
1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

3DHP AND NWI FEATURES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 1 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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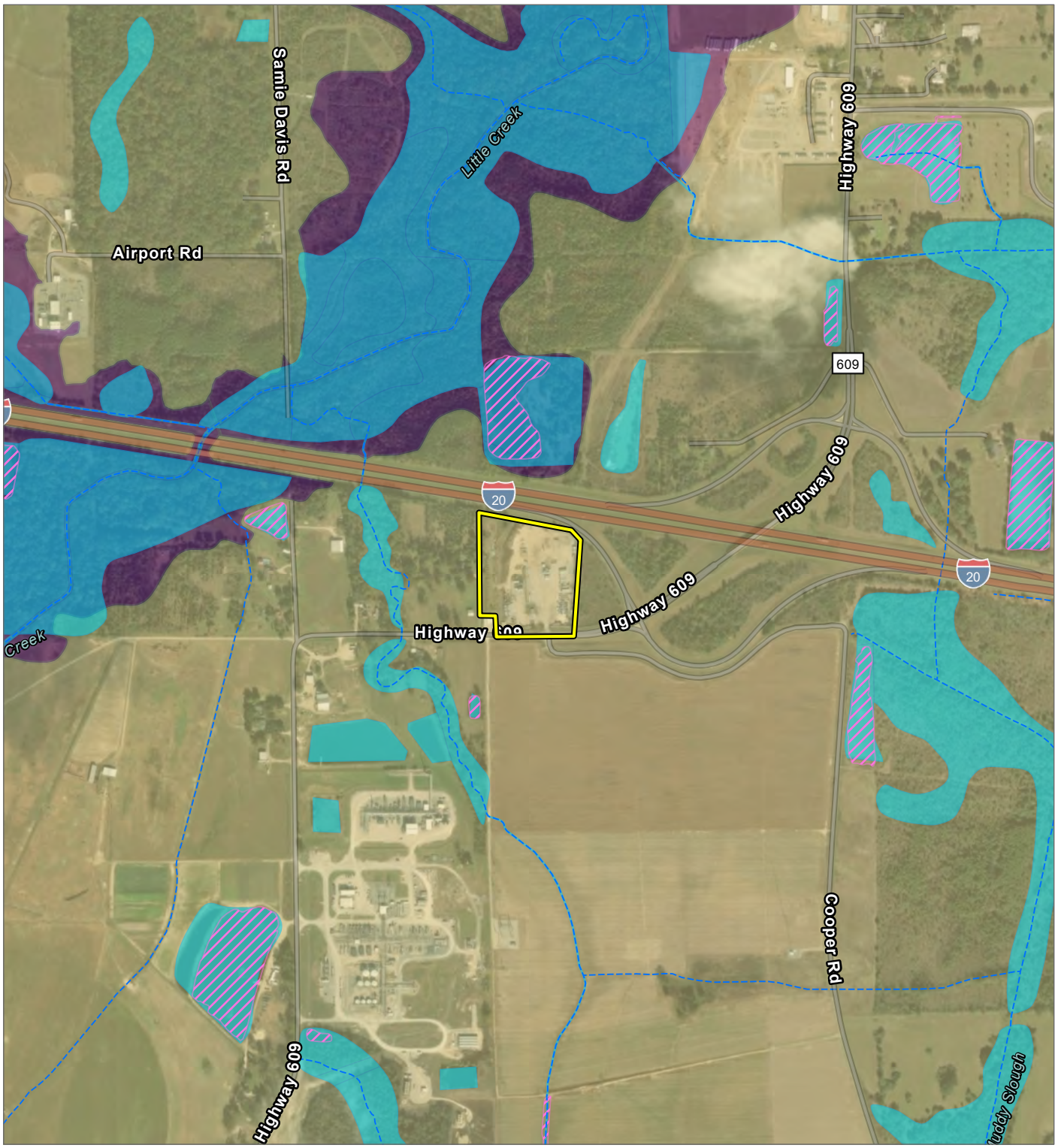

PROJECT BOUNDARY
3DHP FLOWLINE
3DHP WATERBODY
NWI WETLAND
1% ANNUAL CHANCE FLOOD HAZARD
AREA WITH REDUCED RISK DUE TO LEVEE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
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N

0 500 1,000
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

3DHP AND NWI FEATURES MAP

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FIGURE 6
PAGE 2 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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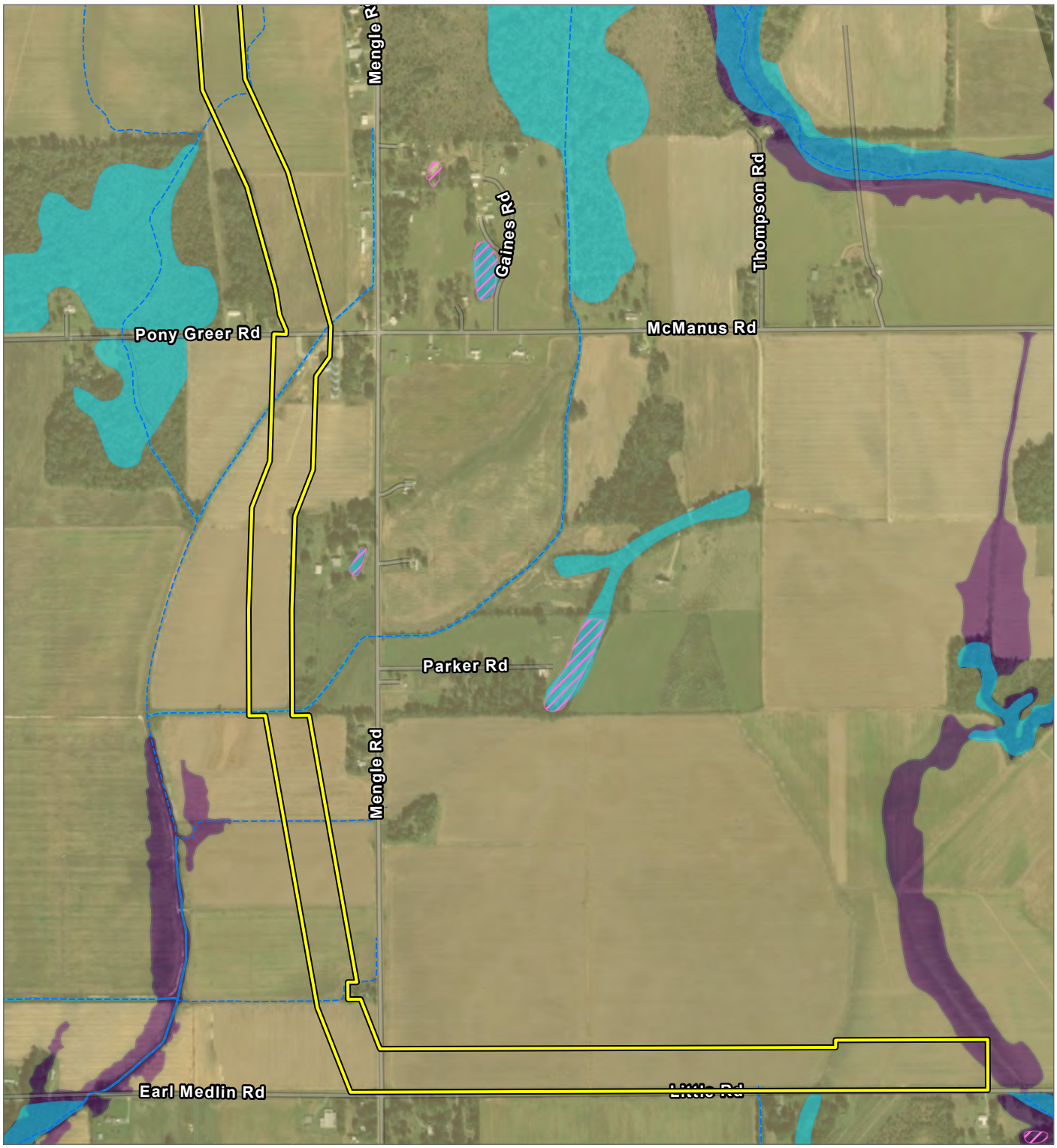

PROJECT BOUNDARY
3DHP FLOWLINE
3DHP WATERBODY
NWI WETLAND
1% ANNUAL CHANCE FLOOD HAZARD
AREA WITH REDUCED RISK DUE TO LEVEE

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3DHP AND NWI FEATURES MAP

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FIGURE 6
PAGE 3 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
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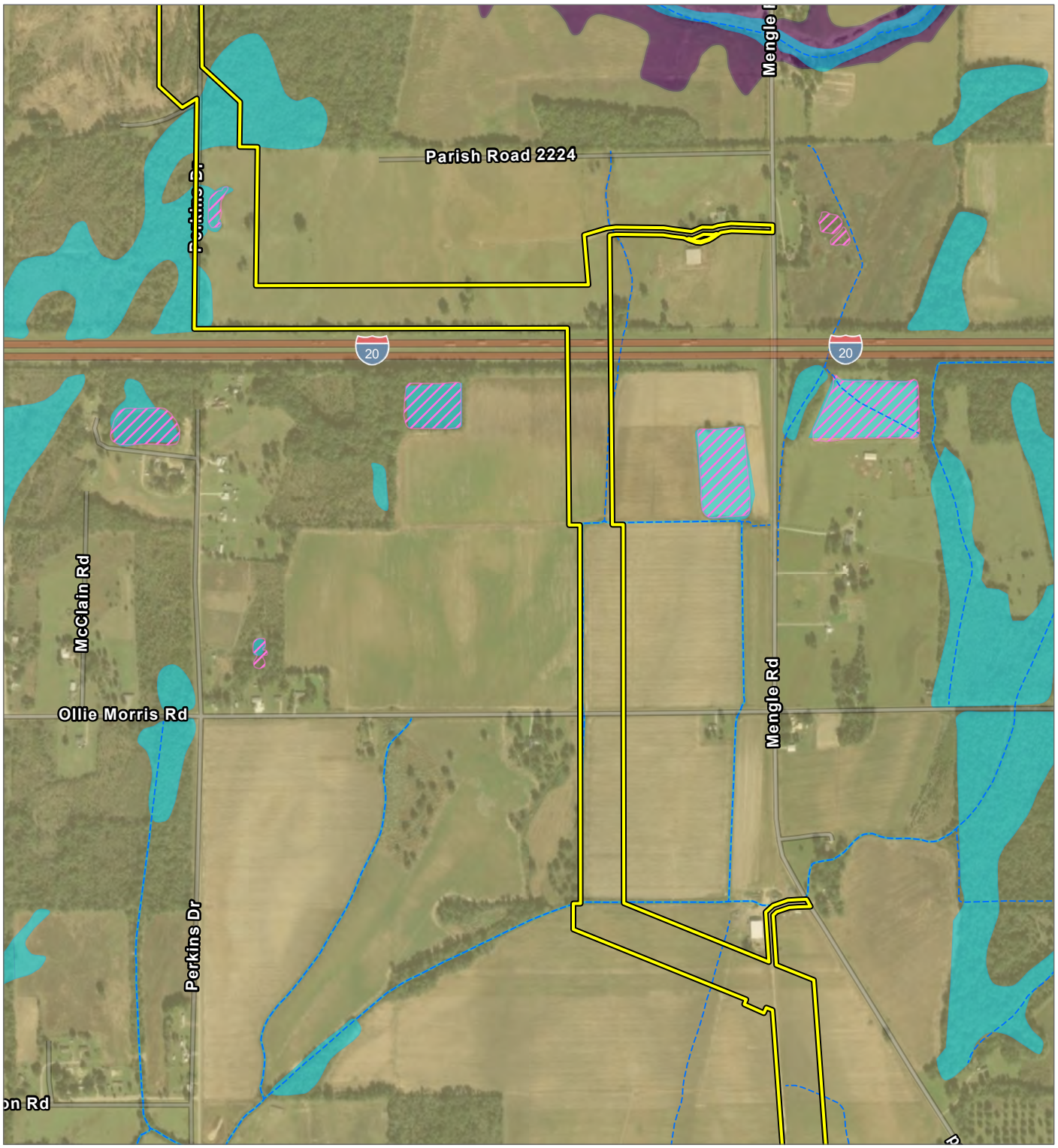

PROJECT BOUNDARY
3DHP FLOWLINE
3DHP WATERBODY
NWI WETLAND
1% ANNUAL CHANCE FLOOD HAZARD
AREA WITH REDUCED RISK DUE TO LEVEE

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

3DHP AND NWI FEATURES MAP

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APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 4 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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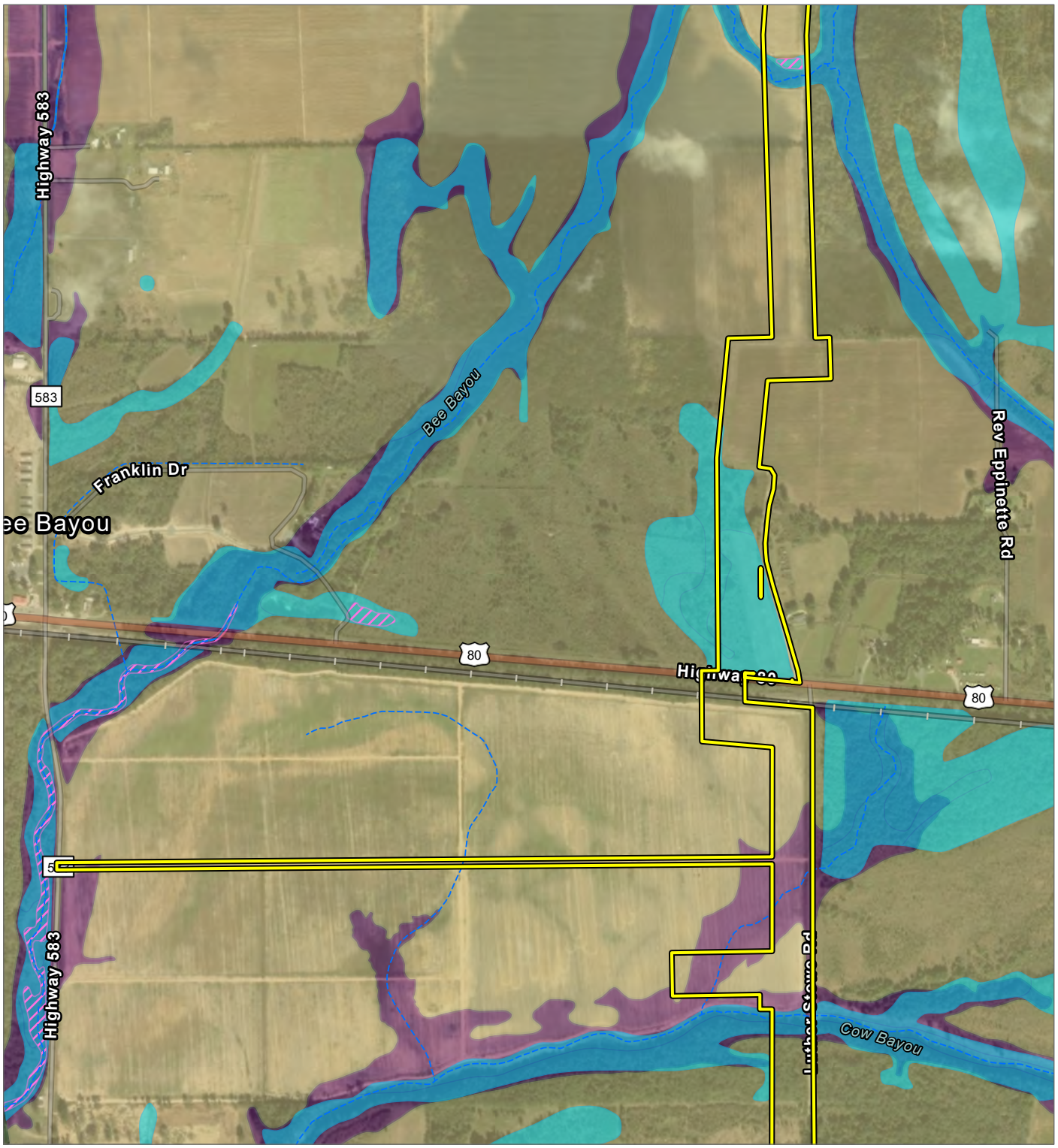

PROJECT BOUNDARY
3DHP FLOWLINE
3DHP WATERBODY
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1% ANNUAL CHANCE FLOOD HAZARD
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BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

3DHP AND NWI FEATURES MAP

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APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 5 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
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COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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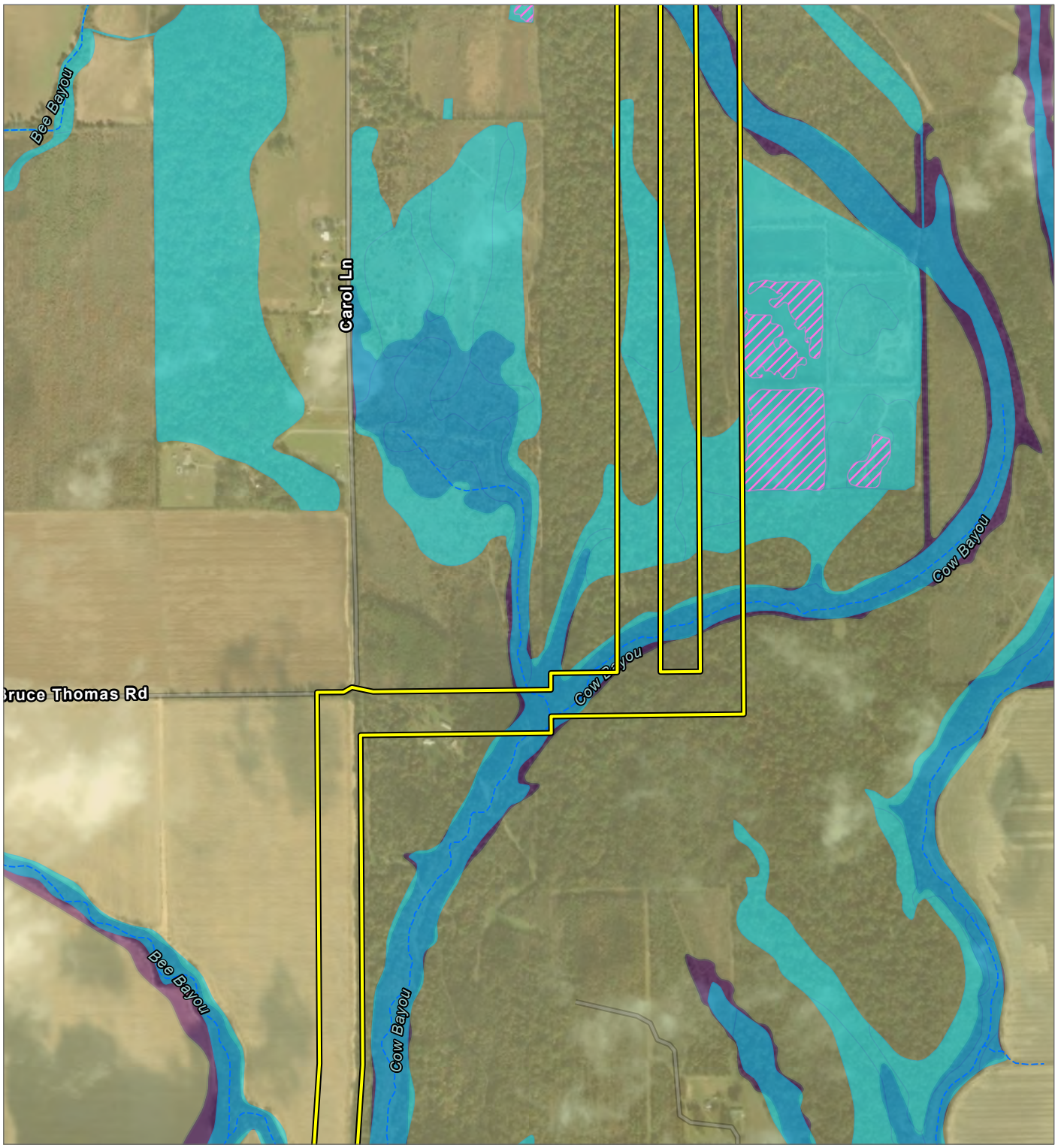

PROJECT BOUNDARY
3DHP FLOWLINE
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NWI WETLAND
1% ANNUAL CHANCE FLOOD HAZARD
AREA WITH REDUCED RISK DUE TO LEVEE

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FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

3DHP AND NWI FEATURES MAP

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FIGURE 6
PAGE 6 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
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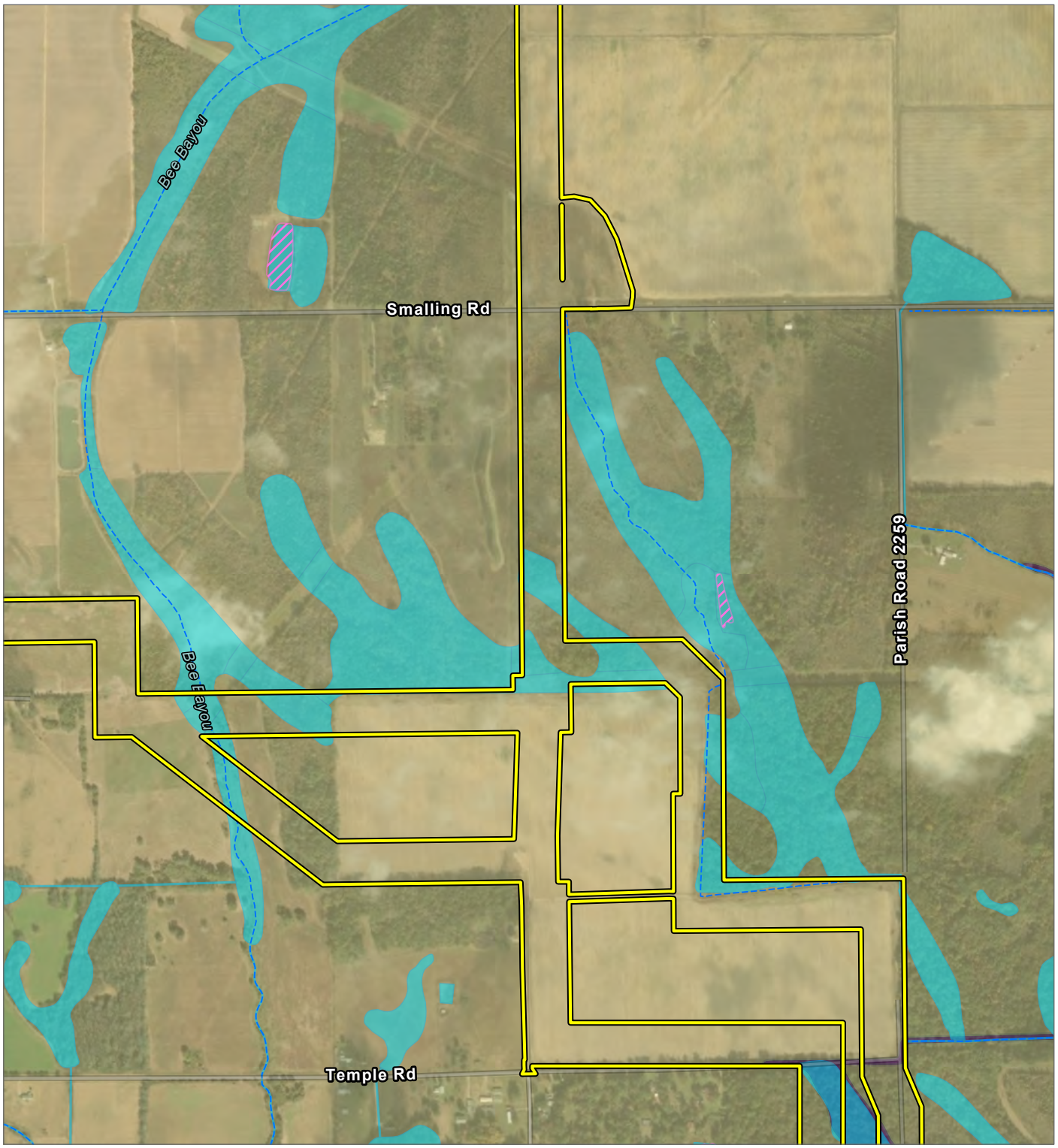

PROJECT BOUNDARY
3DHP FLOWLINE
3DHP WATERBODY
NWI WETLAND
1% ANNUAL CHANCE FLOOD HAZARD
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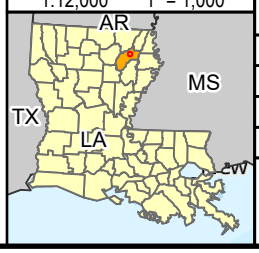

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

3DHP AND NWI FEATURES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
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DATE:	MAY 2026		

FIGURE 6
PAGE 7 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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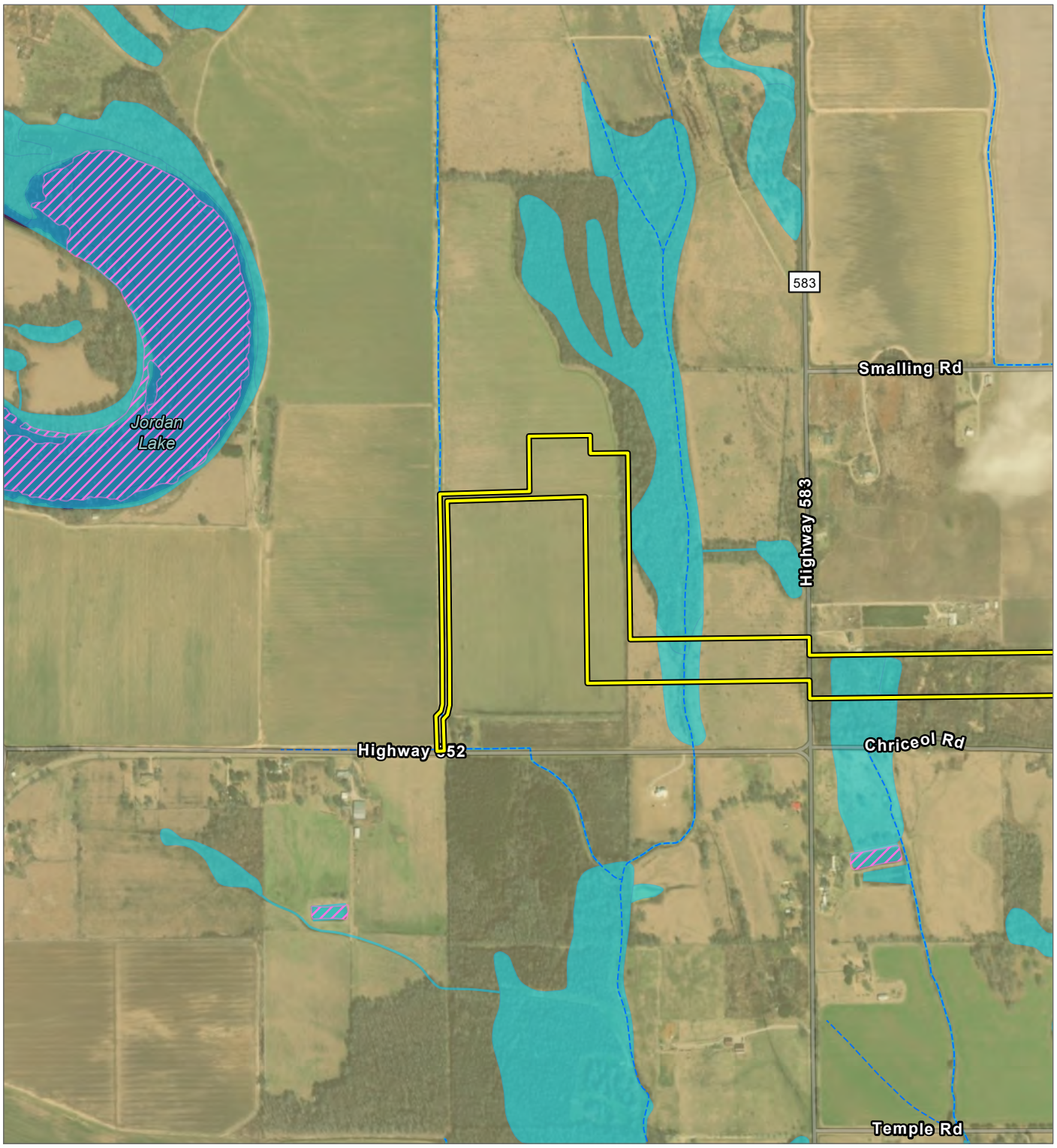

PROJECT BOUNDARY
3DHP FLOWLINE
3DHP WATERBODY
NWI WETLAND
1% ANNUAL CHANCE FLOOD HAZARD
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BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
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N

0 500 1,000
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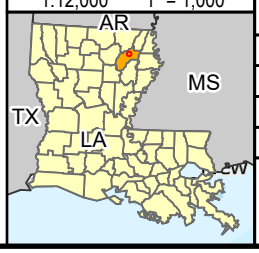

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

3DHP AND NWI FEATURES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 8 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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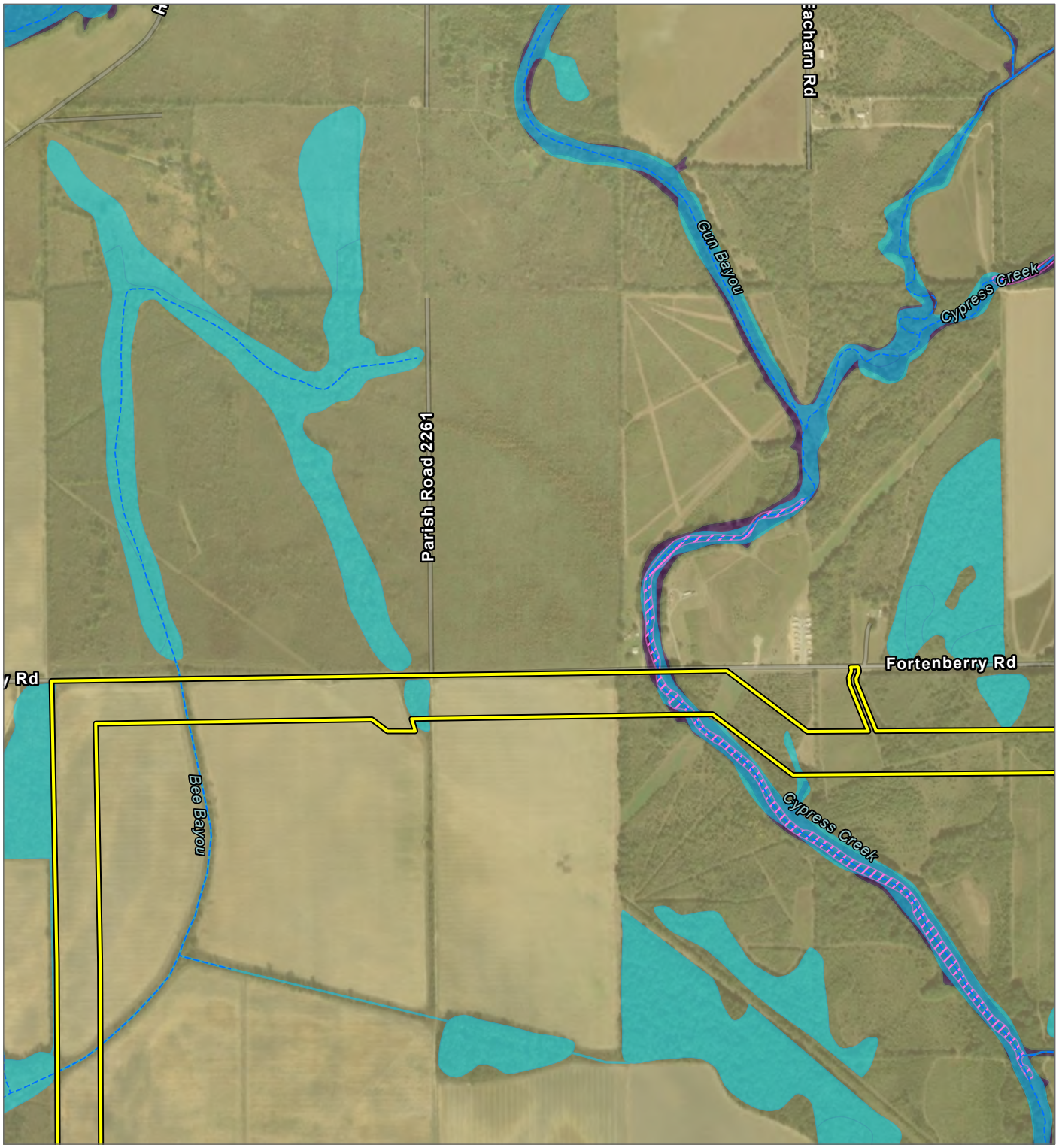

PROJECT BOUNDARY
3DHP FLOWLINE
3DHP WATERBODY
NWI WETLAND
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IMAGERY DATE: 11/05/2025
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N

0 500 1,000
FEET
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

3DHP AND NWI FEATURES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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FIGURE 6
PAGE 9 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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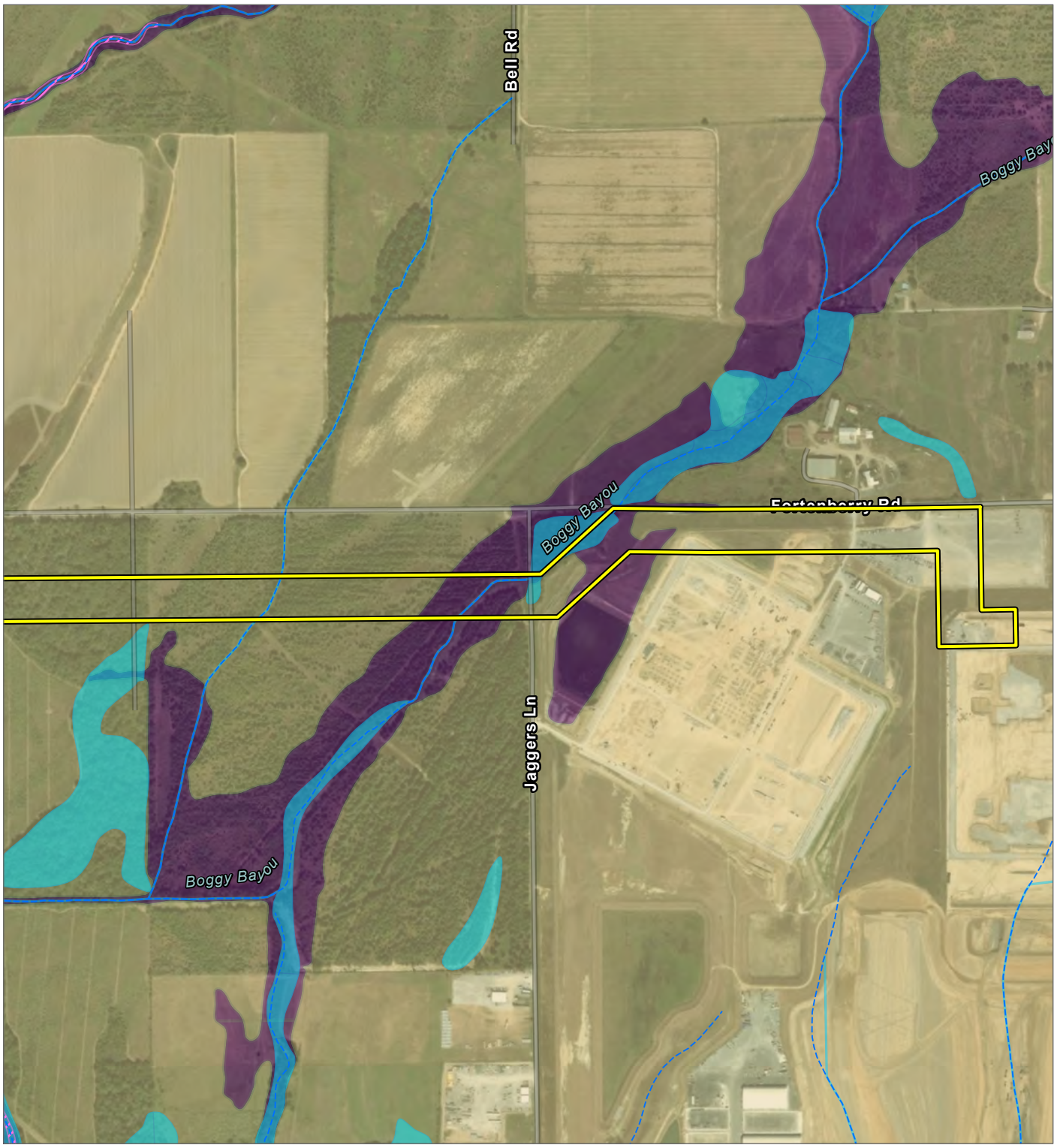

PROJECT BOUNDARY
3DHP FLOWLINE
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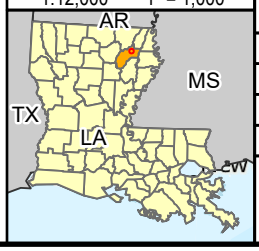

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

3DHP AND NWI FEATURES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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DATE:	MAY 2026		

FIGURE 6
PAGE 10 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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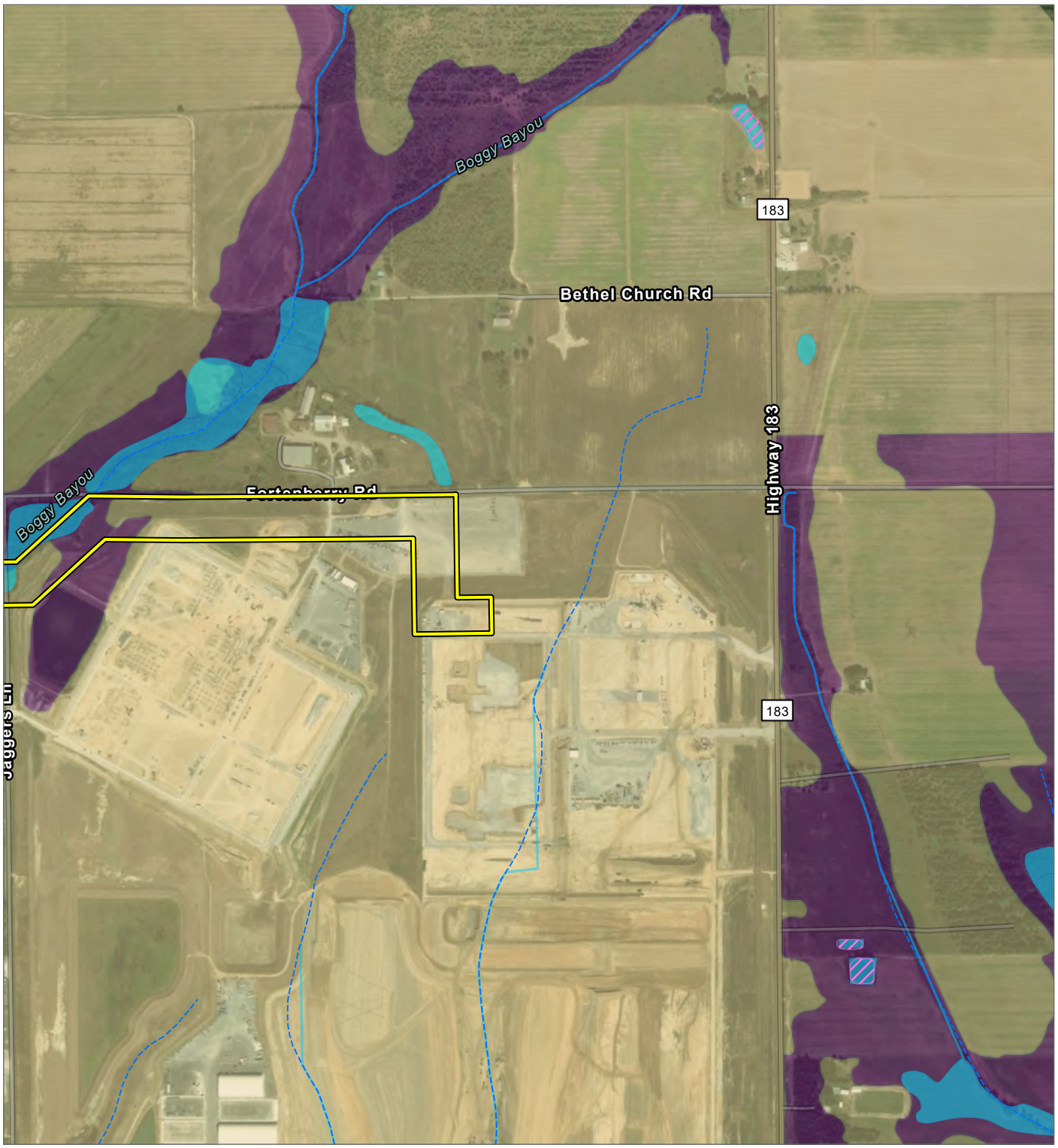

PROJECT BOUNDARY
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NWI WETLAND
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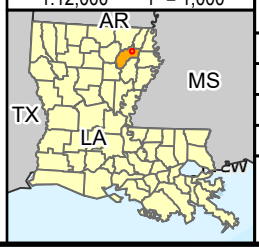

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

3DHP AND NWI FEATURES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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DATE:	MAY 2026		

FIGURE 6
PAGE 11 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
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PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM -- FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 6_NHD AND NWI

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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17

Highway 17

Airport Rd

D-T3F-08

D-T3F-07

Parish Line Rd

PROJECT BOUNDARY
CULVERT
ROADSIDE DITCH (NRPW)

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 1 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 7. DELINEATED RESOURCES 11X17P

PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT
INTERMITTENT STREAM
PEM

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 2 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
CULVERT
AG DITCH (NRPW)

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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FIGURE 7
PAGE 3 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
CULVERT
AG DITCH (NRPW)
AG DITCH (RPW)

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 4 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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D-T2F-03
D-T2F-03
UPL-T2F-102
UPL-T2F-101
UPL-T2F-100
Mengle Rd
Mengle Rd
Mengle Rd
Mengle Rd
Mengle Rd
Pony Greer Rd
Pony Greer Rd
P-T1F-01
P-T1F-01
D-T1F-06
D-T1F-06

PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
CULVERT
AG DITCH (RPW)
POND

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE	FIGURE 7	
APPROVED BY:	A. THOMPSON	PAGE 5 OF 21	
DATE:	MAY 2026		

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PEM-T2F-10
UPL-T2F-10
PEM-T2F-10
Mengle Rd
20
20
20
20
PEM-T2F-01
UPL-T2F-01
D-T2F-14
PEM-T2F-01
D-T2F-02
Mengle Rd
UPL-T2F-103
Mengle Rd
Ollie Morris Rd
Ollie Morris Rd
O'Quin
D-T2F-01
D-T2F-01
Mengle Rd
D-T2F-02
Mer

PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT
AG DITCH (NRPW)
AG DITCH (RPW)
PEM

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 6 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 7. DELINEATED RESOURCES 11X17P

D-T2F-05b
D-T2F-05a
UPL-T2F-105
D-T2F-05a
Luther Stowe Rd
S-T2F-02
PFO-T2F-09
S-T2F-02
Cow Bayou
PFO-T2F-09
UPL-T2F-09
Cow Bayou
Cow Bayou
S-T2F-01
S-T2F-01
P-T2F-03
P-T2F-03
UPL-T2F-02
Luther Stowe Rd
UPL-T2F-03
PFO-T2F-03
PFO-T2F-03
UPL-T2F-04
PFO-T2F-04
PEM-T2F-04
PEM-T2F-04
P-T2F-04
PFO-T2F-04
S-T2F-03
S-T2F-03
D-T2F-04
UPL-T2F-104
D-T2F-04

PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT
AG DITCH (NRPW)
AG DITCH (RPW)
INTERMITTENT STREAM
PERENNIAL STREAM
POND
PEM
PFO

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 7 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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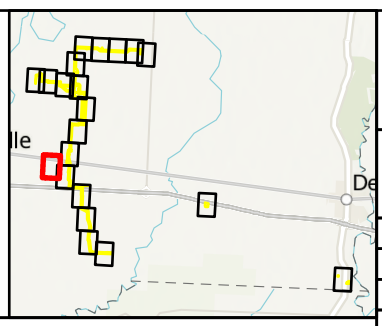
PROJECT BOUNDARY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 8 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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D-T1F-10
D-T1F-10
Bee Bayou
Cow Bayou
D-T1F-09
D-T1F-09
Cow Bayou
Rev Eppinette Rd
D-T1F-08
Cow Bayou
D-T1F-08
PFO-T1F-03
UPL-T1F-03
PEM-T1F-03a
PFO-T1F-03
PEM-T1F-03b
PEM-T1F-03b
Rev Eppinette Rd
Rev Eppinette Rd
Highway 80
D-T1F-07
Canadian Pacific Railway
Highway 80
D-T1F-07
Canadian Pacific Railway
Highway 80
D-T2F-05b
Rev Eppinette Rd

PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT
AG DITCH (NRPW)
AG DITCH (RPW)
ROADSIDE DITCH (RPW)
PEM
PFO

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
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DATE:	MAY 2026		

FIGURE 7
PAGE 9 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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Carol Ln
Bruce Thomas Rd
D-T1F-11
D-T1F-11
D-T1F-11
UPL-T1F-01
PFO-T1F-01
S-T1F-02
S-T1F-02
S-T1F-03
S-T1F-03
S-T1F-04
S-T1F-04
S-T1F-04
PFO-T1F-01
Cow Bayou
Cow Bayou
Cow Bayou
Cow Bayou
Cow Bayou
Cow Bayou
McCormick Rd
McCormick Rd
McCormick Rd
PFO-T1F-02
P-T1F-02
P-T1F-02
PFO-T1F-02
S-T1F-01
S-T1F-01
UPL-T1F-02
Cow Bayou

PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT
AG DITCH (NRPW)
PERENNIAL STREAM
POND
PFO

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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DATE:	MAY 2026		

FIGURE 7
PAGE 10 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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Temple Rd

Temple Rd

Temple Rd

PFO-T1F-04

PFO-T1F-04

PEM-T1F-04

PFO-T1F-01

PEM-T1F-04

UPL-T1F-01

Cow Bayou

PFO-T1F-01

PFO-T1F-01

- PROJECT BOUNDARY
- UPLAND PLOT
- WETLAND PLOT
- CULVERT
- AG DITCH (NRPW)
- PEM
- PFO

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

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APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 11 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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UPL-T1F-06
PFO-T1F-05a
UPL-T1F-05
PFO-T1F-05a
PEM-T1F-05
D-T1F-14
PEM-T1F-05
D-T1F-14
D-T1F-14
PFO-T1F-05b
PFO-T1F-05b
PFO-T3F-01
D-T1F-12
D-T1F-13
D-T1F-12
D-T1F-13
Parish Road 2259
Parish Road 2259
Temple Rd
Temple Rd
Temple Rd

PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT
AG DITCH (NRPW)
AG DITCH (RPW)
PEM
PFO

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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FIGURE 7
PAGE 12 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PFO-T1F-0
Bee Bayou
Bee Bayou
PEM-T2F-16
UPL-T2F-16
PEM-T2F-16
Bee Bayou
D-T2F-06
D-T2F-06
D-T2F-07
D-T2F-07
S-T3F-01
S-T3F-01
UPL-T3F-02
PSS-T3F-02
UPL-T3F-100
PSS-T3F-02
UPL-T3F-01
PFO-T3F-01
PFO-T3F-01
Bee Bayou
S-T3F-01
PEM-T4F-02
UPL-T4F-02
PEM-T4F-02
PFO-T3F-01
PFO-T3F-01
Bee Bayou
Bee Bayou

PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
AG DITCH (NRPW)
AG DITCH (RPW)
PERENNIAL STREAM
PEM
PFO
PSS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N
0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 13 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT

PEM
AG DITCH (NRPW)
AG DITCH (RPW)
PERENNIAL STREAM
PEM
PFO

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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DATE:	MAY 2026		

FIGURE 7
PAGE 14 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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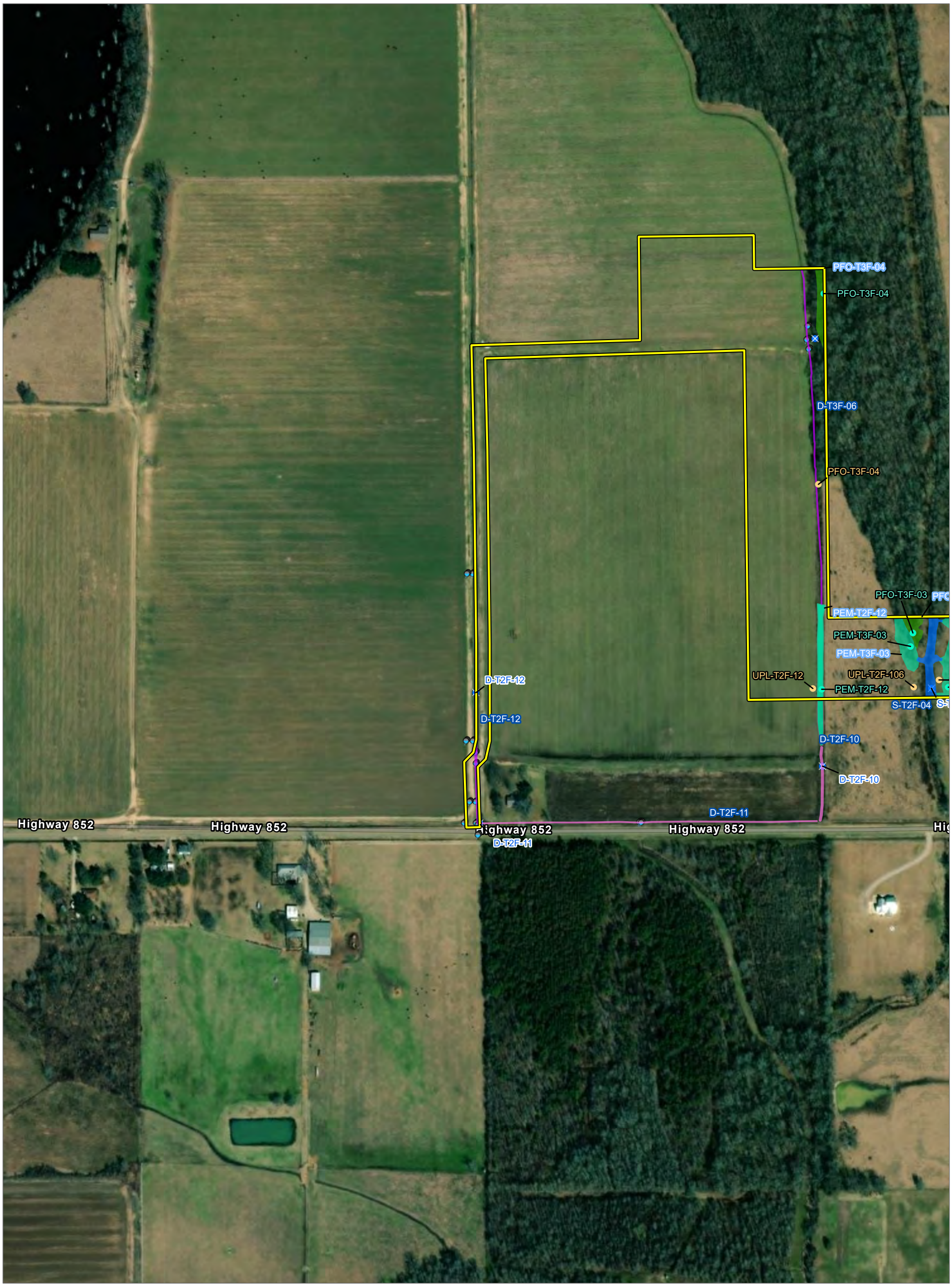
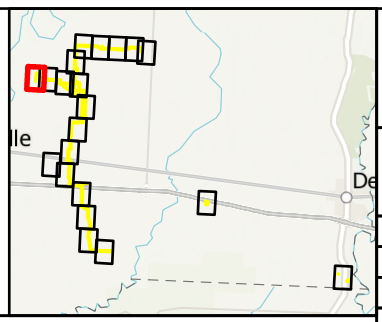


PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT
PEM
AG DITCH (NRPW)
AG DITCH (RPW)
PERENNIAL STREAM
PEM
PFO



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 15 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 7. DELINEATED RESOURCES 11X17P

PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
EPHEMERAL STREAM
PERENNIAL STREAM
PEM
PFO

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 16 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT
AG DITCH (NRPW)
AG DITCH (RPW)
PERENNIAL STREAM
PEM
PFO

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 17 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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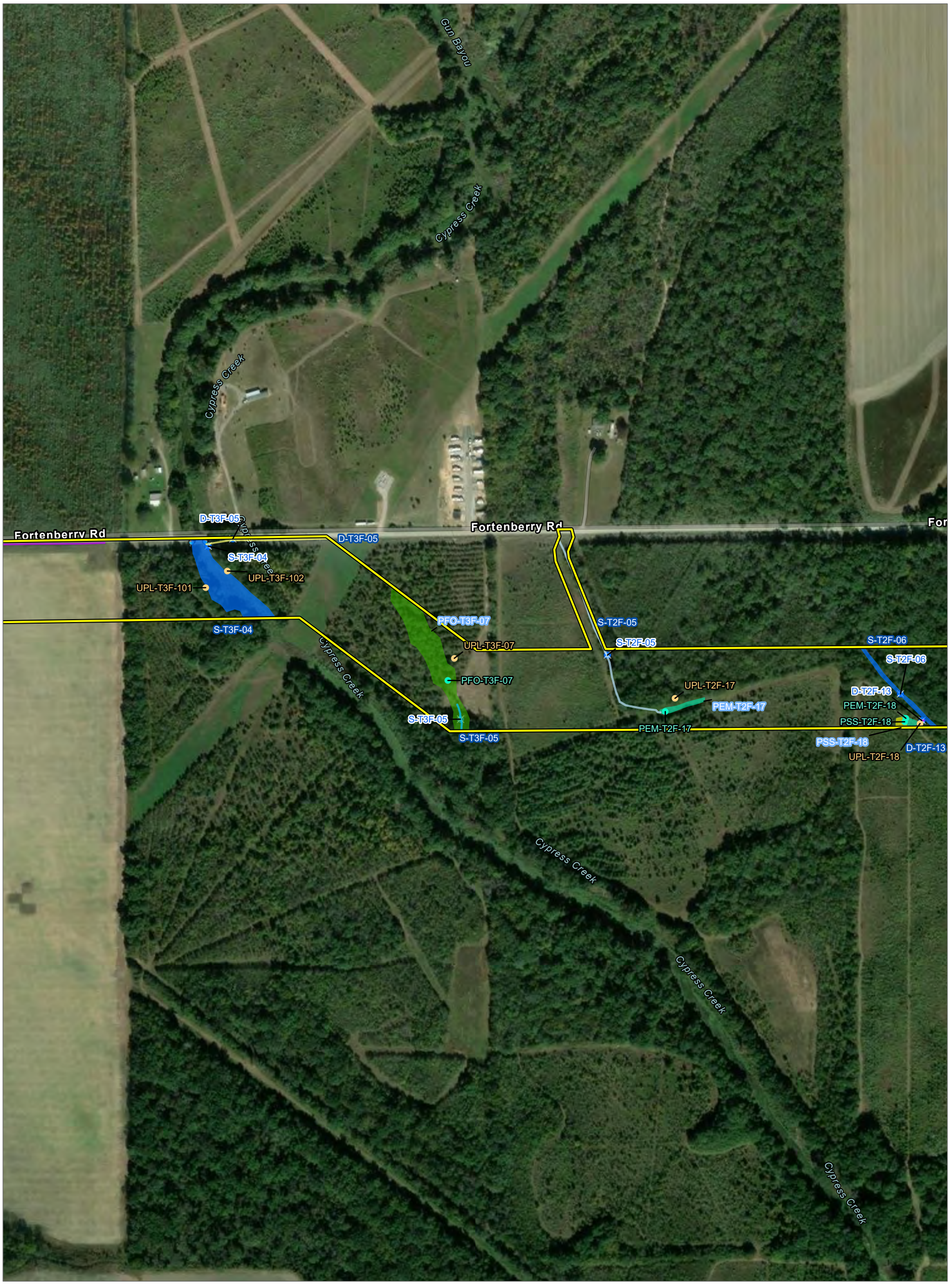
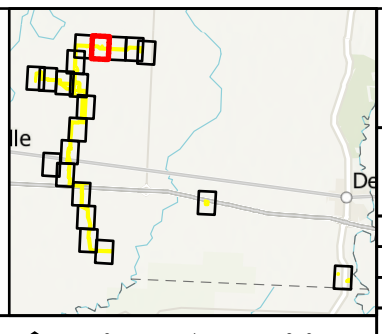


PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT

AG DITCH (NRPW)
EPHEMERAL STREAM
INTERMITTENT STREAM
PERENNIAL STREAM
ROADSIDE DITCH (NRPW)

PEM
PFO
PSS



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 18 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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Parish Road 2256C

Parish Road 2256B

Fortenberry Rd

Fortenberry Rd

Fortenbe

PFO-T2F-19

PFO-T2F-19

S-T2F-10

PFO-T2F-20

PFO-T2F-21

PFO-T2F-21

UPL-T2F-21

S-T2F-07

S-T2F-07

S-T2F-10

S-T2F-11

S-T2F-11

S-T2F-09

UPL-T2F-19

UPL-T2F-20

UPL-T2F-08

PEM-T2F-08

UPL-T2F-108

PEM-T2F-20

PFO-T2F-20

PEM-T2F-20

PEM-T2F-08

Boggy Bayou

Boggy Bayou

Boggy Bayou

Boggy Bayou

Boggy Bayou

- PROJECT BOUNDARY
- STREAM PLOT
- UPLAND PLOT
- WETLAND PLOT
- INTERMITTENT STREAM
- PERENNIAL STREAM
- PEM
- PFO

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 19 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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Boggy Bayou
Boggy Bayou
Boggy Bayou
Fortenberry Rd
Fortenberry Rd
Jaggers Ln
Jaggers Ln
S-T2F-09
PFO-T2F-07
PFO-T2F-07
S-T2F-08
PFO-T2F-05
UPL-T2F-07
UPL-T2F-05
UPL-T2F-06
S-T2F-08
PEM-T2F-06
PEM-T2F-06
PFO-T2F-05
P-T2F-02
P-T2F-01
P-T2F-02
P-T2F-01

PROJECT BOUNDARY
STREAM PLOT
UPLAND PLOT
WETLAND PLOT
CULVERT
INTERMITTENT STREAM
PERENNIAL STREAM
POND
PEM
PFO

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 20 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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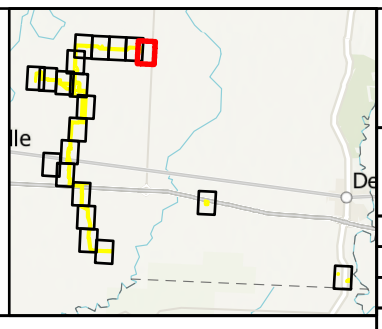
PROJECT BOUNDARY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

DELINEATED RESOURCES MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 7
PAGE 21 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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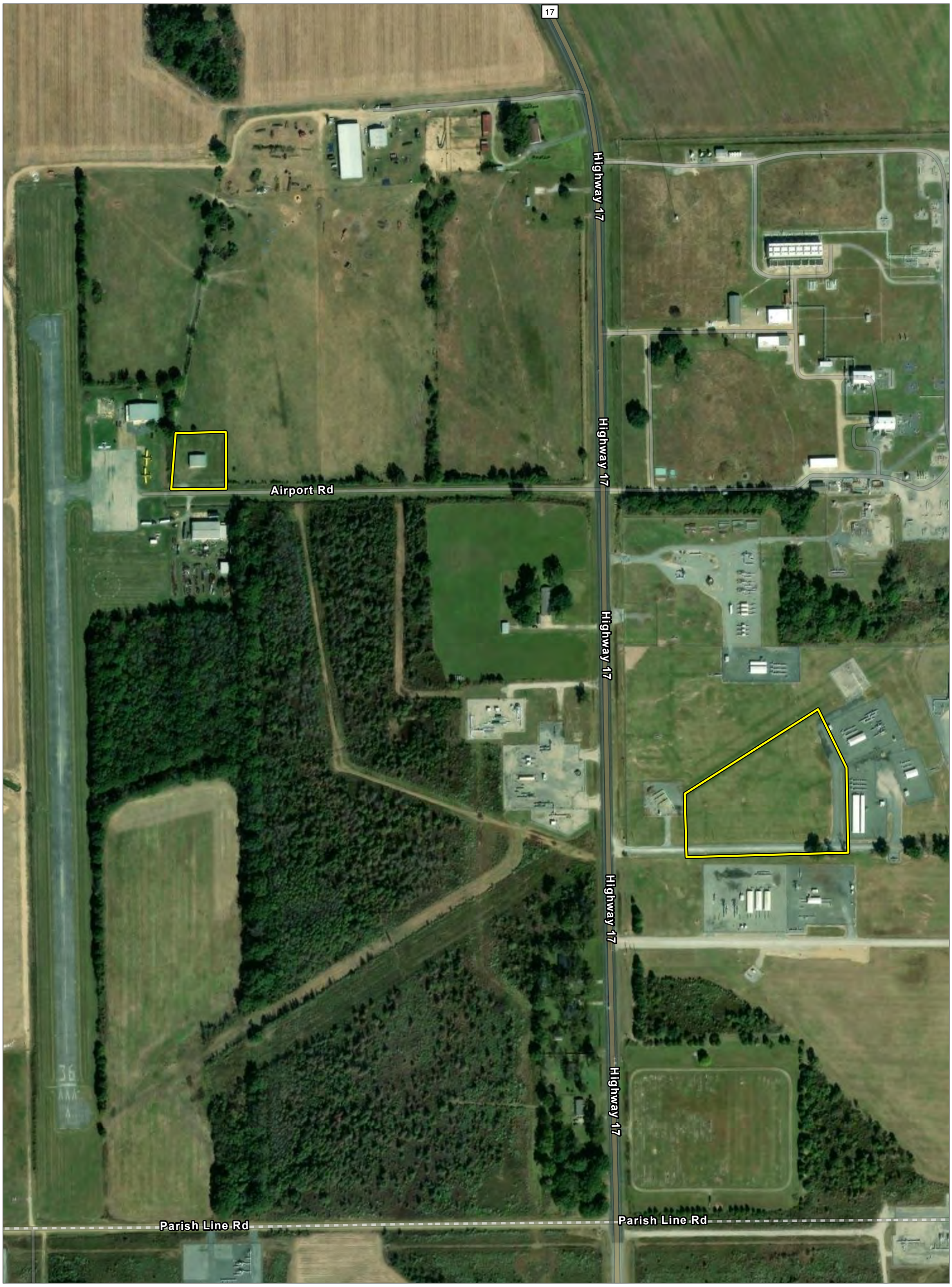

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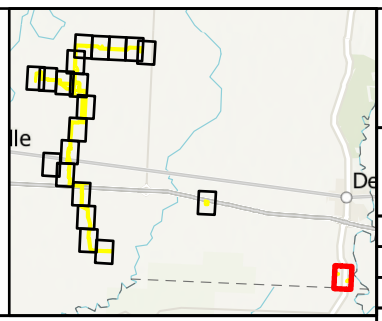

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 8
PAGE 1 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

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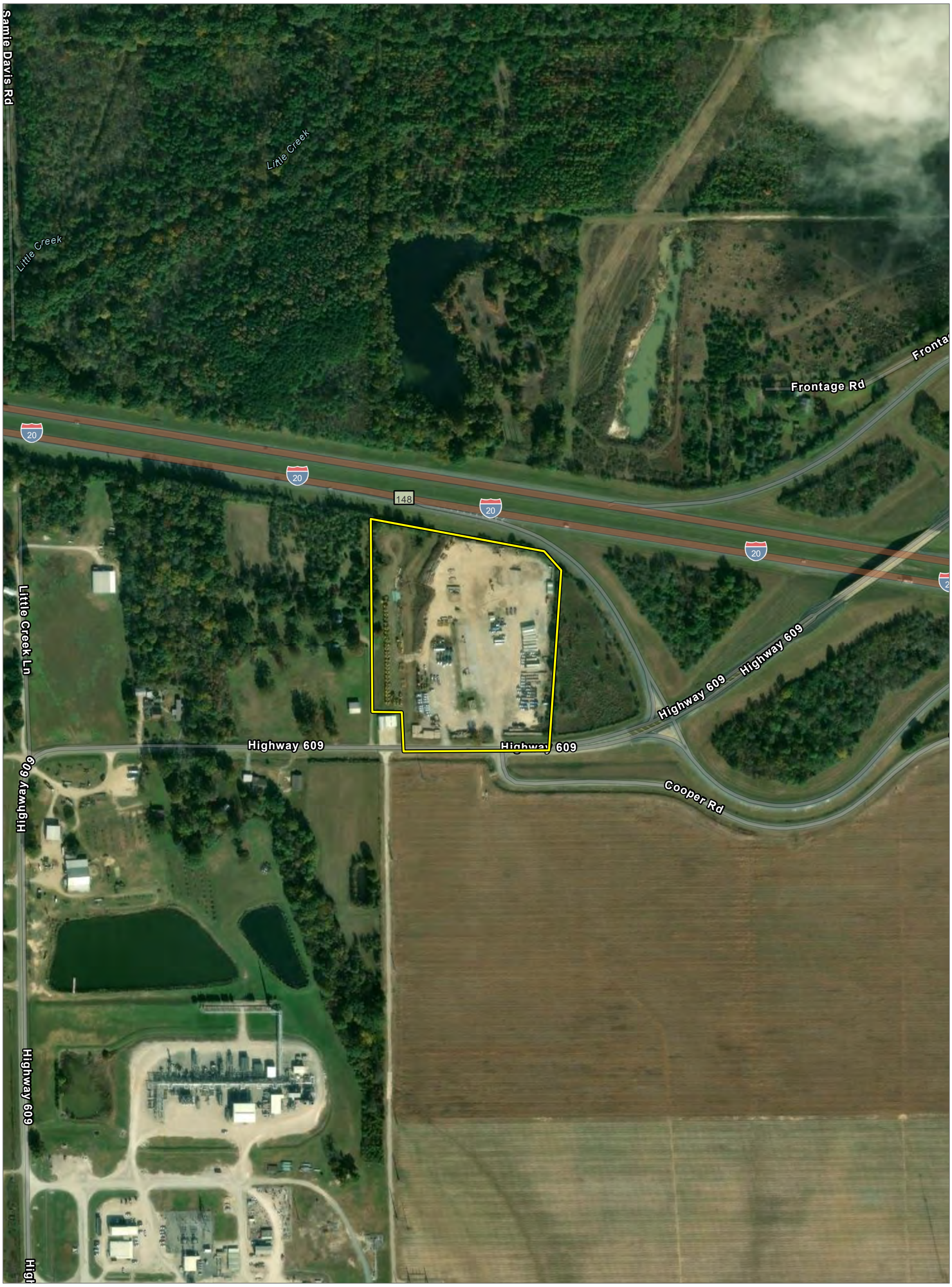
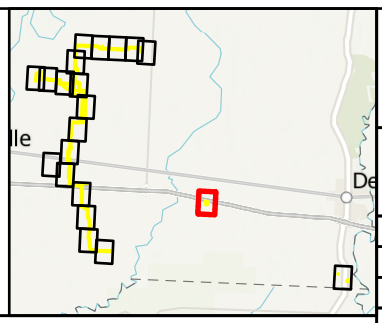


PROJECT BOUNDARY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 8
PAGE 2 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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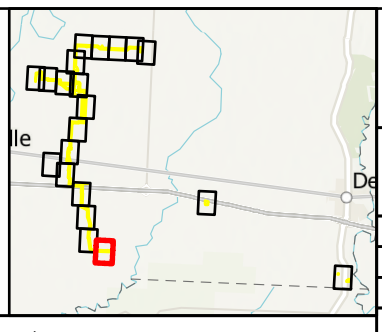
PROJECT BOUNDARY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE	**FIGURE 8**	
APPROVED BY:	A. THOMPSON	**PAGE 3 OF 21**	
DATE:	MAY 2026		

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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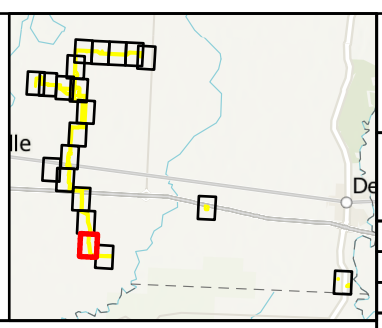
PROJECT BOUNDARY

OBSERVATION



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET

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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 8
PAGE 4 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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Mengle Rd
Mengle Rd
Mengle Rd
Mengle Rd
Mengle Rd
Pony Greer Rd
Pony Greer Rd
O-T3F-01

PROJECT BOUNDARY

OBSERVATION

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 8
PAGE 5 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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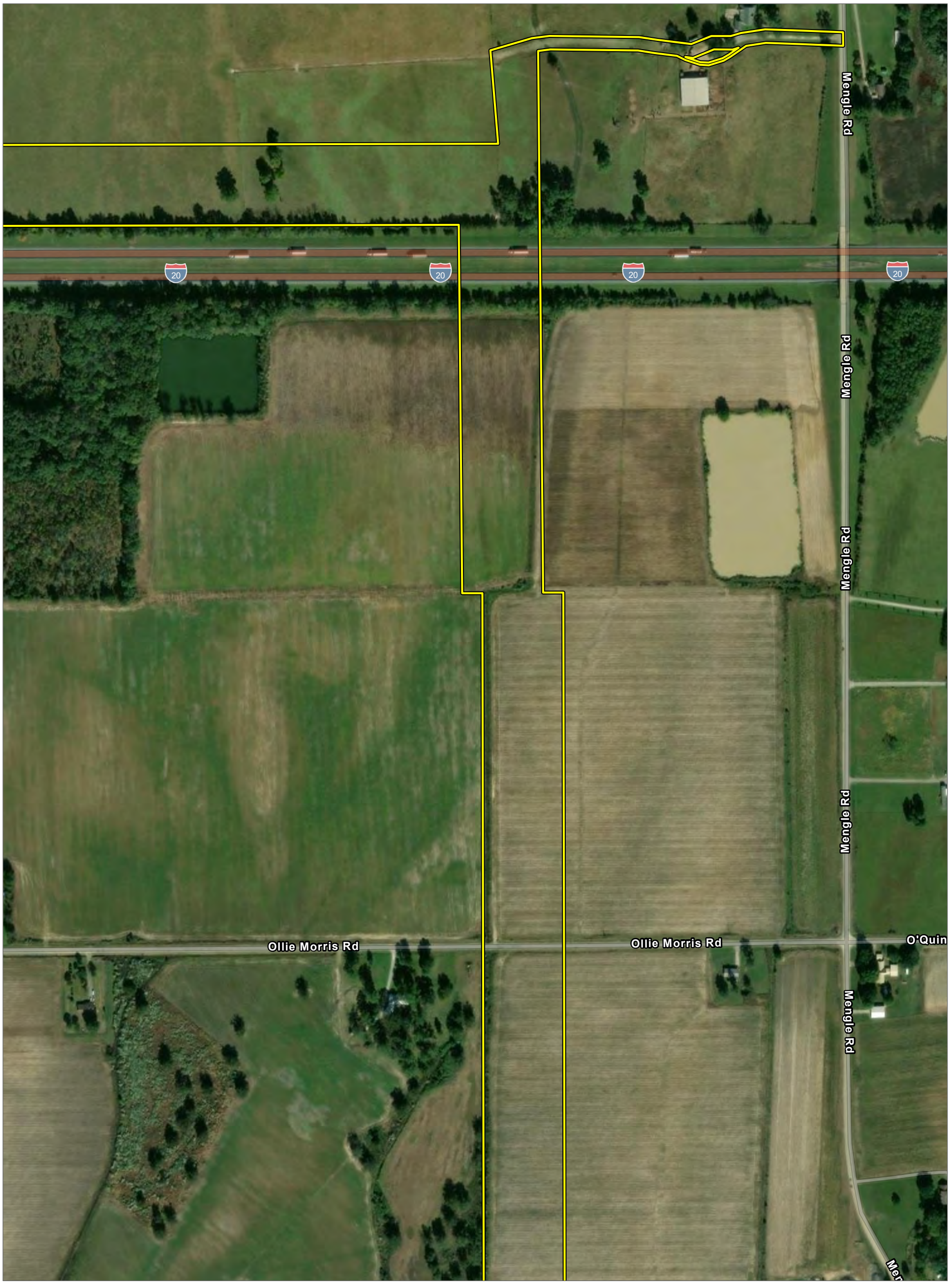
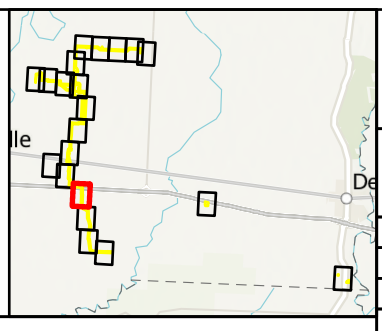


PROJECT BOUNDARY



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0 175 350 FEET
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 8
PAGE 6 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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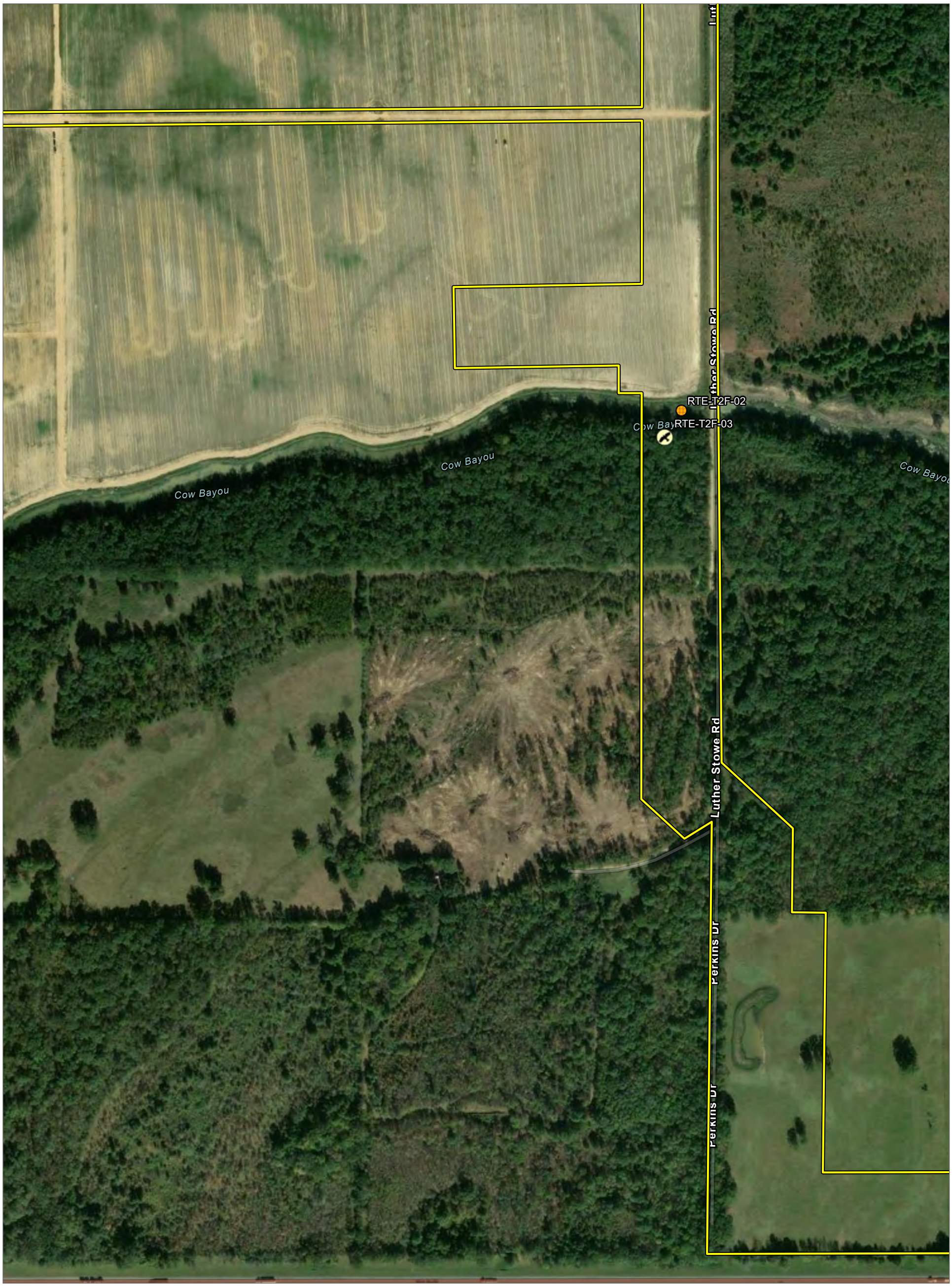
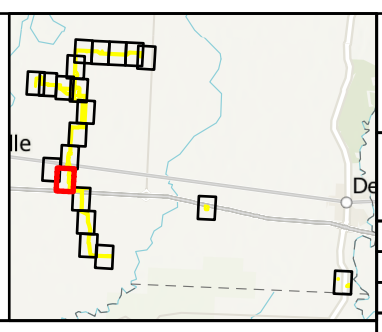


PROJECT BOUNDARY

RTE SPECIES

ALLIGATOR SNAPPING TURTLE HABITAT



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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE	**FIGURE 8**	
APPROVED BY:	A. THOMPSON	**PAGE 7 OF 21**	
DATE:	MAY 2026		

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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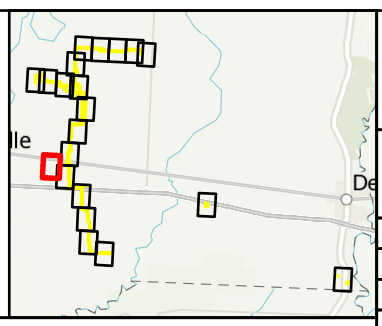
PROJECT BOUNDARY



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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

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DATE:	MAY 2026		

FIGURE 8
PAGE 8 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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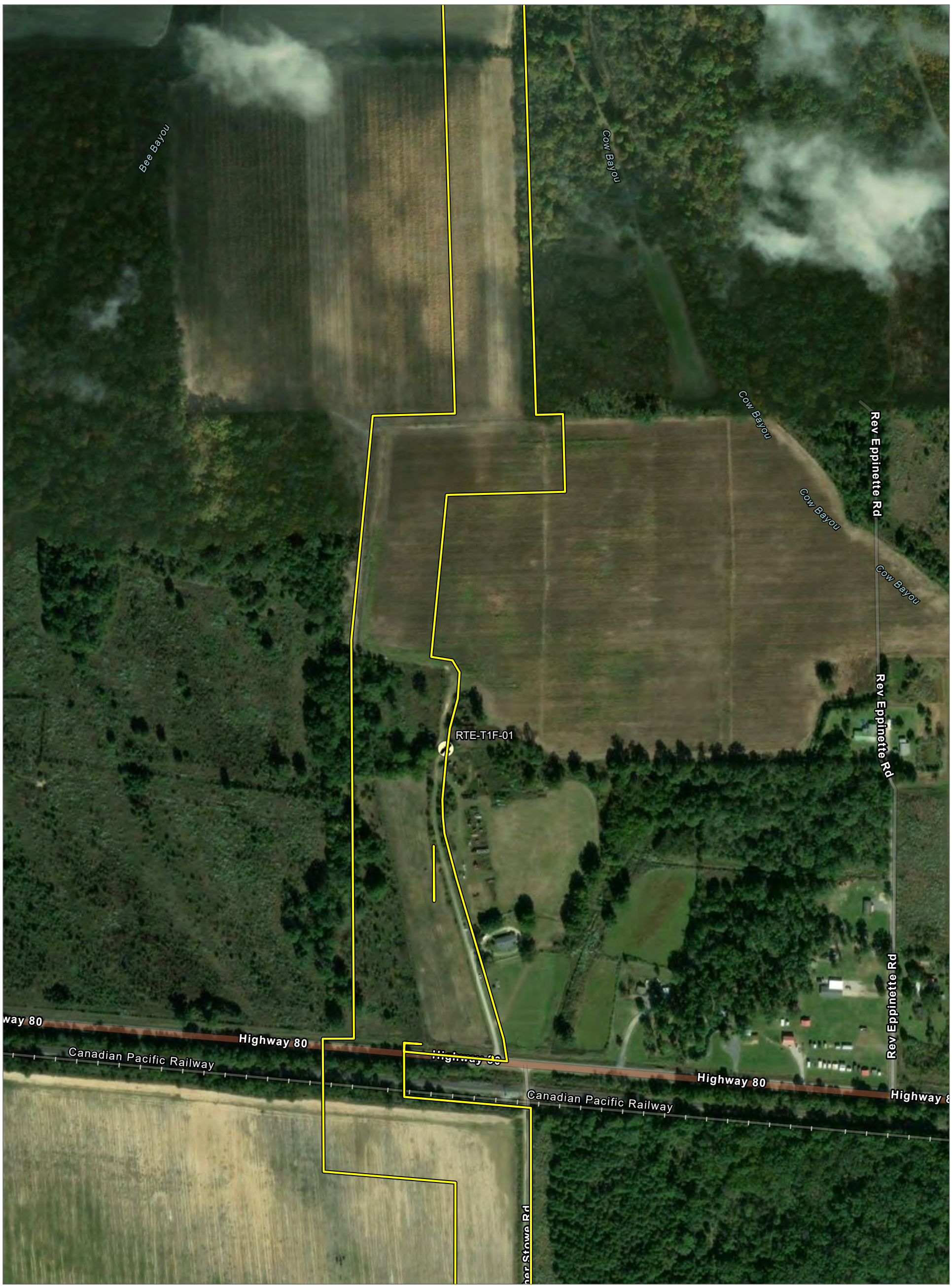

PROJECT BOUNDARY

RTE SPECIES

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
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DATA SOURCES: TRC

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
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RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

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FIGURE 8
PAGE 9 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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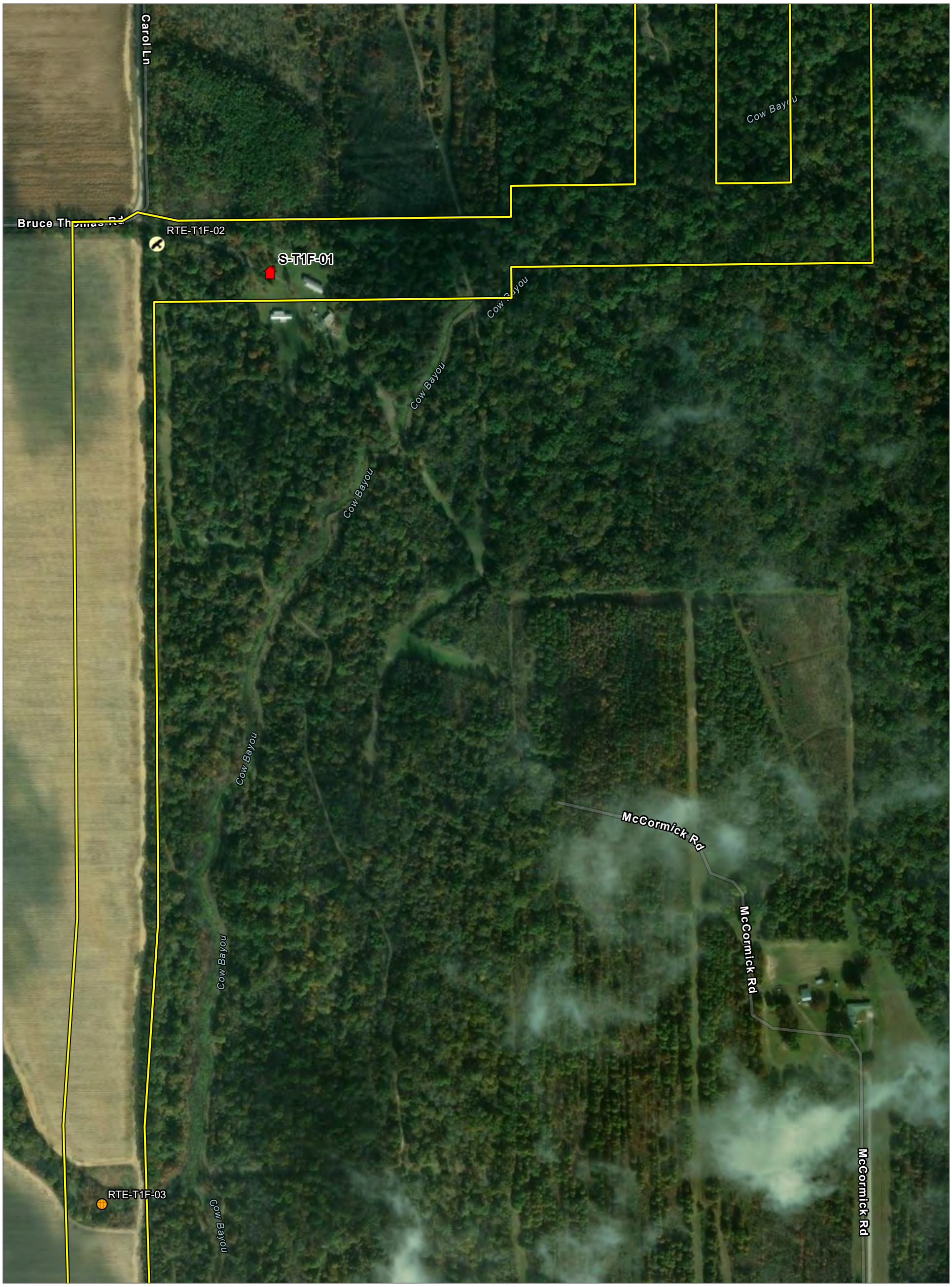

PROJECT BOUNDARY
STRUCTURE OR BUILDING
RTE SPECIES
ALLIGATOR SNAPPING TURTLE HABITAT

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
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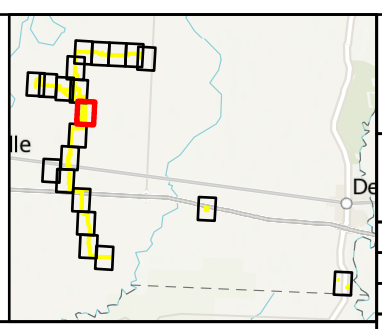

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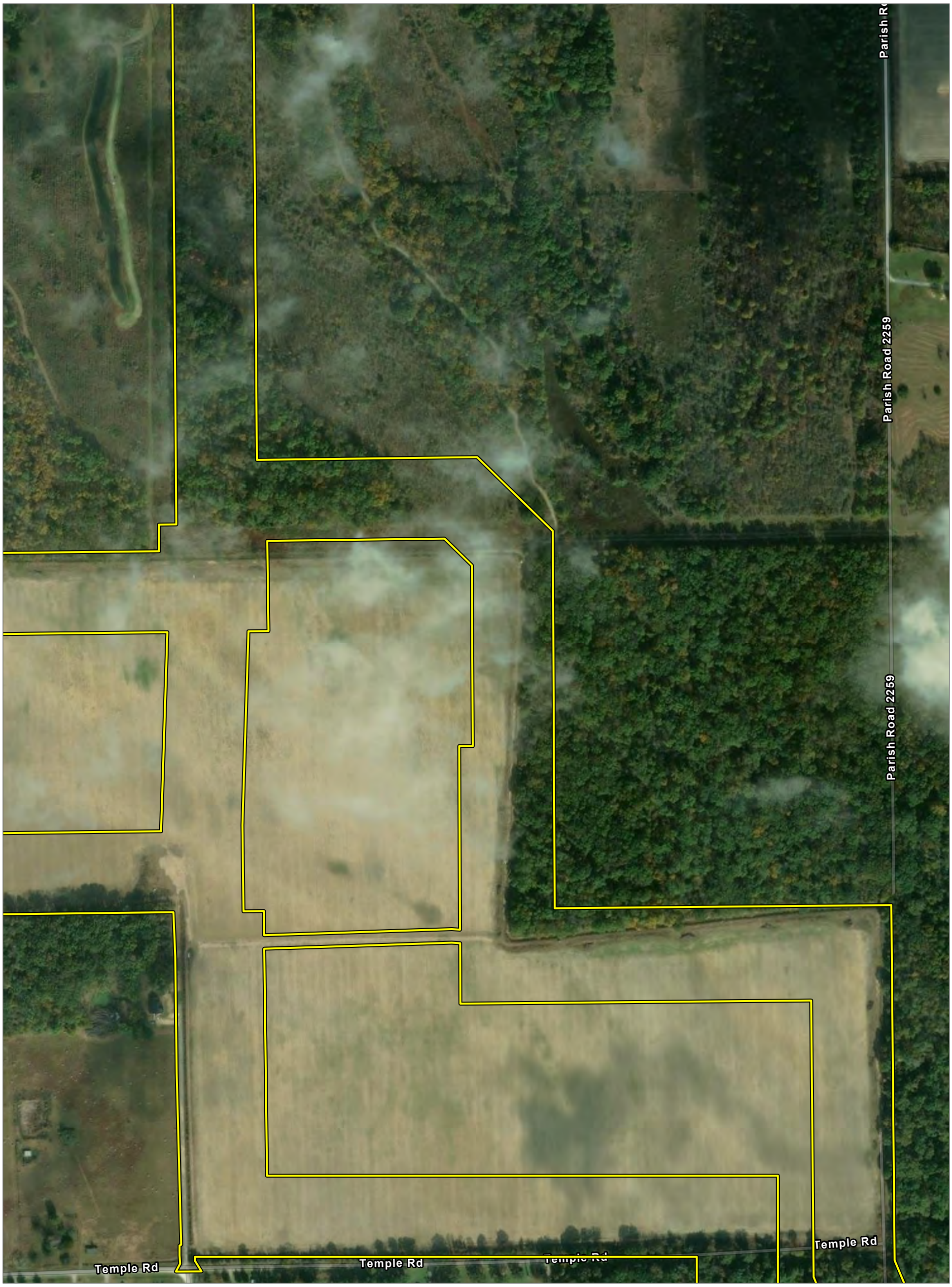

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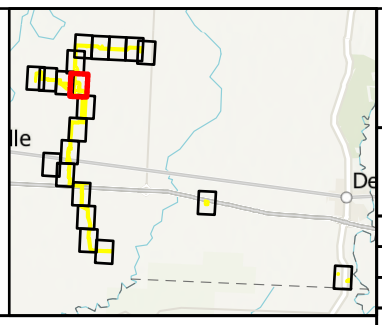

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FIGURE 8
PAGE 12 OF 21

TRC
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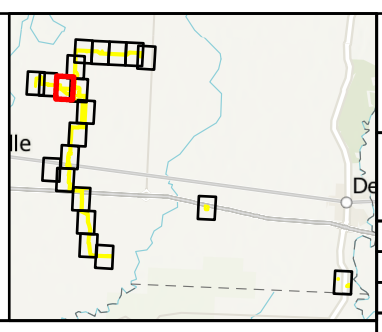

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RICHLAND PARISH, LA

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CHECKED BY:	A. CLINE	**FIGURE 8**	
APPROVED BY:	A. THOMPSON	**PAGE 13 OF 21**	
DATE:	MAY 2026		

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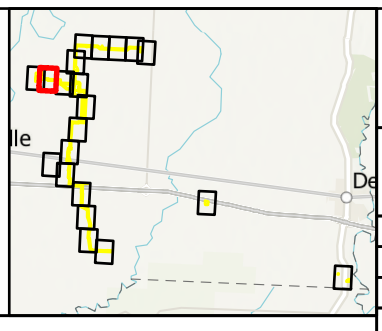

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FIGURE 8
PAGE 14 OF 21

TRC
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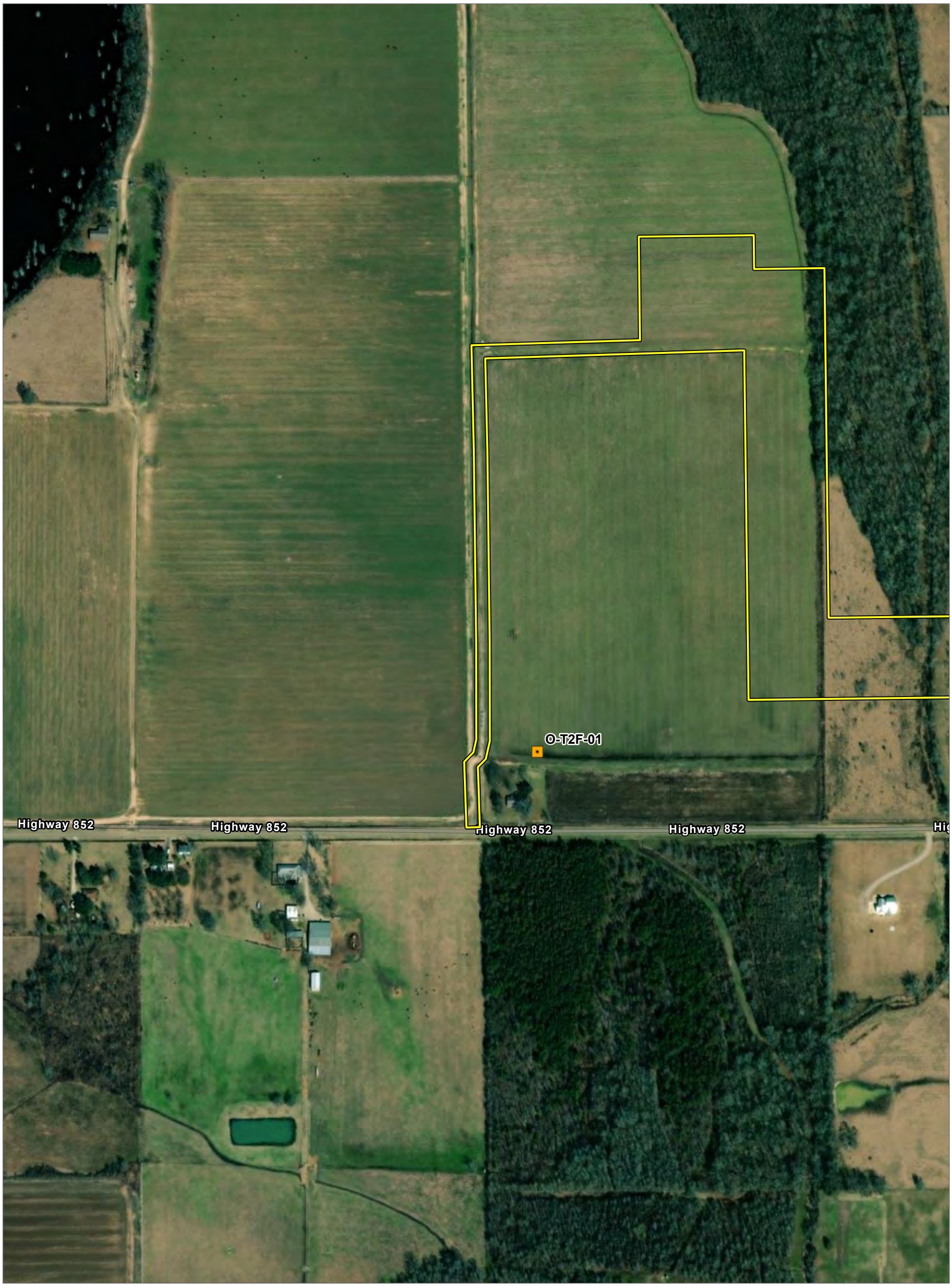
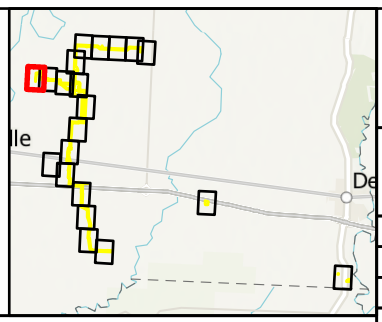


PROJECT BOUNDARY

OBSERVATION



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ETC TIGER PIPELINE, LLC
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FIGURE 8
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TRC
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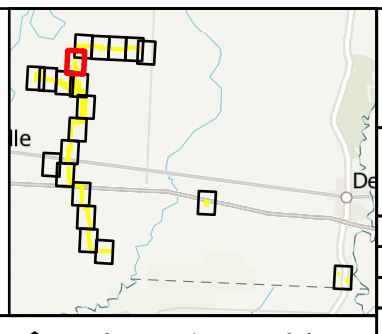

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FIGURE 8
PAGE 16 OF 21

TRC
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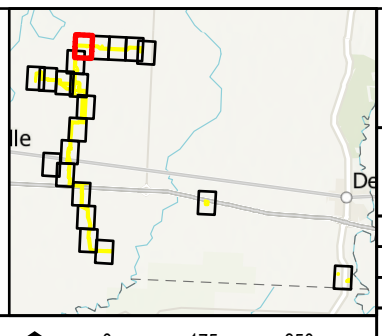

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FIGURE 8
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TRC
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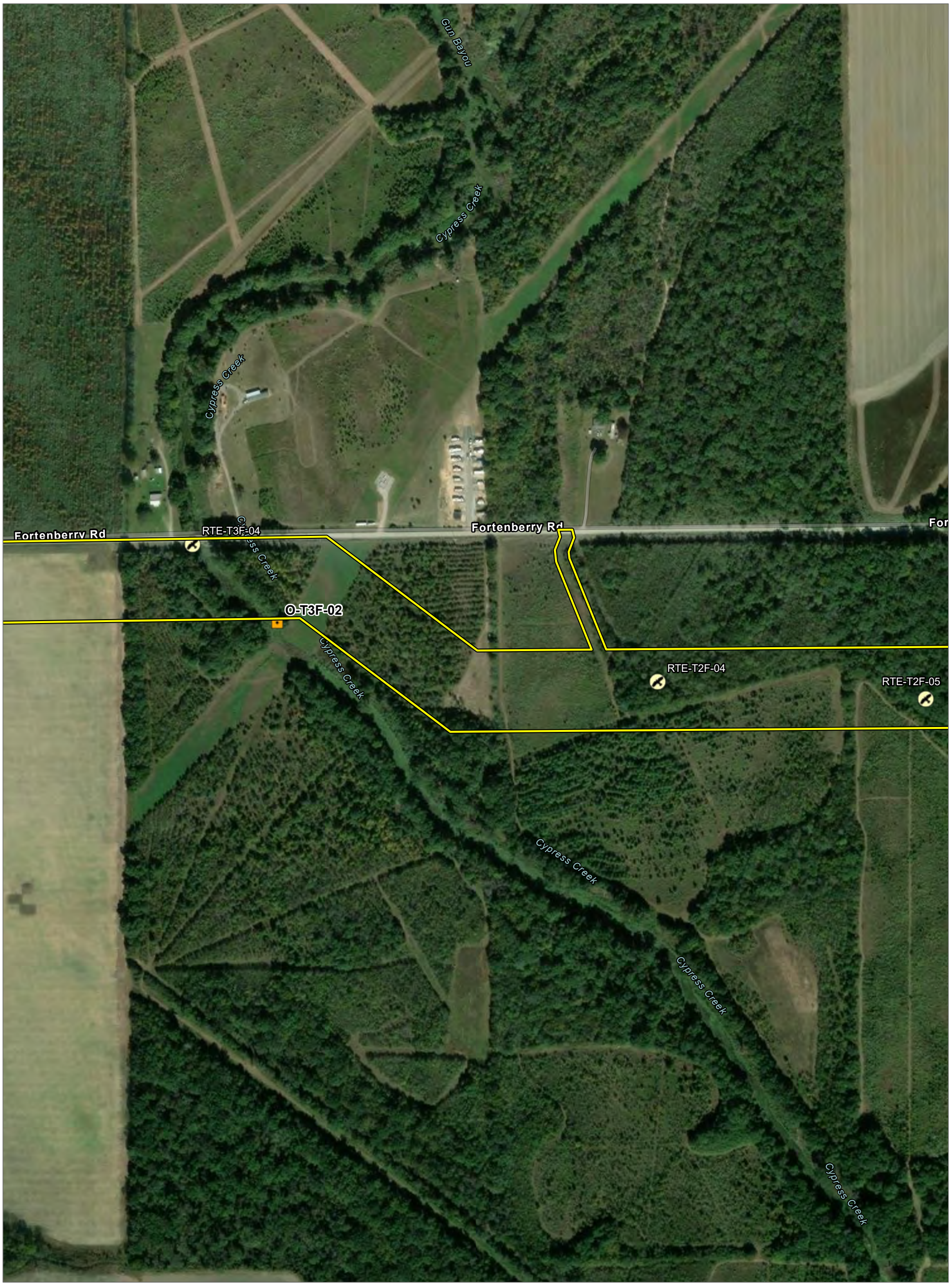

PROJECT BOUNDARY
OBSERVATION
RTE SPECIES

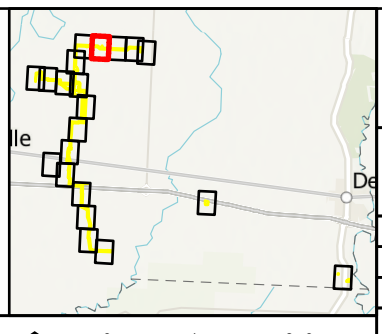

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FIGURE 8
PAGE 18 OF 21

TRC
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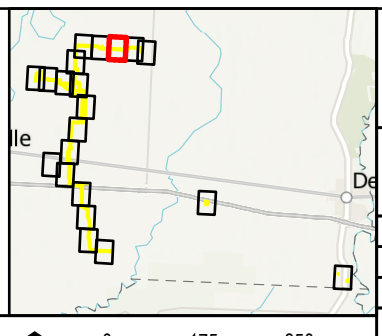

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FIGURE 8
PAGE 19 OF 21

TRC
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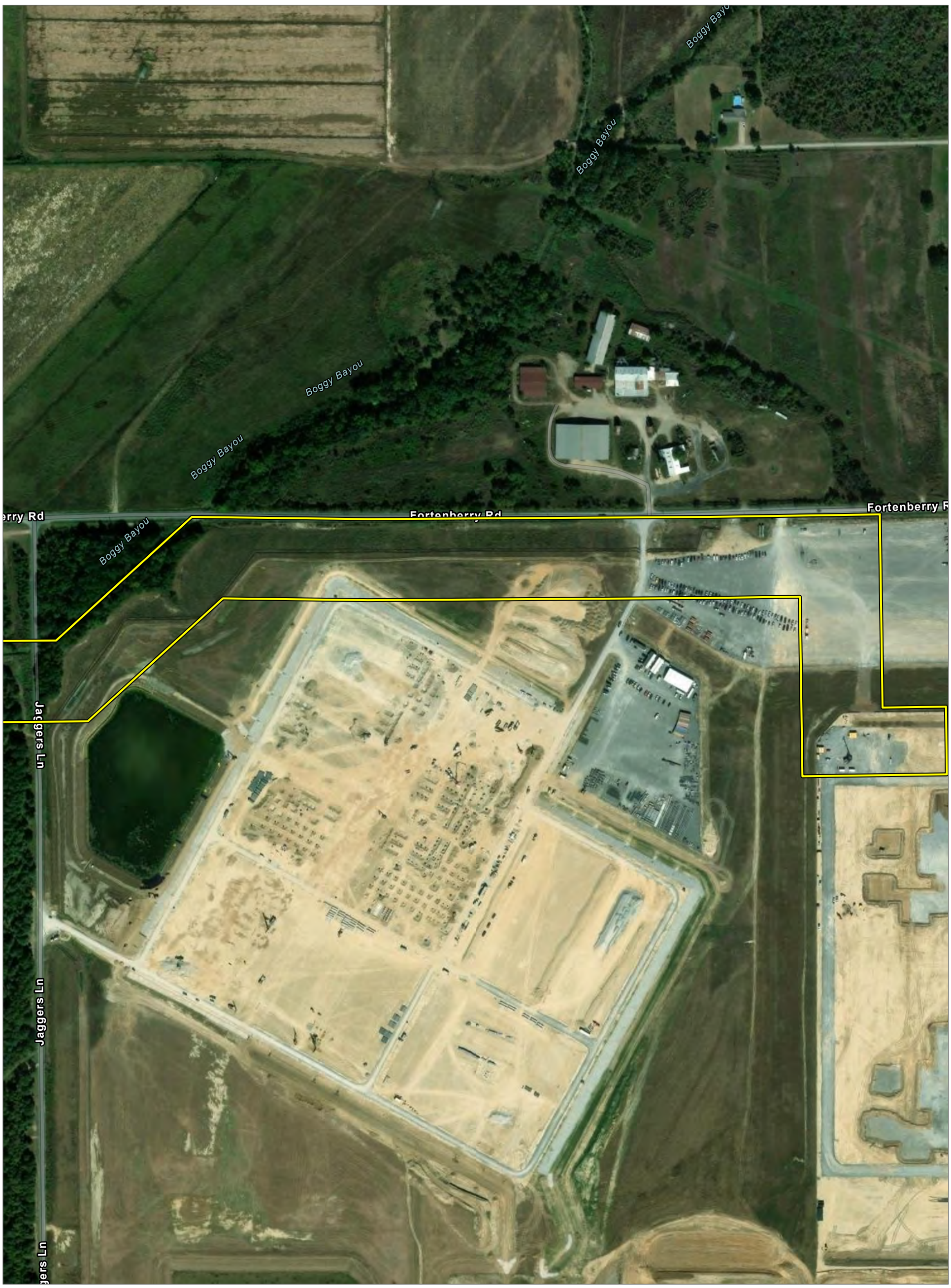

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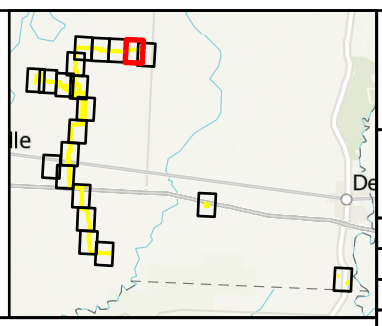

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FIGURE 8
PAGE 20 OF 21

TRC
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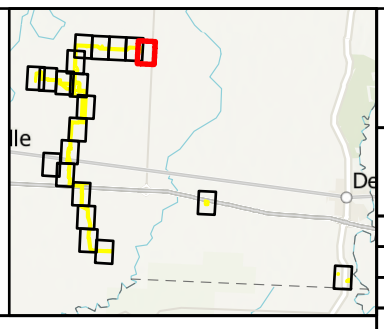

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FIGURE 8
PAGE 21 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
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Appendix B

Wetland Determination Datasheets, Stream Data Forms, and Representative Photographs

(Filed under separate cover as Critical Energy Infrastructure Information)

Appendix C
Antecedent Precipitation Tool Results

(Filed under separate cover as Critical Energy Infrastructure Information)

Appendix D
Observation Point Photograph Log

(Filed under separate cover as Critical Energy Infrastructure Information)

APPENDIX F-2

Listed Species Report

(Portions filed in Critical Energy Infrastructure Information)

Threatened and Endangered Species Habitat Assessment

Franklin Farms Project
Richland Parish, Louisiana

TRC Project No.: 000705831.0000.0000

May 2026

Prepared For:
ETC Tiger Pipeline, LLC
1300 Main Street
Houston, TX 77002

Prepared By:
TRC Environmental Corporation
11767 Katy Freeway, Suite 850
Houston, TX 77079

Prepared by: Morgan Thigpen

Reviewed and Approved by: Ashley Thompson

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APPENDICES

Appendix A: Figures
Appendix B: U.S. Fish and Wildlife Service Official Species List and State Species List
Appendix C: Photolog

ACRONYMS AND ABBREVIATIONS

Notation	Definition
BGEPA	Bald and Golden Eagle Protection Act
ESA	Endangered Species Act
FF	Franklin Farms
IPaC	Information for Planning and Consultation
MBTA	Migratory Bird Treaty Act
MRLC	Multi-Resolution Land Characteristics Consortium
NLCD	National Land Cover Database
NRCS	Natural Resources Conservation Service
Project	Franklin Farms 7(c)
T&E	Threatened and endangered
TRC	TRC Companies, Inc.
USGS	U.S. Geological Survey
USFWS	U.S. Fish and Wildlife Service

1.0 Introduction

ETC Tiger Pipeline, LLC contracted TRC Environmental Corporation (TRC) to support their environmental due diligence review of the proposed Franklin Farms Project (Project) located in Richland Parish, Louisiana (Project). The Project includes the following components:

- Franklin Farms (FF) Lateral - approximately 13.2-mile 36” pipeline lateral
- FF Trio Farm Lateral - approximately 1.7-mile 20” pipeline lateral
- FF Gulf Run Bi Di Meter Station
- FF Gulf South Receipt Meter Station
- FF Bee Bayou Delivery Meter Station
- FF Interconnect Site
- FF Gulf Crossing Receipt Meter Station
- FF MEP Receipt Meter Station
- Franklin Farms Delivery Meter Station
- FF Trio Farm Delivery Meter Station
- MEP Tap Site

This report summarizes the results of a desktop data review and on-site habitat assessment conducted for the Project. The on-site habitat assessment identified potentially suitable habitats for federal and state protected species protected under the Endangered Species Act (ESA), Migratory Bird Treaty Act (MBTA), and the Bald and Golden Eagle Protection Act (BGEPA). This report evaluates potential Project impacts on federally threatened and endangered (T&E) species and state listed species.

2.0 Methods

2.1 Project Location

The Project encompasses approximately 691 acres and is located in Richland Parish, Louisiana approximately four to five miles east of Rayville, Louisiana (Project Area). A map illustrating the general vicinity of the Project Area is shown in **Figure 1** of **Appendix A**. Maps illustrating aerial imagery and topography of the Project Area are shown in **Figures 2 and 3** of **Appendix A**. The Project Area primarily consists of the forested areas, agricultural land, and agricultural ditches.

2.2 Desktop Data Review

Prior to conducting on-site fieldwork, TRC conducted a desktop-level review of readily and publicly available site information including, but not limited to, the following:

- Recent and historic aerial imagery (Google Earth Pro, 2026);
- United States Geological Survey (USGS) 7.5-minute quadrangle topographic maps – Whitney Island South, LA; Bee Bayou, LA; Dunn, LA; Delhi, LA (USGS, 2026);

- USGS National Land Cover Database ([NLCD]; Multi-Resolution Land Characteristics Consortium [MRLC], 2023);
- Natural Resources Conservation Service (NRCS) Web Soil Survey (NRCS, 2026);
- U.S. Fish and Wildlife Service (USFWS) Information for Planning and Consultation (IPaC) project planning tool (USFWS, 2026a);
- USFWS Critical Habitat for T&E Species online viewer (USFWS, 2026b); and
- Louisiana Department of Wildlife and Fisheries (LDWF): Rare Species and Natural Communities by Parish (Louisiana Department of Wildlife and Fisheries, 2026).

Publicly available literature on species ecology and habitat preferences was reviewed to assess the potential for federally listed T&E species to occur within or near the Project Area. Additionally, the USFWS Information Planning and Consultation (IPaC) tool was used to generate an Official Species List for the Project Area (USFWS, 2026a; **Appendix B**).

2.3 Field Investigation

TRC Biologists conducted a field survey for aquatic and biological resources February 23 through May 8, 2026. During the field survey, pedestrian surveys were performed throughout the Project Area to document and assess the suitability of habitats for federal and state listed species. Biologists focused on any unique habitat areas identified during pre-fieldwork desktop analysis.

Observed habitats and species-specific habitat features (e.g., potential bat roosts, human structure habitats, or obligate host plants) were photographed and documented. Habitat data included general habitat descriptions, dominant vegetation, and notes on species-specific habitat features. When possible, observations of any federal or state protected species were photographed and documented. All data points were recorded using a Global Positioning System with submeter accuracy. The results of the field survey are summarized in **Section 3.0**.

3.0 Results

Results of the desktop analysis and the on-site habitat assessment are summarized below. Representative photographs of the Project Area presented in **Appendix C**.

3.1 General Characteristics of Habitat Resources

3.1.1 Ecoregion

The Project Area is located within the Macon Ridge (Level IV ecoregion 73j) ecoregion of the Mississippi Alluvial Plain ecoregion (Level III ecoregion 73; Daigle et al., 2006). The Macon Ridge ecoregion is described as having wide, flat to irregular alluvial terrace with relict patterns of branching channels, irregular braided bars, and interfluves. Low-gradient, channelized streams and canals with silty substrates. The Macon Ridge is characterized mainly by forested wetlands, deciduous forest, cropland with sugarcane, cotton, soybeans and corn, pasture, crawfish aquaculture, and some urban and industrial developments. Wetness and flooding are

the main limitations to farming. Naturally the vegetation within this area included bottomland hardwoods and hardwood flatwoods of willow oak, water oak, Nuttall oak, swamp chestnut oak, sweetgum and hickory. With some upland hardwood forests of white oak, southern red oak, and on drier sites some post oak. In wet areas vegetation included cypress-gum swamps (bald cypress, water tupelo).

Review of the NLCD (MRLC, 2023) found that the Project Area contains ten mapped land cover classifications summarized in **Table 1**. A map illustrating the land cover classifications within the Project Area is provided in **Figure 4** of **Appendix A**.

Table 1. Land Cover Classifications Mapped within the Project Area

Land Cover Classification[1]	Acres within Project Area	Percentage within Project Area
Cultivated Crops	455.42	65.92%
Developed, High Intensity	0.24	0.03%
Developed, Low Intensity	26.70	3.87%
Developed, Medium Intensity	3.39	0.49%
Developed, Open Space	41.69	6.03%
Evergreen Forest	1.30	0.19%
Mixed Forest	3.70	0.54%
Pasture/Hay	11.00	1.59%
Shrub/Scrub	0.22	0.03%
Woody Wetlands	147.17	21.30%
Total	**690.82**	**100.00%**

[1] Land cover classifications from NLCD (MRLC, 2023).

During the field survey, similar land cover types were observed that are consistent with the NLCD land cover classifications described in **Table 1**. Dominant land cover observed during the field survey include woody wetlands, cultivated crops, and developed, open space.

2.2.1 Plant Communities

Plant species diversity and vegetation density in the Project Area is consistent with similar habitats of the surrounding region. Plant species observed within the Project Area during the February/March/May 2026 field survey along with its wetland indicator status (U.S. Army Corps of Engineers [USACE], 2023) and non-native/invasive status (LDWF, 2019) are presented in **Table 2.**

Table 2. Plant Species Observed within the Project Area

Common Name	Scientific Name	Wetland Indicator Status[1]	Non-native/Invasive Status[2]
Tree Stratum			
Red Maple	*Acer rubrum*	FAC	No
Pignut Hickory	*Carya glabra*	FACU	No
Sugarberry	*Celtis laevigata*	FACW	No

Common Name	Scientific Name	Wetland Indicator Status[1]	Non-native/Invasive Status[2]
Common Persimmon	*Diospyros virginiana*	FAC	No
Honey-Locust	*Gleditsia triacanthos*	FAC	No
Eastern Red-Cedar	*Juniperus virginiana*	FACU	No
Sweetgum	*Liquidambar styraciflua*	FAC	No
Osage-orange	*Maclura pomifera*	FACU	No
Water Tupelo	*Nyssa aquatica*	OBL	No
Swamp Tupelo	*Nyssa biflora*	OBL	No
Eastern Hop-Hornbeam	*Ostrya virginiana*	FACU	No
Loblolly Pine	*Pinus taeda*	FAC	No
Northern White Oak	*Quercus alba*	FACU	No
Southern Red Oak	*Quercus falcata*	FACU	No
Overcup Oak	*Quercus lyrata*	OBL	No
Blackjack oak	*Quercus marilandica*	UPL	No
Water Oak	*Quercus nigra*	FAC	No
Cherrybark Oak	*Quercus pagoda*	FAC	No
Willow Oak	*Quercus phellos*	FACW	No
Post Oak	*Quercus stellata*	UPL	No
Southern Dewberry	*Rubus trivialis*	FACU	No
Bagpod	*Sesbania vesicaria*	FAC	No
Winged Elm	*Ulmus alata*	FACU	No
American Elm	*Ulmus americana*	FAC	No
Cedar Elm	*Ulmus crassifolia*	FAC	No
Sapling/Shrub Stratum			
Red Maple	*Acer rubrum*	FAC	No
Groundseltree	*Baccharis halimifolia*	FAC	No
Sugarberry	*Celtis laevigata*	FACW	No
Common Persimmon	*Diospyros virginiana*	FAC	No
Honey-Locust	*Gleditsia triacanthos*	FAC	No
Yaupon	*Ilex vomitoria*	FAC	No
Eastern Red-Cedar	*Juniperus virginiana*	FACU	No
Chinese Privet	*Ligustrum sinense*	FAC	Yes
Sweetgum	*Liquidambar styraciflua*	FAC	No
Osage-Orange	*Maclura pomifera*	FACU	No
Water Tupelo	*Nyssa aquatica*	OBL	No
Northern White Oak	*Quercus alba*	FACU	No
Water Oak	*Quercus nigra*	FAC	No
Cherrybark Oak	*Quercus pagoda*	FAC	No
Black Oak	*Quercus velutina*	NI	No
Saw-Tooth Blackberry	*Rubus argutus*	FAC	No
Dwarf Palmetto	*Sabal minor*	FACW	No

Common Name	Scientific Name	Wetland Indicator Status[1]	Non-native/Invasive Status[2]
Chinese Tallow	*Triadica sebifera*	FAC	Yes
Winged Elm	*Ulmus alata*	FACU	No
American Elm	*Ulmus americana*	FAC	No
Cedar Elm	*Ulmus crassifolia*	FAC	No
Herbaceous Stratum (Groundcover)			
Alligatorweed	*Alternanthera philoxeroides*	OBL	Yes
Broom-Sedge	*Andropogon virginicus*	FAC	No
Narrow-Leaf Carpet Grass	*Axonopus fissifolius*	FACW	No
Shepherd's-Purse	*Capsella bursa-pastoris*	FACU	No
Hairy Bittercress	*Cardamine hirsuta*	FACU	No
Sand Bittercress	*Cardamine parviflora*	FACU	No
Eastern Woodland Sedge	*Carex blanda*	FAC	No
Cherokee Sedge	*Carex cherokeensis*	FACW	No
Cypress-swamp Sedge	*Carex joorii*	OBL	No
Louisiana Sedge	*Carex louisianica*	OBL	No
Shallow Sedge	*Carex lurida*	OBL	No
Hairy-Fruit Chervil	*Chaerophyllum tainturieri*	FAC	No
Indian Wood-Oats	*Chasmanthium latifolium*	FAC	No
Slender Wood-Oats	*Chasmanthium laxum*	FACW	No
Long-Leaf Wood-Oats	*Chasmanthium sessiliflorum*	FAC	No
Bermuda Grass	*Cynodon dactylon*	FACU	Yes
Tall Flat Sedge	*Cyperus eragrostis*	FACW	No
Rusty Flat Sedge	*Cyperus odoratus*	FACW	No
Marsh Flat Sedge	*Cyperus pseudovegetus*	FACW	No
Straw-Color Flat Sedge	*Cyperus strigosus*	FACW	No
Silky Wild Oat Grass	*Danthonia sericea*	FACU	No
Heller's Rosette Grass	*Dichanthelium oligosanthes*	FACU	No
Poorjoe	*Diodia teres*	FACU	No
Virginia Buttonweed	*Diodia virginiana*	FACW	No
Blunt Spike-Rush	*Eleocharis obtusa*	OBL	No
Eastern Daisy Fleabane	*Erigeron annuus*	FACU	No
Asthmaweed	*Erigeron bonariensis*	FACU	No
Dogfennel	*Eupatorium capillifolium*	FACU	No
Evening Trumpet-Flower	*Gelsemium sempervirens*	FAC	No
Blue Mud-Plantain	*Heteranthera limosa*	OBL	No
St. Andrew's-Cross	*Hypericum hypericoides*	FAC	No
Lamp Rush	*Juncus effusus*	OBL	No
Torrey's Rush	*Juncus torreyi*	FACW	No
Eastern Red-Cedar	*Juniperus virginiana*	FACU	No
Henbit Deadnettle	*Lamium amplexicaule*	UPL	No

Common Name	Scientific Name	Wetland Indicator Status[1]	Non-native/Invasive Status[2]
Common Duckweed	*Lemna minor*	OBL	No
Poorman's-Pepperwort	*Lepidium virginicum*	UPL	No
Chinese Privet	*Ligustrum sinense*	FAC	Yes
Japanese Honeysuckle	*Lonicera japonica*	FACU	Yes
seedbox	*Ludwigia alternifolia*	OBL	No
Marsh Primrose-Willow	*Ludwigia palustris*	OBL	No
Creeping Primrose-Willow	*Ludwigia repens*	OBL	No
Trailing Yellow-Loosestrife	*Lysimachia radicans*	OBL	No
Wart-Removing-Herb	*Murdannia keisak*	OBL	Yes
Tiny Mousetail	*Myosurus minimus*	FACW	No
Wand Panic Grass	*Panicum virgatum*	FAC	No
Golden Crown Grass	*Paspalum dilatatum*	FAC	Yes
Bahia Grass	*Paspalum notatum*	FACU	Yes
Vasey's Grass	*Paspalum urvillei*	FAC	Yes
Swamp Smartweed	*Persicaria hydropiperoides*	OBL	No
Savannah-Panic Grass	*Phanopyrum gymnocarpon*	OBL	No
English Plantain	*Plantago lanceolata*	FACU	No
Annual Blue Grass	Poa *annua*	FACU	No
Spiny-Fruit Buttercup	*Ranunculus muricatus*	FACW	No
Hairy Buttercup	*Ranunculus sardous*	FAC	No
saw-Tooth Blackberry	*Rubus argutus*	FAC	No
curly Dock	*Rumex crispus*	FAC	Yes
Dwarf Palmetto	*Sabal minor*	FACW	No
Giant Plume Grass	*Saccharum giganteum*	FACW	No
Canadian Black-Snakeroot	*Sanicula canadensis*	FACU	No
Sawbrier	*Smilax glauca*	FAC	No
Horsebrier	*Smilax rotundifolia*	FAC	No
Spiny-Leaf Sow-Thistle	*Sonchus asper*	FACU	No
Bluejacket	*Tradescantia ohiensis*	FAC	No
Virginia Spiderwort	*Tradescantia virginiana*	FAC	No
Long-Spike Fluff Grass	*Tridens strictus*	FACW	No
White Clover	*Trifolium repens*	FACU	No
Purple-Top Vervain	*Verbena bonariensis*	FAC	No
Brazilian vervain	*Verbena brasiliensis*	FACU	Yes
Brazilian Vervain	*Verbena incompta*	FACU	No
White Vervain	*Verbena urticifolia*	FAC	No
Yellow Crownbeard	*Verbesina occidentalis*	FACU	No
Neckweed	*Veronica peregrina*	FAC	No
Louisiana Vetch	*Vicia ludoviciana*	FACU	No
Woody Vine Stratum			

Common Name	Scientific Name	Wetland Indicator Status[1]	Non-native/Invasive Status[2]
Japanese Honeysuckle	*Lonicera japonica*	FACU	Yes
Saw-Tooth Blackberry	*Rubus argutus*	FAC	No
Southern Dewberry	*Rubus trivialis*	FACU	No
Horsebrier	*Smilax rotundifolia*	FAC	No
Muscadine	*Vitis rotundifolia*	FAC	Yes

[1] UPL = Upland; FACU = Facultative Upland; FAC = Facultative; FACW = Facultative Wetland; OBL = Obligate. Wetland indicator status from USACE National Wetland Plant List (USACE, 2023).

[2] Non-native/invasive status from LDWF Wildlife Action Plan Plants Database (LDWF, 2019).

3.1.2 Soils

Soils within the Project Area were identified using the NRCS Web Soil Survey database (NRCS, 2026). Review of the soil survey data found that a total of 17 soil map units are mapped within the Project Area. **Figure 5** in **Appendix A** illustrates the soils, and **Table 3** provides a summary and abbreviated description of the soils mapped within the Project Area.

Table 3. Soils within the Project Area

Map Unit Name (Symbol ID)	Acreage of Project Area (percentage)	Hydric Rating (Yes/No)[1]	Description[1]
Calhoun-Calloway silt loams, gently undulating (Cc)	6.31 (0.91%)	Yes	All areas are considered prime farmland. Parent material consists of loess or loess-like material. Minor components are non-hydric.
Deerford silt loam (Da)	4.30 (0.62%)	Yes	Not prime farmland. Parent material consists of loess. Minor components are a mix of hydric and non-hydric.
Dexter silt loam, 1 to 3 percent slopes (De)	13.23 (1.92%)	Yes	All areas are considered prime farmland. Parent material consists of thin loess and underlying loamy and sandy sediments on terraces. Minor components are a mix of hydric and non-hydric.
Dexter silt loam, 3 to 5 percent slopes (Df)	3.68 (0.53%)	Yes	All areas are considered prime farmland. Parent material consists of thin loess and underlying loamy and sandy sediments on terraces. Minor components are a mix of hydric and non-hydric.
Egypt silt loam, 1 to 3 percent slopes (Eg)	4.17 (0.60%)	Yes	All areas are considered prime farmland. Parent material consists of mixed loess and terrace sediments. Minor components are a mix of hydric and non-hydric.
Forestdale silty clay loam, 0 to 1 percent slopes, rarely flooded, upland phase (Fr)	224.53 (32.50%)	Yes	All areas are considered prime farmland. Parent material consists of loess. Minor components are a mix of hydric and non-hydric.
Gigger silt loam, 1 to 3 percent slopes (Ge)	6.35 (0.92%)	Yes	All areas are considered prime farmland. Parent material consists of thin loess over terrace sediments. Minor components are a mix of hydric and non-hydric.
Gigger-Gilbert silt loams, gently undulating (Gg)	69.26 (10.03%)	Yes	All areas are considered prime farmland. Parent material consists of thin loess and mixed loess over terrace sediments. Minor components are a mix of hydric and non-hydric.

Map Unit Name (Symbol ID)	Acreage of Project Area (percentage)	Hydric Rating (Yes/No)[1]	Description[1]
Gilbert silt loam (Gk)	69.31 (10.03%)	Yes	All areas are considered prime farmland. Parent material consists of mixed loess over terrace sediments. Minor components are a mix of hydric and non-hydric.
Gilbert-Egypt silt loams, gently undulating (Gm)	166.51 (24.10%)	Yes	All areas are considered prime farmland. Parent material consists of mixed loess and terrace sediments. Minor components are a mix of hydric and non-hydric.
Grenada silt loam, 1 to 3 percent slopes (Gr)	3.35 (0.49%)	Yes	All areas are considered prime farmland. Parent material consists of loess. Minor components are a mix of hydric and non-hydric.
Hebert silt loam, 0 to 1 percent slopes (Hb)	1.28 (0.19%)	Yes	All areas are considered prime farmland. Parent material consists of Holocene silty alluvium. Minor components are a mix of hydric and non-hydric.
Necessity silt loam, 1 to 3 percent slopes (Ne)	13.19 (1.91%)	Yes	All areas are considered prime farmland. Parent material consists of loamy sediments. Minor components are a mix of hydric and non-hydric.
Necessity-Gilbert silt loams, gently undulating (Ng)	83.48 (12.08%)	Yes	All areas are considered prime farmland. Parent material consists of loamy sediments and mixed loess over terrace sediments. Minor components are a mix of hydric and non-hydric.
Perry silty clay loam (Pc)	8.17 (1.18%)	Yes	All areas are considered prime farmland. Parent material consists of clayey alluvium. Minor components are a mix of hydric and non-hydric.
Perry clay (Pd)	7.03 (1.02%)	Yes	All areas are considered prime farmland. Parent material consists of clayey alluvium. Minor components are a mix of hydric and non-hydric.
Rilla silt loam, 0 to 1 percent slopes (Ra)	6.65 (0.96%)	Yes	All areas are considered prime farmland. Parent material consists of silty alluvium. Minor components are a mix of hydric and non-hydric.

[1] Hydric soil rating and soil descriptions from the NRCS Web Soil Survey (NRCS, 2026).

3.2 Threatened, Endangered, and Other Protected Species

The ESA protects plants and animals that are listed as endangered or threatened by the USFWS. Section 9 of the ESA prohibits the taking of endangered wildlife, where “take” is defined as the harassment, harm, pursuit, hunting, shooting, wounding, killing, trapping, capture, or collection, as well as modification or degradation of habitat that results in death or injury of these species (50 Code of Federal Regulations 17.3).

Under Section 7 of the ESA, federal agencies are required to consult with the USFWS if their actions, including permit approvals or funding, could adversely affect a listed (or proposed) species (including plants) or critical habitat. Under Section 10 of the ESA, individuals or project sponsors/owners of projects without a federal nexus (e.g., federal funding, approvals, permits) may apply for an Incidental Take Permit from the USFWS for projects or actions that may adversely affect critical habitat or result in the incidental taking of federally listed T&E species.

The Louisiana Endangered Species Conservation Act, which is enforced by the LDWF,

protects state-listed species by prohibiting the take, possession, or transport of listed endangered or threatened species without authorization.

3.2.1 State and Federally Listed Threatened and Endangered Species

Federal
The USFWS Official Species List for the Project Area identified three federally listed threatened, endangered, or proposed-for-listing species as having the potential to occur within the Project Area and/or be impacted by the Project (**Appendix B**). The USFWS Critical Habitat for T&E Species online viewer (USFWS, 2026b) and the IPaC indicated that no critical habitat occurs within the Project Area. A summary of federally listed T&E species with potential to occur within or near the Project Area and/or be affected by the Project is provided in **Table 4**.

State
The LDWF identified two species with a defined state status have the potential to occur within the Project Area. A summary of state-listed T&E species with potential to occur within or near the Project Area and/or be affected by the Project is provided in **Table 4**.

Table 4. Federally Listed Threatened and Endangered Species with Potential to Occur within the Project Area

Species[1] (*Scientific Name*)	Status[2]	Preferred Habitats	Habitat Suitability and Preliminary Effect Determination
Mammals			
Tricolored bat (*Perimyotis subflavus)*	Federal – FPE State – Not listed	The tricolored bat primarily roost in dead or live tree foliage of deciduous hardwoods. Hibernacula for the species includes caves, mines, tunnels, culverts, or other manmade structures. Maternity colonies may use manmade structures and tree cavities in open sites (USFWS, 2026d).	**Not Likely to Jeopardize the Continued Existence** – The Project Area contains potentially suitable habitat for this species; however, clearing windows will be coordinated with the agencies to minimize impacts during the winter torpor (December 15 – February 15) and roosting season/pup season (March 15 – July 15). Proposed species do not receive statutory protection from "take" under the ESA; however, potential impacts to proposed species must be assessed for federal actions (e.g., funding, permits, etc.). Federal actions must not jeopardize the continued existence of proposed species. Given the scope of work for the Project and mitigation measures it is anticipated that the Project is *Not Likely to Jeopardize the Continued Existence* of the species.
Northern Long-eared Bat (*Myotis septentrionalis*)	Federal – FT (*not identified in IPaC*) State – ST	The northern long-eared bat is primarily associated with hardwood and mixed forested landscapes during the summer, where they forage near trees with a preference for low understory cover. This species typically overwinters in caves or mines (USFWS, 2026c).	**May Impact –** The Project Area contains potentially suitable habitat for this species; however, the USFWS official IPaC did not list this species as one to occur within the Project Area. Nonetheless, clearing windows will be consulted with agencies to minimize impacts during the winter torpor (December 15 – February 15) and roosting season/pup season (March 15 – July 15).
Reptiles			
Alligator snapping turtle (*Macrochelys temminckii*)	Federal – FPT State – SR	The alligator snapping turtle prefers to inhabit large rivers, streams, canals, lakes, and swamps. Favored features for these habitats include high canopy forest areas and structures like tree root masses, stumps, and submerged trees. Alligator snapping turtles have seasonal preferences of spending winters in shallow areas and summer in deeper areas. Hatchlings prefer shallow water and abundant canopy and vegetation (USFWS, 2021).	**Not Likely to Jeopardize the Continued Existence** – The Project Area contains limited suitable habitat for this species within Cow Bayou and a tributary between Cow Bayou and Bee Bayou. These streams will be crossed with a flume method that would require temporary dewatering of the main stream bed, though water would be routed through piping to not disturb downstream flow. BMPs would also be in place to prevent degradation of water quality in the immediate vicinity and downstream of the water crossings. Proposed species do not receive statutory protection from "take" under the ESA; however, potential impacts to proposed species must be assessed for federal actions (e.g., funding, permits, etc.). Federal actions must not jeopardize the continued existence of proposed species. The Project Area does contain suitable habitat that could support the preferred habitat of this species; however, to minimize impacts disturbance/alterations of nesting habitat will be minimized during nesting season (April-June). Additionally, all in-stream work would be temporary in nature and avoid permanent impacts to streams by restoring pre-construction contours. It is anticipated that the Project is *Not Likely to Jeopardize the Continued Existence* of the species.
Insects			
Monarch butterfly (*Danaus plexippus*)	Federal – FPT State – Not Listed	The monarch butterfly is a migratory species that utilizes a variety of habitat types. Except for a non-migratory population in Florida, monarch butterflies in eastern North America (i.e., east of the Rocky Mountains) travel north in the spring from overwintering areas in Mexico to summer breeding areas as far north as Canada, typically over two to three successive generations and breed along the way. In the fall, monarch butterflies migrate back to southern overwintering grounds. Depending upon environmental conditions, eastern North American monarch butterflies may require up to five generations to complete their annual migration cycle. During migration and breeding, adults depend on a diversity of blooming nectar resources (i.e., wildflowers) for feeding. Native milkweeds (primarily *Asclepias* spp.) are obligatory host plants for egg-laying and feeding during the larval life stage. In drier, more arid regions such as the southwestern U.S., migrating monarch butterflies tend to occur more frequently near water sources such as rivers, creeks, roadside ditches, and irrigated canals (USFWS, 2020).	**Not Likely to Jeopardize the Continued Existence –**While the Project Area primarily consists of cropland, developed areas, and woody wetlands; these areas could serve as potential habitat for the monarch butterfly. No milkweed species or monarch butterflies were observed during field surveys. Proposed species do not receive statutory protection from "take" under the ESA; however, potential impacts to proposed species must be assessed for federal actions (e.g., funding, permits, etc.). Federal actions must not jeopardize the continued existence of proposed species. Given the land types crossed, temporary nature of the pipeline construction, and placement of aboveground facilities in already disturbed areas; potential Project impacts/effects to monarch butterflies are anticipated to be minimal; therefore, the Project is *Not Likely to Jeopardize the Continued Existence* of the species.

[1] Species obtained from the USFWS Official Species List for the Project Area (**Appendix B**).
[2] FPE = Federally Proposed Endangered, FPT = Federally Proposed Threatened, FT = Federally Threatened, ST = State Threatened, SR = State Restricted

3.2.2 Bald and Golden Eagles

Bald and golden eagles are federally protected under the BGEPA. The BGEPA prohibits the taking or possession of, and commerce in, bald and golden eagles with very limited exceptions. Under the BGEPA, it is a violation to "take, possess, sell, purchase, barter, offer to sell, transport, export or import, at any time or any manner, any bald eagle commonly known as the American eagle, or golden eagle, alive or dead, or any part, nest, or egg, thereof."

Bald eagles are found year-round in Louisiana and are found primarily near rivers and large lakes where they forage and nest in tall trees or on cliffs near water (Cornell Lab of Ornithology, 2026a). Review of citizen science observation data found a single bald eagle observations within a 10-mile radius of the Project Area, occurring at the Poverty Point Reservoir State Park (eBird, 2026). Bald eagles could utilize Cow Bayou for foraging and could also utilize large trees surrounding the Project Area for nesting; however, no nests were documented during the field survey. Given that no bald eagle nests were observed during the field survey, it is anticipated that the Project would have *No Effect* on bald eagles; however, formal bald eagle nest surveys are recommended to be performed prior to construction given the suitable habitat surrounding the Project Area. Should a bald eagle nest be found within 660 feet of the Project Area, all work should be stopped immediately and coordination with the USFWS should be initiated.

Golden eagles scarcely occur in Louisiana as nonbreeding migrants (Cornell Lab of Ornithology, 2026b). Golden eagles live in open and semi-open country featuring native vegetation across most of the northern hemisphere. They avoid developed areas and uninterrupted stretches of forest and tend to nest on cliffs and steep escarpments in grassland, chapparal, shrubland, forest, and other vegetated areas (Cornell Lab of Ornithology, 2026b). Review of citizen science observation data found that there were no golden eagle occurrences within a 10-mile radius of the Project Area (eBird, 2026). No suitable golden eagle nesting habitat was observed within the Project Area, and the Project Area is outside of the species' currently known breeding range; therefore, it is anticipated that the Project would have *No Effect* on golden eagles.

3.2.3 Migratory Birds

The Project Area contains vegetation that may provide suitable nesting habitat for migratory bird species. The MBTA provides federal protection of migratory birds and prohibits take, killing, possession, and transportation of migratory birds, their eggs, and parts unless specifically permitted. While the MBTA is not currently interpreted to prohibit incidental take of migratory birds (see the April 11, 2025, U.S. Department of Interior memo M-37085), intentional take is still prohibited under the MBTA. The Project does not include the intentional take of migratory birds and is expected to comply with the regulations of the MBTA as they are interpreted within this report.

3.3 Field Observations

Multiple observation points were recorded during the February/March/May 2026 field survey to further document on-site conditions and to support the findings presented in this report.

Observation points are shown in **Figure 6** of **Appendix A** and photographs can be found in **Appendix C.**

4.0 Conclusions

TRC conducted a field survey between February 23 and May 8, 2026 to determine the presence of suitable habitat for federally listed T&E species and state listed species within the Project Area.

The Project Area contains potentially suitable habitat for the alligator snapping turtle, within limited aquatic resources found during field surveys. To minimize impacts disturbance/alterations of nesting habitat will be minimized during nesting season (April-June). Additionally, all in-stream work would be temporary in nature and avoid permanent impacts to streams by restoring pre-construction contours. It is anticipated that the Project is *Not Likely to Jeopardize the Continued Existence* of the alligator snapping turtle. If the scope of work changes, coordination with the USFWS and LDWF would be required for any in-stream activities, riverbank disturbances, or construction scheduled during the nesting season.

The Project Area also contains potentially suitable habitat for the tricolored bat and the northern long-eared bat. Coordination/consultation with the USFWS and/or LDWF will occur to establish clearing windows to minimize impacts to these species. Therefore, it is anticipated that the Project is *Not Likely to Jeopardize the Continued Existence* of the tricolored bat and *May Impact* on the northern long-eared bat.

The Project Area contains potentially suitable habitat for the monarch butterfly, as it primarily consists of cropland, developed areas, and woody wetlands. No milkweed species or monarch butterflies were observed during the field surveys. Therefore, it is anticipated that the Project is *Not Likely to Jeopardize the Continued Existence* of the monarch butterfly.

Bald eagles could utilize Cow Bayou for foraging and could also utilize large trees surrounding the Project Area for nesting; however, no nests were documented during the field survey. Given that no bald eagle nests were observed during the field survey, it is anticipated that the Project would have *No Effect* on bald eagles. TRC recommends formal bald eagle nest surveys to be performed prior to construction given the suitable habitat surrounding the Project Area. Should a bald eagle nest be found within 660 feet of the Project Area, all work should be stopped immediately and coordination with the USFWS should be initiated.

Given the lack of suitable habitat within and adjacent to the Project Area, the lack of documented golden eagle species observations in the vicinity of the Project Area, and that individuals are only expected to occur as non-breeding migrants within the Project Area, the Project is anticipated to have *No Effect* on golden eagles.

While the MBTA is not currently interpreted to prohibit incidental take of migratory birds (see the April 11, 2025, U.S. Department of Interior memo M-37085), intentional take is still prohibited under the MBTA. The Project does not include the intentional take of migratory birds and is expected to comply with the regulations of the MBTA as they are interpreted within this report.

5.0 References

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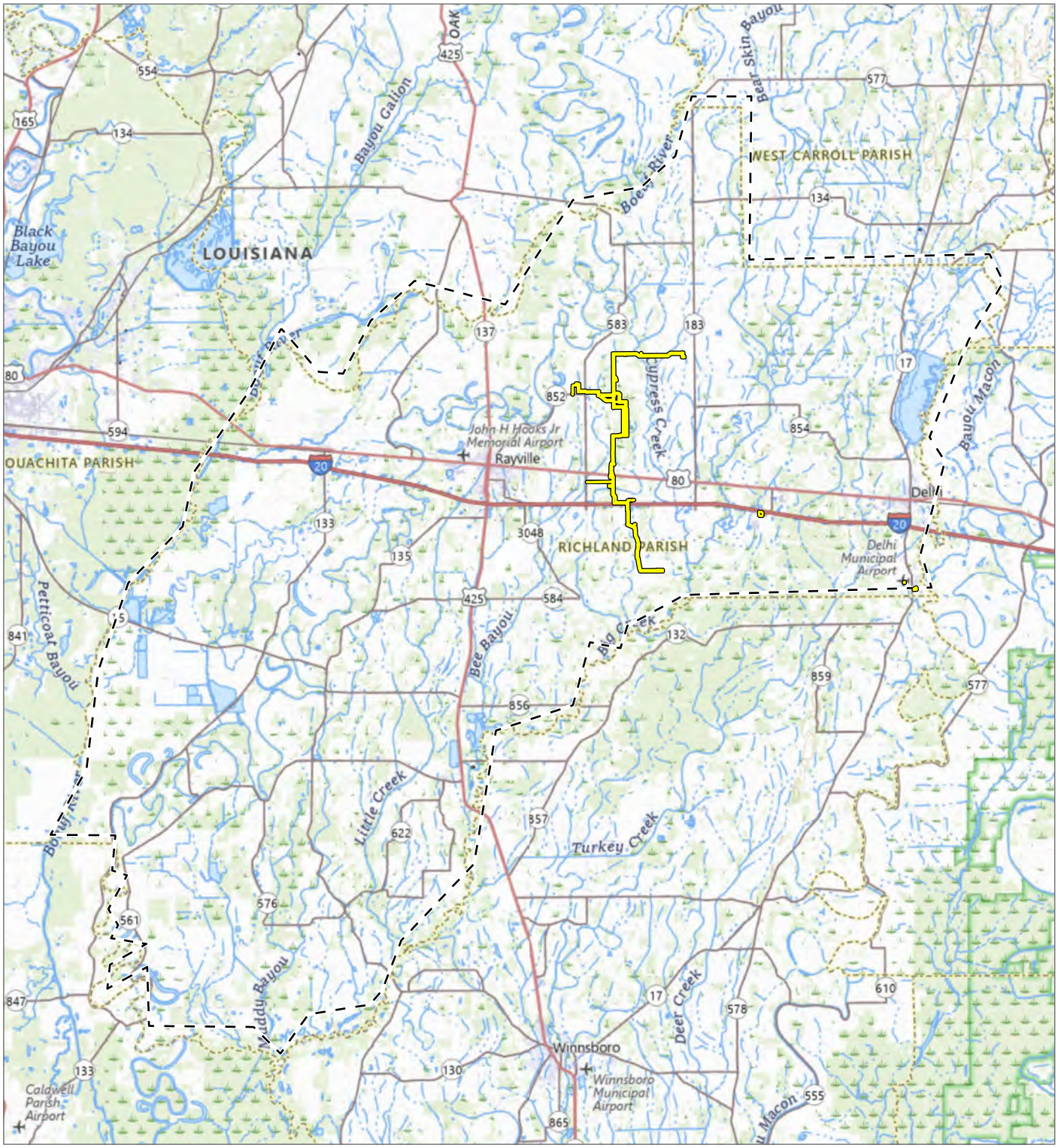
Appendix A
Figures



PROJECT BOUNDARY

RICHLAND PARISH

0 14,000 28,000

FEET

1:336,000 1" = 28,000'

BASE MAP: USGS "THE NATIONAL MAP" ONLINE SERVICE LAYER;
DATA SOURCES: TRC, USGS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

PROJECT VICINITY MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON	**FIGURE 1**	
DATE:	MAY 2026		

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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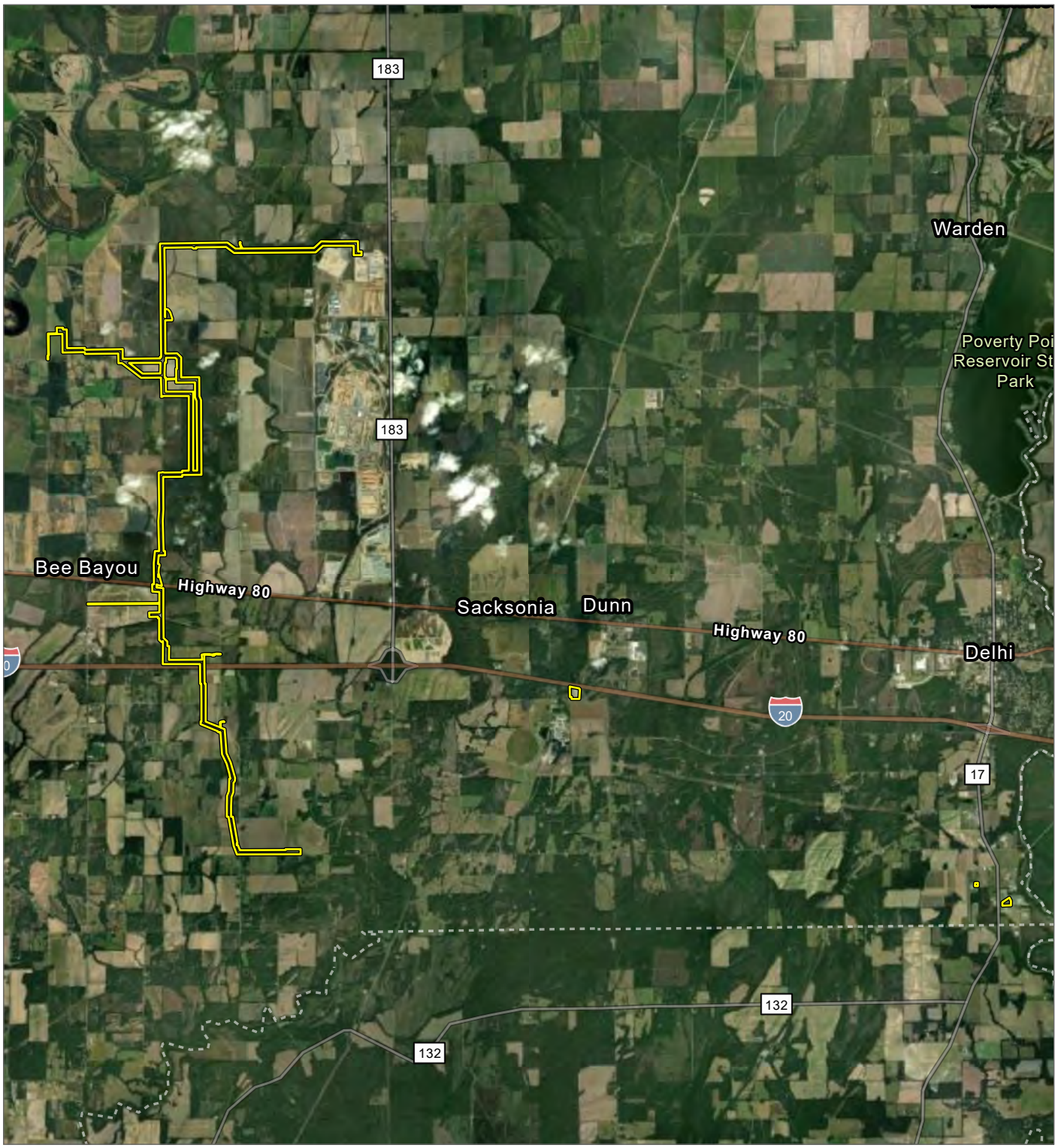

PROJECT BOUNDARY

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, ESRI

N

0 5,000 10,000

FEET
1:120,000 1" = 10,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

PROJECT LOCATION MAP (AERIAL)

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 2

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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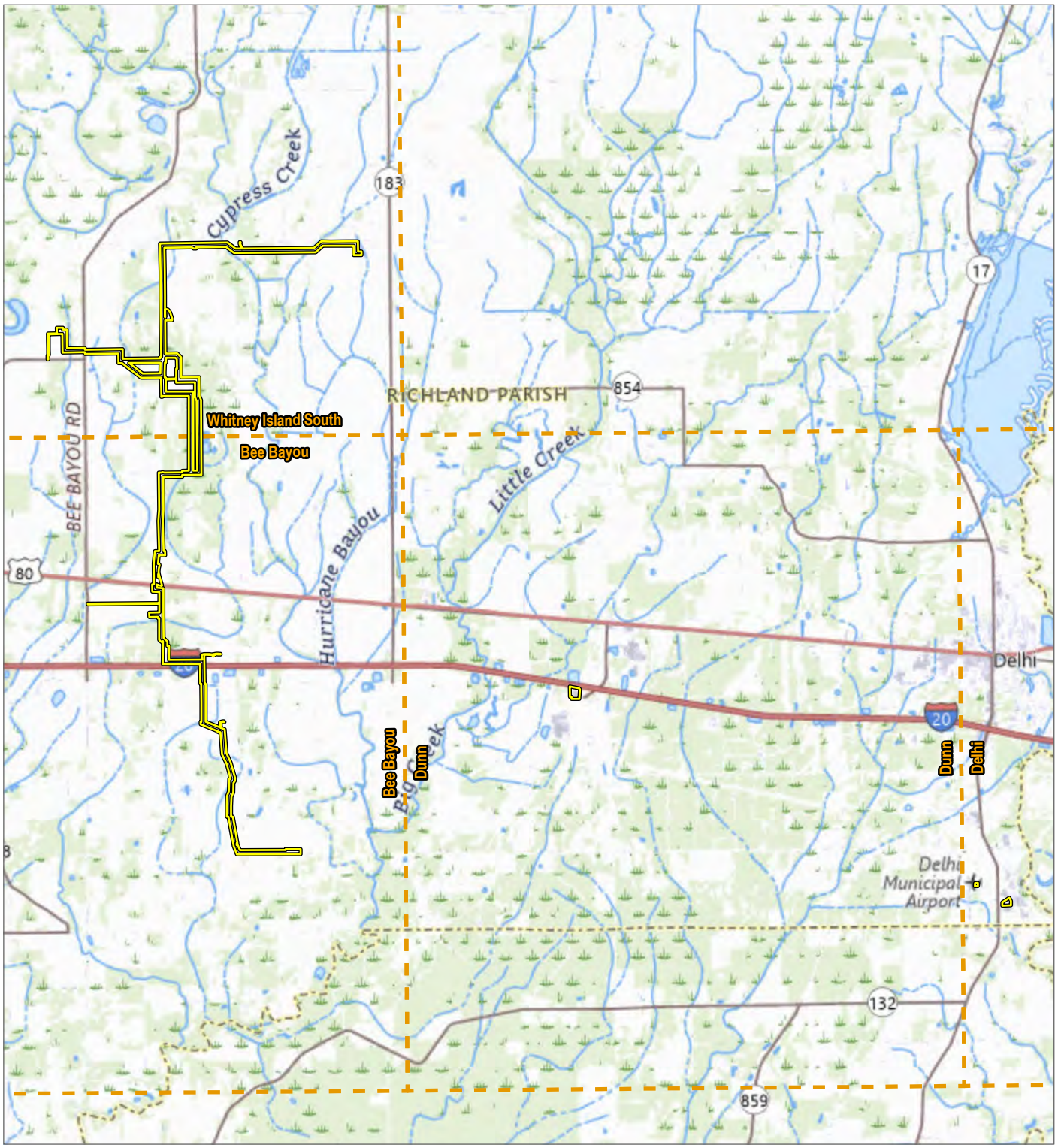

PROJECT BOUNDARY

USGS QUAD BOUNDARIES

0 5,000 10,000

FEET

1:120,000 1" = 10,000'

BASE MAP: USGS "THE NATIONAL MAP" ONLINE SERVICE LAYER; 7.5' QUADS: WHITNEY ISLAND SOUTH, LA (1986), BEE BAYOU, LA (1987), DUNN, LA (1987), DELHI, LA (1986)
DATA SOURCES: TRC, USGS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

PROJECT LOCATION MAP (TOPOGRAPHIC)

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		**FIGURE 3**
DATE:	MAY 2026		

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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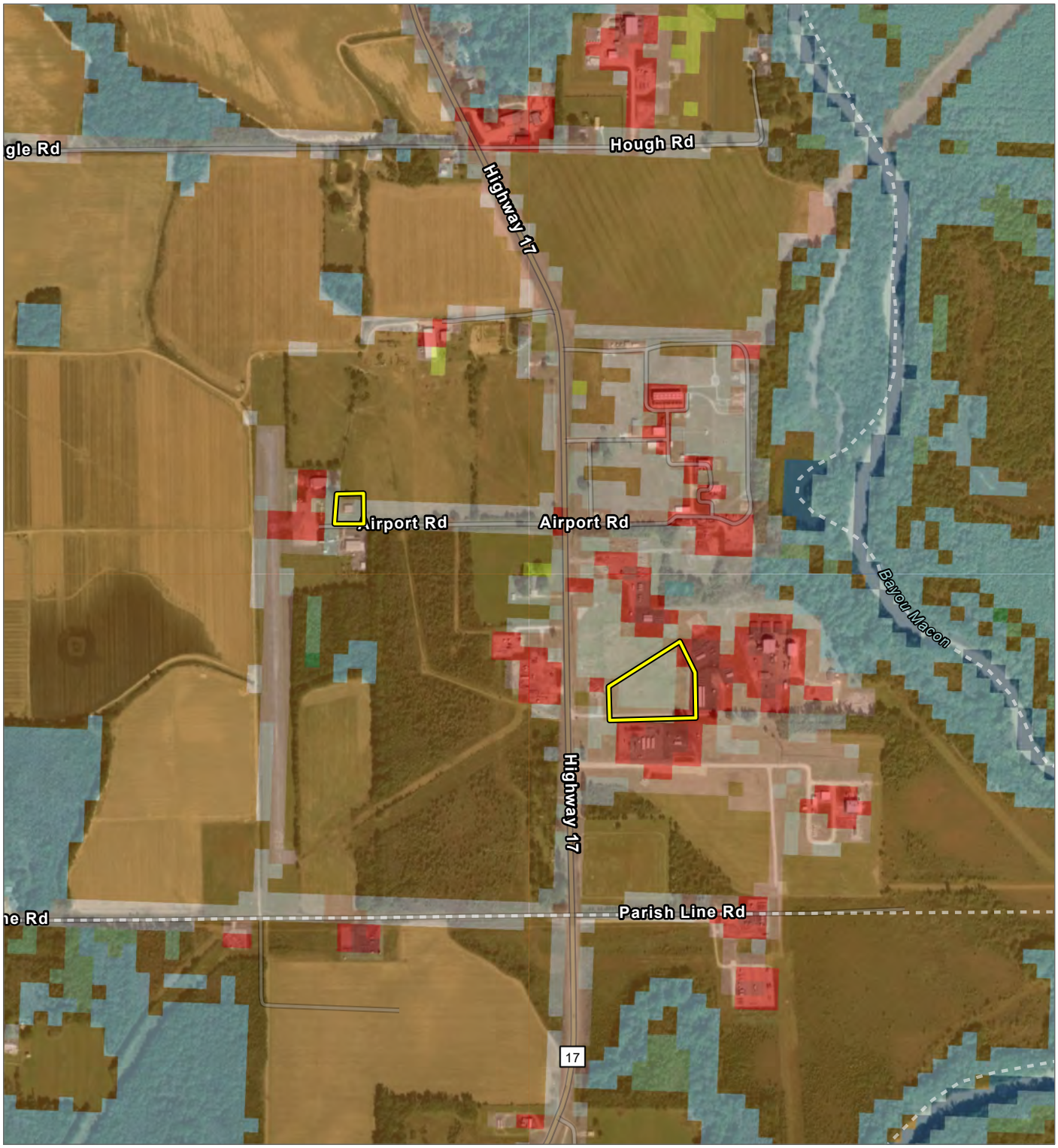

- PROJECT BOUNDARY
- OPEN WATER
- DEVELOPED, OPEN SPACE
- DEVELOPED, LOW INTENSITY
- DEVELOPED, MEDIUM INTENSITY
- DEVELOPED, HIGH INTENSITY
- DECIDUOUS FOREST
- MIXED FOREST
- PASTURE/HAY
- CULTIVATED CROPS
- WOODY WETLANDS
- EMERGENT HERBACEOUS WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000
FEET
1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 1 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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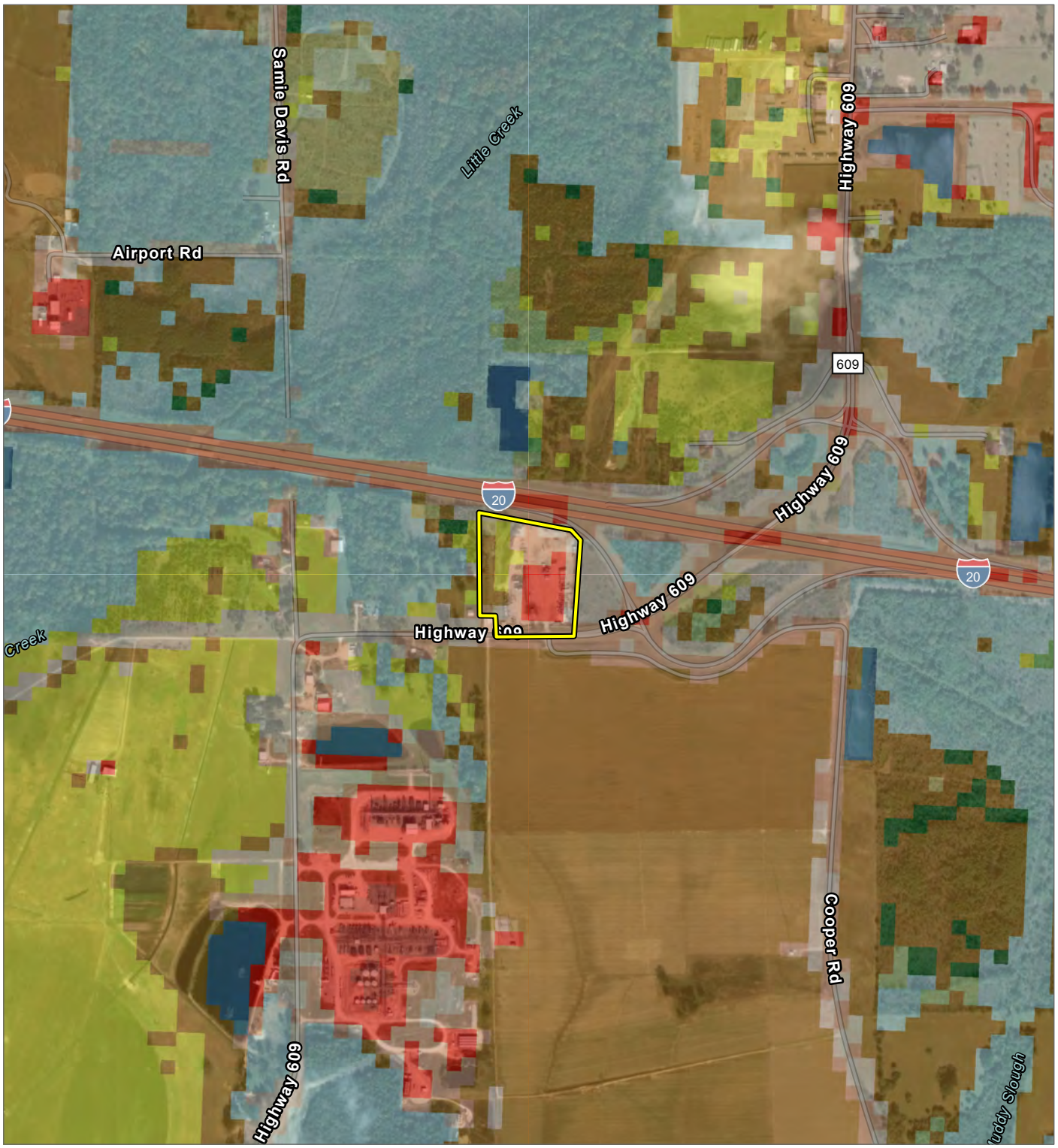

PROJECT BOUNDARY
OPEN WATER
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
DEVELOPED, MEDIUM INTENSITY
DEVELOPED, HIGH INTENSITY
DECIDUOUS FOREST
EVERGREEN FOREST
MIXED FOREST
SHRUB/SCRUB
GRASSLAND/HERBACEOUS
PASTURE/HAY
CULTIVATED CROPS
WOODY WETLANDS
EMERGENT HERBACEOUS

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1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 2 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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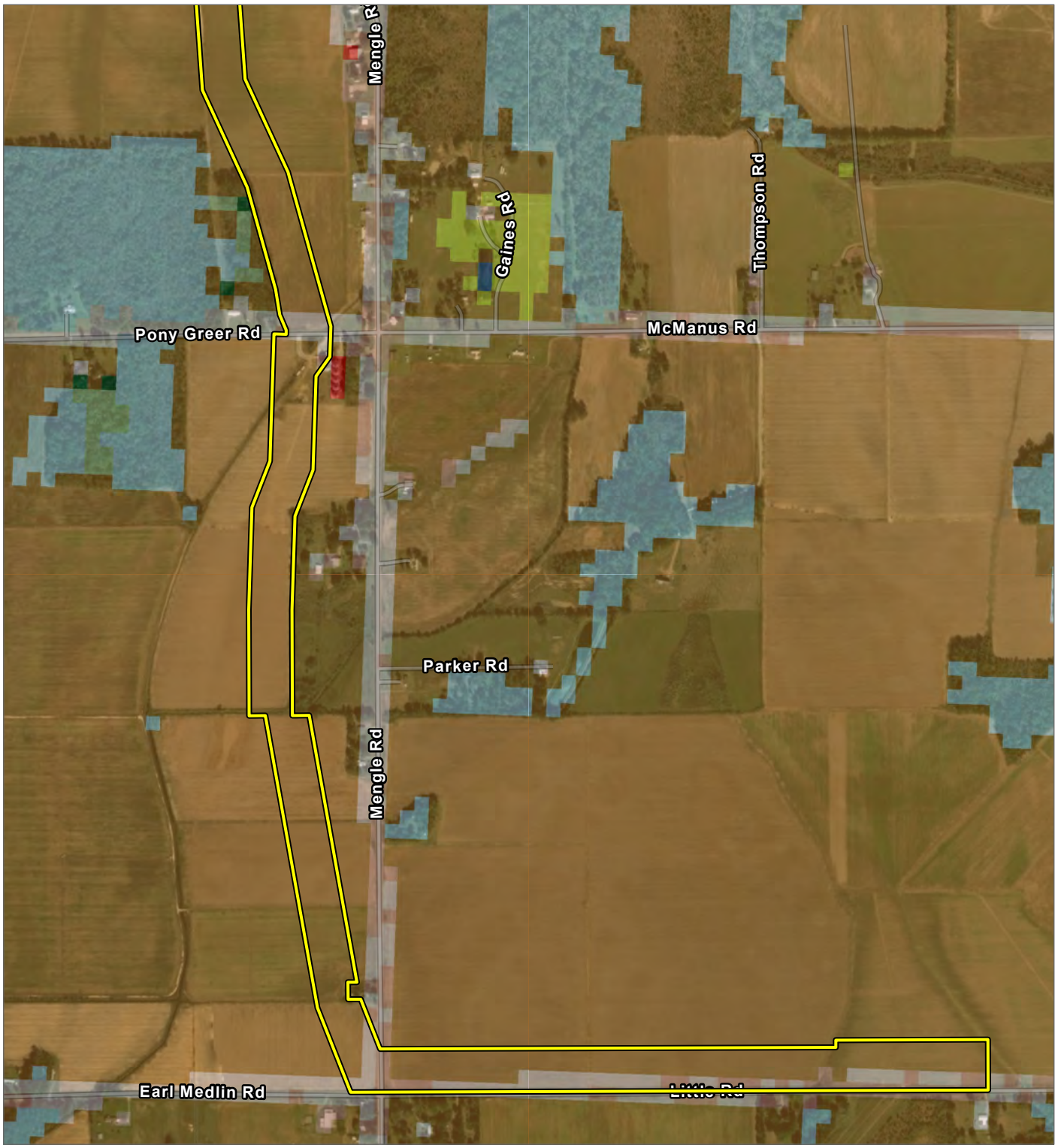

PROJECT BOUNDARY
OPEN WATER
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
DEVELOPED, MEDIUM INTENSITY
EVERGREEN FOREST
MIXED FOREST
PASTURE/HAY
CULTIVATED CROPS
WOODY WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

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1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 3 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET: MAP ROTATION: 0
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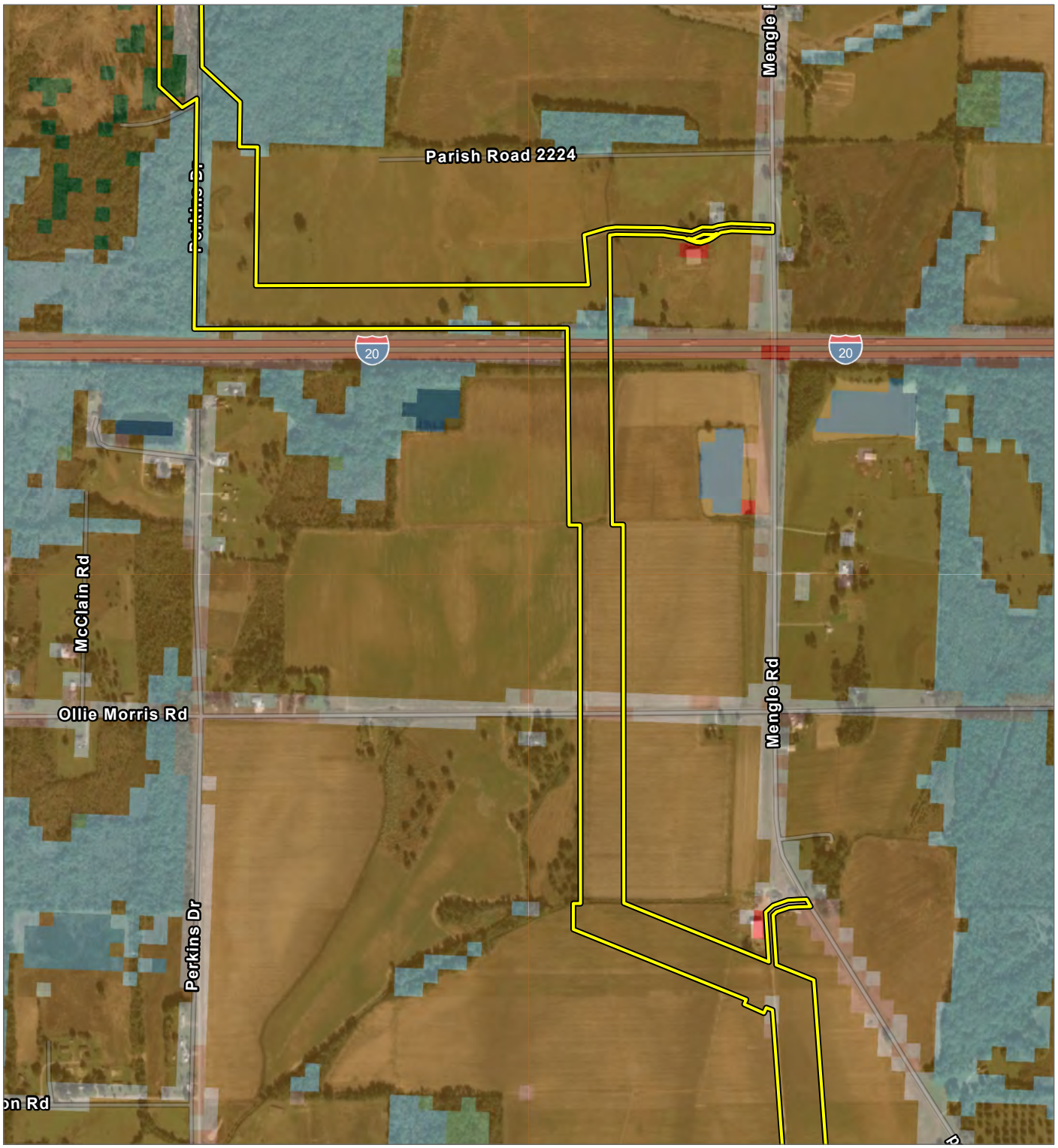

- PROJECT BOUNDARY
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- DEVELOPED, OPEN SPACE
- DEVELOPED, LOW INTENSITY
- DEVELOPED, MEDIUM INTENSITY
- EVERGREEN FOREST
- MIXED FOREST
- GRASSLAND/HERBACEOUS
- CULTIVATED CROPS
- WOODY WETLANDS
- EMERGENT HERBACEOUS WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

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FEET
1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 4 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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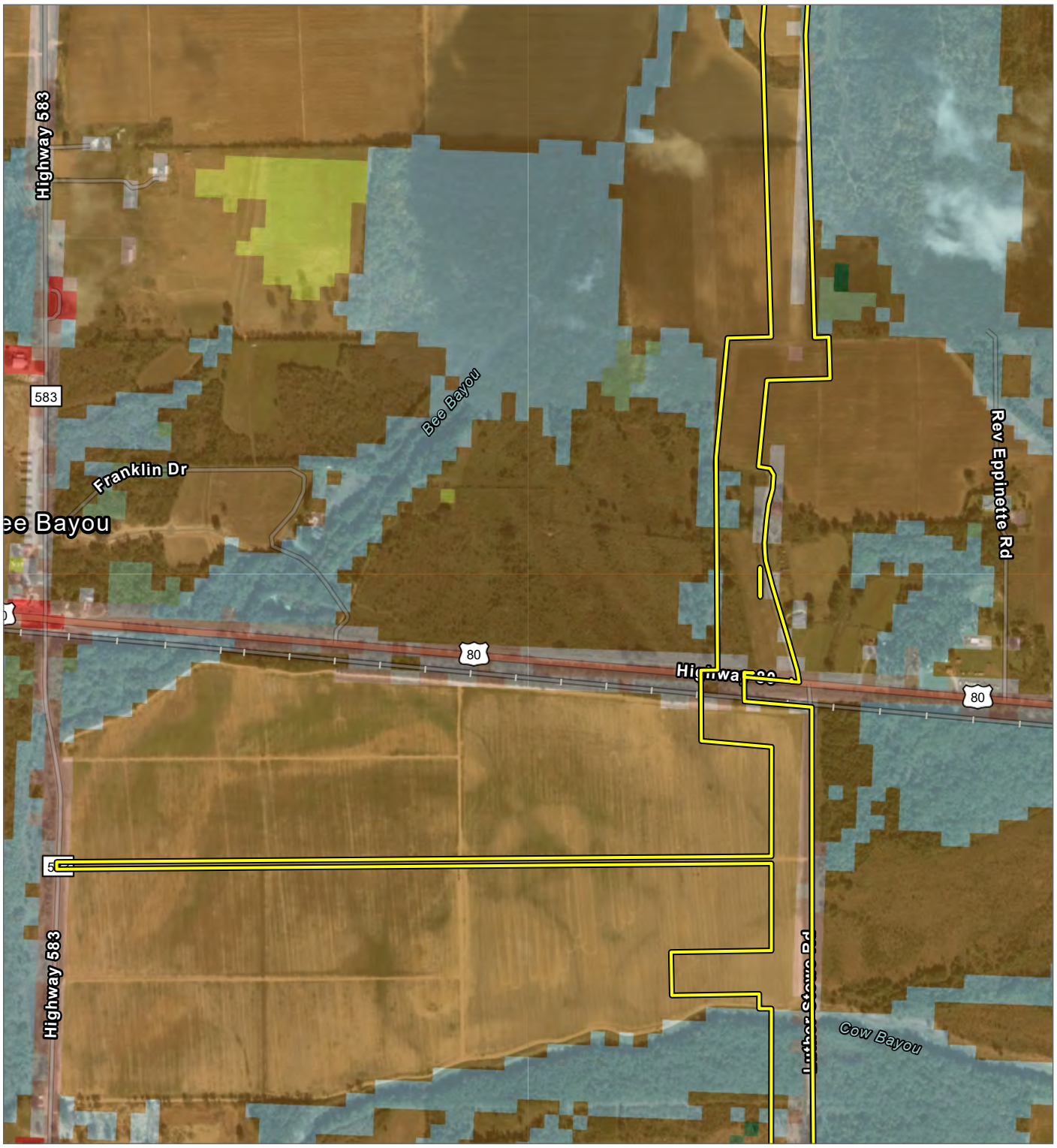

- PROJECT BOUNDARY
- DEVELOPED, OPEN SPACE
- DEVELOPED, LOW INTENSITY
- DEVELOPED, MEDIUM INTENSITY
- EVERGREEN FOREST
- MIXED FOREST
- PASTURE/HAY
- CULTIVATED CROPS
- WOODY WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000
FEET
1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 5 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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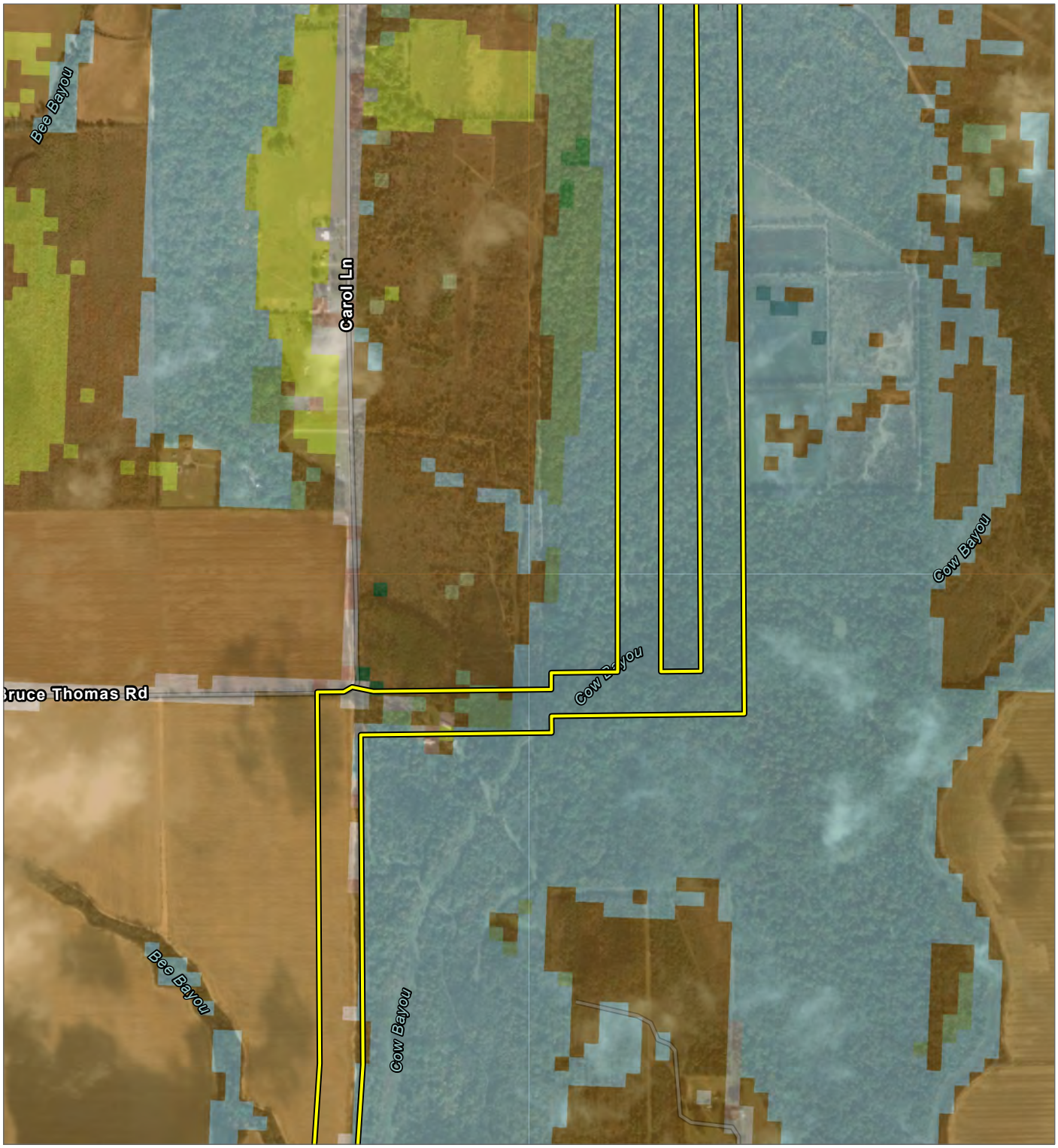


- PROJECT BOUNDARY
- DEVELOPED, OPEN SPACE
- DEVELOPED, LOW INTENSITY
- DECIDUOUS FOREST
- EVERGREEN FOREST
- MIXED FOREST
- SHRUB/SCRUB
- PASTURE/HAY
- CULTIVATED CROPS
- WOODY WETLANDS
- EMERGENT HERBACEOUS WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

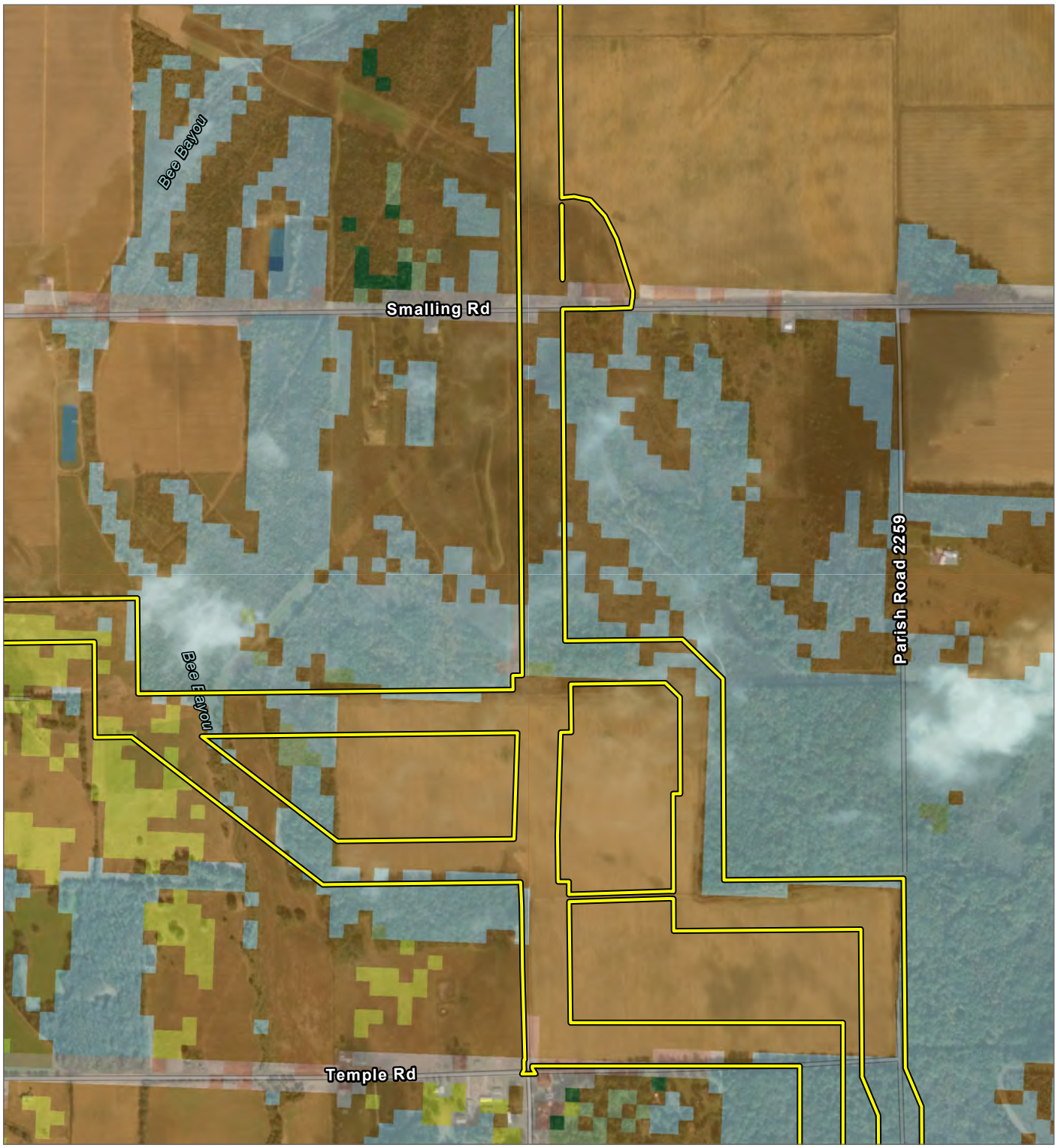
NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 6 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET: MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026 16:43:59 PM -- FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX -- LAYOUT NAME: 4_NLCD



- PROJECT BOUNDARY
- OPEN WATER
- DEVELOPED, OPEN SPACE
- DEVELOPED, LOW INTENSITY
- EVERGREEN FOREST
- MIXED FOREST
- PASTURE/HAY
- CULTIVATED CROPS
- WOODY WETLANDS
- EMERGENT HERBACEOUS WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

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FEET
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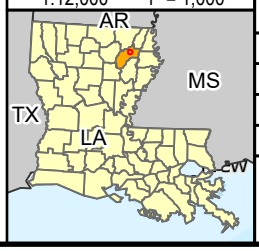

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 7 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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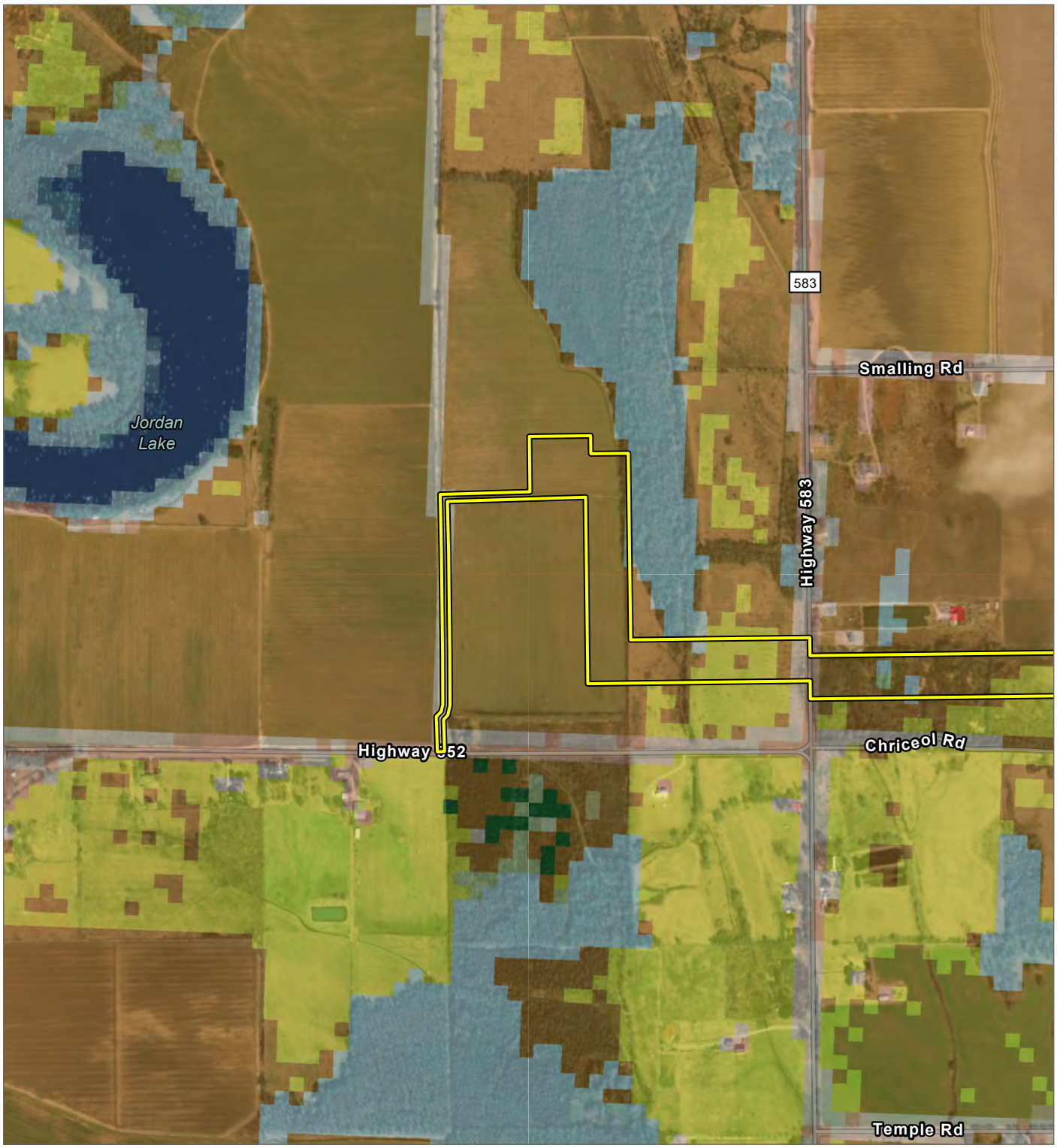
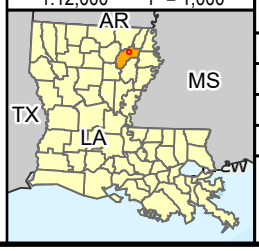


PROJECT BOUNDARY
OPEN WATER
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
DEVELOPED, MEDIUM INTENSITY
EVERGREEN FOREST
MIXED FOREST
SHRUB/SCRUB
GRASSLAND/HERBACEOUS
PASTURE/HAY
CULTIVATED CROPS
WOODY WETLANDS
EMERGENT HERBACEOUS

N

0 500 1,000
FEET
1:12,000 1" = 1,000'



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

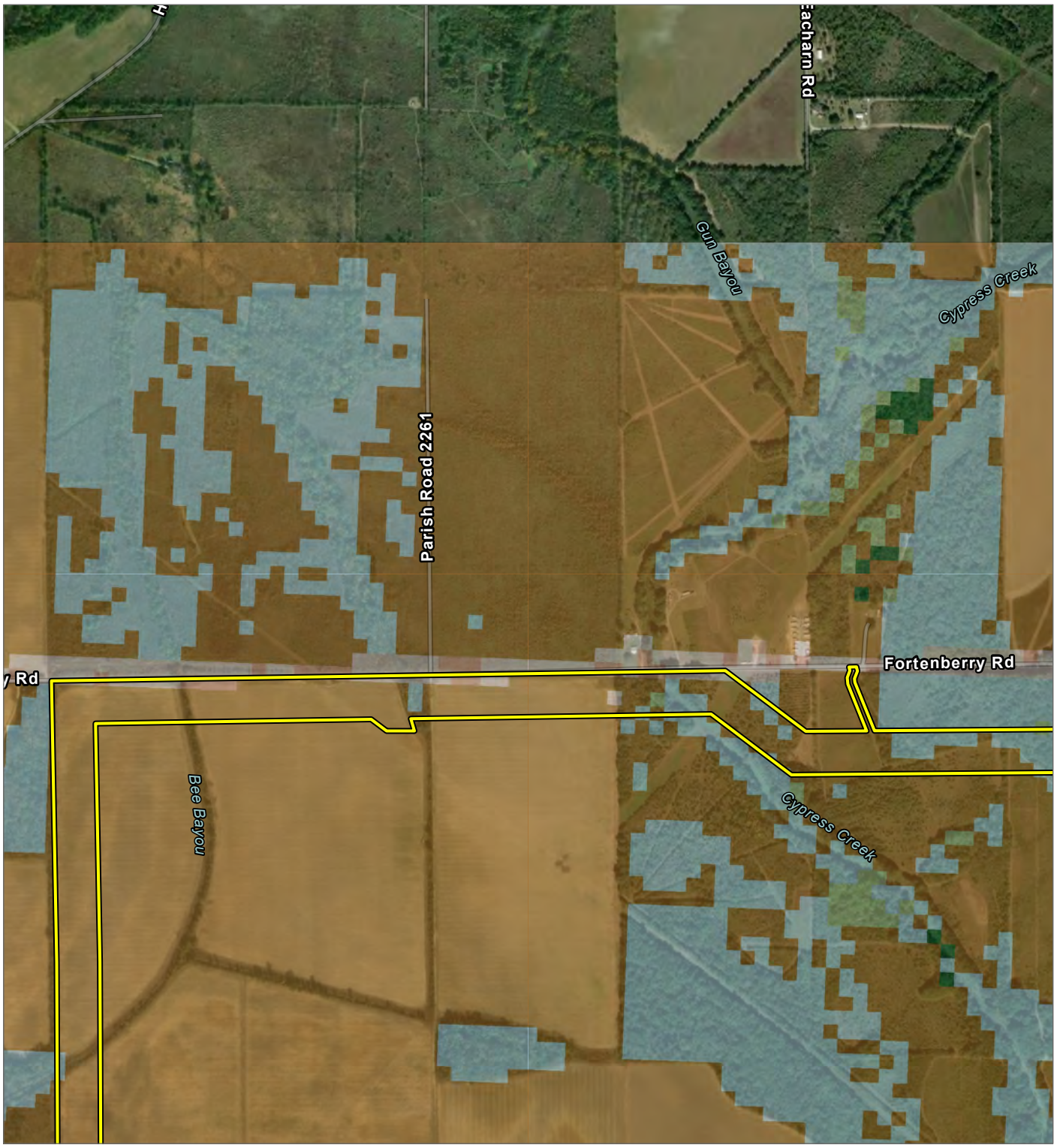
NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 4
PAGE 8 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FRANKLIN_FARMS_BIO_SURVEY_CLONE APPX; LAYOUT NAME: 4_NLCD



PROJECT BOUNDARY
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
EVERGREEN FOREST
MIXED FOREST
CULTIVATED CROPS
WOODY WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000
FEET
1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

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FIGURE 4
PAGE 9 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
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PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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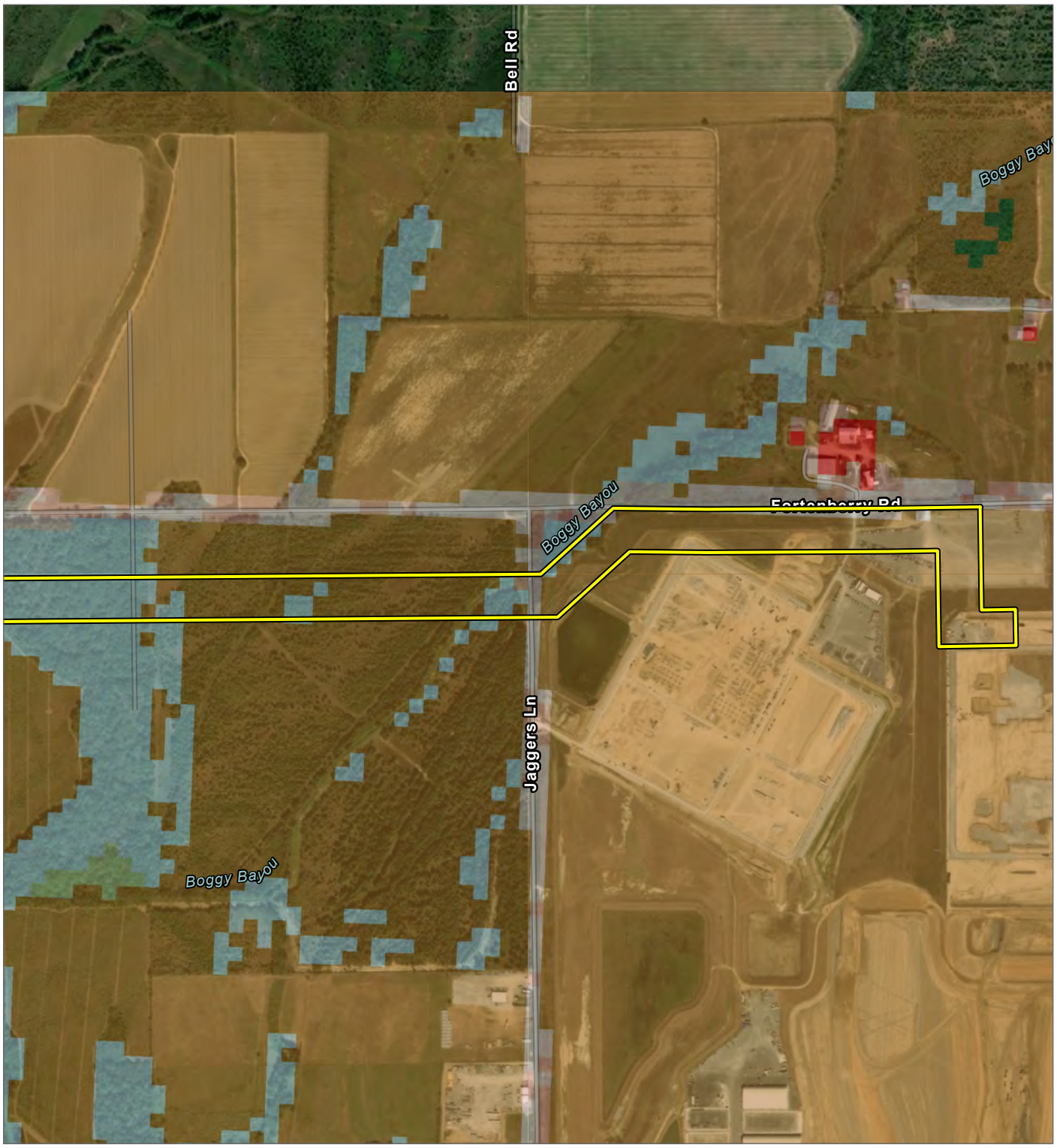

PROJECT BOUNDARY
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
DEVELOPED, MEDIUM INTENSITY
EVERGREEN FOREST
MIXED FOREST
CULTIVATED CROPS
WOODY WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

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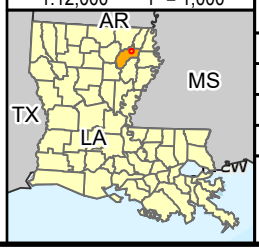

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
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DATE:	MAY 2026		

FIGURE 4
PAGE 10 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET: MAP ROTATION: 0
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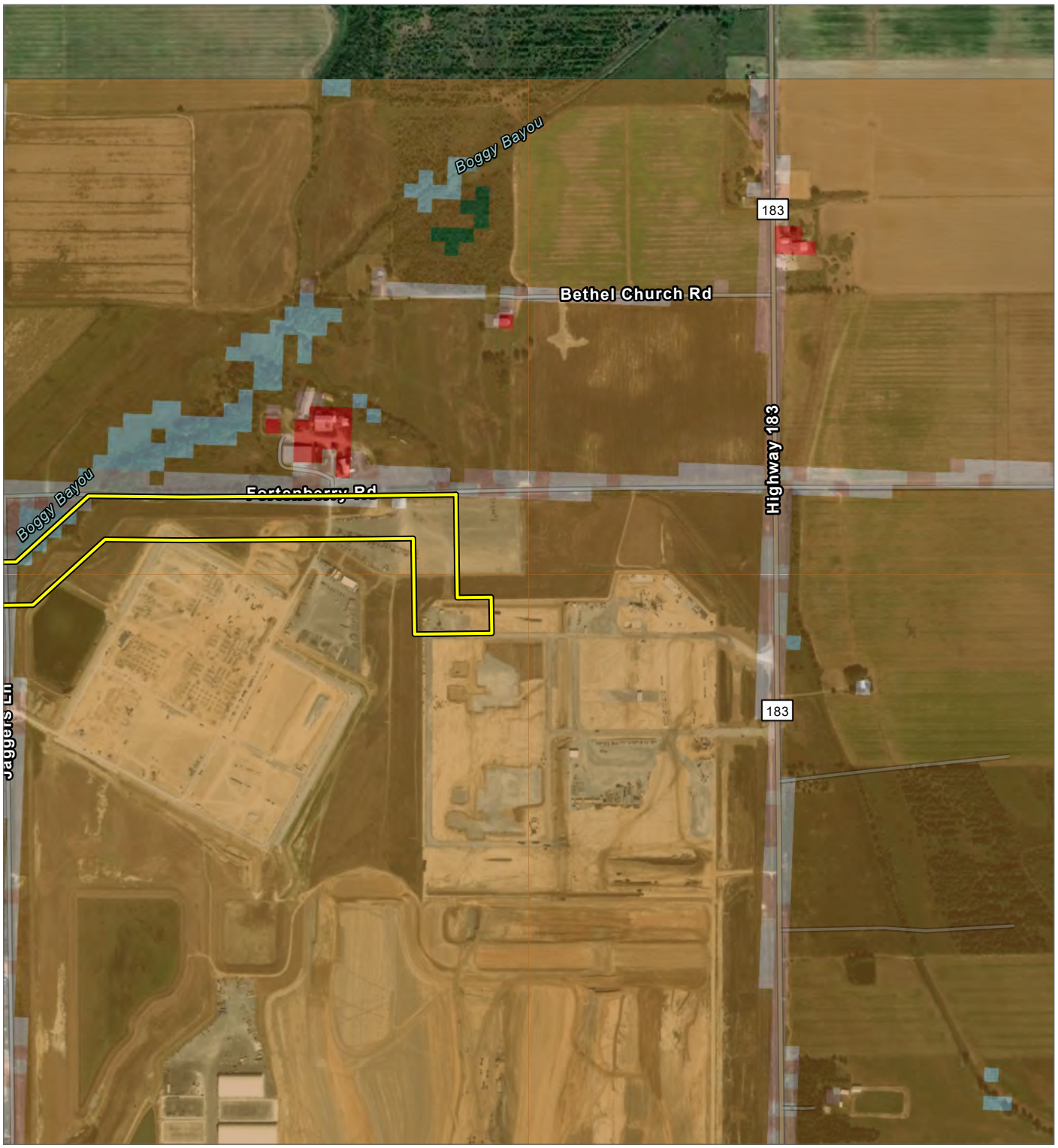
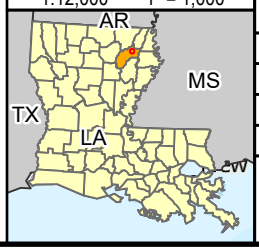


PROJECT BOUNDARY
DEVELOPED, OPEN SPACE
DEVELOPED, LOW INTENSITY
DEVELOPED, MEDIUM INTENSITY
EVERGREEN FOREST
CULTIVATED CROPS
WOODY WETLANDS

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NLCD

N

0 500 1,000
FEET
1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NLCD LANDCOVER MAP

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FIGURE 4
PAGE 11 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PROJECT BOUNDARY

SOIL MAP UNIT

Cc, CALHOUN-CALLOWAY SILT LOAMS, GENTLY UNDULATING

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 1 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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PROJECT BOUNDARY

SOIL MAP UNIT

Gg, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING

Gr, GRENADA SILT LOAM, 1 TO 3 PERCENT SLOPES

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 2 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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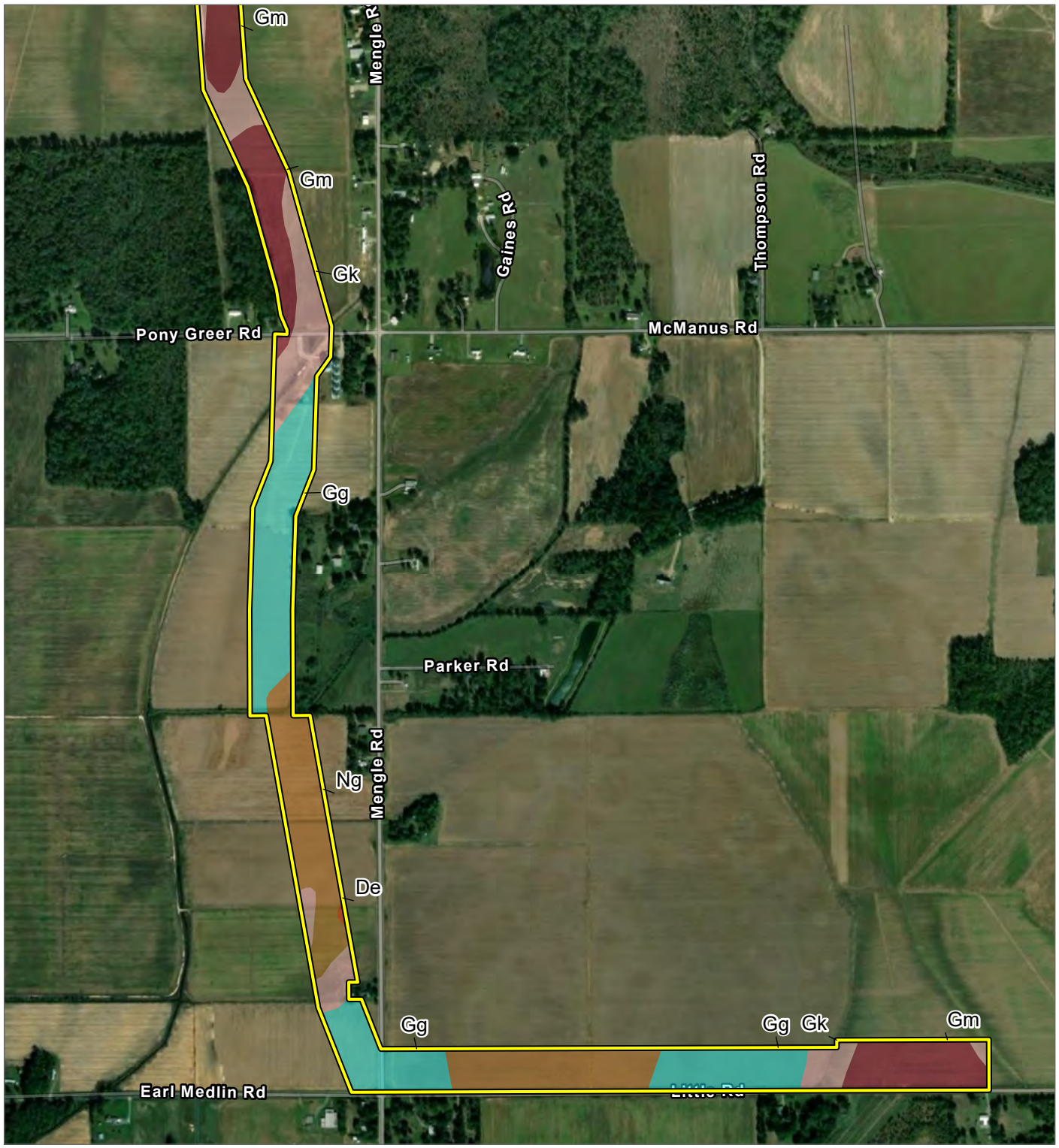

PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gg, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING
- Gk, GILBERT SILT LOAM
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ng, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

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FIGURE 5
PAGE 3 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET: MAP ROTATION: 0
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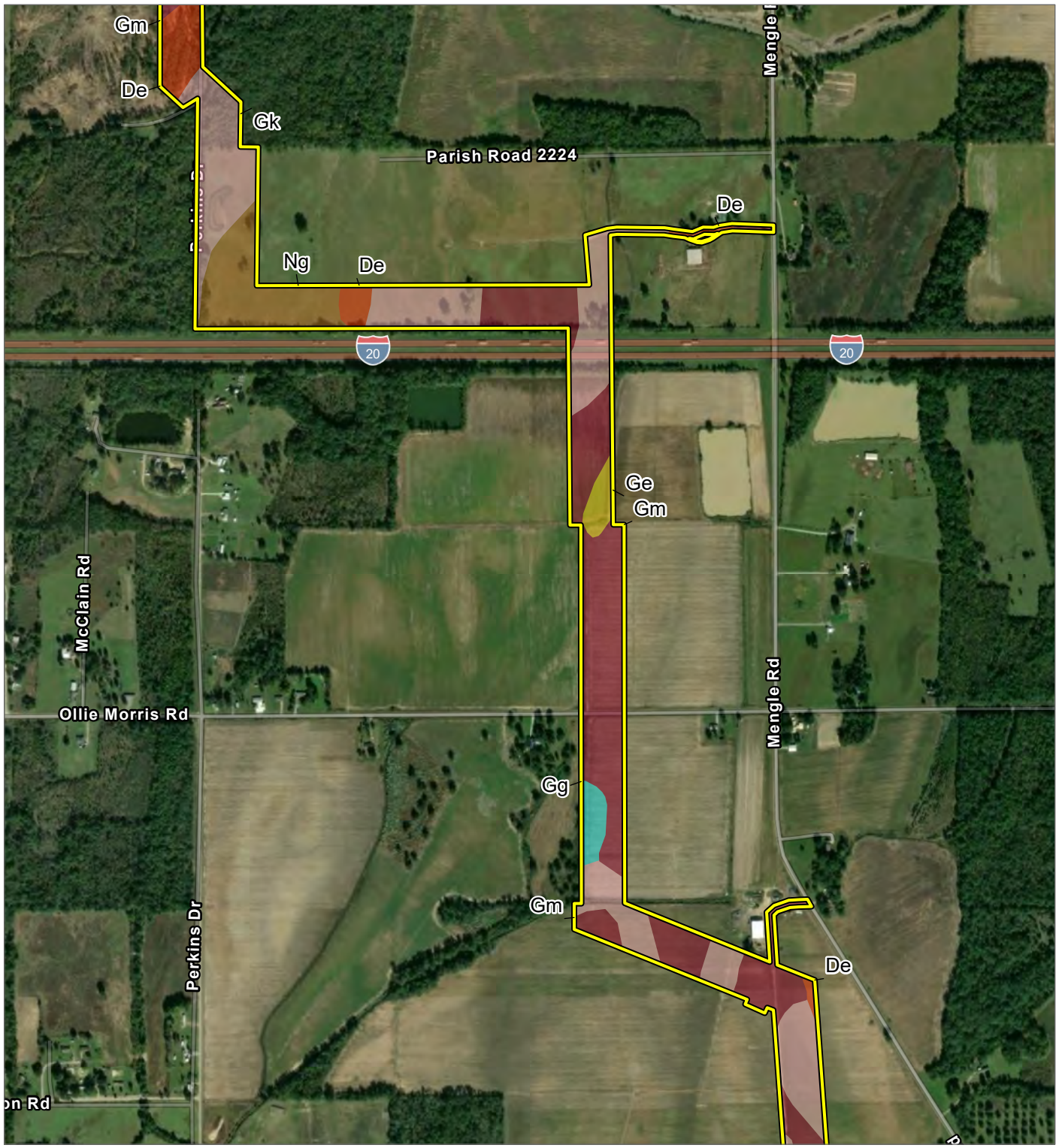

PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Ge, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gg, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING
- Gk, GILBERT SILT LOAM
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ng, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

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DATE:	MAY 2026		

FIGURE 5
PAGE 4 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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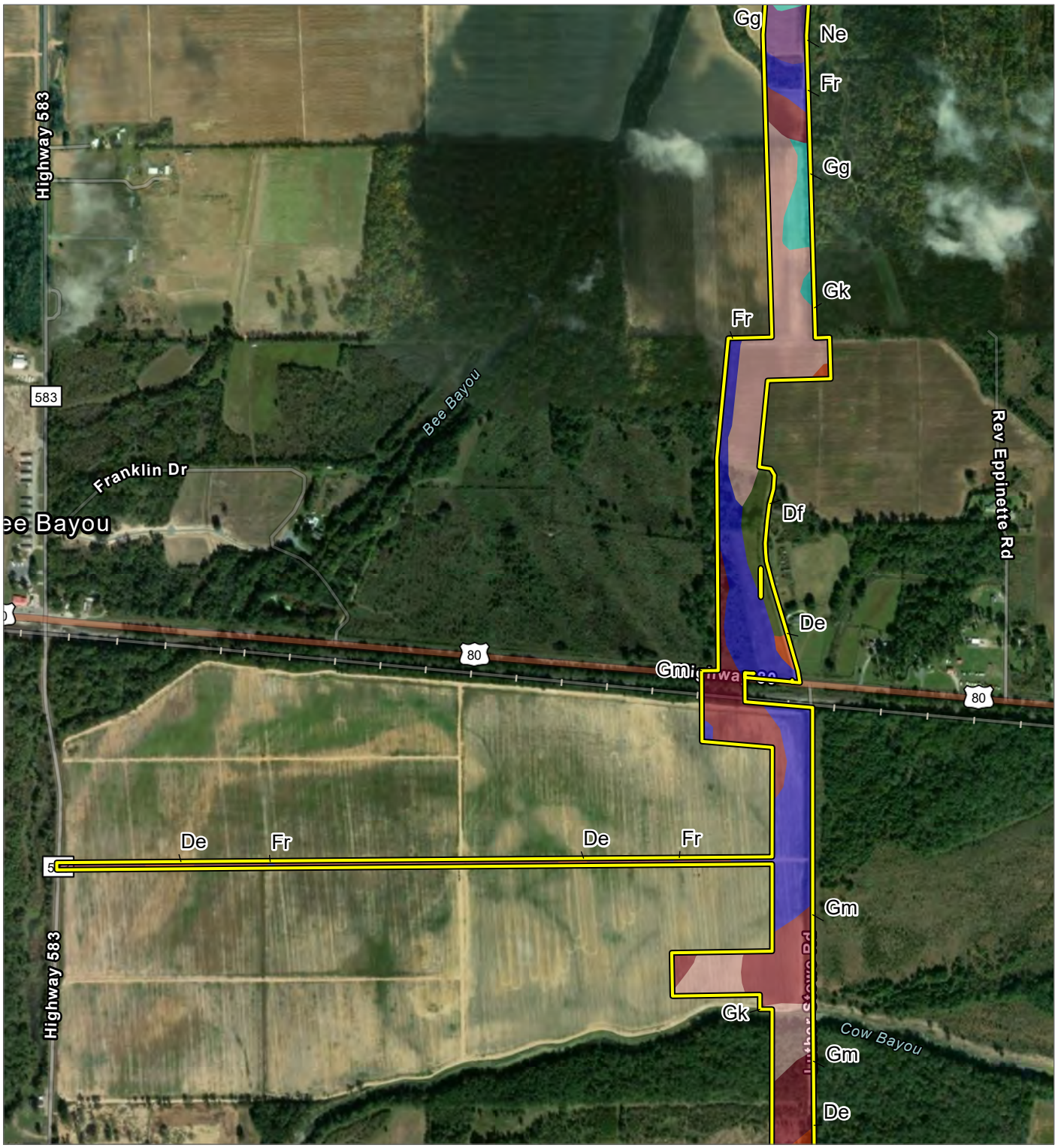

PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Df, DEXTER SILT LOAM, 3 TO 5 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Gg, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING
- Gk, GILBERT SILT LOAM
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ne, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

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CHECKED BY:	A. CLINE		
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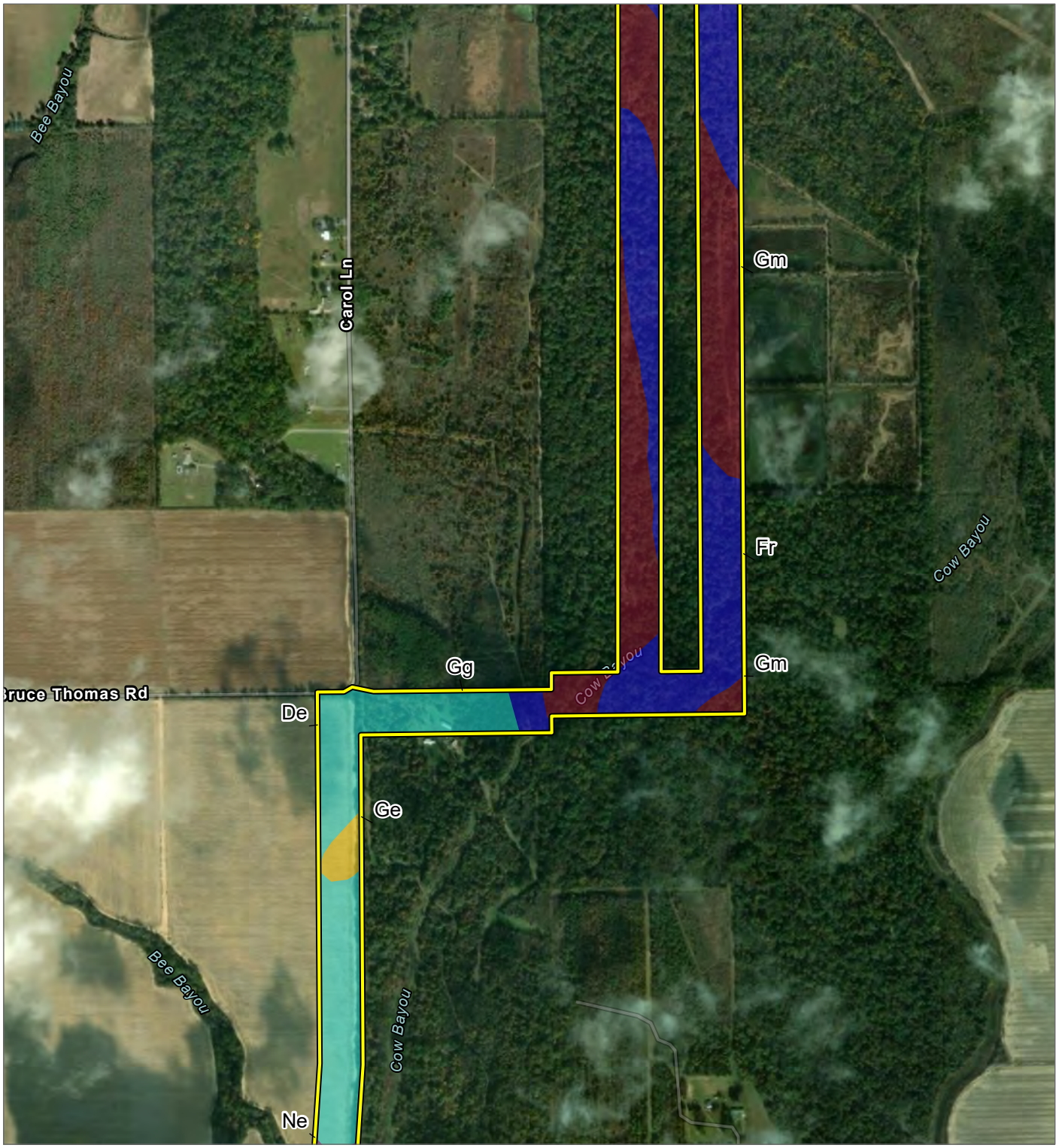
FIGURE 5
PAGE 5 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 1:16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FRANKLIN_FARMS_BIO_SURVEY_CLONE APPX; LAYOUT NAME: 5_NRCS SOILS



PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Ge, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gg, GIGGER-GILBERT SILT LOAMS, GENTLY UNDULATING
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ne, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

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DATE:	MAY 2026		

FIGURE 5
PAGE 6 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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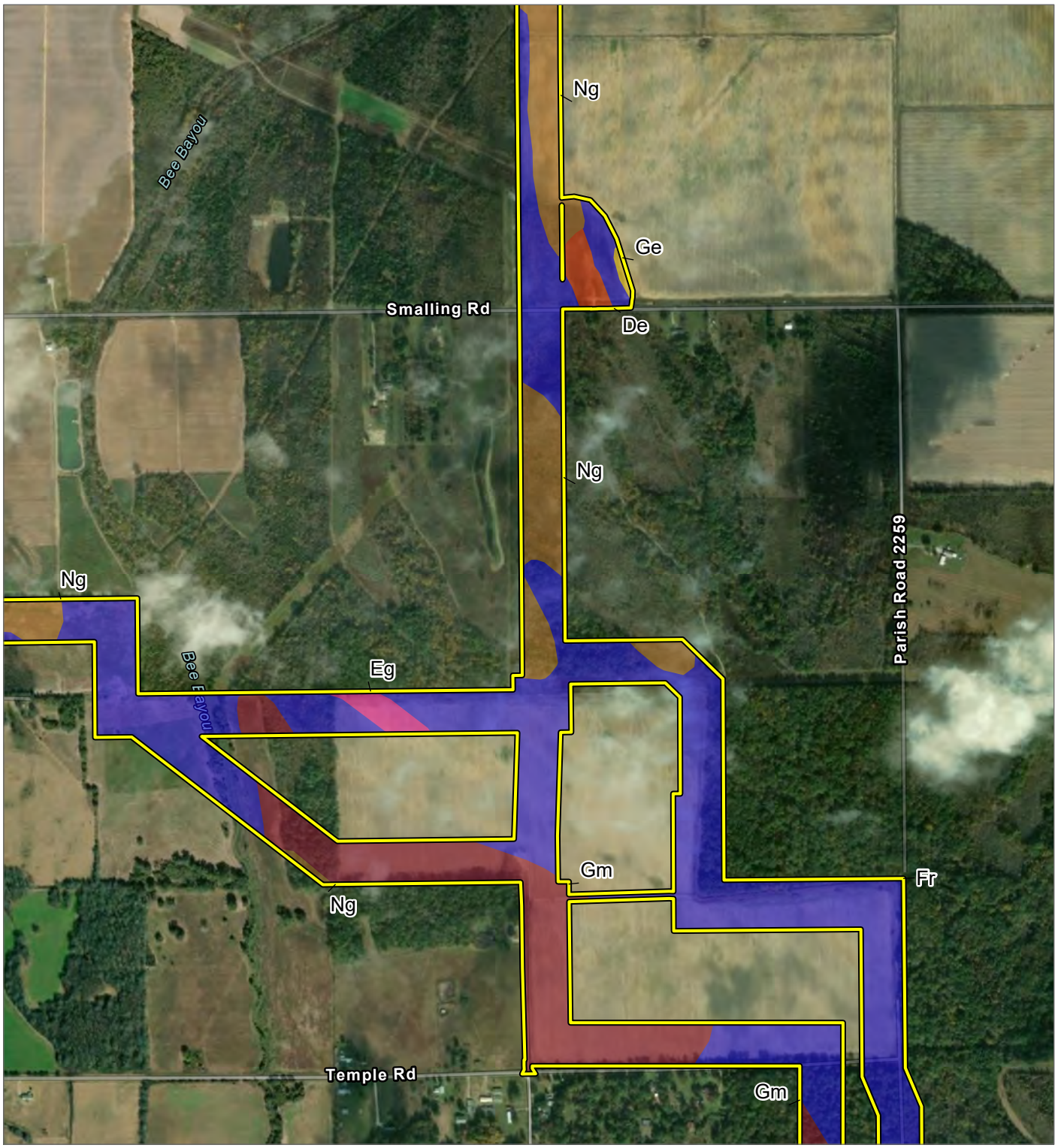

PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Eg, EGYPT SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Ge, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ng, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

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FEET

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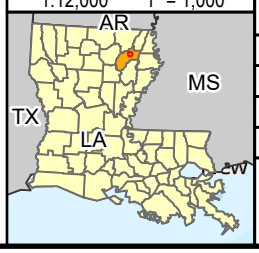

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
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FIGURE 5
PAGE 7 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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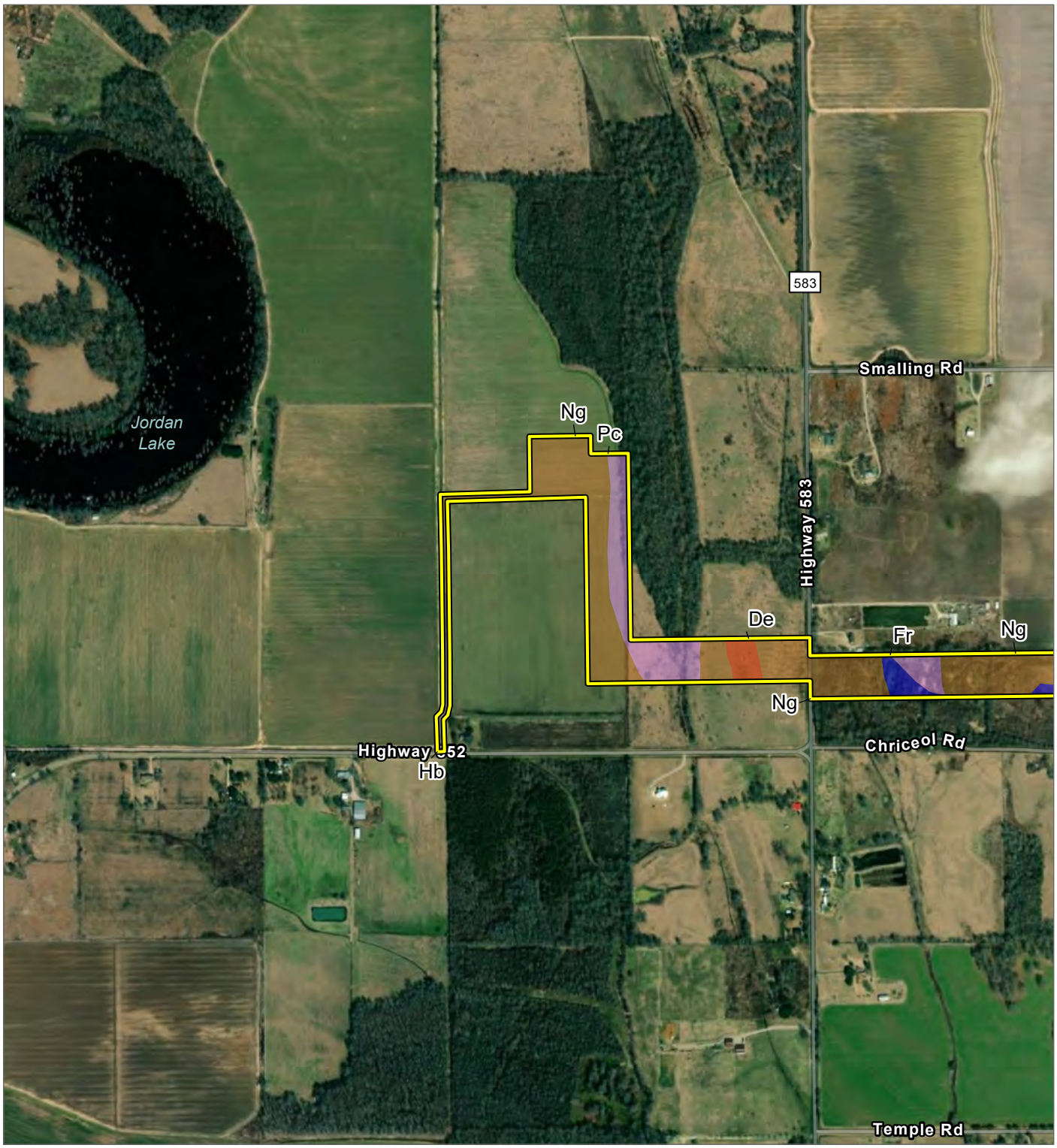

PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Hb, HEBERT SILT LOAM, 0 TO 1 PERCENT SLOPES
- Ng, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING
- Pc, PERRY SILTY CLAY LOAM

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

0 500 1,000

FEET

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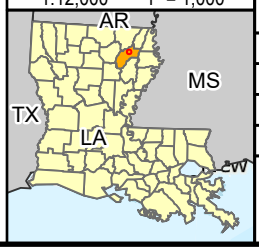

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

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FIGURE 5
PAGE 8 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

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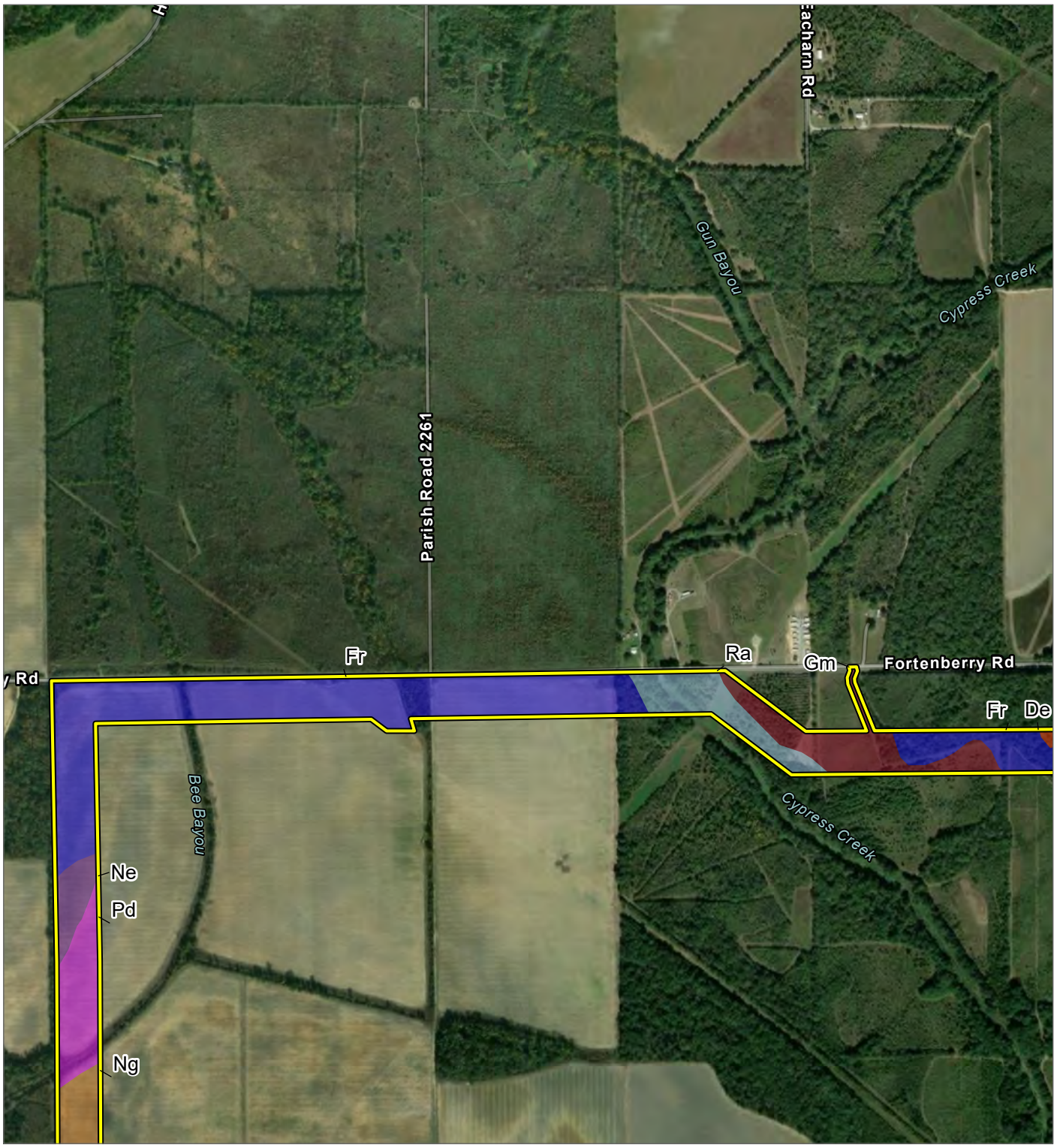

PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ne, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES
- Ng, NECESSITY-GILBERT SILT LOAMS, GENTLY UNDULATING
- Pd, PERRY CLAY
- Ra, RILLA SILT LOAM, 0 TO 1 PERCENT SLOPES

N

0 500 1,000

FEET

1:12,000 1" = 1,000'

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 9 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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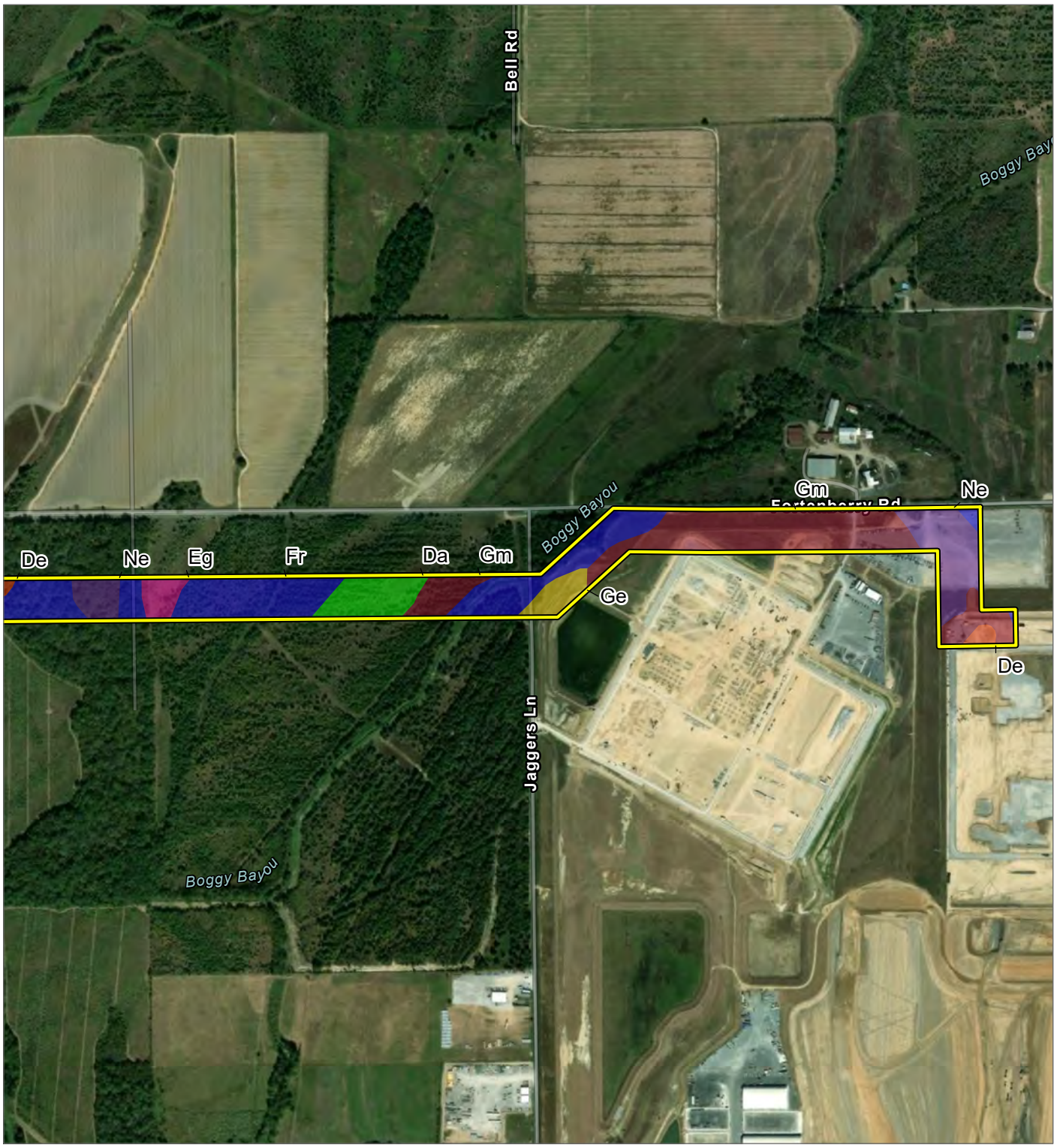


PROJECT BOUNDARY

SOIL MAP UNIT

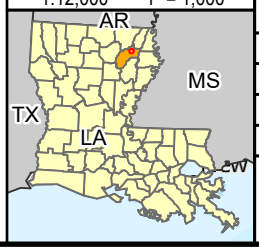
- Da, DEERFORD SILT LOAM
- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Eg, EGYPT SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Ge, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ne, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES

N

0 500 1,000

FEET

1:12,000 1" = 1,000'



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

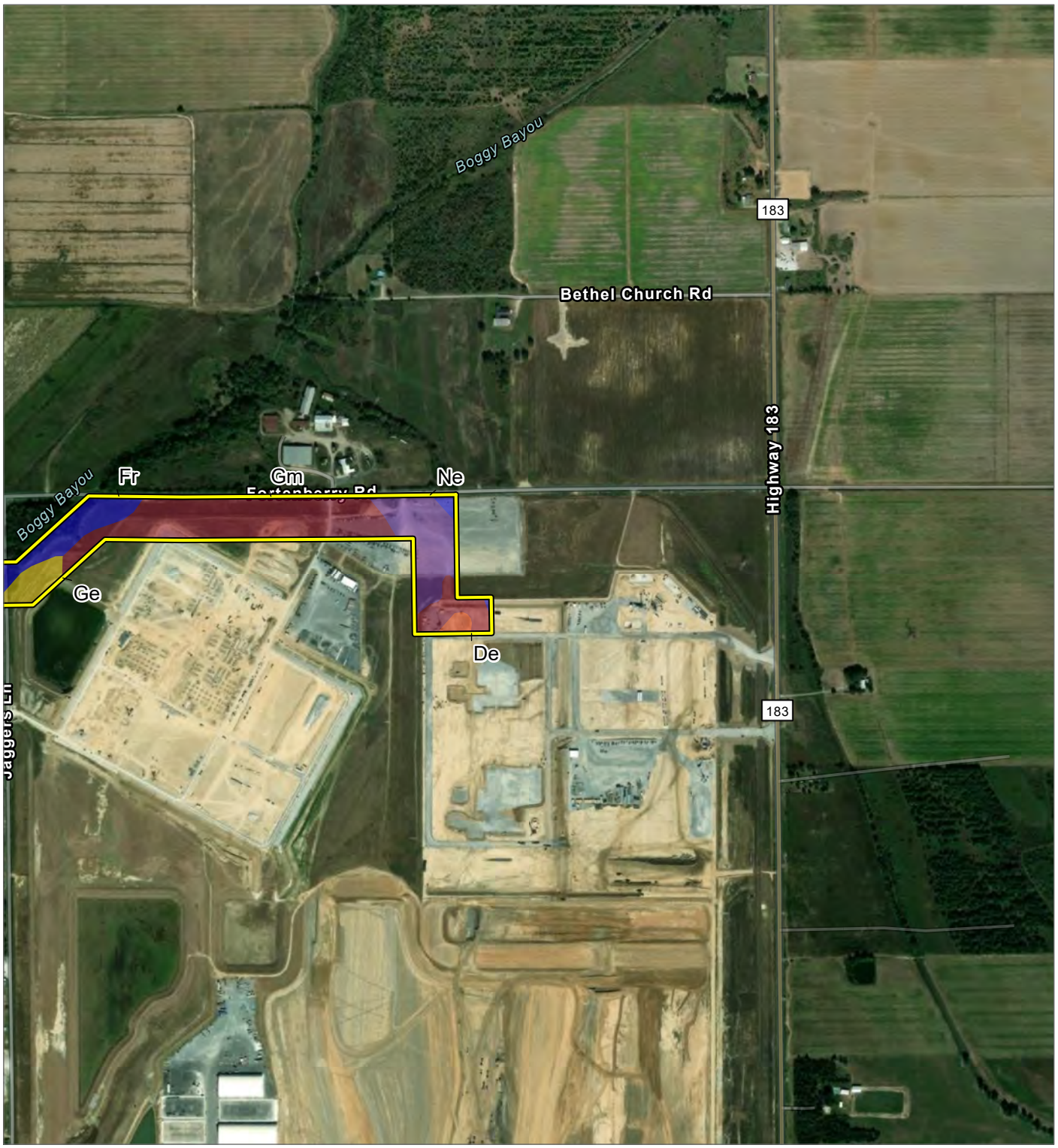
NRCS SOILS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 10 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 1:16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; -- LAYOUT NAME: 5_NRCS SOILS



PROJECT BOUNDARY

SOIL MAP UNIT

- De, DEXTER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Fr, FORESTDALE SILTY CLAY LOAM, 0 TO 1 PERCENT SLOPES, RARELY FLOODED, UPLAND PHASE
- Ge, GIGGER SILT LOAM, 1 TO 3 PERCENT SLOPES
- Gm, GILBERT-EGYPT SILT LOAMS, GENTLY UNDULATING
- Ne, NECESSITY SILT LOAM, 1 TO 3 PERCENT SLOPES

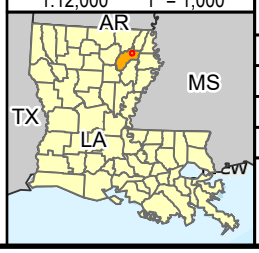
BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC, NRCS

N

0 500 1,000

FEET

1:12,000 1" = 1,000'



ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

NRCS SOILS MAP

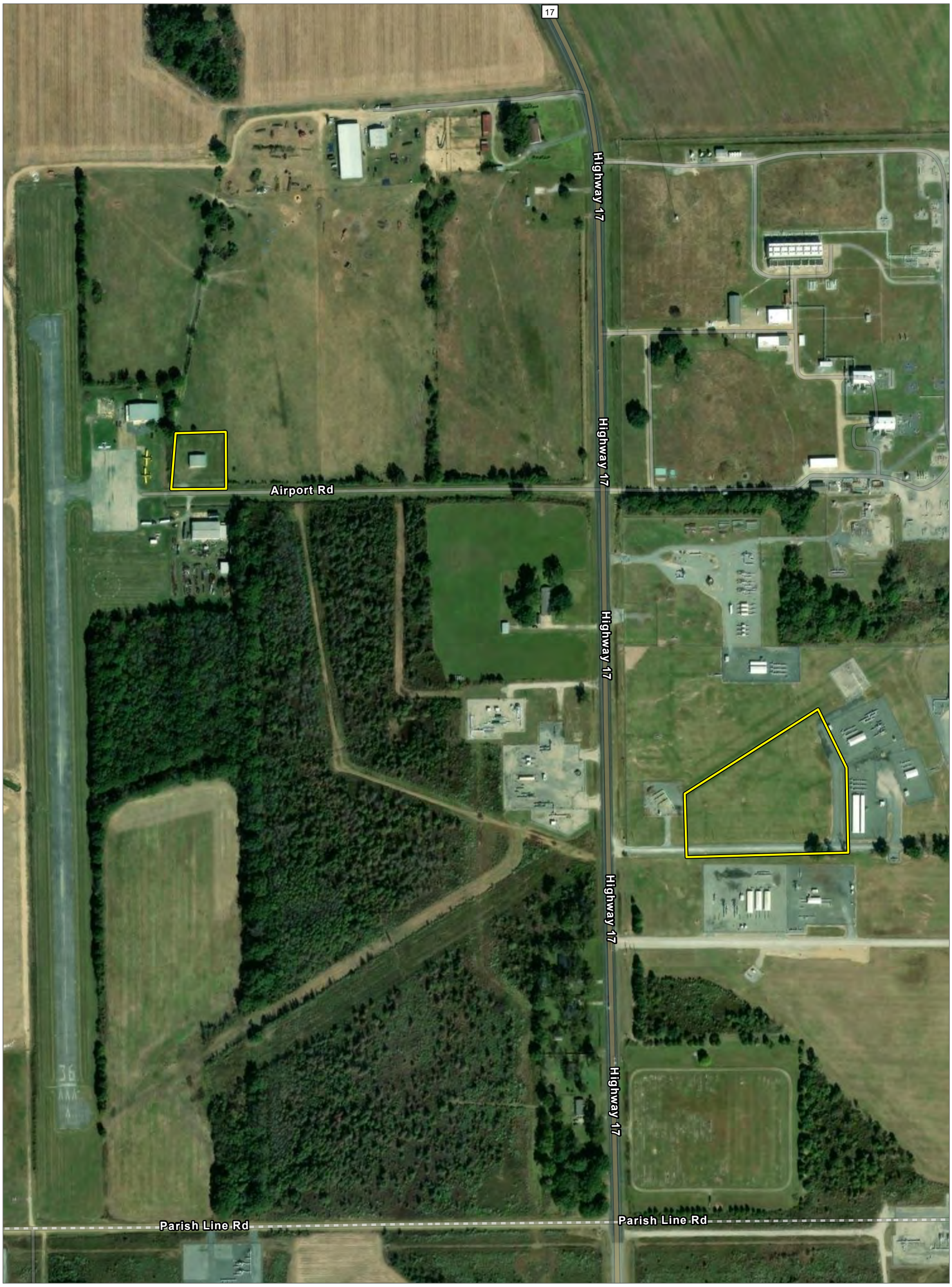
DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 5
PAGE 11 OF 11

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APPX\FRANKLIN_FARMS_BIO_SURVEY_CLONE APPX; -- LAYOUT NAME: 5_NRCS SOILS

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 6. FIELD OBSERVATIONS 11X17P



PROJECT BOUNDARY

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 1 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 6. FIELD OBSERVATIONS 11X17P

Samie Davis Rd
Little Creek
Little Creek
Frontage Rd
Frontage Rd
20
148
20
20
Little Creek Ln
Highway 609
Highway 609
Highway 609
Highway 609
Highway 609
Cooper Rd
Highway 609

PROJECT BOUNDARY

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 2 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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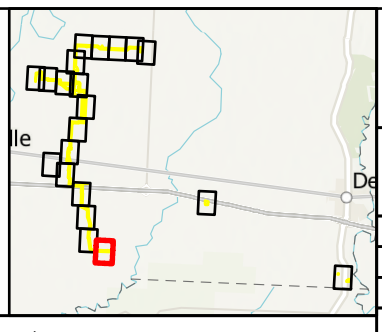
PROJECT BOUNDARY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

N

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE	**FIGURE 6**	
APPROVED BY:	A. THOMPSON	**PAGE 3 OF 21**	
DATE:	MAY 2026		

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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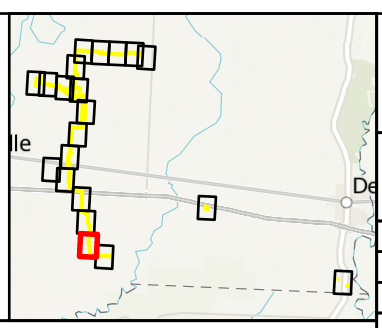
PROJECT BOUNDARY

OBSERVATION



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET

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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 4 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 6. FIELD OBSERVATIONS 11X17P

PROJECT BOUNDARY

OBSERVATION

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

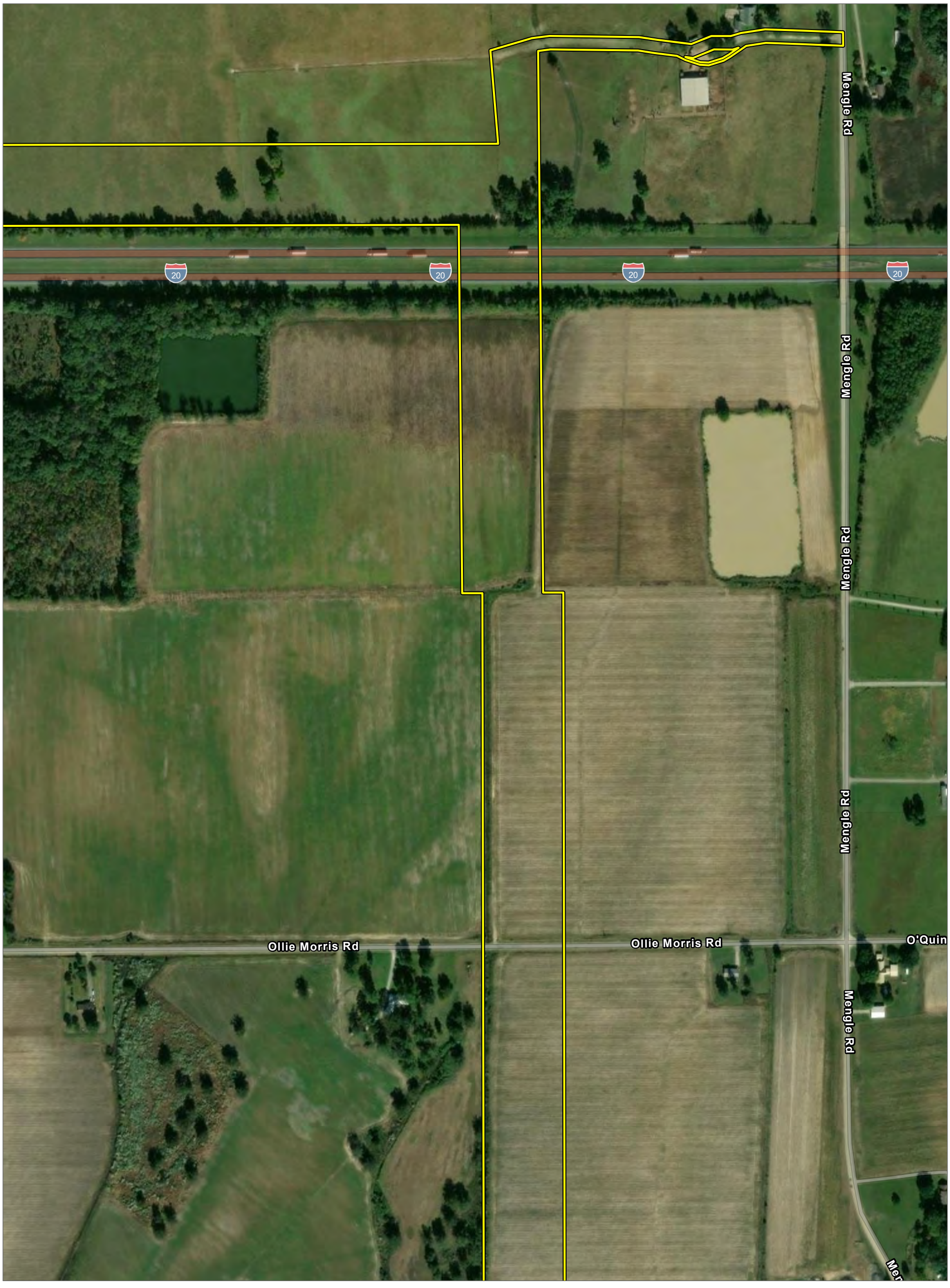
FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

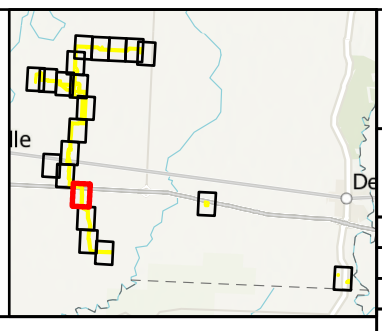
FIGURE 6
PAGE 5 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 6. FIELD OBSERVATIONS 11X17P



PROJECT BOUNDARY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

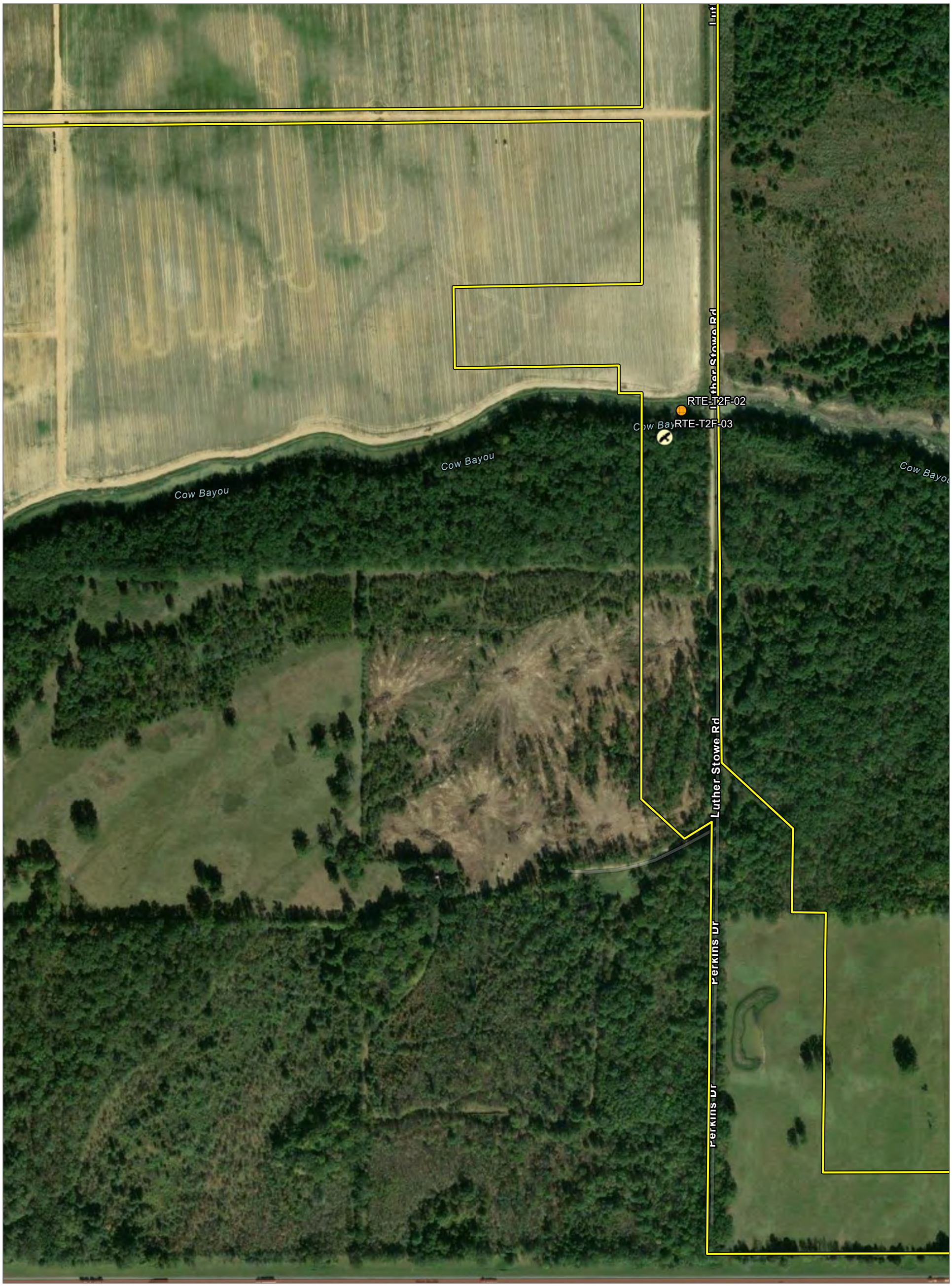
FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 6 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

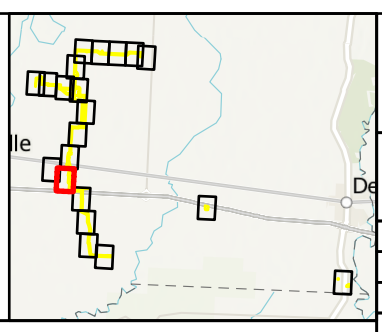
COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 6. FIELD OBSERVATIONS 11X17P



PROJECT BOUNDARY

RTE SPECIES

ALLIGATOR SNAPPING TURTLE HABITAT



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

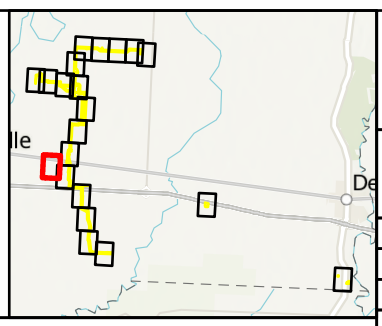
DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 7 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 6. FIELD OBSERVATIONS 11X17P

PROJECT BOUNDARY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

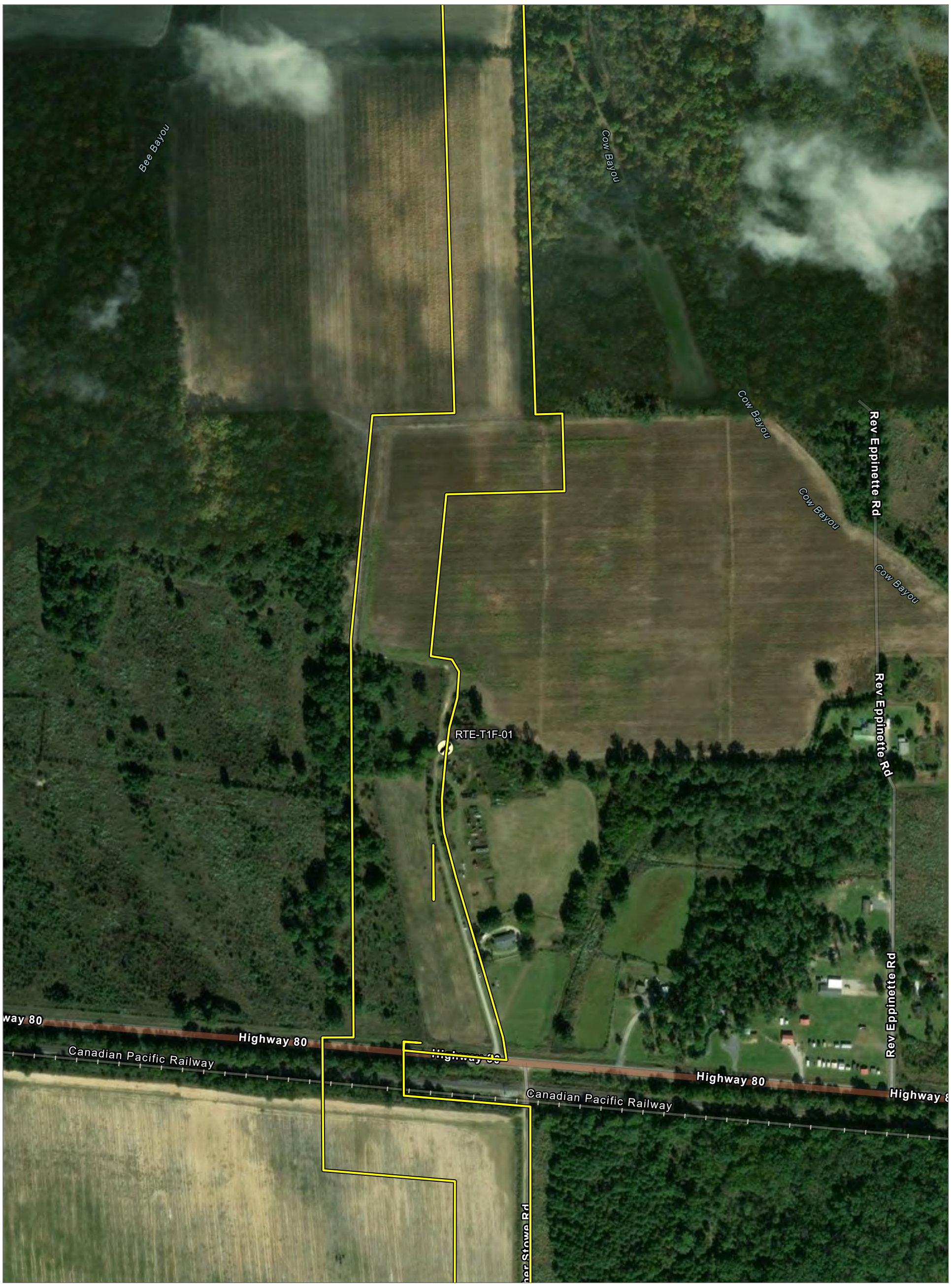
FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 8 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 6. FIELD OBSERVATIONS 11X17P



PROJECT BOUNDARY

RTE SPECIES

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET

1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

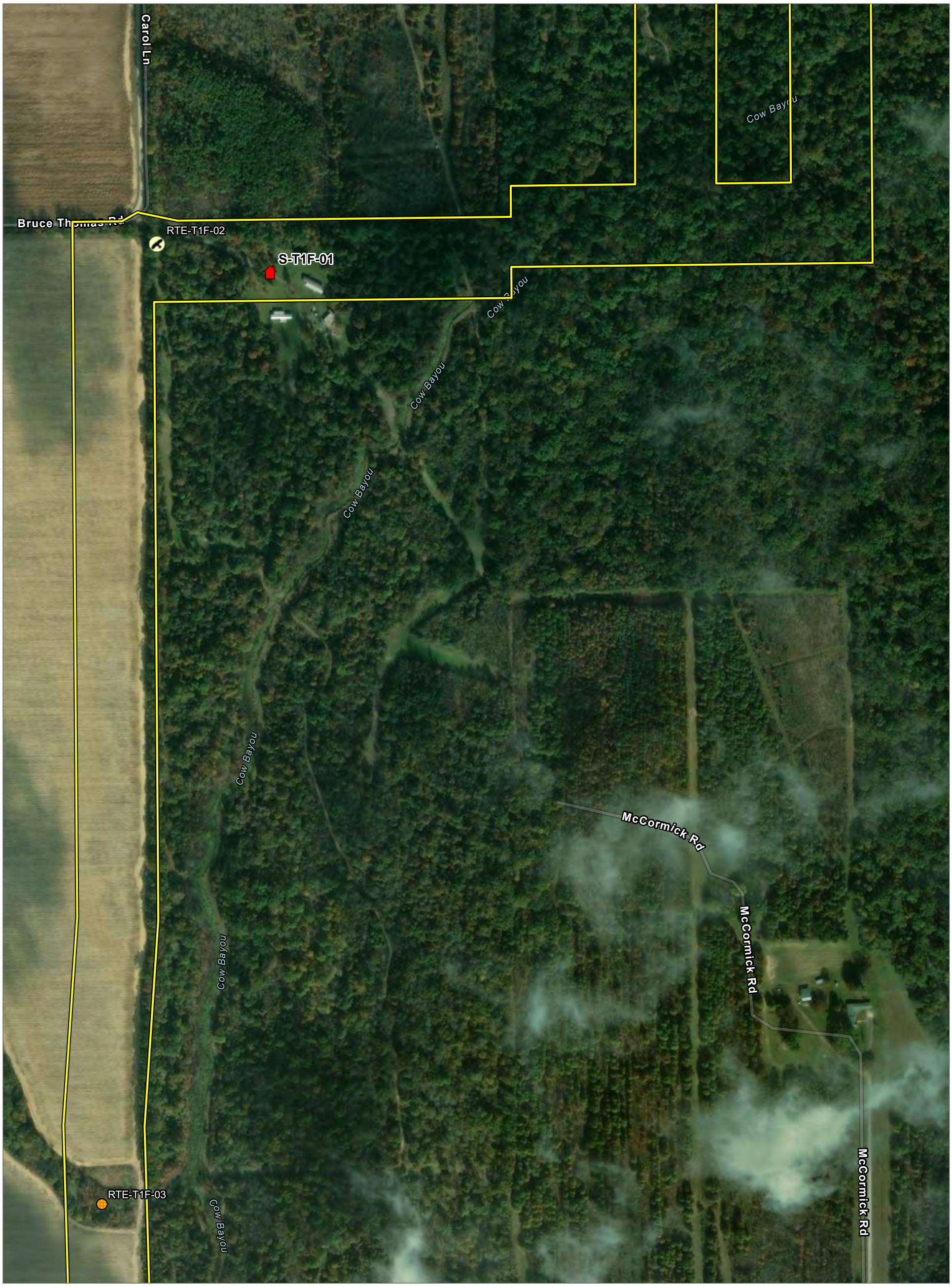
FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 9 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 6. FIELD OBSERVATIONS 11X17P



PROJECT BOUNDARY
STRUCTURE OR BUILDING
RTE SPECIES
ALLIGATOR SNAPPING TURTLE HABITAT

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

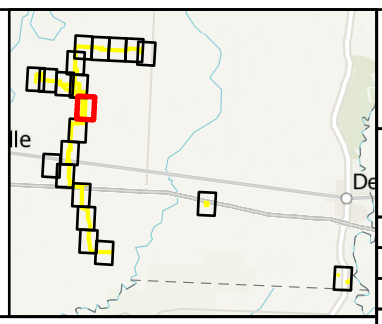
DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 10 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
-- SAVED BY: AGIBSON ON 5/27/2026, 16:43:59 PM; FILE PATH: T:\1-PROJECTS\ENERGY TRANSFER LP\705831_FRANKLIN_FARMS\2-APRX\FRANKLIN_FARMS_BIO_SURVEY_CLONE.APRX; LAYOUT NAME: 6. FIELD OBSERVATIONS 11X17P

PROJECT BOUNDARY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 11 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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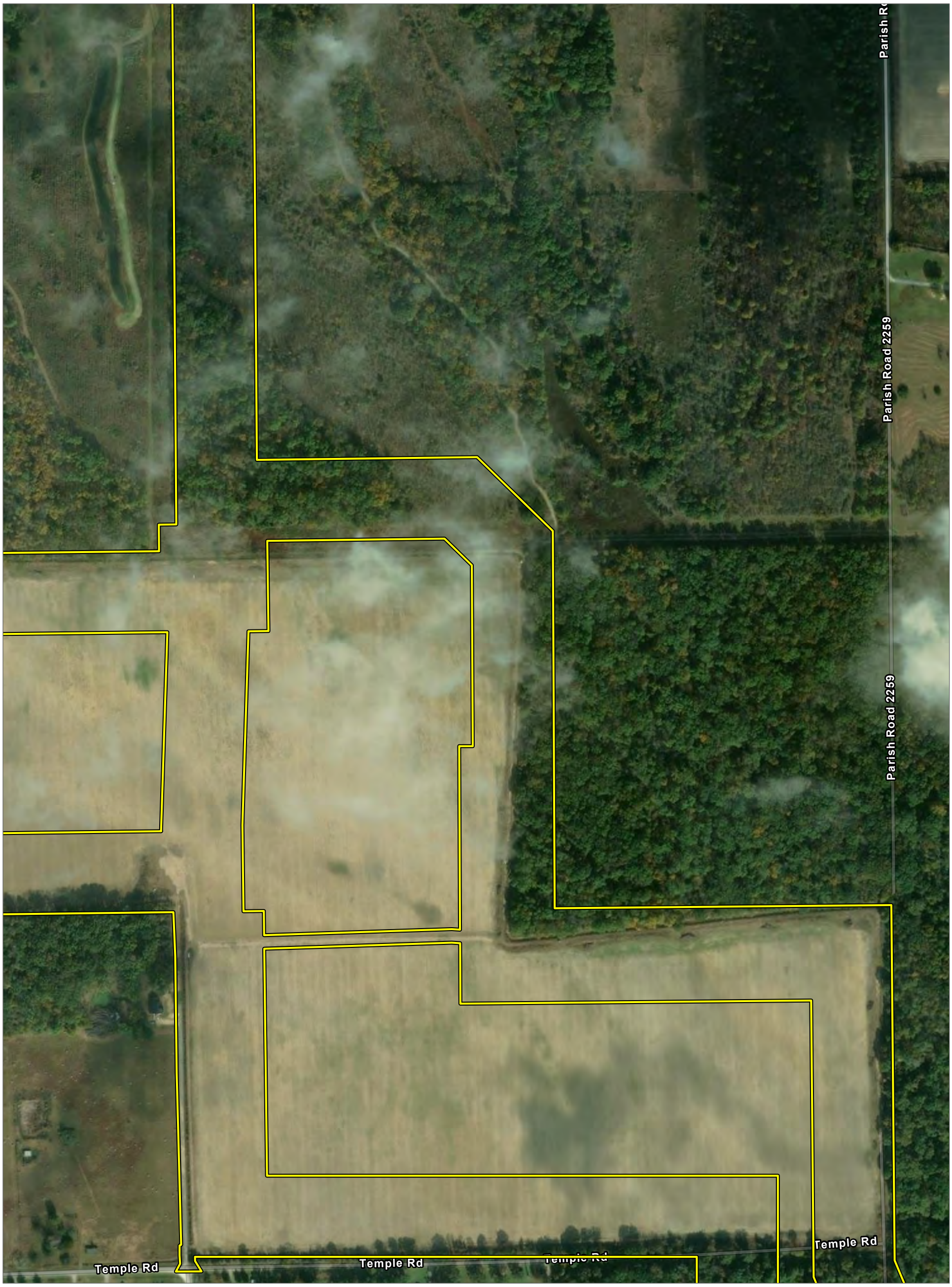
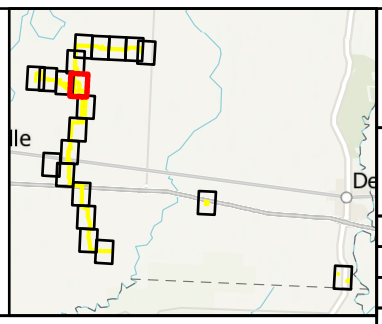


PROJECT BOUNDARY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
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DATA SOURCES: TRC

0 175 350 FEET
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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 12 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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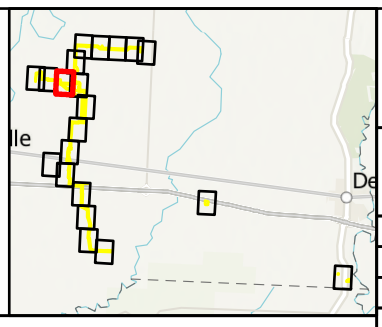
PROJECT BOUNDARY



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DATA SOURCES: TRC

N

0 175 350 FEET

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ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE	**FIGURE 6**	
APPROVED BY:	A. THOMPSON	**PAGE 13 OF 21**	
DATE:	MAY 2026		

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

COORDINATE SYSTEM: NAD 1983 STATEPLANE LOUISIANA NORTH FIPS 1701 FEET; MAP ROTATION: 0
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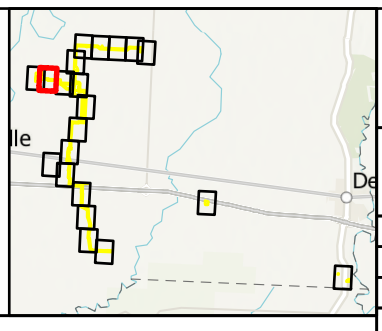
PROJECT BOUNDARY



BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER;
IMAGERY DATE: 11/05/2025
DATA SOURCES: TRC

0 175 350 FEET
1:4,200 1" = 350'

ETC TIGER PIPELINE, LLC
FRANKLIN FARMS PROJECT,
RICHLAND PARISH, LA

FIELD OBSERVATIONS MAP

DRAWN BY:	A. GIBSON	PROJ. NO.:	705831
CHECKED BY:	A. CLINE		
APPROVED BY:	A. THOMPSON		
DATE:	MAY 2026		

FIGURE 6
PAGE 14 OF 21

TRC
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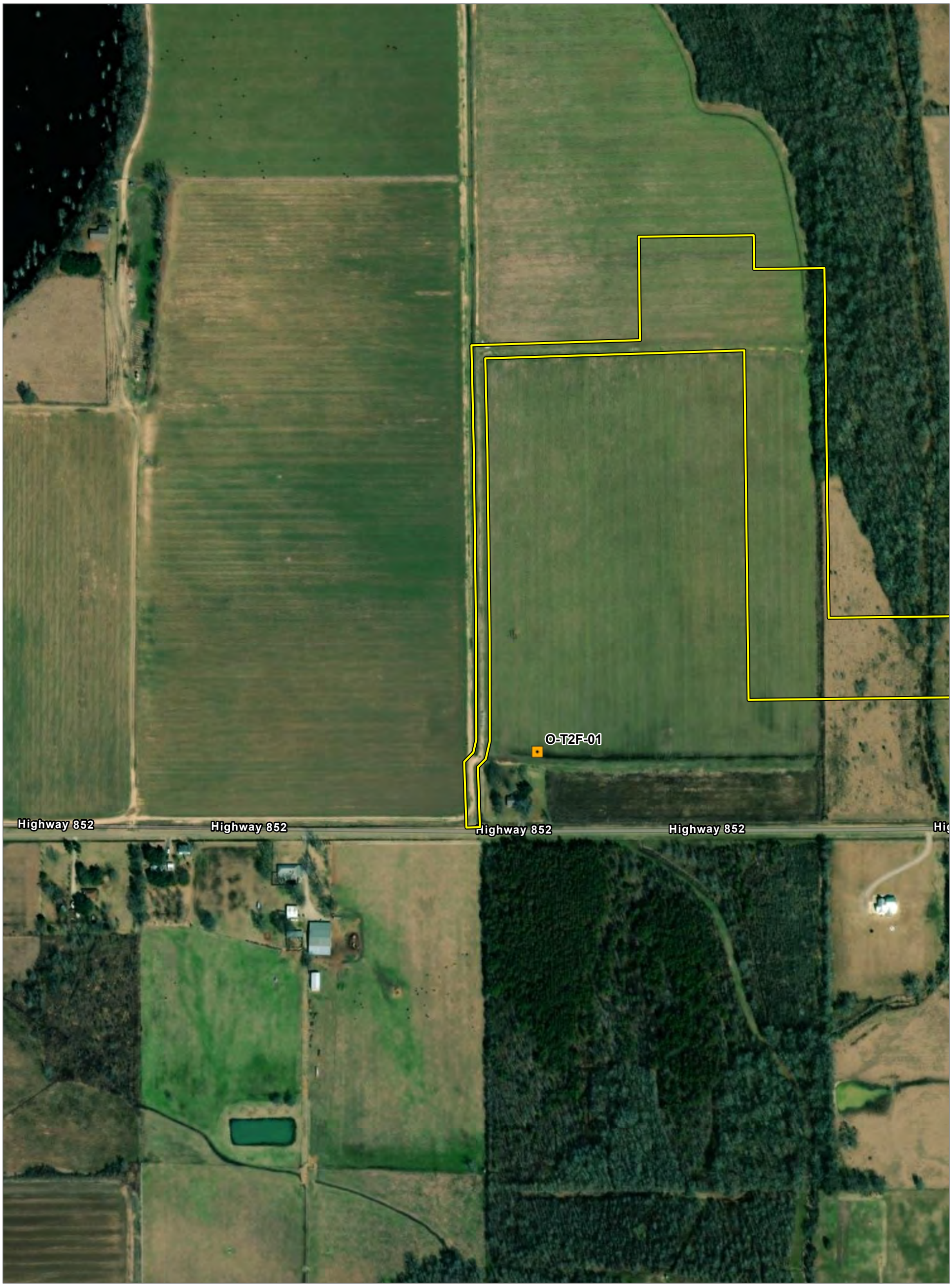

PROJECT BOUNDARY

OBSERVATION

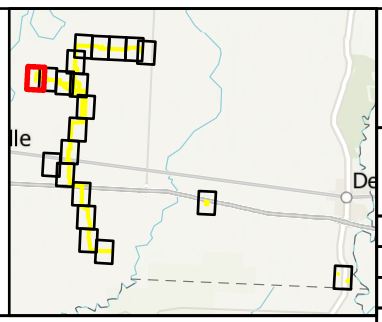

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FIGURE 6
PAGE 15 OF 21

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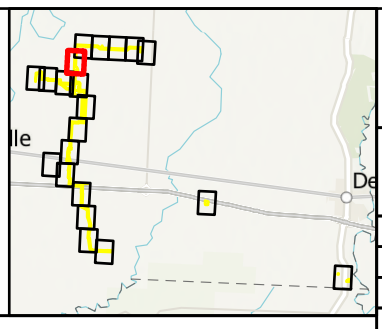

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FIGURE 6
PAGE 16 OF 21

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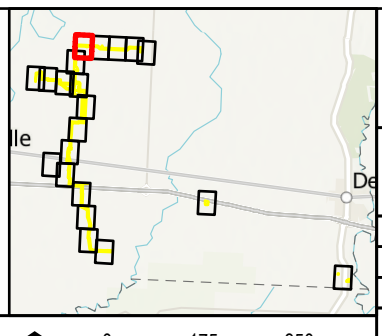

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FIGURE 6
PAGE 17 OF 21

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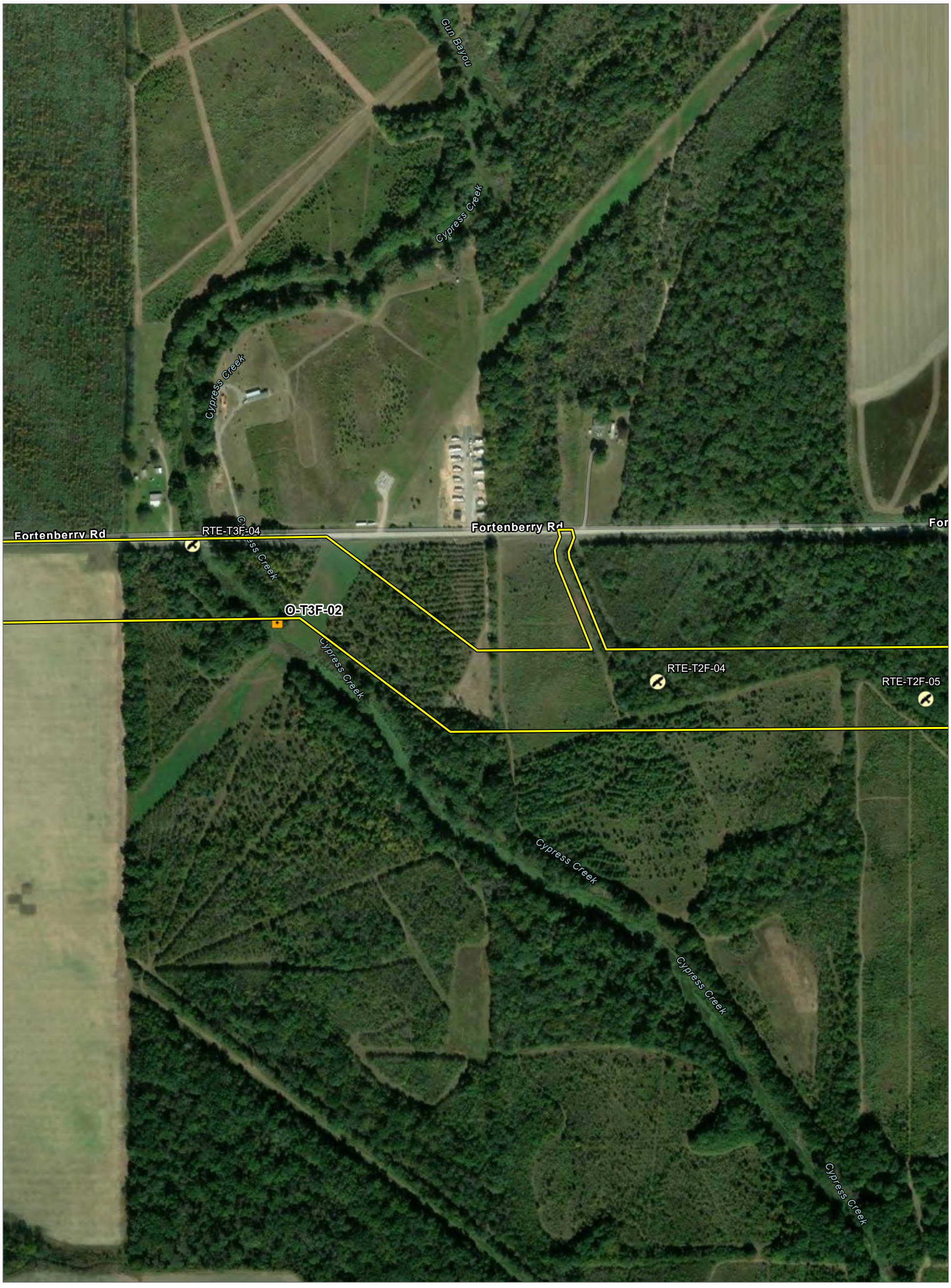

PROJECT BOUNDARY
OBSERVATION
RTE SPECIES

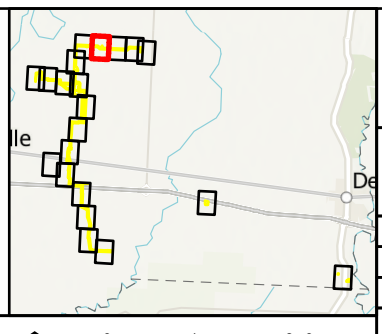

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FIGURE 6
PAGE 18 OF 21

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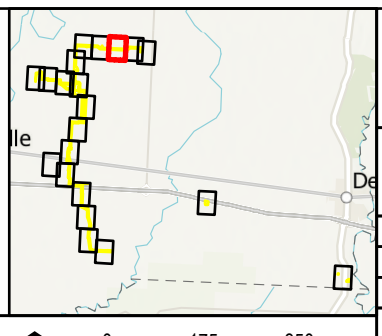

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PAGE 19 OF 21

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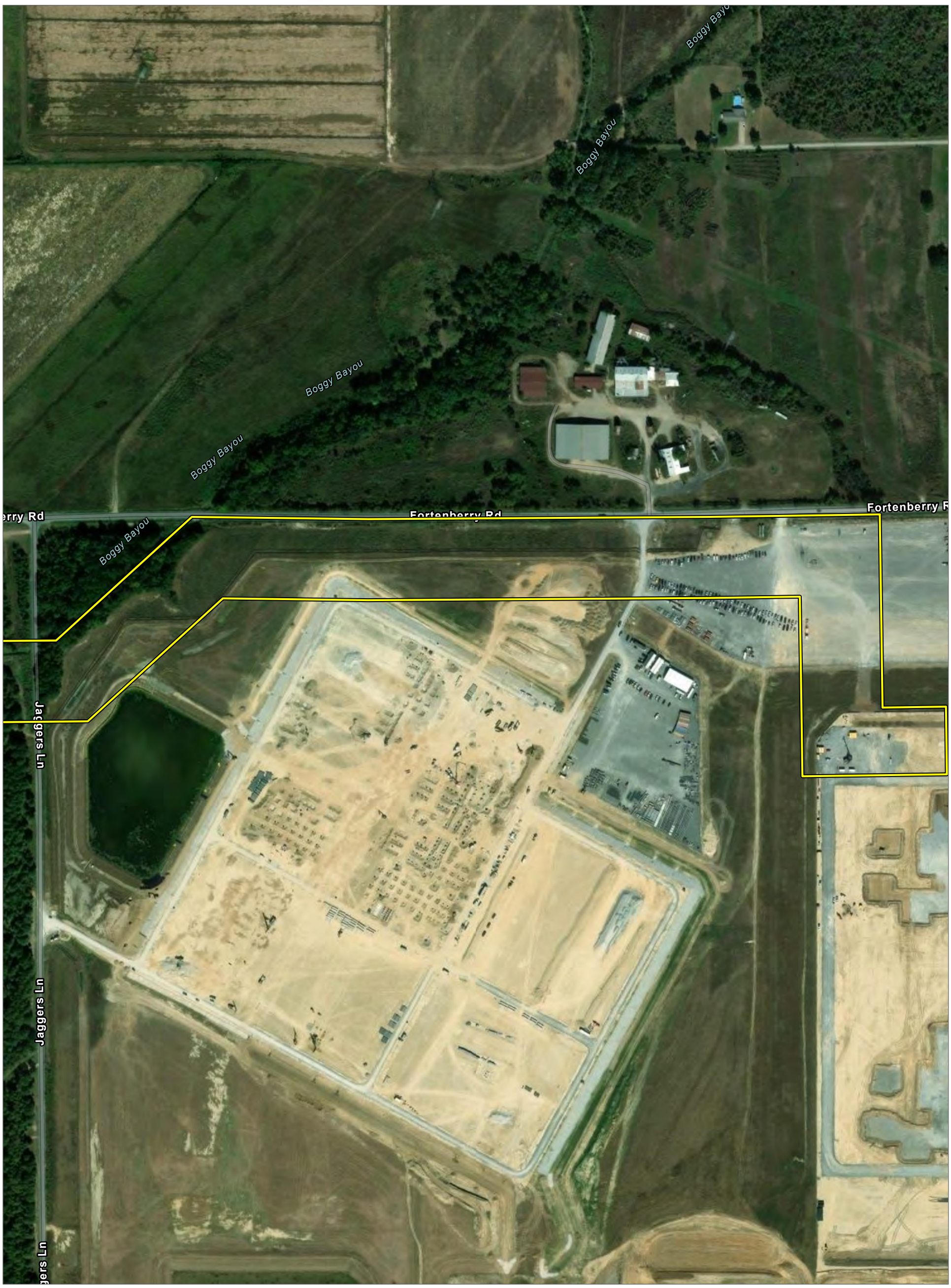

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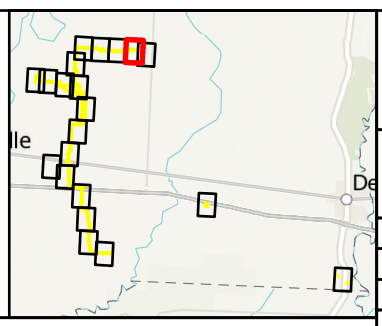

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FIGURE 6
PAGE 20 OF 21

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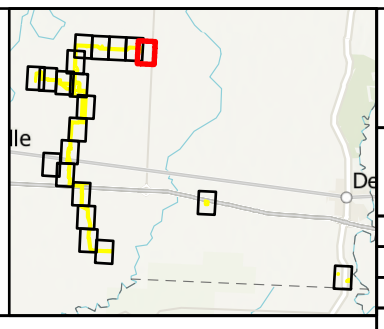

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FIGURE 6
PAGE 21 OF 21

TRC
4545 SHERWOOD COMMON BLVD.
BUILDING 3 - SUITE A
BATON ROUGE, LA 70816
PHONE: 225.216.7483

Appendix B
U.S. Fish and Wildlife Service Official Species List and Determination Key

United States Department of the Interior

FISH AND WILDLIFE SERVICE
Louisiana Ecological Services Field Office
200 Dulles Drive
Lafayette, LA 70506
Phone: (337) 291-3100 Fax: (337) 291-3139

In Reply Refer To: 05/20/2026 21:50:07 UTC
Project Code: 2026-0069606
Project Name: ETC Tiger Pipeline LLC Franklin Farms Project

Subject: List of threatened and endangered species that may occur in your proposed project location or may be affected by your proposed project

To Whom It May Concern:

The enclosed species list identifies threatened, endangered, and candidate species, as well as designated and proposed critical habitat that may occur within the boundary of your proposed project and may be affected by your proposed project. The Fish and Wildlife Service (Service) is providing this list under section 7 (c) of the Endangered Species Act (Act) of 1973, as amended (16 U.S.C. 1531 et seq.). Changes in this species list may occur due to new information from updated surveys, changes in species habitat, new listed species and other factors. Because of these possible changes, feel free to contact our office (337-291-3109) for more information or assistance regarding impacts to federally listed species. The Service recommends visiting the IPaC site or the Louisiana Ecological Services Field Office website (https://www.fws.gov/southeast/lafayette) at regular intervals during project planning and implementation for updated species lists and information. An updated list may be requested through the IPaC system by completing the same process used to receive the enclosed list.

The purpose of the Act is to provide a means whereby threatened and endangered species and the habitats upon which they depend may be conserved. Under sections 7(a)(1) and 7(a)(2) of the Act and its implementing regulations (50 CFR 402 et seq.), Federal agencies are required to utilize their authorities to carry out programs for the conservation of Federal trust resources and to determine whether projects may affect Federally listed species and/or designated critical habitat.

A Biological Assessment is required for construction projects (or other undertakings having similar physical impacts) that are major Federal actions significantly affecting the quality of the human environment as defined in the National Environmental Policy Act (42 U.S.C. 4332(2) (c)).

Bald eagles have recovered and were removed from the List of Endangered and Threatened Species as of August 8, 2007. Although no longer listed, please be aware that bald eagles are protected under the Bald and Golden Eagle Protection Act (BGEPA) (16 U.S.C. 668 et seq.).

The Service developed the National Bald Eagle Management (NBEM) Guidelines to provide landowners, land managers, and others with information and recommendations to minimize potential project impacts to bald eagles, particularly where such impacts may constitute "disturbance", which is prohibited by the BGEPA. A copy of the NBEM Guidelines is available at: https://www.fws.gov/migratorybirds/pdf/management/nationalbaldeaglenanagementguidelines.pdf

Those guidelines recommend: (1) maintaining a specified distance between the activity and the nest (buffer area); (2) maintaining natural areas (preferably forested) between the activity and nest trees (landscape buffers); and (3) avoiding certain activities during the breeding season. Onsite personnel should be informed of the possible presence of nesting bald eagles within the project boundary, and should identify, avoid, and immediately report any such nests to this office. If a bald eagle nest occurs or is discovered within or adjacent to the proposed project area, then an evaluation must be performed to determine whether the project is likely to disturb nesting bald eagles. That evaluation may be conducted on-line at: https://www.fws.gov/southeast/our-services/eagle-technical-assistance/. Following completion of the evaluation, that website will provide a determination of whether additional consultation is necessary. The Division of Migratory Birds for the Southeast Region of the Service (phone: 404/679-7051, e-mail: SEmigratorybirds@fws.gov) has the lead role in conducting any necessary consultation.

Activities that involve State-designated scenic streams and/or wetlands are regulated by the Louisiana Department of Wildlife and Fisheries and the U.S. Army Corps of Engineers, respectively. We, therefore, recommend that you contact those agencies to determine their interest in proposed projects in these areas.

Activities that would be located within a National Wildlife Refuge are regulated by the refuge staff. We, therefore, recommend that you contact them to determine their interest in proposed projects in these areas.

Additional information on Federal trust species in Louisiana can be obtained from the Louisiana Ecological Services website at: https://www.fws.gov/southeast/lafayette

We appreciate your concern for threatened and endangered species. The Service encourages Federal agencies to include conservation of threatened and endangered species into their project planning to further the purposes of the Act. Please include the Consultation Tracking Number in the header of this letter with any request for consultation or correspondence about your project that you submit to our office.

Attachment(s):

- Official Species List
- USFWS National Wildlife Refuges and Fish Hatcheries
- Bald & Golden Eagles
- Migratory Birds

OFFICIAL SPECIES LIST

This list is provided pursuant to Section 7 of the Endangered Species Act, and fulfills the requirement for Federal agencies to "request of the Secretary of the Interior information whether any species which is listed or proposed to be listed may be present in the area of a proposed action".

This species list is provided by:

Louisiana Ecological Services Field Office
200 Dulles Drive
Lafayette, LA 70506
(337) 291-3100

PROJECT SUMMARY

Project Code: 2026-0069606
Project Name: ETC Tiger Pipeline LLC Franklin Farms Project
Project Type: Natural Gas Distribution
Project Description: The proposed Project consists of two new natural gas pipeline laterals, eight new meter stations, and other appurtenant facilities within Richmond Parish, Louisiana.
Project Location:

The approximate location of the project can be viewed in Google Maps: https://www.google.com/maps/@32.4505664,-91.58669576719586,14z

Counties: Richland County, Louisiana

ENDANGERED SPECIES ACT SPECIES

There is a total of 3 threatened, endangered, or candidate species on this species list.

Species on this list should be considered in an effects analysis for your project and could include species that exist in another geographic area. For example, certain fish may appear on the species list because a project could affect downstream species.

IPaC does not display listed species or critical habitats under the sole jurisdiction of NOAA Fisheries[1], as USFWS does not have the authority to speak on behalf of NOAA and the Department of Commerce.

See the "Critical habitats" section below for those critical habitats that lie wholly or partially within your project area under this office's jurisdiction. Please contact the designated FWS office if you have questions.

---

1. NOAA Fisheries, also known as the National Marine Fisheries Service (NMFS), is an office of the National Oceanic and Atmospheric Administration within the Department of Commerce.

MAMMALS

NAME	STATUS
Tricolored Bat *Perimyotis subflavus* No critical habitat has been designated for this species. Species profile: https://ecos.fws.gov/ecp/species/10515	Proposed Endangered

REPTILES

NAME	STATUS
Alligator Snapping Turtle *Macrochelys temminckii* No critical habitat has been designated for this species. Species profile: https://ecos.fws.gov/ecp/species/4658	Proposed Threatened

INSECTS

NAME	STATUS
Monarch Butterfly *Danaus plexippus* There is **proposed** critical habitat for this species. Your location does not overlap the critical habitat. Species profile: https://ecos.fws.gov/ecp/species/9743	Proposed Threatened

CRITICAL HABITATS

THERE ARE NO CRITICAL HABITATS WITHIN YOUR PROJECT AREA UNDER THIS OFFICE'S JURISDICTION.

YOU ARE STILL REQUIRED TO DETERMINE IF YOUR PROJECT(S) MAY HAVE EFFECTS ON ALL ABOVE LISTED SPECIES.

USFWS NATIONAL WILDLIFE REFUGE LANDS AND FISH HATCHERIES

Any activity proposed on lands managed by the National Wildlife Refuge system must undergo a 'Compatibility Determination' conducted by the Refuge. Please contact the individual Refuges to discuss any questions or concerns.

THERE ARE NO REFUGE LANDS OR FISH HATCHERIES WITHIN YOUR PROJECT AREA.

BALD & GOLDEN EAGLES

Bald and Golden Eagles are protected under the Bald and Golden Eagle Protection Act [2] and the Migratory Bird Treaty Act (MBTA) [1]. Any person or organization who plans or conducts activities that may result in impacts to Bald or Golden Eagles, or their habitats, should follow appropriate regulations and consider implementing appropriate avoidance and minimization measures, as described in the various links on this page.

1. The Bald and Golden Eagle Protection Act of 1940.
2. The Migratory Birds Treaty Act of 1918.
3. 50 C.F.R. Sec. 10.12 and 16 U.S.C. Sec. 668(a)

Bald and Golden Eagles are protected under the Bald and Golden Eagle Protection Act and the Migratory Bird Treaty Act (MBTA). Any person or organization who plans or conducts activities that may result in impacts to Bald or Golden Eagles, or their nests, should follow appropriate regulations and implement required avoidance and minimization measures, as described in the various links on this page.

The data in this location indicates that no eagles have been observed in this area. This does not mean eagles are not present in your project area, especially if the area is difficult to survey. Please review the 'Steps to Take When No Results Are Returned' section of the Supplemental Information on Migratory Birds and Eagles document to determine if your project is in a poorly surveyed area. If it is, you may need to rely on other resources to determine if eagles may be present (e.g. your local FWS field office, state surveys, your own surveys).

Any person or organization who plans or conducts activities that may result in impacts to bald or golden eagles, or their habitats, should follow appropriate regulations and consider implementing appropriate conservation measures, as described in the links below. Specifically, please review the "Supplemental Information on Migratory Birds and Eagles".

MIGRATORY BIRDS

The Migratory Bird Treaty Act (MBTA) [1] prohibits the take (including killing, capturing, selling, trading, and transport) of protected migratory bird species without prior authorization by the Department of Interior U.S. Fish and Wildlife Service (Service).

1. The Migratory Birds Treaty Act of 1918.
2. The Bald and Golden Eagle Protection Act of 1940.
3. 50 C.F.R. Sec. 10.12 and 16 U.S.C. Sec. 668(a)

For guidance on when to schedule activities or implement avoidance and minimization measures to reduce impacts to migratory birds on your list, see the "Probability of Presence Summary" below to see when these birds are most likely to be present and breeding in your project area.

NAME	BREEDING SEASON
Chimney Swift *Chaetura pelagica* This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska. https://ecos.fws.gov/ecp/species/9406	Breeds Mar 15 to Aug 25

NAME	BREEDING SEASON
Kentucky Warbler *Geothlypis formosa* This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska. https://ecos.fws.gov/ecp/species/9443	Breeds Apr 20 to Aug 20
Le Conte's Sparrow *Ammospiza leconteii* This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska. https://ecos.fws.gov/ecp/species/9469	Breeds elsewhere
Lesser Yellowlegs *Tringa flavipes* This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska. https://ecos.fws.gov/ecp/species/9679	Breeds elsewhere
Little Blue Heron *Egretta caerulea* This is a Bird of Conservation Concern (BCC) only in particular Bird Conservation Regions (BCRs) in the continental USA https://ecos.fws.gov/ecp/species/9477	Breeds Mar 10 to Oct 15
Prothonotary Warbler *Protonotaria citrea* This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska. https://ecos.fws.gov/ecp/species/9439	Breeds Apr 1 to Jul 31
Red-headed Woodpecker *Melanerpes erythrocephalus* This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska. https://ecos.fws.gov/ecp/species/9398	Breeds May 10 to Sep 10
Wood Thrush *Hylocichla mustelina* This is a Bird of Conservation Concern (BCC) throughout its range in the continental USA and Alaska. https://ecos.fws.gov/ecp/species/9431	Breeds May 10 to Aug 31

PROBABILITY OF PRESENCE SUMMARY

The graphs below provide our best understanding of when birds of concern are most likely to be present in your project area. This information can be used to tailor and schedule your project activities to avoid or minimize impacts to birds. Please make sure you read "Supplemental Information on Migratory Birds and Eagles", specifically the FAQ section titled "Proper Interpretation and Use of Your Migratory Bird Report" before using or attempting to interpret this report.

Probability of Presence (■)

Green bars; the bird's relative probability of presence in the 10km grid cell(s) your project overlaps during that week of the year.

Breeding Season (■)

Yellow bars; liberal estimate of the timeframe inside which the bird breeds across its entire range.

Survey Effort (|)
Vertical black lines; the number of surveys performed for that species in the 10km grid cell(s) your project area overlaps.

No Data (–)
A week is marked as having no data if there were no survey events for that week.

Additional information can be found using the following links:

- Eagle Management https://www.fws.gov/program/eagle-management
- Measures for avoiding and minimizing impacts to birds https://www.fws.gov/library/collections/avoiding-and-minimizing-incidental-take-migratory-birds
- Nationwide avoidance and minimization measures for birds

- Supplemental Information for Migratory Birds and Eagles in IPaC https://www.fws.gov/media/supplemental-information-migratory-birds-and-bald-and-golden-eagles-may-occur-project-action

IPAC USER CONTACT INFORMATION

Agency: Private Entity
Name: Melissa Cross
Address: 11767 Katy Fwy
Address Line 2: Suite 850
City: Houston
State: TX
Zip: 77079
Email mcross@trccompanies.com
Phone: 3612783850

LEAD AGENCY CONTACT INFORMATION

Lead Agency: Federal Energy Regulatory Commission

United States Department of the Interior

FISH AND WILDLIFE SERVICE
Louisiana Ecological Services Field Office
200 Dulles Drive
Lafayette, LA 70506
Phone: (337) 291-3100 Fax: (337) 291-3139

In Reply Refer To: 05/20/2026 21:34:31 UTC
Project code: 2026-0069606
Project Name: ETC Tiger Pipeline LLC Franklin Farms Project

Federal Nexus: yes
Federal Action Agency (if applicable): Federal Energy Regulatory Commission

Subject: Technical assistance for 'ETC Tiger Pipeline LLC Franklin Farms Project'

Dear Melissa Cross:

This letter records your determination using the Information for Planning and Consultation (IPaC) system provided to the U.S. Fish and Wildlife Service (Service) on May 20, 2026, for 'ETC Tiger Pipeline LLC Franklin Farms Project' (here forward, Project). This project has been assigned Project Code 2026-0069606 and all future correspondence should clearly reference this number. **Please carefully review this letter. Your Endangered Species Act (Act) requirements are not complete.**

Ensuring Accurate Determinations When Using IPaC

The Service developed the IPaC system and associated species' determination keys in accordance with the Endangered Species Act of 1973 (ESA; 87 Stat. 884, as amended; 16 U.S.C. 1531 et seq.) and based on a standing analysis. All information submitted by the Project proponent into IPaC must accurately represent the full scope and details of the Project. **Failure to accurately represent or implement the Project as detailed in IPaC or the Northern Long-eared Bat and Tricolored Bat Range-wide Determination Key (Dkey), invalidates this letter.**

Determination for the Northern Long-Eared Bat and Tricolored Bat

Based on your IPaC submission and a standing analysis completed by the Service, you determined the proposed Project will have the following effect determinations:

Species	Listing Status	Determination
Tricolored Bat (*Perimyotis subflavus*)	Proposed Endangered	May affect

Federal agencies must consult with U.S. Fish and Wildlife Service under section 7(a)(2) of the Endangered Species Act (ESA) when an action *may affect* a listed species. Tricolored bat is proposed for listing as endangered under the ESA, but not yet listed. For actions that may affect a proposed species, agencies cannot consult, but they can *confer* under the authority of section 7(a)(4) of the ESA. Such conferences can follow the procedures for a consultation and be adopted as such if and when the proposed species is listed. Should the tricolored bat be listed, agencies must review projects that are not yet complete, or projects with ongoing effects within the tricolored bat range that previously received a NE or NLAA determination from the key to confirm that the determination is still accurate. Projects that receive a may affect determination for tricolored bat through the key, should contact the appropriate Ecological Services Field Office if they want to conference on this species.

Other Species and Critical Habitat that May be Present in the Action Area

The IPaC-assisted determination key for the northern long-eared bat and tricolored bat does not apply to the following ESA-protected species and/or critical habitat that also may occur in your Action area:

- Alligator Snapping Turtle *Macrochelys temminckii* Proposed Threatened
- Monarch Butterfly *Danaus plexippus* Proposed Threatened

You may coordinate with our Office to determine whether the Action may cause prohibited take of the species listed above.

Conclusion

Consultation with the Service is not complete. Further consultation or coordination with the Service is necessary for those species or designated critical habitats with a determination of “May Affect.” A “May Affect” determination in this key indicates that the project, as entered, is not consistent with the questions in the key. Not all projects that reach a “May Affect” determination are anticipated to result in adverse impacts to listed species. These projects may result in a “No Effect”, “May Affect, Not Likely to Adversely Affect”, or “May Affect, Likely to Adversely Affect” determination depending on the details of the project. Please contact our Louisiana Ecological Services Field Office to discuss methods to avoid or minimize potential adverse effects to those species or designated critical habitats.

Action Description
You provided to IPaC the following name and description for the subject Action.

1. Name

ETC Tiger Pipeline LLC Franklin Farms Project

2. Description

The following description was provided for the project 'ETC Tiger Pipeline LLC Franklin Farms Project':

> The proposed Project consists of two new natural gas pipeline laterals, eight new meter stations, and other appurtenant facilities within Richmond Parish, Louisiana.

The approximate location of the project can be viewed in Google Maps: https://www.google.com/maps/@32.4505664,-91.58669576719586,14z

DETERMINATION KEY RESULT

Based on the answers provided, the proposed Action is consistent with a determination of “may affect” for a least one species covered by this determination key.

QUALIFICATION INTERVIEW

1. Does the proposed project include, or is it reasonably certain to cause, intentional take of listed bats or any other listed species?

   Note: Intentional take is defined as take that is the intended result of a project. Intentional take could refer to research, direct species management, surveys, and/or studies that include intentional handling/encountering, harassment, collection, or capturing of any individual of a federally listed threatened, endangered or proposed species?

   No

2. Is the action area wholly within Zone 2 of the year-round active area for northern long-eared bat and/or tricolored bat?

   Automatically answered
   No

3. Does the action area intersect Zone 1 of the year-round active area for northern long-eared bat and/or tricolored bat?

   Automatically answered
   Yes

4. Your project overlaps with an area where northern long-eared bats or tricolored bats may be present and roosting in trees year-round.

   Do you understand that your project may impact bats roosting in trees at any time during the year?

   Yes

5. Does any component of the action involve leasing, construction or operation of wind turbines? Answer 'yes' if the activities considered are conducted with the intention of gathering survey information to inform the leasing, construction, or operation of wind turbines.

   No

6. Is the proposed action authorized, permitted, licensed, funded, or being carried out by a Federal agency in whole or in part?

   Note for projects in Pennsylvania: Projects requiring authorization under Section 404 of the Clean Water Act and/or Section 10 of the Rivers and Harbors Act would be considered as having a federal nexus. Since the U.S. Army Corps of Engineers (Corps) has issued the Pennsylvania State Programmatic General Permit (PASPGP), which may be verified by the PA Department of Environmental Protection or certain Conservation Districts, the need to receive a Corps authorization to perform the work under the PASPGP serves as a federal nexus. As such, if proposing to use the PASPGP, you would answer 'yes' to this question.

   Yes

7. Is the Federal Highway Administration (FHWA), Federal Railroad Administration (FRA), or Federal Transit Administration (FTA) funding or authorizing the proposed action, in whole or in part?

   No

8. Are you an employee of the federal action agency or have you been officially designated in writing by the agency as its designated non-federal representative for the purposes of Endangered Species Act Section 7 informal consultation per 50 CFR § 402.08?

   Note: This key may be used for federal actions and for non-federal actions to facilitate section 7 consultation and to help determine whether an incidental take permit may be needed, respectively. This question is for information purposes only.

   No

9. Is the lead federal action agency the Environmental Protection Agency (EPA) or Federal Communications Commission (FCC)? Is the Environmental Protection Agency (EPA) or Federal Communications Commission (FCC) funding or authorizing the proposed action, in whole or in part?

   No

10. Is the lead federal action agency the Federal Energy Regulatory Commission (FERC)?

    Yes

11. Is FERC reviewing the proposed action under the Natural Gas Act, in whole or in part?

    Yes

12. [Semantic] Is the action area located within 0.5 miles of a known bat hibernaculum or winter roost? Note: The map queried for this question contains proprietary information and cannot be displayed. If you need additional information, please contact your state wildlife agency.

    Automatically answered
    No

13. Does the action area contain any winter roosts or caves (or associated sinkholes, fissures, or other karst features), mines, rocky outcroppings, or tunnels that could provide habitat for hibernating bats?

    No

14. Is the action located in New England (CT, MA, ME, NH, RI, or VT)?

    No

15. Will the action cause effects to a bridge?

    Note: Covered bridges should be considered as bridges in this question.

    No

16. Will the action result in effects to a culvert or tunnel at any time of year?

    No

17. Are trees present within 1000 feet of the action area? **Note:** If there are trees within the action area that are of a sufficient size to be potential roosts for bats answer "Yes". If unsure, additional information defining suitable summer habitat for the northern long-eared bat and tricolored bat can be found in the USFWS' Range-wide Indiana Bat and Northern long-eared bat Survey Guidelines.

    Yes

18. Does the action include the intentional exclusion of bats from a building or building-like structure? **Note:** Exclusion is conducted to deny bats' entry or reentry into a building. To be effective and to avoid harming bats, it should be done according to established standards. If your action includes bat exclusion and you are unsure whether northern long-eared bats or tricolored bats are present, answer "Yes." Answer "No" if there are no signs of bat use in the building/structure. If unsure, contact your local Ecological Services Field Office to help assess whether northern long-eared bats or tricolored bats may be present. Contact a Nuisance Wildlife Control Operator (NWCO) for help in how to exclude bats from a structure safely without causing harm to the bats (to find a NWCO certified in bat standards, search the Internet using the search term "National Wildlife Control Operators Association bats"). Also see the White-Nose Syndrome Response Team's guide for bat control in structures.

    No

19. Does the action involve removal, modification, or maintenance of a human-made building-like structure (barn, house, or other building) **known or suspected to contain roosting bats?**

    No

20. Will the action cause construction of one or more new roads open to the public?

    For federal actions, answer 'yes' when the construction or operation of these facilities is either (1) part of the federal action or (2) would not occur but for an action taken by a federal agency (federal permit, funding, etc.).

    No

21. Will the action include or cause any construction or other activity that is reasonably certain to increase average night-time traffic permanently or temporarily on one or more existing roads? **Note:** For federal actions, answer 'yes' when the construction or operation of these facilities is either (1) part of the federal action or (2) would not occur but for an action taken by a federal agency (federal permit, funding, etc.). .

    No

22. Will the action include or cause any construction or other activity that is reasonably certain to increase the number of travel lanes on an existing thoroughfare?

    For federal actions, answer 'yes' when the construction or operation of these facilities is either (1) part of the federal action or (2) would not occur but for an action taken by a federal agency (federal permit, funding, etc.).

    No

23. Will the proposed Action involve the creation of a new water-borne contaminant source (e.g., leachate pond, pits containing chemicals that are not NSF/ANSI 60 compliant)?

    Note: For information regarding NSF/ANSI 60 please visit https://www.nsf.org/knowledge-library/nsf-ansi-standard-60-drinking-water-treatment-chemicals-health-effects

    No

24. Will the proposed action involve the creation of a new point source discharge from a facility other than a water treatment plant or storm water system?

    No

25. Will the action include drilling or blasting?

    No

26. Will the action involve military training (e.g., smoke operations, obscurant operations, exploding munitions, artillery fire, range use, helicopter or fixed wing aircraft use at night)?

    No

27. Will the proposed action involve the use of herbicides or pesticides (e.g., fungicides, insecticides, or rodenticides)?

    No

28. Will the action include or cause activities that are reasonably certain to cause chronic or intense nighttime noise (above current levels of ambient noise in the area) in suitable summer habitat for the northern long-eared bat or tricolored bat during the active season? Chronic noise is noise that is continuous or occurs repeatedly again and again for a long time. Sources of chronic or intense noise that could cause adverse effects to bats may include, but are not limited to: road traffic; trains; aircraft; industrial activities; gas compressor stations; loud music; crowds; oil and gas extraction; construction; and mining. **Note:** Additional information defining suitable summer habitat for the northern long-eared bat and tricolored bat can be found in the USFWS' Range-wide Indiana Bat and Northern long-eared bat Survey Guidelines.

    No

29. Does the action include, or is it reasonably certain to cause, the use of permanent or temporary artificial lighting within 1000 feet of suitable forested northern long-eared bat or tricolored bat roosting habitat? **Note:** Additional information defining suitable summer habitat for the northern long-eared bat and tricolored bat can be found in the USFWS' Range-wide Indiana Bat and Northern long-eared bat Survey Guidelines.

    No

30. Will the action include tree cutting or other means of knocking down or bringing down trees, tree topping, or tree trimming?

    Yes

31. Is the project related to the production of coal, including projects that support the mining of coal, as well as the production and/or distribution of energy produced from coal?

    No

32. Will the proposed action occur exclusively in an already established and currently maintained utility right-of-way?

    No

33. Does the action include emergency cutting or trimming of hazard trees in order to remove an imminent threat to human safety or property? See hazard tree note at the bottom of the key for text that will be added to response letters

    Note: A "hazard tree" is a tree that is an immediate threat to lives, public health and safety, or improved property.

    No

34. Does the project intersect with the 0- 9.9% forest density category?

    Automatically answered
    Yes

35. Does the project intersect with the 10.0- 19.9% forest density category map?

    Automatically answered
    Yes

36. Does the project intersect with the 20.0- 29.9% forest density category map?

    Automatically answered
    Yes

37. Does the project intersect with the 30.0- 100% forest density category map?

    Automatically answered
    Yes

38. Will the action cause trees to be cut, knocked down, or otherwise brought down across an area greater than 0.5 acre in total extent?

    Yes

39. Does the action area intersect the tricolored bat species list area?

    Automatically answered
    Yes

40. Is the action area located within 0.5-mile of radius of an entrance/opening to any known tricolored bat hibernacula or winter roost?

    Note: The map queried for this question contains proprietary information and cannot be displayed. If you need additional information, please contact your state wildlife agency.

    Automatically answered
    No

41. [Semantic] Is the action area located within 0.25 miles of a culvert that is known to be occupied by northern long-eared or tricolored bats? **Note:** The map queried for this question contains proprietary information and cannot be displayed. If you need additional information, please contact your State wildlife agency.

    Automatically answered
    No

42. Your project overlaps with an area where tricolored bats may be present and roosting in trees year-round.

    Has a presence/probable absence survey for the tricolored bat following the Service's Range-wide Indiana Bat and Northern Long-Eared Bat Survey Guidelines been conducted within the project area? If unsure, answer "No."

    No

43. Your project overlaps with an area where tricolored bats may be present and roosting in trees year-round. Is suitable tricolored bat habitat present within 1000 feet of project activities? Note: If there are trees within the action area that may provide potential roosts for tricolored bats (e.g., clusters of leaves in live and dead deciduous trees, Spanish moss (*Tillandsia usneoides*), clusters of dead pine needles of large live pines) answer "Yes." Additional information defining suitable summer habitat for the northern long-eared bat and tricolored bat can be found in the USFWS' Range-wide Indiana Bat and Northern long-eared bat Survey Guidelines.

    Yes

44. Do you have any documents that you want to include with this submission?

    No

PROJECT QUESTIONNAIRE

Enter the extent of the action area (in acres) from which trees will be removed - round up to the nearest tenth of an acre. For this question, include the entire area where tree removal will take place, even if some live or dead trees will be left standing.

16.1

If tree cutting will be part of the proposed action, at what time of year will tree cutting be conducted?

Outside of maternity season and the hibernation period.

Can the proposed action be modified to ensure that tree cutting will be conducted outside of the season when bats will be present and roosting in trees?

Yes

IPAC USER CONTACT INFORMATION

Agency: Private Entity
Name: Melissa Cross
Address: 11767 Katy Fwy
Address Line 2: Suite 850
City: Houston
State: TX
Zip: 77079
Email mcross@trccompanies.com
Phone: 3612783850

LEAD AGENCY CONTACT INFORMATION

Lead Agency: Federal Energy Regulatory Commission

Rare Species and Natural Communities by Parish

RARE, THREATENED, AND ENDANGERED RANKS AND STATUSES

RARE ANIMALS TRACKING LIST (PDF)

RARE PLANTS TRACKING LIST (PDF)

RARE NATURAL COMMUNITIES TRACKING LIST (PDF)

RARE SPECIES AND NATURAL COMMUNITIES TRACKING LIST (PDF)

RARE ANIMAL SPECIES FACT SHEETS

RARE PLANT SPECIES FACT SHEETS

NATURAL COMMUNITIES FACT SHEETS

Filter by ELEMENT TYPE

<Any>

Filter by GLOBAL RANK

<Any>

Filter by STATE RANK

<Any>

Filter by FEDERAL STATUS

<Any>

Filter by STATE STATUS

<Any>

Filter by PARISH

Richland

Filter by FACT SHEET

<Any>

Filter by IMPERILED OR CRITICALLY IMPERILED

<Any>

Rare Species and Natural Communities by Parish

Alligator Snapping Turtle	Macrochelys temminckii	Reptile	G3	S3	Proposed Threatened	Restricted	Acadia, Allen, Ascension, Avoyelles, Beauregard, Bienville, Bossier, Caddo, Calcasieu, Caldwell, Catahoula, Concordia, De Soto, East Baton Rouge, East Carroll, Franklin, Grant, Iberia, Iberville, Jefferson, La Salle, Lafayette, Lafourche, Livingston, Madison, Morehouse, Natchitoches, Ouachita, Rapides, Red River, Richland, Sabine, St. Charles, St. John the Baptist, St. Landry, St. Martin, St. Tammany, Tangipahoa, Tensas, Terrebonne, Union, Vernon, Washington, West Feliciana, Winn	Yes	Yes

Bald Eagle	Haliaeetus leucocephalus	Bird	G5	S3	Delisted	Delisted	Ascension, Assumption, Avoyelles, Beauregard, Bienville, Bossier, Caddo, Calcasieu, Caldwell, Cameron, Catahoula, Claiborne, Concordia, De Soto, East Baton Rouge, Franklin, Grant, Iberia, Iberville, Jackson, Jefferson, La Salle, Lafourche, Livington, Morehouse, Natchitoches, Orleans, Ouachita, Plaquemines, Pointe Coupee, Rapides, Red River, Richland, Sabine, St. Bernard, St. Charles, St. James, St. John the Baptist, St. Landry, St. Martin, St. Mary, St. Tammany, Tangipahoa, Tensas, Terrebonne, Union, Vermilion, West Baton Rouge, West Feliciana	Yes	
Black Sandshell	Ligumia recta	Mollusk	G4G5	S1			Morehouse, Ouachita, Richland, Washington		Yes

Bottomland Hardwood Forest	Bottomland hardwood forest	Natural Community	G4G5	S4			Avoyelles, Bossier, Caddo, Calcasieu, Caldwell, Catahoula, Concordia, East Baton Rouge, East Carroll, Franklin, Grant, Iberville, Lincoln, Livingston, Madison, Natchitoches, Orleans, Plaquemines, Rapides, Richland, Sabine, St. Landry, St. Martin, St. Tammany, Tangipahoa, Tensas, Union, Vernon, Webster, West Baton Rouge, West Carroll, Winn	Yes	
Cypress Swamp	Cypress swamp	Natural Community	G4G5	S4			Ascension, Bienville, Bossier, Catahoula, Evangeline, Franklin, Iberia, Iberville, Rapides, Richland, St. Landry, St. Martin, St. Mary, Tangipahoa, Vermilion, Webster	Yes	
Ebonyshell	Reginaia ebenus	Mollusk	G4G5	S3			Madison, Morehouse, Richland, St. Tammany, Union		
Mesic Hardwood Flatwoods	Mesic hardwood flatwoods	Natural Community	G1G2?	S2S3			East Carroll, Franklin, Morehouse, Richland, West Carroll		Yes

Mixed Hardwood-loblolly Forest	Mixed hardwood-loblolly forest	Natural Community	G3G4	S3			Allen, Bienville, Bossier, Caddo, Caldwell, Catahoula, Claiborne, East Feliciana, Evangeline, Franklin, Grant, Jackson, La Salle, Lincoln, Natchitoches, Ouachita, Rapides, Richland, Sabine, St. Tammany, Tangipahoa, Union, Vernon, Washington, Webster, West Carroll, West Feliciana, Winn	Yes	
Northern Long-eared Bat	Myotis septentrionalis	Mammal	G2G3	S1	Threatened	Threatened	Avoyelles, Bienville, Bossier, Caldwell, Catahoula, De Soto, Grant, Jackson, La Salle, Morehouse, Natchitoches, Ouachita, Rapides, Richland, Sabine, Union, Vernon, Webster, West Feliciana, Winn		Yes
Northern Red Oak	Quercus rubra	Plant	G5	S1S3			Caddo, Caldwell, De Soto, East Carroll, Morehouse, Ouachita, Rapides, Richland, St. Tammany, Union, Washington, West Carroll		Yes
Purple Coneflower	Echinacea purpurea	Plant	G4	S2			Caddo, Caldwell, La Salle, Morehouse, Natchitoches, Richland, Tensas, Vernon, Washington, Winn		Yes

Pyramid Pigtoe	Pleurobema rubrum	Mollusk	G2G3	S2			Franklin, Madison, Morehouse, Richland, Union		Yes
Silty Hornsnail	Pleurocera canaliculata	Mollusk	G5	S2			Madison, Morehouse, Richland		Yes
Smooth Softshell	Apalone mutica	Reptile	G5	S3			Allen, Avoyelles, Beauregard, Bossier, Caddo, Caldwell, Catahoula, Concordia, De Soto, East Baton Rouge, Franklin, Grant, Iberville, La Salle, Madison, Morehouse, Natchitoches, Ouachita, Pointe Coupee, Red River, Richland, Sabine, St. Landry, St. Martin, Union, Vernon, West Feliciana, Winn		
Southern Crawfish Frog	Lithobates areolatus areolatus	Amphibian	G4T4	S1			Acadia, Allen, Beauregard, Caddo, Ouachita, Rapides, Richland, Vernon, Webster		Yes
Spike	Eurynia dilatata	Mollusk	G5	S2S3			Madison, Morehouse, Richland, Winn		Yes
Timber Rattlesnake	Crotalus horridus	Reptile	G4	S3S4			Ascension, Bienville, Caldwell, Catahoula, East Carroll, Grant, Lafourche, Morehouse, Natchitoches, Ouachita, Richland, Tensas, Vernon, Winn		

Waterbird Nesting Colony	Colonial Waterbird Nesting Area	Animal Aggregation	GNR	SNR			Acadia, Allen, Ascension, Assumption, Avoyelles, Beauregard, Bossier, Caddo, Calcasieu, Caldwell, Cameron, Catahoula, Concordia, Evangeline, Franklin, Grant, Iberia, Iberville, Jefferson, Jefferson Davis, Lafourche, Livingston, Madison, Morehouse, Natchitoches, Orleans, Ouachita, Plaquemines, Pointe Coupee, Rapides, Red River, Richland, Sabine, St. Bernard, St. Charles, St. James, St. John the Baptist, St. Landry, St. Martin, St. Mary, St. Tammany, Tangipahoa, Tensas, Terrebonne, Vermilion, Vernon, Washington, Webster, West Baton Rouge, West Feliciana		

Western Chicken Turtle	Deirochelys reticularia miaria	Reptile	G5T5	S2			Acadia, Allen, Avoyelles, Beauregard, Caddo, Calcasieu, Caldwell, Cameron, Catahoula, Concordia, De Soto, East Carroll, Evangeline, Franklin, Iberia, Iberville, Jefferson Davis, Lincoln, Morehouse, Natchitoches, Ouachita, Pointe Coupee, Rapides, Richland, St. John the Baptist, St. Landry, St. Martin, Union, Vermilion, Vernon, West Baton Rouge, West Carroll, Winn		Yes
Yellowleaf Tinker's-weed	Triosteum angustifolium	Plant	G5	S2			Bossier, Caddo, Morehouse, Richland, West Carroll		Yes

Louisiana Department of Wildlife and Fisheries
PO Box 98000
2000 Quail Drive
Baton Rouge, LA 70898
800.256.2749
225.765.2800
CONTACT US

Administration
Commissions, Task Forces, & Councils
Resources

Appendix C
Photolog

(Filed under separate cover as Critical Energy Infrastructure Information)

APPENDIX G

Cultural Resource Assessment Survey Reports

Submitted as CUI/PRIVILEGED AND CONFIDENTIAL – DO NOT RELEASE

APPENDIX H

Air Quality Calculations (Construction Emissions)

Construction Emissions

Franklin Farms Project
All Emissions are in Tons per Construction Period

Project Total Construction Emissions	NO_X	CO	VOC	PM_{10}	$PM_{2.5}$	SO_2	CO_2	CH_4	N_2O	CO_2e	Total HAPs
Non-Road Equipment - Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral	6.82	2.21	0.61	0.43	0.42	0.02	9,056.98	0.0091	0.14	9,094.66	0.23
Non-Road Equipment - Franklin Farms Gulf Run Bi Di Meter Station	1.31	0.48	0.13	0.09	0.08	0.00	1,525.27	0.0020	0.02	1,531.87	0.05
Non-Road Equipment - Franklin Farms Gulf South Receipt Meter Station	1.31	0.48	0.13	0.09	0.08	0.00	1,525.27	0.0020	0.02	1,531.87	0.05
Non-Road Equipment - Franklin Farms Gulf Crossing Receipt Meter Station	1.31	0.48	0.13	0.09	0.08	0.00	1,525.27	0.0020	0.02	1,531.87	0.05
Non-Road Equipment - Franklin Farms MEP Receipt Meter Station	1.31	0.48	0.13	0.09	0.08	0.00	1,525.27	0.0020	0.02	1,531.87	0.05
Non-Road Equipment - Franklin Farms Delivery Meter Station	1.31	0.48	0.13	0.09	0.08	0.00	1,525.27	0.0020	0.02	1,531.87	0.05
Non-Road Equipment - Franklin Farms Bee Bayou Delivery Meter Station	1.31	0.48	0.13	0.09	0.08	0.00	1,525.27	0.0020	0.02	1,531.87	0.05
Non-Road Equipment - Franklin Farms Trio Farm Delivery Meter Station	1.31	0.48	0.13	0.09	0.08	0.00	1,525.27	0.0020	0.02	1,531.87	0.05
Construction Activity Fugitive Dust				451.34	67.70						
Roadway Fugitive Dust				171.70	37.34						
On-Road Equipment	53.83	183.62	10.24	4.02	2.78	0.13	21370.44	0.67	0.95	21641.29	0.70
Project Total Construction Emissions	**69.81**	**189.20**	**11.77**	**628.11**	**108.84**	**0.18**	**41,104.28**	**0.69**	**1.27**	**41,459.05**	**1.27**

Franklin Farms Project
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral
Construction Emissions - Off-road Engines **Criteria and GHG Emissions**

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Weeks	Hours per Week	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	Emission Factors EF (g/hp-hr) [1] HC	VOC	CO	PM_{10}	$PM_{2.5}$	NOx	SO_2	CO_2	N_2O EF [3] g/MMBtu	VOC [4] tons	CO [4] tons	PM_{10} tons [4]	$PM_{2.5}$ tons [4]	SO_2 [4] tons	NOx [4] tons	CO_2 [4] tons	CH_4 [5] tons	N_2O [6] tons	GHG [7] tons of CO_2e
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	4	10	60	2,400	0.367	0.43	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.023	0.08	0.018	0.018	0.0002	0.30	75.45	0.0003	0.001	75.84
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	9	27	60	14,580	0.371	0.59	0.012	0.013	0.041	0.011	0.010	0.161	0.001	536.793	0.6	0.024	0.08	0.020	0.019	0.0027	0.31	1018.01	0.0004	0.014	1021.73
Dump Truck	Off-highway Trucks	Diesel	400	2	19	60	2,280	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.007	0.02	0.005	0.005	0.0008	0.08	318.39	0.0001	0.004	319.56
Excavator (KOM 210)	Excavators	Diesel	160	5	19	60	5,700	0.371	0.59	0.010	0.011	0.066	0.015	0.015	0.212	0.001	536.802	0.6	0.006	0.04	0.009	0.009	0.0008	0.13	318.40	0.0001	0.004	319.56
Excavator (CAT 329)	Excavators	Diesel	200	4	18	60	4,320	0.371	0.59	0.011	0.011	0.030	0.008	0.008	0.136	0.001	536.798	0.6	0.006	0.02	0.005	0.005	0.0008	0.08	301.64	0.0001	0.004	302.74
Excavator (CAT 336)	Excavators	Diesel	314	12	22	60	15,840	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.060	0.29	0.061	0.059	0.0047	0.77	1736.35	0.0009	0.024	1742.71
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	31	60	1,860	0.371	0.59	0.017	0.018	0.070	0.017	0.016	0.214	0.001	536.779	0.6	0.004	0.02	0.004	0.004	0.0003	0.05	129.87	0.0001	0.002	130.34
Grader	Graders	Diesel	250	1	31	60	1,860	0.371	0.59	0.012	0.013	0.039	0.010	0.010	0.156	0.001	536.794	0.6	0.004	0.01	0.003	0.003	0.0004	0.05	162.34	0.0001	0.002	162.93
HDD Rig	Bore/Drill Rigs	Diesel	300	1	7	60	420	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.013	0.04	0.008	0.008	0.0001	0.16	31.68	0.0002	0.001	31.84
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	1	20	60	1,200	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.036	0.11	0.022	0.022	0.0003	0.45	90.51	0.0005	0.002	90.98
Cranes	Cranes	Diesel	300	1	4	40	160	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.000	0.00	0.000	0.000	0.0000	0.01	12.08	0.0000	0.000	12.14
Generator Sets	Generator Sets	Diesel	200	2	31	40	2,480	0.367	0.43	0.158	0.171	0.517	0.111	0.108	2.094	0.002	530.549	0.6	0.040	0.12	0.026	0.025	0.0004	0.49	124.73	0.0006	0.002	125.37
Pump	Pumps	Diesel	120	1	15	30	450	0.367	0.43	0.174	0.188	0.627	0.143	0.139	2.227	0.002	530.499	0.6	0.005	0.02	0.004	0.004	0.0000	0.06	13.58	0.0001	0.000	13.65
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	2	31	60	3,720	0.433	0.21	0.244	0.266	0.872	0.173	0.168	1.606	0.002	625.783	0.6	0.034	0.11	0.022	0.022	0.0002	0.21	80.29	0.0005	0.003	81.13
Chipper/Shredder	Other Construction Equipment	Diesel	125	1	12	60	720	0.492	0.59	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.009	0.03	0.007	0.007	0.0001	0.12	31.06	0.0001	0.001	31.21
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	31	30	930	0.433	0.21	0.399	0.435	1.380	0.258	0.250	2.650	0.002	625.297	0.6	0.012	0.04	0.007	0.007	0.0001	0.07	16.83	0.0002	0.001	17.00
SideBoom (Cat 561)	Cranes	Diesel	123	2	17	60	2,040	0.367	0.43	0.021	0.023	0.112	0.028	0.027	0.341	0.001	530.977	0.6	0.003	0.01	0.003	0.003	0.0002	0.04	63.15	0.0000	0.001	63.47
SideBoom (Cat 572)	Cranes	Diesel	240	9	17	60	9,180	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.021	0.06	0.015	0.015	0.0015	0.25	554.51	0.0003	0.010	557.30
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	31	30	930	0.481	0.21	0.822	0.889	4.455	0.670	0.650	4.066	0.002	693.429	0.6	0.019	0.09	0.014	0.014	0.0000	0.09	14.63	0.0003	0.001	14.79
Sweeper	Sweepers/Scrubbers	Diesel	50	1	30	30	900	0.408	0.43	0.087	0.092	0.279	0.020	0.020	2.529	0.002	590.091	0.6	0.002	0.01	0.000	0.000	0.0000	0.05	12.59	0.0000	0.000	12.65
Trenching Machines	Trenchers	Diesel	300	0	0	60	0	0.371	0.59	0.032	0.035	0.128	0.028	0.027	0.412	0.001	536.730	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	28	40	1,120	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.003	0.01	0.002	0.002	0.0004	0.03	136.85	0.0000	0.002	137.35
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	3	21	50	3,150	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.008	0.02	0.006	0.006	0.0010	0.10	384.90	0.0001	0.005	386.31
Grapple Truck	Off-highway Trucks	Diesel	350	1	23	60	1,380	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.003	0.01	0.003	0.003	0.0004	0.04	168.62	0.0001	0.002	169.24
Buggy - IC-50/MST-600 (8,800# L	Other Construction Equipment	Diesel	120	1	22	60	1,320	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.004	0.02	0.004	0.004	0.0002	0.05	55.29	0.0001	0.001	55.50
Buggy - Coating	Other Construction Equipment	Diesel	120	1	13	60	780	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.002	0.01	0.002	0.002	0.0001	0.03	32.67	0.0000	0.000	32.79
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	13	60	780	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.002	0.01	0.002	0.002	0.0001	0.03	32.67	0.0000	0.000	32.79
Tack Rig	Other Construction Equipment	Diesel	120	2	13	60	1,560	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.004	0.02	0.005	0.004	0.0002	0.06	65.35	0.0001	0.001	65.59
Hydro excavator Truck	Excavators	Diesel	400	1	21	60	1,260	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.006	0.03	0.006	0.006	0.0005	0.08	175.95	0.0001	0.002	176.59
Welding Machine	Welders	Diesel	40	46	18	60	49,680	0.481	0.21	0.363	0.389	1.431	0.216	0.209	3.158	0.002	694.857	0.6	0.179	0.66	0.099	0.096	0.0009	1.45	319.64	0.0027	0.012	323.00
185 CFM Compressor	Air Compressors	Diesel	65	4	29	60	6,960	0.408	0.43	0.080	0.084	0.388	0.049	0.048	2.706	0.002	590.118	0.6	0.018	0.08	0.011	0.010	0.0004	0.58	126.54	0.0003	0.002	127.18
Light Plant	Signal Boards/Light Plants	Diesel	20	2	11	20	440	0.408	0.43	0.321	0.357	1.518	0.177	0.172	3.776	0.002	589.348	0.6	0.001	0.01	0.001	0.001	0.0000	0.02	2.46	0.0000	0.000	2.47
1/2 Ton Pickup	Off-highway Trucks	Gas	300	40	26	12	12,480	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	20	26	12	6,240	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.018	0.05	0.014	0.013	0.0023	0.22	871.39	0.0003	0.012	874.58
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	15	25	12	4,500	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.013	0.04	0.010	0.010	0.0017	0.16	628.41	0.0002	0.009	630.70
Mechnics Truck	Off-highway Trucks	Diesel	400	2	31	24	1,488	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.004	0.01	0.003	0.003	0.0005	0.05	207.79	0.0001	0.003	208.55
Truck - Semi	Off-highway Trucks	Diesel	450	4	31	36	4,464	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.014	0.04	0.011	0.011	0.0019	0.18	701.30	0.0002	0.010	703.87
Bus	Off-highway Trucks	Diesel	350	1	28	12	336	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	41.06	0.0000	0.001	41.21
Skid Truck	Off-highway Trucks	Diesel	350	1	25	60	1,500	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.004	0.01	0.003	0.003	0.0005	0.05	183.29	0.0001	0.003	183.96
Passenger Van	Off-highway Trucks	Gas	300	4	28	12	1,344	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
																		Construction TOTAL (Tons)	**0.61**	**2.21**	**0.43**	**0.42**	**0.02**	**6.82**	**9,056.98**	**0.0091**	**0.14**	**9,094.66**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

NOTES:

Note 3: Emission factor for N2O from 40 CFR 98 Subpart C.

Note 4: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Note 5: Annual CH4 emissions are calculated using the following calculation (HC emission factor (g/hp-hr) * (1-0.984) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)
This equation is derived in part from subtracting methane from THC to calculate NMHC. THC - CH4 = NMHC; THC - NMHC = CH4; THC - (0.984 * THC) = CH4; THC * (1-0.984) = CH4
0.984 is the ratio of MNHC to THC and 0.9 is ratio for gasoline - from "Conversion Factors for Hydrocarbon Emission Components", July 2010, EPA-420-R-10-015.

Note 6: Emissions estimate for N2O is based on a diesel higher heating value of 138,000 Btu/gallon and a density of 7.05 lb/gallon
Emissions estimate for N2O is based on a gasoline higher heating value of 130,000 Btu/gallon and a density of 6.17 lb/gallon

Note 7: Greenhouse gases (GHG) are converted to carbon dioxide equivalents (CO2) using 40 CFR 98 Subpart A global warming potentials as follows:

CO2	1
CH4	28
N2O	265

Franklin Farms Project
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral
Construction Emissions - Off-road Engines

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	HAP Emissions (Tons) [3]				
								Benzene	Acrolein	1,3-Butadiene	Formaldehyde	Acetaldehyde
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	4	2,400	0.367	0.43	8.33E-04	4.69E-04	4.16E-05	5.66E-03	2.02E-03
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	9	14,580	0.371	0.59	7.83E-04	3.07E-04	2.70E-05	4.50E-03	1.92E-03
Dump Truck	Off-highway Trucks	Diesel	400	2	2,280	0.371	0.59	1.81E-04	7.39E-05	6.22E-06	1.06E-03	4.80E-04
Excavator (KOM 210)	Excavators	Diesel	160	5	5,700	0.371	0.59	2.72E-04	9.86E-05	9.37E-06	1.51E-03	5.75E-04
Excavator (CAT 329)	Excavators	Diesel	200	4	4,320	0.371	0.59	1.78E-04	7.24E-05	6.12E-06	1.04E-03	4.68E-04
Excavator (CAT 336)	Excavators	Diesel	314	12	15,840	0.371	0.59	2.32E-03	8.66E-04	8.05E-05	1.30E-02	5.17E-03
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	1,860	0.371	0.59	1.65E-04	6.21E-05	5.76E-06	9.31E-04	3.72E-04
Grader	Graders	Diesel	250	1	1,860	0.371	0.59	1.18E-04	4.66E-05	4.07E-06	6.82E-04	2.94E-04
HDD Rig	Bore/Drill Rigs	Diesel	300	1	420	0.367	0.43	4.67E-04	2.66E-04	2.33E-05	3.17E-03	1.13E-03
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	1	1,200	0.367	0.43	1.33E-03	7.60E-04	6.67E-05	9.07E-03	3.24E-03
Cranes	Cranes	Diesel	300	1	160	0.367	0.43	1.77E-05	6.64E-06	6.23E-07	9.99E-05	3.93E-05
Generator Sets	Generator Sets	Diesel	200	2	2,480	0.367	0.43	1.27E-03	8.54E-04	7.33E-05	9.53E-03	3.42E-03
Pump	Pumps	Diesel	120	1	450	0.367	0.43	1.55E-04	1.05E-04	8.90E-06	1.16E-03	4.13E-04
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	2	3,720	0.433	0.21	1.29E-03	6.95E-04	6.31E-05	8.66E-03	3.09E-03
Chipper/Shredder	Other Construction Equipment	Diesel	125	1	720	0.492	0.59	3.43E-04	1.93E-04	1.71E-05	2.33E-03	8.32E-04
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	930	0.433	0.21	4.43E-04	2.45E-04	2.17E-05	2.97E-03	1.06E-03
SideBoom (Cat 561)	Cranes	Diesel	123	2	2,040	0.367	0.43	1.35E-04	4.83E-05	4.74E-06	7.45E-04	2.72E-04
SideBoom (Cat 572)	Cranes	Diesel	240	9	9,180	0.367	0.43	8.12E-04	3.05E-04	2.86E-05	4.59E-03	1.80E-03
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	930	0.481	0.21	5.59E-04	4.11E-04	3.49E-05	4.41E-03	1.57E-03
Sweeper	Sweepers/Scrubbers	Diesel	50	1	900	0.408	0.43	1.03E-04	3.61E-05	3.57E-06	5.69E-04	2.02E-04
Trenching Machines	Trenchers	Diesel	300	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	1,120	0.371	0.59	7.77E-05	3.18E-05	2.67E-06	4.56E-04	2.06E-04
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	3	3,150	0.371	0.59	2.18E-04	8.94E-05	7.52E-06	1.28E-03	5.80E-04
Grapple Truck	Off-highway Trucks	Diesel	350	1	1,380	0.371	0.59	9.57E-05	3.92E-05	3.29E-06	5.62E-04	2.54E-04
Buggy - IC-50/MST-600 (8,800# Load	Other Construction Equipment	Diesel	120	1	1,320	0.371	0.59	1.70E-04	6.40E-05	6.44E-06	9.80E-04	3.55E-04
Buggy - Coating	Other Construction Equipment	Diesel	120	1	780	0.371	0.59	1.01E-04	3.78E-05	3.81E-06	5.79E-04	2.10E-04
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	780	0.371	0.59	1.01E-04	3.78E-05	3.81E-06	5.79E-04	2.10E-04
Tack Rig	Other Construction Equipment	Diesel	120	2	1,560	0.371	0.59	2.01E-04	7.57E-05	7.61E-06	1.16E-03	4.19E-04
Hydro excavator Truck	Excavators	Diesel	400	1	1,260	0.371	0.59	2.35E-04	8.78E-05	8.15E-06	1.32E-03	5.24E-04
Welding Machine	Welders	Diesel	40	46	49,680	0.481	0.21	6.13E-03	4.36E-03	3.32E-04	4.37E-02	1.56E-02
185 CFM Compressor	Air Compressors	Diesel	65	4	6,960	0.408	0.43	8.87E-04	3.24E-04	3.25E-05	5.08E-03	1.80E-03
Light Plant	Signal Boards/Light Plants	Diesel	20	2	440	0.408	0.43	8.04E-05	2.79E-05	2.77E-06	4.34E-04	1.55E-04
1/2 Ton Pickup	Off-highway Trucks	Gas	300	40	12,480	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	20	6,240	0.371	0.59	4.95E-04	2.02E-04	1.70E-05	2.91E-03	1.31E-03
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	15	4,500	0.371	0.59	3.57E-04	1.46E-04	1.23E-05	2.10E-03	9.47E-04
Mechnics Truck	Off-highway Trucks	Diesel	400	2	1,488	0.371	0.59	1.18E-04	4.83E-05	4.06E-06	6.93E-04	3.13E-04
Truck - Semi	Off-highway Trucks	Diesel	450	4	4,464	0.371	0.59	3.98E-04	1.63E-04	1.37E-05	2.34E-03	1.06E-03
Bus	Off-highway Trucks	Diesel	350	1	336	0.371	0.59	2.33E-05	9.53E-06	8.02E-07	1.37E-04	6.19E-05
Skid Truck	Off-highway Trucks	Diesel	350	1	1,500	0.371	0.59	1.04E-04	4.26E-05	3.58E-06	6.11E-04	2.76E-04
Passenger Van	Off-highway Trucks	Gas	300	4	1,344	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
							TOTAL (Tons)	**2.16E-02**	**1.17E-02**	**9.89E-04**	**1.41E-01**	**5.27E-02**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

Note 3: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr)[from MOVES5] * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Franklin Farms Project
Franklin Farms Gulf Run Bi Di Meter Station
Construction Emissions - Off-road Engines **Criteria and GHG Emissions**

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Weeks	Hours per Week	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	Emission Factors EF (g/hp-hr) [1] HC	VOC	CO	PM_{10}	$PM_{2.5}$	NOx	SO_2	CO_2	N_2O EF [3] g/MMBtu	VOC [4] tons	CO [4] tons	PM_{10} tons [4]	$PM_{2.5}$ tons [4]	SO_2 [4] tons	NOx [4] tons	CO_2 [4] tons	CH_4 [5] tons	N_2O [6] tons	GHG [7] tons of CO_2e
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0	0	0.367	0.43	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	3	60	180	0.371	0.59	0.012	0.013	0.041	0.011	0.010	0.161	0.001	536.793	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	12.57	0.0000	0.000	12.61
Dump Truck	Off-highway Trucks	Diesel	400	2	3	60	360	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	50.27	0.0000	0.001	50.46
Excavator (KOM 210)	Excavators	Diesel	160	3	17	60	3,060	0.371	0.59	0.010	0.011	0.066	0.015	0.015	0.212	0.001	536.802	0.6	0.003	0.02	0.005	0.005	0.0005	0.07	170.93	0.0000	0.002	171.55
Excavator (CAT 329)	Excavators	Diesel	200	2	17	60	2,040	0.371	0.59	0.011	0.011	0.030	0.008	0.008	0.136	0.001	536.798	0.6	0.003	0.01	0.002	0.002	0.0004	0.04	142.44	0.0000	0.002	142.96
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0	0	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	16	60	960	0.371	0.59	0.017	0.018	0.070	0.017	0.016	0.214	0.001	536.779	0.6	0.002	0.01	0.002	0.002	0.0002	0.03	67.03	0.0000	0.001	67.27
Grader	Graders	Diesel	250	0	0	0	0	0.371	0.59	0.012	0.013	0.039	0.010	0.010	0.156	0.001	536.794	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Cranes	Cranes	Diesel	300	1	8	40	320	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	24.16	0.0000	0.000	24.28
Generator Sets	Generator Sets	Diesel	200	2	14	40	1,120	0.367	0.43	0.158	0.171	0.517	0.111	0.108	2.094	0.002	530.549	0.6	0.018	0.05	0.012	0.011	0.0002	0.22	56.33	0.0003	0.001	56.62
Pump	Pumps	Diesel	120	0	0	0	0	0.367	0.43	0.174	0.188	0.627	0.143	0.139	2.227	0.002	530.499	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0	0	0.433	0.21	0.244	0.266	0.872	0.173	0.168	1.606	0.002	625.783	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0	0	0.492	0.59	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	14	20	280	0.433	0.21	0.399	0.435	1.380	0.258	0.250	2.650	0.002	625.297	0.6	0.004	0.01	0.002	0.002	0.0000	0.02	5.07	0.0001	0.000	5.12
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0	0	0.367	0.43	0.021	0.023	0.112	0.028	0.027	0.341	0.001	530.977	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0	0	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	14	40	560	0.481	0.21	0.822	0.889	4.455	0.670	0.650	4.066	0.002	693.429	0.6	0.011	0.06	0.009	0.008	0.0000	0.05	8.81	0.0002	0.000	8.90
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0	0	0.408	0.43	0.087	0.092	0.279	0.020	0.020	2.529	0.002	590.091	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Trenching Machines	Trenchers	Diesel	300	0	0	0	0	0.371	0.59	0.032	0.035	0.128	0.028	0.027	0.412	0.001	536.730	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	10	170	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.77	0.0000	0.000	20.85
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	20	340	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	41.54	0.0000	0.001	41.70
Grapple Truck	Off-highway Trucks	Diesel	350	1	14	12	168	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.53	0.0000	0.000	20.60
Buggy - IC-50/MST-600 (8,800# L	Other Construction Equipment	Diesel	120	1	14	40	560	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.002	0.01	0.002	0.002	0.0001	0.02	23.46	0.0000	0.000	23.54
Buggy - Coating	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0	0	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Hydro excavator Truck	Excavators	Diesel	400	1	3	60	180	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	25.14	0.0000	0.000	25.23
Welding Machine	Welders	Diesel	40	20	15	60	18,000	0.481	0.21	0.363	0.389	1.431	0.216	0.209	3.158	0.002	694.857	0.6	0.065	0.24	0.036	0.035	0.0003	0.53	115.81	0.0010	0.004	117.03
185 CFM Compressor	Air Compressors	Diesel	65	1	15	60	900	0.408	0.43	0.080	0.084	0.388	0.049	0.048	2.706	0.002	590.118	0.6	0.002	0.01	0.001	0.001	0.0000	0.08	16.36	0.0000	0.000	16.45
Light Plant	Signal Boards/Light Plants	Diesel	20	1	3	20	60	0.408	0.43	0.321	0.357	1.518	0.177	0.172	3.776	0.002	589.348	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.34	0.0000	0.000	0.34
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	17	12	4,080	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	17	12	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	17	24	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
Mechnics Truck	Off-highway Trucks	Diesel	400	1	17	12	204	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.000	0.000	0.0001	0.01	28.49	0.0000	0.000	28.59
Truck - Semi	Off-highway Trucks	Diesel	450	1	17	36	612	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.002	0.01	0.002	0.001	0.0003	0.02	96.15	0.0000	0.001	96.50
Bus	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Passenger Van	Off-highway Trucks	Gas	300	2	17	12	408	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
																		Construction TOTAL (Tons)	**0.13**	**0.48**	**0.09**	**0.08**	**0.00**	**1.31**	**1,525.27**	**0.0020**	**0.02**	**1,531.87**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVESS. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

NOTES:

Note 3: Emission factor for N2O from 40 CFR 98 Subpart C.

Note 4: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Note 5: Annual CH4 emissions are calculated using the following calculation (HC emission factor (g/hp-hr) * (1-0.984) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)
This equation is derived in part from subtracting methane from THC to calculate NMHC. THC - CH4 = NMHC; THC - NMHC = CH4; THC - (0.984 * THC) = CH4; THC * (1-0.984) = CH4
0.984 is the ratio of MNHC to THC and 0.9 is ratio for gasoline - from "Conversion Factors for Hydrocarbon Emission Components", July 2010, EPA-420-R-10-015.

Note 6: Emissions estimate for N2O is based on a diesel higher heating value of 138,000 Btu/gallon and a density of 7.05 lb/gallon
Emissions estimate for N2O is based on a gasoline higher heating value of 130,000 Btu/gallon and a density of 6.17 lb/gallon

Note 7: Greenhouse gases (GHG) are converted to carbon dioxide equivalents (CO2) using 40 CFR 98 Subpart A global warming potentials as follows:

CO2	1
CH4	28
N2O	265

Franklin Farms Project
Franklin Farms Gulf Run Bi Di Meter Station
Construction Emissions - Off-road Engines

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	HAP Emissions (Tons) [3]				
								Benzene	Acrolein	1,3-Butadiene	Formaldehyde	Acetaldehyde
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	180	0.371	0.59	9.67E-06	3.79E-06	3.33E-07	5.56E-05	2.38E-05
Dump Truck	Off-highway Trucks	Diesel	400	2	360	0.371	0.59	2.85E-05	1.17E-05	9.82E-07	1.68E-04	7.58E-05
Excavator (KOM 210)	Excavators	Diesel	160	3	3,060	0.371	0.59	1.46E-04	5.29E-05	5.03E-06	8.08E-04	3.09E-04
Excavator (CAT 329)	Excavators	Diesel	200	2	2,040	0.371	0.59	8.40E-05	3.42E-05	2.89E-06	4.92E-04	2.21E-04
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	960	0.371	0.59	8.51E-05	3.21E-05	2.97E-06	4.81E-04	1.92E-04
Grader	Graders	Diesel	250	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Cranes	Cranes	Diesel	300	1	320	0.367	0.43	3.54E-05	1.33E-05	1.25E-06	2.00E-04	7.86E-05
Generator Sets	Generator Sets	Diesel	200	2	1,120	0.367	0.43	5.76E-04	3.85E-04	3.31E-05	4.30E-03	1.54E-03
Pump	Pumps	Diesel	120	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0.433	0.21	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0.492	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	280	0.433	0.21	1.33E-04	7.37E-05	6.55E-06	8.95E-04	3.19E-04
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	560	0.481	0.21	3.36E-04	2.47E-04	2.10E-05	2.66E-03	9.46E-04
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0.408	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Trenching Machines	Trenchers	Diesel	300	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	170	0.371	0.59	1.18E-05	4.82E-06	4.06E-07	6.93E-05	3.13E-05
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	340	0.371	0.59	2.36E-05	9.65E-06	8.11E-07	1.39E-04	6.26E-05
Grapple Truck	Off-highway Trucks	Diesel	350	1	168	0.371	0.59	1.17E-05	4.77E-06	4.01E-07	6.84E-05	3.09E-05
Buggy - IC-50/MST-600 (8,800# Load	Other Construction Equipment	Diesel	120	1	560	0.371	0.59	7.23E-05	2.72E-05	2.73E-06	4.16E-04	1.50E-04
Buggy - Coating	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Hydro excavator Truck	Excavators	Diesel	400	1	180	0.371	0.59	3.36E-05	1.25E-05	1.16E-06	1.89E-04	7.48E-05
Welding Machine	Welders	Diesel	40	20	18,000	0.481	0.21	2.22E-03	1.58E-03	1.20E-04	1.58E-02	5.66E-03
185 CFM Compressor	Air Compressors	Diesel	65	1	900	0.408	0.43	1.15E-04	4.20E-05	4.21E-06	6.56E-04	2.33E-04
Light Plant	Signal Boards/Light Plants	Diesel	20	1	60	0.408	0.43	1.10E-05	3.80E-06	3.77E-07	5.92E-05	2.11E-05
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	4,080	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
Mechnics Truck	Off-highway Trucks	Diesel	400	1	204	0.371	0.59	1.62E-05	6.62E-06	5.56E-07	9.50E-05	4.29E-05
Truck - Semi	Off-highway Trucks	Diesel	450	1	612	0.371	0.59	5.46E-05	2.23E-05	1.88E-06	3.21E-04	1.45E-04
Bus	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Passenger Van	Off-highway Trucks	Gas	300	2	408	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
							TOTAL (Tons)	**4.42E-03**	**2.73E-03**	**2.21E-04**	**3.03E-02**	**1.12E-02**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

Note 3: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr)[from MOVES5] * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Franklin Farms Project
Franklin Farms Gulf South Receipt Meter Station
Construction Emissions - Off-road Engines **Criteria and GHG Emissions**

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Weeks	Hours per Week	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	Emission Factors EF (g/hp-hr) [1] HC	VOC	CO	PM_{10}	$PM_{2.5}$	NOx	SO_2	CO_2	N_2O EF [3] g/MMBtu	VOC [4] tons	CO [4] tons	PM_{10} tons [4]	$PM_{2.5}$ tons [4]	SO_2 [4] tons	NOx [4] tons	CO_2 [4] tons	CH_4 [5] tons	N_2O [6] tons	GHG [7] tons of CO_2e
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0	0	0.367	0.43	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	3	60	180	0.371	0.59	0.012	0.013	0.041	0.011	0.010	0.161	0.001	536.793	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	12.57	0.0000	0.000	12.61
Dump Truck	Off-highway Trucks	Diesel	400	2	3	60	360	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	50.27	0.0000	0.001	50.46
Excavator (KOM 210)	Excavators	Diesel	160	3	17	60	3,060	0.371	0.59	0.010	0.011	0.066	0.015	0.015	0.212	0.001	536.802	0.6	0.003	0.02	0.005	0.005	0.0005	0.07	170.93	0.0000	0.002	171.55
Excavator (CAT 329)	Excavators	Diesel	200	2	17	60	2,040	0.371	0.59	0.011	0.011	0.030	0.008	0.008	0.136	0.001	536.798	0.6	0.003	0.01	0.002	0.002	0.0004	0.04	142.44	0.0000	0.002	142.96
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0	0	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	16	60	960	0.371	0.59	0.017	0.018	0.070	0.017	0.016	0.214	0.001	536.779	0.6	0.002	0.01	0.002	0.002	0.0002	0.03	67.03	0.0000	0.001	67.27
Grader	Graders	Diesel	250	0	0	0	0	0.371	0.59	0.012	0.013	0.039	0.010	0.010	0.156	0.001	536.794	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Cranes	Cranes	Diesel	300	1	8	40	320	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	24.16	0.0000	0.000	24.28
Generator Sets	Generator Sets	Diesel	200	2	14	40	1,120	0.367	0.43	0.158	0.171	0.517	0.111	0.108	2.094	0.002	530.549	0.6	0.018	0.05	0.012	0.011	0.0002	0.22	56.33	0.0003	0.001	56.62
Pump	Pumps	Diesel	120	0	0	0	0	0.367	0.43	0.174	0.188	0.627	0.143	0.139	2.227	0.002	530.499	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0	0	0.433	0.21	0.244	0.266	0.872	0.173	0.168	1.606	0.002	625.783	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0	0	0.492	0.59	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	14	20	280	0.433	0.21	0.399	0.435	1.380	0.258	0.250	2.650	0.002	625.297	0.6	0.004	0.01	0.002	0.002	0.0000	0.02	5.07	0.0001	0.000	5.12
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0	0	0.367	0.43	0.021	0.023	0.112	0.028	0.027	0.341	0.001	530.977	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0	0	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	14	40	560	0.481	0.21	0.822	0.889	4.455	0.670	0.650	4.066	0.002	693.429	0.6	0.011	0.06	0.009	0.008	0.0000	0.05	8.81	0.0002	0.000	8.90
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0	0	0.408	0.43	0.087	0.092	0.279	0.020	0.020	2.529	0.002	590.091	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Trenching Machines	Trenchers	Diesel	300	0	0	0	0	0.371	0.59	0.032	0.035	0.128	0.028	0.027	0.412	0.001	536.730	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	10	170	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.77	0.0000	0.000	20.85
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	20	340	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	41.54	0.0000	0.001	41.70
Grapple Truck	Off-highway Trucks	Diesel	350	1	14	12	168	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.53	0.0000	0.000	20.60
Buggy - IC-50/MST-600 (8,800# L	Other Construction Equipment	Diesel	120	1	14	40	560	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.002	0.01	0.002	0.002	0.0001	0.02	23.46	0.0000	0.000	23.54
Buggy - Coating	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0	0	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Hydro excavator Truck	Excavators	Diesel	400	1	3	60	180	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	25.14	0.0000	0.000	25.23
Welding Machine	Welders	Diesel	40	20	15	60	18,000	0.481	0.21	0.363	0.389	1.431	0.216	0.209	3.158	0.002	694.857	0.6	0.065	0.24	0.036	0.035	0.0003	0.53	115.81	0.0010	0.004	117.03
185 CFM Compressor	Air Compressors	Diesel	65	1	15	60	900	0.408	0.43	0.080	0.084	0.388	0.049	0.048	2.706	0.002	590.118	0.6	0.002	0.01	0.001	0.001	0.0000	0.08	16.36	0.0000	0.000	16.45
Light Plant	Signal Boards/Light Plants	Diesel	20	1	3	20	60	0.408	0.43	0.321	0.357	1.518	0.177	0.172	3.776	0.002	589.348	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.34	0.0000	0.000	0.34
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	17	12	4,080	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	17	12	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	17	24	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
Mechnics Truck	Off-highway Trucks	Diesel	400	1	17	12	204	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.000	0.000	0.0001	0.01	28.49	0.0000	0.000	28.59
Truck - Semi	Off-highway Trucks	Diesel	450	1	17	36	612	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.002	0.01	0.002	0.001	0.0003	0.02	96.15	0.0000	0.001	96.50
Bus	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Passenger Van	Off-highway Trucks	Gas	300	2	17	12	408	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
																		Construction TOTAL (Tons)	**0.13**	**0.48**	**0.09**	**0.08**	**0.00**	**1.31**	**1,525.27**	**0.0020**	**0.02**	**1,531.87**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVESS. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

NOTES:

Note 3: Emission factor for N2O from 40 CFR 98 Subpart C.

Note 4: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Note 5: Annual CH4 emissions are calculated using the following calculation (HC emission factor (g/hp-hr) * (1-0.984) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)
This equation is derived in part from subtracting methane from THC to calculate NMHC. THC - CH_4 = NMHC; THC - NMHC = CH_4; THC - (0.984 * THC) = CH_4; THC * (1-0.984) = CH_4
0.984 is the ratio of MNHC to THC and 0.9 is ratio for gasoline - from "Conversion Factors for Hydrocarbon Emission Components", July 2010, EPA-420-R-10-015.

Note 6: Emissions estimate for N2O is based on a diesel higher heating value of 138,000 Btu/gallon and a density of 7.05 lb/gallon
Emissions estimate for N2O is based on a gasoline higher heating value of 130,000 Btu/gallon and a density of 6.17 lb/gallon

Note 7: Greenhouse gases (GHG) are converted to carbon dioxide equivalents (CO2) using 40 CFR 98 Subpart A global warming potentials as follows:

CO2	1
CH4	28
N2O	265

Franklin Farms Project
Franklin Farms Gulf South Receipt Meter Station
Construction Emissions - Off-road Engines

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	HAP Emissions (Tons) [3]				
								Benzene	Acrolein	1,3-Butadiene	Formaldehyde	Acetaldehyde
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	180	0.371	0.59	9.67E-06	3.79E-06	3.33E-07	5.56E-05	2.38E-05
Dump Truck	Off-highway Trucks	Diesel	400	2	360	0.371	0.59	2.85E-05	1.17E-05	9.82E-07	1.68E-04	7.58E-05
Excavator (KOM 210)	Excavators	Diesel	160	3	3,060	0.371	0.59	1.46E-04	5.29E-05	5.03E-06	8.08E-04	3.09E-04
Excavator (CAT 329)	Excavators	Diesel	200	2	2,040	0.371	0.59	8.40E-05	3.42E-05	2.89E-06	4.92E-04	2.21E-04
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	960	0.371	0.59	8.51E-05	3.21E-05	2.97E-06	4.81E-04	1.92E-04
Grader	Graders	Diesel	250	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Cranes	Cranes	Diesel	300	1	320	0.367	0.43	3.54E-05	1.33E-05	1.25E-06	2.00E-04	7.86E-05
Generator Sets	Generator Sets	Diesel	200	2	1,120	0.367	0.43	5.76E-04	3.85E-04	3.31E-05	4.30E-03	1.54E-03
Pump	Pumps	Diesel	120	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0.433	0.21	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0.492	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	280	0.433	0.21	1.33E-04	7.37E-05	6.55E-06	8.95E-04	3.19E-04
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	560	0.481	0.21	3.36E-04	2.47E-04	2.10E-05	2.66E-03	9.46E-04
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0.408	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Trenching Machines	Trenchers	Diesel	300	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	170	0.371	0.59	1.18E-05	4.82E-06	4.06E-07	6.93E-05	3.13E-05
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	340	0.371	0.59	2.36E-05	9.65E-06	8.11E-07	1.39E-04	6.26E-05
Grapple Truck	Off-highway Trucks	Diesel	350	1	168	0.371	0.59	1.17E-05	4.77E-06	4.01E-07	6.84E-05	3.09E-05
Buggy - IC-50/MST-600 (8,800# Load	Other Construction Equipment	Diesel	120	1	560	0.371	0.59	7.23E-05	2.72E-05	2.73E-06	4.16E-04	1.50E-04
Buggy - Coating	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Hydro excavator Truck	Excavators	Diesel	400	1	180	0.371	0.59	3.36E-05	1.25E-05	1.16E-06	1.89E-04	7.48E-05
Welding Machine	Welders	Diesel	40	20	18,000	0.481	0.21	2.22E-03	1.58E-03	1.20E-04	1.58E-02	5.66E-03
185 CFM Compressor	Air Compressors	Diesel	65	1	900	0.408	0.43	1.15E-04	4.20E-05	4.21E-06	6.56E-04	2.33E-04
Light Plant	Signal Boards/Light Plants	Diesel	20	1	60	0.408	0.43	1.10E-05	3.80E-06	3.77E-07	5.92E-05	2.11E-05
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	4,080	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
Mechnics Truck	Off-highway Trucks	Diesel	400	1	204	0.371	0.59	1.62E-05	6.62E-06	5.56E-07	9.50E-05	4.29E-05
Truck - Semi	Off-highway Trucks	Diesel	450	1	612	0.371	0.59	5.46E-05	2.23E-05	1.88E-06	3.21E-04	1.45E-04
Bus	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Passenger Van	Off-highway Trucks	Gas	300	2	408	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
							TOTAL (Tons)	**4.42E-03**	**2.73E-03**	**2.21E-04**	**3.03E-02**	**1.12E-02**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

Note 3: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr)[from MOVES5] * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Franklin Farms Project
Franklin Farms Gulf Crossing Receipt Meter Station
Construction Emissions - Off-road Engines — **Criteria and GHG Emissions** — **Construction Emissions - Off-road Engines**

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Weeks	Hours per Week	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	Emission Factors EF (g/hp-hr) [1] HC	VOC	CO	PM_{10}	$PM_{2.5}$	NOx	SO_2	CO_2	N_2O EF [3] g/MMBtu	VOC [4] tons	CO [4] tons	PM_{10} tons [4]	$PM_{2.5}$ tons [4]	SO_2 [4] tons	NOx [4] tons	CO_2 [4] tons	CH_4 [5] tons	N_2O [6] tons	GHG [7] tons of CO_2e
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0	0	0.367	0.43	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	3	60	180	0.371	0.59	0.012	0.013	0.041	0.011	0.010	0.161	0.001	536.793	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	12.57	0.0000	0.000	12.61
Dump Truck	Off-highway Trucks	Diesel	400	2	3	60	360	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	50.27	0.0000	0.001	50.46
Excavator (KOM 210)	Excavators	Diesel	160	3	17	60	3,060	0.371	0.59	0.010	0.011	0.066	0.015	0.015	0.212	0.001	536.802	0.6	0.003	0.02	0.005	0.005	0.0005	0.07	170.93	0.0000	0.002	171.55
Excavator (CAT 329)	Excavators	Diesel	200	2	17	60	2,040	0.371	0.59	0.011	0.011	0.030	0.008	0.008	0.136	0.001	536.798	0.6	0.003	0.01	0.002	0.002	0.0004	0.04	142.44	0.0000	0.002	142.96
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0	0	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	16	60	960	0.371	0.59	0.017	0.018	0.070	0.017	0.016	0.214	0.001	536.779	0.6	0.002	0.01	0.002	0.002	0.0002	0.03	67.03	0.0000	0.001	67.27
Grader	Graders	Diesel	250	0	0	0	0	0.371	0.59	0.012	0.013	0.039	0.010	0.010	0.156	0.001	536.794	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Cranes	Cranes	Diesel	300	1	8	40	320	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	24.16	0.0000	0.000	24.28
Generator Sets	Generator Sets	Diesel	200	2	14	40	1,120	0.367	0.43	0.158	0.171	0.517	0.111	0.108	2.094	0.002	530.549	0.6	0.018	0.05	0.012	0.011	0.0002	0.22	56.33	0.0003	0.001	56.62
Pump	Pumps	Diesel	120	0	0	0	0	0.367	0.43	0.174	0.188	0.627	0.143	0.139	2.227	0.002	530.499	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0	0	0.433	0.21	0.244	0.266	0.872	0.173	0.168	1.606	0.002	625.783	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0	0	0.492	0.59	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	14	20	280	0.433	0.21	0.399	0.435	1.380	0.258	0.250	2.650	0.002	625.297	0.6	0.004	0.01	0.002	0.002	0.0000	0.02	5.07	0.0001	0.000	5.12
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0	0	0.367	0.43	0.021	0.023	0.112	0.028	0.027	0.341	0.001	530.977	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0	0	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	14	40	560	0.481	0.21	0.822	0.889	4.455	0.670	0.650	4.066	0.002	693.429	0.6	0.011	0.06	0.009	0.008	0.0000	0.05	8.81	0.0002	0.000	8.90
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0	0	0.408	0.43	0.087	0.092	0.279	0.020	0.020	2.529	0.002	590.091	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Trenching Machines	Trenchers	Diesel	300	0	0	0	0	0.371	0.59	0.032	0.035	0.128	0.028	0.027	0.412	0.001	536.730	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	10	170	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.77	0.0000	0.000	20.85
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	20	340	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	41.54	0.0000	0.001	41.70
Grapple Truck	Off-highway Trucks	Diesel	350	1	14	12	168	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.53	0.0000	0.000	20.60
Buggy - IC-50/MST-600 (8,800# L	Other Construction Equipment	Diesel	120	1	14	40	560	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.002	0.01	0.002	0.002	0.0001	0.02	23.46	0.0000	0.000	23.54
Buggy - Coating	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0	0	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Hydro excavator Truck	Excavators	Diesel	400	1	3	60	180	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	25.14	0.0000	0.000	25.23
Welding Machine	Welders	Diesel	40	20	15	60	18,000	0.481	0.21	0.363	0.389	1.431	0.216	0.209	3.158	0.002	694.857	0.6	0.065	0.24	0.036	0.035	0.0003	0.53	115.81	0.0010	0.004	117.03
185 CFM Compressor	Air Compressors	Diesel	65	1	15	60	900	0.408	0.43	0.080	0.084	0.388	0.049	0.048	2.706	0.002	590.118	0.6	0.002	0.01	0.001	0.001	0.0000	0.08	16.36	0.0000	0.000	16.45
Light Plant	Signal Boards/Light Plants	Diesel	20	1	3	20	60	0.408	0.43	0.321	0.357	1.518	0.177	0.172	3.776	0.002	589.348	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.34	0.0000	0.000	0.34
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	17	12	4,080	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	17	12	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	17	24	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
Mechnics Truck	Off-highway Trucks	Diesel	400	1	17	12	204	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.000	0.000	0.0001	0.01	28.49	0.0000	0.000	28.59
Truck - Semi	Off-highway Trucks	Diesel	450	1	17	36	612	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.002	0.01	0.002	0.001	0.0003	0.02	96.15	0.0000	0.001	96.50
Bus	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Passenger Van	Off-highway Trucks	Gas	300	2	17	12	408	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
																		Construction TOTAL (Tons)	**0.13**	**0.48**	**0.09**	**0.08**	**0.00**	**1.31**	**1,525.27**	**0.0020**	**0.02**	**1,531.87**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

NOTES:

Note 3: Emission factor for N2O from 40 CFR 98 Subpart C.

Note 4: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Note 5: Annual CH4 emissions are calculated using the following calculation (HC emission factor (g/hp-hr) * (1-0.984) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)
This equation is derived in part from subtracting methane from THC to calculate NMHC. THC - CH_4 = NMHC; THC - NMHC = CH_4; THC - (0.984 * THC) = CH_4; THC * (1-0.984) = CH_4
0.984 is the ratio of MNHC to THC and 0.9 is ratio for gasoline - from "Conversion Factors for Hydrocarbon Emission Components", July 2010, EPA-420-R-10-015.

Note 6: Emissions estimate for N2O is based on a diesel higher heating value of 138,000 Btu/gallon and a density of 7.05 lb/gallon
Emissions estimate for N2O is based on a gasoline higher heating value of 130,000 Btu/gallon and a density of 6.17 lb/gallon

Note 7: Greenhouse gases (GHG) are converted to carbon dioxide equivalents (CO2) using 40 CFR 98 Subpart A global warming potentials as follows:

CO2	1
CH4	28
N2O	265

Franklin Farms Project
Franklin Farms Gulf Crossing Receipt Meter Station
Construction Emissions - Off-road Engines

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	HAP Emissions (Tons) [3]				
								Benzene	Acrolein	1,3-Butadiene	Formaldehyde	Acetaldehyde
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	180	0.371	0.59	9.67E-06	3.79E-06	3.33E-07	5.56E-05	2.38E-05
Dump Truck	Off-highway Trucks	Diesel	400	2	360	0.371	0.59	2.85E-05	1.17E-05	9.82E-07	1.68E-04	7.58E-05
Excavator (KOM 210)	Excavators	Diesel	160	3	3,060	0.371	0.59	1.46E-04	5.29E-05	5.03E-06	8.08E-04	3.09E-04
Excavator (CAT 329)	Excavators	Diesel	200	2	2,040	0.371	0.59	8.40E-05	3.42E-05	2.89E-06	4.92E-04	2.21E-04
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	960	0.371	0.59	8.51E-05	3.21E-05	2.97E-06	4.81E-04	1.92E-04
Grader	Graders	Diesel	250	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Cranes	Cranes	Diesel	300	1	320	0.367	0.43	3.54E-05	1.33E-05	1.25E-06	2.00E-04	7.86E-05
Generator Sets	Generator Sets	Diesel	200	2	1,120	0.367	0.43	5.76E-04	3.85E-04	3.31E-05	4.30E-03	1.54E-03
Pump	Pumps	Diesel	120	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0.433	0.21	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0.492	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	280	0.433	0.21	1.33E-04	7.37E-05	6.55E-06	8.95E-04	3.19E-04
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	560	0.481	0.21	3.36E-04	2.47E-04	2.10E-05	2.66E-03	9.46E-04
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0.408	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Trenching Machines	Trenchers	Diesel	300	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	170	0.371	0.59	1.18E-05	4.82E-06	4.06E-07	6.93E-05	3.13E-05
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	340	0.371	0.59	2.36E-05	9.65E-06	8.11E-07	1.39E-04	6.26E-05
Grapple Truck	Off-highway Trucks	Diesel	350	1	168	0.371	0.59	1.17E-05	4.77E-06	4.01E-07	6.84E-05	3.09E-05
Buggy - IC-50/MST-600 (8,800# Load	Other Construction Equipment	Diesel	120	1	560	0.371	0.59	7.23E-05	2.72E-05	2.73E-06	4.16E-04	1.50E-04
Buggy - Coating	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Hydro excavator Truck	Excavators	Diesel	400	1	180	0.371	0.59	3.36E-05	1.25E-05	1.16E-06	1.89E-04	7.48E-05
Welding Machine	Welders	Diesel	40	20	18,000	0.481	0.21	2.22E-03	1.58E-03	1.20E-04	1.58E-02	5.66E-03
185 CFM Compressor	Air Compressors	Diesel	65	1	900	0.408	0.43	1.15E-04	4.20E-05	4.21E-06	6.56E-04	2.33E-04
Light Plant	Signal Boards/Light Plants	Diesel	20	1	60	0.408	0.43	1.10E-05	3.80E-06	3.77E-07	5.92E-05	2.11E-05
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	4,080	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
Mechnics Truck	Off-highway Trucks	Diesel	400	1	204	0.371	0.59	1.62E-05	6.62E-06	5.56E-07	9.50E-05	4.29E-05
Truck - Semi	Off-highway Trucks	Diesel	450	1	612	0.371	0.59	5.46E-05	2.23E-05	1.88E-06	3.21E-04	1.45E-04
Bus	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Passenger Van	Off-highway Trucks	Gas	300	2	408	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
							TOTAL (Tons)	**4.42E-03**	**2.73E-03**	**2.21E-04**	**3.03E-02**	**1.12E-02**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

Note 3: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr)[from MOVES5] * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Franklin Farms Project
Franklin Farms MEP Receipt Meter Station
Construction Emissions - Off-road Engines **Criteria and GHG Emissions**

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Weeks	Hours per Week	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	Emission Factors EF (g/hp-hr) [1] HC	VOC	CO	PM_{10}	$PM_{2.5}$	NOx	SO_2	CO_2	N_2O EF [3] g/MMBtu	VOC [4] tons	CO [4] tons	PM_{10} tons [4]	$PM_{2.5}$ tons [4]	SO_2 [4] tons	NOx [4] tons	CO_2 [4] tons	CH_4 [5] tons	N_2O [6] tons	GHG [7] tons of CO_2e
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0	0	0.367	0.43	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	3	60	180	0.371	0.59	0.012	0.013	0.041	0.011	0.010	0.161	0.001	536.793	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	12.57	0.0000	0.000	12.61
Dump Truck	Off-highway Trucks	Diesel	400	2	3	60	360	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	50.27	0.0000	0.001	50.46
Excavator (KOM 210)	Excavators	Diesel	160	3	17	60	3,060	0.371	0.59	0.010	0.011	0.066	0.015	0.015	0.212	0.001	536.802	0.6	0.003	0.02	0.005	0.005	0.0005	0.07	170.93	0.0000	0.002	171.55
Excavator (CAT 329)	Excavators	Diesel	200	2	17	60	2,040	0.371	0.59	0.011	0.011	0.030	0.008	0.008	0.136	0.001	536.798	0.6	0.003	0.01	0.002	0.002	0.0004	0.04	142.44	0.0000	0.002	142.96
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0	0	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	16	60	960	0.371	0.59	0.017	0.018	0.070	0.017	0.016	0.214	0.001	536.779	0.6	0.002	0.01	0.002	0.002	0.0002	0.03	67.03	0.0000	0.001	67.27
Grader	Graders	Diesel	250	0	0	0	0	0.371	0.59	0.012	0.013	0.039	0.010	0.010	0.156	0.001	536.794	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Cranes	Cranes	Diesel	300	1	8	40	320	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	24.16	0.0000	0.000	24.28
Generator Sets	Generator Sets	Diesel	200	2	14	40	1,120	0.367	0.43	0.158	0.171	0.517	0.111	0.108	2.094	0.002	530.549	0.6	0.018	0.05	0.012	0.011	0.0002	0.22	56.33	0.0003	0.001	56.62
Pump	Pumps	Diesel	120	0	0	0	0	0.367	0.43	0.174	0.188	0.627	0.143	0.139	2.227	0.002	530.499	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0	0	0.433	0.21	0.244	0.266	0.872	0.173	0.168	1.606	0.002	625.783	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0	0	0.492	0.59	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	14	20	280	0.433	0.21	0.399	0.435	1.380	0.258	0.250	2.650	0.002	625.297	0.6	0.004	0.01	0.002	0.002	0.0000	0.02	5.07	0.0001	0.000	5.12
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0	0	0.367	0.43	0.021	0.023	0.112	0.028	0.027	0.341	0.001	530.977	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0	0	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	14	40	560	0.481	0.21	0.822	0.889	4.455	0.670	0.650	4.066	0.002	693.429	0.6	0.011	0.06	0.009	0.008	0.0000	0.05	8.81	0.0002	0.000	8.90
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0	0	0.408	0.43	0.087	0.092	0.279	0.020	0.020	2.529	0.002	590.091	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Trenching Machines	Trenchers	Diesel	300	0	0	0	0	0.371	0.59	0.032	0.035	0.128	0.028	0.027	0.412	0.001	536.730	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	10	170	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.77	0.0000	0.000	20.85
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	20	340	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	41.54	0.0000	0.001	41.70
Grapple Truck	Off-highway Trucks	Diesel	350	1	14	12	168	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.53	0.0000	0.000	20.60
Buggy - IC-50/MST-600 (8,800# U	Other Construction Equipment	Diesel	120	1	14	40	560	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.002	0.01	0.002	0.002	0.0001	0.02	23.46	0.0000	0.000	23.54
Buggy - Coating	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0	0	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Hydro excavator Truck	Excavators	Diesel	400	1	3	60	180	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	25.14	0.0000	0.000	25.23
Welding Machine	Welders	Diesel	40	20	15	60	18,000	0.481	0.21	0.363	0.389	1.431	0.216	0.209	3.158	0.002	694.857	0.6	0.065	0.24	0.036	0.035	0.0003	0.53	115.81	0.0010	0.004	117.03
185 CFM Compressor	Air Compressors	Diesel	65	1	15	60	900	0.408	0.43	0.080	0.084	0.388	0.049	0.048	2.706	0.002	590.118	0.6	0.002	0.01	0.001	0.001	0.0000	0.08	16.36	0.0000	0.000	16.45
Light Plant	Signal Boards/Light Plants	Diesel	20	1	3	20	60	0.408	0.43	0.321	0.357	1.518	0.177	0.172	3.776	0.002	589.348	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.34	0.0000	0.000	0.34
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	17	12	4,080	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	17	12	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	17	24	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
Mechnics Truck	Off-highway Trucks	Diesel	400	1	17	12	204	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.000	0.000	0.0001	0.01	28.49	0.0000	0.000	28.59
Truck - Semi	Off-highway Trucks	Diesel	450	1	17	36	612	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.002	0.01	0.002	0.001	0.0003	0.02	96.15	0.0000	0.001	96.50
Bus	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Passenger Van	Off-highway Trucks	Gas	300	2	17	12	408	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
																		Construction TOTAL (Tons)	**0.13**	**0.48**	**0.09**	**0.08**	**0.00**	**1.31**	**1,525.27**	**0.0020**	**0.02**	**1,531.87**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVESS. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

NOTES:

Note 3: Emission factor for N2O from 40 CFR 98 Subpart C.

Note 4: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Note 5: Annual CH4 emissions are calculated using the following calculation (HC emission factor (g/hp-hr) * (1-0.984) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)
This equation is derived in part from subtracting methane from THC to calculate NMHC. THC - CH_4 = NMHC; THC - NMHC = CH_4; THC - (0.984 * THC) = CH_4; THC * (1-0.984) = CH_4
0.984 is the ratio of MNHC to THC and 0.9 is ratio for gasoline - from "Conversion Factors for Hydrocarbon Emission Components", July 2010, EPA-420-R-10-015.

Note 6: Emissions estimate for N2O is based on a diesel higher heating value of 138,000 Btu/gallon and a density of 7.05 lb/gallon
Emissions estimate for N2O is based on a gasoline higher heating value of 130,000 Btu/gallon and a density of 6.17 lb/gallon

Note 7: Greenhouse gases (GHG) are converted to carbon dioxide equivalents (CO2) using 40 CFR 98 Subpart A global warming potentials as follows:

CO2	1
CH4	28
N2O	265

Franklin Farms Project
Franklin Farms MEP Receipt Meter Station
Construction Emissions - Off-road Engines

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	HAP Emissions (Tons) [3]				
								Benzene	Acrolein	1,3-Butadiene	Formaldehyde	Acetaldehyde
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	180	0.371	0.59	9.67E-06	3.79E-06	3.33E-07	5.56E-05	2.38E-05
Dump Truck	Off-highway Trucks	Diesel	400	2	360	0.371	0.59	2.85E-05	1.17E-05	9.82E-07	1.68E-04	7.58E-05
Excavator (KOM 210)	Excavators	Diesel	160	3	3,060	0.371	0.59	1.46E-04	5.29E-05	5.03E-06	8.08E-04	3.09E-04
Excavator (CAT 329)	Excavators	Diesel	200	2	2,040	0.371	0.59	8.40E-05	3.42E-05	2.89E-06	4.92E-04	2.21E-04
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	960	0.371	0.59	8.51E-05	3.21E-05	2.97E-06	4.81E-04	1.92E-04
Grader	Graders	Diesel	250	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Cranes	Cranes	Diesel	300	1	320	0.367	0.43	3.54E-05	1.33E-05	1.25E-06	2.00E-04	7.86E-05
Generator Sets	Generator Sets	Diesel	200	2	1,120	0.367	0.43	5.76E-04	3.85E-04	3.31E-05	4.30E-03	1.54E-03
Pump	Pumps	Diesel	120	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0.433	0.21	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0.492	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	280	0.433	0.21	1.33E-04	7.37E-05	6.55E-06	8.95E-04	3.19E-04
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	560	0.481	0.21	3.36E-04	2.47E-04	2.10E-05	2.66E-03	9.46E-04
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0.408	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Trenching Machines	Trenchers	Diesel	300	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	170	0.371	0.59	1.18E-05	4.82E-06	4.06E-07	6.93E-05	3.13E-05
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	340	0.371	0.59	2.36E-05	9.65E-06	8.11E-07	1.39E-04	6.26E-05
Grapple Truck	Off-highway Trucks	Diesel	350	1	168	0.371	0.59	1.17E-05	4.77E-06	4.01E-07	6.84E-05	3.09E-05
Buggy - IC-50/MST-600 (8,800# Load	Other Construction Equipment	Diesel	120	1	560	0.371	0.59	7.23E-05	2.72E-05	2.73E-06	4.16E-04	1.50E-04
Buggy - Coating	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Hydro excavator Truck	Excavators	Diesel	400	1	180	0.371	0.59	3.36E-05	1.25E-05	1.16E-06	1.89E-04	7.48E-05
Welding Machine	Welders	Diesel	40	20	18,000	0.481	0.21	2.22E-03	1.58E-03	1.20E-04	1.58E-02	5.66E-03
185 CFM Compressor	Air Compressors	Diesel	65	1	900	0.408	0.43	1.15E-04	4.20E-05	4.21E-06	6.56E-04	2.33E-04
Light Plant	Signal Boards/Light Plants	Diesel	20	1	60	0.408	0.43	1.10E-05	3.80E-06	3.77E-07	5.92E-05	2.11E-05
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	4,080	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
Mechnics Truck	Off-highway Trucks	Diesel	400	1	204	0.371	0.59	1.62E-05	6.62E-06	5.56E-07	9.50E-05	4.29E-05
Truck - Semi	Off-highway Trucks	Diesel	450	1	612	0.371	0.59	5.46E-05	2.23E-05	1.88E-06	3.21E-04	1.45E-04
Bus	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Passenger Van	Off-highway Trucks	Gas	300	2	408	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
							TOTAL (Tons)	**4.42E-03**	**2.73E-03**	**2.21E-04**	**3.03E-02**	**1.12E-02**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

Note 3: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr)[from MOVES5] * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Franklin Farms Project
Franklin Farms Delivery Meter Station
Construction Emissions - Off-road Engines

Criteria and GHG Emissions

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Weeks	Hours per Week	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	Emission Factors EF (g/hp-hr) [1] HC	VOC	CO	PM_{10}	$PM_{2.5}$	NOx	SO_2	CO_2	N_2O EF [3] g/MMBtu	VOC [4] tons	CO [4] tons	PM_{10} tons [4]	$PM_{2.5}$ tons [4]	SO_2 [4] tons	NOx [4] tons	CO_2 [4] tons	CH_4 [5] tons	N_2O [6] tons	GHG [7] tons of CO_2e
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0	0	0.367	0.43	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	3	60	180	0.371	0.59	0.012	0.013	0.041	0.011	0.010	0.161	0.001	536.793	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	12.57	0.0000	0.000	12.61
Dump Truck	Off-highway Trucks	Diesel	400	2	3	60	360	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	50.27	0.0000	0.001	50.46
Excavator (KOM 210)	Excavators	Diesel	160	3	17	60	3,060	0.371	0.59	0.010	0.011	0.066	0.015	0.015	0.212	0.001	536.802	0.6	0.003	0.02	0.005	0.005	0.0005	0.07	170.93	0.0000	0.002	171.55
Excavator (CAT 329)	Excavators	Diesel	200	2	17	60	2,040	0.371	0.59	0.011	0.011	0.030	0.008	0.008	0.136	0.001	536.798	0.6	0.003	0.01	0.002	0.002	0.0004	0.04	142.44	0.0000	0.002	142.96
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0	0	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	16	60	960	0.371	0.59	0.017	0.018	0.070	0.017	0.016	0.214	0.001	536.779	0.6	0.002	0.01	0.002	0.002	0.0002	0.03	67.03	0.0000	0.001	67.27
Grader	Graders	Diesel	250	0	0	0	0	0.371	0.59	0.012	0.013	0.039	0.010	0.010	0.156	0.001	536.794	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Cranes	Cranes	Diesel	300	1	8	40	320	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	24.16	0.0000	0.000	24.28
Generator Sets	Generator Sets	Diesel	200	2	14	40	1,120	0.367	0.43	0.158	0.171	0.517	0.111	0.108	2.094	0.002	530.549	0.6	0.018	0.05	0.012	0.011	0.0002	0.22	56.33	0.0003	0.001	56.62
Pump	Pumps	Diesel	120	0	0	0	0	0.367	0.43	0.174	0.188	0.627	0.143	0.139	2.227	0.002	530.499	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0	0	0.433	0.21	0.244	0.266	0.872	0.173	0.168	1.606	0.002	625.783	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0	0	0.492	0.59	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	14	20	280	0.433	0.21	0.399	0.435	1.380	0.258	0.250	2.650	0.002	625.297	0.6	0.004	0.01	0.002	0.002	0.0000	0.02	5.07	0.0001	0.000	5.12
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0	0	0.367	0.43	0.021	0.023	0.112	0.028	0.027	0.341	0.001	530.977	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0	0	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	14	40	560	0.481	0.21	0.822	0.889	4.455	0.670	0.650	4.066	0.002	693.429	0.6	0.011	0.06	0.009	0.008	0.0000	0.05	8.81	0.0002	0.000	8.90
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0	0	0.408	0.43	0.087	0.092	0.279	0.020	0.020	2.529	0.002	590.091	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Trenching Machines	Trenchers	Diesel	300	0	0	0	0	0.371	0.59	0.032	0.035	0.128	0.028	0.027	0.412	0.001	536.730	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	10	170	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.77	0.0000	0.000	20.85
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	20	340	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	41.54	0.0000	0.001	41.70
Grapple Truck	Off-highway Trucks	Diesel	350	1	14	12	168	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.53	0.0000	0.000	20.60
Buggy - IC-50/MST-600 (8,800# L	Other Construction Equipment	Diesel	120	1	14	40	560	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.002	0.01	0.002	0.002	0.0001	0.02	23.46	0.0000	0.000	23.54
Buggy - Coating	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0	0	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Hydro excavator Truck	Excavators	Diesel	400	1	3	60	180	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	25.14	0.0000	0.000	25.23
Welding Machine	Welders	Diesel	40	20	15	60	18,000	0.481	0.21	0.363	0.389	1.431	0.216	0.209	3.158	0.002	694.857	0.6	0.065	0.24	0.036	0.035	0.0003	0.53	115.81	0.0010	0.004	117.03
185 CFM Compressor	Air Compressors	Diesel	65	1	15	60	900	0.408	0.43	0.080	0.084	0.388	0.049	0.048	2.706	0.002	590.118	0.6	0.002	0.01	0.001	0.001	0.0000	0.08	16.36	0.0000	0.000	16.45
Light Plant	Signal Boards/Light Plants	Diesel	20	1	3	20	60	0.408	0.43	0.321	0.357	1.518	0.177	0.172	3.776	0.002	589.348	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.34	0.0000	0.000	0.34
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	17	12	4,080	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	17	12	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	17	24	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
Mechnics Truck	Off-highway Trucks	Diesel	400	1	17	12	204	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.000	0.000	0.0001	0.01	28.49	0.0000	0.000	28.59
Truck - Semi	Off-highway Trucks	Diesel	450	1	17	36	612	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.002	0.01	0.002	0.001	0.0003	0.02	96.15	0.0000	0.001	96.50
Bus	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Passenger Van	Off-highway Trucks	Gas	300	2	17	12	408	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
																		Construction TOTAL (Tons)	**0.13**	**0.48**	**0.09**	**0.08**	**0.00**	**1.31**	**1,525.27**	**0.0020**	**0.02**	**1,531.87**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

NOTES:

Note 3: Emission factor for N2O from 40 CFR 98 Subpart C.

Note 4: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Note 5: Annual CH4 emissions are calculated using the following calculation (HC emission factor (g/hp-hr) * (1-0.984) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)
This equation is derived in part from subtracting methane from THC to calculate NMHC. THC - CH_4 = NMHC; THC - NMHC = CH_4; THC - (0.984 * THC) = CH_4; THC * (1-0.984) = CH_4
0.984 is the ratio of MNHC to THC and 0.9 is ratio for gasoline - from "Conversion Factors for Hydrocarbon Emission Components", July 2010, EPA-420-R-10-015.

Note 6: Emissions estimate for N2O is based on a diesel higher heating value of 138,000 Btu/gallon and a density of 7.05 lb/gallon
Emissions estimate for N2O is based on a gasoline higher heating value of 130,000 Btu/gallon and a density of 6.17 lb/gallon

Note 7: Greenhouse gases (GHG) are converted to carbon dioxide equivalents (CO2) using 40 CFR 98 Subpart A global warming potentials as follows:

CO2	1
CH4	28
N2O	265

Franklin Farms Project
Franklin Farms Delivery Meter Station
Construction Emissions - Off-road Engines

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	HAP Emissions (Tons) [3]				
								Benzene	Acrolein	1,3-Butadiene	Formaldehyde	Acetaldehyde
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	180	0.371	0.59	9.67E-06	3.79E-06	3.33E-07	5.56E-05	2.38E-05
Dump Truck	Off-highway Trucks	Diesel	400	2	360	0.371	0.59	2.85E-05	1.17E-05	9.82E-07	1.68E-04	7.58E-05
Excavator (KOM 210)	Excavators	Diesel	160	3	3,060	0.371	0.59	1.46E-04	5.29E-05	5.03E-06	8.08E-04	3.09E-04
Excavator (CAT 329)	Excavators	Diesel	200	2	2,040	0.371	0.59	8.40E-05	3.42E-05	2.89E-06	4.92E-04	2.21E-04
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	960	0.371	0.59	8.51E-05	3.21E-05	2.97E-06	4.81E-04	1.92E-04
Grader	Graders	Diesel	250	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Cranes	Cranes	Diesel	300	1	320	0.367	0.43	3.54E-05	1.33E-05	1.25E-06	2.00E-04	7.86E-05
Generator Sets	Generator Sets	Diesel	200	2	1,120	0.367	0.43	5.76E-04	3.85E-04	3.31E-05	4.30E-03	1.54E-03
Pump	Pumps	Diesel	120	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0.433	0.21	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0.492	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	280	0.433	0.21	1.33E-04	7.37E-05	6.55E-06	8.95E-04	3.19E-04
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	560	0.481	0.21	3.36E-04	2.47E-04	2.10E-05	2.66E-03	9.46E-04
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0.408	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Trenching Machines	Trenchers	Diesel	300	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	170	0.371	0.59	1.18E-05	4.82E-06	4.06E-07	6.93E-05	3.13E-05
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	340	0.371	0.59	2.36E-05	9.65E-06	8.11E-07	1.39E-04	6.26E-05
Grapple Truck	Off-highway Trucks	Diesel	350	1	168	0.371	0.59	1.17E-05	4.77E-06	4.01E-07	6.84E-05	3.09E-05
Buggy - IC-50/MST-600 (8,800# Load	Other Construction Equipment	Diesel	120	1	560	0.371	0.59	7.23E-05	2.72E-05	2.73E-06	4.16E-04	1.50E-04
Buggy - Coating	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Hydro excavator Truck	Excavators	Diesel	400	1	180	0.371	0.59	3.36E-05	1.25E-05	1.16E-06	1.89E-04	7.48E-05
Welding Machine	Welders	Diesel	40	20	18,000	0.481	0.21	2.22E-03	1.58E-03	1.20E-04	1.58E-02	5.66E-03
185 CFM Compressor	Air Compressors	Diesel	65	1	900	0.408	0.43	1.15E-04	4.20E-05	4.21E-06	6.56E-04	2.33E-04
Light Plant	Signal Boards/Light Plants	Diesel	20	1	60	0.408	0.43	1.10E-05	3.80E-06	3.77E-07	5.92E-05	2.11E-05
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	4,080	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
Mechnics Truck	Off-highway Trucks	Diesel	400	1	204	0.371	0.59	1.62E-05	6.62E-06	5.56E-07	9.50E-05	4.29E-05
Truck - Semi	Off-highway Trucks	Diesel	450	1	612	0.371	0.59	5.46E-05	2.23E-05	1.88E-06	3.21E-04	1.45E-04
Bus	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Passenger Van	Off-highway Trucks	Gas	300	2	408	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
							TOTAL (Tons)	**4.42E-03**	**2.73E-03**	**2.21E-04**	**3.03E-02**	**1.12E-02**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

Note 3: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr)[from MOVES5] * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Franklin Farms Project
Franklin Farms Bee Bayou Delivery Meter Station
Construction Emissions - Off-road Engines

Criteria and GHG Emissions

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Weeks	Hours per Week	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	Emission Factors EF (g/hp-hr) [1] HC	VOC	CO	PM_{10}	$PM_{2.5}$	NOx	SO_2	CO_2	N_2O EF [3] g/MMBtu	VOC [4] tons	CO [4] tons	PM_{10} tons [4]	$PM_{2.5}$ tons [4]	SO_2 [4] tons	NOx [4] tons	CO_2 [4] tons	CH_4 [5] tons	N_2O [6] tons	GHG [7] tons of CO_2e
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0	0	0.367	0.43	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	3	60	180	0.371	0.59	0.012	0.013	0.041	0.011	0.010	0.161	0.001	536.793	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	12.57	0.0000	0.000	12.61
Dump Truck	Off-highway Trucks	Diesel	400	2	3	60	360	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	50.27	0.0000	0.001	50.46
Excavator (KOM 210)	Excavators	Diesel	160	3	17	60	3,060	0.371	0.59	0.010	0.011	0.066	0.015	0.015	0.212	0.001	536.802	0.6	0.003	0.02	0.005	0.005	0.0005	0.07	170.93	0.0000	0.002	171.55
Excavator (CAT 329)	Excavators	Diesel	200	2	17	60	2,040	0.371	0.59	0.011	0.011	0.030	0.008	0.008	0.136	0.001	536.798	0.6	0.003	0.01	0.002	0.002	0.0004	0.04	142.44	0.0000	0.002	142.96
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0	0	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	16	60	960	0.371	0.59	0.017	0.018	0.070	0.017	0.016	0.214	0.001	536.779	0.6	0.002	0.01	0.002	0.002	0.0002	0.03	67.03	0.0000	0.001	67.27
Grader	Graders	Diesel	250	0	0	0	0	0.371	0.59	0.012	0.013	0.039	0.010	0.010	0.156	0.001	536.794	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Cranes	Cranes	Diesel	300	1	8	40	320	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	24.16	0.0000	0.000	24.28
Generator Sets	Generator Sets	Diesel	200	2	14	40	1,120	0.367	0.43	0.158	0.171	0.517	0.111	0.108	2.094	0.002	530.549	0.6	0.018	0.05	0.012	0.011	0.0002	0.22	56.33	0.0003	0.001	56.62
Pump	Pumps	Diesel	120	0	0	0	0	0.367	0.43	0.174	0.188	0.627	0.143	0.139	2.227	0.002	530.499	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0	0	0.433	0.21	0.244	0.266	0.872	0.173	0.168	1.606	0.002	625.783	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0	0	0.492	0.59	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	14	20	280	0.433	0.21	0.399	0.435	1.380	0.258	0.250	2.650	0.002	625.297	0.6	0.004	0.01	0.002	0.002	0.0000	0.02	5.07	0.0001	0.000	5.12
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0	0	0.367	0.43	0.021	0.023	0.112	0.028	0.027	0.341	0.001	530.977	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0	0	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	14	40	560	0.481	0.21	0.822	0.889	4.455	0.670	0.650	4.066	0.002	693.429	0.6	0.011	0.06	0.009	0.008	0.0000	0.05	8.81	0.0002	0.000	8.90
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0	0	0.408	0.43	0.087	0.092	0.279	0.020	0.020	2.529	0.002	590.091	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Trenching Machines	Trenchers	Diesel	300	0	0	0	0	0.371	0.59	0.032	0.035	0.128	0.028	0.027	0.412	0.001	536.730	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	10	170	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.77	0.0000	0.000	20.85
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	20	340	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	41.54	0.0000	0.001	41.70
Grapple Truck	Off-highway Trucks	Diesel	350	1	14	12	168	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.53	0.0000	0.000	20.60
Buggy - IC-50/MST-600 (8,800# L	Other Construction Equipment	Diesel	120	1	14	40	560	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.002	0.01	0.002	0.002	0.0001	0.02	23.46	0.0000	0.000	23.54
Buggy - Coating	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0	0	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Hydro excavator Truck	Excavators	Diesel	400	1	3	60	180	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	25.14	0.0000	0.000	25.23
Welding Machine	Welders	Diesel	40	20	15	60	18,000	0.481	0.21	0.363	0.389	1.431	0.216	0.209	3.158	0.002	694.857	0.6	0.065	0.24	0.036	0.035	0.0003	0.53	115.81	0.0010	0.004	117.03
185 CFM Compressor	Air Compressors	Diesel	65	1	15	60	900	0.408	0.43	0.080	0.084	0.388	0.049	0.048	2.706	0.002	590.118	0.6	0.002	0.01	0.001	0.001	0.0000	0.08	16.36	0.0000	0.000	16.45
Light Plant	Signal Boards/Light Plants	Diesel	20	1	3	20	60	0.408	0.43	0.321	0.357	1.518	0.177	0.172	3.776	0.002	589.348	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.34	0.0000	0.000	0.34
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	17	12	4,080	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	17	12	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	17	24	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
Mechnics Truck	Off-highway Trucks	Diesel	400	1	17	12	204	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.000	0.000	0.0001	0.01	28.49	0.0000	0.000	28.59
Truck - Semi	Off-highway Trucks	Diesel	450	1	17	36	612	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.002	0.01	0.002	0.001	0.0003	0.02	96.15	0.0000	0.001	96.50
Bus	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Passenger Van	Off-highway Trucks	Gas	300	2	17	12	408	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
																		Construction TOTAL (Tons)	**0.13**	**0.48**	**0.09**	**0.08**	**0.00**	**1.31**	**1,525.27**	**0.0020**	**0.02**	**1,531.87**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

NOTES:

Note 3: Emission factor for N2O from 40 CFR 98 Subpart C.

Note 4: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Note 5: Annual CH4 emissions are calculated using the following calculation (HC emission factor (g/hp-hr) * (1-0.984) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)
This equation is derived in part from subtracting methane from THC to calculate NMHC. THC - CH_4 = NMHC; THC - NMHC = CH_4; THC - (0.984 * THC) = CH_4; THC * (1-0.984) = CH_4
0.984 is the ratio of MNHC to THC and 0.9 is ratio for gasoline - from "Conversion Factors for Hydrocarbon Emission Components", July 2010, EPA-420-R-10-015.

Note 6: Emissions estimate for N2O is based on a diesel higher heating value of 138,000 Btu/gallon and a density of 7.05 lb/gallon
Emissions estimate for N2O is based on a gasoline higher heating value of 130,000 Btu/gallon and a density of 6.17 lb/gallon

Note 7: Greenhouse gases (GHG) are converted to carbon dioxide equivalents (CO2) using 40 CFR 98 Subpart A global warming potentials as follows:

CO2	1
CH4	28
N2O	265

Franklin Farms Project
Franklin Farms Bee Bayou Delivery Meter Station
Construction Emissions - Off-road Engines

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	HAP Emissions (Tons) [3]				
								Benzene	Acrolein	1,3-Butadiene	Formaldehyde	Acetaldehyde
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	180	0.371	0.59	9.67E-06	3.79E-06	3.33E-07	5.56E-05	2.38E-05
Dump Truck	Off-highway Trucks	Diesel	400	2	360	0.371	0.59	2.85E-05	1.17E-05	9.82E-07	1.68E-04	7.58E-05
Excavator (KOM 210)	Excavators	Diesel	160	3	3,060	0.371	0.59	1.46E-04	5.29E-05	5.03E-06	8.08E-04	3.09E-04
Excavator (CAT 329)	Excavators	Diesel	200	2	2,040	0.371	0.59	8.40E-05	3.42E-05	2.89E-06	4.92E-04	2.21E-04
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	960	0.371	0.59	8.51E-05	3.21E-05	2.97E-06	4.81E-04	1.92E-04
Grader	Graders	Diesel	250	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Cranes	Cranes	Diesel	300	1	320	0.367	0.43	3.54E-05	1.33E-05	1.25E-06	2.00E-04	7.86E-05
Generator Sets	Generator Sets	Diesel	200	2	1,120	0.367	0.43	5.76E-04	3.85E-04	3.31E-05	4.30E-03	1.54E-03
Pump	Pumps	Diesel	120	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0.433	0.21	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0.492	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	280	0.433	0.21	1.33E-04	7.37E-05	6.55E-06	8.95E-04	3.19E-04
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	560	0.481	0.21	3.36E-04	2.47E-04	2.10E-05	2.66E-03	9.46E-04
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0.408	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Trenching Machines	Trenchers	Diesel	300	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	170	0.371	0.59	1.18E-05	4.82E-06	4.06E-07	6.93E-05	3.13E-05
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	340	0.371	0.59	2.36E-05	9.65E-06	8.11E-07	1.39E-04	6.26E-05
Grapple Truck	Off-highway Trucks	Diesel	350	1	168	0.371	0.59	1.17E-05	4.77E-06	4.01E-07	6.84E-05	3.09E-05
Buggy - IC-50/MST-600 (8,800# Load	Other Construction Equipment	Diesel	120	1	560	0.371	0.59	7.23E-05	2.72E-05	2.73E-06	4.16E-04	1.50E-04
Buggy - Coating	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Hydro excavator Truck	Excavators	Diesel	400	1	180	0.371	0.59	3.36E-05	1.25E-05	1.16E-06	1.89E-04	7.48E-05
Welding Machine	Welders	Diesel	40	20	18,000	0.481	0.21	2.22E-03	1.58E-03	1.20E-04	1.58E-02	5.66E-03
185 CFM Compressor	Air Compressors	Diesel	65	1	900	0.408	0.43	1.15E-04	4.20E-05	4.21E-06	6.56E-04	2.33E-04
Light Plant	Signal Boards/Light Plants	Diesel	20	1	60	0.408	0.43	1.10E-05	3.80E-06	3.77E-07	5.92E-05	2.11E-05
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	4,080	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
Mechnics Truck	Off-highway Trucks	Diesel	400	1	204	0.371	0.59	1.62E-05	6.62E-06	5.56E-07	9.50E-05	4.29E-05
Truck - Semi	Off-highway Trucks	Diesel	450	1	612	0.371	0.59	5.46E-05	2.23E-05	1.88E-06	3.21E-04	1.45E-04
Bus	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Passenger Van	Off-highway Trucks	Gas	300	2	408	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations						**TOTAL (Tons)**		**4.42E-03**	**2.73E-03**	**2.21E-04**	**3.03E-02**	**1.12E-02**

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

Note 3: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr)[from MOVES5] * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Franklin Farms Project
Franklin Farms Trio Farm Delivery Meter Station
Construction Emissions - Off-road Engines **Criteria and GHG Emissions**

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Weeks	Hours per Week	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	Emission Factors EF (g/hp-hr) [1] HC	VOC	CO	PM_{10}	$PM_{2.5}$	NOx	SO_2	CO_2	N_2O EF [3] g/MMBtu	VOC [4] tons	CO [4] tons	PM_{10} tons [4]	$PM_{2.5}$ tons [4]	SO_2 [4] tons	NOx [4] tons	CO_2 [4] tons	CH_4 [5] tons	N_2O [6] tons	GHG [7] tons of CO_2e
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0	0	0.367	0.43	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	3	60	180	0.371	0.59	0.012	0.013	0.041	0.011	0.010	0.161	0.001	536.793	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	12.57	0.0000	0.000	12.61
Dump Truck	Off-highway Trucks	Diesel	400	2	3	60	360	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	50.27	0.0000	0.001	50.46
Excavator (KOM 210)	Excavators	Diesel	160	3	17	60	3,060	0.371	0.59	0.010	0.011	0.066	0.015	0.015	0.212	0.001	536.802	0.6	0.003	0.02	0.005	0.005	0.0005	0.07	170.93	0.0000	0.002	171.55
Excavator (CAT 329)	Excavators	Diesel	200	2	17	60	2,040	0.371	0.59	0.011	0.011	0.030	0.008	0.008	0.136	0.001	536.798	0.6	0.003	0.01	0.002	0.002	0.0004	0.04	142.44	0.0000	0.002	142.96
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0	0	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	16	60	960	0.371	0.59	0.017	0.018	0.070	0.017	0.016	0.214	0.001	536.779	0.6	0.002	0.01	0.002	0.002	0.0002	0.03	67.03	0.0000	0.001	67.27
Grader	Graders	Diesel	250	0	0	0	0	0.371	0.59	0.012	0.013	0.039	0.010	0.010	0.156	0.001	536.794	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0	0	0.367	0.43	0.195	0.213	0.631	0.131	0.128	2.615	0.002	530.434	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Cranes	Cranes	Diesel	300	1	8	40	320	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	24.16	0.0000	0.000	24.28
Generator Sets	Generator Sets	Diesel	200	2	14	40	1,120	0.367	0.43	0.158	0.171	0.517	0.111	0.108	2.094	0.002	530.549	0.6	0.018	0.05	0.012	0.011	0.0002	0.22	56.33	0.0003	0.001	56.62
Pump	Pumps	Diesel	120	0	0	0	0	0.367	0.43	0.174	0.188	0.627	0.143	0.139	2.227	0.002	530.499	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0	0	0.433	0.21	0.244	0.266	0.872	0.173	0.168	1.606	0.002	625.783	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0	0	0.492	0.59	0.146	0.159	0.543	0.128	0.124	2.076	0.002	530.587	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	14	20	280	0.433	0.21	0.399	0.435	1.380	0.258	0.250	2.650	0.002	625.297	0.6	0.004	0.01	0.002	0.002	0.0000	0.02	5.07	0.0001	0.000	5.12
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0	0	0.367	0.43	0.021	0.023	0.112	0.028	0.027	0.341	0.001	530.977	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0	0	0.367	0.43	0.019	0.020	0.060	0.014	0.014	0.239	0.001	530.985	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	14	40	560	0.481	0.21	0.822	0.889	4.455	0.670	0.650	4.066	0.002	693.429	0.6	0.011	0.06	0.009	0.008	0.0000	0.05	8.81	0.0002	0.000	8.90
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0	0	0.408	0.43	0.087	0.092	0.279	0.020	0.020	2.529	0.002	590.091	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Trenching Machines	Trenchers	Diesel	300	0	0	0	0	0.371	0.59	0.032	0.035	0.128	0.028	0.027	0.412	0.001	536.730	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	10	170	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.77	0.0000	0.000	20.85
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	17	20	340	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	41.54	0.0000	0.001	41.70
Grapple Truck	Off-highway Trucks	Diesel	350	1	14	12	168	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0001	0.01	20.53	0.0000	0.000	20.60
Buggy - IC-50/MST-600 (8,800# L	Other Construction Equipment	Diesel	120	1	14	40	560	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.002	0.01	0.002	0.002	0.0001	0.02	23.46	0.0000	0.000	23.54
Buggy - Coating	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	14	25	350	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.001	0.00	0.001	0.001	0.0000	0.01	14.66	0.0000	0.000	14.71
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0	0	0.371	0.59	0.032	0.036	0.165	0.037	0.036	0.456	0.001	536.731	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Hydro excavator Truck	Excavators	Diesel	400	1	3	60	180	0.371	0.59	0.017	0.018	0.089	0.019	0.018	0.239	0.001	536.777	0.6	0.001	0.00	0.001	0.001	0.0001	0.01	25.14	0.0000	0.000	25.23
Welding Machine	Welders	Diesel	40	20	15	60	18,000	0.481	0.21	0.363	0.389	1.431	0.216	0.209	3.158	0.002	694.857	0.6	0.065	0.24	0.036	0.035	0.0003	0.53	115.81	0.0010	0.004	117.03
185 CFM Compressor	Air Compressors	Diesel	65	1	15	60	900	0.408	0.43	0.080	0.084	0.388	0.049	0.048	2.706	0.002	590.118	0.6	0.002	0.01	0.001	0.001	0.0000	0.08	16.36	0.0000	0.000	16.45
Light Plant	Signal Boards/Light Plants	Diesel	20	1	3	20	60	0.408	0.43	0.321	0.357	1.518	0.177	0.172	3.776	0.002	589.348	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.34	0.0000	0.000	0.34
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	17	12	4,080	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	17	12	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	17	24	2,040	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.006	0.02	0.004	0.004	0.0008	0.07	284.88	0.0001	0.004	285.92
Mechnics Truck	Off-highway Trucks	Diesel	400	1	17	12	204	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.001	0.00	0.000	0.000	0.0001	0.01	28.49	0.0000	0.000	28.59
Truck - Semi	Off-highway Trucks	Diesel	450	1	17	36	612	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.002	0.01	0.002	0.001	0.0003	0.02	96.15	0.0000	0.001	96.50
Bus	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0	0	0.371	0.59	0.011	0.011	0.031	0.008	0.008	0.136	0.001	536.797	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
Passenger Van	Off-highway Trucks	Gas	300	2	17	12	408	0.000	0.59	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.6	0.000	0.00	0.000	0.000	0.0000	0.00	0.00	0.0000	0.000	0.00
																		Construction TOTAL (Tons)	**0.13**	**0.48**	**0.09**	**0.08**	**0.00**	**1.31**	**1,525.27**	**0.0020**	**0.02**	**1,531.87**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVESS. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

NOTES:

Note 3: Emission factor for N2O from 40 CFR 98 Subpart C.

Note 4: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Note 5: Annual CH4 emissions are calculated using the following calculation (HC emission factor (g/hp-hr) * (1-0.984) * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)
This equation is derived in part from subtracting methane from THC to calculate NMHC. THC - CH_4 = NMHC; THC - NMHC = CH_4; THC - (0.984 * THC) = CH_4; THC * (1-0.984) = CH_4
0.984 is the ratio of MNHC to THC and 0.9 is ratio for gasoline - from "Conversion Factors for Hydrocarbon Emission Components", July 2010, EPA-420-R-10-015.

Note 6: Emissions estimate for N2O is based on a diesel higher heating value of 138,000 Btu/gallon and a density of 7.05 lb/gallon
Emissions estimate for N2O is based on a gasoline higher heating value of 130,000 Btu/gallon and a density of 6.17 lb/gallon

Note 7: Greenhouse gases (GHG) are converted to carbon dioxide equivalents (CO2) using 40 CFR 98 Subpart A global warming potentials as follows:

CO2	1
CH4	28
N2O	265

Franklin Farms Project
Franklin Farms Trio Farm Delivery Meter Station
Construction Emissions - Off-road Engines

Description*	Equipment Category based on MOVES/NONROAD Classification	Fuel Type	Equipment HP	Number of Equipment	Total Working hrs	BSFC [1] lb/hp-hr	Load Factor [2]	HAP Emissions (Tons) [3]				
								Benzene	Acrolein	1,3-Butadiene	Formaldehyde	Acetaldehyde
Chipper/Shredder	Chippers/Stump Grinders (com)	Diesel	125	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Dozers (D6T)	Crawler Tractor/Dozers	Diesel	200	1	180	0.371	0.59	9.67E-06	3.79E-06	3.33E-07	5.56E-05	2.38E-05
Dump Truck	Off-highway Trucks	Diesel	400	2	360	0.371	0.59	2.85E-05	1.17E-05	9.82E-07	1.68E-04	7.58E-05
Excavator (KOM 210)	Excavators	Diesel	160	3	3,060	0.371	0.59	1.46E-04	5.29E-05	5.03E-06	8.08E-04	3.09E-04
Excavator (CAT 329)	Excavators	Diesel	200	2	2,040	0.371	0.59	8.40E-05	3.42E-05	2.89E-06	4.92E-04	2.21E-04
Excavator (CAT 336)	Excavators	Diesel	314	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Front End Loaders	Rubber Tire Loaders	Diesel	200	1	960	0.371	0.59	8.51E-05	3.21E-05	2.97E-06	4.81E-04	1.92E-04
Grader	Graders	Diesel	250	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
HDD Rig	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Bore/Drill Rigs	Bore/Drill Rigs	Diesel	300	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Cranes	Cranes	Diesel	300	1	320	0.367	0.43	3.54E-05	1.33E-05	1.25E-06	2.00E-04	7.86E-05
Generator Sets	Generator Sets	Diesel	200	2	1,120	0.367	0.43	5.76E-04	3.85E-04	3.31E-05	4.30E-03	1.54E-03
Pump	Pumps	Diesel	120	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Tractors/Loaders/Backhoes	Tractors/Loaders/Backhoes	Diesel	149	0	0	0.433	0.21	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Chipper/Shredder	Other Construction Equipment	Diesel	125	0	0	0.492	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Scissor Lift/ Fork Lift	Aerial Lifts	Diesel	125	1	280	0.433	0.21	1.33E-04	7.37E-05	6.55E-06	8.95E-04	3.19E-04
SideBoom (Cat 561)	Cranes	Diesel	123	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
SideBoom (Cat 572)	Cranes	Diesel	240	0	0	0.367	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Steer Loader (299D3)	Skid Steer Loaders	Diesel	98	1	560	0.481	0.21	3.36E-04	2.47E-04	2.10E-05	2.66E-03	9.46E-04
Sweeper	Sweepers/Scrubbers	Diesel	50	0	0	0.408	0.43	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Trenching Machines	Trenchers	Diesel	300	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Fuel Truck 4K GAL	Off-highway Trucks	Diesel	350	1	170	0.371	0.59	1.18E-05	4.82E-06	4.06E-07	6.93E-05	3.13E-05
Water Truck 4K GAL	Off-highway Trucks	Diesel	350	1	340	0.371	0.59	2.36E-05	9.65E-06	8.11E-07	1.39E-04	6.26E-05
Grapple Truck	Off-highway Trucks	Diesel	350	1	168	0.371	0.59	1.17E-05	4.77E-06	4.01E-07	6.84E-05	3.09E-05
Buggy - IC-50/MST-600 (8,800# Load	Other Construction Equipment	Diesel	120	1	560	0.371	0.59	7.23E-05	2.72E-05	2.73E-06	4.16E-04	1.50E-04
Buggy - Coating	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Buggy - Sand Blasting	Other Construction Equipment	Diesel	120	1	350	0.371	0.59	4.52E-05	1.70E-05	1.71E-06	2.60E-04	9.40E-05
Tack Rig	Other Construction Equipment	Diesel	120	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Hydro excavator Truck	Excavators	Diesel	400	1	180	0.371	0.59	3.36E-05	1.25E-05	1.16E-06	1.89E-04	7.48E-05
Welding Machine	Welders	Diesel	40	20	18,000	0.481	0.21	2.22E-03	1.58E-03	1.20E-04	1.58E-02	5.66E-03
185 CFM Compressor	Air Compressors	Diesel	65	1	900	0.408	0.43	1.15E-04	4.20E-05	4.21E-06	6.56E-04	2.33E-04
Light Plant	Signal Boards/Light Plants	Diesel	20	1	60	0.408	0.43	1.10E-05	3.80E-06	3.77E-07	5.92E-05	2.11E-05
1/2 Ton Pickup	Off-highway Trucks	Gas	300	20	4,080	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3/4 Ton Pickup	Off-highway Trucks	Diesel	400	10	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
1 Ton Crew Truck	Off-highway Trucks	Diesel	400	5	2,040	0.371	0.59	1.62E-04	6.62E-05	5.56E-06	9.50E-04	4.29E-04
Mechnics Truck	Off-highway Trucks	Diesel	400	1	204	0.371	0.59	1.62E-05	6.62E-06	5.56E-07	9.50E-05	4.29E-05
Truck - Semi	Off-highway Trucks	Diesel	450	1	612	0.371	0.59	5.46E-05	2.23E-05	1.88E-06	3.21E-04	1.45E-04
Bus	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Skid Truck	Off-highway Trucks	Diesel	350	0	0	0.371	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Passenger Van	Off-highway Trucks	Gas	300	2	408	0.000	0.59	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
							TOTAL (Tons)	**4.42E-03**	**2.73E-03**	**2.21E-04**	**3.03E-02**	**1.12E-02**

* Gasoline 1/2 Ton Pickup Truck and Gasoline Passenger Van included with onroad calculations

NOTES:

Note 1: Brake-specific fuel consumption and zero hour steady state emission factors (EF; g/hp-hr) are from MOVES5. Emission Factors are averaged for the period of construction for Richland Parish, LA.

Note 2: Load factor based on Appendix A of "Median Life, Annual Activity, and Load Factor Values for Nonroad Engine Emissions Modeling", July 2010, EPA-420-R-10-016.

Note 3: Annual emissions are calculated using the following calculation (emission factor (g/hp-hr)[from MOVES5] * horsepower * hours operated * load factor) / (2000 lb/ton * 453.59 g/lb)

Franklin Farms Project
Construction Emissions - Fugitive dust from construction activities

Source/Construction Activity	Area of Land Disturbance (acres):	Months of Construction Activity[1]:	PM2.5 ton/period	PM10 ton/period
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral - 2027	207.00	6	61.72	411.46
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral - 2028	50.00	1	2.48	16.56
Franklin Farms Gulf Run Bi Di Meter Station - 2027	2.20	5	0.55	3.64
Franklin Farms Gulf South Receipt Meter Station - 2027	2.20	4	0.44	2.92
Franklin Farms Gulf Crossing Receipt Meter Station - 2027	2.20	4	0.44	2.92
Franklin Farms MEP Receipt Meter Station - 2027	2.20	4	0.44	2.92
Franklin Farms Delivery Meter Station - 2027	2.20	5	0.55	3.64
Franklin Farms Bee Bayou Delivery Meter Station - 2027	2.20	5	0.55	3.64
Franklin Farms Trio Farm Delivery Meter Station - 2027	2.20	5	0.55	3.64
Total PM Emissions			67.70	451.34
Total(1)	272.40	39	67.70	451.34

Emission Factor Equations[2]

PM10 Emission Factor = 0.42 x (1 - CE) x SAF x SMA CE = control efficiency

PM2.5 Emission Factor = PM10 Emission Factor x Ratio

Project Area Specific PM Emission Factors

Base PM10 Emission Factor(tons/acre-month)	0.42	*Table 4-8 of ([2])
Ratio of PM2.5 to PM10 (Ratio) [2]	0.15	*Section 4.1 of ([2])
Construction Dust Plan Control Efficiency Factor (CE) [3]	50%	*Section 5.7.1 of ([2])
Silt Adjustment Factor (SAF):	5.78	(SAF = Dry Silt Content/ 9%) *Section 5.7.3 of ([2])
Basis (original Studies value, see Note 2)	9%	*Section 5.7.2 of ([2])
Dry Silt Content of Project Soils	52%	Value for silt loam

Soil Moisture Adjustment Factor (per Climate Divisions of Project Area)

Thornthwaite's Precipitation-Evaporation Index[4]			Adjusted Emission Factors (tons/acre-month)	
Basis (Arid Southwest US)	24	Soil Moisture Adjustment (SMA)	**$PM_{2.5}$**	**PM_{10}**
	87.9	0.27	0.0497	0.3313

NOTES:

[1] The entire construction activity duration is expected to be 7 months.

[2] Estimating Particulate Matter Emissions From Construction Operations, Final Report, Eastern Research Group, Inc., September 30, 1999, EPA Contract 68-D7-0068. It is assumed that the factor and methodology for adjusting the factor for Roadway Construction Emissions are similar in nature to the pipeline construction.

[3] Required and effective fugitive dust control measures will be determined at the site during construction activity. Application of water to mitigate fugitive dust will be used as one of the primary control measures. Mitigation of fugitive dust by the use of water application and other measures is reflected in the 50% control efficiency recommended in the referenced EPA Contract 68-D7-0068 Document.

[4] Days of precipitation >0.01 inch per year (P) of 87.9 per Normal NCEI data for Rayville, LA

Franklin Farms Project
Construction Emissions - On-road Vehicles

Description	Category Based on MOVES5 Classification	Number of Equipment	Total Days	Working Days	Miles per Day[3]	Calc. VMT [1]
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral - 2027						
1/2 Ton Pickup	31 - Passenger Truck - Gas	40	182	182	80	582,400
Passenger Van	21 - Passenger Car - Gas	4	182	182	80	58,240
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	30	182	182	490	2,675,400
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	70	182	182	490	6,242,600
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	30	182	182	490	2,675,400
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	20	182	182	490	1,783,600
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	182	182	490	89,180
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral - 2028						
1/2 Ton Pickup	31 - Passenger Truck - Gas	40	28	28	80	89,600
Passenger Van	21 - Passenger Car - Gas	4	28	28	80	8,960
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	15	28	28	490	205,800
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	35	28	28	490	480,200
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	15	28	28	490	205,800
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	10	28	28	490	137,200
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	28	28	490	13,720
Franklin Farms Gulf Run Bi Di Meter Station						
1/2 Ton Pickup	31 - Passenger Truck - Gas	20	119	119	40	95,200
Passenger Van	21 - Passenger Car - Gas	2	119	119	40	9,520
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0	119	119	490	0
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	25	119	119	490	1,457,750
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	10	119	119	490	583,100
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	20	119	119	490	1,166,200
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	119	119	490	58,310
Franklin Farms Gulf South Receipt Meter Station - 2027						
1/2 Ton Pickup	31 - Passenger Truck - Gas	20	102	102	40	81,600
Passenger Van	21 - Passenger Car - Gas	2	102	102	40	8,160
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0	0	0	0	0
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	25	102	102	490	1,249,500
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	10	102	102	490	499,800
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	20	102	102	490	999,600
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	102	102	490	49,980
Franklin Farms Gulf South Receipt Meter Station - 2028						
1/2 Ton Pickup	31 - Passenger Truck - Gas	20	17	17	40	13,600
Passenger Van	21 - Passenger Car - Gas	2	17	17	40	1,360
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0	0	0	0	0
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	12	17	17	490	99,960
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	2	17	17	490	16,660
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	2	17	17	490	16,660
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	17	17	490	8,330
Franklin Farms Gulf Crossing Receipt Meter Station - 2027						
1/2 Ton Pickup	31 - Passenger Truck - Gas	20	102	102	40	81,600
Passenger Van	21 - Passenger Car - Gas	2	102	102	40	8,160
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0	0	0	0	0
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	25	102	102	490	1,249,500
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	10	102	102	490	499,800
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	20	102	102	490	999,600
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	102	102	490	49,980
Franklin Farms Gulf Crossing Receipt Meter Station - 2028						
1/2 Ton Pickup	31 - Passenger Truck - Gas	20	17	17	40	13,600
Passenger Van	21 - Passenger Car - Gas	2	17	17	40	1,360
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0	0	0	0	0
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	12	17	17	490	99,960
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	2	17	17	490	16,660
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	2	17	17	490	16,660
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	17	17	490	8,330
Franklin Farms MEP Receipt Meter Station - 2027						
1/2 Ton Pickup	31 - Passenger Truck - Gas	20	102	102	20	40,800
Passenger Van	21 - Passenger Car - Gas	2	102	102	20	4,080
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0	0	0	0	0
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	25	102	102	70	178,500
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	10	102	102	70	71,400
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	20	102	102	70	142,800
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	102	102	70	7,140
Franklin Farms MEP Receipt Meter Station - 2028						
1/2 Ton Pickup	31 - Passenger Truck - Gas	20	17	17	40	13,600
Passenger Van	21 - Passenger Car - Gas	2	17	17	40	1,360
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0	0	0	0	0
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	12	17	17	490	99,960
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	2	17	17	490	16,660
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	2	17	17	490	16,660
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	17	17	490	8,330
Franklin Farms Delivery Meter Station						
1/2 Ton Pickup	31 - Passenger Truck - Gas	20	119	119	40	95,200
Passenger Van	21 - Passenger Car - Gas	2	119	119	40	9,520
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0	119	119	490	0
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	25	119	119	490	1,457,750
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	10	119	119	490	583,100
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	20	119	119	490	1,166,200
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	119	119	490	58,310
Franklin Farms Bee Bayou Delivery Meter Station						
1/2 Ton Pickup	31 - Passenger Truck - Gas	20	119	119	40	95,200
Passenger Van	21 - Passenger Car - Gas	2	119	119	40	9,520
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0	119	119	490	0
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	25	119	119	490	1,457,750
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	10	119	119	490	583,100
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	20	119	119	490	1,166,200
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	119	119	490	58,310
Franklin Farms Trio Farm Delivery Meter Station						
1/2 Ton Pickup	31 - Passenger Truck - Gas	20	119	119	40	95,200
Passenger Van	21 - Passenger Car - Gas	2	119	119	40	9,520
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0	119	119	490	0
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	25	119	119	490	1,457,750
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	10	119	119	490	583,100
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	20	119	119	490	1,166,200
Fuel Delivery	32 - Light Commerical Truck - Diesel	1	119	119	490	58,310

Emission Factors [2] (g/VMT)

Description	Category Based on MOVES5 Classification	VOC	CO	PM_{10}	$PM_{2.5}$	SO_2	NOx	CO_2	CH_4	N_2O	GHG (CO_2e)	HAPs
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral - 2027												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral - 2028												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms Gulf Run Bi Di Meter Station												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms Gulf South Receipt Meter Station - 2027												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms Gulf South Receipt Meter Station - 2028												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms Gulf Crossing Receipt Meter Station - 2027												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms Gulf Crossing Receipt Meter Station - 2028												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms MEP Receipt Meter Station - 2027												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms MEP Receipt Meter Station - 2028												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms Delivery Meter Station												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms Bee Bayou Delivery Meter Station												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02
Franklin Farms Trio Farm Delivery Meter Station												
1/2 Ton Pickup	31 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Passenger Van	21 - Passenger Car - Gas	0.25	6.33	0.02	0.01	0.00	0.20	322.61	0.02	0.00	323.63	0.01
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	0.22	3.19	0.08	0.06	0.00	1.29	555.53	0.01	0.04	565.18	0.03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	0.24	6.03	0.02	0.01	0.00	0.28	416.44	0.02	0.00	417.72	0.01
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	0.34	1.80	0.33	0.25	0.00	4.34	893.01	0.01	0.08	913.73	0.04
Fuel Delivery	32 - Light Commerical Truck - Diesel	0.16	2.45	0.06	0.04	0.00	1.01	554.59	0.01	0.05	568.43	0.02

Emissions (ton/yr) [4]

Description	Category Based on MOVES5 Classification	VOC ton/yr	CO ton/yr	PM_{10} ton/yr	$PM_{2.5}$ ton/yr	SO_2 ton/yr	NOx ton/yr	CO_2 ton/yr	CH_4 ton/yr	N_2O ton/yr	GHG ton CO_2e /yr	HAPs ton/yr
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral - 2027												
1/2 Ton Pickup	31 - Passenger Truck - Gas	1.54E-01	3.87E+00	1.34E-02	4.82E-03	2.29E-03	1.81E-01	2.67E+02	1.26E-02	1.76E-03	2.68E+02	4.02E-03
Passenger Van	21 - Passenger Car - Gas	1.62E-02	4.06E-01	1.21E-03	4.19E-04	1.27E-02	1.27E-02	2.07E+01	1.12E-03	1.27E-04	2.08E+01	3.52E-04
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	6.54E-01	9.40E+00	2.30E-01	1.66E-01	5.57E-03	3.81E+00	1.64E+03	3.38E-02	1.04E-01	1.67E+03	8.57E-02
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	1.65E+00	4.15E+01	1.44E-01	5.17E-02	2.45E-02	1.94E+00	2.87E+03	1.35E-01	1.88E-02	2.87E+03	4.31E-02
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	7.09E-01	1.78E+01	6.16E-02	2.22E-02	1.05E-02	8.32E-01	1.23E+03	5.79E-02	8.07E-03	1.23E+03	1.85E-02
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	6.68E-01	3.53E+00	6.51E-01	4.85E-01	6.02E-03	8.53E+00	1.76E+03	2.56E-02	1.51E-01	1.80E+03	8.84E-02
Fuel Delivery	32 - Light Commerical Truck - Diesel	1.54E-02	2.41E-01	6.14E-03	4.28E-03	1.84E-04	9.95E-02	5.45E+01	9.98E-04	5.03E-03	5.59E+01	2.00E-03
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral - 2028												
1/2 Ton Pickup	31 - Passenger Truck - Gas	2.37E-02	5.96E-01	2.06E-03	7.42E-04	3.52E-04	2.79E-02	4.11E+01	1.94E-03	2.70E-04	4.13E+01	6.18E-04
Passenger Van	21 - Passenger Car - Gas	2.49E-03	6.25E-02	1.86E-04	6.44E-05	2.73E-05	1.95E-03	3.19E+00	1.73E-04	1.96E-05	3.20E+00	5.42E-05
Light-Duty Diesel Truck	31 - Passenger Truck - Diesel	5.03E-02	7.23E-01	1.77E-02	1.28E-02	4.29E-04	2.93E-01	1.26E+02	2.60E-03	7.98E-03	1.28E+02	6.59E-03
Light-Duty Gasoline Truck	32 - Passenger Truck - Gas	1.27E-01	3.19E+00	1.11E-02	3.98E-03	1.88E-03	1.49E-01	2.20E+02	1.04E-02	1.44E-03	2.21E+02	3.31E-03
Light-Duty Gasoline Vehicle	32 - Passenger Truck - Gas	5.45E-02	1.37E+00	4.74E-03	1.70E-03	8.09E-04	6.40E-02	9.45E+01	4.45E-03	6.21E-04	9.48E+01	1.42E-03
Heavy Duty Diesel Vehicle	52 - Single Unit Short-haul Truck - Diesel	5.14E-02	2

Franklin Farms Project
Construction Emissions - Fugitive Dust from Roadways

Vehicle Type	Daily Onsite Vehicular Traffic	Construction Days	Average Daily Travel Distance per Vehicle (VMT/vehicle/day)		Distance Traveled [1] (VMT)		Estimated Vehicle Weight	
	(Veh)		Paved	Unpaved	Paved	Unpaved	(lbs)	(tons)
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral - 2027								
Dump Trucks	2	182	119	1	43,316	364	20000	10.0
Light Duty Trucks	44	182	79	1	632,632	8,008	5000	2.5
Medium Duty Trucks	41	182	79	1	589,498	7,462	15000	7.5
Water / Fuel Truck	7	182	159	1	202,566	1,274	50000	25.0
Franklin Farms Lateral and the Franklin Farms Trio Farm Lateral - 2028								
Dump Trucks	2	28	119	1	6,664	56	20000	10.0
Light Duty Trucks	44	28	79	1	97,328	1,232	5000	2.5
Medium Duty Trucks	41	28	79	1	90,692	1,148	15000	7.5
Water / Fuel Truck	7	28	159	1	31,164	196	50000	25.0
Franklin Farms Gulf Run Bi Di Meter Station								
Dump Trucks	2	119	79	1	18,802	238	20000	10.0
Light Duty Trucks	22	119	39	1	102,102	2,618	5000	2.5
Medium Duty Trucks	17	119	39	1	78,897	2,023	15000	7.5
Water / Fuel Truck	3	119	79	1	28,203	357	50000	25.0
Franklin Farms Gulf South Receipt Meter Station - 2027								
Dump Trucks	2	102	79	1	16,116	204	20000	10.0
Light Duty Trucks	22	102	39	1	87,516	2,244	5000	2.5
Medium Duty Trucks	17	102	39	1	67,626	1,734	15000	7.5
Water / Fuel Truck	3	102	79	1	24,174	306	50000	25.0
Franklin Farms Gulf South Receipt Meter Station - 2028								
Dump Trucks	2	17	79	1	2,686	34	20000	10.0
Light Duty Trucks	22	17	39	1	14,586	374	5000	2.5
Medium Duty Trucks	17	17	39	1	11,271	289	15000	7.5
Water / Fuel Truck	3	17	39	1	1,989	51	50000	25.0
Franklin Farms Gulf Crossing Receipt Meter Station - 2027								
Dump Trucks	2	102	79	1	16,116	204	20000	10.0
Light Duty Trucks	22	102	39	1	87,516	2,244	5000	2.5
Medium Duty Trucks	17	102	39	1	67,626	1,734	15000	7.5
Water / Fuel Truck	3	102	79	1	24,174	306	50000	25.0
Franklin Farms Gulf Crossing Receipt Meter Station - 2028								
Dump Trucks	2	17	79	1	2,686	34	20000	10.0
Light Duty Trucks	22	17	39	1	14,586	374	5000	2.5
Medium Duty Trucks	17	17	39	1	11,271	289	15000	7.5
Water / Fuel Truck	3	17	39	1	1,989	51	50000	25.0
Franklin Farms MEP Receipt Meter Station - 2027								
Dump Trucks	2	102	79	1	16,116	204	20000	10.0
Light Duty Trucks	22	102	39	1	87,516	2,244	5000	2.5
Medium Duty Trucks	17	102	39	1	67,626	1,734	15000	7.5
Water / Fuel Truck	3	102	79	1	24,174	306	50000	25.0
Franklin Farms MEP Receipt Meter Station - 2028								
Dump Trucks	2	17	79	1	2,686	34	20000	10.0
Light Duty Trucks	22	17	39	1	14,586	374	5000	2.5
Medium Duty Trucks	17	17	39	1	11,271	289	15000	7.5
Water / Fuel Truck	3	17	39	1	1,989	51	50000	25.0
Franklin Farms Delivery Meter Station								
Dump Trucks	2	119	79	1	18,802	238	20000	10.0
Light Duty Trucks	22	119	39	1	102,102	2,618	5000	2.5
Medium Duty Trucks	17	119	39	1	78,897	2,023	15000	7.5
Water / Fuel Truck	3	119	79	1	28,203	357	50000	25.0
Franklin Farms Bee Bayou Delivery Meter Station								
Dump Trucks	2	119	79	1	18,802	238	20000	10.0
Light Duty Trucks	22	119	39	1	102,102	2,618	5000	2.5
Medium Duty Trucks	17	119	39	1	78,897	2,023	15000	7.5
Water / Fuel Truck	3	119	79	1	28,203	357	50000	25.0
Franklin Farms Trio Farm Delivery Meter Station								
Dump Trucks	2	119	79	1	18,802	238	20000	10.0
Light Duty Trucks	22	119	39	1	102,102	2,618	5000	2.5
Medium Duty Trucks	17	119	39	1	78,897	2,023	15000	7.5
Water / Fuel Truck	3	119	79	1	28,203	357	50000	25.0
TOTALS (VMT)					**3,283,768**	**56,392**		
WEIGHTED AVERAGE VEHICLE WEIGHT (tons)							Paved	**6.37**
							Unpaved	**6.37**

Parameter	PM_{10}			$PM_{2.5}$		
	Paved	Unpaved	Total	Paved	Unpaved	Total
Uncontrolled Emission Factor (lb/VMT) [2],[3]	0.08	1.17	--	0.02	0.12	--
Control Efficiency for dust suppression activities	0%	0%	--	0%	0%	--
Controlled Emission Factor (lb/VMT)	0.08	1.17	--	0.02	0.12	--
Fugitive Roadways Emissions (tons) [4]	138.67	33.04	**171.70**	34.04	3.30	**37.34**

[1] VMT = Vehicles x Construction Days x Daily Distance Traveled per Vehicle.

[2] The emission factor for paved roads was estimated using USEPA's AP-42 (1/11) Section 13.2.1 "Paved Roads" and the following:

$$E_{ext} = [k(sL^{0.91} * (W)^{1.02}](1 - \frac{P}{4N})$$

A. Emission factor (E) is calculated using Equation (2) in Section 13.2.1:
B. Table 13.2.1-1 specifies that for PM-10, k is 0.0022 lb/VMT and for PM-2.5, k is 0.00054 lb/VMT.
C. Table 13.2.1-3 specifies that the typical mean silt loading value (sL) for a municipal solid waste landfill is 7.4 g/m^2, which is assumed to represent the construction site condition.
D. Mean # of days of precipitation with >0.01 inches (P) is 135.7 per Normal NCEI data for Waynesburg 1 E, PA.

[3] The emission factor for unpaved roads was estimated using USEPA's AP-42 (11/06) Section 13.2.2 "Unpaved Roads" and the following:

$$E_{ext} = k * \left(\frac{s}{12}\right)^a * \left(\frac{W}{3}\right)^b * [\frac{365 - P}{365}]$$

A. Emission factor (E) is calculated using Equations (1a) and (2) in Section 13.2.2:
B. Table 13.2.2-1 specifies that the typical mean silt content value (s) for a construction site is 8.5%.
C. Table 13.2.2-2 specifies that for PM-10, k is 1.5 lb/VMT, a is 0.9, and b is 0.45, and for PM-2.5, k is 0.15 lb/VMT, a is 0.9, and b is 0.45.
D. Mean # of days of precipitation with >0.01 inches (P) is 87.9per Normal NCEI data for Rayville, LA

[4] Dust suppression by water spray at construction site is conservatively assumed to have a control efficiency of 0%.

Operational Emissions

Emissions Summary
Franklin Farms Pipeline

Location	Source	NO_X		CO		VOC		$PM_{10/2.5}$		SO_2	
		lb/hr	tpy	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy	lb/hr	tpy
Franklin Farms Delivery Meter Station	Emergency Generator	0.53	0.03	25.60	1.28	0.13	0.01	3.31E-03	1.65E-04	4.67E-03	2.34E-04
	Fugitives	--	--	--	--	0.04	0.19	--	--	--	--
	Condensate Tank	--	--	--	--	50.08	0.79	--	--	--	--
Franklin Farms MEP Receipt Meter Station	Emergency Generator	0.53	0.03	25.60	1.28	0.13	0.01	3.31E-03	1.65E-04	4.67E-03	2.34E-04
	Fugitives	--	--	--	--	0.03	0.13	--	--	--	--
	Condensate Tank	--	--	--	--	50.08	0.79	--	--	--	--
Franklin Farms Gulf Crossing Receipt Meter Station	Emergency Generator	0.53	0.03	25.60	1.28	0.13	0.01	3.31E-03	1.65E-04	4.67E-03	2.34E-04
	Fugitives	--	--	--	--	0.03	0.13	--	--	--	--
	Condensate Tank	--	--	--	--	50.08	0.79	--	--	--	--
Franklin Farms Trio Farm Delivery Meter Station	Emergency Generator	0.53	0.03	25.60	1.28	0.13	0.01	3.31E-03	1.65E-04	4.67E-03	2.34E-04
	Fugitives	--	--	--	--	0.04	0.19	--	--	--	--
	Condensate Tank	--	--	--	--	50.08	0.79	--	--	--	--
Franklin Farms Bee Bayou Delivery Meter Station	Emergency Generator	0.53	0.03	25.60	1.28	0.13	0.01	3.31E-03	1.65E-04	4.67E-03	2.34E-04
	Fugitives	--	--	--	--	0.04	0.19	--	--	--	--
	Condensate Tank	--	--	--	--	50.08	0.79	--	--	--	--
Franklin Farms Gulf South Receipt Meter Station	Emergency Generator	0.53	0.03	25.60	1.28	0.13	0.01	3.31E-03	1.65E-04	4.67E-03	2.34E-04
	Fugitives	--	--	--	--	0.03	0.13	--	--	--	--
	Condensate Tank	--	--	--	--	50.08	0.79	--	--	--	--
Franklin Farms Gulf Run Bi Di Meter Station	Emergency Generator	0.53	0.03	25.60	1.28	0.13	0.01	3.31E-03	1.65E-04	4.67E-03	2.34E-04
	Fugitives	--	--	--	--	0.04	0.19	--	--	--	--
	Condensate Tank	--	--	--	--	50.08	0.79	--	--	--	--
Franklin Farms Interconnect Site	Fugitives	--	--	--	--	0.01	0.04	--	--	--	--
Total		**3.70**	**0.19**	**179.17**	**8.96**	**351.78**	**6.76**	**0.02**	**1.16E-03**	**3.27E-02**	**1.64E-03**

Emergency Generator
Franklin Farms Pipeline
Delivery Meter Station

Parameter Name and Variable	Engine Data		Basis/Calculation/Notes
	GEN	Units	
Engine Horsepower	30	hp	Vendor data
Fuel Consumption	327	scfh	
HHV of Natural Gas	1020	Btu/scf	
Heat Input	0.33	MMBtu/hr	Based on fuel consumption and HHV of natural gas
Operating Schedule	100	hr/yr	Normal Operating hours for Non-emergency operations
Emission Factors			
NO_X	8	g/hp-hr	NSPS JJJJ Table 1 - Emergency 25<HP<130
CO	387	g/hp-hr	
VOC	2	g/hp-hr	
$PM_{10/2.5}$	9.91E-03	lb/MMBtu	AP-42 Section 3.2 Uncontrolled Emission Factors Engines
SO_2	10	gr/100 scf	Typical value used
Hourly Emissions Rate			
NO_X	0.53	lb/hr	(hp) x (g/hp-hr) x (1 lb/453.6 g)
CO	25.60	lb/hr	
VOC	0.13	lb/hr	
$PM_{10/2.5}$	0.003	lb/hr	(hp) x (MMBtu/hr) x (lb/MMBtu)
SO_2	0.00	lb/hr	(gr/scf) x (scf/hr) x (lb/7000 gr)
Annual Emissions Rate			
NO_X	0.03	ton/yr	(lb/hr) x (hr/yr) x (1 ton/2,000 lb)
CO	1.28	ton/yr	
VOC	0.01	ton/yr	
$PM_{10/2.5}$	1.65E-04	ton/yr	
SO_2	2.34E-04	ton/yr	

Fugitive Emissions
Franklin Farms Pipeline
Delivery Meter Station

Component Type	Stream Type	Number of Components	Emission Factor (lb/hr per component)	VOC Content (wt%)	Reduction Credit (%)	VOC Emission	
						lb/hr	ton/yr
Valves	Gas/Vapor	53	9.92E-03	5%	0%	0.03	0.11
Flanges/Connectors	Gas/Vapor	401	8.60E-04	5%	0%	0.02	0.08
				Total VOC		**0.04**	**0.19**

NOTES:

Fugitive Emission Factors and Reduction Credits are per TCEQ Technical Guidance Document for Equipment Leak Fugitives, dated October 2000. The emission factors are for total hydrocarbon. Factors for oil and gas prodution operations are used for the natural gas.

Hourly VOC emissions are calculated as follows:

VOC (lb/hr) = (53 Components)*(0.010 lb/hr/component)*(5.00 %) = 0.026 lb/hr

Annual VOC emission rates are calculated as follows:

VOC (tpy) = (0.03 lb/hr)*(8760 hr/yr)/(2,000 lb/ton) = 0.115 tpy

Condensate Tank Emissions
Franklin Farms Pipeline
Delivery Meter Station

Vapor Molecular Weight (lb/lbmol)	M_V	66
Vapor Pressure of Tank at Worst Case Temperature (psia)	P_{VA}	8.04
Ideal Gas Constant ((psia x gal)/(lbmol x ^{0}R))	R	80.27
Temperature of Liquid Surface (Rankine)	T	554.67
Maximum Filling Rate (gal/hr)	FR_M	4200.00

Maximum Short Term Emissions [1]	lb/hr	50.08
Annual Emissions [2]	tpy	0.79

Notes:

[1] Hourly emissions based on equation below from TCEQ Estimating Short Term Emission Rates from Fixed Roof Tanks Document:

$$L_{MAX} = \frac{M_V \, x \, PVA}{R \, x \, T} \text{ x FRM}$$

[2] Annual emissions calculated using TankESP included in Attachment B

Emergency Generator
Franklin Farms Pipeline
MEP Receipt Meter Station

Parameter Name and Variable	Engine Data		Basis/Calculation/Notes
	GEN	Units	
Engine Horsepower	30	hp	Vendor data
Fuel Consumption	327	scfh	
HHV of Natural Gas	1020	Btu/scf	
Heat Input	0.33	MMBtu/hr	Based on fuel consumption and HHV of natural gas
Operating Schedule	100	hr/yr	Normal Operating hours for Non-emergency operations
Emission Factors			
NO_X	8	g/hp-hr	NSPS JJJJ Table 1 - Emergency 25<HP<130
CO	387	g/hp-hr	
VOC	2	g/hp-hr	
$PM_{10/2.5}$	9.91E-03	lb/MMBtu	AP-42 Section 3.2 Uncontrolled Emission Factors Engines
SO_2	10	gr/100 scf	Typical value used
Hourly Emissions Rate			
NO_X	0.53	lb/hr	(hp) x (g/hp-hr) x (1 lb/453.6 g)
CO	25.60	lb/hr	
VOC	0.13	lb/hr	
$PM_{10/2.5}$	0.003	lb/hr	(hp) x (MMBtu/hr) x (lb/MMBtu)
SO_2	0.00	lb/hr	(gr/scf) x (scf/hr) x (lb/7000 gr)
Annual Emissions Rate			
NO_X	0.03	ton/yr	(lb/hr) x (hr/yr) x (1 ton/2,000 lb)
CO	1.28	ton/yr	
VOC	0.01	ton/yr	
$PM_{10/2.5}$	1.65E-04	ton/yr	
SO_2	2.34E-04	ton/yr	

Fugitive Emissions
Franklin Farms Pipeline
MEP Receipt Meter Station

Component Type	Stream Type	Number of Components	Emission Factor (lb/hr per component)	VOC Content (wt%)	Reduction Credit (%)	VOC Emission	
						lb/hr	ton/yr
Valves	Gas/Vapor	36	9.92E-03	5%	0%	0.02	0.08
Flanges/Connectors	Gas/Vapor	283	8.60E-04	5%	0%	0.01	0.05
				Total VOC		**0.03**	**0.13**

NOTES:

Fugitive Emission Factors and Reduction Credits are per TCEQ Technical Guidance Document for Equipment Leak Fugitives, dated October 2000. The emission factors are for total hydrocarbon. Factors for oil and gas prodution operations are used for the natural gas.

Hourly VOC emissions are calculated as follows:

VOC (lb/hr) = (36 Components)*(0.010 lb/hr/component)*(5.00 %) = 0.018 lb/hr

Annual VOC emission rates are calculated as follows:

VOC (tpy) = (0.02 lb/hr)*(8760 hr/yr)/(2,000 lb/ton) = 0.078 tpy

Condensate Tank Emissions
Franklin Farms Pipeline
MEP Receipt Meter Station

Vapor Molecular Weight (lb/lbmol)	M_V	66
Vapor Pressure of Tank at Worst Case Temperature (psia)	P_{VA}	8.04
Ideal Gas Constant ((psia x gal)/(lbmol x ^{0}R))	R	80.27
Temperature of Liquid Surface (Rankine)	T	554.67
Maximum Filling Rate (gal/hr)	FR_M	4200.00

Maximum Short Term Emissions [1]	lb/hr	50.08
Annual Emissions [2]	tpy	0.79

Notes:

[1] Hourly emissions based on equation below from TCEQ Estimating Short Term Emission Rates from Fixed Roof Tanks Document:

$$L_{MAX} = \frac{M_V \, x \, PVA}{R \, x \, T} \text{ x FRM}$$

[2] Annual emissions calculated using TankESP included in Attachment B

Emergency Generator
Franklin Farms Pipeline
Gulf Crossing Receipt Meter Station

Parameter Name and Variable	Engine Data		Basis/Calculation/Notes
	GEN	Units	
Engine Horsepower	30	hp	Vendor data
Fuel Consumption	327	scfh	
HHV of Natural Gas	1020	Btu/scf	
Heat Input	0.33	MMBtu/hr	Based on fuel consumption and HHV of natural gas
Operating Schedule	100	hr/yr	Normal Operating hours for Non-emergency operations
Emission Factors			
NO_X	8	g/hp-hr	NSPS JJJJ Table 1 - Emergency 25<HP<130
CO	387	g/hp-hr	
VOC	2	g/hp-hr	
$PM_{10/2.5}$	9.91E-03	lb/MMBtu	AP-42 Section 3.2 Uncontrolled Emission Factors Engines
SO_2	10	gr/100 scf	Typical value used
Hourly Emissions Rate			
NO_X	0.53	lb/hr	(hp) x (g/hp-hr) x (1 lb/453.6 g)
CO	25.60	lb/hr	
VOC	0.13	lb/hr	
$PM_{10/2.5}$	0.003	lb/hr	(hp) x (MMBtu/hr) x (lb/MMBtu)
SO_2	0.00	lb/hr	(gr/scf) x (scf/hr) x (lb/7000 gr)
Annual Emissions Rate			
NO_X	0.03	ton/yr	(lb/hr) x (hr/yr) x (1 ton/2,000 lb)
CO	1.28	ton/yr	
VOC	0.01	ton/yr	
$PM_{10/2.5}$	1.65E-04	ton/yr	
SO_2	2.34E-04	ton/yr	

Fugitive Emissions
Franklin Farms Pipeline
Gulf Crossing Receipt Meter Station

Component Type	Stream Type	Number of Components	Emission Factor (lb/hr per component)	VOC Content (wt%)	Reduction Credit (%)	VOC Emission	
						lb/hr	ton/yr
Valves	Gas/Vapor	36	9.92E-03	5%	0%	0.02	0.08
Flanges/Connectors	Gas/Vapor	283	8.60E-04	5%	0%	0.01	0.05
				Total VOC		**0.03**	**0.13**

NOTES:

Fugitive Emission Factors and Reduction Credits are per TCEQ Technical Guidance Document for Equipment Leak Fugitives, dated October 2000. The emission factors are for total hydrocarbon. Factors for oil and gas prodution operations are used for the natural gas.

Hourly VOC emissions are calculated as follows:

VOC (lb/hr) = (36 Components)*(0.010 lb/hr/component)*(5.00 %) = 0.018 lb/hr

Annual VOC emission rates are calculated as follows:

VOC (tpy) = (0.02 lb/hr)*(8760 hr/yr)/(2,000 lb/ton) = 0.078 tpy

Condensate Tank Emissions
Franklin Farms Pipeline
Gulf Crossing Receipt Meter Station

Vapor Molecular Weight (lb/lbmol)	M_V	66
Vapor Pressure of Tank at Worst Case Temperature (psia)	P_{VA}	8.04
Ideal Gas Constant ((psia x gal)/(lbmol x ^{0}R))	R	80.27
Temperature of Liquid Surface (Rankine)	T	554.67
Maximum Filling Rate (gal/hr)	FR_M	4200.00

Maximum Short Term Emissions [1]	lb/hr	50.08
Annual Emissions [2]	tpy	0.79

Notes:

[1] Hourly emissions based on equation below from TCEQ Estimating Short Term Emission Rates from Fixed Roof Tanks Document:

$$L_{MAX} = \frac{M_V \, x \, PVA}{R \, x \, T} \text{ x FRM}$$

[2] Annual emissions calculated using TankESP included in Attachment B

Emergency Generator
Franklin Farms Pipeline
Trio Farm Delivery Meter Station

Parameter Name and Variable	Engine Data		Basis/Calculation/Notes
	GEN	Units	
Engine Horsepower	30	hp	Vendor data
Fuel Consumption	327	scfh	
HHV of Natural Gas	1020	Btu/scf	
Heat Input	0.33	MMBtu/hr	Based on fuel consumption and HHV of natural gas
Operating Schedule	100	hr/yr	Normal Operating hours for Non-emergency operations
Emission Factors			
NO_X	8	g/hp-hr	NSPS JJJJ Table 1 - Emergency 25<HP<130
CO	387	g/hp-hr	
VOC	2	g/hp-hr	
$PM_{10/2.5}$	9.91E-03	lb/MMBtu	AP-42 Section 3.2 Uncontrolled Emission Factors Engines
SO_2	10	gr/100 scf	Typical value used
Hourly Emissions Rate			
NO_X	0.53	lb/hr	(hp) x (g/hp-hr) x (1 lb/453.6 g)
CO	25.60	lb/hr	
VOC	0.13	lb/hr	
$PM_{10/2.5}$	0.003	lb/hr	(hp) x (MMBtu/hr) x (lb/MMBtu)
SO_2	0.00	lb/hr	(gr/scf) x (scf/hr) x (lb/7000 gr)
Annual Emissions Rate			
NO_X	0.03	ton/yr	(lb/hr) x (hr/yr) x (1 ton/2,000 lb)
CO	1.28	ton/yr	
VOC	0.01	ton/yr	
$PM_{10/2.5}$	1.65E-04	ton/yr	
SO_2	2.34E-04	ton/yr	

Fugitive Emissions
Franklin Farms Pipeline
Trio Farm Delivery Meter Station

Component Type	Stream Type	Number of Components	Emission Factor (lb/hr per component)	VOC Content (wt%)	Reduction Credit (%)	VOC Emission	
						lb/hr	ton/yr
Valves	Gas/Vapor	53	9.92E-03	5%	0%	0.03	0.11
Flanges/Connectors	Gas/Vapor	401	8.60E-04	5%	0%	0.02	0.08
				Total VOC		**0.04**	**0.19**

NOTES:

Fugitive Emission Factors and Reduction Credits are per TCEQ Technical Guidance Document for Equipment Leak Fugitives, dated October 2000. The emission factors are for total hydrocarbon. Factors for oil and gas prodution operations are used for the natural gas.

Hourly VOC emissions are calculated as follows:

VOC (lb/hr) = (53 Components)*(0.010 lb/hr/component)*(5.00 %) = 0.026 lb/hr

Annual VOC emission rates are calculated as follows:

VOC (tpy) = (0.03 lb/hr)*(8760 hr/yr)/(2,000 lb/ton) = 0.115 tpy

Condensate Tank Emissions
Franklin Farms Pipeline
Trio Farm Delivery Meter Station

Vapor Molecular Weight (lb/lbmol)	M_V	66
Vapor Pressure of Tank at Worst Case Temperature (psia)	P_{VA}	8.04
Ideal Gas Constant ((psia x gal)/(lbmol x ^{0}R))	R	80.27
Temperature of Liquid Surface (Rankine)	T	554.67
Maximum Filling Rate (gal/hr)	FR_M	4200.00

Maximum Short Term Emissions [1]	lb/hr	50.08
Annual Emissions [2]	tpy	0.79

Notes:

[1] Hourly emissions based on equation below from TCEQ Estimating Short Term Emission Rates from Fixed Roof Tanks Document:

$$L_{MAX} = \frac{M_V \, x \, PVA}{R \, x \, T} \text{ x FRM}$$

[2] Annual emissions calculated using TankESP included in Attachment B

Emergency Generator
Franklin Farms Pipeline
Bee Bayou Delivery Meter Station

Parameter Name and Variable	Engine Data		Basis/Calculation/Notes
	GEN	Units	
Engine Horsepower	30	hp	Vendor data
Fuel Consumption	327	scfh	
HHV of Natural Gas	1020	Btu/scf	
Heat Input	0.33	MMBtu/hr	Based on fuel consumption and HHV of natural gas
Operating Schedule	100	hr/yr	Normal Operating hours for Non-emergency operations
Emission Factors			
NO_X	8	g/hp-hr	NSPS JJJJ Table 1 - Emergency 25<HP<130
CO	387	g/hp-hr	
VOC	2	g/hp-hr	
$PM_{10/2.5}$	9.91E-03	lb/MMBtu	AP-42 Section 3.2 Uncontrolled Emission Factors Engines
SO_2	10	gr/100 scf	Typical value used
Hourly Emissions Rate			
NO_X	0.53	lb/hr	(hp) x (g/hp-hr) x (1 lb/453.6 g)
CO	25.60	lb/hr	
VOC	0.13	lb/hr	
$PM_{10/2.5}$	0.003	lb/hr	(hp) x (MMBtu/hr) x (lb/MMBtu)
SO_2	0.00	lb/hr	(gr/scf) x (scf/hr) x (lb/7000 gr)
Annual Emissions Rate			
NO_X	0.03	ton/yr	(lb/hr) x (hr/yr) x (1 ton/2,000 lb)
CO	1.28	ton/yr	
VOC	0.01	ton/yr	
$PM_{10/2.5}$	1.65E-04	ton/yr	
SO_2	2.34E-04	ton/yr	

Fugitive Emissions
Franklin Farms Pipeline
Bee Bayou Delivery Meter Station

Component Type	Stream Type	Number of Components	Emission Factor (lb/hr per component)	VOC Content (wt%)	Reduction Credit (%)	VOC Emission	
						lb/hr	ton/yr
Valves	Gas/Vapor	53	9.92E-03	5%	0%	0.03	0.11
Flanges/Connectors	Gas/Vapor	401	8.60E-04	5%	0%	0.02	0.08
				Total VOC		**0.04**	**0.19**

NOTES:

Fugitive Emission Factors and Reduction Credits are per TCEQ Technical Guidance Document for Equipment Leak Fugitives, dated October 2000. The emission factors are for total hydrocarbon. Factors for oil and gas prodution operations are used for the natural gas.

Hourly VOC emissions are calculated as follows:

VOC (lb/hr) = (53 Components)*(0.010 lb/hr/component)*(5.00 %) = 0.026 lb/hr

Annual VOC emission rates are calculated as follows:

VOC (tpy) = (0.03 lb/hr)*(8760 hr/yr)/(2,000 lb/ton) = 0.115 tpy

Condensate Tank Emissions
Franklin Farms Pipeline
Bee Bayou Delivery Meter Station

Vapor Molecular Weight (lb/lbmol)	M_V	66
Vapor Pressure of Tank at Worst Case Temperature (psia)	P_{VA}	8.04
Ideal Gas Constant ((psia x gal)/(lbmol x ^{0}R))	R	80.27
Temperature of Liquid Surface (Rankine)	T	554.67
Maximum Filling Rate (gal/hr)	FR_M	4200.00

Maximum Short Term Emissions [1]	lb/hr	50.08
Annual Emissions [2]	tpy	0.79

Notes:

[1] Hourly emissions based on equation below from TCEQ Estimating Short Term Emission Rates from Fixed Roof Tanks Document:

$$L_{MAX} = \frac{M_V \, x \, PVA}{R \, x \, T} \text{ x FRM}$$

[2] Annual emissions calculated using TankESP included in Attachment B

Emergency Generator
Franklin Farms Pipeline
Gulf South Receipt Meter Station

Parameter Name and Variable	Engine Data		Basis/Calculation/Notes
	GEN	Units	
Engine Horsepower	30	hp	Vendor data
Fuel Consumption	327	scfh	
HHV of Natural Gas	1020	Btu/scf	
Heat Input	0.33	MMBtu/hr	Based on fuel consumption and HHV of natural gas
Operating Schedule	100	hr/yr	Normal Operating hours for Non-emergency operations
Emission Factors			
NO_X	8	g/hp-hr	NSPS JJJJ Table 1 - Emergency 25<HP<130
CO	387	g/hp-hr	
VOC	2	g/hp-hr	
$PM_{10/2.5}$	9.91E-03	lb/MMBtu	AP-42 Section 3.2 Uncontrolled Emission Factors Engines
SO_2	10	gr/100 scf	Typical value used
Hourly Emissions Rate			
NO_X	0.53	lb/hr	(hp) x (g/hp-hr) x (1 lb/453.6 g)
CO	25.60	lb/hr	
VOC	0.13	lb/hr	
$PM_{10/2.5}$	0.003	lb/hr	(hp) x (MMBtu/hr) x (lb/MMBtu)
SO_2	0.00	lb/hr	(gr/scf) x (scf/hr) x (lb/7000 gr)
Annual Emissions Rate			
NO_X	0.03	ton/yr	(lb/hr) x (hr/yr) x (1 ton/2,000 lb)
CO	1.28	ton/yr	
VOC	0.01	ton/yr	
$PM_{10/2.5}$	1.65E-04	ton/yr	
SO_2	2.34E-04	ton/yr	

Fugitive Emissions
Franklin Farms Pipeline
Gulf South Receipt Meter Station

Component Type	Stream Type	Number of Components	Emission Factor (lb/hr per component)	VOC Content (wt%)	Reduction Credit (%)	VOC Emission	
						lb/hr	ton/yr
Valves	Gas/Vapor	36	9.92E-03	5%	0%	0.02	0.08
Flanges/Connectors	Gas/Vapor	283	8.60E-04	5%	0%	0.01	0.05
				Total VOC		**0.03**	**0.13**

NOTES:

Fugitive Emission Factors and Reduction Credits are per TCEQ Technical Guidance Document for Equipment Leak Fugitives, dated October 2000. The emission factors are for total hydrocarbon. Factors for oil and gas prodution operations are used for the natural gas.

Hourly VOC emissions are calculated as follows:

VOC (lb/hr) = (36 Components)*(0.010 lb/hr/component)*(5.00 %) = 0.018 lb/hr

Annual VOC emission rates are calculated as follows:

VOC (tpy) = (0.02 lb/hr)*(8760 hr/yr)/(2,000 lb/ton) = 0.078 tpy

Condensate Tank Emissions
Franklin Farms Pipeline
Gulf South Receipt Meter Station

Vapor Molecular Weight (lb/lbmol)	M_V	66
Vapor Pressure of Tank at Worst Case Temperature (psia)	P_{VA}	8.04
Ideal Gas Constant ((psia x gal)/(lbmol x ^{0}R))	R	80.27
Temperature of Liquid Surface (Rankine)	T	554.67
Maximum Filling Rate (gal/hr)	FR_M	4200.00

Maximum Short Term Emissions [1]	lb/hr	50.08
Annual Emissions [2]	tpy	0.79

Notes:

[1] Hourly emissions based on equation below from TCEQ Estimating Short Term Emission Rates from Fixed Roof Tanks Document:

$$L_{MAX} = \frac{M_V \, x \, PVA}{R \, x \, T} \text{ x FRM}$$

[2] Annual emissions calculated using TankESP included in Attachment B

Emergency Generator
Franklin Farms Pipeline
Gulf Run Bi Di Meter Station

Parameter Name and Variable	Engine Data		Basis/Calculation/Notes
	GEN	Units	
Engine Horsepower	30	hp	Vendor data
Fuel Consumption	327	scfh	
HHV of Natural Gas	1020	Btu/scf	
Heat Input	0.33	MMBtu/hr	Based on fuel consumption and HHV of natural gas
Operating Schedule	100	hr/yr	Normal Operating hours for Non-emergency operations
Emission Factors			
NO_X	8	g/hp-hr	NSPS JJJJ Table 1 - Emergency 25<HP<130
CO	387	g/hp-hr	
VOC	2	g/hp-hr	
$PM_{10/2.5}$	9.91E-03	lb/MMBtu	AP-42 Section 3.2 Uncontrolled Emission Factors Engines
SO_2	10	gr/100 scf	Typical value used
Hourly Emissions Rate			
NO_X	0.53	lb/hr	(hp) x (g/hp-hr) x (1 lb/453.6 g)
CO	25.60	lb/hr	
VOC	0.13	lb/hr	
$PM_{10/2.5}$	0.003	lb/hr	(hp) x (MMBtu/hr) x (lb/MMBtu)
SO_2	0.00	lb/hr	(gr/scf) x (scf/hr) x (lb/7000 gr)
Annual Emissions Rate			
NO_X	0.03	ton/yr	(lb/hr) x (hr/yr) x (1 ton/2,000 lb)
CO	1.28	ton/yr	
VOC	0.01	ton/yr	
$PM_{10/2.5}$	1.65E-04	ton/yr	
SO_2	2.34E-04	ton/yr	

Fugitive Emissions
Franklin Farms Pipeline
Gulf Run Bi Di Meter Station

Component Type	Stream Type	Number of Components	Emission Factor (lb/hr per component)	VOC Content (wt%)	Reduction Credit (%)	VOC Emission	
						lb/hr	ton/yr
Valves	Gas/Vapor	52	9.92E-03	5%	0%	0.03	0.11
Flanges/Connectors	Gas/Vapor	408	8.60E-04	5%	0%	0.02	0.08
				Total VOC		**0.04**	**0.19**

NOTES:

Fugitive Emission Factors and Reduction Credits are per TCEQ Technical Guidance Document for Equipment Leak Fugitives, dated October 2000. The emission factors are for total hydrocarbon. Factors for oil and gas prodution operations are used for the natural gas.

Hourly VOC emissions are calculated as follows:

VOC (lb/hr) = (52 Components)*(0.010 lb/hr/component)*(5.00 %) = 0.026 lb/hr

Annual VOC emission rates are calculated as follows:

VOC (tpy) = (0.03 lb/hr)*(8760 hr/yr)/(2,000 lb/ton) = 0.112 tpy

Condensate Tank Emissions
Franklin Farms Pipeline
Gulf Run Bi Di Meter Station

Vapor Molecular Weight (lb/lbmol)	M_V	66
Vapor Pressure of Tank at Worst Case Temperature (psia)	P_{VA}	8.04
Ideal Gas Constant ((psia x gal)/(lbmol x ^{0}R))	R	80.27
Temperature of Liquid Surface (Rankine)	T	554.67
Maximum Filling Rate (gal/hr)	FR_M	4200.00

Maximum Short Term Emissions [1]	lb/hr	50.08
Annual Emissions [2]	tpy	0.79

Notes:

[1] Hourly emissions based on equation below from TCEQ Estimating Short Term Emission Rates from Fixed Roof Tanks Document:

$$L_{MAX} = \frac{M_V \, x \, PVA}{R \, x \, T} \text{ x FRM}$$

[2] Annual emissions calculated using TankESP included in Attachment B

Fugitive Emissions
Franklin Farms Pipeline
Lateral and Trio Farm Interconnect Site

Component Type	Stream Type	Number of Components	Emission Factor (lb/hr per component)	VOC Content (wt%)	Reduction Credit (%)	VOC Emission	
						lb/hr	ton/yr
Valves	Gas/Vapor	14	9.92E-03	5%	0%	0.01	0.03
Flanges/Connectors	Gas/Vapor	44	8.60E-04	5%	0%	0.00	0.01
				Total VOC		**0.01**	**0.04**

NOTES:

Fugitive Emission Factors and Reduction Credits are per TCEQ Technical Guidance Document for Equipment Leak Fugitives, dated October 2000. The emission factors are for total hydrocarbon. Factors for oil and gas prodution operations are used for the natural gas.

Hourly VOC emissions are calculated as follows:

VOC (lb/hr) = (14 Components)*(0.010 lb/hr/component)*(5.00 %) = 0.007 lb/hr

Annual VOC emission rates are calculated as follows:

VOC (tpy) = (0.01 lb/hr)*(8760 hr/yr)/(2,000 lb/ton) = 0.031 tpy

APPENDIX I

Sound Study Report

Sound Study Report

Franklin Farm Project
ETC Tiger Pipeline, LLC

May 2026

Prepared For:

Energy Transfer LP
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Prepared By:

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[Attached]

Figure 1 Nearby Noise Sensitive Areas and Measurement Points 1
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1.0 Introduction

The following Sound Study Report has been prepared by TRC for ETC Tiger Pipeline, LLC (ETC) for the Franklin Farms Project (Project), located in Richland Parish, Louisiana.

2.0 Concepts of Environmental Sound

Sounds are generated by a variety of sources (e.g., a musical instrument, a voice speaking, an airplane that passes overhead). Energy is required to produce sound and this sound energy is transmitted through the air in the form of sound waves – tiny, quick oscillations of pressure just above and just below atmospheric pressure. These oscillations, or sound pressures, impinge on the ear, creating the sound we hear. The range of sound pressures that can be detected by a person with normal hearing is very wide, ranging from about 20 micro-pascals (μPa) for very faint sounds at the threshold of hearing to nearly 10 million μPa for extremely loud sounds, such as a jet during take-off at a distance of 300 feet. Because the range of human hearing is so wide, sound levels are reported using "sound pressure levels", which are expressed in terms of decibels. The sound pressure level in decibels is the logarithm of the ratio of the sound pressure of the source to the reference sound pressure of 20 μPa, multiplied by 20.

Table 2.0-1 provides some examples of common sources of sound and their sound pressure levels. All sound levels in this assessment are provided in A-weighted decibels "(dB(A) or dBA). The A-weighted sound level reflects how the human ear responds to sound, by deemphasizing sounds that occur in frequencies at which the human ear is least sensitive to sound (at frequencies below about 100 hertz [Hz] and above 10,000 Hz) and emphasizing sounds that occur in frequencies at which the human ear is most sensitive to sound (in the mid-frequency range from about 200 to 8,000 Hz). In the context of environmental sound, noise is defined as "unwanted sound."

Table 2.0-1 Examples of Common Sound Pressure Levels

Sound Level dB(A)	Common Indoor Sounds	Common Outdoor Sounds
110	Rock Band	Jet Takeoff at 1000 feet
100	Inside NYC Subway Train	Chain Saw at three feet
90	Food Blender at three feet	Impact Hammer (Hoe Ram) at 50 feet
80	Garbage Disposal at three feet	Diesel Truck at 50 feet
70	Vacuum Cleaner at 10 feet	Lawn Mower at 100 feet
60	Normal Speech at three feet	Auto (40 miles per hour [mph]) at 100 feet
50	Dishwasher in Next Room	Busy Suburban Area at night
40	Empty Conference Room	Quiet Suburban Area at night
25	Empty Concert Hall	Rural Area at night

Sound pressure levels are typically presented in community noise assessments utilizing the noise metrics described below and expressed in terms of A-weighted decibels.

- "L_{10}" is the sound level that is exceeded for 10 percent of the time. This metric is a measure of the intrusiveness of relatively short-duration noise events that occurred during the measurement period.
- "L_{50}" is the sound level that is exceeded for 50 percent of the measurement period.
- "L_{90}" is the sound level that is exceeded for 90 percent of the time and is a measure of the background or residual sound levels in the absence of recurring noise events.
- "L_{eq}" is the is the constant sound level which would contain the same acoustic energy as the varying sound levels during the time period and is representative of the average noise exposure level for that time period.
- "L_{MAX}" is the instantaneous maximum sound level for the time period.
- "L_d" is the average of daytime L_{90} background sound levels.
- "L_{dn}" is the day-night sound level.

It is often necessary to combine the sound pressure levels from one or more sources. Because decibels are logarithmic quantities, it is not possible to simply add the values of the sound pressure levels together. For example, if two sound sources each produce 70 dB and they are operated together, their combined impact is 73 dB – not 140 dB as might be expected. Four equal 70 dB sources operating simultaneously result in a total sound pressure level of 76 dB. In fact, for every doubling of the number of equal sources, the sound pressure level goes up another three decibels. A tenfold increase in the number of sources makes the sound pressure level increase by 10 dB, while a hundredfold increase makes the level increase by 20 dB. The logarithmic combination of *n* different sound levels is calculated by the following equation:

$$L_{total}=10*\log_{10}\left(10^{\frac{L_1}{10}} + 10^{\frac{L_2}{10}} +\ldots+10^{\frac{L_n}{10}}\right)$$

Perceived changes in sound level can be slightly more subjective; the average person will not notice a change of 1-2 dB, a three dB increase is just barely perceptible, while a five dB change is clearly noticeable.

3.0 Applicable Standards and Regulations

3.1 Federal Criteria

FERC's guidelines require that construction activity that would or may occur during nighttime hours should be performed with the goal that the activity contributes noise levels below 55 dBA Ldn and 48.6 Leq, or no more than 10 dBA over background if ambient noise levels are above 55 dBA Ldn. Applicants must describe and commit to all reasonable noise mitigation to be implemented during the construction activity to reduce noise impacts at the NSAs to meet the goal.

3.2 State, County, and Local Criteria

No other applicable state or local regulations relating to noise impacts have been identified for the site location.

3.3 Noise Sensitive Areas

The meter stations are located in an area with various nearby noise sensitive areas (NSAs) such as residences, schools, and hospitals. NSAs are defined as areas that may be particularly sensitive to noise and may require additional protection. Table 3.1 lists the most potentially impacted NSAs that are within 0.5 miles of the meter stations and their approximate distance and direction from the sites. These NSAs are also depicted on attached Figures 1-4.

Table 3.1 Nearby Noise Sensitive Areas

NSA	Description	Distance (feet)	Direction
1	Residence	250	SW
2	Residence	550	SE
3	Residence	1,500	SE
4	Residence	1,700	S
5	Residence	1,600	NW
6	Residence	1,300	S
7	Residence	2,200	SW
8	Residence	1,700	S
9	Residence	1,200	S
10	Residence	2,000	SE
11	Residence	2,200	E
12	Residence	2,300	E
13	Residence	1,900	SE
14	Residence	1,600	E
15	Residence	650	NW
16	Residence	1,200	N
17	Residence	2,300	N
18	Residence	2,200	N
19	Residence	2,200	N

4.0 Short Term Ambient Sound Survey

An ambient sound survey was conducted to characterize the existing sound levels at the nearby NSAs. The ambient sound survey was completed on May 6-8, 2026.

Short-term background sound level measurements were taken using a sound level meter that meets the requirements of the American National Standards Institute (ANSI) Standards for Type

I instruments. The sound level meter was field calibrated before and after each monitoring period using an acoustic calibrator following procedures that are traceable to the National Institute of Standards and Technology (NIST).

Short-term ambient sound level measurements were conducted during daytime and nighttime periods at nineteen measurement points for a minimum of ten continuous minutes for each criterion.

The following measurement criteria are provided in Tables 4.1-1 and 4.1-2 for each measurement location and for each measurement period:

- LA_{EQ}, LC_{EQ}, LA_{10}, LA_{50}, and LA_{90}
- Unweighted octave-band analysis (16, 31.5, 63, 125, 250, 500, 1K, 2K, 4K, 8K Hz)

The meter stations and the locations of the measurement points are shown in Figures 1-4.

4.1 Short Term Sound Monitoring Results

The results of the short-term measurements are summarized in Tables 4.1-1, 4.1-2, and 4.1-3 below.

Table 4.1-1 Background Sound Monitoring Results Summary

	LA_{EQ}	LC_{EQ}	LA_{10}	LA_{50}	LA_{90}
Daytime (11 AM – 4 PM)					
MP-1	51.3	57.4	51.0	47.0	45.0
MP-2	50.1	56.5	49.3	46.7	45.0
MP-3	52.2	67.9	54.8	50.6	48.5
MP-4	57.9	65.2	55.4	54.2	53.5
MP-5	48.8	69.2	51.2	46.6	42.8
MP-6	57.6	63.6	57.2	46.8	42.5
MP-7	60.6	70.7	60.5	46.2	43.4
MP-8	64.6	69.1	68.6	50.2	42.3
MP-9	60.3	68.6	61.3	42.9	40.3
MP-10	61.0	68.3	57.1	46.0	41.0
MP-11	45.4	57.7	43.5	38.4	34.3
MP-12	52.7	61.2	42.4	33.5	30.6
MP-13	47.0	60.4	48.9	46.9	40.3
MP-14	59.5	67.7	62.1	51.5	49.5
MP-15	59.8	67.2	63.4	54.5	48.0
MP-16	64.0	70.9	68.3	53.3	50.3
MP-17	66.8	74.3	71.6	54.6	46.9
MP-18	65.1	77.3	68.3	50.9	41.7
MP-19	49.9	62.9	49.0	43.9	42.1

Nighttime (10 PM- 3 AM)					
MP-1	45.2	66.0	47.9	43.1	41.2
MP-2	49.3	54.0	49.4	49.0	48.7
MP-3	55.7	67.9	56.2	55.5	55.0
MP-4	60.7	66.9	63.3	53.7	52.0
MP-5	53.6	63.4	54.0	53.4	53.0
MP-6	61.2	67.1	64.8	51.2	49.4
MP-7	70.2	77.0	73.7	63.8	52.5
MP-8	63.0	67.9	62.4	50.3	37.9
MP-9	61.2	66.2	65.1	41.6	37.3
MP-10	71.0	74.3	76.5	55.0	49.3
MP-11	61.4	73.8	61.6	43.7	37.2
MP-12	57.6	66.0	51.9	40.6	38.1
MP-13	59.0	65.5	58.1	41.9	38.2
MP-14	58.8	64.5	59.7	50.1	48.8
MP-15	52.4	61.3	55.6	45.9	42.5
MP-16	56.2	62.5	57.4	48.4	43.4
MP-17	64.4	69.9	64.3	47.3	40.3
MP-18	66.0	74.8	69.6	58.5	53.0
MP-19	50.0	58.4	53.1	48.4	45.2

Table 4.1-2 Octave Band Analysis Summary

	Octave Band Center Frequency (Hz)									
	16	31.5	63	125	250	500	1000	2000	4000	8000
Daytime (11 AM – 4 PM)										
MP-1	52.0	47.2	47.3	43.8	37.3	36.7	40.8	37.3	36.4	32.7
MP-2	51.0	47.5	47.7	41.2	35.8	33.5	41.1	38.7	34.2	35.1
MP-3	60.3	57.5	62.8	52.7	40.7	46.0	42.9	33.7	33.7	18.3
MP-4	55.0	58.4	54.3	50.6	52.6	50.3	48.8	38.8	39.6	30.8
MP-5	70.0	62.4	53.3	49.2	49.2	37.6	35.7	36.7	35.5	32.4
MP-6	56.8	52.4	51.2	53.5	48.9	47.6	49.4	43.8	38.2	32.7
MP-7	67.3	60.9	64.5	59.9	52.1	45.9	52.3	47.4	39.5	31.3
MP-8	61.1	56.9	56.1	60.0	53.6	55.2	57.8	50.2	41.2	38.0
MP-9	64.2	57.6	55.6	58.5	52.0	51.2	52.3	44.9	37.9	28.8
MP-10	51.1	50.2	57.2	57.6	56.4	52.9	52.8	46.0	39.7	30.1
MP-11	52.8	48.9	47.2	47.2	41.7	35.3	36.7	30.4	29.2	17.9
MP-12	54.6	51.3	51.3	51.6	47.1	46.5	44.5	34.3	26.3	19.8
MP-13	59.3	51.1	47.8	48.6	42.9	36.7	34.4	33.5	33.3	23.6
MP-14	56.2	59.0	59.1	56.3	54.0	49.4	49.8	48.2	41.3	32.4
MP-15	55.7	57.8	59.3	57.8	50.6	51.6	50.5	48.1	39.0	28.0
MP-16	54.8	57.6	63.4	60.2	58.5	56.8	55.8	48.7	41.4	31.3
MP-17	56.4	59.2	65.7	63.9	62.2	59.7	57.8	51.3	45.0	36.1
MP-18	54.4	57.7	67.7	63.5	59.1	55.7	55.7	49.9	42.5	34.7
MP-19	55.1	54.2	54.3	53.4	43.7	42.3	39.5	34.1	34.8	26.6
MP-1	63.5	60.0	55.2	52.7	40.2	27.1	19.8	14.9	40.6	24.7
MP-2	49.3	44.7	45.6	37.0	25.1	26.2	30.5	30.4	47.5	24.9
MP-3	65.2	59.8	57.6	54.0	43.4	46.8	42.9	33.0	50.8	30.3
MP-4	57.6	57.4	57.7	54.5	51.1	51.9	54.2	43.9	43.3	27.4
MP-5	62.8	55.3	49.4	47.0	36.6	33.8	31.1	35.1	51.0	29.0
MP-6	58.8	55.0	55.9	58.5	47.6	49.9	53.9	49.3	47.3	33.7
MP-7	55.9	55.9	69.3	71.7	58.6	60.5	62.3	56.6	48.5	36.3
MP-8	56.3	54.8	62.1	55.9	53.6	54.9	55.2	48.3	46.5	35.3
MP-9	55.6	52.6	54.2	56.1	49.7	52.7	54.1	44.2	36.0	25.8
MP-10	50.9	54.8	59.9	66.2	60.2	61.3	64.5	55.5	43.8	33.9
MP-11	50.6	54.4	57.3	71.0	56.7	48.7	52.2	45.0	42.0	38.2
MP-12	48.4	50.7	54.2	53.9	50.1	48.2	50.7	42.3	36.5	30.2
MP-13	49.5	54.6	54.4	57.4	52.4	50.3	49.6	46.3	41.0	34.6
MP-14	50.5	54.3	54.9	52.2	47.4	47.3	52.5	43.6	43.3	24.6
MP-15	49.6	53.2	53.0	50.0	41.0	42.5	45.3	37.5	30.1	19.4

MP-16	50.5	52.8	52.5	48.1	44.6	45.9	49.3	42.7	36.4	22.7
MP-17	51.4	52.0	59.0	57.6	57.2	57.7	55.3	48.5	41.6	32.4
MP-18	56.7	57.5	65.5	68.2	58.5	56.6	57.8	51.8	47.9	40.7
MP-19	51.2	51.9	51.0	44.9	30.7	33.9	26.6	39.6	41.4	27.6

Using the data collected during the short-term monitoring, a Day-Night average L_{dn} using the measured L_{EQ} sound levels was calculated for each of the MPs in Table 4.1-3.

Table 4.1-3 Ambient Noise Measurements Summary

Site ID			
	Daytime Avg. Ld	Nighttime Avg. Ln	Day-Night Avg. Ldn
MP-1	51.3	45.2	53.2
MP-2	50.1	49.3	55.8
MP-3	52.2	55.7	61.8
MP-4	57.9	60.7	66.8
MP-5	48.8	53.6	59.6
MP-6	57.6	61.2	67.2
MP-7	60.6	70.2	76.0
MP-8	64.6	63.0	69.7
MP-9	60.3	61.2	67.5
MP-10	61.0	71.0	76.8
MP-11	45.4	61.4	67.2
MP-12	52.7	57.6	63.6
MP-13	47.0	59.0	64.8
MP-14	59.5	58.8	65.3
MP-15	59.8	52.4	61.0
MP-16	64.0	56.2	65.0
MP-17	66.8	64.4	71.2
MP-18	65.1	66.0	72.3
MP-19	49.9	50.0	56.4

5.0 Construction Noise Impacts

5.1 Modeling Inputs

Short-term increases in sound levels could occur during construction. Based on the Project equipment usage predictions, a sound level calculation was performed using the Federal Highway Administration's (FHWA) Roadway Construction Noise Model version 1.1 (RCNM; FHWA, 2006), a computer program used to assess construction noise impacts. Table 5.0-1 provides a

preliminary list of equipment that may be used during construction. The dB levels for each piece of equipment were taken from the RCNM model, and if the equipment is not listed in the model, 85 dB is assumed. To be conservative in construction noise calculations, all Project construction equipment was included in the noise model as operating at the same time.

Table 5.0-1 Construction Equipment

Chipper/Shredder	Fuel Truck
Dozers	Water Truck
Dump Truck	Grapple Truck
Excavator	Buggy
Front End Loaders	Hydro excavator Truck
Grader	Welding Machine
Bore/Drill Rigs	Compressor
Cranes	Light Plant
Generator Sets	Pickups
Pump	Mechanics Truck
Tractors/Loaders/Backhoes	Truck - Semi
Portable Pipe Bender	Bus
Scissor Lift/ Fork Lift	Skid Truck
SideBoom	Passenger Van
Skid Steer Loader (299D3)	Fuel Truck
Sweeper	Trenching Machines
Trenching Machines	

5.2 Modeling Results

Table 5.2-1 summarizes the results of the construction noise modeling at the closest NSAs in each direction.

Table 5.2-1 Construction Noise Projections

NSA	Closest MP	Distance (ft)	Existing Ldn	Modeled Leq	Total Leq	Temporary Increase Leq
1	2	250	55.8	75.5	75.5	19.7
2	1	550	53.2	68.7	68.8	15.6
3	1	1,500	53.2	60	60.8	7.6
4	5	1,700	59.6	58.9	62.3	2.7
5	6	1,600	67.2	59.4	67.9	0.7
6	6	1,300	67.2	61.2	68.2	1.0

7	10	2,200	76.8	56.6	76.8	0.0
8	9	1,700	67.5	58.9	68.1	0.6
9	9	1,200	67.5	61.9	68.6	1.1
10	18	2,000	72.3	57.5	72.4	0.1
11	8	2,200	69.7	56.6	69.9	0.2
12	8	2,300	69.7	56.2	69.9	0.2
13	6	1,900	67.2	57.9	67.7	0.5
14	13	1,600	64.8	59.4	65.9	1.1
15	16	650	65.0	67.2	69.2	4.2
16	16	1,200	65.0	61.9	66.7	1.7
17	3	2,300	61.8	56.2	62.9	1.1
18	3	2,200	61.8	56.6	62.9	1.1
19	3	2,200	61.8	56.6	62.9	1.1

As shown in the above tables, there could be significant temporary increases in sound levels at a few of the nearby NSAs during Project construction activities. However, the modeled results are extremely conservative, in that the model assumes that all proposed construction equipment is operating simultaneously and that there will be no sound mitigation from intervening structures and geographic features such as trees and elevation changes. The actual temporary sound level increases will be significantly lower than those predicted by the Roadway Construction Noise Model.

5.3 Noise Mitigation Measures

The construction activities related to the Project could have short-term noise impacts to nearby noise sensitive areas. The following temporary noise mitigation measures could be adopted as needed to reduce temporary Project noise impacts during construction.

- Operate equipment according to manufacturer's recommendations to reduce excess sound;
- Utilize equipment with muffled exhausts;
- Minimize equipment idling;
- Minimize the use of backup beepers;
- Schedule construction activities for periods when nearby residents are less likely to be home; and
- Utilize temporary sound barriers

Predicted sound levels from Project construction exceed the recommended FERC guideline of 55 dBA; however, measured existing background sound levels already exceed this guideline at every NSA except for at NSA 2 and 3.

Sound levels during construction may result in temporary, localized impacts to the nearest noise sensitive receptors. However, the implantation of equipment idling restrictions, and the use of sound control devices such as temporary sound walls will be used to mitigate the impacts to the greatest extent practicable.

6.0 References

Federal Highway Administration (“FWHA”). 2018. Noise Measurement Field Guide. Publication FWHA-HEP-18-066. Available at: https://www.fhwa.dot.gov/ENVIRonment/noise/measurement/fhwahep18066.pdf. Accessed March 2026.

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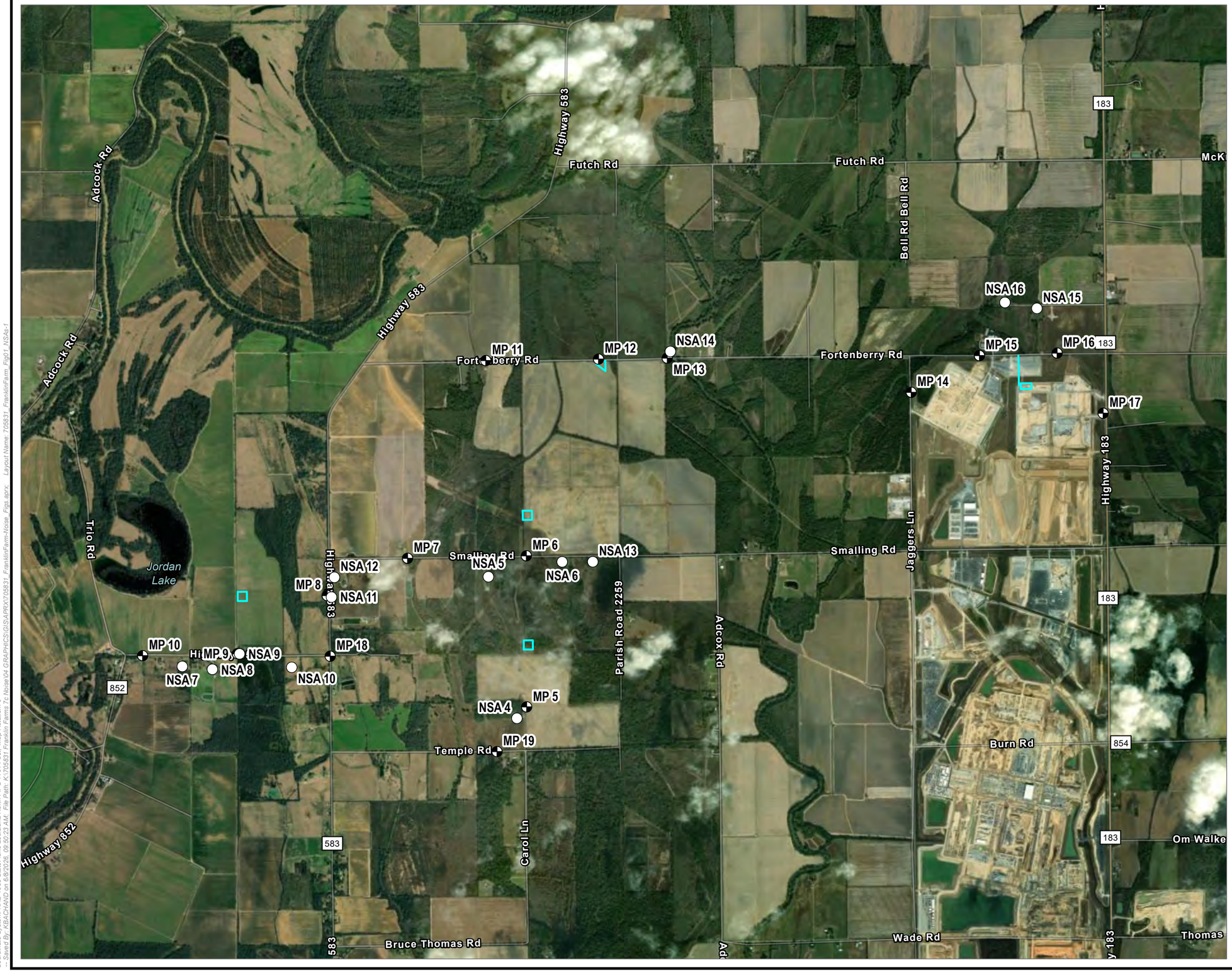

MONITORING LOCATION
NOISE SENSITIVE AREA
SITE LOCATION

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER; 2025
DATA SOURCES: TRC GPS LOCATIONS 2026
ENERGY TRANSFER LP PROJECT LOCATIONS 2026

1:31,200
1" = 2,600'
0 1,300 2,600 FEET

ENERGY TRANSFER LP
FRANKLIN FARMS
RAYVILLE, LA

NOISE SENSITIVE AREAS AND MONITORING LOCATIONS

DRAWN BY:	K. BACHAND	PROJ. NO.:	705831
CHECKED BY:	M. PROKO		
APPROVED BY:	M. FEINBLATT		
DATE:	JUNE 2026		

FIGURE 1
PAGE 1 OF 3

404 WYMAN STREET
SUITE 375
WALTHAM, MA 02451
PHONE: 781.419.7696

Coordinate System: NAD 1983 StatePlane Louisiana North FIPS 1701 Feet. Map Rotation: 0
Saved By: KBACHAND on 6/2/2026 09:52:23 AM. File Path: K:\705831 Franklin Farms\Tx-No\exl\04 GRAPHICS\GIS\APRX\705831_FranklinFarms.aprx; Layout Name: 705831_FranklinFarm_Fig01_NSAs-1

MONITORING LOCATION
NOISE SENSITIVE AREA
SITE LOCATION

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER; 2025
DATA SOURCES: TRC GPS LOCATIONS 2026
ENERGY TRANSFER LP PROJECT LOCATIONS 2026

1:5,400
1" = 450'
0 225 450 FEET

ENERGY TRANSFER LP
FRANKLIN FARMS
RAYVILLE, LA

NOISE SENSITIVE AREAS AND MONITORING LOCATIONS

DRAWN BY:	K. BACHAND	PROJ. NO.:	705831
CHECKED BY:	M. PROKO	**FIGURE 1 PAGE 2 OF 3**	
APPROVED BY:	M. FEINBLATT		
DATE:	JUNE 2026		

404 WYMAN STREET
SUITE 375
WALTHAM, MA 02451
PHONE: 781.419.7696

Coordinate System: NAD 1983 StatePlane Louisiana North FIPS 1701 Feet; Map Rotation: 0
Saved By: KBACHAND on 1/8/2026, 09:50:23 AM; File Path: K:\705831 Franklin Farms Tx_Noise\04 GRAPHICS\GIS\APRX\705831_FranklinFarm_Noise_Figs.aprx; Layout Name: 705831_FranklinFarm_Fig01_NSAs-2

MONITORING LOCATION
NOISE SENSITIVE AREA
SITE LOCATION

BASE MAP: ESRI "WORLD IMAGERY" ONLINE SERVICE LAYER; 2025
DATA SOURCES: TRC GPS LOCATIONS 2026
ENERGY TRANSFER LP PROJECT LOCATIONS 2026

1:3,000
1" = 250'
0 125 250 FEET

ENERGY TRANSFER LP
FRANKLIN FARMS
RAYVILLE, LA

NOISE SENSITIVE AREAS AND MONITORING LOCATIONS

DRAWN BY:	K. BACHAND	PROJ. NO.:	705831
CHECKED BY:	M. PROKO		
APPROVED BY:	M. FEINBLATT		
DATE:	JUNE 2026		

FIGURE 1
PAGE 3 OF 3

404 WYMAN STREET
SUITE 375
WALTHAM, MA 02451
PHONE: 781.419.7696

Coordinate System: NAD 1983 StatePlane Louisiana North FIPS 1701 Feet; Map Rotation: 0
Saved By: KBACHAND on 6/8/2026, 6:56:23 AM; File Path: K:\705831 Franklin Farms\7s Noise\04 GRAPHICS\GIS\ARRX\705831_FranklinFarm_Noise_Figs.aprx; Layout Name: 705831_FranklinFarm_Fig01_NSAs 3

Exhibits G, G-1, G-2

ETC Tiger Pipeline, LLC

Flow Diagram Showing Daily Design Capacity and
Reflecting Operation of Applicant's System with Proposed

Facilities Added and

Flow Diagrams Reflecting Maximum Capabilities

Exhibit G, G-I, G-II and the flow modeling files are designated as **CUI/Critical Energy Infrastructure Information - Do Not Release** and are submitted under Volume II.

Exhibit K

ETC Tiger Pipeline, LLC

Cost of Facilities

ETC TIGER PIPELINE, LLC

Franklin Farms Project
Project Cost Estimate

INDEX

Description	Page
Project Cost Estimate	1
Franklin Farms Project - AFUDC	2

ETC TIGER PIPELINE, LLC

Franklin Farms Project
Project Cost Estimate
February 1, 2028 In-Service Date

Line No.	Description	Laterals	Meter Stations	Total
		(a)	(b)	(c)
1	ROW - Land	$ 5,222,249	$ 105,000	$ 5,327,249
2	ROW - Damages	440,252	-	440,252
3	Land - Purchases	20,000	115,000	135,000
4	Survey	529,281	105,000	634,281
5	Materials	24,921,738	25,384,016	50,305,754
6	Labor	30,593,391	18,456,197	49,049,588
7	Engineering / Environmental / Inspection	8,907,560	4,802,440	13,710,000
8	Line Pack / Gas Loss	160,579	-	160,579
9	Overheads	2,910,112	2,032,162	4,942,274
11	Contingencies	5,334,452	1,465,762	6,800,214
12	Legal / Regulatory / Permitting	330,000	370,634	700,634
13	Sub-total	79,369,614	52,836,211	132,205,825
14	AFUDC	3,111,486	2,071,311	5,182,797
15	Total Estimated Costs	$ 82,481,100	$ 54,907,522	$ 137,388,622

ETC TIGER PIPELINE, LLC

Franklin Farms Project
AFUDC Calculation
February 1, 2028 In-Service Date

Line No.		Monthly Expenditure	Cumulative Monthly Expenditure	Equity During Construction	Cumulative Equity AFUDC	Debt During Construction	Cumulative Debt AFUDC	Total Monthly AFUDC	Cumulative AFUDC
		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
	Estimated PS&I Period								
1	October 2025	$ 13,221	$ 13,221	$ -	$ -	$ -	$ -	$ -	$ -
2	November	1,374,941	1,388,162	-	-	-	-	-	-
3	December	1,983,087	3,371,249	-	-	-	-	-	-
4	January 2026	1,983,087	5,354,336	-	-	-	-	-	-
5	February	3,966,175	9,320,511	-	-	-	-	-	-
6	March	2,644,117	11,964,628	-	-	-	-	-	-
7	April	4,627,204	16,591,832	-	-	-	-	-	-
8	May	6,610,291	23,202,123	-	-	-	-	-	-
9	June	7,932,350	31,134,473	-	-	-	-	-	-
10	July	6,610,291	37,744,764	-	-	-	-	-	-
11	August	3,966,175	41,710,939	-	-	-	-	-	-
12	September	2,644,117	44,355,056	-	-	-	-	-	-
13	October	1,322,058	45,677,114	-	-	-	-	-	-
14	November	661,029	46,338,143	-	-	-	-	-	-
15	December	6,610,291	52,948,434	-	-	-	-	-	-
16	January 2027	1,322,058	54,270,492	-	-	-	-	-	-
17	February	661,029	54,931,521	-	-	-	-	-	-
18	March	661,029	55,592,550	-	-	-	-	-	-
19	April	1,983,087	57,575,637	-	-	-	-	-	-
20	May	1,983,087	59,558,724	-	-	-	-	-	-
21	June	5,288,233	64,846,957	-	-	-	-	-	-
	Estimated Construction Period								
22	July 2027	6,610,291	71,457,248	332,276	332,276	187,447	187,447	519,723	519,723
23	August	9,254,408	80,711,656	370,950	703,225	209,264	396,711	580,214	1,099,937
24	September	10,576,466	91,288,122	419,292	1,122,518	236,536	633,247	655,828	1,755,765
25	October	11,898,525	103,186,647	474,081	1,596,599	267,444	900,691	741,525	2,497,290
26	November	10,576,466	113,763,113	528,869	2,125,468	298,352	1,199,043	827,221	3,324,511
27	December	6,610,291	120,373,404	570,766	2,696,234	321,987	1,521,030	892,753	4,217,265
28	January 2028	7,932,350	128,305,754	619,363	3,315,597	346,170	1,867,200	965,533	5,182,797
	Estimated Post Construction Period								
29	February 2028	1,983,087	130,288,841	-	3,315,597	-	1,867,200	-	5,182,797
30	March	1,322,058	131,610,899	-	3,315,597	-	1,867,200	-	5,182,797
31	April	594,926	132,205,825	-	3,315,597	-	1,867,200	-	5,182,797
32	Total	$ 132,205,825		$ 3,315,597		$ 1,867,200		$ 5,182,797	

	Annual Amortization	
33	Equity	$ 82,890
34	Debt	46,680
35	Total	$ 129,570

	Capitalization	Ratio	Cost	Annual Rate	Monthly Rate
36	Equity	41.79%	14.00%	5.85%	0.49%
37	Debt	58.21%	5.67%	3.30%	0.28%
38	Total	100.00%		9.15%	0.76%

Exhibit N

ETC Tiger Pipeline, LLC

Revenues, Expenses and Income

ETC TIGER PIPELINE, LLC

Franklin Farms Project
Cost of Service
Summary

Line No.	Description	YEAR 1 (a)	YEAR 2 (b)	YEAR 3 (c)
1	Operating Expenses	$ 924,842	$ 952,587	$ 981,165
2	Depreciation Expense	4,117,609	4,117,609	4,117,609
3	Taxes - Other Than Income	2,872,629	2,787,842	2,705,640
4	Return	12,194,298	11,817,536	11,440,775
5	Federal Income Taxes	-	-	-
6	State Income Taxes	-	-	-
7	Total	$ 20,109,378	$ 19,675,574	$ 19,245,189

ETC TIGER PIPELINE, LLC

Franklin Farms Project
Cost of Service
Operation and Maintenance Expenses

Line No.	Acct. No.	Description	YEAR 1 Total	YEAR 1 Labor	YEAR 1 Non-Labor	YEAR 2 Total	YEAR 2 Labor	YEAR 2 Non-Labor	YEAR 3 Total	YEAR 3 Labor	YEAR 3 Non-Labor
			(a)	(b)	(c)	(a)	(b)	(c)	(a)	(b)	(c)
		Transmission Operating Expenses									
		Operation									
1	850	Operation Supervision and Engineering	$ 107,461	$ 63,308	$ 44,153	$ 110,685	$ 65,207	$ 45,478	$ 114,005	$ 67,163	$ 46,842
2	851	System Control and Load Dispatching	60,794	45,908	14,886	62,618	47,285	15,333	64,496	48,704	15,793
3	852	Communication System Expenses	-	-	-	-	-	-	-	-	-
4	853	Compressor Station Labor and Expenses	-	-	-	-	-	-	-	-	-
5	854	Gas for Compressor Station Fuel	-	-	-	-	-	-	-	-	-
6	855	Other Fuel and Power For Compressor Stations	-	-	-	-	-	-	-	-	-
7	856	Mains Expenses	101,942	32,108	69,834	105,000	33,071	71,929	108,150	34,063	74,087
8	857	Measuring and Regulating Station Expenses	174,298	96,127	78,171	179,527	99,011	80,516	184,913	101,981	82,932
9	858	Transmission and Compression of Gas by Others	-	-	-	-	-	-	-	-	-
10	859	Other Expenses	-	-	-	-	-	-	-	-	-
11	860	Rents	15,088	-	15,088	15,541	-	15,541	16,007	-	16,007
12		Total Operation	459,583	237,451	222,132	473,370	244,575	228,796	487,572	251,912	235,660
		Maintenance									
13	861	Maintenance Supervision and Engineering	399	-	399	411	-	411	423	-	423
14	862	Maintenance of Structures and Improvements	-	-	-	-	-	-	-	-	-
15	863	Maintenance of Mains	96,445	801	95,644	99,338	825	98,513	102,319	850	101,469
16	864	Maintenance of Compressor Station Equipment	-	-	-	-	-	-	-	-	-
17	865	Maintenance of Measuring and Regulating Sta. Equip.	15,294	-	15,294	15,753	-	15,753	16,225	-	16,225
18	866	Maintenance of Communication Equipment	-	-	-	-	-	-	-	-	-
19	867	Maintenance of Other Equipment	-	-	-	-	-	-	-	-	-
20		Total Maintenance	112,138	801	111,337	115,502	825	114,677	118,967	850	118,117
21	920-932	Administrative & General Expense	353,121	64,878	288,243	363,715	66,824	296,890	374,626	68,829	305,797
22		Total Operating Expenses	$ 924,842	$ 303,130	$ 621,712	$ 952,587	$ 312,224	$ 640,363	$ 981,165	$ 321,591	$ 659,574

ETC TIGER PIPELINE, LLC

Franklin Farms Project
Cost of Service
Depreciation Expense

Line No.	Description		YEAR 1 (a)		YEAR 2 (b)		YEAR 3 (c)
1	Gross Plant	$	137,388,622	$	137,388,622	$	137,388,622
2	Less: Non-Depreciable Assets		135,000		135,000		135,000
3	Depreciable Plant		137,253,622		137,253,622		137,253,622
4	Depreciation Rate	1/	3.00%		3.00%		3.00%
5	Annual Depreciation Expense	$	4,117,609	$	4,117,609	$	4,117,609

1/ Reflects ETC Tiger's current depreciation rate of 3.00%, as approved by the Commission in the Order Issuing Certificates dated April 7, 2010 in Docket No. CP09-460-000.

ETC TIGER PIPELINE, LLC

Franklin Farms Project
Cost of Service
Other Taxes

Line No.	Description	Detail	YEAR 1	YEAR 2	YEAR 3
		(a)	(b)	(c)	(d)
	Ad Valorem Taxes - Louisiana				
1	Gross Plant	$ 137,388,622			
2	Taxable Costs	137,388,622			
3	Assessment Ratio	25.00%			
4	Taxable Value	34,347,156			
5	Average Rate	8.296%	$ 2,849,440	$ 2,763,957	$ 2,681,038
6	Total Ad Valorem Taxes		2,849,440	2,763,957	2,681,038
	Payroll Taxes				
7	Labor		303,130	312,224	321,591
8	FICA @ 6.200%		18,794	19,358	19,939
9	Medicare @ 1.450%		4,395	4,527	4,663
10	Total Payroll Taxes		23,189	23,885	24,602
11	Total Other Taxes		$ 2,872,629	$ 2,787,842	$ 2,705,640

ETC TIGER PIPELINE, LLC

Franklin Farms Prject
Cost of Service
Working Capital

Line No.	Description	YEAR 1 (a)	YEAR 2 (b)	YEAR 3 (c)
1	Cash Working Capital	$ -	$ -	$ -
2	Materials and Supplies	-	-	-
3	Prepayments	-	-	-
4	Total Working Capital	$ -	$ -	$ -

ETC TIGER PIPELINE, LLC

Franklin Farms Project
Cost of Service
Rate Base and Return

Line No.	Description	YEAR 1 (a)	YEAR 2 (b)	YEAR 3 (c)
	Rate Base			
1	Plant	$ 137,388,622	$ 137,388,622	$ 137,388,622
2	Accumulated Provision for Depreciation	(4,117,609)	(8,235,218)	(12,352,827)
3	Net Plant	133,271,013	129,153,404	125,035,795
4	Accumulated Deferred Income Taxes	-	-	-
5	Working Capital	-	-	-
6	Total Rate Base	$ 133,271,013	$ 129,153,404	$ 125,035,795
	Return			
7	Return on Rate Base	$ 12,194,298	$ 11,817,536	$ 11,440,775
8	Rate of Return	9.15%	9.15%	9.15%

	Capitalization	As of 1/1/2026 (a)	1/ Cost (b)	Weighted Average (c)
9	- Debt	58.21%	5.67%	3.30%
10	- Equity	41.79%	14.00%	5.85%
11	Total	100.00%		9.15%

1/ Reflects the capitalization of Energy Transfer LP, the parent company of ETC Tiger.

ETC TIGER PIPELINE, LLC

Franklin Farms Project
Cost of Service
Income Taxes

Line No.	Description	YEAR 1 (a)	YEAR 2 (b)	YEAR 3 (c)
	Federal Income Tax			
1	Return on Rate Base	$ 12,194,298	$ 11,817,536	$ 11,440,775
	Federal Income Tax Adjustments			
2	Interest Expense	(4,397,943)	(4,262,062)	(4,126,181)
3	Amortization of Equity AFUDC	82,890	82,890	82,890
4	Net FIT Adjustments	(4,315,053)	(4,179,172)	(4,043,291)
5	Return after FIT Adjustments	7,879,245	7,638,364	7,397,484
6	Federal Income Taxes @ 0% (Line 5 x 0.000%)	$ -	$ -	$ -
	State Income Tax			
7	Return Before Federal Income Tax	$ 7,879,245	$ 7,638,364	$ 7,397,484
8	Composite State Tax @ 0% (Line 7 x 0.000%)	$ -	$ -	$ -

ETC TIGER PIPELINE, LLC

Franklin Farms Project

Comparison of Proposed Revenues to Cost of Service

Line No.	Description	Reference	YEAR 1 Total	YEAR 1 Reservation	YEAR 1 Usage	YEAR 2 Total	YEAR 2 Reservation	YEAR 2 Usage	YEAR 3 Total	YEAR 3 Reservation	YEAR 3 Usage
		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
1	Cost of Service		$ 20,109,378	$ 20,109,378	$ -	$ 19,675,574	$ 19,675,574	$ -	$ 19,245,189	$ 19,245,189	$ -
2	Design Determinants	1,000,000 Dt/d		365,000,000 1/	365,000,000 2/		365,000,000 1/	365,000,000 2/		365,000,000 1/	365,000,000 2/
3	Illustrative Unit Rate			$ 0.0551	$ -		$ 0.0539	$ -		$ 0.0527	$ -
	Revenues										
4	-Lateral Segment	1,000,000 Dt/d		365,000,000 1/			365,000,000 1/			365,000,000 1/	
5			Reservation Rate	$ 0.0652		Reservation Rate	$ 0.0652		Reservation Rate	$ 0.0652	
6			Reservation Revenue	23,798,000		Reservation Revenue	23,798,000		Reservation Revenue	23,798,000	
7	-ETC Tiger Location	250,000 Dt/d			91,250,000 3/			91,250,000 3/			91,250,000 3/
8				Usage Rate	$ -		Usage Rate	$ -		Usage Rate	$ -
9				Usage Revenue	-		Usage Revenue	-		Usage Revenue	-
10	-Gulf Run Location	250,000 Dt/d			91,250,000 3/			91,250,000 3/			91,250,000 3/
11				Usage Rate	$ 0.0055 4/		Usage Rate	$ 0.0055 4/		Usage Rate	$ 0.0055 4/
12				Usage Revenue	501,875		Usage Revenue	501,875		Usage Revenue	501,875
13	-Non-ETC Tiger or Gulf Run Locations	500,000 Dt/d			182,500,000 5/			182,500,000 5/			182,500,000 5/
14				Usage Rate	$ 0.0255 4/		Usage Rate	$ 0.0255 4/		Usage Rate	$ 0.0255 4/
15				Usage Revenue	4,653,750		Usage Revenue	4,653,750		Usage Revenue	4,653,750
16				Total Usage Revenue	5,155,625		Total Usage Revenue	5,155,625		Total Usage Revenue	5,155,625
17	Total Lateral Segment		$ 28,953,625	$ 23,798,000	$ 5,155,625	$ 28,953,625	$ 23,798,000	$ 5,155,625	$ 28,953,625	$ 23,798,000	$ 5,155,625
18	Excess Revenues over Cost of Service		$ 8,844,247			$ 9,278,051			$ 9,708,436		

1/ 1,000,000 x 365.
2/ Assumes 100% Load Factor. 1,000,000 x 365.
3/ Assumes 100% Load Factor. 250,000 x 365.
4/ Fixed usage rate of $0.0055 from ETC Tiger or Gulf Run locations. Fixed usage rate of $0.0255 from non-ETC Tiger or Gulf Run locations.
5/ Assumes 100% Load Factor. 500,000 x 365.

ETC TIGER PIPELINE, LLC

Franklin Farms Project
Compressor Fuel Projection
(MMBtu)

Line No.	Description	Existing (a)	Post Construction (b)	
	Base System - Mainline Segment:			
1	Total Compressor Fuel	4,410,866	4,837,809	
2	Total Receipt Quantities	720,561,879	811,811,879	2/
3	Base System Fuel % 3/	0.61%	0.60%	
	Lateral Segment			
4	Total Compressor Fuel 4/		-	
5	Total Receipt Quantities		365,000,000	5/
6	Lateral Segment Fuel % 3/		0.00%	

1/ Represents data from ETC Tiger's latest two Fuel Reimbursement Adjustment Filing in Docket No. RP26-121-000 and RP26-825-000.
2/ Includes 250,000 Dt/d on ETC Tiger's mainline.
3/ Excludes Lost and Unaccounted For Gas.
4/ The new lateral has no compression facilities.
5/ 1,000,000 x 365.

Exhibit Z-1

ETC Tiger Pipeline, LLC

Form of Protective Agreement

PROTECTIVE AGREEMENT

This Protective Agreement ("Protective Agreement"), is made and entered into as of _____, 202___ ("Effective Date"), by and between ETC Tiger Pipeline, LLC, a Delaware limited liability company ("Applicant") and _____ ("Participant," and together with Applicant, the "Parties").

WHEREAS, a certificate application regarding the proposed Franklin Farms Project is currently pending before the Federal Energy Regulatory Commission ("Commission") in Docket No. CP26-___-000 ("Certificate Proceeding"); and

WHEREAS, pursuant to Section 388.112(b) of the Commission's regulations, 18 C.F.R. § 388.112(b), this Protective Agreement will apply to any document provided by Applicant to Participant pursuant to this Protective Agreement that Applicant filed as privileged/proprietary or as Critical Energy Infrastructure Information ("CEII") in the Certificate Proceeding; and

WHEREAS, Participant is submitting this Protective Agreement as part of its request pursuant to 18 C.F.R. § 388.112(b)(iii) for a complete, non-public version of _____ included in the Commission's eLibrary under Accession No(s). _____ ("Protected Materials") subject to the terms of this Protective Agreement; and

NOW, THEREFORE, the Parties agree as follows:

1. This Protective Agreement shall govern the use of all Protected Materials produced by, or on behalf of, Applicant to Participant hereunder. Notwithstanding any order of the Commission approving the certificate application in the Certificate Proceeding, this Protective Agreement shall remain in effect until specifically modified or terminated by (i) written agreement of the Parties or (ii) the Commission.

2. This Protective Agreement applies to the Protected Materials, as defined in the recitals to this Protective Agreement. Notwithstanding anything herein to the contrary, Applicant's delivery of Protected Materials to Participant pursuant to this Protective Agreement shall not affect the Protected Materials' protected status under the Commission's regulations and Freedom of Information Act, 5 U.S.C. 552 ("FOIA").

3. Definitions -- For purposes of this Protective Agreement:

   (a) The term "Notes of Protected Materials" means memoranda, handwritten notes, or any other form of information (including electronic form) which copies or discloses materials described in the Protected Materials. Except as specifically provided in this Protective Agreement, Notes of Protected Materials are subject to the same terms and restrictions as the Protected Materials under this Protective Agreement.

   (b) The term "Non-Disclosure Certificate" shall mean the certificate annexed to this Protective Agreement by which a Reviewing Representative of the Participant who has been granted access to Protected Materials shall certify its understanding that

such access to Protected Materials is provided pursuant to the terms and restrictions of this Protective Agreement, and that such Reviewing Representative has read this Protective Agreement and agrees to be bound by it.

(c) The term "Non-Disclosure Certificate of Marketing/Competitive Duty Personnel" shall mean the attached certificate by which a representative of Participant who is a Marketing/Competitive Duty Personnel pursuant to Paragraph 21 acknowledge that it is prohibited from reviewing Protected Materials that are marked "Not Available to Marketing/Competitive Duty Personnel" pursuant to Paragraph 21, shall certify its agreement that it shall neither have access to, nor disclose, the contents of the Protected Materials that are marked "Not Available to Marketing/Competitive Duty Personnel," any notes or other memoranda, or any other form of information that copies or discloses Protected Materials that are marked as "Not Available to Marketing/Competitive Duty Personnel," and that such Reviewing Representative has read this Protective Agreement and agrees to be bound by it.

(d) The term "Reviewing Representative" shall mean a person who has signed a Non-Disclosure Certificate or a Non-Disclosure Certificate for Marketing/Competitive Duty Personnel, as applicable, and who is: (i) an attorney who has made an appearance in the Certificate Proceeding for Participant; (ii) attorneys, paralegals, and other employees associated for purposes of the Certificate Proceeding with an attorney described in Paragraph 3(d)(i); (iii) an expert or an employee of an expert retained by Participant for the purpose of advising, preparing for or testifying in the Certificate Proceeding; or (iv) employees or other representatives of Participant appearing in the Certificate Proceeding with significant responsibility for this docket.

4. Protected Materials shall be made available under the terms of this Protective Agreement only to Participant's Reviewing Representatives; provided that if the Protective Materials include rates, rate-related provisions, credit support provisions or other highly sensitive information, Applicant may redact the rates, rate-related provisions, credit support provisions or other highly sensitive information from the version of the Protected Materials provided to Participant's Reviewing Representatives. In the event that Applicant redacts any such information, if requested by Participant the Parties shall meet to discuss the terms and conditions under which one or more of Participant's Reviewing Representatives may be provided such redacted information. If no agreement is reached, Participant may submit such dispute to the Commission for resolution.

5. Protected Materials shall remain available to Participant until the later of the date that an order approving the certificate application in the Certificate Proceeding becomes no longer subject to judicial review, or the date that any other Commission proceeding relating to the Protected Materials is concluded and no longer subject to judicial review. If requested to do so in writing after that date, Participant shall, within fifteen (15) days of such request, return the Protected Materials (excluding Notes of Protected Materials) to Applicant, or shall destroy the materials, except that copies of filings, official transcripts and exhibits in the Certificate Proceeding that contain Protected Materials and Notes of Protected Materials may be retained, if they are maintained in accordance with Paragraph 6 below. Within such time period, Participant, if

requested to do so, shall also submit to Applicant an affidavit stating that, to the best of its knowledge, all Protected Materials, including those materials designated "Not Available to Marketing/Competitive Duty Personnel," and all Notes of Protected Materials have been returned or have been destroyed or will be maintained in accordance with Paragraph 6. To the extent Protected Materials are not returned or destroyed, they shall remain subject to this Protective Agreement.

6. All Protected Materials shall be maintained by Participant in a secure place. Access to those materials shall be limited to those Reviewing Representatives specifically authorized pursuant to this Protective Agreement. Participant shall exercise at least the same degree of care with respect to the Protected Materials as Participant uses in handling its own confidential information.

7. Protected Materials shall be treated as confidential by Participant and by the Reviewing Representative in accordance with the Non-Disclosure Certificate or Non-Disclosure Certificate for Marketing/Competitive Duty Personnel executed pursuant to Paragraph 9. Protected Materials shall not be used by Participant or a Reviewing Party except as necessary for the conduct of the Certificate Proceeding, nor shall they be disclosed in any manner to any person except a Reviewing Representative of Participant who is engaged in the conduct of the Certificate Proceeding and who needs to know the information in order to carry out that person's responsibilities in the Certificate Proceeding. Reviewing Representatives may make copies of Protected Materials, but such copies become Protected Materials. Reviewing Representatives may make notes of Protected Materials, which shall be treated as Notes of Protected Materials if they disclose the contents of Protected Materials.

8. (a) A Reviewing Representative may not use information contained in any Protected Materials obtained through the Certificate Proceeding to give Participant, any customer or potential customer of Applicant or any competitor of Applicant a commercial advantage or for any other purpose other than the prosecution or defense of the proceedings conducted under the Certificate Proceeding before the Commission.

(b) Subject to the provisions of Paragraph 21 regarding access to Protected Materials that are Not Available to Marketing/Competitive Duty Personnel, in the event that Participant wishes to designate as a Reviewing Representative a person not described in Paragraph 3(d) above, Participant shall seek agreement from Applicant. If an agreement is reached, that person shall be a Reviewing Representative pursuant to Paragraph 3(d) above with respect to those materials. If no agreement is reached, Participant may submit the disputed designation to the Commission for resolution.

9. (a) A Reviewing Representative shall not be permitted to inspect, participate in discussions regarding, or otherwise be permitted access to Protected Materials pursuant to this Protective Agreement unless that Reviewing Representative has first executed (i) a Non-Disclosure Certificate or (ii) a Non-Disclosure Certificate for Marketing/Competitive Duty Personnel, as applicable; provided, that if an attorney qualified as a Reviewing Representative has executed such a certificate, the paralegals, secretarial and clerical personnel employed by the same entity as the attorney and under the attorney's instruction, supervision or control need not do so. A copy of each Non-Disclosure Certificate and Non-Disclosure Certificate for

Marketing/Competitive Duty Personnel shall be provided to counsel for Applicant prior to disclosure of any Protected Material to that Reviewing Representative.

(b) Attorneys qualified as Reviewing Representatives are responsible for ensuring that persons under their instruction, supervision or control comply with this Protective Agreement.

10. Subject to Paragraph 4 above and to the provisions of Paragraph 21 regarding access to Protected Materials that are Not Available to Marketing/Competitive Duty Personnel, any Reviewing Representative may disclose Protected Materials to any other Reviewing Representative of Participant as long as the disclosing Reviewing Representative and the receiving Reviewing Representative both have executed a Non-Disclosure Certificate. In the event that any Reviewing Representative to whom the Protected Materials are disclosed ceases to be engaged in the Certificate Proceeding, or is employed or retained for a position whose occupant is not qualified to be a Reviewing Representative under Paragraph 3(d), access to Protected Materials by that person shall be terminated. Even if no longer engaged in the Certificate Proceeding, every person who has executed a Non-Disclosure Certificate shall continue to be bound by the provisions of this Protective Agreement and the certification.

11. Subject to Paragraph 16, the Commission shall resolve any disputes arising under this Protective Agreement. Prior to presenting any dispute under this Protective Agreement to the Commission, the Parties shall use their best efforts to resolve it.

12. All copies of all documents reflecting Protected Materials, including the portion of any pleadings, motions, exhibits and other documents that refer to Protected Materials, shall be filed and served in sealed envelopes or other appropriate containers endorsed to the effect that they are sealed pursuant to this Protective Agreement. Such documents shall be marked "PROTECTED MATERIALS" and shall be filed under seal and served under seal upon the Commission and the other Party. Any such documents containing Critical Energy Infrastructure Information shall be additionally marked "Critical Energy Infrastructure Information – Do Not Release", and any such documents containing Privileged Information shall be additionally marked "Privileged Information – Do Not Release", and any such documents containing Proprietary Business Information shall be additionally marked "Proprietary Business Information – Do Not Release". If not done so previously, for anything filed under seal, redacted versions or, where an entire document is protected, a letter indicating such, will also be filed with the Commission and served on the other Party. Counsel shall take all precautions necessary to assure that Protected Materials are not distributed to unauthorized persons.

13. If Participant desires to include, utilize or refer to any Protected Materials or information derived therefrom in any submission during the Certificate Proceeding in such a manner that might require disclosure of such material to other participants in the Certificate Proceeding, Participant shall first notify counsel for Applicant and the Commission of such desire, identifying with particularity each of the Protected Materials. Thereafter, use of such Protected Materials will be governed by procedures determined by the Commission.

14. Nothing in this Protective Agreement shall be construed as precluding Applicant from objecting to the use of Protected Materials on any legal grounds.

15. Nothing in this Protective Agreement shall preclude Participant from requesting the Commission or any other body having appropriate authority, to find that this Protective Agreement should not apply to all or any materials previously designated as Protected Materials pursuant to this Protective Agreement.

16. If the Commission finds at any time in the course of the Certificate Proceeding that all or part of the Protected Materials need not be protected, those materials shall, nevertheless, be subject to the protection afforded by this Protective Agreement for seven (7) business days from the date of issuance of the Commission's determination. Applicant has not waived its rights to seek additional administrative or judicial remedies after any decision respecting Protected Materials or Reviewing Representatives. The provisions of 18 C.F.R. §§ 388.112 and 388.113 shall apply to any requests under the FOIA for Protected Materials in the files of the Commission.

17. Nothing in this Protective Agreement shall be deemed to preclude either Party from independently seeking through discovery in any other administrative or judicial proceeding information or materials produced in the Certificate Proceeding under this Protective Agreement.

18. Applicant does not waive its right to pursue any other legal or equitable remedies that may be available in the event of actual or anticipated disclosure of Protected Materials. Participant agrees that remedies at law may be inadequate in the event of actual or anticipated breach of this Protective Agreement and that Applicant may seek injunctive relief to prevent the actual or anticipated breach without proof of actual damages and Participant shall not unreasonably oppose any such request for injunctive relief.

19. Participant shall not disclose the contents of Protected Materials or any other form of information that copies or discloses Protected Materials to anyone other than in accordance with this Protective Agreement and shall only use such contents and information in connection with the Certificate Proceeding. Any violation of this Protective Agreement and of any Non-Disclosure Certificate executed hereunder shall constitute a breach of this Protective Agreement.

20. In addition to the provisions of Paragraph 4, Applicant may physically mark with the words "Not Available to Marketing/Competitive Duty Personnel" any Protected Materials that Applicant believes in good faith would, if disclosed to Reviewing Representatives without added precautions beyond Paragraphs 1-19 of this Protective Agreement, subject Applicant, or third party(ies), to undue risk of competitive disadvantage or business injury. Such information may include, but is not limited to (a) non-public business development, acquisition, or marketing data, pricing strategies, plans or activities; (b) non-public financial data, or strategic business or financial plans or activities; or (c) negotiations of services, prices and rates, the public disclosure of which Applicant in good faith believes would competitively harm Applicant or one or more third parties (hereinafter "Market Sensitive Information"). Any challenge to such a designation may be made as provided in this Protective Agreement for challenges to designations of Protected Materials.

21. (a) Solely with respect to Protected Materials that have been marked "Not Available to Marketing/Competitive Duty Personnel" and information derived therefrom, and subject to the other terms of this Paragraph 21, a Reviewing Representative shall not be any employee or agent of a receiving Participant whose duties include: (1) the marketing, sale or purchase of natural gas or natural gas transportation or storage services or facilities, or management or supervisory

responsibility thereof; or (2) the development, planning, construction, marketing or operation of a pipeline, or a prospective pipeline, that competes or may compete with Applicant, or management or supervisory responsibility thereof; or (3) the provision of consulting services related to the development, planning, construction, marketing or operation of a pipeline or a prospective pipeline, or the acquisition of or marketing of pipeline capacity that currently competes, or may potentially compete, with Applicant regarding the provision of natural gas transportation or storage services; or (4) management responsibility regarding other business activities in which use of Market Sensitive Information could be reasonably expected to cause competitive harm to Applicant (collectively, "Marketing/Competitive Duties" and persons whose duties or responsibilities include Marketing/Competitive Duties are referred to herein as "Marketing/Competitive Duty Personnel").

(b) Notwithstanding the foregoing, with respect to Privileged Materials that have been marked "Not Available to Marketing/Competitive Duty Personnel" and information derived therefrom, a Reviewing Representative may not be an employee of a Commission-regulated natural gas pipeline, storage facility, or LNG terminal that competes with Applicant. Reviewing Representatives of such a pipeline, storage facility, or LNG terminal, with respect to Privileged Materials that have been marked "Not Available to Marketing/Competitive Duty Personnel", shall be limited to outside counsel and/or consultants, provided such individuals are not engaged in Marketing/Competitive Duties on behalf of such pipeline, storage facility, or LNG terminal.

(c) Notwithstanding the foregoing, a person who otherwise would be disqualified as Marketing/Competitive Duty Personnel may serve as a Reviewing Representative upon agreement of Applicant, in the absence of such agreement, upon entry of an order of the Commission authorizing such person to serve as a Reviewing Representative. Any request for an agreement or order under the preceding sentence shall be subject to the following conditions: (i) the Participant who employs or has retained the person in question must certify in writing to Applicant that the certifying Participant's ability to participate effectively in the Certificate Proceeding would be prejudiced if it was unable to rely on the assistance of the particular Reviewing Representative; (ii) the Participant claiming such prejudice must identify by name and job title the particular Reviewing Representative required and must describe the person's duties and responsibilities for the requesting Participant; (iii) the Participant claiming such prejudice must acknowledge in writing to Applicant that access to the Protected Materials that are Not Available to Marketing/Competitive Duty Personnel shall be restricted only to such access necessary for its participation in the Certificate Proceeding, absent prior written consent of Applicant or authorization of the Commission with opportunity for Applicant to seek review of such decision as provided in this Protective Agreement; (iv) such Participant must acknowledge in writing that any other use of Protected Materials that are Not Available to Marketing/Competitive Duty Personnel shall constitute a violation of this Protective Agreement; and (v) prior to having access to any Protected Materials that are Not Available to Marketing/Competitive Duty Personnel, the Marketing/Competitive Duty Personnel who is authorized to act as a Reviewing Representative must execute and deliver to Applicant a Non-Disclosure Certificate of Marketing/Competitive Duty Personnel acknowledging his or her familiarity with the contents of this Protective Agreement and the particular restrictions set forth in this paragraph regarding such Protected Materials. Such agreement by Applicant referenced above in this paragraph shall not be unreasonably withheld, delayed or conditioned. Provided that Protected Materials are clearly and correctly labeled, compliance with this Protective Agreement shall be the responsibility of the

receiving Participant. Materials marked as "Not Available to Marketing/Competitive Duty Personnel" shall be returned or destroyed at the conclusion of the Certificate Proceeding as otherwise provided in this Protective Agreement.

(d) Notwithstanding the above, in-house and outside counsel for Participant and experts who are not Marketing/Competitive Duty Personnel may serve as a Reviewing Representative of Market Sensitive Information, provided such persons abide by the restrictions in this Protective Agreement and do not disclose such Market Sensitive Information to Marketing/Competitive Duty Personnel. Subject to the preceding sentence, in the event that any person who has been a Reviewing Representative subsequently is assigned to perform any Marketing/Competitive Duties, such Reviewing Representative immediately shall cease accessing Protected Materials that are marked "Not Available to Marketing/Competitive Duty Personnel" or information derived therefrom and shall immediately dispose of any Protected Materials in his/her possession that are marked "Not Available to Marketing/Competitive Duty Personnel" or information derived therefrom and shall continue to comply with the requirements of the Non-Disclosure Certificate and this Protective Agreement with respect to any Protected Materials to which such person previously had access.

IN WITNESS WHEREOF, the Parties hereto have caused this Protective Agreement to be duly executed in several counterparts by their proper officers duly authorized as of the Effective Date.

ETC Tiger Pipeline, LLC

By _____

Title _____

_____

By_____

Title _____

NON-DISCLOSURE CERTIFICATE

I hereby certify my understanding that access to Protected Materials is provided to me pursuant to the terms and restrictions of the Protective Agreement between ETC Tiger Pipeline, LLC and _____ dated _____, 202__ ("Protective Agreement"), that I have been given a copy of and have read, and that I agree to be bound by it. I certify that my duties and responsibilities do not include "Marketing/Competitive Duties" as described in Paragraph 21 of the Protective Agreement. I understand that the contents of the Protected Materials, any notes or other memoranda, or any other form of information that copies or discloses Protected Materials shall not be disclosed to anyone other than in accordance with the Protective Agreement. I recognize that material marked "Not Available to Marketing/Competitive Duty Personnel" or the contents thereof may not be disclosed to those with duties and responsibilities that include "Marketing/Competitive Duties". I acknowledge that a violation of this certificate constitutes a breach of the Protective Agreement.

By: _____

Printed Name: _____

Title: _____

Representing: _____

Date: _____

NON-DISCLOSURE CERTIFICATE OF MARKETING/COMPETITIVE DUTY PERSONNEL

I hereby certify my understanding that access to Protected Materials is provided to me pursuant to the terms and restrictions of the Protective Agreement between ETC Tiger Pipeline, LLC and _____ dated _____, 202__, ("Protective Agreement"), that I have been given a copy of and have read, and that I agree to be bound by it. I acknowledge that my duties and responsibilities include "Marketing/Competitive Duties" as described in Paragraph 21 of the Protective Agreement and, as such, I understand that I shall neither have access to, nor disclose, the contents of the Protected Materials that are marked "Not Available to Marketing/Competitive Duty Personnel," or any notes, other memoranda, or any other form of information that copies or discloses Protected Materials that are marked "Not Available to Marketing/Competitive Duty Personnel." As for Protected Materials that I am permitted to have access to under the Protective Agreement, I understand that the contents of such Protected Materials, any notes or other memoranda, or any other form of information that copies or discloses such Protected Materials shall not be disclosed to anyone other than in accordance with the Protective Agreement. I acknowledge that a violation of this certificate constitutes a breach of the Protective Agreement.

By: _____

Printed Name: _____

Title: _____

Representing: _____

Date: _____